पेटेंट कार्यालय
शासकीय जर्नल

OFFICIAL JOURNAL
OF
THE PATENT OFFICE

निर्गमन सं. 45/2026	शुक्रवार	दिनांक: 06/11/2026
ISSUE NO. 45/2026	FRIDAY	DATE: 06/11/2026

पेटेंट कार्यालय का एक प्रकाशन
PUBLICATION OF THE PATENT OFFICE

INTRODUCTION

In view of the recent amendment made in the Patents Act, 1970 by the Patents (Amendment) Act, 2005 effective from 01st January 2005, the Official Journal of The Patent Office is required to be published under the Statute. This Journal is being published on weekly basis on every Friday covering the various proceedings on Patents as required according to the provision of Section 145 of the Patents Act 1970. All the enquiries on this Official Journal and other information as required by the public should be addressed to the Controller General of Patents, Designs & Trade Marks. Suggestions and comments are requested from all quarters so that the content can be enriched.

(PROF. (DR) UNNAT P. PANDIT)
CONTROLLER GENERAL OF PATENTS, DESIGNS & TRADE MARKS

06th November, 2026

CONTENTS

SUBJECT		*PAGE NUMBER*
JURISDICTION	**:**	**142605-142606**
SPECIAL NOTICE	**:**	**142607-142608**
EARLY PUBLICATION (DELHI)	**:**	**142609-142713**
EARLY PUBLICATION (MUMBAI)	**:**	**142714-142906**
EARLY PUBLICATION (CHENNAI)	**:**	**142907-144322**
EARLY PUBLICATION (KOLKATA)	**:**	**144323-144365**
PUBLICATION AFTER 18 MONTHS (DELHI)	**:**	**144366-144517**
PUBLICATION AFTER 18 MONTHS (MUMBAI)	**:**	**144518-144646**
PUBLICATION AFTER 18 MONTHS (CHENNAI)	**:**	**144647-144926**
PUBLICATION AFTER 18 MONTHS (KOLKATA)	**:**	**144927-144944**
WEEKLY ISSUED FER (DELHI)	**:**	**144945-145119**
WEEKLY ISSUED FER (MUMBAI)	**:**	**145120-145177**
WEEKLY ISSUED FER (CHENNAI)	**:**	**145178-145319**
WEEKLY ISSUED FER (KOLKATA)	**:**	**145320-145337**
PUBLICATION UNDER SECTION 43(2) IN RESPECT OF THE GRANT (DELHI)	**:**	**145338-145371**
PUBLICATION UNDER SECTION 43(2) IN RESPECT OF THE GRANT (MUMBAI)	**:**	**145372-145390**
PUBLICATION UNDER SECTION 43(2) IN RESPECT OF THE GRANT (CHENNAI)	**:**	**145391-145421**
PUBLICATION UNDER SECTION 43(2) IN RESPECT OF THE GRANT (KOLKATA)	**:**	**145422-145428**
PUBLICATION U/S.60 IN RESPECT OF APPLICATION FOR RESTORATION OF PATENT (KOLKATA)	**:**	**145429**
INTRODUCTION TO DESIGN PUBLICATION	**:**	**145430**
DESIGN CORRIGENDUM	**:**	**145431**
REGISTRATION OF DESIGNS	**:**	**145432-145865**

THE PATENT OFFICE
KOLKATA, 06/11/2026
Address of the Patent Offices/Jurisdictions
The following are addresses of all the Patent Offices located at different places having their Territorial Jurisdiction on a Zonal basis as shown below:-

<table>
<tr><td>A</td><td>Office of the Controller General of Patents,
Designs & Trade Marks,
Bouddhik Sampada Bhawan,
Plot No. 32, Sector 14, Dwarka,
New Delhi-110078

Phone: (+91)(11) 2803 4301, 2803 4302
E-mail: cgoffice.in@gov.in</td><td>3</td><td>The Patent Office,
Government of India,
Boudhik Sampada Bhavan,
CP-2, Sector –V, Salt Lake City,
Kolkata- 700 091

Phone: (+91)(33) 2367 1944/87 , 2367 9101
E-Mail: kolkata-patent@nic.in

❖ Rest of India</td></tr>
<tr><td>1</td><td>The Patent Office,
Government of India,
Boudhik Sampada Bhavan,
Plot No. 32., Sector-14, Dwarka,
New Delhi – 110078

Phone: (+91)(11) 2803 4315, 1800 110-989
E.mail: delhi-patent@nic.in

❖ The States of Delhi, Haryana, Himachal Pradesh, Punjab, Rajasthan, Uttar Pradesh and Uttarakhand and the Union Territory of Chandigarh, Jammu and Kashmir and Ladakh.</td><td rowspan="2">4</td><td rowspan="2">The Patent Office,
Government of India,
Intellectual Property Rights Building,
G.S.T. Road, Guindy,
Chennai – 600 032.

Phone: (+91)(44) 2250 2039
E-mail: chennai-patent@nic.in

❖ The States of Andhra Pradesh, Karnataka, Kerala, Tamil Nadu and Telangana and the Union Territories of Lakshadweep and Pondicherry.</td></tr>
<tr><td>2</td><td>The Patent Office,
Government of India,
Boudhik Sampada Bhavan,
Near Antop Hill Post Office,S.M.Road,Antop Hill,
Mumbai – 400 037

Phone: (+91)(22) 2415 9192, 2414 1026, 2415 3651
E-mail: mumbai-patent@nic.in

❖ The States of Chhattisgarh, Goa, Gujarat, Maharashtra and Madhya Pradesh and the Union Territories of Daman and Diu and Dadra and Nagar Haveli.</td></tr>
</table>

Website: www.ipindia.gov.in

All applications, notices, statements or other documents or any fees required by the Patents Act, 1970 (as amended) or by the Patents (Amendment) Rules, 2003 (as amended) will be received either online or offline at the appropriate offices of the Patent Office.

Fees: The Fees may either be paid in cash or may be sent by Bank Draft or Banker's Cheques payable to the Controller of Patents drawn on a scheduled Bank at the place where the appropriate office is situated.

पेटेंट कार्यालय
कोलकाता, दिनांक 06/11/2026

• कार्यालयों के क्षेत्राधिकार के पते

विभिन्न जगहों पर स्थित पेटेंट कार्यालय के पते आंचलिक आधार पर दर्शित उनके प्रादेशिक अधिकार क्षेत्र के साथ नीचे दिए गए है:-

A	कार्यालय : महानियंत्रक, एकस्व, अभिकल्प तथा व्यापार चिहन, बौद्धिक संपदा भवन, प्लॉट सं. 32, सेक्टर- 14, द्वारका, नई दिल्ली- 110 078, भारत, फोन: (+91) (11) 2803 4301, 2803 4302 ई. मेल: cgoffice.in@gov.in	3	पेटेंट कार्यालय, भारत सरकार कोलकाता, (प्रधान कार्यालय) बौद्धिक संपदा भवन, सीपी-2, सेक्टर- v, साल्ट लेक सिटी, कोलकाता-700 091, भारत. फोन: (+91)(33) 2367 1944/87, 2367 9101 ई. मेल: kolkata-patent@nic.in ❖ भारत का अवशेष क्षेत्र
1	पेटेंट कार्यालय, भारत सरकार बौद्धिक संपदा भवन, प्लॉट सं. 32, सेक्टर- 14, द्वारका, नई दिल्ली- 110 078. फोन: (+91)(11) 2803 4315, 1800 110 989 ई. मेल: delhi-patent@nic.in ❖ दिल्ली, हरियाणा, हिमाचल प्रदेश, पंजाब, राजस्थान, उत्तर प्रदेश तथा उत्तराखंड राज्य क्षेत्रों एवं संघ शासित क्षेत्र चंडीगढ़, जम्मू तथा कश्मीर तथा लद्दाख	4	पेटेंट कार्यालय, भारत सरकार इंटेलेक्चुअल प्रॉपर्टी राइट्स बिल्डिंग, इंडस्ट्रियल इस्टेट एसआईडीसीओ आरएमडी गोडाउन एरिया एडजसेन्ट टु ईगल फ्लास्क, जी. एस. टी. रोड, गायन्डी चेन्नई - 600 032. फोन: (+91)(44) 2250 2039 ई. मेल: chennai-patent@nic.in आन्ध्र प्रदेश, कर्नाटक, केरल, तमिलनाडु तथा तेलंगाना राज्य क्षेत्र एवं संघ शासित क्षेत्र लक्षदीप तथा पुडुचेरी
2	पेटेंट कार्यालय, भारत सरकार बौद्धिक संपदा भवन, एंटोप हिल डाकघर के समीप, एस. एम. रोड, एंटोप हिल, मुम्बई- 400 037, फोन: (+91) (22) 2415 9192, 2414 1026, 2415 3651 ई. मेल: mumbai-patent@nic.in छत्तीसगढ़ , गोवा, गुजरात, महाराष्ट्र तथा मध्य प्रदेश राज्य क्षेत्र एवं संघ शासित क्षेत्र दमन तथा दीव, दादरा और नगर हवेली.		

वेबसाइट: www.ipindia.gov.in

पेटेंट अधिनियम, 1970 (यथा संशोधित) अथवा पेटेंट (संशोधन) नियम, 2003 (यथा संशोधित) द्वारा वांछित सभी आवेदन, सूचनाए, विवरण या अन्य दस्तावेज़ या कोई शुल्क ऑनलाइन या ऑफ़लाइन पेटेंट कार्यालय के केवल उपयुक्त कार्यालय में स्वीकृत होंगे।
शुल्क: शुल्क या तो नगद रूप में या Controller of Patents के नाम में देय बैंक ड्राफ्ट या बैंकर का चेक के द्वारा भेजी जा सकती है जो उसी स्थान के किसी अनुसूचित बैंक में प्रदत्त हो जहाँ उपयुक्त कार्यालय स्थित है ।

SPECIAL NOTICE

18 Months publication as required under Section 11A of the Patents Act, 1970 as amended by the Patents (Amendment) Act, 2005.

Notice is hereby given that any person at any time before the grant of Patent may give representation by way of opposition to the Controller of Patents at appropriate office on the ground and in a manner specified under section 25(1) of the Patents (Amendment) Act, 2005 read with Rule 55 of the Patents (Amendment) Rules, 2006.

Notice is also given that if any interested person requests for copies of the complete specification, drawing and abstract of any application already published, the photocopy of the same can be supplied by the Patent Office as per the jurisdiction on payment of prescribed fees of Rs.10/- per page. If any further details are required to be obtained, the same can be provided by the respective Patent Offices on request.

(PROF. (DR) UNNAT P. PANDIT)
CONTROLLER GENERAL OF PATENTS, DESIGNS & TRADE MARKS

SPECIAL NOTICE

Under the new provision of the Patents Act, 1970 as amended by the Patents (Amendment) Act, 2005 and Rules there under, Publication of the matter relating to Patents in the Official Gazette of India Part III, Section 2 has been discontinued and instead The Official Journal of the Patent Office is being published containing all the activities of The Patent Office such as publication of all the patent applications after 18th months , grant of patents & all other information in respect of the proceedings as required under the provisions of the Patents (Amendment) Act, 2005 and Rules thereunder on weekly basis on every **Friday**.

The Journal is uploaded in the website every Friday. So Paper form and CD-ROM form of the Journal are discontinued from 01/01/2009.

SPECIAL NOTICE

Every effort is being taken to publish all the patent applications under section 11(A) of the Patents Act. However, if duplication of publication of any application is found, then earlier date of publication will be taken for the purpose of provisional protection for applicant and Patent Office will grant Patent not before six months from the date of second publication, provided that there is there is no third party representation.

Early Publication:

The following patent applications have been published under section 11A (2) of The Patents (Amendment) Act 2005 and rule 24A of The Patents (Amendment) Rules, 2006. Any person may file representation by way of opposition to the Controller of Patents at the appropriate office against the grant of the patent in the prescribed manner under section 25(1) of the Patents (Amendment) Act 2005 read with the rule 55 of The Patents (Amendment) Rules, 2006:

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202611007970 A

(19) INDIA

(22) Date of filing of Application :27/01/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : GRAIN DISPENSING AND PRESERVATION DEVICE

(51) International classification	:A01F 25/14, A01M 1/20, A01M 7/00, A01M 9/00, A01M 1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Aafrin Gouri

(57) Abstract :
A grain dispensing and preservation device comprises of a circular disc 101 developed to be installed on a surface with a plurality of suction cups 109, a circular slider 103 allowing rotation about a central axis, a circular flap 105 via a link 106 holds a plurality of compartment 104 for storing different types of grains, a sliding plate 108 for dispensing a grain quantity, a camera 115 to capture real-time images for insect detection, a speaker 107 notifies the user if the bowl is improperly positioned, a pair of guiding panels 112 is arranged via hinges 113 on opposing sides of each opening to guide grains during manual filling, an imaging unit 114 detects user approach and unfolds the guiding panels 112, and a motorized sliding door 111 to release an antibacterial packet from a storage chamber 110.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007971 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : TRASH RETRIEVAL AND MANAGEMENT ASSISTIVE DEVICE FOR CLEAN AND HYGIENIC SURFACES

(51) International classification	:E01H 1/12, B65F 1/14, B65F 1/00, B25J 13/08, E01H 1/04	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Muhammad Mubashshir**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A trash retrieval and management assistive device for clean and hygienic surfaces, comprising a body 101 with a handle 102 for manipulation and control of body 101, an input means 103 paired with body 101 for enabling a user to provide input commands, an imaging unit 104 mounted on body 101 to determine position of waste, and classify waste by type, size, shape and color, a gripping means mounted on body 101 via an extendable rod 107 to position gripping means in proximity to detected waste, a twelve bar linkage assembly 108 mounted at end of rod to collect and dispose waste, an odor sensor and a hyperspectral imaging unit 109 disposed on body 101 to detect hazardous or infected debris, a peristaltic-based sprayer unit installed on body 101 to sanitize contaminated areas, and a cleaning module integrated with gripping means for cleaning and drying surface after picking waste.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007972 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : POSTURE CORRECTION DEVICE

(51) International classification	:A61F 5/01, A61B 5/11, A61H 1/02, A61H 3/00, A61F 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Kailash Agarwal

(57) Abstract :
A posture correction device comprises of an elongated member 101 comprising telescopic segments linked via motorized hinge 102 is adapted to be positioned on a user's back portion, a pair of securing means 103, including a strap, an electromagnet 104, and a magnetically responsive element 105, is attached to the upper and lower body portions to secure around the chest and waist, an imaging unit 106 detects the user's back length, prompting the microcontroller to actuate telescopic segments for a secure fit, a motorized hinge 102 to maintain spinal alignment and a spray unit 107 to mist water on the user's face and an extendable rod with an LED(Light Emitting Diode) 109 to illuminate the workspace, the extendable rod uses hydraulic, pneumatic, or electromechanical actuators.

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007973 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TAP MANAGEMENT DEVICE FOR ELDERLY

(51) International classification	:E03C 1/04, E03C 1/05, A47K 1/08, E03C 1/182, A47K 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Kajal Thakuriya

(57) Abstract :
A tap management device for elderly, comprising a platform 101 with a drawer arrangement 102 for size adjustment and positioning on wash basin; an input on platform 101 for user input commands, an imaging unit 104 mounted on platform 101 to determine basin dimensions and detect dirt on user's hands, a securing module 105 on platform 101 with extendable clamping arms and motorized actuators with tactile sensors for securing platform 101, a lever handling unit mounted on platform 101 including a hemispherical member 106 with extendable rod 107 and a twelve-bar linkage arrangement 108 to control water flow; a cleaning solution dispensing module 109 comprising a vessel, an electronic sprayer, an extendable conduit for dispensing solution; a brushing module with a motorized brush 110 with extendable link for cleaning user's hand; and an expandable ring member 112 with extendable pole 113 to distribute and regulate water flow.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/12/2025

(21) Application No.202511121100 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUAL BAND BANDPASS FILTER WITH OPEN-ENDED STUB INTEGRATION FOR PASSBAND ENHANCEMENT

(51) International classification	:H01P 1/203, H01P 7/08, H01P 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NATIONAL INSTITUTE OF TECHNOLOGY-DELHI
Address of Applicant :Plot No. FA7, Zone P1, GT Karnal Road, Delhi-110036, India Delhi Delhi India
(72)**Name of Inventor :**
1)DHARMENDRA KUMAR JHARIYA
2)AMIT KUMAR
3)HARKARANDEEP SINGH
4)HEMANT KUMAR VARSHNEY

(57) Abstract :
A compact dual-band bandpass filter has been proposed in the present invention that provides two distinct passbands with high frequency selectivity, low insertion loss, and improved isolation as shown in figure-1. The filter aims to achieve miniaturization while maintaining stable performance, making it suitable for applications in GNSS, 5G, WLAN, radar, and IoT systems. Additionally, the invention seeks to enhance stopband rejection through the use of step-impedance resonators, open-ended stubs, and controlled even–odd-mode coupling, ensuring reliable dual-band operation without requiring additional resonator units.

No. of Pages : 16 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007966 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVICE FOR DISLODGING AND REMOVING STUCK MUD OR STONES

(51) International classification	:E01H 1/08, E01H 1/00, E01H 1/10, E21D 9/12, E02F 3/88	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Sanjay Choudhary**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A device for dislodging and removing stuck mud or stones comprising a hollow cylindrical housing 101 for tunnel mobility, motorized wheels 102 for movement, a visual monitoring unit 103 for detecting obstructions and identifying their type, a pneumatic plate 104 for dislodging debris, a telescopic rod 105 for extending and expanding the plate 104 to match tunnel surfaces, a vacuum unit 106 for collecting loosened mud through a central hole in the plate 104, a suction pump 106a and flexible pipe 106b for transferring debris, a hydraulic unit 107 for exerting forward force, a flap 108 for ground stabilization during resistance, an AI camera and ultrasonic sensor for debris classification, ball-and-socket joints 109 on the wheels for omnidirectional movement, a light detection resistor and LED unit 111 for illumination in low light, and a user interface for delivering operational alerts and blockage notifications during tunnel cleaning.

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007967 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIN HANDLING DEVICE

(51) International classification	:B65G 65/23, B01F 15/00, B65G 65/00, B62B 3/02, B60P 1/64
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Mridula Purohit

(57) Abstract :
A bin handling device, comprises of a structural frame 101 enables traversal over varying terrain, a holding lift unit 104 allows raising or lowering of a mounted bin, an input means 106 permits user commands for bin handling, a pair of telescopic side arms 107 adjusts according to bin dimensions, a rear-mounted telescopic rod 108 provides vertical height adjustment, a base plate 109 assists with stabilization, tilt correction, and load support, an automatic blender and mixing module 111 enables mixing of bin contents, a sensing module detects bin weight and tilt angle for stabilization, an extendable T-shaped handle 112 offers maneuverability, a microphone or display panel allows command input, a height adjustment module 114 facilitates frame 101 elevation, pneumatic cylinder units and a holding plate 115 perform lifting, an L-shaped telescopic rod 108 supports a blender unit, mixing chambers store fluids, a pump and valve manage fluid flow.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007968 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WALL MOUNTED DOCKING DEVICE

(51) International classification	:G06F 1/16, F16M 13/02, F16B 47/00, H05K 7/20, G06F 1/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Kamini Sharma

(57) Abstract :
A wall mounted docking device, comprising a base panel 101 configured to mounted on a vertical surface, a plurality of suction units 102 integrated into the base panel 101 to facilitate secure attachment of the panel 101 to the surface, a heat-resistant holding plate 103 connected to the front portion of the base panel 101 to support a portable display device, a grasping unit 104 including an extendable arm 105 mounted on a slider 106 and equipped with clamps 107 to accommodate different sizes of portable display devices and securely grip at the device's edges, a pusher unit 108 mounted on an articulated link 109 extending from the base panel 101 to precisely align and gently push a charging connector into a port of the display and a speaker 110 unit is integrated into the front portion of the base panel 101 configured to deliver audio alerts.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007974 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SECURE GAS PIPE CONNECTOR DEVICE

(51) International classification	:F17C 13/04, F16L 37/12, F16L 21/08, F16L 37/084, F16L 19/02	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Jyoti Shekhawat**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A secure gas pipe connector device, comprising a pair of sealing cylinders 101 to grip ends of an inserted gas pipe, a pair of motorized clamps 102 mounted on outer surface of each of the cylinders 101 to maintain stability during operation, a threaded cavity 103 configured with a valve 112 located at an upper section of each of the cylinders 101 to facilitate a fluid communication between both the cylinders 101, a vertical electromagnetic gripper module comprises a telescopic link 108 attached with the pin 104 and an electromagnetic plate 109 provided at a bottom of the link 108 to stabilize the cylinders 101 vertically and aligning it with the gas pipes.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007975 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BRICK HANDLING DEVICE

(51) International classification	:B65G 7/12, B65G 47/90, B65G 61/00, E04G 21/22, B66C 1/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Pramod Kumar Faujdar

(57) Abstract :
A brick handling device comprising an inverted U-shaped structural frame 101, an L-shaped telescopic handle 102 connected to one side of frame via motorized ball and socket joint 104, a pair of segmented pneumatic gripping plates 105 connected to frame 101 via a cam and roller module 106 to grip bricks of varying dimensions, an AI camera 108 mounted on top of frame 101 for identifying brick characteristics, L-shaped telescopic handle comprises foam cushioning 103, an integrated shock absorbing unit including an electromechanical damper, a pneumatic actuator and rubber bushings, activated in response to feedback from a piezoelectric sensor, controlled by a control unit, a motion and a position sensor integrated with telescopic handle 102 to determine handle's angle and orientation, an EMG sensor is embedded in handle 102 to detect signs of muscle fatigue, triggering control unit to lower brick load and issue an alert via a speaker 110.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007976 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FROZEN LAYER BREAKING DEVICE

(51) International classification	:E01H 5/12, E01H 5/10, E01H 5/09, E01H 5/06, F25C 5/18	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Nutan Sharma**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A frozen layer breaking device comprising an elongated body 101 with a user-accessible handle 102 for grip, an input means including microphone 103 and touch display for command input, an imaging unit 104 with optical sensor to detect ice accumulation, a four-bar linkage with position sensors to orient a snow collecting plate 106, a motorized sharp-edged rod 108 with spring-barrel cam for penetrating ice, a curved- panel 111 articulated arm 110 to move dispelled ice, a microcontroller for component coordination, a cylindrical grip with auto-adjusting segments via embedded sensor, a heating unit 114 with nichrome coils and temperature sensor to warm the handle 102, a holographic projector to guide movement, a vibration module on the plate 106 to loosen ice, a magnetic field sensor to optimize rod 108 force, a hydraulic plunger with ultrasonic sensor to actuate panel 111 movement, and a speaker for audio feedback and operational alerts.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007977 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DRAIN GRATE REMOVAL DEVICE

(51) International classification	:E03F 5/04, E03F 5/06, E03F 5/14, E03F 7/10, E03F 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Sudhir Verma

(57) Abstract :
A drain grate removal device comprises a movable frame 101 mounted on an inverted U-shaped rods 102 equipped with motorized wheels 103 for mobility, a horizontal bar 105, between a pair of vertical rods 106, enables height adjustment and carries a horizontal slider 104 for side-to-side motion, a camera 107 captures road images, processed using a machine learning protocol to identify the drain grate's type, position, and orientation, an ultrasonic sensor measures the distance between the grate and the horizontal slider 104, while a capacitive proximity sensor detects the grate's material, an electromagnet 108 on a telescopic link 109 is extended downwards and lift the grate, an suction unit 110 with a flexible sealing ring 110a, vacuum pump 110c, and motorized clamps 110b lifts the non-metal grate with the help of a telescopic pole 111.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007978 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE KEY MANAGEMENT ASSISTIVE DEVICE

(51) International classification	:A61L 2/24, G08B 13/14, G08B 13/24, A61L 2/10, G08B 13/196	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Satish Kumar Jangid**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An adaptable key management assistive device includes a body 101 with a fingerprint-enabled locking unit, extendable pins 202 with hooks 203 for storage, and a suction unit 204 for key stability, RFID scanner 103 for identifying and retrieving keys, an AI-based imaging unit 105 for real-time surveillance and intrusion detection, GPS module for location tracking, a UV-C light strip 205 for key disinfection, an audio output for alerts and controlled via a user interface or input means, the device supports multiple users and uses machine learning to optimize key management and security.

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007979 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : EYELASH APPLICATION DEVICE

(51) International classification	:A41G 5/02, A45D 24/36, A45D 44/00, A45D 24/04, A45D 24/22	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Deeplata Sharma**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An eyelash application device comprising of a spectacle like wearable frame 101 with side supports, configured to rest securely around a user's head, a curved pneumatic plate 102 to extend outward beneath the user's eye and serve as a stable resting platform for fake eyelashes, a motorized trimming unit 102 mounted on an articulated arm to cut eyelashes, an AI (artificial intelligence) imaging unit for making trimming decisions accordingly on the fake eyelashes to be applied, a comb-like structure 104 mounted on a telescopic bar 104 , configured to comb the eyelashes, a dual-substance dispensing unit 111 comprising a pair of reservoirs 111a configured to store hardening agent and glitter to be applied to the eyelashes based on user preferences, a miniature laser projection to emit a beam of light across the fake eyelashes, defining a target length and curvature for the trimming unit 102 to follow.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007980 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WATER LOGGING MANAGEMENT ASSISTIVE SYSTEM

(51) International classification	:E02B 17/00, E02B 7/20, B63B 21/50, E02B 3/20, E02B 3/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Sanjay Choudhary

(57) Abstract :
A water logging management assistive system, comprises a plurality of frames 101 installable on a ground surface in flood-prone areas, a plurality of suction units 102 for underwater anchoring, a monitoring unit 103 for detecting logged water levels and identifying individuals, a pair of expandable plates 104 for dynamic water flow control, a motorized roller 105 with a rope 106 and a floatable member 107 launched via a hydraulic actuator 108 toward a detected individual, a tension sensor within the rope 106 for strain detection and controlled retraction, a plurality of inflatable paddings 109 for buoyant support, a plurality of motorized iris lids 110 for excess water release, an input means in a computing unit for remote control, an audio output module 111 for verbal alerts, a plurality of LEDs 112 activated for illumination.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007981 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED WHEEL HUB PLACEMENT DEVICE

(51) International classification	:B24B 27/033, B25J 5/00, B60B 29/00, G01B 11/24, B62D 65/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Jitendra Kumar Katariya

(57) Abstract :
An automated wheel hub placement device, comprising a frame 100 incorporating a pair of inverted U-shaped members 101 for providing support to the frame 100 on a ground surface, a horizontal bar 102 mounted with each leg of member, each attached with a pair of motorized wheels 103 for maneuvering the frame 100, an input means 104 for input commands, an artificial intelligence-based imaging unit 105 to scan surroundings of the frame 100, a sensing module installed on the frame 100 for determining dimensions of a wheel hub, a rust removal module installed on the frame 100 for removing rust, a gripping means slidably mounted on the frame 100 for grabbing and aligning wheel hub, a fastening unit slidably installed on the frame 100 for grabbing bolts, and a chamber 107 installed on the frame 100 to position and fasten the bolts, a lubricating unit to apply lubricant onto bolts.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007982 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED RAW EATABLES CHOPPING DEVICE

(51) International classification	:B02C 18/06, B02C 18/00, B02C 25/00, B02C 18/14, B02C 18/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Sunil Kumar Jakhar

(57) Abstract :
An automated raw eatables chopping device comprising a body101 with a chamber102 for feeding eatables, an input means including microphone103 and user interface for receiving user commands, an AI-based imaging unit104 for detecting type orientation and size of eatables, a gripping means with motorized clamp105 on an extendable L-shaped rods106 for transferring eatables, a frame108 with adjustable segmented blades107 for chopping as per size, a pusher flap110 with hydraulic actuator109 applying calibrated force, a biosensor in the container detecting bacterial presence, a pressure sensor regulating actuator109 force based on eatable type, an ultrasonic sensor for blades107 sharpness monitoring, a weight sensor in the vessel for measuring chopped quantity, an audio output module 111 for alerts and instructions, a microcontroller integrated with machine learning protocols for real-time analysis and adaptive operation.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007983 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HAIR GROOMING DEVICE

(51) International classification	:A45D 20/12, A45D 19/00, A45D 24/22, A45D 19/02, A45D 2/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Shivangi Giri

(57) Abstract :
A hair grooming device, comprising a curved-shaped member 101 integrated with a conical structure 102, adapted to conform around a user's neck, a shower and dryer assembly integrated with the wearable member 101, an imaging unit 103 installed with member 101, a hair oiling module installed on member 101 to apply hair oil, a massage module mounted on the member 101 to provide targeted head massage, an L-shaped telescopic rod, mounted on the member 101 to position a shower unit 107 over the user's head, a multi-section chamber 105 integrated with the member 101, a sliding unit installed in between the rod 104 and the member 101 to translate the rod 104 to cover entire head area, an air blower 108 integrated into the member 101 via a robotic link 109 to generate a targeted airflow and an L-shaped telescopic link 109 mounted on the member 101 to extend outward.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007984 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PLANTER SUPPORT DEVICE

(51) International classification	:A01G 9/02, A01C 7/20, A01C 7/08, A01G 9/12, A01G 9/029	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Aarushi Kaushik**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
1) A planter support device, comprising: i) a curved-shaped frame 101 comprising a plurality of segmented flexible links, adapted to receive a planter; ii) a pair of vertical rods 102 configured with the frame 101; iii) a plurality of suction units 103 attached on side of the rods 102 to secure the frame 101 over a vertical supporting surface; iv) an imaging unit 104 integrated with a processor installed on the frame 101 in sync with an ultrasonic sensor integrated within the frame 101 to detect width of the planter, based on which the an inbuilt microcontroller actuates the segmented links to adjust the frame 101 dimensions for secure holding of the planter; v) a user-interface inbuilt in a computing unit wirelessly linked with the microcontroller for enabling a user to give input commands regarding trimming of overgrown leaves; and vi) an articulated arm 105 configured with a motorized trimming blade 106 installed on the frame 101 that works in sync with the imaging unit 104 to precisely perform trimming operations on overgrown leaves. 2) The device as claimed in claim 1, wherein a V-shaped member 107 having a hook-shaped bottom is connected to the rods 102 to provide additional structural support to the planter for mounting on vertical surfaces. 3) The device as claimed in claim 1, wherein a non-contact moisture sensor is integrated with frame 101 that works in sync with the imaging unit 104 to monitor moisture level of the soil within the planter. 4) The device as claimed in claim 3, wherein in case the moisture level recedes a threshold limit, the microcontroller actuates an electronic nozzle 108 attached to a water chamber 109 installed over the frame 101 to dispense a controlled amount of water into the planter to maintain optimal soil hydration. 5) The device as claimed in claim 1, wherein a motion sensor is integrated within the frame 101 to detect motion in the vicinity of the planter. 6) The device as claimed in claim 5, wherein upon detection of motion, the microcontroller activates an LED lighting unit installed on the frame 101 to illuminate the surrounding area for visibility of the planter. 7) The device as claimed in claim 4, wherein a level sensor is integrated within the chamber 109 detects available water level, to alert the user via the computing unit when the water level drops below a predefined threshold. 8) The device as claimed in claim 1, wherein a pressure sensor is integrated with the suction units 103 to detect securement of the suction units 103 with the surface, and accordingly the microcontroller regulates the negative pressure of the suction unit for secure gripping. 9) The device as claimed in claim 1, wherein the segmented links are operated by actuators including, but not limited to, hydraulic, pneumatic or electromechanical actuators. 10) The device as claimed in claim 1, wherein a push button is integrated with the frame 101 accessed by the user to active/deactivate the device.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007985 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED EXPANSION JOINT FORMING DEVICE

(51) International classification	:E01C 23/09, E01C 19/48, E04G 21/04, E01C 19/52, E01C 11/02	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Kunal Sharma**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated expansion joint forming device, comprising a pair of U-shaped frames 101 configured with motorized wheels 102 adapted to position over a freshly poured concrete water channel, a touch interactive display panel 103 for receiving user input regarding position, curvature, and depth of the expansion groove, an imaging unit 104 integrated with a processor and an ultrasonic sensor for detecting channel dimensions, an extendable rod 105 configured between the frames 101 and adjusted according to detected dimensions, a laser module projecting a reference path for alignment, a pair of hydraulic bars 106 connected to semi-circular member 107 for forming a curved groove in the concrete surface, a pressure monitoring units in hydraulic bars 106 for force control, a IR (Infrared)-based moisture sensor to monitor concrete surface conditions, a vibration sensor with a vibration damper 108 for smooth operation.

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007990 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ASSISTIVE AERIAL TRAVERSING PRACTICING DEVICE

(51) International classification	:A63B 69/00, G09B 9/00, A63B 24/00, G09B 9/08, A61H 1/02	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Sandeep Kumar Jain**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A assistive aerial traversing practicing device, comprises of a platform 101 configured with a plurality of inverted U-shaped frames 102 on a ground surface, a horizontal bar 103 is radially centrally to connect them, an input means 104 to enable user input for aerial traverse sessions, a monitoring unit detects presence, dimensions, activity, posture, dizziness, or unconsciousness of the user, a plurality of extendable rods 106 for extending/retracting based on the user's height for aerial traverse practice, a hoop 107 to practice aerial traverse movement, a gripping means with a motorized clamp 108 and silicone member ensures hand stability, a motorized swing joint 111 between the rod 106 and bar 103 facilitates arcuate motion for aiding movement and an audio output module 109 provides alerts and instructions, the platform 101 is cushioned for impact absorption.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007992 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED GRAFTING DEVICE FOR FIELD-BASED HORTICULTURE

(51) International classification	:A01G 1/06, A01G 2/35, A01G 2/30, A01G 2/38, A01G 2/00	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Seema Verma**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated grafting device for field-based horticulture, comprises of a housing 101 positioned over a surface, a rotatable imaging unit 102 with an ultrasonic sensor to determine the type of terrain, track wheels 103 to manoeuver the housing 101, a telescopic pole 104, a motorized clamp 105 to acquire a grip over the target plant stem, a telescopic rod 106 with a blade 107 to extend and retract to perform cutting of a plant stem, a pneumatic gripper 108 to retrieve a selected scion from a storage chamber 109, a wrapping assembly 110 comprises an extendable L-shaped bar 110a, a C-shaped frame 110b positioned at a distal end of the bar, a motorized roller 110c wrapped with adhesive sheet 110d configured with the frame 110b and a pair of robotic grippers 110e to securely grip the sheet 110d and wrap around the graft junction.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007994 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED STRAP CLAMP MANUFACTURING DEVICE

(51) International classification	:A44B 99/00, B65B 13/18, B26F 1/38, B26F 1/02, A44B 11/25	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Mohammad Asif Iqbal**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated strap clamp manufacturing device, comprising of a housing 101 adapted to be positioned on a ground surface, plurality of suction cups 103 secures the housing 101 over the surface, a display panel 104 to provide input commands regarding a quantity of strap clamps a user desires to manufacture, a roller 105 wrapped with a to rotate to unwind the sheet and translates towards a conveyor 106, two rolling members 107 rotates and presses the sheet, multiple punching die 108 installed via hydraulic rods 109 extends and apply optimal force on the sheet to create holes, a cutter installed via an extendable bar 115 to extend and precisely cut the clamp sheet, an arm 110 grips the sheet, and positions the sheet over a structure 111, a plate 112 connected via a hydraulic bar 113, to extends and apply force for formation of clamp sheet.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007996 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED WALL PLASTER SMOOTHENING DEVICE

(51) International classification	:E04F 21/08, E04F 21/00, E04F 21/165, G01B 11/24, E04F 21/18	(71)**Name of Applicant :** **1)Vivekananda Global University** Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India (72)**Name of Inventor :** **1)Ankesh Gupta**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated wall plaster smoothening device, comprising a base 100 arranged with a platform 101 atop, a plurality of motorized wheels for maneuvering the platform, an input means 103 paired with the platform, an inbuilt microcontroller for enabling a user to provide input commands, an artificial intelligence-based imaging unit 104 mounted on the platform 101 to scan surroundings of the platform, a monitoring unit 105 installed on the platform, for measuring the wall gap, a 3D surface profiler for surface texture analysis, to detect surface irregularities, a scissor lift arrangement 106 installed in between the base 100 and platform 101 for height adjustment of the platform, a sanding module 107 installed on the platform, configured to collect sand papers from a container 105 mounted on the platform 101 and a sensor suite 108 installed on the sanding module, for detecting real-time pressure and movement of sanding paper.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007998 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIPURPOSE TODDLER ROCKING DEVICE

(51) International classification	:A61B 5/00, A61B 5/024, A61B 5/01, A47D 9/02, A47D 13/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Govind Shay Sharma

(57) Abstract :
A multipurpose toddler rocking device comprising a frame with curved members 101, a seating platform 103 for toddler seating, motorized rollers with straps 105 to secure the platform 103, a handle-bar 106 for toddler support, a wheel assembly 107 for frame mobility, a user –interface for caretaker commands to enhance gait activity, a motorized slider 107b to adjust wheel assembly 107 position enabling dual-activity modes, multiple suction cups to prevent slipping or tipping, a health monitoring module with a PPG sensor to detect heart rate and oxygen saturation, a temperature sensor for body temperature measurement, a display screen to show health status, an IMU sensor to detect abnormal tilting or falling and deploy airbags for protection, a sun sensor to detect sunlight intensity, extendable fabric panels to provide shade, and a counterweight 104 with linear actuators 113 for dynamic balance adjustment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611007999 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : VEHICLE LIFTING AND INSPECTION DEVICE

(51) International classification	:B66F 11/04, B66F 17/00, B66F 7/28, B66F 7/08, B66F 7/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Krishna Nandan

(57) Abstract :
A vehicle lifting and inspection device comprising, a platform 101 adapted to be positioned in vicinity of an inspection pit, an extendable rod 102 configured with a motorized wheel 103 installed on side portions of the platform 101, an imaging unit 104 installed on the platform 101 to detect presence of a vehicle positioned over the pit, a pair of hydraulic actuators 105 to extend and position a horizontal plate 106 mounted over the actuators 105 underneath the vehicle chassis and an extendable clamping units 107 on edge portion of the plate 106, to extend and securely grip side portions of the vehicle chassis.

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611008000 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUCKWEED GATHERING DEVICE

(51) International classification	:E02B 15/10, A01D 44/00, A01M 21/02, A01M 21/04, B63B 35/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)S.V.H Nagendra

(57) Abstract :
A duckweed gathering device comprises a floating platform 101 with a rectangular deck 102 for buoyancy, a propulsion assembly 103 including a propulsion core, multiple propulsion fins, and an RPM sensor to drive and regulate movement, and a vision guided module to capture imagery and adjust navigation, a duckweed collection module 105 with an articulated linkage 105a, a semi-circular collection plate 105b with adjustable curvature, and a motorized hinge joint for adaptive gathering, containment unit 106 with micro-perforated filtration tray separates water from collected duckweed, water storage chamber 107 connects to a flexible suction hose 108 and spray nozzle 109 for washing, controlled by a contamination sensor, cleaning module 110 includes an L-shaped support arm 111, dual-axis slider 112, and hinged cleaning flap 113 for sweeping, storage container 114 with thermal insulation, thermoelectric cooling, mesh drainage, and sensors maintains optimal conditions, robotic arms 115 transfer duckweed.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :13/01/2026

(21) Application No.202611003407 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR ONLINE INSPECTION OF INTERLEAVING POWDER DISTRIBUTION IN FLOAT (FLAT) GLASS MANUFACTURING AND A METHOD THEREOF

(51) International classification	:G01N 15/02, G01N 21/88, G06T 7/00, G01N 21/958, G01N 21/95
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ASAHI INDIA GLASS LTD.
Address of Applicant :(RESEARCH & DEVELOPMENT DEPARTMENT) ASAHI INDIA GLASS LTD, Plot No-190, Sector-4, HSIIDC Growth Centre, Tehsil Bawal, District Rewari-123501, State-Haryana. Haryana India
(72)**Name of Inventor :**
1)Devinder Kumar
2)Ajay Dhiman
3)Amit Kumar

(57) Abstract :
The invention relates to machine-vision inspection of powder-applicated float (flat) glass during production. At least one high-resolution line scan camera, positioned downstream of a powder applicator, is moved across the sheet width by a linear actuator while vision lights illuminate the powder-bearing surface. An image processing subsystem identifies particles using reflectivity, particle size and distribution pattern, determines particle count and percentage powder coverage for scanned segments, and generates a powder-density heat map. Powder density (% powder/m²) is calculated by correlating pixel-intensity values with particle-size distribution and reflectivity characteristics, and compared with predefined acceptance criteria. Widthwise powder-density distribution is correlated with powder-application nozzle positions to identify nozzle abnormalities. The system enables continuous, non-destructive, real-time inspection over substantially the entire glass surface for production quality assessment.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :13/01/2026

(21) Application No.202611003408 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PRINT INSPECTION SYSTEM AND A METHOD FOR AUTO-ADJUSTING PRINT ALIGNMENT IN FLAT GLASS PRINTING MACHINES

(51) International classification	:B41F 15/08, B41F 33/00, G06T 7/00, G01B 11/00, G01N 21/956	(71)**Name of Applicant :** **1)ASAHI INDIA GLASS LTD.** Address of Applicant :(RESEARCH & DEVELOPMENT DEPARTMENT) ASAHI INDIA GLASS LTD, Plot No-190, Sector-4, HSIIDC Growth Centre, Tehsil Bawal, District Rewari-123501, State-Haryana. Haryana India (72)**Name of Inventor :** **1)Devinder Kumar** **2)Ankit yadav** **3)Dharmendra Chaudhary** **4)Ramdas Patil**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a print inspection system and a method for auto-adjusting print alignment in flat glass printing machines. The system of the present invention is designed to automate the detection and correction of stencil-to-glass misalignment during the printing process. It continuously monitors the alignment between the BCP pattern and the glass edge via a Print Inspection System, using image analysis to accurately detect any Print Edge Gap. The system then automatically repositions the printing screen using a three-servo-based mechanism to correct detected deviations without halting the machine. This results in consistent print quality, minimizes edge gap defects, reduces the need for operator involvement, and improves the overall efficiency and reliability of the process. By automating the inspection process, this solution removes operator variability and ensures consistent, reproducible results independent of human skill or attention. Figure 1 is the representative figure.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/06/2026

(21) Application No.202611075362 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELENIUM-ENRICHED FUNCTIONAL NUTRITION POWDER PREPARATION APPARATUS

(51) International classification	:A23N12/02, A23N12/08, A23L5/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SHARDA UNIVERSITY
Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)MS. M SHAHLLAH
2)DR. ZOOBIYA ISLAM
3)DR. MAYURI RASTOGI

(57) Abstract :
Disclosed is an apparatus for preparing a selenium-enriched functional nutrition powder for thyroid-health support. The apparatus comprises ingredient hoppers for receiving plant-based ingredient materials comprising seed material, mushroom material, cereal material, pulse material, and amla material. A cleaning chamber with a perforated retaining basket separates foreign particles, and a drying chamber with ingredient-support trays reduces moisture content of cleaned materials. A roasting chamber thermally treats at least a portion of dried materials, and a grinder converts the materials into powdered ingredient material. A sieve assembly passes a uniform fine-powder fraction, dosing hoppers dispense measured quantities of the fine-powder fraction, and a blender mixes the measured quantities into a homogeneous selenium-enriched powder. Fig. 1 Dated June 17, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202611003516 A
(19) INDIA
(22) Date of filing of Application :13/01/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : A GLASS MOULD RING INSPECTION AND POSITIONING SYSTEM AND A METHOD THEREOF

(51) International classification	:C03B 23/03, G01B 11/24, G01N 3/20, G06T 7/00, C03B 23/023	(71)**Name of Applicant :** **1)ASAHI INDIA GLASS LTD.** Address of Applicant :(RESEARCH & DEVELOPMENT DEPARTMENT) ASAHI INDIA GLASS LTD, Plot No-190, Sector-4, HSIIDC Growth Centre, Tehsil Bawal, District Rewari-123501, State-Haryana. Haryana India (72)**Name of Inventor :** **1)Jasveer Singh** **2)Sudhir Kumar** **3)Naveen Sharma** **4)Vikash Jangra** **5)Rajesh Yadav**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The present invention relates to the windshield glass gravity bending process, which includes both mould centring and glass centring operations. Further, the invention provides an improved system and method for glass mould ring inspection and positioning. The system of the present invention automates quality control for rings by performing precise 2D defect detection, analyzing each item for deformations, irregularities, or shape deviations that could impact bending quality. To ensure consistency, a strict tolerance threshold is applied, flagging any deviation greater than ±5 mm as defective (NG). This automated inspection process replaces traditional manual visual checks, effectively eliminating human error and guaranteeing a consistent level of mould quality throughout production. Figure 1 is the representative figure.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/06/2026

(21) Application No.202611077191 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART LEARNING ENVIRONMENT WITH DYNAMIC KNOWLEDGE GAP PREDICTION AND AUTOMATED CONTENT RECONFIGURATION

(51) International classification	:G06Q50/20, G06N20/00, G06F18/20, G06N5/022
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PROF. (DR.) PALLAVI PANDIT
Address of Applicant :Dean, University School of Education, Rayat Bahra Professional University, Hoshiarpur, Punjab-146104 HOSHIARPUR Punjab India
2)DR. PARUL PANDIT
3)DR. ASHWANI KUMAR
4)MRS. SHIKHA KOUNDAL
5)DR. JYOTSNA
6)MS. SIMSY JASWAL
7)MR. SHUBHAM DALORIA
8)RAYAT BAHRA PROFESSIONAL UNIVERSITY, HOSHIARPUR, PUNJAB

(72)**Name of Inventor :**
1)PROF. (DR.) PALLAVI PANDIT
2)DR. PARUL PANDIT
3)DR. ASHWANI KUMAR
4)MRS. SHIKHA KOUNDAL
5)DR. JYOTSNA
6)MS. SIMSY JASWAL
7)MR. SHUBHAM DALORIA

(57) Abstract :
The present invention discloses a smart learning environment system and method for dynamically predicting knowledge gaps and automatically reconfiguring educational content. The system comprises a data acquisition module for capturing real-time learner interactions, a preprocessing unit for generating structured feature vectors, and a knowledge gap prediction engine employing a hybrid model combining rule-based inference and machine learning techniques. Based on predicted deficiencies, a content reconfiguration module dynamically modifies learning pathways by adjusting difficulty levels and inserting remedial content. The system operates through an integrated architecture ensuring real-time responsiveness and optimized data processing. The invention provides a measurable technical effect in terms of reduced computational latency, improved prediction accuracy, and enhanced system efficiency. The disclosed system enables adaptive learning through structured computational mechanisms, ensuring compliance with technical contribution requirements and improving overall learning outcomes. FIG 01.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :13/07/2026

(21) Application No.202611085437 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WATER-QUALITY TREND ANALYSIS AND FORECASTING KIOSK SYSTEM

(51) International classification	:G01N 33/18, G06Q 10/06, G06Q 50/06, G06Q 50/26, G06Q 10/04	(71)**Name of Applicant :** **1)SHARDA UNIVERSITY** Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India (72)**Name of Inventor :** **1)ANSHUMAN** **2)MANAV MEHTA** **3)TARAN BHUEE** **4)DR. SMITA TIWARI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed is a water-quality trend analysis and forecasting kiosk system comprising a kiosk housing, a data acquisition interface, a user input interface, a display, a server, and a visualization interface. The data acquisition interface receives historical and real-time water-quality data associated with a river-water body. The user input interface receives a selection of a monitoring location, water-quality parameter, or forecast time interval. The server preprocesses the water-quality data by cleaning missing values, removing inconsistent measurements, normalizing measured values, and arranging measured parameters as time-series data. The server generates temporal input features, analyzes trend information, generates forecast values, compares forecast values using error metrics and performance metrics, and generates dashboard data comprising current measured values, historical trend information, forecast values, and comparison information. The visualization interface displays the dashboard data for monitoring and forecasting river-water quality. Fig. 1 Dated July 11, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/01/2026

(21) Application No.202611008004 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FOOT CARE DEVICE

(51) International classification	:A47K 7/04, A47K 7/02, A61H 35/00, A47K 7/08, A61B 5/103
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Deepak Kumar

(57) Abstract :
A foot care device, comprising of a circular disc 101 developed to be installed on a surface, an extendable flap 103 to allow user to place their foot, a biosensor to detect hygiene levels of the foot, a foot scrubbing assembly mounted on the disc 101, comprising a liquid soap dispenser 104, and a foot scrubbing plate 105 arranged with multiple bristles 106, connected to the disc 101 via an extendable rod for cleaning the foot of the user, an imaging unit 107 to detect accumulation of mud or dust on the user's foot, multiple pneumatic pins 108 housed beneath iris lids 110, the iris lids 110 open selectively for outward extension of the pins 108 through the openings to engage the foot region and dislodge accumulated dust, an air blower 111 blows air towards the user's foot to dry the moisture therefrom.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :08/07/2026

(21) Application No.202611084109 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HYBRID MULTI CHAMBER SYSTEM FOR CO-PRODUCING BIOCHAR AND DRIED AGRICULTURAL PRODUCE

(51) International classification	:C10B53/02, C10B53/00, C10B51/00, C10B49/00, C10B47/00, C10B57/10, C10B21/18, F26B23/02	(71)**Name of Applicant :** **1)Sher-e-Kashmir University of Agricultural Science and Technology of Kashmir (SKUAST-K** Address of Applicant :Shalimar, Srinagar, J&K, India SRINAGAR Jammu & Kashmir India (72)**Name of Inventor :** **1)Mohammad Muzamil** **2)Sagar Malik** **3)Saqib Rashid** **4)Athar Bashir** **5)Maliqa Majid** **6)Jagvir Dixit** **7)Gazala Nazir** **8)Immad Ahmad Shah** **9)Syed Imran** **10)Shyam Nath** **11)Masrat Mohiuddin** **12)Rizwan Ul Zama Banday**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A hybrid system (100) for co-producing biochar and dried agricultural produce includes a combustion chamber (102) configured to burn fuel to generate thermal energy. A biochar chamber (104) is arranged above the combustion chamber (102) and configured to receive heat from the combustion chamber (102) to pyrolyse biomass in an oxygen-deficient environment to produce biochar. A drying chamber (106) is arranged above the biochar chamber (104) and includes one or more drying shelves (122) for holding agricultural produce. An energy recycling chamber (108) is configured to circulate hot flue gas from the combustion chamber (102) around the drying chamber (106) to indirectly heat the one or more drying shelves (122). An exhaust system (110) is configured to discharge exhaust gas from the hybrid system (100). (FIG. 3)

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/07/2026

(21) Application No.202611087859 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A CONDUCTIVE CARBON-METAL OXIDE NANOCOMPOSITE, A TWO-STEP PROCESS FOR PREPARATION THEREOF FROM WASTE BIOMASS

(51) International classification	:B82Y30/00, B82Y40/00, H01B 1/04, C09D 11/52, H01G 11/36, C08L 97/02, C01G 49/06, C01G 53/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)UNIVERSITY OF ALLAHABAD
Address of Applicant :Senate House Campus, University Road, Old Katra, Prayagraj (Allahabad), Uttar Pradesh – 211002 Uttar Pradesh India
(72)**Name of Inventor :**
1)Anupam Giri
2)Rishabh Kanojia
3)Sandeep Yadav
4)Santimoy Khilari
5)Madhu Shukla
6)Prerana Verma

(57) Abstract :
The invention relates to conductive nanocomposite materials, biomass-derived carbon materials, and printed electronic and energy-storage components. Disclosed is a conductive carbon-metal oxide nanocomposite comprising a graphitic carbon-rich matrix derived from lignocellulosic waste biomass (wood dust, crop residue, paper waste etc), the matrix having graphitic carbon rich domains, epitaxial (conformal) interface with the metal oxide, and embedded crystalline metal oxide domains distributed therein to provide a porous, wrinkled, nanosheet-like architecture. The nanocomposite is obtainable by a two-step process including room-temperature formation of a biomass-metal oxide composite or biomass-metal hydroxide composite within a lignocellulosic matrix, followed by controlled thermal annealing to convert the biomass into the graphitic carbon-rich matrix while retaining crystalline metal oxide domains. The material may exhibit linear current-voltage behavior and high capacitive performance. Also disclosed are an ink comprising the nanocomposite and printed electrodes or conducting lines deposited on glass or textile fabric, including flexible conductive patterns for powering LEDs or micro-LED circuits.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/06/2026

(21) Application No.202611075363 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LOW-DENSITY POLYETHYLENE INTERLOCK MOLDING APPARATUS

(51) International classification	:B29C 43/32, B29C 43/36, B29C 43/50, B29C 33/00, B29B 17/00, B29B 17/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SHARDA UNIVERSITY
Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)ABDULRAHMAN ABBA GONI
2)DR. SUMAN

(57) Abstract :
Disclosed is a low-density polyethylene interlock molding apparatus comprising an LDPE receiving hopper, a shredding unit, a temperature-controlled melting vessel, a sand pre-warming and metering unit, a hot mixing chamber, and a steel interlock mold. The shredding unit reduces low-density polyethylene waste into flakes, and the melting vessel softens the flakes into a molten LDPE mass. The sand pre-warming and metering unit supplies sieved sand in a pre-warmed state to the hot mixing chamber, where a coated LDPE-sand mixture is formed. The steel interlock mold receives the coated mixture and includes a mold body with an interlocking-profile mold cavity, a clampable mold cover for compression, restricted upper vent passages for discharging entrapped air while blocking sand particles, and a demolding insert movable through the mold cavity to separate a solidified interlocking plastic body. Fig. 1 Dated June 17, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/07/2026

(21) Application No.202611087233 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PEDIATRIC AIRWAY VIBRATION BELT

(51) International classification	:A61H23/00, A61M16/00, A61B5/113, A61B5/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Riya Sharma (PT)
Address of Applicant :Mahatma Gandhi Physiotherapy College, MGPC, Mahatma Gandhi University of Medical Sciences & Technology, Jaipur, RIICO Institutional Area, Mahatma Gandhi Rd, Sitapura, Jaipur 302022, Rajasthan, India Jaipur Rajasthan India
2)Mahatma Gandhi University of Medical Sciences & Technology, Jaipur
(72)**Name of Inventor :**
1)Dr. Riya Sharma (PT)
2)Dr. Eshita Sharma (PT)
3)Dr. Monika Hariramani (PT)
4)Dr. Roop Sharma

(57) Abstract :
The present invention relates to a pediatric airway vibration belt (100) configured to be worn around a thoracic region of a pediatric patient for airway clearance therapy. The pediatric airway vibration belt (100) comprises a multilayer wearable belt structure (102), a plurality of vibration pads (104) positioned across anterior, posterior, left lateral, and right lateral thoracic regions, a respiratory sensing arrangement (106), a control unit (108), a detachable remote controller (110), and a power supply arrangement (112). The respiratory sensing arrangement (106) detects thoracic expansion and contraction, and the control unit (108) processes respiratory signals to identify respiratory phases and selectively activate the one or more vibration pads (104). The pediatric airway vibration belt (100) supports breath-synchronized, continuous rhythmic and time-cycle vibration therapy for mobilization of airway secretions. The invention provides respiration-synchronized, multi-zone airway clearance therapy through a lightweight wearable pediatric-specific belt configuration. Referring Fig.1

No. of Pages : 29 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/07/2026
(21) Application No.202611089106 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : LOCKABLE ENDODONTIC BUR DEPTH GAUGE ASSEMBLY

(51) International classification	:A61C1/00, A61C3/00, A61C19/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SHARDA UNIVERSITY
Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)DR.POOJA KABRA
2)DR.SADAF NAHID

(57) Abstract :
Disclosed is a lockable endodontic bur depth gauge assembly comprising a collar body through which an axial bore is formed for receiving a shank of an endodontic access bur. An axial stop surface is defined by the collar body for contacting a tooth surface during advancement of the bur. A radial passage is formed through the collar body and intersects the axial bore. A locking member is threadedly received in the radial passage and is movable between a released position and a locked position. In the released position, axial displacement of the collar body along the shank is permitted. In the locked position, a selected exposed cutting length of the bur is maintained, whereby penetration of the bur into a tooth is limited by contact between the axial stop surface and the tooth surface. Fig. 1 Dated July 21, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/07/2026

(21) Application No.202611089107 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TWO-WHEELER FALL-RESPONSIVE IGNITION INTERRUPTION SYSTEM

(51) International classification	:B62H 5/20, F02P 5/15, B60R 16/023, B60R 16/02, B60R 25/04	(71)**Name of Applicant :** **1)SHARDA UNIVERSITY** Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India (72)**Name of Inventor :** **1)DR. NEETU FAUJDAR** **2)DR. ANJUM MOHD ASLAM**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed is a two-wheeler fall-responsive ignition interruption system comprising a vehicle-mounted control assembly, first and second inclination sensors, a relay interface coupled with a propulsion-enabling circuit, and a control unit comprising a memory and a processor. First and second inclination values are generated along respective sensing directions and are received by the processor. Fuzzy membership states are assigned to the inclination values and evaluated according to stored fuzzy rules. A fall-related inclination condition is determined from the evaluation. A shutdown signal is generated upon determination of the fall-related inclination condition, and the relay interface is actuated using the shutdown signal to interrupt the propulsion-enabling circuit. Coordinated evaluation of the inclination values permits abnormal vehicle orientation to be distinguished from permissible riding conditions and initiates propulsion interruption without rider intervention. Fig. 1 Dated July 20, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 31 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/07/2026
(21) Application No.202611089108 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : VEGAN AND GLUTEN-FREE BAKED CONFECTIONERY COMPOSITION AND PREPARATION METHOD

(51) International classification	:A23G3/48, A23L19/00, A23L25/00, A21D13/04, A21D13/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SHARDA UNIVERSITY
Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)DR RAHUL SAXENA
2)DR NIMMI AGARWAL

(57) Abstract :
Disclosed is a vegan and gluten-free baked confectionery composition comprising 72 wt% to 80 wt% ripe banana pulp, 14 wt% to 20 wt% desiccated coconut powder, 1.5 wt% to 4 wt% date paste, 1.5 wt% to 4 wt% milled Euryale ferox seed powder, 0.5 wt% to 2 wt% cocoa powder, 0.2 wt% to 1 wt% a food-grade acid-reacting leavening agent, and 0.05 wt% to 0.5 wt% sodium bicarbonate. A preparation method comprises homogenizing the ripe banana pulp and the date paste, incorporating the desiccated coconut powder and the milled Euryale ferox seed powder to form a cohesive batter, maintaining the cohesive batter at room temperature, incorporating the cocoa powder and leavening ingredients, transferring the cohesive batter into a baking vessel, baking the cohesive batter at 180°C for 20 minutes, and cooling the baked confectionery product. Fig. 1 Dated July 20, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/07/2026
(21) Application No.202611089109 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SUBLINGUAL MUCOADHESIVE PATCH AND CASTING APPARATUS

(51) International classification	:A61K9/70, A61K47/10, A61K31/00, A61K47/30, A61K47/32, B29C39/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SHARDA UNIVERSITY
Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)DR. ASHWINI KUMAR
2)VAISHALI SHUKLA

(57) Abstract :
Disclosed is a sublingual mucoadhesive patch composition and a casting apparatus for preparation of the patch composition. The patch composition comprises a polyvinyl alcohol matrix, a therapeutically effective amount of a therapeutic agent dispersed within the polyvinyl alcohol matrix, and a permeation-enhancing and taste-masking amount of menthol dispersed within the polyvinyl alcohol matrix. A saliva-resorbable patch is thereby formed for mucoadhesive contact with a sublingual mucosal surface and systemic absorption of the therapeutic agent. The casting apparatus comprises a casting platform, a formulation vessel, a menthol reservoir, a mixing chamber, a casting head comprising a metering outlet and a spreading edge, and a covered drying chamber. A formulation layer is formed through a casting gap, and solvent is removed while evaporation of menthol is restricted. Fig. 1 Dated July 22, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/01/2026
(21) Application No.202611007969 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CHILD PERCHING DEVICE

(51) International classification	:A47D 15/00, A47D 13/08, G08B 21/02, A47D 13/02, A47D 9/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vivekananda Global University
Address of Applicant :Sector 36, NRI Road, Jagatpura, Jaipur Rajasthan 303012 (India). Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Menka Bhasin

(57) Abstract :
A child perching device comprising a wearable body 101 worn over a caregiver's neck and shoulders for child support, a pair of straps 102 to secure the body 101 , a bowl-shaped curved seat105 with extendable plates106 for carrying the child, an imaging unit104 to detect child's size and posture, a microcontroller to actuate plate movement and motorized hinges for back support, an extendable cushioned sheet with sensors for dynamic posture-based adjustment, a foot-gripping module108 for added leg stability, a voice-interactive unit with microphone 103 and speaker 112 for hands-free alerts and commands, cushioning around the seat105 for comfort, a shock-absorbing unit109 to reduce impact during movement, telescopic rods 110 with grip handles111 for additional body 101 support, an EMG sensor to detect caregiver muscle strain and adjust strap fit, and an IoT module to send real-time safety notifications to the caregiver.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/06/2026

(21) Application No.202611076816 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS UAV SWARM SYSTEM WITH GPS-DENIED LOCALIZATION, MULTI-MODAL SUBSURFACE SENSING AND DIGITAL TWIN-BASED DISASTER RESPONSE

(51) International classification	:G05D 1/10, G01S 13/86, B64C 39/02, G01S 17/86, G01S 13/933
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute Of Technology (NIT), Hamirpur
Address of Applicant :Hamirpur, Himachal Pradesh, India 177005 Himachal Pradesh India
(72)**Name of Inventor :**
1)Arun Kushwaha
2)Rahul Prasad
3)Utkarsh Maurya
4)Nitin Saini
5)Ananay Kaushal
6)Sandeep Kumar Singh
7)Rakesh Sharma

(57) Abstract :
An autonomous UAV system (102) for subsurface disaster response, comprising a UAV airframe (106) supporting a sensor suite consisting of a LiDAR unit, an RGB camera, and an Ultra-Wideband (UWB) ground-penetrating radar (GPR) assembly, a flight controller unit (108) linked to an Inertial Measurement Unit (IMU), and a processor (104) configured to lock timestamps from the IMU with radar pulses to map radar signals to the orientation and position of the UAV airframe, execute a real-time motion compensation loop within a radar data-stream to filter out motion smear, spatially overlay radar-detected anomalies onto surface geometry derived from LiDAR point clouds and visual data from the RGB camera to confirm the presence of buried objects, and calculate a local coordinate frame for anomaly mapping via visual-inertial SLAM and UWB peer-to-peer ranging, functioning in the absence of Global Positioning System (GPS) signals. FIG. 1

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :15/07/2026

(21) Application No.202611086652 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A COMPOSITION AND PROCESS FOR STABILIZATION OF SILVER NANOWIRES

(51) International classification	:B82Y30/00, H01B1/02, H01B1/22, B22F1/054, B22F1/06, B22F5/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY KANPUR
Address of Applicant :INDIAN INSTITUTE OF TECHNOLOGY KANPUR, KALYANPUR, KANPUR – 208016, UTTAR PRADESH, INDIA. KANPUR Uttar Pradesh India
(72)**Name of Inventor :**
1)UMESH PRATAP PANDEY
2)ASHISH GUPTA
3)YASHOWANTA N MOHAPATRA

(57) Abstract :
The present invention relates to a composition comprising silver nanowires dispersed in an aloe vera gel matrix containing 70 wt.% to 98 wt.% water. The silver nanowires possess an average diameter ranging from 20 nm to 200 nm, an average length ranging from 1 µm to 50 µm, and an aspect ratio ranging from 1:300 to 1:1000. The aloe vera gel matrix comprises acemannan-containing inner gel extracted from aloe vera leaves and optionally includes antioxidants selected from ascorbic acid and tocopherol. The invention further relates to a process for preparing the composition by synthesizing silver nanowires through reduction of silver nitrate in ethylene glycol in presence of poly(vinylpyrrolidone), followed by dispersion of the silver nanowires in homogenized aloe vera gel. The composition is useful for storage of silver nanowires for conductive and flexible electronic applications. FIG. 3

No. of Pages : 26 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/07/2026

(21) Application No.202611090207 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Hybrid Multi-Fault Localization in Software Using Artificial Bee Colony Optimization and Chaotic Genetic Algorithm

(51) International classification	:G06F 11/36, G06N 3/00, G06N 3/12, G06N 3/006, G06N 99/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Manipal University Jaipur
Address of Applicant :Manipal University Jaipur Off Jaipur-Ajmer Expressway, Post: Dehmi Kalan, Jaipur-303007, Rajasthan, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Dr. Debolina Ghosh
2)Dr. Jay Prakash Singh

(57) Abstract :
The present invention relates to a method for automated software fault localization using a hybrid Artificial Bee Colony (ABC) and Chaotic Genetic Algorithm (CGA) optimization framework. The method comprises instrumenting a software program, executing test cases, collecting pass/fail outcomes and execution coverage data, and generating feature vectors for program entities. Candidate solutions representing suspiciousness weighting parameters, suspiciousness mapping functions, and multi-fault separation parameters are optimized using the hybrid ABC–CGA framework. The ABC algorithm performs global exploration through employed, onlooker, and scout bee operations, while the CGA performs fine-grained refinement using cooperative sub-populations and chaotic evolutionary search. Candidate solutions are evaluated using objective functions including Top-N accuracy, EXAM score, Mean Average Precision (MAP), and runtime-aware performance metrics to generate optimized suspiciousness rankings. The disclosed method improves fault localization accuracy, enhances multi-fault detection, reduces debugging effort, supports large-scale software systems, and integrates efficiently with automated testing and continuous integration environments.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/07/2026

(21) Application No.202611089760 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-END RIGID GUIDE ASSEMBLY FOR CORE-SPUN YARN PRODUCTION

(51) International classification	:D01H 13/04, D01H 1/36, D01H 7/86, D01H 1/11, D01H 1/16	(71)**Name of Applicant :** **1)Dr. Nandan Kumar** Address of Applicant :Institute of Technical Textiles, Plot 145, HSIIDC, Phase 1, Barhi, Sonipat, Haryana, India (IN) Sonipat Haryana India (72)**Name of Inventor :** **1)Dr. Nandan Kumar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A multi-end rigid guide assembly (10) for delivery of core filaments in core-spun yarn production on a ring spinning frame includes a mounting bracket (20) configured for detachable attachment to a cradle bar (80) of the ring spinning frame, a rigid guide body (25) fixedly secured to or integrally formed with the mounting bracket (20), and a clamping mechanism (30) operable to rigidly fix the mounting bracket (20) on the cradle bar (80) without spring loading. The guide body (25) presents at least two discrete filament-guiding apertures (26a, 26b) each sized and positioned to guide a respective core filament end (F1, F2) toward a nip of front drafting rollers. The guide body (25) has an external profile dimensioned to permit a clearer roller (50) to maintain operative contact with an adjacent rubber drafting roller (60) without obstruction. (FIG. 1)

No. of Pages : 26 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/07/2026

(21) Application No.202611089782 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTEGRATED PHOTOBIOMODULATION, ACOUSTIC RESONANCE, AND MICRO-AEROSOL IONIZATION DELIVERY DEVICE

(51) International classification	:B05B 17/06, A61M 11/00, A61M 11/04, A61M 15/00, A61M 11/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)AMIT KUMAR
Address of Applicant :187A, Near Dussehra Ground, New Colony, Gurgaon, Haryana, 122002 GURGAON Haryana India
(72)**Name of Inventor :**
1)AMIT KUMAR

(57) Abstract :
A delivery device (100) includes a housing defining an airflow pathway, a liquid reservoir (112) disposed in the housing, and a piezoelectric nebulizer (110) arranged to nebulize a liquid substrate from the liquid reservoir (112) into charged micro-aerosol droplets (302). A dual-wavelength LED array (108) is arranged to illuminate the charged micro-aerosol droplets (302). A negative ion generator (210) is arranged to impart a negative electrostatic charge to the charged micro-aerosol droplets (302). A coherent energy circuit board (116) is configured to control pulsing of the piezoelectric nebulizer (110), the dual-wavelength LED array (108), and a fan (114) according to a respiratory cadence. An acoustic resonance system (400) defines acoustic resonance chambers along the airflow pathway. (FIG. 1)

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/07/2026

(21) Application No.202611092935 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR PROLONGED MATERNITY PATIENT ENGAGEMENT AT A MEDICAL HEALTHCARE FACILITY

(51) International classification	:G16H 40/20, G06Q 20/36, G06Q 30/02, G16H 10/60, G06Q 20/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)USEKIWI INFOLABS PRIVATE LIMITED
Address of Applicant :211-212, 2nd Floor, JMD Regent Square, M G Road, Gurugram - 122002, Haryana, India Gurugram Haryana India
(72)**Name of Inventor :**
1)Aditya Sharma
2)Anand Nevatia
3)Pruthvinath Kancherla
4)Swati Sharma

(57) Abstract :
The present disclosure relates to a system (100) and a method (300) for prolonged engagement of a maternity patient at a medical healthcare facility. The system (100) receives service requests from the maternity patient through a customer application and process payments for availed services. Based on the payment and clinical inputs from a doctor, the system (100) generates service-related metadata and computes multiple rewards. A nested wallet structure is activated, and a first wallet stores redeemable cashback for outpatient services while a second wallet accumulates rewards redeemable toward inpatient maternity services. The system (100) notifies the maternity patient to utilize earned rewards for subsequent medical tests and medications, enabling continuous reward accumulation across multiple services. The nested wallet-based reward mechanism facilitates repeated service utilization and ensures prolonged engagement of the maternity patient within the same healthcare facility throughout the maternity care journey.

No. of Pages : 46 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/07/2026
(21) Application No.202611092936 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND A METHOD FOR RETENTION OF PATIENTS AT A MEDICAL HEALTHCARE FACILITY

(51) International classification	:G06Q30/0208, G16H40/20, G06Q10/10, G06Q50/10, G16H80/00, G06F3/048
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)USEKIWI INFOLABS PRIVATE LIMITED
Address of Applicant :211-212, 2nd Floor, JMD Regent Square, M G Road, Gurugram - 122002, Haryana, India Gurugram Haryana India
(72)**Name of Inventor :**
1)Aditya Sharma
2)Anand Nevatia
3)Pruthvinath Kancherla
4)Swati Sharma

(57) Abstract :
The present disclosure relates to a system (100) and a method (300) for retention of patients at a medical healthcare facility. The system enables onboarding of patients via at least one of a Field Consultant (FC) application, Customer application, or hospital staff. The system receives input from the patient on a customer application for availing a first service at the hospital and receives a corresponding payment. Metadata associated with the first service is generated based on input received from a doctor (OPD), and the metadata comprises one or more medical tests and medications. A reward, in the form of redeemable cashback, is computed based on the payment and deposited in a wallet of the patient. The system notifies the patient to use the reward for availing a second service at the hospital, and the reward is redeemable without restriction for subsequent services, thereby promoting continuous engagement and retention. [To be published with Figure 2]

No. of Pages : 46 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :13/07/2026

(21) Application No.202611085438 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NANO-REAGENT ETHANOL TEST CARTRIDGE

(51) International classification	:G01N 21/78, B01L 3/00, G01N 31/22, G01N 33/487, G01N 21/03
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SHARDA UNIVERSITY
Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)MS.SWETA VERMA
2)DR.LALIT PRATAP CHANDRAVANSHI
3)DR. HIMANSHU YADAV
4)DR. RAJENDRA KUMAR SHUKLA
5)DR. BHARTI JAIN

(57) Abstract :
Disclosed is a nano-reagent ethanol test cartridge for colorimetric screening of ethanol in a liquid sample. The cartridge comprises a reaction chamber comprising at least one optically viewable wall portion and adapted to retain a predefined volume of an acidic sensing reagent. A reagent reservoir is formed within the reaction chamber and contains molybdenum trioxide nanoparticles dispersed in a sulfuric acid activation medium. A sample inlet is arranged at an upper portion of the reaction chamber to receive a microlitre-volume portion of the liquid sample. A mixing zone is disposed below the sample inlet and fluidly connected with the reagent reservoir to bring the liquid sample into contact with the dispersed molybdenum trioxide nanoparticles. A viewing window is defined by the optically viewable wall portion and aligned with the mixing zone to display a visible colorimetric transition generated in presence of ethanol. Fig. 1 Dated July 11, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 35 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/07/2026

(21) Application No.202611087592 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A RESOURCE-CONSTRAINED MULTI-LEAD ECG SYSTEM FOR ROBUST HEART RATE VARIABILITY ANALYSIS

(51) International classification	:A61B 5/352, A61B 5/00, A61B 5/024, A61B 5/364, A61B 5/0402
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. ROHIT SAROHA
Address of Applicant :SANTOSH DEEMED TO BE UNIVERSITY, NO.1, SANTOSH NAGAR, PRATAP VIHAR, GHAZIABAD, UTTAR PRADESH - 201009, INDIA Uttar Pradesh India
2)SANTOSH DEEMED TO BE UNIVERSITY
(72)**Name of Inventor :**
1)DR. ROHIT SAROHA
2)DR. SONI SINGH
3)DR. MUNEEB KOSVI
4)DR. CHETNA CHHABRA
5)MR. GAURI MUDGAL
6)MS. PRABHA KUMARI

(57) Abstract :
A multi-lead ECG-based heart rate variability analysis system is disclosed for resource-constrained embedded environments. An analog front-end achieves CMRR ≥100 dB and implements an analog high-pass filter at ≤0.05 Hz, preserving VLF (0.003-0.04 Hz) and LF (0.04-0.15 Hz) HRV content. A digital high-pass filter at ≤0.05 Hz in the DSP pipeline implements SOS-structured fixed-point IIR for numerical stability. A synchronous ADC achieves ENOB ≥14 bits at 250-1000 sps. A signal quality assessment engine computes a Lead Quality Index per lead using TP-segment SNR and six additional metrics; an adaptive lead selection module routes HRV computation to the highest-quality source. VLF power is computed exclusively from long-term recordings (≥18 h); LF and HF from short-term windows (5 min). Non-linear parameters include SD1, SD2, SD1/SD2 ratio, DFAa1/a2, and SampEn. Patient safety complies with IEC 60601-1 CF-type specifications.

No. of Pages : 18 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202611089369 A

(19) INDIA

(22) Date of filing of Application :22/07/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SMART MODULAR MULTI-BED DESICCANT HUMIDITY CONTROL SYSTEM AND METHOD THEREOF

(51) International classification	:B01D 53/26, F24F 3/14, F24F 11/00, G05D 22/02, B01D 53/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Dr. Anusha Vohra
2)Dr. Karan Kumar
3)Dr. Nipun Lamba

(57) Abstract :
The present invention discloses a smart modular multi-bed desiccant humidity control system (100) and a method (200) for continuously maintaining ultra-low relative humidity within a sample chamber (101). Smart modular multi-bed desiccant humidity control system (100) comprises a sample chamber (101), a circulation fan (103), a rotary distributor valve (105), a plurality of slots (106) configured to receive desiccant bed cartridges (107), a shared heat pump manifold (109), a heat pump (110), a vacuum pump (111), and a controller (114). Controller (114) performs predictive cartridge selection, adsorption capacity monitoring, airflow switching, and simultaneous regeneration, thereby enabling uninterrupted humidity control with improved energy efficiency and modular cartridge replacement. Refer: Fig. 1

No. of Pages : 36 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :25/08/2026
(21) Application No.202611102713 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : COMBUSTIBLE PLANT-BASED MOSQUITO-REPELLENT BODY AND METHOD OF MANUFACTURE

(51) International classification	:A01N 65/00, A01N 65/28, A01N 65/26, A01N 65/24, A01N 25/00, A01N 25/20, A01N 25/34, A01P 17/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Pradnya Pravin Tiwad
Address of Applicant :Assistant Professor, Research Guide JJT University Rajasthan Chudela Rajasthan India
2)Ms. Vandana Anuj Kedia
3)Mrs. Suvarna Manoj Singh
4)Ms. Nandini Ramesh Shrote
5)Ms. Sandeepa Manohar Waghmare
6)Ms. Pramila Subhash Yadav
(72)**Name of Inventor :**
1)Dr. Pradnya Pravin Tiwad
2)Ms. Vandana Anuj Kedia
3)Mrs. Suvarna Manoj Singh
4)Ms. Nandini Ramesh Shrote
5)Ms. Sandeepa Manohar Waghmare
6)Ms. Pramila Subhash Yadav

(57) Abstract :
A combustible plant-based mosquito-repellent body includes a base oil, a plant-derived repellent extract, and one or more botanical additives. The base oil, the plant-derived repellent extract, and the one or more botanical additives are blended in fixed ratios to form a homogeneous mixture configured to release an aromatic repellent upon combustion. (FIG. 1)

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/01/2026

(21) Application No.202611003955 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PIEZO-ELECTRIC BASED ENERGY HARVESTING SENSOR SYSTEM

(51) International classification	:H01N30/00, H10N30/30, H02N2/18, H10N30/85, H02M7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY ALLAHABAD
Address of Applicant :Prayagraj - 211004, Uttar Pradesh, India Prayagraj Uttar Pradesh India
(72)**Name of Inventor :**
1)VARUN SINGH
2)RAMA SHANKER
3)SOMIT PRATAP SINGH

(57) Abstract :
The present invention provides a piezoelectric based energy harvesting system (100) comprises a plurality of piezoelectric discs (1), a rectifier unit (2), a voltage boosting unit (3), a microcontroller (4) and a storage unit (5). The plurality of piezoelectric discs (1) is installed on a structural surface configured to convert mechanical vibration and pressure vibration on said structural surface into electrical energy. The plurality of piezoelectric discs (1) is electrically coupled to said rectifier unit (2) to convert an AC current generated from said plurality of piezoelectric discs (1) into DC current. The rectifier unit (2) is electrically coupled to said voltage boosting unit (3) configured to generate an amplified output voltage via said microcontroller (4) which generates an optimized pulse width modulation signal (PWM) to dynamically regulate switching operation of said voltage boosting unit (3) to amplify said output voltage to efficiently charge said storage unit (5). Figure 1 on sheet no. 1 of the drawings may accompany the abstract when published.

No. of Pages : 30 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/07/2026

(21) Application No.202611089459 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MILK SEPARATION APPARATUS WITH ADJUSTABLE SEPARATOR ASSEMBLY

(51) International classification	:A01J 11/06, A01J 11/00, A47J 31/44, A01J 11/10, B01D 17/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Anuradha Sharma
2)Varnita Prabhakar
3)Dr. Bushra Fiza

(57) Abstract :
A milk separation apparatus (100) for separating and selectively dispensing cream and skimmed milk from raw or non-homogenized milk and a method (200) are disclosed. The apparatus (100) comprises a transparent cylindrical container body (102) for holding milk, a removable lid (108) for supporting a removable separator assembly (110), and the separator assembly (110) including an actuator knob (110a) for adjustment, a threaded guide rod (110b) for axial movement, a lock collar (111) for positional locking, and a buoyancy-assisted separator baffle (112). The separator baffle (112) comprises a separator baffle core plate (112b), an annular float ring (112c) for buoyant positioning, a radial expansion wedge/cam mechanism (112d), an expandable peripheral sealing lip (112e) for sealing, and a lower stabilizer hub (112f) for alignment. An anti-vortex drain feature (114) reduces turbulence during dispensing, thereby enabling controlled separation, reliable sealing, hygienic dispensing, and easy cleaning. Referring Fig.1

No. of Pages : 30 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/08/2026
(21) Application No.202611103220 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Reduced Graphene Oxide–P3HT Composite Photodetector

(51) International classification	:G16B40/20, G06N3/0455, G16B5/00, G06N20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Noida International University
Address of Applicant :Noida International University, Plot 1, Yamuna Expy, Sector 17A, Greater Noida, Uttar Pradesh 203201 Uttar Pradesh India
(72)**Name of Inventor :**
1)Ankita Saxena
2)Minakshi Sharma
3)Pooja Singh
4)Pradyut Anand
5)Rahul Kumar Singh

(57) Abstract :
The present invention relates to a computational framework for reliability analysis of graph neural network-based disease prediction systems. The framework evaluates graph neural network models used for analysing genomic, transcriptomic, and biological interaction data. The system extracts model characteristics, analyses optimization and testing procedures, detects potential data leakage conditions, and evaluates relationships between model characteristics and prediction performance. The framework analyses node-level and graph-level prediction tasks and identifies factors influencing model performance. The invention is applicable in bioinformatics, computational genomics, pharmaceutical research, precision medicine, and artificial intelligence-assisted biomedical analysis. The system supports improved evaluation, reproducibility, and reporting practices for graph neural network-based biological prediction models.

No. of Pages : 14 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :25/06/2026

(21) Application No.202611078473 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : METAL COORDINATION COMPLEX

(51) International classification :C07F1/08
(31) Priority Document No :NA
(32) Priority Date :NA
(33) Name of priority country :NA
(86) International Application No :
Filing Date :01/01/1900
(87) International Publication No : NA
(61) Patent of Addition to Application Number :NA
Filing Date :NA
(62) Divisional to Application Number :NA
Filing Date :NA

(71)**Name of Applicant :**
1)SHARDA UNIVERSITY
Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)RAM CHANDRA SINGH
2)AFSAR ALI
3)BHASKAR BANERJEE
4)SRISTI KUMARI
5)RAJANEESH KUMAR VERMA
6)VINAY KUMAR VERMA
7)GARIMA CHOUHAN

(57) Abstract :
Disclosed is a metal coordination complex represented by Formula I, [M(κ²-N,O-L¹)(κ²-N,O-L²)], wherein M is a divalent metal ion and L¹ and L² are first and second monoanionic selenium-bearing Schiff-base ligands. Each ligand comprises an organoselenium group linked through an alkylene chain to an azomethine group and an arylene phenolate donor portion. L¹ coordinates with M through an azomethine nitrogen donor atom and a phenolate oxygen donor atom of the first ligand, and L² coordinates with M through an azomethine nitrogen donor atom and a phenolate oxygen donor atom of the second ligand. The complex provides a defined nitrogen-oxygen bidentate coordination arrangement around the divalent metal ion while incorporating selenium-bearing organic substituents within the ligand framework. Dated June 20, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/07/2026
(21) Application No.202611091550 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AGGLOMERATED NATURAL SWEETENER COMPOSITION AND A PROCESS FOR THE PREPARATION THEREOF

(51) International classification	:A23L 27/00, A23L 27/30, A23L 27/12, A23P 10/20, A23P 10/22, A23L 33/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ARBOREAL BIOINNOVATIONS PRIVATE LIMITED
Address of Applicant :55, 2ND FLOOR, LANE NO. 2, WESTEND MARG SAIDULAJAB, NEW DELHI- 110030, INDIA NEW DELHI Delhi India
(72)**Name of Inventor :**
1)SWATI PANDEY
2)MANISH CHAUHAN
3)SURESH CHANDER CHANDRASEKAR RAJENDRAN

(57) Abstract :
The present invention relates to an agglomerated natural sweetener composition comprising a porous agglomerated matrix formed from a plurality of natural fibres and a synergistic sweetener blend distributed within and coated onto the porous matrix. The sweetener blend comprises at least one of a Stevia rebaudiana extract and a monk fruit extract, a primary taste modulator comprising one of a mushroom extract and a yeast extract, and a secondary botanical taste modulator. The composition is present as fluid-bed agglomerated granules having a particle size ranging from 100 microns to 1200 microns, bulk density from 0.2 g/cm³ to 0.7 g/cm³, porosity from 25% to 75%, and water absorption index from 2 g/g to 5 g/g. The composition exhibits improved dispersibility, rapid dissolution, reduced bitterness, enhanced mouthfeel, and stability under processing conditions. The invention also relates to a fluid-bed agglomeration process for preparing the composition. FIG. 1

No. of Pages : 35 No. of Claims : 22

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/08/2026

(21) Application No.202611103900 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for Advanced Driver Monitoring in Intelligent Vehicles Using Millimeter-Wave Radar

(51) International classification	:B60W40/08, G01S13/88, G06V40/20, G06F18/24, G06N20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Malaviya National Institute of Technology Jaipur
Address of Applicant :Jawahar Lal Nehru Marg, Jhalana Gram, Malviya Nagar, Jaipur, Rajasthan 302017 , India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Anand Mohan
2)Hemant Kumar Meena
3)Mohd Wajid
4)Abhishek Srivastava

(57) Abstract :
ABSTRACT System and Method for Advanced Driver Monitoring in Intelligent Vehicles Using Millimeter-Wave Radar The present subject matter refers method and system (500) for driver drowsiness detection using mmWave radar. The system (500) implements a method for transmitting (101) mmWave signals by a transceiver toward a vehicle driver and capturing reflected radar signals, and generating (102) three-dimensional radar point cloud data by the transceiver and converting the point cloud into a two-dimensional top-View point cloud images. The Top-View point cloud images are processed (104-122) using Graph Fractional Fourier Transform (GFrFT) by a signal processing module implementing Top-View filtration for clutter suppression and Graph Signal Processing (GSP) techniques including Graph Fractional Fourier Transform (GFrFT) to extract discriminative graph-domain features. The driver's alertness state is classified (124) in real time by a lightweight machine learning classifier trained on GFrFT. Fig. 1

No. of Pages : 27 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/07/2026

(21) Application No.202611088636 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EHV-1 BASED RECOMBINANT VECTOR VACCINE CONSTRUCT EXPRESSING VP2 OF INFECTIOUS BURSAL DISEASE VIRUS (IBDV)

(51) International classification	:C12N 15/86, C07K 14/08, A61K 39/00, A61K 39/12	(71)**Name of Applicant :** **1)INDIAN COUNCIL OF AGRICULTURAL RESEARCH** Address of Applicant :Krishi Bhawan, Dr. Rajendra Prasad Road, New Delhi - 110001, Delhi, India New Delhi Delhi India **2)LALA LAJPAT RAI UNIVERSITY OF VETERINARY & ANIMAL SCIENCES** (72)**Name of Inventor :** **1)BERA, Bidhan Chandra** **2)NARANG, Gulshan** **3)VIRMANI, Nitin** **4)KANDPAL, Diksha** **5)ANAND, Taruna** **6)JANGIR, Babu Lal** **7)JOSHI, Vinay G.** **8)CHHABRA, Rajesh**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a recombinant Equine Herpesvirus-1 (EHV-1) based vectored vaccine and methods for its preparation. The invention provides a genetically engineered EHV-1 construct comprising deletion of virulence-associated genes including IR6 and glycoprotein E, and insertion of a codon-optimized VP2 gene of Infectious bursal disease virus. The recombinant genetic construct is generated using bacterial artificial chromosome-based markerless two-step en-passant mutagenesis. The invention further provides a recombinant EHV-1 vector vaccine virus capable of inducing protective immune responses against IBDV infection in chicken.

No. of Pages : 51 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/08/2026

(21) Application No.202611104685 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NEXT-GENERATION CONTROLLED RELEASE PLATFORM FOR ERYTHROMYCIN DERIVATIVES WITH ENHANCED

(51) International classification	:C07H 17/08, A61P 31/04, A61K 31/7048, A61K 9/22, A61K 9/50
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SEEMA MAHAJAN
Address of Applicant :308-C BHAI RANDHIR SINGH NAGAR LUDHIANA PUNJAB Punjab India
2)Vega Mahajan
(72)**Name of Inventor :**
1)SEEMA MAHAJAN
2)Vega Mahajan

(57) Abstract :

The present invention relates to a next-generation controlled-release pharmaceutical 10 platform for erythromycin derivatives. More particularly, the invention provides controlled-release pharmaceutical compositions comprising at least one erythromycin derivative together with one or more pharmaceutically acceptable release-controlling polymers, binders, fillers, lubricants, stabilizers, optional pH-modifying agents, and optional coating materials. The pharmaceutical compositions are formulated to 15 provide prolonged and controlled release of the active pharmaceutical ingredient following oral administration, thereby maintaining therapeutic drug availability over an extended period while reducing fluctuations in drug release. The invention encompasses a variety of controllec ·release technologies including hydrophilic polymer matrices, hydrophobic matrices, polymer-coated pellets, reservoir systems, 20 osmotic delivery systems, multilayer tablets, multiparticulate formulations, lipid-based controlled-release systems, ion-exchange resin systems, microsphere technology, nanoparticle-assisted delivery systems, and combinations thereof. The compositions may be formulated as oral tablets, capsules, bead-filled capsules, pellet-filled capsules, sachets, granules, oral suspension granules, pediatric sprinkle formulations, extended- 25 release pellets, matrix systems, or other modified-release oral dosage forms. The pharmaceutical compositions are designed to improve gastrointestinal stability, enhance physicochemical stability during storage, reduce dosing frequency, improve patient compliance, and provide reproducible release characteristics while remaining suitable for scalable commercial pharmaceutical manufacturing using conventional 30 pharmaceutical processing technologies. The invention further relates to methods of preparing such controlled-relea5:e pharmaceutical compositions

No. of Pages : 23 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/08/2026

(21) Application No.202611104686 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE NEURO-CARE COSMETIC COMPOSITION WITH BIOMIMETIC PHYTOLIPID NANOSTRUCTURES FOR RESTORATION

(51) International classification	:A61Q 19/00, A61Q 19/08, A61K 8/67, A61K 8/64, A61K 8/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SEEMA MAHAJAN
Address of Applicant :308-C BHAI RANDHIR SINGH NAGAR LUDHIANA PUNJAB Punjab India
2)Vega Mahajan
(72)**Name of Inventor :**
1)SEEMA MAHAJAN
2)Vega Mahajan

(57) Abstract :
The present invention relates to an adaptive neuro-care cosmetic composition comprising biomimetic phytolipid nanolamellar nanostructures for improving the cosmetic appearance of keratinous tissues. The composition com;:.rises a biomimetic phytolipid nanoemulsion containing hydrogenated phosphatidylcholine, sunflower phospholipids, phytosterols, ceramide precursors, plant-derived squalane, adaptogenic botanical extracts, ectoin, trehalose, triple molecular weight hyaluronic acid, fermented probiotic lysates, bioactive peptides, natural antioxidants, skin-compatible polysaccharides, silica microspheres, and physiologically acceptable cosmetic carriers. The biomimetic nanolamellar vesicles, having an average particle size of about 70-180 nm, simultaneously encapsulate hydrophilic and lipophilic cosmetic ingredients to enhance formulation stability, improve cosmetic deposition, provide prolonged moisturization, enhance spreadability, improve visible skin smoothness and radiance, support a healthy-looking skin barrier and microbiome, and provide a lightweight, non-greasy sensory profile. The invention further provides a scalable manufacturing process comprising high-shear emulsification, high-pressure homogenization, controlled cooling, incorporation of heat-sensitive cosmetic ingredier,ts, pH adjustment, vacuum deaeration, and packaging to produce a stable, high-performance cosmetic formulation suitable for creams, lotions, seruMs, balms, beard care, hair care, scalp care, lip care, and other topical cosmetic products.

No. of Pages : 18 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/07/2026

(21) Application No.202611091027 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ZONE-BASED DISEASE MANAGEMENT SYSTEM FOR AN AGRICULTURAL FIELD

(51) International classification	:G06Q 10/06, G06Q 50/02, A01M 21/04, G05B 19/418, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Amrita Vishwa Vidyapeetham
Address of Applicant :Amrita Vishwa Vidyapeetham, Delhi NCR Faridabad Campus, Amrita Hospital, Mata Amritanandamayi Marg, Sector 88, Faridabad, Haryana - 121002, India. Faridabad Haryana India
(72)**Name of Inventor :**
1)TARAFDAR, Anirban
2)CHOKKARA, Mounika
3)KOPPU, Neeraj
4)KAMMA, Srikar
5)SINGAM, Hemanth Veera Kumar

(57) Abstract :
The present disclosure provides a disease management system (100) for a field partitioned into a plurality of zones (102) and a disease management method (300) for the field. The system (100) includes a sensor (104) within each zone (102-2) that outputs an image of a plant (106) and an environmental parameter, a processor (108) that populates a risk multiplier (212) as an exponential of a factor of the environmental parameter and entries of a transition matrix (210) on the risk multiplier (212), and an applicator (112) that dispenses an application agent on the plant (106) at a volatility score (218) above a threshold value. Unlike conventional calendar-scheduled broadcast spraying that wastes chemicals on healthy regions, this approach confines the dispensing of the application agent to the zone (102-2) anticipated to escalate, enabling targeted crop protection across the field.

No. of Pages : 33 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/07/2026

(21) Application No.202611091153 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD OF PREPARATION OF 1,8-NAPHTHALIMIDE–NICKEL NANOCLUSTER NANOHYBRID MATERIAL AND ITS APPLICATION AS ANTIBODY-FREE GLUTEN DETECTION

(51) International classification	:G01N 21/64, C09K 11/06, G01N 33/68, G01N 33/58, G01N 33/533
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NAVNEET KAUR
Address of Applicant :Department of Chemistry, Panjab University, Chandigarh-160014, India Chandigarh Chandigarh India
2)NISHA JAIN
(72)**Name of Inventor :**
1)NAVNEET KAUR
2)NISHA JAIN

(57) Abstract :
The present invention relates to a nanohybrid material comprising 1,8-naphthalimide and nickel nanoclusters, a method for synthesizing the same, and its application in sensing for gluten detection. The nanohybrid material includes bovine serum albumin (BSA) stabilized nickel nanoclusters surface-conjugated with a 1,8-naphthalimide derivative to form a fluorescent nanoprobe. The controlled organic-inorganic interface allows for the modulation of fluorescence emission, which is measurable after interaction with gluten proteins. The present invention further provides a reproducible method of chemical synthesis, which includes the preparation of the fluorophore, capping of BSA-capped nickel nanoclusters, and subsequent conjugation. The nanohybrid material is useful for the fast, sensitive, and antibody-free detection of gluten and can be extended for integration in optical sensing arrangements in food and biological samples.

No. of Pages : 23 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/09/2026

(21) Application No.202611105164 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MODULAR EDGE-BASED ARTIFICIAL INTELLIGENCE RETROFIT SYSTEM FOR LEGACY INFRASTRUCTURE

(51) International classification	:G05D 101/10, H04N 7/167, H01M 8/04992, F21S 2/00, G10L 21/0364
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rewright AI Private Limited
Address of Applicant :Flat No.3, Building Bilal Complex, Road/Street-Opposite Sheerwani Hall Baramulla Jammu & Kashmir India 193101 Baramulla Jammu & Kashmir India
(72)**Name of Inventor :**
1)Mr. Ovais Nazir
2)Mr. Aaqib Zahoor
3)Ms. Ayna Zahoor
4)Dr. Shabir Ahmad Sofi

(57) Abstract :
The present invention relates to a modular edge-based artificial intelligence retrofit system (100) for enabling intelligent monitoring, predictive analytics, adaptive learning, explaina-ble artificial intelligence, and decision support in legacy infrastructure (102) without replac-ing existing equipment. The system (100) comprises an embedded computing platform (202), a universal interface manager (104, 302), a data acquisition module (106, 304), a data processing and feature extraction module (306), an edge artificial intelligence inference en-gine (108, 308), an adaptive learning module (110, 310), an explainable artificial intelli-gence module (112, 312), and an intelligent decision support module (114, 314). The system performs protocol detection, operational data acquisition, local artificial intelligence infer-ence, adaptive model optimisation, explainability generation, and communication of action-able recommendations, thereby reducing latency, preserving data privacy, extending the op-erational life of legacy infrastructure, and enabling deployment across multiple application domains through software-based reconfiguration.

No. of Pages : 35 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/07/2026

(21) Application No.202611086966 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ANTI-MOLD BAKERY PRODUCT STORAGE BOX

(51) International classification	:B65D81/24, B65D81/26, B65D81/28, B65D81/18, B65D81/20, B65D85/36
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Dr. Vineeta Bansal
2)Dr. Nikhil Marwah
3)Dr. Vidhi Gupta (PT)

(57) Abstract :
The present invention relates to an anti-mold bakery product storage box (100) for reducing mold formation and preserving freshness of bakery products (102a) during storage. The storage box (100) comprises an outer housing (101) defining a storage chamber (102) with a perforated bakery product support tray (103) and a crumb collection region (104). Labyrinth airflow channels (105) enable passive circulation of moisture-laden air toward a moisture-conditioning cartridge (106) having a first absorbent stage (106a) and a second buffer stage (106b) for humidity regulation. An upper cover (107) with a sloped condensation-diverting surface (108) directs moisture into side collection gutters (109) and an isolated moisture collection sump (114). Humidity-responsive louvers (111) and a cartridge exposure shutter (113) are operated by a humidity-responsive mechanical actuator (112) comprising a hygroscopic bi-layer strip which bends in response to humidity variations to regulate airflow and cartridge exposure, maintaining controlled humidity and reducing mold growth. Referring Fig.1

No. of Pages : 30 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202611086987 A
(19) INDIA
(22) Date of filing of Application :16/07/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : MAGNETIC TIP LATCH ASSEMBLY FOR A MICROPIPETTE

(51) International classification	:B01L 3/02, G01N 35/10, B01L 9/00, B01L 3/00, G01N 35/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Dr. Shaveta Kataria
2)Dr. Shalu Sharma
3)Dr. Tanjul Saxena

(57) Abstract :
The present invention relates to a magnetic tip latch assembly (100) integrated within a micropipette shaft (102) for releasably coupling a disposable pipette tip (200) to a micropipette. The magnetic tip latch assembly (100) comprises a central air-displacement bore (104) for fluid communication, a tip coupling interface (106) for receiving the disposable pipette tip (200), at least one permanent magnetic holding element (108) for magnetic retention, at least one electromagnetic release coil (110) for selective tip release, one or more elongated magnetic pole pieces (112) and a magnetic return sleeve (114) for directing and confining magnetic flux, one or more thermal isolation barriers (116) for reducing heat transfer, and a magnetic field sensor (118) with a controller for verifying tip seating and controlling release. The invention provides low-force tip attachment, substantially zero-power retention, thermally stable pipetting, and controlled contactless tip release. Referring Fig.1

No. of Pages : 28 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/09/2026
(21) Application No.202611106466 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Biodegradable Foldable Paper Sun-Protection Hat with Reinforced Crown, Ventilation and Extended Brim

(51) International classification	:B60C 9/18, A42B 1/201, A42C 1/00, A42B 1/02, A42B 1/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Love Kumar
Address of Applicant :Department of Mechanical Engineering, IIMT College of Engineering, Greater Noida, Uttar Pradesh, India 201310 Uttar Pradesh India
2)Prof.(Dr) Hari Om Sharma
3)Dr. Radha Rani
4)Dr. Shikha Dixit
5)Dr. Rajiv Kumar Sharma
6)Er. Pankaj Sharma
7)Sanjeev Kumar
8)Dr. Bodhisatwa Seal
9)Dr. Shradha Dixit
10)Er. Mohit Agarwal
(72)**Name of Inventor :**
1)Dr. Love Kumar
2)Prof.(Dr) Hari Om Sharma
3)Dr. Radha Rani
4)Dr. Shikha Dixit
5)Dr. Rajiv Kumar Sharma
6)Er. Pankaj Sharma
7)Sanjeev Kumar
8)Dr. Bodhisatwa Seal
9)Dr. Shradha Dixit
10)Er. Mohit Agarwal

(57) Abstract :
The present invention relates to a biodegradable foldable sun-protection hat formed substantially from a biodegradable sheet material. The hat comprises a crown portion configured to cover a user's head, an outwardly extending brim portion configured to reduce direct solar exposure, predetermined folding lines for converting a substantially planar blank into a three-dimensional hat configuration, fixing portions for retaining the assembled configuration, and reinforced structural regions positioned at selected portions of the hat. Ventilation openings may be provided in the crown portion to permit air movement. An environmentally compatible protective coating may optionally be applied to selected surfaces for resistance to moisture and sweat. The sheet may be die-cut to form the crown, brim, fixing portions, ventilation portions and reinforcement regions. The hat may be supplied in a substantially flat-folded condition and subsequently assembled by folding and securing corresponding tabs, slots or adhesive regions. The configuration is suitable for temporary outdoor sun protection.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :05/09/2026

(21) Application No.202611107012 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Dual-Provenance Navigational Join-Graph Index with Generation-Gated Serving, and Declarative Semantic Cascade Dataflow

(51) International classification	:F24H 15/325, D06F 33/06, D06F 33/08, D06F 33/10, G07C 9/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Manjeet
Address of Applicant :C34/1 Second Floor Ardee City , Sector -52 Haryana India
(72)**Name of Inventor :**
1)Arun Kumar

(57) Abstract :
A database system executes a relational join by constructing, as a byproduct of a sort-merge pass it is already performing and at substantially no additional asymptotic cost, a dense positional array mapping each child row ordinal to its parent row ordinal, containing no hash table, with unmatched children written to a reserved sentinel. The array is published to a process-global registry keyed by a direction-normalized identity of catalog column identifiers, together with the mutation generation of each endpoint table at capture time. On a subsequent join, the array is read only when the live mutation generations of **both** endpoint tables exactly equal the recorded generations; otherwise the engine declines and re-executes the join over current data, making a stale serve structurally impossible. The same identity also admits operator-declared relationship indexes, whose asserted cardinality is verified against live data at declaration time and which additionally carry a sorted parent-key array and a compressed-sparse-row fan-out, so a join filtered to a point, range, or membership list of keys is answered by binary search plus a constant-time offset hop in time logarithmic in parent count plus linear in matches — scanning neither table and never opening the child input operator. The system further provides a reactive cascade dataflow in which columns tagged with semantic dimension and measure roles drive an automatically generated aggregation whose result atomically replaces a child table by pointer swap, and whose transformation executes as a WebAssembly module whose linear memory is a borrowed, non-copying view over the parent's memory-mapped storage segment, confined to a shadow output region by a host import surface exposing only output-offset, output-length, and logging imports.

No. of Pages : 24 No. of Claims : 34

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :05/09/2026
(21) Application No.202611107049 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR PREDICTING TIME-TO-ONSET OF STRESS-INDUCED PHYSIOLOGICAL DISORDERS USING AN ALERTNESS-MODULATED STOCHASTIC CUMULATIVE-DAMAGE MODEL

(51) International classification	:H04N 19/52, G07C 9/15, A61P 11/04, A61P 15/02, A61P 15/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Avdhesh Gupta
Address of Applicant :Ajay Kumar Garg Engineering College, Ghaziabad Uttarakhand India
2)Dr. Paras Bhatnagar
3)Dr. Ruchi Nager
4)Dr Nigam Kumar
5)Dr. Anushree Verma
6)Dr. Nikunj Agarwal
(72)**Name of Inventor :**
1)Dr. Paras Bhatnagar
2)Dr. Ruchi Nager
3)Dr Nigam Kumar
4)Dr. Anushree Verma
5)Dr. Nikunj Agarwal
6)Dr. Avdhesh Gupta

(57) Abstract :
A wearable physiological-monitoring system and an associated computer-implemented method are disclosed for predicting, on an individualized basis, the time remaining before onset of a stress-induced physiological disorder in a monitored subject. The system employs a physiological sensor, such as a photoplethysmography or electrocardiographic sensor, to acquire a cardiovascular signal from which discrete, emotionally-triggered stress episodes associated with anger and/or fear are detected by a processor. For each detected episode, the processor computes a numerical alertness index representative of the subject's real-time physiological and cognitive regulatory capacity, derived from heart-rate-variability coherence features extracted from the acquired signal. An estimated physiological-damage magnitude is then assigned to the episode using a predetermined function that decreases as the alertness index increases, reflecting the relationship between effective coping and reduced catecholamine-mediated physiological wear. Successive damage magnitudes are accumulated in memory as a compound stochastic process to obtain a running cumulative-damage value, which is compared against a stored disease-onset threshold. From the subject's estimated stress-episode arrival rate, alertness-dependent damage history, and applicable threshold, the system computes a predicted expected time-to-disease-onset together with a variance value quantifying the statistical uncertainty of that prediction, both of which are continuously updated and presented to the subject or an authorized third party through an output interface. Where warranted, the system may additionally trigger a real-time coping intervention through an optional biofeedback actuator. By coupling a minimal, low-power physiological sensor with a stochastic cumulative-damage computation engine — rather than relying on multiple sensing modalities or cloud infrastructure — the invention enables individualized, statistically grounded, preventive stress-risk monitoring not achievable using conventional threshold-based physiological monitoring devices.

No. of Pages : 18 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/09/2026
(21) Application No.202611105607 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PRESERVATION STRIP FOR MICROBIAL STORAGE AND A METHOD FOR PREPARING THE SAME THEREOF

(51) International classification	:C08L 1/02, C08L 5/00, C07H 3/04, B01D 69/10, C12N 1/04, A01N 1/02, C12M 1/00, C12Q 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Council of Agricultural Research
Address of Applicant :Krishi Bhawan, Dr. Rajendra Prasad Road, New Delhi - 110001, India. New Delhi Delhi India
(72)**Name of Inventor :**
1)VAIYAPURI, Murugadas
2)ANBALAGAN, Muthukumar
3)RAVEENDRAN, Karthika
4)RAVIKUMAR, Neethu Koothuparambu
5)KOOMBANKALLIL, Reshmi
6)DEVADAS, Ammu Lakshmi
7)RAJAN, Preenanka
8)KUSUNUR, Ahamed Basha
9)SUMATHI, Ezhil Nilavan
10)SHEELA, Greeshma Sadanandan
11)JOSEPH, Toms Cheriyath
12)SIVAM, Visnuvinayagam
13)BADIREDDY, Madhusudhana Rao
14)THANGARAJ, Raja Swaminathan
15)RATHORE, Gaurav
16)NINAN, George
17)JENA, Joy Krushna

(57) Abstract :
The present disclosure relates to an engineered preservation strip and a method of preparing the same for preservation and transport of viable microorganisms under refrigerated and non-refrigerated conditions. The engineered preservation strip comprises a biopolymer–saccharide matrix configured to immobilize and stabilize microorganisms while maintaining viability during storage and transportation. The strip may further comprise one or more stabilizing agents, buffering agents, protective nutrients, or antimicrobial balancing components to enhance structural integrity and microbial survival. The method of preparation comprises preparing a biopolymer solution, incorporating at least one saccharide and viable microorganisms therein, casting or molding the composition into strip form, and drying the strip under controlled conditions to obtain a flexible preservation strip. The engineered preservation strip enables convenient handling, reduced contamination risk, prolonged shelf life, improved transport stability, and rapid microbial recovery suitable for diagnostic, industrial, environmental, research, and healthcare applications.

No. of Pages : 33 No. of Claims : 23

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/09/2026

(21) Application No.202611105678 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FITAURA- FIT INTO YOUR AURA

(51) International classification	:G06Q30/0601, G06T19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Pranveer Singh Institute of Technology, Kanpur
Address of Applicant :Pranveer Singh Institute of Technology, National Highway (NH-19), Kanpur, Bhautipratappur, Uttar Pradesh 209305 College Code: 164 Kanpur Uttar Pradesh India
(72)**Name of Inventor :**
1)Mr. Shubham Nigam
2)Prof. Chandra Kumar Dwivedi
3)Dr. Dinesh Kumar Pandey
4)Mr. Sharad Gupta
5)Samriddhi Upadhyay
6)Rakhi Singh
7)Samriddhi Verma
8)Richa Tiwari

(57) Abstract :
FITAURA – Fit into Your Aura is an AI-enabled fashion technology platform designed to enhance the online fashion shopping experience through virtual visualization and intelligent fashion decision-making. The platform's core feature is a 360° AR-based virtual try-on system designed to enable users to visualize apparel themselves before making a purchase. In addition, FITAURA offers digital wardrobe management and personalized fashion suggestions. The system looks at what a user already owns and compares it with new looks at what a user already owns and compares it with new products to find similar items, helping them avoid buying things they don't need or already have. It can also find products that go well with existing clothes and help users find the right items to finish an outfit. By using virtual try-on, smart wardrobe insights, personalized styling, similarity checks, and product discovery, FITAURA aims to make online fashion shopping more informed, tailored and efficient.

No. of Pages : 13 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/09/2026

(21) Application No.202611105724 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Loknirman Sahbhag – From Gram to Growth COMMUNITY-BASED RURAL INFRASTRUCTURE INVESTMENT WITH STRUCTURED RETURN MECHANISM

(51) International classification	:B41J 19/68, B41J 19/36, G01L 23/02, H01H 50/34, B41J 19/18	(71)**Name of Applicant :** **1)Pranveer Singh Institute of Technology, Kanpur** Address of Applicant :Pranveer Singh Institute of Technology, National Highway (NH-19), Kanpur, Bhautipratappur, Uttar Pradesh 209305 College Code: 164 Kanpur Uttar Pradesh India (72)**Name of Inventor :** **1)Dr. Ranjana Tiwari** **2)Ms. Stuti Jain**
(31) Priority Document No	:NA	**3)Mr. Karan Upadhyay**
(32) Priority Date	:NA	**4)Dr. Deepak Shukla**
(33) Name of priority country	:NA	**5)Prof. Ruchika Rastogi**
(86) International Application No Filing Date	: :01/01/1900	**6)Utkarsh Mishra** **7)Vivek Yadav**
(87) International Publication No	: NA	**8)Vijeta**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**9)Vaishnavi Chaudhary**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The invention of the month is a system that allows citizens from rural communities to pool their small contributions in order to finance infrastructure development projects. When individuals contribute financially, they form a substantial amount of money that will be invested in solar power plants, road development, irrigation, provision of water, and internet connection. The invention provides a way for individuals to easily come together, pool their investments, and choose the projects in which they will invest. After projects have been developed, the system monitors the revenue generated and manages the finances.

No. of Pages : 9 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :07/09/2026

(21) Application No.202611107417 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-CAMERA BASED SMART SECURITY SYSTEM WITH SOLAR-POWERED ELECTRIC-SHOCK BARRIER AND AUTOMATIC

(51) International classification	:G08B 13/196, H04N 5/225, H04N 7/18, H04N 5/232, H02J 7/35
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GURMIT SINGH BAGGA
Address of Applicant :FLAT NO. 103, TOWER SUN-8, MIGSUN TWINZ, SECTOR ETA-2, GREATER NOIDA-201310, UTTAR PRADESH, INIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)GURMIT SINGH BAGGA

(57) Abstract :
AnAl-camera based, solar-powered smart security system for building openings is disclosed, providing three integrated embodiments: a smart security window with a built-in wall air-conditioning unit, a smart sliding glass door, and a smart security entry gate. The window and door embodiments each comprise a double-glazed toughened-glass panel assembly having a 29-gauge electric shock wire embedded between an outer and an inner glass panel, camera/sensor based motion detection, solarcharged battery backup, and remote/mobile-application control; the window embodiment further integrates a concealed, wall-mounted air-conditioning unit providing both hot and cold air flow. The entry gate embodiment comprises an AI camera performing face recognition, weapon scanning and blacklist matching, coupled to a control unit that automatically deploys a corresponding number of net assemblies to physically capture one or more detected persons identified as a threat, particularly for use at supermarket and airport access points. All embodiments share a common solar-power, batterybackup, and remote-monitoring architecture enabling continuous, energy-efficient, tamper-alerted operation.

No. of Pages : 16 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/09/2026

(21) Application No.202611106632 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MACHINE LEARNING BASED SYSTEM AND METHOD FOR AUTOMATED CLASSIFICATION OF RICE AND AGRICULTURAL CROPS

(51) International classification	:G06N 3/08, G06N 20/00, G06N 20/20, G06N 3/0985, G06N 3/09
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GNIOT (Greater Noida Institute of Technology), Department of Computer Science and Engineering
Address of Applicant :Plot No. 7, Knowledge Park-II, Greater Noida, Uttar Pradesh 201310. Greater Noida Uttar Pradesh India
(72)**Name of Inventor :**
1)Mr. Rahul Jha
2)Mr. Rupesh Yadav
3)Ms. Vagissha Dwivedi
4)Mr. Satyajeet Kumar Sahu
5)Dr. Dhiraj Gupta
6)Dr. Sanjay Kumar
7)Dr. Arun Kumar Singh
8)Mr. Arjun Singh
9)Dr. Rohit Vaid
10)Dr. Sanjay Pratap Singh Chauhan

(57) Abstract :
The invention relates to an image-based machine-learning system for automated classification of rice varieties and other agricultural crops. The system receives a crop image through a user interface and preprocesses the image by operations including resizing and normalization. The processed image is supplied to an Efficient Net-based convolutional neural network for visual feature extraction and classification. In an embodiment, the system classifies rice-grain images into Arborio, Basmati, Ipsala, Jasmine and Karacadag varieties. A Flask-based backend performs inference and communicates a predicted crop class and associated confidence score to a React-based user interface. Classification queries, results and confidence information are persistently stored in a SQLite database. The modular architecture permits extension to additional agricultural crop classes. The disclosed implementation provides an integrated image-acquisition, preprocessing, neural-network inference, result-delivery and record-storage architecture for rapid agricultural crop classification.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :04/09/2026
(21) Application No.202611106652 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR MONITORING BIOLOGICS AN TEMPERATURE-SENSITIVE PHARMACEUTICAL PRODUCTS

(51) International classification	:G05B 19/4063, B60W 50/04, F41H 11/132, G08B 29/02, E21B 47/009
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sharda University
Address of Applicant :Plot No. 32-34, Knowledge Park-III, Greater Noida - 201310, Uttar Pradesh, India. Greater Noida Uttar Pradesh India
(72)**Name of Inventor :**
1)SUNTHARAM, Velayudham Sathiya
2)AGARWAL, Anushk
3)PANDEY, Tanu
4)PANWAR, Abhay

(57) Abstract :
The present disclosure relates to a system (202) for monitoring biologics and temperature-sensitive pharmaceutical products. The system (202) comprises at least one sensor (204) configured to acquire environmental condition data associated with a pharmaceutical package, a controller (206) configured to execute a predictive spoilage algorithm based on at least one degradation model, a display unit (208) configured to display real-time potency information and dynamically updated expiry dates, and a communication module (210) configured to transmit degradation-related information to an external device. The controller (206) processes temperature data using an Arrhenius-based degradation equation to determine a rate of chemical degradation and an estimated remaining potency of the pharmaceutical product. Further, the controller maintains a digital twin representation indicative of a real-time chemical stability state of the pharmaceutical product and dynamically updates the expiry date according to cumulative thermal exposure experienced during storage and transportation.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :09/09/2026

(21) Application No.202611108442 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR CONTROLLING PATIENT-TRANSPORT APPARATUS MOVEMENT BASED ON MEDICAL-LINE CONSTRAINTS

(51) International classification	:G16H 40/63, A61B 6/00, A61B 5/11, A61G 1/02, A61B 6/03
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Dr. Priya Marwah
2)Dr. Vipin Kumar Goyal
3)Dr. Mukesh Sarna

(57) Abstract :
The present invention relates to a medical-line protected patient transport movement control system (100) and method (200) for safely controlling movement of a patient-transport apparatus (101) while a patient remains connected to a plurality of medical lines and associated medical equipment. The medical-line sensing assemblies (102) monitor line-state parameters, the equipment-position sensing assembly (103) determines the position of connected medical equipment, the stretcher-position sensing assembly (104) determines movement parameters, and the steering-angle sensing member (104a) determines steering angle. The trajectory determination module (105) predicts a future trajectory, while the line-specific constraint determination module (106) determines line-specific movement constraints. The restrictive-constraint selection module (107) selects the earliest predicted critical constraint, and the dynamic safe movement envelope generator (108) defines permissible movement limits. The mechanical movement governor (109) progressively restricts movement approaching the critical constraint, thereby reducing unsafe line-state condition or unsafe connected-equipment conditions and protecting connected medical lines and equipment. Refer: Fig. 1

No. of Pages : 35 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/09/2026

(21) Application No.202611109573 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PRIVACY-GATED MULTIMODAL WORKPLACE SENSING SYSTEM FOR EMPLOYEE SATISFACTION ANALYSIS

(51) International classification	:H04N 7/18, G06V 40/16, H04N 7/14, H04N 7/15, H04N 5/232
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)G.L. Bajaj Institute of Technology and Management
Address of Applicant :PLOT NO :2, KNOWLEDGE PARK - III, GREATER NOIDA, GAUTAM BUDDHA NAGAR, UTTAR PRADESH, 201306 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)Dr. Priyanka R. Naagar

(57) Abstract :
Abstract Disclosed is a privacy-controlled workplace sensing system that generates an anonymous employee-satisfaction warning. A sensor head includes a camera, a microphone array, and a consent actuator coupled to a shutter and an electrical contact. In a privacy position, the shutter occludes an optical aperture and the contact deenergizes the microphone array; in an acquisition position, the aperture is exposed and the array is energized. A volatile buffer receives raw video and audio, while a workplace-messaging connector receives job-related text. An on-premises edge server extracts facial and paralinguistic representations and erases the raw media. A fusion engine combines the representations with textual sentiment to produce and smooth a satisfaction score, calculates a rolling measure, and causes a notification output to emit the anonymous warning when the rolling measure crosses a threshold. Dated September 7, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/09/2026

(21) Application No.202611109579 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INCREMENTAL ACCREDITATION EVIDENCE RECONCILIATION AND DOCUMENT GENERATION SYSTEM

(51) International classification	:G06F 17/30, G06Q 10/10, G06F 9/44, G06F 17/00, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)G.L. Bajaj Institute of Technology and Management
Address of Applicant :PLOT NO :2, KNOWLEDGE PARK - III, GREATER NOIDA, GAUTAM BUDDHA NAGAR, UTTAR PRADESH, 201306 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)Dr. Priyanka R. Naagar

(57) Abstract :
Abstract Disclosed is a distributed accreditation-documentation system having source-system interfaces, a graph memory storing accreditation-clause nodes and relationships to permissible evidence types, and a rule compiler producing executable assertions from accreditation requirements. A change-processing engine receives source records changed after a preceding retrieval and maps each changed record to at least one clause node. Execution circuitry runs associated assertions in parallel to identify satisfied and unsatisfied requirements. A document composer maintains report portions linked to the clause nodes, regenerates portions affected by mapped changes, retains unaffected portions, and assembles an updated accreditation submission package. The package includes an evidence-provenance manifest linking generated content to supporting source records, applicable clauses, acquisition times, and assertion results. Dated September 7, 2026 / Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/09/2026

(21) Application No.202611111068 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART AND SUSTAINABLE AUTOMATED ROOF GARDENING SYSTEM FOR URBAN GREEN SPACES

(51) International classification	:A01G 25/16, A01G 27/00, A01G 9/24, E04D 13/18, E03B 3/02	(71)**Name of Applicant :** **1)Thapar Institute of Engineering & Technology** Address of Applicant :Bhadson Rd, Adarsh Nagar, Prem Nagar Patiala Punjab India (72)**Name of Inventor :** **1)DR. SANGITA ROY** **2)Dr. Sukhdev Singh** **3)Dr. Dwarika Nath Ratha**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a smart and sustainable automated roof gardening system for urban green spaces. The system comprises a water tank, a pipeline network, controllable valves, soil moisture and environmental sensors, sprinklers, and an AI-enabled control system for zone-based automated irrigation. The system regulates water supply based on soil moisture, environmental conditions, and weather forecast data and returns residual pipeline water to the water tank, thereby reducing water wastage and stagnation. The system further integrates rainwater harvesting, selective greywater reuse, solar-powered operation with battery backup, mobile-based monitoring and control, bird and pest deterrence, and nutrient and organic-waste management, thereby providing a resource-efficient and sustainable solution for automated urban rooftop gardening.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/08/2026

(21) Application No.202611104017 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN EMBEDDED DUAL-INTERFACE HARDWARE COMPANION APPARATUS AND METHOD FOR NON-ROOT REAL-TIME BIDIRECTIONAL TELEPHONY AND VOIP AI VOICE AGENT/ASSISTANCE

(51) International classification	:G10L 25/81, G10L 25/84, G10L 21/0272, G10L 25/60, G10L 13/033
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)tanush dubey
Address of Applicant :lucknow alambagh Uttar Pradesh India
(72)**Name of Inventor :**
1)Hemant Dwivedi
2)tanush dubey

(57) Abstract :
An embedded dual-interface companion hardware apparatus and method for non-root real-time bidirectional telephony and VoIP artificial intelligence (AI) voice assistance are disclosed. The system includes an embedded microcontroller configured as a composite USB device presenting simultaneous USB Audio Class (UAC 2.0) and USB Communications Device Class (CDC) endpoints. A host mobile application intercepts cellular and VoIP audio without root privileges, resamples audio bidirectionally between 48 kHz OS audio and 16 kHz AI inference audio using a low-latency Catmull-Rom cubic spline interpolation filter, and streams conversational speech with a dynamic 250ms DTX speech-hold VAD gating window. Synthesized AI speech is injected into active calls via paced CDC streaming with 16KB pre-buffering and cooperative event-loop yielding. Asynchronous notification action intent execution provides automated VoIP answering and sub-100ms idempotent ALSA driver teardown.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/09/2026

(21) Application No.202611106083 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PASSIVE HAND REHABILITATION ORTHOSIS WITH INDIVIDUALLY ADJUSTABLE ELASTIC-ASSISTED FINGER AND THUMB EXTENSION

(51) International classification	:A61F5/01, A61H1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Aarav Jain
Address of Applicant :577, Block 20, Ashiana Umang, Mahapura, Jaipur. Rajasthan. PIN - 302026 Rajasthan India
2)Richa Jain
(72)**Name of Inventor :**
1)Aarav Jain

(57) Abstract :
Disclosed is a system (100) for facilitating assisted grasp-release rehabilitation of a hand of a user. The system (100) comprises a wrist and forearm frame assembly (102), articulated finger linkage assemblies (104), a thumb linkage assembly (106), a tensioning subsystem (108), a retention subsystem (110), an anchor loop arrangement (112), distal anchor posts (114), a per-digit tension adjustment arrangement (116), a hyperextension limiting stop (118), and a strap slot arrangement (120). The tensioning subsystem (108) includes elastic tensioning elements extending between the anchor loop arrangement (112) and the corresponding distal anchor posts (114). During voluntary flexion of the fingers and thumb, the elastic tensioning elements are stretched to store elastic energy. Upon reduction of the voluntary flexion force, the stored elastic energy generates an extension-assisting force through the respective linkage assemblies to assist reopening of the digits. The per-digit tension adjustment arrangement (116) enables independent adjustment of extension assistance for respective digits.

No. of Pages : 39 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/09/2026
(21) Application No.202611111373 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Synthesis of Novel Oxindole–Chalcone Derivatives via Claisen–Schmidt Condensation and Computational Studies Thereof

(51) International classification	:C07D 209/34, C07D 405/12, C07C 45/74, C01D 1/02, C07C 31/08, A61K 31/404, A61K 31/12, A61P 35/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Integral University
Address of Applicant :Dasulli, Kursi Road, Lucknow Lucknow Uttar Pradesh India
(72)**Name of Inventor :**
1)Atif Husain
2)Jamal Akhtar Ansari
3)Firoj Hassan
4)Qazi Inamur Rahman
5)Abdul Rahman Khan
6)Malik Nasibullah

(57) Abstract :
This invention relates to the Synthesis of Novel Oxindole–Chalcone Derivatives via Claisen–Schmidt Condensation and Computational Studies Thereof The present invention relates to the synthesis of novel series of chalcone derivatives containing an oxindole moiety to evaluate their physico-chemical properties, pharmacokinetic profiles and in silico biological potential. The designed derivatives 3a-d were obtained via a condensation reaction between 5-halo-isatin and substituted acetophenones using sodium ethoxide under moderate reaction conditions. The structures of the synthesized chalcone derivatives were confirmed using 1H NMR, 13C NMR and HRMS. The in-silico evaluations were performed using established database platforms, including SwissADME and admetLAB 3.0, to assess their absorption, distribution, metabolism, excretion and toxicity (ADMET) properties. All the compounds exhibited favourable drug-likeness, low predicted toxicity and compliance with Lipinski's rule of five, suggesting their potential as orally active therapeutic agents. Molecular docking studies were also carried out to assess the binding affinities of compounds 3a-d against cancer-related target proteins such as PDGFRα (PDB ID: 9GZH) and VEGFR2 (PDB ID: 4ASD). Compounds 3b and 3d demonstrated the most promising binding energies (–11.973 and –7.726 kcal/mol, respectively), outperforming the reference drug, sunitinib. The docking results revealed key hydrogen bonding and hydrophobic interactions, offering insights into the stability of the compounds within the active sites of the target proteins.

No. of Pages : 22 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202611113264 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DYNAMIC MECHANICAL IV-TUBE GUIDANCE AND TENSION MANAGEMENT SYSTEM

(51) International classification	:A61M 5/14, A61M 5/168, A61M 5/142, A61M 39/28, A61M 39/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Rajasthan India
(72)**Name of Inventor :**
1)Dr. Vijay Mathur
2)Prof. Giriraj Prasad Soni
3)Dr. Ravi Jain

(57) Abstract :
The present invention relates to a dynamic IV-tube tension compensating system (100) configured to support an infusion container and guide an IV tube (T) along an elevated path above a floor. The dynamic IV-tube tension compensating system (100) comprises a vertical support pole (101), a base (101c), and a tube guide assembly (102) having a guide carriage (102a), a rotational coupling (102b), a tension-responsive mechanical mechanism (102c), a mechanical coupling arrangement (102d), a movement limiting mechanism (102e), a mechanical tension-release mechanism (102f), and a mechanical tension indicator (102g). The mechanical coupling arrangement (102d) coordinates vertical and angular movement of a tube guide (104) according to IV tube (T) tension and pulling direction. The system provides passive tension compensation, controlled tension release, movement limitation, and visual tension indication. Referring Fig.1

No. of Pages : 29 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202611113319 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE PATIENT SUPPORT SYSTEM WITH PREDICTIVE PRESSURE REDISTRIBUTION

(51) International classification	:A61G 7/057, A61G 7/015, A61G 7/018, A47C 27/08, A61G 7/05
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Dr. Ashish Jain
2)Dr. Saleem Akhtar Naqvi
3)Dr. Abhishek Chandra

(57) Abstract :
The present invention discloses an adaptive patient support system (100) for reducing prolonged localized loading on a patient body during extended bed-rest conditions. The adaptive patient support system (100) comprises a patient support mattress (101), a pressure sensing layer (102), a segmented support matrix (103), a reconfigurable mechanical actuator assembly (104), a patient position sensing unit (105), a load sensing assembly (106), a controller (107), a power supply unit (108), a bed frame (109), an alert module (110), a communication module (111), a manual override interface (112), an emergency release assembly (113), a thermal and humidity sensing unit (114), and a predictive actuator-response modeling module (115). The predictive actuator-response modeling module (115) predicts spatial pressure responses, evaluates candidate support configurations, determines prediction errors, and updates an actuator-response map for adaptive pressure redistribution. Refer: Fig. 1

No. of Pages : 37 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202611113365 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MODULAR RACK SYSTEM FOR SUPPORTING AND DRYING REUSABLE HEALTHCARE INSTRUMENTS

(51) International classification	:F26B 9/10, A47B 81/00, F26B 25/18, A47B 96/06, A47B 96/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Rajasthan India
(72)**Name of Inventor :**
1)Dr. Bhartendra Sharma
2)Dr. Vipin Kumar Goyal
3)Major Alka

(57) Abstract :
The present invention relates to a modular adjustable rack (100) configured to support reusable healthcare instruments during drying. The modular adjustable rack (100) comprises a rectangular base plate (101) having castor wheels (102), a plurality of vertical T-posts (103) detachably attachable to the base plate (101), a plurality of cylindrical bodies (104) slidably mounted on respective vertical T-posts (103), fixing screws (105), and a plurality of racks (106) detachably received by the cylindrical bodies (104). Each vertical T-post (103) comprises a plurality of circular openings (103a) for selectively positioning the cylindrical bodies (104). Each cylindrical body (104) comprises a T-shaped slot (104a) and cuboidal arrangements (104b) having cuboidal slots (104c) for receiving the racks (106). The modular adjustable rack (100) provides multiple vertically adjustable supporting levels and configurable spacing between the racks (106) for supporting a plurality of instrument trays. Refer: FIG. 1

No. of Pages : 26 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :09/09/2026

(21) Application No.202611108268 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PATIENT-SPECIFIC PEDIATRIC MEDICATION DISPENSING SYSTEM AND METHOD WITH INTEGRATED VERIFICATION AND PHYSICAL DOSE LIMITATION

(51) International classification	:A61M 5/315, G16H 20/17, G16H 20/13, A61J 7/04, A61J 1/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Dr. Roop Sharma
2)Dr. Nikita Pal
3)Dr. Nikhil Marwah

(57) Abstract :
The present invention relates to a pediatric dose verification and limitation system (100) and method (200) for controlled withdrawal of a patient-specific medication quantity into a syringe (114). The system (100) verifies patient identity, patient weight, medication identity, medication concentration, authorized medication order, and syringe identity. A pediatric dose calculation module (112) determines an authorized medication dose and corresponding medication volume. A syringe identification reader (115) enables retrieval of syringe-specific calibration data, based on which a controller (103) determines a required plunger displacement. A motorized plunger stop assembly (116) moves a movable stop member (116a) to a target stop position corresponding to the required plunger displacement. A position sensor (116e) verifies the stop position and a mechanical locking mechanism (116f) mechanically retains the movable stop member (116a). During withdrawal, a plunger rod (114c) contacts the mechanically locked movable stop member (116a), thereby physically preventing further withdrawal beyond the authorized medication volume. Refer Fig. 1

No. of Pages : 35 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :09/09/2026

(21) Application No.202611108315 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADJUSTABLE PATIENT WEIGHING PLATFORM SYSTEM

(51) International classification	:G01G 19/44, A61G 7/05, A61G 7/018, A61G 7/057, A61G 7/015
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahatma Gandhi University of Medical Sciences and Technology, Jaipur
Address of Applicant :Mahatma Gandhi University of Medical Sciences and Technology, Institutional Area, Sitapura, Jaipur, Rajasthan - 302022, India Jaipur Rajasthan India
(72)**Name of Inventor :**
1)Dr. Rupesh Masand
2)Dr. Yashu Saini
3)Dr. Prateek Suren

(57) Abstract :
The present invention relates to an adjustable patient weighing platform system (100) configured for supporting and weighing a patient in different postures. The system (100) comprises a main base frame (101), a weighing platform assembly (102) having a fixed central platform (103), a first extension platform (104) and a second extension platform (105), first and second rack-and-pinion drive assemblies (106a, 106b), a hydraulic vertical lifting assembly (109), and a weighing sensor assembly (110) having load cells (110a). The first and second extension platforms (104, 105) are independently extendable and retractable to selectively provide a compact or enlarged patient-supporting configuration. A controller (115) controls platform movement, vertical lifting and processing of load-cell signals for determining patient weight. Safety barriers (113a, 113b, 113c), a telescopic rear barrier assembly (113c1), and position sensing arrangement (108a) facilitate controlled and safe patient weighing and transfer. Refer: Fig. 1

No. of Pages : 37 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202611113754 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-TENSIONING HOLD-DOWN GARMENT STEAMING RACK SYSTEM WITH INTEGRATED CONDENSATE-DIVERTING GUIDE RAIL

(51) International classification	:D06F 73/00, D06F 71/34, D06F 73/02, D05B 35/06, D06F 71/36
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Chandigarh University
Address of Applicant :Chandigarh University National Highway 05, Chandigarh-Ludhiana Highway, Mohali, Punjab -140413, India. Mohali Punjab India
(72)**Name of Inventor :**
1)Rajat Kamble
2)Shubham Sharma
3)Ashita Chadha

(57) Abstract :
A garment steaming rack holds a hung garment taut during steaming. A hem clamp with spring-biased jaws is carried by a carriage guided along a hollow vertical guide rail. A constant-force spring reel pulls the carriage through a flexible strap, so that as the garment relaxes and lengthens, the carriage descends and the reel takes up the strap while maintaining substantially constant hem tension. The guide rail also defines an internal condensate channel and carries a drip-shedding weir communicating through an inlet aperture with the channel. Condensate is routed to a base gutter and removable reservoir, away from the garment. A tension preset detent provides selectable force settings. The rack is wholly mechanical and unpowered.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :09/09/2026

(21) Application No.202611108516 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An IoT-Based Data-Driven Framework for Real-Time Environmental Impact Assessment of Urban Development Projects

(51) International classification	:G06Q 10/06, G06Q 50/26, H04L 29/08, H04L 12/24, G06Q 10/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ms. Teena Kiran Nichant
Address of Applicant :Assistant Registrar Vivekananda Global University. jaipur Rajasthan India
2)Dr Ravi Kant Pareek
3)Dr. Dilendra Jasutkar
(72)**Name of Inventor :**
1)Ms. Teena Kiran Nichant
2)Dr Ravi Kant Pareek
3)Dr. Dilendra Jasutkar

(57) Abstract :
A system based on Internet of Things (IoT) that involves monitoring of the environmental impact assessment in real time or near real time for urban development projects is described. Environmentally relevant parameters are sensed, processed by an edge or controller layer, communicated via an IoT communication network, stored at a data management facility, and analyzed for purposes of filtering, aggregation, visualization, and threshold evaluation. A dashboard provides the status of monitoring, while an exceeding condition may lead to the generation of alerts, notifications, mitigation, and corrective actions. Environmental reports and outputs in compliance with the requirements can be periodically generated, such as PARIVESHrelated submissions if technically feasible. In the disclosed prototype, an ESP32, a DHT22 temperature and humidity sensor, display, buzzer, LED status indicator, and potentiometers which are only used as exemplary inputs for air quality, water quality, and noise levels, respectively.

No. of Pages : 30 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :10/09/2026

(21) Application No.202611108688 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Apparatus for Single-Cycle Algorithmic Concurrency Resolution and Hardware-Accelerated Memory Defense via Bare-Metal 2D/3D Silicon Matrices

(51) International classification	:G06F9/30, G06F9/38, G06F15/78
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Shubham Tomar
Address of Applicant :A-194 Street no 5 Adarsh Nagar Kota road Rajasthan India
(72)**Name of Inventor :**
1)Shubham Tomar
2)Dr. Manoj Kumar Jain

(57) Abstract :
The present invention relates generally to semiconductor architecture, Deep-Tech Microprocessors, and Real-Time Operating Systems (RTOS). More specifically, the invention relates to shifting complex operating system algorithms out of software execution and into isolated, single-clock-cycle 2D/3D hardware matrices encoded directly into the physical silicon logic gates.

No. of Pages : 6 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202611114005 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A System and Method for Assessing Creditworthiness of Individual Debtors in Personal Loans Using an Integrated Financial, Behavioural, Professional and Digital Footprint Framework

(51) International classification	:G06Q 40/02, G06Q 40/00, G06N 20/00, G06Q 30/02, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Puneet Kumar Gupta
Address of Applicant :The ICFAI University, Rajawala Road Uttarakhand India
2)Dr. Aatm Prakash Rai
3)Dr. Ankita Srivastava
4)Dr. Aishwarya Kumar
(72)**Name of Inventor :**
1)Dr. Puneet Kumar Gupta
2)Dr. Aatm Prakash Rai
3)Dr. Ankita Srivastava
4)Dr. Aishwarya Kumar

(57) Abstract :
A computer-implemented system and method are disclosed for determining the creditworthiness of an individual debtor applying for a personal loan. Debtor data is acquired across four parameter categories — financial status, behavioural traits, professional attributes, and digital footprints — the latter comprising indicators such as e-commerce transaction data, e-wallet usage, mobile/SMS communication data, and social-media activity. A factor-computation module derives a factor score for each category using weights obtained through a validated factor-analytic and/or trained machine-learning procedure, and a scoring engine combines these into a single composite creditworthiness score used to generate a loan-eligibility recommendation. A real-time monitoring module recomputes the score as new data arrives and issues alerts when defined conditions are met, while an adaptive learning module periodically retrains the underlying weights using actual repayment outcomes. A data privacy and security module manages consent, encryption and audit-logging for data acquired from third-party digital sources via an integrated API layer. The system enables more accurate, inclusive and continuously updated credit risk assessment than conventional, traditional-data-only scoring approaches, particularly for debtors with limited formal credit history.

No. of Pages : 22 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/09/2026

(21) Application No.202611105195 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN OF AVIATION-GRADE 7075-T6 ALUMINIUM ALLOY REAR SPROCKET CARRIER FOR BIKE

(51) International classification	:C22C21/00, B22D21/04, C22C21/10, C22F1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Pranveer Singh Institute of Technology
Address of Applicant :Pranveer Singh Institute of Technology, Kanpur. Kanpur Uttar Pradesh India
(72)**Name of Inventor :**
1)Mr. Amir Khan
2)Dr. Shakun Srivastava
3)Mr. Ritesh Dixit
4)Dr. Asif Ahmad
5)Mr. Ashish Kumar Pandey
6)Mr. Kapil Katri

(57) Abstract :
The invention disclosed herein is concerned with the design and development of a lightweight, high strength rear sprocket carrier for motorcycles fabricated from aircraft quality 7075-T6 aluminium alloy. The conventional rear sprocket carrier may be subject to excess weight, stress concentration, fatigue failure and reduced performance under repetitive torque, chain tension, braking and impact loads. The present invention overcomes these limitations by providing a unique structural geometry fabricated from high strength 7075-T6 aluminum alloy. The sprocket carrier is equipped with a precisely machined sprocket mounting interface, a central wheel or bearing interface, and strengthened load bearing areas, intended to efficiently spread mechanical stresses. The optimized design provides adequate structural strength and dimensional stability while minimizing the rotational and unsprung mass. The part is able to transmit drivetrain torque from the rear sprocket to the rear wheel while resisting cyclic torsional, shear and impact loading. Precision CNC machining and optional surface treatment also improve dimensional accuracy, surface quality, corrosion resistance and service life. The proposed aviation-grade 7075-T6 aluminium alloy rear sprocket carrier improves strength-to-weight ratio, fatigue resistance, reliability of the drivetrain, handling of the motorcycle, acceleration and overall performance in comparison with conventional sprocket carrier designs.

No. of Pages : 13 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :07/09/2026

(21) Application No.202611107267 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Wearable Optical-Target, Color-Changing OLED, Audio, Spoken-Message, Haptic and Seven-Chakra-Associated Electronic Biofeedback System.

(51) International classification	:G02B27/01, G06F1/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Notty Noodles Private Limited
Address of Applicant :2020 Kaurali More Rd Tigaon Haryana India
(72)**Name of Inventor :**
1)Sukhbir Singh Bachhal

(57) Abstract :
Abstract A wearable electronic system (100) includes a headset (110) having a downward bridge (112) supporting an infrared optical emitter (114) directed toward a remote target receiver (120). The target receiver (120) includes an optical detector (122), processor (126), and OLED display (124). A mobile application (130) stores seven configurable electronic zone profiles associated with Root, Sacral, Solar Plexus, Heart, Throat, Third Eye and Crown identifiers. Each profile may include an OLED colour state, programmable audio parameter, spoken-message identifier, haptic pattern, target identifier, timing data and sensor parameters. The infrared emitter transmits a coded optical signal and the target validates the signal to generate a sequenced electronic event. The event coordinates the OLED display, audio, spoken output and haptic output. Session identifiers, sequence numbers, acknowledgements and recovery logic synchronize the headset, target and mobile application.

No. of Pages : 30 No. of Claims : 45

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/09/2026

(21) Application No.202611116287 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRICALLY DOPED VERTICAL TUNNEL FIELD EFFECT TRANSISTOR BIOSENSOR SYSTEM FOR LABEL-FREE BREAST CANCER CELL DETECTION

(51) International classification	:G01N 27/414, H01L 29/78, G01N 33/487, G01N 33/48, G01N 27/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Mukesh Kumar Bind
Address of Applicant :Assistant Professor, Galgotias University, Greater Noida, Uttar Pradesh. Greater Noida Uttar Pradesh India
2)Mr. Bandi Venkata Chandan
3)Dr. Kaushal Kumar Nigam
4)Mr. Ashish Patle
5)Dr. Dharmender
6)Parvin Akhter

(72)**Name of Inventor :**
1)Dr. Mukesh Kumar Bind
2)Mr. Bandi Venkata Chandan
3)Dr. Kaushal Kumar Nigam
4)Mr. Ashish Patle
5)Dr. Dharmender
6)Parvin Akhter

(57) Abstract :
The present invention provides an electrically doped vertical tunnel field-effect transistor biosensor system (100) for label-free electrical discrimination of breast-cell samples. The system includes a drain contact (1), vertical silicon channel (2), source contact (3), gate dielectric (4), drain polarity-gate elements (5A, 5B), control-gate elements (6A, 6B), and source polarity-gate elements (7A, 7B). A source-side sensing nanocavity receives a biological sample. Source and drain carrier regions are induced electrostatically, avoiding implanted abrupt junctions. Dielectric constant and fixed interface charge of the sample modulate source-region electrostatics, source-channel electric field, tunnelling-barrier width and band-to-band tunnelling generation, producing changes in drain current and related electrical parameters. The resulting response enables differentiation of non-tumorigenic and malignant breast-cell samples and remains evaluable under partial cavity filling, non-uniform biomolecule profiles and variable interface-charge conditions. Refer to Figure 1.

No. of Pages : 38 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202611113229 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A DETECTOR AND METHOD FOR MEASURING JOINT ORBITAL ANGULAR MOMENTUM SPECTRA OF ENTANGLED PHOTON PAIRS

(51) International classification	:G01J9/02, G01J9/00, G01J9/04, G02B27/00, G01F1/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY KANPUR
Address of Applicant :Dean, Research and Development Office, 2nd Floor, New Faculty Annexe Building, IIT Kanpur, Kalyanpur, Kanpur – 208016, Uttar Pradesh, India Uttar Pradesh India

(72)**Name of Inventor :**
1)Suman Karan
2)Anand Kumar Jha

(57) Abstract :
The invention relates to quantum optics and photonic measurement of orbital angular momentum entanglement. The invention provides a detector and a corresponding method in which an entangled two-photon state generated in a Laguerre-Gaussian basis is introduced into a Franson-type two-photon interferometer having signal-side and idler-side Mach-Zehnder type interferometers, each with an image-rotator branch and a longer reference branch. Imaging optics preserve and relay a Fourier transform of the crystal plane to multimode-fiber-coupled single-photon counting modules, and coincidence visibility is measured as a function of independently controlled signal-arm and idler-arm rotation angles. A processing arrangement reconstructs the joint orbital angular momentum spectrum by applying a two-dimensional Fourier transform to the coincidence visibility, treated as a two-photon angular coherence function Fourier-conjugate to the orbital angular momentum basis, thereby enabling efficient determination of joint probabilities, including non-zero off-diagonal orbital angular momentum elements, summed over radial mode indices.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/10/2026

(21) Application No.202611122005 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Gravity-Fed Water Defluoridation System Using Manganese Dioxide-Coated Activated Alumina.

(51) International classification	:C02F 1/28, B01J 20/08, C02F 101/14, B01J 20/06, B01J 20/04	(71)**Name of Applicant :** **1)Mr. K.K. GARG.** Address of Applicant :M/s C & I Systems, Kota, Rajasthan (India)- 324003. Kota Rajasthan India **2)Er. ASHOK PATNI.** **3)Er. AKANSHA JAIN**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Mr. K.K. GARG.**
(33) Name of priority country	:NA	**2)Er. ASHOK PATNI.**
(86) International Application No Filing Date	: :01/01/1900	**3)Er. AKANSHA JAIN**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

[0030] The present invention is related to a Gravity-Fed Water Defluoridation System Using Manganese Dioxide-Coated Activated Alumina. A gravity-fed water treatment system and process for defluoridation of groundwater uses manganese oxide-coated activated alumina (MCAA) as a packed-bed adsorption medium. In a disclosed preparation, activated alumina is impregnated with an aqueous manganese nitrate tetrahydrate solution, maintained for approximately 2–4 hours, dried at approximately 100–120°C and calcined at approximately 300–400°C for approximately 3–4 hours to form manganese oxide on the alumina. The MCAA is loaded into a gravity-fed treatment vessel, optionally together with sand/gravel and activated carbon, so that groundwater passes through the treatment media under gravitational head without continuous pressurized pumping. Laboratory data supplied by the applicant report reduction of approximately 20 ppm fluoride to approximately 1.5 ppm in about 8 hours and approximately 0.5 ppm in about 10 hours under a disclosed test condition. The medium can be regenerated using alkaline desorption followed by rinsing and acid neutralization. The system is intended for modular decentralized treatment of fluoride-contaminated groundwater as depicted in Figures 1-6.

No. of Pages : 30 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202611131202 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HPV16 AND HPV18 L1-SPECIFIC DNA APTAMERS AND A GOLD NANOPARTICLE-BASED LATERAL FLOW DEVICE

(51) International classification	:C12N 15/115, G01N 33/569, G01N 33/558, G01N 33/532, G01N 33/574
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Pavika Lal
Address of Applicant :Associate Professor, Department of Obstetrics and Gynaecology, GSVM medical college Kanpur, UP, 208002, INDIA Kanpur Uttar Pradesh India
2)Dr. Bandana Sharma
3)Dr. Garima Gupta
4)Dr. Alok Raghav
(72)**Name of Inventor :**
1)Dr. Pavika Lal
2)Dr. Bandana Sharma
3)Dr. Garima Gupta
4)Dr. Alok Raghav

(57) Abstract :
DNA aptamers (SEQ ID NO:1 and SEQ ID NO:2) for the L1 major capsid proteins of human papillomavirus types 18 and 16 respectively, functional variants thereof having at least 90% sequence identity, gold nanoparticle conjugates of the said aptamers, and a paper-based lateral flow device incorporating the conjugates. The aptamers are selected computationally from the L1 pentamer structures and are attached to gold nanoparticles of 8 nm to 15 nm diameter through 5′-terminal thiol groups. The device comprises a sample pad, a conjugate pad, a nitrocellulose membrane having a test region and a control region, and an absorbent pad, preferably in a two-lane configuration providing separate detection pathways for HPV16 L1 and HPV18 L1. Methods of preparing the conjugate and of manufacturing the device are also provided. FIG. 4.

No. of Pages : 51 No. of Claims : 18

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202611127644 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD FOR SYNTHESIS, CHARACTERIZATION AND IN VITRO SCREENING OF SUBSTITUTED CHALCONE DERIVATIVES

(51) International classification	:G01N 30/90, G01N 30/94, C07C 49/84, G01N 30/88, G01N 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sanjivani College of Pharmaceutical Education & Research (Autonomous)
Address of Applicant :Sanjivani College of Pharmaceutical Education & Research (Autonomous), Savitribai Phule Pune University, Kopargaon, Maharashtra 423603, India Delhi India
2)Dr. Agasti Ware
3)Ms. Vaishnavi Deshmukh
4)Mr. Sarthak Sanap
5)Ms. Shruti Shelke
(72)**Name of Inventor :**
1)Dr. Agasti Ware
2)Ms. Vaishnavi Deshmukh
3)Mr. Sarthak Sanap
4)Ms. Shruti Shelke

(57) Abstract :
[041] Disclosed is a research-stage method for synthesis, structural characterization and comparative in vitro screening of substituted chalcone derivatives. Selected benzaldehyde and acetophenone precursor pairs are subjected to base-mediated Claisen–Schmidt condensation, followed by isolation, purification and documentation of the resulting chalcones. Thin-layer chromatography is used for reaction monitoring, while complementary spectroscopic techniques are used to assess structural consistency. Products satisfying predefined analytical criteria are advanced to a controlled cell-viability assay comprising a concentration series and vehicle controls. Assay measurements are linked to corresponding precursor, batch and characterization records to support preliminary structure–activity evaluation. The method provides traceability between chemical preparation and biological assessment without presuming anticancer efficacy of untested compounds. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :06/09/2026

(21) Application No.202611107117 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTERACTIVE AI-BASED SYSTEM FOR REAL-TIME GENERATION OF PERSONALIZED MUSIC FOR EMOTIONAL REGULATION

(51) International classification	:G06Q 10/0637, G10D 13/01, G10D 1/00, G10D 7/00, C09D 159/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ahana Gandhi
Address of Applicant :9 Jiwan Nagar Bank Colony Karnal 132001 Haryana Karnal Haryana India
(72)**Name of Inventor :**
1)Ahana Gandhi

(57) Abstract :
The present invention relates to an artificial-intelligence-assisted computer-implemented system and method for generating personalized musical compositions and/or soundscapes based on a user-specified emotional or functional goal state. The system comprises a user interface for receiving an emotional or functional goal, an intent-processing module for generating a machine-processable representation of the goal, an audio-parameter mapping engine for converting the representation into a multidimensional audio-parameter profile, an audio-generation engine for generating or computationally assembling an audio soundscape according to the profile, and an audio playback module for substantially immediate playback. The audio parameters may comprise tempo, rhythm, harmonic characteristics, instrumentation, timbre, spectral characteristics, dynamic range, musical-event density, ambient sound characteristics and transition characteristics. The system may receive pre-session, during-session and/or post-session user feedback and associate the feedback with parameters used for generating the soundscape. A personalization engine may thereafter modify subsequent parameter profiles according to previous user responses, thereby providing an adaptive feedback loop for progressively personalized soundscape generation. The system is preferably implemented as a lightweight web-based application enabling personalized and repeatable sound-based mental-wellness sessions without requiring musical composition expertise.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :25/09/2026

(21) Application No.202611115006 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : “An adaptive board-based system with a machine learning”

(51) International classification	:G06Q 40/00, G06Q 40/06, G06N 20/00, G06Q 40/02, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Thapar Institute of Engineering and Technology
Address of Applicant :Thapar Institute of Engineering and Technology, P.O. Box No. 32, Bhadson Road, Patiala, Punjab-147004 Patiala Punjab India
(72)**Name of Inventor :**
1)Dr. Samya Muhuri
2)Vansh Chambyal
3)Vandan Arora

(57) Abstract :
The present invention provides an adaptive board-based system with a machine learning. The platform transforms financial education into an interactive simulation specifically created for children below the age of 15. The system comprises a 16-tile modular grid representing financial concepts, a stochastic Dice Roll Engine, and a Decision-Making Engine providing branching choices. A hybrid scoring engine combines rule-based formulas with a Random Forest model to analyze user decision patterns and risk tolerance. The process involves initializing user states, stochastic progression via dice rolls, and real-time updates to savings, expenses, and investments based on user choices. Data is transmitted to the ML Model for behavioral analysis, allowing the system to dynamically adjust complexity. Built in Flutter, the invention provides an experiential simulation environment for building positive financial behaviors.

No. of Pages : 68 No. of Claims : 19

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :05/09/2026

(21) Application No.202611106933 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Computational Sustainability Prompt Optimization System

(51) International classification	:G06F40/30, G06N3/04, G06N20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SHARDA UNIVERSITY
Address of Applicant :PLOT NO. 32, 34, KNOWLEDGE PARK III, GREATER NOIDA, UTTAR PRADESH 201310 GREATER NOIDA Uttar Pradesh India
(72)**Name of Inventor :**
1)KUMKUM MISHRA
2)DEVESH KUMAR GOLA
3)HIMANSHU DHAKA
4)KAPIL KUMAR

(57) Abstract :
Computational Sustainability Prompt Optimization System Abstract Disclosed is a computational sustainability prompt optimization system comprising a prompt input interface, a memory, and a processing unit. The processing unit determines a token count, prompt type, and task type for an input prompt and determines a Green Carbon Score from the token count, a prompt-type weighting factor, and a task-type weighting factor. Candidate prompt formulations are generated when the Green Carbon Score satisfies an optimization condition. For each candidate prompt formulation, the processing unit determines a respective Green Carbon Score and preservation information representing semantic similarity and task preservation. One candidate prompt formulation is selected according to the respective Green Carbon Score and the preservation information and is provided as an optimized prompt for inference by a Large Language Model. Fig. 1 Dated September 3, 2026 Kumar Tushar Srivastava IN/PA-3973 Agent for the Applicant

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/07/2026
(21) Application No.202617087766 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PROCESSING METHOD, COMMUNICATION DEVICE AND STORAGE MEDIUM

(51) International classification :H04W 72/231
(31) Priority Document No :NA
(32) Priority Date :NA
(33) Name of priority country :NA
(86) International Application No :NA
Filing Date :NA
(87) International Publication No : NA
(61) Patent of Addition to Application Number :NA
Filing Date :NA
(62) Divisional to Application Number :NA
Filing Date :NA

(71)**Name of Applicant :**
1)SHENZHEN TRANSSION HOLDINGS CO., LTD.
Address of Applicant :Unit 1, 24th Floor, Transsion Building, No. 8 Xianyuan Road, Xili Community, Xili Street, Nanshan District Shenzhen, Guangdong 518000 China
(72)**Name of Inventor :**
1)SHEN, Xingya
2)LI, Xin

(57) Abstract :
Disclosed in the present application are a processing method, a communication device and a storage medium. The method comprises: a terminal device processing UL resource muting on the basis of the indication of first information. By means of the technical solution of the present application, in an SBFD gNB-to-gNB CLI scenario, the processing mechanism for UL resource muting is improved, thereby supporting an improvement in uplink transmission performance.

No. of Pages : 26 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/09/2026
(21) Application No.202611106155 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A TRANSFORMABLE WHEEL-PROPELLER HYBRID UNMANNED VEHICLE

(51) International classification	:G05D 1/228, B60K 6/42, B60W 20/00, G08G 5/57, H02J 105/37
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sharda University
Address of Applicant :Plot No. 32-34, Knowledge Park-III, Greater Noida - 201310, Uttar Pradesh, India. Greater Noida Uttar Pradesh India
(72)**Name of Inventor :**
1)PANDEY, Bishwash Kumar
2)KUMAR, Abhay
3)RAI, Anurag
4)SINGH, Shubham Kumar
5)SINGH, Rajneesh Kumar

(57) Abstract :
A transformable wheel-propeller hybrid unmanned vehicle (100) incorporates a structure (102) with a main vehicle body (102A) supported by a structural frame (102B) that supports multiple wheel-propeller modules (104). Each wheel-propeller module (104) encompasses a ground wheel (104A) driven by ground propulsion motors (104B) for ground movement, and an aerial propeller (104D) driven by aerial propulsion motors (104C) for aerial movement. A transformation unit (106) is connected with each wheel-propeller module (104), transforming the wheel-propeller module (104) between a ground mode and an aerial mode, while a locking unit (108) secures the wheel-propeller module (104) in the selected mode. A mechanical transmission unit (110) incorporates a torque transmission unit (110A) for transmitting or modifying torque to the ground wheel (104A), a bearing unit (110B) for supporting rotation, and a suspension unit (110C) connected between the wheel-propeller modules (104) and structure (102) for accommodating mechanical loading during ground locomotion.

No. of Pages : 31 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/09/2025
(21) Application No.202521030455 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : "ARTIFICIAL INTELLIGENCE-DRIVEN SYSTEM FOR FLIGHT DATA ANALYSIS TO ASSESS TRAINEE PILOT PERFORMANCE AND ENHANCE TRAINING EFFECTIVENESS"

(51) International classification	:G09B0019160000, G09B0019000000, G06N0005040000, G06F0040400000, G06N0020000000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JUMPS AUTOMATION LLP
Address of Applicant :K103-Indraprastha 5, Prahlad Nagar, Satellite, Ahmedabad, Gujarat 380015, INDIA Ahmedabad Gujarat India
(72)**Name of Inventor :**
1)ROHAN CHANDAK

(57) Abstract :
ARTIFICIAL INTELLIGENCE-DRIVEN SYSTEM FOR FLIGHT DATA ANALYSIS TO ASSESS TRAINEE PILOT PERFORMANCE AND ENHANCE TRAINING EFFECTIVENESS A standalone, air-gapped debriefing workstation (130), coupled to a flight simulator (110) solely by a dedicated network link (120), fetches completed-sortie recording artefacts by pull-only, staged transfer without modifying the simulator. A telemetry decoding module (134) converts the binary flight recording into engineering-unit time-series stored as write-once compressed columnar files (144) queried in place by an embedded analytical engine (145); a flight-phase segmentation module (136) partitions each sortie into its constituent flight phases from taxi to go-around and detects approach and manoeuvre episodes. Curated analytics, including per-phase performance fingerprints, deviation events, motifs, red flags and instructor annotations, reside in a transactional relational store (142) alongside a vector index (146). Locally hosted language models (174) answer instructor questions and author sortie narratives from evidence assembled by a deterministic, temporally scoped retrieval engine (172); a fact-lock verification module (178) checks every numeral, unit and direction-bearing token of the generated text against the retrieved evidence, with bounded regeneration and deterministic template fallback upon failure. A local audio-analysis subsystem (160) fuses speech transcripts, vocal-stress measures and malfunction reaction latency with the telemetry-derived flight-phase timeline. The system thereby enables verified, whole-sortie, instructor-grade debriefing entirely within an offline computing environment. Fig 2

No. of Pages : 54 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2025

(21) Application No.202521091754 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ONBOARD CHARGER CUM INVERTER SYSTEM

(51) International classification	:B60L53/20, H02J7/02, H02M7/48, H02M5/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Matter Motor Works Private Limited
Address of Applicant :301, PARISHRAM BUILDING, 5B RASHMI SOC., NR. MITHAKHALI SIX ROADS, NAVRANGPURA AHMEDABAD, GUJARAT, INDIA - 380010 Ahmedabad Gujarat India
(72)**Name of Inventor :**
1)KUMAR PRASAD TELIKEPALLI
2)SHIVAM GARG
3)DARSHAN PANCHAL
4)MOHIT JHA
5)ANKUR SRIVASTAVA

(57) Abstract :
ABSTRACT ONBOARD CHARGER CUM INVERTER SYSTEM The present disclosure describes an onboard charger cum inverter system (100) for an electric vehicle, comprising at least one input interface (102) configured to receive an alternating current (AC) input, at least one power conversion unit (104), at least one transformer module (106), at least one direct power path (108) connected between the at least one input interface (102) and the at least one power conversion unit (104), at least one switching unit (110) operatively coupled to the at least one transformer module (106) and the at least one direct power path (108) and a controller (112) communicatively coupled to the at least one switching unit (110). Further, the controller (112) is configured to control the at least one switching unit (110) and establish electrical coupling between the at least one input interface (102) and the at least one power conversion unit (104) based on an input voltage. FIG. 1

No. of Pages : 31 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2025

(21) Application No.202521091755 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTEGRATED ONBOARD CHARGER CUM INVERTER SYSTEM

(51) International classification	:B60R16/02, H02J3/32, H02M7/797
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Matter Motor Works Private Limited
Address of Applicant :301, PARISHRAM BUILDING, 5B RASHMI SOC., NR. MITHAKHALI SIX ROADS, NAVRANGPURA AHMEDABAD, GUJARAT, INDIA - 380010 Gujarat India
(72)**Name of Inventor :**
1)KUMAR PRASAD TELIKEPALLI
2)SHIVAM GARG
3)DARSHAN PANCHAL
4)MOHIT JHA
5)ANKUR SRIVASTAVA

(57) Abstract :
ABSTRACT INTEGRATED ONBOARD CHARGER CUM INVERTER SYSTEM The present disclosure describes an integrated onboard charger cum traction inverter system (100) for an electric vehicle. The system (100) comprises at least one input interface (102) configured to receive an Alternating Current (AC) input from at least one external power source, at least one phase measurement circuit (104) configured to generate phase characteristic signals corresponding to the received AC input, at least one phase detection unit (106) configured to determine a phase type based on the phase characteristic signals, at least one control unit (108) communicatively coupled to the phase detection unit (106), and at least one power conversion unit (110) electrically coupled to at least one traction motor (112) and at least one battery (114). Further, the control unit (108) dynamically reconfigures a switching topology of the power conversion unit (110) according to the determined phase type to facilitate optimized charging and traction inverter operation. FIG. 1

No. of Pages : 45 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2025

(21) Application No.202521091511 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR RUNTIME CONTROL AND CONTAINMENT OF PROBABILISTIC ARTIFICIAL INTELLIGENCE INFERENCE

(51) International classification	:G06F21/53, G06N20/00, G06F21/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Upasna Tiwari
Address of Applicant :D5-903, Rohan Abhilasha, Lohegaon Wagholi Road, Wagholi, Pune, Maharashtra-412207, India Pune Maharashtra India
2)Narendra Tiwari
(72)**Name of Inventor :**
1)Upasna Tiwari
2)Narendra Tiwari

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR RUNTIME CONTROL AND CONTAINMENT OF PROBABILISTIC ARTIFICIAL INTELLIGENCE INFERENCE A system and method for runtime control and containment of probabilistic artificial intelligence inference are disclosed. An inference interface (100) receives or accesses computational data associated with an inference process. A governance evaluation engine (110) evaluates the data against machine-executable governance constraints and generates a governance result. An execution-state controller (120) determines an execution condition, and a containment controller (130), according to the execution condition, modifies or constrains an operation associated with the inference process. A controlled output-release interface (150) controls release according to completion of governance evaluation and applicable containment. Embodiments provide control of exposed intermediate computational data, retrieval restriction, agent and tool invocation containment, state-dependent resource isolation (140), execution-state hysteresis, and cryptographically linked execution provenance including optional Merkle-tree evidence aggregation. The architecture couples governance evaluation to computer execution so that an inference, invocation, resource or output path can be constrained before controlled release. Figure-4

No. of Pages : 33 No. of Claims : 22

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/10/2025

(21) Application No.202521094376 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEMS AND METHODS FOR REAL-TIME UNDERGROUND INTRUSION DETECTION USING FIBER OPTIC AND SEISMIC SENSOR FUSION

(51) International classification :G01V1/00
(31) Priority Document No :NA
(32) Priority Date :NA
(33) Name of priority country :NA
(86) International Application No :
Filing Date :01/01/1900
(87) International Publication No : NA
(61) Patent of Addition to Application Number :NA
Filing Date :NA
(62) Divisional to Application Number :NA
Filing Date :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
Address of Applicant :INDIAN INSTITUTE OF TECHNOLOGY BOMBAY POWAI, MUMBAI-400076, MAHARASHTRA, INDIA Mumbai Maharashtra India
(72)**Name of Inventor :**
1)BHARATH CHANDRA SHEKAR
2)DIP SHANKAR NANDA

(57) Abstract :
ABSTRACT SYSTEMS AND METHODS FOR REAL-TIME UNDERGROUND INTRUSION DETECTION USING FIBER OPTIC AND SEISMIC SENSOR FUSION A system (10) for detecting subterranean excavation in a monitored zone (18). A distributed acoustic sensing fiber optic cable (14) permanently installed in the zone (18) is coupled to an interrogator (16) feeding a first digitizer and recording unit (50). An areal array of geophone sensors (12), not permanently installed, feeds a second digitizer and recording unit (50) on a common time base. An event detector fixes an approximate position along the cable (14), where the array (12) is emplaced. A central processing unit (32), constituted by an inference unit (52) fed by both recording units (50), comprises a pick reconciliation fit (46), a coherence stack (24) giving a localisation estimate output (26), an event catalogue (22) of localised microseismic events (44), a robust estimator giving a trajectory and a corridor output (28), and a confirmation gate (42) whose confirmation decision output (34) drives a graphical user interface (54). [[FIGURE 8]

No. of Pages : 53 No. of Claims : 17

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/01/2026
(21) Application No.202621006156 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD FOR PREPARING A MULTI-LAYER LOW-EMISSIVITY THIN FILM COATING

(51) International classification	:C03C17/36, C03C17/34, E06B3/67, B32B9/00, B32B7/02, B32B17/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR
Address of Applicant :Palaj, Gandhinagar-382055, Gujarat, India Gandhinagar Gujarat India
(72)**Name of Inventor :**
1)HEMANI PRIYANKA
2)PANDA EMILA

(57) Abstract :
ABSTRACT The present disclosure relates to a method for preparing a multilayer low-emissivity thin-film coating on a substrate. The method is simple, scalable, and enables the formation of a film coating that permits the transmission of visible light while reflecting infrared (IR) radiation. Consequently, the coated glass substrate is suitable for use in insulated windows for buildings, automobiles, and other glazing applications, thereby reducing heat transfer through the glass and lowering the interior temperature of the enclosed space.

No. of Pages : 37 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :15/07/2025
(21) Application No.202521067569 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PROCESS OF RECYCLING OF LITHIUM IRON PHOSPHATE BATTERY

(51) International classification	:H01M0004580000, H01M0010540000, H01M0010052500, C01D0015080000, C22B0026120000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Masood Khajenoori
Address of Applicant :A66, H-Block, MIDC Pimpri, Pune, Maharashtra State, India 411018. Pune Maharashtra India
(72)**Name of Inventor :**
1)Dr. Masood Khajenoori

(57) Abstract :
"PROCESS OF RECYCLING OF LITHIUM IRON PHOSPHATE BATTERY" The present invention relates to a hydrometallurgical process 100 for recycling spent lithium iron phosphate (LFP) batteries. The process includes sequential phases of preprocessing 105, single stage leaching 110, carbon recovery 115, copper recovery 120, aluminium recovery 125, iron phosphate recovery 130, iron recovery 135, and lithium phosphate recovery 140. The black mass obtained from spent LFP batteries is subjected to acid peroxide leaching, followed by sequential separation of graphite, copper, aluminium compounds, iron phosphate, iron hydroxide, and lithium phosphate. The process eliminates the need for roasting, calcination, pyrolysis, solvent extraction, and ammonia-based purification, thereby reducing energy consumption, chemical usage, processing time, and operating cost. Copper and iron phosphate are recovered substantially at room temperature, while lithium phosphate is directly precipitated from a comparatively dilute lithium containing solution without energy intensive concentration. (FIG. 1 for Publication)

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :20/11/2025

(21) Application No.202521114082 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ERROR COMPENSATING COLLET FOR TAPPING ON CNC MACHINE

(51) International classification	:B23G1/16, B23G1/50, B23G5/06, G05B19/404
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mrs. Pradnya Pradip Khare
Address of Applicant :"Sharveya" S.No.19/1a House No.953, Saswad Road Hadapsar, Pune, Maharashtra, India Hadapsar Maharashtra India
2)Pradip Ramchandra Khare
(72)**Name of Inventor :**
1)Mrs. Pradnya Pradip Khare
2)Pradip Ramchandra Khare

(57) Abstract :
ERROR COMPENSATING COLLET FOR TAPPING ON CNC MACHINE The present invention relates to an error-compensating collet for tapping operations on CNC machines. It provides an error-free and burr-free tapping process by incorporating a pre-vertical pressing mechanism that ensures precise initiation of thread cutting or thread rolling. The invention achieves an absolutely accurate and uniform thread profile due to its integrated compensating element within the tap-holding system, enabling rapid and effective error correction during operation. The error-compensating collet is versatile and compatible with standard collet holders. Additionally, the tool holders of the present invention can be employed for various machining processes such as drilling, milling, and multi-operation machining. Thus, the present invention offers a cost-effective error-compensating collet that overcomes the limitations of conventional CNC tapping systems, ensuring accurate, burr-free, and consistently uniform thread production.

No. of Pages : 66 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/01/2026
(21) Application No.202621006477 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTOMATED AGRICULTURE DRONE FOR SPRAYING LIQUID FERTILIZERS

(51) International classification	:B64U50/00, B64U40/00, G05D1/00, G05D1/248, B64D1/18, B64C39/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)St. Vincent Pallotti College of Engineering and Technology
Address of Applicant :Wardha Road, Gavsi-Manapur, Nagpur, Maharashtra, India - 441108 Maharashtra India
(72)**Name of Inventor :**
1)RANGARI, Rohit Umesh
2)UPARKAR, Priti S.
3)BRAMHE, Manoj V.

(57) Abstract :
The present invention relates to a automated agriculture drone (101) for precision spraying of liquid fertilizers. The drone comprises a drone body (101) in an Octa-X configuration providing stability and redundancy, a power module (102) connected to brushless motors (1, 2) via ESCs (5, 12) for propulsion, and an APM flight controller (103) managing flight, navigation, and spraying operations. A GPS navigation system (106) provides precise flight paths, while a spraying mechanism (105) including a pump assembly (13), nozzle (19), and liquid storage bottle (9) delivers variable-rate fertilizer according to pre-mapped field conditions. A communication module (107) enables remote monitoring and control via mission planning software (108) on a mobile application. The drone autonomously initiates flight, scans and maps the field, applies fertilizer precisely, and returns to base, logging spraying data. Safety features include obstacle avoidance, fail-safe landing, and automatic return upon low battery, enabling efficient, eco-friendly, and cost-effective agricultural operations.

No. of Pages : 16 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/09/2025
(21) Application No.202521092530 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : TOPICAL COMPOSITION FOR THE TREATMENT OF CHRONIC NEUROPATHIC PAIN

(51) International classification	:A61K 31/197, A61K 31/704, A61K 9/06, A61K 9/107, A61K 47/32, A61K 47/10, A61P 25/02, A61P 21/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)LYKA LABS LIMITED
Address of Applicant :Plot 4801/B & 4802/A, GIDC Industrial estate, Ankleshwar -393002, Dist.-Bharuch, Gujarat. India. Ankleshwar Gujarat India
(72)**Name of Inventor :**
1)GANDHI, Kunal Narendra
2)SAMANT, Rajan Shantaram
3)MAHAJAN, Anil Arun
4)DHOPATKAR, Arvind Tanaji
5)BISANDRE, Krunal Uttam
6)MALI, Ganesh Shivaji

(57) Abstract :
Stable Topical pharmaceutical compositions, wherein the Pregabalin is 7 – 10% w/w and Thiocolchicoside content is 0.05 to 0.5 % w/w of the total composition for Peripheral Neuropathic pain accompanied by muscle spasms, Low back pain with muscle spasm and neuropathic pain (due to nerve root irritation) (lumbosacral dorsalgia) and cervical spondylosis with peripheral neuropathic pain and muscle spasms. are described herein. The invention additionally provides a process to prepare the Pregabalin and Thiocolchicoside composition. The compositions described herein are provided in topical dosage forms for localized administration of Pregabalin and Thiocolchicoside.

No. of Pages : 36 No. of Claims : 19

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/09/2025
(21) Application No.202521092803 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR CONTEXT-HARVESTING-BASED AUGMENTED SEARCH EXECUTION

(51) International classification	:G06F0016332000, G06F0016330000, G06F0016220000, G06F0016350000, G06N0005020000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)UJJWAL RAO
Address of Applicant :A2, Song of Joy, S. No. 64, Kharadi Maharashtra India
(72)**Name of Inventor :**
1)UJJWAL RAO

(57) Abstract :
A system (800) and method (600) for context-harvesting-based augmented search are disclosed. A Context Harvesting and Query Synthesis Module (100) embeds proprietary documents in a private space and prompts a generative model for provenance-tagged queries. A Delta Computation and Prioritization Module (200) embeds the knowledge base in a public space and, querying that space only, classifies each query by similarity threshold as already cited or a missing reference in a prioritized second set. A Recursive and Refined Query Synthesis Module (300) cross-validates refined queries from concepts absent in those documents. A Search Execution Module (400) dispatches only that set through heterogeneous connectors, harvesting full text. A Ranking, Filtering and Traceability Module (500) de-duplicates and ranks results by a tunable semantic-lexical hybrid, logging every query, provenance and decision in an auditable ledger. Confining external calls to identified gaps cuts network load, yielding reproducible, auditable evidence for regulated industries such as healthcare.

No. of Pages : 66 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/01/2026

(21) Application No.202621010273 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR DECENTRALIZED WEB 3.0–ENABLED IIOT-BASED AUTONOMOUS INTELLIGENT CASTING PROCESS CONTROL AND OPTIMIZATION

(51) International classification	:H04L 67/12, G06F 17/00, B22D 46/00, G05B 19/418, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PARUL UNIVERSITY (Parul Institute of Engineering and Technology)
Address of Applicant :Parul Institute of Engineering and Technology, Parul University, P.O. Limda Tal. Waghodia, Dist. Vadodara, Gujarat - 391760, INDIA VADODARA Gujarat India
(72)**Name of Inventor :**
1)Shukla, Dr. Minal
2)Vadhel, Dr. Pravin
3)Sata, Amit
4)Singh, Payal

(57) Abstract :
ABSTRACT A SYSTEM AND METHOD FOR DECENTRALIZED WEB 3.0–ENABLED IIOT-BASED AUTONOMOUS INTELLIGENT CASTING PROCESS CONTROL AND OPTIMIZATION The invention relates to industrial process control and optimization, particularly for casting operations. A sensor network (1) comprising various industrial sensors, including K-type thermocouples and Hall-effect sensors, acquires critical process parameters and transmits real-time data via an embedded IIoT gateway (2) using lightweight protocols. The data is immutably recorded on a blockchain-based distributed ledger (3) employing consensus algorithms and decentralized storage, and processed by a smart contract automation layer (4) that triggers control actions on detecting anomalies. Further analysis is performed by an AI-driven analytics module (5) using machine learning models and integrated with a decentralized digital twin module (6) to simulate process dynamics in real time. A Web 3.0–enabled DApp (7) provides secure, permissioned monitoring by stakeholders, ensuring autonomous and dynamic control of the casting process.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :20/02/2026

(21) Application No.202621019649 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ACCELERATED CITRUS BIOMASS CONVERSION USING ENZYME–ß- CYCLODEXTRIN SYSTEM FOR LOW-ALCOHOL BIOACTIVE CLEANER PRODUCTION

(51) International classification	:C11D 3/386, C12P 1/00, C12N 9/00, C08B 37/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GREEN NATURAL PRODUCTS PRIVATE LIMITED
Address of Applicant :Unit No - A-6, kohinur industrial complex, opp. hindalco, Mahanagar, Taloja, Navi Mumbai Maharashtra India
(72)**Name of Inventor :**
1)Prakul Rakesh
2)Divya Priyadarshani

(57) Abstract :
ACCELERATED CITRUS BIOMASS CONVERSION USING ENZYME–ß CYCLODEXTRIN SYSTEM FOR LOW-ALCOHOL BIOACTIVE CLEANER PRODUCTION The present invention relates to a process for reducing citrus biomass conversion time in the manufacture of a cleaning composition. Citrus biomass comprising citrus peels is combined with water and an optional carbohydrate source, and contacted with one or more biomass degrading enzymes selected from pectinase, cellulase, hemicellulase and xylanase, together with a cyclodextrin compound selected from alpha-, beta- and gamma-cyclodextrin and derivatives thereof, under conditions sufficient to degrade citrus peel cell wall structures and reduce free limonene concentration. The enzyme releases limonene trapped within citrus peel oil glands, while the cyclodextrin simultaneously forms inclusion complexes with the released limonene, suppressing its antimicrobial and enzyme-inhibitory activity without physical removal of terpenes. The combined treatment reduces citrus biomass conversion time from about 75–90 days to about 20–45 days, reduces residual limonene by at least about 80–90%, and yields a naturally clarified, low-alcohol (below about 5% v/v) bioactive liquid composition having a pH of about 3.0 to 5.0, suitable as a natural cleaning base, without requiring external clarifying agents or energy-intensive terpene-removal steps.

No. of Pages : 13 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/04/2026
(21) Application No.202621052731 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : POLARITY-CONTROLLED DUAL-MAGNET LATCHING RELAY AND METHOD OF OPERATING THEREOF

(51) International classification	:H01H50/54, H01H50/20, H01H50/44, H01H50/64
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BG Electricals and Electronics India Ltd
Address of Applicant :M-137, Waluj MIDC, Aurangabad, Maharashtra-431136, India Aurangabad Maharashtra India
(72)**Name of Inventor :**
1)KAPSE, Mr. Sachin Rajendra
2)MISHRA, Mr. Mahesh Shyamlal
3)KULKARNI, Mr. Abhay Prabhakar
4)Mr. Mohammad Zafeer Mohammad Habib

(57) Abstract :
The present invention relates to a polarity-controlled dual-magnet latching relay (100) with central moving core (17). The latch relay (100) comprises a base (03), coil assembly (01) configured to receive a direct current control signal, yoke (02) surrounding at least a portion of the coil assembly (01), a movable central core (17) configured to reciprocate movement in response to magnetic flux generated by the coil assembly (01), a pair of permanent magnets configured to establish a bistable magnetic holding arrangement, a motion transfer mechanism comprising a connecting pin (14), a rotor (06) pivotably mounted on a rotor pin (05), and a slider (09) coupled to movable central core (17), and the pair of permanent magnets retain the movable central core (17) in either of first and second stable positions after removal of the pulse signal, thereby maintaining the switching state without continuous energization of the coil assembly (01).

No. of Pages : 29 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/05/2026

(21) Application No.202621066901 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A COLLAR FOR TRACKING A WILD ANIMAL

(51) International classification	:H04W 4/14, A01M 29/16, H04W 4/029, G01S 19/14, G08B 25/10	(71)**Name of Applicant :** **1)St. Vincent Pallotti College of Engineering and Technology** Address of Applicant :Wardha Road, Gavsi-Manapur, Nagpur, Maharashtra, India - 441108 Maharashtra India (72)**Name of Inventor :** **1)DHAGE, Sandhya Narhari** **2)JIBHKATE, Aayush Nitin** **3)TARAPURE, Rajas Rahul** **4)THOTE, Sarthak Govind** **5)DANGE, Sujal Anant**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A collar for tracking a wild animal comprises a GPS module (3) configured to output latitude and longitude coordinates, a microcontroller (1) connected to the GPS module (3) and configured to execute geofencing logic defining at least one virtual boundary and to detect an intrusion event based on the coordinates, and a cellular communication module (2) connected to the microcontroller (1) via a UART interface and configured to generate an intrusion alert by sending an SMS message and initiating an automated voice call. A power regulation unit (4) is connected to a battery pack (5) to supply required voltage, and a collar enclosure (6) houses the components along with GSM and GPS antennas (7). A wildlife monitoring system comprises the smart animal collar, a cloud backend (8) configured to store location data and intrusion events, and a mobile application (9) configured to display live animal locations, geofence alerts and movement history, wherein the system provides remote monitoring without remote control.

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :09/06/2026
(21) Application No.202621071509 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ASSOCIATION-BASED CANONICAL DISCOVERY ARCHITECTURE FOR COLLECTIVE MEMBER SEO AUTHORITY

(51) International classification	:G06Q30/0601, G06Q10/40, G06Q30/02, G06F16/958, G06F16/957
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TALENTGLOW LLP
Address of Applicant :4/1201 SAGAR DARSHAN, SECTOR 18, NERUL, NAVI MUMBAI 400706, INDIA NAVI MUMBAI Maharashtra India
(72)**Name of Inventor :**
1)KRISHAN JAGTIANI
2)JAYESH RAJESH GADDAM

(57) Abstract :
ABSTRACT The present invention relates to a computer-implemented association-canonical discovery architecture (100) configured to confer collective search-engine authority upon the members of an institutional body — a trade association, professional body, government body or locality association — by hosting the structured digital storefronts of its members under a single canonical domain operated by or on behalf of that institution. The platform comprises an Institutional Onboarding Module (103) configured to register institutional bodies and configure their canonical hosts; a Member Database (108) storing, for each member, a linkage to the institutional body, membership or certification metadata, and a structured business profile; a Canonical Content Generation Engine (104) configured to resolve a network request's host identifier to the corresponding institutional context and dynamically render a member directory, individual member storefront pages and product pages under uniform resource locators rooted at the institutional canonical host; a Structured-Data Emitter (105) generating canonical link elements and JSON-LD structured data attributing publisher identity to the institution; and a Compounding Authority Engine (106) that increases the aggregate discoverability of every member as additional member content is published, through concurrent crawl-budget concentration, internal link-equity flow, topical-cluster formation, institutional trust inheritance and structured-data uniformity. Member discoverability thereby becomes an automatic, compounding by-product of institutional membership rather than the result of individual member marketing effort, while each member retains independent commercial identity and direct buyer communication.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202621071510 A
(19) INDIA
(22) Date of filing of Application :09/06/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : PROXIMITY-NODE MERCHANT DISCOVERY PLATFORM WITH PHYGITAL CONVERSION LAYER

(51) International classification	:G06Q30/00, G06Q20/36, G06Q20/08, G06Q10/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TALENTGLOW LLP
Address of Applicant :4/1201 SAGAR DARSHAN, SECTOR 18, NERUL, NAVI MUMBAI 400706, INDIA Navi Mumbai Maharashtra India
(72)**Name of Inventor :**
1)KRISHAN JAGTIANI ·
2)JAYESH RAJESH GADDAM

(57) Abstract :
ABSTRACT The present invention relates to a computer-implemented proximity-based merchant discovery and phygital commerce platform (100) configured to facilitate location-aware discovery of merchants, structured product and service presentation, buyer-merchant interaction, and transaction enablement within a unified digital ecosystem. The platform comprises a Buyer Computing Device (101) configured to communicate with a Merchant Discovery Platform (102) comprising a Geo-Discovery Engine (103), a Commerce Interaction Layer (104), and a Reservation Engine (105). The Merchant Discovery Platform (102) communicates with a Spatial Index (106) and a Merchant Database (107) maintaining a plurality of geo-verified Merchant Geo Nodes (108) associated with corresponding physical business locations. The Geo-Discovery Engine (103) identifies, retrieves and ranks merchant nodes based on one or more geographical, contextual and search parameters. The Commerce Interaction Layer (104) enables product-level buyer interactions comprising direct purchase, product reservation without immediate payment, shopping cart management, direct merchant communication and inquiry-based engagement. The Reservation Engine (105) manages reservation creation, monitoring, expiry and release of reserved products. The platform further facilitates merchant-controlled pricing visibility, location verification, direct buyer-to-merchant communication and direct merchant payment settlement, thereby providing an integrated digital-to-physical commerce infrastructure for enhancing merchant discoverability, buyer engagement and location-based commercial interactions.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :09/06/2026
(21) Application No.202621071536 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CLOSED-LOOP SUPPLY-SIDE CONTENT WITH SCRUBBED CANONICAL ROUTING AND FIELD-LEVEL VISIBILITY MATRIX

(51) International classification	:G06Q30/06, G06F16/23, G06F21/62, G06F16/958
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TALENTGLOW LLP
Address of Applicant :4/1201 SAGAR DARSHAN, SECTOR 18, NERUL, NAVI MUMBAI 400706, INDIA NAVI MUMBAI Maharashtra India
(72)**Name of Inventor :**
1)KRISHAN JAGTIANI ·
2)JAYESH RAJESH GADDAM

(57) Abstract :
ABSTRACT The present invention relates to a computer-implemented closed-loop supply-side content platform in which a Supply-Side Participant (101) submits structured content through a Supply-Side Digital Storefront (102), persisted as a single canonical record within a Canonical Content Repository (103) by an atomic write. A Field-Level Visibility Matrix (104) declares, for each field of the record, the output surfaces on which it may be disclosed and any value-transformation rule applicable, and is maintained as an append-only structure governed exclusively through a versioned migration pathway, with a continuous-integration enforcement mechanism rejecting any modification outside that pathway. A Dual-Projection Rendering Engine (105) applies the visibility matrix to the canonical record to generate an Authenticated Business-to-Business Projection (106), disclosing the supply-side participant's identity, contact and commercial terms to verified institutional members, and a Public Canonical Projection (107), search-engine-indexable and structurally omitting such sensitive fields. Because both projections derive from the same canonical record, structural drift between them is impossible. A Two-Variable Routing Engine (113) resolves, from a public enquiry, the enquirer's approximate geographical location and a subject-matter or destination context drawn from the canonical record, identifies and ranks eligible credentialed institutional members accordingly, applies a deterministic tie-break sequence, and routes the enquiry to the top-ranked member's private communication channel, recording the decision in an append-only ledger.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202621071537 A

(19) INDIA

(22) Date of filing of Application :09/06/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : RETAILER-ANONYMISED INSTITUTIONAL PRODUCT DISCOVERY WITH SILENT GEOLOCATION ROUTING

(51) International classification	:G06Q10/10, G06F16/958, G06Q30/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TALENTGLOW LLP
Address of Applicant :4/1201 SAGAR DARSHAN, SECTOR 18, NERUL, NAVI MUMBAI 400706, INDIA NAVI MUMBAI Maharashtra India
(72)**Name of Inventor :**
1)KRISHAN JAGTIANI
2)JAYESH RAJESH GADDAM

(57) Abstract :
ABSTRACT The present invention relates to a computer-implemented retailer-anonymised institutional product discovery platform configured to enable certified retailers in design-driven commercial categories to publish structured product content to a shared, search-engine-indexable discovery surface while structurally suppressing retailer identity from every public-facing output. A Structured Design Repository (104) persists, for each product record, a discovery-relevant attribute class (301) and a routing-only attribute class (302) comprising a retailer-identification attribute. A Public Page Generation Engine (105) auto-generates publicly accessible discovery pages (106) from the discovery-relevant attribute class alone, having no programmatic access to the routing-only class, with structural suppression further enforced by build-time static-analysis checks. Emitted structured data attributes publisher identity to an institutional body rather than to the individual retailer. Upon a buyer's expression of interest, a Silent Routing Engine (109) extracts the buyer's approximate location from network-level request metadata, identifies and ranks eligible certified retailers by proximity, records the routing decision in an append-only ledger (110), and routes the buyer to a private communication channel associated with the nearest eligible retailer, whose identity is disclosed to the buyer for the first time only within that private channel. A two-tier suppression mechanism independently governs retailer identity and any upstream supplier identity, enabling design-driven retailers to achieve public discoverability without compromising competitive confidentiality.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/09/2025
(21) Application No.202521082522 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : APPARATUS AND METHOD FOR LIQUID-RESISTANCE DISCHARGE AND REENERGIZATION OF CAPACITOR BANKS

(51) International classification	:H01G2/00, H02J7/00, H02J3/1828
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rohiet Babbajeeh Jagtap
Address of Applicant :Flat No. 21, Building No. B, Sarita Kunj, Near Ladies ITI, Kasarwadi, Pune City, Pune, Maharashtra, - 411034 Maharashtra India
(72)**Name of Inventor :**
1)Rohiet Babbajeeh Jagtap

(57) Abstract :
ABSTRACT APPARATUS AND METHOD FOR LIQUID-RESISTANCE DISCHARGE AND RE-ENERGIZATION OF CAPACITOR BANKS The present disclosure relates to an apparatus and method for discharging and re-energizing a capacitor bank (114). The apparatus comprises a liquid-resistance discharge unit (204) including a vessel (206) containing a liquid resistive medium (308) and two electrodes (208), a discharge switching device (202), a voltage sensing arrangement (220), and a control unit (222). The control unit (222) selectively connects the liquid-resistance discharge unit (204) across the capacitor bank following disconnection, monitors residual voltage, and permits re-energization when residual voltage reaches or falls below predetermined threshold. A current-limiting impedance (210) is connected during an initial portion of re-energization and subsequently bypassed by a bypass switching device (212). Conductivity, temperature, and liquid level are monitored by sensing arrangements (214, 216, 218), and conductivity of the liquid resistive medium (308) is conditioned using a liquid-conditioning arrangement (226). Effective discharge resistance is adjustable through the liquid resistive medium (308) and the electrodes (208). FIG. 2

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/06/2026

(21) Application No.202621079250 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : UNIFIED EXPORTERS INFRASTRUCTURE: CAPABILITY-DECLARED NATIONAL DISCOVERY WITH AI RAW-MATERIAL VISUALISATION AND DEMAND-INTELLIGENCE ROUTING

(51) International classification	:G06Q 30/02, G06Q 10/06, G06Q 30/06, G06Q 50/04, G06T 19/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TALENTGLOW LLP
Address of Applicant :4/1201 SAGAR DARSHAN, SECTOR 18, NERUL, NAVI MUMBAI 400706, INDIA Navi Mumbai Maharashtra India
(72)**Name of Inventor :**
1)KRISHAN JAGTIANI ·
2)JAYESH RAJESH GADDAM

(57) Abstract :
ABSTRACT The present invention relates to a computer-implemented capability-declared national discovery platform hosted under the canonical domain of a national trade-promotion association, on which every verified member exporter (511-514) is persisted as a structured node carrying a declared capability profile within an Exporter Capability Database (516) comprising machine inventory, manufacturing capacity, currently available capacity, minimum order quantity, lead time, cluster location and certifications. A Multi-Layer Capability Taxonomy (530-534), governed by a curator-maintained extensible vocabulary, structures each exporter's offering into a discrete, canonically addressable tuple. An Artificial-Intelligence Visualisation Pipeline (540-547) generates photorealistic finished-product renderings from a raw-material photograph or video, removing the content-production barrier facing smaller exporters. A Canonical Content-Node Assembly Engine (555) combines the taxonomy tuple, the rendered imagery, the capability profile and a currently available capacity declaration into a structured, search-engine-indexable canonical page. A Precision Multi-Parameter Matching Engine (570-574) applies a weighted freshness, capacity-fit, certification-match, cluster-affinity and responsiveness scoring function to return a ranked shortlist of matched exporters, each with a direct messaging deeplink, responsive to a structured buyer query. A Freshness-Aware Capacity Lifecycle (601-606) classifies each capacity declaration and weights matching accordingly. A Demand-Intelligence Layer (580-590), built upon an append-only event ledger (585) and a plurality of automated detectors, identifies unmet buyer demand and geo-origin demand spikes for exporter notification and institutional or governmental policy use. A Smart Routing Engine (611) disaggregates a buyer enquiry exceeding a single exporter's capacity into a multi-exporter consortium fulfilment plan.

No. of Pages : 29 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2025
(21) Application No.202521091756 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD AND SYSTEM FOR CONTROLLING OPERATIONS OF MOTOR IN ELECTRIC VEHICLE

(51) International classification	:B60L50/51, H02P23/28, H02P25/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Matter Motor Works Private Limited
Address of Applicant :301, PARISHRAM BUILDING, 5B RASHMI SOC., NR. MITHAKHALI SIX ROADS, NAVRANGPURA AHMEDABAD, GUJARAT, INDIA - 380010 Ahmedabad Gujarat India
(72)**Name of Inventor :**
1)KUMAR PRASAD TELIKEPALLI
2)SHIVAM GARG
3)DARSHAN PANCHAL
4)MOHIT JHA
5)ANKUR SRIVASTAVA

(57) Abstract :
ABSTRACT METHOD AND SYSTEM FOR CONTROLLING OPERATIONS OF MOTOR IN ELECTRIC VEHICLE The present disclosure describes a system (100) for dynamically controlling operation of an electric motor in an electric vehicle. The system (100) comprises at least one electric motor (102) configured for operation in a single-phase mode and a three-phase mode, a plurality of sensing units (104) configured to generate operational data of the electric motor, at least one control unit (106) communicatively coupled to the plurality of sensing units (104), and at least one phase switching unit (108) electrically connected between the at least one control unit (106) and the at least one electric motor (102). Further, the at least one control unit (106) is configured to determine an operating phase configuration based on the operational data and switch the at least one electric motor (102) between a single-phase mode and a three-phase mode during vehicle operation. FIG. 1

No. of Pages : 38 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/05/2026

(21) Application No.202621062490 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SYSTEM AND METHOD FOR PERFORMING COMPUTER-VISION-BASED EVALUATION AND CONDITION ASSESSMENT OF ELECTRONIC DEVICES

(51) International classification	:G06T7/00, G06V10/82, G06V30/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)ONEASSIST CONSUMER SOLUTIONS PRIVATE LIMITED

Address of Applicant :3RD FLOOR, TOWER C, TIMES SQUARE, ANDHERI - KURLA ROAD, OPP. MITTAL INDUSTRIAL ESTATE, MAROL, ANDHERI EAST, MUMBAI, MAHARASHTRA 400059 Mumbai Maharashtra India

(72)**Name of Inventor :**

1)Subrat Pani

2)Gagan Maini

3)Saurabh Chandra

(57) Abstract :

The present invention provides an automated system and method to evaluate, classify, and generate a validated condition report for an electronic device using computer vision. The system (100) receives a video captured via a guided script enforcing a predefined sequence of device surfaces and screen displays. Frames are extracted using a configurable frame-skip interval and corrected for geometric rotation based on metadata. A first trained object detection model (112A) classifies extracted frames into canonical surface segments and identifier-bearing segments. A frame quality assessment model (112B) tags valid frames and filters quality deficiencies. Valid frames are processed to detect physical defects across surfaces using a cross-segment detector and to extract identifiers directly via an OCR engine. Video authenticity is verified by evaluating temporal continuity, identifier consistency, and synthetic manipulation. A weighted aggregation function computes a consolidated device-level score, producing a structured report with annotated evidence heatmaps. FIG. 1

No. of Pages : 42 No. of Claims : 17

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202621081162 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND A METHOD OF HYBRID RENEWABLE HIGH-TEMPERATURE THERMAL PROCESSING

(51) International classification	:F03G 6/06, F24S 23/70, F24S 20/20, F24S 23/71, F03G 6/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Shrikant Gopichand Patole
Address of Applicant :B 2004, Arkade Earth, Kanjur Village Road, Near Nitco tiles, Kanjurmarg east, Mumbai 400042 Maharashtra India
2)Debarshi Dey
3)Soma Dey
(72)**Name of Inventor :**
1)Shrikant Gopichand Patole
2)Debarshi Dey

(57) Abstract :
Disclosed is a system (100) for off-grid high-temperature thermal processing. The system (100) comprises a high-temperature thermal processing chamber (102), a Scheffler concentrating solar thermal dish (104), a compound parabolic concentrator (106), a vertical axis wind turbine (108), a lithium ferro phosphate battery energy storage system (110), a phase change material thermal buffer (112), a silicon carbide or molybdenum disilicide heating element (120), an adaptive energy management system (124), and a power controller (130). The compound parabolic concentrator (106) corrects focal drift and provides uniform thermal flux into the chamber (102). The phase change material thermal buffer (112) stores and releases heat to stabilize chamber temperature during renewable-energy intermittency. The adaptive energy management system (124) dynamically coordinates solar thermal energy, wind-generated electrical energy, stored electrical energy, thermal buffer discharge, and electrical heating to maintain substantially continuous high-temperature processing with improved energy continuity, thermal stability, and reduced fuel dependency. Fig. 1 is the reference figure.

No. of Pages : 31 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2025
(21) Application No.202521090515 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A MICROFLUIDIC DEVICE TO GENERATE MICROBUBBLES

(51) International classification	:B01L0003000000, C12M0003060000, A61B0017220000, B01F0023231000, B01F0025433000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR
Address of Applicant :Palaj, Gandhinagar-382055, Gujarat, India Gandhinagar Gujarat India
(72)**Name of Inventor :**
1)DALVI, SAMEER VISHVANATH
2)KHAN, AAQIB HAROON
3)SHETH, JINIOUS KAUSHALBHAI
4)PARAKH, PUSHKAR DEEPAK

(57) Abstract :
ABSTRACT The present disclosure provides a microfluidic device and a method for the production of microbubbles with narrow-size distribution for use as ultrasound contrast agents and related biomedical applications. The disclosed device enables the controlled formation of microbubbles to obtain microbubbles with high production rates. The device is compact, portable, and cost-effective compared to conventional systems. The disclosed device and method may be particularly suitable for ultrasound imaging, targeted drug delivery, and biotechnology research applications. Figure 1

No. of Pages : 37 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :10/04/2026
(21) Application No.202621046354 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD FOR PREPARING A FORMALDEHYDE-FREE LIGNIN-BASED BIOADHESIVE

(51) International classification	:C09J197/02, C08H7/00, C07G1/00, C08H8/00, C09J11/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RENNOWASTE PRIVATE LIMITED
Address of Applicant :7, Tulsidham, Phase 3, Hake Nagar, Tilaknagar, Latur, Maharashtra, India - 413531 Maharashtra India
(72)**Name of Inventor :**
1)CHAUHAN, Aditya Singh
2)AGRAWAL, Prajwal
3)DABRAL, Anuj Kumar

(57) Abstract :
The present invention relates to a method for preparing a formaldehyde-free lignin-based bioadhesive. The invention aims to develop a lignin-based, environmentally friendly bioadhesive capable of replacing petroleum-derived synthetic adhesives such as phenol formaldehyde and urea formaldehyde resins used in plywood and engineered wood products. Lignin cake obtained from lignocellulosic biomass is dissolved in an alkaline solution of 0.1–1.5 N sodium hydroxide followed by centrifugation to remove cellulose and ash impurities. Sulphuric acid of 0.5–1.5 M is added to adjust pH 2 at 60–95°C for two hours to precipitate purified lignin which is filtered and dried. The purified lignin is solubilized in sodium hydroxide solution and modified with chloroacetic acid at 65–85°C for 1–3 hours to obtain carboxymethylated lignin. Glyoxal is added at 40–60°C to form a crosslinked network, followed by neutralization with acetic acid and addition of calcium carbonate to produce a formaldehyde-free lignin-glyoxal bioadhesive for plywood applications.

No. of Pages : 16 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :05/05/2026
(21) Application No.202621057035 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD OF TREATING POLYAMIDE SUTURES FOR ENHANCED PROPERTIES, AND ENHANCED POLYAMIDE SUTURES

(51) International classification	:A61B 17/04, A61B 17/06, A61L 17/00, A61L 17/14, A61L 17/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ADVANCED MEDTECH SOLUTIONS PRIVATE LIMITED
Address of Applicant :P-21-22, 25-26 & 34-35, GIDC Manjusar, Tal-Savli, Dist. Vadodara, Gujarat-391775, India Vadodara Gujarat India
(72)**Name of Inventor :**
1)DHAMA, Devendra Kumar
2)KAUSHIK, Haridutt
3)MISTRY, Viral
4)MALKHANIA, Satish

(57) Abstract :
The present invention provides an innovative method for treating medical use polyamide sutures in order to enhance its pliability, tissue holding strength, and tensile strength. The method also lowers the suture memory, smooth knot run down, and higher fray resistance.

No. of Pages : 16 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/07/2026

(21) Application No.202621092228 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD AND SYSTEM FOR USER INPUT DETECTION AND DISPLAY SYSTEM USING FPGA-CONTROLLED HEXADECIMAL KEYPAD INTERFACE WITH DEBOUNCING TECHNIQUE

(51) International classification	:H03M 11/20, G06F 3/023, G06F 3/02, H04M 1/23, H03M 11/22	(71)**Name of Applicant :** **1)RATHKANTHIWAR, Shubhangi Vikas** Address of Applicant :Yeshwantrao Chavan College of Engineering, Hingna Rd, Wanadongri ct, Nagpur, Maharashtra, India - 441110 Maharashtra India (72)**Name of Inventor :** **1)RATHKANTHIWAR, Shubhangi Vikas**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

This invention presents a hexadecimal keypad interface built using an FPGA (FieldProgrammable Gate Array) platform. The system uses a 4x4 matrix keypad (16 keys labeled 0–9 and A–F), interfaced with the Nexys 4 DDR FPGA board and programmed using Verilog HDL. This invention presents a hexadecimal keypad interface system implemented on an FPGA using Verilog. A 4x4 matrix keypad, capable of inputting hexadecimal values (0–9, A–F), is interfaced with the Nexys 4 DDR FPGA board. The system utilizes a matrix scanning technique to efficiently detect key presses, with a dedicated debounce logic to eliminate false signals from mechanical bounce. The detected key is encoded into its hexadecimal equivalent and displayed on a 7-segment display. The design showcases the power of FPGA-based solutions in developing fast, reliable, and scalable user input systems suitable for embedded applications and digital electronics.

No. of Pages : 28 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :13/04/2026
(21) Application No.202621047125 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVICE AND METHOD FOR DETECTION OF TARGET COMPOUNDS IN A BREATH SAMPLE

(51) International classification	:G01N21/64, G01N33/497, G01N21/77, G01N21/78, G08B21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MIT School of Bioengineering, Sciences & Research
Address of Applicant :MIT-ADT University, Pune 412201, Maharashtra, India Pune Maharashtra India
2)VEM Tooling India Pvt. Ltd.
(72)**Name of Inventor :**
1)Pooja Rana
2)Navya Sethu
3)Renu Vyas
4)Vijaysarathi
5)Preetam Bala
6)Marc Weinman
7)Gopal Kabre

(57) Abstract :
ABSTRACT DEVICE AND METHOD FOR DETECTION OF TARGET COMPOUNDS IN A BREATH SAMPLE Disclosed is a portable fluorescence-based sensing device (100) incorporating a metal–organic framework (MOF) coated transparent substrate (20) enclosed within a compact handheld enclosure (50). When a user exhales into the device, breath containing Cannabidiol (CBD) and equivalent compounds interacts with the MOF surface, resulting in a characteristic fluorescence emission. An integrated light source (30) illuminates the MOF-coated substrate (20), and the emitted fluorescence is captured using a colour sensor (40) operating in the visible wavelength range. The detected optical signal is processed electronically and displayed in real time. Upon detection of the target compound, a visual indication is provided on the display and an audible alert is generated through a buzzer. The device enables rapid, non-invasive, and qualitative detection of cannabinoids without the need for laboratory-based analysis or complex instrumentation. Reference figure: Figure 1

No. of Pages : 20 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/07/2026

(21) Application No.202621092776 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SPLINT DEVICE FOR MANDIBULAR DISTRACTION OSTEOGENESIS

(51) International classification	:A61B 17/00, A61B 90/00, A61B 17/17
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ANCHLIA, Sonal
Address of Applicant :4, "swa'' bungalows, Tapovan circle, DistAhmedabad, Gujarat, India - 382424 Gujarat India
2)KHAN, Zibran
3)GUPTA, Mudit
4)KUMAR, Jitendra
5)DHUVAD, Jigar
6)GOSAI, Hrushikesh
7)SHAH, Shalin
8)ANWAR, Shameen
(72)**Name of Inventor :**
1)ANCHLIA, Sonal
2)KHAN, Zibran
3)GUPTA, Mudit
4)KUMAR, Jitendra
5)DHUVAD, Jigar
6)GOSAI, Hrushikesh
7)SHAH, Shalin
8)ANWAR, Shameen

(57) Abstract :

The present invention is related to a splint device for mandibular distraction osteogenesis designed for guided and parallel placement of shanz pins (50) during mandibular distraction osteogenesis. The device comprises a mandibular splint body (10) contoured along the inferior border of the mandible (60), incorporating integrated parallel drill sleeve ports (20) arranged in fixed parallel orientation to control drill trajectory and maintain parallel Shanz pin (50) alignment. Each drill sleeve port (20) has an internal diameter of approximately 2.4 mm with an open-entry angulation of approximately 60°, with an approximate width of 3 mm. The device incorporates osteotomy guide markings (40) for corticotomy orientation and a predefined inferior alveolar nerve reference marking (30) indicating safe surgical limits. The splint body (10) maintains a minimum width of approximately 5 mm at the port region for structural stability. The device is digitally designed using computed tomography-based virtual surgical planning and fabricated by three-dimensional printing, compatible with standard Shanz pins (50) and mandibular distractor devices (70).

No. of Pages : 21 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/07/2026

(21) Application No.202621092827 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : "AGENTIC ARTIFICIAL INTELLIGENCE DATA ORCHESTRATION PLATFORM USING GOAL-ORIENTED AUTONOMOUS WORKFLOW AGENTS"

(51) International classification	:G06N 20/00, G06N 5/04, G06N 5/02, G06Q 10/06, G06F 16/2455
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Abhilash Rao Mesala
Address of Applicant :San Francisco, California, USA U.S.A.
(72)**Name of Inventor :**
1)Abhilash Rao Mesala

(57) Abstract :
An intelligent data orchestration platform and computer-implemented method for autonomous goal-oriented workflow execution are disclosed. The platform comprises an objective interpretation engine configured to receive high-level natural language business objectives and extract target data requirements. A multi-agent orchestration ecosystem comprising domain-specific autonomous artificial intelligence agents collaboratively decomposes the objectives into operational milestones and formulates a dynamic execution graph without static pipeline schedules. Specialized agents autonomously discover enterprise data sources, negotiate schema discrepancies, dynamically synthesize transformation algorithms, and validate data integrity. A distributed execution and monitoring engine executes the workflow while embedding autonomous self-healing capabilities to detect runtime anomalies, resolve schema drift, and reconstruct execution pathways in real time, ensuring continuous, reliable data delivery without human engineering intervention.

No. of Pages : 25 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/06/2025
(21) Application No.202521053503 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI INTEGRATED & AUTOMATED POLLEN DRONE FOR VOLATILE ORGANIC COMPOUNDS AND METHOD THEREOF

(51) International classification	:A01H0001020000, G05D0001000000, B64U0101300000, B64C0039020000, A01H0005020000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JAYKUMAR V. CHOUDHARY
Address of Applicant :E 304, Dsk Vidyanagari Phase 2, Pashan Sus Road, Baner, Survey No 47, Pune - 411045, Maharashtra India Pune Maharashtra India
(72)**Name of Inventor :**
1)JAYKUMAR V. CHOUDHARY

(57) Abstract :
ABSTRACT AN AI INTEGRATED & AUTOMATED POLLEN DRONE FOR VOLATILE ORGANIC COMPOUNDS AND METHOD THEREOF The present invention relates to the field of agricultural automation, precision farming, and pollination technology. More particularly the present invention provides an automated drone system integrated with AI vision module for real time and autonomous identification and detection of flower type and gender of flower for collection of pollen from male flowers using the dispenser arm and dispersion of pollen to targeted female flowers and performs precised targeted pollination.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/07/2026
(21) Application No.202621085274 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR PLATFORM-NATIVE CANONICAL BUSINESS REPRESENTATION WITH MULTI-ECOSYSTEM DIGITAL DISCOVERY

(51) International classification	:G06Q30/06, G06F16/95, G06Q10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TALENTGLOW LLP
Address of Applicant :4/1201 SAGAR DARSHAN, SECTOR 18, NERUL, NAVI MUMBAI 400706, INDIA Navi Mumbai Maharashtra India
(72)**Name of Inventor :**
1)KRISHAN JAGTIANI
2)JAYESH RAJESH GADDAM

(57) Abstract :
ABSTRACT The present invention relates to a computer-implemented Platform-Native Canonical Discovery Architecture (600) in which any participating business entity, professional, merchant or artisan achieves structured digital visibility and buyer discoverability without owning or maintaining an independent website, mobile application, or digital-marketing infrastructure. The platform comprises a Canonical Business Node (601), created once by the participating entity, persisting identity, product/service, communication, and visibility-control data on the shared platform; an Ecosystem Participation Interface (602) applying institutional or proximity-governed rendering and disclosure rules to the same canonical node across multiple discovery contexts; a Canonical Node Assembly Engine (603) generating machine-readable, search-engine-indexable structured content under canonical domains associated with each ecosystem; a Direct Communication Layer (604) enabling verified buyer-to-entity messaging embedded within the canonical node's rendered output; and a Network-Density Compounding Mechanism (605) wherein each additional node joined to the platform increases discovery utility for all pre-existing nodes. The architecture serves as a foundational common layer from which a plurality of distinct discovery-ecosystem embodiments are derived — including proximity-based local discovery, association-canonical institutional discovery, anonymised institutional sourcing, visibility-matrix-governed professional discovery, and capability-declared export discovery — all operating upon the same underlying canonical node without duplication or external-website dependency.

No. of Pages : 21 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/07/2026
(21) Application No.202621093356 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Bio-formulation with insecticidal activity for control of rugose spiralling whitefly in Cocos nucifera

(51) International classification	:A01N 65/00, A01N 65/08, A01N 65/40, A01P 7/04, A01N 43/90, A01N 25/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Gujarat Natural Farming Science University
Address of Applicant :Near New Court, Pavagadh Road, Panchmahal, Halol, Gujarat, India - 389350 Halol Gujarat India
(72)**Name of Inventor :**
1)Chenabhai Gamnabhai Panchal
2)Dr. Gopalbhai Dhirajbhai Vadodariya
3)Dr. Alpesh Ashokbhai Bhimani
4)Dr. Raghunandan B L
5)Dr. Prabhu Nayaka
6)Dr. Chandulal Karamshi Timbadia

(57) Abstract :
The present invention discloses a synergistic bio-formulation with insecticidal activity for controlling rugose spiralling whitefly in Cocos nucifera. The composition comprises fermented sap of Borassus flabellifer as a biologically derived solvent matrix and an aqueous leaf extract of Adhatoda vasica containing quinazoline alkaloids, including vasicine and vasicinone. Fermentation-derived ethanol, organic acids, and metabolites improve solubilization, stability, foliar adhesion, and trans-cuticular penetration of the bioactive compounds, thereby enhancing pesticidal efficacy. The formulation is prepared by controlled fermentation of palm sap, aqueous extraction of A. vasica leaves, clarification, blending in optimized proportions, and pH adjustment under mildly acidic conditions. Field studies demonstrate effective suppression of rugose spiralling whitefly and other sap-sucking pests at practical application rates. The biodegradable, eco-compatible formulation provides a sustainable alternative to synthetic insecticides for integrated pest management and natural farming systems.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/06/2026

(21) Application No.202621070114 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SINGLE-USE DISPENSING CAP FOR INTRODUCING ACTIVE INGREDIENTS INTO A FLUID CONTAINER

(51) International classification	:B65D51/28, G16H20/10, G06K19/07, B65D81/32, A61J7/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SWETA DEVNANI
Address of Applicant :201A, Rising Sun Apartment, Juhu Tara Road, Vile Parle (W), Mumbai – 400049, Maharashtra, India Mumbai Maharashtra India
(72)**Name of Inventor :**
1)SWETA DEVNANI

(57) Abstract :
A single-use dispensing cap (200) engageable with the mouth of a fluid container comprising a body (202) divided by a partition wall (214) into a first chamber (218) and a second chamber (220), each holding a respective active ingredient; at least one breakable seal (210) closing lower ends of the chambers; and an actuator button (206) coupled to at least one actuator stem (212) carrying piercing members. A single compressive stroke of the actuator button (206) ruptures the seal substantially simultaneously for both chambers, discharging the contents into the host fluid. A friction-engagement formation at the lower end of the body engages container mouths independently of thread profile. A Near Field Communication tag optionally carried by the body enables dosage tracking and co-factor bioavailability information on a portable computing device.

No. of Pages : 40 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :15/07/2026
(21) Application No.202621086704 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A GRID-CONNECTED THREE-PHASE POWER CONVERSION SYSTEM USING MULTI-MODE DC–DC CONVERTER MODULES

(51) International classification	:H02M7/42, H02M1/08, H02J3/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY, SURAT
Address of Applicant :Ichchhanath Surat- Dumas, Road, Keval Chowk, Surat, Gujarat 395007, India Gujarat India
(72)**Name of Inventor :**
1)Dr. Jammala Venkataramanaiah
2)Pankaj Darji
3)Gulshan Yadav
4)Dr. Yellasiri Suresh

(57) Abstract :
A GRID-CONNECTED THREE-PHASE POWER CONVERSION SYSTEM USING MULTI-MODE DC–DC CONVERTER MODULES The present invention relates to a grid-connected three-phase DC–AC power conversion system with an integrated architecture for efficient power conversion. The system employs multiple converter units, each based on a two-switch DC–DC configuration with actively controlled semiconductor devices, preferably IGBTs, enabling bidirectional current flow and improved switching performance. The converter units operate as phase modules and are directly interfaced with the three-phase grid. A modulation technique is utilized to control the switching actions based on modified sinusoidal reference signals, ensuring that the output voltages inherently track the desired waveforms. This enables the direct synthesis of a three-phase sinusoidal output without requiring additional output-filtering components, thereby enhancing power density and overall system compactness. Consequently, it provides improved efficiency, compact design, and reliable harmonic performance for grid-connected applications.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/08/2026

(21) Application No.202621094390 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADVANCED SUBMERSIBLE AIR-POWERED MIXING APPARATUS AND SYSTEM FOR WASTEWATER TREATMENT

(51) International classification	:C02F 3/20, B01F 3/04, C02F 3/12, C02F 3/22, C02F 1/24	(71)**Name of Applicant :** **1)AWATECH SOLUTIONS (I) PRIVATE LIMITED** Address of Applicant :E/4B,OJAS APPARTMENT NIRMA CIRCLE NEHRUNAGAR CROSS ROAD AMBAWADI AHMEDABAD Pin 380015 AHMEDABAD Gujarat India
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Vinitkumar Patel**
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a submersible air-powered mixing apparatus for wastewater treatment applications. The apparatus comprises a vertically extending circulation chamber, a diffuser support plate, a plurality of diffuser assemblies, an air distribution manifold, an air supply conduit, one or more lower intake regions, and a support frame configured for installation within a wastewater treatment basin. Compressed air supplied through the air supply conduit is distributed to the diffuser assemblies, which generate air bubbles creating an air-lift effect within the circulation chamber. The air-lift effect induces wastewater from lower regions of the basin to flow upwardly through the circulation chamber and discharge outwardly from an upper region thereof. The discharged wastewater generates horizontal circulation currents while the air-lift effect simultaneously generates vertical circulation currents, thereby providing efficient multidirectional mixing, solids suspension, oxygen distribution, and dead-zone elimination. The apparatus operates without submerged rotating mechanical components, resulting in reduced maintenance requirements, improved reliability, enhanced treatment efficiency, and lower operating costs.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :10/04/2026

(21) Application No.202621045905 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR NON-INTRUSIVE ULTRASONIC ASSESSMENT OF COMPOSITE HEARTH LINING IN BLAST FURNACE

(51) International classification	:G01N29/04, G01N29/44, G01N29/06, G01N29/11, G01B17/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY, BHILAI
Address of Applicant :Kutelabhata, Bhilai- 491002, Chhattisgarh, India Bhilai Chattisgarh India
(72)**Name of Inventor :**
1)GEETHA, Ganesh Kolappan
2)GANGOPADHYAY, Soumya

(57) Abstract :
A method and system (100) for non-intrusive ultrasonic assessment of composite hearth lining in blast furnaces during operation. The system (100) employs an ultrasonic transducer assembly (102) configured in a matrix arrangement for pulse-echo measurements through the external furnace shell. A high-gain ultrasonic pulser-receiver unit (104) compensates for attenuation through composite refractory structures. Signal acquisition and processing module (106) is configured for capturing and processing time-domain echo signals. Data analysis unit (108) is configured to compute time-of-flight, B-Scan, and Total Focusing Method (TFM) based computation on the processed signals, to eliminate false alarm. TFM signal processing extracts weak backwall echoes from the refractory-molten interface. Residual lining thickness is calculated from time-of-flight measurements using calibrated acoustic velocities. The system (100) enables circumferential mapping of thickness distribution, wear rate estimation, and detection of anomalous penetration zones without requiring furnace shutdown or invasive access, supporting predictive maintenance in steel plant operations.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/08/2026

(21) Application No.202621094388 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A HIGH-PERFORMANCE ORGANIC , SOLID & MUNICIPAL WASTE DEPACKAGING SYSTEM AND METHOD FOR SEPARATING ORGANIC OR SOLID MATERIAL FROM PACKAGED WASTE

(51) International classification	:B09B 3/00, B09B 5/00, B02C 18/14, B02C 18/22, B02C 13/286	(71)**Name of Applicant :** **1)AWATECH SOLUTIONS (I) PRIVATE LIMITED** Address of Applicant :E/4B,OJAS APPARTMENT NIRMA CIRCLE NEHRUNAGAR CROSS ROAD AMBAWADI AHMEDABAD Pin 380015 GUJARAT Ahmedabad Gujarat India (72)**Name of Inventor :** **1)Vinitkumar Patel**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a high-performance organic waste depackaging system and method for separating biodegradable and organic material from packaged waste streams. The system comprises a feed hopper, a processing chamber, a rotatable shaft assembly, a plurality of paddle assemblies, a drive mechanism, a screening assembly, an organic material discharge outlet, and a packaging residue discharge outlet. Packaged waste is introduced into the processing chamber, wherein the rotating paddle assemblies generate controlled agitation, compression, rubbing, and separating actions to open packaging materials and release the organic contents. The liberated organic material passes through the screening assembly and is recovered as a substantially homogeneous organic fraction, while packaging residues are separately discharged. The system minimizes fragmentation of packaging materials, reduces contamination of the recovered organic material, and improves overall recovery efficiency. The invention is particularly suitable for processing food waste, supermarket returns, agricultural waste, industrial organic waste, and other packaged biodegradable materials for resource recovery, composting, anaerobic digestion, and renewable energy generation applications.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/08/2026

(21) Application No.202621094389 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED FIXED-FILM ACTIVATED SLUDGE (IFAS) BIOCELL SYSTEM COMPRISING MODULAR FIXED-FILM CAGE MEDIA FOR BIOLOGICAL WASTEWATER TREATMENT

(51) International classification	:C02F 3/20, B01F 3/04, C02F 3/12, C02F 3/22, C02F 1/24	(71)**Name of Applicant :** **1)AWATECH SOLUTIONS (I) PRIVATE LIMITED** Address of Applicant :E/4B,OJAS APPARTMENT NIRMA CIRCLE NEHRUNAGAR CROSS ROAD AMBAWADI AHMEDABAD Pin 380015 GUJARAT Ahmedabad Gujarat India
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Vinitkumar Patel**
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a submersible air-powered mixing apparatus for wastewater treatment applications. The apparatus comprises a vertically extending circulation chamber, a diffuser support assembly, a plurality of diffuser units, an air distribution manifold, an air inlet conduit, wastewater intake regions, and a structural support frame. Compressed air supplied through the air inlet conduit is distributed to the diffuser units, which release air bubbles into the surrounding wastewater to generate an air-lift effect within the circulation chamber. The air-lift effect induces wastewater to flow upwardly through the circulation chamber and discharge outwardly from an upper region thereof, thereby producing simultaneous vertical and horizontal circulation currents throughout a wastewater treatment basin. The apparatus promotes efficient mixing, solids suspension, oxygen distribution, sludge control, and elimination of stagnant zones. The system operates without submerged rotating mechanical components such as motors, impellers, shafts, or gearboxes, thereby reducing maintenance requirements, improving operational reliability, enhancing treatment efficiency, and lowering overall operating costs.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/08/2026

(21) Application No.202621098988 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TOOL GUIDE ASSEMBLY WITH ADJUSTABLE CUTTING LENGTH

(51) International classification	:A61B34/30, A61B17/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Merai Newage Private Limited
Address of Applicant :Survey No. 1574, Bilakhia House, Chala Muktanand Marg, Vapi, Valsad-396191 Gujarat, India Gujarat India
(72)**Name of Inventor :**
1)SAHANE, Pradip Vishnu
2)DESAI, Indrajit Anandrao

(57) Abstract :
ABSTRACT TITLE OF THE INVENTION: TOOL GUIDE ASSEMBLY WITH ADJUSTABLE CUTTING LENGTH The present disclosure pertains to an assembly (100) including a guide (110) coupled to a robotic arm (10) and a stopper assembly (120) movably disposed on a distal end (110b) of the guide (110). The guide (110) includes at least one longitudinal slot (130) and a plurality of indexing notches (140) disposed towards the distal end (110b) of the guide (110). The at least one longitudinal slot (130) extends transversely through the guide (110). The plurality of indexing notches (140) is disposed along a length of the at least one longitudinal slot (130). The stopper assembly (120) includes a stopper frame (170) movably disposed towards the distal end (110b) of the guide (110) and a locking member (190) coupled to the stopper frame (170). The locking member (190) is configured to removably couple the stopper frame (170) with a selected indexing notch (140) of the plurality of indexing notches (140). Fig. 1a

No. of Pages : 35 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/08/2026

(21) Application No.202621098989 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TOOL GUIDE ASSEMBLY WITH ADJUSTABLE CUTTING LENGTH

(51) International classification	:A61B 34/30, A61B17/34, B26B29/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Merai Newage Private Limited
Address of Applicant :Survey No. 1574, Bilakhia House, Chala Muktanand Marg, Vapi, Valsad-396191 Gujarat, India Gujarat India
(72)**Name of Inventor :**
1)SAHANE, Pradip Vishnu
2)DESAI, Indrajit Anandrao

(57) Abstract :
TITLE OF INVENTION: TOOL GUIDE ASSEMBLY WITH ADJUSTABLE CUTTING LENGTH The present disclosure pertains to an assembly (100) including a guide (110) coupled to a robotic arm (10) and a stopper assembly (120). The guide (110) includes at least one longitudinal slot (130) and a plurality of locking slots (140). The at least one longitudinal slot (130) extends transversely through the guide (110). The plurality of locking slots (140) is disposed towards the distal end (110b) of the guide (110) along a length of the at least one longitudinal slot (130). The stopper assembly (120) includes a stopper frame (170) and a locking member (190). The stopper frame (170) is movably disposed on a distal end (110b) and is removably coupled to a selected locking slot (140) of the plurality of locking slots (140). The locking member (190) is received within an opening (172) defined by the stopper frame (170) and engages with a selected locking slot (140). Fig. 1a

No. of Pages : 37 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :14/08/2026
(21) Application No.202621098990 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : TOOL GUIDE ASSEMBLY WITH ADJUSTABLE CUTTING LENGTH

(51) International classification	:A61B34/30, A61B17/17, A61B17/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Merai Newage Private Limited
Address of Applicant :Survey No. 1574, Bilakhia House, Chala Muktanand Marg, Vapi, Valsad-396191 Gujarat, India Gujarat India
(72)**Name of Inventor :**
1)DESAI, Indrajit Anandrao
2)SAHANE, Pradip Vishnu
3)BHOSALE, Subhodh Jalindar

(57) Abstract :
TITLE OF THE INVENTION: TOOL GUIDE ASSEMBLY WITH ADJUSTABLE CUTTING LENGTH The present disclosure pertains to an assembly (100, 200, 300) including a guide (110, 210, 310) coupled to a robotic arm (10) and a stopper assembly (120, 220, 320) movably disposed on a distal end (110b, 210b, 310b) of the guide (110, 210, 310). The guide (110, 210, 310) includes at least one longitudinal slot (130, 230, 330) and a plurality of blind holes (140, 240, 340) disposed towards the distal end (110b, 210b, 310b) of the guide (110, 210, 310). The at least one longitudinal slot (130, 230, 330) extends transversely through the guide (110, 210, 310). The plurality of blind holes (140, 240, 340) disposed along a length of the at least one longitudinal slot (130, 230, 330). The stopper assembly (120, 220, 320) is removably coupled to a selected blind hole (140, 240, 340) of the plurality of blind holes (140, 240, 340). Fig. 1a

No. of Pages : 58 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/08/2026

(21) Application No.202621104634 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR GENERATING A TRAVEL PASS TO STORE AND TRACK USER TRAVEL DATA

(51) International classification	:G06F21/60, H04W4/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MYTRAVEL JUNCTION INDIA PRIVATE LIMITED
Address of Applicant :FLAT # 20, BLDG # C, SHRIRAM NAGAR CHS, NEAR GAIKWAD NAGAR, AUNDH, PUNE, MAHARASHTRA 411007, INDIA PUNE Maharashtra India
(72)**Name of Inventor :**
1)PRAVIN UTTAMRAO PATIL

(57) Abstract :
A system and method for generating a travel pass to store and track user travel data is disclosed. The system includes receiving, inputs from at least one of a user interface of a user device and dynamically obtained data, wherein inputs from the user interface of the user device comprises Personally Identifiable Information; and wherein the dynamically obtained data, comprises transaction data and location tracker data of the user, wherein the transaction data is obtained from booking systems. System facilitates, real-time tracking of the user by location tracker data from a safety monitoring device. System anonymizes, the received Personally Identifiable Information to maintain the user privacy while facilitating the sharing of the transaction data and the location tracking data with stakeholders. System enables, real-time availability of the received transaction data and location tracker data from a server to at least one of stakeholders and third parties for actionable insights. FIG. 3

No. of Pages : 27 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/07/2026

(21) Application No.202621092338 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR REAL-TIME GEOSPATIAL AIR POLLUTION MONITORING AND DECISION SUPPORT

(51) International classification	:G01N 33/00, G06N 20/00, G06N 5/04, G01N 15/06, G06N 5/02	(71)**Name of Applicant :** **1)Mr. Kshitij S. Bairagi** Address of Applicant :Assistant Professor, S.P. College Pune, Maharashtra 411030 PUNE Maharashtra India **2)Dr. Piyush R. Telang** (72)**Name of Inventor :** **1)Mr. Kshitij S. Bairagi** **2)Dr. Piyush R. Telang**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A system (100) for real-time geospatial air pollution monitoring and decision support includes heterogeneous sensing sources (102) configured to acquire environmental observations, edge computing nodes (104) configured to receive the environmental observations from the heterogeneous sensing sources (102) and to perform validation, anomaly detection, and adaptive calibration on the environmental observations, and a cloud geospatial processing engine (106) configured to receive validated environmental observations from the edge computing nodes (104) and to generate high-resolution pollution concentration maps. The system (100) further includes artificial intelligence engines (108) having an environmental digital twin (110) configured to maintain a real-time computational representation of pollutant concentrations and to simulate pollutant movement, and a pollution source attribution engine (112) configured to estimate a probable origin of detected pollutants based on spatial pollutant signatures and meteorological transport pathways. An explainable decision support engine (120) generates location-specific mitigation strategies based on outputs of the artificial intelligence engines (108). (FIG. 1)

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/07/2026

(21) Application No.202621092427 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR PRODUCTION OF RENEWABLE AROMATICS FROM CATALYTIC CRACKING OF WASTE COOKING OIL

(51) International classification	:C10G 11/05, C10G 11/18, C10G 69/04, C10G 69/06, B01J 23/755
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MAULANA AZAD NATIONAL INSTITUTE Of TECHNOLOGY, BHOPAL
Address of Applicant :Link Road No. 3, Near Mata Mandir, Bhopal - 462003, Madhya Pradesh, India Madhya Pradesh India
(72)**Name of Inventor :**
1)Sumit H. Dhawane
2)Rehan Khan
3)Ramkishor Anant

(57) Abstract :
A PROCESS FOR PRODUCTION OF RENEWABLE AROMATICS FROM CATALYTIC CRACKING OF WASTE COOKING OIL The present invention relates to a thermo-catalytic process for the production of renewable aromatic hydrocarbons and hydrogen from waste cooking oil (WCO) using an indigenously developed bimetallic catalyst. The process comprises subjecting waste cooking oil to catalytic cracking, deoxygenation, cyclization, and aromatization reactions in a high-pressure reactor under optimized conditions of catalyst loading, temperature, reaction time, nitrogen pressure, and agitation speed. The process enables simultaneous production of aromatic hydrocarbons and hydrogen in a single integrated system while minimizing coke formation. Under optimized conditions comprising 0.5 wt.% catalyst loading, 450°C reaction temperature, 80 minutes reaction time, 9 bar nitrogen pressure, and 400 RPM agitation speed, the process achieves an aromatics yield of about 78.22%, hydrogen yield of about 52.8%, and coke formation of about 1%. The invention provides an energy-efficient, environmentally sustainable, and economically viable route for converting waste cooking oil into valuable aromatic chemicals and clean hydrogen.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :15/09/2026

(21) Application No.202621110822 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BANKING SAFE MODE– ONE -TAP FRAUD PREVENTION SYSTEM

(51) International classification	:G06Q20/40, G06N20/00, G06Q20/38, G06F21/31
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)jayesh shrikrushna hage
Address of Applicant :Adarsh colony , gaurakshan road , akola Maharashtra India
2)Prof.Kishor Pathak
3)Adhirath Bharat Deshmukh
4)Shreya Nalesh Dange
5)Vinayak Vilasrao Deshmuhk
6)Vaishnavi Sanjay Chiwate
(72)**Name of Inventor :**
1)jayesh shrikrushna hage
2)Prof.Kishor Pathak
3)Adhirath Bharat Deshmukh
4)Shreya Nalesh Dange
5)Vinayak Vilasrao Deshmuhk
6)Vaishnavi Sanjay Chiwate

(57) Abstract :
The proposed Banking Safe Mode – One-Tap Fraud Prevention System is a real-time financial security platform designed to prevent digital fraud, social engineering, and coerced financial transactions. The system continuously monitors suspicious activities such as suspicious calls, repeated OTP requests, and transaction-related events to evaluate the user's financial risk. A cumulative risk scoring mechanism analyzes these events within a defined time window and compares the resulting score with a user-specific threshold. When the risk exceeds the threshold, the system automatically activates Safe Mode and blocks outgoing transactions. A discreet banking server-error response is provided to conceal the security intervention during potential coercive situations. The system integrates secure authentication, UPI-based transaction processing, risk logging, and administrative monitoring using a mobile frontend, Node.js/Express backend, and SQLite database. The proposed solution provides a proactive, responsive, and scalable approach to enhancing digital financial safety and reducing potential fraud-related losses.

No. of Pages : 6 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :08/09/2026

(21) Application No.202621107803 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A VALVE SYSTEM

(51) International classification	:F16K11/056, F16K27/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INTERVALVE POONAWALLA PVT. LTD.
Address of Applicant :94/1 Manjri, Off Soli Poonawalla Road, Pune, Maharashtra, India-412307 Pune Maharashtra India
(72)**Name of Inventor :**
1)Sanjay Dhondaapa Waghmare
2)Basavaraj Uddi
3)Mukesh Patil
4)Vivek Kondapur
5)Veeranagouda Mali Patil

(57) Abstract :
ABSTRACT A VALVE SYSTEM The present invention discloses a valve system is described. The valve system comprising a ball supported by a support system. The support system comprising a spherical ball (104) position at the bottom of the ball and a groove (106). The groove (106) is configured to receive the spherical ball (104), wherein the spherical ball (104) nis free to rotate in the groove (106). The valve system further comprising a sealing arrangement (108) comprising a seat supported by a spring. [To be published with Figure 1]

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/09/2026
(21) Application No.202621112052 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF BLUE COPPER HYDROXIDE NANOFLUID

(51) International classification	:B82Y30/00, B82Y40/00, C01G 3/02, B01J 13/00, C07C 323/25, C08L 29/04, C07C 229/16, C07C 59/265
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Prof. JIGNASU P. MEHTA
Address of Applicant :Maharaja Krishnakumarsinhji Bhavnagar University Mahatma Gandhi Campus Bhavnagar, India Bhavnagar Gujarat India
2)Prof. DINESH R. GODHANI
(72)**Name of Inventor :**
1)Dr. KAHAN PANDYA

(57) Abstract :
ABSTRACT A PROCESS OF PREPARATION OF BLUE COPPER HYDROXIDE NANOFLUID The present invention is a process of preparation of blue copper hydroxide nanofluid. The present invention is a process of preparation of blue copper hydroxide nanofluid having nanoparticles size between 1 nm to 100 nm and Z-average of 28nm±10 nm. The present invention is related to process of preparation of blue copper hydroxide nanofluid that uses multi-layer capping process that coordinates copper ions (Cu2+) from an aqueous precursor solution using a dual-chelating matrix of Citric Acid and EDTA disodium Salt at an elevated temperature of 70°C±5°C to prevent premature precipitation and agglomeration.

No. of Pages : 18 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/09/2026
(21) Application No.202621112064 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN COMPOUND DISASTER RESILIENT FLOOD MANAGEMENT AND OFFLINE EMERGENCY RESPONSE SYSTEM

(51) International classification	:G06N20/00, G06Q10/04, G06N3/092, G06N5/02, G06Q50/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DKTE Society's Textile and Engineering Institute
Address of Applicant :Rajwada, Ichalkaranji, Maharashtra 416115 Maharashtra India
2)Sandipkumar Sagare
3)Ahilya Krushnakant Hapate
4)Prachi Nitin Chougule
5)Pranoti Rahul Athawale
6)Rakhi Vidyasagar Adgane
7)Sayka Salim Arab
(72)**Name of Inventor :**
1)Sandipkumar Sagare
2)Ahilya Krushnakant Hapate
3)Prachi Nitin Chougule
4)Pranoti Rahul Athawale
5)Rakhi Vidyasagar Adgane
6)Sayka Salim Arab

(57) Abstract :
AI-Driven Compound Disaster Resilient Flood Management and Offline Emergency Response System This invention describes an Artificial Intelligence and Reinforcement Learning Based Compound Disaster Resilient Flood Prediction, Infrastructure Survivability Assessment, Adaptive Evacuation Optimization, Decentralized Emergency Communication, and Personalized Explainable Crisis Management System for infrastructure-compromised environments. The system integrates multimodal environmental intelligence including hydrological, seismic, thermal, geospatial, aerial, and mobile telemetry data for identifying and predicting compound disaster scenarios comprising floods, landslides, infrastructure collapse, and secondary hazard events. A hybrid analytical architecture employing convolutional neural networks, long short-term memory models, graph neural networks, and reinforcement learning dynamically performs spatial-temporal hazard analysis, survivability assessment, and evacuation optimization. The system further enables decentralized emergency communication through localized mesh networking and aerial broadcasting mechanisms during communication failures. Personalized explainable emergency guidance is generated according to user-specific vulnerability and accessibility conditions, thereby enabling adaptive, autonomous, and survivability-oriented disaster management within geographically isolated and communication-degraded operational environments.

No. of Pages : 30 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/09/2026

(21) Application No.202621112068 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-INTEGRATED SOLAR THERMOELECTRIC SMART CHILLBOX WITH ADAPTIVE THERMAL ZONING AND INVENTORY MONITORING

(51) International classification	:F25B 21/02, H02J 7/35, F25D 29/00, F25B 21/04, F25B 49/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DKTE Society's Textile and Engineering Institute
Address of Applicant :Rajwada, Ichalkaranji, Maharashtra 416115 Maharashtra India
2)Dr. Mrs. A. D. Kadage
3)Dr. S. R. Prasad
4)Ms. Suchita Ghatte
5)Rushikesh Sutar
6)Siddhi Bhasme
7)Stuti Kale
(72)**Name of Inventor :**
1)Dr. Mrs. A. D. Kadage
2)Dr. S. R. Prasad
3)Ms. Suchita Ghatte
4)Rushikesh Sutar
5)Siddhi Bhasme
6)Stuti Kale

(57) Abstract :
AI-Integrated Solar Thermoelectric Smart Chillbox with Adaptive Thermal Zoning and Inventory Monitoring This invention describes an artificial intelligence-based solar-powered thermoelectric smart chillbox for off-grid cold-chain management of temperature-sensitive payloads. The system integrates photovoltaic energy harvesting, rechargeable energy storage, thermoelectric cooling, adaptive thermal zoning, and predictive cooling control for maintaining controlled preservation conditions under varying environmental and power conditions. A multi-sensor inventory intelligence architecture utilizing weight, infrared, ultrasonic, and proximity sensing mechanisms generates real-time spatial inventory awareness for detecting payload absence, improper placement, overcrowding, and compartment occupancy conditions. The system further incorporates pulse-width modulated cooling regulation and a critical payload preservation protocol configured to prioritize essential storage regions during low-power conditions. An artificial intelligence-based control unit dynamically optimizes cooling operation, energy allocation, and compartment-specific preservation according to payload sensitivity and environmental conditions. The invention provides portable, refrigerant-free, energy-efficient, and context-aware cold-chain preservation suitable for medical, pharmaceutical, emergency, and remote transportation applications.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/09/2026
(21) Application No.202621112956 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BINOCULAR PUPILLOMETRY APPARATUS WITH FAIL-SAFE CONTROL OF OCULAR LIGHT EXPOSURE

(51) International classification	:A61B 3/11, A61B 3/113, A61B 3/117, A61B 3/00, A61B 3/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PARAS AJITKUMAR MEHTA
Address of Applicant :SF-9C, LOTUS AURA, OPP. LILLERIA, SAMA SAVLI ROAD Gujarat India
(72)**Name of Inventor :**
1)PARAS AJITKUMAR MEHTA

(57) Abstract :
An apparatus (100) for directing a light stimulus into an eye, such as a binocular pupillometer, bounds the light delivered to the eye by independent layers of protection. Each light emitter (140) is driven through a switching element (176) whose control terminal is held off by a passive pull-down (178) whenever its control line is undriven. A controller (150) validates the complete stimulus timeline against an exposure envelope before any light is emitted and refuses, rather than modifies, a breaching timeline; checks each emission again as it is issued; and arms, before each emission, a watchdog independent of the controlling thread that switches the emitter off at a deadline and latches a fault until explicitly reset. In a binocular embodiment two light-excluding channels with off-axis infrared illumination record both pupils on one clock, quantify a relative afferent defect with an efferent-asymmetry check, and verify their own stimulus-to-frame synchronisation.

No. of Pages : 40 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/09/2026
(21) Application No.202621112978 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A COMPUTER IMPLEMENTED EVALUATION SYSTEM

(51) International classification	:G06Q 10/06, G06N 5/02, G06Q 50/18, G06N 5/04, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
Address of Applicant :Powai, Mumbai - 400076, Maharashtra, India MUMBAI Maharashtra India
(72)**Name of Inventor :**
1)KUMARESAN, P

(57) Abstract :
A computer-implemented method, computing system, and non-transitory computer-readable medium receive, by an input interface (104), an evaluation request (202) for an evaluated item (102). A concept gate (106) blocks evaluation until the evaluation request (202) includes a declared intended function and a declared principal. One or more processors generate or select a machine-readable evaluation contract (110) identifying product-quality dimensions, enterprise-readiness dimensions, evidence requirements, scoring rules, and stakeholder weights. Evidence from an evidence repository (112) is validated for selected dimensions, an unsupported score (214) is rejected, and a scoring engine (114) computes dimension scores. A stakeholder-weighting module (116) applies stakeholder weights to generate externality-adjusted scores (218). An anti-closure filter (108) requires a falsifiable provocation (222) before release, and an output module (118) outputs an auditable quality assessment (224). FIGURE 1

No. of Pages : 85 No. of Claims : 22

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/09/2026
(21) Application No.202621112982 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SPRINKLER ENERGY HARVESTING SYSTEM

(51) International classification	:A01G 25/09, B05B 3/04, F03B 17/06, B05B 3/02, B05B 3/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rajiv Dey
Address of Applicant :Hno 528/A1 Samdhariya colony, Kanchaghar, besides Yashroop school Madhya Pradesh India
(72)**Name of Inventor :**
1)Arko Chandra
2)Varsha Shree
3)Soma Das
4)Sudeep Kumar
5)Rajiv Dey
6)Sandeep Kumar
7)Lalit Tanwar

(57) Abstract :
The present invention relates to a sprinkler energy harvesting system comprising of a S/Sigmoid shaped sprinkler 1 whose shaft 12 is connected to a number of multi-stage spur gear mechanism 2, 3, 4, 5, and 6 on two opposite sides which are used to up-convert the generated RPM of the sprinkler and provide it to the two 10 DC-motors 7 connected to the base plate 8. The base plate 8 is connected to the bush 11, and the shaft 12 of the sprinkler rotates inside the bush 11. The nozzle 9 and 10 is connected at the two outlets of the S-shaped sprinkler 1, which spreads the water uniformly over a large distance. This in combination with the circular motion of the sprinkler system, distributes the water uniformly in a large circular 15 region. Additionally, it converts kinetic energy of water and generates electrical energy using the combination of sprinkler 1, gear mechanism 2, 3, 4, 5, and 6 and DC motors 7. Ref to: Fig. 1, Fig. 2, Fig. 3 and Fig. 4

No. of Pages : 9 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/08/2026

(21) Application No.202621099451 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR COMMUNICATION-AWARE TRANSPRECISION FUSED MULTIPLY-ADD WITH CANONICALIZED COMPUTATION

(51) International classification	:G06Q 90/00, G06Q 50/40, G06Q 50/50, G06Q 50/60, C03C 27/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY INDORE
Address of Applicant :Khandwa Road, Simrol, Indore-453552, Madhya Pradesh, India Madhya Pradesh India
(72)**Name of Inventor :**
1)Santosh Kumar Vishvakarma
2)Mukul Lokhande
3)Mohd Faisal Khan

(57) Abstract :
A SYSTEM AND METHOD FOR COMMUNICATION-AWARE TRANSPRECISION FUSED MULTIPLY-ADD WITH CANONICALIZED COMPUTATION
The invention relates to low-precision numerical processing for AI accelerators, neural processing units, inference engines, and compute fabrics. Disclosed is a system and method in which operands received in compressed communication formats including INT4, FP4, Posit-(4,1), and FP2 are decoded and canonicalized into a common internal compute representation, namely a canonical E4M2 compute token for INT4, FP4, and Posit-(4,1), with FP2 internally typecast to E1M1. A reconfigurable exponent unit and reconfigurable mantissa multiplier share arithmetic resources to support scalar and SIMD operation, enabling fused multiply-add computation (A × B) + C across homogeneous and mixed-format combinations, followed by BF16 accumulation. By separating communication precision from execution precision, the invention reduces data movement cost while improving flexibility and utilization over fixed-format low-bit arithmetic. Approximate FP8 modes E4M3 and E5M2 may also be supported through the same framework.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :08/09/2026

(21) Application No.202621107937 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SUSTAINABLE WIRELESS POWER BEAMING SYSTEM FOR AIRBORNE PLATFORMS

(51) International classification	:B64C 39/02, H02J 50/20, H02J 7/35, H02J 50/10, H02J 50/90	(71)**Name of Applicant :** **1)Indian Institute Of Technology, Indore** Address of Applicant :Simrol, Khandwa Road, Indore-453552, Madhya Pradesh, India. Indore Madhya Pradesh India (72)**Name of Inventor :** **1)GAUTAM, Sumit** **2)MISHRA, Devesh**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a sustainable wireless power beaming system for airborne platforms supporting smart aerial infrastructure and long-endurance unmanned aerial vehicle (UAV) operations. The system comprises a ground-based solar power generation unit, a wireless transmission tower, and an airborne receiving platform including a wireless power receiving module, an energy management controller, one or more drone docking stations, an onboard buffer battery, and an electric propulsion system. Electrical energy generated by the solar power generation unit is transmitted to the airborne receiving platform through adaptive wireless power transmission modes selected according to transmission distance and operational requirements. The energy management controller intelligently regulates and distributes the received power between the onboard buffer battery, the electric propulsion system, and the drone docking stations to ensure efficient power utilization and uninterrupted operation. The renewable aerial energy infrastructure enables continuous airborne operation while supporting simultaneous wireless charging of multiple UAVs.

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621112869 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD FOR PREPARING WOUND DRESSING AND GEL INFUSED WITH ZINC NANOPARTICLES

(51) International classification	:C01G9/02, B82Y40/00, A61L15/28, A61K9/70, A61P17/02, A61L15/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)LAHOTI, Ronith Amol
Address of Applicant :8th Floor, "6 Prabhat" Building, Next to Lizzat Papad Office, 6th Lane, Prabhat Road, Pune, Maharashtra, India - 411004 Maharashtra India
2)LAHOTI, Smita Amol
(72)**Name of Inventor :**
1)LAHOTI, Ronith Amol
2)LAHOTI, Smita Amol

(57) Abstract :
The present invention is related to a method for preparing wound dressing and gel infused with zinc nanoparticles. The zinc nanoparticles (ZnNPs) are synthesized through a green, plant-based process using tea extract as a reducing and stabilizing agent, yielding particles in the size range of 18–42 nanometers. These nanoparticles are incorporated into a hydroxypropyl methylcellulose (HPMC) polymer matrix to produce either a flexible, breathable film or a spreadable gel. The preparation involves dissolving HPMC in water, dispersing ZnNPs uniformly, and casting the mixture to form a dressing, or preparing a gel without guar gum. The resulting formulations exhibit broad-spectrum antimicrobial activity, anti-inflammatory effects, and enhanced wound healing properties by promoting fibroblast proliferation and collagen deposition. Unlike conventional synthetic dressings, the material is eco-friendly, biocompatible, and degrades naturally within 7–21 days, reducing biomedical waste. This invention provides a cost-effective, sustainable, and scalable solution for clinical, emergency, home, veterinary, and cosmetic wound care applications.

No. of Pages : 26 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621112875 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An AI-enabled Battery Container Management System for Utility Scale Energy Storage Systems

(51) International classification	:H02J3/32, H02J13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Truboard CleanTech Private Limited
Address of Applicant :24, 2402, Lightbridge, Gladys Alwares Road, Off Pokhran Road No. 2, Hiranandani Meadows, Hiranandani Meadows, Thane, Maharashtra, 400610, India. Maharashtra India

(72)**Name of Inventor :**
1)Sudhakar Sundaray
2)Uppuluri Ranjith Kumar
3)Samrat Saha

(57) Abstract :

The present invention relates to a Make in India Battery Container Management System (BCMS) for utility-scale Battery Energy Storage Systems (BESS). The BCMS functions as an intelligent supervisory controller that integrates the Energy Management System (EMS), Battery Management System (BMS), WECON gateway, cloud platform, and operator interface into a unified architecture. Based on operating commands received from the EMS, the BCMS controls battery charging and discharging operations through the BMS while continuously acquiring and communicating critical battery parameters, including voltage, current, temperature, and SOC. The system enables secure cloud connectivity, real-time remote monitoring, and control through a web-based or remote desktop interface, eliminating the need for physical site access. Advanced analytics platforms, such as Microsoft Power BI, process the collected operational data to generate actionable insights, performance trends, predictive maintenance indicators, and incorporates intelligent notification mechanisms that automatically generate SMS and email alerts whenever system parameters exceed predefined operating thresholds, thereby improving system safety, reliability, and operational efficiency. Currently, most utility-scale BESS containers deployed in India rely on imported control hardware and associated subsystems. The proposed invention focuses on the indigenous development of a cost-effective, secure, and scalable BCMS hardware platform for deployment within each BESS container. By localizing the control and monitoring infrastructure, the invention reduces import dependency, supports the Government of India's "Make in India" initiative, enhances customization for Indian grid applications, and contributes to the growth of a self-reliant energy storage ecosystem.

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621112926 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Portable Multi-Analyte Device with Stabilized Reagent Matrix for Detection of Milk Adulterants

(51) International classification	:G01N 21/78, G01N 33/04, G01N 33/52, G01N 31/22, C12Q 1/54
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Deccan Education Society's Fergusson College (Autonomous), Pune
Address of Applicant :Fergusson College (Autonomous), Shivajinagar, Pune, Maharashtra, India, 411004 Pune Maharashtra India
2)Dr. Dhanashri Godbole
3)Shardul Vishwad
(72)**Name of Inventor :**
1)Dr. Dhanashri Godbole
2)Shardul Vishwad
3)Madhur Chaware
4)Shishir Raut

(57) Abstract :
The present invention relates to a portable device (100) for detection of multiple milk adulterants. The device (100) comprises a base layer (01), a porous cellulose-based paper layer (02), an agarose-glycerol stabilization layer (03), a reagent layer (04) having spatially separated test zones (06–13), and a covering layer (05) having sample inlet wells (14). The stabilization layer (03) comprises 0.5–3% (w/v) agarose and 5–10% (v/v) glycerol and immobilizes respective detection reagents at corresponding test zones (06–13). The test zones detect urea, starch, glucose, sodium chloride, detergent, soap, hydrogen peroxide, and neutralizing adulterants through respective colorimetric responses. A smartphone captures images of the colorimetric responses, and a machine-learning analysis module comprising a trained convolutional neural network processes image-derived features to classify an adulterant and determine an adulteration concentration category. The invention further provides a method for preparing the reagent-containing agarose-glycerol stabilization layer by controlled reagent incorporation, localized deposition, and drying.

No. of Pages : 37 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/09/2026
(21) Application No.202621112941 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Real-Time Object Detection and Tracking System Using Edge AI

(51) International classification	:G06V 10/764, G06T 7/20, G06N 3/04, G06K 9/00, G06V 10/774	(71)**Name of Applicant :** **1)Pravin Shegade** Address of Applicant :18/9, Ganeshnagar, Thergaon, Pune-411033, Maharashtra, India Maharashtra India (72)**Name of Inventor :** **1)Pravin Shegade**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT [505] The present invention introduces an advanced real-time object detection and tracking system that integrates edge-AI-based inference mechanisms with predictive motion modelling and adaptive multi-object association protocols, facilitating continuous visual analysis, dynamic trajectory determination, and robust identity confirmation while maintaining seamless on-device integration and operational accuracy for consistent detection-and-tracking applications. [510] The comprehensive detection framework employs adaptive neural-inference algorithms and frame-stream processing protocols, utilising embedded computational accelerators and energy-efficient inference systems to ensure timely object identification, enhanced motion understanding, and optimal tracking reliability while maintaining continuous visual monitoring capabilities across distributed edge devices. [515] The integrated methodology combines multi-frame visual ingestion techniques with artificial-intelligence-driven pattern recognition systems, leveraging variable-confidence detection signals and multi-factor association indicators to optimise tracking procedures and analysis workflows for maximum identification accuracy and minimal association uncertainty during critical real-time operations. [520] The novel responsive detection architecture features engineered high-throughput frame-processing components with specialised object-fingerprinting protocols, enabling complex multi-stage tracking verification while ensuring detection consistency and performance optimisation across various edge applications without compromising system reliability. [525] The innovative design incorporates strategic validation mechanisms for enhanced object identification and operational security, utilising optimised multi-model systems and adaptive association technology to ensure legitimate identity assignment while maintaining functionality across diverse deployment environments and scene scenarios. [530] Implementation methodology emphasises scalable device integration and efficient tracking sequences, implementing interactive monitoring measures and predictive motion algorithms to achieve superior trajectory determination, enhanced object identification, and identity-switch prevention while ensuring technological simplicity during edge operation. [535] The system demonstrates exceptional adaptability through comprehensive integration of object-identification protocols and intelligent tracking technologies, validating its effectiveness across various scene configurations and operational scenarios while maintaining consistent detection performance and operational efficiency under diverse conditions. [540] The developed framework enables sustainable and reliable detection and tracking of objects through streamlined, edge-AI-powered inference systems, providing significant advantages over traditional cloud-dependent approaches through low-latency validation mechanisms, predictive identification protocols, and improved identity assignment while maintaining superior tracking accuracy during critical real-time processing.

No. of Pages : 13 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621113065 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : “AN ORAL TABLET FORMULATION COMPRISING A CINNAMALDEHYDE-DERIVED SCHIFF BASE AND PREPARATION METHOD THEREOF”

(51) International classification	:A61P 3/10, A61K 9/20, A61K 31/11, C07C 249/02, A61K 47/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Mohib Khan
Address of Applicant :Oriental college of Pharmacy, Plot No. 3,4,5, Sector 2, Sanpada (W), Behind Sanpada Railway Station, Sanpada,Navi Mumbai Navi Mumbai Maharashtra India
(72)**Name of Inventor :**
1)Dr. Mohib Khan
2)Ms. Sayed Anam Kamal Ahmed
3)Dr. Sayyed Mateen

(57) Abstract :
ABSTRACT “AN ORAL TABLET FORMULATION COMPRISING A CINNAMALDEHYDE-DERIVED SCHIFF BASE AND PREPARATION METHOD THEREOF” The present invention relates to a cinnamaldehyde-derived Schiff base and an oral pharmaceutical formulation thereof for potential management of diabetes mellitus. The invention particularly relates to the synthesis of (E)-3-phenyl-N-(4-hydroxy-2-nitrophenyl) prop-2-en-1-imine by condensation of cinnamaldehyde with 4-amino-2-nitrophenol. The cinnamaldehyde-derived Schiff base is prepared by conventional reflux and microwave-assisted synthesis, wherein the microwave-assisted process provides rapid reaction completion. The invention provides a cinnamaldehyde-derived Schiff base and its tablet formulation for application in management of diabetes mellitus.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621113068 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A NOVEL PANCH PHORAN-BASED EFFERVESCENT GRANULE COMPOSITION FOR THE MANAGEMENT OF RHEUMATOID ARTHRITIS

(51) International classification	:A61K 9/46, A61P 19/02, A61K 9/16, A61K 36/53, A61K 36/23
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Trupti Dubey
Address of Applicant :Research Scholar, Faculty of Pharmacy, Kalabhavan Campus, The Maharaja Sayajirao University of Baroda, Vadodara, Gujarat, India-390001 Vadodara Gujarat India
2)Dr. Kirti V. Patel
(72)**Name of Inventor :**
1)Dr. Trupti Dubey

(57) Abstract :
ABSTRACT TITLE: A NOVEL PANCH PHORAN-BASED EFFERVESCENT GRANULE COMPOSITION FOR THE MANAGEMENT OF RHEUMATOID ARTHRITIS The present invention discloses a novel oral effervescent granule composition for supportive management of rheumatoid arthritis, comprising, per approximately 2000-milligram unit sachet, 600 milligrams of an equi-proportional Panch Phoran aqueous dry powder extract blend of Cuminum cyminum, Brassica nigra, Trigonella foenum-graecum, Nigella sativa and Foeniculum vulgare, together with 217 milligrams of citric acid, 435 milligrams of tartaric acid, 738 milligrams of sodium bicarbonate and 10 milligrams of hydroxypropyl methylcellulose E5 as binder, prepared by wet granulation using two-percent ethanol as granulating vehicle. The granules exhibit good pharmaceutical flow properties with bulk density of 0.49 grams per millilitre, angle of repose of 25 degrees and Carr's index of 13 percent, and in a collagen-induced rheumatoid arthritis rat model produce approximately 42 percent reduction of arthritis score, 38 percent reduction of paw oedema and normalisation of serum inflammatory markers, providing a safe, economical, patient-friendly herbal dosage form.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621113161 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ENERGY MANAGEMENT OF AN ENERGY MICROGRID FOR ELECTRIC VEHICLE CHARGING INFRASTRUCTURE

(51) International classification	:H02J3/28, H02J3/32, H02J3/38, B60L53/54, B60L53/68
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ROHAN BHAGAWAN BHARGAV
Address of Applicant :B-104, Avishi Apartment CHS,Sector 11, Plot no 52, Near Gautam Budh Vihar, Kamothe, Navi Mumbai, Panvel, Maharashtra Navi Mumbai Maharashtra India
2)ALPANA BHAGAWAN KAMBLE
(72)**Name of Inventor :**
1)ROHAN BHAGAWAN BHARGAV

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR ENERGY MANAGEMENT OF AN ENERGY MICROGRID FOR ELECTRIC VEHICLE CHARGING INFRASTRUCTURE The present disclosure relates to system and method for energy management of an energy microgrid for electric vehicle charging infrastructure. The energy microgrid (100) for supplying electrical power to an electric vehicle charging infrastructure (114), comprising a renewable energy generation system (102), a hydrogen production system (104), a hydrogen storage system (108), a fuel cell generation system (110), a battery energy storage system (112), and a centralized energy management controller (116). The centralized energy management controller (116) receives renewable-energy availability, hydrogen-storage, battery operating, and charging-demand information to control allocation of renewable electrical power for vehicle charging and hydrogen production, and control the fuel cell generation system (110) and battery energy storage system (112) for supplying electrical power to the electric vehicle charging infrastructure (114). FIG. 1 will be the reference figure.

No. of Pages : 31 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621112942 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for AI-Driven Financial Fraud Detection and Risk Assessment Using Real-Time Transaction Analysis and Predictive Behavioural Risk Models

(51) International classification	:G06Q 40/00, G06N 20/00, G06N 5/04, G06Q 20/40, G06Q 40/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Nitin Bodade
Address of Applicant :Add-1: 701 Shreeji Pinnacle apartment, Samta Nagar, Nashik, Maharashtra, India, PIN - 422006 Add-2: UGC Patent, New Panval, MH, India. Add2: 501 Bighorn Pass, Denton, Texas - 76210 Maharashtra India
(72)**Name of Inventor :**
1)Nitin Bodade

(57) Abstract :
ABSTRACT [505] The present invention introduces an advanced AI-driven financial fraud detection and risk assessment platform that integrates real-time transaction analysis with predictive behavioural risk modelling mechanisms, facilitating continuous monitoring of financial activity, dynamic anomaly identification, and robust risk confirmation while maintaining seamless institutional integration and operational accuracy for consistent fraud-prevention applications. [510] The comprehensive detection framework employs adaptive machine-learning algorithms and transaction-stream processing protocols, utilising embedded computational processing arrays and resource-efficient inference systems to ensure timely fraud identification, enhanced risk understanding, and optimal decisioning reliability while maintaining continuous behavioural monitoring capabilities across distributed financial channels. [515] The integrated methodology combines multi-source signal ingestion techniques with artificial-intelligence-driven pattern recognition systems, leveraging variable-confidence risk signals and multi-factor fraud indicators to optimise detection procedures and adjudication workflows for maximum decisioning accuracy and minimal false-positive uncertainty during critical financial operations. [520] The novel responsive detection architecture features engineered high-throughput transaction-processing components with specialised behavioural-fingerprinting protocols, enabling complex multi-stage risk verification while ensuring detection consistency and performance optimisation across various financial applications without compromising system reliability. [525] The innovative design incorporates strategic validation mechanisms for enhanced fraud identification and financial security, utilising optimised multi-model systems and adaptive scoring technology to ensure legitimate transaction classification while maintaining functionality across diverse financial environments and payment scenarios. [530] Implementation methodology emphasises scalable channel integration and efficient detection sequences, implementing interactive monitoring measures and predictive modelling algorithms to achieve superior risk determination, enhanced fraud identification, and unauthorised transaction prevention while ensuring technological simplicity during financial monitoring. [535] The system demonstrates exceptional adaptability through comprehensive integration of behavioural identification protocols and intelligent detection technologies, validating its effectiveness across various financial fraud configurations and transaction scenarios while maintaining consistent detection performance and operational efficiency under diverse conditions. [540] The developed framework enables sustainable and reliable protection of financial operations through streamlined, AI-powered detection and risk-scoring systems, providing significant advantages over traditional rule-based approaches through adaptive validation mechanisms, predictive identification protocols, and improved risk assignment while maintaining superior detection accuracy during critical financial transaction processing.

No. of Pages : 11 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621112943 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Enabled Smart Modular Educational Furniture System with Digital Twin Management

(51) International classification	:G06Q 50/20, G05B 15/02, G05B 19/418, G09B 5/08, G09B 5/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Kamil Khan Ajmal Khan
Address of Applicant :Abeda Inamdar Senior College, Camp, Pune - 411001, Maharashtra, India. Maharashtra India
2)Veena Kiran Gandhi
3)Fatima Rafique Shaikh
4)Mrs. Huma Mannan Sayyad
(72)**Name of Inventor :**
1)Kamil Khan Ajmal Khan
2)Veena Kiran Gandhi
3)Fatima Rafique Shaikh
4)Mrs. Huma Mannan Sayyad

(57) Abstract :
System and Method for an AI-Enabled Smart Modular Educational Furniture System with Digital Twin Management ABSTRACT [505] The present invention introduces an advanced AI-enabled smart modular educational furniture system that integrates reconfigurable furniture modules with embedded sensing mechanisms and digital-twin management protocols, facilitating real-time configuration characterisation, dynamic space optimisation, and robust utilisation confirmation while maintaining seamless educational integration and operational accuracy for consistent learning-environment applications. [510] The comprehensive furniture framework employs adaptive layout algorithms and occupancy-processing protocols, utilising embedded sensor arrays and energy-efficient control systems to ensure timely configuration identification, enhanced space understanding, and optimal utilisation reliability while maintaining continuous environmental monitoring capabilities across distributed learning zones. [515] The integrated methodology combines multi-source sensing techniques with artificial-intelligence-driven pattern recognition systems, leveraging variable-context occupancy signals and multi-factor utilisation indicators to optimise reconfiguration procedures and management workflows for maximum learning effectiveness and minimal resource uncertainty during critical educational activities. [520] The novel responsive furniture architecture features engineered modular reconfigurable components with specialised asset-fingerprinting protocols, enabling complex multi-stage configuration verification while ensuring layout consistency and performance optimisation across various educational applications without compromising system reliability. [525] The innovative design incorporates strategic validation mechanisms for enhanced asset identification and environmental security, utilising optimised multi-sensor systems and adaptive control technology to ensure legitimate configuration assignment while maintaining functionality across diverse educational environments and pedagogical scenarios. [530] Implementation methodology emphasises scalable environment integration and efficient reconfiguration sequences, implementing interactive monitoring measures and pattern-recognition algorithms to achieve superior configuration determination, enhanced utilisation identification, and space-underuse prevention while ensuring technological simplicity during educational operation. [535] The system demonstrates exceptional adaptability through comprehensive integration of asset-identification protocols and intelligent management technologies, validating its effectiveness across various classroom configurations and pedagogical scenarios while maintaining consistent utilisation performance and operational efficiency under diverse conditions. [540] The developed framework enables sustainable and reliable management of educational furniture through streamlined, AI-powered sensing and digital-twin systems, providing significant advantages over traditional static-furniture approaches through adaptive validation mechanisms, configuration-driven identification protocols, and improved space assignment while maintaining superior utilisation accuracy during critical educational activities.

No. of Pages : 13 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/09/2026
(21) Application No.202621113169 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : THE ROLE OF BLOCKCHAIN IN ENHANCING SECURITY AND TRANSPARENCY IN COMMERCE

(51) International classification	:G06Q 20/38, H04L 9/06, H04L 9/32, G06Q 20/06, G06Q 20/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Yogesh V Torawane
Address of Applicant :Associate Professor, Pratap College, Amalner, Jalgaon, Pin:425401, Maharashtra, India Maharashtra India
2)Dr. Kalpana M. Gholap
3)Usha Shivnath Derle
4)Mr. Aarav
5)Dr. S. Vijayalakshmi
6)Bipin Shivaji Ghorpade
7)Dr. Radhika R
8)Mrs. N. Jeyabharathi
9)Mr. Sathish R
10)Dr. R. Arun
11)Dr. A. Muthulakshmi
12)Prof. Dr. Harikumar Pallathadka
(72)**Name of Inventor :**
1)Dr. Yogesh V Torawane
2)Dr. Kalpana M. Gholap
3)Usha Shivnath Derle
4)Mr. Aarav
5)Dr. S. Vijayalakshmi
6)Bipin Shivaji Ghorpade
7)Dr. Radhika R
8)Mrs. N. Jeyabharathi
9)Mr. Sathish R
10)Dr. R. Arun
11)Dr. A. Muthulakshmi
12)Prof. Dr. Harikumar Pallathadka

(57) Abstract :
The present invention provides a blockchain-based commerce system configured to enhance security, integrity, traceability, and transparency of commercial transactions. The system authenticates commercial participants using cryptographic credentials, receives transaction information, generates cryptographic representations, validates transaction integrity and authorization, and submits validated transactions to a distributed blockchain network. A consensus mechanism verifies transaction acceptance before recording the transaction in a cryptographically linked distributed ledger. A smart-contract module automatically evaluates predefined commercial conditions and generates corresponding transaction-state transitions. An access-control module selectively exposes transaction information to authorized participants, while a monitoring module generates verifiable transaction histories and audit records. The system may further communicate with external commerce, payment, inventory, and logistics platforms through an interoperability interface, thereby providing blockchain-based security and transparency while supporting integration with existing commercial infrastructure.

No. of Pages : 21 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621113178 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN INTELLIGENT ACADEMIC MENTORING AND ADAPTIVE INTERVENTION RECOMMENDATION SYSTEM AND METHOD

(51) International classification	:G06Q 50/20, G06Q 10/06, G06N 5/02, G09B 7/00, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Sandeep Awachar
Address of Applicant :Associate Professor, College of Engineering and Technology, Akola- 444104, Maharashtra Maharashtra India
2)Dr. Kalpana Swapnil Gilda
3)Mrs. Sharwari S. Dixit
4)Prof. K. P. Barabde
5)Rutuja Jay Deshmukh
6)Prof. Priya Ashish Tayade
7)Prof. Renuka Pramod Arbat
8)Arti Vinaysingh Gautam
(72)**Name of Inventor :**
1)Dr. Sandeep Awachar
2)Dr. Kalpana Swapnil Gilda
3)Mrs. Sharwari S. Dixit
4)Prof. K. P. Barabde
5)Rutuja Jay Deshmukh
6)Prof. Priya Ashish Tayade
7)Prof. Renuka Pramod Arbat
8)Arti Vinaysingh Gautam

(57) Abstract :
An AI-driven academic mentoring system and method are disclosed for converting heterogeneous, time-dependent academic signals into context-sensitive and executable mentoring interventions. A data-ingestion engine normalizes academic events and a learner-state engine generates a temporal learner representation including learner-relative trends, prerequisite relationships, intervention history, missing-data indicators and uncertainty. A risk-and-need estimator determines an academic risk, actionable need and contributing factor. A context-and-constraint engine forms a feasibility mask from timetable, resource, policy, consent, frequency and accessibility constraints. Feasible intervention candidates are evaluated by a recommendation engine that predicts individualized outcome uplift relative to a reference action and performs constrained multi-objective ranking using burden, urgency, confidence, fatigue, fairness and resource criteria. A mentor interface provides explainable recommendations and records human validation. Post-intervention outcomes are measured and used to update a learner-state parameter, intervention-effect model or recommendation policy, thereby providing a closed-loop adaptive mentoring architecture. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/09/2026

(21) Application No.202621113179 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EFFECT OF FEEDING DRIED DISTILLERS GRAIN WITH SOLUBLES (DDGS) ON MILK PRODUCTION AND COMPOSITION OF CROSSBRED COW

(51) International classification	:A23K 10/00, A23K 10/30, A23K 10/38, A23K 20/00, A23K 20/10, A23K 20/20, A23K 50/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Mahatma Phule Krishi Vidyapeeth, Department of Animal Husbandry and Dairy Science

Address of Applicant :Post Graduate Institute, Rahuri, Taluka Rahuri, District Ahilyanagar, Maharashtra – 413722, India. Maharashtra India

(72)**Name of Inventor :**

1)Mr. Rushikesh Ramchandra Kore

2)Dr. Dilip Kundlik Deokar

3)Mr. Rushikesh Dilip Sathe

4)Dr. Samir Ashok Dhage

5)Dr. Ulhas Shivaji Gaikwad

6)Dr. Mahendra Gorakh Mote

7)Dr. Bharat Murlidhar Bhalerao

8)Dr. Amardeep Tabaji Lokhande

9)Dr. Sunil Bhikaji Adangale

10)Mr. Shingade Sumit Vilas

(57) Abstract :

The present invention relates to feeding dried distillers grain with soluble (DDGS) to crossbred cows through a nutritionally balanced complete ration in which DDGS partially replaces conventional concentrate while the roughage fraction remains constant. In a preferred formulation, the ration comprises, on a dry matter basis, 25.90% concentrate mixture, 11.10% DDGS, 35.00% jowar kadbi, 15.00% green maize, 10.00% lucerne, 2.00% mineral mixture and 1.00% salt. The 11.10% DDGS corresponds to replacement of 30% of the baseline concentrate requirement. Experimental feeding demonstrated improved dry matter and nutrient digestibility, increased milk yield, milk fat, milk protein and total solids, and reduced enteric methane production compared with a control ration. The formulation further reduced feeding cost and improved net economic return while maintaining body weight and milk pH substantially unaffected. The invention therefore provides a practical DDGS-based ration for productive, economical and environmentally improved dairy cattle feeding under controlled crossbred cow production conditions.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113188 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ENVIRONMENTALLY ADAPTIVE WEARABLE MULTIMODAL EOG-EEG SYSTEM FOR REAL-TIME THREE-CLASS ADHD CLASSIFICATION

(51) International classification	:A61B 5/0476, A61B 5/00, A61B 5/374, A61B 5/16, A61B 5/0478	(71)**Name of Applicant :** **1)National Institute of Technology Raipur** Address of Applicant :G.E. Road, Raipur, Chhattisgarh 492010 Raipur Chattisgarh India (72)**Name of Inventor :** **1)Mr. Rohit Sharma** **2)Dr. Suman Kumar Saha** **3)Dr. Deboraj Muchahary**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A wearable multimodal EOG-EEG system (100) for real-time classification of attention-deficit/hyperactivity disorder severity is disclosed. The system (100) comprises a spectacle assembly (1) carrying an electrooculography sensing module (2), a headband assembly (3) carrying an electroencephalography sensing module (4), and a differential amplification module (5) for generating five physiological channels comprising horizontal EOG, vertical EOG, FP1, FPZ and FP2. A microcontroller unit (6) receives the channels and executes a subject-specific calibration module (9), a context-aware adaptive filtering module (10), an ambient noise processing module (11), a windowing module (12), a normalization module (13) and a TinyML classification module (14). A digital MEMS microphone provides ambient noise information for generating a scaled ambient noise reference and dynamically adjusting adaptive filter step size. A memory module (8) stores metaheuristic-optimized adaptive filter weights and a pre-trained TinyML convolutional neural network model. An output indication module (15) visually indicates a healthy, moderate ADHD or severe ADHD class.

No. of Pages : 41 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113216 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A RESEARCH STUDY ON LEVERAGING OPEN-SOURCE LLM FALCON 7B AND LANGCHAIN FOR ADVANCED CHATBOT DEVELOPMENT

(51) International classification	:G06F 16/332, G06N 20/00, G06F 16/33, G06F 40/30, G06F 16/338	(71)**Name of Applicant :** **1)Mr. SANKHADEEP DEBDAS** Address of Applicant :Student Computer Science Engineering Amity University Chhattisgarh, Kharora, Raipur-Chhattisgarh – 493225, India Raipur Chattisgarh India **2)Dr. SATYASIS MISHRA** **3)Dr. SRIKANTA MOHAPATRA** **4)Mr. SWASTIK PANDA** **5)Miss. PRATIKSHA PODDAR** **6)Dr. REVA HARI KULKARNI**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Mr. SANKHADEEP DEBDAS**
(33) Name of priority country	:NA	**2)Dr. SATYASIS MISHRA**
(86) International Application No Filing Date	: :01/01/1900	**3)Dr. SRIKANTA MOHAPATRA** **4)Mr. SWASTIK PANDA**
(87) International Publication No	: NA	**5)Miss. PRATIKSHA PODDAR**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)Dr. REVA HARI KULKARNI**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to an artificial intelligence based conversational chatbot system (100) integrating an open-source Falcon 7B large language model with a LangChain orchestration framework for processing natural-language user queries and generating corresponding conversational responses. The system comprises a user interaction module (110), input processing module (120), prompt management module (130), LangChain orchestration module (140), Falcon 7B language-model module (150), conversational context module (160), response generation module (170), and output interface (180). A received user query is processed and converted into a model input, optionally incorporating selected conversational context. The LangChain orchestration module (140) coordinates communication between the prompt management module (130) and Falcon 7B language-model module (150), which performs language-model inference and generates output tokens. The generated output is processed and delivered as a natural-language response through the output interface (180). The architecture provides an extensible software arrangement for implementing chatbot applications using an open-source language model.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113262 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : POLYMER-FATTY ACID IONIC LIQUID-BASED METHOTREXATE HYDRATE-LOADED TRANSDERMAL PATCH AND METHOD OF PREPARATION THEREOF

(51) International classification	:A61K 9/70, A61K 47/32, A61K 31/519, A61K 47/10, A61K 47/34	(71)**Name of Applicant :** **1)SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY, SURAT** Address of Applicant :SVNIT CAMPUS, S.V.R COLL OF ENGG AND TECH, DUMAS ROAD, ICHCHHANATH, Surat, Gujarat, 395007, India Surat Gujarat India **2)Dr. Naved Malek** **3)Ms. Vidhi M. Joshi** **4)Dr. Naina Raje** **5)Prof. Mousumi Chakraborty**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr. Naved Malek**
(33) Name of priority country	:NA	**2)Ms. Vidhi M. Joshi**
(86) International Application No Filing Date	: :01/01/1900	**3)Dr. Naina Raje**
(87) International Publication No	: NA	**4)Prof. Mousumi Chakraborty**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The invention relates to transdermal drug delivery, and discloses a single-layer, free-standing transdermal patch composition comprising methotrexate hydrate, choline linoleate as a fatty acid-based ionic liquid, and a physically crosslinked hydrophilic polymer matrix of polyvinyl alcohol and polyethylene glycol-400 formed by freeze-thaw cycling. Compared with conventional methotrexate formulations and multi-layer adhesive patches, the disclosed patch provides integrated drug loading, ionic liquid-assisted permeation, flexibility, self-healing, and skin adherence without a separate adhesive layer, while maintaining stability and controlled release. The method includes preparing an aqueous polymer solution, adding choline linoleate and methotrexate hydrate, casting, and subjecting the cast mixture to repeated freezing and thawing to form polyvinyl alcohol-rich junction zones. The invention is principally useful for sustained transdermal administration of methotrexate.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2026
(21) Application No.202621113276 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for English Speaking Practice and Performance Assessment

(51) International classification	:G09B 19/06, G10L 15/06, G10L 15/22, G10L 15/18, G10L 15/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Jayavelu Damodhrakrishnan
Address of Applicant :Assistant Professor, Department of Structural Engineering, Sanjivani College of Engineering, Kopargaon Maharashtra India
2)Dr. Sunita Rajendra Dandwate
(72)**Name of Inventor :**
1)Dr. Jayavelu Damodhrakrishnan
2)Dr. Sunita Rajendra Dandwate

(57) Abstract :
ABSTRACT The present invention provides an English speaking practice and performance assessment system which includes machine learning models that are used to intelligently analyze spoken audio to derive accurate assessment insights for aid in language learning in various application areas. The disclosed system includes a speech acquisition module, a speech preprocessing module, a linguistic feature extraction module, a machine learning model management module, a performance assessment engine, an explainable feedback module, a practice exercise generation module, a progress tracking and reporting module, a security management module and an institutional integration module. The system automatically captures audio from different sources, cleans, transcribes, extracts and selects linguistic features from it, trains and optimizes one or more machine learning models. The performance assessment engine provides more accurate and efficient pronunciation scores, fluency scores, grammar accuracy scores, vocabulary range scores and overall speaking competence estimates. The invention also includes automatic model selection, hyperparameter optimization, continual performance monitoring, proficiency drift detection and intelligent model retraining to ensure assessment accuracy in the presence of changes in learner accent and proficiency. Moreover, the disclosed system can generate assessment results that are understandable by estimating prediction confidence, highlighting pronunciation and grammar errors and summarizing the assessment into an interpretable recommendation, which enhances the transparency and learner confidence in the assessment results. The invention also features secure communications protocols, encrypted audio processing, authentication systems, role-based access control and cloud based distributed computing architectures that ensure scalability and security, and institutional interoperability. The speaking practice framework disclosed in the present invention greatly improves the assessment accuracy, learning efficiency, adaptability, computational performance, and intelligent feedback capabilities of speaking practice systems, while addressing the shortcomings of existing systems relying on manual scoring and static rule-based evaluation processes.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2026
(21) Application No.202621113277 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for Metal Oxide Nanoparticle Composite Water Treatment

(51) International classification	:C02F 1/28, C02F 1/72, C02F 1/32, C02F 1/30, C02F 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. P. Dinesh Kumar
Address of Applicant :Assistant Professor, Department of Computer Science and Engineering, Dnyanshree Institute of Engineering and Technology, Sajjangad, Satara, Maharashtra 415013, India Maharashtra India
2)Dr. M. Velammal
3)Dr. E. Bementa
4)Dr. S. Lourduraj
5)Dr. V. Hemalatha
6)Dr. C. Aswathy
7)Dr. S. Rayappan
8)Dr. J. Vijaya Rajasekaran
9)Dr. K. Premalatha
10)Dr. M. Earnest Stephen Gnanadoss
(72)**Name of Inventor :**
1)Dr. P. Dinesh Kumar
2)Dr. M. Velammal
3)Dr. E. Bementa
4)Dr. S. Lourduraj
5)Dr. V. Hemalatha
6)Dr. C. Aswathy
7)Dr. S. Rayappan
8)Dr. J. Vijaya Rajasekaran
9)Dr. K. Premalatha
10)Dr. M. Earnest Stephen Gnanadoss

(57) Abstract :
ABSTRACT A system and method are suggested of treating water containing supported metal oxide nanoparticle composites. Representing a thrilling new technology, the system includes feed conditioning, iron oxide adsorption cartridge, optional titanium dioxide photocatalytic cartridge, downstream particle retention, analysis sampling and a controlled product outlet. Thanks to the separation realized with the cartridges, both the adsorption and the photocatalyzing effect can be qualified and maintained separately. A controller keeps track of operating conditions and reroutes water if conditions are not present or if they're exceeded. Regeneration is done by a separated circuit with residuals collected and verifying before reuse. Prior developments in oxide adsorption, magnetic composites and supported photocatalysis, as well as future characterisation of materials and hydraulic testing of the reactor, contaminant analysis and particle release assessment, and repeated operation using the same support are identified. All benefits being claimed are engineering hypotheses and need to be validated through experimentation. The recommended thinking approach does not include complete removal statements: measurable treatment performance, long term material containment, and residual accountability. There is no report of any experimental results for the suggested system.

No. of Pages : 10 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2026
(21) Application No.202621113294 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Computer-Implemented Veterinary Surgical Management System for Integrated Patient, Diagnostic and Multimedia Record Processing

(51) International classification	:G16H 10/60, G16H 40/20, G16H 15/00, G06F 19/00, G16H 50/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Maharashtra Animal & Fishery Sciences University, Nagpur
Address of Applicant :Futala Lake Road, Seminary Hills, Nagpur – 400006 Udgir Maharashtra India
(72)**Name of Inventor :**
1)Dr. S.S. Pitlawar
2)Dr. P.S. Masare
3)Dr. A.D. Patil
4)Dr. S.B. Swami
5)Dr. V.B. Dongre

(57) Abstract :
11. ABSTRACT Title of Invention: Computer-implemented veterinary surgical management system for integrated patient, diagnostic and multimedia record processing The invention relates to a computer-implemented veterinary clinical and surgical management system comprising a processor, memory, central data store, patient registration module, outpatient/inpatient routing module, diagnostic and surgical record module, multimedia association module and report-processing module. The system assigns a case identifier to a registered animal and associates outpatient or inpatient episodes, clinical information, diagnostic tests, surgical procedures, images, videos and reports with the case identifier. The multimedia module stores media metadata and a storage reference linked to the relevant case or episode. The report-processing module retrieves associated records, validates their case relationship and generates a structured clinical report or preview. A role-control module may restrict operations according to user category. The system provides coordinated processing and retrieval of heterogeneous veterinary clinical records through persistent identifier relationships. Presented as representative figure of abstract Fig 1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2026
(21) Application No.202621113358 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A FOOD COMPOSITION AND THE METHOD THEREOF

(51) International classification	:A23L 7/10, A23L 11/00, A23L 25/00, A23D 9/00, A23L 27/14, A23L 29/30, A23L 33/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)AKSHAYA SANJAY KADAM
Address of Applicant :D/O Sanjay Sayajirao Kadam, Govind Arcade, Old R.T.O. Sadarbazar, Satara, Maharashtra, India -415001 Satara Maharashtra India
2)RENUKA CHANDRASHEKHAR MAINDE
3)REKHA SUSHIL KUMAR SHARMA
4)SABIHA ANWAR VALI
(72)**Name of Inventor :**
1)AKSHAYA SANJAY KADAM
2)RENUKA CHANDRASHEKHAR MAINDE
3)REKHA SUSHIL KUMAR SHARMA
4)SABIHA ANWAR VALI

(57) Abstract :
"A FOOD COMPOSITION AND THE METHOD THEREOF" Disclosed is a food composition with good nutritional properties and palatability comprises broken wheat, rice flour, barnyard millet, finger millet, green gram, flax seed, peanut wherein the said composition is made without pearl millet. Also provided is method of manufacturing the food composition.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2026
(21) Application No.202621113360 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : NETWORK-BASED SYSTEM FOR CONSENT-CONTROLLED TRADITIONAL KNOWLEDGE MANAGEMENT

(51) International classification	:G06F21/62, G06Q10/06, G06F16/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
Address of Applicant :Powai, Mumbai - 400076, Maharashtra, India Mumbai Maharashtra India
(72)**Name of Inventor :**
1)ANAND, Shubham
2)SONAR, Rajendra M
3)JAIN, Karuna

(57) Abstract :
A network-based traditional knowledge management system (600) includes a community user device (610), an assessor device (620), an application server (640) having a processor (642), a secure traditional knowledge repository (690), a consent/protocol database (700), an access-control module (680), an assessment engine (650), a classification engine (660), a strategy recommendation engine (670), and an audit-log module (710). The access-control module (680) authorizes access to a traditional knowledge profile (910) based on a consent record (930) and a community protocol, determines a sacred/non-sacred flag (920), and applies protected sacred-knowledge handling (742) for sacred traditional knowledge (110). After authorization, the assessment engine (650) evaluates traditional knowledge data using a traditional knowledge assessment tool (130), the classification engine (660) generates a maturity level (960) from a maturity score (950), the strategy recommendation engine (670) generates a selected strategy (970), and the audit-log module (710) generates an audit trail (990). Figure 6

No. of Pages : 92 No. of Claims : 25

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113378 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TRIGLYCERIDE BASED DE-BONDABLE OPTICALLY CLEAR ADHESIVE FOR DISPLAYS

(51) International classification	:C09J 175/14, C09J 175/16, C08G 18/67, C09J 175/04, C08F 2/48
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Institute of Chemical Technology
Address of Applicant :Prof. Shashank T. Mhaske Department of Polymer and Surface Engineering Institute of Chemical Technology Nathalal Parekh Marg, Matunga, Mumbai- 400 019 MAHARASHTRA, INDIA Matunga, Mumbai Maharashtra India
(72)**Name of Inventor :**
1)SHOURYA GUPTA
2)KARAN W. CHUGH
3)PROF. SHASHANK T. MHASKE

(57) Abstract :
Present invention describes triglyceride based de-bondable optically clear adhesive for displays. Further it provides a solvent-free, biobased UV-curable optically clear polyurethane acrylate adhesive composition and a process for its preparation. The adhesive comprises 20 wt% to 29 wt% of an N,N-bis(2-hydroxyethyl) soybean oil fatty amide (HEOA) polyol precursor, 20 wt% to 22 wt% of isophorone diisocyanate, 2 wt% to 6 wt% of dipentaerythritol pentaacrylate, 1 wt% to 4 wt% of 2-hydroxyethyl acrylate, 30 wt% to 45 wt% of 2-ethylhexyl acrylate, and 1 wt% to 2 wt% of a photoinitiator, with an "NCO/OH" stoichiometric ratio of 1.2 to 1.8. The process utilizes isopropyl alcohol and titanium isopropoxide to achieve an HEOA yield of 92% to 95% with suppressed chromophores. The cured adhesive is 100% amorphous, exhibits visible transmittance greater than 95%, a Yellowness Index below 3.0, lap shear strength of 2.242 to 2.294 MPa, and enables non-destructive thermal debonding at 120°C after UV overexposure for circular substrate recovery.

No. of Pages : 34 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113391 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR ISOLATING 2,5-BIS(CARBOXYMETHYL)-1,4-BENZOQUINONE FROM THE LEAF OF MORINDA PUBESCENS Sm

(51) International classification	:C07C 46/10, C07C 46/00, C07C 50/28, C07C 50/34, C07C 46/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MR. OMKAR BHARAT TIPUGADE
Address of Applicant :Krishna Institute of Pharmacy, Krishna Vishwa Vidyapeeth (Deemed to be University), Karad – 415539, District Satara, Maharashtra, India Karad Maharashtra India
2)DR. JYOTIRAM ARJUN SAWALE
3)DR. NAMDEO RAMHARI JADHAV
4)MS. PRITI SHRIKANT SHIRKE
(72)**Name of Inventor :**
1)MR. OMKAR BHARAT TIPUGADE
2)DR. JYOTIRAM ARJUN SAWALE
3)DR. NAMDEO RAMHARI JADHAV
4)MS. PRITI SHRIKANT SHIRKE

(57) Abstract :
"A PROCESS FOR ISOLATING 2,5-BIS(CARBOXYMETHYL)-1,4-BENZOQUINONE FROM THE LEAF OF MORINDA PUBESCENS Sm." Disclosed is a process for isolating 2,5-bis(carboxymethyl)-1,4-benzoquinone from Morinda pubescens Sm. leaf. Fresh leaves, collected before flowering, are shade-dried and powdered, and the powder is Soxhlet-extracted with AR grade methanol (about 64.7°C, about 8 hours). The methanolic extract is concentrated on a rotary evaporator (about 50°C, about 2 hours), reconstituted in methanol–distilled water (about 1:4), and successively fractionated with petroleum ether, chloroform and ethyl acetate. The methanolic fraction, selected by comparative TLC profiling, is purified by column chromatography over silica gel (60–120 mesh) using a chloroform–methanol gradient of increasing polarity. Chromatographically similar fractions (43–48, about 50 mL each) are pooled, concentrated, and crystallised by refrigeration to yield the target compound as a white amorphous powder. The isolated compound is characterised by ^{1}H NMR, ^{13}C NMR and DEPT-135 spectral data confirming a symmetrical, highly oxygenated benzoquinone bearing two carboxymethyl substituents, not previously reported from Morinda pubescens Sm. or from any other source. (Figure 1)

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113440 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PLASTIC BASED OBSTETRIC FORCEPS FOR ASSISTED VAGINAL DELIVERY WITH ENHANCED MATERNAL AND NEONATAL SAFETY

(51) International classification	:A61B17/44, A61B17/28, A61B17/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Rajendrasingh Pardeshi
Address of Applicant :Jijai Maternity and Nursing Home, 12-Chaitanya CHS, Near Gajanan Maharaj Temple, Garkheda, Chhatrapati Sambhaji Nagar, 431009, Maharashtra Aurangabad Maharashtra India
2)Dr. Kalpana Rajendrasingh Pardeshi
(72)**Name of Inventor :**
1)Dr. Rajendrasingh Pardeshi

(57) Abstract :
ABSTRACT PLASTIC BASED OBSTETRIC FORCEPS FOR ASSISTED VAGINAL DELIVERY WITH ENHANCED MATERNAL AND NEONATAL SAFETY
The present disclosure relates to obstetric forceps. The obstetric forceps (100) comprising first and second forceps members (102, 104), each having a fetal-contacting blade portion (106), an elongated substantially linear shank portion (108), and a handle portion (110). Each blade portion (106) comprises an elongated closed-loop configuration defining an internal opening (112), with an arcuate upper portion (114) and side portions (116) converging towards a lower junction (118) connected to the shank portion (108). The first and second forceps members (102, 104) comprise Nylon. FIG. 1 will be the reference figure.

No. of Pages : 27 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113442 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE HYBRID-NANOFLUID THERMAL MANAGEMENT SYSTEM FOR ELECTRIC-VEHICLE LITHIUM-ION BATTERY PACKS WITH ZONED COLD-PLATE FLOW CONTROL AND IN-LINE COLLOIDAL STABILITY CONDITIONING

(51) International classification	:H01M 10/625, H01M 10/615, H01M 10/6556, H01M 10/613, B60H 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Vikramsingh Ravindrasingh Parihar

Address of Applicant :61, Mahesh Nagar, near Sainagar, Amravati, Maharashtra, India, 444607 Maharashtra India

2)Sumit Ganesh Wani

3)Dr. Akshat Singh Jhala

4)Snehal Hanumant Darekar

5)ADITYA RAJENDRA WANKHADE

6)Prabhakar Kisan Sakhare

7)Akshay Pradip Bhingade

8)Shweta Dheeraj Gunjal

9)Amol Angadrao Khot

10)Dhiraj Manoj Shribate

(72)**Name of Inventor :**

1)Sumit Ganesh Wani

2)Dr. Akshat Singh Jhala

3)Snehal Hanumant Darekar

4)ADITYA RAJENDRA WANKHADE

5)Prabhakar Kisan Sakhare

6)Akshay Pradip Bhingade

7)Shweta Dheeraj Gunjal

8)Amol Angadrao Khot

9)Dhiraj Manoj Shribate

10)Vikramsingh Ravindrasingh Parihar

(57) Abstract :

An adaptive battery thermal management system for electric vehicles circulates a hybrid nanofluid through a plurality of independently controllable cold-plate zones thermally coupled to a lithium-ion battery pack. The hybrid nanofluid contains at least two dissimilar nanoparticle populations dispersed in a base coolant to improve heat transfer while maintaining acceptable hydraulic performance. Battery temperature sensors, flow and pressure sensors, and one or more coolant-condition sensors provide thermal, hydraulic, and colloidal-stability information to a controller. The controller regulates pump speed and zonal coolant distribution to limit maximum cell temperature and temperature nonuniformity while reducing auxiliary power. A colloidal-stability module determines an agglomeration condition from optical, ultrasonic, electrical, dielectric, viscosity-related, or pressure-drop information. When conditioning is required, a controlled fraction of coolant is routed through a side-stream re-dispersion chamber while main battery cooling continues. A safety supervisor overrides optimization during thermal anomalies and may prioritize affected-zone cooling, heat rejection, and battery-power derating. The system supports normal driving, fast charging, high-load operation, low-temperature warm-up, condition-based coolant maintenance, and thermal-anomaly response.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/06/2026
(21) Application No.202621077135 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Functionalization of Titanium Dioxide Nanoparticles Using Moss Extract and Method Thereof

(51) International classification	:C01G23/053, B82Y40/00, A61K36/11, B82Y30/00, B01J13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ashwini Dilip Banginwar
Address of Applicant :Dada Ramchand Bakhru Sindhu Mahavidyalaya, K. T. R. Marg, Panchpaoli, Nagpur, Maharashtra, India 440017 Nagpur Maharashtra India
2)Dr. Susmita Abhijit Mandavgane
3)Arti Ashok Rahangdale
4)Nikita Satish Gupta
5)Rani Shivshankar Meshram
6)Mayuri Surajmal Jain
(72)**Name of Inventor :**
1)Ashwini Dilip Banginwar
2)Dr. Susmita Abhijit Mandavgane
3)Arti Ashok Rahangdale
4)Nikita Satish Gupta
5)Rani Shivshankar Meshram
6)Mayuri Surajmal Jain

(57) Abstract :
The present invention relates to a method for functionalization of titanium dioxide (TiO2) nanoparticles using an extract of the moss Ditrichum pusillum as a natural bioactive functionalizing and stabilizing agent. The method comprises preparing a moss extract; preparing a titanium precursor mixture comprising titanium(IV) isopropoxide and ethyl alcohol; adding the moss extract and an acid to the titanium precursor mixture to form a moss-containing TiO2 gel; ageing and dehydrating the gel; and calcining the dehydrated material to obtain moss extract-functionalized TiO2 nanoparticles. Bioactive constituents derived from the moss are associated with the TiO2 nanoparticles and contribute to modification of their surface and optical properties, resulting in enhanced light absorption and photocatalytic activity. The functionalized nanoparticles exhibit enhanced degradation efficiency toward dyes, organic pollutants, and other contaminants present in wastewater. The invention provides an eco-friendly and sustainable approach for preparation of photocatalytic TiO2 nanomaterials useful in water purification, wastewater treatment, and environmental remediation applications.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/09/2026

(21) Application No.202621105382 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SOLAR PANEL SELF-RECHARGING SYSTEM WITH INTEGRATED BATTERY MANAGEMENT

(51) International classification	:H02S40/38, H02J7/35, G05F1/67, H01M10/46, H01M10/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Worldone Energies Private Limited
Address of Applicant :6th Floor, Supreme Forest Avenuetower, Nagpur, Maharashtra, India Maharashtra India
(72)**Name of Inventor :**
1)Murtaza Hamzabhai Kothawala

(57) Abstract :
The present invention relates to a solar panel self-recharging system with integrated battery management. The proposed system comprises a photovoltaic panel array (101) connected to a maximum power point tracking charge controller (102), a main rechargeable battery pack (103) supervised by an integrated battery management system unit (104), an auxiliary energy storage element (105) that exclusively powers control electronics, and a prioritising power switching and conversion network (106) coordinated by a microcontroller (107). Harvested energy is preferentially directed to maintain the auxiliary storage, ensuring continuous monitoring and automatic resumption of main battery charging after low-irradiance or deep-discharge periods without external intervention. The system belongs to the field of photovoltaic energy storage, advances prior discrete arrangements by eliminating recovery failures and reducing self-discharge of primary storage, and is principally used for autonomous off-grid, portable and remote power applications.

No. of Pages : 11 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/09/2026
(21) Application No.202621113487 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Human-Powered Vapour Compression Desalination Apparatus with Flywheel-Assisted Heat Recovery

(51) International classification	:C02F 1/04, B01D 1/26, B01D 1/28, C02F 103/08, B01D 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rungta College Of Engineering and Technology
Address of Applicant :Rungta Educational Campus, Kohka- Kurud Road, Bhilai, Chhattisgarh- 490024 Chattisgarh India
2)Rungta International Skills University
(72)**Name of Inventor :**
1)Prof. Dr. Debshree Bhattacharya
2)Prof. Priyanka Verma
3)Prof. Rubi Kombo
4)Prof. Monika Gupta
5)Prof. Shikha Verma
6)Prof. Shweta Katre
7)Prof. Dr.Abha Sinha
8)Prof. Dr.C Ramesh Kumar
9)Prof. Dr Jayant Damodar Supe

(57) Abstract :
ABSTRACT: Human-Powered Vapour Compression Desalination Apparatus with Flywheel-Assisted Heat Recovery This invention describes a manually operated desalination apparatus for producing distilled water from saline or brackish feedwater without electrical power, reverse-osmosis membranes, or external fuel. The apparatus employs a manually actuated rotary drive coupled to a flywheel and a speed-increasing transmission for operating a miniature vapour compressor. Feedwater is introduced into an evaporative region maintained at reduced pressure, whereby water vapour is generated at a reduced evaporation temperature. The vapour is mechanically compressed to increase its pressure and condensation temperature and is subsequently condensed within a heat-recovery condenser thermally associated with incoming feedwater and the evaporative region. Latent heat released during condensation is thereby recycled to support continued evaporation. Condensed distillate is separately collected while concentrated brine is retained and periodically discharged. Flywheel energy storage smooths intermittent human input and stabilizes compressor operation, enabling compact household-scale vapour-compression desalination in off-grid environments.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :15/07/2026

(21) Application No.202621086826 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR PERSISTENT ELECTRICITY CUT-OFF AND CONTROLLED POWER RESTORATION FOR ENERGY-SAVING APPLICATIONS

(51) International classification	:G01R19/00, H04W4/38, G05B19/048, H01H47/00, H01H50/32	(71)**Name of Applicant :** **1)MAULANA AZAD NATIONAL INSTITUTE Of TECHNOLOGY, BHOPAL** Address of Applicant :Link Road No. 3, Near Mata Mandir, Bhopal - 462003, Madhya Pradesh, India Madhya Pradesh India
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)V. Kavin Ishwar**
(33) Name of priority country	:NA	**2)Dr. Suresh Kumar Gawre**
(86) International Application No	:	
Filing Date	:01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

A SYSTEM AND METHOD FOR PERSISTENT ELECTRICITY CUT-OFF AND CONTROLLED POWER RESTORATION FOR ENERGY-SAVING APPLICATIONS Disclosed is an energy-efficient system and method employing an ESP32-based client node integrated with sensor modules to continuously monitor electrical and environmental conditions, transmitting sensor data via the ESP-NOW protocol to a server node. The server node incorporates deterministic decision logic with safety interlocks and timing constraints to verify the received data and, upon detecting a power cut-off condition, issues a relay actuation command to a relay-based power switching unit connected to a high-voltage mains supply. A modular clamp-based latch mechanism then mechanically secures the relay in a de-energized state independent of a continuous control signal, with an infrared sensor configuration providing engagement feedback. The invention further enables controlled power restoration once pre-established safety criteria are fulfilled.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/08/2026
(21) Application No.202621103449 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR THE DEVELOPMENT OF IN-SITU HIGH ENTROPY ALLOY COATING ON METALLIC MATERIALS

(51) International classification	:C23C24/04, C23C24/06, C22C30/00, C23C24/10, C23C24/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Maulana Azad National Institute of Technology, Bhopal
Address of Applicant :Department of Materials and Metallurgical Engineering, Maulana Azad National Institute of Technology Bhopal, Bhopal 462003, Madhya Pradesh, India Bhopal Madhya Pradesh India
(72)**Name of Inventor :**
1)Ansari Mohd Farhan Mohd Saleem
2)ANANT, Ramkishor

(57) Abstract :
The present invention relates to a novel, cost-effective, and scalable process for the in-situ synthesis of high entropy alloy (HEA) coatings on metallic substrates using pulsed Tungsten Inert Gas (P-TIG) arcing. The process of the present invention allows in-situ alloying through rapid melting, elemental diffusion, and solidification, resulting in the formation of a dense, metallurgically bonded HEA coating that exhibit significantly enhanced hardness, wear resistance, and corrosion resistance compared to the uncoated substrate. The process eliminates the need for pre-alloyed powders, vacuum environments, or high-energy laser/plasma systems making it suitable for large-scale industrial applications in aerospace, defence, marine, and advanced engineering components.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :15/09/2026

(21) Application No.202621110666 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A FUNCTIONAL MIXING CHAMBER FOR MULTI-INGREDIENT SCREW-BASED ADDITIVE MANUFACTURING SYSTEMS

(51) International classification	:B29C 64/321, B29C 64/393, B01F 15/04, B01F 15/02, B29C 64/336
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur
Address of Applicant :Dumna Airport Road, P. O. Khamaria, Jabalpur-482005, Madhya Pradesh, India Jabalpur Madhya Pradesh India
(72)**Name of Inventor :**
1)Rahul Soni
2)Ponappa K.
3)Puneet Tandon

(57) Abstract :
The present invention relates to a smart functional mixing chamber (100) for multi-ingredient screw-based additive manufacturing systems. The functional mixing chamber (100) comprises a chamber body (110) defining an internal mixing region, material inlets (120), internal flow-directing members (140), a central outlet (130), and a movement arrangement configured to move the chamber body (110) in an X-Y plane relative to a plurality of fixed material dispensing hoppers. Under a digitally defined recipe, the chamber (100) sequentially positions itself relative to selected hoppers to collect predetermined quantities of different ingredients, with the quantity monitored by a load-cell arrangement (170). The collected ingredients are internally mixed using a motor-driven rotating blade assembly and the resulting homogenized mixture is supplied directly to a downstream screw-based extrusion system.

No. of Pages : 22 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621114216 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A phytochemical composition comprising Azadirachta indica and Camellia sinensis extracts for chemoprevention of mammary carcinogenesis

(51) International classification	:A61K 36/00, A61K 36/58, A61K 36/82, A61P 35/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Pravin Badhe
Address of Applicant :Flat No 708 Ragini C Durvankur Shrushti, Near JSPMs college Narhe Pune Maharashtra India
(72)**Name of Inventor :**
1)Dr Pravin Badhe
2)Dr Saranya Punniyakotti
3)Ashwini Badhe
4)Tapasya Dwivedi
5)Dr Puneet

(57) Abstract :
The present invention relates to a phytochemical composition comprising polar extracts of Azadirachta indica (neem) and Camellia sinensis (green tea) and a process for preparation thereof. In a preferred embodiment, ethanolic extracts of neem and green tea are combined in an approximately 3:1 weight ratio. The plant materials are subjected to sequential extraction comprising non-polar defatting followed by ethanolic extraction, concentration and combination of the resulting polar extracts. The composition contains phytochemical classes including polyphenols, catechins, flavonoids, terpenoids and limonoids. Experimental evaluation demonstrated antioxidant and anti-mitotic activities and acute oral tolerability at tested doses. In a DMBA-induced mammary carcinogenesis model, administration of the composition was associated with reductions in tumor volume, tumor weight, tumor burden and tumor incidence, together with improvements in selected hematological, oxidative-stress and histopathological parameters. The composition is suitable for further development into phytopharmaceutical and nutraceutical formulations for chemopreventive applications.

No. of Pages : 30 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :20/08/2026

(21) Application No.202621100451 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM FOR PERFORMANCE-BASED IDENTIFICATION OF OPTIMAL MULTI-FEATURE PEPTIDE ENCODING COMPOSITIONS USING A WEIGHTED INTERACTION GRAPH

(51) International classification	:G16B30/10, G16B40/20, G06N20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Technology Raipur
Address of Applicant :G.E. Road, Amanaka, Raipur, Chhattisgarh 492010 Raipur Chattisgarh India
(72)**Name of Inventor :**
1)Dr. Deepak Singh
2)Aditya Kumar

(57) Abstract :
The present invention relates to a computer-implemented system (100) for performance-based identification of multi-feature peptide or protein encoding compositions. A biological sequence data acquisition module (1) receives labelled sequences; a feature encoding module (2) generates multiple complete feature matrices; and a weighted graph construction module (3) represents the feature matrices as nodes. A classification performance evaluation module (4) evaluates individual and concatenated feature sets using performance metrics including accuracy, F1-score, and AUC. An edge weight assignment module (5) assigns performance-derived edge weights representing complementary relationships between feature sets. A feature composition selection module (6) evaluates weighted edges and subgraphs to identify two-feature, category-wise, and higher-degree compositions. An output and visualization module (7) provides an interpretable selected composition without requiring training of a graph neural network. The selected composition is usable for peptide or protein classification and bioactivity prediction.

No. of Pages : 36 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621113966 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SOXHLET EXTRACTION APPARATUS WITH CAPACITIVE SENSOR-BASED CYCLE DETECTION AND RELAY-CONTROLLED HEATING SYSTEM

(51) International classification	:B01D 11/04, B01D 11/02, B01D 11/00, G01N 27/22, G05D 23/19	(71)**Name of Applicant :** **1)Somavanshi Deepak B.** Address of Applicant :Associate Professor, Department of Pharmacognosy, Krishi Nagar, Malegaon Camp, Bhaygaon Road, Malegaon, Maharashtra – 423105 Maharashtra India **2)Bairagi Vinod A.** **3)Thore Nilesh Rameshwar** **4)Mandale Chetana Sadashiv** **5)Pagar Pallavi Ramdas** **6)Mhaske Nikita Yuvraj** **7)Matkar Omkar Mangesh**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Somavanshi Deepak B.**
(86) International Application No Filing Date	: :01/01/1900	**2)Bairagi Vinod A.** **3)Thore Nilesh Rameshwar**
(87) International Publication No	: NA	**4)Mandale Chetana Sadashiv**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**5)Pagar Pallavi Ramdas** **6)Mhaske Nikita Yuvraj**
(62) Divisional to Application Number Filing Date	:NA :NA	**7)Matkar Omkar Mangesh**

(57) Abstract :

AUTOMATED SOXHLET EXTRACTION APPARATUS WITH CAPACITIVE SENSOR-BASED CYCLE DETECTION AND RELAY-CONTROLLED HEATING SYSTEM The present invention provides a system and method for automating Soxhlet extraction with verified siphoning-cycle detection and extraction-responsive heater control. The system consists of a Soxhlet assembly, an external multi-zone capacitive sensing collar, a siphon-return monitoring module, an optical extract-monitoring module, a thermal-control module, a safety-interlock module and a processor. The processor registers a successful extraction cycle if the ordered filling-and-emptying sequence is sensed by spatially separated sensing zones and the solvent passage is confirmed by the return-monitoring module. A cycle-specific optical value is determined from returned solvent and compared to preceding values to identify an extraction plateau. Heater power is controlled by temperature, confirmed cycle time and extraction progress and is switched off via residual-heat compensation. Heater-present, coolant-flow, low-solvent and over-temperature measurements give proven fail-safe isolation. The disclosed arrangement eliminates false cycling events, improves endpoint accuracy and reproducibility, and reduces unnecessary heating, solvent exposure, energy consumption and thermal degradation of thermolabile constituents.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :23/09/2026
(21) Application No.202621114033 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Manual Toothbrush with Independently Deformable Multi-Zone Peripheral Soft-Tissue Protection System

(51) International classification	:A46B 9/04, A46B 15/00, A46B 5/00, A61C 17/22, A46D 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr Apurva Prashant Deshpande
Address of Applicant :"Mahalasa" Bungalow, Plot no 82, Faj Housing Colony, Madel, Thivim, North Goa, Goa. 403502 Goa India
2)Mr Keshav L Dukle
3)Mr Atharva Prashant Deshpande
(72)**Name of Inventor :**
1)Dr Apurva Prashant Deshpande
2)Mr Keshav L Dukle
3)Mr Atharva Prashant Deshpande

(57) Abstract :
Manual Toothbrush with Independently Deformable Multi-Zone Peripheral Soft-Tissue Protection System This invention describes a manual toothbrush configured to reduce accidental contact trauma to intraoral soft tissues while maintaining conventional tooth-cleaning performance. The toothbrush includes a substantially rigid bristle-supporting head carrying a plurality of bristle tufts and a plurality of discrete compliant protective regions positioned selectively around impact-prone portions of the head. A distal protective region and opposing lateral protective regions are independently deformable such that accidental contact at one region causes localized deformation substantially without requiring deformation of the remaining regions. The proximal transition of the toothbrush head remains substantially free from a corresponding protective region to maintain a compact configuration and facilitate posterior access. Controlled deformation may be provided through reduced-thickness flexural regions, internal compliant cavities, differentiated stiffness characteristics, and optional displacement-limiting formations. The protective regions terminate below the operative bristle tips such that routine plaque removal remains principally performed by the bristles.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621114037 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-MODAL WEARABLE SYSTEM FOR ADAPTIVE PANCREATIC CANCER RISK STRATIFICATION

(51) International classification	:A61B 5/00, A61B 5/145, A61B 5/1455, G16H 50/30, A61B 5/0205	(71)**Name of Applicant :** **1)Rajarambapu Institute of Technology** Address of Applicant :Rajaramnagar, Islampur, Tal- Walwa, Dist. Sangli, Maharashtra 415414 Maharashtra India (72)**Name of Inventor :** **1)Dr. Varsha T. Lokare** **2)Prakash M. Jadhav** **3)Aarati R. Gavade** **4)Monica S. Patil** **5)Snehal S. Patil** **6)Dipak R. Patil** **7)Roshani Shaikh** **8)Onkar Thalpate** **9)Pooja Patil** **10)Purna Shah**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Multi-Modal Wearable System For Adaptive Pancreatic Cancer Risk Stratification This invention describes a multi-modal wearable-assisted system for pancreatic cancer risk stratification using clinical information and continuously acquired physiological information. A textile-integrated wearable sensing arrangement acquires conductivity, pH and glucose-related physiological measurements and associates the measurements with dynamically determined reliability information. Patient-specific baselines are maintained for the measured parameters, and deviations from the baselines are processed together with cross-sensor corrections, temporal characteristics and persistence information. A multi-rate temporal fusion process synchronizes continuously acquired wearable information with episodically acquired demographic, laboratory, medical-history and symptom information. A machine-learning inference engine generates a pancreatic-risk state using confidence-weighted multimodal information. The generated risk state further controls operation of the wearable sensing arrangement by modifying sampling frequency, confirmation sensing and data transmission. Persistent abnormalities may consequently initiate increased acquisition while low-risk conditions permit reduced sensing activity, thereby providing longitudinal risk monitoring with adaptive use of sensing, processing, communication and power resources.

No. of Pages : 29 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621114044 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONTEXT-AWARE PORTABLE DRINKING BOTTLE WITH MULTI-PARAMETER POTABILITY VERIFICATION AND SAFETY RESTRICTION

(51) International classification	:G01N 33/18, G01N 27/06, B65D 23/00, G08B 21/18, B65D 23/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DKTE Society's Textile and Engineering Institute
Address of Applicant :Rajwada, Ichalkaranji, Maharashtra 416115 Maharashtra India
2)Mrs. Varsha Toshniwal
3)Dr. Mrs. J. P. Kharat
4)Swaraj Shivaji Shinde
5)Abhijeet Ulhas Patil
(72)**Name of Inventor :**
1)Mrs. Varsha Toshniwal
2)Dr. Mrs. J. P. Kharat
3)Swaraj Shivaji Shinde
4)Abhijeet Ulhas Patil

(57) Abstract :
Context-Aware Portable Drinking Bottle with Multi-Parameter Potability Verification and Safety Restriction This invention describes a portable drinking water bottle for in-situ potability assessment of contained liquid is disclosed. The bottle performs context-aware multi-parameter evaluation using turbidity, conductivity-derived total dissolved solids, pH, temperature, storage-duration, environmental-exposure, and motion-history data. A controller applies motion-compensated sensor fusion and staged decision logic to distinguish transient agitation-induced sediment suspension from persistent contamination conditions. When dynamically unsettled liquid is detected, the system initiates delayed reverification after a settling interval before issuing a safety classification. The bottle provides externally visible condition outputs including safe, cautionary, conditionally safe after settling, and unsafe states. In selected embodiments, unsafe-state indication persists during reduced-power or battery-depleted conditions, and a mechanical drinking restriction mechanism inhibits or discourages access to the drinking interface when unsafe liquid conditions are determined. The arrangement enables autonomous, application-independent potable-water verification for field, travel, emergency, industrial, military, rural, and low-infrastructure environments.

No. of Pages : 29 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :23/09/2026
(21) Application No.202621114071 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : "UPPER PULL-OUT CORE ROD TABLET TOOLING PUNCH"

(51) International classification	:B30B 11/02, A61J 3/10, B30B 15/02, B30B 11/22, B30B 11/34
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M/S PARLE TABLETING TECHNOLOGIES PRIVATE LIMITED
Address of Applicant :PE-40, BOIGIDC, RASULPURA GAV, SANAD-II, WASNA, RASULPUR, SANAD GIDC, AHMEDABAD, GUJRAT AHMEDABAD Gujarat India
(72)**Name of Inventor :**
1)MR. JURGEN BOECKX
2)SAGAR ASHVINKUMAR PADHIAR
3)MR. RAMCHANDRA VITHAL PARAB

(57) Abstract :
ABSTRACT UPPER PULL-OUT CORE ROD TABLET TOOLING PUNCH The invention discloses an upper pull-out core rod tablet tooling system for producing tablets having a central hole. The system comprises an upper punch (4), a die (5), and a lower punch (6) having a central bore configured to receive a core rod (7), the core rod (7) being inserted into the lower punch (6) from the punch tip and projecting upward during compression to form a central hole in the tablet. The core rod (7) is axially retained by a channel-shaped U-type holder (9) secured to the turret by a locking bush (11), with a washer (10) positioned therebetween, the retention arrangement permitting rotation of the lower punch (6) and the core rod (7). A scraper (8) and a milled chamber remove powder build-up from the core rod (7). The invention improves alignment, reduces wear, and is suitable for high-speed pharmaceutical and confectionery tablet production. Reference Figure: FIG. 3

No. of Pages : 21 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621114247 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR PREDICTIVE TERRAIN-ADAPTIVE TIRE PRESSURE MANAGEMENT

(51) International classification	:B60C 23/04, B60C 23/00, B60C 23/06, B60C 23/02, B60C 23/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lakshmi Narain College of Technology Excellence
Address of Applicant :Khajuri Khurd, Raisen Road, Bhopal, Madhya Pradesh, India - 462022 Madhya Pradesh India
(72)**Name of Inventor :**
1)SATYARTH TIWARI

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR PREDICTIVE TERRAIN-ADAPTIVE TIRE PRESSURE MANAGEMENT The present invention provides a system and process for artificial intelligence enabled terrain adaptive tire pressure management. Forward-facing cameras capture images in RGB and infrared modes of terrain in front of the vehicle. The edge processing module classifies the terrain, computes a confidence value, determines the terrain boundary and computes a look-ahead distance. Wheel sensors measure the actual pressure, temperature inside and on the surface and tread wear. The vehicle sensors measure speed and loads. The predictive controller computes the available pressure transition time, computes the pressure envelope and computes the pressure transition time using a pneumatic transition model. If the desired pressure cannot be achieved before terrain contact, a safe pressure that can be achieved is determined. Independently controlled valves inflate/deflate the respective tires using the rotary air unions attached to the axles while the vehicle is in motion. Pressure feedback confirms achievement and a failsafe response is triggered if the safe pressure cannot be reached. Fig 3

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621114248 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR FOOD FRESHNESS PREDICTION AND ADAPTIVE PRESERVATION

(51) International classification	:G06N 20/00, G06N 3/08, F25D 29/00, G01N 33/02, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lakshmi Narain College of Technology (MCA)
Address of Applicant :LNCT Campus, Kalchuri Nagar, Raisen Road, P.O. Kolua, Bhopal, Madhya Pradesh, India - 462022 Madhya Pradesh India
(72)**Name of Inventor :**
1)Dr. Swagatika Lenka

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR FOOD FRESHNESS PREDICTION AND ADAPTIVE PRESERVATION The present invention relates to a system and method for artificial-intelligence-enabled food freshness prediction and adaptive preservation. A food-storage enclosure receives a food item, while heterogeneous sensors acquire food-condition and environmental data including temperature, humidity, gaseous, weight and optical information. A processing unit identifies a food category, determines sensor-specific relevance and reliability, and combines present and historical measurements to generate a temporal food-condition representation. An artificial-intelligence model processes the representation to determine a Food Freshness Index, spoilage probability, Remaining Freshness Period and deterioration trajectory. Based on the predicted condition, a controller operates one or more environmental devices to modify cooling, humidity, ventilation, air circulation or atmospheric composition. Updated sensor data acquired after the preservation action is used to evaluate the food's response. The controller retains, modifies or replaces a subsequent action according to whether the preceding action improved the deterioration trajectory, thereby providing food-specific, feedback-controlled preservation and extending usable freshness. Fig 3

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114252 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for Consent-Governed Cross-Entity Animal Identity and Health Data Management

(51) International classification	:H04L 9/32, H04L 29/06, G16H 10/60, G06F 21/62, G06F 21/00	(71)**Name of Applicant :** **1)Petara Pet Solutions Private Limited** Address of Applicant :FL-C4-1102 NR BAPDEV GHAT,THE LAKE DISTRICT SN- 1516,Khondhwa KH,Pune City,Pune-411048,Maharashtra Pune,Maharashtra Maharashtra India (72)**Name of Inventor :** **1)Janmejay Kaushikkumar Mistry**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A system and method for consent-governed cross-entity management of animal identity, health, behavioural, and care information is disclosed. Each registered animal is assigned a unique, persistent, non-reassignable identifier encoded in a machine-readable portable token carrying a cryptographically signed copy of core identity data, enabling offline identity resolution without specialised hardware. Animal data is organised into hierarchical tiers governed by custodian consent enforced at the data layer. Two consent paths are provided: a standing revocable grant for entities in an established service relationship, and a 30-minute OTP-gated session for any party the custodian admits. A profile completeness gate requests only absent or expired fields at each service interaction. Authorised entities accumulate longitudinal behavioural observations accessible to all subsequently authorised entities. Custodial authority transfers through a dual channel verified sequence while the animal's record persists intact. Field-level change history is retained differentially by data class.

No. of Pages : 23 No. of Claims : 19

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621114073 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Smart Paper Strip for Rapid Screening of Multiple Milk Adulterants

(51) International classification	:G01N 31/22, G01N 33/04, C12Q 1/54, C12Q 1/58, G01N 33/52
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vishal Dnyaneshwar Chavhan
Address of Applicant :At. Post Goul (KD) Tq. Pusad Dist. Yavatmal, Maharashtra 445215 Maharashtra India
2)Sarika Sunil Rathod
3)Shalini Dhirendra Rai
4)Ashok Annaraya Muchandi
5)Shubham Sakharam Kshirsagar
6)Chetan Yashwantrao Kadam
7)Deepika Kartikchand Vishwas
8)Pankaj Yashwant Thorat
9)Prathmesh Sudhir Gogate
10)Siddhi Kailas Vyas
(72)**Name of Inventor :**
1)Vishal Dnyaneshwar Chavhan
2)Sarika Sunil Rathod
3)Shalini Dhirendra Rai
4)Ashok Annaraya Muchandi
5)Shubham Sakharam Kshirsagar
6)Chetan Yashwantrao Kadam
7)Deepika Kartikchand Vishwas
8)Pankaj Yashwant Thorat
9)Prathmesh Sudhir Gogate
10)Siddhi Kailas Vyas

(57) Abstract :
A Smart Paper Strip for Rapid Screening of Multiple Milk Adulterants This invention describes a paper-based system for rapid preliminary screening of milk for multiple adulterants, particularly starch, urea and detergent. The system comprises a porous paper substrate defining spatially separated reagent-bearing test zones configured to receive a milk sample through direct application or capillary transport. A starch-responsive zone includes an iodine–potassium iodide reagent system that develops a blue to blue-black colour in the presence of starch. A urea-responsive zone includes urease and a pH-sensitive indicator configured to generate a visible colour response following enzymatic hydrolysis of urea and consequent formation of ammonia. A detergent-responsive zone includes an indicator composition capable of producing a visually distinguishable response in the presence of detergent or surfactant. The developed colour responses may be interpreted directly or compared with a reference colour chart, thereby enabling portable, low-cost and instrument-free preliminary screening of milk adulteration.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621114125 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-based pest detection and Automated Targeted Semiochemical Control system

(51) International classification	:A01M 1/02, A01M 1/20, G06N 20/00, G06N 3/08, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Shivam Satyawan Madrewar

Address of Applicant :404/45 B, Shivanjali, Laxmi Garden, Bijapur Road, Jath, Sangli, Maharashtra. Maharashtra India

2)Sushma Satyawan Madrewar

(72)**Name of Inventor :**

1)Purva Ravindra Wankhade

2)Pallavi Jayesh Saindane

3)Dr. Kedarnath

4)Dr. Sharanabasappa

5)Ashwini S Koute

6)Dr. Sangram Sahebrao Dhumal

7)Dr. Ramniwas

8)Dr. Deepak Deshmukh

9)Dr. Ravindra Dadabhau Pawar

10)Dr. Ravidra Ashok Karande

11)Dr. Anuja Suresh Ingale

12)Janhavi Hemant Rane

13)Mahesh Subhash Chandankar

14)Shivam Satyawan Madrewar

(57) Abstract :

A system and method are disclosed for artificial-intelligence-based multimodal detection, identification and population quantification of agricultural insect pests, and for automated, demand-based targeted semiochemical control thereof. Solar-powered field sensor nodes fuse long-wave infrared thermal imaging, visible/near-infrared imaging, bio-acoustic sensing and environmental data through a trained edge artificial-intelligence model to yield a confidence-scored pest identification and calibrated population or activity estimate. Upon sufficiently confident detection of a target pest, the AI engine actuates a smart trap having a pest-specific semiochemical cartridge and an electronically metered dispensing mechanism, releasing a controlled quantity of semiochemical according to the estimated pest population or activity level. Captured insects are counted and the capture data are fed back to adjust subsequent control cycles, thereby providing a closed-loop detect-identify-quantify-control-verify-adjust operation. The architecture is reconfigurable for different target pests and semiochemicals, as demonstrated by working examples involving Holotrichia spp. and Helicoverpa armigera, thereby reducing reliance on broad-spectrum chemical pesticides.

No. of Pages : 33 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :23/09/2026
(21) Application No.202621114136 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR DEFLECTOMETRIC MEASUREMENT OF SPECTACLE LENSES MOUNTED IN A FRAME

(51) International classification	:G01M 11/02, G02C 13/00, G01M 11/00, G01B 11/25, G01B 11/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PARAS AJITKUMAR MEHTA
Address of Applicant :SF-9C, LOTUS AURA, OPP. LILLERIA, SAMA SAVLI ROAD Gujarat India
(72)**Name of Inventor :**
1)PARAS AJITKUMAR MEHTA

(57) Abstract :
An apparatus (100) measures spectacle lenses (142, 144) while they remain glazed in the frame (140). A display panel (110) shows a ladder of phase-shifted fringe patterns (112) whose periods are held as lengths upon the panel; a holder (120) presents the frame at a first distance (d1); and a camera (130), focused upon the lens plane (126) at a second distance (d2), images the lenses through the fringes. A processor (156) recovers phase at every pixel, resolves the ambiguity across the ladder without propagating along any path, so that both lenses and the opaque bridge between them fall upon one absolute scale, subtracts a reference acquired with the holder empty, and differentiates the resulting slope field into mean power and two cross-cylinder components. The curl of the slope field is reported as a quality figure, the fringe modulation as a scattering-defect channel, and the measured field is read where the wearer's pupils look. (FIG. 1)

No. of Pages : 59 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621114366 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SanVie -A System and Method for Automated Three Dimensional Digital Occlusal Plane Analysis for Dental Applications

(51) International classification	:A61C 9/00, A61C 19/04, A61C 7/00, A61C 19/045, A61C 19/05
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. V Srishti
Address of Applicant :PG Resident, Prosthodontics, Crown & Bridge and Implantology, Maitri College of Dentistry and Research Centre, Anjora, Durg, Chhattisgarh, India, 491001 Chattisgarh India
2)Dr. Sanjeev Singh
(72)**Name of Inventor :**
1)Dr. V Srishti
2)Dr. Sanjeev Singh

(57) Abstract :
The present invention relates to a software application named the Digital Occlusal Plane Analyser, comprising an automated three-dimensional digital occlusal plane analysis system for dental applications. The system receives a patient-specific three-dimensional representation of dentition and processes the representation to identify individual teeth, occlusal regions, and clinically relevant dental landmarks. A plurality of three-dimensional landmark or occlusal surface coordinates is validated according to geometric and anatomical reliability and used to computationally derive a representative occlusal plane. A reference plane is established within a common three-dimensional coordinate system using anatomical, dental, craniofacial, standardized, or treatment-related reference information. A comparative analysis module determines angular and linear relationships including occlusal cant, inclination, pitch, roll, yaw, vertical discrepancy, and regional asymmetry. Outlier landmarks may be excluded or computationally down-weighted. A visualization module superimposes detected landmarks, the calculated occlusal plane, reference geometry, and deviation information on the digital dental model. The system enables reproducible digital assessment, treatment planning, treatment comparison, and dental rehabilitation.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114398 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SUSPENSION SYSTEM FOR A VEHICLE

(51) International classification	:B60G 11/10, B60G 11/04, B60G 11/12, B60G 11/113, B60G 11/107
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VE COMMERCIAL VEHICLES LTD
Address of Applicant :102, INDUSTRIAL AREA 1, PITHAMPUR- 454775, DIST. DHAR, MP, INDIA DHAR Madhya Pradesh India
(72)**Name of Inventor :**
1)Krishna Srikanth Achanta
2)Suyash Gupta
3)Bhushan R Kukekar

(57) Abstract :
A SUSPENSION SYSTEM FOR A VEHICLE ABSTRACT A suspension system (100) for a vehicle is disclosed. The suspension system (100) comprises a plurality of housings (102), a plurality of leaf spring assemblies (104), each connected with at least one housing of the plurality of housings (102), each leaf spring assembly comprising a plurality of leaves (114), at least one wear plate (106) disposed within each of the plurality of leaf spring assemblies (104), at least one bolster (108) assembled between each end of the plurality of leaf spring assemblies (104) and an axle (116) of the vehicle, a pair of V-stay assemblies (110) coupled between an upper portion of each housing of the plurality of housings (102) and the axle (116) of the vehicle, and a torque rod assembly (112) coupled between a lower portion of each housing of the plurality of housings (102) and the axle (116) of the vehicle.

No. of Pages : 33 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114410 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A COMPOSITION FOR SOIL ENHANCEMENT, A METHOD OF PREPARING THE COMPOSITION, AND A METHOD OF ENHANCING SOIL

(51) International classification	:C05F11/08, C05F3/00, C09K17/52, C05G5/12, C05G5/30, B01J2/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BAOOL, Yogita
Address of Applicant :CAMPRO, Sagar Developer, Q Group, Unit No. B/3, Survey No. 352 B-1, Western Express Highway, Next to JIO Petrol Pump, Viva compound, Nalasopara East - Pelhar, Maharashtra - 401208, India. Maharashtra India

(72)**Name of Inventor :**
1)BAOOL, Ravindra
2)KASHTE, Ganesh

(57) Abstract :
The present invention discloses a composition for soil enhancement, a method of preparing the composition, and a method of enhancing soil. The composition comprises a porous carbonaceous biochar material, a charging medium comprising a bovine urine component and a bovine manure component impregnating pores of the biochar material, at least one microorganism associated with the biochar material, and a binder, the composition being in the form of a compressed unit-dose body configured to disintegrate in response to soil moisture to release the charging medium and the at least one microorganism into soil over an extended period. The method of preparing comprises contacting the biochar material with the charging medium, admixing the charged biochar material with the at least one microorganism and the binder, and compressing the mixture into the unit-dose body without requiring a specified temperature condition before or during or after compression to minimize adverse effects on that retain viability of the at least one beneficial microorganism. The method of enhancing soil comprises placing the compressed unit-dose body in soil at or near a root zone of a plant and exposing the compressed unit-dose body to soil moisture. Figure 1 is the representative figure.

No. of Pages : 36 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :23/09/2026
(21) Application No.202621114147 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR MULTI-VENDOR NETWORK CONFIGURATION COMPLIANCE ANALYSIS

(51) International classification	:H04L 41/0893, H04L 9/40, H04L 41/14, G06F 40/295, H04L 41/0893, H04L 9/40, H04L 41/14, G06F 40/295
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JOMON JOSEPH DANIEL
Address of Applicant :Block No 428/856, Subash Teakdy, Ulhasnagar, Maharashtra, India - 421004 Ulhasnagar Maharashtra India
2)GINI ELIZABETH KUNJACHAN
(72)**Name of Inventor :**
1)JOMON JOSEPH DANIEL
2)GINI ELIZABETH KUNJACHAN

(57) Abstract :
A method for analysing configuration compliance of a multi-vendor network under credential isolation includes authenticating (302), by a collection agent (120) within a trust boundary, locally to network enforcement devices (132, 134, 136, 138) occupying different enforcement layers, and transforming (304) their device configuration data into a vendor-neutral rule representation within the trust boundary such that device authentication secrets remain local and only the vendor-neutral rule representation is conveyed to an evaluation environment. The method includes deriving (306), by a natural-language policy engine at an input stage, a structured zone-pair access policy from an unstructured natural-language document (406); constructing (308) a unified path model determining an ordered sequence of enforcement points; mapping (310) address ranges to externally supplied zone definitions by longest-prefix-match; computing (312) an aggregate effective access state across the full path and comparing it against the policy; and attributing (314) a non-compliance to a device identifier, a configuration-element identifier and a location. Responsive to a locally detected change, only an affected rule is re-normalised and only an affected zone pair is re-evaluated (316), while device authentication secrets remain within the trust boundary. FIG. 1

No. of Pages : 30 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621114151 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR EVALUATING RIGHT TO HEALTH OF TRANSGENDER PERSONS IN INDIA

(51) International classification	:G16H 10/60, G06Q 40/08, G06Q 50/22, G06Q 10/06, G06Q 50/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Anu Mishra
Address of Applicant :Research Scholar, School of Law, Christ (Deemed to be University), Lavasa, Pune, Lavasa Campus. Maharashtra India
2)Dr. Somu C S
(72)**Name of Inventor :**
1)Anu Mishra
2)Dr. Somu C S

(57) Abstract :

11. ABSTRACT The present invention provides a computer-implemented system and method for evaluating healthcare access and Right-to-Health dimensions of transgender persons in India. The system receives heterogeneous healthcare-access data from a plurality of authorized sources and classifies the data into predefined dimensions comprising availability, accessibility, affordability, non-discrimination, quality, continuity, gender-affirming healthcare access, counselling access, insurance or financial protection and confidentiality. A normalization module converts heterogeneous data into standardized machine-readable parameter values, while a validation module detects incomplete, inconsistent, invalid or stale data. An assessment engine generates a multidimensional Right-to-Health assessment vector and optionally a composite assessment value and confidence value. A gap-detection engine identifies healthcare-access deficiencies using configurable thresholds. A privacy module separates or pseudonymizes personally identifiable information, while an audit module generates an electronically verifiable record of assessment inputs, transformations and model versions. Geographic and temporal processing modules enable spatial and longitudinal assessment. The system thereby provides a standardized, auditable and privacy-preserving technical mechanism for evaluating healthcare-access conditions of transgender persons at individual, facility and population levels.

No. of Pages : 11 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621114152 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ANALYSIS OF CHALLENGES AND OPPORTUNITIES FOR WOMEN IN THE LEGAL PROFESSION

(51) International classification	:G06Q 10/06, G06Q 10/10, G06Q 50/18, G06Q 10/00, G06F 16/34
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Minoti Madhu Bhatija
Address of Applicant :Research Scholar, School of Law, Christ (Deemed to be University), Lavasa, Pune, Lavasa Campus. Maharashtra India
2)Dr. Somu Chikthrahlly S
(72)**Name of Inventor :**
1)Minoti Madhu Bhatija
2)Dr. Somu Chikthrahlly S

(57) Abstract :
ABSTRACT The present invention discloses a computer-implemented system and method for analysis of challenges and opportunities for women in the legal profession. The system receives heterogeneous data from professional, institutional, workplace, educational, judicial, survey, and other data sources. A preprocessing and normalization architecture converts the received data into standardized analytical variables. Challenge and opportunity analysis modules identify measurable factors associated with recruitment, remuneration, retention, career progression, leadership, workplace conditions, professional development, mentorship, and related parameters. An analytical engine processes the variables using configurable weighting parameters and generates one or more composite indicators. A comparative analysis module compares the indicators across career stages, geographic regions, institutions, professional categories, and time periods. A visualization module presents the results through interactive dashboards, matrices, and trend representations, while a reporting module generates analytical reports. Privacy and security functionality may provide anonymization, pseudonymization, aggregation, encryption, and access control. The system enables repeatable, data-driven and longitudinal analysis of professional challenges and opportunities affecting women in the legal profession.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114438 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Hybrid Cyber-Safety Framework for Dual-Pipeline Message-URL Analysis, Indian DLT Sender Authenticity, and Explainable Risk Assessment

(51) International classification	:H04L 29/06, H04L 9/40, G06F 21/57, G06F 21/55, G06F 16/955	(71)**Name of Applicant :** **1)Dr. Krupa Bhavsar** Address of Applicant :32, Hingalaj Society, B/H, Old Dharam Cinema and Gayatri Temple, Mehsana, Gujarat - 384002, India. Mehsana Gujarat India **2)Dr. Ajay Patel** (72)**Name of Inventor :** **1)Dr. Krupa Bhavsar** **2)Dr. Ajay Patel**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Title of Invention Hybrid Cyber-Safety Framework for Dual-Pipeline Message-URL Analysis, Indian DLT Sender Authenticity, and Explainable Risk Assessment Abstract Hybrid Cyber-Safety Framework for Dual-Pipeline Message-URL Analysis, Indian DLT Sender Authenticity, and Explainable Risk Assessment operates by acquiring message, sender-identification, URL, and message-with-link inputs (1001). Embedded URLs are extracted, separated, and routed with message data through respective analysis paths (1002). Message data is normalized, feature-represented, classified, and contextually analysed (1003)-(1004), while sender data is verified using sender-format, Indian DLT-style pattern, and trusted sender-header analysis (1004). URL data is validated, feature-extracted, classified for phishing, and assessed for domain authenticity, typo-squatting, and impersonation (1005)-(1006). Dual-pipeline outputs are retained and correlated (1007)-(1008) to compute a consolidated cyber-risk score and level, with informational context limiting message contribution. Evidence-linked risk output is generated from contributing indicators (1009), followed by multilingual safety guidance and cyber-response assistance (1010). Analysis records, history, and analytics are maintained (1011). Figure 8: Overall Operational Flowchart of the Hybrid Cyber-Safety Framework illustrating the main working steps.

No. of Pages : 46 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114464 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ARTIFICIAL INTELLIGENCE ENABLED INSTITUTIONAL PERFORMANCE AND STRATEGIC MARKETING MANAGEMENT SYSTEM

(51) International classification	:G06Q 30/02, G06Q 10/06, G06N 20/00, G06Q 50/20, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Shailesh Tripathi
Address of Applicant :Professor, Balaji Institute of Management & HRD Sri Balaji University Pune, Maharashtra, Pin Code: 411033 Maharashtra India
2)Dr. Anju Khandelwal
3)Dr. Nidhi Kumari
4)Dr. Shailesh Kumar
5)Dr. Apoorva Sharma
6)Amar Nath Dutta
7)Shalan Pralhad Lad
(72)**Name of Inventor :**
1)Dr. Shailesh Tripathi
2)Dr. Anju Khandelwal
3)Dr. Nidhi Kumari
4)Dr. Shailesh Kumar
5)Dr. Apoorva Sharma
6)Amar Nath Dutta
7)Shalan Pralhad Lad

(57) Abstract :
The present invention discloses an artificial intelligence enabled institutional performance and strategic marketing management system (100) that integrates institutional performance monitoring with strategic marketing management for educational institutions. The system (100) comprises a data acquisition module (110) that ingests heterogeneous data from academic, administrative, and marketing sources; a data preprocessing and integration unit (120) that harmonizes multimodal data; and an artificial intelligence engine (130) having a performance analytics sub-module (132), a predictive modeling sub-module (134), and a natural language processing sub-module (136). An institutional performance analysis module (140) evaluates key performance indicators while a strategic marketing management module (150) generates personalized outreach campaigns. A recommendation and decision support engine (160) delivers actionable insights via a user interface (170), a communication interface (180) provides secure connectivity, and a feedback loop (190) enables continual reinforcement learning.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114465 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NOVEL 1,3,4-OXADIAZOLE DERIVATIVES, PROCESSES FOR THEIR PREPARATION, AND THEIR ANTIPROLIFERATIVE ACTIVITY

(51) International classification	:A61P 35/00, C07D 413/12, A61K 31/4245, C07D 413/14, C07D 271/113
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Suvarna Nandkishor Vakare
Address of Applicant :Sinhgad Technical Education society's Sinhgad College of Pharmacy, Off Sinhgad Road, Vadgaon (Bk), Pune-411041 Maharashtra India
2)Rajesh Bhimgonda Patil
(72)**Name of Inventor :**
1)Suvarna Nandkishor Vakare
2)Rajesh Bhimgonda Patil
3)Bhoomendra Atmaram Bhongade
4)Sandeep Prabhu Gadhwe
5)Sakshi Prashant Mukkawar

(57) Abstract :
This specification discloses eleven new N-phenyl-1,3,4-oxadiazol-2-amine compounds, DZ001 through DZ011, differing in the aryl, heteroaryl or benzyloxy group carried at the 5-position, made by combining a substituted hydrazide with phenylisothiocyanate to give a thiosemicarbazide intermediate, which was then cyclised in presence of 2-iodoxybenzoic acid (IBX) and triethylamine (TEA). Compounds were characterized by FT-IR, 1H-NMR and ESI-mass spectrometry. Molecular docking against the ATP-binding pocket of cyclin-dependent kinase 4 (CDK4) and in vitro screening against HCT116 cell lines by the MTT assay identified DZ008 and DZ004 as compounds showing notable antiproliferative activity against HCT116 cells for further development as antiproliferative agents, while other compounds are moderately active.

No. of Pages : 15 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621114510 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRONIC BRAKE PRESSURE CONTROL UNIT FOR AN ANTI-LOCK BRAKING SYSTEM OF A VEHICLE

(51) International classification	:B60T 8/40, B60T 8/48, B60T 17/02, B60T 8/42, B60T 8/34
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ENDURANCE TECHNOLOGIES LIMITED
Address of Applicant :Endurance Technologies Limited, E-93, M.I.D.C. Industrial Area, Waluj, Chh. Sambhajinagar – 431136 (formerly Aurangabad), Maharashtra, India Chh. Sambhajinagar Maharashtra India
(72)**Name of Inventor :**
1)Sandip D. Kale
2)Kamalkishor B. Kakade
3)Moresh R. Deshpande

(57) Abstract :
ABSTRACT Title: Electronic Brake Pressure Control Unit for an Anti-Lock Braking System of a Vehicle The electronic brake pressure control unit (100) for an Anti-Lock Braking System, in accordance with the disclosed invention, comprises of, a body (10), solenoid valves (15A, 15B), a cover (20), a pump assembly (30), an accumulator assembly (40), a motor (50) and an electronic control module (65). The disclosed invention, has a simple and compact construction, its pump assembly uses fewer parts to provide effective braking performance, is light weight, is easy to manufacture, can be economically provided in low-capacity two wheeled vehicles, has higher mechanical efficiency and reduced machining requirement for its to manufacturing. (Refer Fig. 2)

No. of Pages : 30 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114536 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A DUAL-COMPARTMENT THERMALLY ISOLATED SMART FOOD DELIVERY BOX

(51) International classification	:F25D 29/00, F25D 11/00, A47J 39/02, F25D 31/00, B65D 81/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur
Address of Applicant :Dumna Airport Road, P. O. Khamaria, Jabalpur-482005, Madhya Pradesh, India Jabalpur Madhya Pradesh India
(72)**Name of Inventor :**
1)Dr. Matadeen Bansal
2)Aanya Jain
3)Ishank Kumar
4)Akshay Panwar
5)Anushree Arora
6)Vivek Konadi
7)Pratyush Singh

(57) Abstract :
A dual-compartment thermally isolated smart food delivery box (100) comprises an outer enclosure (102) defining a hot compartment (104) and a cold compartment (106), thermally isolated by a sealed partition wall (112) having XPS insulation core (114) between aluminium foil layers (116, 118). The hot compartment (104) comprises an active heating arrangement (120) having a self-regulating PTC heater (122) thermally coupled to aluminium heat-spreader layer (124), and forced-air circulation arrangement (130) having a DC fan (132). The cold compartment (106) comprises reusable frozen gel cooling packs (128) for passive cooling. A sensing and monitoring unit (134) provides temperature, humidity and battery-voltage information to a control unit (144), which controls the PTC heater (122) and DC fan (132) through a switching arrangement (148). The control unit (144) pre-heats hot compartment (104), maintains temperature through closed-loop feedback, generates a deterministic Food Quality Score, and disables PTC heater (122) upon a low-battery condition.

No. of Pages : 37 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114544 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONTROL SYSTEM AND METHOD FOR DISTRIBUTION NETWORK SWITCHING USING COMMUNICATION-PATH-INDEPENDENT EVIDENCE SELECTION

(51) International classification	:H02J 13/00, H04L 12/24, H04L 45/24, H02H 7/26, H02J 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Shrey Shaileshbhai Shah
Address of Applicant :70, Tekra Faliya, Sultanpura, Jhagadia, Dist: Bharuch, Gujarat - 393110, India. Bharuch Gujarat India
2)Reenaben Shaileshbhai Shah
(72)**Name of Inventor :**
1)Shrey Shaileshbhai Shah
2)Reenaben Shaileshbhai Shah

(57) Abstract :
ABSTRACT CONTROL SYSTEM AND METHOD FOR DISTRIBUTION NETWORK SWITCHING USING COMMUNICATION-PATH-INDEPENDENT EVIDENCE SELECTION A control system (100) for an electric power distribution network receives, through a first communication path, data indicating an abnormal condition of a feeder section (210), and maintains confidence values for competing candidate causes comprising a physical electrical event and a compromise or failure of that path. While the confidence values fail an authorisation criterion for a candidate switching operation, a switching command is withheld and suspected communication dependency elements of the first path are identified from a stored communication-topology data structure. For candidate evidence sources, a discrimination value between the candidate causes and an independence value relative to those elements are determined; a source satisfying both criteria is selected and an additional measurement therefrom updates the confidence values. An authenticated switching command changes the state of a switching device (200) only where the updated confidence values and an electrical operating constraint are satisfied. FIG. 1 Block diagram of the control system and its relationship to the electric power distribution network and to the field devices

No. of Pages : 36 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114546 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Vertical-Axis Wind Energy Harvesting Apparatus with Wind-Responsive Paired Wing Assemblies

(51) International classification	:F03D3/00, F03D3/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BREEZE POWER SOLUTIONS PRIVATE LIMITED
Address of Applicant :Flat No. 1101, Eastwoods, SN-29, HN 10/1,10/2 & 10/4, N.I.B.M, Pune - 411048 (Maharashtra, India) Pune Maharashtra India
(72)**Name of Inventor :**
1)ZAREER AGA
2)ANOSH AGA

(57) Abstract :
The present invention relates to a vertical-axis wind energy harvesting apparatus (100) comprising a central shaft (107), a central body (103), a plurality of radial arms, wing assemblies, a gearing system (104), and a generator (106). Each wing assembly comprises curved, concave first and second blades (101, 102) interconnected by a synchronizing linkage mechanism (105) and pivotally movable about axes positioned towards rear edges of the blades. During a torque-producing portion of rotation, the blades open to form sail-like wind-receiving surfaces that generate rotational torque. During a return portion of rotation, the blades move towards a closed position, thereby reducing their wind-receiving area and return-side drag. The blades move under influence of wind without depending upon gravity. The apparatus is configured for unidirectional rotation and receives wind from horizontal directions without yaw adjustment. Rotational energy is transmitted through the central shaft, gearing system and generator for conversion into electrical energy. Fig. of Abstract: Fig. 1

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114612 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TICKETING AND MANAGEMENT SYSTEM FOR TRANSIT VEHICLES

(51) International classification	:G07B 15/02, G07B 15/00, G07B 11/00, G07B 15/06, G07B 15/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lakshmi Narain College of Technology Excellence
Address of Applicant :Khajuri Khurd, Raisen Road, Bhopal, Madhya Pradesh, India-462022. Bhopal Madhya Pradesh India
(72)**Name of Inventor :**
1)Rajdeep Shrivastava

(57) Abstract :
A ticketing and management system for transit vehicles comprises a kiosk 101 hosts a display panel 102 that captures a passenger's destination and travel choices, an imaging unit 103 integrated with a NIR sensor captures secure facial biometric data to tie the passenger's identity to the transaction, a printer module 104 dispenses a physical transit ticket, allowing the passenger to proceed to the vehicle's entrance, a gate assembly 201 utilizes an OCR (Optical Character Recognition) module 204 on a scanning plate 203 to read the ticket data and display clearance details, an artificial intelligence-based imaging camera 205 records boarding faces, counts passengers, and monitors for structural irregularities, and a control unit continuously aggregates all this information querying scheduling databases, generating secure payment codes, matching gate-side faces to kiosk data, and cross-verifying real-time passenger counts.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114645 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART KRISHI SAHAYAK: An Agricultural Intelligence and Automation System for Multi-Stakeholder Decision Support

(51) International classification	:G06Q 50/02, G06K 9/62, G10L 15/26, G06Q 10/06, G06K 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Trupti Shah
Address of Applicant :Vidyavardhini's College of Engineering and Technology, Vasai(w) K. T. Marg. Maharashtra India
2)Dr. Maniroja M
(72)**Name of Inventor :**
1)Trupti Shah
2)Dr. Maniroja M

(57) Abstract :

Abstract: The present invention provides Smart Krishi Sahayak, an intelligent, multi-modal agricultural intelligence and automation system for providing real-time agricultural information and decision support to farmers and other agricultural stakeholders. The system receives voice, text, and crop image inputs and integrates real-time weather data, mandi market price data, crop information, and plant disease information through a centralized backend processing engine. Natural language processing, speech-to-text, computer vision, and AI-based processing are employed to analyse user inputs and generate relevant agricultural advisories. The system further incorporates a GPS-based clustering mechanism for identifying similar agricultural issues within a predefined geographical region and generating localized community alerts. A web-based interface provides multilingual interaction, weather information, market intelligence, crop-related information, disease detection, and advisory recommendations. The system thereby provides an integrated platform for real-time agricultural intelligence, localized information, disease diagnosis, market analysis, and data-driven decision support.

No. of Pages : 22 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/09/2026
(21) Application No.202621113488 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Passive Mechanical Rockfall Interception Apparatus with Progressive Slope-Creep Warning and Energy Absorption

(51) International classification	:E01F 7/04, G08B 21/10, G01L 1/22, E01F 7/06, G01L 5/04	(71)**Name of Applicant :** **1)Rungta College Of Engineering and Technology** Address of Applicant :Rungta Educational Campus, Kohka- Kurud Road, Bhilai, Chhattisgarh- - 490024 Chattisgarh India **2)Rungta International Skills University**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Prof. Dr Farhan Khan**
(32) Priority Date	:NA	**2)Prof. Dr. Satyadharma Bharti**
(33) Name of priority country	:NA	**3)Prof. Dr. Payal Nene**
(86) International Application No Filing Date	: :01/01/1900	**4)Prof. Dr. Jyoti Upadhyay** **5)Prof. Dr Vikas Dilliwar**
(87) International Publication No	: NA	**6)Prof. Chandra Shekhar Nagendra**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**7)Prof. Mahendra Kumar Soni** **8)Prof. Dr. Rupesh Mishra**
(62) Divisional to Application Number Filing Date	:NA :NA	**9)Prof. Dr Jayant Damodar Supe**

(57) Abstract :
ABSTRACT: Passive Mechanical Rockfall Interception Apparatus with Progressive Slope-Creep Warning and Energy Absorption This invention describes a passive mechanical apparatus for intercepting rockfall and indicating progressive slope movement in unstable terrain. The apparatus employs a ground-anchored high-tensile hybrid cable net connected to telescopic strain nodes configured to respond mechanically to variations in cable tension. Each strain node incorporates a spring-biased telescopic arrangement that accommodates gradual strain while maintaining predetermined net tension. Progressive slope creep produces controlled extension of the strain node and, upon reaching a calibrated displacement threshold, mechanically releases a visually detectable warning member without requiring electrical power, sensors, batteries, or communication infrastructure. The strain nodes further incorporate graduated shear-pins configured to fail sequentially at predetermined loading thresholds during sudden rockfall impact, thereby permitting staged telescopic displacement and progressive absorption of impact energy. The apparatus accordingly combines rockfall interception, mechanical pre-failure slope-creep indication, self-tensioning, and controlled impact-energy dissipation for remote roads, railway cuttings, mine slopes, and mountainous terrain.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113489 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MOBILE CARRIER PLATFORM-ASSISTED RIS-ENABLED COMMUNICATION AND WIRELESS POWER TRANSFER SYSTEM AND METHOD

(51) International classification	:H02J 50/20, H02J 50/90, H02J 50/10, H02J 50/80, H02J 50/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology Indore
Address of Applicant :Khandwa Road, Simrol, Madhya Pradesh - 453552, India. Simrol Madhya Pradesh India
(72)**Name of Inventor :**
1)RAMABADRAN, Swaminathan
2)GHOSH, Saptarshi
3)GAUTAM, Sumit
4)DAS, Sourosmita
5)MISHRA, Devesh

(57) Abstract :
An integrated system (100) and method (300) provide adaptive wireless communication and wireless power transfer across infrastructure-deficient or disaster-affected environments. The system (100) comprises a mobile carrier platform (MCP) (102) carrying a reconfigurable intelligent surface (RIS) with a plurality of controllable electromagnetic reflecting elements, memory (106), and one or more processors (104). The one or more processors (104) execute a closed-loop control sequence to jointly optimize target MCP positioning, RIS phase-shift configurations, wireless power transfer (WPT) delivery, and energy allocation based on real-time channel conditions, position metrics, and system power reserves. The configured RIS dynamically modifies incident electromagnetic propagation to steer wireless signals, establishing or maintaining high-quality communication links between obstructed source and destination nodes. Concurrently, a WPT subsystem transmits directed electromagnetic power toward targeted receiving units (112), while an energy harvesting and management subsystem captures and redistributes ambient and transmitted energy to sustain continuous operation. By coordinating platform positioning, surface reflect array configurations, power delivery, and energy management, the system provides rapid, self-sustaining wireless connectivity and power distribution.

No. of Pages : 31 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113508 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A TRANSACTIONAL FEATURE DRIFT ISOLATION AND SELECTIVE ENSEMBLE RECONFIGURATION SYSTEM AND METHOD

(51) International classification	:G06N 20/20, G06F 9/46, G06F 17/30, G06N 5/04, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vikramsingh Ravindrasingh Parihar
Address of Applicant :61, Mahesh Nagar, near Sainagar, Amravati, Maharashtra, India, 444607 Maharashtra India
2)Abhay A. Dande
3)Dr. Mahendra A. Pund
4)Dr. Amruta Deshmukh
(72)**Name of Inventor :**
1)Abhay A. Dande
2)Dr. Mahendra A. Pund
3)Dr. Amruta Deshmukh
4)Vikramsingh Ravindrasingh Parihar

(57) Abstract :
A transaction-controlled system (100) adapts a fixed-size machine-learning ensemble without permitting a batch to alter the state that predicts it. An active inference store (130) supplies a read-only state, and a commit controller (140) durably records batch identity, state identity, predictions and decisions before enabling adaptation. A semantic registry (120) groups transformed descendants of each source feature. A drift analyzer (150) forms availability-marked distribution and relevance evidence, while a state engine (160) maintains drift memory and hysteretic feature states. A learner-impact router (170) maps changed features to affected learner slots and schedules keep, update, retrain or replacement actions within compute, memory and write budgets. A parent-bound candidate state (180) is activated only at a later batch boundary by a conditional atomic pointer change. The system provides state-isolated adaptation, bounded training resources, deterministic recovery and auditable prediction lineage for sensor and record streams. FIG. 1 is the representative drawing.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2026
(21) Application No.202621113527 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : “Graphene Quantum Dot-Engineered Graphene Oxide Ethylene Sensing Device and Method”

(51) International classification	:C01B32/198, B82Y15/00, B82Y30/00, G01N27/30, G01N27/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology Bombay.,
Address of Applicant :having place of business at IITB, Powai, Mumbai, Maharashtra, India 400076 mumbai Maharashtra India
(72)**Name of Inventor :**
1)Julfikar Ali Sarkar
2)Mohammed Aslam

(57) Abstract :
ABSTRACT “Graphene Quantum Dot-Engineered Graphene Oxide Ethylene Sensing Device and Method” The present invention relates to a method and ethylene sensing device (100) for sensing ethylene. The ethylene sensing device (100) includes a substrate (101), an electrode structure (102) disposed on the substrate (101) and including first and second electrodes separated by at least one electrode gap and a graphene quantum dot-engineered graphene oxide sensing layer (104) disposed in electrical contact with the first and second electrodes and extending across the at least one electrode gap to define a conductive pathway between the first and second electrodes, where the sensing layer (104) includes graphene oxide sheets (105) and graphene quantum dots (106) incorporated into a matrix to form a zero-dimensional/two-dimensional heterostructure having defect-associated ethylene adsorption sites and graphene quantum dot/graphene oxide interfaces, and where adsorption of ethylene at the defect-associated ethylene adsorption sites redistributes charge at the graphene quantum dot/graphene oxide interfaces and modulates an electrical current through the conductive pathway. FIG. 1

No. of Pages : 67 No. of Claims : 30

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113533 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A TRI-ACTION POLYHERBAL HAIR OIL COMPOSITION FOR DANDRUFF, HAIR FALL AND PREMATURE HAIR GRAYING

(51) International classification	:A61Q 7/00, A61Q 5/00, A61K 36/9068, A61K 8/92, A61K 8/9789
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Shaikh Reshmabano
Address of Applicant :Assistant Professor, Gramonnati Mandal's Arts, Commerce and Science College, Narayangao, Junner Pune, Maharashtra, India. Maharashtra India
2)Dr. Jamadar Rasul
3)Prof. Pokale Sanjaykumar
(72)**Name of Inventor :**
1)Dr. Shaikh Reshmabano
2)Dr. Jamadar Rasul
3)Prof. Pokale Sanjaykumar

(57) Abstract :
The present invention relates to a tri-action polyherbal hair oil composition comprising mustard oil and coconut oil as dual carrier base oils infused with curry leaves (Murraya koenigii), ginger (Zingiber officinale), hibiscus flowers (Hibiscus rosa-sinensis), kalonji seeds (Nigella sativa), fenugreek seeds (Trigonella foenum-graecum) and onion (Allium cepa) through controlled thermal extraction. The composition simultaneously addresses dandruff through antifungal phytoconstituents, hair fall through follicle-strengthening and anti-inflammatory bioactives, and premature graying through melanocyte-protective antioxidants and melanogenesis-promoting compounds. The formulation is prepared exclusively from edible, plant-based, generally recognised as safe ingredients, is non-irritant, exhibits acceptable physicochemical stability and provides a cost-effective, safe herbal alternative to synthetic hair care products for regular topical scalp application across all age groups.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113579 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR OUTDOOR AIR REPLENISHMENT WITH ADAPTIVE VECTOR DISTRIBUTION AND MULTI-STAGE FILTRATION

(51) International classification	:B01D 46/00, F24F 8/108, B01D 46/42, B01D 46/44, F24F 13/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BIG UMBRELLA VENTURE LLP
Address of Applicant :Sr. No. 15/2, Off Tushar Industries, Gujarwadi Road, Katraj, Pune 411046, Maharashtra Pune Maharashtra India
(72)**Name of Inventor :**
1)Neelkanth Suresh Shinde
2)Anirudha Chhatrapati Kshirsagar
3)Nilesh Devadas Shinde

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR OUTDOOR AIR REPLENISHMENT WITH ADAPTIVE VECTOR DISTRIBUTION AND MULTI-STAGE FILTRATION The present disclosure relates to an outdoor air replenishment system (100) configured to generate a localized fluidic clean-air region (128). The system comprises suction vents (104), a pre-filter (106), a particulate filtration stage (108), a gas-phase pollutant treatment stage (110), a fine post-filtration stage (112), a central blower (114), a pressurized plenum (116), delivery channels (118), second air-moving assemblies (120), and an adaptive vector distribution assembly (122). The second air-moving assemblies independently accelerate treated air through respective exhaust outlets (126), while the adaptive vector distribution assembly controls exit velocity, horizontal direction, vertical direction and spread. A wind sensor (130) and control system (134) modify the air discharge based on wind conditions. An optional onboard small language model (136) may further enable predictive operation based on environmental and traffic data, the system remaining operable through conventional automation without the small language model (136). The system is configured to produce a measurable air-quality improvement within the first minute of operation. (Referring to FIG. 1)

No. of Pages : 36 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113609 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AGENTIC ARTIFICIAL INTELLIGENCE CONTROLLED MARKETPLACE ELECTRONIC NOTARISATION SYSTEM WITH HARDWARE ENFORCED POLICY GATING AND TAMPER EVIDENT CRYPTOGRAPHIC EVIDENCE CHAINING, AND METHOD THEREOF

(51) International classification	:G06Q50/18, H04L9/32, G06F21/60, G06N3/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sankar Narayan KARASUDHA PATNAIK
Address of Applicant :A3 802 Nyati Elan Near JSPM College Wagholi Pune Maharashtra 412207 Maharashtra India
2)Tarun Sharma
3)Shalabh Gupta
4)Gunjan Dawar Sharma
(72)**Name of Inventor :**
1)Sankar Narayan KARASUDHA PATNAIK
2)Tarun Sharma
3)Shalabh Gupta
4)Gunjan Dawar Sharma

(57) Abstract :
ABSTRACT AGENTIC ARTIFICIAL INTELLIGENCE CONTROLLED MARKETPLACE ELECTRONIC NOTARISATION SYSTEM WITH HARDWARE ENFORCED POLICY GATING AND TAMPER EVIDENT CRYPTOGRAPHIC EVIDENCE CHAINING, AND METHOD THEREOF An electronic notarisation system (100) is disclosed, comprising a hardware processor (202), a cryptographic accelerator engine (216) computing a document fingerprint hash, a hardware security module (210) retaining a non extractable notarial sealing private key, a hardware real time clock and monotonic counter (218), a write once read many storage controller (226) and a graphics and neural processing accelerator (220). An agentic artificial intelligence orchestration engine (300) routes tasks to specialised agents (306 to 320) whose tool invocation authority is confined by a governed tool registry (322) and a tool permission table (324). A notary marketplace dispatch engine (400) derives an eligible notary subset (414) by a hard eligibility gate filter (404) before applying a multi factor ranking module (416). A policy gated state machine controller (120) defines hard gates (518), confidence gates (520) and human in the loop gates (522), and a hardware watchdog and gate enforcement circuit (228) asserts an inhibit signal causing the hardware security module (210) to withhold the sealing private key until every gate is satisfied. A cryptographic evidence graph module (124) chains each event record by a previous event hash pointer (622), a monotonic counter value (626) and a hardware generated signature (628), whereby alteration is detectable by a chain verification module (630). A privacy preserving verification service (712) releases a permitted attribute set (716) to a verifier client device (708) whilst withholding a withheld evidence set (718). A corresponding method (800) is also disclosed. Reference Figure: FIG. 1

No. of Pages : 22 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2026
(21) Application No.202621113633 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ACTUATION MECHANISM FOR COMBINED BRAKING SYSTEM

(51) International classification	:B60T 7/10, B60T 11/04, B62L 3/02, B60T 7/08, B62L 3/08	(71)**Name of Applicant :** **1)ENDURANCE TECHNOLOGIES LIMITED** Address of Applicant :E-92, MIDC Industrial Area, Waluj, Chh. Sambhajinagar – 431136 (formerly Aurangabad), Maharashtra, India Chh. Sambhajinagar Maharashtra India (72)**Name of Inventor :** **1)Kamalkishor B. Kakade** **2)Priyanka D. Arak** **3)Abhijit Sonawane**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT Title: Actuation Mechanism for Combined Braking System The actuation mechanism (100) for combined braking system, in accordance with the disclosed invention, is comprises of a brake lever (5), a secondary lever (10), a double torsion spring (15), fasteners (20B, 20N, 30), a lever mounting body (25A), and a fastening cap (25B). The disclosed actuation mechanism (100), provides an actuation mechanism for Combined Braking System that has a simple construction with fewer overall number of parts, provides an actuation mechanism for Combined Braking System that can be easily manufactured and provided on the vehicle, provides an actuation mechanism for Combined Braking System that is easier to maintain during the rated life of a vehicle, and provides an actuation mechanism for Combined Braking System with reduced overall economic cost associated with its provision in a vehicle. (Refer Fig. 2)

No. of Pages : 20 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621113730 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ENHANCING INFORMAL DIGITAL ENGLISH LEARNING ENGAGEMENT AMONG HIGHER EDUCATION LEARNERS USING RESILIENCE AND FLOW ANALYTICS

(51) International classification	:G06Q 50/20, G09B 19/06, G09B 5/06, G09B 7/02, G
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. SANJU PANDURANG JADHAV
Address of Applicant :Associate Professor and Head, Department of English, M S Kakade College, Someshwarnagar, Tal-Baramati, Pune - 412306, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Dr. SANJU PANDURANG JADHAV

(57) Abstract :
The present invention relates to a computer-implemented system and method for enhancing informal digital English learning engagement among higher education learners using resilience and flow analytics. The system receives interaction events generated during digital English learning activities and processes learner behaviour associated with task continuation, interruption, repeated attempts, recovery, performance, and activity transitions. A resilience analytics engine generates a learner resilience representation based on behavioural responses to learning difficulty and interruption. A flow analytics engine evaluates the relationship between activity challenge and learner capability together with interaction characteristics to generate a flow representation. An engagement state generation module combines the resilience and flow representations to determine a dynamic learner engagement state. An adaptive learning module subsequently modifies learning content, task difficulty, activity sequence, feedback, or re-entry activities according to the determined state. The system continuously updates learner representations using subsequent interaction events, thereby providing an adaptive digital English learning environment capable of supporting sustained participation, recovery from learning interruptions, and progression through appropriately calibrated learning challenges.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621113732 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LOW-COST URBAN VEHICULLAR AIR POLLUTION CONTROL DEVICE

(51) International classification	:B60H 3/06, B01D 46/00, B01D 53/04, B01D 46/42, B60H 1/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEDICAPS UNIVERSITY
Address of Applicant :AB Road Rau Pigdambar, Indore - 453331, Madhya Pradesh, India Indore Madhya Pradesh India
(72)**Name of Inventor :**
1)Dr. Ruchir Lashkari
2)Dr. Panneerselvam Siluvainathan
3)Sachin Solanki
4)Akshat Ambekar
5)Vansh Pal
6)Yashraj Gupta
7)Aditya Verma
8)Heet Shah

(57) Abstract :
The present invention relates to a low-cost urban vehicular air-pollution control device (100) configured for localized treatment of polluted ambient air. The device comprises a weather-resistant housing (101), protective inlet grille (110), coarse particulate pre-filter (120), nanofiber fine-particle filter (130), biochar adsorption stage (140), zeolite adsorption stage (150), contained calcium-hydroxide treatment stage (160), blower (170), solar panel (180), rechargeable battery (190), and electrical arrangement (200). During operation, the blower (170) draws ambient air through the sequential treatment stages. The inlet grille (110) restricts relatively large debris, while the coarse pre-filter (120) removes larger particulate matter. The nanofiber filter (130) subsequently captures fine particulate matter including PM2.5. The biochar (140) and zeolite (150) stages provide adsorption-based treatment for selected gaseous contaminants, followed by the contained calcium-hydroxide stage (160) for selected acidic gases. The removable treatment modules permit individual servicing, while solar-assisted power supports outdoor operation at roadside locations traffic intersections.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :23/09/2026
(21) Application No.202621113876 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Matrix-Adaptive ICP-MS Method for Simultaneous Trace-Level Determination of Elemental Impurities in Nutraceutical Products

(51) International classification	:G01N 1/44, G01N 27/62, G01N 1/28, G01N 27/64, G01N 27/626
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lakshmi Narain College of Pharmacy
Address of Applicant :Kalchuri Nagar, Raisen Road, Bhopal, Madhya Pradesh, India - 462022 Madhya Pradesh India
(72)**Name of Inventor :**
1)Dr. Shivakant Shukla
2)Akshay Namdev
3)Dr. Surendra Kumar Jain
4)Dr. Gopal Patel

(57) Abstract :
ABSTRACT A Matrix-Adaptive ICP-MS Method for Simultaneous Trace-Level Determination of Elemental Impurities in Nutraceutical Products The present invention relates to a matrix adaptive inductively coupled plasma mass spectrometric method for the simultaneous determination of trace level amounts of chromium, nickel, copper, arsenic, selenium, cadmium, tin, antimony, lead and mercury in nutraceuticals. A nutraceutical is digested by microwave digestion in a sealed container containing nitric acid, hydrogen peroxide and water with a return condensate path for minimizing volatilization losses of elemental species. A digest is then separated into measurement and matrix-reference portions wherein the target elements are selectively extracted from the matrix-reference portion by a chelating agent resulting in an analyte-free matrix that serves as a basis for a calibration. An internal standard matrix load index and rhodium and yttrium internal standards are used to compensate for matrix interference in the individual analyte responses. In case of matrix index or internal standard recovery out of range, the sample is automatically diluted and measured again until the calibration and validation criteria are fulfilled. Fig. 3

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621113877 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD, A FLIGHT CONTROL SYSTEM AND AN AGRICULTURAL UAV FOR ROBUST FLIGHT CONTROL

(51) International classification	:G05D 1/10, G05D 1/00, G05B 13/04, B64C 39/02, G05D 1/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lakshmi Narain College of Technology
Address of Applicant :Kalchuri Nagar, Raisen Road, Bhopal, Madhya Pradesh, India – 462021 Bhopal Madhya Pradesh India
(72)**Name of Inventor :**
1)Dr. Govind Prasad Pandiya
2)Dr. Amrita Pahadia

(57) Abstract :
ABSTRACT A METHOD, A FLIGHT CONTROL SYSTEM AND AN AGRICULTURAL UAV FOR ROBUST FLIGHT CONTROL The present invention relates to a method, a flight control system, and an agricultural unmanned aerial vehicle (UAV) for robust flight control. The system employs a two-dimensional discrete Roesser state-space model comprising horizontal and vertical state sub-vectors evolving along respective independent index directions. The model incorporates uncertainties associated with payload mass variation, center-of-gravity displacement, fluid sloshing, actuator effectiveness, and environmental disturbances. Linear Matrix Inequality (LMI)-based controller synthesis employing an asymmetric Lyapunov formulation determines a robust control gain for maintaining flight stability over predefined uncertainty bounds. The robust control gain is determined offline and stored in an onboard flight controller. During operation, a sensor suite acquires flight-state information, and the onboard flight controller applies the robust control gain to generate control inputs for UAV actuators. The present architecture thereby facilitates robust flight stability and trajectory control during agricultural operations under dynamically varying payload and environmental conditions. Figure. 3

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621113878 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A HETEROLEPTIC BIS(TRIAZAPENTADIENE) PALLADIUM COMPLEX WITH SEQUENTIAL ORGANONITRILE-DERIVED LIGAND FORMATION

(51) International classification	:C07F 15/00, C07C 257/10, C07D 213/22, B01J 31/22, C07F 9/94, C07C 311/48, C07F 5/02, C07C 19/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lakshmi Narain College of Technology Excellence
Address of Applicant :Khajuri Khurd, Raisen Road, Bhopal, Madhya Pradesh, India - 462022 Bhopal Madhya Pradesh India
(72)**Name of Inventor :**
1)Dr. Vivek Gupta

(57) Abstract :
ABSTRACT A HETEROLEPTIC BIS(TRIAZAPENTADIENE) PALLADIUM COMPLEX WITH SEQUENTIAL ORGANONITRILE-DERIVED LIGAND FORMATION The present invention is directed towards a heteroleptic bis(triazapentadiene) palladium complex, and a sequential organonitrile-derived ligand formation process. The heteroleptic bis(triazapentadiene) palladium complex comprises a palladium atom bound by first and second constitutionally distinct bidentate triazapentadiene ligands bearing a nitrile-derived substituent group on each respective ligand, to create an asymmetric coordination environment. The sequential organonitrile-derived ligand formation process comprises contacting a palladium precatalyst, a first formamidinium compound, and a first organonitrile compound to yield a mono(triazapentadiene) palladium intermediate, isolating or characterizing the mono(triazapentadiene) palladium intermediate, reduction of any remaining first organonitrile compound, and then contacting the intermediate with a second formamidinium compound and a constitutionally different second organonitrile compound to yield the second ligand. Reactant stoichiometry, reaction conditions and choice of a counteranion for the palladium compound selectively promotes the heteroleptic complex at the expense of homoleptic complexes and statistical mixtures of ligands. FIG 3

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621113895 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A NOVEL DEEP EUTECTIC SOLVENT ASSISTED EXTRACTION COUPLED WITH GREEN CHROMATOGRAPHY FOE SELECTIVE AND SUSTAINABLE DETERMINATION OF BIOACTIVE COMPONENTS

(51) International classification	:G01N 30/02, B01D 11/04, G01N 30/06, G01N 30/88, G01N 30/72
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Jaya Vasant Gade
Address of Applicant :Department of Analytical Chemistry, SNDT Women's University, Sir Vithaldas Vidyavihar, Juhu Tara Road, Santacruz (West), Mumbai-400049, India. Mumbai Maharashtra India
(72)**Name of Inventor :**
1)Dr. Jaya Vasant Gade

(57) Abstract :
A NOVEL DEEP EUTECTIC SOLVENT ASSISTED EXTRACTION COUPLED WITH GREEN CHROMATOGRAPHY FOE SELECTIVE AND SUSTAINABLE DETERMINATION OF BIOACTIVE COMPONENTS ABSTRACT The present invention relates to a sustainable analytical method for the selective extraction and determination of bioactive components using a deep eutectic solvent assisted extraction process integrated with green chromatographic analysis. The method employs a tailor-made deep eutectic solvent system formulated from environmentally benign hydrogen bond donors and acceptors to enhance the solubility, recovery, and selectivity of target bioactive compounds from complex matrices. Extraction conditions, including solvent composition, solid-to-liquid ratio, temperature, time, and agitation, are optimized to achieve high extraction efficiency with reduced consumption of hazardous organic solvents. The resulting extract is directly or minimally processed prior to chromatographic separation using an eco-friendly mobile phase and optimized operating conditions designed to minimize solvent use, waste generation, and energy demand. The proposed approach provides reliable qualitative and quantitative determination of selected bioactive constituents with satisfactory sensitivity, precision, accuracy, and reproducibility. Integration of deep eutectic solvent extraction with green chromatography offers a simple, economical, and environmentally responsible alternative to conventional sample preparation and analytical procedures. The invention is applicable to pharmaceutical, herbal, food, nutraceutical, and natural product matrices requiring selective, efficient, and sustainable bioactive component analysis for routine quality control and research.

No. of Pages : 14 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/09/2026
(21) Application No.202621111223 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR PRODUCTION SCHEDULING

(51) International classification	:G05B 19/418, G06Q 10/06, G06Q 50/04, G06N 3/12, G06Q 10/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INVENTALITY
Address of Applicant :202, Swanand Apartment, Walvekar Nagar, Parvati, Block 21, Laxman B. Walvekar Road, Pune- 411009, Maharashtra India Pune Maharashtra India
(72)**Name of Inventor :**
1)DHANANJAY KARANDIKAR

(57) Abstract :
ABSTRACT Title: A SYSTEM AND METHOD FOR PRODUCTION SCHEDULING The present invention relates to a method and system for optimization of production, more particularly the disclosure relates to a system and method for scheduling production in a manufacturing and process industries. The system and method thereof of the present invention provide an integrated hybrid production scheduling system and method that combines rigorous mixed-integer linear programming (MILP) formulation with heuristic genetic algorithm (GA) techniques to generate high-quality, production schedules for manufacturing and process plants. The system and method thereof is configured to operate in three distinct ways: (a) GA-only mode, (b) MILP-only mode, and (c) sequential GA-then-MILP mode. In GA-then-MILP, the GA determines the number and values of time points, which are then passed to MILP for further optimization. Ref. Fig.: FIG. 1
No. of Pages : 200 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/09/2026

(21) Application No.202621111404 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIMODAL PHYSIOLOGICAL AND BEHAVIOURAL SIGNAL FUSION SYSTEM FOR REAL-TIME ANXIETY CLASSIFICATION

(51) International classification	:A61B 5/00, A61B 5/16, A61B 5/0205, A61B 5/1455, A61B 5/0476
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rajarambapu Institute of Technology
Address of Applicant :Rajaramnagar, Islampur, Tal- Walwa, Dist. Sangli, Maharashtra 415414 Maharashtra India
(72)**Name of Inventor :**
1)Ms. Priyanka N. Jadhav
2)Ms. Gautami G. Shingan
3)Mr. Amol M. Jagtap
4)Mr. Prasad Koli

(57) Abstract :
Multimodal Physiological and Behavioural Signal Fusion System for Real-Time Anxiety Classification This invention describes a multimodal physiological and behavioural signal-based system and method for real-time anxiety detection and classification. The system acquires 5-channel electroencephalography signals, facial expression video, galvanic skin response, and blood oxygen saturation data and synchronously processes the acquired information to obtain modality-specific neural, behavioural, and autonomic physiological features. A Reliability-Guided Adaptive Fusion framework determines a reliability value for each sensing modality based on a Modality Reliability Index and temporal stability of the corresponding measurements. The reliability values are normalized to generate adaptive modality-specific fusion weights, thereby reducing the contribution of noisy, unstable, obstructed, or degraded modalities during feature fusion. A unified multimodal representation generated by the adaptive fusion framework is processed using a machine-learning classifier to identify anxiety-related conditions including Generalized Anxiety Disorder, Social Anxiety Disorder, Panic Disorder, and Obsessive-Compulsive Disorder. The system enables objective, computationally efficient, non-invasive, and continuous anxiety assessment.

No. of Pages : 28 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202621114719 A
(19) INDIA
(22) Date of filing of Application :24/09/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : CROP MONITORING AND MANAGEMENT SYSTEM

(51) International classification	:A01G 25/16, A01G 25/09, G06Q 50/02, G01N 33/24, A01C 23/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)LNCT University
Address of Applicant :JK Town Sarvadharam C Sector, Kolar Road, Bhopal, Madhya Pradesh India -462024. Bhopal Madhya Pradesh India
(72)**Name of Inventor :**
1)Dr. Ankit Malviya
2)Dr. Hradesh Shivhare
3)Dr. Amol Barve

(57) Abstract :
A crop monitoring and management system, comprising a plurality of bodies 101, a user interface for enabling a farmer to input essential crop-specific information and personal preferences such as crop type, growth stage, irrigation schedule and target yield, a motorized slider 201 to translate a L-shaped articulated bar 202, a sensing unit 203 to determine soil parameters, a motorized circular sliding unit 204 installed along the middle section of the bodies 101 to translate and position a monitoring module to determine crop growth assessment, health monitoring, and climate analysis, a climate-adaptive control module integrated with the bodies 101 comprises an anemometer, a plurality of pyranometer gauges, rain sensors, barometric pressure sensors and works in sync with the sensing unit 203 and the control unit sent data to the computing unit to facilitate the farmer to manage fertilizer application, irrigation scheduling and spraying activities based on comprehensive environmental monitoring.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114745 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FUSION-SENSOR PREDICTIVE MANHOLE AND DRAINAGE OVERFLOW MONITORING SYSTEM AND METHOD

(51) International classification	:G01W 1/14, G01D 21/02, G01F 1/84, G01F 23/296, G01W 1/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Jaydeep Sanjay Lokhande
Address of Applicant :Pune Maharashtra India
2)Suraj More
3)Bharati Dixit
4)Sheetal Kulkarni
5)Shamla Mantri
6)MangalPatil
(72)**Name of Inventor :**
1)Suraj More
2)Jaydeep Sanjay Lokhande
3)Bharati Dixit
4)Sheetal Kulkarni
5)Shamla Mantri
6)MangalPatil

(57) Abstract :
A low-cost, edge-deployable multi-sensor system and method predict, ahead of a selectable time horizon, the probability of overflow at an individual manhole or drainage structure. Sensors including a water-level sensor and one or more of a rainfall, flow, environmental, gas, or vibration sensor are sampled on a shared clock and reduced to a fused, temporally-engineered feature vector. When a wide-area link is available, a cloud-hosted model computes the overflow probability from the fused feature vector; when the link is unavailable, a co-resident on-device fallback estimator computes the same quantity from the same feature vector, allowing uninterrupted operation. A compact fixed-point payload transmits the predicted probability together with a per-sensor health bitmask and a flag identifying which inference path produced the estimate, enabling a downstream system to distinguish genuine predicted events from degraded-sensor artifacts or fallback-mode estimates without added communication overhead. A related bench-scale apparatus safely generates labeled training data using a transparent tank, independent controlled-inflow and simulated-rainfall pathways, an adjustable outflow obstruction, and an independent ground-truth reference, avoiding exposure of personnel or electronics to a live sewage environment.

No. of Pages : 16 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :25/09/2026

(21) Application No.202621114841 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ROBOTIC SURGICAL SYSTEM, METHOD, AND MARKER ASSEMBLY FOR TRACKING BONE POSITION

(51) International classification	:A61B34/30, A61B34/20, A61B90/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MERAI NEWAGE PRIVATE LIMITED
Address of Applicant :Survey No. 135/139, Bilakhia House, Muktanand Marg, Chala, Vapi - 396191, Gujarat, India Vapi Gujarat India
(72)**Name of Inventor :**
1)SAHANE, Pradip Vishnu
2)DESAI, Indrajit Anandrao

(57) Abstract :
A robotic surgical system (100) includes a robot cart (102) having a robotic arm (106) and a robotic controller (206) for controlling the robotic arm, a vision cart (104) having a surgical display console (110), an optical tracking system (116), and a hardware processor (202), and a marker assembly (120) attachable to a bone (122), including at least three marker balls trackable by the optical tracking system. The hardware processor (202) receives position data, determines an initial reference geometry of the marker balls, receives a selection of a first marker ball as an origin marker and a second marker ball as a direction reference marker, and determines a tool frame toward the direction reference marker. The hardware processor (202) tracks the bone (122) using the tool frame, to generate instructions for the robotic controller (206) to move the robotic arm (106) relative to the bone. FIG. 2

No. of Pages : 58 No. of Claims : 17

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :25/09/2026
(21) Application No.202621114915 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : “Smart and Portable Hexa Bottle with Grinder”

(51) International classification	:A47J 31/42, A47J 31/40, A47J 31/44, A47J 31/46, A47J 31/00	(71)**Name of Applicant :** **1)Vrushali Parag Bhamare** Address of Applicant :101-D, Rajkalp Complex, Sunil Nagar, Dombivli East 421201 Maharashtra India (72)**Name of Inventor :** **1)Vrushali Parag Bhamare** **2)Ankita Harishchandra Kor** **3)Vishnu Amlesh Kanojiya** **4)Parth Sanjay Kochale** **5)Harshita Dada Gorad** **6)Purva Vedkumar Bhosale**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
Abstract: The present invention relates to a portable multi-compartment drinking and beverage preparation system comprising a substantially hexagonal main bottle body (101) having six separate liquid compartments (102) arranged around a central chamber (104). Separating walls (103) maintain separation between the respective liquid compartments and the central chamber. One or more selected liquids or ingredients can be selectively transferred from the liquid compartments to the central chamber for preparation. A detachable electrically operated grinder/mixing assembly comprising a grinder body (105), rotating shaft, blade assembly (106) and coupling arrangement (107) is selectively coupled to the central chamber for mixing selected liquids and/or grinding suitable ingredients. The grinder/mixing assembly is powered by a rechargeable electrical power arrangement including an ON/OFF switch (111) and charging interface (112). The bottle further comprises a removable cup-like cap (108) configured to function as a drinking or serving vessel when detached, and a bottom/base assembly (110) accommodating associated electrical components. The system thereby provides an integrated portable arrangement for separately storing, selectively preparing, mixing or grinding and consuming selected liquids or ingredients

No. of Pages : 30 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :25/09/2026

(21) Application No.202621115061 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PORTABLE GOLD PURITY ESTIMATION DEVICE

(51) International classification	:G06N 20/00, G01N 27/26, G01N 27/416, G01N 27/49, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur
Address of Applicant :Dumna Airport Road, P. O. Khamaria, Jabalpur-482005, Madhya Pradesh, India Jabalpur Madhya Pradesh India
(72)**Name of Inventor :**
1)Tejas P Biradar
2)Budida Ajay
3)Battula Venkata Sai Mallik Jay Aditya
4)Akash Kesav Shanmugaraj
5)Shreya Bhondwe
6)Giddaluri Suseel Kumar
7)Dr. K. Ponappa

(57) Abstract :
A portable gold purity estimation device (100) to determine purity and material characteristics of a test sample using combined electrochemical and density-based analysis. The device (100) comprises an electrochemical sensing module (102) configured to acquire a time-series electrochemical signature of test sample and a density measurement module (114) configured to obtain weight measurements in air and when submerged in water for determining a density value. An analog-to-digital conversion module (124) and a microcontroller (126) acquire and process measurement data. A signal-processing and feature-extraction module (128) extracts features, and a machine-learning classification module (130) classifies test sample into a material class. A decision-fusion module (132) combines classification with density value to cross-validate measurements and generate a counterfeit or anomalous sample indication when measurements disagree. A wireless communication module (134) transmits a material classification, purity estimate, density value and confidence score to a companion mobile application (136).

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :25/09/2026

(21) Application No.202621115089 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR REAL-TIME CEREBRAL PERFUSION MONITORING AND ISCHEMIC-RISK PREDICTION DURING NEUROSURGERY

(51) International classification	:A61B 5/00, A61B 5/026, A61B 5/02, G16H 50/30, A61B 5/0295
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Vishal Kishanbhai Chavda
Address of Applicant :37 Nilkamal Tenament, Opposite Galaxy Cinema, Behind Surbhi Hotel, NH-8, Naroda, Ahmedabad-382330, India Gujarat India
2)Chakravarthi Avula
3)Umamaheshwar Kopperi
4)Monika Garg
(72)**Name of Inventor :**
1)Dr. Vishal Kishanbhai Chavda
2)Chakravarthi Avula
3)Umamaheshwar Kopperi
4)Monika Garg

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR REAL-TIME CEREBRAL PERFUSION MONITORING AND ISCHEMIC-RISK PREDICTION DURING NEUROSURGERY A system and method for real-time cerebral-perfusion monitoring and ischemic-risk prediction during neurosurgery are present. The system receives cerebral blood-flow, tissue-oxygenation, electrophysiological, systemic hemodynamic, and surgical-event data and temporally synchronizes the data. A regional mapping processor associates the synchronized data with cerebral vascular territories and establishes patient-specific perfusion-reserve profiles. A differential-response processor compares a post-event response of an affected territory with its perfusion-reserve profile and a contemporaneous response of an unaffected territory to distinguish a surgically induced regional perfusion deficit from a systemic physiological variation. An ischemic-risk prediction engine determines a regional ischemic-risk value and a remaining reversible-ischemia intervention window based on a magnitude, rate, and persistence of the deficit. A guidance interface generates a graded output identifying the affected territory and intervention window. A response-evaluation processor evaluates post-intervention data against an expected perfusion-recovery trajectory and updates the patient-specific profile, thereby providing closed-loop neurosurgical decision support. Figure 3

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :25/09/2026
(21) Application No.202621115098 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : LEGAL COMPLIANCE AND CASE ANALYSIS DEVICE

(51) International classification	:G06Q 50/18, G06Q 10/06, G06F 17/30, G06Q 10/10, G06F 16/2458
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sanjeev Agrawal Global Educational University, Bhopal
Address of Applicant :Sahara Bypass Road, Katara Hills Extension, Bhopal, Madhya Pradesh 462022 Madhya Pradesh India
2)Dr. Prashant Jain
(72)**Name of Inventor :**
1)Dr. Sunil Kumar Pandey
2)Mrs Deepali Dilip Bhole

(57) Abstract :
ABSTRACT LEGAL COMPLIANCE AND CASE ANALYSIS DEVICE A legal compliance and case analysis device is disclosed. The device receives authoritative legal sources and organizational evidence or case materials, validates source authority, jurisdiction, effective period and version, and extracts normalized legal requirements, issues, holdings, exceptions and factual propositions. A mapping engine relates compliance requirements to supporting evidence and case issues to supporting, adverse or distinguishing authorities. A conflict, gap and drift detector identifies missing evidence, inconsistent records, stale or superseded authorities, unsupported propositions and legally material factual differences. A confidence and support engine generates a verified analytical state only when configured source-validity and evidence-support conditions are satisfied; otherwise, a review-required, additional-evidence or remediation state is generated. Updated authorities, evidence or remediation records can trigger selective re-analysis, while an audit logger preserves source versions, mappings, confidence factors and state transitions. The device thereby provides traceable, version-aware and confidence-qualified computer-assisted legal analysis.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :25/09/2026
(21) Application No.202621115099 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR AI-BASED CYBERSECURITY POLICY ANALYSIS AND COMPLIANCE

(51) International classification	:H04L 29/06, H04L 9/40, G06F 21/57, G06N 20/00, G06F 21/55
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sanjeev Agrawal Global Educational University, Bhopal
Address of Applicant :Sahara Bypass Road, Katara Hills Extension, Bhopal, Madhya Pradesh 462022 Madhya Pradesh India
2)Dr. Prashant Jain
(72)**Name of Inventor :**
1)Dr. Sunil Kumar Pandey
2)Soniya Dhantole

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR AI-BASED CYBERSECURITY POLICY ANALYSIS AND COMPLIANCE An AI-based cybersecurity policy analysis and compliance system is disclosed. Cybersecurity policy sources are transformed into normalized control objects retaining source-clause lineage and stored in a policy graph representing applicability, dependency, exception and version relationships. Technical evidence is acquired from authorized configuration and security sources and qualified according to provenance, freshness, completeness, scope and integrity before being mapped to corresponding controls. A compliance engine determines a control-satisfaction state together with a compliance confidence and withholds a verified-compliant state when evidence quality or confidence is insufficient. The system further detects policy conflict, coverage gaps and configuration drift, selectively re-evaluates controls affected by changes, prioritizes remediation, and verifies post-remediation evidence before closing a gap. Policy versions, evidence references, decisions and remediation events are stored in an integrity-protected audit lineage, thereby providing traceable, evidence-qualified and continuously revalidated cybersecurity compliance assessment.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :25/09/2026

(21) Application No.202621115100 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE EDGE-AI DEVICE FOR OFFLINE MULTI-DISEASE SCREENING AND REFERRAL

(51) International classification	:G16H 50/20, A61B 3/14, G16H 50/30, A61B 5/00, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sanjeev Agrawal Global Educational University, Bhopal
Address of Applicant :Sahara Bypass Road, Katara Hills Extension, Bhopal, Madhya Pradesh 462022 Madhya Pradesh India
2)Dr. Prashant Jain
(72)**Name of Inventor :**
1)Dr. Kiran Pachlasiya

(57) Abstract :
ABSTRACT PORTABLE EDGE-AI DEVICE FOR OFFLINE MULTI-DISEASE SCREENING AND REFERRAL A portable edge-AI device for offline multi-disease screening and referral is disclosed. The device acquires a conjunctival image, a retinal fundus image, and an electrocardiogram waveform and processes the inputs locally using respective disease-classification models to generate anaemia, diabetic-retinopathy, and cardiac-arrhythmia risk indicators. Associated confidence values are evaluated by a decision-support module that performs confidence-responsive fusion and generates a single urgency-stratified referral recommendation. The screening and referral workflow executes on an edge-computing unit having an on-device AI hardware accelerator without requiring transmission of physiological data to a remote server or continuous network connectivity. The referral recommendation is communicated through icon-based visual elements and synchronized regional-language audio for use by non-specialist healthcare workers. A de-identified screening record may be stored locally and opportunistically synchronized when connectivity becomes available, and a modular interface supports additional screening modalities.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :25/09/2026

(21) Application No.202621115208 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FLUSH-ACTIVATED TOILET-BOWL CLEANING SYSTEM WITH DISPENSER TRAY, LEVER-ACTUATED VALVE, SPRING RETURN, O-RING AND CONTROLLED DOSING

(51) International classification	:E03D 9/03, E03D 9/02, E03D 5/10, E03D 5/09, E03D 5/01	(71)**Name of Applicant :** **1)Kalpesh Manseta** Address of Applicant :S/O: Vasantrai Manseta,103, Kailash Appartment, Near lskon Temple, Harinagar Gotri Road, Vadodara, Gotri, Gujarat,390021 Gujarat India **2)Pahal Kalpesh Manseta**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Kalpesh Manseta**
(33) Name of priority country	:NA	**2)Pahal Kalpesh Manseta**
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Flush-Activated Toilet-Bowl Cleaning System with Dispenser Tray, Lever-Actuated Valve, Spring Return, O-Ring and Controlled Dosing. This invention describes a flush-activated toilet-bowl cleaning system comprises a chemical-storage dispenser tray mountable on an H-tray of an existing toilet flush tank, a valve positioned within a hole of the dispenser tray, an O-ring arranged around the hole, a spring positioned at a bottom portion of the valve, and a washer and screw retaining the spring relative to the valve. The valve is normally maintained in a closed position by the spring and engages the O-ring to restrict unintended chemical leakage. Operation of a side lever handle causes an inner lever stick of the flushing mechanism to push the valve upward, thereby opening the tray hole and releasing a predetermined quantity of cleaning chemical into water contained within the flush tank. Upon completion of the actuation, the spring returns the valve to the closed position. The chemical mixes with the flush water and is delivered into the toilet bowl during flushing without requiring electrical power.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :25/09/2026
(21) Application No.202621115288 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD FOR OPERATING HYBRID ENERGY STORAGE SYSTEM IN ELECTRIC VEHICLE

(51) International classification	:B60L50/40, B60L50/50, B60W10/26, B60W20/13, H02J7/34, B60L58/12	(71)**Name of Applicant :** **1)VIBHAVARI R. JADHAV** Address of Applicant :At Malewadi Post Tandulwadi Tal Walwa District Sangli, Maharashta, India, 415411 415411 Maharashtra India **2)DR. VIJAY P. MOHALE** **3)DR. SURENDRA RATHOD**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)VIBHAVARI R. JADHAV**
(33) Name of priority country	:NA	**2)DR. VIJAY P. MOHALE**
(86) International Application No Filing Date	: :01/01/1900	**3)DR. SURENDRA RATHOD**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT METHOD FOR OPERATING HYBRID ENERGY STORAGE SYSTEM IN ELECTRIC VEHICLE The present disclosure relates to a method for operating a Hybrid Energy Storage System (HESS) in an Electric Vehicle (EV) (110). The method comprises simulating the HESS based on a plurality of operating parameters corresponding to a plurality of environmental factors associated with an atmospheric condition. The method further comprises monitoring a state of charge (SoC) and a state of health (SoH) of a battery during simulation of the HESS. The method further comprises determining a power allocation through a machine learning between the battery and a super-capacitor based on the plurality of operating parameters and the plurality of environmental factors. The method further comprises controlling a power flow between the battery and the super-capacitor based on the power allocation and the plurality of environmental factors.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :25/09/2026
(21) Application No.202621115289 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVICE FOR MONITORING AND CONTROLLING FIELD EQUIPMENT AND A METHOD THEREOF

(51) International classification	:G05B 19/042, G05B 23/02, G05B 15/02, G05B 19/418, G05B 19/048
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHERISE (INDIA) PRIVATE LIMITED
Address of Applicant :201, PARIJAT APPARTMENTS, NS ROAD 10, JUHU, VILE PARLE (WEST), MUMBAI 400049, MAHARASHTRA INDIA MUMBAI Maharashtra India
(72)**Name of Inventor :**
1)PARIMAL SHAH

(57) Abstract :
Described herein is a device (102) for monitoring and controlling field equipment (104). The device (102) comprises an input interface (102-1) operatively connected to one or more sensing or signalling components (104-1), an output interface (102-2) operatively connected to one or more controllable components (104-2), a first processing unit (102-3), and a second processing unit (102-4). The first processing unit (102-3) receives one or more physical input signals, maps the physical input signals to a corresponding named operational event based on a configurable event taxonomy, generates an event record, and controls the one or more controllable components (104-2). The second processing unit (102-4) communicates with a remote computing system (106) through a communication network (108), associates time information with the event record, stores the event record in a non-volatile memory (102-5) during network unavailability, transmits stored event records upon restoration of connectivity, and communicates received commands to the first processing unit (102-3) for execution. FIG. 1A will be the reference figure

No. of Pages : 40 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/09/2026
(21) Application No.202621115330 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-INTERROGATING ADAPTIVE SMART CHAIR WITH DYNAMIC MECHANICAL CONTACT TOPOLOGY INTELLIGENCE

(51) International classification	:A47C 7/62, A47C 31/12, A61B 5/00, A61B 5/11, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rajarambapu Institute of Technology
Address of Applicant :Rajaramnagar, Islampur, Tal- Walwa, Dist. Sangli, Maharashtra 415414 Maharashtra India
(72)**Name of Inventor :**
1)Jayashree Sudhir Awati
2)Mohammadali Wahid Ibushe
3)Ankita Hanmant Jadhav
4)Shreyas Sudhir Awati

(57) Abstract :
Self-Interrogating Adaptive Smart Chair with Dynamic Mechanical Contact Topology Intelligence This invention describes an intelligent adaptive chair configured to actively determine mechanical interaction between an occupant and the chair through controlled structural interrogation. The chair periodically introduces low-amplitude mechanical excitation into selected structural regions and acquires distributed mechanical responses including amplitude, phase, resonance, damping, propagation delay and spatial correlation. An edge-based inference system generates a dynamic mechanical signature and reconstructs a changing contact-topology representation indicative of mechanical coupling between occupant-contact regions of the chair. A user-specific mechanical operating envelope is progressively developed to distinguish normal movement from sustained or abnormal mechanical states. The same structural interrogation architecture is further usable for detecting changes in structural integrity, sensor condition and actuator performance of the chair. Based on detected ergonomic deviations, selected support regions may autonomously vary stiffness, compliance or support configuration. Subsequent mechanical interrogation verifies whether an adjustment improved the detected mechanical state, thereby providing a closed-loop adaptive ergonomic seating system.

No. of Pages : 30 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/09/2026

(21) Application No.202621111585 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MULTI-SENSING ENDOSCOPIC SYSTEM

(51) International classification	:A61B1/00, A61B5/00, A61B1/273, A61B5/01, A61B5/053, A61B5/03
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH BHOPAL
Address of Applicant :Bhopal Bypass Road, Bhauri, Bhopal – 462066, Madhya Pradesh, India Bhopal Madhya Pradesh India
(72)**Name of Inventor :**
1)Mitradip Bhattacharjee
2)Dhanranjan Kumar
3)Vedika Jakate

(57) Abstract :
The present invention discloses a multi-sensing endoscopic system (101) for acquiring physiological information from a gastrointestinal tract. The system (101) comprises an elongated endoscopic body integrated with a pressure sensor (102), a temperature sensor (103), an optical signal generator (104), an optical signal detector (105), a circuitry peripheral (106), a data transmission module, and a processing unit. The pressure sensor (102) generates electrical signals corresponding to pressure variations, the temperature sensor (103) generates electrical signals corresponding to temperature variations, and the optical signal generator (104) and the optical signal detector (105) cooperate to generate electrical signals corresponding to reflected optical signals. The circuitry peripheral (106) conditions the generated electrical signals for wireless transmission through a wireless communication protocol. The processing unit processes the received pressure, temperature, and optical information and further employs a learning-based anomaly detection model to generate physiological monitoring information associated with the gastrointestinal tract.

No. of Pages : 36 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/09/2026

(21) Application No.202621111753 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Spatially Convergent Forced-Air Thermal Treatment System for Extrusion-Based 3D Food Printing

(51) International classification	:A23P 30/20, A23P 1/12, A21C 11/16, A23P 30/25, A23P 1/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur
Address of Applicant :Dumna Airport Road, P. O. Khamaria, Jabalpur-482005, Madhya Pradesh, India Jabalpur Madhya Pradesh India
(72)**Name of Inventor :**
1)Rahul Soni
2)Ponappa K.
3)Puneet Tandon

(57) Abstract :
A spatially convergent forced-air thermal treatment system for extrusion-based 3D food printing, comprising an open-frame three-dimensional food printing system (102) having a build platform (104) and a movable food extrusion assembly (106) with a syringe-based extrusion mechanism (108) and an extrusion nozzle (110), an externally positioned stationary forced-air heating device (112) having a heating arrangement (114) and integrated fan/blower (116), an outlet connection (118) having an extender/distribution arrangement (120), first and second flexible heated-air transfer pipes (122, 124), and a heated-air distribution module (126) having first and second inlet ports (128, 130) and an internal airflow distribution arrangement (132). Four spatially separated hot-air-discharge vents (134A-134D) direct heated-air streams toward a localized thermal treatment region (138) associated with a deposited edible food structure (140). The heating device (112) remains stationary while module (126) and vents (134A-134D) move with extrusion assembly (106), providing localized forced-convective thermal treatment during layer-by-layer food printing.

No. of Pages : 27 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/09/2026

(21) Application No.202621115438 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE REMOTELY OPERATED MULTIFUNCTION INDUSTRIAL ROBOTIC PLATFORM WITH INTERCHANGEABLE TOOL MODULES

(51) International classification	:B25J 9/16, B25J 13/08, B25J 9/00, B25J 19/02, G05B 19/418
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DKTE Society's Textile and Engineering Institute
Address of Applicant :Rajwada, Ichalkaranji, Maharashtra 416115 Maharashtra India
2)Mr. A.V.Sutar
3)Mr. Meet Nagesh Kalal
4)Mr. Mohammad Zaid Sameer Pailwan
5)Mr. Tejam Bhagat Sing
6)Miss. Janhavi Shrikant Sawant
7)Mr. Shubham Prakash Kamble
8)Dr. V. V. Kheradkar

(72)**Name of Inventor :**
1)Mr. A.V.Sutar
2)Mr. Meet Nagesh Kalal
3)Mr. Mohammad Zaid Sameer Pailwan
4)Mr. Tejam Bhagat Sing
5)Miss. Janhavi Shrikant Sawant
6)Mr. Shubham Prakash Kamble
7)Dr. V. V. Kheradkar

(57) Abstract :
Portable Remotely Operated Multifunction Industrial Robotic Platform with Interchangeable Tool Modules This invention describes a portable remotely operated multifunction industrial robotic platform configured to perform machining, fastening, inspection, material handling, maintenance, cleaning, measurement, and related industrial operations using interchangeable tool modules. The platform includes an omnidirectional mobile chassis, independently controlled drive motors, a rechargeable power supply, a programmable controller, wireless communication, autonomous navigation assistance, and obstacle and collision sensing. An adaptive stabilization arrangement automatically levels and stabilizes the platform before performance of an industrial operation. A precision vertical motion assembly provides controlled tool positioning using a self-locking lead screw and controlled actuator. A universal modular tool interface enables rapid mechanical and electrical coupling of different operational modules and automatically identifies an installed module. Programmable depth control, force sensing, vibration monitoring, machine-health diagnostics, overload protection, emergency stopping, data storage, and remote monitoring facilitate safe, accurate, and repeatable operation across diverse industrial environments.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/09/2026

(21) Application No.202621115493 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-SOURCE PREDICTIVE VECTOR-OUTBREAK INTELLIGENCE SYSTEM USING WEATHER, BREEDING-SITE HISTORY, IMAGERY, DISEASE DATA AND TREATMENT HISTORY

(51) International classification	:G06N 20/00, G06Q 50/02, G01W 1/10, G06N 5/04, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sachiin Shettyy
Address of Applicant :Flat No. 2903, ONE ICC, Island City Centre, G. D Ambedkar Road, Dadar East, Mumbai 400014 Mumbai Maharashtra India
(72)**Name of Inventor :**
1)Sachiin Shettyy

(57) Abstract :
ABSTRACT MULTI-SOURCE PREDICTIVE VECTOR-OUTBREAK INTELLIGENCE SYSTEM USING WEATHER, BREEDING-SITE HISTORY, IMAGERY, DISEASE DATA AND TREATMENT HISTORY A multi-source predictive vector-outbreak intelligence system integrates weather data, historical breeding-site information, imagery, disease-event information and vector-control treatment history. A computing system normalizes the heterogeneous data into common spatial and temporal representations and generates predictive features associated with environmental suitability, breeding-site persistence, imagery-derived conditions, disease events and previous treatment outcomes. A predictive engine determines spatial vector-risk values and identifies emerging or persistent risk clusters. A response-zone engine generates geographical treatment zones and machine-readable vector-control instructions. The instructions may be transmitted to a vector-control apparatus having positioning and treatment-control functionality. Actual treatment information and post-treatment observations are received and used to determine treatment effectiveness and update subsequent vector-risk values. The system thereby provides an adaptive closed-loop technical architecture for predictive vector surveillance, prioritisation, treatment coordination and treatment-effectiveness monitoring.

No. of Pages : 36 No. of Claims : 48

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/09/2026
(21) Application No.202621115494 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DISEASE-SURVEILLANCE-TRIGGERED VECTOR-RESPONSE SYSTEM USING PATIENT EXPOSURE-LOCATION INFORMATION AND CLUSTER ANALYSIS

(51) International classification	:G16H 40/67, G16H 50/20, H04W 4/029, G16H 10/60, G16H 50/80
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sachiin Shettyy
Address of Applicant :Flat No. 2903, ONE ICC, Island City Centre, G. D Ambedkar Road, Dadar East, Mumbai 400014 Mumbai Maharashtra India
(72)**Name of Inventor :**
1)Sachiin Shettyy

(57) Abstract :
ABSTRACT DISEASE-SURVEILLANCE-TRIGGERED VECTOR-RESPONSE SYSTEM USING PATIENT EXPOSURE-LOCATION INFORMATION AND CLUSTER ANALYSIS The present invention relates to a disease-surveillance-triggered vector-response system configured to convert disease-related exposure-location information into a geographically targeted vector-control response. The system receives disease-event records comprising exposure locations and temporal information, converts the exposure locations into geographical exposure coordinates, assigns exposure-location confidence values and performs spatial-temporal processing of multiple exposure events to identify a disease-related geographical cluster. A cluster confidence value is determined and compared with a predetermined response-trigger condition. Upon satisfaction of the response-trigger condition, a response-zone generation module generates a geographically bounded vector-response zone. A response-control module generates a machine-readable vector-control instruction comprising the response zone and at least one treatment parameter and communicates the instruction to a vector-control apparatus. The vector-control apparatus determines its geographical position and enables treatment within the authorised response zone. Treatment execution is verified using geographical position, machine operating information and treatment-flow information. Post-treatment information may be used to generate subsequent response instructions. The invention thereby provides a technical closed-loop architecture linking disease surveillance, exposure-location processing, spatial-temporal cluster analysis, geographical response-zone generation, automated vector-control operation and treatment verification.

No. of Pages : 31 No. of Claims : 48

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/09/2026
(21) Application No.202621112321 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : "A STABLE ORAL OLEOGEL CAPSULE FORMULATION COMPRISING TICAGRELOR-LOADED OLEOGEL AND PREPARATION METHOD THEREOF"

(51) International classification	:A61K 9/48, A61K 31/519, A61P 7/02, A61K 47/38, A61K 9/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bharati Vidyapeeth (Deemed to be University), Poona College of Pharmacy, Pune
Address of Applicant :Poona College of Pharmacy, Erandwane, Bharati Vidyapeeth (Deemed to be University), Pune Pune Maharashtra India
(72)**Name of Inventor :**
1)Dr. Nilesh Deshmukh
2)Dr. Atmaram Pawar
3)Dr. Ravindra Kamble
4)Ms. Aayushi Yelgurwar

(57) Abstract :
ABSTRACT "A STABLE ORAL OLEOGEL CAPSULE FORMULATION COMPRISING TICAGRELOR-LOADED OLEOGEL AND PREPARATION METHOD THEREOF" The present invention generally relates to the field of pharmaceutical formulations, and more particularly to a stable oral oleogel capsule formulation comprising ticagrelor-loaded oleogel, along with pharmaceutically acceptable inert excipients for oral delivery and sustained release of ticagrelor, useful in the management of thrombotic cardiovascular conditions. The invention provides a gastro-retentive oral delivery system capable of exhibiting floating behaviour in the gastric environment and providing prolonged and controlled release of ticagrelor. The invention further provides a method for preparing the same.

No. of Pages : 29 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621115721 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ARTIFICIAL INTELLIGENCE BASED SYSTEM AND METHOD FOR AUTOMATED HUMAN BLOOD GROUP DETECTION

(51) International classification	:G06T 7/00, G01N 33/80, G06V 10/764, G06V 10/774, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Munnuru Varshitha
2)Kummari Harini
3)Gundu Chandrika
4)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT AN ARTIFICIAL INTELLIGENCE BASED SYSTEM AND METHOD FOR AUTOMATED HUMAN BLOOD GROUP DETECTION An artificial intelligence based system (100) and method (200) for automated human blood group detection are disclosed. A test card assembly (110) carries reaction zones (114) charged with anti-A, anti-B and anti-D reagents and a control, together with a colour reference marker (116). An image acquisition module (120) captures an image of the card, which a preprocessing module (130) normalises against the marker (116) and rectifies. A deep learning inference module (140) emits, in a single forward pass, a bounding box and agglutination class for each reaction zone (114) and, independently, a whole-card group label. A decision module (150) maps the zone-wise classes to an ABO-Rh designation, and reports the designation only when it is concordant with the whole-card label and a calibrated aggregate confidence exceeds a threshold, otherwise raising a re-test flag. [to be published with fig.1]

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621115722 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI GOVERNANCE PROXY SYSTEM FOR REAL-TIME SENSITIVE DATA PROTECTION IN LLM INTERACTIONS

(51) International classification	:G06F 21/62, G06N 20/00, G06F 21/55, G06N 5/02, H04L 51/046	(71)**Name of Applicant :** **1)Symbiosis International (Deemed University)** Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India (72)**Name of Inventor :** **1)Mr. Harshit Divekar** **2)Ms. Lakshmi Srujana** **3)Mr. Raj Aryan** **4)Mr. Sai Prashanth Mallellu**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AI GOVERNANCE PROXY SYSTEM FOR REAL-TIME SENSITIVE DATA PROTECTION IN LLM INTERACTIONS An AI governance proxy system (100) for real-time sensitive data protection in large language model interactions comprising a frontend portal module (110) with a chat console interface (112) and a CISO dashboard interface (114); a hybrid detection engine (130) combining a regex pattern detector (132) for structured sensitive data with a named entity recognition detector (134) for contextual PII; a policy engine module (140) with department-scoped governance rules and a stricter-wins conflict resolution unit (146); a risk scoring module (150) computing a composite score from PII severity, policy violation severity, prompt context risk, department risk profile, and target model risk; a decision engine (160) selecting APPROVE, REDACT, BLOCK, or ESCALATE enforcement actions; an LLM router module (170) for multi-provider routing; and an audit and receipt module (180) generating tamper-evident governance receipts for regulatory compliance. The system implements preventive governance by executing all enforcement decisions before prompt data reaches external LLM providers. [to be published with fig.1]

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/09/2026

(21) Application No.202621115495 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-ENABLED MULTI-FACTOR WORKFORCE AUTHORISATION AND MACHINE ENABLEMENT SYSTEM

(51) International classification	:G07C 9/00, G06K 9/00, G06F 21/32, G05B 19/042, G05B 19/418
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sachiin Shettyy
Address of Applicant :Flat No. 2903, ONE ICC, Island City Centre, G. D Ambedkar Road, Dadar East, Mumbai 400014 Mumbai Maharashtra India
(72)**Name of Inventor :**
1)Sachiin Shettyy

(57) Abstract :
ABSTRACT AI-BASED MULTI-FACTOR WORKFORCE AUTHORISATION AND MACHINE ENABLEMENT SYSTEM The invention relates to an AI-based multi-factor workforce authorisation and machine enablement system for controlling a field machine. A workforce-authorisation terminal acquires facial information, liveness information, fingerprint information, breath-analysis information, behavioural or gait information, geographical location and time associated with a worker-verification event. A processor-based authorisation engine correlates the facial and fingerprint information with an authorised worker identity and determines an authorisation state based on predetermined mandatory conditions. Upon successful verification, the authorisation engine generates an electronically authenticated machine-enable instruction communicated through a machine-control interface to a controller of the field machine, thereby enabling physical operation. Failure of a mandatory condition prevents valid machine enablement. The system may additionally verify protective equipment, identify and weigh material issued to the worker, record exposure hours and transmit authorisation and machine-operation records to a supervisory platform.

No. of Pages : 33 No. of Claims : 30

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/09/2026

(21) Application No.202621115496 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : UNIVERSAL INTELLIGENT RETROFIT SYSTEM AND METHOD FOR INSTRUMENTING, CHARACTERISING, AUTHENTICATING, MONITORING AND CONTROLLING HETEROGENEOUS SPRAYING EQUIPMENT

(51) International classification	:B05B 12/12, B05B 12/08, B05B 12/00, A01M 7/00, B05B 13/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sachiin Shettyy
Address of Applicant :Flat No. 2903, ONE ICC, Island City Centre, G. D Ambedkar Road, Dadar East, Mumbai 400014 Mumbai Maharashtra India
(72)**Name of Inventor :**
1)Sachiin Shettyy

(57) Abstract :
ABSTRACT UNIVERSAL INTELLIGENT RETROFIT SYSTEM AND METHOD FOR INSTRUMENTING, CHARACTERISING, AUTHENTICATING, MONITORING AND CONTROLLING HETEROGENEOUS SPRAYING EQUIPMENT The present invention provides a universal retrofit system for heterogeneous spraying and fogging equipment comprising a machine-independent control unit and a configurable machine-specific adapter layer. The system performs an installation-time characterisation of an individual spraying machine having unknown or undocumented discharge characteristics by measuring discharge at a plurality of operating conditions against a reference quantity. A machine-specific transfer function relating sensor output to calibrated discharge quantity and a tolerance band are derived from the characterisation measurements. The transfer function, tolerance band and capability declaration are stored as a digitally signed machine profile associated with a digital identity of the machine. During field operation, the controller applies the machine profile to sensor measurements to determine calibrated discharge volume and may regulate discharge where control capability exists. The controller further provides operator authentication, task validation, geographic verification, physical or procedural interlocking, fault detection, evidence capture, dual-path volume reconciliation, tamper detection and cryptographically protected event logging. Observation-only, no-flow-sensor, no-power and no-coverage modes permit operation with machines having limited capabilities. The machine profile is portable to a replacement controller, thereby separating machine calibration from controller hardware and enabling common fleet-level monitoring across heterogeneous spraying equipment.

No. of Pages : 29 No. of Claims : 40

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/09/2026
(21) Application No.202621115498 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ARTIFICIAL INTELLIGENCE ENABLED FINANCIAL TRANSACTION MANAGEMENT AND MONITORING DEVICE

(51) International classification	:G06Q 20/40, G06Q 40/04, G06N 20/00, G06Q 40/02, G06Q 40/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Pranati Rohit Tilak
Address of Applicant :Vice President and Dean, Tilak Maharashtra Vidyapeeth, Main Campus, Gultekadi, Pune, Maharashtra, India-411037 Pune Maharashtra India
2)Dr. Bhagyashree Deshpande
3)Dr. Manisha Shedge
4)Dr. Divya Lakhani
5)Dr. Manisha Jagtap
6)Dr. Rupa Rawal
7)Dr. Sneha Joshi
8)Dr. Jyoti Harchekar
9)Ms. Sayli Bapat
10)Ms. Prachi Joglekar
(72)**Name of Inventor :**
1)Dr. Pranati Rohit Tilak
2)Dr. Bhagyashree Deshpande
3)Dr. Manisha Shedge
4)Dr. Divya Lakhani
5)Dr. Manisha Jagtap
6)Dr. Rupa Rawal
7)Dr. Sneha Joshi
8)Dr. Jyoti Harchekar
9)Ms. Sayli Bapat
10)Ms. Prachi Joglekar

(57) Abstract :
ABSTRACT TITLE: ARTIFICIAL INTELLIGENCE ENABLED FINANCIAL TRANSACTION MANAGEMENT AND MONITORING DEVICE The present invention discloses an artificial intelligence enabled financial transaction management and monitoring device (100) comprising a data acquisition module (110) for collecting transaction data from heterogeneous sources, an AI-based transaction analysis engine (120) employing machine learning sub-module (122), deep learning sub-module (124), and natural language processing unit (126) for multi-dimensional risk assessment, a real-time monitoring module (130) for continuous transaction stream processing, a fraud detection and anomaly identification module (140) with pattern recognition unit (142) and anomaly scoring unit (144), a user authentication and verification module (150) with biometric and multi-factor authentication, an alert and notification module (160), a compliance and reporting module (170), a central processing unit (180), a database and storage module (190), a communication interface module (200), and an encryption and security module (210). The device employs an ensemble learning framework with adaptive feedback-driven retraining for improved fraud detection accuracy. [Figure 1]

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621115723 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED SMART TRAFFIC MANAGEMENT SYSTEM WITH DYNAMIC SIGNAL OPTIMIZATION

(51) International classification	:G08G 1/01, G08G 1/07, G08G 1/08, G08G 1/017, G08G 1/081
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Ms. Yagna Subhash. K
2)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT AI-BASED SMART TRAFFIC MANAGEMENT SYSTEM WITH DYNAMIC SIGNAL OPTIMIZATION The present invention discloses a system (100) for real-time monitoring and dynamic control of urban traffic. The system (100) comprises a traffic data acquisition unit (110) with surveillance cameras (112) and IoT sensors (114), an AI processing and analysis engine (120) with vehicle detection (122), traffic density estimation (124), and traffic flow analysis (126) modules, a dynamic traffic signal controller (130) with a signal timing optimization algorithm (132), a cloud server and storage unit (140), a congestion prediction module (150) using machine learning algorithms, an emergency vehicle prioritization module (160), an accident detection module (170), and a cloud-based monitoring dashboard (180). The system integrates computer vision, deep learning, IoT, and cloud computing to dynamically optimize signal timings based on real-time vehicle density, predict congestion, prioritize emergency vehicles, detect accidents, and provide remote traffic monitoring, thereby reducing waiting time, fuel consumption, and environmental pollution while improving urban transportation efficiency. [to be published with fig.1]

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621115724 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR REAL-TIME COLLABORATIVE CODE EDITING WITH SANDBOXED EXECUTION

(51) International classification	:G06F 21/53, G06F 9/455, G06F 8/41, G06F 9/54, G06F 8/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Kurikala Hasini
2)Mokshita VP
3)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR REAL-TIME COLLABORATIVE CODE EDITING WITH SANDBOXED EXECUTION The present invention discloses a system (100) and method for real-time collaborative code editing with integrated sandboxed execution. The system comprises a frontend module (110) with a browser-based code editor component (112) supporting multiple programming languages, a real-time synchronization module (120) employing a cloud-hosted database interface (122) for low-latency event-driven synchronization of code and user presence across connected clients with a local storage fallback module (124) for offline operation, and a sandboxed execution engine (130) comprising an API gateway (132) and a container orchestrator (134) that spawns short-lived, resource-isolated containers from language-specific container images (136) to safely execute user-submitted code. A resource isolation controller (138) enforces security constraints including disabled network access, capped memory, limited CPU, and read-only file systems on each container. The system enables secure, portable, and lightweight collaborative coding suitable for educational, workshop, and development use. [to be published with fig.1]

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621115725 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ON-DEVICE MOBILE DREAM ANALYSIS SYSTEM WITH MULTI-DIMENSIONAL EMOTION DETECTION AND ADAPTIVE PERSONALISATION

(51) International classification	:G10L 25/63, G06F 17/27, G06F 17/30, G06F 3/01, G10L 15/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Ms. Riya Shastri
2)Ms. Meenakshi Vedala
3)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT ON-DEVICE MOBILE DREAM ANALYSIS SYSTEM WITH MULTI-DIMENSIONAL EMOTION DETECTION AND ADAPTIVE PERSONALISATION An on-device mobile dream analysis system (100) integrating dual-input dream capture (110) supporting typed text and voice recording, a nine-dimensional emotion detection engine (120) employing a dream-specific lexicon (122) with negation window detection (124), intensifier scaling (126), bigram pattern matching (128), and valence-arousal mapping (129), a dream graph visualisation engine (130) rendering interactive force-directed scene graphs through a WebView canvas renderer (136), a psychological interpretation engine (140) performing automated symbol extraction (142), archetype detection (144), and theme classification (146) based on established psychological frameworks, a predictive modelling engine (150) implementing a first-order Markov chain for scene prediction from personal transition history, and an on-device personalisation engine (160) employing Bayesian co-occurrence tracking (162) and theme prior updating (164) to incrementally adapt analytical models to individual dream patterns. All core processing executes entirely on the mobile device without cloud dependency, ensuring privacy preservation of intimate dream content. [to be published with fig.1]

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/09/2026
(21) Application No.202621115505 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD AND SYSTEM FOR CONTROLLING GENERATION OF CRANIOFACIAL APPROXIMATION FROM CRANIOFACIAL SOURCE DATA

(51) International classification	:G06K 9/00, G16H 30/20, G16H 50/20, A61B 5/00, G16H 30/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DENKEN INTEGRATED TECHNOLOGIES PRIVATE LIMITED
Address of Applicant :Jai Vishwabharti Ho Soc, Colony At Ch. Sambhajinag, Aurangabad City Aurangabad Maharashtra India
(72)**Name of Inventor :**
1)AMRITA SANJEEV KIRTANE
2)JISHNU JITESH JAISWAL
3)BAGATE NITIN DATTATRYA
4)RITESH BANSILAL JAISWAL

(57) Abstract :
A method (800) and system (100) for controlling a craniofacial approximation from craniofacial source data includes receiving case input data comprising subject attributes and accessing a reconstruction repository (110) storing source-linked reconstruction records associating craniofacial landmarks with facial soft-tissue values and source-scope metadata. A branch-specific landmark support dataset is generated from a selected subset of the source-linked reconstruction records and landmark support states. For each facial feature family, a feature support state is determined, a bounded feature parameter is generated for a supported state or high-uncertainty state, and generation is suppressed for an abstention state. An evidence matrix (320) and a constrained render specification are generated. A generative craniofacial rendering model (224) generates craniofacial image variants (318). A provenance record is stored, a review state is received, and release of a selected candidate craniofacial image variant is controlled based on the review state and provenance record.

No. of Pages : 138 No. of Claims : 18

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/09/2026
(21) Application No.202621115509 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Multimodal Postpartum Uterine Recovery Monitoring System with Electrohysterography Belt and Blood-Loss Pad

(51) International classification	:A61B 5/00, A61B 5/024, A61B 5/0205, A61B 5/0245, A61B 5/03	(71)**Name of Applicant :** **1)PARUL UNIVERSITY (PARUL INSTITUTE OF NURSING)** Address of Applicant :PARUL INSTITUTE OF NURSING, PARUL UNIVERSITY, P.O. LIMDA TAL. WAGHODIA, DIST. VADODARA, GUJARAT, 391760, INDIA Vadodara Gujarat India (72)**Name of Inventor :** **1)DR. SEETA DEVI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT A multimodal postpartum uterine recovery monitoring system comprises an abdominal uterine sensing belt (100) with electrohysterography electrodes (102) for continuous uterine activity monitoring, a smart postpartum blood-loss monitoring pad (200) with a sensing layer (203) for real-time blood loss quantification, and a maternal physiological wristband (300) with a photoplethysmography sensor (302) for heart rate and heart rate variability monitoring. A central processing unit (400) synchronizes data from all components. A nurse/midwife clinical assessment input interface (500) receives uterine tone and fundal position entries. A multimodal analysis module (600) integrates all data streams. A Uterine Recovery Index module (700) calculates a composite postpartum recovery score. A trend-based comparison module (800) compares current measurements against stored baseline measurements. A safety/alert module (1000) generates early-warning alerts displayed on a nurse/midwife dashboard (900), enabling proactive clinical reassessment and timely intervention during the critical postpartum period.

No. of Pages : 56 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/09/2026
(21) Application No.202621115510 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTISENSOR PREDICTIVE SYSTEM FOR PERSONALIZED NON-PHARMACOLOGICAL LABOUR PAIN COPING SUPPORT

(51) International classification	:A61B 5/00, A61B 5/024, A61B 5/0205, A61B 5/11, A61B 5/0245
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PARUL UNIVERSITY (PARUL INSTITUTE OF NURSING)
Address of Applicant :PARUL INSTITUTE OF NURSING, PARUL UNIVERSITY, P.O. LIMDA TAL. WAGHODIA, DIST. VADODARA, GUJARAT, 391760, INDIA Vadodara Gujarat India
(72)**Name of Inventor :**
1)DR. SEETA DEVI

(57) Abstract :
ABSTRACT A wearable multisensor system for personalized non-pharmacological labour pain coping support comprises a uterine activity sensor monitoring contraction onset, duration, interval, fundal dominance, and temporal patterns; a maternal physiological sensor measuring heart rate, heart rate variability, electrodermal activity, and respiratory rate; and a maternal coping feedback interface receiving discrete three-state responses indicating adaptive coping, increasing difficulty, or need for assistance. A central processing unit synchronizes sensor data streams. A multisensor fusion module integrates uterine activity, physiological responses, and maternal feedback to generate a combined coping demand representation. A predictive processing module analyzes recent contraction patterns and maternal responses to estimate upcoming coping-support windows before distress escalates. An adaptive interventionselection module selects personalized interventions from breathing guidance, tactile stimulation, audio therapy, and optional virtual reality based on estimated coping state, maternal preferences, and previous effectiveness. Intervention-delivery modules deliver selected interventions. An intervention-response evaluation module compares maternal data before, during, and after intervention delivery to assess effectiveness and adapt subsequent intervention selection across successive contractions, providing proactive closed-loop personalized labour pain coping support.

No. of Pages : 42 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621115726 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ENSEMBLE MACHINE LEARNING BASED ADVERTISING SALES PREDICTION

(51) International classification	:G06Q 30/02, G06N 20/20, G06Q 30/00, G06N 20/00, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Mrigank Pandey
2)Dr. Rajanikanth Aluvalu
3)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR ENSEMBLE MACHINE LEARNING BASED ADVERTISING SALES PREDICTION The present invention discloses a system (100) and method for predicting product sales from multi-channel advertising expenditure data using ensemble machine learning techniques. The system (100) comprises a data ingestion module (110), a data preprocessing module (130), a multi-model training engine (140) configured to simultaneously train linear regression, decision tree, random forest, and gradient boosting regressors, a model evaluation and selection module (150) for automated comparative ranking, a hyperparameter optimisation module (160) employing grid search with cross-validation, a model serialisation module (170), an interactive web-based prediction interface (180) with budget input slider controls and real-time sales prediction display, and a budget optimisation engine (190). The system enables non-technical marketing stakeholders to conduct data-driven budget allocation planning through browser-based real-time sales forecasting. [to be published with fig.1]

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621115727 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED MEDICINE REMINDER AND ADHERENCE MONITORING SYSTEM FOR HEALTHCARE MANAGEMENT

(51) International classification	:G06Q 50/22, G16H 10/60, G16H 20/10, G16H 20/13, G16H 40/67
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Ms. Nikitha Pendlimadugu
2)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT AUTOMATED MEDICINE REMINDER AND ADHERENCE MONITORING SYSTEM FOR HEALTHCARE MANAGEMENT The present invention discloses an automated medicine reminder and adherence monitoring system (100) for healthcare management. The system comprises a user interface module (110), a medicine data management module (120), a reminder scheduling engine (130), a notification and alert module (140), a medication adherence tracking module (150), a database storage unit (160), a server and backend processing unit (170), and a dashboard display module (180). Users enter medicine details including name, dosage, frequency, and timing through the user interface module (110). The reminder scheduling engine (130) processes stored medicine data to generate scheduled reminders at predetermined times. The notification and alert module (140) delivers timely alerts through visual and auditory notifications. The medication adherence tracking module (150) monitors dose intake confirmation and generates adherence statistics displayed on the dashboard display module (180), thereby improving medication compliance and patient safety. [to be published with fig.1]

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621115728 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-FUELED MENTAL WELLNESS PLATFORM WITH INTEGRATED EMERGENCY SUPPORT SYSTEM

(51) International classification	:G06Q 50/30, G16H 20/70, G06Q 50/22, G16H 50/30, A61B 5/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Sneha Gandhi
2)Aparna Velpuri
3)Poulomi Chatterjee
4)Prof. Rajanikanth Aluvalu
5)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT AI-FUELED MENTAL WELLNESS PLATFORM WITH INTEGRATED EMERGENCY SUPPORT SYSTEM The present invention relates to an AI-fueled mental wellness and emergency support system (100) that integrates mood tracking (120), journaling (130), AI-based emotional analysis (140), wellness activities (150), emergency SOS alerts (160), appointment booking (170), and community support (180) within a unified platform. The system employs natural language processing and sentiment analysis to analyze user emotional inputs including mood check-ins and journal entries, identify emotional patterns, and generate personalized wellness recommendations. An emergency SOS module (160) enables users to instantly alert pre-configured trusted contacts during emotionally critical situations. The platform further includes an appointment booking module (170) for scheduling consultations with mental health professionals and a moderated community support module (180) for peer interaction. A centralized database (190) securely stores all emotional data, and a security module (200) provides authentication, encryption, and access control to protect sensitive user information. [to be published with fig.1]

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/09/2026

(21) Application No.202621115511 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUAL-SENSOR BREASTFEEDING MONITORING SYSTEM WITH TIME-SYNCHRONIZED MATERNAL-INFANT SIGNAL CORRELATION

(51) International classification	:A61B 5/01, A61B 5/00, A61B 5/11, A61B 5/0205, A61B 5/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PARUL UNIVERSITY (PARUL INSTITUTE OF NURSING)
Address of Applicant :PARUL INSTITUTE OF NURSING, PARUL UNIVERSITY, P.O. LIMDA TAL. WAGHODIA, DIST. VADODARA, GUJARAT, 391760, INDIA Vadodara Gujarat India
(72)**Name of Inventor :**
1)MRS. TAKHELLAMABAM KIRANMALA CHANU
2)DR. SEETA DEVI
3)MS. SANDHYA KUMARI RAJBHAR

(57) Abstract :
ABSTRACT A breastfeeding monitoring system comprises a maternal breast-side sensor device positioned externally on the breast to detect biophysical changes associated with milk removal, including impedance, strain, and temperature measurements, and an infant-side sensor device positioned externally on the infant's cheek or jaw region to detect sucking activity, swallowing events, and jaw movements. Both sensor devices wirelessly transmit timestamped signals to a signal synchronization and processing module that time-aligns the maternal and infant signal streams and identifies maternal milk-removal events and infant feeding events. A maternal-infant event correlation module determines temporal overlap between the identified events to identify effective milk-transfer events. A milk-transfer estimation module integrates the correlated events, feeding patterns, and session duration to calculate estimated milk intake. A Feeding Sufficiency Index module compares the estimated milk transfer with an individualized motherinfant feeding baseline established through repeated sessions and generates categorical feedback selected from Adequate, Monitor, or Assessment Recommended, enabling noninvasive, objective, real-time assessment of breastfeeding adequacy without interfering with natural feeding dynamics

No. of Pages : 40 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/09/2026

(21) Application No.202621115512 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SENSOR-ASSISTED FINGER CAP SYSTEM FOR BREAST SELF-EXAMINATION AND LONGITUDINAL CHANGE DETECTION

(51) International classification	:A61B 5/00, A61B 5/11, A61B 5/22, A61B 5/103, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PARUL UNIVERSITY (PARUL INSTITUTE OF NURSING)
Address of Applicant :PARUL INSTITUTE OF NURSING, PARUL UNIVERSITY, P.O. LIMDA TAL. WAGHODIA, DIST. VADODARA, GUJARAT, 391760, INDIA Vadodara Gujarat India
(72)**Name of Inventor :**
1)DR. SEETA DEVI
2)MRS. TAKHELLAMABAM KIRANMALA CHANU
3)MS. SANDHYA KUMARI RAJBHAR

(57) Abstract :
ABSTRACT A breast self-examination system comprises a wearable finger cap device with a pressure sensor array measuring palpation force and tissue-response signals, a signal conditioning circuit, a microcontroller unit, and a wireless communication module. A mobile application communicates with the finger cap device and provides guided breast self-examination with standardized anatomical breast-region mapping, real-time sensor monitoring, and digital breast-mapping functionality. The system generates pressure-normalized tissue-response values for each anatomical region and calculates a Palpation Quality Index based on examination coverage, pressure consistency, and sequence completion. The system stores technically adequate examinations as a personal baseline profile and performs longitudinal change detection by comparing subsequent examinations against the baseline. A Breast Tissue Change Index combines tissue-response change magnitude, duration, regional asymmetry, spatial extent, and examination quality. User-reported breast symptoms are integrated with sensor findings to generate referral recommendations for clinical evaluation when persistent or progressive changes are detected.

No. of Pages : 49 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/09/2026

(21) Application No.202621115513 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTEGRATED POSTPARTUM HEMORRHAGE DETECTION SYSTEM COMBINING CONDUCTIVE-MESH BLOOD-LOSS SENSING WITH MACHINE-LEARNING RISK PREDICTION AND TWO-TIER ALERT FUSION

(51) International classification	:A61B 5/00, A61B 5/1455, A61B 5/01, A61B 5/145, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PARUL UNIVERSITY (PARUL INSTITUTE OF NURSING)
Address of Applicant :PARUL INSTITUTE OF NURSING, PARUL UNIVERSITY, P.O. LIMDA TAL. WAGHODIA, DIST. VADODARA, GUJARAT, 391760, INDIA Vadodara Gujarat India
(72)**Name of Inventor :**
1)DR. SEETA DEVI

(57) Abstract :
ABSTRACT An integrated postpartum hemorrhage detection system combines a conductive-mesh absorbent cushion pad (300) with embedded spatially indexed nodes for continuous electrical blood-loss sensing, a wristband vital-sign monitor (400) acquiring heart rate, blood oxygen saturation and body temperature, and a bedside external module (500) that receives triggered node patterns from the pad (300), computes cumulative blood-loss volume estimates accounting for superabsorbent polymer absorption properties and patient weight, and generates local alerts. A machine-learning inference service (111) trained on fifteen binary or ordinal antenatal and intrapartum clinical risk factors using Adaptive Synthetic Sampling achieves receiver operating characteristic area under the curve of 0.74 and minority-class recall of 0.86. A web application (112) fuses the predicted risk probability with measured blood-loss trends through a two-tier alert architecture that escalates alerts when both prediction and measurement signals are positive and de-escalates when measurement contradicts prediction, reducing false-alarm burden while maintaining high clinical sensitivity for deployment in resource-constrained obstetric settings.

No. of Pages : 48 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621115729 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN EVENT PLANNING AND AUTOMATED VENDOR NEGOTIATION SYSTEM

(51) International classification	:G06Q 10/06, G06Q 10/10, G06Q 30/06, G06Q 30/00, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Smaran Jaianand
2)Sujay Indupuru
3)Pramit Panigrahi
4)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT AI-DRIVEN EVENT PLANNING AND AUTOMATED VENDOR NEGOTIATION SYSTEM An AI-driven event planning and vendor negotiation system (100) that transforms natural language event requirements into structured procurement workflows. The system comprises an intent parsing module (110) that extracts event attributes into validated JSON using a large language model, a vendor discovery module (120) that queries location-based APIs to rank and deduplicate vendors per service category, a planning engine (140) that assigns and iteratively revises budget allocations as live quotations arrive, a communication agent (130) that autonomously contacts vendors via WhatsApp and voice telephony while managing conversation state, and a structured data extraction module (150) that normalizes informal vendor responses into comparable quotation records with confidence scoring. The integrated pipeline eliminates manual vendor searching and quote collection, achieving significantly reduced planning time, improved quote retrieval rates, lower budget estimation error, and higher structured quote completeness compared to conventional approaches. [to be published with fig.1]

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/09/2026
(21) Application No.202621115773 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : LONGITUDINAL STUDENT SENTIMENT AND INSTITUTIONAL INTERVENTION EVALUATION SYSTEM

(51) International classification	:G06Q 50/20, G06N 20/00, G06F 16/35, G06F 16/33, G
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mr. Yogesh Bhaurao Desale
Address of Applicant :Flat No. 04, Prathmesh Apartment, Vishwasindhu Nagar, Mandal Shivar, Shirpur Tal – Shirpur Dist – Dhule 425405 Maharashtra India
2)Miss. Kiran Ashok Borase
3)Miss. Gitanjali Kailas Patil
4)Priyanka Virupalsing Girase
5)Harshada Sangramsing Rajput
6)Pooja Dnyaneshwar Wadile
7)Shaikh Gulnaz Moinuddin
8)Shaikh Altamash Ahmed Mohd Zahur
9)Roshani Jijabrao Kuwar
(72)**Name of Inventor :**
1)Mr. Yogesh Bhaurao Desale
2)Miss. Kiran Ashok Borase
3)Miss. Gitanjali Kailas Patil
4)Priyanka Virupalsing Girase
5)Harshada Sangramsing Rajput
6)Pooja Dnyaneshwar Wadile
7)Shaikh Gulnaz Moinuddin
8)Shaikh Altamash Ahmed Mohd Zahur
9)Roshani Jijabrao Kuwar

(57) Abstract :
The present invention relates to a computer-implemented longitudinal student sentiment and institutional intervention evaluation system configured to analyse student feedback across multiple academic periods and determine changes in persistent institutional concerns following corrective institutional actions. The system semantically associates differently worded feedback expressions with persistent feedback themes, determines theme-level sentiment, maps the feedback to institutional entities and academic contextual events, and generates longitudinal sentiment trajectories. Institutional corrective actions are maintained in an intervention registry and computationally matched with corresponding feedback themes. Pre-intervention and post-intervention feedback is evaluated after response-volume normalization and repeated-submission control. Academic contextual events including examination periods, admission cycles, semester transitions, result periods, and other institutional events are incorporated into the evaluation to distinguish temporary variation from persistent concerns. The system further identifies unresolved and emerging issues and generates confidence-bearing explainable assessments indicating improvement, partial improvement, persistence, deterioration, or insufficient evidence associated with an institutional intervention.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/09/2026
(21) Application No.202621115887 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ELASTIC 3D POROUS SCAFFOLDS WITH TUNEABLE MODULUS AND BIOCOMPATIBILITY FOR BONE AND CARTILAGE ENGINEERING

(51) International classification	:A61L 27/56, A61L 27/58, A61L 27/50, A61L 27/54, A61F 2/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
Address of Applicant :Powai, Mumbai - 400076, Maharashtra, India Mumbai Maharashtra India
(72)**Name of Inventor :**
1)SHARMA, Kamendra Prakash
2)KAPOOR, Shobhna
3)V, Rikna T
4)PAULSON, Esther

(57) Abstract :
Provided herein are three-dimensional porous scaffolds with tunable mechanical properties fabricated using PEI–ceria core–shell nanoparticles. Variation in ceria content and crosslinking agent with PEGDE enabled mechanical modulation, yielding compressive modulus and stress values. These scaffolds show potential for load-bearing tissue engineering applications.

No. of Pages : 28 No. of Claims : 18

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/09/2026

(21) Application No.202621115900 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SHAKE-FLASK FED-BATCH PROCESS FOR ENHANCED PRODUCTION OF RECOMBINANT ?-SYNUCLEIN AND ITS MUTANTS THEREOF

(51) International classification	:C12P 21/00, C12P 21/02, C07K 14/47, C12N 15/00, C12N 15/70, C12N 1/21
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
Address of Applicant :Powai, Mumbai - 400076, Maharashtra, India Mumbai Maharashtra India
(72)**Name of Inventor :**
1)MAJI, Samir Kumar
2)DAVDA, Bhumi
3)MANNA, Shouvik
4)SHAW, Ranjit

(57) Abstract :
The present invention provides a fed-batch cultivation process for enhanced production of recombinant α-synuclein (α-Syn) and mutant variants thereof using a shake-flask based bacterial expression system. The invention provides a defined feeding strategy, comprising a carbon source and selected nutritional, chelating, inorganic salt, and trace metal components, which is continuously supplied during a high-cell density cultivation phase. The process comprises sequential cultivation and feeding stages followed by induction of recombinant protein expression, thereby sustaining cellular growth and metabolic activity during the production phase. The continuous feeding strategy enables substantially increased biomass accumulation and recombinant α-Syn production compared with conventional batch cultivation.

No. of Pages : 45 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/09/2026
(21) Application No.202621115919 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI Powered Age Adaptive Cognitive Assessment and Dementia Care Framework

(51) International classification	:A61B 5/16, G16H 50/20, G16H 50/30, A61B 5/00, G09B 19/00	(71)**Name of Applicant :** **1)Vitteshkumar Dattatray Gaikwad** Address of Applicant :opposit pathan baba darga 83 bhavani peth maddi vasti Solapur Maharashtra 413002 India Maharashtra India **2)Sakshi Mallinath Revaje** **3)Chaitali Haresh Mhetras** **4)Sneha Naganath Khairate** **5)Vijayalaxmi Vitthal Kamble**
(31) Priority Document No	:NA	**6)Pallavi Rajendra Metri**
(32) Priority Date	:NA	**7)Anuja Anandrao Birajdar**
(33) Name of priority country	:NA	(72)**Name of Inventor :**
(86) International Application No	:	**1)Vitteshkumar Dattatray Gaikwad**
Filing Date	:01/01/1900	**2)Sakshi Mallinath Revaje**
(87) International Publication No	: NA	**3)Chaitali Haresh Mhetras**
(61) Patent of Addition to Application Number	:NA	**4)Sneha Naganath Khairate**
Filing Date	:NA	**5)Vijayalaxmi Vitthal Kamble**
(62) Divisional to Application Number	:NA	**6)Pallavi Rajendra Metri**
Filing Date	:NA	**7)Anuja Anandrao Birajdar**

(57) Abstract :
The present invention relates to an AI-powered age-adaptive cognitive assessment and dementia care framework configured to support cognitive assessment, personalized cognitive engagement, memory assistance, and caregiver monitoring for elderly individuals experiencing cognitive difficulties. The framework comprises a user interface, an AI-based assessment and personalization module, cognitive assessment modules, adaptive cognitive gaming modules, memory assistance modules, a performance monitoring module, and a caregiver support module. The AI-based assessment and personalization module analyzes user responses, interaction patterns, performance, and age-related parameters to determine suitable cognitive activities and dynamically adjust their difficulty and selection. The cognitive assessment modules evaluate cognitive domains including memory, attention, recognition, reasoning, and problem-solving, while the adaptive gaming modules provide personalized cognitive activities based on the assessed performance. The performance monitoring module maintains user activity and assessment records and generates progress insights for authorized caregivers or family members. The framework further provides simplified navigation, accessible interface elements, clear instructions, and gamification features for elderly users. The invention provides an integrated, adaptive, and supportive technology framework for cognitive assessment and dementia care management and is not intended to replace professional medical diagnosis or treatment.

No. of Pages : 27 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/09/2026
(21) Application No.202621115921 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A FOOTWEAR BASED NAVIGATION SYSTEM FOR VISUALLY IMPAIRED INDIVIDUALS AND METHOD THEREOF

(51) International classification	:A61H 3/06, A43B 3/00, G01C 21/20, G08B 6/00, G01C 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur
Address of Applicant :Dumna Airport Road, P. O. Khamaria, Jabalpur - 482005, Madhya Pradesh, India Jabalpur Madhya Pradesh India
(72)**Name of Inventor :**
1)JADHAV, Sanyogita
2)GUPTA, Suryansh
3)PATEL, Malhar
4)RODGE, Pritesh
5)MENON, Shreya
6)BHUTADA, Ved
7)Dr. Amrita Bhattacharjee

(57) Abstract :
The present invention relates to a navigation aid system for visually impaired individuals comprising a waist-mounted unit worn via a belt structure, incorporating a depth camera to capture three-dimensional spatial data of the surrounding environment, an embedded processing unit for real-time obstacle detection and position determination, and a wireless transmitter. The system further includes a footwear sleeve unit worn over the user's footwear, featuring a wireless receiver, a controller, and a plurality of laterally spaced haptic actuators (left, center, and right) located on a dorsal region. The wireless transmitter sends obstacle position data to the footwear sleeve unit, where the controller selectively activates the haptic actuators to generate directional vibration feedback and varies vibration intensity based on obstacle distance. (Figure 1)

No. of Pages : 22 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/09/2026
(21) Application No.202621115933 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD FOR TREATING A THERMALLY DEGRADED MATERIAL

(51) International classification	:C08L 71/02, C09K 3/00, B05D 3/02, C08J 7/04, C08J 7/00	(71)**Name of Applicant :** **1)D. Y. Patil International University** Address of Applicant :D. Y. Patil International University, J - Block, Sector 29, Akurdi, Pune, Maharashtra, India - 411044 Pune Maharashtra India (72)**Name of Inventor :** **1)Ms. Labhini Lalit Rahangdale** **2)Ms. Taniya Ramesh Rangari** **3)Mr. Suryapratap Ray**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT A method (100) for treating a thermally degraded material (308), comprising positioning the thermally degraded material (308) on a support surface (406), preparing a stabilizing composition (202) comprising polyethylene glycol (PEG) (230) and acetone (220), applying the stabilizing composition (202) to the thermally degraded material (308). The acetone (220) facilitates the penetration of the polyethylene glycol (230) into the thermally degraded material (308). The method (100) further comprises drying the applied stabilizing composition (202) under ambient conditions. The acetone (220) evaporates from the stabilizing composition (202) and the polyethylene glycol (230) remains in the thermally degraded material (308) to internally consolidate the thermally degraded material (308). The method (100) further comprises obtaining a treated material (704) having reduced brittleness and fiber fragmentation and a non-tacky surface. FIG. 1

No. of Pages : 36 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/09/2026

(21) Application No.202621116042 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RADIATION DOSIMETRY APPARATUS AND METHOD FOR DETERMINING AN ABSORBED RADIATION DOSE

(51) International classification	:G01T 1/02, G01T 1/10, G01T 1/11, G01T 1/105, G01T 1/115
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)D. Y. Patil International University
Address of Applicant :D. Y. Patil International University, J - Block, Sector 29, Akurdi, Pune, Maharashtra, India - 411044 Pune Maharashtra India
(72)**Name of Inventor :**
1)Dr. Martina Saran
2)Dr. Jyoti Shakya
3)Prof. Purushottam Dinanath Sahare

(57) Abstract :
A radiation dosimetry apparatus (100) and method (400) for determining absorbed radiation dose using a thin-film optically stimulated luminescence (OSL) dosimeter (202) comprising cerium-doped sodium lithium phosphate ($NaLi_2PO_4:Ce^{3+}$) phosphor are disclosed. The thin-film OSL dosimeter (202) receives ionizing radiation comprising carbon-ion radiation to generate a radiation-induced trapped-charge state. An optical stimulation unit (203) optically stimulates the phosphor to release at least a portion of the trapped charge as luminescence. An optical detection unit (204) detects the emitted luminescence, and one or more processing units (205) determine absorbed radiation dose based on the detected luminescence and predetermined dose-response calibration information (208). The carbon-ion radiation may comprise C^{6+} ions, including an 80 MeV carbon-ion beam. The apparatus (100) may further include an opaque cover (207) for reducing ambient-light exposure and may be implemented with a badge holder (102), clip (108), and transparent labelling window (112) for personal dosimetry. FIG. 1

No. of Pages : 35 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/09/2026
(21) Application No.202621116125 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Intelligent Multi-Source Digital Forensics Platform for Automated Evidence Acquisition, Correlation, and Cyberattack Reconstruction

(51) International classification	:H04L 29/06, H04L 9/40, H04L 9/32, G06F 21/62, G06F 21/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vatsal Uday Choksi
Address of Applicant :Vidyabharti Campus, At & Post Umrakh (Baben), Taluka Bardoli, District Surat, Gujarat - 394345, India. Gujarat India
(72)**Name of Inventor :**
1)Vatsal Uday Choksi

(57) Abstract :
An intelligent multi-source digital forensics platform is disclosed for acquiring, preserving, correlating, and reconstructing digital evidence associated with a cyberattack. The platform includes forensic acquisition interfaces coupled to endpoint, network, cloud, memory, storage, identity, and application evidence sources; a write-protected acquisition gateway; a cryptographic provenance engine; an evidence normalization engine; a temporal correlation engine; an attack reconstruction engine; and an immutable evidence repository. Acquired evidence is preserved as source-faithful forensic objects and is assigned cryptographic hashes, trusted timestamps, source identifiers, and chain-of-custody metadata. Heterogeneous artifacts are normalized into event records while retaining links to corresponding original evidence. The temporal correlation engine compensates for clock offsets and associates events using time, entity, process, network, credential, and artifact relationships. The reconstruction engine produces a causally ordered attack graph with confidence values and supporting evidence references. The platform thereby enables verifiable, contamination-resistant, and reproducible cyberattack reconstruction across heterogeneous digital environments during forensic review.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/09/2026
(21) Application No.202621115595 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR DYNAMIC GAME NARRATIVE ADAPTATION BASED ON PLAYER BEHAVIOUR

(51) International classification	:A63F 13/798, A63F 13/79, A63F 13/35, A63F 13/58, A63F 13/795
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)AMOGHA ABHIJIT KHARE
Address of Applicant :Department of Artificial Intelligence and Data Science, Vishwakarma Institute of Information Technology, 666, Upper Indiranagar, Bibwewadi, Pune, Maharashtra, India – 411037 Pune Maharashtra India
2)HETAVI VINODKUMAR MODI
3)DEVENDRA ANIRUDDHA MOHARIR
4)BASAVKIRAN HANAMANT KARAJANAGI
5)ABHISHEK KALYAN NAWALE
6)SNEHAL ANAND GAVHANE
7)MOHAMMAD MURTAZA WAHEED TAMBOLI
8)AADI PRAVEEN PARKHI
9)AVADHOOT BOREKAR
(72)**Name of Inventor :**
1)AMOGHA ABHIJIT KHARE
2)HETAVI VINODKUMAR MODI
3)DEVENDRA ANIRUDDHA MOHARIR
4)BASAVKIRAN HANAMANT KARAJANAGI
5)ABHISHEK KALYAN NAWALE
6)SNEHAL ANAND GAVHANE
7)MOHAMMAD MURTAZA WAHEED TAMBOLI
8)AADI PRAVEEN PARKHI
9)AVADHOOT BOREKAR

(57) Abstract :
ABSTRACT A SYSTEM AND METHOD FOR DYNAMIC GAME NARRATIVE ADAPTATION BASED ON PLAYER BEHAVIOUR The present disclosure relates to a system and method for dynamic game narrative adaptation based on player behaviour. The system comprises a game engine for executing an interactive game environment, a telemetry acquisition controller for acquiring gameplay events, a behavioural inference controller for determining latent behavioural attributes, a persistent behavioural state store for maintaining a longitudinal player state vector across gameplay sessions, a bounded narrative repository storing predefined narrative nodes having eligibility conditions and narrative constraints, and a constrained narrative director for identifying and evaluating eligible narrative nodes based on the game state, the longitudinal player state vector and behavioural relationships, and selecting a narrative node for execution to modify a subsequent game event. Gameplay telemetry resulting from the selected narrative node is used to update the longitudinal player state vector for subsequent narrative-node selection. FIG. 1 will be the reference figure.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/09/2026
(21) Application No.202621115658 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Sustainable Thermal Insulation Composite Material Comprising Cow Dung, Fly Ash and Phenolic Resin

(51) International classification	:C04B 18/08, C04B 18/30, C04B 18/04, E04B 1/80, E04B 1/76
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Shrikrushna Babasaheb Bhosale
Address of Applicant :Flat no 106, Tirupati Vilas, Sangola Road Maharashtra India
2)Suraj Satyavan Gele
(72)**Name of Inventor :**
1)Shrikrushna Babasaheb Bhosale
2)Suraj Satyavan Gele

(57) Abstract :
[0043] The increasing generation of agricultural and industrial waste has created a need for sustainable materials that can be utilized in value-added engineering applications. Conventional thermal insulation materials may involve relatively high manufacturing costs, energy-intensive processing, or environmental concerns associated with production and disposal. There is therefore a need for a waste-based composite material capable of providing suitable thermal insulation characteristics while utilizing readily available waste resources. [0044] The present invention provides a thermal insulation composite material comprising cow dung, fly ash, and phenolic resin. Cow dung is utilized as an agricultural waste-based constituent, fly ash is utilized as an industrial waste-based filler, and phenolic resin is utilized as a polymeric binder or matrix. The constituent materials are processed and mixed in predetermined proportions to form a substantially homogeneous mixture, which is subsequently molded under pressure and cured to obtain a consolidated composite material. [0045] The developed composite material is evaluated using different material compositions and thicknesses to determine its thermal performance. The performance is characterized in terms of temperature distribution, thermal resistance, and thermal conductivity. Thermal analysis, including ANSYS steady-state thermal analysis, is used to evaluate the temperature distribution through the composite when one surface is subjected to an elevated temperature. [0046] The developed composite material provides a sustainable thermal-insulation material by utilizing agricultural and industrial waste and reducing heat transfer through the material. The composite is suitable for selected industrial thermal-insulation applications, including applications where reduction of heat transfer and temperature transmission is required

No. of Pages : 15 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/09/2026

(21) Application No.202621116254 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR CAUSALITY-GATED DRUG INTERACTION DETECTION AND PHARMACOVIGILANCE MONITORING

(51) International classification	:G16H 20/13, G16H 40/67, G16H 20/10, G16H 10/60, A61J 7/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Sachin Annasaheb Nitave
Address of Applicant :Principal, Dr. J. J. Magdum Trusts Anil Alias Pintu Magdum Memorial Pharmacy College, Dharangutti, Kolhapur, Maharashtra – 416106 Maharashtra India
2)Dr. Swapnil Ashok Phade
3)Vrushabh Sunil Patil
4)Bahubali Nemgonda Patil
5)Miss.Rutuja Mahadev Sathe
6)Miss.Ankita Sanjay Sathe
(72)**Name of Inventor :**
1)Dr. Sachin Annasaheb Nitave
2)Dr. Swapnil Ashok Phade
3)Vrushabh Sunil Patil
4)Bahubali Nemgonda Patil
5)Miss.Rutuja Mahadev Sathe
6)Miss.Ankita Sanjay Sathe

(57) Abstract :
The present disclosure relates to a system and method for causality-gated drug-interaction detection and pharmacovigilance monitoring Authenticated medication administration events are received from pharmaceutical dispensing, packaging, or administration devices and correlated with physiological measurements, laboratory-result trajectories, and adverse-event observations. A processor checks real administration, creates patient specific active-exposure intervals and finds pharmacologically relevant overlap between administered drugs. The overlap is compared temporally to deviations from a rolling, patient-specific baseline. Sensor artefacts, disease-related variations, non-adherence and other competing causal factors are evaluated, before a causality-gated interaction state is produced. If further evidence is required, the processor adaptively adjusts the acquisition of selected physiological signals. An auditable evidence record maintains information about source reliability/integrity, causal parameters, exposure intervals, verified events and . The interaction state is converted into a machine verifiable instruction by a digitally signed safety-control token, which selectively allows, pauses or subjects operation of a pharmaceutical dispensing, administration or monitoring interface to authorised verification, thereby improving safety.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/09/2026

(21) Application No.202621116259 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A REMOTELY CONTROLLED MEDICINE DISPENSING APPARATUS WITH INTEGRATED TELECONSULTATION AND A METHOD FOR OPERATION THEREOF

(51) International classification	:A61J 7/04, G16H 20/13, G16H 40/67, A61J 1/03, A61B 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Medyseva Technologies Private Limited
Address of Applicant :18-A, Electronic Complex, Pardesipura, Indore- 452010, Madhya Pradesh, India Indore Madhya Pradesh India
(72)**Name of Inventor :**
1)KASLIWAL, Vishesh
2)SHUKLA, Vasundhara
3)JAIN, Mehul

(57) Abstract :
A remotely controlled medicine dispensing apparatus (100) with integrated teleconsultation comprises a main housing (102) containing a modular dispensing unit (104) with compartments for prefilled medicine strip cartridges (116). An integrated diagnostic suite (118) on a patient interface panel (126) comprises a photoplethysmography sensor (120) and an infrared temperature sensor (122) capturing physiological measurements without contact. A patient interface (126) comprising a touch screen display (128), a video capture component (130) and audio components (132) is coupled to a telemedicine and computing unit (136). A connectivity and communication module (144) establishes secure communication with a backend services infrastructure (146). An intelligent processing module (160) receives a digital prescription, cross-verifies it against real-time inventory data and authorizes dispensing. A multilingual conversational AI interface (158) performs symptom capture and clinical summarization. A data privacy and security module (178) records patient consent and encrypts data at rest and in transit. FIG. 12

No. of Pages : 61 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/09/2026
(21) Application No.202621116288 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI Based Economic Recession Early Warning System Using Real Time Retail Commerce and Consumer Spending Data

(51) International classification	:G06Q 40/06, G06Q 30/0201, G06N 20/00, G06Q 10/04, G06F 18/21, G06N 3/0442
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ms. Harshala Chaudhari
Address of Applicant :HOD, Information Technology, Bharatiya Vidya Bhavan's M.M. College of Arts, N.M. Institute of Science, H.R.J. College of Commerce, Bhavan's College (Empowered Autonomous), PIN Code – 400058, Maharashtra, India Mumbai Maharashtra India
2)Dr. Anilkumar Nilangi
3)Dr. Sandipa Chatterjee
4)Mr. Rahul Ramshiromani Jaiswal
5)Mr. Raghu Babu K V M
6)Mrs. Nirali Mehta
7)Prof. Ruchi Mishra
8)Dr. Atul Rawal
9)Mr. Sharanabasavareddy Patil
10)Dr Basavaprabhu Nidagundi
(72)**Name of Inventor :**
1)Ms. Harshala Chaudhari
2)Prof. Ruchi Mishra
3)Dr. Atul Rawal
4)Mr. Sharanabasavareddy Patil
5)Dr Basavaprabhu Nidagundi
6)Dr. Anilkumar Nilangi
7)Dr. Sandipa Chatterjee
8)Mr. Rahul Ramshiromani Jaiswal
9)Mr. Raghu Babu K V M
10)Mrs. Nirali Mehta

(57) Abstract :
This paper presents a novel machine learning framework for the automated prediction and early detection of systemic economic recessions utilizing high-frequency retail commerce and consumer spending telemetry. Traditional macroeconomic forecasting relies predominantly on delayed, aggregated national accounting metrics—such as gross domestic product and labor market surveys—which inherently lack the temporal agility required to identify business cycle inflection points before severe contractions materialize. To address these critical vulnerabilities, this research proposes an autonomous, data-driven architecture centered on temporal deep learning networks. The system integrates a dynamic algorithmic filtration and harmonization pipeline designed to isolate authentic baseline consumer demand from complex statistical noise, calendar seasonality, and inflationary price distortions. A defining innovation of this framework is its microeconomic feature engineering tensor, which mathematically quantifies early-stage behavioral adaptations—such as accelerated category trade-downs, shrinking discretionary basket density, and an increasing reliance on revolving credit for non-durable staples—long before consumer financial exhaustion registers in official macroeconomic indicators. By continuously evaluating these non-linear sequential trajectories against historical cyclical footprints, the architecture synthesizes massive, high-dimensional transactional data into a forward-looking, probabilistic recession risk index. Translating chaotic point-of-sale telemetry into immediate, actionable macroeconomic intelligence empowers central banks and fiscal policymakers to deploy targeted countercyclical interventions proactively rather than reactively. Ultimately, this framework facilitates a necessary paradigm shift in economic management, transitioning the discipline from retrospective observation to mathematically grounded foresight, thereby mitigating the amplitude and socioeconomic friction of entrenched cyclical downturns.

No. of Pages : 22 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/09/2026
(21) Application No.202621115682 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : pH-Sensitive Mucoadhesive Polymeric Microbeads Comprising Tamarind Seed Polysaccharide for Colon-Targeted Drug Delivery

(51) International classification	:A61K 47/36, A61P 1/04, A61K 31/606, A61K 9/16, A61K 9/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)shivshri kale
Address of Applicant :At/post- Mhapan, Chavanwadi Maharashtra India
(72)**Name of Inventor :**
1)shivshri kale
2)Mr. ashay omkar madekar
3)dr. rohan barse.
4)Dr. Vijay Jagtap

(57) Abstract :
The present invention relates to a pH-sensitive mucoadhesive pharmaceutical composition for colon-targeted delivery of mesalamine for the management of ulcerative colitis. The invention provides mesalamine-loaded microbeads comprising Tamarind Seed Polysaccharide (TSP) and sodium alginate as a pH-sensitive mucoadhesive polymeric matrix, with calcium chloride as an ionic crosslinking agent. The microbeads are prepared by a simple, aqueous, solvent-free ionotropic gelation technique, providing a reproducible and cost-effective manufacturing process. The developed formulation exhibits minimal drug release under simulated gastric conditions and sustained drug release under intestinal and colonic pH conditions due to pH sensitive swelling of the TSP-sodium alginate matrix. The optimized formulation demonstrated high drug entrapment efficiency, satisfactory particle characteristics, enhanced mucoadhesion, controlled swelling, suitable ex vivo permeation, tissue compatibility, and acceptable physicochemical stability. Histopathological evaluation confirmed the biocompatibility of the formulation with colonic tissue. The invention provides a natural polymer-based colon-targeted drug delivery system that enhances localized delivery of mesalamine, prolongs residence at the site of inflammation, and offers a simple, economical, and industrially scalable approach for the treatment of ulcerative colitis.

No. of Pages : 29 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/09/2026
(21) Application No.202621115720 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SECURE RESILIENT CROSS-PLATFORM FRAMEWORK FOR END-TO-END ENCRYPTED REAL-TIME DEVICE TRACKING

(51) International classification	:H04L 29/06, H04L 9/32, H04L 29/08, H04L 9/08, H04L 12/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Symbiosis International (Deemed University)
Address of Applicant :Symbiosis Knowledge Village, Gram Lavale, Tal Mulshi, Pune 412115, Maharashtra, India Pune Maharashtra India
(72)**Name of Inventor :**
1)Mr. Gunda Ravi Partha Sarathi
2)Ms. Juveriya Begum
3)Mr. Sai Prashanth Mallellu

(57) Abstract :
ABSTRACT SECURE RESILIENT CROSS-PLATFORM FRAMEWORK FOR END-TO-END ENCRYPTED REAL-TIME DEVICE TRACKING The present invention discloses a secure, resilient cross-platform framework (100) for end-to-end encrypted real-time device tracking. The framework comprises a mobile client module (110) with a GPS location acquisition unit (112), a foreground service manager (114), a triple-redundant service resilience engine (116) employing broadcast receivers, alarm managers, and work managers, and a local AES-256-GCM encryption module (118). A backend server module (120) provides hybrid REST API and WebSocket communication with OAuth 2.0 and JWT authentication, operating as a secure relay without decrypting location payloads. A cloud database module (130) enables scalable storage through a NoSQL database service. A web dashboard module (140) provides interactive real-time map visualization with encrypted payload decryption at the authorized destination. The framework ensures true end-to-end encryption, uninterrupted persistent background tracking through multi-layered resilience mechanisms, and cross-platform accessibility. [to be published with fig.1]

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/09/2026

(21) Application No.202621113457 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Alkaline-Responsive Core-Shell Superabsorbent Polymer Composition for Staged Internal Curing of Cementitious Materials

(51) International classification	:C04B 24/38, C04B 28/02, C04B 40/00, C04B 28/04, C08F 251/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lakshmi Narain College of Technology
Address of Applicant :Kalchuri Nagar, Raisen Road, Bhopal, Madhya Pradesh, India – 462021 Madhya Pradesh India
(72)**Name of Inventor :**
1)Dr. Naveen Thakur
2)Dr. Prashant Kumar
3)Dr. Bikram Prasad
4)Dr. Rajeev Singh Parihar
5)Dr. Shubhum Prakash
6)Dr. Rajesh Kumar Misra

(57) Abstract :
ABSTRACT An Alkaline-Responsive Core-Shell Superabsorbent Polymer Composition for Staged Internal Curing of Cementitious Materials The present invention is directed to an alkaline-responsive core-shell superabsorbent polymer composition that is designed to facilitate staged internal curing of cementitious systems. The core-shell structure includes core-shell particles in which the core is a water-reservoir core made up of a graft-crosslinked cellulose-starch polymer network, while the shell is an ion-selective, diffusion controlling outer shell having a higher crosslink density as compared to the core. The hydration nucleating particles are incorporated into or on the outer shell. The outer shell prevents premature ionic deswelling and release of water in an alkaline cement pore solution environment, while allowing controlled water release when the relative humidity inside the cementitious system decreases. This enables the nucleating particles to enable localized hydration of the surroundings of the shrinking particles, leading to refinement of the remaining pores. Different types of particle sizes, shell thickness or crosslink density facilitate water release at various stages. FIG 3

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/09/2026
(21) Application No.202621113458 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD, AN OWNER MODULE, A LUGGAGE MODULE AND A SYSTEM FOR WIRELESS LUGGAGE SECURITY

(51) International classification	:A45C 13/24, G08B 13/14, A45C 5/14, A45C 13/18, H04W 4/029	(71)**Name of Applicant :** **1)Lakshmi Narain College of Technology** Address of Applicant :Kalchuri Nagar, Raisen Road, Bhopal, Madhya Pradesh, India – 462021 Madhya Pradesh India (72)**Name of Inventor :** **1)Dr. Nikita Shivhare** **2)Dr. Tripti Saxena** **3)Dr. Aakriti Jain** **4)Nandita Tiwari**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT A METHOD, AN OWNER MODULE, A LUGGAGE MODULE AND A SYSTEM FOR WIRELESS LUGGAGE SECURITY The present invention relates to a method, owner module, luggage module, and system for wireless luggage security. The owner module and luggage module establish a wireless session and generate ranging measurements representing relative separation. Inertial measurement units generate motion data from which time-correlated owner-motion and luggage-motion vectors are derived. The system identifies coupled movement when the vectors satisfy a correlation criterion and detects unauthorized divergence when a separation rate exceeds a threshold and the vectors remain uncorrelated for a persistence interval. Upon detecting unauthorized divergence, synchronized alerts are activated at both modules. An authenticated custody-transfer mode permits legitimate separation according to an authorization profile specifying an authorization period and permitted movement envelope, while motion and tamper monitoring remain active. Detection of tampering, authorization expiry, or movement outside the permitted envelope cancels the custody-transfer mode, activates a luggage-side alert, stores a security-event record, and transmits the record when communication is available. Fig. 3

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/09/2026

(21) Application No.202621116292 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR SEMANTIC CACHING OF RESPONSES CORRESPONDING TO INPUT INSTRUCTIONS AND A METHOD THEREOF

(51) International classification	:G06F16/2455, G06F12/0871, G06F11/30, G06F12/0815
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rahul Bhaurao Diwate
Address of Applicant :Pune Maharashtra India
2)Nishanthi Chandrashekharan Naidu
3)Mr. Onkar Vijayrao Chandure
4)Dr. Ganesh Digambarrao Goindwar
5)Dr. Sushant Ashokrao Patinge
6)Miss. Radhika Ashokrao Dahane
7)Dr. Sandip Adikumar. Kahate
8)Vishwakarma Institute of Technology
(72)**Name of Inventor :**
1)Rahul Bhaurao Diwate
2)Nishanthi Chandrashekharan Naidu
3)Mr. Onkar Vijayrao Chandure
4)Dr. Ganesh Digambarrao Goindwar
5)Dr. Sushant Ashokrao Patinge
6)Miss. Radhika Ashokrao Dahane
7)Dr. Sandip Adikumar. Kahate

(57) Abstract :
ABSTRACT A SYSTEM FOR SEMANTIC CACHING OF RESPONSES CORRESPONDING TO INPUT INSTRUCTIONS AND A METHOD THEREOF The present disclosure relates to a system (100) for semantic caching of responses corresponding to input instructions. The system comprises an application platform (102) implemented using one or more hardware processors and a memory, wherein the application platform (102) is configured to perform semantic analysis, similarity-based cache retrieval, response execution, and dynamic cache management. The application platform (102) includes an analyzer module (104), a comparator module (106), a cache manager module (108), a cache memory (108a), an execution module (110), and a monitoring module (112), which collectively cooperate to enable efficient retrieval and management of cached responses corresponding to semantically similar input instructions. FIG. 1 will be the reference figure.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/09/2026

(21) Application No.202621116334 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : VISIONAID EXAM SYSTEM.

(51) International classification	:A61H 5/00, A61B 3/032, G09B 19/00, G09B 23/28, A61B 3/028
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MUNDHE TEJAS SURYKANT
Address of Applicant :SARDAR PATEL INSTITUTE OF TECHNOLOGY, BHAVAN'S CAMPUS, MUNSHI NAGAR, ANDHERI WEST, MUMBAI-400058, MAHARASHTRA, INDIA . Maharashtra India
2)DHURGUDE TEJAS RAVINDRA
3)DIVEKAR SUSHANT SADASHIV
4)AGA SAJID
5)SAWANT RUPALI
(72)**Name of Inventor :**
1)MUNDHE TEJAS SURYKANT
2)DHURGUDE TEJAS RAVINDRA
3)DIVEKAR SUSHANT SADASHIV
4)AGA SAJID
5)SAWANT RUPALI

(57) Abstract :
ABSTRACT VISIONAID EXAM SYSTEM The present invention relates to a portable six-key tactile keyboard specifically designed to facilitate independent computer-based examinations for visually impaired and blind individuals. The invention addresses the limitations of conventional QWERTY keyboards, which are complex and unsuitable for efficient examination interaction by users with visual impairments. The proposed keyboard comprises six ergonomically arranged tactile keys, each assigned a dedicated examination function, including navigation between questions, navigation among answer options, answer confirmation, and examination submission. The simplified hardware interface minimizes cognitive effort, reduces the learning curve, and enables intuitive operation solely through tactile interaction. The keyboard is configured to communicate with an accessible examination system that provides real-time audio guidance through text-to-speech technology, allowing users to receive auditory feedback for every action performed. A processing unit interprets the input generated by each tactile key and transmits corresponding commands to the examination application, thereby enabling seamless interaction without reliance on conventional keyboards or external assistance. The compact and portable design makes the device suitable for deployment in educational institutions, competitive examinations, certification tests, and other computer-based assessment environments. By integrating a purpose-built tactile input device with an accessible examination platform, the invention promotes independence, accessibility, examination integrity, and equal participation for visually impaired students. The proposed six-key examination keyboard provides an efficient, reliable, and user-friendly hardware solution that significantly enhances the examination experience while reducing dependency on scribes or assistive personnel. The invention represents a practical advancement in assistive educational technology by providing a dedicated hardware interface optimized for accessible digital examinations.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :23/09/2026
(21) Application No.202621114153 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : An Integrated AI and Nanotechnology-Enabled Framework for Smart Education, FinTech, Digital Governance, Blockchain Security, Management and Multidisciplinary Research

(51) International classification	:H04L 9/32, H04L 9/06, G06N 20/00, G06N 5/02, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mr. Atul Kumar Singh
Address of Applicant :Assistant Professor, Department of Radiology and Imaging Technology, Dr. D. Y. Patil School of Allied Health Sciences, Dr. D. Y. Patil Vidyapeeth, Pune (Deemed to be University), Sant Tukaram Nagar, Pimpri, Pune 411018 Maharashtra India
2)Dr. Chethan Chandra S Basavaraddi
3)Dr. Sapna S Basavaraddi
4)Dr. Dipak M. Sanki
5)Dr. Pradnya A. Gajbhiye (Ramteke)
6)Dr. Kamatam Kiran
7)Dr. Desh Deepak Singh
8)Padasanaboina Sona
9)Shweta Pethe
10)Dr. Atul Ladana
11)Venissa N Shenoy
12)Dr. Utpaljit Deori
13)Dr. Anuj Sangwan
14)Dr. Katepogu Praveen
15)Dr. Deepak Chandrasekharan
(72)**Name of Inventor :**
1)Mr. Atul Kumar Singh
2)Dr. Chethan Chandra S Basavaraddi
3)Dr. Sapna S Basavaraddi
4)Dr. Dipak M. Sanki
5)Dr. Pradnya A. Gajbhiye (Ramteke)
6)Dr. Kamatam Kiran
7)Dr. Desh Deepak Singh
8)Padasanaboina Sona
9)Shweta Pethe
10)Dr. Atul Ladana
11)Venissa N Shenoy
12)Dr. Utpaljit Deori
13)Dr. Anuj Sangwan
14)Dr. Katepogu Praveen
15)Dr. Deepak Chandrasekharan

(57) Abstract :
ABSTRACT The present invention relates to an integrated artificial intelligence (AI) and nanotechnology-enabled framework for providing secure, intelligent, and adaptive services across smart education, financial technology (FinTech), digital governance, blockchain security, management, and multidisciplinary research. The framework comprises an AI processing engine, nanotechnology-enabled sensing and authentication components, a secure blockchain layer, a data integration module, and domain-specific analytical modules. The system collects and processes heterogeneous data, generates predictive and decision-support outputs, provides secure identity and transaction management, and enables knowledge sharing across multiple domains. The framework further incorporates privacy-preserving mechanisms, intelligent automation, anomaly detection, and adaptive learning to improve security, efficiency, transparency, and research collaboration.

No. of Pages : 4 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2026

(21) Application No.202621114157 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Process for Preparing Cocoa-Based Nutraceutical Compositions Comprising a Dual-Immobilized C-Phycocyanin Complex

(51) International classification	:A23G1/32, A23G1/48, A23G1/56
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr. Megha Ramchandra Shinge

Address of Applicant :D/O Ramchandra Tukaram Shinge, 463 KH-2 Eward, Shahuroad Shahupuri, Kolhapur, PO: Kolhapur, Sub District: Karvir, Distict: Kolhapur- 416001, Maharashtra, India Kolhapur Maharashtra India

2)Sensoact Innovations Pvt. Ltd.

(72)**Name of Inventor :**

1)Dr. Megha Ramchandra Shinge

2)Asmita Moghe

3)Rushabh Runwal

4)Tanvi Joshi

5)Tanvi Purohit

(57) Abstract :

A Process for Preparing Cocoa-Based Nutraceutical Compositions Comprising a Dual-Immobilized C-Phycocyanin Complex The present invention relates to a process for preparing cocoa-based nutraceutical compositions comprising a dual-immobilized C-phycocyanin complex, wherein about 0.5 to 1.0 g of C-phycocyanin is sequentially immobilized using about 1.0 to 2.0 wt.% sodium alginate and about 1.0 to 3.0 wt.% gum arabic at a sodium alginate to gum arabic ratio of about 1:1 to 1:2 to obtain a substantially stable bioactive complex. The process comprises incorporating the dual-immobilized C-phycocyanin into a cocoa matrix comprising about 30–40 wt.% cocoa solids, 15–25 wt.% cocoa powder, 5–20 wt.% cocoa liquor and 20–25 wt.% cocoa butter, followed by blending below about 30°C and homogenization not exceeding about 35°C, and processing into one or more product forms, exhibiting enhanced physicochemical stability, antioxidant activity, bioactive retention, gastrointestinal bioaccessibility and prolonged storage stability.

No. of Pages : 61 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/09/2026

(21) Application No.202621115331 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADJUSTABLE SWAPPABLE-BATTERY MULTIFUNCTION AGRICULTURAL VEHICLE FOR WEEDING SPRAYING AND SEED SOWING

(51) International classification	:A01M 7/00, A01B 49/06, A01M 21/04, A01B 51/02, A01B 49/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rajarambapu Institute of Technology
Address of Applicant :Rajaramnagar, Islampur, Tal- Walwa, Dist. Sangli, Maharashtra 415414 Maharashtra India

(72)**Name of Inventor :**
1)Prof. Sachin P. Jadhav
2)Prof. Sachin S. Kumbhar
3)Mr. Atharv Pardeshi
4)Ms. Prasanna Tomake
5)Ms. Vaishanavi Salunkhe

(57) Abstract :

Adjustable Swappable-Battery Multifunction Agricultural Vehicle for Weeding, Spraying and Seed Sowing This invention describes a compact multifunction agricultural vehicle configured to selectively perform weeding, pesticide spraying and seed sowing operations using interchangeable attachments mounted on a common mobile platform. The vehicle incorporates a removable and replaceable battery arrangement enabling a discharged battery to be exchanged with a charged battery to reduce operational downtime. An adjustable structural arrangement permits variation of the working width and operating height of the vehicle according to crop height, furrow width, row spacing and field conditions. The vehicle is further configured to transport agricultural materials and equipment having a combined load of approximately 100 kg. Detachable working attachments permit an operator to selectively configure the vehicle for weeding, spraying or sowing without requiring separate agricultural machines. The battery may be charged through an electrical supply and optionally through a solar charging arrangement, thereby providing a versatile, low-emission agricultural platform particularly suitable for small and marginal farming operations.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :08/10/2026

(21) Application No.202621120645 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD AND SYSTEM FOR AI-BASED MULTIDIMENSIONAL STUDENT PROFILING AND PERSONALIZED DECISION SUPPORT

(51) International classification	:G09B 7/00, G09B 7/02, G09B 7/04, G06Q 50/20, G09B 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GAVHANE Jyoti Nagnath
Address of Applicant :SKYi Songbirds Tower-6, 201, Near Daulat Lawns, Taluk Mulshi, Bhugaon, Pune, Maharastra, India, 412115 Pune Maharashtra India
(72)**Name of Inventor :**
1)GAVHANE Jyoti Nagnath
2)PAGARE Reena Dinesh

(57) Abstract :
The present invention relates to a system and method for AI-based multidimensional student profiling and personalized decision support. The method (400) comprises receiving (402) student data including Adversity Quotient (AQ) assessment responses, constructed textual responses, academic information, and skill or competency information. The AQ responses are processed (404) using an Item Response Theory model to estimate latent AQ characteristics. The textual responses are processed (406) using natural language processing to generate numerical representations for determining (408) a cognitive representation. An integrated computational student representation comprising AQ, cognitive, skill or competency, and academic information is generated (410). A multidimensional retrieval vector is generated (412) therefrom and used to retrieve (414) information items from a knowledge repository based on similarities. Personalized educational or career decision-support information is generated (416) based on the integrated representation and retrieved information items.

No. of Pages : 41 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114704 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Precision Medical Automatic Syrup Dispenser

(51) International classification	:A61J 1/20, A61J 7/00, B67D 1/00, A61J 1/00, B67D 1/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHINMAY KOPARDE
Address of Applicant :93/B,GUNVANTI NAGAR, RAO COLONY,TALEGAON DABHADE,PUNE Maharashtra India
2)Manthan Prashant Vedpathak
3)Tanishq Amol Borse
4)Nilesh Ashok Mamdyal
5)Shruti Gajanan Chhatre
6)Pramod Kale
(72)**Name of Inventor :**
1)CHINMAY KOPARDE
2)Manthan Prashant Vedpathak
3)Tanishq Amol Borse
4)Nilesh Ashok Mamdyal
5)Shruti Gajanan Chhatre
6)Pramod Kale

(57) Abstract :
The Precision Medical Automatic Syrup Dispenser is a device designed for children and elderly people. It dispenses syrup accurately and automatically using IoT technology. Because of this, the problem of overdosing is avoided, human errors are eliminated, and the medicine is always given at the correct time. Along with this, proper hygiene is also maintained. Precise medication dosing is crucial to ensure the safety of the patient, especially for liquid medicines where conventional measuring cups, spoons, and syringes may lead to measurement errors, spillage, and irregular dosing. These problems become more acute when handling viscous or thick pharmaceutical syrups. This project proposes a Precision Medical Syrup Dispenser, an automated, weight-based medication dispensing system, to dispense accurate and repeatable doses of liquid medicine. A load cell sensor is used in this system to continuously measure the weight of the medication container, while an ESP-based microcontroller processes the information from the sensor and controls the dispensing mechanism using a closed-loop feedback system. The system achieves the target dose by continuously comparing the measured weight with the medication quantity it has to dispense and automatically controls the fluid delivery mechanism. The proposed system comprises five major functional subsystems, including (1) weight sensing and measurement, (2) ESP-based control and processing, (3) precision fluid delivery, (4) mechanical support framework, and (5) user interface with integrated safety features. Besides these, the system requires additional components, including a Real-Time Clock (RTC), motor driver, voltage regulator, display, battery, and load cell, to be able to operate smoothly and offer an intuitive interface to the user. This system can enhance the dosing accuracy, especially for viscous medicines, because it continuously monitors the medication instead of dispensing a given amount of liquid medicine by time or dispensing volume. The proposed dispenser offers an automated and compact solution for medication administration and can be integrated into home healthcare, hospitals, and assisted-care environments. This project demonstrates how sensor-based closed-loop systems can be used in synergy with mechanical dispensing systems to achieve automation.

No. of Pages : 6 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202621114718 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A synergistic polar extract composition comprising punica granatum peel and curcuma longa rhizome and a process for preparation thereof

(51) International classification	:A61K 36/00, A61K 36/185, A61K 36/9066, A61P 35/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Pravin Badhe
Address of Applicant :Flat No 708 Ragini C Durvankur Shrushti, Near JSPMs college Narhe Pune Maharashtra India
(72)**Name of Inventor :**
1)Dr Pravin Badhe
2)Pratap Shivaji Dabhade
3)Dr Saranya Punniyakotti
4)Mrs Ashwini Badhe
5)Dr. Manjushri Pratap Dabhade

(57) Abstract :
A SYNERGISTIC POLAR EXTRACT COMPOSITION COMPRISING PUNICA GRANATUM PEEL AND CURCUMA LONGA RHIZOME AND A PROCESS FOR PREPARATION THEREOF The present invention relates to a botanical polar extract composition comprising Punica granatum peel extract and Curcuma longa rhizome extract in a weight ratio ranging from 1:1 to 2:1. The botanical materials are separately subjected to polar solvent extraction, followed by filtration, concentration under reduced pressure and removal of residual solvent. In an exemplary embodiment, extraction is performed using 95% ethanol at a plant-material-to-solvent ratio of approximately 1:3 for approximately 24 hours at room temperature. The resulting extracts contain polyphenolic, tannin, flavonoid and curcuminoid constituents. The combined composition demonstrates antioxidant activity in DPPH and total-antioxidant assays, biological activity in cytotoxicity and antimitotic screening models, and reduction of tumour weight, tumour volume and tumour burden in a DMBA-induced mammary carcinogenesis model in female Sprague-Dawley rats. The composition may be incorporated with pharmaceutically or nutraceutically acceptable excipients into suitable dosage forms. The invention further provides a reproducible process for preparation of the combined botanical extract composition.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :10/09/2026
(21) Application No.202623108790 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT-BASED CATTLE HEALTH AND HEAT MONITORING SYSTEM

(51) International classification	:H04W4/38, H04L67/12, G06N3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filed on	:202221038997 :01/01/1900
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)AREETE BUSINESS SOLUTIONS PRIVATE LIMITED
Address of Applicant :605, Baner Business Bay, Pashan - Sus Rd, Pune Bangalore Highway Pashan Exit, behind Audi, off Mumbai Highway, Baner Annex, Baner, Pune-411045, Maharashtra India Pune Maharashtra India
(72)**Name of Inventor :**
1)SRINIVAS SUBRAMANIAN
2)ANNAREDDY VENKATA SIVA REDDY
3)SARVESH DESHPANDE
4)PRASHANT SHARMA
5)AKSHAY DHUMAL
6)ADITYA KOLTE

(57) Abstract :
ABSTRACT IOT-BASED CATTLE HEALTH AND HEAT MONITORING SYSTEM A system (200) for monitoring cattle health and heat conditions is disclosed. Collar units (10) secured on cattle each comprise sensors (11, 12, 13, 14), a controller (16), a communication module (18) and a power supply unit (19), the controller (16) converting sensor output signals into digital data packets. A portable computing device (40) executes a mobile gateway application (60), separate from a dashboard or mobile application (44), comprising a scan and connect module (61) which scans through a short range wireless transceiver (41) for a collar unit (10) within range, a local telemetry buffer (62) which holds received digital data packets unaltered in a data storage (43), and a synchronisation module (63) which transmits them over a communication network (50) to a data ingestion module (21) of a server (20). An artificial intelligence module (30) processes the digital data packets and generates alerts.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/05/2025

(21) Application No.202541047325 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A synergistic formulation to enhance testosterone levels

(51) International classification	:A61K36/48, A61K36/9068, A61K31/198, A61K31/315, A61K47/24, A61K47/36, A61P5/26, B01D11/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Padma Herbals Private Limited
Address of Applicant :Sreevihar, Kanakkaparambil house, Annanad P.O, Chalakudy, Thrissur - 680309, Kerala, India Thrissur Kerala India
(72)**Name of Inventor :**
1)Dr. Sreerag Gopi

(57) Abstract :
A synergistic formulation to enhance testosterone levels and a method of preparation [0078] The present invention discloses synergistic formulation to enhance testosterone levels. The formulation comprises 23%-27% w/w saponin enriched Trigonella Foenum Graecum (fenugreek) extract powder, 24%-26% w/w Kaempferia parviflora (black ginger) extract powder, 24%-26% w/w, zinc methionine, 6%-8% w/w lecithin and 35%-40% gum Arabic. The formulation comprising extracts of bioactive components exhibits synergistic effects in enhancing the levels of testosterone through multiple pathways. The invention also discloses a unique In Situ 360 method for preparation of the formulation. The formulation aids in enhancing muscle mass, endurance and performance. The formulation disclosed in the present invention exhibits bio efficacy, safety and is cost-effective. (FIGURE 1)

No. of Pages : 33 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :10/07/2025
(21) Application No.202541065942 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD FOR PREPARING A CONCRETE COMPOSITION USING METALLIC FIBRES AND BINDING MATERIALS

(51) International classification	:A61P0043000000, C04B0028020000, C04B0111340000, C08G0063183000, C04B0020100000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Government Engineering College, Hassan
Address of Applicant :Government Engineering College, Hassan, Under Department of Collegiate and Technical Education (DCTE), GoK, Dairy Circle, B M Road, Hassan - 573201, Karnataka, India. Hassan Karnataka India
(72)**Name of Inventor :**
1)THEJASWINI R M

(57) Abstract :
The present disclosure pertains to a method (100) for preparing a concrete composition. The method (100) may include preparing (102) a concrete mixture, mixing (104) a predefined amount of metallic fibres in the prepared concrete mixture to prepare a fibre concrete mixture, and adding (106) a predefined ratio of a binding material to the fibre concrete mixture to prepare the concrete composition. The method may include mixing a predefined amount of cement with a predefined ratio of raw materials and a predetermined ratio of water to prepare the concrete mixture. The metallic fibres may be coated with carbon and may be formed of a predefined geometry. The method of preparing the concrete composition using metallic fibres and the binding materials, enhances the strength, durability, and crack resistance of the concrete structures.

No. of Pages : 23 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/07/2025

(21) Application No.202541069780 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR AGRICULTURAL MANAGEMENT

(51) International classification	:G06Q50/02, G06N20/00, G06N20/20, G06V20/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Government Degree College, Yadgir
Address of Applicant :Government Degree College, Yadgir, Chittapur Road, Beside, RTO Office, Chittapur Road, Yadgir - 585202, Karnataka, India. Yadgir Karnataka India
(72)**Name of Inventor :**
1)BASAVAPRASAD B

(57) Abstract :
The present disclosure provides a system (102) and a method for agricultural management. The system (102) receives images associated with a predetermined area and records parameters associated with the predetermined area based on the received images. The system (102) analyzes, via an artificial intelligence (AI) engine (214), the recorded parameters to determine a condition associated with the predetermined area. The system (102) in response to a determination that the condition exceeds a threshold range, disperse a quantity of spraying material to the predetermined area for functioning of the predetermined area. The system (102) subsequently enables dispersal of germinating material to the predetermined area and monitor growth of the germinating material along the predetermined area. The system (102) generates via the AI engine (214), recommendations associated with functioning of the predetermined area based on the monitored growth and the determined condition.

No. of Pages : 30 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/08/2025

(21) Application No.202541074197 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR REAL-TIME DETECTION OF RETAIL CHECKOUT ANOMALIES USING AI AND POS INTEGRATION

(51) International classification	:G06V 40/20, G06V 20/40, G06F 16/20, G06Q 20/20, G06V 20/52, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Proglint Software Solutions Private Limited
Address of Applicant :Prince Info City I, 3rd Floor, #286/1, Rajiv Gandhi Salai, Nehru Nagar, Perungudi, Chennai, Tamil Nadu 600096, India Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Sriram Sudhakar

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR REAL-TIME DETECTION OF RETAIL CHECKOUT ANOMALIES USING AI AND POS INTEGRATION The present invention relates to a system and method for real-time detection of non-scan events, product switching, and bottom-of-the-basket (BOB) losses in retail environments using AI-powered video analytics integrated with point-of-sale (POS) systems. The system employs strategically positioned cameras and edge computing devices running deep learning models to monitor customer behaviour and item movement during checkout. By cross-referencing visual data with POS transaction logs, the system identifies anomalies such as unscanned items (111) or product mismatches (114). Detected events trigger real-time alerts with contextual video clips and metadata (119), enabling immediate intervention by store personnel. A centralized dashboard aggregates incidents for review, reporting, and training, while supporting analytics on shrinkage trends. The system is scalable, modular, and designed for seamless integration with existing retail infrastructure. It ensures enhanced loss prevention, operational efficiency, and customer transparency without disrupting the checkout experience. Fig.1

No. of Pages : 32 No. of Claims : 16

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :14/09/2025
(21) Application No.202541077570 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : EARLY YEARS LEARNING SYSTEMS AND METHODS DESIGNED FOR OBJECTIFIED DIGITAL LEARNING EXPERIENCE

(51) International classification	:G06Q 50/20, G09B 5/06, G09B 7/02, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Shinde Amruta Sayaji
Address of Applicant :409, Nidhi apartment , #40, Netaji Road Frazer town, Bangalore – 560005, Karnataka, India Bangalore Karnataka India
2)2. Shinde Shivangi Sayaji
(72)**Name of Inventor :**
1)Shinde Amruta Sayaji
2)2. Shinde Shivangi Sayaji

(57) Abstract :
ABSTRACT EARLY YEARS LEARNING SYSTEMS AND METHODS DESIGNED FOR OBJECTIFIED DIGITAL LEARNING EXPERIENCE. Figure 1. illustrates the objectified digital learning system and methods, the system comprises structured teaching methods and teaching materials for interactive learning between teacher and students through IOT enabled virtual platform, the learning system utilizes specially designed assessment techniques, involving adaptive learning technologies for covering all the core Montessori education methods comprising mathematical learning, language learning, sensory learning and cultural learning and like, by utilizing multimedia instructional methods comprising digital books, video applications, 2D / 3D digitalized learning materials. Dated this 01st September 2026 (Santhanam L) IN/PA No. 3125 Patent Agent for Applicant

No. of Pages : 65 No. of Claims : 25

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/09/2025

(21) Application No.202541090882 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for Creating and Participating in structured Social Media Challenges

(51) International classification	:G06F 16/22, G06F 16/27, G06F 16/40, H04L 9/32, G06Q 10/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VINAYAK YADAV
Address of Applicant :Lakshith Golden Nest, Plot No. 105/B 11th Main, Vinayaka Temple, Vinayaka Layout 9th block, 2nd Stage, Nagarbhavi Bangalore Karnataka India 560072 Bangalore Karnataka India
(72)**Name of Inventor :**
1)VINAYAK YADAV

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR CREATING AND PARTICIPATING IN STRUCTURED SOCIAL MEDIA CHALLENGES Embodiments disclose a system and method for structured social media challenges. A server instantiates a challenge object. A user device presents the challenge, receives a join or competing entry, and transmits media with a challenge identifier. The server stores the media in association with the challenge. A client application displays associated submissions together and accepts ratings, comments and invitations. Popularity points and title tags may be awarded on defined actions. In an embodiment the association is a directed acyclic graph indexed by materialized paths so a subtree is read by prefix match. Engagement counters are held in an ephemeral store and updated by atomic increments, separate from topological writes. The server emits an attach-intent event and releases the user device before commit, computes engagement velocity, and sends a title payload with a keyed message-authentication code for local verification. FIG. 1

No. of Pages : 38 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :25/12/2025

(21) Application No.202541092294 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEMS AND METHODS FOR UNIFIED GOVERNANCE MANAGEMENT AND ACCOUNTABILITY THROUGH BLOCKCHAIN-BACKED, FEDERATED DATA ORCHESTRATION

(51) International classification	:H04L0009000000, H04L0009320000, G06Q0050260000, G06F0016270000, G06Q0020120000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sharath KashyapR
Address of Applicant :Skanda, No.144, 9th Cross,6th block, 2nd Stage, Nagarabhavi, Bangalore-560072, Indian Nagarabhavi Karnataka India
(72)**Name of Inventor :**
1)Sharath KashyapR

(57) Abstract :
ABSTRACT SYSTEMS AND METHODS FOR UNIFIED GOVERNANCE MANAGEMENT AND ACCOUNTABILITY THROUGH BLOCKCHAIN-BACKED, FEDERATED DATA ORCHESTRATION The present disclosure relates to systems and methods for unified governance management and accountability through blockchain-backed, federated data orchestration. The system integrates a synchronisation layer configured to capture and reconcile changes across a plurality of ministry ERP systems, citizen engagement portals, and operational data lakes. A central API gateway coupled with middleware authenticates and authorises user requests, translates them into native ERP queries, and ensures resilient federated execution. Returned data is aggregated by an analytics module to compute multidimensional performance indicators, including citizen sentiment, grievance resolution, fiscal efficiency, and operational resilience, which are synthesised into a social return-on-investment score. A grievance redressal engine classifies and routes grievances, validates responses against service-level agreements, and publishes performance rankings on gamified leaderboards. A real-time visualisation console dynamically renders authorised dashboards for officials, while all queries, directives, and actions are immutably logged on blockchain, ensuring transparency, accountability, and non-repudiation across governance processes.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/09/2025
(21) Application No.202541092981 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SCREW-JACK BASED OSTEOTOMY REALIGNMENT DISTRACTOR AND METHOD OF USE THEREOF

(51) International classification	:A61B 17/66, A61B 17/02, A61B 17/17, A61B 17/56, A61B 17/70
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MANIPAL ACADEMY OF HIGHER EDUCATION
Address of Applicant :MADHAV NAGAR, MANIPAL, 576104, KARNATAKA, INDIA Karnataka India
(72)**Name of Inventor :**
1)SHIVAM THAKER
2)DEEPTHI PARAJ
3)ANIL K BHAT

(57) Abstract :
A method of operating a screw-jack based osteotomy realignment distractor comprises inserting a retention housing (101) at an osteotomy site, positioning adjustable bone jaws (125, 127) against the bone ends and securing them via threaded stabilisers (129, 131), engaging a placement device having handles (145, 153) and distractor heads (139) to achieve initial gross distraction, inserting a threaded screw (123) through a screw-jack entry port (103) into a spirally grooved fixed block (107) and rotating the screw to drive a sliding box (111) through a connector (119), parallel connector blocks (117), and a dynamic joint (115), thereby achieving fine distraction, locking the bone ends via internal friction while applying a fixation plate, and withdrawing the screw (123) to collapse and extract the retention housing (101). (FIG. 5)

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202541094668 A
(19) INDIA
(22) Date of filing of Application :01/10/2025
(43) Publication Date : 06/11/2026

(54) Title of the invention : BATTERY MANAGEMENT SYSTEM FOR MANAGING STATE OF HEALTH OF A BATTERY

(51) International classification	:G06N0020000000, H01M0010420000, H02J0007000000, H04N0019132000, H01M0010480000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Government Polytechnic Mirle
Address of Applicant :Government Polytechnic Mirle, Under Department of Collegiate and Technical Education, GoK Saligrama Taluk, Mysore District - 571603, Karnataka, India. Mysore Karnataka India
(72)**Name of Inventor :**
1)PARAMESHA R

(57) Abstract :
The present disclosure pertains to a battery management system (102). The battery management system (102) includes a processor (104) to receive, by a receiving module (114), battery operating data indicative of a condition of the battery; determine, by a determination module (116), aging indicators associated with degradation mechanisms of the battery based on the battery operating data; calculate, by a DAIS calculation module (118), a Dynamic Aging Influence Score (DAIS) based on a weighted combination of the aging indicators; estimate, by a SOH estimation module (120), a State of Health (SOH) of the battery based on the DAIS and the battery operating data; determine, by an event determination module (122), an event associated with the battery based on the estimated SOH; and transmit, by a transmission module (124), an alert signal to a device associated with the battery based on the determined event.

No. of Pages : 45 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :22/10/2025
(21) Application No.202541102057 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR SEMANTIC TOPIC AND SENTIMENT ANALYSIS OF CUSTOMER CONVERSATIONS

(51) International classification	:G06F0040300000, G10L0015180000, H04L0051216000, G10L0015260000, G06N0003080000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Immensitas Private Limited
Address of Applicant :2nd floor, Municipal No.9, Ward No.63, Adugodi Madiwala Range, Block 3, Prestige Blue Chip Software Park, Hosur Main Road, Dairy Colony, Bengaluru, Karnataka - 560029, India Karnataka India
(72)**Name of Inventor :**
1)Rahul R
2)Anirudh Srinivasan

(57) Abstract :
A computer-implemented system 100 for semantic topic and sentiment analysis of customer conversations is disclosed. The computer-implemented system 100 comprises one or more processors 102A-N, a memory 104, and a customer data platform (CDP) 106 storing customer profiles 108A-N associated with conversational transcripts 110A-N. The one or more processors 102A-N process audio recordings to generate transcribed text segments and dense vector representations using a domain-adaptive transformer fine-tuned through masked language modelling and next sentence prediction. Uniform Manifold Approximation and Projection (UMAP) reduces the dimensionality of the vector representations, which are grouped into topic clusters using unsupervised clustering. A representative conversation is selected based on proximity to a cluster centroid. A large language model generates a semantic topic label, and a joint sentiment classifier determines a binary sentiment label. The resulting topic-sentiment information is integrated into customer profiles for downstream predictive models. FIG. 1

No. of Pages : 42 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2025

(21) Application No.202541103908 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR DETERMINING AN EXERCISE-CAPACITY INDICATOR ASSOCIATED WITH A USER

(51) International classification	:G16H 50/30, C04B 35/00, G01S 3/84, B61K 9/04, F17B 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ultrahuman Healthcare Private Limited
Address of Applicant :AM Chambers, 2nd & 3rd Floor, Survey No 49/1, 49/3 Garvebhavipalya, 7th Mile, Hosur Main Road, Bommanahalli, Bangalore, Bangalore South, Karnataka-560068, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)KUSHWAHA, Ayush
2)SHREY, Shwetank
3)PURANAM, Akash Raja
4)AGARWAL, Aman
5)SINGHAL, Vatsal
6)KUMAR, Mohit

(57) Abstract :
The present disclosure relates to system and method for determining an exercise-capacity indicator associated with a user. The method comprises receiving a set of physiological data associated with the user, the set of physiological data being captured by one or more sensors of a wearable device. The method further comprises determining a set of features based at least on the set of physiological data. The method further comprises identifying an exercise modality based on an analysis of the received set of physiological data and the determined set of features. The method further comprises assigning a weight to each feature of the set of features based at least on the identified exercise modality. The method furthermore comprises determining the exercise-capacity indicator associated with the user based on the identified exercise modality and the weighted set of features. FIG. 2

No. of Pages : 55 No. of Claims : 29

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2025
(21) Application No.202541103910 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A MODULAR HEALTH TRACKING BAND

(51) International classification	:A61B 5/00, A44C 5/14, G06F 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ultrahuman Healthcare Private Limited
Address of Applicant :AM Chambers, 2nd & 3rd Floor, Survey No 49/1, 49/3 Garvebhavipalya, 7th Mile, Hosur Main Road, Bommanahalli, Bangalore, Bangalore South, Karnataka-560068, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)KUSHWAHA, Ayush
2)SHREY, Shwetank
3)PURANAM, Akash Raja
4)AGARWAL, Aman
5)SINGHAL, Vatsal
6)KUMAR, Mohit

(57) Abstract :
The present disclosure relates to a modular health tracking band [100] and a wearable device [200]. The modular health tracking band [100] comprising at least one modular electromyograph (EMG) unit [108] comprising a set of sensor modules, arranged on an inner surface of the modular health tracking band [100]. The set of sensor modules further comprises a plurality of electrode pads, and one or more sensor module docking frames. The plurality of electrode pads and the one or more sensor module docking frames are configured to hold the set of sensor modules in contact with the skin of the user. The modular health tracking band [100] further comprising at least one display unit [110]. The display unit [110] is arranged on a lateral surface of the modular health tracking band [100]. Fig. 1

No. of Pages : 49 No. of Claims : 27

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2025

(21) Application No.202541105208 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-DRIVEN SYSTEM AND METHOD FOR ACCELERATED TRIAGE AND DIAGNOSTICS OF PHYSICAL ASSET INCIDENTS REPORTED THROUGH HETEROGENEOUS COMMUNICATION CHANNELS.

(51) International classification	:H04L0041060400, G16H0040200000, G06N0005040000, H04L0041060000, H04W0036300000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)LetUSense Solutions Private Limited
Address of Applicant :VIVA 306, SJR VERITY, Kasavanahalli,Off Sarjapur Road. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Sheeba Shoukat

(57) Abstract :
Title: An Ai-Driven System and Method for Accelerated Triage and Diagnostics of Physical Asset Incidents Reported Through Heterogeneous Communication Channels. AI-driven system and method for accelerating incident resolution concerning physical assets (such as monitor, coffee machine, server) or a location (meeting room or restroom) in enterprises, residential properties, and commercial buildings by minimizing reliance on subjective human input is disclosed. FIG.1

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2025

(21) Application No.202541105538 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYRINGE

(51) International classification	:A61M 5/00, A61M 5/31, A61M 5/50	(71)**Name of Applicant :** **1)MUDUNURI, Venkata Rama Krishna Raju** Address of Applicant :Villa no.38 Iris by Sriven Avenues, Narsingi Hyderabad, Telangana 500075, India. Telangana India (72)**Name of Inventor :** **1)MUDUNURI, Venkata Rama Krishna Raju**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention discloses a syringe (100) and a method of administering a fluid using the syringe (100). The syringe (100) comprises a barrel (102) having a proximal end (104) and a distal end (106) and defining a bore (108) configured to receive a fluid. A plunger (200) is slidably received within the bore (108) in sealing engagement with an inner surface (110) of the barrel (102), with a proximal end (204) of the plunger (200) extending outwardly from the proximal end (104) of the barrel (102). A loop (300) is disposed at the proximal end (204) of the plunger (200) to receive a thumb of a user, the loop (300) having a thumb-contacting surface (304) comprising a grip-enhancing material, and a pad (320) attached to the proximal end (204) distributes pressure applied to the loop (300) during depression of the plunger (200). At least one finger flange (400) extends from an outer surface (112) of the barrel (102) to receive at least one finger of the user and includes a grip-enhancing surface (402) that resists slippage of the finger. In the method, the user receives the thumb in the loop (300) and the fingers on the at least one finger flange (400), draws the fluid into the bore (108) by withdrawing the loop (300), and expels the fluid by depressing the loop (300) while the pad (320) distributes the applied pressure. Figure 1 is the representative figure.

No. of Pages : 61 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/10/2025
(21) Application No.202541105663 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SOUND INSULATED POD

(51) International classification	:H04W0088080000, H01L0021673000, F04D0029320000, G06F0001160000, H01L0033120000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)FEATHERLITE OFFICE SYTEMS (P) LTD
Address of Applicant :#16/A, Millers Tank Bund Road, Vasanthnagar, Bangalore – 560052, Karnataka, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)Mr. DHIREN GOPAL

(57) Abstract :
TITLE: “A SOUND INSULATED POD” ABSTRACT 5 10 15 A sound insulated pod (100) includes a frame (110), plurality of sidewalls (200), flooring (500), and ceiling (400). The sidewalls (200) are supported by the frame (110) and extend between proximal and distal ends. Each sidewall (200) includes a first layer (210) facing the pod interior and slots (212) having predefined depths. A second layer (220) disposed adjacent to the first layer (210). A third layer (230), disposed adjacent to the second layer (220) and opposite the first layer (210), includes projections (232), each having a base portion (234) and tip portion (236). The flooring (500) connects to the frame (110) and distal ends of the sidewalls (200), while the ceiling (400) connects to the frame (110) and proximal ends of the sidewalls, thereby forming an acoustic enclosure. FIG. 4b is the representative figure

No. of Pages : 26 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2025

(21) Application No.202541106840 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ESTIMATING BIOLOGICAL MUSCLE AGE OF A USER

(51) International classification	:H04H 60/52, G06F 21/31, H04H 60/45, H04L 67/306, H04N 19/162	(71)**Name of Applicant :** **1)Ultrahuman Healthcare Private Limited** Address of Applicant :AM Chambers, 2nd & 3rd Floor, Survey No 49/1, 49/3 Garvebhavipalya, 7th Mile, Hosur Main Road, Bommanahalli, Bangalore, Bangalore South, Karnataka-560068, India Bangalore Karnataka India (72)**Name of Inventor :** **1)KUSHWAHA, Ayush** **2)SHREY, Shwetank** **3)PURANAM, Akash Raja** **4)AGARWAL, Aman** **5)SINGHAL, Vatsal** **6)KUMAR, Mohit**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A system (100) and a method (200) for estimating biological muscle age of a user are disclosed. The system (100) includes a wearable device (102) having a set of electromyography (EMG) electrodes (104), and a processor (106). The processor (106) receives a set of surface electromyography (sEMG) signals associated with muscles of the user from the set of EMG electrodes (104), processes the received set of sEMG signals to extract a set of EMG features, and estimates, via a first trained model (108), the biological muscle age of the user based at least on the set of EMG features. FIG. 1

No. of Pages : 38 No. of Claims : 29

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :05/11/2025

(21) Application No.202541107259 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INSECTICIDAL COMPOSITION AND A PROCESS FOR ITS PREPARATION

(51) International classification	:C12Q 1/00, C12P 1/00, G11B 5/65, C12N 1/00, C08G 63/78
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CROPNOSYS (INDIA) PRIVATE LIMITED
Address of Applicant :9th Floor, Arliga Northstar (Beside RMZ Galleria Mall), Opposite Police Station, Yelahanka, Bengaluru-560064, Karnataka, India Bengaluru Karnataka India
2)GODREJ AGROVET LIMITED
(72)**Name of Inventor :**
1)PULLAIAH NARAHARI
2)PRASANTH KUMAR MAVATI
3)REKHA PUTTASWAMY
4)ONKAR SUNIL NALAWADE

(57) Abstract :
ABSTRACT The present disclosure relates to an insecticidal composition and a process for its preparation. The insecticidal composition comprises Flonicamid as a first insecticide, Spinosad as a second insecticide, and a third insecticide selected from the group consisting of Fipronil, Diafenthiuron, Tolfenpyrad, Bifenthrin and Pyriproxifen. The insecticidal composition of the present disclosure is effective against broad-spectrum insect species and control of sucking pest, thrips and lepidopterans. The process for the preparation of the insecticidal composition is simple, economic and environment friendly.

No. of Pages : 51 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :21/11/2025

(21) Application No.202541115099 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : miRNA PROFILE OF STROKE SERA PRECONDITIONED HUMAN UMBILICAL CORD - MESENCHYMAL STEM CELL DERIVED EXOSOMES

(51) International classification	:C12N 5/0775, A61K 35/28, C12N 1/02, C12N 5/02, C12N 5/095
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JAWAHARLAL INSTITUTE OF POSTGRADUATE MEDICAL EDUCATION AND RESEARCH (JIPMER)
Address of Applicant :Jawaharlal Institute of Postgraduate Medical Education and Research (JIPMER), Dhanvantri Nagar, Puducherry – 605 006, India Puducherry Pondicherry India
(72)**Name of Inventor :**
1)Prof. Sunil K Narayan
2)Simy Grace Cherian

(57) Abstract :
The present invention provides a method for isolating exosomes from mesenchymal stem cells. The method includes obtaining conditioned media from mesenchymal stem cells preconditioned with diseases sera cultured in exosome-depleted medium, performing a first centrifugation of the conditioned media at 500 × g for 10 minutes at 4°C to remove cell debris, performing a second centrifugation of the supernatant at 2000 × g for 30 minutes at 4°C to remove apoptotic bodies and larger vesicles, adding polyethylene glycol 4000 at a concentration of 8.5% and sodium chloride at a concentration of 0.4 M to the supernatant, incubating the mixture overnight at 4°C to allow vesicle precipitation, and performing a final centrifugation at 3220 × g for 60 minutes at 4°C to pellet the exosomes. The method eliminates the requirement for ultracentrifugation equipment while maintaining high recovery efficiency and preserving exosome morphological integrity. Microarray profiling of hUCMSC-derived exosomes identified differentially expressed miRNAs following pretreatment with ischemic stroke patient serum. Bioinformatic analysis revealed enrichment of target genes involved in neuroprotection, angiogenesis, inflammation, and neuronal repair pathways. Figure 1

No. of Pages : 36 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :21/11/2025
(21) Application No.202541115321 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR LOWER LIMB REHABILITATION AND A METHOD OF OPERATING THE SAME

(51) International classification	:A61B 5/00, A61B 5/11, A61B 5/103, A61B 5/04, A61B 5/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :Kattankulathur, Chennai-603203, Tamil Nadu, India Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)ARJUNAN SRIDHAR
2)VASUDEVAN ARJUN
3)SARAVANAN DINESHA
4)SURESHBABU CHANDRA PRAKASH

(57) Abstract :
ABSTRACT The present disclosure relates to a system (100) for lower limb rehabilitation and a method of operating the same. The system (100) comprises a frame (120), a pedal (102) resisted by a resilient member (104), a force sensor (106), a position sensor (110), a controller (114), a simulator (116) and a user interface (118). The force sensor (106) generates a force signal indicative of the force applied by a lower limb of a user to the pedal (102), and the position sensor (110) generates a position signal indicative of the position of the pedal (102). The controller (114) processes the force signal into force data, which the user interface (118) displays during a rehabilitation session, while the simulator (116) updates a simulated task, such as a driving task, based on the position signal. The system (100) thereby quantifies lower limb effort during an engaging, goal-oriented exercise.

No. of Pages : 50 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :10/12/2025

(21) Application No.202541124935 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A DEVICE FOR PAINLESS SURGICAL SUTURE REMOVAL WITH SKIN PROTECTION TIP

(51) International classification	:A61B 17/00, A61B 17/32, A61B 90/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MANIPAL ACADEMY OF HIGHER EDUCATION
Address of Applicant :MADHAV NAGAR, MANIPAL, 576104, KARNATAKA, INDIA Karnataka India
(72)**Name of Inventor :**
1)CHRISTOPHER SAM THOMAS
2)MOHAMMAD SAMEER AHMED
3)SAHAJANYA SRIKANTH

(57) Abstract :
An atraumatic surgical scissor device (100) is disclosed, configured to cut a suture thread (111) while preventing contact between a cutting element and a skin surface. The device (100) comprises a body (101), at least one cutting blade (103), an atraumatic tip guard (105) extending distally beyond the at least one cutting blade (103) and comprising a reflective surface, an illumination unit (107) mounted on the body (101) adjacent to the atraumatic tip guard (105), and a power source (109) coupled to the illumination unit (107). The reflective surface reflects light from the illumination unit (107) toward the suture thread (111), while the atraumatic tip guard (105) elevates the suture thread (111) away from the skin surface and shields the skin surface from the at least one cutting blade (103), such that a single protective element performs an elevating, a shielding, and a light-redirecting function without a separate instrument. (FIG.1)

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/12/2025

(21) Application No.202541125919 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A combustion system for a two-stroke internal combustion engine and method thereof

(51) International classification	:F01L 7/02, F02B 75/02, F02B 23/08, F02B 23/10, F02B 75/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology (IIT Madras)
Address of Applicant :The Dean, Industrial Consultancy & Sponsored Research (IC&SR), Indian Institute of Technology Madras, Sardar Patel Road, IIT P.O, Chennai Tamil Nadu India
2)Indian Institute of Technology (IIT Palakkad)
(72)**Name of Inventor :**
1)Anand Thirumalai Nallan Chakravarthy
2)Nithin Joseph Tito

(57) Abstract :
The present invention relates to a two-stroke internal combustion engine (100) with uniflow scavenging, comprising an asymmetrical wedge-shaped combustion chamber (208), a pre-chamber (300), a rotary intake valve (400) and a sealing arrangement (500), and a method thereof. The combustion chamber (208), formed by the cylinder head (204) and the hemispherical face of the rotary intake valve (400), provides an enlarged volume on the pre-chamber (300) side. The pre-chamber (300) receives the charge from the rotary intake valve (400) through an auxiliary inlet pipe (304) and is enriched by a fuel injector (306). The rotary intake valve (400) fluidly connects to the combustion chamber (208) through a curved passage (402) during scavenging, reducing short-circuiting. To prevent high-pressure gas leakage, the sealing arrangement (500) comprises dual labyrinth seals (502a, 502b) with split stator rings (506a, 506b). The present invention minimizes charge short-circuiting and improves combustion and thermal efficiency.

No. of Pages : 36 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/12/2025

(21) Application No.202541126124 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEMS AND METHODS FOR DETECTING AND LOCALIZING AN EVENT ASSOCIATED WITH AN UNDERGROUND PIPELINE

(51) International classification	:G01M 3/24, F17D 5/06, F17D 5/02, G01H 9/00, G08B 29/18	(71)**Name of Applicant :** **1)INDIAN INSTITUTE OF TECHNOLOGY MADRAS (IIT Madras)** Address of Applicant :The Dean, Industrial Consultancy & Sponsored Research [IC&SR], Indian Institute of Technology Madras, Sardar Patel Road, IIT P.O, Chennai- 600 036, Tamil Nadu, India. Chennai Tamil Nadu India **2)Gail (India) Limited**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Balaji Srinivasan**
(33) Name of priority country	:NA	**2)Sheetal Kalyani**
(86) International Application No Filing Date	: :01/01/1900	**3)Anand Venganallur Padmanabhan** **4)Neethu Sasikumar**
(87) International Publication No	: NA	**5)Arsath Abbasali Ayubali**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)Nandhakishore Chitra Sundararajan** **7)Tanjore Puli Yuvaraj**
(62) Divisional to Application Number Filing Date	:NA :NA	**8)Lalit Kumar Maurya** **9)Sanjeev Kumar**

(57) Abstract :

ABSTRACT SYSTEMS AND METHODS FOR DETECTING AND LOCALIZING AN EVENT ASSOCIATED WITH AN UNDERGROUND PIPELINE A method for detecting and localizing an event associated with an underground pipeline is provided. The method includes transmitting one or more sensing pulses through an optical fiber deployed along the underground pipeline. The method includes receiving backscattered sensing signals generated in response to propagation of the one or more sensing pulses through the optical fiber. The method includes deriving, from the backscattered sensing signals, distributed acoustic sensing (DAS) data and distributed temperature sensing (DTS) data. The method includes simultaneously processing the DAS data and the DTS data to identify one or more acoustic signatures and one or more temperature signatures associated with the event. The method includes detecting the event, using an artificial intelligence (AI) module, based on the simultaneous processing of the identified one or more acoustic signatures and the identified one or more temperature signatures. The method includes determining, using the AI module, a location of the event along the underground pipeline. Figure 1

No. of Pages : 43 No. of Claims : 19

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015295 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CLAY POT MAKING ASSISTIVE DEVICE

(51) International classification	:B28B 17/00, B28B 1/00, B28B 11/24, B28B 1/02, B28B 1/087
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Nivrity Sinha

(57) Abstract :
A clay pot making assistive device, comprising of a platform 101 positioned on a fixed surface and configured with plurality of suction cups 102 to secure the platform 101 on the surface, a touch interactive display panel 103 installed on the platform 101 to provide command for making a clay pot along with specifying dimension of the pot, a rotary actuator 105 coupled with the plate 104 to rotate slightly to allow the user to perform shaping of the clump by applying pressure on the limp using finger, a microphone 106 embedded in the platform 101, to provide command regarding need of water, an electronically controlled sprayer 107 configured with a chamber 108 installed on the platform 101 to spray water on the lump of clay being shaped, a holographic projection unit 109 installed on the platform 101 to project three dimensional projection in space depicting step by step demonstrations.

No. of Pages : 18 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015296 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BATHING ASSISTIVE DEVICE

(51) International classification	:A47K 3/12, A61H 33/00, A47K 3/28, A47K 3/022, A47K 17/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Bhagyashree Nadig Y S

(57) Abstract :
A bathing assistive device, comprises of a platform 101 configured with plurality of suction cups, a vertical extendable pole 103 configured with a seat 104 for accommodation of a user, an artificial intelligence-based imaging unit 105 to determine presence and height of the user standing in vicinity of the platform 101, a microphone 106 to provide input voice commands, a motorized swivel arm 107 to provide a controlled rotational movement to the seat, a weight sensor to detect weight of the user sitting over the seat, a telescopic rod 108 to extend and position a suction unit 109 over the bathtub surface to establish a secure vacuum seal over the bathtub surface, a tilt sensor to detect any unintended tilting or instability of the platform 101, an electromagnetic lock 110 to temporarily restrict rotational movement of the seat until stability is restored.

No. of Pages : 18 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015297 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : RESTING AID DEVICE FOR FOUR-WHEELER VEHICLE

(51) International classification	:A61H 3/04, A61G 5/10, B60N 2/90, B60N 2/22, A61H 3/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Shashikala A R**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A resting aid device for four-wheeler vehicle, comprises of a frame 101 with plurality of suction cups 102 affixed on backrest of a vehicle's seat, an artificial intelligence-based imaging unit 103 detect presence of a user on seat, imaging unit 103 activates a pair of motorized rollers 104 allow user to secure straps over user's body portion, a pair of permanent magnets 105 embedded within free ends of straps and a pair of electromagnets 106 disposed on frame 101, electromagnets 106 secure straps by attracting permanent magnets 105, a primary air allow user to comfortably rest over seat, a U-shaped cushioned structure 108 provide tilting movement to structure 108 towards user's neck portion and an oxygen sensor installed over platform to detect oxygen levels within vehicle, imaging unit 103, a pair of telescopic pushers 110 installed over frame 101 to extend and break window of vehicle.

No. of Pages : 19 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015298 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NAIL PAINT BOTTLE HOLDING DEVICE

(51) International classification	:A45D 29/00, B05C 13/02, B65G 47/91, B25J 15/06, A45D 29/05
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Pallavi R

(57) Abstract :
A nail paint bottle holding device comprises of, a circular base 101 installed on a fixed surface and configured with plurality of suction cups 102 to secure base 101, an inverted U-shaped frame 104 accessed by a user to align a nail paint bottle, a microphone 105 embedded on base 101 to provide voice command, a proximity sensor configured in frame 104 to detect positioning of bottle, an imaging unit 106 integrated with a processer to capture and process images of bottle, a pair of electromagnetic pushers 107 installed in frame 104, to extend and press bottle from each side, a level sensor installed on frame 104 to monitor level of nail paint, and a motorized ball and socket joint 109 installed between frame 104 and link 103, to tilt frame 104 to allow flow of paint toward bottle's opening.

No. of Pages : 16 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641015299 A
(19) INDIA
(22) Date of filing of Application :11/02/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : PET CARRYING ASSISTIVE DEVICE

(51) International classification	:A01K 29/00, A01K 27/00, A01K 15/02, A01K 11/00, G06K 19/077	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Krishnakant Bhole**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A pet carrying assistive device, includes a housing 101 with multiple wheels 102 for easy manoeuvrability, a T-shaped handle 103 attached via a telescopic rod 104 for optimal gripping, an imaging unit 105 works with an ultrasonic sensor to detect the user's height, allowing the handle 103 to adjust accordingly, an RFID module verifies the pet's identity using RFID tags on the pet's collar, a motorized lid 107 allows secure accommodation of the pet inside the housing 101, a temperature and humidity sensors monitor internal conditions, while motorized iris pores 108 regulate airflow. A gyroscopic and tilt sensor continuously detects angular displacement and tilt variations, and a hydraulic actuator 109 at each corner of the housing 101 to ensure a stable and level surface for the pet.

No. of Pages : 21 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015300 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRONIC GADGET HOLDING DEVICE

(51) International classification	:A47C 7/62, A61B 8/00, A47C 7/72, A61B 8/08, A47C 1/032	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. K Prakash**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An electronic gadget holding device, comprising a cylindrical body 101 configured with a motorized clamping unit 102 adapted to be secured over a frame portion of a chair, clamping unit 102 to extend and grip frame to provide support to body 101 on chair, an imaging unit 103 to detect position of the chair, an ultrasonic sensor to detect height of a user sitting over the chair, a telescopic rod 104 to get optimally aligned in proximity to the seated user, a pair of horizontal arms 105 connected to each other via motorized swivel joint 106, an IR (infrared) proximity sensor installed over body 101 to detect positioning of user relative to rod 104 and a pair of horizontal arms 105 connected to each other via motorized swivel joint 106 to provide rotational movement to arms 105 allowing user to accommodate an electronic gadget over holder 110.

No. of Pages : 16 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015301 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED FOOD-GRADE MOLDING PLATES MANUFACTURING DEVICE

(51) International classification	:A21C 15/00, A21C 9/08, A21C 11/16, A21C 11/10, A21C 9/06	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)John Bennet**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated food-grade molding plates manufacturing device, comprises of a platform 101, a sheet feeding unit with roller-driven feeding unit 105 delivery and ultrasonic thickness sensor for uniformity, a handling means with robotic arm 115, quick-lock clamps 116, and motorized ball-spline guiding assembly 117 for precise die and punch placement, a clamping module with motorized clamps 106 mounted on extendable L-shaped rod 107 via motorized ball-and-socket joints with adaptive pressure control and laser alignment sensors for accurate sheet positioning, a pressing module with frame 112, electromagnetic strips 113, hydraulically operated bar 114, and heating unit for cavity formation, a trimming module with motorized deburring tool 108 on extendable L-shaped pole 109 with rotary joint for providing rotary motion to remove excess material, a polishing unit with motorized buffing pad 110 on articulated linkage 111, a sensor suite with tactile, force, vibration, and optical sensors, a speaker for audible alerts.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015302 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED INSECT-REPELLENT DEVICE FOR WINDOWS

(51) International classification	:A01M 29/12, E06B 9/68, E06B 9/52, E06B 9/40, A01M 29/34	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. M Sudharsan**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated insect-repellent device for windows, comprises of an elongated body 101 with a plurality of suction cups 102 on top portion, for adhering to a window panel of a premise, an imaging unit to detect approaching insects from outside of the window, a primary motorized roller 104 coiled with an insect repellent sheet 109, to unwind the sheet 109 over surface of the window, that is translated towards an open lower portion in view of preventing the insects to enter into premise, a humidity sensor 105 for detecting humid temperature inside the premise, a plurality of motorized iris lids 106 to allow the air to flow through the sheet 109, a secondary motorized roller 107 coiled with a water-resistant fabric, mounted within the body 101, a rain sensor 108 for preventing rain droplets to enter through the window for a comfortable ambience.

No. of Pages : 15 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015303 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CRICKET STUMP AND BAIL MANUFACTURING DEVICE

(51) International classification	:A01G 23/06, A01K 67/00, A63H 33/04, A41H 43/00, A01K 67/033	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. N Baskar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A cricket stump and bail manufacturing device, comprises of a platform 101 with a storage unit 102 for larger and smaller rounded wooden blocks, an input module for user-specified manufacturing commands, a transferring unit with grippers 105 on robotic links 106 to move blocks from storage to a holding unit, and a carving assembly with a blade 107 on a dual-axis slider 109 for precise carving, the input module includes a microphone 103 or user-interface linked to a microcontroller, the storage unit 102 has chambers with weight sensors to alert users via a speaker 112, when blocks are low, the holding unit features U-shaped clamps 110 on sliders adjusting to block dimensions via laser sensors, a reservoir collects finished stumps and bails, an inspection module with imaging unit and LiDAR ensures quality, the device automates precise, user-defined production of cricket stumps and bails.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015304 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED BOLUS MASSAGE THERAPY DEVICE

(51) International classification	:A61H 7/00, A61H 9/00, A61H 23/02, A61H 15/00, A61H 33/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Chinmayi G V**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated bolus massage therapy device, comprising, a housing 101 configured with suction cups 102 for adhering on fixed surface, a display panel 103 to provide input commands, a first imaging unit 104 to detect presence of user's hand, a motorized slidable flap 107 for getting translated upwards, a speaker 108 for producing audible guidance, a second imaging unit for detection of filling, a pair of nozzles 110 for dispensing water, a heating unit 112 to provide heat, a first and second robotic arms 113, 114 to transfer rice and hallow in a tray 116, a compartment 117 for storing cloths, a robotic link 118 for grabbing cloth, first robotic arm 113, transfers mixture to cloth, a third robotic arm 119 to form bolus pouch, a receptacle 120 stored with heated oil, a platform for allowing a patient to lie down, a robotic limb 121 for grabbing pouch.

No. of Pages : 20 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015305 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FARM PRODUCE COLLECTION DEVICE

(51) International classification	:A01D 46/30, G06Q 50/02, A01D 46/26, B25J 19/04, G01N 21/359
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Abhilash B S

(57) Abstract :
A farm produce collection device, comprises of a plurality of cylindrical shaped bodies 101 interconnected by means of plurality of links 102, a user-interface installed in a computing unit to provide input command, an imaging unit 103 integrated with a processer to capture and process images of surrounding, a motorized wheels 104 attached with bodies 101 a first and last body 101, to provide movement to the device, a NIR module installed on one of the body 101 to monitor ripeness a motorized gripper 106 installed in continuation to the lid 105 to open and collect the farm produce, and a plurality of iris pores 107 configured with each the bodies 101 to open and allow circulation of air in each the bodies 101.

No. of Pages : 17 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015306 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SACK FILLING AND SEALING DEVICE

(51) International classification	:B65B 43/30, B65B 1/04, B65B 43/54, B65B 43/46, B65B 1/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Ganesh Kumaar K Y

(57) Abstract :
An automated sack filling and sealing device comprising a hollow cuboidal frame 101 installed on a ground surface, an imaging unit 102 to detect a pile of grain, a plurality of motorized wheels 103 to position frame 101 in proximity to pile, a plurality of motorized clippers 105 to clip the sacks for securing, a pair of motorized gripper 107 to grip mouth portion of sack, a pair of motorized slider 108 to provide upward movement to channel 106, a motorized hinge joint 109 to provide angular movement to channel 106 for transferring grain in sack, a robotic link 110 to align a sack sealing unit 111 with the mouth portion to carrying out sealing of sack, a speaker 112 to notify user for packaging, an air blower 113 to blow air in mouth portion of sack to keep sacks inflated for allowing grains to settle properly in sacks.

No. of Pages : 19 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015307 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED HAIR GROOMING DEVICE

(51) International classification	:B26B 19/38, A45D 44/00, B26B 19/48, B26B 19/44, B26B 19/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Shahul Hammed

(57) Abstract :
An automated hair grooming device comprises of a frame 101 to position the user's head portion in proximity to the frame 101, plurality of suction cups 104 to secure the frame 101 over a vertical surface, a microphone 105 to provide input voice commands, an inbuilt microcontroller processes the input commands, an electronic nozzle 106 to transfer the hair nourishing oil from the chamber 107 into a hollow extendable cylindrical body 108, an artificial intelligence-based imaging unit 110 to determine positioning of the user's head, a plurality of soft conical members 111 to receive the hair nourishing oil, a motorized iris lid 112 to open and dispense the hair nourishing oil over the user's hair, a motorized hair massaging unit 113 to ensure even distribution of the nourishing oil, a motorized dual-axis lead screw arrangement 115 to provide controlled vertical and horizontal movement to the body.

No. of Pages : 19 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015308 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CHAIR-MOUNTED ASSISTIVE FOOD ITEM CHOPPING DEVICE

(51) International classification	:A47J 47/00, A47J 43/07, A47J 44/00, A47B 77/08, A47J 43/044
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Mahaboob Pasha

(57) Abstract :
A chair-mounted assistive food item chopping device comprising a platform 101 with a L-shape telescopic bar 102 to extend and position motorized clamping unit 103 to grip rod for securing platform 101 mover chair, a microphone 104 to provide input commands for securing platform 101 with chair, an imaging unit 105 to determine supporting rod of chair, a motorized dual axis slider 107 to slide particular chopping board of surface friction to user for position food over board to perform chopping, a motorized vacuum unit 110 to remove particles ensuring a clean chopping surface, a motorized grinding unit 109 to rotate at a predefined speed for sharpening knife, a first aid compartment 111 providing medical supplies for accidental injury during chopping, a LDR to detect ambient light intensity a plurality of LED lights 112 for optimal visibility during operation.

No. of Pages : 17 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015309 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BEVERAGE CONTAINER COVERING DEVICE

(51) International classification	:B67B 3/20, B65B 7/28, B67B 7/16, B67B 3/00, B67B 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Likhith S R

(57) Abstract :
A beverage container covering device, comprises of a platform 101 with a holding means which includes at least one pair of clippers 103 with multiple suction unit 104 to grip a container 102, a pressure sensor along with a tactile sensor with the holding means to regulating the holding by preventing damage to the container 102, an input module on the platform 101 provide input commands for opening/closing of the container 102, a motorized hinge joint 106 between platform 101 and holding means open/close the container 102, an inspection module including imaging unit 107 and ultrasonic sensor, mounted on the platform 101 to detect dimension of the container 102, a drawer arrangement 108 to extend/retract based on the detected dimensions of the container 102 to cover the container 102 entirely, a sealing arrangement 109 within the platform 101 airtight the container 102 after the container 102 is closed.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015310 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED BORE WELL MAKING DEVICE

(51) International classification	:E21B 7/04, E21B 7/06, E21B 7/08, E21B 44/00, E21B 47/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Afroj Alam

(57) Abstract :
An automated bore well making device, comprising a housing 101, an input means to provide input commands regarding initiation of drilling of a bore well, an imaging unit 104 to determine the exact location, alignment, and orientation of the bore well drilling site, a pair of caterpillar wheels 105 to provide mobility over uneven terrain, a DTH (Down-The-Hole) drilling rig 106 integrated with the housing 101 via an L-shaped hydraulic rod 107 to perform pilot drilling and rock penetration at the bore well site, a cylindrical body 108 having an open top comprising a handle 109 at its upper portion, that to engaged by a hook at the end of a hydraulic L-shaped bar 110, a shutter 112 to control the discharge of collected mud and debris and a drag and hoe 113 via a L-shaped hydraulic link 114 configured to grasp solid rocks.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015311 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED WATCH MAINTENANCE DEVICE

(51) International classification	:G04B 47/00, G04B 47/06, B08B 1/04, G04D 7/00, G04D 7/12	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Paruchuru Jyothi Satya Siva Naga Swaroop**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated watch maintenance device comprising a housing 101 with a plurality of suction cups 102 for securing housing 101 on surface, a motorized slidable flap 103 to accommodate a watch inside housing 101, a microphone 104 to provide input voice commands for cleaning watch, an imaging unit 105 to determine dimensions of watch, deposition of dirt/dust, broken glass, incorrect time, hands and battery status of watch, a pair of motorized clamps 106 to grip watch for securing position, an electronically controlled nozzle 108 for spraying cleaning liquid stored in a vessel 110 onto watch, a motorized brush 111 to rotate for scrubbing and removing dust, a pair of robotic arms 113 for grabbing suitable spare parts including but not limited to new battery cells and hands of watch, that are of stored in a container 114 installed within housing 101, to replace with damaged parts, for repairing watch.

No. of Pages : 18 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641015312 A
(19) INDIA
(22) Date of filing of Application :11/02/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED BADMINTON RACQUET MAINTENANCE DEVICE

(51) International classification	:A63B 67/187, A63B 69/40, A63B 47/02, A63B 69/00, A63B 67/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Varalakshmi T

(57) Abstract :
An automated badminton racquet maintenance device, comprises of a housing 101 positioned on a ground surface and configured with a platform 102 to be placed with a badminton racquet, multiple suction cups 117 arranged underneath housing 101 each via an extendable link 107 to adhere the hosing over surface, an input means 103 integrated with the housing 101 for enabling a user to give input commands for carrying out maintenance of the badminton racquet, an imaging unit 104 mounted within the for detecting presence of the badminton racquet on the platform 102, a telescopically operated gripper 105 configured on the platform 102 for changing orientation of the racquet on platform 102 in a manner to position head of the racquet between multiple curved-shaped members 106 installed on the platform 102, and an L-shaped telescopically operated link 118 to extend for pushing the members 106 against the head of racquet.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015313 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED LAWN MAINTENANCE DEVICE

(51) International classification	:A01D 34/00, A01D 34/63, A01B 45/02, A01D 34/64, A01D 43/063
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Anju Mathew

(57) Abstract :
An automated lawn maintenance device, comprising a body 101 for supporting the functional modules, an inspection module having artificial intelligence-based imaging unit 103 and proximity sensor for detecting rock-like objects on the lawn surface, a digging module having a trowel 104 mounted on articulated arm 105, a depth and a FSR for excavating soil around detected rock-like objects, a cutting module having rotary blades 106 mounted on robotic arm 107 for detaching rock-like objects adhered to soil, an object removal unit including a gripper 108 installed on robotic link 109 for grasping and removing rock-like objects, an artificial grass dispensing and levelling module having a drag-and hoe arrangement 110 with a gripper, a storage reservoir 113 and a levelling plunger 114 for filling lawn voids created after rock-like object removal.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015314 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SLAG CLEANING DEVICE

(51) International classification	:F27D 25/00, F23J 1/06, F27D 3/15, B60B 19/00, B60S 1/68	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)P Harika**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A slag cleaning device, comprises of a housing 101 mounted on omnidirectional wheels 102 for mobility, a cleaning plate 103 with a longitudinal cavity 104, the cleaning plate 103 including rectangular panels to vary cavity width based on slag characteristics, an L-shaped extendable rod 105 coupled to the plate 103 via a ball-and-socket joint 106 for multi-axis articulation, an AI-enabled camera 107 detects slag location and density, while an ultrasonic sensor measures thickness and density of slag deposits, a proximity sensor detects relative distance between the cleaning plate 103 and the serrated slat bed, a tilt sensor monitors inclination of the cleaning plate 103, a machine learning module processes operational data to optimize cleaning parameters over time, a speaker 108 and microphone 109 for audio alerts and voice commands respectively, and a communication module for remote monitoring and control of operations.

No. of Pages : 17 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015315 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED INSULATION LAYER SMOOTHENING DEVICE

(51) International classification	:G05D 1/02, E04F 21/00, G06T 7/00, E04G 23/02, E04F 21/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Divya Nair

(57) Abstract :
An automated insulation layer smoothening device comprises of, a platform 101 positioned on a floor surface, a user-interface installed in a computing unit wirelessly linked with device to provide input command, a rotatable imaging unit 102 integrated with a processer to capture and process a series of images, a plurality of motorized wheels 103 arranged underneath platform 101 to manoeuvre platform 101, an ultrasonic sensor installed on platform 101 to detect height of selected area, a telescopically operated rod 104 installed between each wheel 103 and platform 101, to extend and lift platform 101, a sharp blade 106 attached with arm 105 to remove excess insulation, for providing a finishing to insulation, a Peltier unit integrated in blade 106 to generate heat effect, and a vacuum based cleaning unit 107 arranged on platform 101 to extract residues and transfer into a waste chamber 108.

No. of Pages : 19 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015316 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HYDROGRAPHIC FILM TRANSFERRING ASSISTIVE DEVICE

(51) International classification	:G03H 1/22, G03B 21/11, G03B 21/132, G03B 21/00, G03B 31/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Hasan Hussain S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A hydrographic film transferring assistive device comprises of, a platform 101 with an elongated handle 102, equipped with a microphone 103 that allows the user to provide voice commands, an artificial intelligence-based imaging unit 104 to detect the shape of an object and motorized hinge joints 105 for angular adjustments, ensuring the platform 101 conforms to the object's shape, A set of suction units 106 generates negative pressure to hold the object securely in place, Once held, the device activates a holographic projection unit 107, synced with a speaker 108, to project a 3D visual guide on how to dip the object for film transfer, and a LiDAR sensor detects the user's hand dimensions and a drawer arrangement 109, adjusting the handle 102 for better comfort, while an optical sensor ensures correct alignment by triggering a red LED 110 light to indicate misalignment.

No. of Pages : 16 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015317 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED BRUSH BRISTLES TRIMMING DEVICE

(51) International classification	:A46D 3/04, A46D 1/00, A46D 1/04, A01G 3/08, A46D 3/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Geetha A**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated brush bristles trimming device, comprising a platform 101 with plurality of supporting legs for supporting at a height from ground surface, a touch interactive display panel 102 for providing input commands regarding trimming of bristles of a brush, an imaging unit 103 to detect presence, physical appearance, color and texture of brush bristles, a pair of robotic grippers 104 to grip, align brush underneath blade at optimum angle, a plurality of motorized pop-out balls 108 to rotate for aligning brush at optimum orientation, a trimming to trim the bristles, a primary blade has a segregated or straight edge, a secondary blade has a series of teeth that interlock with the primary blade's teeth to move rapidly to cuts bristles of brush, a vacuum-based cleaning unit 107 for withdrawing the cut bristles that are further stored in a receptacle.

No. of Pages : 16 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641015318 A

(19) INDIA

(22) Date of filing of Application :11/02/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED CYLINDRICAL CLAY OVEN MAKING DEVICE

(51) International classification	:B28B 11/14, B25J 15/06, B28B 11/16, B28B 11/12, B25J 9/16	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. S Sakthivel**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated cylindrical clay oven making device, comprising a platform 101 on a ground surface, a display panel 102 to provide input command, an imaging unit 103 to capture and process images, motorised wheels 104 to maneuver platform 101 to position near to drum, robotic gripper 105 to grip drum and place on a rotatable plate 106, multiple suction units 114 to hold drum, a first robotic arm 107 to cut top, bottom and selected portion of drum, motorised cutter 108 to create multiple openings, second arm 109 to carry out smoothing of cut edges of openings, grinding unit 110 to allow the grippers 105, speaker 111 to notify user, a chamber 112 to store thermal insulting material, third arm 113 to press and settle insulation material, a container 115 having suction pump to extract mixture, clippers 116 to align free end of pipe, fourth arm 117 to level mixture.

No. of Pages : 20 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015319 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED LED SHEET CUTTING DEVICE

(51) International classification	:B26D 7/01, B26D 5/00, H01L 33/00, B26D 5/08, B26D 7/06	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. M Rajeshwari**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated LED sheet cutting device comprises of a platform 101 positioned on a fixed surface, a touch interactive display panel 102 to provide input command for cutting a LED (light emitting diode) sheet, an inbuilt microcontroller processes the command, an artificial intelligence based imaging unit 103 to detect exact location of the sheet with respect to the platform 101, plurality of motorized wheels 104 to manoeuvre the platform 101, a motorized slider 105 to provide movement to plurality of telescopically operated rods 106 towards/away from each other, a motorized blade 107 to extend in downward direction for positioning the blades 107, a robotic arm 108 to align an active positive and negative terminals of a testing unit 109, a robotic gripper 110 to grip the defective strip and store in a waste box 111, a LiDAR (light detection and ranging) module to detect thickness of the sheet.

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015320 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED WEARABLE CHAIN FABRICATION DEVICE

(51) International classification	:B21L 9/06, A44C 27/00, A44C 25/00, A44C 5/00, G06F 3/0488	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Josephine R**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated wearable chain fabrication device, comprising a platform 101, with supporting legs 102 for fixed positioning, a chamber 103 for storing chain components, a touch interactive display panel 104 for providing input commands, an AI-based imaging unit 105 paired with a processor captures and processes images to detect user's neck dimensions, an L-shaped telescopic rod 106 with a motorized clipper 107 as an end-effector, a platform 101 slot 108 with pneumatic pins 109 that adjust to the chain component size, enabling folding through the slot 108, a proximity sensor detects chain components on the slot 108, a motorized clamp 110 to pull, fold, and connect them, a cylindrical hollow member 111 attached to the platform 101 via a telescopic bar 112 checks the folding of chain components, with pressure sensors and touch sensors, detects irregularities like excessive pressure or scratches and triggers a speaker 113 to alert the user.

No. of Pages : 18 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015321 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PLASTIC COMB MANUFACTURING DEVICE

(51) International classification	:B29B7/10, B29B17/00, B29C51/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Sanaskar Verma

(57) Abstract :
A plastic comb manufacturing device, comprising housing 101 with a chamber 102 stored with plastic waste, a touch enabled screen 103 to give input commands, a microcontroller processes the input commands, a motorized lid 104 for dispensing determined amount of plastic waste in a cylindrical hollow member 105, a motorized cutter 106 for grinding plastic paste, plurality of heating units 107 for melting plastic waste, a rotatory arrangement 108 configured with multiple dies 109 for positioning a particular die 109, an iris unit for dispensing liquid plastic within the die 109, a weight sensor for detecting liquid plastic within die 109, a Peltier unit 110 for cooling the molten metal, a robotic arm 111 for extracting the comb from the die 109, a motorized grinding unit configured a robotic link for smoothening edges of the comb, a speaker 113 to notifying the user to take away the comb.

No. of Pages : 17 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015322 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TIRE REPAIRING ASSISTIVE DEVICE

(51) International classification	:B60C 23/04, B60C 19/00, B60C 25/05, B29C 73/24, G01S 17/931
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S Saravana Kumar

(57) Abstract :
A tire repairing assistive device, comprising an elongated body 101 installed with a handle 102 to acquire a firm grip by a user, a microphone 103 embedded in handle 102 to provide voice command for performing repairing operation on torn tire, a speaker 104 to notify user to translate distal end of body 101 through surface of torn tire fusion of the tire's material an imaging unit 105 synced with a LiDAR module to scan tire, a laser emitter 106 to emit concentrated laser beam on torn area to align with a cuboidal frame 107 installed on distal end, an air blower 108 housed in frame 107 for clearing torn area, a glue dispenser 109 to dispense an adhesive solution to bond and seal area, a roller 110 unroll patches from roller 110 and a pair of robotic grippers 111 to unroll patches from roller 110.

No. of Pages : 18 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015323 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : OBJECT HANGING ASSISTIVE DEVICE FOR A BICYCLE

(51) International classification	:B62J 99/00, B62J 11/00, B62J 11/19, B62J 3/00, B62J 6/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Attar Mahay Sheetal**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An object hanging assistive device for a bicycle, comprising a plate 101 with a motorized C-shaped clamp 102 to be positioned in proximity to a mirror handle of a bicycle, an ultrasonic sensor in sync with an imaging unit 103 for determining dimensions of handle, an inflatable member 104 to securely mount clamp 102 onto handle, a microphone 105 to provide voice commands for assistance in hanging items, an extendable L-shaped link 105 for extending to deploy plate 101, a plurality of hooks 106 to hang bag, a slider 107 to adjust the gaps between hooks 106 for optimal positioning, an electromagnetic strip 108 for securing position of the bags, an extendable bar 109 to extend/retract for providing additional support, a motorized pivot joint for alignment of plate 101, an anemometer 110 is installed on plate 101 to measure wind speed.

No. of Pages : 18 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015324 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED CARPET EDGE TRIMMING DEVICE

(51) International classification	:B25J 9/16, G05D 1/02, A47L 11/40, B26D 5/00, B26D 7/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Katta Sri Ram

(57) Abstract :
An automated carpet edge trimming device comprising a platform 101 installed with a carpet, a user-interface installed with a computing unit wirelessly linked with the device to provide input command for trimming, a rotatable artificial intelligence based imaging unit 102, to generate a map of the carpet, a computing unit displays map of carpet to select portion of carpet to be trimmed, a plurality of motorized wheels 103 for maneuvering the platform 101, a robotic link 104 to align a holding unit 105, with edge of the carpet, a LiDAR module to detect thickness of the carpet, a robotic arm 106 to align motorized cutter 107 to cut extra portion from the carpet, a laser scanner to scan the surface to detect depressions/elevations, and a laser emitter 110 to project a beam to virtually guide user to find depressions/elevations on surface.

No. of Pages : 19 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015325 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED RING MANUFACTURING DEVICE

(51) International classification	:A44C 27/00, B25J 15/06, B21F 45/00, B26D 7/01, B26D 5/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Neha Arora**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated ring manufacturing device, comprising a platform 101 positioned on a fixed surface and configured with plurality of suction cups 102 to secure the platform 101 on the surface, a touch display panel 103 installed on the platform 101 to provide input command for manufacturing rings along with specifying dimension of the rings to be manufactured, a motorized roller 104 rolled with a wire, installed on the platform 101 to rotate in a pre-defined direction for unrolling estimated length of the wire, a robotic link 105 installed on the platform 101 to position a motorized cutter 106 attached with the link 105, in contact with the wire unrolled from the roller 104 , a semi-circular member 107 positioned on the platform 101 integrated with a motorized drawer arrangement 108 to enable the member 107 to adjust its dimension in accordance to dimension specified by the user.

No. of Pages : 16 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015326 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED PRE-HEATED METAL COIL EXPANSION DEVICE

(51) International classification	:B21C 51/00, B21C 47/26, B21C 47/24, H05B 6/36, G06F 3/041
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Kumari K

(57) Abstract :
An automated pre-heated metal coil expansion device comprises of, a housing 101 configured with plurality of motorized wheels 102 adapted to be maneuvered over a ground surface, a touch interactive display panel 103 enables a user to provide input specifications regarding an expansion length, an imaging unit 104 detect presence of pre-heated metal coil positioned in vicinity of housing 101 , an inverted L-shaped telescopic grippers 105 to extend and grip metal coil in a secured manner to prevent unintended movement of metal coil, an inverted L-shaped telescopic rod 106 configured with a V-shaped link 107 to detection of metal coil, a motorized slider 108 to provide sliding movement to rod, and an electromagnet spring 109 to generate a magnetic field to extract a controlled outward force on metal coil, a LED lights 110 ensures optimal visibility during operation.

No. of Pages : 17 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015327 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED DESIGN EMBOSSING DEVICE

(51) International classification	:B31F 1/07, B41F 19/06, B65H 35/00, B26D 7/01, B26D 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Preeti Khatri

(57) Abstract :
An automated design embossing device comprises of a platform 101 positioned to secure the platform 101 on the surface, a motorized roller 103 rolled to rotate in a pre-defined direction to unwrap the sheet, a robotic link 104 to carry out cutting of the sheet into pieces, via notarized cutter 105, of a user-specified size, a robotic gripper 106 to grip one of the piece of the sheet and place on a plate 107 arranged on the platform 101, plurality of clipper 108 to clip corners of sheet to secure sheet, a plurality of pins 109 evenly to extend retract in order to form a die of specified design, a robotic arm 110 to press sheet via a flap, and an artificial intelligence based imaging unit 112 integrated with a processer, to continuously capture and process images of the sheet during embossing.

No. of Pages : 19 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015328 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED FASTENERS SORTING AND STORING DEVICE

(51) International classification	:B25J 15/06, B65G 47/91, B65G 1/137, B07C 5/36, B07C 5/342	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Nithya B A**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated fasteners sorting and storing device, comprising of a platform 101 positioned on a fixed surface, a plurality of suction cups 102 to secure platform 101 on surface, a display panel 103 to provide input command for storing fasteners, a speaker 104 to generate audible instructions to position fasteners, a primary robotic gripper 105 to grip and place fasteners on a slanted plate 106, a plurality of slots 108 each of different dimensions carved on the plate 106 to allow fasteners fall in section of chamber 112, an imaging unit 109 to detect deviating of any of the fasteners, a secondary robotic gripper 110 to grip the deviated fastener and store in the chamber 112, a plurality of balls 111 to facilitate smooth rolling of the fasteners along the plate 106 during sorting process, a vibrating unit to generate vibration that helps to transfer the fastener into the sections.

No. of Pages : 19 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015329 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CAKE BASE BOARD CRAFTING DEVICE

(51) International classification	:A21C 15/00, A21B 3/13, G06F 3/042, A21C 15/04, G01B 11/24	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Annette Christinal**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A cake base board crafting device, comprising of a platform 101 installed with a chamber 102 stored with multiple card boards and positioned on a ground surface, multiple supporting rods 103 are arranged underneath the platform 101 for supporting the platform 101 at a height on the surface, a touch interactive display panel 104 mounted on the platform 101 providing initial commands regarding crafting of a cake base board along with specifying dimensions of the board, an imaging unit 107 installed on the platform 101 for capturing and processing multiple images of surroundings, respectively, to determine dimensions of the board, an inverted L-shaped telescopic bar 109 assembled on the platform 101 to extend for positioning a circular ring 110 attached with a free-end of the bar 109 as an end-effector over the board, and a hammering unit 113 is assembled on the platform 101 providing forceful hitting movement on board.

No. of Pages : 16 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015330 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED HOLE CARVING DEVICE FOR BASINS

(51) International classification	:B44B 3/06, B25J 15/06, B44B 1/06, B23Q 17/24, B44B 1/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Sushasini**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated hole carving device for basins comprising of a platform 101 positioned on a fixed surface configured with a plurality of suction cups 102 to secure platform 101 on surface, a touch interactive display panel 103 to provide command for carving hole in basin, an imaging unit 104 to detect placement of the basin, a pair of robotic grippers 105 to grip the basin and position over a holding unit 106 installed on platform 101, a weight sensor to detect placement of basin, a plurality of suction units to secure the basin, a LiDAR (Light Detection and Ranging) module to generate a map of basin where hole is to be made, and a robotic link 107 to position a motorized drilling unit 108 to make hole on selected spots of basin.

No. of Pages : 15 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015331 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BEVERAGE HOLDING DEVICE

(51) International classification	:B60N 3/10, B67D 1/08, B67D 1/00, A47G 23/02, A47B 88/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sindhu G

(57) Abstract :
A beverage holding device comprises of a U-shaped body 101 installed with a pair of motorized extendable clamps 102, a microphone 103 for receiving voice command, an artificial intelligence-based imaging unit 104 for detecting dimension of the fixed surface, a motorized drawer arrangement 105 integrated within each of clamps 102 to provide required extension/retraction to the clamp 102, a plate 106 attached by pair of hinge joints 107 having cavity for holding beverage container and to deploy plate 106 accommodating the container within cavity, an ultrasonic sensor for detecting dimensions of container, an iris lid 108 to close lid 108 for securely accommodating container, a timer module to keep track of time of the container within the cavity, an L-shaped telescopically operated rod 109 to extend and position a coaster gripped by means of a clipper 110 mounted on the rod 109 over the beverage container.

No. of Pages : 20 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015332 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPUTER PERIPHERAL OPERATING ASSISTIVE DEVICE

(51) International classification	:G06F3/033, G06F3/0354, G06F1/16, G06F3/039, H04N23/60, G06F3/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Aadhithyan K A

(57) Abstract :
A computer peripheral operating assistive device comprises of, a frame 101 configured to be positioned in proximity to a mouse peripheral, an ultrasonic sensor is mounted for determining dimensions of mouse, a drawer arrangement for increasing/decreasing dimensions of frame 101, a plurality of suction units for securely adhering to mouse's surface, a motorized roller 104 coiled with a strap 105, installed for unwrapping an appropriate length of strap 105, a pair of motorized clippers 106 for properly accommodating mouse within straps 105, a pair of curved flaps 108 for enabling a user to place user's hands, a microphone 110 for receiving voice commands regarding pressing of desired buttons of the mouse, a sliding unit 109 for allowing user to press desired buttons, and an extendable L-shaped pin 103, to extend/retract for positioning member 111 to user-specified button.

No. of Pages : 15 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015333 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED ROAD SAFETY MIRROR MANAGEMENT DEVICE

(51) International classification	:B25J 13/08, B25J 9/16, G05D 1/02, G05D 1/00, E01F 9/619
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sathyanayaran Gardasu

(57) Abstract :
An automated road safety mirror management device, comprising a platform 101, with plurality of plates 102, designed for positioning on road safety mirror, a user-interface in a wirelessly linked computing unit allows input command, an AI-based imaging unit 103 with a processor detects the mirror's housing curvature, a motorized hinge joint 104 between consecutive plates 102, to adjust the shape of platform 101, a plurality of suction units 105 for securing the platform 101, a dust sensor detects dust on mirror, an electronically controlled spryer 106, with a water chamber 107 to spray water on mirror, a first robotic arm 108 with a micro finger 109 for cleaning of mirror, a rain sensor detects rainfall, a wiping unit 110 joined via a motorized pivot joint 111 provides movement for wiping, a second robotic arm 112 positions a curved shield 113 attached on mirror's housing, to prevent rainfall accumulation, ensuring visibility.

No. of Pages : 18 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015334 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : STORAGE DEVICE FOR UMBRELLA

(51) International classification	:A45B 25/24, A45B 25/18, A45B 25/00, A47G 25/12, A45B 25/28	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Kamalraj D**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A storage device for umbrella comprising of a hollow cuboidal body 101 installed with plurality of plates 102 for segregating the body into multiple sections, a microphone 103 for receiving voice command regarding storage of umbrella, an inbuilt microcontroller linked with the microphone 103 to process the commands, an imaging unit 104 mounted within each of the sections, each paired with a processor for detecting dimensions of the umbrella , a motorized slider 105 installed with each of the plates 102 to translate the plates 102 in view of adjusting sizes of the section for securely accommodating umbrella, a non-contact moisture sensor installed within each of the section for detecting presence of moisture on the umbrella, a plurality of iris pores 106 configured in each section and connected to an air blower 107 mounted to blow air drying the umbrella and storing the umbrella in a proper manner.

No. of Pages : 16 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641015338 A

(19) INDIA

(22) Date of filing of Application :11/02/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : MOBILITY AID DEVICE FOR BEDSIDE

(51) International classification	:A61H 23/02, A61G 7/05, A47C 21/04, A61G 7/057, A61H 1/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Bora Upendra Rao**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A mobility aid device for bedside comprises of, an L-shaped extendable body 101 constructed with a plurality of telescopic rods 102 installed on a bed frame underneath a mattress, a plurality of suction cups 104 for adhering body 101 with frame, a microphone 103 mounted on body 101 for receiving voice command, an ultrasonic sensor installed on body 101 for detecting dimensions of user's bed, a temperature sensor installed on body 101 for detecting temperature of user's hand, a multiple Peltier units mounted on handle 109 for providing required heating/cooling effect to user's hand, a multiple vibration units 105 installed on body 101 to provide gentle vibrations, an air blower 106 mounted on body 101 in for blowing cool air towards user, and an imaging unit 107 installed on body 101 paired with a processor for detecting unusual pattern of user.

No. of Pages : 18 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641015339 A

(19) INDIA

(22) Date of filing of Application :11/02/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : CHILD HAIRCUTTING AND ACCOMMODATION DEVICE

(51) International classification	:A45D 44/00, A45D 44/22, A45D 44/02, A45D 44/14, A45D 44/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. V Ramalatha

(57) Abstract :
A child haircutting and accommodation device comprises of a hollow cuboidal body 101 to accommodate a child over a seating unit inside the body 101, a telescopic legs 102 to position the body 101 at a height from the surface, a microphone 103 for receiving voice commands for performing haircut over the child, an ultrasonic sensor to detect height of the individual, an imaging unit 104 to detect body 101 dimensions of the child, a motorized hinges 105 with lateral sides to tilt and a C-shaped curvature at mid-portion the child is accommodated inside the body and head portion atop the curvature, a temperature sensor to detect ambient temperature of the body 101, a Peltier units to maintain temperature inside the body 101, a robotic link 107 with a motorized brush 108 for wiping detected hairs from the child's face or neck via bristles fabricated over the brush 108.

No. of Pages : 18 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015340 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BAMBOO STRIPS MANUFACTURING DEVICE

(51) International classification	:B27J 1/00, B27M 3/00, B27M 1/08, B27L 7/06, B65G 47/90
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Akhila R Udupa

(57) Abstract :
A bamboo strips manufacturing device, comprises of a housing 101 adapted to be positioned over a ground surface, a U-shape plate 102 to position a bamboo over the plate 102 for cutting the bamboo to make bamboo strips, a display panel 103 to provide input specifications regarding a requirement of desired thickness of bamboo strips, an imaging unit 104 to determine presence of the bamboo over the plate 102, a telescopic rod 105 to align the plate 102 with respect to a hollow cylindrical body 106, a pair of telescopic grippers 107 by means of a motorized dual axis lead screw arrangement 108 to grip the bamboo in a secured manner, a plurality of sharp edge blades 109 via a motorized circular sliding unit 110 to provide movement to the blades 109 as per dimension of the required strips to ensure precise and uniform splitting of the bamboo into strips.

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015341 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FOOD ITEM GRILLING DEVICE

(51) International classification	:A47J 37/04, A47J 37/06, A47J 37/07, A47J 37/08, A47J 37/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sonal Manik Gursahani

(57) Abstract :
A food item grilling device comprising a cuboidal housing 101 to positioned over ground surface, Omni-directional wheels 104 to maneuver over surface, a clamping units 105 to extend and securely grip first end of skewers to grill food, a plurality of telescopic rods 106 to extend and align rings 107 relative to grippers, an imaging unit 109 to determine cooking pattern of food, a motorized slider 110 to provide rotational movement to lids 108 for grilling food, a butter dispensing unit 111 to dispense over food, a motorized brush 113 to spread butter over surface of food, an electronic nozzle 115 to dispense optimal amount of seasoning, an infrared temperature to detect grilling temperature of food, a motorized air blower 117 unit to increase combustion and raise temperature to an optimal grilling level, a motorized canopy to get deployed to shield housing 101 from the rain.

No. of Pages : 18 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015342 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : INFANT RESTING DEVICE

(51) International classification	:A47D 9/02, A47D 13/02, A47D 13/10, A47D 13/06, A47D 15/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Ghazala Zameer**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An infant resting device, comprises of a curved-shaped platform 101 configured with an oval shaped frame 102 to provide support to the platform 101 over a ground surface to accommodate an infant to rest over the platform 101, an imaging unit 103 to detect presence of the infant over the platform 101, an air compressor 104 linked with an inflatable member 105 to allow the infant to comfortably rest over the platform 101, a motorized crank and rocker assembly 106 to provide continuous impart vertically upward and downward motion to the platform 101 for a soothing movement to the infant, a microphone 107 to provide input commands to provide a gentle movement to the baby, a moisture sensor to detect moisture over the platform 101, a motorized air blower 108 to direct a controlled flow of air over the platform 101 for quick drying.

No. of Pages : 17 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015343 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED DRIED NOODLES PREPARATION DEVICE

(51) International classification	:A21C 9/04, A21C 9/08, A21C 9/06, A21C 11/16, A23L 7/109	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Jessie Rezin Grand**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated dried noodles preparation device, comprising a housing 101 positioned over ground surface, a touch interactive display panel 102 receives user input specifications, a heating unit 103 integrated with a circular pan 104, an L-shaped telescopic rod 105 installed on ceiling portion of the housing 101, attached with a cylindrical reservoir 106 that stores batter, an AI-based imaging unit 107 installed in sync with the rod 105, a processor integrated with the imaging unit 107 and rod 105 to position the reservoir 106 over the pan 104, an electronic nozzle 108 to dispense batter, a motorized circular sliding unit 109 provides circular motion to the rod 105 for formation of continuous noodle strands, a circular lid 110 attached with the pan 104, and a motorized hinge 111 that provides movement to the lid 110 in a manner to cover the pan 104.

No. of Pages : 18 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015344 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED FOAM INSULATION APPLICATION DEVICE

(51) International classification	:B05B 13/04, B05B 12/12, E04F 21/08, E04G 21/00, E04G 21/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Soumya G D

(57) Abstract :
An automated foam insulation application device, comprises of a platform 101 positioned on a ground surface with a display panel 102 to provide input command regarding spraying foam insulation on a wall, an imaging unit 103 to detect exact location of wall with respect to platform 101, plurality of motorized wheels 104 to position platform 101 in close proximity to wall, a first robotic arm 105 to position an air blower 106 for detaching dust particles that is extracted by a suction unit 107 and transfers in a box 108, a second robotic arm 109 to align a laser heating unit 111 to eliminate moisture from wall, a chamber 110 stored with polyurethane, a suction pump to extract polyurethane from chamber 110 to deliver to a nozzle 112 held by a robotic gripper 113 to spray polyurethane on wall surface.

No. of Pages : 17 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015345 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED TOY STUFFING DEVICE

(51) International classification	:A63H 3/36, A63H 9/00, A63H 3/00, A63H 3/02, A63H 3/28	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Karthik M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated toy stuffing device comprising of a first and second section 102, 103, of a housing 101, a hopper 104 installed over first section 102 to accommodate stuffing material within first section 102, a touch interactive display panel 105 to provide input specifications of toy to be stuffed, an inbuilt microcontroller linked with display panel 105 to process input commands, a motorized cutting unit 106 to rotate and cut clumps into a finer material allowing stuffing of toys, a pair of telescopic grippers 107 installed to extend and retrieve toy stored within a chamber 108, an inlet opening of toy in contact along with a conduit 109 with a motorized vacuum pump 110 integrated within conduit 109 generates suction force ensuring uniform and efficient filling of toy, a platform 111 installed within second section 103 to position stuffed toy and a weight sensor to measure weight of stuffed toy.

No. of Pages : 16 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :11/02/2026
(21) Application No.202641015346 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE HYGIENE MANAGEMENT DEVICE FOR PET ANIMALS

(51) International classification	:A01K 29/00, A01K 13/00, A01K 27/00, A61L 9/14, A01K 15/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Mehbooba Sultana

(57) Abstract :
A wearable hygiene management device for pet animals, comprising a wearable body 101 is accommodated over a back portion of a pet animal, a primary strap 103 for securing the body 101 around torso portion of the pet animal, a secondary strap 104 wrapped around neck portion of the pet animal, a multiple clippers 106 to securely grip the diaper, a moisture sensor for detecting moisture on the member 105, a weight sensor for detecting excreta on the member 105, an electronic sprayer 108 with a fragrance liquid to release a pleasant fragrance in the surrounding, a temperature sensor for detecting body 101 temperature of the pet animal, a motion sensor for detecting movement of the pet animal, a vibration units 110 to produce gentle vibrations in view of encouraging movement of the pet animal.

No. of Pages : 16 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/02/2026

(21) Application No.202641015347 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED INSECTS REPELLING DEVICE

(51) International classification	:A01M 1/20, A01M 7/00, A01M 29/12, A01M 13/00, A01M 1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Ramesh Sengobil

(57) Abstract :
An automated insects repelling device, comprises of a circular platform 101 to be positioned on a ground surface, an imaging unit 102 to detect presence of insect infestation zones, plurality of motorized wheels 103 to position the platform 101 in proximity to the insect infestation zone, a circular chamber 104 to store insecticide, ULV fog to generate fog by atomizing insecticide into tiny droplets and is released by a nozzle 106 by means of a hollow extendable rod 107 to deter the insects, an air blower 111 to blow air towards the nozzle 106 while releasing fog to enhance dispersion and coverage of the fog, an ultrasonic emitter 108 to generate ultrasonic waves that cause discomfort to the insects encouraging them to leave the infested zone, an electronically controlled sprayer 109 configured with a container 110 to spray a fragrance solution on the insect infestation zone to reduce odor.

No. of Pages : 17 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015812 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : HANDHELD NASAL PASSAGE CONGESTION RELIEF ASSISTIVE DEVICE

(51) International classification	:A61M 11/00, A61M 15/08, A61B 1/233, A61M 11/06, A61B 5/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Niveditha S V

(57) Abstract :
A handheld nasal passage congestion relief assistive device comprising a cylindrical body 101 with a handle 102 to gripped for positioning a distal end in proximity to a nostril of user, a microphone 103 to provide input commands for extracting mucus from the user's nose, an imaging unit 104 to determine shape of the user's nostrils, an expandable cylindrical member 105 to expand and conform to shape of the nostril, a vacuum unit 106 to draw the mucus from the nostrils and collected in a chamber 107, an electronic sprayer 109 for spraying the saline liquid from vessel 110 into nasal passage for loosening mucus to make extraction easier, a cloth attached to clean passage, a vibrating unit for generating gentle vibrations to loosen mucus based on congestion levels and mucus thickness, an electronic valve 112 for spraying a sterilizing solution.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015813 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TOOTHBRUSH CLEANING DEVICE

(51) International classification	:A46B 17/06, A46B 13/02, A46B 15/00, A61C 17/22, A61L 2/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Nagma Fariyan

(57) Abstract :
A toothbrush cleaning device, comprises a housing 101 configured with a plurality of chambers 102 for holding individual toothbrushes, an input module 103 to provide input commands regarding cleaning of the toothbrushes, a dispensing module 104 to dispense cleaning solution over the toothbrushes, a cleaning module 105 to clean the toothbrushes, an inspection module 106 to monitor entire cleaning operation, an alert module 107 to alert the user in case any discrepancy is detected via the inspection module 106, a hinged lid 108 and electromagnets for opening and allows the user to place the toothbrush within the chamber.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015814 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE EXERCISING ASSISTIVE DEVICE

(51) International classification	:A63B 23/035, A63B 23/12, A61H 1/02, A63B 24/00, A63B 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. K Krishna Kumar

(57) Abstract :
An adaptable exercising assistive device, comprises of a U-shaped frame 101 of the device is configured with extendable legs 102 for ground support, with suction unit 103 integrated to securely grip surfaces during use, an input means 104 paired with frame 101 and connected to the microcontroller allows the user to provide workout instructions,, an imaging unit 105 installed on the frame 101 to determine the user's height, multiple vertical extendable members 106, installed on the inner lateral walls of the frame 101, hold adjustable footrest 107 and extendable handles for user support, and oscillating units within each member provide calibrated motion for guided exercises, a sensing module integrated into the frame 101 detects body movements, posture, vital health parameters, and steps, and n inflatable sheet 108 arranged at the lower portion of the frame 101 and linked to an inflating unit 109 inflates to prevent falls upon detecting imbalance.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015815 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PIZZA CUTTING AND SERVING DEVICE

(51) International classification	:A21C 15/04, B26D 5/00, B26D 5/08, B26D 7/01, A21C 9/06	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Bipasha Maity**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A pizza cutting and serving device, comprises of a circular frame 101 composed of a plurality of triangular shaped plates 102 is placed on a horizontal surface to carry a pizza, a user gives input commands regarding a number of slice of pizza the user desires through the microphone 106, an artificial intelligence-based imaging unit 105 captures images and sends data to a processor to detect the pizza's presence on the frame 101, an articulated arm 107 positions a motorized blade 108 coupled with the articulated arm 107 which spins to cut the pizza as per user input, an extendable clipper 109 is positioned on each of the plate to serve the pizza to an auxiliary plate placed in vicinity of the frame 101, a temperature sensor detects the temperature of the pizza and a heating element maintains the temperature of the pizza.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015816 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONCRETE COMPACTING ASSISTIVE DEVICE

(51) International classification	:E04G 21/06, E04G 21/10, E01C 19/38, E04G 21/08, B28B 1/087
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Nakul R

(57) Abstract :
A concrete compacting assistive device, comprises an elongated body 101 having a proximal end installed with a handle 102 that is accessed by a user to acquire grip for position a rectangular plate 115 attached with a distal end of the body 101, on a surface spread with concrete mixture, a microphone 103 embedded in the handle 102 that is accessed by the user to provide voice command for levelling and compaction of the mixture, multiple vibrating unit 107 installed on the plate 115 to generate vibration that is transferred from the plate 115 to the mixture, and multiple electronically controlled sprayer 108 installed on the plate 115 to spray water on the concrete mixture, and a pair of robotic arm 110 installed on the plate 115 and each attached with tweezers 111 to pick debris from mixture via the tweezers 111.

No. of Pages : 18 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015817 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE INFUSER ASSISTIVE DEVICE FOR HYDRATION TRACKING

(51) International classification	:A47J 31/40, A47J 27/21, A47J 43/27, A47J 43/044, A47J 31/44	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Hashmat Fida**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An adaptable infuser assistive device for hydration tracking, comprising a hollow body 101 equipped with a threaded lid 102 for opening/closing to allow a user to feed liquid contents, a central chamber 103 to be infused within the liquid contents, and closed bottom for secure holding, an input means 104 for receiving user's specifications for infusing the ingredients within the liquid contents, a plurality of motorized iris-operated lids 105 to regulate release of ingredient flavors into the liquid, as per the user's specifications, a motorized stirrer 106 to stir the liquid for mixing any settles particles of ingredients evenly with the liquid, a filtration module 108 for selectively filter ingredient particles during consumption.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015818 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPACT LIFTING ASSISTIVE DEVICE FOR ARTICLES

(51) International classification	:B66F 11/04, B62D 63/02, B62B 3/02, B62D 63/04, F16M 11/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Robin Rohit Vincent

(57) Abstract :
A compact lifting assistive device comprises of, a triangular platform 101 with a T-shaped extendable handle 102, a plurality of motorized omnidirectional wheels 103, and a microphone 104 for user commands, an artificial intelligence-based imaging unit 105 captures and processes object dimensions for lifting, an L-shaped flap 106 with a drawer mechanism and motorized hinge joint 108, enabling tilting and adjustment for object positioning, Extendable L-shaped pins 109 with a suction cup 110 secure the object, while a motorized pivot joint 110 controls flap 106 movement for lifting, a hydraulically operated pole 112 provides additional support by extending or retracting in sync with the pivot joint, a gyroscopic sensor ensures stability and balance adjustments during external vibrations, a touch interactive display panel 113 alerts users to malfunctions, a tactile sensor detects the object's hardness and regulates pressure application, ensuring secure handling without damage.

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015819 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE FACE AND HEAD PROTECTION ASSISTIVE DEVICE

(51) International classification	:A61B 5/01, A42B 3/04, G08B 6/00, A61B 5/00, A41D 13/05	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Babu D**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A wearable face and head protection assistive device, comprises of a wearable body 101, a plurality of temperature detectors in the inner lining monitor skin temperature, a thermal regulation module 107 adjusts heating or cooling effects of device, strain detectors to activate shape memory alloy wires for expansion or contraction to maintain a secure fit, a haptic alert module with vibrating units 103 provides obstacle detection and navigation cues using feedback from proximity detectors and GPS, bio modules coupled with PPG monitor hydration for real-time assessment of hydration levels, fatigue and stress to trigger alerts, a UV sterilization unit that disinfects the inner lining in the user's absence, an olfactory module detects foul odors and activates an aromatic liquid diffusing unit for enhancing air quality, and LEDs (Light Emitting Diode) 104 provide visual alerts for the user regarding operations of the device.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015820 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED MULCH SHEET INSTALLATION DEVICE

(51) International classification	:A01G 13/02, A01G 13/00, G05D 1/02, A01G 13/10, A01C 11/02	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Devi S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated mulch sheet installation device, comprising a rectangular frame 101 configured with multiple motorized omnidirectional wheels 102 for maneuvering the frame 101 , a microphone 103 is installed on the frame 101 to provide input commands regarding covering of the filed, an imaging unit 105 installed on the frame 101 for capturing and processing multiple images in vicinity of the frame 101, respectively to determine dimensions of the plants, an extendable leg 104 arranged in between the frame 101 and wheel, to extend/retract for adjusting height of the frame 101 as per the determined dimensions of the plants, a container is installed on the frame 101 for storing curved structures, are sequentially grabbed and secured above the plants at varying distances and a motorized roller 107 coiled with a mulch sheet, installed across a front portion of the frame 101 for rotating to unroll the sheet.

No. of Pages : 17 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015821 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED COMBUSTION DEVICE FOR FUELS

(51) International classification	:F23N 5/24, F23N 1/02, F23N 5/12, F23N 5/00, F23N 5/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Senthil Kumar H

(57) Abstract :
An automated combustion device for fuels comprises of a frame 101 shaped into a pyramid structure with slanted walls 103 and quadrangular top opening, input means 105 paired with inbuilt microcontroller for receiving user commands, motorized hinged door 106 for loading and unloading fuel, fuel moisture compensator with humidity sensor and heating elements 107 for drying fuel when moisture exceeds threshold, flame distributing unit 108 with sensors and stirring rod 108b for ensuring even combustion, wind deflection unit with panels 109 responsive to wind direction for maintaining flame stability, ember detection and quenching unit 110 employing imaging and mist sprays to safely extinguish remaining embers, sensor array monitoring CO, CO_2, $PM_{2.5}$ and oxygen levels to detect suboptimal combustion and trigger corrective adjustments, wheels 112 with swivel joints and braking by tilt sensing to ensure device stability and mobility.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015822 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE FOOT CARE DEVICE

(51) International classification	:A61H 39/04, A61H 23/02, A61H 15/00, A45D 29/05, A47K 7/04	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Niveditha A S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A wearable foot care device comprises of a wearable body 101 worn around the foot, a plurality of patterned bristles 102 for exfoliation and massage of the sole, a touch sensor to detect user interaction, a plurality of vibrating unit 103 to deliver massage upon touch detection, a foot scrubber unit 104 providing abrasion to remove hardened skin, calluses, and dead cells, a nail and toe scrubber module 105 for cleaning toenails, nail grooves, alert module 107 including speakers, buzzers, or LED indicators to notify the user of discrepancies, a cylindrical abrasive roller 104a mounted on a slider track 104b for multi-directional scrubbing motion, a bracket 105a with bristles 105b for precise nail and toe scrubbing, an ultrasonic sensor to detect solution levels, a heating unit 108 to regulate solution temperature, and a fragrance diffuser 109 releasing aromatic solutions during cleaning for enhanced comfort and hygiene.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015823 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE SEATING ASSISTIVE DEVICE

(51) International classification	:A47C 7/62, A47C 1/032, A47C 1/035, A47C 7/46, A47C 1/024	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Madhusudana M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An adaptable seating assistive device, comprising a base frame 101 with upper and lower circular rings 102, 103, multiple extendable rods 104 extend or retract for height adjustment based on user height detected by an AI-based imaging unit 106 mounted on the base frame 101, install an inflatable spherical ball seat 107 on the upper circular ring and integrate an inflating unit to regulate air pressure for comfortable seating, embed a sensing module within the ball seat 107 to measure real-time weight distribution and user posture and regulate air pressure via a microcontroller accordingly, attach a motorized, adjustable backrest 108 to the upper ring 102 via a hinge joint 109 to respond to posture, mount a pair of inverted U-shaped adjustable hand rests to lateral sides, include a leg support assembly connected to the lower ring 103 for leg and foot stability during sitting.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015824 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED TURF AERATION AND SOIL REVITALIZATION DEVICE

(51) International classification	:A01B 45/02, A01B 45/04, A01B 45/00, G05D 1/02, A01G 20/35
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Asif Ahmed Najar

(57) Abstract :
An automated turf aeration and soil revitalization device comprises of, a body 101 configured with a plurality of motorized omnidirectional wheels 102 for manoeuvring the body 101, a microphone 103 is integrated in the body 101 for to provide input voice commands, an imaging unit 104 is installed on the body 101 for capturing and processing multiple images of the body 101, a plurality of extendable pins 106 for extending/retracting to create holes in the ground surface, an ultrasonic sensor is installed on the body 101 for determining uneven surface of the turf, a motorized cutter 107 mounted on the body 101 by means of an extendable L-shaped link 108 to extend/retract for positioning the cutter 107 onto the uneven surfaces, in view of enabling the cutter 107 to cut uneven lengths of the turf and evenly trims surface of the turf of the field.

No. of Pages : 17 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015825 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED ANCHORING DEVICE FOR WATER VESSELS

(51) International classification	:B63B 21/50, B63B 21/22, B63B 21/24, G01V 1/38, B63B 21/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Nisha Robin Rohit

(57) Abstract :
An automated anchoring device for water vessels comprises of a primary anchor 101 equipped with a shank 102 and flukes 103, functioning to provide hold in diverse seabed environments, a monitoring unit 105 incorporating a microcontroller, sonar module, imaging unit, and geophysical, chemical sensors for detecting undersea semimetal conditions, flukes 103 that adjust via shape memory alloy hinges 104 and hydraulic actuators 106 to optimize grip strength, a high holding power anchor 107 with multiple broad flukes 103 arranged in triangular layout to augment anchoring stability, a sensor suite with strain gauges, piezoelectric sensors, accelerometers distributed across anchor 107, flukes 103, shank 102, and chain to detect stress, micro cracks, deformations, a locking unit 109 engaging upon detection of vibrations, a navigation unit 110 using GPS and inertial navigation to monitor anchor 107 dragging and position, computing unit enabling real time visualization of holding power, stress levels, drag risks, with predictive alerts.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015826 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SAFETY ENABLED MAINTENANCE ASSISTIVE DEVICE FOR STRUCTURAL REPAIR AND INSPECTION

(51) International classification	:G01S 17/89, B25J 5/00, G01S 17/86, E04G 23/02, G01S 17/931
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Teja Sirapu

(57) Abstract :
A safety enabled maintenance assistive device for structural repair and inspection, comprises of a frame 101 having a first portion 102 and a second portion 103, a securing means mounted on the first portion 102 to secure the frame 101 on an edged wall, a pair of U-shaped extendable members 104 adjustable via a LiDAR module 105 while multiple suction units for adherence of the frame 101 on the edge, multiple motorized wheels 112 on for lateral movement, a gyroscopic sensor for real-time motion monitoring of the frame 101, a vertical arrangement controlled by an AI-based imaging unit 110 for adjusting height of frame 101, a weather protection module with weather detection unit on the frame 101 for user protection from extreme weather by expanding shield.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015827 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : STEEL FILTER HOLES CARVING ASSISTIVE DEVICE

(51) International classification	:B44B 3/06, B23B 41/00, B23Q 17/24, B25J 11/00, B23Q 17/22	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Vijayakumar Adaickalam**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A steel filter holes carving assistive device, comprising of a platform 101 to accommodate a circular steel filter frame with multiple wheels 102, an imaging unit 103 for detecting dimension of filter, a L-shaped telescopic rods 104 extends/retracts for clasping steel filter via a motorized clamps 105 attached with rods 104, a microphone 106 to receive voice commands regarding crafting of holes over marking carved, a robotic link 107 with a drilling unit 108 for carving holes on steel filter, a motorized via a L-shaped pneumatic bar 110 to extend positioning rollers 109 for carving of holes over entire inner portion, a robotic arm 111 with a motorized scrubbing unit 112 for providing a shine to steel filter, a motorized disc 113 installed on platform 101 via a telescopically operated pole 114 and provided with bristles to rotate for scrubbing filter via bristles for cleaning filter.

No. of Pages : 17 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015828 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED CABLE PULLING DEVICE

(51) International classification	:H02G 1/08, H02G 1/06, B25J 13/08, B25J 9/16, F16L 1/11	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Nasa Dhanraj**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated cable pulling device, comprises of a platform 101 adapted to be positioned over a ground surface, a touch interactive display panel 102 to provide input specifications regarding pulling of a, an imaging unit 103 to determine presence of the electrical conduit, a plurality of motorized wheels 104 to maneuver and position the platform 101 towards the detected electrical conduit, an inverted L-shaped telescopic rod 105 configured with a motorized gripper 106 to extend and position the gripper 106 towards the conduit, followed by actuation of the gripper 106 to extend and retrieve the cables from the conduit, a multi-meter unit 107 to measure voltage and current in the retrieved cable, a motorized C-shape clamping unit 108 to retrieve a rubber insulator stored within a chamber 109 installed over the platform 101 and to engage the rubber insulator on the cable for safety purposes.

No. of Pages : 17 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015829 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED TOOTHPICK MANAGEMENT DEVICE

(51) International classification	:A47G 21/12, A61C 15/02, G07C 9/00, A61C 15/04, G07F 17/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Harish Kumar K S

(57) Abstract :
An automated toothpick management device, comprising a hollow body 101 configured with a storage chamber to store multiple toothpicks, an input means including a microphone 104 for enabling a user to provide commands for insertion or retrieval of toothpicks, a lid 105 attached via an electromagnetic strip 106 that opens upon user command for storing or refilling toothpicks, a dispensing arrangement including a motorized slider 107 with raised edges, a plate 108 installed via an extendable rod 109 for adjustable positioning, and a motorized iris aperture 110 to dispense a single toothpick, a rod 109 for controlled adjustment, a motorized hinge 112 acting as a gate to release one toothpick at a time, an audio output module and a UV lamp 113 for sterilizing toothpicks with a timer for controlled activation.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015830 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ICE CREAM SCOOPING AND DISPENSING DEVICE

(51) International classification	:A23G 9/28, A23G 9/22, A23G 9/04, A23G 9/12, A47J 43/28	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Jayakrishna Udupa H**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An ice cream scooping and dispensing device, comprises of a cylindrical hollow body 101 with a handle 102 gripped by a user, a microphone 103 receives an input voice commands from the user regarding a dimension of a portion of the ice cream to be scooped, a pulley arrangement 104 expand/contract dimension of the body 101, a pair of motorized hinged flaps 105 installed on side portion of the body 101 via a motorized slider 106 to translate the flaps 105 towards the end portion of the body 101 to tilt and cover the end portion the body 101 to prevent the scooped portion from falling and an extendable pusher 107 move downward and apply pressure on the scooped portion for releasing scooped ice cream.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015831 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DETOX WATER PREPARATION DEVICE

(51) International classification	:C02F 1/00, A47J 31/44, A47J 31/40, A47J 31/46, C02F 1/52
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Deepthi S

(57) Abstract :
A detox water preparation device, comprising a housing 101 configured to support all operational components, an imaging unit 103 synchronized with a proximity sensor to detect user presence, a motorized sliding door 104 that opens to allow ingredient placement into a multi-section chamber 105, an input means linked to a microcontroller for receiving user commands, an iris lid 107 to dispense selected ingredients onto a cutting platform 108, robotic grippers 109 and an articulated arm 110 with a sharp-edge blade 111 for gripping and cutting ingredients, a robotic link 112 for placing and filling an infuser chamber, a slidable tray 201 for bottle placement, an electronic valve 202 for dispensing additives from a multi-section reservoir 203, a robotic limb 204 to position the bottle, and an iris hole 205 for inserting the infuser chamber into the bottle to allow automated infusion of selected ingredients and additives to prepare detox water.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015839 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FOOD HANDLING DEVICE

(51) International classification	:B65G 47/90, A47J 43/28, A47J 37/04, B65B 25/06, G01G 19/52
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Anitha S Yadav

(57) Abstract :
A food handling device, comprising of an oval-shaped plate 101, a holding unit 102 integrated at the center of the plate 101 via an extendable bar 103 that extends and retracts to position the holding unit 102 for receiving and supporting a food item, a weight sensor to determine presence of the food item within the holding unit 102, wherein upon detection the holding unit 102 extends or retracts to maintain a stable supporting position, multiple extendable rods a having a motorized clipper 105 at the end, and a motorized roller 106 wrapped with a sheet 107 arranged in between each pair of the rods 104, the roller 106 being adapted to unwrap the sheet 107, the end of the sheet 107 being grasped by the clippers 105, and the rods 104 extend simultaneously so as to pull and position the sheet 107 for wrapping around the food item.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015840 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : OIL SEPARATION DEVICE

(51) International classification	:B01D 17/02, B01D 17/032, B01D 17/038, B01D 17/025, B01D 21/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sivaramakrishnan S

(57) Abstract :
An oil separation device, comprising a lid plate 101 removably attach to a rim of an auxiliary food container, an expandable linkage 102 defining an internal passage for fluid transfer, a hemispherical collector chamber 103 including a mesh layer 104 at the base contact a surface of food to permit collection of floating oil, a suction assembly including an extendable suction conduit 106 to extract oil, an oil storage reservoir 107 receive and store the extracted oil, an AI camera 110 monitor oil-separation operations, a vibration unit 108 impart controlled vibrations to prevent clogging of the mesh layer 104, a speaker 111 provide audio alerts regarding device status, completion of oil separation, and food-quality conditions.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015841 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COAL BRIQUETTE PRODUCTION DEVICE

(51) International classification	:C10L5/04, C10L5/10, C10L5/06, C10L5/02, F24F11/00, C10B53/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Abdul Majid

(57) Abstract :
A coal briquette production device comprises of a housing 101, a hopper 102 receiving coal dust, multiple spraying nozzles 103 with binder solution storage chamber 104 dispensing a binder solution, a motorized stirrer 106 mixing the coal dust and binder solution, an auger screw barrel 108 to transfer prepared mixture, a dispensing nozzle 109 dispensing prepared mixture, a slider 110 carrying multiple molds 111 to form briquettes, multiple pneumatic pusher plates 112 with pneumatic pins 113 creating holes in the briquettes, a conveyor unit 114 transferring briquettes to packaging unit, multiple hot air dryers 115 removing moisture, a plastic sheet dispensing roller 116 providing plastic sheet, a telescopic heating pad 117 sealing the packaging, a dispatch conveyor 118 releasing the briquettes, a pair of clippers 119 wrapping briquettes with plastic sheet, a motorized blade 121 cutting plastic sheet, a speaker 123 generating alerts, and display panel 124 receiving user inputs.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015842 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : UPPER LIMB SUPPORT DEVICE

(51) International classification	:A61B 5/107, A61G 13/12, A61B 6/04, A61B 5/00, A61G 7/075
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sreelatha P K

(57) Abstract :
An upper limb support device, comprising a pair of L-shaped frame 101 attached to each other via an extendable horizontal rod 102, an imaging unit 104 installed on the frame 101 to detect distance between the user's hands, a plate 105 mounted over each of the frames 101, a motorized clamping unit 106 to secure the edge table surface, a motorized suction unit 107 attached adjacent to the clamping unit 106 to securely grip the surface, an L-shaped extendable link 110 attached to a flap 111 installed on front portion of each of the frames 101, a microphone 103 to receive a support structure for an object positioned in front of the user, an electromagnet 112 attached on ends of each of the flaps 111 to generate magnetic fields to secure the flaps 111 with each other.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015843 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED POTATO STRIPS PREPARATION AND FRYING DEVICE

(51) International classification	:A23N 12/02, A23L 19/18, A23N 7/02, A23N 7/00, A23L 19/12	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Srinivasan T R**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated potato strips preparation and frying device, comprises of a housing 101 with gravity-assisted hopper 102 for feeding the potatoes, a shutter 103 for controlled potato release into the device, and pair robotic grippers 105 for grasping the potatoes, a peeling unit 110 on an articulated limb 109 peels potato skin, an electronic valve 112 and imaging unit 113 cleans the peeled potatoes, with waste being removed by a suction unit 115, a robotic arm 117 transfers potatoes to a shaping unit with cutting grid and pneumatic pusher 120 for cutting the potatoes, a pair of robotic links 121 dislodge stuck strips, a frying basket 123 mounted on an L-shaped extendable link 122 collects and fries strips, with motorized ball and socket joint enabling dispensing, a colour sensor ensures optimal frying before automatic serving, a speaker 114 installed on housing 101 for providing real time alerts to the user.

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015844 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DUMBBELLS HOLDING AND MANAGEMENT DEVICE

(51) International classification	:A63B 21/072, A63B 71/06, A63B 21/075, A63B 71/00, A63B 24/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. D Vijayasree

(57) Abstract :
A dumbbells holding and management device, comprises a pair of members 101 configured to hold a triangular frame 102, a set of L-shaped rods 103 having gripper 104 for securely gripping the triangular frame 102, a plurality of motorized C-shaped clamps 105 to accommodate and retain a dumbbell, an imaging unit 108 to determine a height and posture of a user and to identify a selected dumbbell for dispensing, a sliding unit 109 to provide translation to the clamps 105 for positioning and dispensing the designated dumbbell to the user, a plurality of suction cups 110 to securely adhere the members 101 on the surface, an air blower 111 via an articulated link 112 for drying the surface of the dumbbell prior to dispensing.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015845 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRONIC GADGET HANDLING AND CHARGING DEVICE

(51) International classification	:B25J15/06, B25J9/16, B25J19/02, B25J19/04, B08B3/02, B08B1/00, B25J15/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Faizan Anwar Khan

(57) Abstract :
An electronic gadget handling and charging device, comprising a hollow body 101 installed over surface, a plurality of suction cups arranged on rear portion of body 101 to adhere securely, a pair of robotic grippers 102 mounted on respective vertical sliders 103 within body 101, to extend and hold gadget in fixed position, a camera 104 integrated within body 101 to detect dust, fingerprints, or debris on gadget surface, an electronic nozzle 105 connected with tank 106 storing cleaning fluid to dispense fluid onto gadget, a motorized spindle 107 integrated via an articulated limb 108 to clean gadget screen, a wireless charging unit 109 to supply power to gadget, motorized clippers 110 mounted externally via articulated links 111 to hold gadget outside the body 101, a speaker 112 to generate audible notifications, a lighting element 113 to provide illumination under low light, and a touchscreen panel 114 to display notifications.

No. of Pages : 18 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015846 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : KEY STORAGE AND RETRIEVAL DEVICE

(51) International classification	:E05B 19/00, G06F 17/30, G07C 9/00, E05B 19/26, G06F 3/023
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. K Krishna Kumar

(57) Abstract :
A key storage and retrieval device, comprising a housing 101 divided into left and right compartments 102, 103 for storing primary and secondary keys respectively, an input module104 including a microphone or touch display for user selection of keys, a primary key handling module with a barrel cam and electromagnetic clamp 106 to secure and retrieve the primary key through a selectively actuated iris hole 107 with IR sensor control, a multi-key storage module with extendable rods 108 and clamping units 109 for individual secondary key retrieval through multiple iris lids110, an inspection module 111 with imaging and ultrasonic sensors for accurate key positioning, biometric authentication for secure access, tamper-protection with vibration sensor and camera to block unauthorized access and capture evidence, and suction cups 114 to prevent slippage during operation, enabling automated, selective, and secure key retrieval.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015847 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MOSQUITOES REPELLING DEVICE

(51) International classification	:B25J 15/06, A01M 1/02, A01M 1/22, A01M 1/06, A01M 29/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Francis V Annareddy

(57) Abstract :
A mosquitoes repelling device comprising a platform 101 to be positioned on surface, suction cups 113 to stabilize platform 101, an input means to allow the user to issue commands, a motorized slider 103 arranged in parallel within a chamber 104 for translating a plurality of motorized clamps 105 to hold mosquito coils, a robotic gripper 106 to grasp the mosquito coil translated outward from the chamber 104, an articulated arm 107 installed over a rigid vertical member 108 to lift and position the mosquito coil for burning, an ignitor 109 arranged with a robotic link 110 to ignite the coil, an imaging unit 111 installed via a spherical joint over the platform 101 to determine the coil's position, a motorized sliding unit 112 integrated with the articulated arm 107 for accurate ignition and controlled smoke circulation, a suction unit 114 with waste box 115 for ash removal, and a speaker 116 to provide audible alerts.

No. of Pages : 29 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015848 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SNACK PREPARATION ASSISTIVE DEVICE

(51) International classification	:A21C 1/14, A21C 1/06, A21C 1/02, A47J 43/044, A47J 43/07	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Virupaksha Goud G**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A snack preparation assistive device, comprises of a housing 101 having an input means 102 to provide input command regarding preparation of a mixture for making snacks, an electronic valve 103 arranged with each of a primary and a secondary chamber 104 for dispensing of the flour and water into the container 105 for mixing, an electronic nozzle 106 for dispensing the spices within the mixture for flavouring, a motorized kneading paddle 108 integrated within container 105 for kneading and thoroughly mixing the flour and water, a pusher 109 arranged within the container 105 to push the mixed dough through a door 110 integrated with the container 105, a motorized roller 112 integrated with the platform 111 to roll the mixed dough into a thin sheet for further processing, and a sharp-edge blade 114 arranged with the platform 111 via an articulated bar, to cut the sheet into pieces.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015849 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : LIQUID SOAP MAKING AND DISPENSING DEVICE

(51) International classification	:A47K 5/12, A47K 5/14, A47K 5/122, A47K 5/16, A47K 5/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Pushpalatha M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A liquid soap making and dispensing device, comprising a body 101 with a handle 117 for easy handling during cleaning, a microphone 102 installed on body 101 to receive voice commands for initiating soap preparation, a motorized gate 103 opens to place a soap bar into a chamber 104, where a rubberized clamp 106 moves the soap along a grater 107, an iris lid 110 to transfer the graded soap into a box 109, where water is added via a primary nozzle and mixed by a stirrer 113, the resulting liquid soap is pumped to a scrubbing pad 114 with optical sensor to detect the stubborn stain, multiple secondary nozzles for uniformly distributing the soap, a multiple pneumatic pins 116 for removing the stains, an AI camera to detect stains, optimize cleaning, and a speaker 118 to guide and provide alert for operational anomalies.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015850 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED MEDICINE PREPARATION AND DISPENSING DEVICE FOR PETS

(51) International classification	:A61D 7/00, A01K 29/00, A61J 7/00, A01K 5/02, A01K 15/02	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Suhail Manzoor**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

an automated medicine preparation and dispensing device for pets comprises a body 101 having a proximal and a distal end and equipped with a handle 102 for comfortable gripping by a user, the user inputs commands via an input means to prepare and dispense medicine, an iris lid 104 arranged underneath a tablet storing chamber 105 stored with tablets and releases tablets into the crushing container 106 where multiple of suction pads 107 hold them firmly in place, then a hydraulic actuator 108 with a rubberized pad 109 crushes the tablets, after crushing, an electronic valve 110 integrated within a tank 111 sored with the milk or other liquid releases milk or liquid, a motorized stirrer 112 rotates for uniform mixing, a peristaltic pump 114 integrated with the container 106 passes mixture through a conduit 115 to a curved extendable member 116 to the pet's mouth.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015851 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE STABILITY ASSISTIVE DEVICE FOR TRAVEL IN PUBLIC TRANSPORTATION

(51) International classification	:B62K 21/26, B62J 99/00, B60R 11/02, B60R 11/00, B62K 21/16	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Aadil Ferooz**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An adaptable stability assistive device for travel in public transportation, comprises of a body 101 paired with an input means 102 for enabling a user to provide input commands for attachment/detachment onto a surface, a monitoring unit 103 scanning surroundings to detect type, shape and distance of the surface from the body 101, a clamping arrangement securely clamp onto the detected surface, in view of providing a temporary stability for the user, a handle bar assembly 104 attached horizontally beneath the lower section, via an extendable rod 105 and a motorized slider assembly 106, to absorb shocks and the slider assembly 106 provide flexibility during sudden movements during transit, a sensing module measuring real-time orientation of the body 101 and environmental conditions, to detect sudden movements and imbalances, to trigger regulation of the clamping arrangement for ensuring secured adhering to the surface.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015852 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE DOOR SECURITY ASSISTIVE DEVICE

(51) International classification	:E05B 47/00, G07C 9/00, E05B 45/06, G08B 13/14, G08B 13/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Deasri Aditya

(57) Abstract :
A portable door security assistive device comprising a frame 101 with primary 102 and secondary portions 103, a securing means 104 to grip a door's lower edge based on door thickness detected by a first monitoring unit 105, a circular base 111 rotates via a lead screw arrangement 112 to apply pressure on the ground surface, restricting door movement, a second monitoring unit 113 scans RFID tags and captures facial details for secured authentication, a motion sensor monitors door opening and closing to ensure secure locking after each cycle, an input means receives emergency commands for unlocking the door, an adjustable inverted L-shaped rod 106 with suction cup 110 for door adhesion, expandable segments 108 form a U-shaped grip, an imaging unit 115s provide face recognition and motion detection, and a gripping arrangement adapts to different ground surfaces for enhanced stability.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015853 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE HYGIENE MAINTENANCE DEVICE FOR BASINS

(51) International classification	:A47K 5/12, A47K 1/09, A47L 15/00, A47L 15/44, A47K 7/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Shabeena

(57) Abstract :
An adaptable hygiene maintenance device for basins comprises of a housing 101 with plurality of suction units 102 for mounting on a vertical support, a plurality of storage chambers 103 for storing multiple soap bars, an ultrasonic sensor for detecting soap size, shape and remaining quantity, a scraping module for scraping soap bars into flakes, a container 108 for receiving the scraped flakes via a motorized iris lid 111 with a motorized stirrer 114 for blending flakes and water, an artificial intelligence-based imaging unit 115 for user recognition, gesture detection and unclean areas, a dispensing unit for releasing the cleaning liquid mixture, a cleaning unit having a motorized brush 117 for cleaning basin's unclean area, and an electronic sprayer 120 for spraying a solution.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015854 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TAPE MASKING AND SURFACE PROTECTION DEVICE

(51) International classification	:H01R 43/28, H01R 43/00, H01H 11/00, H01R 43/20, B65H 35/07
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Afroz Pasha

(57) Abstract :
A tape masking and surface protection device comprises of a platform 101, an imaging unit 102 determining exact location, orientation, and dimensions of sockets, switches, wires, and surfaces to be masked, multiple telescopic rods 103 with a motorized wheel 104 providing multidirectional movement of platform 101, a storage chamber 105 storing different tape sizes, a robotic gripper 106 grasp and position a masking tape roll, a tape dispensing unit including a housing, a spool 107 with a rigid bar 108 dispensing tape, a sliding unit 109 move the dispensing unit for precise positioning, a pair of articulated arms 110 guide the tape through a slit 113 carved on the unit onto the surface, an articulated bar 111 having a blade 112 cut the tape and position over the surface, a spherical joint 114 provide rotation, a speaker 115 provide audible alerts regarding tape status.

No. of Pages : 22 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015855 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : METALLIC SHEET ENGRAVING DEVICE

(51) International classification	:B44B 3/06, B44B 3/00, B44B 5/00, B23Q 7/04, B41C 1/055	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Sachin V Raikar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A metallic sheet engraving device, comprises of a housing 101 with a plurality of suction cup 102 to securely adhere the housing 101 over the surface, an input means to provide input commands initiation of engraving operation, a chamber 104 to store metallic sheets, pneumatic pusher 105 with vertical slider 106 to pushes metallic sheets through a cavity carved on the chamber 104, a pair robotic arms 107 to grasp and position the sheets onto a conveyor 108 which transports the sheets further, an engraving arrangement 109 includes a pair of motorized cylindrical members 109a constructed with multiple extendable plates adjust the members 109a by pneumatic rod 109b and a heating element to thermally engraves user-defined patterns onto the sheets, a pair of robotic links 110 to place the engraved sheets near an outlet slit for user access.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015856 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PET-FOOD PRODUCTION DEVICE

(51) International classification	:A23N 17/00, A01K 5/00, B01F 15/02, A01K 5/02, A23K 40/25
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Subhashini R

(57) Abstract :
A pet-food production device comprising of a housing 101, a grinding tank 102 placed to reduce input food item into a fine powder and discharge the paste onto a first conveyer 105, a motorized sieving roller 106 with an opening 107 configured to receive the powder from the first conveyor and filter the powder, a mixing tank 109 placed to mix the received fine powder with a nutritional fluid via a spiral mixing rod 110 , a perforated vessel 112 to remove excess water and discharge the paste onto a Scout and Russel unit 113 for further operations, a molding unit 114 for receiving the paste from the Scout and Russel unit 113, a hydraulic press 115 to extrude the paste through a die 116, and a motorized cutter to segment the discharged paste into pellets.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015857 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CEMENT BOUNDARY FORMING DEVICE

(51) International classification	:E01C 19/48, E04G 21/10, B28B 13/02, E01C 19/20, E01C 19/50
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Madhavi T

(57) Abstract :
A cement boundary forming device, comprising a housing 101 over a ground surface, an input module 102 accessed by a user to provide input commands regarding cement boundary to be formed, a gripping means access a multi-sectioned chamber 105 arranged within housing 101 to retrieve a bar, a platform 106 integrated having an iris hole 107 at a centre portion to receive bar and stabilize bar, a hollow cylindrical shaping unit 108 mounted at a ceiling portion of housing 101 via a hydraulic piston 109 for positioning shaping unit 108 over rod, a cement filling module dispense mortar mixture into a gap associated with shaping unit 108, a design rod module form structural patterns of cement boundary, a transfer module lift, position, and remove completed cement boundary, an inspection module monitor entire operation and a speaker 120 alert user in case any discrepancy is detected.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015858 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CHOCOLATE MAKING KIOSK DEVICE

(51) International classification	:A23G 1/20, A23G 1/04, A23G 1/22, A23G 9/22, A23G 1/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Tintu Vijayan

(57) Abstract :
A chocolate making kiosk device comprising of a housing 101, a plurality of chocolate chambers 102 each configured to store and melt different types of chocolate using integrated heating element 104, a sectioned ingredient storage unit 105 with individual dispensers for varied ingredient types, a central mixing unit 108 having a motorized stirrer for blending chocolates and ingredients, a freezing module 109 utilizing a Peltier unit 112 to rapidly solidify the mixture onto a tray, a cutting module with a robotic arm and motorized blade to shape solidified chocolate, a packaging module 114 equipped to wrap and seal chocolates, a storage chamber 114a with a cushioned base and level sensor for holding final products, an imaging unit for monitoring internal operations, a user interface through a touch-enabled display panel, enabling automated, personalized chocolate production and delivery.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015867 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FOAM SHAPING AND CUTTING DEVICE

(51) International classification	:B26F 3/12, B26F 3/00, B29C 44/34, B26D 7/01, B26F 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Yamanappa

(57) Abstract :
A foam shaping and cutting device, comprises of a platform 101 with suction cups 102 to adhere the platform 101 over the surface, a plate 104 on each lateral side of the platform 101 to support foam blocks, an input means 103 linked to a microcontroller for input commands regarding initiation of a foam shaping and cutting operation, a rotatable imaging unit 105 for determining the alignment and orientation of foam blocks, a vertical slider 107 on each plate 104 that translates a motorized roller 108 wrapped with a Nichrome wire for vertical cutting, a pneumatic link 110 ensures synchronized movement of the roller 108 with the slider 107, a pair of circular sliding units 109 enable rotational motion of foam blocks during cutting to achieve predetermined profiles, and a temperature sensor monitors the temperature of the Nichrome wire to maintain safe operating conditions.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015868 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NUT EXTRACTION AND STORAGE DEVICE

(51) International classification	:B23P 19/06, B25J 13/08, B65G 1/04, B25B 13/48, B65G 1/137	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Riyazulla Rahman J**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A nut extraction and storage device, comprises of a platform 101, an input means for enabling a user to provide input command regarding initiation of a nut extraction and storage operation, an imaging unit 106 installed over the platform 101 to determine the distance between the platform 101 and a metallic sheet, a plurality of extendable rods 107 each having a motorized wheel 108 to position the platform 101 in proximity to the metallic sheet, a hydraulic articulated arm 109 having a circular disc 110 with an expandable pulley arrangement 111 configured to vary the diameter of the disc 110, a plurality of vibrating units to provide adaptive vibrations for loosening the nuts from the sheet, a pair of robotic grippers 112 to grasp the extracted nut and transfer the nuts into a storage chamber 113.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015869 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GOLD EXTRACTION AND STORAGE DEVICE

(51) International classification	:A01G 25/16, A01G 25/09, A01G 27/00, G01N 33/24, G01N 1/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. M Akheela Khanum**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A gold extraction and storage device, comprises of a housing 101 having an input means 102 for enabling a user to provide input command regarding initiation of a gold extraction operation, an imaging unit 103 installed over the housing 101 to detect a soil mixture collection zone, a pair of track wheels 104 arranged underneath the housing 101 to enable locomotion, an electronic nozzle 105 connected with a water tank 106 arranged with the housing 101, to dispense water over soil mixture regions, an electronic valve 111 connected with a water reservoir 112 arranged over the housing 101 via an articulated arm 113, to dispense the water over the plate 109, and a pair of robotic grippers 114 integrated over the housing 101, each gripper having a curved member 115 to press and drag the soil mixture spread on the circular plate 109.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015870 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED DEVICE FOR RECYCLING PLASTIC WASTE INTO UNIFORM PELLETS

(51) International classification	:B29B 17/00, B29B 17/04, B29B 17/02, B09B 3/00, B29K 105/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Jyoti Sharma

(57) Abstract :
An automated device for recycling plastic waste into uniform pellets, comprises of an enclosure 101 having a chamber 102 holding waste, multiple blades for cutting plastic waste, a conveyor belt 103 transporting fragments, a tank 104 storing cleaning solution, a basket 105 retaining fragments, a sliding unit 106 moving basket 105 within tank 104, a discharge gate releasing cleaned fragments onto conveyor, multiple heated air blowers 109 drying cleaned fragments, an induction furnace 110 converting plastic fragments into molten form, a hopper 113 guiding fragments into a melting unit, a spiral blade 116 mixing molten plastic with colorants, a nozzle 118 with each receptacle dispensing colorants, an iris-style outlet 119 shaping molten plastic, a hydraulic piston 121 with a plate 115 applying pressure on molten mixture, a cutting arrangement cut extruded material, a transfer belt 123 receiving freshly extruded pellets, multiple water sprinklers spray water onto pellets.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015871 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CEDAR CHEST INSTALLATION DEVICE

(51) International classification	:B25J 13/08, B25J 9/16, B25J 15/08, B25J 15/00, A41H 43/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Sharon M

(57) Abstract :
A cedar chest installation device, comprises of a stable platform 101 with multiple chamber 107 for pins and caps, a microcontroller-controlled input module allows user selection of tufting parameters, an imaging unit 102 maps foam surface in real-time extendable rods 103 with omnidirectional wheel 104 position the platform 101 precisely, a motorized drilling head 110 with depth-control sensor creates accurate perforations, a first robotic gripper 106 retrieves pins placing them in perforations, a pneumatic pusher 108 with pressure sensor presses pins securely a second robotic gripper 109 handles caps and collaborates with the pusher 108 for sealing weight sensors monitor inventory triggering speaker 112 alerts vibration and ultrasonic sensors ensure quality and synchronization a communication module enables remote monitoring via Wi-Fi or GSM and the microcontroller orchestrates all operations ensuring precise tufting and sealing with minimal user input.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015872 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GOLF COURSE MAINTENANCE DEVICE

(51) International classification	:A01G 25/09, E01H 1/08, A01D 43/063, A01G 20/43, A01D 34/63
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dayalan J

(57) Abstract :
A golf course maintenance device comprising of a chassis 101 configured to traverse course surfaces, a mowing assembly 102 to facilitate easy extension and retraction of the rotary blade 103, a debris-collection assembly with the suction pipe 106 coupled with a viper at a front region of the chassis 101 to guide debris into a suction pipe 106, an irrigation assembly to water dry patches on the course, a turf-puncture to puncture turf to release trapped moisture and air, a surface-finishing assembly to smooth and level surfaces following mowing and debris collection.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/02/2026

(21) Application No.202641015873 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED COLOR APPLICATION DEVICE FOR WOODEN OR METALLIC PANELS

(51) International classification	:B05C 11/10, B05C 1/08, B05C 13/02, B05C 1/02, B05B 13/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Gaman Palem

(57) Abstract :
An automated color application device for wooden or metallic panels comprises of a housing 101 having an inlet 102 and an outlet 103, an input means 104 for receiving user commands, a platform 105 attached for placement of wooden or metallic panels, a motorized hinged door 106 for panel entry and exit, an imaging unit 107 to determine panel dimensions, a gripping means for securely handling panels during the coloring operation, a color dispensing unit 109 positioned at the inlet 102 releasing color liquids onto panels via primary and secondary motorized rollers 109b layered with foam members soaked in color liquids, clamping units 110 for handling and directing panels towards a brushing module having sets of brushes 111 for proper color application, and a drying module 112 positioned on the outlet 103 for directing air to dry colored panels, an audio output module 113 for providing audible alerts.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015874 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : POT MANUFACTURING DEVICE

(51) International classification	:A01G 9/02, A01G 9/10, B27N 3/08, B27N 1/00, B27N 3/04	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Yusaf Harun K**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A pot manufacturing device, comprises of a housing 101 equipped with a hopper 102, shredding unit for cutting the waste wood, grinding unit for pulverize the fragments into powder, and a mixing unit that blends wood powder with binding agents using a spiral-blade mixer and weight sensor to regulate the proportion of wood powder and binding agents, a molding unit with hydraulic press 111 to shape the mixture and AI-enabled inspection to detect dimensional errors and surface defects in the molded pots, followed by automated baking, painting, and resin coating of the flower pot, an articulated arm 123 transfers pots between stations, a dust-collection duct 125 and air filter 126 to maintain a clean environment, ensuring efficient, precise, and high-quality pot production.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015875 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FIRE EXTINGUISHER HANDLING AND PROTECTION DEVICE

(51) International classification	:A62C 13/76, A62C 13/78, A62C 13/64, A62C 37/50, A62C 13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Abhishek M

(57) Abstract :
A fire extinguisher handling and protection device, comprising a plate 101 with a cushioned handle102 mounted on a motorized hinges for user manipulation, a holding arrangement with segmented plates 104,suction cups 105, electromagnetic locking, and sensors for secure cylinder 115 alignment and gas level monitoring, a gas spraying unit 107 with electronically controlled valves, a main supply pipe 108, a mini pump 109, and a flow sensor for precise gas discharge, an AI-based fire detection unit 110 with smoke, thermal, and gas sensors, a covering unit 111 with adjustable protective plates and ultrasonic sensors coupled with laser measurement for dynamic user shielding, a fitting unit 114 with a regulator and extendable rods 116 for safe cylinder 115 refilling, and a microcontroller with a communication module for real-time alerts, ensuring efficient fire suppression and user safety.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/02/2026
(21) Application No.202641015876 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SNOW REMOVAL DEVICE FOR VEHICLE

(51) International classification	:E01H 5/06, E01H 5/09, E01H 5/04, E01H 5/10, E01H 5/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Ramanaiah G**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A snow removal device for vehicle, comprises of a pair of vertical rods 101 configured with a horizontal frame 102 adapted to be mounted on each ends of a roadway, an extendable pin is attached on ends of the rods 101 to penetrate into the roadway surface to affix the frame 102 over the roadway, an imaging unit 104 installed on the frame 102 to detect presence of snow layer accumulated over a rooftop of a vehicle, an articulated arm 105 attached with a curved-shaped plate 106 suspended from the frame 102 to perform removal of the accumulated snow layer from the vehicle rooftop, a heating unit embedded across the inner surface of the plate 106 for melting the snow layer accumulated on the vehicle rooftop, and a motorized slider 107 configured between the articulated arm 105 and frame 102 for removing the snow layer from the vehicle rooftop effectively.

No. of Pages : 18 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017042 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONCRETE BUCKET LIFTING DEVICE

(51) International classification	:E04G 21/04, E04G 21/02, B66C 23/36, B66F 9/06, B66F 9/075
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dayalan J

(57) Abstract :
A concrete bucket lifting device comprising a L-shaped frame 101 having a base portion and an upright portion, the frame 101 supported on a plurality of wheels 102, a lifting assembly mounted on the frame 101, configured to lift the concrete bucket, a tilting assembly integrated with the frame 101, configured to tilt the concrete bucket for dumping the concrete, an AI (artificial intelligence) enabled camera 103 mounted on the frame 101 and coupled to a microcontroller embedded in the frame 101, configured to monitor the bucket and alert a user upon detection of abnormal conditions.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017043 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BALLOON MAKING DEVICE

(51) International classification	:B29C 33/72, A63H 27/10, B29C 33/44, B08B 3/02, B29C 33/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. P S Joan Kingsly

(57) Abstract :
A balloon making device, comprises of a housing 101 integrated with a conveyor 102 configured to transport a plurality of glass molds 103 along a sequential operational path, multiple vertical telescopic linkages 104 coupled to the glass molds 103, configured to extend and retract a respective mold vertically, a cleaning solution storage tank 105 arranged within the housing 101 to receive the glass molds 103 via the telescopic linkages 104 for removal of dirt and impurities, the rubber solution storage compartment 106 is provided with multiple burners 111 positioned beneath the compartment 106, the burners 111 configured to maintain the rubber solution at an optimal temperature and viscosity, and a level sensor is integrated with the storage tank 110 and the rubber solution compartment 106 to continuously monitor fluid levels.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017044 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RANGOLI MAKING DEVICE

(51) International classification	:B41L 13/04, B41F 15/08, B41F 15/36, B43L 1/00, G06F 3/0488
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)D Dhanya

(57) Abstract :
A rangoli making device comprises a main body 101 which is equipped with motorized wheels 102 to move the body 101 to the predefined rangoli creation area, an internal stencil storage chamber 103 holds stencils in rack slots, a touch-enabled display 111 presents a library of rangoli designs and receive user inputs for stencil selection, a user selects rangoli designs, stencils, and colors through the touch-enabled display 111, an articulated arm 104 moves along guided rail 106 to place the stencil using a clamp 105, selected color powder is drawn from a plurality of color storage compartments 107 by a pump 108 into an extendable conduit 110 to a nozzle 109 for dispensing color through a motorized valve, an AI camera 113 ensures alignment correction, a speaker 116 alerts on powder misplacement or misalignment, a suction unit 114 removes excess powder into a sealed waste compartment 115.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017045 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE EXTENSION BOARD MANAGEMENT DEVICE

(51) International classification	:G06F 1/16, G03B 17/08, G03B 17/56, H05K 5/06, H01R 13/52
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Tigulla Rajitha

(57) Abstract :
An adaptable extension board management device, comprising of an adjustable body 101 configured to securely accommodate a standard extension board, an iris lid 102 for dynamically sealing around the main cable of the extension board, in view of preventing entry of dust, debris and moisture, multiple wire handling unit for winding and unwinding of the wire, to ensure the wires are untangled, an imaging unit 105 for detecting approaching water/liquid droplets towards the body 101, a covering module for deploying a waterproof sheet 107 over an open top portion of the body 101 to prevent contact of plugs and sockets of the extension board with approaching water/liquid droplets, an input means for receiving user commands for to switching on/off the power supply in the extension board, to trigger a pressing module for manipulating a main switch of the extension board.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017046 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EYEWEAR MAINTENANCE AND ORGANIZING DEVICE

(51) International classification	:G02C 13/00, G01N 21/88, G02C 11/00, G01N 21/958, G01N 21/95
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S Pravinthraja

(57) Abstract :
An eyewear maintenance and organizing device, comprises of a housing 101 adapted to accommodate an eyewear, a laser sensor configured to detect dimensions of the eye wear, a drawer arrangement 102 to extend/retract based on the detected dimensions of the eyewear to securely hold the eyewear, an inspection module configured to detect scratches on lenses and structure cracks on the eyewear, at least one lighting element 103 configured to provide corresponding notification regarding the detected scratches, a crack restoration module configured to dispense glue over the detected structural crack on the eyewear, and a dust detection and cleaning module configured to detect dust concentration on lens and clean the detected dust.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017047 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED ROASTING DEVICE FOR BARBEQUE FOODS

(51) International classification	:A47J 37/06, A47J 37/04, A23N 12/12, A47J 37/00, A23N 12/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Jayanthi Kamalasekaran**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated roasting device for barbeque foods, comprises of a body 101 installed with a plurality of chambers 102 for storage of different food ingredients, a user-input interface to provide commands for selecting desired barbecue dishes, a mixing bowl 105 for receiving measured quantities of ingredients, an iris lid 108 to dispense a regulated amount of the ingredients within the mixing bowl 105, a motorized rod 109 mounted inside the bowl 105 equipped with spiral-shaped blades 110, the rod 109 being configured to rotate at variable speeds and directions to achieve uniform mixing of the ingredients, a roasting tank 111 installed within the body 101 to hold fuel sticks, a heat-resistant mesh 113 for holding the ingredients during roasting, a motorized ball-and-socket joint 107 coupled to the mixing bowl 105 for transferring mixed ingredients over the mesh 113 and an AI camera 114 to monitor roasting progress.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017048 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED AGRICULTURAL WASTE REPROCESSING AND MOLDING SYSTEM

(51) International classification	:C04B38/00, B09B3/00, F26B7/00, B02C13/00, B65G3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Sandhya L

(57) Abstract :
An automated agricultural waste reprocessing and molding system, configured to transform raw agricultural waste into user specified solid products, comprising a framework 101 with bucket elevation module 102 for lifting waste into a grating chamber 103, integrated with a grinding unit 105 to pulverize the waste into fine powder, a container 106 with motorized iris lid 107 for receiving powder, a bleach spraying unit108 for decolorization, a dispensing module 109 with L shaped duct transfers mixture onto a cylindrical tank 110 with sieving plate 111, assisted by a scraping unit112 having air blower and blade for drying and removal of powder and a motorized stirrer 115 forms mixture conveyed to a basin117 via conveyor belt 116, where water dispensing arrangement 118 rehydrates the mixture into slurry, the slurry is guided by an electronic nozzle 121 to a molding assembly119 with detachable platform120 and mould die for shaping solid products.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017049 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GRAIN STORING AND DISPENSING DEVICE

(51) International classification	:A47J47/01, A47J47/04, B65D83/06, B07B1/28, G05B19/042
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Kalpana K

(57) Abstract :
A grain storing and dispensing device comprising a cuboidal body 101 divided into multiple compartments 102 for storing different grains, imaging unit 103 integrated with body 101 for detecting user presence, motorized sliding door 104 at top of each compartment 102 for loading grains, vibratory mesh panel 105 within each compartment 102 for sieving impurities, display unit 106 on body 101 for selecting grain type and quantity, iris-controlled lid 107 beneath each compartment 102 for dispensing into tray 109, motorized sliding unit 108 for moving water reservoir 110 with electronic nozzle 111 to wash grains, plurality of iris apertures 112 for draining excess water, air blower 113 for drying grains, L-shaped extendable bar 114 for pushing grains toward outlet 121, a storage chamber 116 with motorized sliding gate 117 for releasing preservative and a camera 118 to detect insects and suction cups 122 with extendable poles 123 for mounting.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017050 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PHONE CHARGING HOLDER

(51) International classification	:H02J 7/00, H04M 1/04, H02J 7/02, H04M 1/02, H04M 1/72409
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. T Ramesh

(57) Abstract :
A phone charging holder comprising a U-shaped frame 101 constructed from overlapping plates, structured to accommodate mobile phones of varying sizes, a plurality of suction cups 103 for adhering the frame 101 to a fixed surface, at least one imaging unit 107 in synchronization with at least one laser sensor for detecting phone dimensions, multiple sliders 102 for translating the overlapped plates to securely hold the phone, a temperature sensor configured to monitor internal heat during charging, wherein a microcontroller commands at least one Peltier unit with heat sink to regulate temperature beyond a threshold, an alerting unit for prompting the user upon full charge, a vibration sensor for detecting phone vibrations, a cleaning unit 104 with microfiber brushes 104b for dust removal; and an iris hole 105 with elastomeric diaphragm 106 for securing the charging cable, preventing bending stress, and ensuring stable connection.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017051 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SEATING ASSISTIVE DEVICE

(51) International classification	:A47C 7/62, A61H 23/02, A47C 31/12, B60N 2/28, A47C 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Alina Raheen

(57) Abstract :
A seating assistive device comprises of a seat 101 suspended by a four-hook arrangement converging at a single point, at least one electromagnetic clamping unit attaching the seat 101 to a support structure, at least one imaging unit 103 to identify a user profile, a safety harness system configure to automatically deploy upon detection of a child, a vibration massage unit to provide massage, a recliner arrangement 105 to provide support and comfort, a foldable tray 107 for supporting user articles, at least one accelerometer and an inflatable configured to deploy upon detecting irregular motion or instability, a set of load cells for detecting user presence and weight, a pivot assembly 109 for rotating backrest 106 relative to the seat 101 base, a linear actuator to adjust the angular position of the backrest 106, and a locking unit to fix the backrest 106 at the desired angle.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017052 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-BALANCING FITNESS DEVICE

(51) International classification	:A63B 24/00, A63B 21/00, A63B 23/04, A63B 71/06, A63B 22/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Mohan Gowda G S

(57) Abstract :
A self-balancing fitness device, comprising of a hemispherical dome-shaped body 101 adapted to be positioned over a ground surface, a vertical extendable pole 102 configured with a T-shaped frame 103, to secure the user's thigh portion over the frame 103 via a harness strap, an imaging unit 104 work in sync with an ultrasonic sensor to detect dimension of the user, the rod to extend/retract as per the user's height for proper securement of the harness strap, a stability control unit to maintain balance of the body, a tilt sensor work in sync with the imaging unit 104 to detect inclination of the body to provide a corrective balancing forces, and a display panel 105 to select a type of operational mode including assist mode for providing maximum support, sport mode for allowing controlled mobility while assisting balance, and free mode requiring the user to maintain balance independently.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017053 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : VEGETABLE PEELING AND CUTTING DEVICE

(51) International classification	:A47J 17/14, A47J 17/16, A23N 7/00, A47J 17/18, A23N 7/02	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)G S Akshay**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A vegetable peeling and cutting device, comprises of a cylindrical body 101 with plurality of extendable rods 102 for accommodating vegetable to be cut or peeled, an imaging unit 103 in sync with the ultrasonic sensor for detecting height of the vegetable, a first and second motorized expandable pulley arrangement 104 for expanding and contracting the lid 105 and body 101, a motorized peeler 106 through extendable bar 107 for peeling the surface of vegetable, a motorized circular slider 108 having motorized sharp blades 109 for cutting the vegetables, an electronic valve 112 for controlled flow of water for washing the vegetable, a sensing module including a temperature sensor, an odor senor, a gas senor and a moisture sensor for detecting internal environmental conditions and assess the freshness of the stored vegetables.

No. of Pages : 20 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017054 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS ROAD LINE MARKING DEVICE

(51) International classification	:E01C 23/22, E01C 23/20, E01C 23/16, E01C 23/24, G05D 1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Karthik M H

(57) Abstract :
An autonomous road line marking device, comprises of a body 101 equipped with motorized wheels 102 for movement over road surfaces, a user-input interface for selecting and initiating marking patterns, a chamber 104 storing waste glass connected via a hollow passage 106 to a mixing container 105 housing a motorized grinding unit 107 with sharp blades to pulverize the glass into reflective particles, a vessel 108 storing thermoplastic powder melted by a heating unit 109 regulated by a temperature sensor and dispensed through a motorized discharge valve 110, a conduit tube 113 dispensing ground glass powder onto freshly applied thermoplastic strips, a motorized brush 114 to ensure uniform spreading of reflective glass powder, an AI-powered camera 115 for real-time surface mapping and navigation, length sensors for consistent road marking length, an IoT module for remote monitoring and control, and a controller coordinating all components for seamless autonomous operation.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641017055 A
(19) INDIA
(22) Date of filing of Application :16/02/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-CLEANING BASED HYDROGRAPHIC PRINTING DEVICE

(51) International classification	:B41J 29/17, G03G 15/01, B41F 31/02, G03G 21/10, G03G 21/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Chithambar Gupta Vankadari**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A self-cleaning based hydrographic printing device, comprises of a housing 101 equipped with a conveyer belt assembly 102 for receiving and translating an object towards the housing 101, an input means 103 for receiving user commands regarding a specific design to be applied onto the object, an imaging unit 104 installed on the housing 101 to determine dimensions of the object, a colour dispensing unit installed on a peripheral portion of the housing 101 for dispensing calibrated amount of colour liquids onto the water surface, based on the user's specified design, a drainage assembly 109 installed in the housing 101 for releasing used water and debris into a recycling module linked with the drainage assembly 109, and a cleaning assembly arranged in the housing 101 to clean inner surface of the housing 101 after completing drainage of the used water.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017056 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CEILING FAN CLEANING DEVICE

(51) International classification	:A47L 13/20, F04D 25/08, A47L 13/42, A47L 11/40, F04D 29/70
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Anu Joseph

(57) Abstract :
A ceiling fan cleaning device comprises a detachable C-shaped frame 101 mounted onto an extendable rod 102 via a horizontal panel 103, user grips and engages the frame with ceiling fan’s blade by a handle 104, an imaging unit 105 monitors ceiling fan's position on which the extendable rod 102 adjusts position of the frame near fan, a set of laser sensors scan blade’s dimensions for allowing a set of motorized hinges 106 to adjust the frame 101, a mop attachment is interchangeable between a C-shaped mop and a rotating circular mop 201 via a locking arrangement to clean dust while a motorized slider 112 moves the frame across the fan’s blade, a rotating brush 202 along the perimeter of the brush 202 dislodge dust, a suction vacuum unit 108 collects dust into a dust collection compartment 109, a sanitizer spray nozzle 110 connected to a reservoir 111 sprays.

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017057 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TREE STABILIZING DEVICE

(51) International classification	:A01G 23/099, A01G 13/02, A01G 23/095, A01G 23/00, A01G 17/14	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. E Dhanasekar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A tree stabilizing device, comprising a mobile housing 101 mounted on a plurality of motorized wheels 102 to enable movement across uneven terrain, a plurality of artificial intelligence cameras 103 continuously monitoring surrounding trees, a vertical slider 104 move vertically and telescopically extend and retract a rod 105, a motorized gripping unit 106 mounted at an end of rod 105, gripping unit 106 being connected via a ball-and-socket joint 107 to enable 360-degree rotation, a wrapping rope 108 coiled on a roller 109 mounted on housing 101 and integrated with gripping unit 106 configured to wrap rope 108 around a tree trunk and a supporting wooden log to straighten and stabilize tree, an ultrasonic sensor mounted on housing 101 in synchronization with cameras 103 to determine position, distance, and center of tree trunk or log to grip tree trunk and enable movement of wheels 102 synchronously for straightening tree trunk.

No. of Pages : 16 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017058 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED MANHOLE COVERS HANDLING DEVICE

(51) International classification	:E02D 29/14, E02D 29/12, B66F 19/00, E02F 9/20, E02F 9/24	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Shwetha A**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated manhole covers handling device, comprises of a body 101 with motorized wheels 102 for independent movement, an in-built microcontroller receives user commands via an input means 103, an artificial intelligence-based imaging unit 104 detects the position and type of manhole covers, triggering movement of wheels 102 towards manhole cover, a cleaning module, comprising motorized rollers 106, a drill chuck 108 with a drill bit 109, and a vacuum unit, removes debris and hardened asphalt and collect in a vessel, the lifting assembly, equipped with an electromagnetic member 114 for metallic covers and a hook 115 for concrete covers, engages and lifts the cover, an audio output module 116 provides audible alerts and operational instructions to the user and a communication module establishes a wireless connection with the computing unit.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017059 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ORGANIZING ASSISTIVE DEVICE FOR CAR SEATS

(51) International classification	:B60R 7/04, B60N 3/10, B60R 11/02, B60R 11/00, B60N 2/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. V Vamsi Krishna

(57) Abstract :
An organizing assistive device for car seats comprising a housing 101 equipped with multiple suction mounting clamps 102 for secure attachment to rear side of a car seat, a gimbal assembly 103 positioned between clamps 102 and housing 101 to stabilize against vehicular motion, a plate 104 mounted on top portion of housing 101 via a motorized slider 105 for securely holding electronic gadgets, actuated through an input module 106, a primary and secondary compartment arranged on lateral sides for storing personal items, a foldable tray a motorized hinge joint that deploys and locks into position for user convenience, one or more utility hooks 107 embedded along lower edges for hanging lightweight items, and an inflatable member positioned at lower frontal edge to automatically inflate in response to sudden deceleration, enhancing user safety and comfort during travel.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017060 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS PEA SHELLING AND PACKAGING DEVICE

(51) International classification	:A23N 15/10, B65B 25/04, A01F 25/14, A47J 43/24, A23N 15/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Swetha Rajagopal**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An autonomous pea shelling and packaging device, comprises of a housing 101 configured with an inlet for pea pod insertion and internally divided compartments, a cutting unit to slice open each pod longitudinally without damaging the peas inside, a pair of horizontally movable plates 102, each mounted on an extendable rod with a vertical slider 103 positioned to engage and spread open the flap 104 of the cut pod, a scoop-shaped flap 104 disposed below the plates 102 via a linear to push peas from within the pod toward a pea collection chamber, a peel collection tray to slide outward from the housing 101 for easy removal of waste pod shells, a robotic arm 105 to pick and remove defective peas and deposit the defective peas in the tray, a washing module arranged within the chamber for rinsing and draining the peas.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017061 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PAINTING ASSISTIVE DEVICE

(51) International classification	:B44D 3/00, B44D 3/02, B44D 3/04, B44D 3/22, B44D 2/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Mohammed Mujeer Ulla

(57) Abstract :
A painting assistive device, comprises of a frame 101 constructed from a horizontal painting platform connected to a vertical back support through a motorized hinge joint 102 with multi position locking for angular adjustment, a set of extendable legs 103 with locking units for stable height positioning, a brush holder 104 mounted on extendable rod 105 with motorized iris holes 106 for accommodating different brushes, a cleaning chamber 107 with vibrating units for automatic brush cleaning, a color palette 108 integrated with storage chamber 109 s and electronically controlled nozzle 110 for regulated paint dispensing, an imaging unit 112 detecting brush strokes color usage and canvas coverage, an artificial intelligence based color module providing blending recommendations and a holographic projection unit 115 for real time simulation of color module along with a collapsible cushioned seat 114 with adjustable height and angle ensuring user comfort and precision.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017073 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WIRE WRAPPING DEVICE

(51) International classification	:H02G 1/12, H01B 13/26, H02G 1/14, G01R 31/02, H01B 13/012
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Mahaboob Pasha

(57) Abstract :
A wire wrapping device, comprises of an elongated body 101 with a handle 102 to be held by a user, a user-interaction module to provide input commands for wrapping of a wire, a wrapping module to wrap insulating tape uniformly around a wire, a cutting module to sever the insulating tape upon completion of the wrapping, a robotic gripper 112 secures the tape's starting end, while an inspection module detects completion of wrapping, a sensing module to detect insulation defects, a leakage clamp meter to monitor abnormal leakage currents and a speaker 114 to generate an alert upon detection of defects and moisture content.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017074 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PIZZA MAKING DEVICE

(51) International classification	:A21C 1/14, A21C 9/04, A21C 9/06, A21C 9/08, A21C 1/02	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Joseph Michael Jerard**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A pizza making device comprises of a housing 101 with a hinged access opening secured via electromagnetic locks to provide access while placing a pizza/pizza ingredient, a mixing chamber 102 within the housing 101 to mix and knead the materials, an articulated arm 103 with a vacuum gripper 104 extending from the top surface of the housing 101 to retrieve and transfer the dough onto a platform 105, a hydraulic piston bar 106 with a pressing plate 107 to press onto the dough on the platform 105, a conveyor belt 108 to the platform 105 to receive and transfer the pizza base, a topping dispensing assembly within the housing 101 to dispense toppings onto the base placed on the conveyor, an oven 109 placed subsequent to the conveyor to bake the pizza and an artificial intelligence enabled camera 110 integrated within the housing 101 to monitor the pizza making operations.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017075 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE FROSTING PREPARATION AND DISPENSING ASSISTIVE DEVICE

(51) International classification	:A23G 9/28, A23G 9/22, B67D 7/74, A47J 43/07, A23G 9/04	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Bikram Sarkar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An adaptable frosting preparation and dispensing assistive device comprises a handheld housing 102 configured to accommodate modules for preparing and dispensing frosting onto food items, a mixing chamber 103 to receive raw ingredients, blended using a motorized stirrer 104, with weight sensors monitoring ingredient and frosting levels to notify the user when refilling is required, a container 105 linked to the mixing chamber 103 via a conduit 106 equipped with a solenoid valve 107 and a flow sensor to regulate transfer volume, a nozzle arrangement mounted on the housing 102 includes multiple interchangeable nozzles 108 on a rotatable gun-barrel arrangement 109, with a moisture sensor detecting residue or moisture buildup for cleaning alerts, a dispensing unit with linear actuator 110 and plunger 111 applies pressure to guide frosting through the aligned nozzle, while pressure and flow monitoring sensors detect obstruction, the microcontroller coordinates operations, ensuring precision and user-friendly frosting application.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017076 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS REPAIRING AND MAINTENANCE DEVICE FOR VEHICLE CHASSIS

(51) International classification	:B60S 5/00, G05D 1/02, B23P 6/00, B62D 65/00, B62D 65/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Keren Lois Daniel

(57) Abstract :
An autonomous repairing and maintenance device for vehicle chassis, comprises of a platform 101 with multiple motorized wheels 102 for providing movement to the device, an input means for enabling a user to provide input commands regarding the operation, an artificial intelligence-based unit to detect and monitor the surroundings for scanning the vehicle for visual inspection and repairing monitoring, a sensing module to detect the condition of the chassis before repairing, a cleaning unit for clearing the residues within the chassis surface, a smoothening unit for smoothening out the surface, a welding unit 111 for depositing filler materials onto the prepared cavities, a painting unit for spraying colored fluid onto the discolored sections, a heating unit to heat the misaligned sections and an alignment correction module to realign the softened section guided by a strain gauge feedback.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017077 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE SURFBOARDING BALANCE CONDITIONING DEVICE

(51) International classification	:A63B69/00, A63B22/16, A63B24/00, A63B26/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Divyashree D

(57) Abstract :
An adaptive surfboarding balance conditioning device comprises of a base structure 101 with an inclined ramp 102 having a motorized hinge 103 for angle adjustment, a platform 104 to accommodate a user, a plurality of air bladders 105 to simulate wave-like instability for balance training, multiple weight sensors for detecting user's presence and weight distribution, a 6 Degree of freedom motion deck 106 adjusting tilt angle of the platform, a safety harness 107 with a motorized slider 108 for securing user's body over the platform, multiple edge-grip modulators 109 for additional grip during excessive sway, an audio output module 110 providing audible alerts to the user and a display panel 111 to display real-time balance metrics including sway amplitude and weight distribution.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017078 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COLUMN CAVITY CREATING DEVICE

(51) International classification	:B23Q 11/10, B23Q 17/24, B23Q 11/14, B23Q 11/12, B23D 47/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Anju Mathew

(57) Abstract :
A column cavity creating device, comprises of a housing 101, an imaging unit 103 with a proximity sensor for user detection, a sliding door 104 for loading a metal column, a pair of holding units 105 to secure the metal column, a cutting assembly with a rotary cutting blade 107 on an articulated hydraulic link 108 performs the vertical cut while an RPM sensor monitors rotational speed of the cutting blade 107, a pneumatic pusher 109 provides counter-force to the metal column ensuring stability during cutting, an electronically controlled nozzle 110 dispenses cooling water from a reservoir 111 onto the cutting blade 107 and the press head to dissipate heat while a level sensor determines water level in reservoir 111, a display unit or a microphone for user input and a communication module for remote access and control.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017079 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MRI (MAGNETIC RESONANCE IMAGING) SAFETY SCREENING SYSTEM

(51) International classification	:G01R 33/28, A61B 5/055, G06K 9/00, G01R 33/56, G01R 33/48	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. D R Denslin Brabin**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A MRI (Magnetic Resonance Imaging) safety screening system comprises of an enclosure 101 internally divided into a first, second and third section 104 positioned inside the MRI room, a microphone 105 providing input commands regarding opening of enclosure 101, a primary motorized hinged door 106 to enter inside the first section 102 for changing clothes, a facial recognition camera capturing facial features of user, a biometric electronic lock 108 accommodating user's belongings into locker 107, a cloth hanging chamber 109 hanging the clothes, a display panel 110 prompting questions regarding metallic implants, an X-ray scanner 113 detecting the presence and absence of concealed metallic objects, a steam generator 114 releasing steam over clothes, a motorized air blower 115 drying the clothes, a UV disinfection light 116 disinfecting clothes, a Peltier unit 117 regulating thermal environment, a secondary motorized hinged door 118 to exit the enclosure 101.

No. of Pages : 23 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017080 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WIRE PROCESSING AND REFURBISHMENT DEVICE

(51) International classification	:B21F 11/00, H01R 43/28, B23P 23/04, B23P 19/04, B23K 9/133
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sridevi S

(57) Abstract :
A wire processing and refurbishment device comprises of a housing 101 with a roller to feed wire through an opening at a frontal portion of the housing 101, a microphone 103 receiving input commands regarding refurbishment operations of the wire, an imaging unit 104 inspect the cracks present in the wire, a MIG welding module restore the detected cracks, a wire bending unit including a pair of clamping jaws 110 receive, secure and bend the wire into the user-specified shape, a cutter module including includes a motorized shear blade 112 trim the wire to a threshold length, a nozzle 113 dispense a rust removal agent via a conduit 115, a brush 116 scrub the wire and remove the rust from the wire, a cooling unit including at least one fan 119 and coolant circulation path to dissipate heat generated during MIG welding and rust removal processes.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017081 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONDUIT ELBOW MAKING DEVICE

(51) International classification	:B21D 7/16, F16L 43/00, B21D 7/00, B21D 7/025, B21D 7/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Krishna Mishra

(57) Abstract :
A conduit elbow making device comprising a housing 101 equipped with an inlet for receiving pipes by a user, an input module installed on housing 101 and accessed by user to provide input commands for producing conduit elbows, a defect detection module installed within housing 101 and configured to detect cracks or irregularities along with material type of pipe, a cutting module installed within housing 101 and configured to cut pipe into a user-defined length and remove defective sections upon detection by defect detection module, and a bending unit 107 mounted within housing 101 and configured to bend pipes at desired angles, device enabling automated and precise production of conduit elbows with integrated inspection, cutting, and bending functionalities to enhance efficiency and accuracy in pipe processing.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017082 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED PROCESSING DEVICE FOR METALLIC SHEETS

(51) International classification	:B23Q 17/24, B23Q 17/20, B23Q 17/22, B21D 1/02, B21B 38/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Battula Bhavya

(57) Abstract :
An automated processing device for metallic sheets comprises of a frame 101 configured with multiple legs 102 for stabilization, an input means 103 to provide commands for levelling, thickness adjustment, trimming, or surface finishing operations on a metallic sheet, an imaging unit 104 to assess surface conditions including dimensions, irregularities, dust, or rust, a primary and secondary set of motorized rollers via a worm gear rack jack assembly 107 for vertical gap adjustment based on sheet dimensions, a heating unit 108 to provide calibrated heating, a cutting unit 109 for trimming excess edges, a cleaning module 110 for removing corroded surfaces, and a sensing module monitoring real-time temperature, pressure, and excess materials to regulate all processing operations.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017086 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ROOF SHEET PROCESSING DEVICE

(51) International classification	:E04D 15/04, E04D 15/00, E04G 21/32, E04D 15/02, B29C 51/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)K lekhya Sree

(57) Abstract :
A roof sheet processing device, comprising a platform 101 adapted to be placed over a surface, a plurality of suction cups 102 is installed beneath the platform 101 to stabilize the platform 101 over the surface during the operation, an input module 103 paired with the platform 101 and accessed by a user to provide input commands regarding shaping a roof sheet along with desired contours and dimensions, a conveyor belt 104 mounted over the platform 101, the conveyor belt 104 configured to transfer a roof sheet for further processing, a moulding unit installed over the platform 101, the moulding unit configured to receive the sheet and apply controlled shaping and pressure to form the desired contours and dimensions, a rotatable imaging unit 116 mounted over the platform 101, the imaging configured to detect surface defects including rust, dents, and scratches on the sheets.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017087 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE BED-MOUNTED MOBILITY ASSISTIVE DEVICE FOR IMPAIRED AND ELDERLY INDIVIDUALS

(51) International classification	:A61H 3/04, A61G 7/10, A61H 3/00, A61G 7/015, A61G 7/05	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. T Lalitha**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An adaptable bed-mounted mobility assistive device for impaired and elderly individuals, comprising a U-shaped frame 101 having secure mounts 102 on each leg stabilized on a ground surface by an Archimedean spiral jack assembly 103, a plurality of clamp 104ing modules are arranged on each leg through articulated links 105 and motorized clamp 104s attached via extendable bar 106 with motorized ball and socket joints 107 for gripping the bed front panel, a monitoring unit with an artificial intelligence-based imaging unit 108 enables posture and movement detection, a handle grip 109 mounted on an adjustable strap 110 with motorized hinge joints is deployed via a bar-car hood assembly, an input means installed on the frame 101 to provide input commands, an audio output module 112 paired with frame 101 to give alerts and a laser projection unit 113 installed on the frame 101 for projecting laser beams.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017088 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RESTING ASSISTIVE DEVICE

(51) International classification	:G01W 1/14, E04H 15/00, E04H 15/58, E04H 15/06, A47C 7/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Megala G

(57) Abstract :
A resting assistive device, comprising an X-shaped frame 101 positioned over beach surface with extendable pneumatic pins for stable anchoring, a motorized roller 103 wrapped with synthetic sheet 104 for creating smooth resting surface, an input means 102 for triggering deployment of resting surface on the sand, an optical sensor for monitoring surface conditions, a rain sensor for detecting rainfall and actuating extension of plurality of fabric panels 105 mounted on L-shaped bars 106 via drawer arrangement to shield user from rain, a solar panel 107 mounted on articulated limb 108 supplying power to battery, a storage compartment 109 for securing personal belongings and a speaker 110 for providing audible alerts with a communication module is integrated within the microcontroller ensuring automated safe efficient and convenient resting assistance with minimal user intervention.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017089 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GROUNDNUT SHELLING DEVICE

(51) International classification	:A23N 5/08, A23N 5/00, A47J 43/26, A23N 5/01, A23N 5/06	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Sharanya S Shalin**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A groundnut shelling device, comprises of a housing 101 positioned on a fixed surface, upon activation, a user initiates groundnuts shelling operation by a microphone 103, a sliding tray 104 extends outward for groundnut placement and retracts back, a dual-axis lead screw arrangement 105 moves a pressing module 106 above the groundnuts, a vertical rod 106a with a pneumatic plunger 106b extends a pressing plate 106c downward to crack the groundnut, a separation unit 107 comprising a V-shaped pin 107a extends through an extendable rod 107b and aligns via an optical sensor on which an electromagnetic spring 107c drives the V-shaped pin 107a to split the shell, an air blower 110 pushes hulls through a motorized flap door 108 into a storage compartment 109, a sensing unit comprising a weight sensor monitors load accumulation in the chamber and the compartment 109, a speaker 117 issues alerts if weight exceeds.

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017090 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS FOOTWEAR SOLE RESTORATION DEVICE

(51) International classification	:A47L 23/22, A47L 23/02, A47L 23/26, A43D 1/02, A43D 1/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Sushmitha S S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An autonomous footwear sole restoration device, comprises of a housing 101 having a pair of sections 102 configured to support a footwear, an input means operatively linked with an inbuilt microcontroller for receiving user commands regarding sole restoration, an inspection unit installed on the housing 101 to determine dimensions, tread patterns, dust and debris accumulation, embedded sharp objects and wear condition of the footwear soles, a gripping arrangement integrated within each section 102 for providing stable support to the footwear, a cleaning module arranged on inner walls of each section 102 via a two-axis motorized slider 108, including a motorized brush 109 mounted on an articulated link 110 for scrubbing, a spraying unit for cleaning accumulated dust and debris from the soles, an extracting arrangement for removing sharp objects from the soles, and a waste compartment 119 arranged underneath the sections 102 for collecting debris and water efficiently.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017091 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FARM PRODUCE PREPARATION DEVICE

(51) International classification	:A23N 12/02, A01G 9/24, B65G 1/137, B65G 1/04, A01K 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Poonam Nilesh Yadav

(57) Abstract :
A farm produce preparation device configured with a tray 101 comprising a plurality of storage compartments 102 configured to store farm produce, motorized sliding panel 103 configured to selectively open and close the compartments 102 for controlled access and storage, a container 104 configured to facilitate washing or rinsing of the farm produce placed on the tray 101, an imaging unit 115 configured to detect the presence, position, and condition of the farm produce placed on or within the compartments 102 of the tray 101, and a sensing module to detect quality and quantity parameters of the farm produce and transmit corresponding data to a connected user-interface for enabling a user to take appropriate actions.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017092 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE MOVEMENT SUPPORTING DEVICE

(51) International classification	:A61H 3/00, A61H 3/04, G03B 17/56, F16M 11/14, A61H 1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Impa B H

(57) Abstract :
An adaptable movement supporting device comprises of a frame 101 having a pair of vertical L-shaped rods 102 to be positioned on ground surface, a curved link 103 supporting the rods 102, a U-shaped member 104 providing stability to the frame 101, a plurality of suction units 105 affixing the frame 101 on the surface, a pair of bars 106 providing support to the user, an imaging unit 107 detecting the height of user, a motorized sliding unit 108 orienting the bars 106 in proximity to the user's hand, a cushioning material 109 for comfortable grip, a spherical joint 110 for optimal detection of the user's height, an extendable plate 111 providing a seating support for the user, a plurality of wheels 113 with an extendable pole 114 for movement of the frame 101, braking pads securing the frame 101 in position and a lighting element 115 providing illumination.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017093 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HANDHELD SUMP RESTORATION DEVICE

(51) International classification	:E02D 19/10, E02D 31/02, B05C 11/10, E02D 31/00, E02D 19/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)B Ramana Reddy

(57) Abstract :
A handheld sump restoration device, comprises of an elongated body 101 with a handle 128 for the user to grip and manipulate, a multi-sectioned storage chamber 104 mounted at the distal end 103 to store resin–hardener mixture and fiber mat sheets, input module to provide operational commands by user, a stabilization module anchors the device securely against sump walls, an inspection module to detect cracks, measure depth, classify severity, and identify surface irregularities over the sump base, a surface cleaning module to smoothen the defective sump surface, a suction unit 115 smoothens the defective sump surface, a dispensing module on the body 101 discharges resin–hardener mixture over defects, a spreading module with a motorized soft-silicone roller 118 and extendible linkage ensures uniform material application with pressure feedback, a fiber-mat application module mounted on to grip, align, and place a fiber mat sheet over the coated defective surface.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017094 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TRANSFORMABLE HEALTH-ASSISTIVE RESTING DEVICE

(51) International classification	:A47C 20/04, A47C 7/62, A47C 17/86, A47C 17/04, A61B 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Muthuraju V

(57) Abstract :
A transformable health-assistive resting device comprises of a rectangular panel 101 segmented into multiple sections 102 with hinge 105 joints to switch between flatbed and chair configurations, a drawer arrangement 107 to extend or retract according to user height, an input module 109 to receive user commands for preferred configurations, an extendable bar 110 with a curved member 111 to assist in tilting sections 102 during chair configuration, supporting rods 103 with stabilizing plates 104 for stability, multiple pressure sensors to monitor air pressure within the inflatable member, a health monitoring module 112 to detect pulse rate and blood flow, a vibrating unit 113 for therapeutic relief, a motorized bar-car hood arrangement 114 to deploy a desk surface, a temperature regulation module with Peltier unit, fan 116, and aroma diffusers for cooling, heating, and stress relief, and a speaker 118 to deliver posture guidance and health alerts.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017095 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE PAW-CLEANING DEVICE

(51) International classification	:A01K 29/00, A01K 13/00, A46B 13/02, B25J 19/02, A61B 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Vivek Bongale

(57) Abstract :
A wearable paw-cleaning device comprises of a wearable body 101 worn on a foot of a pet animal, motorized hinges 102 configured to adjust shape and dimensions of the body 101 to accommodate paws of varying sizes, a looped strap 103 wound on a motorized roller 104 to allow extension of the strap 103 around the foot, an imaging unit 105 integrated with a soil detection sensor to determine presence of mud, soil, or debris on the paw surface, a cleaning unit to clean the paw surface detected via the imaging unit 105, a plurality of iris lids 113 allowing drainage of water, mud, and debris from the paw, an air blower 114 generating a stream of warm air for drying the paw post cleaning, and a plurality of UV lights 115 sterilizing and sanitizing the paw after drying to eliminate microbial contaminants.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017096 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIRD FEEDER DEVICE

(51) International classification	:A01K 39/012, A01K 39/014, A01K 39/01, A01K 39/04, A01K 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Shashikala A R

(57) Abstract :
A bird feeder device, comprises of platform 101 having a concave upper surface with radial grooves, configured to collect dispensed grains and enable access from multiple directions, a hanging module comprising an inverted U-shaped extendable rod 102 and a circular ring 103 with curved grippers 104 to securely suspend the platform 101 from an external mounting structure, a motorized lid 107 positioned at an inlet of each compartment, configured to enable controlled refilling of grains, a solenoid-based dispensing valve 108 at an outlet of each compartment to direct dispensed grains onto the platform 101, an imaging camera 109 mounted on the storage 105 to detect the presence of birds on or near the platform 101, and a canopy arrangement 112 installed on the platform 101 to deploy a fabric configured with the links in a manner to avoid rain.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017097 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EGG BEATING AND INGREDIENT MIXING DEVICE

(51) International classification	:A47J 43/10, A47J 43/14, A23L 15/00, B01F 15/00, B01F 11/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Kummara Sruthi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An egg beating and ingredient mixing device a base platform 101 resting on a surface, a pair of elongated side supports 103 extending upward from opposite ends of the base platform 101, a circular top housing 104 mounted across the upper ends of the side supports 103, multiple fastening members 105 securing the side supports 103 with the base platform 101 and the top housing 104, an imaging unit 106 to determine a spatial position and alignment of an auxiliary bowl, a robotic gripper 107 grasp the bowl placed by the user, a pair of motorized clamps 108 integrated with the housing 104 connected to a DC motor for driving interchangeable mixing tools, and a dual-axis slider translate the link 109 carrying the motorized clamps 108, thereby enabling complete surface coverage of the auxiliary bowl during mixing operations.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017098 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ROPE CLIMBING TRAINING DEVICE

(51) International classification	:A63B 69/00, A63B 24/00, A63B 29/02, A63B 21/00, A63B 23/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Suhel Khan

(57) Abstract :
A rope climbing training device comprises a support frame 101 with an upper portion and a base portion, upon activation, a proximity sensor integrated with the frame 101 detects the user's presence on a plate 104 positioned on base portion, a motorized pulley 102 mounted on the upper portion of the frame 101 rotates to dispense or retract the climbing rope 103 as required, a pair of motorized slider rails 107 positioned on the base portion adjust the plate's position to vary rope 103 angle and climbing difficulty, a plurality of suction units 105 on an underside of the base portion of the support frame 101 maintain firm floor contact throughout, a 360-degree AI camera 108 mounted on the frame 101 monitors posture and movement, corrections are projected via a holographic unit as floating 3D guidance cues.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017099 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CLIMBING TRAINING ASSISTIVE DEVICE

(51) International classification	:A63B 69/00, A63B 24/00, A63B 71/06, A63B 23/04, G06F 3/042	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Smitha S P**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A climbing training assistive device comprising a first frame 101 adapted to be positioned over a ground surface, wherein an L-shaped extendable rod configured with a suction unit 103 is installed on sides of first frame 101 adapted to provide support to first frame 101 over surface, an imaging unit 104 integrated with a processor captures multiple images in vicinity of first frame 101, wherein based on captured images, an inbuilt microcontroller determines positioning of first frame 101 over surface, a second frame configured with multiple plates 107 attached on front portion of first frame 101, a touch interactive display unit 108 installed on first frame 101 to receive input commands, a plurality of extendable link 109 configured with a rectangular block 110 on each plate, and a motorized swivel joint 111 configured between first frame 101 and second frame to adjust orientation of second frame based on user input.

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017100 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED VEHICLE BALANCING ROD MAINTENANCE DEVICE

(51) International classification	:G01M 17/007, G01M 1/02, G01M 1/16, B23P 6/00, G01M 1/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Vijayalakshmi P

(57) Abstract :
An automated vehicle balancing rod maintenance device, comprises of a platform 101 equipped with holding means to securely hold the balancing rod, an input module to enable a user to provide input commands regarding repairing of the balancing rod, an inspection module, mounted on an articulated arm to detect misalignment, cracks, and corrosion on the rod's surface, an alignment correction unit to apply controlled forces to restore alignment of the balancing rod, a crack repairing module to prepare surface, repair cracks and smoothen repaired areas respectively, a rust treatment module to remove rust and apply an anti-rust protective coating on the rod's surface.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017101 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SHEET CLEANING DEVICE

(51) International classification	:G07D 11/00, B65H 7/06, B65H 7/02, B65H 5/06, B65H 5/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Roshni M Balakrishnan**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated sheet cleaning device comprising the automated sheet cleaning device comprising a housing 101 paired with an input means for receiving user commands regarding sheet handling operations, a chamber 102 integrated with a weight sensor to store multiple sheets and detect real-time weight of the stored sheets, a transferring means installed along the inner walls of the housing 101 to grip and transport the sheets across to a platform 116, an artificial intelligence based imaging unit 103 to scan the position and dimensions of the sheet being transferred, a sensing module identify the type of sheet, a vibrating unit to dislodge minute particles from the sheet's surface, a water spraying arrangement for spraying water onto the sheet's surface and remove dust, a drying unit to remove moisture from the sheet and a container 104 for receiving dried an clean sheets handled by a gripping assembly.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017102 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-TUMBLER HOLDING AND CONVEYING DEVICE

(51) International classification	:B65G 7/12, G03B 17/56, B65G 47/90, B65D 25/28, B62B 3/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Bhavana S

(57) Abstract :
A multi-tumbler holding and conveying device comprising a circular plate 101, extendable rod 102 arranged on plate 101 via motorized ball and socket joint 103, a handle 104 at end of rod to adjust plate 101 height, plurality of expandable tumbler holding modules 105 arranged over plate 101, including a C-shaped rigid member, a circular ring with an iris lid, and an L-shaped extendable supporting bar 106 integrated with ring to adjust grip, an imaging unit 107 installed over plate 101 to monitor tumbler positioning, extendable flaps 108 arranged on each side of plate 101 to partially cover tumblers, an inflatable pad integrated underneath plate 101 to cushion tumblers, a suction unit 110 over plate 101 to remove spilled liquid, a six-bar linkage arrangement 111 for smooth deployment and retraction of flaps 108, a gyroscope integrated with each module to detect angular movement and speaker 112 for audible feedback.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017103 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SEAT BELT ENGAGEMENT DEVICE

(51) International classification	:B60R 22/48, B60R 22/18, B60R 22/46, B60R 22/00, B60R 21/015
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S Siraj Ahmed

(57) Abstract :
A seat belt engagement device, comprises of a hollow body 101, an imaging unit 102 integrated with the hollow body 101 to determine the position and orientation of a seat belt tongue and detect a malfunction of the underlying seat belt locking means on which the hollow body 101 is mounted, a laser sensor integrated with the body 101 and in sync with the imaging unit 102 to detect dimensions of the locking means, a locking unit 103 integrated within the body 101 to engage the seat belt tongue to hold it securely in position, an extendable plate 106 arranged on each lateral side of the body 101, to extend or retract for providing ergonomic hand support without obstructing neighbouring seats, and an electromagnetic spring 109 is integrated between each articulated limb 108 and its respective clipper to provide controlled gripping force and resilient retraction upon release.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :16/02/2026
(21) Application No.202641017104 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DRAWSTRING INSERTION DEVICE FOR GARMENTS

(51) International classification	:A41H 37/00, D05B 35/06, A41H 43/00, A41H 43/02, D05B 35/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Sai Dhivya

(57) Abstract :
A drawstring insertion device for garments comprising of a housing 101 adapted to positioned on a ground surface, plurality of suction cups 102 to secure the housing 101 over the surface, a touch interactive display unit 109 to provide input specifications regarding insertion of drawstrings into garments, an extendable gripper 103 to work in sync with an imaging unit 104 to retrieve the garment from the chamber, a motorized dual- axis slider 105 configured with plurality of articulated clamps 106 positions the garment towards the articulated clamps 106 to grip the garment, a spiral-shaped guiding ring 107 enabling the garment to be inserted onto the guiding ring 107 in a clockwise manner to facilitate the drawstring insertion process, plurality of drawstrings to remove the garment from the guiding ring 107 by rotating into anti-clockwise such that during removal of the garment the drawstring is getting inserted into the garment.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/02/2026

(21) Application No.202641017105 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HONEYCOMB WAX EXTRACTION AND CANDLE MAKING DEVICE

(51) International classification	:C11C 5/00, C11C 5/02, B01D 11/02, A01K 59/04, A01K 59/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Kuppala Saritha**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A honeycomb wax extraction and candle making device, comprising a housing 101, an input means to provide commands, an imaging unit 102 to detect presence of user near housing 101, motorized sliding door 103 to allow placement of honeycombs into chamber 104, pneumatic pusher 105 to press honeycombs for honey extraction, suction unit 106 to withdraw extracted honey through conduit 107 into storage unit 114, robotic gripper 108 to grasp residual honeycomb and place it into water reservoir 109 for wax purification, a camera 110 to monitor floating wax and direct extraction unit via an articulated link 111, a wax melting unit with a bowl above a water box 112 for uniform melting of wax, a robotic link 113 to place wicks into mold 115 cavities, a robotic arm 116 to pour melted wax into mold 115 to form candles, and suction cups 118 to adhere housing 101 to surface.

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017681 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TAP INSTALLATION DEVICE FOR BOTTLES OR CONTAINERS

(51) International classification	:B23P 19/04, B25J 15/08, B23P 19/00, B25J 15/04, B23B 41/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Shivabasamma Beli**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A tap installation device for bottles or containers comprises of a housing 101, an input means 102 for initiating tap installation operation, an imaging unit 103 determines the distance between the housing 101 and the bottle, extendable rods 104 enable positioning of the housing 101 near the bottle, a motorized wheel 105 aids movement, robotic arms 106 grasps and positions the bottle within the housing 101, a drilling unit 107 forms a hole in the bottle for tap installation, a pair of robotic grippers 109 holds the bottle securely during drilling, an articulated bar 110 having a clamp 111 grasps a tap and a sealing component from a multi-sectioned chamber 112 and positions them in the drilled hole, an electronic nozzle 113 dispenses adhesive from a tank 114 over the tap and sealing component to secure the tap, a speaker 115 provides audible alerts, a flow sensor monitors adhesive usage.

No. of Pages : 22 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017682 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SHOCK-ABSORBING SUPPORT DEVICE FOR WASHING MACHINE LEGS

(51) International classification	:D06F 39/12, D06F 37/20, D06F 39/00, D06F 37/22, D06F 33/02	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Akkamahadevi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A shock-absorbing support device for washing machine legs, comprises of a frame101 integrated with an imaging unit 103 and ultrasonic sensors to detect dimensions of each leg and adjust the frame101 accordingly for secure placement and balance, corner-mounted shock-absorbing units 104 to reduce vibrations, noise, and structural stress, a plurality of suction cups 105 connected via extendable links 106 with pressure sensors to maintain optimal grip on the floor without causing damage, a camera 107 and dust sensor to detect debris or dust beneath or along the machine, a motorized brush to clean the detected areas automatically, a microphone 108 for receiving user voice commands, omnidirectional wheels109 controlled by the microcontroller for multidirectional movement, telescopic frame segments 102 for size adjustment of the frame101 , a speaker111 for audio feedback, providing stability, hygiene, noise reduction, and enhanced user interaction for improved washing machine performance and maintenance.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017683 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RESCUE SYSTEM FOR EMERGENCY SIGNALLING AND VICTIM DETECTIONS DURING NATURAL CALAMITIES

(51) International classification	:B63C 9/02, B63C 9/00, G01F 23/292, B63C 9/01, B63C 9/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Madhusudhan M V**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A rescue system for emergency signaling and victim detections during natural calamities comprising, a column 101 integrated with water level sensors for detecting water level within the surroundings, an artificial intelligence based imaging unit 102 with a proximity sensor to scan the vicinity for individuals in need of help, a shielding module having five panes to form a hull-like structure for diverting the water flow, a lifting unit arranged with each panes to adjust the height of the panes, an inflatable balloon 104-like structure to be inflate by an inflating unit to generate SOS signals, an emergency light 105 to aid in survivor detection and a transmission module to send SOS signals through a wireless communication interface.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017684 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MANHOLE COVER HANDLING AND CLEANING DEVICE

(51) International classification	:E02D 29/14, B08B 1/04, B08B 1/00, E02D 29/12, B66F 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Venkatesha Raju K

(57) Abstract :
A manhole cover handling and cleaning device, comprises of a main body 101 supporting extendable rods 104 and motorized wheel 105 for maneuvering and positioning over a manhole cover, a motorized sliding gate 106 at the base of the body 101 to allow an electromagnetic flap 107, connected via an articulated link 108, to securely engage and lift the manhole cover, an imaging unit 103 for determining exact location and alignment, a cleaning unit including an electronic nozzle 109 mounted on a robotic gripper and a motorized brush 111 mounted on an articulated bar 112 for thorough cleaning of the manhole cover and surrounding inner walls, a flow sensor and RPM sensor for regulating fluid dispensing and cleaning intensity, a speaker 113 for real-time audible alerts, and a communication module for wireless remote monitoring and control.

No. of Pages : 23 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017685 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BLADE MANUFACTURING DEVICE

(51) International classification	:B26D 7/06, B26D 7/02, B26D 3/08, B26D 7/01, B26D 3/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)B Ramana Reddy

(57) Abstract :
A blade manufacturing device, comprises of a housing 101 securely fixed to a surface, a first motorized conveyer belt 102 arranged within the housing 101 to facilitate the transportation of a container through operations, a blade 103 coupled to a rod 104 via a ball and socket joint to cut the container into sections, a pair of motorized rollers 105 arranged subsequently to the first conveyer belt 102 to receive and flatten the cut container sections into sheet form, multiple articulated arms 114, each coupled with an adjustable clamping unit to grip and stabilize the container during cutting, a second conveyer belt 106 arranged subsequently to the rollers 105 to receive transport the flattened sheets for further operations, and a piston 107 coupled to a pressing plate 108 integrated with multiple blade-molding units arranged above the second conveyor belt to press onto the flattened sheets to cut and form blades.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017686 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED CHARCOAL CUBES MANUFACTURING DEVICE

(51) International classification	:B30B 11/24, C10L 5/44, B30B 11/22, C10B 53/02, B01J 2/20	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Santhosh M B**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated charcoal cubes manufacturing device comprises of a housing 101 configured with a hopper 102 for receiving charcoal powder, an input means operatively connected with an inbuilt microcontroller for enabling a user to provide input commands, an extrusion unit linked with the hopper 102 for precise molting of the charcoal powder, a dispensing arrangement linked with the extrusion unit for controlled transfer of the molten charcoal mixture to a plate 104, a gripping means to acquire grip of suitable molding tray from a container 103 corresponding to the user-specified size of charcoal cubes, a plurality of motorized iris lids 106 arranged on the plate 104 configured to dispense the charcoal mixture into the molding trays, a collection unit adapted to collect the charcoal cubes from the molding trays.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017687 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DRUM RUBBERS MANUFACTURING DEVICE FOR TWO-WHEELERS

(51) International classification	:B26D 7/18, B26D 5/12, B26D 7/01, B26D 3/16, B29D 30/26	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Thatimakula Sai Kumar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A drum rubbers manufacturing device for two-wheelers device comprises of housing 101, an input means 102 for user commands, an imaging unit 103 to detect user presence, a motorized sliding gate 104 for placing waste tyre tube, a motorized cutter 106 with a hydraulic piston for controlled cutting, a robotic gripper 107 for handling cut rubber sheets, a pneumatic pusher 109 with a frame-mounted contact plate 110 and articulated arms 111 for pressing and shaping, suction cups 112 for stable placement, a suction unit 115 for waste removal, a frame storage chamber 114 for different drum rubber designs, position and RPM sensors for precise cutting, tactile and ultrasonic sensors for material detection, a weight sensor for waste monitoring, and a speaker 116 for audible notifications, the device automates cutting, shaping, pressing, and storing of drum rubbers from waste tyre tubes, ensuring accurate, consistent production with user-friendly operation and waste management.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017688 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SUSTAINABLE BIOLUMINESCENT LIGHTING AND AIR PURIFICATION SYSTEM

(51) International classification	:B01D53/84, B01D53/74, B01D53/34, B01D53/62, B01D53/46, B01D53/34, B01D53/14, F24F 8/175
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Venkatesha Raju K

(57) Abstract :
A sustainable bioluminescent lighting and air purification system, comprising of a body 101 adapted to contain a marine water solution for simultaneous light emission and carbon dioxide absorption, a canopy 102 for preventing entry of foreign debris while allowing oxygen release into the atmosphere, an air intake pump 105 connected to an intake conduit 106 and exhaust conduit, the intake conduit 106 being equipped with a filter for introducing air into the marine water solution, a motorized roller 107 comprising a plurality of flaps 108, to rotate the roller 107 for maintaining circulation of the water and for mixing nutrients within the solution, a fertilizer injection assembly directing the fertilizer downward into the marine water for controlled nutrient delivering, and a sensor suite for determining water levels, monitoring biomass concentration and filter concentration in the solution.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017689 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SNACK PREPARATION AND STORAGE DEVICE

(51) International classification	:A47J 37/12, A47J 37/10, A47J 37/00, G07F 9/10, G07F 17/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Anjana D

(57) Abstract :
An automated snack preparation and storage device, comprises of a housing 101 having an input means 102 to provide input commands regarding initiation of snack preparation operation, an imaging unit 103 installed over the housing 101 by means of a spherical joint to determine the position of the frying utensil, a pair of sliding units 106 integrated within the housing 101 to translate a meshed sheet 107 to position the sheet 107 over the frying utensil for shaping and filtering the mixture uniformly, multiple clippers 108 integrated on each lateral side of the meshed sheet 107 to grasp the rim of the frying utensil for securing the sheet 107 in place, and an articulated arm 112 integrated on the housing 101 to position an electronic nozzle 109 to dispense the mixture through the meshed sheet 107 for uniform shaping and delivery over the utensil.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641017690 A
(19) INDIA
(22) Date of filing of Application :17/02/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : METAL SHEET BENDING DEVICE

(51) International classification	:B21D5/00, B21D5/04, B21D43/00, B21D43/10, B21D5/00, B21D5/04, B21D43/00, B21D43/10	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Arunkumar K Bhavikatti**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A metal sheet bending device, comprises of a housing 101 securely mounted on a surface, a motorized rotatable platform 103 arranged within the housing 101 to receive a metal sheet from the storage chamber 102 and rotate the sheet to facilitate bending on different sides, a pick-and-place assembly 104 integrated within the housing 101 to selectively retrieve a metal sheet from the storage chamber 102 and accurately place the sheet onto the platform 103 for bending operations, a hydraulic piston 105 coupled with a pressing panel to exert a downward pressing force onto the sheet positioned on the platform 103 to grip the sheet securely, a linear actuator 107 coupled to a plate 108, and placed adjacent to the platform 103 to extend upward from a base portion of the housing 101 to apply an upward force onto a portion of the metal sheet extending beyond the platform 103.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017691 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED KNOTTING DEVICE FOR PRODUCING NETS FOR SPORTS AND INDUSTRIAL APPLICATIONS

(51) International classification	:B65H 69/04, B65H 69/06, D04G 5/00, A63B 61/00, B65H 69/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Riya Sanjesh**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated knotting device for producing nets for sports and industrial applications, comprises of a housing 101 with an input means connected to a microcontroller for receiving user commands, a chamber 103 configured to hold multiple spools of strings, a gripping arrangement with motorized grippers 104 for aligning string ends, and a knotting assembly with rotatable plates 105 , semi-circular member 109 configured to grasp and pull free ends of the strings to align the ends precisely onto a platform arranged in the housing 101, a motorized spindle 106 with pneumatic pins 107 that selectively extends/retracts to define a base structure for adjusting net dimensions in real-time, a motorized slider 108 and C-shaped hooks 110 for forming symmetric knots and loops a storage unit with a motorized roller 114 spindle 106 for synchronizing together to prevent tension variations in the net formation.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017692 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PAPER SHEET FORMING DEVICE

(51) International classification	:D21G 9/00, D21B 1/32, B02C 18/00, G07D 11/00, B02C 18/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. M Renukadevi

(57) Abstract :
A paper sheet forming device, comprises of a housing 101 that includes an input means for initiation of the paper sheet forming and handling operation, an imaging unit 103 for detecting user presence, a sliding door 104 for paper sheet intake, cutting rollers 106 for shredding, a sliding gate for transfer to a water tank, and a robotic link with a sieve for fibre extraction, the container 108 integrates a motorized stirrer 107 and a pusher 109 with iris apertures 120 for compression and water removal, with pulp further dispersed by a second stirrer 111 in a reservoir 112,, a pair of robotic grippers 113 and an arm 114 align and dip mold deckle units into the reservoir 112, for sheet formation, followed by heating and drying and wireless communication modules ensure precision, monitoring, feedback, and remote access, providing an efficient, automated, eco-friendly paper recycling operation.

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017693 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CORRUGATED SHEET MANUFACTURING DEVICE

(51) International classification	:B21D 13/04, B31F 1/20, B31F 1/28, B25J 15/00, B25J 15/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)D Dhanya**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A corrugated sheet manufacturing device, comprises of a housing 101 installed with a display unit 102 to provide input specifications regarding a type of sheet corrugation to be manufactured, a robotic gripper 103 installed within the housing 101 to extend towards a chamber 105 installed within the housing 101 and positions over a die block 106 installed within the housing 101, a pair of motorized clippers 107 mounted over the die block 106 to grip side portions of the pattern plate to secure the pattern plate over the die block 106, a robotic clamp installed within the housing 101 to retrieve a sheet material stored within a compartment 108 installed within the housing 101, and a hydraulic unit 109 configured with a pressing die 110, suspended from ceiling portion of the housing 101 to emboss corrugated patterns corresponding to the plate design over the sheet material.

No. of Pages : 18 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017694 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : THERMAL-MANAGEMENT BASED CARRYING ASSISTIVE DEVICE FOR ITEMS

(51) International classification	:F25B 21/02, B65G 7/12, F25B 21/04, B62B 5/06, B65G 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Heena Firdose

(57) Abstract :
A thermal-management based carrying assistive device for items comprises of a frame 101 with a base plate 102 and at least four flaps 103 via a motorized hinge joint 104 for securely accommodating and retrieving items, an electromagnetic locking unit 105 for securely locking the flaps 103 in position, a pair of handles 106 via a gimbal assembly for manipulating and handling of the frame 101, an artificial-intelligence based imaging unit 107 for determining dimensions of the items, an inflatable member 108 for automatically adjusting inflation, a Peltier unit 109 for providing active heating and cooling effect to the stored items, a weight sensor for monitoring the weight of stored items, a plurality of LED 110 lights with a temperature sensor for monitoring handle 106 surface temperature, an audio output module for providing audible alerts and operational instructions to the user.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017695 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE TOE CORRECTION DEVICE

(51) International classification	:A61F 5/01, A61H 3/00, A61H 1/02, A61F 5/02, A43B 7/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. V Leelambika

(57) Abstract :
A wearable toe correction device comprising a hollow cylindrical body 101 to be worn around user's toe, lateral foot plate 102 providing anchoring support to cylindrical body 101 via connecting frame 103, multiple gel-infused patches for relieving pressure on foot, flexible fastening strap 104 attached to foot plate 102 to wrap over dorsal portion of user's foot for securing body 101 and plate 102 in stable position, a microphone 105 for receiving verbal commands, monitoring module integrated with body 101 to detect toe alignment parameters, set of tensile corrective wires 107 arranged in a bi-directional crisscross pattern, routed through a motorized tensioning roller 108 mounted between frame 103 and plate 102, to adjust alignment of toe, an air inflation unit comprising multiple inflatable members providing compression massage to toe by rhythmic inflation and deflation and speaker 109 configured with body 101 to deliver real-time voice alerts.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017696 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED PATTERN PRINTING DEVICE FOR FABRICS

(51) International classification	:B41F 15/08, D06P 5/00, B41F 15/36, D06P 5/30, B41L 13/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Manjula H M

(57) Abstract :
An automated pattern printing device for fabrics comprises of a housing 101 with multiple suction cups 102 attached for placement over a surface, an input means provided with the platform to specify input specifications for printing a label onto a fabric, an adjustable inlet plate 103 installed on the housing 101 via a sliding means 104 to enable a user access while positioning the fabric while printing, a monitoring unit for monitoring the presence and dimensions of the placed fabric, a handling means to grip the fabric and position it onto a platform installed in the housing, a stencil positioning unit positioned along the inner walls of housing 101 to retrieve and align appropriate stencil with an ink application module for delivering ink over the stencil and a pressing unit suspended from the ceiling to secure the stencil for applying force onto it to imprint the pattern over the fabric.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017697 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS ALGAE CULTIVATION SYSTEM

(51) International classification	:A01G 33/00, C12M 1/00, C12M 1/04, C12N 1/12, C12M 1/12	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)T Tamilselvan**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An autonomous algae cultivation system comprises an annular hollow body 101 to store water, plurality of motorized rollers 102 with flaps 103 to maintain uniform water flow, one inlet valve 104 to allow the flow of water into the body 101, a fertilizer dispensing module to dispense fertilizer within water, a set of oxygen pumps 105 with air filters 106 to infuse oxygen into the cultivation environment, an exhaust valve 107 with an exhaust fan 108 to remove gases out from the body 101, a segregating module to separate biomass from the water and a sensing module with an artificial intelligence based imaging unit 109 to monitor the cultivation of algae in real time.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017698 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CEMENT FILLING AND WALL REPAIRING DEVICE

(51) International classification	:E04G 23/02, E04G 23/08, E04B 2/84, E04G 23/00, E04B 1/68
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Divya Nair

(57) Abstract :
A cement filling and wall repairing device, comprises of a housing 101 for accommodating functional components, an imaging unit 102 for determining distance between housing 101 and cracked wall areas, a plurality of extendable rods 103 having motorized wheels 104 for positioning the housing 101 in vicinity to wall surface and cracks, an extendable rectangular-shaped gripping frame 105 integrated with the housing 101 via an articulated link 106 to position the frame 105 over detected inter-brick joints and wall surface cracks, an electronic nozzle 108 with a cement storage tank 109 via an articulated arm for dispensing cement over wall, a robotic limb 110 having a levelling flap 111 at the end for uniformly levelling the dispensed cement over the wall surface, a lighting element for providing illumination, a speaker 112 for providing audible notifications to the user.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017699 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CHAIN MANUFACTURING DEVICE

(51) International classification	:B21L 9/06, B21L 11/00, B21L 21/00, B21L 1/00, B21L 9/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Sunitha B J

(57) Abstract :
A chain manufacturing device comprising a housing 101 fixed over a surface, a rod storage chamber 102 placed within the housing 101 to store multiple rods, a clamping unit 103 attached with an articulated arm 104, mounted on a slider rail 105 within the housing 101, a molding assembly to receive and position the rods for forming chain links, a hydraulic piston bar 106 extending downward from the ceiling of the housing 101 to apply controlled pressure to bend and shape the rods within the molding assembly and a pair of gripper units 106 coupled to a robotic arm 107 positioned adjacent to the molding assembly to hold and secure the rods during chain link formation.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017700 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BRICK GAP FILLING DEVICE

(51) International classification	:E04G 23/02, E04G 21/20, E04G 21/22, B28B 11/24, E01C 23/09
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Bhavan Kumar M

(57) Abstract :
A brick gap filling device comprising a housing 101, an input means linked with a microcontroller to provide input commands regarding cleaning and crack filling operations, an imaging unit 102 to determine position of the housing 101 with respect to a wall surface, multiple extendable rods with motorized wheels 104 to maneuver the housing 101, an articulated arm 105 with a scraper roller 106 to remove dried cement from brick gaps, an extendable plate 107 arranged to collect and transfer the removed cement into a storage chamber 113, a plurality of heating to provide heat for curing the collected cement mixture, an electronic valve 110 connected to a water reservoir 109 to supply water through an expandable conduit 111 for mixing the collected cement, a stirrer 112 located to mix the collected cement and water and an electronic nozzle 114 to dispense cement over cracks and damaged wall surfaces.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017709 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SOY SAUCE PRODUCTION SYSTEM

(51) International classification	:A23L 27/50, A23L 11/00, A23L 11/50, A23L 11/20, A23P 30/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Zafar Ali Khan

(57) Abstract :
An automated soy sauce production system, comprises of an enclosure 101 positioned on a ground surface, a display unit 102 is installed within the enclosure 101 to provide input specifications regarding production of soy sauce, an imaging unit 104 installed within the enclosure 101 to detect accommodation of soybeans into the chamber, a nozzle 105 attached with a water container 106 to dispense water into the first chamber 103 to rinse the raw soybeans, a valve attached on base of the first chamber 103 to dispense the rinsed water from the first chamber 103 and dispose into a connected waste bin attached underneath the first chamber 103, a pair of motorized stirrer blades 114 installed within the compartment 110 to mix, and an electronic spout 115 attached each of multiple sections of a vessel 116 and connected to the compartment 110 via multiple ducts 117 to forming a homogenous mixture.

No. of Pages : 21 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017710 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEED PLUCKING DEVICE

(51) International classification	:A01M 21/02, A01D 34/00, A01M 21/04, A01M 21/00, B25J 13/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Jijina M T

(57) Abstract :
A weed plucking device comprising a detachable frame 101, a robotic arm 103 with a gripping jaw 104, an AI-enabled camera 102 for weed identification and guides to position robotic arm 103 for precise removal, an ultrasonic sensor integrated with the gripping jaw 104 measures weed depth to adjust arm 103 penetration, a weed collection chamber 105 with a motorized flap 106 activated by a proximity sensor to store extracted weeds, a level sensor monitoring fill capacity, cliff sensors to detect drops and ledges, prompting alerts, an infrared thermal sensor to identify heat signatures resembling animals, halting operations for safety, a speaker 107 provides audio notifications and a communication module enables remote monitoring and control of operations.

No. of Pages : 16 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017711 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE SUPPORT ASSISTIVE DEVICE FOR OPERATIONS ON BUILDING EDGES

(51) International classification	:E04G 21/18, G01N 29/04, F16M 13/02, E04G 21/10, F16M 11/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Blesson S

(57) Abstract :
An adaptable support assistive device for operations on building edges, comprises of a U-shaped adjustable frame 101 that fits the edged surface using an AI-based imaging unit 102 with an ultrasonic sensor to measure dimensions of building's edge and adjust size of frame 101, a suction mounts 103 at the frame's 101 ends securely grip the surface, while synchronized motorized wheels 105 enable bi-directional movement by temporarily releasing suction, a sensing module detects surface material, cracks, moisture, and applied force to adapt suction, a motorized pulley 109 with a rope 108 and hook 107 secures a user wearing a harness, controlled via user commands through an input interface, an audio module provides alerts and instructions for safe operation.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017712 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CORNER GUARD TAPE APPLICATION DEVICE

(51) International classification	:B65H 35/07, B65H 35/00, B65H 37/00, B65H 35/06, B26D 7/18	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Vinitha C**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A corner guard tape application device comprises of a housing 101 with a handle 102 for manual operation, a motorized roller 104 configured to store and dispense tape, a pressing roller 105 to apply pressure to the tape against surfaces for secure adhesion, a motorized cutting blade 106 coupled to a telescopic rod 107 for precise severing of tape, a robotic arm 108 with a gripper 109 to guide the tape's leading edge for accurate application, an AI-enabled camera 110 to monitor surface conditions and tape application, a motorized cleaning brush 111 to remove dust and debris before tape application, an ultrasonic sensor to monitor the application path and adjust dispensing parameters, a proximity sensor to track tape edge position for accurate guidance, a LiDAR sensor to detect obstacles and surface irregularities.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017713 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED LOADING AND UNLOADING DEVICE FOR HEAVY CONTAINERS

(51) International classification	:B60P 1/64, B65G 67/24, B65G 67/60, B60P 1/44, B65G 67/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Jinesh V N

(57) Abstract :
An automated loading and unloading device for heavy containers comprising a platform 101, an input means 102 for receiving user commands to load or unload container, a monitoring unit 103 for determining position and alignment of the container, a motorized slider 104 equipped with an extendable bar 106 for supporting the frontal portion of the container, a clamping unit 107 to acquire a hold of the container, a loading assembly 108 to securely lift and position the container onto the platform, an audio output module 105 for providing audible alerts and operational instructions, a pair of expandable clamps 107a each mounted on an extendable pole 107b with a motorized ball and socket joint 107c for precise lateral securing, a flap 108a via a hydraulically operated rod 108b for supporting the frontal portion of the container, an imaging unit coupled with IR and ultrasonic sensors for dimension detection and alignment verification.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017714 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COCONUT SHELL CHARCOAL PRODUCTION SYSTEM

(51) International classification	:C10B53/02, B01J20/20, C01B32/30, C10L5/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S V Niveditha

(57) Abstract :
A coconut shell charcoal production system comprises an enclosure 101 having a hopper 102 to hold coconut shells for charcoal production, user inputs through a touch interactive display unit 103 regarding initiation of charcoal block formation, an extendable pusher 104 pushes the shells into a L-shaped cylindrical hollow body 105 attached on end of the hopper 102 where a motorized spiral auger blade 106 shreds and moves the shells into a furnace 107 connected with end of the cylindrical hollow body 105, a plurality of heating elements 108 inside the furnace 107 carbonizes shells, produced charcoal is discharged through a motorized iris unit 109 into a chamber 110, where a pair of motorized grinders 111 grinds it into powder, the powder is dispensed into a molding assembly 114 by an electronically operated valve 113 via a conduit 115 and form the charcoal powder into solid charcoal blocks.

No. of Pages : 20 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017715 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WALKING ASSISTIVE DEVICE

(51) International classification	:A61H 3/04, A61H 3/00, B62D 57/032, B62B 3/02, B62D 57/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Santhosh Kumar K L

(57) Abstract :
A walking assistive device comprises a pair of U-shaped frames 101 equipped with two vertical legs 103 and an upper curved portion 104 with hand grips 105 gripped by a user, an imaging unit 109 captures and detects terrain and user position on which the frames 101 constructed with extendable rods 102 adjust height through the extendable rods 102, a plurality of crossbars 106 connecting the pair of U-shaped frames 101 to maintain stability and load distribution, an omnidirectional wheels 107 with one of the legs 103 deploy and allow user for mobile movement, a suction cup 108 positioned on other leg enhance grip on uneven terrain, a magnetic suspension arrangement 110 mounted between each leg 103 and the wheels 107 and suction cups 108 absorbs shocks for stable movement, a sensing modules monitor vital parameters including EMG sensor, blood pressure sensor, heart rate sensor, and pulse oximetry sensor.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017716 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : STEM LETTUCE PROCESSING SYSTEM

(51) International classification	:A23N 7/00, A23N 15/00, A23N 7/02, A23N 15/04, A23N 7/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. V Chandrasekar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A stem lettuce processing system, comprising an enclosure 101 positioned over a ground surface, a chamber 102 stored with stem lettuce, a display unit 103 provide inputs regarding processing of stem lettuce, multiple clamps 104 retrieve stem lettuces from chamber 102, an imaging unit 105 determine positioning of stem lettuces held by clamps 104, multiple cutting blades 106 precisely cut lettuce leaves and roots, a slider 108 provide sliding movement to clamps 104 towards a peeling drum 109 to dispose lettuce stems into drum 109 to peel outer layers of lettuce stems, a nozzle 110 attached to a water container 111 dispense water into drum 109 for cleaning lettuce stems, a pusher 113 push stems towards a hollow body 114 with a rotary cutter 115 to cut and slice stems into elongated lettuce strips and simultaneously translates lettuce strips towards a vessel 116, multiple heating elements 117 dry lettuce stem strips.

No. of Pages : 20 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017717 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : GRAIN STORAGE AND PACKAGING ASSISTIVE DEVICE

(51) International classification	:A01F 25/14, B65B 1/32, B65B 1/04, B65B 1/30, A01F 25/16	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. S P Anandaraj**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A grain storage and packaging assistive device comprising a platform 101 mounted with caster wheels 102 for smooth maneuverability over a surface, a touch interactive display unit 103 for user input, a microcontroller processing input commands to actuate a motorized roller 107 wrapped with polybag installed on the platform 101, an artificial intelligence-based imaging unit 104 capturing and analyzing surroundings to detect polybag unwrapping, an articulated robotic arm 105 retrieving the polybag and placing it into a cylindrical hollow body 106 , motorized sliders 108 with clamps 109 positioning the polybag for manual grain filling, an IR-based level sensor monitoring grain level, a motorized clamping unit 110 with articulated arm retrieving and applying a zipper to seal the polybag, a light-dependent resistor monitoring ambient illumination activating an LED under low-light conditions.

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017718 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-CLEANING GARBAGE CHUTE DEVICE

(51) International classification	:E04F 17/10, B65G 11/20, A61L 2/10, B65F 1/14, B65F 1/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Pakruddin B**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A self-cleaning garbage chute device comprises of a segmented duct 101 extending vertically across multiple floors of a building, a floor-level vent 102 for waste disposal and selective opening during emergency evacuation, multiple ultraviolet sterilization lights 103 sterilizing inner walls of duct 101, multiple disinfectant spraying nozzles 104 disinfecting storage tanks, an exhaust fan 106 controlling odor, inflatable air cushions 107 and an inflatable mattress 108 for safe evacuation of individuals during emergencies, motors 109 to divert the landing location of the duct 101 to an open area for safe deployment of inflatable mattress 108, light emitting diode indicator lights 110 to signal availability and status of device, a motorized latch door 111 to open in a limited manner for controlled garbage disposal and to unlock fully under emergency conditions, a speaker 112 providing audible alerts and evacuation instructions.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017719 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE SAFETY DEVICE FOR INDUSTRIAL USE

(51) International classification	:A41D 13/08, A41D 19/015, A41D 13/05, A41D 19/00, A44C 5/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Gnanakumar G**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A wearable safety device for industrial use, comprises of a wearable body 101 comprising a palm portion, a back portion to enclose a user's hand, multiple finger portions and a wrist portion defining an interior cavity for insertion of the user's hand, a retractable wristband 102 coupled to the wrist portion of the body 101 to secure the body 101 on the user's hand, a thermoregulation unit integrated in the body 101 to maintain real-time temperature of the user's hand by facilitating airflow through nanotubes 104, multiple nanotubes 104 embedded within the body 101 to circulate air for maintaining real-time temperature of the user's hand, a duct 105 having an inlet on the back portion of the body 101 for extracting air via a micro-impeller that is transferred to the nanotubes 104, and a health monitoring unit integrated in the wrist band for monitoring user's vital health parameters.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017720 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FOREST FIRE PREDICTION AND SUPPRESSION SYSTEM

(51) International classification	:A62C 3/02, G08B 17/12, A62C 99/00, G08B 17/00, G08B 17/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Srabana Pramanik

(57) Abstract :
A forest fire prediction and suppression system, comprises of a body 101 for installation at one or more locations within a forest area, an imaging unit 103 and thermal scanner to detect thermal hotspots, incipient fires, and fire-prone zones, a water cannon 105 to dispense water over detected fire-prone regions using a high-pressure nozzle, a user interface wirelessly integrated with a microcontroller via a primary communication module for remote monitoring and control, a secondary communication module ensures uninterrupted network continuity, an odor sensor to identify smoke and VOCs, triggering preemptive suppression of fires, a speaker 107 for audible alerts, a solar panel 108 for sustainable power generation and a rechargeable battery to store energy to store the generated solar energy and supply power to the components.

No. of Pages : 16 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017721 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FRUIT POWDER PRODUCTION DEVICE

(51) International classification	:A23N 12/02, A47J 43/24, A47J 19/02, B08B 3/12, B02C 18/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Madhavi T

(57) Abstract :
A Fruit powder production device comprising a cuboidal enclosure 101 located over a ground surface with a hopper, a touch interactive display unit 103 located within the enclosure 101 to provide input commands, an ultrasonic wave generator integrated to a chamber 104 to emit waves for cleaning the fruits, a water container to deliver water via a motorized pump for cleaning the fruits, a telescopic pusher 108 to push the cleaned fruits towards a vessel 110, a motorized cutter 111 to cut the cleaned fruits, a motorized flap 112 to dispense the fruits into the housing 113, a motorized shredder 114 to shred the cut fruits and a motorized grinder 115 to grind the shredded fruits into fine powder.

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017722 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE HANDHELD DEVICE FOR TESTING FRUITS QUALITY

(51) International classification	:G01N 33/02, G01N 21/359, G01N 21/35, G01N 21/3563, G01N 33/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Kalpana K Harish

(57) Abstract :
A portable handheld device for testing fruits quality, comprises an elongated body 101 attached with a handle 102 gripped by a user, a near-infrared (NIR) spectroscopy sensor configured to detect ethylene content, moisture level, chemical composition, and freshness of fruits, an RGB (Red Green Blue) camera 105 configured to capture surface details including color, texture, defects, fungal spots, and damaged areas of the fruits, an odor sensor configured to detect volatile gases, foul odors, and ripening conditions of the fruit, a firmness tester configured to contact the surface of the fruit for detecting firmness and juiciness, a microcontroller configured to receive input data and process the data, to determine freshness, ripeness, and overall quality of the fruit and a display panel 108 for presenting real-time analysis results and recommendations to a user regarding the fruits.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017723 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : READING MATERIALS MANAGEMENT DEVICE

(51) International classification	:G06Q 10/08, G06K 7/10, B65G 1/137, G06F 17/30, G06F 17/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Selvaraj Poornima**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A reading materials management device comprising a housing 101 with a plurality of shelves 102 for storing multiple reading materials, an input means paired with the housing 101 with an inbuilt microcontroller to provide input commands regarding operations, a monitoring unit installed within the housing 101 linked with a database to detect the position of the reading materials, multiple reed sensors mounted along the shelves 102 to detect the positions of each shelf segment and update the segments based on book genres, a handling module arranged within the housing 101 to move laterally along the shelves 102 and transfer specified reading materials from the shelves 102 and a clamping arrangement installed within the cavity to grip the specified reading material and transfer them accordingly.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017724 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : OUTDOOR ENVIRONMENTAL SYSTEM FOR AIR PURIFICATION AND NOISE CONTROL

(51) International classification	:F24F 3/16, B01D 53/84, A01G 9/02, F24F 3/14, A01G 9/24	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Santhosh M B**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An outdoor environmental system for air purification and noise control comprises of a honeycomb structured body 101 installed around an outdoor environment, a genetically modified moss 102 absorbing harmful gases and releasing oxygen through photosynthesis, a chamber 104 storing nutrient gel, multiple nano-tubes 103 with electronic valves 105 dispensing nutrients, a hopper 106 connected with a central conduit 107 capturing and recharging groundwater, humidifiers maintaining humidity levels around moss 102, multiple specially designed acoustic metamaterials 108 absorbing and controlling noise, a collection funnel 109 collecting excess moss 102 biomass falling due to excessive growth, a communication module for establishing a wireless connection to a computing unit for receiving notifications and alerts.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017725 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COPPER FILTER DRIER FABRICATION DEVICE

(51) International classification	:F26B 25/00, F26B 9/06, B25J 15/06, F26B 5/14, F26B 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Praveena K N

(57) Abstract :
A copper filter drier fabrication device, comprises of a base platform 101 developed to be positioned over a working surface , an L-shaped member 102 integrated with a cylindrical chuck 103 configured to hold copper filter driers , a user-input interface 104,to allow a user to provide commands for filter drier manufacturing , a robotic arm 105 to pick copper filter driers from a storage chamber 106 installed on the platform 101 and engage with the cylindrical chuck 103 , a bending arrangement 107 configured to bend copper filter driers to a user-defined shape and a storage unit 111 provided with the platform 101 for holding the copper filter drier cores, the robotic arm 105 picks and inserts the cores inside the copper filter driers to ensure precise alignment and insertion.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017726 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PIPE RESHAPING DEVICE

(51) International classification	:B21D 3/14, B21D 3/00, B21D 3/16, B21D 3/10, B21D 3/05	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Chintamreddy Meghana**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A pipe reshaping device, comprising a platform 101, an input means operatively linked with a microcontroller to receive user commands, an imaging unit 103 in sync with an ultrasonic sensor to determine the pipe's position and dimensions, a pair of hydraulic arms 104 to grasp and place the pipe onto holding units 105 integrated with a slider 106 provided on the platform 101, a tactile sensor to determine the pipe's material type, a heating gun 107 to heat the pipe for material softening, a pneumatic pusher 109 to precisely move the pipe along the slider 106, a shaping unit 110 to reshape the pipe into a user-desired profile, an angle sensor to determine the angle and alignment of the pipe, an eddy current sensor to detect defects in the pipe, and a lighting element 113 integration with the platform 101 to provide illumination.

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017727 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MEDICAL ASSISTANCE SYSTEM

(51) International classification	:A61G 12/00, G16H 40/67, G16H 40/20, A61B 5/00, G16H 40/63
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Priyanka Niranjan Savadekar

(57) Abstract :
A medical assistance system comprising a narrow elongated housing 101 to be positioned over a median rail of a roadway, a plurality of artificial intelligence based imaging unit 102 mounted over the median rail for capturing and processing the captured images to detect accidents on the roadway, a plurality of motorized wheels 103 located beneath the housing 101 to maneuver the housing 101, a motorized lid 104 allowing the user to retrieve medications from a first aid box 105 installed within the housing 101 and a Global Positioning System (GPS) integrated within the housing 101 to provide real time location to navigate the housing 101 towards the accident location.

No. of Pages : 16 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017728 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WALKING STICK MANAGEMENT ASSISTIVE DEVICE

(51) International classification	:A45B 3/00, A45B 9/02, A45B 3/04, A45B 9/00, A61B 5/1172	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Lakshmisha S K**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A walking stick management assistive device comprising a housing 101 to encase a walking stick equipped with a biometric scanner 104 for user authentication, and a solenoid lock to control access, an imaging unit 109 to detect cracks and deformities, a pair of motorized clipper to securely hold walking sticks of varying sizes, a biometric scanner 104 to perform authentication for providing access to only authorized users, a sensor suite to monitor surface of walking stick for detecting anomalies such as cracks, deformities, moisture levels and rust deposition onto the walking stick surface and an internal UV-C light 108 sterilizes the interior after extraction.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017729 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RAIL CYCLE STABILIZATION DEVICE

(51) International classification	:B61K 13/00, B61K 13/04, B61J 3/12, B61K 7/16, B61K 7/20	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Thatimakula Sai Kumar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A rail cycle stabilization device comprises of a U-shaped connector frame 101 connecting a pair of clamps 102 gripping the front fork of cycle, a first roller 103 running along a first rail, a second roller 105 engaging with an opposite rail to provide additional support, an AI camera 107 monitoring rail track, detect obstacles, curvature and slope of the rail, a twelve-bar linkage unit 108 attaching clamps 102 to connector frame 101, a spring positioned to maintain continuous contact with the rail, a pneumatic gripper 110 to selectively clamp onto a rail for preventing derailment, a set of LED lights 112 to illuminate in low-light conditions, a microphone 113 to accept rider voice commands, and speaker 109 to issue alerts.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017730 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMOKING ARTICLE DISPOSAL DEVICE

(51) International classification	:A24F 13/00, A24F 19/00, A24F 13/16, A24F 13/18, A24F 13/06	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Seerat Ul Nisa**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A smoking article disposal device, comprises of a cylindrical member 101 to place over a surface, a plurality of suction cups 102 is installed beneath the member 101 to affix the member 101 over the surface, an inspection module installed over the member 101 to detect an approaching object is a smoking article, a motorized iris lid 104 disposed at an upper portion of the member 101 to open that the approaching object is the smoking article, an extinguishing module installed inside the member 101 to mechanically and thermally suppress combustion of the smoking article, an iris hole 108 provided adjacent to the iris lid 104 for permitting entry of ash to collect the ash, a suction unit 109 disposed below the iris hole 108 to extract ash without affecting user movement and a circular plate 110 to press the smoking article manually for extinguishing residual heat and smoke.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017731 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : GARMENT ENGAGEMENT AND TRACKING DEVICE

(51) International classification	:A41H 1/02, G06K 7/10, G06Q 30/02, G06Q 30/06, G06K 19/077
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S Aarif Ahamed

(57) Abstract :
A garment engagement and tracking device, comprises of a U-shaped frame 101 adapted to be affixed with a garment, a circular plate 102 connected at each distal end of the frame 101 , each plate 102 is equipped with a suction unit 103 to apply clamping force over the garment, an inspection module 104 includes at least one imaging unit 104a and a fabric detection sensor 104b disposed on the plates 102 to identify fabric type and thickness, and an NFC module 105 embedded within the frame 101 to store garment-related data including size, brand, care instructions, and price, a LiDAR-based scanning module installed on the frame 101 to capture the consumer's body dimensions, and a sensing module disposed on the frame 101 to detect consumer interaction with the garment for analytics and inventory management.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017732 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PULLEY BELT REPLACEMENT DEVICE

(51) International classification	:F16H 7/24, F16H 7/08, B65G 45/12, F16H 7/12, B65G 23/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Mithun S

(57) Abstract :
A pulley belt replacement device, comprises of a platform 101 coupled to motorized wheels 103 configured to facilitate repositioning of the platform 101, a belt storage unit 104 hold replacement belts for installation on an auxiliary pulley, a plunger assembly engage and remove an existing belt from the auxiliary pulley, a scraping plate 109 clean pulley surface by removing dust and debris prior to replacement belt installation, a clamping assembly retrieve a replacement belt from the belt storage unit 104 and install the replacement belt onto the auxiliary pulley, an AI (artificial intelligence) camera 113 monitor the belt replacement operations, pulley alignment and belt seating, a plurality of LED (light emitting diode) lights 114 illuminate the working area during low-light conditions under control of a light sensor and a speaker 115 provide audio alerts, operational instructions, and status notifications to the user.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017733 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MAINTENANCE DEVICE FOR ELECTRIC PARTS

(51) International classification	:B08B 5/02, B08B 5/04, B60S 5/00, B23Q 11/00, B08B 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Prajeesh C B

(57) Abstract :
A maintenance device for electric parts comprises of a platform 101 with motorized wheels 102 for mobility, a body 103 capable of vertical movement using a scissor lift arrangement 104, an input module 105 for receiving user commands, an inspection module 106 to detect distance and position the device near electric parts, a cleaning module for dispensing pressurized air and dielectric fluid and mechanically removing contaminants, a dust collector module 115 with a suction unit 115d to capture and transfer dust into a reservoir 115b, a sensing unit to detect wire cuts, corrosion, and deformation, a speaker for real-time audio alerts, a sensing suite to assess cleanliness and trigger additional cleaning if dust remains.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017734 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WHEEL POLISHING DEVICE

(51) International classification	:B24B 49/12, B24B 49/16, B24B 49/00, B24B 49/04, B24B 29/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Aswathy N Rajan

(57) Abstract :
A wheel polishing device, comprises of a cylindrical body 101 adapted to be placed over a surface, an input module to provide input commands regarding polishing of a wheel, an inspection module to detect distance between a wheel placed in proximity to the body 101, a clamping to grip, lift, and position the wheel within the body 101, a motorized shaft 105 equipped with a circular member 106 installed within the body 101 configured to receive the gripped wheel, a laser sensor synchronized with the inspection module to detect dimensions of a centre bore of the wheel, an expandable pulley arrangement integrated within the member 106 extendable or retractable based on detected dimensions, a ceramic media dispensing and recovery unit to dispense ceramic media into the body 101 for polishing the wheel, a camera 107 for detecting alignment, and a cleaning module for post-polishing cleaning.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017735 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GROUND LEVELLING DEVICE

(51) International classification	:E04G 21/10, E01C 19/42, E04F 21/24, E01C 19/40, E02F 3/84	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Shwetha A**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A ground levelling device comprising a platform 101 with an input means linked with a microcontroller enabling a user to provide input, an imaging unit 102 mounted over the platform 101 via an L-shaped articulated bar 103 to detect the location of a concrete block and uneven surfaces, a pair of L-shaped hydraulic bars 106 with clamp 107 107 arranged over the platform 101 to lift and reposition the concrete blocks, a suction unit 108 with an articulated arm 109 placed within the platform 101 to extract debris and other waste to a connected waste box 110 for hygienic disposal, a drilling unit 111 placed over the platform 101 to drill one or more holes into the concrete block, a robotic arm 112 mounted over the platform 101, a primary high-pressure electronic nozzle 114 mounted on an articulated limb 115 connected to a reservoir 116 containing polyurethane foam solution and a motorized cutter 117 placed over the platform 101 to remove excess polyurethane foam extending above the block surface level.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :17/02/2026

(21) Application No.202641017736 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONDUIT BENDING DEVICE

(51) International classification	:B21D 7/024, B21D 7/025, B21D 7/16, B21D 7/06, B21D 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Shreenidhi B

(57) Abstract :
A conduit bending device comprising a frame 101 with an ergonomic handle 102 to facilitate the lifting of the frame 101, a curved cavity 103 integrated in the frame 101 to receive a conduit to be bent, a ratchet unit coupled with the curved cavity 103 to apply a bending torque to the conduit, a clamping attached to the frame 101 y motorized hinges to grip the conduit while bending, a plurality of rollers coupled to the cavity 103 to grip and advance the conduit into the bending position, an artificial intelligence camera 106 integrated with the frame 101 to detect surface defects within the conduit and a speaker 105 mounted on the frame 101 to generate audio alerts during operation.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017737 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SHOE CLEANING DEVICE

(51) International classification	:A47L 23/02, A47L 23/05, A47L 23/22, A47L 23/26, A47L 23/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S Shanthi

(57) Abstract :
A shoe cleaning device comprising a cuboidal body 101 opened from top positioned over ground surface with plurality of inverted U-shaped frames to hold shoe, an artificial intelligence-based imaging unit 102 to detect shoe, a pair of clippers within body 101 to grip shoe, an optical dust detection unit to identify dust or debris, an articulated arm 103 with an electronically operated nozzle 104 connected to an air compressor 105 via a pipe directs compressed air over shoe, an articulated link 106 with an electronically operated valve 107 dispenses cleaning solution from a cleaning solution chamber 108 on shoe tread, a micro-actuator Stewart assembly 109 with a brush head having bristles 110 providing motion to scrub tread, a motorized suction-based vacuum unit 111 to remove residual dust, a motorized air blower 113 to provide drying, a deodorizing spray to eliminate odor and a UV-C sterilization lamp 115 for sterilization.

No. of Pages : 22 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :17/02/2026
(21) Application No.202641017738 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FOOD BALL FORMING DEVICE

(51) International classification	:A23P 1/08, A21C 9/06, A21C 15/00, A21C 9/04, A21C 11/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Shankar J

(57) Abstract :
A food ball forming device comprises of a housing 101 having a bowl 102 to hold the food mixture, upon activation, a plurality of motorized iris cavities 104 integrated in the plate 103 open for forming food balls, a robotic arm 105 extending from an upper portion of the housing 101 then extends and lowers the plate 103 to compress the mixture into the cavities to form spherical balls, a coating unit 106 with spray nozzles 106b connected to a plurality of storage chambers 106a then coats the food balls, a scraper assembly 107 repositions balls using a scraping plate 107a connected to an articulated arm 107b, after coating the robotic arm 105 places the food balls on a tray 111 and then a motorized slider 112 slides out the tray 111 for user access.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018380 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : LEAF SPRING REFURBISHMENT DEVICE

(51) International classification	:B23P 19/06, B23P 19/00, B23P 19/04, B25B 27/30, B23P 19/10	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)M Pushpalatha**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A leaf spring refurbishment device, comprises of a platform 101 configured with a chamber 102 stored leaf spring nuts, an input module 105 installed on the platform 101 to provide input commands regarding refurbishment of leaf spring, an inspection module 113 to detect distance of the spring from the alignment zone, measure dimensional deformation, an aligning unit installed over the platform 101 to apply controlled lateral force on a deformed leaf spring to restore original camber, a guiding unit arranged on the platform 101 to lift, guide, and transfer nuts from the chamber 102 for further processing, a nut installation unit 109 disposed over the platform 101 to grip, align, and fasten nuts on the spring, and a lubrication and surface conditioning unit integrated with the platform 101 to clean, lubricate, and protect the surface of the leaf spring during and after refurbishment.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018381 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : HOSE CONNECTOR SEALING DEVICE

(51) International classification	:F16L 55/18, B62D 63/02, B62D 63/04, F16L 53/38, F16L 55/168
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Ashishika Singh

(57) Abstract :
A hose connector sealing device a platform 101, a set of motorized omnidirectional wheels 102 placed over the bottom of the platform 101 with an extendable rod 103 for adjusting the height of the platform 101, a pair of clamping unit 104 mounted on the platform 101 to seal a hose with a connector, a storage chamber 105 placed on the platform 101 containing the connector, an imaging unit 106 placed on the frame to detect hose and connector parameters, a heating unit installed within the platform 101 to heat the hose and improve flexibility during connection, a sensing module to monitor the sealing parameters.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018382 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CLAY CRAFTING DEVICE

(51) International classification	:B44B 3/06, B44B 1/06, B44B 3/00, B28D 1/00, B44C 1/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Ayesha Taranum

(57) Abstract :
A clay crafting device, comprises of a platform 101 adapted to be placed over a surface, an input module accessed by a user to provide input commands regarding crafting operations, a holding means configured to hold the clay tableware, a crafting module configured to manipulate a plurality of carving, cutting, and shaping tools over the clay tableware, a depth control unit configured to detect and regulates carving depth in real time, a cleaning module configured to remove clay residue, dust, and fragments generated during crafting, an inspection module configured to monitor entire carving operation and triggers an alert module that includes, but not limited to a speaker 109 in case any discrepancy is detected.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018383 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : LAMINATED WRAPPING MATERIAL CUTTING DEVICE

(51) International classification	:B65B 11/02, B65B 11/04, B65B 11/00, B65B 61/06, B26D 7/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Shashikala A R

(57) Abstract :
A laminated wrapping material cutting device, comprising a platform 101 accommodating operational components, an input means 102 operatively linked with a microcontroller to enable a user to provide input command regarding initiation of cutting operation of laminated wrapping material, a motorized roller 103 wrapped with a laminated sheet and integrated with a U-shaped frame 104 installed over the platform 101, to roll and unroll the laminated wrapping material during cutting operations, a pair of robotic arms 105 with rubberized ends integrated over the platform 101 to grasp the portion of the laminated wrapping material unwrapped by the roller 103 for accurate alignment and positioning before cutting, an articulated link 106 integrated with the platform 101 and coupled with a hydraulic plunger 107 having a sharp edge blade 108 at the end to cut the laminated wrapping material for producing precise cuts as per input specifications.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018384 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SECURED ACCESS CONTROLLED LOCKING DEVICE FOR BICYCLE PEDALS

(51) International classification	:E05B 71/00, B62H 5/20, B62M 3/08, B62H 5/00, G07C 9/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Debasmita Mishra**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A secured access controlled locking device for bicycle pedals comprising a frame 101 with a clamping unit to secure a bicycle pedal, an authenticating means located within the frame 101 to provide access to authorized users stored in a database, a monitoring unit installed on the frame 101 to scan the bicycle for detecting the dimensions of the bicycle pedal, a locking arrangement mounted on the frame 101 to lock the pedal, an acoustic sensor installed within the frame 101 to detect tampering sounds and theft attempts, a deterrent unit mounted on the frame 101 to release non-harmful fog upon theft detection, an ultrasonic deterrent unit located within the frame 101 to emit species specific frequencies to repel stray animal within the detected safety range and an adjustable traction pad 103 mounted over the upper surface of the frame 101 to adjust as per the user's shoe treat.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018385 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SPECTACLES CLEANING DEVICE

(51) International classification	:G02C 13/00, G02C 11/00, G02C 11/08, B08B 1/00, B08B 11/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dhannya J

(57) Abstract :
A spectacles cleaning device comprising a housing 101 with a curved flap 102 on upper portion of the housing 101 via motorized hinge 103 for user access, an input means for user's input commands regarding opening/closing of the housing 101, an imaging unit 107 to detect spectacle dimensions, and inflatable C-shaped clamps 108 with inflatable member for scratch-free holding, a dust sensor for detecting dust on spectacles, a motorized brush 112 via an articulated limb 113 to clean spectacles surface, electronic sprayer 115 for spraying the cleaning solution, and heated cotton wiping cloth 123 for cleaning temple portion of the spectacles, a microfibre cloth 118 on an articulated pole 117 removes stubborn stains, while a UV light 119 sterilizes the spectacles, a moisture sensor activates an air blower 120 for drying, a Wi-Fi or GSM modules, ensuring convenient, hands-free spectacle maintenance and hygiene.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018386 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BALL BEARING REMOVAL ASSISTIVE DEVICE

(51) International classification	:B25B 27/06, B25B 27/073, B25B 11/00, F16C 43/04, B25B 27/067	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Priyanka Gour**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A ball bearing removal assistive device comprises of a platform 101, a T-shaped handle 102 to enable a user to position the platform 101, an imaging unit to determine position of ball bearings, a plurality of extendable bars 103 each having a suction cup at an end to secure the platform 101, a motorized circular sliding unit to provide sliding movement to a plurality of telescopic rods 105 attached to the sliding unit over gaps between balls of the bearing, an extendable pad 106 attached with each of the telescopic rods 105 for securing the pads within the ball bearing gaps, a hydraulic actuator 107 extend on a center portion of the structure to apply a pushing force, while simultaneously causing the telescopic rods 105 to retract in synchronization with the pads to exert a pulling force over the ball bearings, thereby precisely removing the ball bearing from the structure.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018387 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED RUST-CLEANING DEVICE FOR METAL FITTINGS OF GARMENTS

(51) International classification	:B24B 27/033, C23G 3/00, D06F 39/02, B24C 1/08, B08B 3/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Pradeep Bhaskar

(57) Abstract :
An automated rust-cleaning device for metal fittings of garments, comprises of a housing 101 having a monitoring unit for measuring dimensions and determine type of the metal fitting, along with accumulation of rust, multiple concentric sealing rings 105 mounted in the housing 101 , each via a first extendable rod 106 for extending to align a suitable sealing ring over the metal fitting, a frame 107 mounted on the housing 101 via a second extendable rod 108, adapted to extend for positioning the frame 107 around the metal fitting in case the determined type is a zipper, a chemical dispensing unit mounted on the housing 101 for spraying chemical solution in a controlled manner, and a rust removal unit installed on the housing 101 for positioning over the accumulated rust, for dissolving and removing the rust from the surface of the metal fittings.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018388 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVICE FOR STORING DOCUMENTS

(51) International classification	:H04N 1/00, G06V 30/10, G06K 9/20, H04N 1/04, G07F 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Tanveer Ahmed

(57) Abstract :
A device for storing documents, comprises of a housing 101 segregated into multiple chambers for securely holding categorized documents, a platform 102 for receiving the documents manually from a user, a set of gripping pads 103 installed at the base of the platform 102 to provide stability and minimize slippage, an imaging unit working in sync with an infrared (IR) sensor for detecting the presence of the documents on the platform 102, an inspection unit for carrying out document inspection and digitization unit further comprising an optical character recognition (OCR) module for converting scanned content into editable and searchable digital data, a categorization unit incorporating handwriting recognition and an AI-driven suggestion engine configured to analyze extracted text and metadata, and an alignment module configured to pick and place documents into the designated chamber.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018389 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : RAIL ANCHOR FASTENING DEVICE

(51) International classification	:E01B 29/32, E01B 29/24, E01B 29/28, E01B 29/16, E01B 29/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Suma N G

(57) Abstract :
A rail anchor fastening device comprises of a platform 101, an input means 102 enables user commands for initiating the fastening operation, an imaging unit 103 detects the precise location of the rail anchor to be engaged, an articulated bar 104 with a wire claw 105 removes stones and debris near the rail anchor before fastening, a hydraulic articulated arm 106 applies force for fastening, a motorized slider 108 adjusts the hydraulic arm's position for precise alignment, a pressure sensor monitors and regulates the fastening force, a pair of track wheels 109 facilitates movement along the railway track, a speaker 110 provides audible alerts upon completion, a laser profilometer 111 captures high-resolution three-dimensional rail surface measurements to detect deformities, a ground penetrating radar sensor detects subsurface anomalies beneath the rail track.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018390 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DOUGH SHEET STRIP SLICING AND HANDLING DEVICE

(51) International classification	:A21C 9/08, A21C 9/06, A21C 3/02, A21C 11/10, A21C 9/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dumpala Swetha

(57) Abstract :
A dough sheet strip slicing and handling device comprises of a housing 101 provides structural support, an input means 103 linked with a microcontroller enables user commands for operation initiation, a pneumatic pusher 104 within a container 105 stores and pushes dough, a motorized sliding door 106 allows dough transfer onto a platform 107, a pair of extendable plates 109 flank the platform 107, one plate equipped with motorized clamps 110 mounted on sliding unit 111, an articulated arm 112 grasps wire and positions it into clamps 110 and iris lids 114 for stable suspension, a motorized roller 115 rolls dough across the wire arrangement, forming and slicing dough sheets into strips, an electronic valve 117 dispenses a non-stick flouring substance from a tank 118 over the dough sheet during rolling to prevent sticking, a motorized sliding gate 119 opens to allow collection of sliced dough strips into a user-accessible tray.

No. of Pages : 22 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018391 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPRESSOR PIPE JOINING DEVICE

(51) International classification	:B23K 1/008, B23K 3/08, B23K 3/00, B23K 1/00, B23K 1/018
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Nagaraja S R

(57) Abstract :
A compressor pipe joining device comprises of an elongated extendable body 101 having a handle 102 for gripping, an AI camera 104 detecting exact location of a compressor pipe, a pair of gripping unit 105 on a plate 103 holding the compressor pipe and retrieving a brazing rod from a storage chamber 106, a motorized ball-and-socket joint 107 providing multi-directional flexibility, a heating unit 108 installed on the plate 103 for controlled flow of heating gas towards the brazing rod, a shielding panel to protect adjacent components during brazing, a holding hoop at the handle 102 for ergonomic handling, a cushioned padding 109 to provide comfort to the user, a display panel 110 to present operational data, performance parameters, and user inputs.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018392 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WIRE SURFACE CLEANING DEVICE

(51) International classification	:B08B 1/04, B08B 1/00, H01B 13/00, B08B 1/02, H02G 1/12	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Alamelu Mangai J**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A wire surface cleaning device comprising a circular body 101 to clamp onto a wire, segmented portions 102, articulated arms 104 with clamps securing the wire, a sensing unit and imaging unit 105 monitoring alignment, cleaning unit 106 removing contaminants from the wire surface, a guiding rail 106b enabling controlled rotational and angular movement of the cleaning unit 106 around the wire, a collection container 107 receiving debris and waste material, a pair of L-shaped brackets 108 supporting the body 101 in vertical orientation, a brush 106a in the cleaning unit 106 with an actuator 106c and pivot joint regulating cleaning pressure, an air jet assembly 109 spraying water regulated by a solenoid valve 111, a vibration cleaning module 112 with a piezoelectric actuator generating vibrations to loosen contaminants, and an accelerometer 113 monitoring vibration parameters while force sensors regulate cleaning pressure to prevent surface abrasion

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018393 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED TEA BAG MANIPULATION DEVICE FOR TEA PREPARATION

(51) International classification	:A47J 31/40, A47J 31/44, A23F 3/06, A47J 31/52, B25J 9/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)S Sai Dhivya

(57) Abstract :
An automated tea bag manipulation device for tea preparation comprises of a body 101 with multiple motorized wheels 102 for movement, a multi-sectioned chamber 103 storing tea bags and sugar separately, a load sensor 104 monitoring weight of content, a microphone 105 providing user commands for releasing a specific quantity of sugar and tea infusion, an artificial intelligence-based imaging unit 106 determining presence and dimensions of an auxiliary vessel, a motorized gripper 107 with a rubberized member 108 to safely grip the tea bags from the chamber 103, a motorized rotary assembly 110 with the gripper 107 dipping the tea bag into the auxiliary vessel, an electronic nozzle 112 on an extendable L-shaped conduit 113 releasing the user-specified quantity of sugar into the auxiliary vessel, a vibrating unit 114 mixing the dispensed and infused materials within the liquid content and an audio output module 111 providing audible alerts.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018394 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CORN ROASTING DEVICE

(51) International classification	:A47J 37/06, A47J 37/04, A47J 37/07, A23N 12/08, A23N 12/12	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. A K Sampath**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A corn roasting device, comprising a curved frame 101 provided to securely support the ear of corn during roasting, a user inputs the roasting command through a user interface integrated with a computing unit and a communication module transmits the command to the microcontroller, a LiDAR sensor measures the distance between the corn and the heat source, a pair of telescopic rods 102 extends and a motorized hinge joints 103 connected between the rods 102 rotate the rods 102 for proper orientation, the user then manipulates a pair of handles 104 to position the corn for roasting, a pair of foldable thermal shields 105 deploy to protect the user's hands, a tactile sensor detect kernel firmness and regulate pressure, an AI (artificial intelligence) camera 106 checks roasting uniformity, and an articulated arm 107 cleans the corn surface.

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018395 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED POULTRY MANURE MANAGEMENT DEVICE

(51) International classification	:A01K 31/04, A01K 29/00, A01K 45/00, A01K 31/00, A01K 1/01	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Pooja P P**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated poultry manure management device comprising a housing 101 divided into a primary part and a secondary part, a platform 102 configured with housing 101 and divided into multiple sections, each section arranged to collect manure, a telescopic rod 103 integrated between each of sections for accurate positioning during manure discharge caused by birds, an imaging unit104 installed over platform 102 to determine presence of manure accumulation over respective section, a flipping arrangement 105 integrated within primary part of housing 101 via an articulated bar 106 to engage and flip respective section determined by imaging unit 104, a robotic link 109 integrated with a scrapping unit 110 to remove residual manure, an iris lid integrated within primary part to allow manure entry into secondary part, and a cleaning unit 111 to clean platform 102 sections by dispensing water and scrubbing after flipping operations.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018396 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PIPELINE MAINTENANCE DEVICE

(51) International classification	:F16L 55/18, F16L 55/32, F16L 55/40, F16L 101/30, F16L 55/165
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Shwetha A

(57) Abstract :
A pipeline maintenance device comprises of a platform 101, a set of motorized track wheels 102 for controlled movement of the platform 101, an AI-enabled camera 103 detecting exact location of pipe, a vibration-damping module 104 detecting structural defects of pipe, an articulated arm 105 gripping a pipe piece and align with the portion of pipeline requiring repair, a robotic link 107 with a rotating cutter 108 for accurate and controlled cutting of the pipeline, a compartment 109 containing an adhesive solution, an electronically controlled nozzle 110 dispensing adhesive solution over the cut pipe piece, adjustable extended rods 111 to adjust the platform 101 for accessing pipes from varying heights, a touch-interactive display panel 112 to display operational data, real-time sensor data and provide input command, a speaker 113 providing auditory alerts.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018397 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WIRE STRAND SEPARATION AND INSERTION DEVICE

(51) International classification	:H02G 1/12, H01R 43/28, H02G 1/14, H01B 13/012, H02G 1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Chintamreddy Meghana

(57) Abstract :
A wire strand separation and insertion device, comprising a main frame 101 supporting platform 102 configured to hold and guide a primary wire, a gripping assembly 103 mounted on the platform for selectively engaging and rotating the primary wire, a clamping unit 104 coupled to an articulated arm 105 for separating strands of the primary wire to create gap for insertion, a gripping bracket 106 mounted on a robotic arm 107 for guiding secondary wire into the gap, a storage compartment 110 for holding plurality of secondary wire strands, an AI imaging unit for monitoring wire alignment and defects in synchronization with capacitive and optical sensors, an IMU sensor for detecting tilt and instability, LED lights and a light sensor for illumination control, a speaker 109 for audible alerts, and a communication unit for remote monitoring and control, ensuring precise, efficient, and safe wire strand insertion with minimal user intervention.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018398 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTABLE STABILITY AND USER PROTECTION ASSISTIVE DEVICE FOR MOPED VEHICLES

(51) International classification	:B62J 27/00, B62H 1/12, B62H 1/02, B60R 3/00, B62H 1/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Nayana R**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An adaptable stability and user protection assistive device for moped vehicles, comprises of a platform 101 constructed with multiple expandable sections 102 via motorized hinge joints 103 for adjusting dimensions to fit over moped footboard, a securing means with motorized clamps 104 mounted on extendable L-shaped bar 105 and suction cups 106 to stabilize platform 101, a leg protection module with segmented thermoplastic polyurethane panels 107 deployed by a pantograph-linkage assembly 108 to provide adjustable leg coverage, a pair of motorized wheels 109 connected through a twelve bar linkage assembly 110 deploys when imbalance is detected by an integrated IMU, a temperature regulation module with a gel padding to ensure comfort, an imaging unit 111 to monitor surroundings to detect hazards, and an ultrasonic deterrent unit deters approaching animals by adjusting ultrasonic frequency and intensity.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018399 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED VEHICLE CHAIN MAINTENANCE DEVICE

(51) International classification	:B60S 5/00, G01S 17/931, B25J 19/02, B25J 9/16, B25J 19/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Amreen Khanum D

(57) Abstract :
An automated vehicle chain maintenance device comprises of a body 101 to be positioned adjacent to a chain-driven vehicle, a user-input interface present with the body 101 to provide commands regarding initiating chain maintenance operation, an artificial intelligence-based camera 102 with an integrated LiDAR sensor mounted on the body 101 to detect the type and dimensions of the vehicle, a motorized clamp 103 attached to the body 101 via a spiral jack arrangement 104 to engage with a portion of the vehicle, a six-bar gripper 105 mounted on an articulated arm 106 installed on the body 101 to hold the rear tire, a reciprocating quick arrangement 107 coupled with the arm to rotate the tire for cleaning and lubrication, a cleaning assembly attached with the body 101 to perform targeted cleaning of the vehicle chain.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018419 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BEARING ALIGNMENT DEVICE

(51) International classification	:B25J 15/06, F16C 43/04, F16C 43/06, B23P 19/00, B25J 15/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Bindhu M C**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A bearing alignment device comprising a platform 101, an input means 102 for initiating the bearing alignment process, a robotic gripper 103 to grasp bearing balls from a multi-sectioned storage chamber 104 and place them into slots on an alignment unit 105, a first articulated arm 107 to grasp an initial alignment ring from the storage chamber 104 and position it on the alignment unit 105 to receive the balls, suction cups 109 secure the platform 101 to the surface, position sensors monitor movement of the articulated arms 107, 108 for accurate placement, the alignment unit 105 comprises a slotted framework 106 for precise accommodation of balls and rings, a speaker 111 notifies the user upon completion, and weight sensors detect the quantity of balls or rings in the storage chamber 104 to ensure proper assembly.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018420 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : KEY MANAGEMENT DEVICE

(51) International classification	:G06K 9/00, G06F 21/32, G07C 9/00, E05B 19/00, E05B 47/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Vidyasree S M

(57) Abstract :
A key management device, comprises of a plate 101 equipped with suction cups 102 for adhering to a wall surface, an imaging unit 103 for detecting user's presence and facial recognition, a plurality of vertical sliders 104 translates extendable bars 105, with each bar 105 terminating in a U-shaped extendable member 106 for securely hanging keys, a first electromagnet 107 and a second electromagnet 108 to lock the keys, a fingerprint scanner 110 for biometric authentication of users, an IoT module facilitates schedule management and alert notifications, while a communication module allows remote monitoring and access through a user interface and a speaker 111 provides audible feedback during operation.

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018421 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ROAD SURFACE CLEANING DEVICE

(51) International classification	:E01H 1/05, E01H 1/08, E01H 1/00, E01H 1/10, E01H 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Nakul R

(57) Abstract :
A road surface cleaning device, comprises of a body 101 equipped with motorized wheels 102 for maneuvering over road surface, a touch-interactive display unit 103 for user input regarding cleaning of road surface, an AI-based imaging unit 104 for real-time debris detection on road surface, an articulated boom arm 105 with a bucket 106 collects coarse debris, while a motorized brush 107 mounted on an extendable rod 108 sweeps fine dust and dirt particles as detected by a moisture sensor, a motorized spray nozzle assembly 109 connected to a water chamber 110 sprays water for thorough cleaning after sweeping, a moisture sensor for segregating collected waste into wet and dry compartments, and an infrared proximity sensor to detect obstacles during operation and based on which the microcontroller adjusts the components to avoid collisions while maintaining cleaning efficiency.

No. of Pages : 18 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018422 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRONIC GADGET HOLDING DEVICE

(51) International classification	:B60R 11/02, G03B 17/56, B60R 11/04, F16M 11/12, F16M 11/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Ranjitha P

(57) Abstract :
An electronic gadget holding device, comprising a rectangular frame 101 developed to positioned on a ground surface and pivotally installed with a platform 103 via a hinge joint, a microphone 104 for receiving voice commands from the user regarding accommodation of an electronic gadget, a curved member 105 to support and hold the gadget, an imaging unit 106 to detect dimensions of the gadget, a motorized drawer arrangement to extend and retract the platform 103 and frame 101, a pair of linear actuators 107 to extend or retract to adjust an inclination angle of the platform 103, a camera 108 to detect user posture and viewing angle, a motorized spindle 109 to unrolls the sheet over the gadget screen and a plurality of air blowers 110 integrated with the platform 103 are activated to cool the gadget.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018423 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WRITING ASSISTIVE DEVICE FOR SECURE AND EFFICIENT JOURNALING

(51) International classification	:G06V 30/10, G06K 9/00, G06V 30/224, G06F 21/32, G06V 30/144
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S Nithya

(57) Abstract :
A writing assistive device for secure and efficient journaling, comprising a diary frame 101 having multiple bonded writable surfaces 102 for allowing manual writing, an elastic strap 104 equipped with a electromagnets for secure locking, a biometric scanner 103 for authenticating user, an artificial intelligence-based imaging unit 107 in sync with an OCR module for monitoring content in real-time, detecting errors, monitoring page occupancy and enables to convert written content into a document, a motorized slider 108 for translating a vessel 109 stored with whitener liquid towards detected writing error, an electronic valve 110 arranged on vessel 109 via an extendable L-shaped conduit 111 to dispense whitener liquid onto error, a motorized sliding unit 112 connected to extendable L-shaped link 113 for positioning of a suction unit 114 for turning of writable surface 102 and an audio output module 116 for providing audible alerts.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018424 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE SCALP MASSAGER AND OIL DISPENSER DEVICE

(51) International classification	:A61H23/02, A61H15/00, A61H7/00, A61H23/00, A61M37/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Syed Mohsin Abbasi

(57) Abstract :
An adaptive scalp massager and oil dispenser device comprises of a circular disc 101 forms the base of the device, supporting the plurality of rubberized members 102 each ending with a soft pad 104 for scalp contact, a vibrating unit 103 generates adjustable oscillatory motion to provide consistent massage, a microphone 105 captures voice commands for hands-free control, an electronic nozzle 106 connected to an oil chamber 107 dispenses oil through iris lids 108 carved on the pad 104, motorized hinge 109 adjust the position and angle of the rubberized members 102 to optimize contact, flexible members 102 conform to scalp contours for uniform pressure, a Peltier heating unit 110 warms the oil for enhanced absorption, flow sensors monitor oil delivery to ensure precision, and a speaker 111 provides audible feedback indicating oil dispensing status, all working together to deliver an adaptive and personalized scalp massage and oiling experience.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018425 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INFANT RESTING DEVICE

(51) International classification	:A47D 13/06, A47D 9/02, A47D 15/00, A47D 13/02, A47D 9/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Fathima Shana**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An infant resting device a platform 101 to be positioned over a ground surface, an artificial intelligence-based imaging unit 102 detecting the presence and dimensions of infant, a motorized drawer arrangement 103 that expands and contracts for proper accommodation of the infant over the platform 101, an air compressor 104 connected to an inflatable member 105 to inflate the inflatable member 105 for cushioning effect to the infant, a L-shaped rod 106 to extend position a motorized dome shaped canopy structure 107, a plurality of extendable curved links 108 connected with mesh net 109 within the motorized dome shaped canopy structure 107 to cover for creating a protective cover over the infant, a Peltier unit 110 heating or cooling the platform 101 surface and a fragrance sprayer 111 releasing a safe and controlled amount of deodorizing or fragrant substance.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018426 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SEED DRYING AND INSPECTION DEVICE

(51) International classification	:A01C 1/00, F26B 9/06, F26B 25/00, F26B 21/00, F26B 25/18	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Salma Frirdose**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated seed drying and inspection device, comprises of a housing 101 positioned over a surface, a supporting unit including a vertical rod 102 and clamp 103 to engage housing with surface, a plurality of removable horizontal tray 104 made of food-safe material to hold seeds or food items for drying, an inspection module installed with each tray 104 to detect defective or spoiled seeds, and a defective seed collection module including at least one gripper 106 mounted on an articulated arm 107 to remove and place defective seeds into a chamber 108, an airflow module comprising air blowers 109 to direct controlled airflow for drying seeds, and a vibrating unit mounted on each tray 104 to disperse seeds for uniform drying, a weather sensing module to detect real-time environmental conditions, and a protection assembly 110 including an L-shaped extendable rod and plate to shield tray 104 from adverse weather.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018427 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BLADE MAINTENANCE DEVICE

(51) International classification	:A47L 11/40, B08B 1/00, B08B 1/04, A61C 17/22, A61B 17/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Selvaganesh R

(57) Abstract :
A blade maintenance device, comprises a housing 101, an input means 102 operatively to enable a user to provide input commands regarding initiation of blade maintenance operation, an imaging unit 103 to determine presence of a user in the vicinity of the housing, a motorized sliding unit 106 to adjust position of the grippers 105, a hammering unit 107 to gently strike the blade, a speaker 109 to provide audible notifications regarding inability to perform maintenance, an electronic nozzle 110 connected to solution tank 111 to dispense the solution onto the affected blade surface, a motorized brush 112 to scrub the blade for removal of rust, a camera 113 to determine blade deformation and accordingly actuate an articulated hydraulic arm 114 with a broad clamp 115 to correct the blade orientation, a dual-axis slider 116 to translate an articulated link 117 having a grinding stone 118 to sharpen the blade.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018428 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ABDOMINAL PAIN RELIEF DEVICE

(51) International classification	:A61F 7/00, A61H 15/00, A61H 7/00, A61H 23/02, A61F 7/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. V Nagendramma

(57) Abstract :
An abdominal pain relief device comprises of a body 101 positioned over a waist portion of a user, an imaging unit 102 to detect the dimension of the user's waist portion using a laser sensor, a pair of motorized rollers 103 configured with straps 104 to secure the body around the waist by unrolling and locking the straps 104 with electromagnets, a sensing module to detect physiological conditions, a gel pad 106 with an Peltier module 107 to provide localized heating or cooling therapy based on processed data, a pair of extendable rods 108 with motorized massage nodes 109 to deliver massaging actions to the abdominal region, an odor sensor to detect ambient or user-released odors, a motorized aroma diffuser 110 to release aroma upon odor detection, a hydration monitoring module 111 to monitor hydration status, and a speaker 112 to remind the user to consume fluids upon dehydration detection.

No. of Pages : 23 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018431 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ERGONOMIC COOLER SUPPORT DEVICE

(51) International classification	:F25D 25/02, F25D 23/06, F25D 23/00, A47B 96/00, A47B 96/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Priyanki Dutta

(57) Abstract :
An ergonomic cooler support device, comprises of an adjustable frame 101 comprising telescopic rods, configured to adapt to varying cooler dimensions, multiple plates arranged in a drawer-like configuration, mounted on the frame 101 to deploy to securely accommodate the size of the cooler, a motorized roller 103 attached on each side of the frame 101, each carrying an insulating sheet 104 to deploy and cover the cooler to protect users from electrical hazards during operation, a pair of articulated arms 105 equipped with grippers, mounted on the frame 101 to assist in the deployment and retraction of the insulating sheets 104 from the motorized rollers 103, each insulating sheet 104 includes suction cups 106 at the edges to securely attach to the cooler, and multiple cooling fans 107 integrated with the frame 101 via a ball and socket joint to dissipate heat generated during operation to prevent overheating.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018432 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PIPE BENDING DEVICE

(51) International classification	:B21D 7/16, B21D 7/024, B21D 7/00, B21D 7/025, B21D 7/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Karthik M H

(57) Abstract :
A pipe bending device comprises of a housing 101, a microphone 102 receiving input command regarding initiation of bending operation, a pair of robotic grippers 103 grasp a pipe and place over platform 105, a pneumatic link 106 with a clamp grasping a circular stopper and position at one end of the pipe, the electronic valve 109 connected to a box 110 via the extendable L-shaped bar 111 to dispense the required material in vicinity to the pipe, the extendable limb 112 with a wedge-shaped plate 113 to align and place the material within the pipe, multiple heating elements 115 within a tray 116 to heat the pipe, a pair of articulated arms 117 for pressing the pipe on opposite sides to bend the pipe, the electronic nozzle 119 dispense water on the pipe for cooling of the pipe, a speaker 120 provide audible feedback and alerts to the user.

No. of Pages : 25 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018433 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : VEGETABLE WASHING AND STERILIZING DEVICE

(51) International classification	:A23N 12/02, A47J 43/24, A23L 5/20, A23N 12/06, A23L 5/30	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Jayavadivel Ravi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A vegetable washing and sterilizing device, comprises of compartmentalized housing 101 designed to securely fix to a surface, a first compartment 102 integrated with the housing 101 stores and supplies water for washing and sterilization, connected via a water supply conduit 109 to a second compartment 103 configured for washing vegetables, the second compartment 103 features a plurality of pneumatic pins 104 at its base, each equipped with a motorized brush 105 that rotates to thoroughly clean the outer surfaces of vegetables, a motorized valve 107 integrated between the walls of the second and third compartment 106 s for sterilization and houses a Peltier heating unit that heats water to boiling temperatures to eliminate pathogens, and a temperature sensor is integrated within the third compartment 106 and configured to continuously monitor the water temperature.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641018434 A

(19) INDIA

(22) Date of filing of Application :18/02/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : BANANA HARVESTING ASSISTIVE DEVICE

(51) International classification	:A01D 46/30, A01D 46/24, A01D 46/26, A01D 46/22, A01D 46/253
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Vetrimani Elangovan

(57) Abstract :
A banana harvesting assistive device, comprises of a wearable curved body 101 configured to be secured on the user's back, a pair of motorized rollers 102 coiled with looped strap 103 to pass over the user's shoulders for secure and comfortable attachment of the device to the user, a pair of extendable L-shaped links 104 integrated with the body 101, having an extendable rectangular plate 105 to receive a banana bunch, a robotic gripper 106 integrated on a curved-shaped member 107 to grasp a banana bunch stem detected via an imaging unit and laser sensor integrated with the body 101, an articulated arm 108 having a sharp motorized blade 109 to cut the banana bunch stem grasped by the gripper 106, and multiple robotic arms 110 integrated with the curved-shaped member 107 to grip and stabilize the banana bunch placed by the gripper 106 on the plate 105.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018435 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED GIFT WRAPPING DEVICE

(51) International classification	:B65B11/00, B65B11/02, B65B59/00, B65B11/04, B65B61/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Libin Jacob

(57) Abstract :
An automated gift wrapping device, comprises of a C-shaped hollow cylindrical housing 101 mounted on a platform 102, an imaging unit to detect dimension of an auxiliary gift box, a paper dispensing assembly comprising of a primary roller 103 rotatably for controlled unrolling of wrapping paper, a secondary roller 105 for dispensing adhesive tape, a cutting assembly comprising of a linear slider 106 and a cutting blade 108 for slicing the wrapping paper or adhesive, a plate 109 comprising a non-slip surface for supporting the auxiliary gift box, a gyro sensor for stabilizing the plate 109 and a load cell for detecting box weight, a pair of articulated arms for gripping free-edges of the gifting paper and the tape, a perimeter slider track 112 to carry repositionable panels, and a set of clamping unit to guide accurate folding.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018436 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CONVEYOR BELT JOINING DEVICE

(51) International classification	:B27F 7/38, B65H 37/04, B27F 7/36, B27F 7/17, B27F 7/19
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Kirubakaran N

(57) Abstract :
A conveyor belt joining device, comprises, a body 101 adapted to be placed over a surface, an input module 102 to allow a user to provide input commands regarding joining of a conveyor belt, an inspection module 103 configured to detect alignment gaps, misalignments, any missed staples and delamination over the conveyor belt, and a stapling module configured to securely join the conveyor belt, preventing separation and maintaining consistent tension across the belt. The stapling belt comprises a chamber 104 to store and dispense staple pins, a stapling arrangement configured to precisely position and drive staples into the conveyor belt to ensure secure joining, and a waste collection compartment 109 to store used staples or other debris generated during the stapling operation.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018437 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : POPCORN MAKING DEVICE

(51) International classification	:A23L 1/18, A23L 7/183, B01F 15/06, A23L 7/187, A23L 7/191
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. P Sudha

(57) Abstract :
A popcorn making device comprising a hollow cylindrical cooking chamber 101 with a base-mounted heating unit 105 that provides uniform heat for optimal kernel popping, a touch-enabled display panel 103 for user interaction, a microphone 102 to receive voice commands, an AI-enabled camera 104 to monitor popping progress and ingredient mixing, a plurality of sealed ingredient storage compartments 106 to store popcorn ingredients, a dispensing assemblies featuring motorized iris pore 107 with dispensing conduit 108 connected to a dispenser 109 to dispense ingredients according to programmed recipes, a stirring assembly comprises a robotic arm 110 equipped with a stirring member 111 to ensure thorough mixing and prevent burning, a motorized hinged access lid 112 secured via electromagnetic locking unit 113 to enable popcorn dispensing, a plurality of pneumatically actuated pins 114 for efficient prepared popcorn dispensing and a speaker 115 to deliver audio alerts and status notifications.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018438 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PLASTIC BALL MANUFACTURING DEVICE

(51) International classification	:A63B 45/00, B29C 45/17, B29C 45/14, B29C 45/00, B29C 45/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Vinayak Raju Kage

(57) Abstract :
A plastic ball manufacturing device, comprises of a housing 101 , an input means for enabling a user to provide input command regarding initiation of a plastic ball forming operation , a pair of robotic grippers 102, each configured to grasp a mold from a chamber 103 , an electronic nozzle 104 connected to a reservoir 105 storing melted plastic material, to dispense the melted plastic over the molds positioned by the grippers 102 , a plurality of electromagnets 106 integrated on each of the molds, ensuring secure attachment and precise alignment for the molding operation, a circular slider 107 move the molds, a heat gun 108 via an articulated limb 109 to apply focused heat over the molds during the molding process, a robotic linkage 110 to grasp the mold after the molding operation and place the mold within a water tank 118 for cooling of the mold.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018439 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TOOL COATING DEVICE

(51) International classification	:B05C 11/10, B05B 15/12, B05D 3/04, B05B 13/02, B05B 12/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dumpala Swetha**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A tool coating device comprises of a housing 101 equipped with multiple suction cups 102 to adhere the housing 101 over the surface, a robotic gripper 104 to grasp a tool from a multi-sectioned storage chamber 105 and place the tool on a platform 106 for coating and drying operation, an electronic sprayer 107 attached with each section of a multi-sectioned reservoir 108 dispenses a controlled quantity of coating color, guided by a flow sensor for uniform application, multiple air blowers 109 to direct regulated airflow for drying the coated tool, a temperature sensor and a Peltier unit regulate airflow temperature during drying process, an input means 103 for providing input commands regarding initiation of coating operation and a speaker 111 to provide audible alerts to the user regarding completion of coating and drying operations.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018440 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : METAL STORAGE AND POURING DEVICE

(51) International classification	:B22D 41/12, B22D 39/00, B22D 39/06, B22D 39/02, B22D 39/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Mahaboob Pasha

(57) Abstract :
A metal storage and pouring device, comprises of a frame 101 , a plurality of telescopic support rods 104, each equipped with a motorized wheel 105 configured to provide mobility to the frame 101 , a molten metal reservoir 102 configured to receive, contain, and store molten metal, a precision positioning assembly configured to support, position, and orient the reservoir 102 during metal receiving and dispensing operations and an AI (artificial intelligence) camera 103 configured to monitor the operations in real-time, thereby enabling a microcontroller embedded in the frame 101 to adjust the operational parameters accordingly.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018441 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED FUELING DEVICE FOR VEHICLES

(51) International classification	:B67D 7/42, B67D 7/04, B67D 7/32, B67D 7/06, B60K 15/05	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Raja Jitendra Nayaka**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An automated fueling device for vehicles, comprising a housing 101 configured with multiple chambers 102 storing various fuel types, an input means operatively connected to a microcontroller, enabling users to specify fuel quantity for dispensing into a vehicle's fuel tank, a monitoring unit employing AI-based imaging unit 104, LiDAR, and RFID to identify engine type and locate the fuel tank inlet, nozzles 110 linked to each chambers 102, manipulated by a handling means with a motorized clamp 105 and robotic gripper 106 on an articulated linkage 107 to align and insert the nozzles 110 upon detecting an open fuel cap, a vapour recovery assembly with sensors, vacuum unit 108, and hoses 109 to capture fuel vapours during dispensing, a flow meter 111 to regulate fuel quantity, and machine learning protocols to optimize operations, ensuring precise, efficient, and safe fueling for diverse vehicles.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018442 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONDUIT MANUFACTURING DEVICE

(51) International classification	:B32B1/08, B29D23/00, B32B3/26, B29C53/56, B67D7/08, B67D7/06, B67D7/78, B67D7/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Saiqa Khan N

(57) Abstract :
A conduit manufacturing device, comprises of a housing 101 , an input means 102 to enable a user to provide input command regarding initiation of conduit manufacturing operation , a motorized roller 103 coiled with a wire, arranged within the housing 101 for feeding the wire during the wrapping process , an articulated bar 104 having a grasping tool 105 at the end to apply melted high-density polyethylene stored within a chamber arranged within the housing 101 over the wire, a robotic gripper to grasp an end of the wire during rotation of the roller 103 for ensuring proper alignment and feeding of the wire during the wrapping process, a motorized spindle 108 for adjusting conduit position and orientation during the wrapping process for precise alignment and a robotic arm 110 to grasp the final conduit from the roller 103 and place within a user accessible container 111.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018443 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PAPER STAPLING DEVICE

(51) International classification	:B65H 37/04, B25C 5/16, B25C 5/15, B25C 5/02, B27F 7/19	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Jeevitha V K**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A paper stapling device comprising a frame 101, a microphone 104 to provide input voice command, an AI-enabled camera 102 monitors dimensions and position visual feedback, a pair of telescopic links 106 coupled with motorized ball-and-socket joint 107 support clamping unit 105 to ensure proper alignment and secure gripping of paper stack, a staple magazine assembly 108 stores and sequentially feeds staples, a motorized sliding unit 109 directs individual staples along a predetermined path toward paper stack, a pneumatic pusher unit 110 forcibly ejects staple pins to secure the papers, a linear actuator 111 connected to an articulated arm 112 equipped with a gripping plate 113 featuring multiple retractable engagement pins 114 for staple removal, a speaker 103 to deliver audio alerts and status notifications.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018444 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GLOVE DISPENSING DEVICE

(51) International classification	:A61B 42/50, A41D 19/04, A41D 19/00, G07F 11/00, B65G 1/137	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Rakheeba Taseen**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A glove dispensing device comprising a frame 101 integrated with a plurality of glove-holding slots 102 configured to securely retain individual disposable gloves, a pair of pneumatic pins 103 to retract and enable seamless glove dispensing, a microphone 104 to receive voice commands, an AI enabled camera 105 configured to monitor glove dispensing operations and provide real-time feedback, a motorized slider rail to sequentially advance subsequent gloves forward after dispensing of the front-most glove, an articulated arm 106 equipped with a V-shaped plate 107 to engage and hold open the glove opening, an air blower unit 108 to inflate the glove and facilitate user hand insertion, a glove securing assembly to secure the glove around the user's wrist, a suction unit 111 to extract residual air from within glove for optimal conformity and a speaker 118 to deliver audio alerts and status notifications.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018445 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED TILE MANUFACTURING DEVICE

(51) International classification	:B28B 13/02, B28B 17/00, B28B 3/02, B28B 3/00, B28B 11/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Gopalakrishna N

(57) Abstract :
An automated tile manufacturing device comprises of a housing 101 with a hopper 102 for feeding raw clay material, an input means 103 for receiving user commands for tile manufacturing, a chamber 104 with a grinding unit 105 for crushing and refining raw clay, a moisture control module 106 for adjusting the clay's moisture, a conveyer belt 107 for transporting the clay mixture, a pressing module 109 suspended from the ceiling to compress the clay into tiles of consistent shape and density, a designing unit 110 for molding and applying onto the tiles, a drying module with multiple air dispensing nozzles 111 to dry and solidify the molded tiles, an imaging unit 113 for real-time inspection of tile shape and design accuracy, and an audio output module 112 for providing operational alerts and instructions to the user.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018446 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NUT DISPENSING AND TESTING DEVICE

(51) International classification	:G01B 11/02, G01B 11/00, G01B 11/24, B65G 59/06, G01B 5/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Delsy Jyothi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A nut dispensing and testing device comprising a housing 101, a pair of extendable pins 102 integrated within a chamber 103 storing a stack of nuts arranged inside the housing 101, an iris lid 104 placed at the base of the chamber 103 to open and allow the dispensing of the nuts onto a platform 105 placed within the housing 101, an imaging unit 106 mounted within the housing 101 to detect the dimensions of the nuts placed on the platform 105, a magnetic plate 110 with an iris hole 111 arranged within the housing 101 through an articulated bar 112, a robotic gripper 113 placed within the housing 101 to grasp the nuts after processing and place them in a compartment 114 having multi-sectioned storage compartment 114 arranged within the housing 101 and a waste compartment 114 arranged within the housing 101 to store defective nuts.

No. of Pages : 16 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018447 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SEALING DEVICE FOR A LOWER DOOR GAP

(51) International classification	:E06B 7/16, E06B 7/23, E06B 7/215, E06B 7/20, F16J 15/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)B Ramana Reddy

(57) Abstract :
An automated sealing device for a lower door gap comprising a platform 101 configured with plurality of expandable plates 102 for size adjustment, an artificial intelligence-based imaging unit 103 installed on platform 101 for determining dimensions of door, a pair of motorized pop out balls 104 arranged underneath platform 101 for translating platform 101 underneath door, a plurality of suction units including a plurality of grooves 108 and an extendable pin 109 arranged underneath platform 101 for securing platform 101 underneath door, a pair of cylindrical members 106 on lateral sides, layered with sheet for sealing under-door gaps, a vibrating unit 107 integrated within platform 101 for dislodging dirt, an imaging camera 110 installed on cylindrical members 106, to generate alerts requiring urgent repairs and a moisture removal module including a flap 111 installed on the cylindrical member for detecting and removing moisture.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :18/02/2026
(21) Application No.202641018448 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BANGLE SHAPING AND PRECISION POLISHING DEVICE

(51) International classification	:B24B 41/06, B24B 21/00, B24B 9/00, B24B 21/18, B24B 19/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Padma T G**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A bangle shaping and precision polishing device, comprises of a housing 101, an input means 102 to provide input commands regarding initiation of bangle shaping and polishing operations, a pair of extendable pins 103 integrated within a chamber 104 to extend and retract to allow dispensing of one bangle at a time from the stack, an iris lid 105 to open and allow dispensing of the bangle onto a shaping unit 106 to enable precise shaping, a pair of belt lifting arrangements 107 to raise or lower a plate 108, a robotic gripper 109 configured to securely hold the bangle and align bangle against the unit to apply targeted corrective force to a uniform circular shape, an electronic nozzle 110 connected to a tank 111 storing polishing lubricant, to dispense the lubricant on the bangle, a motorized pad roller 112 via an articulated pole 113 to polish the bangle.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018449 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED DEVICE FOR MANUFACTURING GLASSES FROM WASTE BOTTLES

(51) International classification	:C03B 33/033, G02C 13/00, B25J 9/16, B65G 47/90, B25J 15/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Nishad P M

(57) Abstract :
An automated device for manufacturing glasses from waste bottles comprising a housing 101, an input means linked with a microcontroller to provide user input commands, a robotic griper integrated within the housing 101 to grasp a bottle from a storage chamber 103 placed within the housing 101 and place the bottle over a rotary platform 104 within the chamber 103, a robotic arm 105 integrated over the platform 104 to hold the bottle, an articulated link 109 within the housing 101 with a glass cutter 106 at the end to mark a cut line on the bottle, a robotic link 107 integrated within the housing 101 to grasp the bottle from platform 104 post-cutting and place within a hot water reservoir 110 and a cold water reservoir 111 and an articulated arm 112 integrated within the housing 101 to grasp the broken portion of the bottle from the reservoir.

No. of Pages : 15 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :18/02/2026

(21) Application No.202641018450 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BATTERY TERMINAL REPLACEMENT DEVICE

(51) International classification	:B23P 19/04, H01M 2/30, H01M 50/55, H01M 2/20, H01M 50/562	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Shet Reshma Parakash**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A battery terminal replacement device, comprises of a platform 101 to provide structural support, a jaw based centering unit 102 integrated with the platform 101, configured to securely hold and align the battery for subsequent operations, a motorized cutter blade 103 mounted on the platform 101 via an expandable arm 104, configured to cut a battery charging terminal, a hollow cylindrical molding member 105 coupled to an articulated arm 106, mounted on the platform 101 and configured to position the molding member 105 on the cut terminal surface for forming a new terminal, and a molten lead dispensing module 107 comprises a molten lead storage unit 107a arranged on the platform 101 and connected to a dispensing nozzle 107b via a flexible conduit 107c integrated with the platform 101 to dispense molten lead into the molding member 105 to form a new terminal.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037180 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : INCENSE STICK MAKING DEVICE

(51) International classification	:A47G 33/00, A61Q 13/00, B30B 11/00, B30B 11/22, A61L 9/12	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Chinmayi G V**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An incense stick making device comprising a housing 101, a hopper 102 mounted on housing to receive raw material, an iris aperture provided at base of hopper for dispensing husks and shell into a storage chamber 103 integrated within housing 101, an iris hole provided at a base of storage chamber with a conduit to control flow of sliced material from the chamber into a collection unit, a telescopic rod attached to a spatula 104 mounted within housing 101 to pick up material from collection unit and deposit it onto a conveyor belt 105 provided within housing, a magnetic separator furnished over conveyor belt to transport material while removing magnetic contaminants, an imaging unit 106 disposed within housing 101 for monitoring transferring of material, a scoop 107 installed within housing to scoop material powder from conveyor and a mixing unit 108 positioned within housing 101 for receiving material from scoop.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037181 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BRAKE DRUM MAINTENANCE DEVICE

(51) International classification	:B25J 19/02, B25J 13/08, B25J 9/16, F16D 65/10, B25J 19/04	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Ashishika Singh**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A brake drum maintenance device, comprises of a housing 101, an input means allows a user to provide commands to initiate brake drum maintenance operations, a motorized sliding gate 104 integrated with housing 101 to allow placement of the brake drum, a three-claw centering unit 106 integrated with platform 105, securely centers the brake drum on a platform 105, an articulated arm 107 with a motorized impact wrench 108 loosens and tightens wheel hub bolts, a robotic arm 109 with an electromagnet grasps and stores the loosened bolts, a camera 111 in sync with a dust sensor detects dust and debris on the brake drum, a motorized brush 114 cleans the brake drum surface, an eddy current sensor integrated with three-claw centering unit 106 detects cracks and alerts the user, a position sensor monitors the articulated arm 107.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037182 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AIR-VENT CLEANING DEVICE

(51) International classification	:B08B 5/02, B60S 1/60, A47L 9/28, B60S 1/56, A47L 9/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Tigulla Rajitha

(57) Abstract :
An air-vent cleaning device comprising an elongated body 101 with handle 102 adapted to be held by a user, a rod extending from one end of the body 101 to reach vent slats, a soft microfiber cleaning element 103 rotate and clean surface of the air vent, a microphone 104 to receive voice commands from the user, an imaging unit 105 to capture multiple images of the air vent and provide positional feedback, a motorized finger coupling arrangement 106 to allow flexible movement of the rod during cleaning, a pair of electronic nozzles 107 to dispense cleaning solution, a pressure sensor to regulate cleaning force, a vibration unit 110 to dislodge residual debris, a motorized air blower 111 to dry the vent, a motorized suction unit 112 to collect dust and debris, and a cap 113 to remove collected debris.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037183 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : POULTRY HEALTH MANAGEMENT DEVICE

(51) International classification	:A01K 45/00, A01K 29/00, A01K 31/00, A01K 39/012, A01K 31/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Akhila R Udupa

(57) Abstract :
A poultry health management device comprises a housing 101 adapted to be positioned over a ground surface, a plurality of motorized wheels 102 installed beneath the housing 101 to maneuver the housing 101 towards poultry, an artificial intelligence based imaging unit 103 integrated with a processor installed on the housing 101 captures multiple images, an inbuilt microcontroller processes the captured images to detect presence of poultry, multiple electronic nozzles 104 mounted on the housing 101 and connected to corresponding storage reservoirs 105 through conduits adapted to store water, nutrient liquid and medicated liquid, wherein based on detected health conditions, the microcontroller accordingly actuates the electronic nozzle 104 to dispense water, nutrient liquid or medicated liquid over multiple tray 106 installed on front portion of the housing 101 to feed the poultry for health improvement.

No. of Pages : 20 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037184 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BROKEN FLOOR TILE SEALING DEVICE

(51) International classification	:E04F 21/22, E04F 21/20, E04G 23/08, E04F 21/00, E01C 23/09
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Anju Mathew

(57) Abstract :
A broken floor tile sealing device, comprises of a housing 101 with a plurality of motorized wheels 102 configured on base of the housing 101 to maneuver the housing 101, an artificial intelligence based imaging unit 103 installed on housing 101 to detect position of broken floor tile, a motorized clamp 104 installed on front portion of housing 101 to grip and hold the tiles, an electronically operated nozzle 105 connected with a chamber 106 stored with a waterproof sealant, installed within housing 101, to dispense the waterproof sealant over tile floor surface and a telescopically operated bar configured with a roller 107 installed on front portion of the housing 101 evenly level the sealant over the floor surface.

No. of Pages : 21 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037185 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : HANDWRITING ASSISTANCE DEVICE

(51) International classification	:G06F 3/042, B43K 29/00, G06F 3/0354, B43K 29/08, G06F 3/0346
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. T Naresh Naidu

(57) Abstract :
A handwriting assistance device comprises of a a cylindrical body 101 configured to be held by a child while writing, a cylindrical plate 102 integrated with an iris aperture 103 configured to securely hold a writing instrument, a pair of parallel cylindrical members 104 configured to hold child's fingers in a correct writing position, a pair of inflatable air bags 106 operable to adjust shape and size based on pressure sensor embedded within the plate 102, an imaging camera 108 configured to monitor handwriting, track writing alignment, and detect errors, a holographic projection unit 109 configured to display interactive visual guides and tutorials for handwriting.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037186 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WOOD BLOCK PATTERNING DEVICE

(51) International classification	:B65G 47/90, B27B 31/00, B27C 5/06, B27C 5/02, B25J 15/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Babu D

(57) Abstract :
An automated wood block patterning device comprising a housing 101, a motorized sliding gate 103 to place a wood block over a motorized conveyor 104 for conveying the wood block toward a processing region, a robotic arm 105 to grasp and position the wood block over a platform 106, a pair of articulated arms 107 arranged over the platform 106 to grasp and reposition the wood block, a motorized cutting unit 108 integrated at a ceiling portion via a dual-axis slider 109 for cutting predefined patterns, a robotic gripper 110 to transfer cut portions into a chamber 111, a shaping unit within the chamber 111 to shape the cut portions based on user input, an electronic nozzle 115 to dispense adhesive into patterned cavities, and an air blower 118 to accelerate drying and bonding.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037187 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVICE FOR RESTORING OLD CNG CYLINDERS

(51) International classification	:F17C 13/02, F17C 13/00, B05B 13/04, F17C 13/04, F17C 5/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Chandrashekhar C

(57) Abstract :
A device for restoring old CNG cylinders, comprises of a housing 101, an input means to receive input from a user regarding restoration of CNG cylinders, an artificial intelligence camera 102 to monitor cylinder alignment and device performance, a pair of motorized rollers 104 to rotate the cylinder upon receiving cylinder manually, a rack-and-pinion arrangement 106 to adjust spacing between the rollers 104 based on cylinder size, a magnetic proximity sensor to detect the cylinder size and actuate the rack-and-pinion arrangement 106, a spray-nozzle assembly 107 being supplied from a storage chamber 110 containing paint-removal liquid, a grinder assembly 111 having a sandpaper-like outer surface for removing old paint, a pair of robotic linkages 113 to pick, transfer, and place the cylinder to subsequent processing stages, a set of inspection sensors for detecting bends, a painting assembly for coloring the cylinders.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037188 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COOKING SAFETY DEVICE

(51) International classification	:F24C 15/20, A47B 77/08, A47J 37/06, F24C 7/08, A47J 36/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. P S Joan Kingsly

(57) Abstract :
A cooking safety device, comprises of a U-shaped frame 101 positioned on a kitchen platform to house cookware with suction lining 102 to secure frame, an imaging unit 104 paired with an ultrasonic sensor detects cookware size and dimensions, multiple vertical panels extend or retract via a timing belt telescopic arrangement for splash protection, a cleaning assembly detects and cleans splashes and stains, a transparent shield sheet 106 covers the user's face via motorized folding and linkage, a rotatable motorized fan 114 for precise airflow control to remove smoke, fumes and fog around cooking area, a voice module enables voice-based interaction, a built-in LED lights 116 adjust to ambient brightness and a user interface enables real-time control and alerts.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037189 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BLADE SHARPENING DEVICE

(51) International classification	:B24B 3/36, B24B 49/12, B24B 49/00, B24B 51/00, B24B 3/54	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Bora Upendra Rao**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A blade sharpening device, comprises of a platform 101, an input means 102 operatively for enabling a user to provide input commands regarding blade sharpening parameters and process initiation or termination, an articulated arm 103 to grasp and stabilize the blade, prior initiation of sharpening operation, a motorized slider 104 translates arms 103 along a predetermined path to align and position blade accurately over a sharpening stone 105 arranged on platform, an optical sensor 106 in synchronization with an imaging unit 107 to detect sharpness of blade edge, an electronic sprayer 108 is actuated to dispense a controlled amount of fluid directly onto blade and sharpening stone 105, to reduce friction prevent overheating, remove metal debris, and ensure uniform material removal from blade edge during sharpening process, a suction unit 110 to withdraw residual water and metal particles generated during sharpening process.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037190 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : STONE ENGRAVING DEVICE

(51) International classification	:B44B 3/06, B44B 1/06, B44B 3/00, B44B 1/00, B28D 1/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Pushpalatha M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A stone engraving device comprising a housing 101, an input means operatively for enabling a user to provide input commands regarding initiation of stone scanning, selection of design patterns, and engraving execution operation, an imaging unit 104 to determine exact location, orientation, shape, and dimensional positioning of a stone relative for guiding engraving operation, a pair of hydraulic arms 105 to securely grasp the stone from opposite sides for stabilizing and rotating the stone during cutting and engraving processes, an articulated arm 107 having an engraving tool 108 via a dual-axis slider 109 for controlled linear and lateral positioning of the engraving tool 108 to execute selected design patterns, a projection unit 110 to project a selected design pattern onto the stone surface for visual confirmation and guiding, and an articulated bar 111 having a sandpaper 112 at the end to smooth and level stone surface after completion of engraving.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037191 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SUGARCANE PROCESSING DEVICE

(51) International classification	:A01D 45/10, A23N 7/00, A23N 15/04, A23N 15/00, A23N 7/02	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Bhuvaneshwari Patil**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A sugarcane processing device comprising a housing 101 with inlet and outlet access openings, a motorized conveyor belt 103 and incorporating a plurality of vacuum suction cups 104 to transport and immobilize sugarcane stalks, a telescopic bar 105 coupled to a motorized cutter blade 106 to transect sugarcane stalks into segments of predetermined length, a pair of robotic arms 108 each coupled with a gripper 109 and mounted on a linear slider rail 110, a thermal processing chamber 107 to apply heat treatment to soften sugarcane segments, a peeling module comprising a pair of counter-rotating motorized rollers 116 arranged vertically and integrated with pneumatic pins 117 to remove outer fibrous layers, an AI-enabled camera 118 to monitor operations and adjust processing parameters, an interactive display panel 119 and a speaker 120, for operational alerts and user interaction.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037192 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ATMOSPHERIC WATER GENERATION AND DISTRIBUTION DEVICE

(51) International classification	:E03B 3/28, B01D 5/00, C02F 1/28, C02F 1/44, C02F 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Harishkumar

(57) Abstract :
An atmospheric water generation and distribution device, comprises of a housing 101, an air intake assembly to draw ambient air into the housing 101 and remove particulates therefrom, a condensation chamber 104 to condense water from the ambient air, a water collection tray 108 to collect condensed water and channel the water via a sediment filter 109 at a narrow end, a storage tank 110 for delivering water for drinking, household use, and groundwater recharge, a filtration assembly to provide potable water at a dispensing unit provided with the housing 101, the thermoelectric cooling plates `105 to prevent ice formation and maximize water condensation, a dust sensor to detect contamination levels of the air filter 103 and trigger maintenance alerts and a pair of solar panels 117 to ensure continuous operation during low-light or nighttime conditions.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037193 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED DEVICE FOR MAINTENANCE OF DRY OR DEGRADED TREE MATERIAL

(51) International classification	:A01G 23/099, A01G 23/095, A01G 23/08, A01G 23/00, A01G 23/087
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Hashmat Fida

(57) Abstract :
An automated device for maintenance of dry or degraded tree material, comprises of a housing 101 adapted to be positioned adjacent a palm tree, an input means 102 for enabling a user to provide voice input commands regarding initiation, of palm trunk cutting, an imaging unit 105 to determine an exact location, height of rotted palm trunks, a pair of extendable members 106 to extend outwardly toward the base of the tree, a plurality of hinged plates 107 to form a continuous collection surface for receiving cut palm trunks, an articulated arm 108 having a sharp-edge blade 109 to cut detected dry or rotted palm trunks, a suction unit 111 to collect the cut palm trunks, a robotic arm 112 to collect the cut palm material from a box 113 and a motorized stirrer 115 to break down and mix the collected palm trunks for further handling or storage.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037194 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIRD PROTECTION DEVICE

(51) International classification	:A01M 29/32, A01M 29/16, B64C 39/02, A01M 29/06, B64D 47/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S Nithya

(57) Abstract :
A bird protection device, comprises of a wearable body 101 to be worn by a bird, a primary opening 102 to expose the head of the bird for clear vision, a secondary opening 103 to allow extension of the wings for flight, a plurality of electromagnets 104 to engage with each other to securely fasten the jacket onto the bird, an imaging module 105 to detect environmental threats, a plurality of U-shaped links 110 to extend and retract the links to form a deployable canopy structure for covering the bird in response to detected threats, a plurality of pneumatic pins 106 to extend outward to deter approaching predatory birds upon detection of threats, a cushioned member 107 to provide comfort and protection to the bird during flight, and a speaker 109 to provide audible alerts regarding threat detection, or other safety notifications.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037195 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WALNUT KERNEL SEPARATION DEVICE

(51) International classification	:A23N 5/00, A23N 5/08, B07C 5/36, B07C 5/342, B07C 5/02	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Mohamed Shakir**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A walnut kernel separation device, comprises of a housing 101, a hopper 102 to store walnuts, including an iris hole 103 for controlled sequential release of walnuts onto a conveyor belt 104 to transport the walnuts to a subsequent processing stage, an imaging unit 107 to inspect the walnuts for quality, a solenoid-operated striker 109 to tap the walnuts for acoustic quality assessment, a clamp 112 to transfer defective walnuts into a cavity 114 provided, a platform 115 comprising an inverted L-shaped bar 116 having a sharp edged plate 117 for cracking the walnut, a set of telescopic clamping unit 119 to separate kernels from shell of the walnuts, a motorized ball and socket joint 118 for enabling multi-axis movement for cracking the walnut and the multi-sectioned container 120 to monitor fill levels of kernels and shells and trigger alerts when a section is full.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037196 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : METALLIC POT MANUFACTURING DEVICE

(51) International classification	:B21D 51/26, B21D 51/18, B21D 22/20, B21D 51/38, B21D 22/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sathayanarayana Garadasu

(57) Abstract :
A metallic pot manufacturing device, comprises of a platform 101 which supports operational components, an input means 103 to receive user input commands for initiating a shaping process, a robotic gripper 104 to grasp a metallic sheet from a chamber 105 and place the metallic sheet between a pair of rotatable dies 106, one of the dies 106 comprises a pot mold and the dies 106 rotate synchronously to shape the metallic sheet into a pot structure, a sliding unit 107 integrated between each of the dies 106 and the platform 101 to maintain a predetermined distance between the dies 106 during shaping, a motorized pressing roller 108 to apply controlled pressure on the metallic sheet to refine the surface of the pot, and an articulated arm 109 grasps the shaped pot and place the pot onto a user-accessible storage panel 110.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037197 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SPRING FABRICATION DEVICE

(51) International classification	:B21F 35/00, B21F 35/02, B21F 33/04, B21F 3/04, B21F 3/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Pushpalatha M

(57) Abstract :
A spring fabrication device comprises of a platform 101, a set of extendable rods 102 to adjust height, a microphone 103 receiving dimensional and operational commands, an articulated arm 104 retrieving rods and discarding defective ones, a motorized extendable cylindrical member 105 with an expandable pulley for forming springs, an aligning unit with a motorized linear slider 106 and pivotable bracket 107 with adjustable lock for rod alignment, robotic grippers 108 for handling formed springs, an imaging unit 109 detecting defects and impurities, a storage chamber 110 holding multiple rods, a speaker 111 for alerts, wheels 112 powered by DC motors for mobility, a box 113 with a fill-level sensor for storing defective rods, and a vibration damping layer to minimize operational vibrations.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037198 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FRUIT PROCESSING DEVICE

(51) International classification	:A23N 7/00, A23N 7/02, A23N 7/08, A47J 17/14, A23N 7/10	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Paruchuru Jyothi Satya Siva Naga Swaroop**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A fruit processing device, comprises a platform 101 for receiving a fruit and processing the received fruit for peeling and seed separation, an inlet 102 and outlet 103 to enable entry of unprocessed fruit and exit of processed fruit, an articulated arm 104 having a blade 105 for cutting and peeling of the fruit, a clamp 106 to grip and position the fruit during peeling operations, an ultrasonic sensor for detecting proximity, alignment of the blade 105 to ensure accurate peeling, an AI-enabled camera 107 and a color sensor to detect ripeness, texture of the fruit and dynamically adjust the blade 105 pressure, cutting speed, a detachable compartment 108 for collecting waste material including peel, seeds, and other discarded fruit parts, a motorized ball and socket joint 109 to rotate the fruit during peeling, and a cleaning module 110 to remove fruit residue from the blade 105.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037199 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CHILD-SAFE ROCKING DEVICE

(51) International classification	:A47D 13/10, A63G 13/06, A47D 9/02, A63G 9/00, A63G 9/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Geetha C J

(57) Abstract :
A child-safe rocking device, comprising a curved base frame 101 extending longitudinally and installed with a pair of seating units 102 mounted on opposite sides of curved base frame 101, to secure child, a pair of side hand-grip bars 107 extending upwardly from frame 101 to enable user to maintain balance during rocking operation, an adjustable spring arrangement 108 positioned at both ends of frame 101, coupled with a load sensor to detect weight of the child and automatically adjust spring resistance, a balance terrain stabilization unit 109 integrated with frame 101, configured to detect uneven ground conditions and adjust frame 101's orientation in real-time to maintain stability and a C-shaped gripper 110 mounted on frame 101 via a telescopic rod 111 along a slider 112, synchronized with a tilt sensor to secure a child when an unsafe tilt is detected.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037204 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED DECONTAMINATING DEVICE FOR CARGO HOLDS

(51) International classification	:B08B 1/04, B08B 1/00, B08B 3/02, B65G 1/04, B08B 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Josephine. R

(57) Abstract :
An automated decontaminating device for cargo holds, comprises of a body 101 configured with a suspension assembly on a chassis area and equipped with a pair of motorized track wheels 102 for stable mobility over uneven surfaces, an input means paired with body 101 to provide input for cleaning a cargo hold area, a monitoring unit mounted on body 101, for detecting contaminated areas, a dispensing unit installed on body 101, to selectively dispense cleaning liquid and water over contaminated areas, a scrubbing unit installed on body 101, to dislodge the accumulated stains and dust and a waste extraction arrangement mounted on body 101, to remove waste generated during cleaning of the cargo hold area.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641037205 A

(19) INDIA

(22) Date of filing of Application :26/03/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : BREAD PREPARATION DEVICE

(51) International classification	:A21C 1/14, A47J 37/00, A21C 1/06, A21C 1/02, A21C 1/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Rafeeq Ahmed**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A bread preparation device comprises of a housing 101, motorized sliding access panels 102 for manual loading and unloading of ingredients and finished bread, an interactive touchscreen display panel 103 for user input, multiple ingredient storage compartments 104 storing ingredients, iris valve 105 with central supply tube 107 dispensing ingredient, agitator for dough formation, an articulated transfer arm 110 with gripper transferring dough between stages, flavor and additive storage reservoirs 115 dispensing additives and flavorings, a flavoring integration chamber 113 with blending blades 114 for mixing, compression plates with hydraulic pistons shaping dough into loaf configurations, motorized fans and heating elements 119 maintaining optimal temperature and airflow, a baking oven 120 baking fermented dough, a bread storage tray 121 receiving baked bread, a slicing blade 123 for uniform slicing, a cooling fan accelerating cooling of the baked bread.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037206 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BISCUIT PREPARATION DEVICE

(51) International classification	:A21C 1/14, A21C 1/06, A21C 1/00, A21C 1/02, A21C 11/16	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Pritha Sanyal**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A biscuit preparation device, comprising a housing 101, an input means 102 for enabling a user to provide input commands regarding initiation of biscuit preparation operation, an electronic valve 103 to dispense the raw material into a mixing container within the housing 101, an electronic nozzle 105 connected to each section of a multi-sectioned compartment 106 to dispense the liquids as specified by the user, a kneading panel 107 to mix and knead the dough to a homogeneous consistency, a robotic arm 108 to securely grasp a user-specified biscuit mold from a box 109, and place the mold over a forming platform 110, a spout 111 to dispense the prepared dough over the mold, a detachable motorized roller 114 with an embossing pattern, to imprint pattern on biscuit, and a robotic clamp 116 to remove the mold and grasp and place finished biscuit into a user-accessible tray 117.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037207 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : GAIT-ASSIST DEVICE FOR PROVIDING REAL-TIME OCCUPATIONAL THERAPY

(51) International classification	:A61H 3/00, A61H 1/02, A63B 24/00, A63B 23/02, A63B 23/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Pushpalatha M

(57) Abstract :
A gait-assist device for providing real-time occupational therapy, comprises of a platform 101 configured to support a user during rehabilitation, a plurality of suction cups 104 beneath platform 101 to secure and adjust platform 101 according to uneven floors, an imaging unit 106 on platform 101 to detect body dimensions of the user, a plurality of handrails 107 attached to platform 101 for providing real-time ergonomic support to user during rehabilitation exercise, a pair of C-shaped waist support units 113 comprising tilt and force sensors, mounted on handrails 107, for dynamically stabilizing the user's torso and pelvis, a pair of motorized clamping units 115 on a pair of motorized sliders 114 that is on platform 101 to secure and guide the user's legs upon detecting imbalance, a VR module 117 on platform 101 to display interactive scenarios aligned with the user's line of sight and provide movement guidance.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037208 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PIPELINE LEAK MONITORING AND REPAIRING DEVICE

(51) International classification	:F16L 55/18, F17D 5/06, F17D 5/02, F16L 55/168, F16L 55/165	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. S Fakruddin Ali Ahmed**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A pipeline leak monitoring and repairing device, comprises of a platform 101 developed to be positioned adjacent to a pipeline network, an ultrasonic leakage detection sensor for non-contact detection of leakages in the pipe, a pipe cutting arrangement to perform precise cutting and trimming of pipe sections, a first robotic arm 105 equipped with a motorized gripper 106, for picking up PVC pipe sections to positioning the pipe section with the carved hole, a second robotic arm 108 integrated with a glue-dispensing unit 109 to secure the pipe section with the carved hole, a third robotic arm 110 integrated with a motorized brush 111, to evenly distribute adhesive over the pipe, a microcontroller to monitor, coordinate operations of sensors, robotic arms, an AI-enabled camera 112 to capture real-time images videos of pipe surface and a speaker 113 for providing audible voice alerts and notifications to the user.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037209 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MODULAR ORAL-CARE STORAGE DEVICE

(51) International classification	:A61C 15/04, A47K 1/09, A46B 15/00, A61C 15/00, A45C 11/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Kuppala Saritha

(57) Abstract :
A modular oral-care storage device, comprises of a body 101 to be mount on a wall or mirror, a cylindrical frame 103 to adjust height and angle of the frame 103, a plurality of gripping units 105 to secure oral-care items, a plurality of circular cavities 109 for receiving toothbrush handles, a set of hinged compartments 112 for storing toothpaste, floss, and accessories, an interface based camera 113 to detect toothbrush usage frequency, brushing duration and brushing angle, a microphone 102 to enable voice-based interaction for operating the hinged compartments 112 or selecting oral-care items, a speaker 114 to deliver audio alerts, reminders, or brushing guidance to the user, a UV-C (Ultraviolet) sterilization module 111 to activate automatically after the toothbrush is placed inside, and a locking element 110 to secure the toothbrush upon insertion and release it upon removal.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037210 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED JADE GLASS MANUFACTURING DEVICE FROM RECYCLING WASTE GLASS

(51) International classification	:C03C6/04, C03B25/00, C03C1/00, C03C3/118, C03C6/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Bora Upendra Rao

(57) Abstract :
An automated jade glass manufacturing device from recycling waste glass, comprises of a framework 101 with a conveyer belt assembly 102 for receiving waste glass pieces, a hyperspectral imaging unit 103 for identifying glass type and colour characteristics, a handling unit for transferring the waste glass into a cleaning chamber for decontamination, a cooling and fracturing unit for cryogenic exposure and mechanical impact to produce fracture-clean feedstock, a grinding unit 107 for chemical composition determination, an additive dosing unit for dispensing colour-correction additives and nucleating agents, a melting furnace with a plasma-assisted electric torch 110 and mechanical stirring element 111, a pattern injection assembly 102 for generating veining patterns, a compression unit 114 with water sprayers 114a and motorized rollers 114b, and an artificial intelligence-based imaging unit 116 triggering a water-jet cutter 117.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037211 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED HEAVY DUTY DRAINAGE CHANNEL INSTALLATION DEVICE

(51) International classification	:E02F 5/10, E02F 5/02, E02B 11/00, E02B 11/02, E02F 5/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Ramana Reddy

(57) Abstract :
An automated heavy duty drainage channel installation device comprises of a platform 101 with a wheel assembly 102, a microphone 103 for receiving installation commands, an AI-based imaging unit 104 determining drainage channel position and dimensions, a trenching unit forming a trench, a drilling unit creating anchor holes, and a gripping assembly positioning channel sections, an articulated linkage 105 with a motorized clamp 106 and a rotatable hollow member 107 for fastening anchors, an electronic nozzle 108 connected to a concrete chamber 109, and a quality inspection unit detecting voids, misalignment, spacing, and slope, a speaker 110 provides alerts, suction units stabilize the device, and a disc-based drilling assembly removes debris, a robotic arm 114 installs accessories, an inclined conveyor 118 transfers excavated material, and a hydraulic arm 122 on a two-axis slider 121 places heavy channel sections.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037212 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : THREAD-SPOOL MANAGEMENT DEVICE

(51) International classification	:B65H 67/06, B65H 67/04, B65H 67/08, D05B 47/04, B65H 63/036
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sathayanarayana Garadasu

(57) Abstract :
A thread-spool 109 management device, comprises of a plate 101, a plurality of motorized clamps 102 arranged via an articulated bar 103 to grasp a surface for stabilizing the plate 101, an extendable rod 106 having a rotatable disc 107 to hold a thread spool 109 of a specific color for thread retrieval, an input means 110 for enabling a user to provide commands regarding spool selection, a projection unit 111 to project real-time information including color families, low-stock alerts, a robotic arm 112 to manipulate thread spools 109, an imaging unit 113 to determine whether similar thread spools 109 are in use, an extendable flap 115 via a hinge joint 116 to cover the thread spool 109 for protection, the rotatable disc 107 to enable smooth positioning of each thread spool 109 and a speaker 117 provides audible alerts regarding low-stock spools or operational guidance for the user.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641037213 A

(19) INDIA

(22) Date of filing of Application :26/03/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : FOOT ALIGNMENT AND WELLNESS CARE DEVICE

(51) International classification	:A61H 15/00, A61H 23/02, A61H 1/02, A61H 7/00, A61H 39/04	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Mehbooba Sultana**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A foot alignment and wellness care device, comprising a platform 101 configured to support foot of a user, a sensing unit detects foot pressure and alignment, an alignment-correction module to modify foot tilt, pitch, roll and arch elevation, one or more IMUs (inertial measurement units) to detect foot rotation, tilt, gait phase, and biomechanical deviations, stride consistency, and biomechanical deviations, a vibrating unit 105 to generate vibration cues when abnormal posture or gait is detected, a multi-zone massaging unit having rollers 110 arranged beneath the toes, and metatarsal zones to perform adaptive therapeutic massage, a hygiene and self-cleaning module mounted on a robotic link to remove sweat, moisture, dust, and debris from the device, an air blower 111 to maintain skin comfort and reduce microbial build-up.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037214 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FARM PRODUCE QUALITY DETECTION DEVICE

(51) International classification	:G01N 33/02, G01N 33/00, G01N 1/10, G01N 1/02, G01N 1/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Sharon M

(57) Abstract :
A farm produce quality detection device, comprises of a body 101 having a proximal end includes a handle 102 configured to be gripped by a user, an inspection unit to determines freshness, ripeness, moisture content, chemical composition, surface defects, a motorized sliding gate 103 to allow the user to place a biological sample or solution within a chamber 104 for analysis of the sample, an electronic nozzle 105 to dispense the EDTA solution within the chamber 104 for preservation of the biological sample or solution, a sensing unit to determine freshness, ripeness, quality, and safety of the agricultural product, a microphone 107 to receive voice commands to initiate scanning, freshness testing, a proximity sensor to detect the presence of a biological sample, a flow sensor to precisely regulate the volume of EDTA dispensed into the chamber 104 and a speaker 109 to provide audible feedback.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037215 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM FOR PRODUCING TENNIS BALLS USING RECYCLED TIRE MATERIAL

(51) International classification	:B29B 17/04, B29B 17/02, B09B 3/00, B29B 17/00, A63B 45/00	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Praveena K N**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A system for producing tennis balls using recycled tire material comprises of a tire intake and processing assembly receiving whole waste tires, an inclined conveyor belt 102 moving tires to the shredder 103, a shredder 103 cutting tires into rubber chunks, a magnetic separator drum 104 extracting metallic reinforcements, a bucket elevator transporting rubber chunks, a vessel 106 storing liquid nitrogen, a nozzle 107 dispensing liquid nitrogen onto the rubber chunks, a hammer 108 pulverizing frozen rubber into fine powder, a tank 110 with heating coils 111 for melting, a helical stirrer 112 for homogenizing, an electronically controlled valve 113 for regulating flow, a heated extruder screw transporting molten composite, a spherical mold 116 with hydraulic rods opening and closing mold 116 halves, Peltier units solidifying the injected composite, a fiber coating unit with a roller conveyor 117, an adhesive spray nozzle 118, a fiber sprinkling chamber 120 sprinkler fibers.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037216 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CIGARETTE BUTT RECYCLING DEVICE

(51) International classification	:A24F 19/14, A24F 19/10, A24F 19/00, B65F 1/14, A24C 5/39
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Suma N G

(57) Abstract :
A cigarette butt recycling device, comprising a housing 101 constructed with a hopper 102 positioned at an upper portion of housing 101, for a gravity-assisted feeding of cigarette butts, a shutter 103 attached underneath hopper 102 for facilitating a flow of cigarette butts out of hopper 102, a bucket conveyor 104 positioned underneath hopper 102 to receive cigarette butts from hopper 102, each bucket being configured to individually collect a measured quantity of cigarette butts during upward translation of conveyor 104, and upon reaching an upper discharge position, each bucket tilts automatically to dispense collected cigarette butts into a container 105 arranged within housing 101, container 105 being pre-filled with an acetone solution, for dissolving cellulose-acetate filter material from cigarette butts to form an acetone–cellulose solution, multiple heating elements 108 attached within heating tank 107 to elevate temperature of acetone–cellulose solution.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037217 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED BANGLE FABRICATION DEVICE

(51) International classification	:B21C 47/24, B21C 47/34, B21D 43/28, B21D 5/08, B21D 43/05
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. T Lalitha

(57) Abstract :
An automated bangle fabrication device, comprises of a housing 101, an input means 102 for receiving user commands regarding bangle fabrication, an imaging unit 103 to determine hand dimensions, a motorized roller 105 coiled with a metal strip 106 arranged to unroll the strip 106 based on detected dimensions while an articulated arm 109 over a rotatable cylindrical member for coiling, a motorized cutter 107 to cut the strip 106 at a specified length, a robotic arm 109 to transfer the coiled strip 106 onto a panel 108, a pair of robotic clamps 110 to hold the strip 106 while a welding torch 111 performs TIG welding, a laser engraving tool 115 for finishing operations, a suction unit 116 to remove debris, an articulated gripper 117 to place the finished bangle, and a pair of robotic grippers 119 to polish the bangle prior to user collection.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037218 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CORK STOPPER PRODUCTION DEVICE

(51) International classification	:B27J 1/00, B67B 1/03, B27K 7/00, B67B 1/04, B27J 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Jeevitha V K

(57) Abstract :
A cork stopper production device, comprises of a housing 201, an input means 205 operatively linked with a microcontroller for enabling a user to provide input commands regarding initiation of cork production operations, a motorized sliding gate 202 positioned with the housing 201 to open and allow a user to place raw cork bark over a motorized conveyor 101 installed within the housing 201, the conveyor 101 configured to transport the bark into a water tank 102 arranged within the housing 201 for boiling, multiple hydraulic pistons 109 suspended from a ceiling portion of the housing 201, each 109 integrated with a cylindrical die 110 configured to engage a corresponding cavity 116 carved on the punching member 107 pressing the die 110 into the cavity 116 shears and, an imaging unit 111 installed within the member 107 to determine whether each cork stopper is conforming or defective.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037219 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ARTIFACT MAKING DEVICE

(51) International classification	:B02C 18/00, B02C 23/00, C04B 33/132, B02C 23/02, B02C 18/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Geetanjali Diwani

(57) Abstract :
An artifact making device, comprises of a housing 101 having a funnel 102 to receive a clay, an input means paired with the housing 101 and communicatively coupled with a control unit enables a user to provide input specification regarding artifacts making, a shredder integrated within the funnel 102 shreds waste clay into small pieces, a grinding unit 106 positioned downstream of the shredder converts the small pieces into fine clay powder, a mixing unit 108 installed within the housing 101 blends the clay powder with water and additives to form a homogenous clay paste, a molding unit 112 installed below the mixing unit 108 shapes the clay paste into pre-defined artifacts, a heating unit 114 positioned downstream of the molding unit 112 uniformly hardens the molded artifacts.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037220 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CUSTOMIZED ICE-CREAM PREPARATION DEVICE

(51) International classification	:A23G 9/04, A23G 9/22, A23G 9/28, A23G 9/32, A23G 9/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. S MD Azash

(57) Abstract :
A customized ice-cream preparation device, comprises of a cylindrical housing 101, an input means mounted on the housing 101 enables a user to provide input commands regarding flavour selection, mixture ratios, shapes and freezing parameters, a flavour storage and dispensing unit 103 installed within the housing 101 stores and dispenses predetermined quantities of flavour powders, a mixing unit 104 positioned below the flavour storage and dispensing unit 103 blends a liquid base with the dispensed flavour powders to form a homogeneous mixture, a molding unit 105 installed below the mixing unit 104 receives the homogeneous mixture and shapes it within one or more molding cavities 105a into user-selected shapes, a Peltier unit with a temperature sensor maintains precise freezing conditions for uniform ice cream formation, a twisting arrangement 109 gently rotates and loosens the frozen ice cream for easy extraction.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037221 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED BALL MANUFACTURING SYSTEM

(51) International classification	:B29C 41/14, A63B 45/00, A63B 41/08, B29C 41/22, B29C 41/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Nasa Dhanraj

(57) Abstract :
An automated ball manufacturing system, comprising a framework 101 with an input means 102 coupled with a computing unit for enabling a user to prepare a ball of user-specified dimensions, a primary chamber 103 arranged in the framework 101 for preparing a latex compound by mixing natural rubber latex with chemical additives, a bladder formation assembly arranged over the primary chamber 103 for forming spherical rubber bladders by dipping ceramic formers 105 in the latex compound, a removal and valve installation unit for extracting the prepared bladders and installing an inflation valve, a rubber sheet storage and dispensing unit 116 with a motorized cutter 117 for preparing panels, a mold assembly 119 for sequentially placing panels, a hydraulic press molding unit for thermal bonding of panels, and an imaging unit 128 for real-time quality analysis and adaptive process control of the ball manufacturing operations.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037222 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED JELLY PREPARATION DEVICE

(51) International classification	:A23L 21/10, A23P 30/10, A23L 29/20, A21D 13/00, A23L 29/206	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Ramana Reddy**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated jelly preparation device, comprising a framework 101 configured with a flour preparation module 102 to produce a homogenous slurry of an ingredient, an alkaline solution preparation station 103 to form an alkaline solution with an exact concentration, a precision mixing 104 and alkalization chamber 105 to uniformly mix the slurry and the alkaline solution under controlled conditions to initiate the irreversible gelation process, a molding assembly 106 to deposit the gel-initiating mixture into molds with precise volume and shape, a multi-zone thermal gelation chamber 107 to apply a precise, controlled temperature profile to complete the gelation process and develop the optimal texture, a demolding assembly 108 to remove solidified jelly from the molds without any damage or tearing.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037223 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MOBILITY SUPPORT DEVICE

(51) International classification	:A61H 3/00, A61H 3/04, A61G 5/10, A61G 5/14, A61G 5/04	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Libin Jacob**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A mobility support device, comprises an enclosure 101, an L-shaped frame 103 mounted on a two-axis lead screw arrangement 104 installed over a ceiling portion, an input means 102 for receiving movement commands, a lifting unit including a motorized sliding block 106 for controlled motion, a pair of curved rods 107 operated by a hydraulic bar 108 for positioning, a swivel joint 109 for 360-degree rotational adjustment, a pair of flaps 110 with cushioned padding for user support, an imaging unit 105 for detecting user presence, posture, and environment, a microphone 112 for voice command and an emergency stop button 111 for user safety, enabling intelligent assisted mobility.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037241 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED LONG-STEM VEGETABLE STRAPPING AND PACKAGING DEVICE

(51) International classification	:B65B 13/18, B65B 13/02, B65B 13/32, B65B 27/10, B65B 13/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Asif Ahmad Najar

(57) Abstract :
An automated long-stem vegetable strapping and packaging device, comprising a housing 101, a motorized sliding gate 102 to open and allow a user to place raw vegetable bundles onto a conveyor 103, a motorized conveyor 103 to transport the bundles through the device for inspection, alignment, and subsequent processing, an imaging unit 104 to determine alignment, type, size, and orientation of the bundles, an inverted U-shaped frame 105 having an articulated arm 106 to assist in holding, guiding, and positioning the vegetable bundles during strapping, a strapping unit to wrap a strap around each aligned bundle and cut the strap, an inspection unit to detect whether the bundles match pre-fed quality parameters, a wrapping unit to wrap and seal each strapped bundle within a polybag, and an articulated clamp 115 to grasp the wrapped bundle and place it within a user-collection tray 116.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037242 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BREAD LOAF PREPARATION DEVICE

(51) International classification	:B26D 7/06, B26D 7/32, B26D 7/08, B26D 5/08, B26D 3/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Rakheeba Taseen

(57) Abstract :
A bread loaf preparation device, comprises a housing 101 incorporating a sliding access lid 116 to facilitate manual loading of a bread loaf, a motorized conveyor belt 106 to translate a detachable storage tray 108, a motorized precision cutting blade 111 coupled to the housing 101 via a robotic arm 112 to execute precise slicing operations to cut uniform bread slices from the loaf, a butter application platform 113 configured to receive individual bread slices via an articulated transfer arm, the platform 113 attached to a butter storage reservoir 114 via a flexible conduit 115 terminating in a mist spray nozzle configured to apply an aerosolized butter coating a toasting panel 107 to receive the butter-applied bread slice and toast the bread slice to a desired level of browning and a collection chamber 105 to receive the finished toasted bread slice.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037243 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PROCESSING AND SLICING DEVICE FOR FARM PRODUCE

(51) International classification	:B26D 3/28, B26D 7/06, B26D 7/32, B26D 1/00, B26D 1/03	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Abhishek M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A processing and slicing device for farm produce comprising a housing 101 to be placed over a surface, a user-interaction module installed with the housing 101, a conveyor belt 103 arranged beneath the sliding lid 102 to transport the farm produce forward, a slicing unit to perform primary horizontal slicing of the farm produce based on the type, a vertical slicer chamber 107 installed within the housing 101 and containing a plurality of slicing blades 108 arranged in a stacked manner along corner-mounted sliding rails, a multi-layer mesh strip-cutting unit disposed within the slicing and stripping unit to further cut the stripped farm-produce segments into strips of selectable thickness and a dispensing outlet ocated beneath the slicing and stripping unit to deliver processed farm produce material into an auxiliary container 114.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037244 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-FUNCTIONAL FRAME HANDLING DEVICE

(51) International classification	:B25J 15/06, B65G 47/91, G01B 17/00, B65G 47/90, B65G 1/04	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Dr. Umme Jahanara**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A multi-functional frame handling device, comprises of a horizontal telescopic rod 101 configured to extend and retract to accommodate auxiliary frames of varying dimensions, a pair of frame holders 102 with each frame holder being connected to a motorized ball-and-socket joint 103 for adjustable angular and height positioning, an imaging unit 104 for detecting dimension of the frames that is to be accommodated, a pair of clamping units 105 comprising multiple plates connected via motorized hinge joints and integrated with suction cups, configured to securely grip flat surfaces or fold around poles and corners, a plurality of ultrasonic sensors configured to detect surface type and frame dimensions in real time, a plurality of pneumatic plates 106 being expandable and retractable to match the width and thickness of the frame for secure and damage-free clamping of the frame.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037245 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AQUATIC MOBILITY AND TRAINING ASSISTIVE DEVICE

(51) International classification	:A63B 24/00, A63B 69/12, A61B 5/11, A63B 69/00, A61B 5/0205	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Smitha S P**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An aquatic mobility and training assistive device comprises of an extendable frame 101 to accommodate user's body, an imaging unit detecting body dimensions, posture, and movement, a drawer arrangement 103 adjusting the frame 101 size, motorized rollers 104 with retractable straps 105 connected to a wearable harness securing the user, a training module for adapting swimming lessons, difficulty levels and routines, a holographic projection unit 106 displaying guidance and stroke corrections, a vital-sign unit monitoring physiological parameters and adjusting training intensity, a motorized propeller assembly 108 providing thrust, vertical handles 110 to steer propeller, a motion-tracking unit detecting posture and stroke accuracy, inflatable sheets with air valves and CO_2 cartridges for safety, and a speaker 112 for audible instructions and a microphone 111 for voice commands.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037246 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WATER RESCUE DEVICE

(51) International classification	:B63C 9/02, B63C 9/00, B63C 9/22, B63C 9/01, B63C 9/08	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Amreen Khanum D**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A water rescue device comprises of a vertical plate 101 forming a rigid structural base, multiple mounting members 102 with motorized hinge joints 103 and a suction unit 104 for secure attachment, an inspection unit 105 detecting individuals in water and providing real-time distance, speed, and direction data, a universal joint 106 orienting a front throwing unit toward a detected target, a front cuboidal housing 108 with a motorized flap 109 and a spring-based pusher 111 for launching a rescue cabinet 114, a motorized roller 112 with an attached rope 113 for free deployment and controlled retrieval, an AI camera 115 on a motorized ball-and-socket joint for tracking and low-light visibility, a rescue cabinet 114 containing a folded inflatable buoy with a CO_2 cartridge 107 for rapid inflation, and a wireless communication module for transmitting real-time alerts and rescue data.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/03/2026
(21) Application No.202641037247 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : EGG PROCESSING DEVICE

(51) International classification	:A23L 15/00, A47J 43/14, A23J 1/09, A01K 43/00, B65G 47/90
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Suma N G

(57) Abstract :
An egg processing device comprises of a housing 101 with a motorized iris-style aperture 102 configured to receive eggs, a motorized conveyor belt 103 containing cradle slots 104 with retention grippers 105 for holding and transporting eggs, a articulated transfer arm 106 designed to retrieve eggs and deliver them to selected processing units, an egg boiling unit 107 performing controlled boiling operations, a rotating drum peeling unit 108 that applies rotational agitation for shell removal, a pair of precision clamping arms 109 that removes remaining shell fragments, an egg cracking unit 110 that cracks eggs and dispenses contents into a collection container 111, a pair of waste collection chambers 112 that stores discarded shells, a AI-enabled camera 114 that monitors processing in real time, a touchscreen display, a speaker 116, a microphone 117 for remote operation.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641037248 A

(19) INDIA

(22) Date of filing of Application :26/03/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : VEHICLE LUG NUT REMOVAL DEVICE

(51) International classification	:B60S 5/00, B23P 19/06, B25J 15/00, B25J 15/08, B08B 5/02	(71)**Name of Applicant :** **1)Presidency University, Bangalore** Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)Smitha S P**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A vehicle lug nut removal device, comprises an expandable base platform 101, a plurality of clamping units 102 coupled to the expandable base platform 101 establishes secure attachment between the base platform 101 structure and an underside frame of a vehicle, at least one pair of hydraulic lifting actuators 104 attached to the expandable base platform 101 provides vertical elevation adjustment to lift the vehicle for wheel access, a pressurized air blower unit 106 coupled to a robotic arm 107 attached to the expandable base platform 101 directs high-velocity air streams toward wheel surfaces to dislodge accumulated dirt and debris, an interchangeable wrench tool 108 coupled to an articulated arm 109 mounted on the expandable base platform 101 engages and removes lug nuts from wheels to facilitate wheel replacement operations.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037249 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LED MANAGEMENT DEVICE

(51) International classification	:G09F 9/33, H05K 7/20, F21V 21/22, F21V 29/00, F21V 29/503
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Bindhu M C

(57) Abstract :
An LED management device, comprises of a vertically oriented base 101, a motorized cylindrical roller 102 attached to a vertical side of the base 101 and storing a flexible protective sheet deployable via L-shaped telescopic rods 103 having clamping units at distal ends to cover an LED panel, a dual-axis lead screw arrangement 104 supporting a horizontal clamping plate 105 configured to move horizontally and vertically for adjustable positioning of the LED panel, a plurality of inverted L-shaped extendable frames 106 to grip the LED panel securely from three sides, a thermal management assembly comprising a thermally insulated chamber 107, Peltier cooling and heating unit 108, coolant conduits 109, peristaltic pump 110, and mini air circulation fans 111 to maintain desired operational temperature, and an AI camera 112 via a motorized ball-and-socket joint to monitor user position for automatic LED facing and activation.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/03/2026

(21) Application No.202641037250 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CASSAVA PLANT HARVESTING DEVICE

(51) International classification	:A01D 33/08, A01D 33/10, A01D 17/10, A01D 27/04, A01D 25/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Presidency University, Bangalore
Address of Applicant :Itgalpur Rajanakunte, Yelahanka, Bengaluru, Karnataka Pin-560119, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Vetrimani Elangovan

(57) Abstract :
A cassava plant harvesting device comprising an open housing 101, a vertical telescopic support bar 104 mounted to a front portion of the housing 101 through a horizontal pivot shaft 105, a pair of clamping units 106 to securely grip a cassava stem, a hydraulic actuator 107 to the telescopic support bar 104 to raise the clamping units 106 to pull the cassava plant from the ground, a motorized cutter 108 coupled to a robotic arm 109 to sever a leafy upper portion of the cassava plant from the stem, a pair of motorized counter-rotating rollers 110 to feed the pulled cassava stem, and a motorized rotary blade 111 subsequent to the rollers 110 and configured to cut the cassava stem into segments of predetermined length.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :24/05/2026
(21) Application No.202641065355 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CONTINUOUS FLOW REACTOR SYSTEM FOR IONIC LIQUID MEDIATED SYNTHESIS OF BETA GALBUTENONE

(51) International classification	:B01J 19/24, B01J 19/00, B01J 19/18, C07C 49/04, C07C 45/58
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY
Address of Applicant :Vellore Institute of Technology, Tiruvalam Road, Katpadi, Vellore, Tamil Nadu, India -632014 Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)Ekbal Ahmed Choudhury
2)Abdulla Ajmal
3)Dr. S. Sarveswari
4)Dr. V. Vijayakumar
5)Dr. Chidambaram Rishinaradamangalam Ramaswami

(57) Abstract :
ABSTRACT CONTINUOUS FLOW REACTOR SYSTEM FOR IONIC LIQUID MEDIATED SYNTHESIS OF BETA GALBUTENONE A continuous flow reactor system (100) synthesizes Beta Galbutenone (34) by cyclization of dehydrolinalool (12) in a reaction medium at a controlled temperature. A first reactor stage (10) receives a first feed stream comprising the dehydrolinalool (12) and a second feed stream comprising an ionic liquid (14) forming the reaction medium, and cyclizes the dehydrolinalool (12) to a cyclic ketone intermediate (16). A second reactor stage (20) in series receives the cyclic ketone intermediate (16) without isolation and an acetalizing reagent (22), such as methanol, and converts the cyclic ketone intermediate (16) into an acetal (24). A third reactor stage (30) receives the acetal (24) without isolation and allyl alcohol (32), and performs a rearrangement reaction at an elevated temperature to produce Beta Galbutenone (34). A layer separation unit (40) separates a crude product stream (42) from an ionic liquid stream (44)for recovery and reuse. (FIG. 3) Dated This .... day of October 2026

No. of Pages : 38 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641081016 A

(19) INDIA

(22) Date of filing of Application :01/07/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : SECURE CLOUD STROAGE SYSTEM WITH ENCRYPTION AND ACCESS CONTROL.

(51) International classification	:H04L 29/06, G06N 20/00, G06F 21/62, G06F 21/60, H04L 9/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, 534007. Andhra Pradesh India
(72)**Name of Inventor :**
1)V Naga Siva Rama Murthy
2)Dr P Sudhakar
3)N Madhavi Latha
4)S.Anantha Kranthi

(57) Abstract :
A secure cloud storage system ensures safe storage, access, and management of digital data. Data is encrypted using advanced cryptographic algorithms before being uploaded to the cloud. Access control mechanisms define and enforce user permissions based on roles or policies. Multi-factor authentication enhances security during login and data access. Real-time monitoring detects unauthorized access attempts or anomalies. Integration with enterprise applications enables seamless data management and sharing. Historical access logs support auditability, compliance, and regulatory requirements. Data redundancy and backup mechanisms prevent data loss and ensure reliability. Machine learning algorithms detect unusual patterns and potential security threats. Visualization dashboards provide insights into storage usage, access trends, and security status. Scalability allows deployment across individual, organizational, or multi-tenant environments. Privacy and security measures protect sensitive personal or corporate information. Automated alerts notify administrators of potential breaches or policy violations. Energy efficient encryption and storage reduce operational costs.

No. of Pages : 10 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202641081018 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RANSOMWARE DETECTION AND PREVENTION USING AI.

(51) International classification	:H04L 29/06, G06F 21/55, G06N 20/00, G06F 21/56, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)N Madhavi Latha
2)Dr M L S N Swarajya Lakshmi
3)Dr, Shameena Begum
4)J Neeraja

(57) Abstract :
An Al-based ransom ware detection and prevention system safeguards digital devices and networks from malicious attacks. System and file activity data are continuously monitored in real time. Machine learning and deep learning algorithms detect abnormal behaviors indicative of ransom ware, such as unusual file encryption, access patterns, or process activity. The system classifies potential threats with high accuracy and generates instant alerts. Preventive actions, including process isolation, file backup, and access restriction, are executed automatically. Historical data improves model performance and reduces false positives. Integration with enterprise and cloud systems allows centralized monitoring and protection. Real-time dashboards provide actionable insights on threat status and mitigation measures. Automated reporting supports compliance, audits, and security governance. Predictive analytics identify emerging ransom ware patterns and enable proactive defense. Multi-layered security ensures protection across endpoints, networks, and cloud storage. Privacy and data integrity measures safeguard sensitive information. The system adapts continuously to evolving attack strategies. Scalability allows deployment in individual, organizational, or distributed network environments

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202641081028 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ECO-FRIENDLY CONSTRUCTION BRICKS MADE FROM INDUSTRIAL CHEMICAL WASTE.

(51) International classification	:C04B18/06, C04B18/04, C04B14/10, C04B14/04, C04B14/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)M.Durga Bhavani
2)B.Ganesh
3)CH.Veerottam Kumar
4)CH.Bhagya Lakshmi

(57) Abstract :
Eco-friendly construction bricks are developed using treated industrial chemical waste as a primary raw material. The waste materials are processed and stabilized to ensure safe handling and structural integrity. Binding agents and additives improve strength, durability, and uniformity of the bricks. The manufacturing process reduces dependence on conventional clay resources. Controlled curing enhances mechanical properties and dimensional stability. The bricks exhibit adequate compressive strength for construction applications. Reduced water absolution improves resistance to weathering and environmental damage. Utilization of industrial waste minimizes landfill disposal and environmental pollution. The production process requires lower energy compared to traditional fired bricks. Lightweight characteristics facilitate easy handling and transportation. The bricks provide thermal insulation and improved energy efficiency in buildings. Standard molding techniques ensure compatibility with existing construction practices. The approach supports sustainable waste management and resource recycling. Cost-effective production makes the bricks suitable for large-scale adoption.g

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :01/07/2026
(21) Application No.202641081031 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : NANO-COATED CORROSION RESISTANT REINFORCEMENT STEEL FOR BUILDINGS

(51) International classification	:C08K3/22, C08K3/20, C08K3/34, C08K9/02, C09D7/61, C09D7/60, C09D7/40, C09D7/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, india-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)K.Soma Sekhar
2)CH.Bhagya Lakshmi
3)D.Leela Prasad
4)T.Ganga Sai Kumar

(57) Abstract :
Nano-coated reinforcement steel enhances corrosion resistance and structural durability in construction. A protective nanomaterial layer is applied to reinforcement bars using advanced coating techniques. The nano-coating forms a dense barrier against moisture, oxygen, and corrosive agents. Improved adhesion ensures long-term stability under varying environmental conditions. The coating reduces rust formation in reinforced concrete structures. Enhanced resistance to chemical exposure increases lifespan in coastal and industrial environments. The nano-layer maintains the mechanical strength and flexibility of steel. Reduced coiTosion minimizes structural degradation and maintenance requirements. The coated reinforcement improves bonding performance with concrete. The solution supports use in bridges, buildings, tunnels, and marine structures. Manufacturing processes are compatible with existing reinforcement production methods. The coating thickness is optimized to balance protection and cost. Environmental benefits include reduced material replacement and lower lifecycle emissions. Quality control measures ensure uniform coating and performance consistencyO8%na

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202641081035 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CHEMICAL-BASED WATERPROOFING ADMIXTURE FOR SUSTAINABLE STRUCTURES.

(51) International classification	:C04B 28/02, C04B 40/00, C04B 14/06, C04B 24/26, C04B 28/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)CH.Bhagya Lakshmi
2)K.Soma Sekhar
3)M.Durga Bhavani
4)CH.Veerottam Kumar

(57) Abstract :
A chemical-based waterproofing admixture enliances the durability and sustainability of concrete structures. The admixture is incorporated into cementations materials during mixing. Reactive compounds form insoluble crystals within the concrete matrix. These ciystals block capillary pores and micro-cracks to prevent water penetration. Improved wateipiroofing reduces moisturerelated deterioration and corrosion. The admixture enhances resistance to chemicals, chlorides, and sulfate attacks. Reduced permeability increases structural lifespan and reliability. The solution is suitable for basements, foundations, roofs, and water-retaining structures. Compatibility with standard concrete mixes ensures easy implementation. The self-sealing action activates in the presence of moisture. Reduced maintenance requirements lower lifecycle costs. The admixture improves structural integrity without affecting workability significantly. Environmental benefits include reduced material usage and improved sustainability. Quality control ensures consistent performance and durability.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202641081036 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART AIR POLLUTION ABSORBING CONCRETE USING PHOTOCATALYTIC MATERIALS

(51) International classification	:C04B22/06, C04B28/02, C04B103/10, C04B103/12, B01J35/39, B01J35/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)CH.VEEROTTAM KUMAR
2)D.LEELA PRASAD
3)K.SOMA SEKHAR
4)T.Swathi

(57) Abstract :
Smart air pollution absorbing concrete incorporates photo catalytic materials to reduce environmental pollutants. Photo catalysts such as titanium dioxide are embedded within the concrete matrix. When exposed to sunlight, the material initiates photo catalytic reactions. These reactions break down harmful pollutants including nitrogen oxides and organic compounds. The surface converts pollutants into less harmful substances. The system improves air quality in urban environments and high-traffic areas. The concrete maintains structural strength while providing environmental benefits. The photo catalytic action is continuous and requires minimal maintenance. Self-cleaning properties reduce surface dirt accumulation. The solution is suitable for pavements, facades, tunnels, and road baniers. Compatibility with conventional concrete manufacturing processes is ensured. The technology supports sustainable and eco-friendly infrastructure development. Reduced pollution contributes to improved public health. Long-term performance enhances durability and cost-effectiveness.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641081038 A

(19) INDIA

(22) Date of filing of Application :01/07/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : LOW-COST CHEMICAL TREATMENT FOR SOIL STABILIZATION IN ROAD CONSTRUCTION

(51) International classification	:E01C 3/04, E01C 3/00, E02D 3/12, E01C 3/06, C09K 17/40	(71)**Name of Applicant :** **1)RAMACHANDRA COLLEGE OF ENGINEERING** Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India (72)**Name of Inventor :** **1)K.Kanaka Lakshmi** **2)T.Swathi** **3)Dr.K.Sundara Kumar** **4)K.SOMA SEKHAR**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A low-cost chemical treatment improves soil stability for road construction applications. Chemical additives are mixed with weak or expansive soil to enhance engineering properties. The treatment reduces plasticity and improves load-bearing capacity. Stabilized soil exhibits increased compressive and shear strength. The process minimizes swelling and shrinkage in moisture-varying conditions. Improved compaction characteristics enhance road base performance. The chemical formulation reduces permeability and erosion. The solution supports construction on poor or problematic soil conditions. Reduced material replacement lowers construction costs significantly. The method is compatible with conventional road construction practices. Faster curing improves construction efficiency and project timelines. Environmental benefits include reduced use of natural aggregates. Field performance ensures durability under traffic loading. Quality control enables consistent stabilization results.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641081044 A

(19) INDIA

(22) Date of filing of Application :01/07/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-CLEANING BUILDING SURFACE COATING USING NANO CHEMISTRY.

(51) International classification	:C09D 5/16, E04F 13/02, C09D 7/12, C09D 5/00, E04B 1/64	(71)**Name of Applicant :** **1)RAMACHANDRA COLLEGE OF ENGINEERING** Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India (72)**Name of Inventor :** **1)B.Ganesh** **2)M.Durga Bhavani** **3)T.Ganga Sai Kumar** **4)K.Kanaka Lakshmi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No	:NA	
Filing Date	:NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

A self-cleaning building surface coating utilizes Nano-chemistry to maintain clean and durable structures. Nan op articles are incorporated into a protective coating applied to exterior surfaces. The nano-layei' creates hydrophobic and photo catalytic properties. Water droplets roll off the surface, carrying dirt and contaminants. Photo catalytic reactions break down organic pollutants under sunlight exposure. The coating reduces dust accumulation and staining. Improved surface protection enhances resistance to weathering and environmental damage. The solution is suitable for glass, concrete, metal, and painted surfaces. Reduced maintenance requirements lower cleaning costs and water usage. The nano-coating maintains transparency and aesthetic appearance. Strong adhesion ensures long-term durability of the coated surface. The formulation is compatible with standard application methods such as spraying or brushing. Environmental benefits include reduced chemical cleaning agents. The coating enhances lifespan and performance of building facades.r

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641081045 A
(19) INDIA
(22) Date of filing of Application :01/07/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : CHEMICAL-BASED FIRE RESISTANT CONTRUCTION MATERIAL

(51) International classification	:C04B28/26, C04B28/24, E04B1/94, E04B1/92, E04B1/62, B32B13/04, B32B13/14, C04B35/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr.K.Sundara Kumar
2)T.Ganga Sai Kumar
3)T.Swathi
4)B.Ganesh

(57) Abstract :
A chemical-based fire resistant construction material enhances safety and structural protection against fire hazards. Specialized chemical additives are incorporated into the base construction material. These additives improve thermal stability and reduce flame spread. The material forms an insulating barrier when exposed to high temperatures. Reduced heat transfer delays structural damage during fire incidents. The formulation minimizes smoke generation and toxic gas emission. Enhanced fire resistance increases evacuation time and safety. The material maintains mechanical strength under elevated temperatures. Compatibility with conventional construction methods ensures easy adoption. The solution is suitable for walls, ceilings, and structural components. Improved durability reduces maintenance and replacement costs. The chemical composition supports lightweight and efficient design. Environmental benefits include reduced fire damage and resource loss. Quality control ensures consistent fire-resistant perfoimance.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :01/07/2026
(21) Application No.202641081048 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WASTE PLASTIC AND CHEMICAL ADDITIVE REINFORCED ASPHALT PAVEMENT.

(51) International classification	:C08L 95/00, C04B 26/26, E01C 7/35, E01C 7/32, E01C 7/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)T.Ganga sai kumar
2)Dr.K.Sundara Kumar
3)K.Kanaka Lakshmi
4)D.Leela Prasad

(57) Abstract :
Waste plastic and chemical additives are utilized to reinforce asphalt pavement for improved performance. Processed plastic waste is blended with bitumen to enhance binding properties. Chemical additives improve adhesion, flexibility, and durability of the asphalt mix. The modified pavement exhibits higher resistance to rutting and deformation. Improved tensile strength reduces cracking under traffic loads. Enhanced water resistance minimizes damage from moisture infiltration. The solution supports recycling of plastic waste, reducing environmental pollution. Lower maintenance requirements extend pavement service life. The modified asphalt perfonus effectively under varying temperature conditions. The method is compatible with conventional road construction techniques. Reduced material consumption lowers overall construction costs. Improved load-bearing capacity supports heavy traffic conditions. Environmental benefits include waste utilization and reduced landfill disposal. Quality control ensures uniform mixing and consistent pavement performance.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202641081050 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HEAVY METAL REMOVAL SYSTEM FOR CONSTRUCTION SITE WASTEWATER TREATMENT.

(51) International classification	:C01B17/42, C02F1/52, C02F9/00, C02F1/62, C02F1/58
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)T.Swathi
2)K.Kanaka Lakshmi
3)B.Ganesh
4)Dr.K.Sundara Kumar

(57) Abstract :
A heavy metal removal system treats construction site wastewater to reduce environmental contamination. Waste water containing metals such as lead, chromium, and cadmium is collected for processing. Chemical treatment agents facilitate precipitation and separation of dissolved metals. Filtration and sedimentation stages enhance removal efficiency. The system reduces toxicity and improves water quality for safe discharge, pH adjustment improves effectiveness of metal removal reactions. The treated water can be reused for construction activities. Reduced pollutant discharge supports environmental compliance. Tlic process minimizes sludge volume and simplifies disposal. Modular design allows easy installation at construction sites. The system operates with low energy and maintenance requirements. Monitoring mechanisms ensure consistent treatment performance. Compatibility with existing wastewater treatment practices is maintained. The solution supports sustainable water management in construction projects

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202641081051 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SOLAR-POWERED AUTOMATIC WATER PUMP WITH THERMAL EFFICIENCY ENHANCEMENT.

(51) International classification	:F04D 15/00, F04B 49/06, F04B 17/03, A01G 25/16, H02J 3/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Victor Babu Penumala
2)Ch Lakshmi
3)P Devadass
4)Praveena Eda

(57) Abstract :
A solar-powered automatic water pump system improves water pumping efficiency using enhanced thermal management. Photovoltaic panels convert solar energy into electrical power for pump operation. Thermal efficiency enhancement mechanisms optimize panel performance under varying temperature conditions. Intelligent controllers regulate power flow based on solar intensity. The system automatically activates and deactivates pumping demand. Energy storage components enable operation during low Temperature regulation improves overall energy conversion efficiency. without reliance on conventional grid electricity. Sensors monitor water level, flow rate, and system performance. Automated protection prevents dry running and overheating of the pump. The design supports agricultural irrigation and rural water supply applications. Reduced energy losses improve pumping capacity and operational reliability. The system minimizes operational costs and carbon emissions. Modular construction allows easy installation and scalability. according to water sunlight conditions. The pump operatespa

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641081122 A

(19) INDIA

(22) Date of filing of Application :01/07/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : PHISHING WEBSITE DETECTION USING MACHINE LEARNING

(51) International classification	:H04L 29/06, G06N 20/00, H04L 9/40, G06F 21/55, G06F 16/951	(71)**Name of Applicant :** **1)RAMACHANDRA COLLEGE OF ENGINEERING** Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, 534007, Andhra Pradesh India (72)**Name of Inventor :** **1)Nagendra Kumar Yakkala** **2)S.Anantha Kranthi** **3)S Venkat Swamy** **4)Dr J RANGA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A phishing website detection system uses machine learning to identify fraudulent websites and prevent online scams. Website URLs, content, metadata, and user interaction patterns are collected for analysis. Feature extraction techniques identify suspicious characteristics such as domain age, SSL usage, and link structure. Machine learning models classify websites as legitimate or phishing with high accuracy. Real-time monitoring allows immediate detection of malicious sites. Alerts and notifications are sent to users or administrators upon identifying threats. Historical data improves model performance and reduces false positives. Integration with browsers, email clients, and security platfoiTns ensures seamless protection. Data analytics provide insights into phishing trends and attack strategies. Automated reporting supports compliance and cyber security audits. Privacy and security measures protect sensitive user and browsing information. Predictive models anticipate emerging phishing techniques and adapt dynamically. Multi-platform compatibility ensures protection across devices and networks. The system reduces financial losses, data breaches, and reputational risks.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202641081123 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-FACTOR AUTHENTICATION USING BEHAVIORAL BIOMETRICS

(51) International classification	:G06F 21/32, G06N 20/00, H04L 29/06, H04L 9/32, H04L 9/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr G Chamundeswari
2)P V Kishore Kumar
3)P Anusha
4)Dr Shameena Begum

(57) Abstract :
A multi-factor authentication system enhances security using behavioral biometrics in addition to traditional credentials. User interactions such as typing patterns, mouse movements, touchscreen gestures, and device usage behavior are captured. Machine learning algorithms analyze these behavioral ti□aits to create unique user profiles. Authentication decisions are made by combining behavioral biometrics with passwords, OTPs, or token-based methods. Real-time monitoring detects anomalies and potential unauthorized access attempts. Alerts notify administrators of suspicious activities or login attempts. Historical behavioral data improves model accuracy and reduces false positives. Integration with enterprise applications and web platforms enables seamless deployment. Privacy and encryption protocols protect sensitive user infoimation. Adaptive learning allows the system to evolve with changes in user behavior over time. The system supports multi-device access while maintaining consistent security standards. Automated logging provides traceability and auditability for compliance. Scalability ensures applicability across individual, organizational, and enterprise systems. Energy-efficient processing allows continuous monitoring without excessive computational load.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/07/2026

(21) Application No.202641081124 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED INTRUSION DETECTION SYSTEM FOR NETWORK SECURITY

(51) International classification	:H04L 29/06, G06N 20/00, G06F 21/55, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING,NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. M L S N Swarajya Lakshmi
2)G. Navya Sree
3)Dr J RANGA
4)S Venkat Swamy

(57) Abstract :
An Al-based intrusion detection system (IDS) monitors network traffic to detect and prevent unauthorized access or cyber-attacks. Data packets from network nodes are collected and analyzed in real time. Machine learning and deep learning algorithms identify anomalies, suspicious patterns, and potential threats. The system classifies activities as normal or malicious with high accuracy. Alerts and notificaLious are generated instantly for detected intrusions. Historical network data improves predictive capabilities and detection accuracy. Integration with security information and event management (SIEM) platforms ensures centralized monitoring. Automated logging and reporting facilitate compliance and audit requirements. Real-time dashboards provide actionable insights into network status and threat trends. The system supports multiple protocols, devices, and distributed network environments. Threat severity assessment prioritizes critical interventions. Privacy and security measures protect sensitive network data. Continuous learning adapts the IDS to evolving attack strategies. Scalability allows deployment across enterprise, cloud, or industrial networks.

No. of Pages : 11 No. of Claims : 0

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :15/07/2026

(21) Application No.202641086421 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT-ENABLED PIEZOELECTRIC ENERGY HARVESTING SYSTEM FOR MOBILE CHARGING WITH DC-DC POWER REGULATION AND SUPERCAPACITOR STORAGE

(51) International classification	:H02N2/18, H02J7/00, H02J7/34, H02M3/158
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)A. Kathiravan
Address of Applicant :Associate Professor, Department of Electrical and Electronics Engineering, Rajiv Gandhi College of Engineering and Technology, Puducherry, 607403 Puducherry Pondicherry India
2)S. Suguna
3)S. Sivakumar
4)P. Dhoshibane
5)K.Vishal
6)S. Kathiravan
(72)**Name of Inventor :**
1)A. Kathiravan
2)S. Suguna
3)S. Sivakumar
4)P. Dhoshibane
5)K.Vishal
6)S. Kathiravan

(57) Abstract :
IOT-enabled piezoelectric energy harvesting system for mobile charging with DC-DC power regulation and supercapacitor storage is the proposed invention. The proposed invention a piezoelectric-based energy harvesting station designed to capture ambient mechanical vibrations and convert them into electrical energy for mobile charging applications. The system utilizes piezoelectric crystals that generate a voltage in response to mechanical stress. A microcontroller continuously monitors the voltage and current output from the piezoelectric crystals and transmits real-time data to an IoT platform for remote monitoring and performance analysis. The harvested energy is conditioned through a variable input, constant-output DC-DC converter which stabilizes the voltage and ensures efficient power delivery despite fluctuating input levels.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :15/07/2026

(21) Application No.202641086430 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING AND IMAGE PROCESSING-BASED SYSTEM FOR EARLY SKIN CANCER PREDICTION WITH TARGETED NANOPARTICLE DRUG DELIVERY

(51) International classification	:G06V10/82, G06N3/0455, G16H50/20, A61K9/51
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.Md Sirajul Huque
Address of Applicant :Assistant Professor, Department of Computer Science and Engineering, Guru Nanak Institutions Technical Campus, Hyderabad, Telangana, 501506 Hyderabad Telangana India
2)Lavanyaa E
3)Dr Thallapally Sreelatha
4)Mohanavalli M
5)Annu Karnwal
6)Pushpa V
7)Dr Thirunahari Ugandhar
8)Dr P Agilan
9)S.Devisri
10)Anvesh Perada
11)Dr. Dhanusha.C
12)S. Antony dhas

(72)**Name of Inventor :**
1)Dr.Md Sirajul Huque
2)Lavanyaa E
3)Dr Thallapally Sreelatha
4)Mohanavalli M
5)Annu Karnwal
6)Pushpa V
7)Dr Thirunahari Ugandhar
8)Dr P Agilan
9)S.Devisri
10)Anvesh Perada
11)Dr. Dhanusha.C
12)S. Antony dhas

(57) Abstract :
Machine Learning and Image Processing-Based System for Early Skin Cancer Prediction with Targeted Nanoparticle Drug Delivery is the proposed invention. The proposed invention discloses a machine learning and image processing-based system for early skin cancer prediction integrated with targeted nanoparticle drug delivery for precision oncology. High-resolution dermoscopic or smartphone-acquired skin images undergo adaptive preprocessing, artifact removal, color normalization, and attention-based lesion segmentation to accurately isolate suspicious regions. A Vision Transformer-based multimodal learning model analyzes lesion morphology together with optional clinical parameters to classify skin cancer types and estimate malignancy risk. Explainable artificial intelligence generates visual diagnostic maps to improve interpretability and clinical reliability. Based on the predicted cancer characteristics, an intelligent optimization engine selects biodegradable ligand-functionalized nanoparticles encapsulating anticancer drugs and determines personalized formulation parameters including particle size, drug loading, dosage, and controlled-release profile. Sequential lesion monitoring combined with reinforcement learning continuously refines treatment recommendations according to therapeutic response.

No. of Pages : 14 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097308 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED DIGITAL MARKETING CAMPAIGN OPTIMIZATION PLATFORM

(51) International classification	:G06Q 30/02, G06N 20/00, G06Q 30/00, G06Q 10/06, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. S Naga Padma
2)Dr. M Sasi Bhushan
3)B. Aruna
4)A Anuradha Das

(57) Abstract :
An automated digital marketing campaign optimization platform enhances campaign performance using intelligent analytics. The system collects data from advertisements, social media, email campaigns, and website interactions. Analytical models evaluate engagement, conversion, and reach metrics. Machine learning algorithms optimize targeting, bidding, and content strategies. Real-time adjustments improve campaign efficiency and ROT. Predictive analytics forecast audience behavior and campaign outcomes. Automated reporting provides actionable insights for marketers. Visualization dashboards present performance trends clearly. The system segments audiences for personalized marketing approaches. Integration with existing marketing tools ensures seamless deployment. Continuous learning enhances optimization accuracy over time. Reduced manual effort streamlines campaign management. The platform supports multi-channel marketing strategies. Improved decision-making increases customer engagement and conversions.

No. of Pages : 14 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097311 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED FINANCIAL FRAUD DETECTION SYSTEM FOR BUSINESSES

(51) International classification	:G06Q 40/00, G06N 20/00, G06Q 40/02, G06Q 20/40, G06Q 30/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Ch Lakshmi
2)P Devadass
3)Dr D Sridhar
4)Victor Babu Penumala

(57) Abstract :
An AI-based financial fraud detection system identifies suspicious transactions in business operations. The system collects financial data from payments, invoices, and accounts. Data preprocessing ensures accuracy and consistency of inputs. Machine learning models analyze transaction patterns and anomalies. Predictive algorithms detect unusual financial behavior in real time. The system flags potential fraud risks automatically. Automated alerts notify administrators for further investigation. Historical data improves detection accuracy over time. The approach reduces financial losses and operational risks. Visualization dashboards present fraud indicators clearly. Integration with accounting and enterprise systems ensures seamless operation. Continuous monitoring enhances financial security. The solution supports multiple business sectors. Automated analysis reduces manual auditing effort.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097313 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART INVENTORY OPTIMIZATION SYSTEM USING PREDICTIVE ANALYTICS

(51) International classification	:G06Q 10/08, G06Q 10/06, G06N 20/00, G06Q 10/087, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)V. Soundarya
2)A. Durga Madhavi
3)Sk. Basheera
4)V Vineetha Naga Sai Ganga Devi

(57) Abstract :
A smart inventory optimization system improves stock management using predictive analytics. The system collects data from sales, demand patterns, and supply chains. Analytical models identify trends and seasonal variations. Predictive algorithms forecast future inventory requirements. The system determines optimal stock levels automatically. Real-time monitoring prevents overstocking and stock shortages. Automated alerts notify users of replenishment needs. The approach improves supply chain efficiency. Visualization dashboards present inventory performance metrics. Integration with enterprise resource planning systems is supported. Continuous learning enhances forecasting accuracy over time. The solution reduces storage costs and wastage. Optimized inventory improves order fulfillment rates. Automated analysis minimizes manual planning effort.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097314 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED PRICING STRATEGY OPTIMIZATION USING DATA SCIENCE

(51) International classification	:G06Q 30/02, G06N 20/00, G06Q 10/06, G06Q 40/04, G06Q 30/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Y Divya
2)Sk. Basheera
3)V. Soundarya
4)A. Durga Madhavi

(57) Abstract :
An automated pricing strategy optimization system enhances business profitability using data science techniques. The system collects historical sales, competitor pricing, demand patterns, and market trends. Data preprocessing ensures accuracy, consistency, and relevance of inputs. Analytical models identify pricing sensitivity and optimal price points. Machine learning algorithms predict revenue and demand responses to price adjustments. The system generates dynamic pricing strategies automatically. Real-time insights enable rapid adaptation to market fluctuations. Visualization dashboards present pricing performance metrics and trends. Integration with sales and e-commerce platforms ensures seamless implementation. Continuous learning improves prediction and strategy accuracy over time. Scenario analysis supports decision-making under different market conditions. Automated alerts notify users of significant deviations or opportunities. Reduced manual effort streamlines pricing management. Optimized pricing improves revenue, market competitiveness, and customer satisfaction.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097315 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED STARTUP SUCCESS PREDICTION SYSTEM

(51) International classification	:G06N 20/00, G06Q 10/06, G06Q 40/06, G06Q 40/00, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)V Vineetha Naga Sai Ganga Devi
2)B. Aruna
3)Dr. S Naga Padma
4)Dr. M Sasi Bhushan

(57) Abstract :
An Al-based startup success prediction system evaluates the potential of new business ventures using intelligent analytics. The system collects data from market trends, financial projections, team experience, and business models. Data preprocessing ensures accuracy and consistency of inputs. Machine learning models analyze historical startup performance and relevant success factors. Predictive algorithms estimate the likelihood of business success or failure. Real-time insights assist investors, incubators, and entrepreneurs in decision-making. Visualization dashboards present performance indicators and risk assessments clearly. The system identifies strengths, weaknesses, and opportunities for improvement. Automated scoring ranks startups based on predicted success probability. Continuous learning enhances prediction accuracy over time. Integration with business intelligence platforms enables seamless operation. Scenario analysis supports strategic planning under different market conditions. Reduced manual evaluation effort improves efficiency and objectivity. Data-driven insights optimize resource allocation and funding decisions

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097316 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BUSINESS PERFORMANCE DASHBOARD USING REAL-TIME DATA ANALYTICS

(51) International classification	:G06Q 10/06, G06Q 40/00, G06Q 30/02, G06Q 10/00, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)B. Aruna
2)Y Divya
3)V Vineetha Naga Sai Ganga Devi
4)V. Soundarya

(57) Abstract :
A business performance dashboard monitors and analyzes enterprise metrics using real-time data analytics. The system collects operational, financial, sales, and customer data from multiple . sources. Data preprocessing ensures accuracy, consistency, and relevance of inputs. Analytical models identify trends, patterns, and key performance indicators. Real-time visualization displays actionable insights for managers and decision-makers. Automated alerts notify users of critical deviations or anomalies. Predictive analytics forecast future performance based on current trends. Integration with enterprise resource planning and business software ensures seamless operation. Interactive dashboards allow drill-down analysis for detailed reporting. The system reduces manual reporting effort and errors. Continuous monitoring enhances responsiveness to business changes. Performance metrics support strategic planning and operational optimization. Scalable architecture accommodates organizations of various sizes. Data-driven insights improve productivity, efficiency, and decision-making.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097317 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT CRM SYSTEM WITH MACHINE LEARNING INSIGHTS

(51) International classification	:G06Q 30/02, G06N 20/00, G06Q 10/06, G06Q 30/00, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)A Anuradha Das
2)V Vineetha Naga Sai Ganga Devi
3)Y Divya
4)Sk. Basheera

(57) Abstract :
An intelligent CRM system enhances customer relationship management using machine learning insights. The system collects data from sales, support interactions, and customer engagement channels. Data preprocessing ensures accuracy, consistency, and relevance of inputs. Machine learning models analyze behavior patterns, preferences, and interaction history. Predictive algorithms forecast customer needs and potential chum. Automated insights enable personalized marketing and service strategies. Real-time dashboards present customer analytics and actionable metrics. Integration with sales, marketing, and support platforms ensures seamless operation. Segmentation algorithms group customers for targeted campaigns. Continuous learning improves prediction accuracy over time. Alerts notify teams of high-value opportunities or at-risk clients. Data-driven decision-making improves customer retention and satisfaction. Reduced manual analysis enhances operational efficiency. Scalable design supports businesses of various sizes

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097318 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED SUPPLY CHAIN DEMAND PREDICTION SYSTEM

(51) International classification	:G06Q 10/06, G06N 20/00, G06Q 10/08, G06Q 10/04, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)G.Prabhu
2)A Anuradha Das
3)Dr. K Sri Durga
4)Dr. P S V Padma Latha

(57) Abstract :
An AI-based supply chain demand prediction system forecasts product requirements using intelligent analytics. The system collects data from sales, inventory, logistics, and market trends. Data preprocessing ensures accuracy and consistency of inputs. Machine learning algorithms identify patterns and predict future demand. Predictive models optimize inventory levels and replenishment schedules. Real-time insights support proactive supply chain decisions. Visualization dashboards display demand trends and key performance metrics. Automated alerts notify managers of potential stock shortages or overstocking. Integration with enterprise resource planning systems ensures seamless operation. Continuous learning improves prediction accuracy over time. The approach reduces operational costs and improves resource utilization. Multiproduct and multi-location scalability is supported. Scenario analysis assists in planning for market fluctuations. Reduced manual forecasting enhances efficiency and decision-making.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097319 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART BUDGET PLANNING TOOL USING DATA ANALYTICS

(51) International classification	:G06Q 40/00, G06Q 40/02, G06Q 10/06, G06Q 40/06, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Ch. Sumathi Devi
2)G.Prabhu
3)Dr. P S V Padma Latha
4)Dr. K Sri Durga

(57) Abstract :

_A smart b_udget planning tool enhances financial management using data analytics. The system collects data from income, expenses, and financial transactions. Data preprocessing ensures accuracy and consistency of inputs. Analytical models categorize expenditures and identify spending patterns. Predictive algorithms forecast future financial requirements and cash flows. The system generates optimal budget allocations automatically. Real-time insights assist users in decision-making and expense control. Visualization dashboards present financial metrics and trends clearly. Automated alerts notify users of overspending or budget deviations. Integration with banking and accounting platforms ensures seamless operation. Continuous learning improves accuracy of forecasts over time. Scenario analysis supports planning for different financial conditions. Reduced manual effort streamlines financial management. The solution enhances savings, resource allocation, and operational efficiency.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097320 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE ANALYTICS SYSTEM FOR RETAIL DEMAND FORECASTING

(51) International classification	:G06Q 10/08, G06N 20/00, G06Q 10/06, G06Q 30/02, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant : NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 9949995754 deanresearch@rcee.ac.in Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. K Sri Durga
2)Dr. P S V Padma Latha
3)Ch. Sumathi Devi
4)Dr. S Naga Padma

(57) Abstract :
A predictive analytics system forecasts retail demand using historical and real-time data. The system collects sales, inventory, promotions, and market trend data. Data preprocessing ensures consistency, accuracy, and relevance. Analytical models identify patterns and seasonality in customer demand. Machine learning algorithms predict future product requirements. The system supports inventory optimization and supply chain planning. Real-time insights enable dynamic adjustments to stock levels. Visualization dashboards present demand trends and actionable metrics. Automated reporting reduces manual forecasting effort. Integration with point-of-sale and ERP systems ensures seamless operation. Predictive insights improve product availability and customer satisfaction. Continuous learning enhances prediction accuracy over time. The approach reduces overstocking, stockouts, and operational costs. Multi-product and multi-store scalability is supported.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097321 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED RECRUITMENT AND CANDIDATE RANKING SYSTEM

(51) International classification	:G06N 20/00, G06Q 10/06, G06Q 10/10, G06F 16/2457, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-I6 Bypass Road, Vatluru (V) Eluru Di Andhra Pradesh India 534007 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. PS V Padma Latha
2)Dr. K Sri Durga
3)G.Prabhu
4)Ch. Sumathi Devi

(57) Abstract :
An Al-based ecruitment and candidate ranking system streamlines hiring .using analytics. The system collects candidate data from resumes, applications, and assessments. Natural language processing algorithms extract relevant skills, experience, and qualifications. Machine learning models evaluate and score candidates based on predefined job criteria. Predictive analytics forecast candidate suitability and potential performance. The system generates a ranked shortlist for recruiters automatically. Real-time insights assist in informed hiring decisions. Data visualization dashboards present candidate comparisons and metrics clearly. Automated filtering reduces manual screening effort. Integration with human resource management systems ensures seamless operation. Continuous learning improves ranking accuracy over time. The system enhances fairness and objectivity in recruitment. Optimized candidate selection reduces hiring time and costs. Scalable architecture supports multiple job roles and industries.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/08/2026
(21) Application No.202641097322 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CUSTOMER SENTIMENT ANALYSIS SYSTEM USING SOCIAL MEDIA DATA

(51) International classification	:G06N 20/00, G06Q 30/02, G06F 16/35, G06N 5/04, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 9949995754 deanresearch@rcee.ac.in Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. M Sasi Bhushan
2)Ch. Sumathi Devi
3)A Anuradha Das
4)G.Prabhu

(57) Abstract :
A customer sentiment analysis system evaluates opinions and feedback from social media platforms. The system collects textual, visual, and interaction data from multiple online sources. Data preprocessing ensures consistency, accuracy, and relevance. Natural language processinc, algorithms analyze sentiment polarity and emotional tone. Machine learning models classify customer opinions into positive, negative, or neutral categories. Predictive analytics detect emerging trends in customer perception. Real-time insights support marketing, product development, and customer engagement strategies. Visualization dashboards display sentiment trends and key metrics. The system identifies potential issues before they escalate. Automated reporting reduces manual data analysis effort. Integration with business intelligence tools ensures seamless operation. Continuous learning improves prediction and classification accuracy. Insights help optimize brand image and customer satisfaction. Reduced response time enhances operational efficiency.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097324 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT BUSINESS ANALYTICS PLATFORM FOR DECISION MAKING

(51) International classification	:G06Q 10/06, G06N 20/00, G06Q 30/02, G06Q 10/04, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 9949995754 deanresearch@rcee.ac.in Andhra Pradesh India
(72)**Name of Inventor :**
1)Sk. Basheera
2)K.Lavanya
3)A. Durga Madhavi
4)Y Divya

(57) Abstract :
An intelligent business analytics platform enhances organizational decision-making using advanced analytics. The system collects data from sales, operations, finance, and customer interactions. Data preprocessing ensures consistency, accuracy, and relevance of inputs. Analytical models identify patterns, trends, and key performance indicators. Predictive algorithms forecast future business outcomes and risks. Real-time dashboards present actionable insights for managers and decision-makers. Scenario analysis supports strategic planning under varying conditions. Automated reporting reduces manual effort and errors. Integration with enterprise systems enables seamless data flow and operation. Continuous learning improves prediction and analysis accuracy over time. The platform identifies operational inefficiencies and improvement opportunities. Visualization tools provide clear and interactive representations of complex data. Data-driven insights enhance productivity, profitability, and competitive advantage. Scalable architecture accommodates businesses of different sizes.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097325 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BUSINESS RISK ASSESSMENT TOOL USING AI AND DATA SCIENCES

(51) International classification	:G06Q 10/06, G06N 20/00, G06Q 40/00, G06Q 40/06, G06Q 40/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 Bypass Road, Vatluru (V) Eluru Dt Andhra Pradesh India 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)A. Durga Madhavi
2)Dr. S Naga Padilla
3)Dr. M Sasi Bhushan
4)B. Aruna

(57) Abstract :
A business risk assessment tool evaluatespotential risks using Al and data science techniques. The system collects financial, operational, and market-related data. Data preprocessing improves reliability and consistency of inputs. Machine learning models analyze patterns associated with business risks. Predictive algorithms estimate probability and impact of potential threats. The system identifies financial, operational, and strategic risk factors. Real-time analytics support proactive decision-making. Visualization dashboards present risk indicators clearly. Automated alerts notify users of high-risk conditions. The tool supports planning and mitigation strategies. Integration with enterprise systems enables seamless data exchange. Continuous learning improves assessment accuracy over time. The solution reduces uncertainty in business operations. Data-driven insights enhance organizational resilience

No. of Pages : 13 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097331 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DIGITAL CUSTOMER LOYALTY MANAGEMENT SYSTEM WITH PREDICTIVE INSIGHTS

(51) International classification	:G06Q 30/02, G06N 20/00, G06Q 30/00, G06N 5/04, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 9949995754 deanresearch@rcee.ac.inh Tamil Nadu India
(72)**Name of Inventor :**
1)K.Lavanya
2) V. Soundarya
3)M.Lakshmi Prasanna
4)K.Pravallika

(57) Abstract :
A digital customer_ _loyalty management_system_enhances_retention_using_predictiye analytics. The system collects customer interaction and transaction data. Analytical models evaluate purchasing behavior and engagement levels. Predictive algorithms identify customers likely to churn. Personalized reward strategies are generated automatically. The platform supports loyalty points, offers, and incentive programs. Real-time insights improve customer engagement decisions. Data visualization dashboards present loyalty performance metrics. The system segments customers based on behavioral patterns. Automated notifications enhance communication with customers. Integration with sales and CRM platforms ensures seamless operation. Continuous learning improves prediction accuracy over time. The solution helps increase repeat purchases and brand loyalty. Reduced manual effort improves operational efficiency.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097379 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED PREDICTIVE MAINTENANCE MODEL USING STATISTICAL ANALYSIS FOR MACHINES

(51) International classification	:G06Q 10/00, G05B 23/02, G06N 20/00, G06Q 10/06, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)B Pathrisamma
2)Dr S Jagan Mohan Rao
3)Dr B V Manikanta
4)Dr A Chiranjeevi

(57) Abstract :
An AI-based predictive maintenance model improves reliability of industrial mechanics using statistical analysis. The system collects operational -data from sensors—monitoring_ _machine performance. Parameters such as vibration, temperature, and load are analyzed continuously. Statistical techniques identify patterns indicating potential equipment failure. Machine learning algorithms predict maintenance requirements in advance. Early detection reduces unexpected breakdowns and downtime. The model provides real-time alerts for preventive action. Data driven insights support efficient maintenance scheduling. Historical data improves prediction accuracy over time. The system minimizes maintenance costs and resource usage. Integration with existing monitoring infrastructure is supported. Automated analysis reduces manual inspection requirements. Improved equipment health enhances operational productivity. The solution supports various industrial and mechanical applications.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097393 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MATHEMATICAL OPTIMIZATION MODEL FOR ENERGY EFFICIENT MECHANICAL DRIVE SYSTEMS

(51) International classification	:G05B 13/04, G06F 17/50, G05B 17/02, G05B 13/02, G06F 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 9949995754 deanresearch@rcee.ac.in Tamil Nadu India
(72)**Name of Inventor :**
1)T Vinoditha
2)Praveena Eda
3)Dr S V B Subrahrnanyeswara Rao
4)M.Lakshmi Prasanna

(57) Abstract :
A—mathematical optimization_model improves energy efficiency of mechanical drive systems. The model analyzes parameters such as speed, torque, and load conditions. Optimization algorithms determine optimal operating points for minimum energy consumption. The system evaluates performance of motors, gears, and transmission components. Real-time data inputs enhance accuracy of optimization results. The model reduces power losses in mechanical drive operations. Adaptive control adjusts system parameters based on varying loads. Simulation tools validate efficiency improvements before implementation. The approach supports industrial machinery and automated systems. Reduced energy consumption lowers operational costs. The model improves performance and reliability of mechanical drives. Integration with monitoring systems enables continuous optimization. Data analysis identifies inefficiencies and corrective measures. The solution supports sustainable and energy-efficient industrial operations.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/08/2026
(21) Application No.202641097398 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : HYBRID RENEWABLE ENERGY MEHANICAL STORAGE SYSTEM

(51) International classification	:H02J15/20, H02J3/38, H02J101/24, H02J101/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-I6 Bypass Road, Vatluru (V) Eluru Dt Andhra Pradesh India 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1) Dr S Jagan Mohan Rao
2)B Pathrisamma
3)Dr B Prasad Babu
4) L L S Maneesha

(57) Abstract :
A hybrid renewable_energy_mechanical-storage-system stores-energy generated-from-multiple renewable sources. The system integrates solar, wind, or other renewable inputs. Excess energy is converted into mechanical form for storage. Mechanical storage mechanisms such as flywheels or weighted systems retain energy. Stored mechanical energy is converted back into electrical power when required. Intelligent control optimizes energy charging and discharging cycles. The system improves reliability of intermittent renewable sources. Energy losses are minimized through efficient conversion mechanisms. Sensors monitor system performance and operating conditions. The design supports grid-connected and standalone applications. Modular architecture allows scalability for various capacities. Reduced dependence on batteries enhances system lifespan. The solution improves energy utilization and stability. Environmental benefits include clean energy storage and reduced emissions.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097400 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MAGNETIC FIELD WIRELESS POWER TRANSMISSION PROTOTYPE

(51) International classification :H02J50/12
(31) Priority Document No :NA
(32) Priority Date :NA
(33) Name of priority country :NA
(86) International Application No :NA
Filing Date :NA
(87) International Publication No : NA
(61) Patent of Addition to Application Number :NA
Filing Date :NA
(62) Divisional to Application Number :NA
Filing Date :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 Bypass Road, Vatluru (V) Eluru Dt Andhra Pradesh India 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)PRAVEENA EDA
2)Kurnara Swamy
3) K Jeeyana Iyothi
4)T Vinoditha

(57) Abstract :
A magnetic-field-based wireless-power transmission prototype transfers electrical energy_without physical connections. The system utilizes inductive coupling between transmitter and receiver coils. Alternating current in the transmitter coil generates a magnetic field. The receiver coil captures magnetic flux to produce electrical output. Resonant frequency tuning improves power transfer efficiency. The prototype supports short-range wireless energy transmission applications. Power conditioning circuits stabilize the received output. The system reduces dependence on wired connections and connectors. Alignment mechanisms improve coupling between coils. Real-time monitoring ensures safe and stable operation. The design supports charging of electronic devices and sensors. Reduced mechanical wear improves system reliability. Compact construction allows easy integration into various platforms. Energy losses are minimized through optimized coil design.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097404 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT MARKET TREND PREDICTION USING BIG DATA ANALYTICS

(51) International classification	:G06Q 30/02, G06N 20/00, G06Q 10/06, G06Q 10/04, G06Q 40/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 9949995754 deanresearch@rcee.ac.in Andhra Pradesh India
(72)**Name of Inventor :**
1)K.Pravallika
2)Dr S V B Subrahmanyeswara Rao
3)K.Layanya
4)K Jeeyana Jyothi

(57) Abstract :
An_ intelligentmarket trend_prediction system analyzes large,scale .data using big data analytics. The system collects information from sales, social media, and market reports. Data preprocessing ensures accuracy and consistency of inputs. Analytical models identify emerging market patterns and trends. Predictive algorithms forecast future demand and opportunities. The system supports strategic planning and investment decisions. Real-time analytics provide updated market insights. Visualization dashboards present trends in an understandable format. The approach enhances competitiveness for businesses. Automated analysis reduces manual effort in market evaluation. Integration with enterprise systems enables seamless deployment. Continuous learning improves prediction accuracy over time. The solution supports multi-industry applications. Data-driven insights help reduce business risks.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097405 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-POWERED EMPLOYEE PERFORMANCE EVALUATION PLATFORM

(51) International classification	:G06Q 10/06, G06N 20/00, G06Q 10/10, G06Q 10/00, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 Bypass Road, Vatiuru (V) Eluru Dt Andhra Pradesh India 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)P Devadass
2)Victor Babu Penumala
3)Ch Lakshmi
4)Dr D Sridha

(57) Abstract :
An AI=powered employee_performance evaluation platform analyzes workforce productivity using intelligent algorithms. The system collects data from task completion, attendance, and performance metrics. Machine learning models evaluate individual and team performance trends. Automated scoring mechanisms provide objective assessment results. The platform identifies strengths and areas for improvement. Real-time analytics support informed managerial decisions. Performance dashboards present clear and actionable insights. The system enables continuous monitoring of employee productivity. Predictive analytics forecast future performance outcomes. Integration with human resource management systems ensures seamless operation. Automated reports reduce manual evaluation effort. The solution promotes fair and transparent assessments. Adaptive algorithms adjust evaluation criteria dynamically. Improved performance tracking enhances organizational efficiency.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097406 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED SALES FORECASTING SYSTEM USING MACHINE LEARNING

(51) International classification	:G06N 20/00, G06Q 30/02, G06Q 10/06, G06Q 10/08, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 9949995754 deanresearch@rcee.ac.in Andhra Pradesh India
(72)**Name of Inventor :**
1)L L S Maneesha
2)Dr A Chiranjeevi
3)Dr S Jagan Mohan Rao
4)Dr B Prasad Babu

(57) Abstract :
An automated sales forecasting system predicts future sales using machine learning techniques. The system collects historical sales and market data. Data preprocessing improves accuracy and consistency of inputs. Machine learning models analyze seasonal and trend patterns. Predictive algorithms estimate future demand and revenue. The system adapts to changing market conditions dynamically. Real-time analytics provide updated forecasting results. Automated reporting supports business planning and inventory management. The approach reduces forecasting errors and manual effort. Visualization dashboards display sales trends clearly. Integration with enterprise systems enables seamless deployment. Continuous learning improves prediction accuracy over time. The solution supports multiple product categories and regions. Improved forecasting enhances operational efficiency and profitability.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097407 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART CUSTOMER BEHAVIOR PREDICTION USING DATA ANALYTICS

(51) International classification	:G06Q 30/02, G06N 20/00, G06Q 10/06, G06N 5/04, G06Q 10/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 9949995754 deanresearch@rcee.ac.in Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr B Prasad Babu
2)L L S Maneesha
3)Dr A Chiranjeevi
4)Dr S Jagan Mohan Rao

(57) Abstract :
A smart customer behavior prediction system analyzes consumer data using advanced analytics. The system collects information from transactions, browsing patterns, and interactions. Data preprocessing improves quality and consistency of inputs. Analytical models identify trends and purchasing preferences. Predictive algorithms forecast future customer behavior. The system segments customers based on behavior patterns. Real-time insights support targeted marketing strategies. Recommendation mechanisms enhance customer engagement and satisfaction. The approach improves demand forecasting and inventory planning. Automated analysis reduces manual effort in data interpretation. Visualization dashboards present actionable business insights. Integration with business platforms enables seamless deployment. Continuous learning improves prediction accuracy over time. The solution enhances decision-making and customer retention.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097408 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED BUSINESS DECISION SUPPORT SYSTEM FOR SMALL ENTERPRISES

(51) International classification	:G06Q 10/06, G06N 20/00, G06Q 30/02, G06Q 10/00, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)M.Lakshmi Prasanna
2)K.Pravallika
3)Kumara Swamy
4)K.Lavanya

(57) Abstract :
An Al-based business decision support system assists small enterprises in strategic planning. The system collects business data from sales, finance, and operations. Machine learning algorithms analyze trends and performance indicators. Predictive models forecast demand and revenue patterns. The system identifies opportunities for cost reduction and growth. Real-time analytics provide actionable business insights. Automated recommendations support informed decisionmaking. Data visualization tools present clear performance dashboards. The system adapts to changing market conditions dynamically. Risk analysis helps minimize fmancial uncertainties. Integration with existing business software ensures seamless operation. Historical data improves accuracy of predictions over time. The solution enhances operational efficiency and productivity. Scalable design supports various small enterprise requirements.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097409 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED ROBOTIC ARM MOVEMENT OPTIMIZATION USING LINEAR PROGRAMMING

(51) International classification	:B25J 9/16, G05B 13/04, B25J 9/00, G05B 19/418, G05B 19/416
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr D Sridhar
2)B Suneetha
3)Victor Babu Penumala
4)Ch Lakshmi

(57) Abstract :
An automated robotic-arm movernent-optimization system enhances operational-efficiency-using linear programming. The method analyzes task requirements and workspace constraints. Linear programming algorithms determine optimal movement paths. The system minimizes travel time and energy consumption. Joint angles and motion sequences are optimized for precision. Collision avoidance constraints ensure safe robotic operation. The approach improves productivity in automated manufacturing environments. Real-time adjustments adapt to dynamic task conditions. Reduced mechanical wear increases robot lifespan. The model supports multiaxis robotic arm configurations. Optimization improves positioning accuracy and repeatability. Integration with control systems enables automated execution. Simulation validates optimized movement before deployment. Reduced idle motion enhances overall system efficiency

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097410 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING AND MATHEMATICAL MODEL FOR PREDICTING MECHANICAL FAILURE

(51) International classification	:G06N 20/00, G05B 23/02, G06Q 10/00, G06Q 10/06, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, INDIA-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr A Chiranjeevi
2)Dr B Prasad Babu
3)L L S Maneesha
4)Dr P Raja Sekhar

(57) Abstract :
A_predictive system combines machine learning and_mathematical modeling _to forecast mechanical failures. Operational data from sensors is continuously collected and processed. Parameters such as vibration, temperature, and load are analyzed. Mathematical models evaluate stress and degradation patterns. Machine learning algorithms identify hidden correlations in performance data. The system predicts potential failures before they occur. Early warnings support preventive maintenance planning. Real-time monitoring improves reliability of mechanical equipment. Historical data enhances prediction accuracy over time. The approach reduces unexpected downtime and repair costs. Automated analysis minimizes manual inspection requirements. Integration with industrial monitoring systems is supported. Improved equipment health increases operational efficiency. The model adapts to varying operating conditions dynamically

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :12/08/2026
(21) Application No.202641097411 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MATHEMATICAL SIMULATION OF HEAT TRANSFER IN INDUSTRIAL MECHANICAL SYSTEMS

(51) International classification	:G06F 17/50, G06F 19/00, G06F 119/08, G05B 17/02, G06F 30/23
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr P Raja Sekhar
2)Dr B V Manikanta
3)Venu Gopal Mutyala
4)CH Dharani

(57) Abstract :
A mathematical simulation system analyzes heat transfer in industrial mechanical systems. The model evaluates conduction, convection, and radiation effects. Thermal equations simulate temperature distribution across components. Input parameters include material properties and operating conditions. Computational methods predict heat flow and thermal gradients. The simulation identifies overheating zones in machinery. Optimization techniques improve cooling and thermal management. Real-time data integration enhances simulation accuracy. The approach supports design of heat exchangers and mechanical equipment. Reduced thermal stress improves component lifespan. The model minimizes energy losses in industrial processes. Visualization tools display temperature profiles clearly. The system reduces reliance on physical testing. Scalable architecture supports various industrial applications.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097412 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT CNC TOOL PATH OPTIMIZATION USING MATHEMATICAL ALGORITHMS

(51) International classification	:G05B 19/4093, G05B 19/4097, G05B 19/416, G05B 19/18, G05B 19/19	(71)**Name of Applicant :** **1)RAMACHANDRA COLLEGE OF ENGINEERING** Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 Andhra Pradesh India (72)**Name of Inventor :** **1)Venu Gopal Mutyala** **2)CH Dharani** **3)Dr P Raja Sekhar** **4)Dr B V Manikanta**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An intelligent CNC tool path optimization system enhances machining efficiency using mathematical algorithms. The method analyzes workpiece geometry and machining constraints. Optimization techniques generate efficient tool paths with reduced travel distance. The system minimizes idle movements and machining time. Mathematical models improve cutting sequence planning. Adaptive algorithms adjust tool paths based on machining conditions. Reduced tool wear improves tool life and productivity. The approach enhances surface finish and dimensional accuracy. Collision avoidance mechanisms ensure safe machining operations. Real-time data integration supports dynamic optimization. The system reduces energy consumption during machining. Compatibility with various CNC machines ensures flexible deployment. Simulation features validate tool paths before execution. Automated optimization reduces programming effort and human error.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097413 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MATHEMATICAL MODELING OF AERODYNAMIC DRAG REDUCTION IN VEHICLES

(51) International classification	:G06F 17/50, G06F 30/15, G06F 30/28, B62D 35/00, G06F 119/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr S V B Subrahmanyeswara Rao
2)M.Lakshmi Prasanna
3)Praveena Eda
4)P Devadass

(57) Abstract :
A mathematical modeling approach reduces aerodynamic drag in vehicles to improve efficiency. The model analyzes airflow behavior around vehicle structures. Computational equations evaluate pressure distribution and drag forces. Design parameters such as shape, surface curvature, and frontal area are optimized. The system predicts drag reduction through mathematical simulations. Optimization algorithms refine aerodynamic profiles for improved performance. Reduced drag enhances fuel efficiency and energy consumption. The model supports cars, trucks, and electric vehicles. Simulation results guide structural design improvements. Real-time analysis assists in evaluating design variations. The approach minimizes wind resistance and turbulence formation. Improved aerodynamics enhances vehicle stability at high speeds. Computational methods reduce reliance on extensive physical testing. The solution supports environmentally friendly vehicle design.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097415 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATIC REFRIDGERATION SYSTEM USING ECO-FRIENDLY REFRIGERANTS

(51) International classification	:F25B 49/02, F25B 1/00, F25D 29/00, F25B 13/00, F25B 49/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)G Suneetha
2)K Jeevana Jyothi
3)T Vinoditha
4)CH Lakshmi

(57) Abstract :
An automatic refrigeration system operates using environmentally friendly refrigerants to reduce ecological impact. The system maintains controlled cooling for storage and preservation applications. Eco-friendly refrigerants provide effective heat transfer with low global warming potential. Automatic control mechanisms regulate temperature and cooling cycles. Sensors monitor internal conditions and adjust compressor operation accordingly. The system optimizes energy consumption during varying load conditions. Efficient heat exchangers enhance cooling performance. Automated protection prevents overcooling and system overload. The design supports domestic, commercial, and industrial refrigeration needs. Reduced emissions improve environmental sustainability. Intelligent control improves operational reliability. Low maintenance requirements reduce operational costs. The system ensures stable cooling performance under different conditions. Modular construction allows flexible installation and scalability.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097418 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HIGH EFFICIENCY SOLAR STEAM GENERATION SYSTEM USING NANO FLUIDS

(51) International classification	:F22B 1/00, F24S 20/20, F24J 2/04, F24S 23/70, F24S 10/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)K Jeevana Jyothi
2)CH Lakshmi
3)P Balakrishna
4)Kumara Swamy

(57) Abstract :
A high efficiency solar steam generation system utilizes nanofluids to enhance thermal absorption. Solar collectors capture sunlight and transfer heat to the working fluid. Nanoparticles dispersed in the fluid improve heat transfer characteristics. Increased thermal conductivity boosts steam generation efficiency. The system converts solar energy into steam for industrial and domestic applications. Optimized heat exchangers maximize energy utilization. Temperature sensors regulate operating conditions for consistent performance. Insulated components reduce heat losses during operation. The system operates using renewable solar energy. Enhanced evaporation rates reduce energy consumption. Modular design allows scalability for different capacity requirements. The solution supports applications such as sterilization, drying, and power generation. Reduced dependence on fossil fuels lowers emissions. Low maintenance requirements improve operational reliability.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097420 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WASTE PLASTIC TO FUEL CONVERSION REACTOR WITH HEAT RECOVERY

(51) International classification	:C10G 1/10, C10B 53/07, B01J 19/00, C10G 1/00, B01J 19/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Kumara Swamy
2)G Suneetha
3)G Bindu Pavani
4)P Balakrishna

(57) Abstract :
A waste plastic to fuel conversion_reactor transforms plastic waste into usable fuel. The system utilizes a controlled thermal degradation process to break down plastic materials. A sealed reactor chamber ensures safe and efficient conversion. Heat recovery mechanisms capture excess thermal energy generated during operation. Recovered heat is reused to improve overall system efficiency. The process reduces dependence on external energy sources. Condensation units convert vapors into liquid fuel products. Non-condensable gases are recycled to support reactor heating. Temperature sensors maintain optimal conversion conditions. The system minimizes harmful emissions through controlled processing. Modular design supports scalability for various waste volumes. The solution reduces plastic waste accumulation in the environment. Produced fuel can be used for industrial• or heating applications. Energy-efficient operation lowers operational costs.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097426 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LASER-BASED PRECISION MEASUREMENT TOOL FOR MECHANICAL ALIGNMENT

(51) International classification	:G01B 11/27, G01B 11/00, G01B 11/26, G01C 15/00, G01B 11/275
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)CH Lakshmi
2)P Balakrishna
3)G Suneetha
4)G Bindu Pavani

(57) Abstract :
A laser,based precision _measurement_tool enables_accurate mechanical alignment of industrial components. A laser source projects a reference beam for alignment measurement. Optical sensors detect deviations between target surfaces and reference points. The system calculates angular and linear misalignment with high precision. Real-time feedback assists in adjusting mechanical components accurately. Digital processing enhances measurement accuracy and reliability. The tool reduces manual measurement errors and setup time. Compact design allows easy portability and on-site use. The system supports alignment of shafts, couplings, and machinery. Visualization displays provide clear alignment indicators and data. Automated calibration ensures consistent measurement performance. Data logging enables tracking of alignment history. The solution improves equipment efficiency and reduces wear. Compatibility with various industrial environments ensures flexible deployment.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097437 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NANO-COATED HEAT RESISTANT MATERIAL FOR INDUSTRIAL MACHINERY

(51) International classification	:B32B 15/04, C23C 4/129, C23C 4/10, C09D 7/61, C09D 5/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)P Balakrishna
2)G Bindu Pavani
3)K.Pravallika
4)Dr S V B Subrahmanyeswara Rao

(57) Abstract :
A nano-coated heat resistant material enhances thermal protection for industrial machinery components. A nanostructured coating is applied to surfaces exposed to high temperatures. The nano-layer acts as a thenrial barrier, reducing heat transfer. Improved heat resistance protects machinery from thermal degradation. The coating maintains mechanical strength under elevated operating conditions. Enhanced resistance to oxidation and thermal fatigue increases component lifespan. The material reduces wear and surface damage caused by high temperatures. Strong adhesion ensures durability under continuous industrial operation. The solution is suitable for engines, turbines, boilers, and processing equipment. Reduced heat loss improves energy efficiency of machinery. The coating is compatible with standard industrial materials and surfaces. Application methods include spraying, dipping, or coating techniques. Lower maintenance requirements reduce operational downtime and costs. Environmental benefits include improved efficiency and reduced energy consumption.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097438 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-LUBRICATING NANO-COATED BEARINGS FOR HIGH-SPEED MACHINES

(51) International classification	:F16C 33/10, F16C 33/66, F16C 33/12, F16C 33/14, F01D 25/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)CH Dharani
2)Dr P Raja Sekhar
3)B Pathrisamma
4)Venu Gopal Mutyala

(57) Abstract :
Self-lubricating nano-coated_bearings improve_performance of-high-speed rotating-machinery. A nanostructured coating is applied to bearing surfaces to reduce friction. The coating provides inherent lubrication without external lubricants. Reduced friction minimizes heat generation during high-speed operation. Enhanced wear resistance increases bearing lifespan. The nanolayer maintains stability under high loads and rotational speeds. Improved thermal properties prevent overheating of machine components. The bearings operate efficiently in environments with limited lubrication. Reduced maintenance requirements lower operational costs. The solution is suitable for motors, turbines, and precision equipment. Strong adhesion ensures durability of the nano-coating. The design supports energy-efficient operation of high-speed machines. Compatibility with existing bearing designs allows easy integration. Environmental benefits include reduced lubricant consumption and waste.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097439 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SOLAR THERMAL HYBRID DRYER FOR AGRICULTURAL PRODUCTS

(51) International classification	:F24S 20/20, F26B 9/06, F26B 21/00, F26B 3/28, F24J 2/04	(71)**Name of Applicant :** **1)RAMACHANDRA COLLEGE OF ENGINEERING** Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING NH-16 BYPASS ROAD, VATLURU (V), ELURU DT, ANDHRA PRADESH-534007 Andhra Pradesh India (72)**Name of Inventor :** **1)G Bindu Pavani** **2)T Vinoditha** **3)CH Lakshmi** **4)G Suneetha**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No	:NA	
Filing Date	:NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

A solar thermal hybrid dryer improves drying efficiency for agricultural products. Solar collectors capture thermal energy to heat drying air. A hybrid backup source ensures continuous operation during low sunlight. Controlled airflow distributes heat uniformly across drying chambers. Temperature and humidity sensors monitor drying conditions in real time. The system reduces moisture content while preserving product quality. Energy-efficient design minimizes dependence on conventional fuels. Improved drying speed reduces post-harvest losses. The dryer supports fruits, vegetables, grains, and herbs. Adjustable trays allow flexible loading capacity. Insulated chambers enhance heat retention and efficiency. The system operates with low maintenance requirements. Reduced drying time improves productivity for farmers. Environmental benefits include renewable energy utilization and reduced emissions.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :12/08/2026

(21) Application No.202641097440 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART GEAR DESIGN OPTIMIZATION USING COMPUTATIONAL MATHEMATICAL ALGORITHMS

(51) International classification	:G06F 17/50, G06F 30/17, F16H 55/17, F16H 55/08, G06F 119/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMACHANDRA COLLEGE OF ENGINEERING
Address of Applicant :RAMACHANDRA COLLEGE OF ENGINEERING, NH-16 BYPASS ROAD, VATLURU (V), ELURU Dt, ANDHRA PRADESH, 534007 Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr B V Manikanta
2)Venu Gopal Mutyala
3)CH Dharani

(57) Abstract :
A smart gear design optimization system improves mechanical performance using computational mathematical algorithms. The method analyzes gear parameters such as tooth profile, module, and pressure angle. Optimization algorithms determine configurations that reduce stress and wear. Computational modeling evaluates load distribution across gear teeth. The system minimizes friction and improves transmission efficiency. Simulation tools assess performance under varying operating conditions. Material properties are incorporated to enhance durability. Automated calculations reduce design time and human error. The optimized design improves noise and vibration characteristics. The system supports spur, helical, and bevel gear configurations. Energy losses in gear transmission are reduced. Iterative optimization refines design accuracy. Integration with CAD tools enables seamless implementation. The solution enhances reliability and lifespan of gear systems.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :19/08/2026

(21) Application No.202641100342 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : "A Precision Intermediate Rotary Endodontic File System for Enhanced Apical Sizing and Obturation"

(51) International classification	:A61C 5/48, A61C 5/46, G06F 16/10, G06F 16/178, G06F 16/182
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sneha Priya K
Address of Applicant :No. 362, 3rd A Main, 1st Stage, 2nd Block, HBR layout, Bangalore Karnataka India
(72)**Name of Inventor :**
1)Sneha Priya K

(57) Abstract :
The present invention relates to a rotary endodontic instrument system comprising intermediate nickel-titanium (NiTi) files designed to enhance apical precision during root canal treatment. Conventional rotary systems utilize standardized ISO tip size increments, which may result in suboptima1 apical fit and require manual adjustment of obturation materials. The disclosed system introduces intermediate rotary file sizes positioned between standard ISO sizes, including but not limited to sizes 26, 31, and 36, configured for use as finishing instruments following conventional canal shaping. These files provide improved incremental enlargement, enabling better apical control and reducing the risk of over-instrumentation. The invention further includes corresponding gutta-percha cones matched to the intermediate file sizes, facilitating enhanced adaptation and minimizing the need for chairside modification. Additionally, a visual identification system incorporating standard color coding with supplementary markings is provided to distinguish intermediate sizes. The proposed system is compatible with existing rotary endodontic workflows and aims to improve obturation quality, clinical efficiency, and treatment outcomes.

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/08/2026

(21) Application No.202641104553 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A NANOEMULSION FORMULATION FOR CURCUMIN PRESERVATION AND A METHOD OF PREPARING THEREOF

(51) International classification	:A61Q 13/00, A23B 11/10, A23B 11/30, A23B 11/40, A23B 11/50
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MANIPAL ACADEMY OF HIGHER EDUCATION
Address of Applicant :MADHAV NAGAR, MANIPAL, KARNATAKA, INDIA-576104 Karnataka India
(72)**Name of Inventor :**
1)JITENDRA CARPENTER
2)ANIRUDDHA BHALCHANDRA PANDIT
3)VIRENDRA KUMAR SAHARAN
4)K VAMSHI KRISHNA GOUD

(57) Abstract :
The present invention relates to a nanoemulsion formulation (100) comprising curcumin (101), a lipid phase (103), and dairy milk (105) forming a continuous aqueous phase, wherein the curcumin (101) is encapsulated within fat globules of the lipid phase (103) dispersed in the dairy milk (105), and wherein casein proteins inherently present in the dairy milk (105) stabilize the fat globules to maintain dispersion stability of the nanoemulsion formulation (100). The present invention further relates to a method (200) of preparing the nanoemulsion formulation (100) comprising dissolving the curcumin (101) in the lipid phase (103) at an elevated temperature to form a curcumin-lipid phase mixture, combining the curcumin-lipid phase mixture with the dairy milk (105) to form a coarse emulsion, and subjecting the coarse emulsion to ultrasonic homogenization to form the nanoemulsion formulation (100). FIG.1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/09/2026
(21) Application No.202641106441 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Green Synthesis of Carbon Quantum Dots for Antibacterial Applications

(51) International classification	:B82Y 15/00, G06N 10/20, G06N 10/60, C07C 51/15, G06N 10/80
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RV College of Engineering
Address of Applicant :Mysore Road, RV Vidyaniketan Post, Bengaluru–560059, Karnataka, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Neeta Shivakumar
2)M S Shritha

(57) Abstract :
Title: Green Synthesis of Carbon Quantum Dots for Antibacterial Applications A carbon quantum dot composition for antibacterial application comprising carbon quantum dots derived from Areca catechu husk through a hydrothermal synthesis process. The carbon quantum dots exhibits a crystallite size of 2–5 nm. The carbon quantum dot exhibits antibacterial activity with a minimum inhibitory concentration of about 22.5 ppm against Bacillus spp. and about 30 ppm against Pseudomonas spp. The carbon quantum dots exhibits a minimum bactericidal concentration of about 30 ppm against Bacillus spp. and Pseudomonas spp. FIG.1A

No. of Pages : 16 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :07/09/2026

(21) Application No.202641107316 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MODULAR SURGICAL INSTRUMENT PLATFORM WITH ORIENTATION-DEPENDENT LOCKING COUPLING MECHANISM

(51) International classification	:A61B 19/00, A61B 17/00, A61B 90/50, A61B 17/16, A61B 90/00	(71)**Name of Applicant :** **1)MANIPAL ACADEMY OF HIGHER EDUCATION** Address of Applicant :MADHAV NAGAR, MANIPAL, KARNATAKA, INDIA-576104 Karnataka India (72)**Name of Inventor :** **1)SAHAJANYA SRIKANTH** **2)JANAKI NEELIYATH OTHAYOTH** **3)LAKSHMI NEELIYATH OTHAYOTH**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a modular surgical platform and a method of coupling an insert module thereto. The platform comprises a base platform (101) comprising a central coupling hub (103), a locking element (111) biased toward the central coupling hub (103) by a compression spring (131), a lock indicator (105), and an insert module (115) comprising a coupling collar (117) having an indexing key feature (119) and a locking recess (123). The locking recess (123) receives the locking element (111) only after the indexing key feature (119) has first engaged an indexing slot (125). The method comprises positioning, aligning, inserting, advancing, biasing, and displaying confirmation of the locking engagement. FIG.1B

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :08/09/2026
(21) Application No.202641108112 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WOODEN EYEWEAR FRAME WITH AN INTEGRATED CONTINUOUS CURVE

(51) International classification	:G02C 5/22, G02C 5/14, G02C 11/00, G02C 5/20, G02C 1/04	(71)**Name of Applicant :** **1)HATCHPALETTE VENTURES PRIVATE LIMITED** Address of Applicant :BUILDING NO. 552/R/10, CRESENT VILLA(H) ONCHIYAM VILLAGE, NADAPURAM ROAD, MADAPALLY COLLEGE PO, VDAKARA, KOZHIKODE Kozhikode Kerala India (72)**Name of Inventor :** **1)AKHIL PRAKASH**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT WOODEN EYEWEAR FRAME WITH AN INTEGRATED CONTINUOUS CURVE The present invention relates to an eyewear frame (100) comprising a front portion (1) formed entirely from a single continuous wood blank. The front portion has a pair of eye openings (3), a bridge (4) connecting the eye openings (3), and a curved region (6) extending rearwardly and continuously from a rim portion of the front portion (1) to a hinge connection point, wherein the natural grain of the wood extends continuously through the curved region (6) without interruption by a glue line, a kerf, or a mechanical joint. A pair of temple portions (2), each shaped from a single piece of wood, are pivotally connected to the curved regions (6) of the front portion (1) using mechanical hinges (5). The frame is produced through a progressive carving and shaping process from solid wood blanks, eliminating separate attached end pieces while maintaining structural integrity and a compact overall profile.

No. of Pages : 13 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :24/09/2026

(21) Application No.202641114431 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR CONTROLLING LASER SCANNING SUBSYSTEMS IN A MULTI-LASER CONTROL SYSTEM

(51) International classification	:G02B26/08, G05D3/00, G06F3/00, G11B20/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SOCIETY FOR INNOVATION & DEVELOPMENT
Address of Applicant :Innovation Centre, Indian Institute of Science Campus, Bengaluru 560012, Karnataka, India Bengaluru Karnataka India
2)Ministry of Heavy Industries
(72)**Name of Inventor :**
1)KUMAR S CHANDRA, Shishir
2)S, Bharath
3)DAGALE, Haresh
4)B, Saumya
5)K H, Thulasi Raman

(57) Abstract :
Disclosed herein is a system (110) and a method (800) for controlling parameters of plurality of laser scanning subsystems in a multi-laser control system. The method (800) includes receiving input command data (302) comprising a pattern to be formed, coordinate information and laser control information for the lasers (104). The method (800) includes extracting coordinate information and laser control information from the input command data (302) for each laser (104). Further, the method (800) includes generating a laser actuation signal for each laser (104) based on the extracted laser control information and generating an updated coordinate value by selectively applying an optical distortion compensation transformation to the extracted coordinate information. Additionally, the method (800) includes controlling the parameters of the lasers (104) based on the laser actuation signal and updated coordinate value.

No. of Pages : 47 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/09/2026

(21) Application No.202641116332 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Hydraulically Pressed Hybrid Composite Interlocking Masonry Block with Cementitious Protective Layers and a Thermally Insulating Core

(51) International classification	:E04C 1/41, E04C 1/40, E04B 2/02, E04C 1/00, B28B 3/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Jobin P James
Address of Applicant :Padinjarekkara House , Karikkode ,Okkal P. O Kerala India
(72)**Name of Inventor :**
1)Jobin P James

(57) Abstract :
Abstract [0001] The present invention relates to a hydraulically pressed hybrid composite interlocking masonry block comprising a thermally insulating inner core and cementitious protective layers provided on upper and lower surfaces of the block. The block is formed by hydraulic pressing to obtain a compact and dimensionally stable structure and comprises complementary interlocking projections and recesses configured to mechanically engage adjacent blocks during construction. The cementitious protective layers comprise cement and sand and provide enhanced resistance to water absorption, mechanical damage and environmental exposure, while the inner core provides thermal insulation and reduces the quantity of cementitious material required for forming the block. In a preferred embodiment, the composite block comprises approximately 10% cement, 20% sand and 70% core material, with the cementitious protective layers having a thickness of approximately 15–20 mm. The core material may comprise cellular cementitious material, soil, agricultural waste-derived material, industrial residue-derived material, mineral-based material or combinations thereof. Hydraulic pressing provides compaction and structural integrity, while the interlocking configuration facilitates alignment and mechanical engagement between adjacent blocks and can reduce dependence on conventional joining materials. The composite block exhibits a compressive strength greater than 8 MPa and may have a substantially smooth cementitious surface suitable for direct painting, thereby reducing or eliminating conventional plaster and putty application. The hybrid construction combines the water-resistant and protective characteristics of the cementitious layers with the thermal-insulating characteristics of the inner core, while enabling utilization of locally available and waste-derived materials to reduce cement consumption and construction material cost.

No. of Pages : 17 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/09/2026
(21) Application No.202641117168 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : "MXene-Reinforced UV-Resistant Crosslinked Polyethylene (XLPE) Insulation Material for Outdoor High Voltage Electrical Applications"

(51) International classification	:H01B 3/44, C08K 3/14, C08L 23/06, C08J 3/24, C08K 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CENTRAL POWER RESEARCH INSTITUTE
Address of Applicant :Central Power Research Institute, Materials Technology Division, Prof. C.V.Raman Road, PB No 8066, Bangalore- 560080, Karnataka, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. Moumita Naskar
2)Ankita Deb

(57) Abstract :
ABSTRACT MXene-Reinforced UV-Resistant Crosslinked Polyethylene (XLPE) Insulation Material for Outdoor High Voltage Electrical Applications The present invention relates to a novel ultraviolet (UV)-resistant crosslinked polyethylene (XLPE) nanocomposite insulation material reinforced with an optimized concentration (0.5 wt%) of two-dimensional $Ti_3C_2T_x$ MXene nanosheets for outdoor electrical insulation applications, including covered conductors, aerial bundled cables, power cables, cable accessories and other outdoor electrical insulation systems. The nanocomposite is prepared by melt compounding followed by peroxide crosslinking, resulting in uniform dispersion of the MXene nanosheets throughout the XLPE matrix. Unlike conventional carbon black-filled XLPE, the developed MXene-reinforced XLPE exhibits superior resistance to UV-induced degradation while maintaining excellent electrical insulation and mechanical properties after accelerated ultraviolet ageing. The layered structure of $Ti_3C_2T_x$ MXene effectively attenuates incident ultraviolet radiation and acts as a diffusion barrier against oxygen, thereby reducing photo-oxidative degradation of the polymer matrix. {FIGURE 1}

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/09/2026

(21) Application No.202641117272 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED ARRANGEMENT FOR MULTI-AXIS ALIGNMENT OF MULTIPLE SOURCES WITH RESPECT TO AN AFM PROBE

(51) International classification	:G01Q 60/24, G01Q 60/38, G01Q 10/04, G01Q 10/06, G01Q 20/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MAKR MICROSYSTEMS PRIVATE LIMITED
Address of Applicant :A003, Vajram Tiara, Doddaballapur Main Road, Yelahanka, Bangalore North-560064, Karnataka, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dorbala Sai Naga Bharghava
2)Mohanasundaram S M
3)Sai Shiva Teja Panchadhara
4)Ashwin Lal

(57) Abstract :
AN INTEGRATED ARRANGEMENT FOR MULTI-AXIS ALIGNMENT OF MULTIPLE SOURCES WITH RESPECT TO AN AFM PROBE The invention provides an integrated optical arrangement for multi-axis alignment of an optical beam with a cantilever probe in an atomic force microscope (AFM). The integrated arrangement 1 includes a solid housing 2 having at least two hollow positioning frames, each configured to accommodate a light source and a photosensitive detector, respectively. Each positioning frame is independently movable along substantially perpendicular X-axis and Y-axis directions by corresponding actuators 11 and integrated flexure members 7. The flexure members 7 provide elastic restoring force and substantially linear movement while reducing backlash, mechanical play, sliding friction and parasitic angular movement. The light source is configured to direct an incident optical beam onto the cantilever probe, while the photosensitive detector is configured to receive a reflected optical beam corresponding to cantilever deflection. The monolithic unibody arrangement optionally includes an optical slot 9 for viewing the cantilever probe. Fig. 1

No. of Pages : 20 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/10/2026

(21) Application No.202641117796 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR GENERATING LOCATION-ADAPTIVE LANGUAGE-LEARNING CONTENT BASED ON GEOGRAPHIC CONTEXT AND LEARNER PROFICIENCY

(51) International classification	:G09B 19/06, G09B 5/06, G06Q 50/20, G06F 17/30, G09B 7/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Balamurugan S
Address of Applicant :2/32 Thamaraipulam West Kallimedu - Post Vedharanyam - Taluk Nagappattinam - 614809 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. S. Balamurugan
2)Dr. D. Pradeek
3)Dr. T. Sriram
4)Dr. Shankarraman C
5)Dr. V. Siva Guru Vignesh

(57) Abstract :
A computer-based system and method for generating and delivering language-learning content that is adaptive based on both the learner's real-time geographic location and personal proficiency profile are disclosed. The system classifies the learner's real-time location as a semantic category using a predefined classification system; and retrieves a learner proficiency profile that includes a vocabulary-response record and an error-pattern record. Candidate content items stored in a repository and tagged with location categories and difficulty levels are scored and filtered as a function of both the classified location category and proficiency profile; thereby, content that addresses the learner's recorded error patterns within the learner's current environmental context is preferentially revealed. Selected content is further adapted in linguistic complexity and presented through text, augmented-reality overlay, or audio narration. The learner's interaction with the presented content is captured and used to update the proficiency profile; thereby establishing a closed feedback loop that progressively improves content relevance. (Figure 1)

No. of Pages : 8 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/10/2026

(21) Application No.202641117798 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR RESOLVING FICTIONAL PLACE NAMES IN LITERARY TEXT TO REAL-WORLD GEOGRAPHIC COORDINATES

(51) International classification	:G06F 17/30, G06F 17/27, G06F 17/00, G06F 7/00, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Balamurugan S
Address of Applicant :2/32 Thamaraipulam West Kallimedu - Post Vedharanyam - Taluk Nagappattinam - 614809 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. S. Balamurugan
2)Dr. S. R. Karpagam
3)Dr. R. Malathi
4)Dr. R. Rohini
5)Dr. T. Sathyaseelan

(57) Abstract :
A computer-based system and method for resolving fictional, renamed, or ambiguous place names appearing in literary text to real-world geographical axioms is disclosed. Using name-identity recognition, the system detects candidate place name expressions in narrative text; and extracts from the text surrounding each expression a contextual reference matrix containing references to relative direction, distance or travel-time, terrain, climate, and historical or cultural cross-references. Candidate real-world locations are retrieved from a standardized list and historical map database; and a resolution engine calculates the degree of similarity between the contextual reference matrix and the geographical specification of each candidate. Candidates are ranked and assigned a credibility score; and the highest-ranked candidate, along with supporting text evidence, is published as the resolved geographical axiom; This enables applications such as literary cartography, curriculum development, and place-based literary tourism. (Figure 1)

No. of Pages : 7 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :09/10/2026
(21) Application No.202641121407 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INFRARED HEATING SYSTEM FOR PREHEATING A POLYMER POWDER BED

(51) International classification	:B29C 64/153, B29C 64/295, B29C 64/255, B29C 64/209, B29C 64/35	(71)**Name of Applicant :** **1)Society for Innovation & Development** Address of Applicant :Innovation Centre, Indian Institute of Science Campus, Bengaluru 560012, Karnataka, India Bengaluru Karnataka India **2)Ministry of Heavy Industries** (72)**Name of Inventor :** **1)KH, Thulasi Raman** **2)DIWAKAR, J.E** **3)DAGALE, Haresh** **4)CHATTERJEE, Kaushik**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT OF THE INVENTION AN INFRARED HEATING SYSTEM FOR PREHEATING A POLYMER POWDER BED An infrared heating system (100) for preheating a polymer powder bed (104) in an additive manufacturing apparatus may be disclosed. The infrared heating system (100) may include a mounting plate (302) with a central opening (204) positioned over a powder bed (104) and a plurality of infrared emitters (102) mounted on the mounting plate (302) along a perimeter surrounding the central opening (204). The plurality of infrared emitters (102) may be adapted to irradiate the polymer powder bed (104) by infrared radiation. Each infrared emitter (102) may include a carbon fibre filament enclosed within a transparent quartz tube (702) to redirect the infrared radiation downward toward the polymer powder bed (104). The infrared heating system (100) may further include a mounting fixture (106) coupled to the mounting plate (302). The infrared heating system (100) may further include a thermal regulation subsystem (400) operatively coupled to the plurality of infrared emitters (102).

No. of Pages : 51 No. of Claims : 19

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :13/10/2026

(21) Application No.202641122517 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AZURE CLOUD BASED EDUTECH TUTOR APP

(51) International classification	:G06Q 50/20, G06N 20/00, G09B 5/14, G09B 5/08, G09B 7/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
Address of Applicant :SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES, SAVEETHA NAGAR, THANDALAM,CHENNAI, TAMIL NADU, INDIA-602105 Tamil Nadu India
(72)**Name of Inventor :**
1)Juthvesh Chadalavada
2)Dr.A.Rama
3)Dr.R.Karthik Raja
4)Dr Ramya Mohan

(57) Abstract :
In an era where digital learning is rapidly expanding and students require personalized academic support, the demand for intelligent, scalable, and cloud-powered tutoring systems has become increasingly significant. The Azure Cloud-Based EdUTech Tutor App is a smart educational platform developed to deliver personalized learning experiences through artificial intelligence and Microsoft Azure Cloud infrastructure.The application enables students to access structured course materials, interactive assessments, and real-time AT tutoring assistance. By analyzing student performance data, the system generates adaptive learning paths, identifies academic strengths and weaknesses, and provides targeted improvement recommendations. The platform also offers performance analytics dashboards, secure cloud-based data storage, and intelligent chatbot-based doubt clarification.

No. of Pages : 6 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/10/2026

(21) Application No.202641122866 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART VANILA FLOWER DETECTION AND MOBILE NOTIFICATION SYSTEM FOR TIMELY POLLINATION

(51) International classification	:A01H 1/02, G06K 9/00, G06T 7/00, G06K 9/62, G06K 9/46
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JEPPIAAR INSTITUTE OF TECHNOLOGY (JIT)
Address of Applicant :PRINCIPAL, JEPPIAAR INSTITUTE OF TECHNOLOGY (JIT), KUNNAM, SUNGUVARCHATRAM, KANCHIPURAM, SRIPERUMBUDUR, TAMIL NADU, INDIA-631604 Tamil Nadu India
(72)**Name of Inventor :**
1)Aadhira N
2)Bavadharini M A
3)Dr.D.Paulraj
4)S.Parthasarathi
5)K.Seejamini
6)Arun K
7)Ragavan P
8)Nandetha S
9)Deepika M
10)SANJITH S
11)Velvizhi K
12)Shree Tharani R
13)Renuga T
14)Rithika L

(57) Abstract :
ABSTRACT OF THE INVENTION Vanilla cultivation depends strongly on timely pollination because the flowers remain receptive for a short period. Manual identification of newly opened and pollination-ready flowers can be difficult when many plants are present. The proposed Smart Vanilla Flower Detection and Mobile Notification System uses a camera and artificial intelligence to identify vanilla flowers and determine their readiness for pollination. YOLO-based object detection and image-processing techniques are used to analyse captured images. When a flower is identified as ready, the system sends a notification to the farmer's mobile phone. The farmer then reaches the plant, verifies the flower, and performs manual pollination. Thus, the system supports the farmer without using a robotic arm or automated pollination mechanism.

No. of Pages : 8 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/10/2026

(21) Application No.202641122971 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RES-NET: AN ESP32-LORA-BASED OFFLINE AI SYSTEM FOR DISASTER MONITORING, RISK ASSESSMENT, EARLY WARNING AND EMERGENCY COMMUNICATION

(51) International classification	:G08B 21/10, H04W 84/18, H04W 4/38, H04W 4/90, G08B 17/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. S . NITHYA
2)S.P. LAVANYA
3)HARSHITHA S
4)DINESH BALAJI D

(57) Abstract :
RES-NET: An ESP32-LoRa-based offline Al system for disaster monitoring, risk assessment, early warnings, and emergency communication. RES-NET (Resilient Edge-Al Environmental Intelligence and Early-Warning Network) is an ESP32-S3 and LoRa-based disaster monitoring system designed to detect hazards and deliver warnings even when Internet or cellular networks are unavailable. Distributed multi-sensor nodes monitor conditions associated with fire, hazardous gas leakage, flooding, smoke, temperature changes. The ESP32-S3 processes sensor readings through validation, filtering, preprocessing and feature extraction. An Edge-Al classification mechanism identifies the hazard and a risk assessment module calculates a severity score to determine normal, warning or critical conditions. GPS associates each event with geographical coordinates, while LoRa transmits hazard details, risk score and location to a local gateway. Audible, visual or electronic alerts can notify users and responsible personnel. When connectivity is available, event data can be displayed through a dashboard or mobile application. By combining local intelligence, longrange communication and distributed monitoring, RES-NET supports scalable early warning and emergency response in remote, industrial and disaster-prone environments

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/10/2026

(21) Application No.202641122972 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TWIN FORGE AI INTELLIGENT DIGITAL TWIN PLATFORM SYSTEM AND METHOD FOR MULTOMODAL MANUFACTURING

(51) International classification	:G06Q 10/06, G06F 19/00, G05B 19/418, G06Q 50/04, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA College of Engineering and Technology,Dindigul Tamil Nadu 624622 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.P. ANITHA CHRISTY ANGELIN
2)V.S Kabilan
3)K Santhosh Kumar
4)G.S Santhosh Kumaar

(57) Abstract :
A manufacturing-memory and quality-assurance system is disclosed for capturing how accepted products are made and reusing that process knowledge during later production. A plurality of acquisition modules record an ordered manufacturing-step sequence, step timing and multiple process-data strands including vibration, temperature, mass, visual and optionally acoustic data. The data are synchronized to process steps and linked to quality outcomes. Accepted manufacturing cycles are used to generate a product-specific multimodal reference profile, referred to as a Golden Manufacturing DNA. During a subsequent build, live data are aligned and compared with the reference to produce strand-level deviation values and a fused risk indication. A localization module identifies the step and data strand responsible for a deviation and can alert the worker before completion of the unit. Reference profiles may be versioned under controlled approval, and each manufactured unit may retain a process trace for training, root-cause analysis and traceability.

No. of Pages : 30 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/10/2026

(21) Application No.202641122974 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTONOMOUS MEDICATION DISPERSION WITH WEARABLE HEALTH MONITORING SYSTEM

(51) International classification	:A61B 5/00, A61J 7/04, G16H 40/67, A61B 5/0205, G16H 20/13	(71)**Name of Applicant :** **1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY** Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India (72)**Name of Inventor :** **1)R. AMBIKA** **2)P. L. RAMANATHAN** **3)G. KRISHNA KUMAR** **4)S. VIKASH** **5)B. LOGESH** **6)K. SELVANAMBI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT OF THE INVENTION SMART HEALTH COMPANION FOR INTEGRATED HEALTH MONITORING AND MEDICATION ADHERENCE Smart Health Companion system developed to monitor a person's health and support regular medication use. The system uses an ESP32-based processing unit along with health and activity sensors to collect information such as heart rate, blood oxygen level, blood pressure, stress level, and physical activity. A non-contact millimeter-wave radar sensor is used to detect the presence, movement, and activities of the person without physical contact. The system also uses a radio-frequency identification (RFID) system to identify the prescribed medicine or medicine container and check the medicine event against the set medication schedule. An automatic servo-based medication assistance mechanism helps to dispense the medicine at the scheduled time. A display unit provides medicine reminders, health information, and alerts. The data analyze module is to analyzes the collected health and activity information and identifies unusual changes & patterns that may require attention. The system also provides a monitoring interface through which the person or an authorized caregiver can view health information, activity status, medication status, medication adherence, and health alerts. By combining health monitoring, contactless activity monitoring, medicine identification, medication assistance, and predictive analysis in one system, Our system provides continuous support for health monitoring and medication adherence, particularly for elderly persons and individuals who need assistance or remote monitoring at home.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :14/10/2026
(21) Application No.202641122975 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : RUBBERSENSE AI: AI-ENABLED SMART SMOKE-HOUSE AUTOMATION SYSTEM FOR RUBBER SHEET PROCESSING

(51) International classification	:G05B 15/02, G05B 19/418, G05B 19/042, A62C 2/24, F24F 11/00	(71)**Name of Applicant :** **1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY** Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India (72)**Name of Inventor :** **1)DHARSHANA DEEPTHI L** **2)NAGASUDHAN T** **3)PANDI ISHWARYA T**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
RUBBERSENSE At: AI-ENABLED SMART SMOKE-HOUSE AUTOMATION SYSTEM FOR RUBBER SHEET PROCESSING An automatic control and monitoring system for rubber smoke-houses is provided, comprising a data-processing and control mechanism based on Internet-of-Things and Artificial Intelligence technologies, along with environmental sensors including temperature, humidity, smoke-density, and airflow sensors that collect data on a real-time, batch-wise basis. The collected data is used to train AI/ML models that identify optimal processing conditions for each stage of smoking and drying. Based on the resulting predictions, the system automatically regulates the smoke-house environment through actuators including a burner, exhaust fan, and air damper, using relay and motor-driven switching elements, without requiring continuous manual supervision. The system further comprises a wireless interface for remote monitoring, batch-record storage, and alert generation, together with a display unit for presenting real-time process data, thereby enabling a low-cost, retrofit-friendly automation solution for improving rubber-sheet quality consistency, fuel efficiency, and process traceability in MSME rubber-processing operations.6 MAIL: ipr@p

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/10/2026

(21) Application No.202641122976 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : M-FUSE: PORTABLE MACHINE HEALTH MONITORING DEVICE

(51) International classification	:A01G 25/16, G01H 1/00, G01M 13/00, G08C 17/02, G01M 13/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India
(72)**Name of Inventor :**
1)J. SANGEETHA
2)DR.N.PANDEESWARI
3)A.GEETHA
4)KAMAL P
5)MADHUMITHA J
6)KEERTI VASAN V

(57) Abstract :
The present invention relates to a portable machine-health monitoring device configured as a removable slap-band structure for monitoring the operational condition of machinery, particularly stationary electric motors and irrigation pump systems. The device comprises a flexible slap-band body having a rigid sensor island, vibration sensing unit, machine-temperature sensing unit, ambient-temperature sensing unit, magnetic-field sensing unit, active attachment-verification mechanism, processing unit, memory unit, power source, status indicators and wireless communication interface.A distinguishing feature of the invention is the active attachmentverification mechanism, wherein a miniature mechanical or piezoelectric excitation actuator generates a controlled mechanical excitation and a vibration sensing unit measures the resulting response to determine whether the device is sufficiently coupled to the machine before acquired vibration data is relied upon for machine-health assessment. This allows the device to distinguish poor attachment from actual machine-related abnormal behavior. improving

No. of Pages : 29 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :14/10/2026

(21) Application No.202641122988 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FABRICATION AND CHARACTERIZATION OF DUPLEX YSZ/PVA-BIOACTIVE GLASS COATINGS ON MEDICAL GRADE AZ31B Mg ALLOYS FOR ORTHOPEDIC IMPLANT APPLICATIONS

(51) International classification	:A61L 27/04, A61L 27/30, A61L 31/02, A61L 27/34, A61F 2/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAJALAKSHMI ENGINEERING COLLEGE
Address of Applicant :RAJALAKSHMI ENGINEERING COLLEGE, RAJALAKSHMI NAGAR, THANDALAM, CHENNAI, TAMIL NADU, INDIA-602105 Tamil Nadu India
(72)**Name of Inventor :**
1)DR. K. SEETHALAKSHMI
2)P. SARALA
3)DR.S. MOHANDOSS

(57) Abstract :
ABSTRACT OF THE INVENTION: Magnesium alloys are promising materials for temporary orthopedic implants because their density (1.74.- 2.0 g/cm3) and elastic modulus (41-45 GPa) closely match those of natural bone, minimizing stress-shielding effects compared to conventional titanium or stainless-steel implants. To overcome this problem, a duplex coating was developed on medical-grade AZ31B magnesium alloy using two coating techniques. First, yttriastabilised zirconia (YSZ) ceramic layer was deposited by electrophoretic deposition (EPD) to improve corrosion protection and mechanical stability. Then, a polyvinyl alcohol (PVA) and bioactive glass (BG) composite layer was applied over the YSZ layer by dip coating. The YSZ layer acts as a protective bather, while PVA improves coating flexibility and adhesion, and bioactive glass provides bioactivity for bone integration. The coated samples will be characterised using SEM, EDS, XRD and FTIR to study their .morphology, composition, phase structure and chemical bonding. Corrosion resistance, coating adhesion, surface wettability and bioactivity in vitro will also be assessed. The developed duplex YSZ/PVA&#xFFFD;BG coating is intended to provide better corrosion protection, surface stability and bioactive performance for biodegradable AZ31B magnesium alloy used in orthopaedic implants.

No. of Pages : 4 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :15/10/2026

(21) Application No.202641123504 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ADAPTIVE TWO STAGE RETRIEVAL AUGMENTED GENERATION IN AUTOMATED INCIDENT RESPON

(51) International classification	:G06Q 10/06, G06Q 10/10, G06Q 10/00, G06N 5/04, G06F 17/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)THE PRINCIPAL, SNS COLLEGE OF TECHNOLOGY
Address of Applicant :SNS KALVI NAGAR, SATHY MAIN ROAD, NH-209, VAZHIYAMPALAYAM, SARAVANAMPATTI POST, COIMBATORE, TAMIL NADU, INDIA-641035 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.M.Shobana
2)Dr.A.Sumithra
3)Dr.B.Vinodhini
4)S.Priyadarsini
5)Jeremiah Paul
6)R.Karthik
7)A.M.Kannan
8)R.Jashwant

(57) Abstract :
ABSTRACT: The present invention relates to an Al-powered Digital Business Twin system designed to __assistsmall and-medium-sized-enterprises-in-rrialiagin-g□their bu□siness operations through a unified digital platform. The system collects information from different business activities, including customer management, communication, tasks, projects, accounting, documents, and other operational sources. The collected information is processed, organized, and stored in a centralized business environment. Artificial intelligence is applied to analyze the available information, identify business trends and operational patterns, detect potential issues, and generate business summaries, recommendations, and predictive information. The system also supports automation of selected routine activities, task assignments, notifications, and predefined workflows. A unified dashboard provides users with business information, performance details, insights, alerts, and decision-support information. By combining business data integration, Al-based analysis, predictive analytics, workflow automation, and decision support in a single platform, the system assists SMEs in reducing fragmented information, minimizing repetitive work, monitoring business operations, and supporting informed business management.

No. of Pages : 9 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/10/2026

(21) Application No.202641124343 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A REACTOR SYSTEM FOR OZONE-BASED WASTEWATER TREATMENT WITH ZERO-LIQUID DISCHARGE

(51) International classification	:C02F 1/78, C02F 1/72, C02F 9/00, C02F 3/12, C02F 101/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BEEMARAO THANGAIYAN
Address of Applicant :1/67, EAST STREET, THIRUVALI VILLAGE, SIRKALI TALUKA, MAYILADUTHURAI DISTRICT, TAMIL NADU, INDIA MAYILADUTHURAI Tamil Nadu India
(72)**Name of Inventor :**
1)BEEMARAO THANGAIYAN

(57) Abstract :
A reactor system for ozone-based wastewater treatment with zero-liquid discharge is disclosed. The reactor system comprises continuous stirred tank reactor(CSTR) vessels among other modules. CSTR vessels comprises: a)first CSTR vessel for receiving wastewater influent from and exposing wastewater influent to high-energy radiation, ozone, and purified oxygen to initiate molecular decomposition of wastewater contaminants; b)second CSTR vessel comprises enhanced oxidation of decomposed contaminants received from first CSTR vessel; c)third CSTR vessel performs advanced molecular decomposition of residual contaminants received from second CSTR vessel; d)fourth CSTR vessel performs continued oxidation of residual contaminants received from third CSTR vessel; e)fifth CSTR vessel performs converting remaining molecular fragments from fourth CSTR vessel into simpler oxide forms; f)sixth CSTR vessel performs final oxidation of residual contaminants from fifth CSTR vessel; g)seventh CSTR vessel performs removal of trace contaminants from sixth CSTR vessel; h)eighth CSTR vessel performs stabilizing reactive species. The vapours are collected and condensed. FIG. 1

No. of Pages : 32 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/10/2026

(21) Application No.202641124432 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ADAPTIVE ACOUSTIC OUTPUT MANAGEMENT IN A LISTENING ENVIRONMENT

(51) International classification	:H04S 7/00, H04R 3/00, H04R 5/04, H04S 1/00, H04R 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Circuit House Technologies Private Limited
Address of Applicant :2nd Floor, Sai Sharan Arcade, 2616, 27th Main Road, 1st Sector, HSR Layout, Bengaluru, Karnataka – 560102 India Bengaluru Karnataka India

(72)**Name of Inventor :**
1)Vaibhav Gupta
2)Manojit Ballav

(57) Abstract :

SYSTEM AND METHOD FOR ADAPTIVE ACOUSTIC OUTPUT MANAGEMENT IN A LISTENING ENVIRONMENT ABSTRACT A system (110) and method for adaptive acoustic output management in a listening environment are disclosed. The method comprises receiving audio data and listening environment characteristics; establishing an Acoustic Comfort Budget based on at least one of the listening environment characteristics, audio data, acoustic measurements, listener-related information, content characteristics, device characteristics, or historical listener preferences, the Acoustic Comfort Budget representing a maximum preferred aggregate acoustic output; determining acoustic contributions of each of a plurality of audio output devices (108A, 108B, 108C, …, 108N) to an aggregate acoustic output; and determining respective output allocations based on the respective acoustic contributions such that the resulting aggregate acoustic output is maintained within the Acoustic Comfort Budget. The method may further comprise content-aware allocation and dynamically updating the output allocations during audio reproduction. Figure 6

No. of Pages : 52 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :16/10/2026

(21) Application No.202641124433 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR PERSISTENT ACOUSTIC INTELLIGENCE MANAGEMENT AND CROSS-DEVICE SYNCHRONIZATION

(51) International classification	:G06F 3/16, H04S 7/00, H04R 3/12, H04R 3/00, H04N 21/43
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Circuit House Technologies Private Limited
Address of Applicant :2nd Floor, Sai Sharan Arcade, 2616, 27th Main Road, 1st Sector, HSR Layout, Bengaluru, Karnataka – 560102 India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Vaibhav Gupta
2)Manojit Ballav

(57) Abstract :
SYSTEM AND METHOD FOR PERSISTENT ACOUSTIC INTELLIGENCE MANAGEMENT AND CROSS-DEVICE SYNCHRONIZATION ABSTRACT A system (110) and method for persistent acoustic intelligence management and cross-device synchronization are disclosed. The method includes obtaining acoustic observations during a first playback session involving a first playback system (102) and an acoustic environment, generating a first acoustic profile based on the acoustic observations, and persistently storing the first acoustic profile in a non-volatile memory of a wireless audio device (106) in association with a first playback-device identifier and a first acoustic-environment identifier. The method further includes maintaining a plurality of acoustic profiles, each associated with a playback-device identifier, an acoustic-environment identifier, a profile version, and a confidence value. In response to a subsequent connection with a second playback system (112), a candidate acoustic profile corresponding to the second playback system (112) and an acoustic environment is identified and provided from the non-volatile memory to the second playback system (112) to initialize audio optimization. Figure 7

No. of Pages : 57 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/10/2026

(21) Application No.202641125728 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN AND DEVELOPMENT OF AN AI AND IOT ENABLED SMART BATTERY MANAGEMENT AND EV LIFECYCLE INTELLIGENCE PLATFORM

(51) International classification	:G01R 31/36, G01R 31/392, H02J 7/00, H01M 10/48, G01R 31/367
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. VANKADARA SAMPATH KUMAR
Address of Applicant :ASSOCIATE PROFESSOR, ELECTRICAL AND ELECTRONICS ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L&T BY-PASS, CHINNIYAMPALAYAM POST, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)S. ARSHAD IBRAHIM
3)R. MADESH
4)J. MITHUN
5)C. GOWTHAM
6)M. JANANI
7)Dr. G. SUNDAR
(72)**Name of Inventor :**
1)DR. VANKADARA SAMPATH KUMAR
2)S. ARSHAD IBRAHIM
3)R. MADESH
4)J. MITHUN
5)C. GOWTHAM
6)M. JANANI
7)Dr. G. SUNDAR

(57) Abstract :
The proposed EVOS ◆ Al and IoT Enabled Smart Battery Management and EV Lifecycle Intelligence Platform consists of a smart battery management unit and a cloud-based monitoring platform. The smart BMS unit uses an ESP32 microcontroller to continuously monitor important battery parameters such as voltage, current, temperature, and operating status. The collected battery data is transmitted through Wi-Fl using the MOTT communication protocol to the backend platform for real-time monitoring and storage. The system provides battery condition monitoring, safety alerts, maintenance tracking, battery history, and lifecycle information through a web-based interface. Al-based analysis is incorporated to support battery health assessment, anomaly identification, predictive maintenance, and remaining useful life estimation. This enables continuous observation of battery performance, improves safety and reliability, and supports datadriven battery lifecycle management for electric vehicle applications.◆◆t,'◆/◆◆

No. of Pages : 10 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/10/2026

(21) Application No.202641125765 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MODULAR AUTONOMOUS MOBILE ROBOT FOR SMART WAREHOUSE AUTOMATION

(51) International classification	:B65G 1/04, B65G 1/137, G05D 1/02, B25J 9/16, B25J 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SARANATHAN COLLEGE OF ENGINEERING
Address of Applicant :VENKATESWARA NAGAR, PANJAPPUR, THIRUCHIRAPPALLI, TAMIL NADU, INDIA-620012 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. M. Shanmugavalli
2)S. Ahamed Niyash
3)SU. Amirtha Gown
4)K. Jeganathan
5)N. Karthikram
6)F. Shirly Siforal
7)K. Thanusri Nachiya

(57) Abstract :
The present invention relates to a modular autonomous mobile robot platform intended for smart warehouse automation. The system comprises a mobile robotic base having a drive system, a controller, navigation and distance sensing units, a power supply, and a standardized modular interface configured to receive interchangeable task-specific attachments. The modular interface allows the same autonomous mobile base to perform different warehouse operations by attaehing modules such as bin handling, pallet handling, conveyor transfer, picking, scanning, or inspection attachments. The controller receives information from the sensors and controls the drive motors according to the selected operating mode and attached module. The proposed configuration reduces the requirement for separate dedicated mobile robots for different material-handling operations and provides a scalable platform for warehouse automation. The system emphasizes modularity, ease of maintenance, reconfigurability, low-cost implementation, and adaptability to different warehouse tasks

No. of Pages : 8 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/10/2026

(21) Application No.202641125886 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WIRELESS SENSOR BASED INTELLIGENT TRAFFIC MONITORING AND EMERGENCY ALERT SYSTEM

(51) International classification	:G08G 1/01, G08G 1/017, G08G 1/0967, G08G 1/00, G08G 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No	:NA
Filing Date	:NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number	:NA
Filing Date	:NA
(62) Divisional to Application Number	:NA
Filing Date	:NA

(71)**Name of Applicant :**
1)Durai Chinnappan J
Address of Applicant :Assistant Professor Department of Information Technology, IndraGanesan College of Engineering, Timchirappalli -620 012, Tamilnadu, India Tamil Nadu India
2)Uma R
3)M. Banupriya
4)A Syed Mazhar
5)G Raja Ramesh
6)M.Saranya
7)Dr. S Rajeswari
8)Dr.M.Baritha Begumn
(72)**Name of Inventor :**
1)Durai Chinnappan J
2)Uma R
3)M. Banupriya
4)A Syed Mazhar
5)G Raja Ramesh
6)M.Saranya
7)Dr. S Rajeswari
8)Dr.M.Baritha Begumn

(57) Abstract :
The present invention provides a wireless sensor based intelligent traffic monitoring and emergency alert system comprising a plurality of wireless sensor nodes having vehicle detection sensors a microcontroller/edge processing unit wireless communication module gateway unit traffic density calculation module traffic classification module emergency event detection module location module emergency alert generation module central server/cloud platform and monitoring dashboard The wireless sensor nodes acquire vehicle and traffic information from a monitored roadway and transmit the information to the gateway for processing. The system determines vehicle count, traffic density and congestion level and identifies abnormal conditions including sudden vehicle stoppage and emergency events. Upon validation of an emergency event, the system determines the event location and automatically generates an alert containing event type, location, timestamp and traffic status. The alert is transmitted to a remote monitoring platform and predefined emergency recipients through a wireless communication network.

No. of Pages : 8 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :22/10/2026

(21) Application No.202641125935 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT BASED SMART DENTAL DISEASE DETECTION AND MONITORING SYSTEM

(51) International classification	:A61B 1/24, A61B 5/00, A61C 19/04, G16H 50/30, A61C 19/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)S. PAVITHRA
Address of Applicant :DEPARTMENT OF BIOMEDICAL ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BYPASS ROAD, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)DEEPIKA R
3)Dharani S
4)Janani K
5)Pooja Sri R

(72)**Name of Inventor :**
1)S. PAVITHRA
2)DEEPIKA R
3)Dharani S
4)Janani K
5)Pooja Sri R

(57) Abstract :
According to WHO statistics ,approximately 3.5 billion people worldwide were affected oral diseases in 2019, including 2 billion people with permanent-tooth decay and 514 million children with primary- tooth decay in 2026, approximately 3.7 billion people worldwide are affected by oral diseases according to the WHO .A recent global analysis estimated the oral diseases causes increased from 3.62 billion in 2019 to 3.73 billion in 2023 .IoT- based smart dental disease detection and monitoring system is designed to monitor dental health in females, males and children. Dental problems such as tooth decay and gum disease can occur due to oral hygiene, dietary habits and hormonal changes. This project involves the development of an IoT-based smart system for the detection and monitoring of dental diseases as a preliminary measure for screening the issues related to oral health like dental caries, plaque, gum disease, and mouth ulcers. The suggested system uses a mini-intraoral camera, 405 nm LED, photodiode, temperature sensor, ESP32 microcontroller, power supply circuit, and BLE-based mobile app. Intraoral camera is used to take pictures of the teeth and gums and optical sensors are used to gather information relating to fluorescence. ESP32 receives the data collected from sensors and communicates with the mobile application. Application shows the data captured and screening results to the user. KEYWORD: Smart Dental Monitoring, Oral Health, Dental Caries, Plaque Detection, Gum Health, Mouth Ulcers.s and sensor re

No. of Pages : 9 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126287 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HEXAGONAL MULTI-ZONE DIVERGENT STEAM DISTRIBUTION CHAMBER WITH VARIABLE PERFORATION

(51) International classification	:A23L 5/10, A23L 3/16, A23N 12/10, A23L 17/60, A23L 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NANDHA ENGINEERING COLLEGE
Address of Applicant :NANDHA ENGINEERING COLLEGE ERODE-PERUNDURAI MAIN ROAD, VAIKKAALMEDU, ERODE, TAMILNADU, INDIA-638052 Tamil Nadu India
2)TECHNOLOGY INNOVATION HUB (TIH), IIT BOMBAY
(72)**Name of Inventor :**
1)Dr. M. EASWARAMOORTHI
2)K. PRADEEP KUMAR
3)P. SANDHIYADEVI

(57) Abstract :
A steam parboiling apparatus for turmeric (Curcuma longa) comprises a hollow chamber (101) of hexagonal cross-section divided by three internal partition members (103) into three wedge-shaped compartments arranged approximately 1200 apart. A central perforated steam-distribution pipe (104) and six peripheral perforated steam distribution pipes (102) positioned at the six corners provide seven steam-discharge locations. Steam-discharge openings (106) progressively increase in diameter (1_05) from a lower inlet end toward an upper distal end to compensate for pressure reduction along each pipe, and each opening has divergent geometry for lateral steam dispersion. Each pipe is connected to an independently controllable steam supply line having a flow-control valve (107). Each compartment receives steam from the central pipe and three peripheral pipes, permitting regulation according to packing density and providing substantially uniform steam distribution and penetration during turmeric parboiling.

No. of Pages : 24 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126291 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUAL-RECEIVER SECURE LONG RANGE PERSONNEL MONITORING SYSTEM USING LORA AND AES ENCRYPTION

(51) International classification	:H04L 29/06, G08B 25/10, H04W 4/90, H04L 9/32, H04L 9/08	(71)**Name of Applicant :** **1)DR. C. RAMYA** Address of Applicant :DEPARTMENT OF EEE, SNS COLLEGE OF TECHNOLOGY, VAZHIYAMPALAYAM, COIMBATORE, TAMIL NADU, INDIA-641035 Tamil Nadu India **2)JOSHUA .J** **3)MARIPRASATH .D** **4)MARTIN PHILIP .A** **5)MONISH .S** **6)SANJAI .D**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)DR. C. RAMYA**
(33) Name of priority country	:NA	**2)JOSHUA .J**
(86) International Application No	:NA	**3)MARIPRASATH .D**
Filing Date	:NA	**4)MARTIN PHILIP .A**
(87) International Publication No	: NA	**5)MONISH .S**
(61) Patent of Addition to Application Number	:NA	**6)SANJAI .D**
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

ABSTRACT: The present invention relates to a dual-receiver secure long-range personnel monitoring system using LoRa communication and AES encryption. A transmitter unit comprising a controller�RElamodule�infrared_prcscncc_scnsor,_GPS_module,_and emergency input collects available personnel identification, presence, location, and emergency-status information. The controller forms a structured data packet and encrypts the packet using AES before transmitting it through a LoRa module. First and second independent receiver stations receive the transmitted packet and process the data for presentation on LCD displays and/or a PC-based monitoring interface. The interface can decrypt the packet using a corresponding key, display personnel identification and GPS coordinates, highlight emergency status, and log received records. The dual-receiver arrangement supports parallel monitoring and redundant reception, reducing dependence on a single receiver station. The system provides an integrated architecture for secure long-range personnel monitoring, with performance depending on the selected hardware, radio configuration, and operating environment.144699 cramya.

No. of Pages : 10 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126305 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : OPEN ARMS - ACCESSBILITY ROUTE MAPPING SERVICE

(51) International classification	:G06F 17/30, G01C 21/36, G01C 21/34, G06Q 50/00, H04W 4/029
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College
Address of Applicant :. #42 Avadi — Vel Tech Road, Poonamallee - Avadi High Rd, Vel Nagar, Chennai, Tamil Nadu 600062 Tamil Nadu India
2)JohnPaul S M.E
3)Agilesh S
4)Manoj K N
5)Mohammed Rifaah Umar R
6)SanjayKumar P
(72)**Name of Inventor :**
1)JohnPaul S M.E
2)Agilesh S
3)Manoj K N
4)Mohammed Rifaah Umar R
5)SanjayKumar P

(57) Abstract :
People with disabilities may encounter various challenges when navigating the world, as information about places, facilities, and routes can be hard to find or understand. Open Arms is a mobile application that aims to address these issues by providing easily accessible information. Therefore, the project aims to facilitate the process of finding accessible places and facilities for people with disabilities and contribute to a stronger community through active participation and information sharing among users. Open Arms will provide map-based location services based on GIS and OpenStreetMap, with a focus on accessibility, crowdsourced data and information from users. The application will allow users to find accessible places, facilities, and routes and provide an opportunity to share, update, and contribute to the data, which will include information on ramps, entrances, elevators, parking, and much more. This information can subsequently be used by people with disabilities to find out more about potential places of interest to them and help them navigate. The application will offer a simplified, intuitive interface, from which users will be able to access and use the information provided. Overall, the Open Arms project aims to provide easier access to the information on places and facilities available for people with disabilities and offer an opportunity to participate in crowdsourcing and share the data within the community. Using the application, people with disabilities will be able to travel more safely and experience fewer challenges related to finding the particular facility they need. Furthermore, the open-source platform will allow people with and without disabilities to engage more actively in contributing to the development and maintenance of the database, which will ultimately lead to a positive impact on the community.

No. of Pages : 8 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126352 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE THERMAL MANAGEMENT AND ACTIVE CELL BALANCING SYSTEM FOR AN ELECTRIC VEHICLE BATTERY MANAGEMENT SYSTEM

(51) International classification	:B60L58/24, B60L58/22, H02J7/56, H01M10/42, H01M10/48	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No	:NA	
Filing Date	:NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(71)**Name of Applicant :**
1)S. SHARMILA
Address of Applicant :Department of EEC, SNS College of Technology, Vazhiyampalayam, Coimbatore 641035, Tamil Nadu India
2)AJAI ANDRAY .X
3)SARAN .B
4)ANUCHAND BALAJI.C
5)DURKA HARISH.P
(72)**Name of Inventor :**
1)S. SHARMILA
2)AJAI ANDRAY .X
3)SARAN .B
4)ANUCHAND BALAJI.C
5)DURKA HARISH.P

(57) Abstract :
The present invention relates to a predictive battery management system for an electric vehicle battery pack comprising a battery sensing and data acquisition unit, an embedded BMS processing and control system, a battery state estimation and monitoring module, a temperature prediction module, a thermal-risk evaluation module, a cell imbalance detection module, a battery actuation and protection unit, and a monitoring and visualization interface. Battery operating parameters including individual cell voltages, battery current, and cell temperature are acquired from the battery pack and processed through the embedded battery management system. A Random Forest temperature prediction module analyzes the battery operating parameters to predict short-term temperature behavior, while a thermal-risk evaluation module evaluates the predicted thermal condition. A cell imbalance detection module identifies voltage differences between series connected cells. The outputs are processed by a BMS decision and control logic to determine corresponding control actions, and the active cell balancing and preventive cooling units perform balancing and thermal control actions. The monitoring and visualization interface provides system information for display and data logging, thereby enabling integrated monitoring, predictive thermal management, and active cell balancing of the battery pack.

No. of Pages : 12 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126355 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GREEN-SYNTHESIZED NANOZYME PAPER MICROFLUIDIC SENSOR FOR P-NITROPHENOL DETECTION

(51) International classification	:G01N 21/78, G01N 33/52, G01N 21/31, G01N 33/18, B01L 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SAVEETHA ENGINEERING COLLEGE
Address of Applicant :SAVEETHA NAGAR, THANDALAM, CHENNAI, TAMIL NADU, INDIA-602105 Tamil Nadu India
(72)**Name of Inventor :**
1)K ANITHA
2)Dr T Aravind
3)Dr S Praveen Kumar
4)G Dinesh Ram
5)M Deepa Lakshmi

(57) Abstract :
The invention presents a portable paper-based microfluidic sensor for rapid detection of p-nitrophenol (PNP) in water samples. The device integrates a sample inlet, preconcentration zone, nanozyme-functionalized reaction zone, detection zone and internal reference zone. A green-synthesized biowastederived nanozyme enables colorimetric PNP detection using passive capillary flow. The sensor requires approximately 20 [IL sample and provides a response within 5 mM. It offers a detection range of 0.05-1.0 mM, R2 0.98 and an estimated LOD of 0.02 mM. Smartphone-based RGB image analysis enables quantitative estimation of PNP concentration. The integrated preconcentration and internal reference zones enhance analyte interaction and improve the reliability of smartphone-based colorimetric measurement. The integration of biowaste-derived nanozyme, paper microfluidics, localized colorimetric detection and digital image analysis provides a multifunctional sensing architecture. The device provides a pump-free, disposable and scalable sensing platform for decentralized PNP monitoring in agricultural and environmental water sources.

No. of Pages : 8 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126361 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : 3D BIOPRINTED ARTIFICIAL SKIN SUBSTITUTE USING NATURAL POLYMERS AND ANTIBACTERIAL NANOPARTICLES FOR

(51) International classification	:A61P 17/02, A61K 38/00, G06Q 50/00, G06Q 50/22, G16H 20/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)S. PAVITHRA
Address of Applicant :Sri Shakthi Institute of Engineering & Technology, Coimbatore, 641062 Tamil Nadu India
(72)**Name of Inventor :**
1)AKILESH .S
2)ASHA C.T
3)KALAIVANI .S
4)VIGNESH. E

(57) Abstract :
Burn injuries cause significant morbidity and mortality each year, making them a major global public health concern. The World Health Organization (WHO) estimates that burn-related injuries claim the lives of over 180,000 people each year, with low- and middle-income nations accounting for the majority of these deaths. Both the epidermal and dermal layers of the skin can be destroyed by severe burns, which can lead to problems like infections, excessive fluid loss, delayed wound healing, and the production of permanent scars. The need for affordable, biocompatible, and effective skin substitutes that can speed up healing and improve patient recovery is highlighted by the rising prevalence of burn injuries. Nowadays, biomaterials including collagen, chitosan, alginate, gelatin, and synthetic polymers are frequently used to create artificial skin substitutes. These materials are fabricated utilising methods such as freeze-drying, electrospinning, and 3D bioprinting to create wound dressings or tissue-engineered scaffolds. Although these techniques have shown promising results, they have a number of disadvantages, including as high production costs, insufficient antibacterial efficacy, limited mechanical strength, the potential for immunological reactions, and limited accessibility in low-resource environments. The current study suggests employing lotus stem cellulose, dragon fruit peel pectin, and moringa-mediated silver nanoparticles to create a 3D-bioprinted smart skin substitute for the treatment of burn wounds. Lotus stem cellulose improves the scaffold's mechanical strength and structural integrity, dragon fruit peel pectin functions as a natural hydrogel matrix, and silver nanoparticles made from moringa provide antibacterial qualities. Using 3D bioprinting technology, these naturally occurring biomaterials are combined to create a biocompatible bio-ink that is subsequently printed into a porous skin scaffold. It is expected that the manufactured scaffold will hasten wound healing, encourage tissue regeneration, lower the risk of infection, and offer a sustainable, affordable, and environmentally friendly substitute for traditional artificial skin.

No. of Pages : 10 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126428 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED NEEDLE BEAM PLASMA IONIZATION (NBPI) SYSTEM AND METHOD FOR SIMULTANEOUS CONTROL OF AIR

(51) International classification	:B01D 53/32, C02F 1/461, A61L 9/22, C02F 1/467, C02F 1/46
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)G S N V K S N SWAMY UNDI
Address of Applicant :FLAT NUMBER 502, VENKATA SAI SUKRUTHI APARTMENT, SHILPA AVENUE COLONY, ROAD 5A, MIYAPUR, HYDERABAD, TELANGANA Telangana India
(72)**Name of Inventor :**
1)G S N V K S N SWAMY UNDI

(57) Abstract :
An integrated Needle Beam Plasma Ionization pollution-control system provides coordinated air and water treatment using needle-electrode modules, process sensors and electrical-discharge feedback. An air module charges or treats airborne contaminants and is followed by collection or neutralisation, while a water module applies a localized highfield electrochemical treatment and is followed by contaminant separation. A supervisory control arrangement regulates each active module from both treatment-effect feedback and voltage/current/arc feedback, and applies medium-specific operating limits including ozone constraints for air and conductivity/flow constraints for water. Needle electrodes can be divided into independently energized banks. Multiple air and water modules can operate concurrently or independently, be connected in series or parallel, and be fault-isolated without stopping unaffected modules. The architecture supports electrostatic collection, 20 electrocoagulation, oxidation, reduction, electrodeposition and microbial load reduction depending on operating configuration.

No. of Pages : 25 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126437 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MOSQUITO-SELECTIVE MOLECULAR AGENT HAVING SUBSTANTIALLY NO DETECTABLE HUMAN ODOR AND REDUCED BIOLOGICAL ACTIVITY TOWARDS NON-MOSQUITO ORGANISMS

(51) International classification	:A01P 7/04, A01N 63/50, A01N 25/00, A01N 43/90, A01N 65/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PERI COLLEGE OF ARTS AND SCIENCE
Address of Applicant :PERI KNOWLEDGE PARK, MANNIVAKKAM, CHENNAI, TAMIL NADU, INDIA-600048 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. Kannirajalingam T
2)Authibhuvaneswari T
3)Robert Wilson M
4)Dinesh D
5)Pravin Kumar K
6) Keerthivasan D

(57) Abstract :
The present invention relates to the development of an innovative Mosquito-Selective Molecular Agent (MSMA) formulated to specifically manage mosquito populations while reducing negative impacts on humans, advantageous insects, animals, flora, and other non-target species. Traditional mosquito management techniques frequently include chemical pesticides that might emit disagreeable scents, lead to ecological pollution, and impact non-target organisms. To address these constraints, the suggested invention presents a chemically characterized molecular agent founded on a multi-tiered biological selectivity framework. The suggested agent is engineered to selectively identify and engage with biological targets that are particular to or abundant in mosquitoes, such as receptors, enzymes, ion channels, brain signaling proteins, and metabolic pathways. Upon engaging with the chosen target, the molecular agent aims to provoke critical biological changes, such neurological impairment, paralysis, metabolic disruption, or cessation of cellular energy synthesis, ultimately leading to the incapacitation or demise of the mosquito. A significant novel characteristic of the proposed invention is its designed capability to attain efficient mosquito management at levels beneath the human scent perception threshold, thereby offering a largely scentless use. Moreover, the molecular agent is engineered to demonstrate diminished biological efficacy against non-target organisms by selective molecular identification and focused biological intervention. The innovation moreover suggests the incorporation of controlled-release carriers, automated mosquito traps, and ecological mosquito management methods to sustain effective concentrations while reducing superfluous chemical exposure. The suggested technology may be utilized in residential structures, healthcare facilities, academic institutions, communal areas, and regions susceptible to mosquitoes. The concept seeks to offer an environmentally sustainable and sophisticated substitute for traditional broad-spectrum mosquito control drugs by integrating mosquito-specific molecular mechanisms, human odor invisibility, less toxicity to non-target organisms, and controlled-release technologies. The stated design objectives of selectivity, safety, and odorlessness necessitate experimental verification

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126439 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT NON-LETHAL ELEPHANT DETERRENCE SYSTEM USING DYNAMIC LED FOREST-FIRE SIMULATION

(51) International classification	:G06K 9/00, A01M 29/16, G08B 17/00, G08B 17/12, G08B 17/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PERI COLLEGE OF ARTS AND SCIENCE
Address of Applicant :PERI KNOWLEDGE PARK, MANNIVAKKAM, CHENNAI, TAMILNADU - 600048, INDIA 7299977711 periinnovationincubation@gmail.com Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. Kannirajalingam T
2)Authibhuvaneswari T
3)Shahul hameed M
4)Dharshini M S
5)Sahana Sri S
6)PAVITHRA V

(57) Abstract :
The present invention relates to an intelligent, non-lethal system for reducing elephant intrusion into residential areas, agricultural lands, plantations, resorts and other human-use locations situated near forests, wildlife corridors and hill stations. The proposed system combines animal detection, image-based or thermal identification, movement-direction analysis, adaptive control and a dynamic LED-based simulated forest-fire display. When an elephant approaches a predetermined protected zone, one or more sensors detect its presence and transmit the detected information to a processing unit. The processing unit analyses the movement, approximate distance and direction of the elephant and determines an appropriate deterrence response. A programmable LED display then generates a dynamic visual representation of a forest fire through controlled variations in brightness, colour, flickering and flame-like movement, without producing actual flames, smoke or harmful heat. The system may further include interconnected LED units, solar power, rechargeable energy storage, a camera, communication modules and an event-recording system. Based on the movement of the elephant, different display units may be activated sequentially to create a directional visual boundary. The system can automatically reduce or terminate the visual stimulus when the elephant moves away from the protected zone. The invention provides an environmentally safer, automated and potentially scalable approach to human-elephant conflict management.

No. of Pages : 11 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126451 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUST-REPELLENT INTELLIGENT FAN SYSTEM WITH ADAPTIVE ELECTROSTATIC CONTROL AND AUTOMATED SELF-CLEANING MECHANISM

(51) International classification	:F04D 25/08, F04D 29/70, F04D 25/06, F04D 29/32, F04D 29/00	(71)**Name of Applicant :** **1)PERI Institute of Technology** Address of Applicant :PERI INSTITUTE OF TECHNOLOGY PERI KNOWLEDGE PARK, MANNIVAKKAM, CHENNAI, TAMILNADU - 600048, INDIA Tamil Nadu India (72)**Name of Inventor :** **1)Dr. K A Sundararaman** **2)Dinesh S** **3)Arshan A S** **4)Pravin M** **5)Gopesh S** **6)Harimurugan R** **7)Rohit V**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The current invention pertains to a novel Dust-Repellent Intelligent Fan System aimed at reducing dust buildup on fan blades and related parts via a synergistic approach that includes electrostatic particle management, sophisticated dust-repellent surface treatment, intelligent dust surveillance, and automated self-cleaning mechanisms. Traditional ceiling and table fans are perpetually subjected to airborne dust, leading to surface pollution, diminished aerodynamic performance, heightened maintenance needs, and possible unbalance after extended use. The suggested innovation has a regulated electrostatic particle-handling system that affects electrically charged airborne dust particles and channels them into a removable dust-collection chamber. The fan blades feature a unique low-adhesion, dust-repellent coating that reduces particle adherence and simplifies dust elimination. A sophisticated sensor configuration perpetually observes dust levels, fan functionality, and dust-collection efficacy. A microcontroller-driven adaptive control system analyzes the gathered data and autonomously modifies electrostatic operational settings and fan-speed conditions to enhance dust-management efficacy. A notable characteristic of the innovation is its automatic self-cleaning system, which employs regulated fan-speed variation and electrostatic-field modification to facilitate the detachment of loosely adhered dust particles, minimizing the need for regular hand cleaning. The system additionally has an energy-conserving operational approach, a removable collection chamber, and an optional wireless monitoring capability. The suggested innovation seeks to deliver a more hygienic, energy-efficient, low-maintenance, and smart air circulation system appropriate for residential properties, educational facilities, workplaces, healthcare institutions, commercial venues, and industrial settings.

No. of Pages : 14 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126452 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TERRAIN-ADAPTIVE INTELLIGENT AGRICULTURE SYSTEM WITH SMART IRRIGATION AND PREDICTIVE MONITORING FOR HILLY REGIONS

(51) International classification	:A01G 25/16, G06Q 50/02, A01G 25/00, A01G 25/02, G01W 1/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PERI INSTITUTE OF TECHNOLOGY
Address of Applicant :PERI KNOWLEDGE PARK, MANNIVAKKAM, CHENNAI, TAMIL NADU, INDIA-600048 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. K A Sundararaman
2)Vishvamalar K
3)Sivaranjani R
4)Swathi G
5)Ramya S
6)Shivani G
7)Nishanthini V

(57) Abstract :
The current invention pertains to an intelligent agricultural system that adapts to terrain, aimed at enhancing cultivation efficiency, water management, and crop surveillance in hilly and mountainous areas. Traditional farming methods in mountainous regions suffer numerous obstacles, such as uneven terrain, inconsistent water allocation, inadequate irrigation systems, erratic weather patterns, and issues in ongoing field surveillance. To address these constraints, the suggested innovation presents a cohesive Internet of Things (IoT)-driven agricultural management framework that includes smart sensors, automated irrigation, anticipatory environmental monitoring, and adaptive water allocation based on terrain. The suggested system utilizes a network of dispersed soil moisture sensors, temperature and humidity sensors, water-level sensors, rainfall sensors, and slope-monitoring sensors to perpetually evaluate agricultural conditions at varying altitudes. A sophisticated control unit analyzes the gathered data and autonomously manages irrigation via electronically operated valves and pumps. A notable characteristic of the invention is its terrain-responsive irrigation system, which takes into account variations in field elevation, soil moisture levels, rainfall intensity, and water supply to deliver regulated irrigation to specific crop areas. Additionally, the system integrates a forecasting water management algorithm, a gravity-driven irrigation system, and a preemptive alert system for high soil moisture and possible slope instability. A solar-powered wireless monitoring system enables functionality in isolated agricultural areas with minimal electricity and communication resources. Agriculturalists can oversee field conditions from a distance and obtain irrigation suggestions and environmental notifications via a mobile application. The suggested innovation seeks to curtail superfluous water usage, lessen physical labor, enhance crop oversight, and promote sustainable agricultural yield in difficult mountainous terrains.y

No. of Pages : 17 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126465 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A CUSTOM RISC-V INSTRUCTION SET ARCHITECTURE FOR ENERGY-EFFICIENT SPIKING AND CONVOLUTIONAL NEURAL N

(51) International classification	:G06N 3/04, G06F 9/30, G06N 3/063, G06N 3/08, G06F 9/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MADHINI MURUGESH
Address of Applicant :Assistant Professor, Department of Electronics and Communication Engineering S.A. Engineering College, Chennai. 600077. Tamil Nadu, India madhinibalamurali@saec.ac.in 7448802062 Tamil Nadu India
2)Chithra Alagappan
3)Dr. Ramya Mohanraj
4)S.A. Engineering College
(72)**Name of Inventor :**
1)Madhini Murugesh
2)Chithra Alagappan
3)Dr Ramya Mohanraj

(57) Abstract :
The present invention relates to a customized Reduced Instruction Set Computer—V (RISC-V) instruction set architecture for accelerating spiking neural network (SNN) and convolutional neural • network (CNN) inference on resource-constrained edge computing platforms. The architecture - comprises a quantization unit, an IM2COL transformation unit, a zero-detection unit, processing elements, a Sparse Convolutional Neural Network (SCNN) array unit and an accumulator. Custom RISC-V instructions coordinate data preparation and processing operations to improve execution efficiency. The zero-detection unit identifies zero-valued input or weight elements to avoid unnecessary computations and reduce energy consumption. The SCNN array unit performs parallel computations, while the accumulator combines partial results to accelerate neural network inference and reduce processing latency. By integrating these components with customized instruction-level control, the proposed architecture aims to reduce computational overhead and improve the efficiency of neural network execution on resource-constrained edge platforms. The proposed architecture is applicable to edge artificial intelligence, intelligent sensing, image and pattern recognition, autonomous systems, wearable devices and Internet of Things applications where energy efficiency, low latency and limited hardware resources are important considerations. Keywords: RISC-V Architecture, Custom Instruction Set, Spiking Neural Networks, Convolutional Neural Networks, Energy-Efficient Computing, Edge Computing

No. of Pages : 8 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126489 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A TREE SHAPE WEARABLE CONDUCTIVE FABRIC PATCH ANTENNA FOR ISM BAND APPLICATIONS

(51) International classification	:H01Q 9/04, H01Q 1/38, H01Q 1/27, H01Q 1/24, H01Q 1/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. T. ANNALAKSHMI
Address of Applicant :PROFESSOR, DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING, S.A. ENGINEERING COLLEGE, CHENNAI, TAMIL NADU, INDIA-600077 Tamil Nadu India
2)S.A. ENGINEERING COLLEGE
(72)**Name of Inventor :**
1)DR. T. ANNALAKSHMI

(57) Abstract :
Tree Shape Wearable Conductive Fabric Patch Antenna for ISM Band Applications In today's world, women safety has become a great concern in most societies. In this paper, we present an innovative conductive fabric antenna that can be used for tracking purpose to ensure safety of women travelling alone. Precise location of the travelling woman can be tracked by their friends or family and immediate help could be dispatched in case of any emergency. The novelty of the designed antenna is that it is made in attractive shape and can be easily assimilated into clothes. The designed antenna operates in ISM band with center frequency of 2.4GHz. It provides a reasonable return loss of -17.8dB with maximum gain of 5.4 dB at 2.4 GHz. Due to its appealing shape and flexibility, the antenna can be worn without being obtrusive and can blend with the design of dresses.ronm

No. of Pages : 9 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126510 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE ELECTRONIC SYSTEM FOR EARLY DETECTION OF UNSAFE DRINKING WATER CONDITIONS

(51) International classification	:G01N 33/18, G01N 27/06, C02F 1/00, C02F 1/66, G06Q 50/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)B.R.TAPAS BAPU
Address of Applicant :S.A.ENGINEERING COLLEGE PROFESSOR AND HEAD DEPARTMENT OF ECE POONAMALLE-AVADI MAIN ROAD THIRUVERKADU CHENNAI-600077 TAMILNADU INDIA Tamil Nadu India
2)S.A.ENGINEERING COLLEGE
(72)**Name of Inventor :**
1)B.R.TAPAS BAPU

(57) Abstract :

Access to safe drinking water is an essential requirement for maintaining public health. Conventional water-quality assessment methods generally involve laboratory testing, specialized equipment, trained personnel, sample transportation, and relatively longer processing time. Such approaches may not be convenient for rapid preliminary assessment at remote or resource-constrained locations. The proposed invention provides a portable electronic system for early detection of potentially unsafe drinking-water conditions using a multi-parameter sensor arrangement integrated with an embedded processing unit. The system may include sensors for measuring pH, turbidity, electrical conductivity, total dissolved solids, temperature, and other configurable water-quality parameters. The outputs of the sensors are supplied to signal-conditioning and data-acquisition circuits. The conditioned signals are digitized and processed by a microcontroller or embedded processor. The processing unit performs calibration, filtering, compensation, parameter estimation, and threshold-based evaluation. The individual sensor results are then combined using a configurable decision algorithm to classify the tested water condition.The resulting status is presented through,a local display and may additionally be indicated using LEDs, a buzzer, or other alert mechanisms. In an optional implementation, the measured parameters and corresponding water-quality status are transmitted wirelessly to a mobile application, computer, or cloud-based monitoring platform for data logging and remote observation.The proposed system is designed as a portable, rapid, low-cost, user-friendly preliminary screening device that provides an early warning when one or more monitored water-quality parameters indicate a potentially unsafe condition. The system is intended to assist users in identifying samples that require further investigation or laboratory confirmation before consumption.

No. of Pages : 13 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2026

(21) Application No.202641126532 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS MULTI-UAV HIGHWAY PATROLLING SYSTEM WITH CORRIDOR DOCK STATIONS, RELAY HANDOFF AND TWO-STAGE EDGE-AI INCIDENT VERIFICATION

(51) International classification	:G08G 1/017, G08G 1/01, G08G 1/00, H04N 7/18, B64C 39/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREEJA VIJAY THELAKKAT
Address of Applicant :S.A. ENGINEERING COLLEGE, ASSOCIATE PROFESSOR, DEPARTMENT OF ECE, POONAMALLE-AVADI MAIN ROAD,THIRUVERKADU,CHENNAI,ITNA TAMILNADU-600 077. Tamil Nadu India
2)PRADEEP SHANMUGAM
3)S.A.ENGINEERING COLLEGE
(72)**Name of Inventor :**
1)SREEJA VIJAY THELAKKAT
2)PRADEEP SHANMUGAM

(57) Abstract :
In India highway patrolling in India is a layered arrangement. Incident management is run by the central highway authority and electronic monitoring on national highways, and state police and transport departments add their own enforcement. Despite the deployment of these systems the response still depends on vehicles that must physically reach the scene. Camera coverage is limited to installed pOints, and detection can produce false alerts. The invention is concerned with keeping a whole highway under continuous, intelligent aerial watch using a coordinated fleet of drones (UAVs), so that incidents are detected, confirmed and reported faster and more reliably than with today's patrol vehicles and fixed cameras. A highway patrol system comprises a plurality of unmanned aerial vehicles (UAVs) stationed at automated docking stations spaced along a highway corridor, each station recharging or exchanging UAV batteries. A scheduler launches UAVs and hands each patrol segment over to a relief UAV before the active UAV's battery is depleted, so that every segment remains under continuous observation. The scheduler assigns patrol routes and dwell times according to a risk score computed from traffic density, weather, time of day and incident history. Each UAV analyses sensor data onboard to detect a candidate incident. A confirmation module verifies the candidate against an independent second source, such as thermal imaging, radar, roadside sensors or vehicle telematics, and may direct a UAV closer to the event, before an alarm is raised, thereby. reducing false dispatches. Only event metadata and anonymized short clips are transmitted, • reducing bandwidth use and privacy exposure.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127107 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : THERMO-RESPONSIVE HYDROGEL KNEE SUPPORT SYSTEM WITH ZONAL EXPANSION AND THRESHOLD-BASED COMPRESSION CONTROL

(51) International classification	:A61F 5/01, A61F 5/00, A61F 13/06, A61F 5/058, A61F 5/055
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY, ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA-604308 Tamil Nadu India
(72)**Name of Inventor :**
1)B. SHARMILA
2)RAJESHKUMAR. K
3)Dr. K. PUVIYARASI
4)Dr. S. RAMYA BHARATHI

(57) Abstract :
Title: THERMO-RESPONSIVE HYDROGEL KNEE SUPPORT SYSTEM WITH ZONAL EXPANSION AND THRESHOLD-BASED COMPRESSION CONTROL
The present invention relates to a thermo-responsive hydrogel knee support system 5 comprising an outer flexible orthopedic support layer (1) configured to surround a knee joint. Thermo-responsive hydrogel compartments (2) are positioned at predetermined zonal compression areas (6) and retained within a hydrogel containment enclosure (7). A directional constraint layer (3) cooperates with a compression transfer interface (8) to direct hydrogel expansion toward the knee joint. An inner comfort layer (4) 10 provides a skin-contact surface, while an adjustable fastening system (5) maintains the position of the support assembly. The hydrogel changes volume in response to a predetermined temperature condition, generating localized mechanical compression without powered control equipment. When the temperature decreases, the hydrogel returns toward its lower-volume state, reducing the additional compression generated 15 by expansion.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127112 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A UNIT-LEVEL PHARMACEUTICAL IDENTIFICATION SYSTEM USING THERMOFORMING- COMPATIBLE ALPHANUMERIC CODING WITH ENGINEERED NON-DEFORMATION PLACEMENT AND BUFFER ZONES

(51) International classification	:G06V 30/224, G06K 19/06, G09F 3/03, G16H 40/67, G09F 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY, ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA - 604308. Tamil Nadu India
(72)**Name of Inventor :**
1)JESIMA JAITHUN.Y
2)RAJESHKUMAR. K
3)B. SHARMILA
4)Dr. K. PUVIYARASI

(57) Abstract :
This invention relates to a unit-level pharmaceutical identification system using thermoforming-compatible alphanumeric coding with engineered non-deformation placement and buffer zones. It typically includes a Front Forming Film (1), a Thermoformed Dome Region (2), a TabID-U Alphanumeric Code (3), a Coding Zone (4), an Engineered Buffer Zone (5), an Inter-Unit Separation Edges (6), a Sealing Backing Foil (7), a Centralized Digital Database (8), a User Application Interface (9), and an Information Display Module (10). This system ensures persistent, unit-level drug identification and instant digital verification after blister separation without code distortion or manufacturing line overhauls, significantly reducing medication errors and enhancing patient safety. This unit-level identification system is applicable across hospital inpatient dispensing, retail pharmacy cut-strip management, elderly home care, emergency medicine triage, clinical trials, and over-the-counter medicine tracking to prevent medication errors and enable instant digital verification of loose tablets.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127114 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A NEW DEEP LEARNING COLOR SPACE SCRAMBLING TECHNIQUE FOR HEALTHCARE IMAGES CLOUD SETTINGS

(51) International classification	:G16H 10/60, G06F 21/60, G06F 21/62, G16H 40/67, H04L 29/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.G.Amjad Khan
Address of Applicant :G.PULLA REDDY ENGINEERING COLLEGE (AUTONOMOUS), NEAR PASUPULA VILLAGE, KURNOOL, NANDYAL, MAIN ROAD, DINNEDEVARAPADU, ANDHRA PRADESH, INDIA-518007 Andhra Pradesh India
2)K. Lakshmi Devi
3)Dr. T. SAMRAJ LAWRENCE
4)E.Muthuramalingam
5)Athappan V
6)A R Ashok kumar
(72)**Name of Inventor :**
1)Dr.G.Amjad Khan
2)K. Lakshmi Devi
3)Dr. T. SAMRAJ LAWRENCE
4)E.Muthuramalingam
5)Athappan V
6)A R Ashok kumar

(57) Abstract :

Abstract The information and intelligence of wisdom medicine have gradually been attained by utilizing big data, cloud computing, artificial intelligence, and other technologies. However, information security issues have become more crucial due to the transmission and storage of massive amounts of medical images on the cloud. The risk of revealing, stealing, or otherwise tampering with patients' private information has emerged as a significant roadblock to the advancement of medical science. It is imperative to address the critical issue of protecting patient data in the cloud environment. Because they require less processing power and are JPEG-compatible, color space-based scrambling algorithms (CSSA) are becoming more popular for encrypting multimedia data. Traditional ways need a colorfill image as input to decrease block size and increase security. Due to the absence of color images, CSSA approaches are insufficient for secure transmission and storage in fields like medical image processing. This study aims to merge CSSA image encryption applications with Multilayer Perceptron (MLP)-based medical image encryption techniques. We developed a noise-based data augmentation method for medical image analysis to alleviate data limitations. Security analysis and temporal complexity are used to examine the effectiveness of our proposed (MLP-CSSA) deep learning model for encrypting medical images. The results demonstrate a high level of security achieved by encrypting both black-and-white and color medical images. A comparison is provided between many encryption techniques. The proposed encryption method MLP-CSSA beats the most recent existing techniques for encrypting medical photos. Keywords:Deep learning, Medical images, Colour space-based scrambling algorithms (CSSA), JPEG standard, Multilayer Perceptron's (MLP), encryption techniques

No. of Pages : 7 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127117 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI- LAYER KERATIN- BASED TRANSDERMAL DRUG DELIVERY PATCH WITH CONTROLLED RELEASE AND ENHANCED WOUND HEALING DERIVED FROM HUMAN HAIR WASTE

(51) International classification	:A61K 9/70, A61F 13/02, A61K 47/42, A61L 26/00, A61L 15/32, A61L 15/44, A61P 17/02, C07K 14/47
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA - 604308. Tamil Nadu India
(72)**Name of Inventor :**
1)RAJESHKUMAR. K
2)B. SHARMILA
3)Dr. K. PUVIYARASI
4)M. VASANTHKUMAR

(57) Abstract :
This invention relates to a multi-layer keratin-based transdermal drug delivery patch with controlled release and enhanced wound healing derived from human hair waste. It typically includes a Gel Base (1), a pH-Sensitive Indicator System (2), a Humectants (3), a Preservatives (4), a Stabilizing and Buffering Agents (5), a Neutralizing Agents (6), a Flavoring and Sweetening Agents (7), and an Applicator System (8). This multilayer patch delivers dual-action care by combining rate-controlled drug delivery with active, keratin-stimulated tissue regeneration using a cost-effective, fully biodegradable biopolymer upcycled from human hair waste. The advantages of this multi-layer patch translate directly to diabetic foot ulcers, severe burn care, postsurgical wounds, chronic pain management, hormonal therapies, and cosmetic dermatology by offering a seamless, non-invasive treatment that accelerates tissue repair while delivering precise, sustained drug therapy without skin irritation. Figure 1. Shows the cross-sectional view of patch showing internal layers

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127122 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIOADHESIVE SMART HERBAL HYDROGEL CONTAINING TRIDAX PROCUMBENS EXTRACT FOR RAPID WOUND HEALING AND HEMOSTASIS

(51) International classification	:A61L 26/00, A61K 36/28, A61K 9/06, A61L 15/44, A61K 36/185
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA - 604308. Tamil Nadu India
(72)**Name of Inventor :**
1)R. MOHAMED SUHAIL
2)M. NETHRA
3)M. VASANTH KUMAR
4)P. TITUS

(57) Abstract :
This invention relates to a bioadhesive smart herbal hydrogel containing tridax procumbens extract for rapid wound healing and hemostasis. It typically includes a Tridax procumbens Hydroalcoholic Extract (1), a Primary Bioadhesive Polymeric Matrix (2), a Secondary Hydrogel Base Polymer (3), an Antimicrobial Synergistic Additives (4), a Soothing & Hydrating Agent (5), a Secondary Regenerative Bioactive (6), a Pharmaceutically Acceptable Plasticizers (7), a Crosslinking Agents & Stabilizers (8), a pH & Viscosity Modifiers (9), an Aqueous Processing Vehicle (10). This smart hydrogel rapidly stops bleeding, accelerates tissue repair, and protects against infection using Tridax procumbens bioactives while maintaining a moist environment for painless dressing changes. This smart herbal hydrogel is applied for the rapid management and treatment of acute traumatic cuts, surgical incisions, burn wounds, diabetic ulcers, and chronic skin lesions across clinical and home-care settings.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127123 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR CREDIT RISK SCORING OF FARMER MICROLOANS USING LAND RECORD, ENCUMBRANCE AND YIELD CAPACITY PARAMETERS

(51) International classification	:G06Q 40/02, G06Q 40/00, G06Q 10/06, G06Q 50/02, G06Q 40/03
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHRIST UNIVERSITY
Address of Applicant :CHRIST (DEEMED TO BE UNIVERSITY), KANMANIKE, KUMBALGODU, BANGALORE, KARNATAKA 560074 Karnataka India
(72)**Name of Inventor :**
1)BARATH KUMAAR T R
2)BHUVANA JAYABALAN

(57) Abstract :
A system and method for credit risk scoring of farmer microloans are disclosed. The system receives applicant identity and collateral or guarantor details, a land document issued by the Village Administrative Officer, land status showing any existing jewel loan, repayment records of the previous year, and the soil and crop type. A verification module checks identity details and confirms that the land document is valid for the current year, while an encumbrance status and a repayment history parameter are derived from the records. A yieldcapacity parameter is computed from the soil and crop type using a stored suitability matrix. The parameters are standardized into a feature vector, to which stored regression weights are applied to obtain a risk score. A repayment-based suggestion is generated and presented with the score to a loan officer, who approves, rejects or refers the application. Farmers without credit bureau history can be assessed consistently.

No. of Pages : 21 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127126 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SUSTAINABLE DISASTER RISK MANAGEMENT SYSTEM USING MARINE MICROALGAE- BASED ECO- FRIENDLY FIRE EXTINGUISHER

(51) International classification	:A62C 3/00, A62C 35/68, A62C 37/38, A62C 37/40, A62C 37/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY, ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA-604308 Tamil Nadu India
(72)**Name of Inventor :**
1)S. SASI
2)Dr. K. GURU
3)E. MADHANKUMAR
4)J. SHARMILA
5)S. LAVANYA

(57) Abstract :
Title: SUSTAINABLE DISASTER RISK MANAGEMENT SYSTEM USING MARINE MICROALGAE-BASED ECO-FRIENDLY FIRE EXTINGUISHER 5 The present invention relates to a sustainable fire suppression system employing a marine microalgae-derived bio-fluid. A cultivation unit (1) produces biomass that is processed by a harvesting and extraction unit (2) to obtain biological constituents. A formulation unit (3) blends the constituents with a liquid carrier and stabilizing 10 additives. A pressurized storage container (4) retains the formulation, while a pressurizing assembly (5) provides the required discharge pressure. A discharge valve and hose assembly (6) conveys the formulation to an adjustable discharge nozzle (7). A fire detection and activation assembly (8), an institutional risk management interface (9) and a performance monitoring and assessment unit (10) support activation, 15 deployment records and performance evaluation. The system is intended to support fire suppression and environmental management, subject to validation of fire performance, electrical safety, storage stability and environmental characteristics.INE

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641127128 A
(19) INDIA
(22) Date of filing of Application :26/10/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : FLORAGUARD - MOBILE APPLICATION FOR LEAF ANOMALY RECOGNITION

(51) International classification	:G06N 3/08, G06N 20/00, G06Q 50/02, G06N 3/04, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEPCO SCHLENK ENGINEERING COLLEGE (AUTONOMOUS)
Address of Applicant :MEPCO SCHLENK ENGINEERING COLLEGE (AUTONOMOUS), MEPCO SCHLENK ENGINEERING COLLEGE (PO), SIVAKASI, TAMIL NADU, INDIA-626005 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.N.MALATHY
2)Dr. M. BLESSA BINOLIN PEPSI

(57) Abstract :
The Plant Leaf Disease Recognition App is a comprehensive mobile application designed to assist plant enthusiasts in identifying and managing diseases affecting their plants. This app leverages the capabilities of smartphones, offering a user-friendly interface and a streamlined process for disease identification and expert guidance. The app begins by initiating the Recognition App, which grants users access to the device's camera. Users can then capture images of affected or unaffected plant leaves, which are subsequently uploaded to the app for analysis. The app employs advanced algorithms of machine and deep learning to compare the captured images against a vast database of known plant diseases, resulting in accurate disease identification within seconds. Alongside disease identification, users can access detailed information about the identified disease, including its symptoms, causes, and recommended remedies. To enhance users' knowledge and plant care abilities, the app facilitates communication with plant experts. Users have the option to connect with plant experts and seek further information and guidance regarding the identified disease and its treatment. Additionally, the app provides access to government sites, enabling users to gather authoritative information about plant diseases and their remedies. Overall, the Plant Leaf Disease Recognition App empowers users to efficiently identify plant diseases, access relevant disease information, and seek expert guidance. By combining cutting-edge technology with user-friendly features, this app simplifies plant care and promotes healthier plants, fostering a greener and more sustainable environment.

No. of Pages : 13 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127133 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FOOTPRINT- BASED AUTOMATED ANIMAL INDENTIFICATION AND TIMER- CONTROLLED FEEDING SYSTEM.

(51) International classification	:A01K 5/02, A01K 29/00, A01K 11/00, A01K 5/00, A01K 5/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA - 604308. Tamil Nadu India
(72)**Name of Inventor :**
1)S. DHIVYA LAKSHMI
2)Dr. R. SUBRAMANIYAN
3)Dr. A. SUPHALAKSHMI
4)Dr. R. SATHYA JANAKI

(57) Abstract :
A footprint-based automated animal identification and timer-controlled feeding system is disclosed. A footprint sensing platform (1) detects paw characteristics and supplies sensor data through a footprint acquisition unit (2). An animal identification module (3) compares the acquired footprint with reference profiles in a registered animal database (4). A central controller (5) obtains schedule information from a timer and scheduling module (6) and evaluates a feeding authorization result from a feeding authorization module (7). An automated food dispensing unit (8) releases a predetermined quantity under actuation of a dispensing actuator (9) only when identity, time, and feedinghistory conditions are satisfied. A feeding-record storage unit (10) records animal identity, feeding time, quantity, and status, thereby pporting individualized scheduled feeding and prevention of repeated dispensing.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127134 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND A PROCESS FOR SYNCHRONISED AUTONOMIC-NOCICEPTIVE PHENOTYPING WITH INDIVIDUALISED RESONANCE BREATHING CONDITIONING

(51) International classification	:A61B 5/00, A61B 5/08, A61B 5/0205, A61N 1/365, A61B 5/0402	(71)**Name of Applicant :** **1)Vinayaka Mission's Medical College and Hospital Vinayaka Mission's Research Foundation (DU)** Address of Applicant :Dean, Vinayaka Mission's Medical College and Hospital, Keezhakasakudimedu, Kottucherry(P.0) Karaikal, Puducherry,India-609 609. Pondicherry India **2)Dr. K. Cheran** **3)Dr. K. Prathiba** **4)Dr. K. Sriram**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr. K. Cheran**
(33) Name of priority country	:NA	**2)Dr. K. Prathiba**
(86) International Application No	:NA	**3)Dr. K. Sriram**
Filing Date	:NA	**4)Dr. Manoj Kumar Mishra**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

A system (100) for synchronised autonomic—nociceptive phenotyping comprises an autonomic acquisition module (110) with electrocardiogram, continuous noninvasive arterial pressure and respiratory sensors; a nociceptive stimulation module (120) with a closed-loop motorised pressure algometer (122), a pinprick actuator (124), a temperature-controlled conditioning immersion unit (126) with immersion-triggered timing, and a subject response input (128); a common clock (132) that time-stamps all signals, stimuli and responses; and a processing unit (130). The processing unit computes pressure pain threshold, temporal summation, conditioned pain modulation, cardiac vagal and baroreflex sensitivity indices, classifies them against stored reference intervals and smallest detectable change values, and computes a composite index that includes the change in vagal activity between two sessions. A paced breathing conditioning device (160) tunes an individual resonance breathing rate, verifies the achieved rate, logs valid sessions and provides an attention-matched control mode. A process of operating the system is also disclosed.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127137 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LECITHIN-FREE BIOACTIVE NANOSTRUCTURED LIPID CARRIERS (NLC) COMPRISING A SYNERGISTIC PHYTOCHEMICAL MATRIX FOR THE DUAL MODULATION OF THE INSULIN- ANDROGEN AXIS IN PMOS WOMEN

(51) International classification	:A61K 36/54, A61K 31/352, A61P 3/10, A61K 47/44, A61K 36/185
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY, ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA-604308 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr S RAMYA BHARATHI
2)B SHARMILA
3)Dr K PUVIYARASI
4)Dr P TITUS

(57) Abstract :
Title: LECITHIN-FREE BIOACTIVE NANOSTRUCTURED LIPID CARRIERS (NLC) 5 COMPRISING A SYNERGISTIC PHYTOCHEMICAL MATRIX FOR THE DUAL MODULATION OF THE INSULIN- ANDROGEN AXIS IN PMOS WOMEN This invention relates to a lecithin-free bioactive nanostructured lipid carriers (NLC) comprising a synergistic phytochemical matrix for the dual modulation of the insulinandrogen axis in PMOS women. It typically includes a Saw Palmetto Oil (1), a 10 Flaxseed Oil (2), a Spearmint Oil (3), a Berberine (4), a Quercetin (5), a Licorice Extract (6), a Cinnamon Extract (7), a Non-Ionic Surfactant (8), and a Purified Water (9). The primary advantage of this platform is its all-active, non-allergenic design that replaces inert carrier lipids with a functional 5-a-reductase-inhibiting lipid scaffold, allowing a single synchronized dose to bypass hepatic first-pass metabolism, protect 15 sensitive phytochemicals from gastric degradation, and simultaneously target both the metabolic root cause and external dermatological symptoms of PMOS. The applications of this platform include the oral systemic management of Polyendocrine Metabolic Ovarian Syndrome (PMOS) and associated insulin resistance, as well as topical dermo-pharmaceutical applications for treating androgenetic alopecia, 20 hormonal acne vulgaris, and hirsutism via targeted local 5-a-reductase inhibition. Figure 1. Shows the schematic cross-sectional view of the lecithin-free bioactive Nanostructured Lipid Carrier (NLC) depicting its structural architecture and encapsulated phytochemical matrix

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127143 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BLOOD PRESSURE MONITORING.

(51) International classification	:A61B 5/0225, A61B 5/022, A61B 5/02, A61B 5/021, A61B 5/00	(71)**Name of Applicant :** **1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION** Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India (72)**Name of Inventor :** **1)Dr. T. ARUNPRASATH** **2)Dr. M. PALLIKONDA RAJASEKARAN** **3)VINAYAGA S** **4)NITHISHKUMAR** **5)OMPRASAD S** **6)BHARATHIRAJA M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The titled invention discloses a digital blood pressure monitoring system comprising a blood pressure sensor (1), cuff (2), air pump (3), valve (4), signal conditioning unit (5), microcontroller (6), display unit (7), buzzer alert system (8), power supply (9) and loT module (10). The system operates using an oscillometric measurement principle, wherein the cuff (2) is inflated and gradually deflated while the blood pressure sensor (1) detects pressure oscillations associated with arterial blood flow. The detected signal is amplified and filtered by the signal conditioning unit (5) and processed by the microcontroller (6) to determine systolic pressure, diastolic pressure and pulse rate. The measured values are displayed through the display unit (7), while abnormal readings activate the buzzer alert system (8). Previous readings may be stored for trend analysis and the loT module (10) may transmit data for remote monitoring. The system provides an automated, compact and user-friendly arrangement for blood pressure monitoring.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127146 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMA PATIENT MONITORING

(51) International classification	:A61B 5/11, A61B 5/00, G08B 21/04, G16H 40/67, A61G 7/05	(71)**Name of Applicant :** **1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION** Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 9600914308 jaijat07@gamil.com Tamil Nadu India (72)**Name of Inventor :** **1)Dr.T. ARUNPRASATH** **2)Dr. M. PALLIKONDA RAJASEKARAN** **3)V. KRISHNAMOORTHY**
(31) Priority Document No	:NA	**4) T JAI SURYA**
(32) Priority Date	:NA	**5)BALAMUKUNDHAN S S**
(33) Name of priority country	:NA	**6)JINO NICKSON N**
(86) International Application No	:NA	**7)HARIHARAN S**
Filing Date	:NA	**8)SHEIK MOHAMMED IMRAN RJ**
(87) International Publication No	: NA	**9)JAVITH ABDUL SALAM B**
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The titled invention discloses a coma patient monitoring system comprising a patient bed (1), MPU6050 (2) gyroscope and accelerometer sensor module, an ESP32 microcontroller (3) with built-in Wi-Fi, a buzzer (4) for local audible alerts, and a Blynk IOT (5) mobile application for remote real-time monitoring and power supply (5). The system captures MPU6050 (2) three-axis acceleration and three-axis angular velocity data from the patient continuously. A firmware-based threshold and pattern detection algorithm running on the ESP32 (3) processes this data to identify abnormal or clinically significant motion events. Upon detection of movements exceeding predefined thresholds, the system simultaneously triggers the buzzer (4) for immediate local auditory alert and transmits notification data to the Blynk IOT platform (5) via Wi-Fi, enabling caregivers to receive real-time alerts on their smartphones regardless of physical proximity. This invention useful in intensive care units (ICUs), home-based palliative care setups, rehabilitation centers, neurological wards, and rural healthcare environments where continuous human surveillance of coma patients is difficult or impractical.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127149 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LOW-COST NON-INVASIVE CONTINUOUS BLOOD PRESSURE MONITORING SYSTEM

(51) International classification	:A61B 5/021, A61B 5/0225, A61B 5/022, A61B 5/00, A61B 5/02	(71)**Name of Applicant :** **1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION** Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India (72)**Name of Inventor :** **1)Dr. T. ARUNPRASATH** **2)Dr. M. PALLIKONDA RAJASEKARAN** **3)JUSTIN DHEEBAN A** **4)HEMNATH S** **5)HARIHARAN P** **6)PRABHU RAJ M** **7)VISHAL B** **8)JEYA GANESH KUMAR B**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a low-cost non-invasive continuous blood pressure monitoring system comprising an optical pulse sensing unit (1) connected to an embedded microcontroller unit (2). A signal processing and estimation unit (3) processes acquired pulse waveform characteristics to calculate estimated systolic and diastolic blood pressure values. A visual display unit (4) presents the estimated measurements, while a wireless communication unit (5) transmits monitoring data to a connected remote platform. An alert generation unit (6) evaluates estimated values against predefined limits and generates notifications when configured abnormal conditions are detected. A rechargeable power supply (7) provides electrical energy to the electronic components housed within a wearable protective enclosure (8). The integrated system supports repeated non-invasive blood pressure estimation, local measurement presentation, and remote monitoring without requiring repeated cuff inflation during the optical monitoring cycle.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127157 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS MULTI-SENSOR WEARABLE EMERGENCY DETECTION AND PROTECTION SYSTEM FOR WOMEN AND CHILDREN.

(51) International classification	:G08B 21/02, G08B 25/10, G08B 25/01, G08B 25/00, G08B 25/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)A. LAWRENCE SAHAYA SUNDAR
2)Dr. K. VIJAYAKUMAR

(57) Abstract :
Title: AUTONOMOUS MULTI-SENSOR WEARABLE EMERGENCY DETECTION AND PROTECTION SYSTEM FOR WOMEN AND CHILDREN 5 The present invention relates to an autonomous multi-sensor wearable emergency detection and protection system for women and children. The system comprises a wearable enclosure (1), embedded processing unit (2), motion sensing unit (3), location determination unit (4), wireless communication unit (5), manual emergency 10 activation unit (6), tamper detection unit (7), local indication unit (8), memory unit (9), and power supply and management unit (10). The embedded processing unit (2) correlates movement events, geographical location changes, device tampering, and manual distress inputs according to configured safety criteria. The system classifies detected conditions into different emergency states and initiates emergency 15 communication upon satisfaction of a predetermined high-risk assessment criterion. The wireless communication unit (5) transmits emergency status and available location information to registered recipients and authorized emergency-response services. The invention provides an autonomous wearable safety arrangement that reduces dependence on continuous smartphone operation and supports personal 20 emergency assistance for women and children.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127160 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A COMPACT ELECTIC SKATE- SCOOTER WITH INTEGRATED WIRELESS CHARGING SYSTEM.

(51) International classification	:B62K 15/00, B62K 3/00, A63C 17/14, B62M 6/40, B62M 6/45
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 9600914308 jaijat07@gamil.com Tamil Nadu India
(72)**Name of Inventor :**
1) Dr. K. VIJAYAKUMAR
2)A. LAWRENCE SAHAYA SUNDAR

(57) Abstract :
The present invention relates to a compact electric skate-scooter for short-distance personal transportation. The device comprises a deck and base frame (7), wheel assemblies (5), a handlebar (1), a steering stem (3), a folding mechanism and locking arrangement (4), a rechargeable battery (6), a control switch and motor control unit (2), an electric hub motor (8), and a braking arrangement (9). The rechargeable battery supplies electrical energy through the control unit to drive a wheel and propel the device. The folding mechanism permits the steering structure to move into a storage position. A contactless charging interface (10) is configured to receive inductively transferred electrical energy from a compatible charging transmitter and supply regulated charging energy to the battery. The charging transmitter can be supplied by solar-generated electrical power through suitable power conditioning circuitry. The device is intended for compact, electrically powered personal mobility in suitable short-distance transportation environments.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127164 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT- BASED SMART WASTE SEGREGATION SYSTEM.

(51) International classification	:B07C 5/34, B07C 5/36, B09B 3/00, B07C 5/344, B09B 5/00	(71)**Name of Applicant :** **1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION** Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India (72)**Name of Inventor :** **1)Dr. S. MUTHUVEL** **2)MANISHA .S** **3)MADHUMITHA S** **4)MANASHA B** **5)MANAALI R** **6)MEENAKSHI M** **7)MAHALAKSHMI P** **8)MARIA ROBY**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The titled invention discloses an loT-based smart waste segregation system for automatically detecting and separating wet, dry and metal waste. The system comprises a waste input hopper (1), IR sensor (2), ultrasonic sensor (3), proximity sensor (4), raindrop sensor (5), stepper motor (6), motor driver (7), servo motor (8), waste collection containers (9), Arduino Uno microcontroller (10), ESP8266 wireless connectivity module (11), buzzer (12), rechargeable battery (13) and base platform (14). The sensors detect the presence and characteristics of waste and provide inputs to the Arduino Uno microcontroller (10). Metal waste is detected through proximity sensing, wet waste through moisture detection and dry waste based on the absence of moisture and metal properties. The motors automatically direct the detected waste into corresponding containers. The ESP8266 module sends real-time Telegram notifications regarding the detected waste category, thereby reducing manual effort and improving waste management efficiency.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127167 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT- BASED AUTOMATIC SMART IRRIGATION SYSTEM.

(51) International classification	:A01G 25/16, G05B 19/418, A01G 27/00, G05B 19/042, G05D 27/02	(71)**Name of Applicant :** **1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION** Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India (72)**Name of Inventor :** **1)Dr. MAYANDI KALIMUTHU** **2)M. CHANDRA VARDHAN REDDY** **3)M. VENU GOPAL REDDY** **4)M. RAGHAVENDRA SWAMY REDDY**
(31) Priority Document No	:NA	**5)M. NANDA KISHOR**
(32) Priority Date	:NA	**6)M. SUMANTH**
(33) Name of priority country	:NA	**7)MANYAM TIRUMALA SRINIVAS**
(86) International Application No	:NA	**8)S. SUBIN**
Filing Date	:NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The titled invention discloses an loT-based automatic smart irrigation system comprising an Arduino controller (1), ESP8266 Wi-Fi module (2), soil moisture sensor (3), flow sensor 5 (4), rain sensor (5), DHT11 sensor (6), water level sensor (7), LCD display (8), selection button (9), confirm button (10), relay module (11), water pump (12), red LED indicator (13), green LED indicator (14), buzzer (15) and battery connection (16). The system provides dynamic sensor assignment through the selection button (9) and confirm button (10), allowing different sensors to be assigned to corresponding GPIO connections. The 10 Arduino controller (1) monitors the soil moisture condition and automatically controls the water pump (12) through the relay module (11). When the soil becomes dry, the pump is activated and is automatically switched OFF when the required moisture condition is reached. The ESP8266 Wi-Fi module (2) provides loT connectivity for real-time sensor data logging and monitoring through Google Sheets, thereby supporting flexible and 15 efficient irrigation management.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127169 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT MONITORING AND RESCUE COORDINATION SYSTEM

(51) International classification	:G08B 21/10, G01W 1/10, G01W 1/14, G06Q 50/26, G01W 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)R. PRABAVATHI
Address of Applicant :Department of -Information Technology, Sri Sairam Engineering College Sai Leo Nagar, West Tambaram Chennai Tamil Nadu India Chennai — 600044 Tamil Nadu India
2)M SURYA
3)P LEELA JANCY
4)P SUTHAHAR
5)SURYA P
6)MANI S
7)MADHAN A
8)BALAMURUGAN S
9)P SUBHA
(72)**Name of Inventor :**
1)R. PRABAVATHI
2)M SURYA
3)P SUBHA
4)P LEELA JANCY
5)P SUTHAHAR
6)SURYA P
7)MANI S
8)MADHAN A
9)BALAMURUGAN S

(57) Abstract :
FLOODSENSE is an AI-powered flood monitoring and alert system focused on flood-prone regions surrounding lakes. It identifies vulnerable low-lying areas using topographical data and monitors water level fluctuations from rainfall, river inflow, and environmental changes. Strategically placed sensors, connected to Arduino boards and SIM-based communication, detect rising water levels and trigger alerts. The system integrates Al-driven predictive analytics to forecast flood impact based on rainfall predictions, enabling early warnings. A companion mobile app delivers real-time flood alerts, weather updates, and evacuation guidance. It also notifies government agencies and enables users to connect with nearby volunteers for assistance. With features like safe route recommendations, shelter information, and seasonal weather forecasts, FLOODSENSE ensures year-round utility. By combining IoT, Al, and mobile technology, the system enhances community preparedness, speeds up emergency response, and minimizes flood-related risks.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127170 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPREHENSIVE SPATIOTEMPORAL OPTIMIZATION FRAMEWORK FOR DYNAMIC LOAD BALANCING AND RESILIENT VEHICLE

(51) International classification	:G06Q 10/08, G06Q 10/06, G01C 21/34, G08G 1/00, G06Q 10/04	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No	:NA	
Filing Date	:NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(71)**Name of Applicant :**
1)P. SUTHAHAR
Address of Applicant :DEPARTMENT OF INFORMATION TECHNOLOGY SANTHOSH NAGAR 2ND CROSS STREET, MADANANTHAPURAM, PORUR, CHENNAI-600125 9952160940 pv.suthahar@gmail.com Tamil Nadu India
2)MADHAN A
3)BALAMURUGAN S
4)ABHISEKH H
5)R SAMPATH
6)R PRABAVATHI
7)SURYA M
8)SURYA P
(72)**Name of Inventor :**
1)P. SUTHAHAR
2)MADHAN A
3)BALAMURUGAN S
4)ABHISEKH H
5)R SAMPATH
6)R PRABAVATHI
7)SURYA M
8)SURYA P

(57) Abstract :
SPATIOTEMPORAL OPTIMIZATION FRAMEWORK FOR DYNAMIC LOAD MATCHING AND RESILIENT VEHICLE ROUTING. The present invention discloses a system and method for dynamic load matching and resilient vehicle routing in stochastic logistics environments. The system partitions a service area into four geographic zones and maintains multi-tiered priority queues for each zone. An Effective Priority Score (EPS) incorporates waiting-time aging to reduce priority starvation. Available fleet capacity is represented as a virtual aggregate Container Size (VC) during an initial allocation phase, followed by assignment to physical vehicles. A Circle Boundary Heuristic addresses high-density order clusters at zone intersections using localized Hamiltonian cycle optimization. The system further integrates load-dependent fuel consumption. re4l-time Nearest Capable Vehicle (NCV) selection for new arrivals, and manifest splitting for vehicle failures. These mechanisms support adaptive allocation, routing and recovery while maintaining delivery-priority information.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127171 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AROUND BASED HEARTRATE AND TEMPERATURE MEASURING.

(51) International classification	:A61B 5/024, A61B 5/01, A61B 5/00, A61B 5/0245, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. T. ARUNPRASATH
2)S. SHANMUGA PRIYA
3)NISHANT KUMAR
4)K JOY KUMAR
5)SHYAMSUNDAR KUMAR
6)P. MANIKANTA
7)S. SANRUTHEP

(57) Abstract :
Title: ARDUINO BASED HEARTRATE AND TEMPERATURE MEASURING The titled invention discloses an Arduino based heart rate and temperature measuring 5 system for real-time monitoring of physiological parameters, comprising a heart rate sensor (1), a temperature sensor (2), an Arduino microcontroller (3), a power supply (4) and an LCD display (5). The heart rate sensor (1) detects variations in blood flow using Photoplethysmography and provides pulse signals for determining heart rate in beats per minute, while the temperature sensor (2) measures body temperature. The Arduino 10 microcontroller (3) receives and processes the sensor signals and provides the calculated physiological values to the LCD display (5) for immediate visual feedback. The system performs local sensing, processing and display without dependence on external devices or cloud connectivity. The arrangement provides a compact, low-cost and non-invasive solution suitable for personal health monitoring, fitness tracking, educational applications, 15 basic healthcare monitoring and embedded system prototyping.

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127172 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED MULTILINGUAL SYMPTOM TRIAGE AND HEALTHCARE GUIDANCE PLATFORM WITH EMERGENCY ESCALATION AND CONVERSATIONAL DECISION SUPPORT

(51) International classification	:G06N 20/00, G16H 40/20, G16H 10/60, G16H 80/00, G16H 50/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SURYA M
Address of Applicant :DEPARTMENT OF INFORMATION TECHNOLOGY, SRI SAIRAM INSTITUTE OF TECHNOLOGY, SAI LEO NAGAR, WEST TAMBARAM, CHENNAI, INDIA-600062 Tamil Nadu India
2)KISHORE KUMAR G
3)NANDHINI J M
4)VIDYAA V
5)AKSHAYA A
6)MADHAN A

(72)**Name of Inventor :**
1)SURYA M
2)KISHORE KUMAR G
3)NANDHINI J M
4)VIDYAA V
5)AKSHAYA A
6)MADHAN A

(57) Abstract :
AI-BASED MULTILINGUAL SYMPTOM TRIAGE AND HEALTHCARE GUIDANCE PLATFORM WITH EMERGENCY ESCALATION AND CONVERSATIONAL DECISION SUPPORT ABSTRACT The invention describes a multilingual (SEV-AI), AI-based healthcare symptom triage and guidance system, which allows for access to healthcare decision support via a smartphone app. It includes a chatbot chat interface that allows users to report their symptoms via text or voice and receive advice in multiple regional languages, using Natural Language Processing (NLP) to interpret and collect additional symptoms from users through dynamic follow-up interactions. An Al symptom analysis engine processes the user data collected via NLP, using machine learning models and clinical decision rules, to assign a severity level to each user's reported symptoms. Using the severity levels assigned by the Al symptom analysis engine, the system classifies symptom data into one of three urgency levels: self-care guidance, recommendation to see a doctor, or emergency alert. In addition to symptom classification and referral, the SEV-AI platform contains a variety of other modules, such as location-based healthcare discovery, tele consult services, medication reminders, exploration of laboratory reports using OCR and integration with IoT health monitoring devices and insurance assistance. The invention provides scalable, low-bandwidth platform for healthcare assistance to improve early decision-making regarding health and to enhance access to healthcare services for rural and underserved populations.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127177 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TRIBITE- EVERY BITE COUNTS.

(51) International classification	:G16H 20/60, G06Q 30/06, G06Q 50/22, G06Q 50/12, G09B 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India
(72)**Name of Inventor :**
1)KOUSHIK NAIDU TUNGALA
2)Dr. K. VIJAYAKUMAR
3)A. LAWRENCE SAHAYA SUNDAR

(57) Abstract :
The titled invention discloses a smart personalized meal delivery system for providing customized and nutritionally balanced meals according to individual fitness goals. The system includes a user interface module (1), processing unit (2), meal planning module (3), meal preparation unit (4), packaging module (5), delivery module (6), and an optional feedback and tracking system (7). User information including personal parameters, fitness goals, dietary preferences and restrictions is processed to determine calorie and macronutrient requirements. Customized meal plans are generated and corresponding meals are prepared using standardized recipes and controlled portions. The meals are packaged, labelled, stored and delivered according to subscription schedules. Feedback and progress information may be used to update future meal plans. The system thereby integrates personalized nutrition planning with actual meal preparation and scheduled delivery in a single end-to-end system.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127183 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MULTI- SENSOR BASED ASSISTIVE GUIDANCE SYSTEM FOR DEAF AND HEARING- IMPAIRED INDIVIDUALS.

(51) International classification	:G01S 15/931, G09B 21/00, G01S 5/18, G08B 21/02, G01S 15/86
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. T. ARUNPRASATH
2)VEMULAPATI PADMAJA
3)CHINTALA KAMAKSHI
4)EDAGOTTI SREELATHA
5)M. SWATHI

(57) Abstract :

The titled invention discloses a multi-sensor based assistive guidance system for deaf and hearing-impaired individuals comprising an environmental sound sensor (1), ultrasonic sensor (2), camera module (3), GPS receiver (4), microcontroller (5), Al processing unit (6), wireless communication module (7), vibration motor (8), led display (9) and smartphone application. The system continuously monitors environmental sounds, nearby obstacles, surrounding objects and navigation information. The Al processing unit (6) classifies important sounds such as vehicle horns, emergency sirens, alarms, doorbells and human speech, while camera-based processing identifies surrounding objects. Ultrasonic sensing detects nearby obstacles and GPS provides navigation information. The detected information is communicated through customized vibration patterns, visual indications and smartphone notifications. The system may be implemented as smart glasses, a wristband, belt or other portable wearable device. The integrated arrangement improves environmental awareness, personal safety, independent mobility and emergency response for deaf and hearing-impaired individuals.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127223 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART FOLLOW- ME SUTICASE USING ESP32- BASED BLUETOOH RSSI TRACKING WITH OBSTACLE- PRIORITY CONTROL.

(51) International classification	:G05D 1/02, A45C 5/14, A45C 5/04, A45C 15/00, A45C 13/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. K. VIJAYAKUMAR
2)A. LAWRENCE SAHAYA SUNDAR

(57) Abstract :
The titled invention discloses a smart autonomous suitcase comprising an ESP32- based controller (1), Bluetooth- communication interface (2), authorized Bluetooth-enabled user device (3), DC geared motors (4), L298M motor driver (5), motorized wheels (6), HC-SR04 ultrasonic obstacle sensor (7), rechargeable battery (8), LM2596 buck converter (9), and suitcase body (10). The ESP32-based controller (1) identifies the authorized user device (3) and obtains Received Signal Strength Indicator (RSSI) information to determine a relative proximity condition and generate corresponding io motor-control signals. PWM control may be used to regulate motor operation according to the RSSI condition. The ultrasonic obstacle sensor (7) monitors the travelling path, wherein an obstacle condition has higher priority than the RSSI-based following command and causes the controller (1) to stop the motorized wheels (6). The system further performs start-up safety verification and fail-safe stopping when the authorized user device (3) is not detected for a predefined condition, thereby providing autonomous and locally controlled transportation of personal luggage.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127226 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI- BASED MULTI-SOURCE FORENSIC EVIDENCE CORRELATION AND INTELLIGENCE SYSTEM

(51) International classification	:G06F 17/30, H04L 9/32, G06F 16/901, H04L 9/40, H04W 4/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India
(72)**Name of Inventor :**
1)PADAMATI SAI GANESH YADAV
2)RAYAPUDI NAGA SAI LAKSHMI KOUSHIK
3)CHILIPIREDDY PAVAN KUMAR REDDY
4)RANGAPPAGARI CHAITANYA REDDY
5)Dr. K. VENKATESH
6)Dr. N. SUBBULAKSHMI

(57) Abstract :
The titled invention discloses a mobile forensic intelligence system for processing digital evidence obtained from authorized mobile devices. The system collects heterogeneous forensic artifacts, including messages, call records, location data, files, applications, and web history, and converts them into a common normalized evidence structure while preserving original data, timestamps, hashes, and provenance. A cross-source correlation engine identifies relationships among evidence items based on temporal, contextual, entity, and location attributes. A reliability and provenance mechanism provides explainable assessment of evidence integrity and supporting information. The system further generates an evidence relationship graph and an automated chronological incident timeline. An Al-assisted analysis component identifies potentially significant patterns and generates evidence-linked forensic reports while distinguishing factual evidence, correlations, and Al interpretations. The invention improves the consistency, traceability, and efficiency of mobile forensic investigation and evidence analysis.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127229 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : REAL- TIME VEHICLE DETECTION AND COLLISION PREVENTION SYSTEM FOR U-TURN ROAD SECTIONS.

(51) International classification	:G08G 1/16, G08G 1/01, G01S 15/931, G08G 1/0967, G08G 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. ABHA DARGAR
2)Dr. SHASHI KANT DARGAR
3)SHAIK.SAMEER BASHA

(57) Abstract :

Title: REAL-TIME VEHICLE DETECTION AND COLLISION PREVENTION SYSTEM FOR U-TURN ROAD SECTIONS 5 The titled invention discloses real-time vehicle detection and collision prevention system for U-turn road sections comprising an Arduino Uno (1), ultrasonic sensors (2), LCD modules with I2C modules (3), LED indicators (4), solar panel (5), ACS7 12 current sensor module (6), potentiometers (7), resistors (8) and voltage regulation and 10 power section (9). The ultrasonic sensors (2) positioned on respective sides of a Uturn detect approaching vehicles and provide distance information to the Arduino Uno (1). The Arduino Uno (1) processes the detected information to determine vehicle speed and U-turn risk and provides corresponding output through the LED indicators (4) and LCD modules (3). The LED indicators (4) provide safe, caution and danger 15 indications, while the LCD modules (3) display numerical distance and speed information. The solar panel (5) supplies electrical power through the voltage regulation and power section (9), thereby providing a solar-powered arrangement for continuous monitoring and real-time warning of approaching vehicles at U-turn road sections.

No. of Pages : 25 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127230 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COLLAPSIBLE BARRIER GATE WITH SHRINKABLE AND FOLDABLE DESIGN FOR EASY PORTABILITY

(51) International classification	:E01F 13/04, E01F 13/06, E01F 13/02, E01F 13/00, E01F 13/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :
Title: COLLAPSIBLE BARRIER GATE WITH SHRINKABLE AND FOLDABLE DESIGN FOR EASY PORTABILITY 5 The present invention relates to a collapsible barrier gate having an expandable barrier structure and a foldable supporting arrangement for temporary access restriction. The barrier gate comprises a base foldable frame (1), detachable wheels (2), a main barrier frame (3), and an extensible barrier frame (4) carrying interconnected cross bars (7). 10 Foldable central hinges (6) permit angular movement of the cross bars during expansion and contraction of the barrier structure. An intermediate locking pin (5) retains the barrier frames at a selected extension position. Wheel hubs (8) receive the detachable wheels (2), while wheel insertion locks (9) retain the wheels during transportation. The base foldable frame (1) can be folded to reduce the storage 15 footprint, and the detachable wheels (2) can be removed when required. The arrangement provides an adjustable, foldable, and portable mechanical barrier suitable for temporary access control in institutional, commercial, industrial, and public environments. %&#xfffd;-

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127231 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TELESCOPIC STACK LADDER WITH LED ILLUMINATED FOOTSTEPS

(51) International classification	:E06C1/12, E06C1/383, F21V33/00, F21Y115/00, H02J7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :
Title: TELESCOPIC STACK LADDER WITH LED ILLUMINATED FOOTSTEPS 5 The present invention relates to a telescopic stack ladder with LED illuminated footsteps comprising a stack ladder base (1), extendable columns (2) and stackable footsteps (3) configured for compact storage and extended climbing access. The stackable footsteps (3) incorporate LED cluster arrays (5) arranged to illuminate the climbing surfaces under insufficient lighting conditions. A power charging port (4) is 10 connected to a charging circuit for recharging a battery accommodated within a battery compartment (7). Slider contact terminals (6) engage corresponding conductive tracks to maintain electrical continuity between relatively movable ladder sections. The integrated structural and electrical arrangement facilitates guided extension, compact retraction and illumination of the climbing surfaces using rechargeable electrical 15 power. The invention provides an integrated access ladder construction combining stackable footsteps, guided telescopic movement and battery-powered illumination

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127232 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SOLAR-POWERED EYE RINSING SPRAY SHOWER WITH INTEGRAL WATER STORAGE AND DRAIN OUTLET COLLECTING TANKS

(51) International classification	:A61H 35/02, A47K 3/28, B05B 1/18, A61H 33/04, A61H 33/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :
Title: SOLAR-POWERED EYE RINSING SPRAY SHOWER WITH INTEGRAL WATER STORAGE AND DRAIN OUTLET COLLECTING TANKS 5 The present invention relates to a solar-powered eye rinsing spray shower apparatus comprising a movable base (1), a bottom deck (2), a middle deck (3), a wash sink (4), an angular spray shower (5), a solar panel (6), a water flow control (7), a water pump (8), a battery port (9), and a drain water outlet (10). The apparatus incorporates a fresh 10 water storage tank and a separate wastewater collection tank within a common supporting structure. The solar panel (6) supplies electrical energy to a rechargeable battery through a charging circuit. The water pump (8) transfers fresh water to the angular spray shower (5), while the water flow control (7) regulates water delivery. The wash sink (4) directs discharged water into the wastewater collection tank through a 15 dedicated drainage passage. The drain water outlet (10) permits controlled removal of collected wastewater. The integrated arrangement provides portable eye rinsing with solar-powered water delivery and separate wastewater collection.n?

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127233 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EMERGENCY FIRST AID SERVING ROBOT WITH ACCIDENT SPOT LOCATION REACH AND LIVE INTERACTIVE FACILITY WITH PHYSICIAN

(51) International classification	:B25J 5/00, B25J 11/00, B25J 9/16, B25J 13/08, B25J 19/02	(71)**Name of Applicant :** **1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION** Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India (72)**Name of Inventor :** **1)Dr.S. SHASI ANAND** **2)Dr. P. JAYAKUMAR**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No	:NA	
Filing Date	:NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

Title: EMERGENCY FIRST AID SERVING ROBOT WITH ACCIDENT SPOT LOCATION REACH AND LIVE INTERACTIVE FACILITY WITH PHYSICIAN

5 The present invention relates to an emergency first-aid serving robot for providing medical assistance within industrial premises. The robot comprises traction wheels (1), a line-following sensor (2), a medicine placing rack (3), a utility storage rack (4), location identifier controls (5), a remote antenna (6), an LCD monitor (7), a live image 10 camera (8), a battery power port (9), a front door (10), and a proximity sensor (11). An accident activation arrangement associated with a machine bay initiates robot dispatch and identifies the designated destination. The robot follows a predefined floor path using the line-following sensor (2) and stops at the designated location through destination detection and controller-operated stopping. First-aid materials stored in the 15 racks are accessible through the front door (10). The live image camera (8), LCD monitor (7), and remote antenna (6) facilitate live audiovisual interaction between personnel at the accident location and a remotely located physician. The invention provides an integrated mobile arrangement for first-aid material delivery and remote medical guidance in industrial environments

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127234 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MOVABLE FLOOR DRYING AIR BLOWER WITH HEIGHT AND ANGULAR ADJUSTER DUCT

(51) International classification	:F26B 21/00, F24H 3/04, F26B 25/00, F26B 9/06, F26B 21/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :

Title: MOVABLE FLOOR DRYING AIR BLOWER WITH HEIGHT AND ANGULAR ADJUSTER DUCT 5 The present invention relates to a movable floor drying air blower comprising a movable chassis (1) supporting a hot air blower (2) mounted on a blower bed (7). A height adjustment jack (3) supports vertical positioning of the blower bed, while a linear actuator (8) provides controlled elevation adjustment relative to the chassis. An air 10 duct (4) communicates with the hot air blower and includes an angularly adjustable mounting arrangement for directing heated airflow towards a selected floor region. A control panel (5) provides operating controls for the blower and the actuator. A handle (6) facilitates movement of the apparatus between drying locations. The integrated arrangement enables adjustment of blower elevation and airflow direction to facilitate 15 drying of wet floor surfaces.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127235 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PH- RESPONSIVE DUAL-LAYER POLYMERIC DURG DELIVERY TABLET FOR TARGETED BREAST CANCER THERAPY.

(51) International classification	:A61K 9/28, A61K 9/20, A61K 31/704, A61K 31/337, A61K 45/06, A61P 35/00, B05D 1/02, B30B 11/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA - 604308. Tamil Nadu India
(72)**Name of Inventor :**
1)R. MOHAMED SUHAIL
2)M. VASANTH KUMAR
3)S. NARMATHA
4)SATHISH. I

(57) Abstract :
This invention relates to a pH-Responsive Dual-Layer Polymeric Drug Delivery Tablet for Targeted Breast Cancer Therapy. It typically includes an Outer Protective Polymeric Coating (1), a pH-Sensitive Inner Core Matrix (2), a Primary Anticancer Payload (3), a Phytochemical Sensitizing Agent (4), a Controlled-Release Rate-Controlling Matrix Modifier (5), a Pharmaceutical Tableting Excipient Complex (6), and a Biocompatible Solvent & Coating Vehicle Base (7). The pH-responsive dual-layer tablet delivers maximum chemotherapeutic efficacy by selectively targeting the acidic breast cancer microenvironment while keeping the drug encapsulated during transit, drastically reducing systemic side effects and cardiotoxicity. The pH-responsive duallayer tablet is primarily applied in targeted oral chemotherapy for breast cancer, as well as adapted localized therapies for colon, pancreatic, and ovarian cancers, inflammatory bowel disease management, and site-specific antibiotic delivery.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127239 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COLORIMETRIC PH-RESPONSIVE DENTAL GEL FOR EARLY DETECTION OF ENAMEL DEMINERALIZATION

(51) International classification	:A61Q 11/00, A61C 19/04, A61K 49/00, A61K 8/19, A61K 6/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA - 604308 Tamil Nadu India
(72)**Name of Inventor :**
1)RAJESHKUMAR. K
2)B. SHARMILA
3)Dr. K. PUVIYARASI
4)M. VASANTHKUMAR

(57) Abstract :
Title: COLORIMETRIC PH-RESPONSIVE DENTAL GEL FOR EARLY DETECTION 5 OF ENAMEL DEMINERALIZATION This invention relates to a Colorimetric pH-Responsive Dental Gel for Early Detection of Enamel Demineralization. It typically includes a Gel Base (1), a pHSensitive Indicator System (2), a Humectants (3), a Preservatives (4), a Stabilizing and Buffering Agents (5), a Neutralizing Agents (6), a Flavoring and Sweetening 1.0 Agents (7), and an Applicator System (8). The colorimetric pH-responsive dental gel provides a non-invasive, radiation-free, affordable, and rapid real-time visual diagnostic tool that identifies early enamel demineralization directly on tooth surfaces, enabling timely preventive intervention before permanent cavity formation occurs. The colorimetric pH-responsive dental gel is applicable in dental clinics, 15 paediatric dentistry, home-based oral health monitoring systems, rural health screening camps, and school preventive healthcare programs for-the early detection and risk assessment of enamel demineralization. Figure 1. Shows the structure of colorimetric dental gel application

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127257 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SIMPLE DENTAL MEDICINE APPLYING DEVICE FOR ACCURATE ORAL DRUG DELIVERY.

(51) International classification	:A61M 31/00, A61C 19/06, A61J 7/00, A61C 17/02, A61C 17/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :Assistant Professor, Pharmaceutics, TAKSHASHILA UNIVERSITY Ongur, Tindivanam Taluk, Villupuram District Tamilnadu India 604305 Tamil Nadu India
(72)**Name of Inventor :**
1)RAJESHKUMAR. K
2)B. SHARMILA
3)Dr. K. PUVIYARASI
4)JESIMA JAITHUN.Y

(57) Abstract :
The present invention discloses a simple dental medicine applying device for accurate oral drug delivery. The invention is designed to provide hygienic, controlled, and precise application of oral medicines onto affected areas inside the mouth. The device mainly consists of a medicine storage chamber (2), a medicine flow channel (5), a soft silicone applicator tip (1), a pressing mechanism (4), and an ergonomic handle body 1.0 (3). The medicine stored inside the medicine storage chamber (2) is delivered through the soft silicone applicator tip (1) via the medicine flow channel (5) using the pressing mechanism (4). The invention helps reduce medicine wastage, prevents contamination, improves treatment efficiency, and enhances patient comfort. The soft silicone applicator tip (1) safely contacts sensitive oral tissues without causing injury or irritation. The device is portable, reusable, low-cost, and easy to use for children, elderly patients, and normal users. The invention is suitable for hospitals, dental clinics, and household oral care applications.

No. of Pages : 14 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127258 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DIGITAL CONTROLLED HOT STERILIZING CHAMBER FOR SURGICAL TOOLS WITH MOTORIZED LOADER AND RETRACTION TRAY

(51) International classification	:A61L 2/07, A61L 2/26, A61L 2/06, A61L 2/20, A61L 2/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :
Title: DIGITAL CONTROLLED HOT STERILIZING CHAMBER FOR SURGICAL TOOLS WITH MOTORIZED LOADER AND RETRACTION TRAY 5 The titled invention discloses a digital controlled hot sterilizing chamber for surgical tools with a motorized loader and retraction tray comprising a portable base (1) on which the entire unit is mounted, a sterilizing chamber (2) filled with steam and kept in a closed condition, and a front door (3) that can be closed or opened when necessary. Slider rails (4) are fixed inside the chamber (2), and tools holding trays (6) are placed 10 over the rails (4), each tray (6) being fixed at its bottom side with traction motors (5) that move the trays (6) over the rails (4) to slide forward or reverse so as to move the trays (6) inside or outside the chamber (2). Each tray (6) is provided with steam passing grates (7) at the bottom side for circulation of steaM inside the tray (6). The status and operating parameters of the chamber (2), such as temperature, steam 15 pressure, and time limit, are shown on an LCD screen (8), and the movement of the trays (6), the setting temperature of the chamber (2), and the time limit are controlled by a control panel (9). The motorized retraction of the tray (6) outside the hot chamber (2) enables easy and safe pickup of the sterilized tools without the user inserting their hands inside the hot chamber. /body

No. of Pages : 15 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127259 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE EYE RINSING WATER SHOWER WITH AUTOMATIC HEIGHT ADJUSTMENT SINK

(51) International classification	:A61H 35/02, A47K 3/28, B05B 1/18, E03C 1/04, A61H 33/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :
Title: PORTABLE EYE RINSING WATER SHOWER WITH AUTOMATIC HEIGHT ADJUSTMENT SINK 5 The present invention relates to a portable eye rinsing water shower with automatic height adjustment. The apparatus comprises a movable base (1), a pedestal (2) and a linear extension column (3) supporting a rinsing assembly. A face eye position detection camera (4) detects the eye position of a user and communicates position 10 data to a height adjustment interface (5). The interface controls vertical movement of the extension column (3) to align a sink (6) and dual-flow shower nozzles (7) with the detected eye region. A water inlet (8) supplies rinsing water to the nozzles, while a drain water outlet (9) discharges collected water from the sink (6). The apparatus provides portable installation, automatic height positioning and integrated water 15 collection for eye rinsing applications.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127260 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FIRE SPOT DETECTION GUIDED FIRE EXTINGUISHER ROBOT WITH AUTOMATIC TRACTION

(51) International classification	:A62C 27/00, B25J 5/00, B25J 11/00, G05D 1/02, A62C 37/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA, PIN - 626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr.P. JAYAKUMAR

(57) Abstract :
The present invention relates to a fire spot detection guided fire extinguisher robot with automatic traction for supporting firefighting operations in hazardous environments. The robot comprises a master camera (1), a slave controller (2), a beacon light (3), traction wheels (4), and a line-following sensor (5) mounted on a supporting frame. A 10 utility storage box (6) houses control electronics, while a hose reel (7) supports storage and deployment of an extinguishing hose. A fire extinguisher holder receptacle (8) retains a fire extinguisher cylinder (9) during movement. The slave controller (2) coordinates camera-based fire detection with navigation feedback and traction commands. The integrated arrangement supports guided movement towards a 15 designated fire-response location and transportation of firefighting equipment.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127261 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WHEELCHAIR WITH PRESSURE-SENSING TYING STRAPS FOR TRANSPORTING PSYCHOLOGICALLY DISTURBED PATIENTS

(51) International classification	:A61G 5/10, A61G 5/12, A61G 5/00, A61B 5/00, A61B 5/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :
Title: WHEELCHAIR WITH PRESSURE-SENSING TYING STRAPS FOR TRANSPORTING PSYCHOLOGICALLY DISTURBED PATIENTS 5 The present invention relates to a wheelchair with pressure-sensing tying straps for supervised transportation of patients requiring postural support. The apparatus comprises a wheelchair frame (1), a sitting rest (2), a footrest assembly (3), and a pressure-sensing strap assembly (4) incorporating knee-supporting straps (5), thigh- 10 supporting straps (6), hand-supporting straps (7), and back-supporting straps (8). A microcontroller interface (9) receives signals from pressure-sensing elements (12) through electrical signal connections (13) and compares measured pressure values with predetermined upper pressure limits. Pressure abnormality indicators (10) provide location-associated alerts when monitored strap pressure exceeds the 15 applicable limit. An electrical power supply (14) powers the monitoring arrangement, and releasable fastening assemblies (15) facilitate controlled strap release. The apparatus supports caregiver awareness of strap-pressure conditions during supervised patient transportation.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127262 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : STEAM CHAMBER FOR POST-SURGERY TOOLS STERILIZING WITH SPRING LOADED RETRACTION RECOVERY TRAY

(51) International classification	:A61L 2/07, A61B 50/33, A61B 19/02, A61L 2/26, A61L 2/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :
Title: STEAM CHAMBER FOR POST-SURGERY TOOLS STERILIZING WITH SPRING LOADED RETRACTION RECOVERY TRAY 5 The titled invention discloses a steam sterilization chamber having a manually operable spring loaded recovery tray for surgery tools. The chamber includes a steam passing chamber drum (1), fixed base (2), steam producing chamber (3), steam vent (4), front door (5), surgery tools holding tray (6), perforated base (7), spring loader rails (8), push lock (9) and release lock (10). The surgery tools are placed on the perforated 10 base (7) of the tray (6), which is manually pushed inward along the spring loader rails (8) until the push lock (9) engages and retains the tray inside the chamber. After sterilization, the externally positioned release lock (10) is pressed to release the push lock (9). The spring loader rails (8) then assist the tray (6) in moving outward, allowing convenient recovery of the sterilized surgery tools while reducing the need for direct 15 hand access into the heated chamber.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127267 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUAL- LAYER FAST SISSOLVING SYNTHETIC ORAL FILM FOR IMMEDIATE AND SUSTAINED COUGH RELIEF.

(51) International classification	:A61K 31/485, A61K 9/70, A61K 47/38, A61P 11/14, A61K 47/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TAKSHASHILA UNIVERSITY
Address of Applicant :TAKSHASHILA UNIVERSITY ONGUR, TINDIVANAM TALUK, VILLUPURAM DISTRICT, TAMIL NADU, INDIA - 604308. 8220844786 jaijat07@gmail.com Tamil Nadu India
(72)**Name of Inventor :**
1)RAJESHKUMAR. K
2)B. SHARMILA
3)Dr. K. PUVIYARASI
4) M. VASANTHKUMAR

(57) Abstract :
The titled invention discloses a dual-layer fast dissolving synthetic oral film for immediate and sustained cough relief. The oral film is designed to rapidly dissolve in the oral cavity without the need for water, providing quick relief from dry cough, throatirritation, and allergic cough conditions. The first layer contains a fast-acting antitussive agent such as Dextromethorphan for immediate cough suppression, while the second layer contains soothing and prolonged-release agents for sustained throat comfort and anti-inflammatory action. The formulation improves patient compliance, especially among pediatric and geriatric patients, and offers portability, accurate dosing, and taste masking advantages over conventional cough syrups and tablets. The invention provides a novel, patient-friendly, and highly effective alternative for cough management with enhanced therapeutic outcomes.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127271 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SURGICAL SCISSORS AND KNIVES FRICTION CLEANING MACHINE WITH INTEGRAL GERMICIDE LIQUID SPRAY.

(51) International classification	:B08B 3/02, A61L 2/18, A61B 90/70, A61L 2/22, B08B 13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMILNADU, INDIA - 626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr.P. JAYAKUMAR

(57) Abstract :
The titled invention discloses surgical scissors and knives friction cleaning machine with integral germicide liquid spray to the easily and efficiently clean the surgical scissors and knives. This machine comprise of an outer protective enclosure (1) enclosing the operational components. A platform (2) is placed on the front end of the machine configured to receive and support surgical scissors and knives placed by healthcare workers during the cleaning cycle. A germicide liquid storage tank (3) and a water storage tank (4) are provided on the top of the enclosure to supply cleaning fluids through a liquid spray nozzles array (5) , positioned to directly supply pressurized streams of water and germicidal liquid onto the machine. A drive motor (6) coupled to a gear box (7) delivers controlled rotational torque to a pair of contra-rotating rollers (8), each fitted with an abrasive cloth belt (9) to grip and scrub both sides simultaneously. A waste water outlet (10) is located at the lower portion of the enclosure (1), and a centralized control panel (11) manages the operational parameters including fluid delivery speed, roller rotation, and spray duration.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127274 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRIFIED SURGICAL ASPIRATOR DEVICE WITH DIGITAL FLUID MEASUREMENT AND IOT MONITORING

(51) International classification	:A61M 1/00, A61M 27/00, A61B 90/00, A61B 5/00, A61B 5/145
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :

Title: ELECTRIFIED SURGICAL ASPIRATOR DEVICE WITH DIGITAL FLUID MEASUREMENT AND IOT MONITORING The titled invention discloses electrified surgical aspirator device with digital fluid 5 measurement and IOT monitoring to measure the quantity of body fluid and display the real time data on external computer and mobile through IOT control interface. The device features a portable base platform (1) housing dual body fluid collecting containers (2), weight sensor (6), a vacuum pump (5), an LCD display (4), and an IOT control interface (3). Prior to surgery, the Body fluid collecting containers (2) are 1.0 prefilled with water. During operation, the vacuum pump (5) suctions body fluids into the containers (2) while the weight sensor (6) continuously measure the quantity. An onboard microcontroller (3.1) processes these signals to calculate net body fluid volumes in real time, displaying local metrics on the LCD display (4) and transmitting data via Wi-Fi (3.2) and GSM modules (3.3) to a secure cloud server (3.4), web 15 dashboard monitor (3.5), and mobile application to the outside medical team. This invention transforms standard surgical aspiration into a smart, data-driven medical process that eliminates manual estimation errors and enhances safety and also helps the medical team to monitor the real time data of the fluid loss.

No. of Pages : 18 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127276 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ATTENDANCE CHECKING TERMINAL WITH FACE INDENTIY FOR EMERGENCY ALERT AND EVACUATION.

(51) International classification	:G07C 1/10, G06K 9/00, G07C 9/00, G06Q 50/26, H04N 7/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION
Address of Applicant :KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION, ANAND NAGAR, KRISHNANKOIL, SRIVILLIPUTTUR (VIA), VIRUDUNAGAR (DT), TAMIL NADU, INDIA-626126 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S. SHASI ANAND
2)Dr. P. JAYAKUMAR

(57) Abstract :
Title: ATTENDANCE CHECKING TERMINAL WITH FACE IDENTITY FOR EMERGENCY ALERT AND EVACUATION 5 The present invention relates to an attendance checking terminal with facial identity verification and emergency alert functionality. The apparatus comprises a movable kiosk (1) supporting a desk (2) carrying an entry input terminal (3), an overhead face identity camera (4), an LCD screen (5), an loT interface (6), and a control panel (7). A 10 processing arrangement compares captured facial features with enrolled identity records and generates time-stamped attendance events. The apparatus maintains occupancy information associated with recorded entry and confirmed exit events. Upon receiving an authenticated emergency alert, the apparatus displays evacuation instructions and transmits available attendance information through the loT interface. 15 The system identifies unconfirmed evacuation status for verification by authorized safety personnel. The integrated arrangement supports relocatable attendance monitoring and coordinated emergency information management in buildings and institutional facilities.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127303 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART RING SYSTEM FOR MONITORING PHYSIOLOGICAL AND MOVEMENT PATTERNS DURING STRUCTURED TEAM ACTIVITIES

(51) International classification	:A61B 5/00, A44C 9/00, A61B 5/0205, A61B 5/02, A61B 5/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SARANATHAN COLLEGE OF ENGINEERING
Address of Applicant :VENKATESWARA NAGAR, PANJAPPUR, THIRUCHIRAPPALLI, TAMIL NADU, INDIA-620012 Tamil Nadu India
(72)**Name of Inventor :**
1)S. Srimathi
2)S. Amirthavarshini
3)S. Celeste Hephzibah
4)A. Kanimozhi
5)C. Kethciyal Elizabeth
6)A. Meena Rojini
7)S. Tejasvini Blessy

(57) Abstract :
This invention discloses a finger-worn ring for collecting time-stamped physiological and movement signals during structured team activities such as group discussions, presentations and problemsolving exercises. The ring houses an optical pulse sensor positioned against the finger, a motion sensor, a microcontroller with a wireless communication module, and a power source, all within a compact band enclosure. The microcontroller samples the sensors and timestamps the readings, which are transmitted to a companion phone or computer. The companion software checks signal quality, marks and excludes um-eliable segments, divides the readings into time windows, and extracts features such as pulse-rate changes and movement intensity. A participant or facilitator may mark the start and end of an activity and optional events within it, such as a presentation or a group decision, so that the extracted features are aligned to an activity timeline for the participant's review. An optional separately worn badge may contribute synchronised speech-activity features. The system is intended as a research and self-reflection tool and does not itself determine personality traits or leadership ability from sensor readings alone.

No. of Pages : 8 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127307 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : POST- QUANTUM CRYPTOGRAPHY- BASED SECURE COMMUNICATION SYSTEM FOR IOT.

(51) International classification	:H04L 9/08, H04L 9/32, H04L 29/06, H04L 9/40, H04L 9/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRI AMARAAVATHI COLLEGE OF ARTS AND SCIENCE
Address of Applicant :Sri Amaraavathi College of Arts and Science, Thirumalainathanpatti, Velliyanai, Karur - 639 118, Tamil Nadu, India Tamil Nadu India

(72)**Name of Inventor :**
1)A. Joshua Sundar Raja
2)Dr. C. Menaka
3)R. Veerasamy
4)V. Saravanan
5)M. Suresh
6)S. Ramesh

(57) Abstract :

A Post-Quantum Cryptography-Based Secure Communication System for Internet of Things (loT) is designed to provide secure and reliable communication between interconnected smart devices and network services. The system addresses emerging security, challenges caused by the increasing computational capabilities of quantum computers, which may compromise conventional cryptographic techniques. Quantum-resistant cryptographic algorithms are incorporated to protect sensitive data transmitted among loT devices, gateways, cloud platforms, and communication networks. The proposed system supports secure device authentication, encrypted data transmission, key establishment, and message integrity verification. A security framework is used to authenticate participating devices before allowing communication and to establish protected communication channels for authorized data exchange. The system can further detect unauthorized access attempts and protect against data manipulation and interception during transmission. Lightweight and scalable security mechanisms may be applied to support loT devices with limited computational and memory resources. The framework is applicable to smart homes, healthcare systems, industrial automation, smart cities, connected vehicles, zit-A. other loT environments. By integrating post-quantum cryptographic techniques with loT communication infrastructure, the system improves resistance against future quantum computing-based attacks.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127313 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LOW-TEMPERATURE SYNTHESIS OF DEFECT-TUNED BATTERY-GRADE HARD CARBON FROM FOOD & VEGETABLE WASTE FOR LITHIUM-ION BATTERIES

(51) International classification	:H01M 10/0525, H01M 4/587, C01B 32/05, H01M 4/583, H01M 4/133
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHRIST UNIVERSITY
Address of Applicant :HOSUR ROAD, BENGALURU, KARNATAKA, INDIA-560029 Karnataka India
(72)**Name of Inventor :**
1)AROCKIA STEVAN RAJ V
2)ABHILASH KP

(57) Abstract :
The invention relates to the preparation of hard carbon anode material for rechargeable lithiumion batteries from heterogeneous wet food and vegetable waste. The method addresses the conversion of food waste, comprising cooked food and rice residues, vegetable residues, vegetable peels and other biodegradable food-derived matter into a valuable, high-performance hard carbon electrode material. The collected waste is segregated, washed, dried and sizereduced, followed by hydrothermal treatment at 200 °C to transform the heterogeneous precursor into a relatively homogenised carbon-rich intermediate as biochar. The biochar is subsequently subjected to controlled pyrolytic carbonisation below 1000 °C without chemical activation or external heteroatom doping, thereby producing an amorphous, disordered hard carbon with optimal defect concentration suitable for lithium-ion storage. It exhibits high capacity and high-rate capability. The process provides a waste-to-wealth route based on controlled thermal processing without reliance on petroleum-pitch precursors or hightemperature graphitisation.

No. of Pages : 26 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127317 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN EXPERIMENTAL INVESTIGATION OF PARTIAL REPLACEMENT ON BAGASSE AND RICE HUSK ASH IN FLY ASH BRICKS.

(51) International classification	:C04B 14/10, C04B 28/00, C04B 18/16, C04B 28/02, C04B 28/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.MG. Rajith Kumar
Address of Applicant :DEPARTMENT OF CIVIL ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L AND T BYPASS ROAD, COIMBATORE - 641062. Tamil Nadu India
2)Arun A
3)Abdul Rahuman M
4)Logeshwaran V
5)Sabari T
(72)**Name of Inventor :**
1)Dr.MG. Rajith Kumar
2)Arun A
3)Abdul Rahuman M
4)Logeshwaran V
5)Sabari T

(57) Abstract :
This project involves developing new construction materials by adding the bentonite, marble dust, and material derived from rice husks to the traditional concrete. The project seeks to come up with sustainable and cost-effective construction material that can reduce reliance on traditional resources and ensure efficient use of industrial and agricultural waste. • Bentonite is one of the natural clay materials whose presence in concrete can affect the workability and water retention capability of the concrete. Marble dust that is produced from the process of cutting and polishing marbles can be used as a fine filler or as a substitute for selected concrete constituents. Rice husk is one of the agricultural waste products, and its ashes can be used in concrete as a supplementary cementitious material due to its silica content. in •the proposed project, these materials are mixed in selected proportions with a view to studying their effects on the performance of the produced concrete. These materials will be weighed, mixed with cement, fine aggregate, coarse aggregate and water, then placed into appropriate moulds for compaction and curing. Evaluation of the strength and performance of the concrete produced is carried out using compressive, split tensile strength, flexural strength and water absorption tests. The results of the test are then compared with those of traditional concrete in order to identify the combination of alternative materials that gives the best results. This project is expected to help reduce wastage in the construction industry, encourage reuse of readily available materials, and ensure construction methods that are environmental friendly. The use of bentonite, marble dust and material made from rice husk could prove advantageous if used in optimum quantities. Nevertheless, the effectiveness will only be proven through testing.

No. of Pages : 3 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127318 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED SECURE ACCESS CONTROL SYSTEM FOR CLOUD APPLICATIONS

(51) International classification	:H04L 29/06, H04L 9/40, G06N 20/00, G06F 21/62, G06F 21/31
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRI AMARAAVATHI COLLEGE OF ARTS AND SCIENCE
Address of Applicant :SRI AMARAAVATHI COLLEGE OF ARTS AND SCIENCE, THIRUMALAINATHANPATTI, VELLIYANAI, KARUR, TAMIL NADU, INDIA-639118 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. M. Usha Ravichandran
2)Dr. K. Kumar
3)D. Chandrabalan
4)P. Rajalakshmi
5)M. Yogeshwaran
6)R. Kesavardhini

(57) Abstract :
An AI-Based Secure Access Control System for Cloud Applications is designed to provide intelligent and adaptive protection for cloud-based resources and services. The system addresses security limitations associated with traditional password-based and static access control mechanisms. Artificial Intelligence techniques are used to analyze access requests and identify potential security risks. Multiple parameters, including user identity, login behavior, device information, location, access time, and network characteristics, are evaluated. A machine learning module processes historical and real-time access data to identify normal and abnormal activities. An intelligent risk assessment mechanism generates a dynamic risk score for each access request. The calculated risk score is used to determine the appropriate access decision. Access may be granted, denied, temporarily restricted, or subjected to additional authentication based on the identified risk level. The system supports anomaly detection for identifying suspicious user behavior and unauthorized access attempts. Continuous monitoring can also be performed throughout an active user session. Any significant deviation from established behavioral patterns can trigger an automated security response. An adaptive authorization engine dynamically updates user permissions according to contextual and risk-related information.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127322 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVELOPMENT OF LOW-COST PCM- BASED THERMAL INSULATION CONCRETE FOR ENERGY EFFICIENT BUILDINGS

(51) International classification	:C04B 14/18, C04B 28/04, C04B 28/00, C04B 28/02, C04B 111/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRE ADETHYA. V
Address of Applicant :DEPARTMENT OF CIVIL ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BYPASS ROAD, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)Saranya. C.V
3)Ayyanar. S
4)Nagenthiran. S
5)Suriya. C
6)Vasugi Prakashram. V.P
(72)**Name of Inventor :**
1)SRE ADETHYA. V
2)Saranya. C.V
3)Ayyanar. S
4)Nagenthiran. S
5)Suriya. C
6)Vasugi Prakashram. V.P

(57) Abstract :
ABSTRACT The innovation of the present invention relates to the development of a phase change material (PCM)-based thermal insulation concrete using paraffin wax and expanded perlite as the principal thermal energy storage components. The invention presents a potential energy-efficient alternative to conventional concrete for sustainable building applications. The process involves heating paraffin wax with an approximate melting temperature of 38�C until it melts, followed by incorporating the molten wax into porous expanded perlite to produce a PCM�perlite composite. The composite is then incorporated into M20-grade concrete prepared using Portland Pozzolana Cement (PPC), fine aggregate, coarse aggregate, and water. The PCM�perlite composite replaces a portion of the coarse aggregate at replacement levels of 5%, 10%, and 15%, while conventional concrete is prepared as a control mix. The fresh concrete is manually mixed, cast into 150 mm x 150 mm x 150 mm cube moulds, and cured for the specified periods. The developed composite concrete is intended to absorb and release thermal energy during the melting and solidification of the paraffin wax, thereby helping regulate temperature fluctuations in buildings. The proposed material is designed for potential use in energy-efficient building components, particularly where thermal energy storage and reduced indoor temperature variations are desirable. The invention provides an approach to incorporating a low-cost phase change material into concrete while utilizing lightweight expanded perlite as a carrier for paraffin wax. This technology has the potential to support energy-efficient construction, reduce building cooling demand, and promote sustainable building-material development.

No. of Pages : 4 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127330 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE THERMAL ENERGY COUPLED INFERENCE MIGRATION FOR HETEROGENEOUS EDGE AI PROCESSORS

(51) International classification	:G06N 5/04, G06N 3/04, G06F 9/50, G06F 9/48, G06F 1/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VALLI PRIYADHARSHINI K
Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF CSE, K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY, SAMAYAPURAM, TRICHY, TAMIL NADU, INDIA-621112 Tamil Nadu India
2)Dr. B. Gunasundari
3)Nisha Soms
4)Dr. N.B. Mahesh Kumar
5)S Saranya

(72)**Name of Inventor :**
1)VALLI PRIYADHARSHINI K
2)Dr. B. Gunasundari
3)Nisha Soms
4)Dr. N.B. Mahesh Kumar
5)S Saranya

(57) Abstract :

ABSTRACT Heterogeneous edge Al processors integrate CPUs, GPUs, and Neural Processing Units (NPUs) to execute Al inference tasks efficiently. However, sustained workloads can increase temperature, energy consumption, and latency, leading to thermal throttling and reduced performance. This paper proposes a Predictive Thermal Energy Coupled Inference Migration (PTEC-IM) framework for thermal-aware and energy-efficient inference scheduling. An LSTM-based thermal prediction model forecasts processor temperatures using runtime features such as utilization, power consumption, frequency, and current temperature. A Multi-Objective Cost-Aware Migration (MOCAM) algorithm selects the optimal processor by balancing energy consumption, inference latency, thermal risk, and migration overhead. Additionally, Dynamic Voltage and Frequency Scaling (DVFS) adjusts processor frequencies based on predicted thermal conditions and workload demands, while hysteresis-based migration reduces unnecessary task switching. The framework aims to minimize energy consumption, prevent thermal throttling, and satisfy inference deadlines. Its performance can be evaluated against conventional scheduling methods using energy per inference, latency, peak temperature, migration overhead, and deadline satisfaction rate.

No. of Pages : 7 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127334 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CHLORAMINE DETECTION DEVICE WITH COLOR-CODE INDICATION AND AUTOMATIC ALARM TO RO PLANT/ CONTROL ROOM

(51) International classification	:A61M 1/16, C02F 1/44, G08B 21/18, A61M 1/14, A61M 1/36	(71)**Name of Applicant :** **1)ANCY YOUCHERISTA** Address of Applicant :Department of Dialysis Technology, Nehru Institute of Health Sciences Thirumalayampalayam ,Coimbatore Tamil Nadu India 641105 Tamil Nadu India **2)Dr.T.DhanaLakshmi** **3)Isha parveen** **4)Adishda** **5)sidheeq** **6)kaviyavarshini**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)ANCY YOUCHERISTA**
(33) Name of priority country	:NA	**2)Dr.T . DhanaLakshmi**
(86) International Application No	:NA	**3)Isha parveen**
Filing Date	:NA	**4)Adishda**
(87) International Publication No	: NA	**5)sidheeq**
(61) Patent of Addition to Application Number	:NA	**6)kaviyavarshini**
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The present invention discloses a chloramine detection device with color-code indication and automatic alarm capability for Reverse Osmosis (RO) plants and hospital control rooms, particularly tailored for hemodialysis applications. The device comprises an inline optical flow cell, LED-photodiode sensing architecture, an embedded microcontroller, a color-coded visual display panel (Green, Yellow, Red), and a multi-channel alarm and telemetry communication interface. The system continuously monitors feed and product water in real time for chloramine breakthrough. Upon detecting chloramine levels approaching or exceeding medical safety thresholds (0.1 mg/L), the device triggers visual color changes, activates local audio sirens, transmits emergency alerts to the central control room dashboard, and energizes a relay to trip RO plant operation or divert water flow. The invention provides an objective, automated, and continuous safety barrier against chloramine toxicity in clinical settings.

No. of Pages : 8 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641127337 A

(19) INDIA

(22) Date of filing of Application :26/10/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEBMIND AI: RAG-POWERED WEBSITE INTELLIGENCE AND CHATBOT PLATFORM

(51) International classification	:G06N 20/00, G06F 9/54, G06N 5/02, G06N 5/04, G06F 16/951
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. M.DEEPA
Address of Applicant :SRI, SHAKTHI INSTITUTE OF ENGINEERING AND THECHNOLOGY, SRI SHAKTHI NAGAR, L AND T BYPASS, CHINNIYAMPALAYAM POST, COIMBATORE - 641062 +91-422-23-69900 9976641131 archanagurusamy648@gmail.com Tamil Nadu India
2)Archana Gurusamy
3)Anika.V
4)Anisabarvin.A
5)Hemalatha. P. N

(72)**Name of Inventor :**
1)DR. M.DEEPA
2)Archana Gurusamy
3)Anika.V
4)Anisabarvin.A
5)Hemalatha. P. N

(57) Abstract :
WebMind-Al is an Al-powered web application developed to provide intelligent and efficient user interactions. It integrates artificial intelligence thrOugh the Grok Al API to process user inputs and generate relevant responses. The system offers a user-friendly web interface for easy access and interaction. Its backend manages application logic and communication between different system components. MongoDB is used to store and manage application data efficiently. The application aims to enhance usability and productivity through AI-assisted functionalities. It provides a flexible platform for accessing Al-powered services through the web. The modular architecture supports maintainability and future improvements. The WebMind-Al is an AI-powered web application developed to provide intelligent and efficient user interactions. It integrates artificial intelligence through the Grok Al API to process user inputs and generate relevant responses. The system offers a user-friendly web interface for easy access and interaction. Its backend manages application logic and communication between different system components. MongoDB is used to store and manage application data efficiently. The application aims to enhance usability and productivity through AI-assisted functionalities. It provides a flexible platform for accessing Al-powered services through the web. The modular architecture supports maintainability and future improvements. WebMind-Al is designed to deliver a smooth and accessible user experience. It combines web technologies, database management, and artificial intelligence in a unified system. The application supports efficient information processing and response generation. Its design allows for potential expansion with additional Al-based features. Overall, WebMind-Al aims to make Al-assisted web interactions more accessible and efficient.

No. of Pages : 6 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127351 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LEARNFOCUS- A COLLABORATIVE LEARNING AND MENTORSHIP PLATFOEM

(51) International classification	:G06Q 50/20, G09B 7/00, G09B 7/02, G09B 7/04, G09B 5/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MS. VIJAYALASHMI S
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L AND T BYPASS, CHINNIYAMPALAYAM POST, COIMBATORE -641062 +91-422-2369900 8220240828 vijayalakshmicse@siet.ac.in Tamil Nadu India
2)Prithika Sri S
3)Redhu Darsini G
4)Rithika K P

(72)**Name of Inventor :**
1)MS. VIJAYALASHMI S
2)Prithika Sri S
3)Redhu Darsini G
4)Rithika K P

(57) Abstract :
LearnFocus is a web-based collaborative learning and mentorship platform designed to improve the online learning experience. It connects students and mentors through study rooms, learning resources, and real-time communication. Mentors can create courses, upload educational materials, and manage student assessments. The platform includes an Al-based course assistance feature to help organize learning content. An integrated online compiler enables students to write and execute programs in multiple programming languages. Visible and hidden test cases support automated coding assessment and evaluation. Progress tracking helps students and mentors monitor-learning performance. LearnFocus combines learning, mentorship, coding practice, and assessment in a single platform.

No. of Pages : 6 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127358 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BUYWISE

(51) International classification	:G06F 21/57, G06F 11/36, G06F 9/44, G06F 11/30, G06F 8/41
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Kiruba S
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L AND T BY- PASS, CHINNIYAMPALAYAM POST, COIMBATORE-641062. Tamil Nadu India
2)Keerthana S
3)Lathika P
4)Mousika G
(72)**Name of Inventor :**
1)Kiruba S
2)Keerthana S
3)Lathika P
4)Mousika G

(57) Abstract :

The present invention describes an intelligent system for monitoring and analyzing software dependencies using automated software analysis techniques. Software projects often rely on multiple external libraries and packages, which may contain security vulnerabilities, outdated versions, or compatibility issues that can affect the reliability and safety of applications. Identifying these issues manually requires considerable time and technical expertise. The proposed system addresses this challenge by providing an automated solution that enables developers to upload software dependency files and receive clear, structured insights into the health of their projects. The system begins by accepting software project dependency files such as "package.json" and performs file validation and dependency extraction to identify the packages used in the project. The extracted dependency information is then processed and analyzed by comparing package versions and available vulnerability information to identify potential security risks, outdated dependencies, and other dependency-related issues. Furthermore, the system performs automated risk assessment by evaluating the detected dependency issues and classifying them into different risk levels, such as Safe, Warning, High, and Critical. Based on the analysis results, the system calculates an overall project health score that provides developers with an overview of the dependency status and software project reliability. The system also supports GitHub repository monitoring, enabling developers to analyze dependencies directly from their repositories and re-evaluate project health when dependency-related changes are detected through GitHub webhook integration. The results are presented through a user-friendly web interface that displays project details, dependency information, detected vulnerabilities, outdated packages, risk classifications, and health scores in an organized manner. Additionally, the system maintains project and scan records in a database, allowing users to securely access previous analysis results and generate reports for documentation and project maintenance. The proposed solution significantly reduces the time and effort required for manual dependency checking and helps developers identify potential software risks at an early stage. By integrating automated dependency analysis with a web based monitoring and reporting architecture, the invention provides an efficient and accessible platform for improving software dependency management and project health evaluation. This system can be further extended with features such as support for multiple programming languages, software bill of materials generation, license analysis, real-time notifications.

No. of Pages : 14 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127370 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SMART SLING BAG WITH INTEGRATED BEVERAGE TEMPERATURE REGULATION AND PORTABLE POWER MANAGEMENT SYSTEM

(51) International classification	:F25B 21/02, F25B 21/04, F25D 11/00, A45F 3/04, B65D 81/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahalakshmi G
Address of Applicant :FOOTWEAR DESIGN & DEVELOPMENT INSTITUTE (FDDI), PLOT NO E1, 7TH MAIN ROAD, SIPCOT INDUSTRIAL PARK, IRUNGATTUKOTTAI, TAMILNADU-602117 Tamil Nadu India
2)Preethi Selvakumar
3)Vishva Kumar M
4)Priyadharshini R
5)FOOTWEAR DESIGN AND DEVELOPMENT INSTITUTE, CHENNAI
(72)**Name of Inventor :**
1)Mahalakshmi G
2)Preethi Selvakumar
3)Vishva Kumar M
4)Priyadharshini R

(57) Abstract :
The present invention relates to a smart sling bag incorporating an integrated beverage temperature conditioning and portable power management system configured to selectively heat or cool a beverage container during transportation and personal use. The invention comprises a sling bag enclosure (11) having a beverage compartment (14,a personal storage compartment (12), an electronics compartment (13) and an adjustable carrying strap (15), wherein the beverage compartment is configured to accommodate beverage containers of varying sizes, capacities and material compositions. A thermoelectric temperature-conditioning assembly comprising a Peltier module (6), thermal dissipation assembly (7) and cooling fan (8) is thermally coupled to the beverage compartment and configured to actively transfer thermal energy toward or away from a beverage container to facilitate controlled heating and cooling operations. An electronic control architecture (10) and regulates operation of the temperature-conditioning assembly in accordance with userdefined settings. Electrical power is supplied through a rechargeable battery (9), which additionally functions as a portable power source for external electronic devices. The system further incorporates wireless communication functionality enabling interaction with a mobile application for monitoring and control of beverage-conditioning operations. The invention provides a compact, lightweight and wearable utility platform integrating active beverage temperature conditioning, personal storage and portable power management within a single self-contained architecture.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127443 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT AUTO-MODULATING RESISTANCE DEVICE WITH REAL-TIME MOVEMENT ANALYSIS FOR PHYSIOTHERAPY REHABILITATION

(51) International classification	:A63B 24/00, A61B 5/11, A61H 1/02, A63B 21/005, A63B 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY
(72)**Name of Inventor :**
1)SATHISH S
2)DR. P.G.MAHESH KUMAR
3)DR.P.V.HARI HARA SUBRAMANYAN

(57) Abstract :
TITLE: INTELLIGENT AUTO-MODULATING RESISTANCE DEVICE WITH REAL-TIME MOVEMENT ANALYSIS FOR PHYSIOTHERAPY REHABILITATION APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention discloses a wearable resistance device for rehabilitation exercises, comprising a cuff housing selectively engageable resistance modules, an electromechanical engagement and locking mechanism, a motion-sensing unit, a processing and control unit, a feedback system, a user interface, and a rechargeable power supply. The motion-sensing unit includes an accelerometer and a gyroscope configured to acquire movement data including angular displacement, range of motion, angular velocity, repetition count, movement duration, and movement consistency. The processing and control unit compares the data with predefined therapeutic criteria and generates a control decision to retain, reduce, or increase resistance. The engagement mechanism operates in response to the control decision to modify the resistance configuration. The feedback system provides visual, auditory, or haptic indications, while the user interface presents exercise information and permits configuration of therapeutic parameters. Movements are reassessed to provide closed-loop adaptation of rehabilitation load according to measured movement performance.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127444 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : REAL-TIME ADAPTIVE NEUROVISUAL PERTURBATION-RECOVERY SYSTEM FOR PREDICTIVE FUNCTIONAL STABILITY ASSESSMENT

(51) International classification	:A61B 5/00, A61B 5/11, G09B 19/00, A61B 5/16, A61B 3/113	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India **2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)DR. P.G.MAHESH KUMAR**
(33) Name of priority country	:NA	**2)DR.P.V.HARI HARA SUBRAMANYAN**
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

TITLE: REAL-TIME ADAPTIVE NEUROVISUAL PERTURBATION- RECOVERY SYSTEM FOR PREDICTIVE FUNCTIONAL STABILITY ASSESSMENT APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention discloses a real-time adaptive neurovisual perturbation-recovery system for predictive functional stability assessment. A visual stimulus display unit presents controlled micro-level visual perturbations during ongoing task performance. A sensor assembly comprising eye-tracking, pupil and eyelid, peripheral detection, physiological, posture, and environmental sensors acquires the user's responses. A signal conditioning unit filters, synchronizes, and normalizes the signals. Processing circuitry extracts descriptors of perturbation-response and recovery and compares them with an individualized baseline. Recovery analysis circuitry determines stabilization latency, recovery slope, residual instability, and correction efficiency. Predictive stability circuitry generates a functional stability score, instability-risk level, recovery profile, or overload indication. A feedback controller adjusts subsequent perturbations according to preceding recovery responses, and an output unit provides graded alerts, recommendations, or predictive stability maps. The system thereby operates as a closed adaptive sequence that predicts functional instability before overt deterioration in task performance.

No. of Pages : 28 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127445 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : STABILIZED PAPER-BASED MICROFLUIDIC DEVICE FOR NON-INVASIVE POINT-OF-CARE ANTI-MÜLLERIAN HORMONE SCREENING

(51) International classification	:B01L 3/00, G01N 33/74, C12Q 1/68, G01N 33/76, G01N 33/53
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)DR. A JOSEPHINE
2)MRS. POORNIMA R VARMA
3)MRS. RACHANA RAVEENDRAN
4)DR. D. DINESH ROY
5)DR. V. SUREKA VARALAKSHMI

(57) Abstract :
TITLE: STABILIZED PAPER-BASED MICROFLUIDIC DEVICE FOR NON-INVASIVE POINT-OF-CARE ANTI-MÜLLERIAN HORMONE SCREENING APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses a stabilized paper-based microfluidic analytical device for minimally invasive point-of-care detection of Anti-Müllerian Hormone (AMH) from a finger-prick blood sample for ovarian reserve screening. The device comprises a cellulose-fiber paper substrate patterned with hydrophobic wax barriers defining capillary-driven microfluidic channels, a sample inlet configured to receive not more than 20 μL of blood, and an integrated plasma-separation membrane having a vertical wicking architecture for filtering red blood cells without centrifugation. A downstream test zone comprises AMH-specific synthetic DNA aptamers stabilized in a trehalose-sucrose matrix and configured to selectively bind AMH. A control zone comprises a pH-sensitive indicator for confirming proper test execution. Gold nanoparticle conjugates provide a colorimetric response corresponding to AMH concentration. The resulting color gradient is visually interpretable and digitizable using a smartphone camera and interpretation algorithm. A fiducial marker facilitates standardized image capture, while a plastic backing strip provides mechanical rigidity, enabling portable point-of-care ovarian reserve screening.

No. of Pages : 24 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127446 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE SEGMENTAL EXPIRATORY MAPPING DEVICE FOR REGION-SPECIFIC POSITIVE EXPIRATORY PRESSURE REHABILITATION

(51) International classification	:A61B 5/087, A61B 5/097, A61M 16/00, A61B 5/08, A61B 5/091	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India **2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY** (72)**Name of Inventor :** **1)TAMIZHNITHI MP** **2)DR. P.G.MAHESH KUMAR** **3)DR.P.V.HARI HARA SUBRAMANYAN**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

TITLE: ADAPTIVE SEGMENTAL EXPIRATORY MAPPING DEVICE FOR REGION-SPECIFIC POSITIVE EXPIRATORY PRESSURE REHABILITATION APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention relates to an adaptive segmental expiratory mapping and dynamic airflow modulation device for region-specific positive expiratory pressure (PEP) rehabilitation. The device comprises a mouthpiece, an internal conduit, pressure and flow sensors, a processing chamber containing a microcontroller, a dynamic airflow modulation chamber, a directional turbulence-conditioning chamber, a pressure-stabilization chamber, a distal atmospheric outlet, and a therapeutic guidance indicator. The sensors measure expiratory parameters including pressure trajectory, airflow magnitude, expiratory duration, waveform stability, oscillatory characteristics, and breath-to-breath variation. The microcontroller processes the measured parameters to generate a multidimensional expiratory pattern, compares with stored therapeutic criteria, and generates control signals to adjust airflow resistance and geometry through mechanically actuated elements. The turbulence-conditioning and pressure-stabilization chambers facilitate controlled airflow conditioning and discharge. The therapeutic guidance indicator provides visual feedback regarding expiration adequacy, excessive effort, and expiratory instability, enabling a closed-loop rehabilitation process through respiratory monitoring, adaptive airflow modulation, patient guidance, and subsequent-breath reassessment.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127447 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SOLAR-INTEGRATED LOWER LIMB REHABILITATION SYSTEM WITH ADAPTIVE ROBOTIC ASSISTANCE

(51) International classification	:A61H 3/00, A61H 1/02, A63B 23/04, A61B 5/11, A61F 2/68	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(31) Priority Document No	:NA	**2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY**
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)SATHISH S**
(86) International Application No Filing Date	: :01/01/1900	**2)DR. P.G.MAHESH KUMAR** **3)DR.P.V.HARI HARA SUBRAMANYAN**
(87) International Publication No	: NA	**4)DR. PARTHASARATHY RANGANATHAN**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
TITLE: SOLAR-INTEGRATED LOWER LIMB REHABILITATION SYSTEM WITH ADAPTIVE ROBOTIC ASSISTANCE APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT A solar-powered adaptive robotic gait rehabilitation system for energy-efficient lower-limb assistive therapy is disclosed. The system comprises a supportive frame with a harness or body-support arrangement and lockable mobility wheels, a photovoltaic energy-harvesting unit, a rechargeable energy-storage unit, and a power-management controller that regulates charging, determines energy-state variables, and distributes regulated power. A lower-limb robotic actuator assembly linked to the hip and/or knee regions of a patient generates controlled assistive torque during selected portions of the gait cycle. A biomechanical sensor system comprising foot-pressure, joint-angle, and load sensors detects gait-phase characteristics. An adaptive control unit identifies gait events, determines the magnitude and timing of assistance, coordinates actuator output with available energy, and executes therapist-assist, autonomous adaptive-assist, and energy-saver modes selectable through a user interface. The system modifies assistance in relation to patient performance and real-time energy availability, enabling portable lower-limb gait rehabilitation without continuous dependence on a conventional electrical supply.

No. of Pages : 27 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127453 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR ONE-POT SYNTHESIS OF GEMINAL TRIFLUOROMETHYL ALCOHOLS

(51) International classification	:C07C 33/46, C07C 29/143, C07C 49/84, C07C 45/67, C07C 45/63
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology (IIT) Hyderabad
Address of Applicant :IITH Road, Near NH65, Sangareddy, Kandi, Telangana 502285, India Sangareddy Telangana India
(72)**Name of Inventor :**
1)Natte, Kishore
2)Bhatt, Tarun
3)Yaswanth, Kokkiripati
4)Kalevaru, Venkata Narayana
5)Wohlrab, Sebastian

(57) Abstract :
ABSTRACT A PROCESS FOR ONE-POT SYNTHESIS OF GEMINAL TRIFLUOROMETHYL ALCOHOLS The present invention provides a chemoselective and efficient process for the synthesis of geminal trifluoromethyl alcohols from trifluoromethyl ketones. More particularly, the present invention provides a process for one-pot synthesis of geminal trifluoromethyl alcohols from readily available trifluoromethyl ketones by integrating carbon–carbon bond formation and reduction in a single reaction system. Moreover, said process produces products that exhibit high chemoselectivity, enabling selective reduction of trifluoromethyl ketones in the presence of other sensitive functional groups such as aldehyde, ketone, and nitro, etc.

No. of Pages : 58 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127454 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART ERGONOMIC HANDBAG WITH AI-BASED LOAD MANAGEMENT

(51) International classification	:A45F 3/04, A45C 13/00, A45C 13/30, A45C 15/00, A45C 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)DR. INDU PURUSHOTHAMAN
2)DR. N. ARUNAGIRINATHAN
3)DR. C.VALLI NACHIYAR
4)DR. K. S SRIDEVI SANGEETHA
5)DR. K. SANGEETHA

(57) Abstract :
TITLE: SMART ERGONOMIC HANDBAG WITH AI-BASED LOAD MANAGEMENT APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses an adaptive intelligent handbag system for ergonomic weight management and personalized load distribution. The system comprises a handbag body having internal compartments, load sensors for continuously monitoring carried weight and compartment-wise load distribution, shoulder straps having pressure sensors for detecting shoulder loading, and inertial measurement sensors for monitoring posture, movement and walking patterns. An artificial intelligence processing module receives and processes the sensor information to determine an ergonomic load index, assess musculoskeletal strain and posture-related injury risk, and provide personalized safe weight recommendations based on a user profile and carrying history. An adaptive internal partitioning mechanism comprising movable dividers and micro-motors automatically adjusts compartment arrangement according to detected load distribution. A dynamic alerting system provides vibration, visual and mobile alerts corresponding to detected carrying conditions. A personalized mobile dashboard presents weight, load distribution, shoulder pressure, posture and fatigue information with ergonomic recommendations and historical records, thereby facilitating continuous ergonomic monitoring and adaptive load management during use.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127455 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED OROFACIAL–RESPIRATORY BIOFEEDBACK SLEEP SYSTEM FOR CORRECTING MOUTH BREATHING AND IMPROVING FEEDING EFFICIENCY IN CHILDREN

(51) International classification	:A61B 5/08, A61M 21/00, A61M 21/02, A61B 5/00, A61B 5/087	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India **2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)MAHMUDAA KN**
(32) Priority Date	:NA	**2)DR. P. V. HARI HARA SUBRAMANYAN**
(33) Name of priority country	:NA	**3)DR. P.G.MAHESH KUMAR**
(86) International Application No Filing Date	: :01/01/1900	**4)DR PARTHASARATHY RANGANATHAN**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

TITLE: AN INTEGRATED OROFACIAL–RESPIRATORY BIOFEEDBACK SLEEP SYSTEM FOR CORRECTING MOUTH BREATHING AND IMPROVING FEEDING EFFICIENCY IN CHILDREN APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention relates to an integrated orofacial–respiratory biofeedback sleep system for facilitating nasal breathing behavior and improving orofacial functional control in children. The system comprises a therapeutic sleep-support pillow having contoured head, cervical, and lateral support regions, respiratory sensing elements, positional monitoring sensors, a signal processing and control unit, feedback modules, a removable orofacial training component, and a caregiver and clinician interface. The respiratory sensing elements acquire signals indicative of nasal or mouth breathing, while the positional monitoring sensors detect head and neck position. The control unit processes the acquired signals, classifies breathing states, and initiates graded non-obstructive tactile or vibrotactile feedback upon detecting predefined mouth-breathing patterns. Subsequent respiratory signals are reassessed to modify, repeat, or cease feedback according to the detected response and clinician-defined safety thresholds. The interface provides breathing mode, episode count, positional information, feedback activation history, and session trends, thereby enabling integrated respiratory monitoring and supervised orofacial training.

No. of Pages : 29 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127456 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A CLOSED-LOOP NEUROPOSTURE REHABILITATION SYSTEM FOR LONGITUDINAL REHABILITATION TRACKING

(51) International classification	:A61B 5/11, A63B 24/00, A61H 1/02, A61B 5/00, G16H 20/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY
(72)**Name of Inventor :**
1)CHINNU. N
2)DR. P.G.MAHESH KUMAR
3)DR. P. V. HARI HARA SUBRAMANYAN
4)DR. PARTHASARATHY RANGANATHAN

(57) Abstract :
TITLE: A CLOSED-LOOP NEUROPOSTURE REHABILITATION SYSTEM FOR LONGITUDINAL REHABILITATION TRACKING APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention discloses a closed-loop neuroposture rehabilitation system for personalized postural correction and longitudinal rehabilitation tracking. The system comprises a wearable support assembly having shoulder straps and an adjustable stabilizing strap supporting a centrally located control module, and an inertial measurement unit having an accelerometer and a gyroscope fixed in a reproducible relation to a user's trunk. The control module comprises a processor, memory, drift-intelligence engine, decision and feedback controller, feedback actuator, and rehabilitation tracker. The processor establishes a personalized baseline from a therapist-approved reference posture and compares real-time posture data with the baseline to characterize postural drift based on magnitude, duration, recurrence frequency, and cumulative persistence. The decision and feedback controller provides graded vibration, auditory, visual, or multimodal corrective biofeedback and evaluates user response. The rehabilitation tracker records correction latency, drift reduction, maintenance duration, recurrence interval, and compliance, while rehabilitation thresholds are progressively modified according to user improvement for longitudinal rehabilitation.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127457 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DIAPHRAGM-GUIDED PEDIATRIC FEEDING AND RESPIRATORY COORDINATION SYSTEM

(51) International classification	:A61M 16/00, A61B 5/08, A61B 5/087, A61B 5/00, A61B 5/113
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY
(72)**Name of Inventor :**
1)POOJASREE .M
2)DR. P. V. HARI HARA SUBRAMANYAN
3)DR P. G. MAHESH KUMAR
4)DR PARTHASARATHY RANGANATHAN

(57) Abstract :
TITLE: DIAPHRAGM-GUIDED PEDIATRIC FEEDING AND RESPIRATORY COORDINATION SYSTEM APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention discloses a diaphragm-guided pediatric feeding and respiratory coordination system configured to monitor respiratory movement and adaptively regulate feeding flow. A wearable diaphragm sensing assembly detects thoracoabdominal movement and generates a respiratory signal, which is processed to determine respiratory cycle timing, respiratory rate, movement amplitude, pauses, irregularity, and respiratory stability. A respiratory-pattern analysis controller classifies the respiratory condition according to predetermined or individualized respiratory-phase and stability rules and generates control signals for feeding regulation. An adaptive feeding interface includes a feeding reservoir, standardized connection, and controllable flow-regulation mechanism having an electronically controllable actuator coupled to an adjustable valve or equivalent flow-modulating element. The system selectively permits, reduces, restricts, or interrupts feeding flow according to the respiratory condition. A caregiver interface provides respiratory and feeding-flow information, while manual override, fail-safe, and wireless communication functions provide control, safety, and monitoring. The closed-loop system supports individualized suck-swallow-breathe coordination during pediatric feeding.

No. of Pages : 27 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127458 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART ORAL-MOTOR TRAINING PACIFIER WITH RESPIRATORY MONITORING FOR NICU INFANTS

(51) International classification	:A61J 17/00, A61B 5/08, A61B 5/087, A61B 5/00, A61M 1/00	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India **2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)SAFINA AMRIN A**
(33) Name of priority country	:NA	**2)DR. P. V. HARI HARA SUBRAMANYAN**
(86) International Application No Filing Date	: :01/01/1900	**3)DR P. G. MAHESH KUMAR** **4)DR PARTHASARATHY RANGANATHAN**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

TITLE: SMART ORAL-MOTOR TRAINING PACIFIER WITH RESPIRATORY MONITORING FOR NICU INFANTS APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention discloses a smart oral-motor training pacifier system for neonatal intensive care unit (NICU) infants. The system comprises a therapeutic oral-motor nipple having predetermined surface features of grooves, ridges, raised tactile zones or variable wall thickness for structured oral stimulation during non-nutritive sucking, and a nipple connector and shield assembly with ventilation openings and a sealed compartment isolated from infant contact. Sucking-related sensors detect sucking pressure, rhythm, frequency, duration, burst length and pause intervals, while a respiratory-sensing arrangement acquires respiratory cycle timing, pauses and irregularities concurrently. A processing and control module performs signal conditioning and synchronized temporal analysis, correlating sucking events with respiratory cycles to evaluate suck-respiration coordination. A rechargeable power source, a caregiver feedback system providing real-time alerts, and a caregiver display interface complete the system, converting pacifier use into an integrated therapeutic and monitoring session for supervised oral-motor training and coordination assessment.

No. of Pages : 31 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127467 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A TERAHERTZ (THz) LOGIC METADEVICE AND METHOD OF PERFORMING PARALLEL LOGIC OPERATIONS

(51) International classification	:G02F 3/00, G02F 3/02, H03K 19/14, H01L 31/0232, G02F 1/015
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD
Address of Applicant :IIT Hyderabad Road, Near NH-65, Sangareddy, Kandi, Telangana, 502284, India. Hyderabad Telangana India
(72)**Name of Inventor :**
1)YOGESH KUMAR SRIVASTAVA
2)PREM PAL
3)VANLAL RINFELA
4)BHAWANA ANDOLA

(57) Abstract :
The present invention discloses a terahertz (THz) logic metadevice (100) and a method (400) of performing parallel logic operations. The metadevice (100) includes a dielectric substrate (110), a subwavelength planar resonator array (120) having resonant unit cells (122), and a photoactive semiconductor layer (130). A first input channel (140) selects between orthogonal linear polarization states of incident THz radiation to produce different resonant responses. A second input channel (150) controls optical excitation of the photoactive semiconductor layer (130) to modify a THz transmission response. A logic-decision module evaluates normalized transmission amplitudes within multiple operational frequency sub-bands, compares the amplitudes with respective thresholds, and assigns binary logic outputs. Different combinations of polarization and optical excitation generate distinct transmission responses, enabling simultaneous execution of different Boolean logic functions, including NAND, OR, and NOR, across the operational frequency sub-bands. TO BE PUBLISHED WITH FIGURE 1

No. of Pages : 42 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127509 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A NANOCOMPOSITE AND A PROCESS FOR ITS PREPARATION

(51) International classification	:C02F 1/30, C02F 1/28, C02F 1/72, B01J 35/10, B01J 20/30	(71)**Name of Applicant :** **1)SRM UNIVERSITY** Address of Applicant :Amaravati, Mangalagiri, Andhra Pradesh-522502, India Guntur Andhra Pradesh India (72)**Name of Inventor :** **1)CHAURASIYA, ARBIND** **2)SELVASEMBIAN, RANGABHASHIYAM**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT A NANOCOMPOSITE AND A PROCESS FOR ITS PREPARATION The present disclosure provides a nanocomposite. The nanocomposite exhibits high adsorption capacity, enhanced surface area, mesoporous structure, and visible-light-driven photocatalytic activity, enabling efficient removal and degradation of methylene blue dye from contaminated water. The present disclosure also provides a process for the preparation of a nanocomposite. The process is simple, scalable, and environmentally sustainable. FIGURE 1

No. of Pages : 56 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127524 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ADAPTIVE WEARABLE SYSTEM FOR INTELLIGENT POSTURE CORRECTION WITH MULTIZONE HAPTIC FEEDBACK AND TEMPERATURE CONTROLLED THERMAL THERAPY

(51) International classification	:A61B 5/11, A61F 7/00, A61B 5/00, A61H 23/02, A61N 1/36	(71)**Name of Applicant :** **1)P Brindha** Address of Applicant :8/46, Kamaraj Nagar, IInd Street, P.N.Road, Tirupur Tamil Nadu India **2)R. SARATHY** **3)J.PIRAJIN**
(31) Priority Document No	:NA	**4)S.TAMILSELVAN**
(32) Priority Date	:NA	**5)S.R.SYED MOHAMMED**
(33) Name of priority country	:NA	(72)**Name of Inventor :**
(86) International Application No Filing Date	: :01/01/1900	**1)P Brindha** **2)R. SARATHY**
(87) International Publication No	: NA	**3)J.PIRAJIN**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**4)S.TAMILSELVAN** **5)S.R.SYED MOHAMMED**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present Invention discloses a wearable posture monitoring and therapeutic feedback device designed for real-time postural correction and controlled thermal therapy. The system integrates a sensor input subsystem comprising an accelerometer sensor, an NTC thermistor, and a push button to acquire physical posture parameters and thermal status at the edge. A central microcontroller, preferably an ESP32-C3, performs local edge processing on the acquired sensor data to minimize operational latency and assess postural alignment. The device features an actuator subsystem comprising a MOSFET driver bank operatively coupled to drive multiple vibration motors for instantaneous haptic posture correction alerts, and a flexible heating element configured to deliver localized heat therapy. To ensure skin-safe thermal operation, a dedicated thermal cutoff switch is integrated directly with the heating element to interrupt power during over-temperature conditions. A user interface subsystem comprising an OLED display and an RGB LED indicator provides real-time visual metrics, operating modes, and alert statuses. Power is managed through a self-contained power supply subsystem including a rechargeable Li-Po battery, a protected TP4056 charging circuit, and a 5V boost converter providing regulated power flow to system modules. The invention provides a compact, energy-efficient, and fail-safe wearable architecture combining active biomechanical feedback with ergonomic thermal relief.

No. of Pages : 24 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127527 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR RELIABILITY-ADAPTIVE PREDICTION OF AIRBORNE DISEASE RISK USING HEALTH DATA

(51) International classification	:G16H 50/30, G06N 20/00, G16H 50/20, A61B 5/00, G16H 10/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.S.John Joseph
Address of Applicant :Head of the Department & Assistant Professor (SL-G) Department of AI&DS, SRM Bharathidasan College of Engineering & Technology, SRM Bharathidasan Campus, Pudukkottai District. Tamil Nadu, India Tamil Nadu India
2)Mrs. A.Kavipriya
3)Mrs. A.Joyce
(72)**Name of Inventor :**
1)Dr.S.John Joseph
2)Mrs. A.Kavipriya
3)Mrs. A.Joyce

(57) Abstract :
The present invention relates to a system and method for reliability-adaptive prediction of airborne disease risk using health-related data and machine learning techniques. The system is designed to assess the reliability of input data before and during the disease risk prediction process. It comprises a data input module, a preprocessing module, a data reliability assessment module, a prediction module, and an output module. The data input module receives health-related information, including demographic details, body temperature, respiratory symptoms, and other relevant health parameters. The preprocessing module processes the received data by identifying missing, invalid, or inconsistent values. The data reliability assessment module evaluates the completeness, freshness, and consistency of the input parameters and assigns corresponding reliability scores. Based on these scores, the prediction module adjusts the contribution of individual parameters to the prediction process using a machine learning model. When the available data do not satisfy a predefined reliability condition, the system identifies the need for updated or additional information. The output module provides an airborne disease risk prediction together with information indicating the reliability of the input data and the uncertainty associated with the prediction. The invention aims to improve the dependability of airborne disease risk assessment by incorporating input data reliability into the prediction process.

No. of Pages : 5 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127534 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE HYDRAULIC LIFT SYSTEM

(51) International classification	:B66F 11/04, B66F 7/28, B66F 7/08, B60P 1/44, A61G 3/06	(71)**Name of Applicant :** **1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI** Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India (72)**Name of Inventor :** **1)AWANI BHUSHAN**
(31) Priority Document No	:NA	**2)VELMATHI G.**
(32) Priority Date	:NA	**3)SANDIP SAHA**
(33) Name of priority country	:NA	**4)KISLAY ANAND**
(86) International Application No Filing Date	: :01/01/1900	**5)ANOOP C A** **6)ABHAY SONY**
(87) International Publication No	: NA	**7)ROSHAN SAJI**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A portable lift system (102) is disclosed. The system (102) includes a base support (114) having a mobility mechanism (114-1) and a wheel locking mechanism (114-2) restricting movement of the base support (114). A lifting platform (104) supported on the base support (114) is movable between a lowered position and a raised position, and includes a ramp (104-1) for providing access. A hydraulic lift mechanism (110) coupled to the lifting platform (104) includes a hydraulic power mechanism (110-1) and a height stabilization mechanism (110-2). A power supply (112) and a user input mechanism (106) are coupled to the hydraulic power mechanism (110-1). Actuation of the user input mechanism (106) moves the lifting platform (104) toward the raised position, the height stabilization mechanism (110-2) retaining the lifting platform (104) at an elevation reached.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127536 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIO-DEGRADABLE ODOUR NEUTRALISING SANITARY WRAPPER

(51) International classification	:B65D 65/46, A61F 13/15, A61L 15/42, A61F 13/472, A61F 13/84	(71)**Name of Applicant :** **1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI** Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India (72)**Name of Inventor :** **1)MANUSHRI JADHAV MANOHAR** **2)SUGUMARAN V**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Embodiments of the present disclosure relate to a biodegradable odour-neutralising sanitary wrapper (100) for hygienic containment of a used sanitary pad (108). The wrapper (100) includes an outer layer (102) forming a biodegradable fluid-resistant structural envelope, a rice husk ash layer (104) disposed inwardly of the outer layer, an inner functional layer (106) comprising a porous biodegradable membrane, and an adhesive flap mechanism (110). Residual moisture and odour molecules from the used sanitary pad (108) permeate through the inner functional layer (106) and activate the rice husk ash layer (104) to adsorb volatile odour-causing compounds and reduce microbial odour generation. The adhesive flap mechanism (110) seals the wrapper (100) into an airtight, leak-resistant containment unit for safe storage or transport until disposal. The biodegradable materials and agricultural waste-derived deodorising medium reduce plastic waste and also support sustainable sanitary waste management in resource-limited environments.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127540 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LIQUID-HANDLING CONTAINER WITH THERMAL REGULATION, SPLASH CONTROL AND SURFACE CLEANING

(51) International classification	:F25B 21/02, B65D 25/28, F25B 21/04, G01P 15/09, B65D 25/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)SHRREYA VISHWANATHAN
2)RAMA PRABHA K.P

(57) Abstract :
A liquid-handling container (102) includes a bucket body (110), a handle assembly (106), a cooling assembly (112) with a thermoelectric module and a phase change material layer, a motion sensing assembly (108) with an accelerometer and a gyroscope, an adaptive cover assembly (114), a self-cleaning assembly (120) with a piezoelectric film and a superhydrophobic nanocoating layer, a drainage assembly (116) with a one-way valve, and a control unit (104). The control unit (104) activates the thermoelectric module upon a handle temperature exceeding a threshold value, and actuates the cover assembly (114) and a vibration alert upon a movement of the container (102) exceeding a threshold speed. The piezoelectric film generates a micro-vibration to dislodge particulate matter, and the valve drains a liquid under gravity.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127542 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Plant-Based Sausage with Banana Pseudostem-Floret Matrix and Method of Preparation

(51) International classification	:A23J 3/00, A23J 3/16, A23L 13/00, A23L 13/40, A23L 19/00, A23P 30/20, A23B 4/06, A23L 29/256
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T)
Address of Applicant :National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T), Pudukkottai Road, Thanjavur-613005, Tamil Nadu, India. Thanjavur Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. R. Jagan Mohan
2)Keerthana Priya. R
3)Surendharan S
4)Sreelakshmi. V
5)Varun Aakash Ganeshvel
6)Dr. Dakshayani. R
7)Karthika R

(57) Abstract :
The present disclosure proposes a plant-based meat analogue sausage (200) using banana pseudostem and banana floret in an 80:20 proportion as a principal meat-replacement base. The process involves cleaning, steaming, grinding or size reduction, combining the banana pseudostem and banana floret, and blending the resulting base with soy protein isolate, carrageenan, tapioca starch, and cashew paste to obtain a fibrous, protein-containing, bindable, and shape-retaining meat analogue matrix. The formulation is subjected to cold extrusion and shaping to obtain the plant-based meat analogue sausage (200), followed by freezing at approximately −18 °C until further processing. The banana pseudostem contributes fibrous texture, soy protein isolate provides a protein source, carrageenan supports binding and structural integrity, and tapioca starch and cashew paste contribute to texture, mouthfeel, and shape retention.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127543 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A System and Method for Cross-Transition Causal Fingerprint-Based Validation and Selective Recalibration of Digital-Twin Parameters

(51) International classification	:G06N 5/04, G06N 20/00, G06F 16/23, G06N 5/02, G06F 16/27
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vellore Institute of Technology
Address of Applicant :Vellore Institute of Technology, Tiruvalam Road, Katpadi, Vellore-632014, Tamil Nadu, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)Bala Ganesh N
2)Mohammed Kaif K
3)Gowtham S
4)Pravin G

(57) Abstract :
The present disclosure proposes a system (100) and method for cross-transition causal fingerprint-based validation and selective recalibration of digital-twin parameters for a physical asset (108). Sensors (110) acquire synchronized data during naturally occurring operating-state transitions. A causal-fingerprint generation module (118) generates machine-specific causal fingerprints and compares them with responses from a digital-twin model (112) to identify physical–digital discrepancies. A candidate-parameter identification module (120) identifies a candidate parameter, and a parameter-locking module (124) maintains the candidate parameter locked during cross-transition causal verification. A cross-transition verification module (122) evaluates the candidate across independent transitions and excludes competing causes. Upon causal consistency, a selective recalibration module (126) recalibrates the verified parameter while maintaining unrelated validated parameters unchanged. An independent validation module (128) validates the recalibrated model using an independent transition before acceptance, rejection, or restoration.

No. of Pages : 60 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127544 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Self-healing IoT sensor system and Method for Pre-ADC Fault Quarantine, Power-Domain Isolation, and Energy-Conditioned Requalification

(51) International classification	:H04W 4/38, H04L 67/12, H04L 29/08, H04L 29/06, H03M 1/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vellore Institute of Technology
Address of Applicant :Vellore Institute of Technology, Tiruvalam Road, Katpadi, Vellore-632014, Tamil Nadu, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)Tamil Priya D
2)MS Shabreena

(57) Abstract :
The present disclosure proposes a self-healing edge Internet of Things (IoT) sensor system (100) comprises a primary precision sensing domain (102), a heterogeneous low-power fallback sensing domain (108), an analog-to-digital converter (114), a pre-ADC multi-parameter integrity fabric (116), and an always-on Sensor Recovery Fabric (118). The integrity fabric (116) evaluates amplitude, noise, bias, response latency, current, and reference response before digitization. Upon confirmed persistent degradation, the Sensor Recovery Fabric (118) actuates an analog measurement-quarantine circuit (120) and a primary power-domain isolation circuit (122), while maintaining fallback sensing. An energy-harvesting subsystem (124), an energy-storage element (126), a recovery-energy firewall (128), and an energy-arbitration power matrix (130) provide protected energy for requalification through a micro-pulse recovery circuit (132). A cross-sensor verification circuit (134) and hardware restoration verifier (136) authorize restoration only after satisfactory electrical, signal, reference-response, and cross-sensor verification.

No. of Pages : 82 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127545 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : STUDIES ON THE TREATMENT OF AN INDUSTRIAL PHARMACEUTICAL WASTEWATER USING NATURAL COAGULANTS AND WITH BIOAUGMENTATION THEREOF

(51) International classification	:C02F 1/52, C02F 1/56, C02F 103/34, C02F 9/00, C02F 1/66
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)T SUBBULAKSHMI
Address of Applicant :No. 29, Lakshmi Nagar, Nandhivaram, Guduvancherry Tamil Nadu India
2)A. Sathiya Priya
3)Dr. S. Sundararaman
(72)**Name of Inventor :**
1)T SUBBULAKSHMI
2)A. Sathiya Priya
3)Dr. S. Sundararaman

(57) Abstract :
Treatment of pharmaceutical wastewater continues to pose several challenges, in spite of several advancements in the treatment processes of the above wastewater. One of the recent challenges, is the development of sustainable treatment technologies using naturally available materials. Use of natural coagulants (NCs), which are locally available, either alone, or in combination thereof, or, in combination of chemical coagulants (CCs), for the treatment of pharmaceutical wastewater, is the recent trend. However, the full potential of the above has not been fully addressed, in terms of the evaluation of locally available NCs and in achieving the maximum removal efficiency (%) of COD of the above wastewater. The present experimental study is an attempt in that direction. In this experimental study the treatment of chosen wastewater was carried out in three stages namely: (i) Acclimatization and treatment of the chosen pharmaceutical wastewater (without/with bioaugmentation) using four types of locally available NCs, namely, Neem, Moringa, Mango and Peepal leaves – based powder form ; (ii) Evaluation of the relative performance of the above NCs, in terms of the significant parameters such as : pH , turbidity , BOD and COD ; (iii) To suggest the treatment process, based on the 'best performance' of the NC. Five HRTs (2,4,6,8,10 hours) and two finenesses of the powder of the NCs (150 and 300 µm), were considered in the experimental programme. Extensive experiment investigations were carried out. Based on the results and discussion and inferences derived thereof, following are the salients conclusions : (i) The 'acclimatization' 15 study leading to attainment of steady state on the 100th day, is the lowest, for the treatment of pharmaceutical wastewater and that the ratio of 1 : 3 (raw wastewater : secondary sewage) is the best ratio for 'acclimatizing' the wastewater; (ii) Based on overall performance of the four NCs, in terms of turbidity, BOD and COD removal efficiencies, and the two finenesses considered, the performance of 'Moringa Leaves Powder' (NC2) is the best, with bioaugmentation, as the COD removal efficiency achieved is 94% ; (iii) There is a reduction in the 'recalcitrant' nature of the treated wastewater, as evident from the results of sophisticated analytical studies, and it may be attributed to partial removal or transformation of such compounds in the raw wastewater, by the treatment process adopted in this study ; and (iv) 'Moringa leaves powder' as a NC in combination with bioaugmentation is recommended for the treatment of the pharmaceutical wastewater (industrial type and 'antibiotic' in nature)

No. of Pages : 8 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127553 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-BASED PLATFORM FOR FUND MANAGEMENT, EXPENDITURE ANALYSIS, AND TRANSPARENCY MONITORING

(51) International classification	:G06Q 40/00, G06Q 10/10, G06Q 40/02, G06Q 10/06, G06Q 40/04	(71)**Name of Applicant :** **1)RNS Institute of Technology** Address of Applicant :RNS Institute of Technology, Dr. Vishnuvardhan Road Post, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098, INDIA Bengaluru Karnataka India **2)Reethu M** **3)Raksha N** **4)Khushboo B** **5)Dhusyanth S** **6)Purushottam Vats** **7)Prabhanjan Kashyap** **8)Dr. Pavithra GS**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	(72)**Name of Inventor :** **1)Reethu M** **2)Raksha N** **3)Khushboo B** **4)Dhusyanth S** **5)Purushottam Vats** **6)Prabhanjan Kashyap** **7)Dr. Pavithra GS**
(86) International Application No	:	
Filing Date	:01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :
Financial accountability is a problem shared across nearly every sector. Individuals, NGOs, public institutions, and private enterprises all depend on the same fundamental capability: knowing not just where money went, but whether it accomplished anything. Conventional accounting tools—spreadsheets, static ledgers, legacy ERP systems—record disbursements faithfully but evaluate outcomes not at all, creating what development finance practitioners call the "Impact Gap" [3][4]. Human audits, meanwhile, realistically cover only a fraction of total transactions, leaving institutional ledgers exposed to duplicate invoicing, price manipulation, and unauthorized vendor disbursements that can persist undetected across multiple reporting cycles [1][2]. This paper presents FundSight AI, a full-stack financial management platform built on a React frontend, a Python Flask API, and a Supabase (PostgreSQL) database governed by Role-Based Access Control (RBAC). The system operates across three user modes—Personal, Institutional, and Business—each tailored to its context while sharing a common AI analytics core built on two ensemble machine learning models: an Isolation Forest for real-time anomaly detection, and a Random Forest classifier for Fund Efficiency Scoring. The platform further incorporates budget forecasting, automated PDF reporting, and interactive dashboards. Validation against a synthetic dataset of 10,000 transaction records produced an anomaly detection F1-score of 0.91, a Fund Efficiency AUC-ROC of 0.95, sub-second API response latency, and a measurable reduction in administrative overhead—results that establish FundSight AI as a viable model for transparent, outcome-oriented financial management.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127559 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DIGITAL PLATFORM FOR SKILL CERTIFICATION AND VERIFICATION

(51) International classification	:G06Q 10/10, H04L 9/32, G06Q 50/20, G06Q 10/06, G06Q 20/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RNS Institute of Technology
Address of Applicant :RNS Institute of Technology, Dr. Vishnuvardhan Road Post, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098, INDIA Bengaluru Karnataka India
2)Navaneeth R
3)Mohammed Nehal
4)Manvanth R
5)Manoj C
6)Kushal Gowda T S
7)Dhanush J
8)Ms. Vani Zunjarwad
(72)**Name of Inventor :**
1)Navaneeth R
2)Mohammed Nehal
3)Manvanth R
4)Manoj C
5)Kushal Gowda T S
6)Dhanush J
7)Ms. Vani Zunjarwad

(57) Abstract :
In the rapidly evolving global job market, the proliferation of fraudulent credentials and the administrative delays associated with traditional certificate verification pose significant challenges to both employers and educational institutions. This project introduces a Digital Platform for Skill Certification and Verification, a streamlined web-based solution designed to enhance the credibility and accessibility of professional achievements. The platform provides a centralized ecosystem where organizations can issue digital certificates and users can manage their professional portfolios. By utilizing unique digital identifiers and a secure verification architecture, the system eliminates the need for manual background checks and reduces the risk of credential tampering. The implementation focuses on a user-centric interface that allows for real-time validation of skills, ensuring that recruiters can verify candidate claims with a single click. This project demonstrates how digital transformation can restore trust in professional certification, offering a scalable and efficient alternative to legacy verification methods.

No. of Pages : 11 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127569 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SafeLink : An AI-Powered Multi-Modal Emergency Response Progressive Web Application for Urban Women's Safety

(51) International classification	:H04W 4/90, G06Q 50/26, G16H 40/67, H04W 4/029, G08B 25/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)RNS Institute of Technology

Address of Applicant :RNS Institute of Technology, Dr. Vishnuvardhan Road Post, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098, INDIA Bengaluru Karnataka India

2)Koushik P Gowda

3)Karan H Gowda

4)Jeevan Gowda H M

5)Kethan Hrshikesh N U

6)Kushwanth B S

7)Niranjan

8)Ms. Vinutha D

9)Dr. Chandana C

(72)**Name of Inventor :**

1)Koushik P Gowda

2)Karan H Gowda

3)Jeevan Gowda H M

4)Kethan Hrshikesh N U

5)Kushwanth B S

6)Niranjan

7)Ms. Vinutha D

8)Dr. Chandana C

(57) Abstract :

The present invention discloses a system and method for an AI-powered emergency response platform designated as "SafeLink," implemented as a Progressive Web Application (PWA) for urban women's safety. The system consolidates six critical safety functionalities: (a) one-tap SOS with auto-countdown, GPS tracking, medical profile transmission, and SMS alerts via Fast2SMS; (b) multi-provider AI threat analysis using Google Gemini 2.0 Flash and Groq Llama 3.3 70B in a cascading fallback architecture for real-time severity classification (LOW/MEDIUM/CRITICAL); (c) an interactive geospatial crime heatmap with 20+ geo-referenced zones on Leaflet.js/OpenStreetMap; (d) accelerometer-based shake detection (3 shakes at 30 m/s² threshold) and triple-keystroke panic shortcuts for zero-UI SOS activation; (e) a configurable fake call simulator with realistic vibration patterns for discreet escape; and (f) a timed safety check-in system with automatic missed-checkin alerts. The architecture employs React 19, Vite 8, Node.js/Express.js, and Supabase PostgreSQL with Row-Level Security, leveraging native Web APIs including Geolocation, DeviceMotion, Web Speech, and Vibration APIs.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127587 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEMS AND METHODS FOR AUTOMATED PROCESS CONTROL OF A CAPACITIVE DEIONIZATION SYSTEM FOR WATER PURIFICATION

(51) International classification	:C02F 1/469, B01D 61/58, C02F 1/42, B01D 61/48, C02F 1/00	(71)**Name of Applicant :** **1)Indian Institute of Technology Tirupati (IIT Tirupati)** Address of Applicant :Indian Institute of Technology, Tirupati, Venkatagiri Road, Yerpedu Post, Tirupati District Andhra Pradesh – 517619, India Tirupati Andhra Pradesh India (72)**Name of Inventor :** **1)MALIYEKKAL, Shihabudheen Mundampra** **2)G, Vajitha** **3)KARAPOOLA, Sareena**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
Same as uploaded

No. of Pages : 52 No. of Claims : 40

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127589 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR SARCASM-AWARE ASPECT-LEVEL SENTIMENT ANALYSIS OF INFORMAL TEXT DATA

(51) International classification	:G06N 3/04, G06N 3/08, G06N 20/00, G06F 16/35, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India
2)DR. MAMATHA BALIPA
3)AARTHIK K
4)A SAMVED RAO
(72)**Name of Inventor :**
1)DR. MAMATHA BALIPA
2)AARTHIK K
3)A SAMVED RAO

(57) Abstract :
Disclosed herein is a system (100) for sarcasm-aware aspect-level sentiment analysis comprising a user interface (102) on a user device (104), a communication network (106), and a processing unit (108) comprising an input processing module (110) receiving informal text, a preprocessing module (112) normalizing data and embedding emojis, a shared transformer encoder module (114) generating a contextual embedding, a sarcasm detection module (116) computing a sarcasm confidence score, an aspect extraction module (118) identifying opinion targets, an aspect-sentiment classification module (120) determining polarity, a joint optimization module (122) dynamically coupling sarcasm confidence with sentiment re-weighting, a fine-tuning module (124) adaptively optimizing the encoder, a logging module (126), an output module (128), and a database (130).

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641127590 A

(19) INDIA

(22) Date of filing of Application :26/10/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : MOTION-RESPONSIVE IONTOPHORETIC DRUG DELIVERY SYSTEM

(51) International classification	:A61N 1/30, A61N 1/32, A61N 1/04, A61M 37/00, A61N 1/05	(71)**Name of Applicant :** **1)NITTE (DEEMED TO BE UNIVERSITY)** Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India **2)VINUTHA M** (72)**Name of Inventor :** **1)VINUTHA M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed herein is a motion-responsive iontophoretic drug delivery system (100), comprising a wearable drug delivery patch (102) transdermally deliver a therapeutic agent through skin of a patient, further comprises a plurality of sensors (104) generate motion data, a drug reservoir (110) contain therapeutic agent, a plurality of electrodes (112) apply iontophoretic current, a current control circuit (114) regulate the iontophoretic current, a microcontroller (118) receive and process motion data further comprises a data acquisition module (120) receive motion data, a pre-processing module (122) performing noise reduction, normalization, and signal conditioning, a feature extraction module (124) extract movement parameters, a pain estimation module (126) estimate corresponding pain level, a drug delivery determination module (128) determine drug delivery parameter, an output module (136) transmit control signals, a communication unit (138) configured to facilitate data communication within the system (100), a user device (140) receive and display output through a user interface (142).

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127591 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A BIO-INSPIRED CRANIAL IMPLANT APPARATUS AND METHOD OF MANUFACTURING THEREOF

(51) International classification	:A61F 2/28, A61L 27/34, A61B 17/68, A61L 27/30, A61F 2/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA - 575018, INDIA Karnataka India
2)ARYAN VASISHTA
3)DR. VIDYASAGAR SHETTY
(72)**Name of Inventor :**
1)ARYAN VASISHTA
2)DR. VIDYASAGAR SHETTY

(57) Abstract :
Disclosed herein is a cranial implant apparatus (100), comprising a base structure (102) providing a biodegradable structural core, a coating layer (104) disposed over the base structure (102), and a surface geometry (106) formed on the coating layer (104) and comprising a plurality of scale elements arranged in an overlapping pattern over a shared-wall hexagonal mesh structure, the surface geometry (106) being a single continuous geometric feature concurrently configured to resist bacterial biofilm formation via a surface topography of the scale elements and to disperse an impact force and reduce a boundary micro-movement via the mesh structure, together with a plurality of fixation screws (108) positioned at a vertex shared by at least three hollow hexagonal cells so as to distribute a fixation load across those cells, and a method of manufacturing the apparatus (100) by three-dimensional printing.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127592 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : REAL-TIME END-EFFECTOR POSITION, ORIENTATION, AND TRAJECTORY VALIDATION SYSTEM

(51) International classification	:B25J 9/16, B25J 13/08, G05B 19/418, B25J 9/00, B25J 19/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India
2)DR. MURALIDHARA
3)MS. RAJASHREE NAMBIAR P
4)RAJAT R PANDYA
5)LIKHITHRAJ T ACHARYA
6)G KSHMA PAI
7)VAIBHAV A KULAL
(72)**Name of Inventor :**
1)DR. MURALIDHARA
2)MS. RAJASHREE NAMBIAR P
3)RAJAT R PANDYA
4)LIKHITHRAJ T ACHARYA
5)G KSHMA PAI
6)VAIBHAV A KULAL

(57) Abstract :
Disclosed herein is a real-time end-effector position, orientation and trajectory validation system (100), the system (100) comprises a cubical device (102) moves synchronously with the parallel manipulator, which comprises a structural frame (104), a plurality of reference nodes (106), a plurality of sensors (108) senses measurement data, an end-effector interface (110) to be coupled to the end-effector of the parallel manipulator to facilitate validation, a communication network (112) establishes a communication link, a processing unit (114) to process real-time data, which further comprises a data input module (116) receives real-time data, a pre-processing module (118) filters and compensates the received data, a position and orientation estimation module (120) determines a real-time position and orientation of the end-effector, a trajectory validation module (122) compares the determined real-time position and orientation, an error estimation module (124) computes positional error, orientational error, and trajectory deviation, an output module (126) transmits validation results.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127593 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PERSONALITY PROFILE PREDICTION SYSTEM

(51) International classification	:G06Q 50/00, G06N 20/00, G06N 3/04, G06N 3/08, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India
2)DR. CHINMAI SHETTY
3)P DEVDAT
4)PRAMUKH A NAYAK
5)PRATHAM PRABHAKAR
(72)**Name of Inventor :**
1)DR. CHINMAI SHETTY
2)P DEVDAT
3)PRAMUKH A NAYAK
4)PRATHAM PRABHAKAR

(57) Abstract :
Disclosed herein is a personality profile prediction system (100) that comprises comprising a user interface (102) integrated within a user device (104) configured to acquire text-based social media posts, a communication network (106), and a processing unit (108) comprising a text acquisition module (110), a text pre-processing module (112), a vector conversion module (114), a feature extraction module (116), a behavioural trend analysis module (118), a personality prediction module (120), a personality classification module (122), a confidence score generation module (124), a personality type determination module (126), a report generation module (128), an output module (130), a feedback module (132), and a cloud database (134). The system processes textual communication data to predict a personality profile, generate confidence scores and a structured personality assessment report, and utilizes user feedback to facilitate refinement of personality prediction parameters and improve prediction reliability.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127612 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : TRAJECTORY PREDICTION FOR VEHICLES

(51) International classification	:G08G 1/01, G08G 1/00, G08G 1/0967, G06K 9/00, G08G 1/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY MADRAS (IIT MADRAS)
Address of Applicant :The Dean, Industrial Consultancy & Sponsored Research [IC&SR], Indian Institute of Technology Madras, Sardar Patel Road, IIT P.O., Chennai, Tamil Nadu 600036, India Tamil Nadu India
(72)**Name of Inventor :**
1)VANAJAKSHI, Lelitha Devi
2)GOSWAMI, Atmadip

(57) Abstract :
ABSTRACT TRAJECTORY PREDICTION FOR VEHICLES A method for predicting a trajectory of a host vehicle includes receiving motion history data of the host vehicle generated based on first sensor data 5 captured by a first sensor, receiving motion history data of a plurality of neighbouring traffic participants generated based on second sensor data captured by a second sensor, determining interactions among the plurality of neighbouring traffic participants based on the motion history data of the plurality of neighbouring traffic participants by a trajectory model, 10 determining social context information by weighting the interactions based on relevance of each neighbouring traffic participant relative to the host vehicle by the trajectory model, combining the social context information with the motion history data of the host vehicle in accordance with a gating value to obtain combined motion information by the trajectory model, 15 generating a predicted future trajectory of the host vehicle based on the combined motion information by the trajectory model. Fig. 4 53

No. of Pages : 64 No. of Claims : 36

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202641127622 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR CROSS-SCRIPT RETRIEVAL AND LOCALISATION OF WORDS IN DOCUMENT IMAGES

(51) International classification	:G06V 30/10, G06F 17/30, G06V 30/224, G06K 9/62, G06V 30/413
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VIEWMYRECORDS PRIVATE LIMITED
Address of Applicant :No. 137, 2nd Main 4th Cross R K Layout 2nd Stage, Padmanabhanagar, Bangalore, Karnataka- 560070, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)JOSHI, Milind Madhav
2)KATTI, Ninad Sudarshan
3)NAIR, Karthik R

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR CROSS-SCRIPT RETRIEVAL AND LOCALISATION OF WORDS IN DOCUMENT IMAGES The present disclosure relates to a system (100) and a method (200) for locating words recorded in a first script in document images (140) by a search query (156) in a second script. An optical character recognition engine (110) produces word tokens (142), each comprising recognised text, pixel coordinates of a bounding region (144) and a confidence value. A phonetic key generator (118) converts each word token (142) into a script-neutral phonetic key using a set of phonetic classes (120) having a merged class of graphemes exceeding a substitution error frequency threshold. A phonetic index (122) stores the keys, n-grams, coordinates and confidence values. A query module (124) converts the search query (156) into a query phonetic key (158), and selects and ranks word tokens (142) by n-gram similarity and confidence. A localisation module (128) overlays a visual highlight (146) on the document image (140) at said coordinates. Figure 1

No. of Pages : 41 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127625 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Stabilizing the Sperm Cells using Herbal Extract by Cryopreservation Technology

(51) International classification	:A01N 1/02, C12N 5/076, A61K 36/482, A01N 1/00, A61D 19/04	(71)**Name of Applicant :** **1)D Abilash** Address of Applicant :27-1034, Telephone Colony Andhra Pradesh India **2)S Rithish** (72)**Name of Inventor :** **1)Dr D Abilash** **2)Dr TB Sridharan** **3)Dr R Babujanarthanam** **4)Dr Dhanesh Gandhi** **5)Dr. R Parameswari** **6)Prof K Bhaskar Reddy** **7)Mr Madhan Kumar P** **8)Mr S Rithish**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a concentration optimized herbal extender in sperm cells cryopreservation comprising an aqueous flower extract of Senna auriculata L. for maintaining sperm cells stability during freezing, ultra-low-temperature storage, and thawing process. The invention addresses cellular damage associated with conventional sperm cryopreservation, including oxidative stress, loss of motility, reduced viability, disruption of plasma membrane integrity, and DNA fragmentation. The disclosed extender incorporates a relatively low concentration of Senna auriculata L. flower extract, preferably less concentration of the extender formulation, in combination with cryoprotective, buffering, osmotic, nutritional, and cellular stabilizing components. In a representative embodiment, the formulation comprises glycerol, dimethyl sulfoxide, Tris, fructose, sodium citrate dihydrate, trehalose, taurine, and gentamicin. The herbal extract contains reported bioactive constituents, including flavonoids, phenolic compounds, vitamin-E related compounds, fatty acids, and amino acid derived compounds, which may contribute to antioxidant activity and membrane stabilization. Comparative experimental results disclosed in the accompanying document indicate that the formulation containing less concentration (5 mL) of extender exhibited better preservation characteristics than the tested formulations containing 10 mL and 15 mL respectively. The lower concentration formulation demonstrated sperm motility of 42.83%, progressive motility of 30.92%, viability of 49.66%, functional plasma membrane integrity of 70.41%, and DNA fragmentation of 9.00% under the reported experimental conditions. These findings support the concentration dependent nature of the formulation's observed cryoprotective performance. The invention further encompasses methods for preparing the herbal extract, incorporating the extract into a cryopreservation medium, contacting spermatozoa with the formulated extender, freezing and storing the resulting mixture at ultra-low temperatures, and evaluating sperm quality following thawing process. The disclosed composition has potential applications in sperm banking, fertility preservation, assisted reproductive technology laboratories, and reproductive biotechnology research. The invention provides a basis for developing standardized herbal cryopreservation formulations intended to support the preservation of sperm quality during cryogenic processing.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127628 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PitPulse: An IoT System and Method for Early Warning of Clogging in Rainwater Recharge Pits from the Rate at Which Water Drains After Each Rain

(51) International classification	:E03F 5/10, E03F 1/00, E03B 3/03, G01W 1/14, E03B 3/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JOSEPHINE RUTH FENITHA
Address of Applicant :ASSISTANT PROFESSOR, Department of Information Technology, S.A. Engineering College, Veeraragavapuram, Chennai-600077. Chennai Tamil Nadu India
2)SRUTHI A.N
3)ELAKIYA M
4)S.A. ENGINEERING COLLEGE
(72)**Name of Inventor :**
1)JOSEPHINE RUTH FENITHA
2)SRUTHI A.N
3)ELAKIYA M

(57) Abstract :
ABSTRACT A system and method warn when a rainwater recharge pit is clogging with silt. A field unit at the pit records rainfall and the water level in the pit every few minutes. After each rain, the fall in water level is fitted to give the pit's drain rate. Because a pit also drains slowly when the surrounding soil is already wet, the drain rate is corrected using a soil-wetness index worked out from the rain of the preceding days. The corrected rate is tracked against the total rain received since the pit was last cleaned, and the further rain the pit can take before it needs cleaning is predicted. Owners and the local body receive alerts and a map of pit health, so that pits are cleaned before they stop recharging groundwater.

No. of Pages : 12 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127646 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WATER RECLAMATION

(51) International classification	:C02F 1/00, E03C 1/122, E03C 1/12, C02F 1/44, C02F 1/52	(71)**Name of Applicant :** **1)TANMULA PVT. LTD.** Address of Applicant :No. 405, Opposite to Cauvery Hostel, Sir M.V. Road, Vijayshreepura, Manasagangothri Mysore Karnataka India (72)**Name of Inventor :** **1)ADARSH S V** **2)LAXMIPRASAD S V** **3)K S LOKESH**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Accordingly embodiments herein disclose a system (100) and a method (200) for water reclamation. The system (100) comprises at least one plumbing fixture (110) that produces waste water (10) during a sanitation process, where the waste water (10) comprises solid waste (20) and liquid waste (40). The system (100) further comprises a water treatment subsystem (120) that treats the waste water (10) to produce recycled water (30).

No. of Pages : 48 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127647 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AIR-ENGINEERED APPARATUS AND PROCESS FOR SECONDARY AND TERTIARY REFINING OF REFUSE DERIVED FUEL AND MUNICIPAL SOLID WASTE WITHOUT THERMAL ENERGY FUEL

(51) International classification	:C10L 5/46, C10L 5/48, B09B 3/00, F23G 5/027, F23G 5/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SAMTHUR KALIMUTHU SIVAKUMAR
Address of Applicant :No. 79, 1st Main Road, Shakthi Nagar, Mudichur, Mannivakkam, Tambaram, Kancheepuram, Chennai – 600048, Tamil Nadu, India Tamil Nadu India
(72)**Name of Inventor :**
1)SAMTHUR KALIMUTHU SIVAKUMAR

(57) Abstract :
AN AIR-ENGINEERED APPARATUS AND PROCESS FOR SECONDARY AND TERTIARY REFINING OF REFUSE DERIVED FUEL AND MUNICIPAL SOLID WASTE WITHOUT THERMAL ENERGY FUEL An air-engineered apparatus and method for the non-thermal refining of refuse derived fuel comprises a perforated rotary drum (110) with internal lifters, located within a ventilated chamber (100). The material is retained for a selectable batch duration while rotation causes repeated cascading. Directed supply air (120) promotes moisture evaporation and loosens adhered non-combustible impurities, and progressive agitation releases particles that pass through the drum perforations to collection conveyors (190). Moisture-laden air is extracted through selectable exhaust arrangements (130), optionally employing diversion plates (170). The apparatus may include adjustable air hoods (125), internal air-recirculating HVLS fans (160), variable-speed rotation, sensor-based pressure and moisture controls, and alternative screw-flight or flanged-door discharge systems. Drum dimensions, fan numbers and fan positions are configurable for different material quantities and chamber geometries.

No. of Pages : 42 No. of Claims : 49

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127652 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Multi-Modal Edge-Computing System for Real-Time Classification of Illness Narrative Typologies Using Acoustic and Linguistic Feature Analysis

(51) International classification	:G10L 15/26, G10L 15/18, G10L 15/06, G10L 15/22, G10L 15/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. V. Karunanithi
Address of Applicant :Assistant Professor, Department of Language Culture and Society, SRM Institute of Science and Technology, SRM Nagar, Kattankulathur, Chengalpattu, Tamil Nadu, 603203, India. Tamil Nadu India
2)Ms. Gayathri K
(72)**Name of Inventor :**
1)Dr. V. Karunanithi
2)Ms. Gayathri K

(57) Abstract :
The following ten claims are preliminary and are framed around the technical apparatus and processing workflow. [Claim 1] Independent apparatus claim We claim a multi-modal computational apparatus for real-time classification of illness narrative typologies, comprising an audio capture interface configured to receive spoken patient narrative data; at least one processor and memory; an automatic speech recognition engine configured to generate a textual representation of the spoken narrative; a linguistic feature extraction module configured to derive narrative-related linguistic features from the textual representation; an acoustic feature extraction module configured to derive acoustic characteristics from the spoken patient narrative data; a feature-fusion module configured to generate a combined feature representation from the linguistic and acoustic features; a multi-class classification engine configured to generate classification scores for a plurality of illness narrative typologies; a Narrative Typology Index generation module configured to convert the classification scores into a structured output; and an Electronic Health Record integration interface configured to provide the structured output to an authorized clinical record system. [Claim 2] Linguistic feature extraction The apparatus as claimed in Claim 1, wherein the linguistic feature extraction module is configured to calculate one or more features selected from temporal-marker density, syntactic coherence, agency-pronoun ratio, causal conjunction density, grammatical voice distribution, and transformation-oriented keyword density. [Claim 3] Acoustic feature extraction The apparatus as claimed in Claim 1, wherein the acoustic feature extraction module is configured to determine one or more acoustic measurements selected from speaking rate, micro-pause frequency, pause duration, fundamental frequency variation, and vocal jitter, subject to signal-quality requirements. [Claim 4] Multi-modal feature fusion The apparatus as claimed in Claim 1, wherein the feature-fusion module associates linguistic and acoustic feature vectors with corresponding speech segments and applies at least one of feature normalization, weighting, missing-value handling, or feature encoding to generate the combined feature representation. [Claim 5] Narrative typology classification The apparatus as claimed in Claim 1, wherein the multi-class classification engine is configured to generate respective classification scores for restitution, chaos, and quest narrative profiles using a trained predictive model operating on the combined feature representation. [Claim 6] Narrative Typology Index The apparatus as claimed in Claim 1, wherein the Narrative Typology Index generation module generates a structured data record comprising a predicted narrative profile, respective class scores, a confidence or uncertainty indicator, at least one supporting feature summary, and a processing timestamp. [Claim 7] Edge-computing implementation The apparatus as claimed in Claim 1, wherein the processor is configured to execute at least the acoustic feature extraction, feature-fusion, and narrative classification operations on an edge-computing device located within or proximate to a clinical speech acquisition environment [Claim 8] Confidence-aware reporting The apparatus as claimed in Claim 1, further comprising a quality assessment module configured to evaluate at least one of audio quality, available speech duration, feature completeness, or classification confidence and to flag, defer, or suppress automatic reporting of the Narrative Typology Index when a predefined reporting condition is not satisfied. [Claim 9] EHR integration and clinician communication support The apparatus as claimed in Claim 1, further comprising a clinician communication support module configured to generate one or more configurable communication prompts based on the predicted narrative profile and associated confidence information, wherein the EHR integration interface is configured to present or transfer the Narrative Typology Index and associated metadata through an authorized interface, subject to configured access controls and review requirements. [Claim 10] Computer-implemented classification method The present invention relates to a multi-modal edge-computing apparatus for real-time classification of illness narrative typologies from spoken patient narratives. The system includes an audio recording interface, an automatic speech recognition system, a linguistic feature extraction component, an acoustic feature extraction component, a feature fusion component, a multi-class classification system, and an Electronic Health Record integration component. Linguistic attributes such as temporal-marker frequency, syntactic consistency, agency-pronoun ratios, causal conjunction frequency, and transformation-focused keyword frequency are integrated with acoustic metrics like speaking pace, pause occurrence, fundamental frequency variation, and vocal jitter. A predictive model analyzes the integrated feature representation to produce classification scores for restitution, chaos, and quest narrative profiles. A Narrative Typology Index is created that includes class scores, confidence details, and summaries of supporting features. The device can conduct specific processing tasks locally and deliver confidence-aware results via an approved EHR interface. Customizable clinician communication suggestions and assessment systems may also be offered. The invention facilitates organized computational analysis of illness stories while maintaining clinician supervision.

No. of Pages : 29 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127653 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN SOIL AND CROP DATA ANALYTICS FRAMEWORK FOR INTELLIGENT IRRIGATION PREDICTION AND OPTIMIZATION

(51) International classification	:A01G 25/16, G06Q 50/02, G06N 20/00, G06Q 10/04, G06N 3/08	(71)**Name of Applicant :** **1)K.S.R. College of Engineering (Autonomous)** Address of Applicant :KSR Kalvi Nagar, Tiruchengode, Namakkal, Tamil Nadu, 637215, India. Tamil Nadu India (72)**Name of Inventor :** **1)The Principal** **2)Dr. Malatthi Sivasundaram** **3)Mr. V. Gopinath** **4)Ms. S. Hemalatha**
(31) Priority Document No	:NA	**5)Mrs. V. Sowmitha**
(32) Priority Date	:NA	**6)Ms. A. Priyadharshini**
(33) Name of priority country	:NA	**7)Mrs. R. Deebika**
(86) International Application No Filing Date	: :01/01/1900	**8)Mr. V. Karthi** **9)Mrs. P. Rathika**
(87) International Publication No	: NA	**10)Mr. K. Vinoth**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**11)Rohith R** **12)Suresh Kumar V**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention discloses the development of an artificial intelligence-driven framework (100) for soil and crop data analytics facilitates intelligent irrigation prediction and optimization. This framework integrates a soil and environmental data acquisition module (101), a crop data acquisition module (102), a data preprocessing and integration module (103), a feature extraction module (104), an artificial intelligence-based irrigation prediction engine (105), a water requirement estimation module (106), an irrigation optimization engine (107), an irrigation control module (108), a monitoring and alert module (109), and a farmer interface and reporting module (110). The system processes soil moisture measurements, environmental parameters, crop characteristics, and historical irrigation records to forecast irrigation demand. The forecasted demand is subsequently translated into an estimated water volume, and an optimized irrigation schedule is formulated based on crop requirements, water availability, and operational constraints. The framework facilitates irrigation monitoring, automated or farmer-approved control mechanisms, alert generation, and feedback-driven model refinement to enhance agricultural water management practices. FIG.1

No. of Pages : 17 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127654 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING AND IOT-BASED CLOUD PLATFORM FOR INTELLIGENT SMART CITY MONITORING AND PREDICTIVE MANAGEMENT

(51) International classification	:G06N 20/00, H04L 29/08, H04L 12/24, G06N 5/04, H04L 29/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Vijayalakshmi Palaniappan

Address of Applicant :Associate Professor, Department of Computer Science and Engineering, Knowledge Institute of Technology, Salem, Tamil Nadu, 637504, India. Tamil Nadu India

2)Sakthivel Subramaniam

3)Sowndarya S

4)Anitha Venkatachalam

5)Pavithra G

6)Ramya D

7)Sowndharya Thiruppathi

8)Sasikumar Palanivel

9)Eswari Alias Petchiammal G

10)Subha N

11)Ravikumar K

12)Gowdhami K

(72)**Name of Inventor :**

1)Vijayalakshmi Palaniappan

2)Sakthivel Subramaniam

3)Sowndarya S

4)Anitha Venkatachalam

5)Pavithra G

6)Ramya D

7)Sowndharya Thiruppathi

8)Sasikumar Palanivel

9)Eswari Alias Petchiammal G

10)Subha N

11)Ravikumar K

12)Gowdhami K

(57) Abstract :

The present invention discloses the development of an integrated cloud-based platform (100) that leverages machine learning and Internet of Things (IoT) technologies to enable intelligent monitoring and predictive management of smart cities. The platform is composed of several interconnected components: IoT sensor devices (102), a data acquisition and communication module (104), a data preprocessing module (106), a cloud storage and integration module (108), a machine learning analytics engine (110), a predictive management module (112), an anomaly detection module (114), a decision support and resource optimization module (116), an alert and notification module (118), and a smart city administrator interface (120). The system aggregates and consolidates comprehensive urban infrastructure data encompassing traffic management, environmental monitoring, water distribution systems, energy utilization and waste management operations. Through advanced machine learning analytics, the platform identifies operational trends, recognizes abnormal situations and forecasts potential service interruptions. The platform delivers ranked management recommendations and transmits alerts to designated administrators, thereby facilitating proactive infrastructure oversight, enhanced resource distribution and synchronized urban service administration. FIG.1

No. of Pages : 17 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127655 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Facile synthesis, characterization and photocatalytic behavior of Zn2V2O7 nanostructure

(51) International classification	:C01G 31/00, C01G 31/02, B01J 35/00, B01J 23/06, B01J 23/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Dr. S. Arulmurugan
2)Dr. Helen P. Kavitha

(57) Abstract :

Abstract: In this paper, the structure, morphology, optical and photocatalytic behavior of Zn2V2O7 nanostructure are reported. The facile thermal decomposition method was used for the synthesis of Zn2V2O7 nanostructure. Commercially available ammonium metavanadate (NH4VO3) and synthesized zinc oxide (ZnO) powders were used as vanadium and zinc source respectively. The structure and morphological investigation was performed by powder X-ray diffraction (XRD) analysis, Fourier transform infrared spectroscopy (FT-IR) spectroscopy, scanning and transmission electron microscopies. The obtained results indicate that Zn2V2O7 particles were formed at 500 °C along with trace amount of V2O5 particles. The ZnO distorted cubic particles were destroyed as they are completely converted as Zn2V2O7 nanoparticles. The optical property of Zn2V2O7 was analyzed through solid state UV-Visible absorption spectroscopy. Further, the synthesized Zn2V2O7 exhibits excellent photocatalytic degradation behavior towards acid blue - 113 (AB113) than that of pure V2O5 nanoparticles, under visible light irradiation.

No. of Pages : 5 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127656 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN AYURVEDIC ADAPTIVE HEALTH SCALE FOR PERSONALIZED NUTRITIONAL REGULATION

(51) International classification	:A61B 5/00, A61B 5/0205, A61B 5/053, A61B 5/145, A61B 5/11
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Ms. V R MEHA VENKAT
2)Ms. KAVYALAKSHMI S G
3)Ms. M. SANDHYA
4)MR. L HARISH
5)MS. V MAHALAKSHMI
6)MR. MEGHA SHYAM J
7)Dr. E. SARASWATHI

(57) Abstract :

Abstract: The present invention deals with the field of intelligent health monitoring technology, specifically the invention of an intelligent health monitoring device in the form of an ‘AI-integrated Ayurvedic smart health scale.’ The device has the ability to collect various physiological parameter values, including body weight and related biometric parameter values, and analyze them in terms of the Ayurvedic Prakriti-Vikriti analysis system. Unlike traditional weighing scales, which only provide information about the physical parameter values, the present invention uses a computational module to analyze the physiological parameter values to derive the Dosha imbalance indicators. These imbalance indicators are then sent to the artificial intelligence-based dietary recommendation system for dynamic adaptation of the nutrient requirement values for the individual. Thus, the present invention creates a closed-loop adaptive system for health care, combining traditional Ayurvedic medicine with modern machine-learning-based dietary management for preventive health care.

No. of Pages : 10 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127657 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Intelligent Automatic Side-Stand Retraction and Safety Control System for Two-Wheelers Based on Motion, Tilt, and Ignition State Detection

(51) International classification	:B62H 1/02, B62H 1/00, B62H 1/06, B62H 1/04, B62H 1/12	(71)**Name of Applicant :** **1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS** Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India **2)Easwari Engineering College**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr. V. Vinodkumar**
(33) Name of priority country	:NA	**2)Dr. V. Gokula Krishnan**
(86) International Application No Filing Date	: :01/01/1900	**3)Dr. V. Thanigaivelan** **4)Dr. T. Mothilal**
(87) International Publication No	: NA	**5)Dr. A. Jemila Percy**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)Dr. R. Thirunavukkarasu** **7)Dr. S.Ilangovan**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Abstract: The present invention relates to a safety control system for two-wheelers designed to prevent accidents caused by improper deployment of the side stand during vehicle operation. The system comprises a mechanical side-stand assembly integrated with an electromechanical actuator, a sensor module including tilt, motion, and ignition state detection units, and an embedded control unit configured to process real-time data. The system continuously monitors the operational state of the vehicle and automatically retracts the side stand upon detecting motion or unsafe tilt conditions while the stand remains deployed. Additionally, the system is capable of initiating engine cut-off as a preventive safety measure. The invention provides a real-time, automated mechanical intervention mechanism that enhances rider safety and reduces the risk of accidents associated with side-stand misuse.

No. of Pages : 5 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127658 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IoT-Based Intelligent Hybrid Cooling System for Photovoltaic Panels with RealTime Efficiency Optimization

(51) International classification	:H02S 40/42, G05D 23/19, H02J 7/35, H02S 50/00, H02J 3/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Ms. VARSHA SRI K J
2)Ms. H NIVEDHA
3)Dr.Srisabarimani.K

(57) Abstract :
ABSTRACT The invention concerns an IoT-Based Intelligent Hybrid Cooling System for Photovoltaic Panels with realtime efficiency optimization, which is intended for continuous monitoring and automatic temperature control of solar photovoltaic panels to optimize the energy conversion efficiency. The proposed system is made up of temperature sensors, environmental sensors, a hybrid cooling system, an embedded controller equipped with intelligent optimization algorithms, an IoT communication module, a monitoring platform on the cloud, a power management module, and a backup power system. The sensors continuously monitor real-time parameters like panel temperature, solar irradiance, humidity, voltage, current and power output. The embedded controller analyzes the operating conditions based on the collected data and automatically switches on the cooling technique that is most suitable for keeping the PV panel at an optimum temperature. The hybrid cooling system cools the product by including both passive and active cooling to reduce thermal stress and to minimise efficiency losses due to high heat generated during high solar radiation. This smart algorithm allows the system to adapt and take real-time decisions based on the temperature fluctuations and cooling performance demands, minimizing energy wastage. The IoT integration enables remote monitoring of the panel performance, cooling status and energy generation data via a wireless network. The system has the capability to detect abnormal conditions, forecast efficiency degradation, and alert for timely maintenance. A further improvement is the incorporation of solar-powered operation, battery backup, and long-range wireless connectivity, allowing for reliable deployment in remote PV installations. The proposed system offers a scalable, cost-effective, and energy-efficient approach to precision solar management, optimizing PV performance, enhancing durability, and ensuring maximum power output by intelligently optimizing cooling strategies in real-time.

No. of Pages : 7 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127659 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A DEEP LEARNING-BASED SYSTEM FOR DETECTION AND CLASSIFICATION OF BRAIN TUMOR USING MRI IMAGES

(51) International classification	:G06N 3/04, G06N 3/08, A61B 5/055, G06V 10/764, G01R 33/56
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Dr. S. Immaculate Joy
2)Dr. S. Devi Priya

(57) Abstract :

Abstract: Recent development in medical image processing and artificial intelligence, particularly to a deep learning-based system and method for classifying brain tumors from magnetic resonance imaging (MRI) images is presented in this work. The system employs a convolutional neural network (CNN) comprising multiple convolutional layers and batch normalization layers to extract discriminative features from MRI images. Image preprocessing and outlier handling are incorporated to improve data consistency, while data augmentation techniques, including flipping, rotation, and zooming, are used to expand the training dataset and improve model generalization. The processed images are supplied to the CNN for feature learning and classification. The proposed system is intended to address variations in tumor appearance and limitations arising from imbalanced training datasets. The invention provides a computational framework for automated brain tumor image classification and may assist medical professionals in image-based assessment.

No. of Pages : 10 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127660 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FIRELINE: Deployable Digital Breadcrumb Navigation & Rescue System for Firefighters

(51) International classification	:G08B 21/02, G08B 21/04, G08B 25/10, H04W 4/90, G08B 17/10	(71)**Name of Applicant :** **1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS** Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India **2)Easwari Engineering College** (72)**Name of Inventor :** **1)VEDHARAMAN P** **2)Dr. Reshma P Vengaloor** **3)ABHINANDA B** **4)DHEERAJ B** **5)SRIRAAM B** **6)P NETHRA** **7)Dr.Aarthi Elaveini M** **8)Dr. Praveenkumar B** **9)Dr. Karthiga R** **10)Jacintha Vijayakumar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Firefighters work in smoke-filled spaces where visibility, satellite positioning and conventional communication is less. FIRELINE is a deployable digital-breadcrumb safety system which supports route retracing, environmental awareness and emergency reporting. Fireline combines Path Linker nodes, a firefighter-worn wristband and a separate command-Centre dashboard. ESP32 helps in controlling LoRa communication, environmental sensing and node-status reporting. The wristband has an OLED display, an MPU6050 which is an inertial measurement unit and a manual SOS control; motion patterns are used to detect the fall of firefighters. This combines sensor-threshold rules, motion-event checks, link-quality monitoring and priority-based message handling. The node transfers itself to a different path as when the node fails due to unexpected occurrences.

No. of Pages : 7 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127661 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : OPTIMIZING CROP YIELD-AN IOT AND ML-BASED SOIL TESTING AND CROP RECOMMENDATION SYSTEM

(51) International classification	:G06Q 50/02, A01C 21/00, A01B 79/02, A01G 7/00, A01B 79/00	(71)**Name of Applicant :** **1)Karpagam Academy of Higher Education** Address of Applicant :Pollachi Main Road, Eachanari Post Coimbatore Tamil Nadu India Tamil Nadu India **2)Karpagam Institute of Technology** **3)Veerasamy K** **4)Prathosh S** (72)**Name of Inventor :** **1)Veerasamy K** **2)Prathosh S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Agriculture remains a vital sector for economic stability and food security, particularly in developing countries where a significant portion of the population depends on farming for livelihood. However, conventional agricultural practices are increasingly challenged by soil nutrient imbalance, excessive fertilizer usage, climate variability, and delayed detection of plant diseases, all of which contribute to reduced crop productivity and environmental degradation. Addressing these challenges requires intelligent, data-driven solutions capable of providing precise, real-time agronomic insights. This work presents an integrated Internet of Things (IoT) and Machine Learning (ML)–based soil testing and crop recommendation system aimed at optimizing crop yield while promoting sustainable soil management. The proposed system employs a distributed network of low-cost soil and environmental sensors to continuously monitor key parameters including soil temperature, moisture content, pH level, and macronutrient concentrations such as nitrogen, phosphorus, and potassium (NPK). These parameters are collected in real time and transmitted to an embedded processing unit, where preliminary data preprocessing and feature normalization are performed. The processed sensor data is analyzed using a Random Forest–based predictive model to generate accurate crop suitability assessments and nutrient recommendations tailored to localized soil and climatic conditions. To enhance model robustness and predictive performance, the system incorporates ensemble learning strategies including Stacking and Adaptive Feature Fusion, enabling optimized feature representation and improved generalization across diverse agricultural environments. Experimental evaluation demonstrates that the proposed approach significantly outperforms existing baseline algorithms, achieving a prediction accuracy of up to 99.5%. In addition to soil analysis and crop recommendation, the system integrates a plant health monitoring module based on Convolutional Neural Networks (CNNs). Images captured by embedded cameras are analyzed to detect early symptoms of plant diseases, nutrient deficiencies, and stress conditions. Early identification allows for timely intervention, reducing crop loss and preventing the spread of diseases. Unlike standalone disease detection tools, the proposed framework correlates plant health analysis with soil and environmental data, providing a comprehensive understanding of crop conditions. The system further features a user-oriented decision-support interface that delivers actionable insights such as fertilizer dosage recommendations, suitable crop selection, and real-time disease alerts. Edge-level processing capabilities reduce latency and dependency on continuous internet connectivity, enabling reliable operation in rural and resource-constrained regions. The modular and scalable design allows deployment across both small-scale and large-scale farming environments. Overall, the proposed IoT- and ML-driven agricultural intelligence system provides a comprehensive solution for precision farming by integrating real-time soil sensing, advanced machine learning analytics, and deep learning–based plant health monitoring. By enabling accurate nutrient management, early disease detection, and adaptive crop guidance, the system enhances crop productivity, supports sustainable soil practices, and holds significant potential to improve the livelihoods of farmers under changing climatic conditions.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641127667 A
(19) INDIA
(22) Date of filing of Application :27/10/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : Cross-Chain Federated Blockchain-IoT System for Decentralized Carbon Credit Management

(51) International classification	:G06N 20/00, H04L 9/06, H04L 9/32, H04L 29/06, H04L 9/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mohan Babu University
Address of Applicant :IPR Cell, Mohan Babu University (Erstwhile Sree Vidyanikethan Engineering College), Sree Sainath Nagar, A. Rangampet, Tirupati, Andhra Pradesh, India - 517102 Tirupati Andhra Pradesh India
(72)**Name of Inventor :**
1)Mr. Praveenkumar Naidu Rayanki
2)Dr. C.Madhusudhana Rao

(57) Abstract :
The present invention relates to a Cross-Chain Federated Blockchain-IoT System for Decentralized Carbon Credit Management configured to provide secure, transparent, and privacy-preserving monitoring, verification, and trading of carbon credits. The system comprises Internet of Things (IoT) sensing devices, federated learning modules, interoperable blockchain networks, smart contracts, and cross-chain communication mechanisms. IoT devices continuously collect carbon-emission and environmental data, while federated learning enables collaborative model training without transmitting raw organizational data. Encrypted model updates are securely aggregated to improve emission analysis and prediction accuracy. Validated environmental information is recorded on interoperable blockchain networks, where smart contracts automatically verify carbon-credit eligibility, execute trading transactions, and maintain immutable transaction records. Cross-chain communication enables seamless interaction among multiple blockchain platforms, improving interoperability and scalability across different organizations and regulatory authorities. The proposed framework enhances transparency, preserves enterprise privacy, reduces transaction latency, supports regulatory compliance, and provides an efficient platform for sustainable carbon-credit management across industrial, commercial, and environmental monitoring applications.

No. of Pages : 26 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127668 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SmartPantry AI- An Image-Based Personalized Recipe Generation System Using Refrigerator Inventory

(51) International classification	:G06N 3/08, G06Q 10/08, G06N 20/00, G06N 3/04, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mohan Babu University
Address of Applicant :IPR Cell, Mohan Babu University (Erstwhile Sree Vidyanikethan Engineering College), Sree Sainath Nagar, A. Rangampet, Tirupati, Andhra Pradesh, India - 517102 Tirupati Andhra Pradesh India
(72)**Name of Inventor :**
1)Mrs. K. Jagadeeswari
2)Mr. Mulukutla Sai Krishna
3)Ms. Naru Sowmya
4)Mr. Marri Ajay Kumar
5)Ms. Muralidhran Snavya

(57) Abstract :
The present invention relates to an Image-Based Personalized Recipe Generation System Using Refrigerator Inventory configured to automatically identify food ingredients from refrigerator or pantry images and generate personalized recipes using artificial intelligence. The system comprises an image acquisition module, image preprocessing module, YOLOv8-based ingredient detection module, structured inventory generation module, user preference management module, prompt engineering module, large language model-based recipe generation module, backend services, and a microservice architecture. The uploaded refrigerator image is processed to accurately detect available food ingredients, which are converted into a structured inventory. The system combines the detected ingredients with user preferences including dietary restrictions, cuisine type, health conditions, spice level, and cooking time to generate a context-aware prompt for the large language model. The generated output includes personalized recipes, cooking instructions, ingredient lists, and health-related recommendations tailored to the available ingredients and user requirements. The proposed invention eliminates manual ingredient entry, reduces food wastage, improves meal planning efficiency, enhances user convenience, supports healthy dietary practices, and provides a scalable intelligent cooking assistant suitable for smart kitchen environments, restaurants, food service platforms, and household applications.

No. of Pages : 22 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127669 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Hybrid PDF Knowledge Extraction and Contextual Retrieval System Using RAG–CAG Framework

(51) International classification	:G06F 17/30, G06F 16/33, G06F 17/27, G06F 16/338, G06F 16/332
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mohan Babu University
Address of Applicant :IPR Cell, Mohan Babu University (Erstwhile Sree Vidyanikethan Engineering College), Sree Sainath Nagar, A. Rangampet, Tirupati, Andhra Pradesh, India - 517102 Tirupati Andhra Pradesh India
(72)**Name of Inventor :**
1)Mr.P.Venkateswarlu Reddy
2)Mr. Mappeti Sri Harsha Tej
3)Ms. Pamula Pranathi
4)Ms. Nelimela Varalakshmi
5)Ms. Mudadla Keerthana

(57) Abstract :
The present invention relates to a hybrid PDF knowledge extraction and contextual retrieval system utilizing Retriever-Augmented Generation (RAG) and Cache-Augmented Generation (CAG) for intelligent processing of unstructured PDF documents. The invention provides an integrated framework for extracting textual content from uploaded PDF files, preprocessing and segmenting the extracted information, generating semantic vector embeddings, and storing the embeddings within a vector database for efficient similarity-based retrieval. Upon receiving a user query, the system initially evaluates a cache repository to determine whether a previously generated response is available. In the event of a cache hit, the stored response is immediately returned, thereby reducing computational overhead and response latency. In the event of a cache miss, the system activates the Retriever-Augmented Generation module to retrieve contextually relevant document segments from the vector database and generate an accurate natural language response using a large language model. The generated response is subsequently stored within the cache for future reuse. By integrating semantic embedding generation, vector database indexing, intelligent caching, contextual retrieval, and natural language response generation within a unified framework, the present invention improves document retrieval accuracy, reduces redundant computations, minimizes response time, enhances scalability, and provides an efficient solution for knowledge extraction from large collections of PDF documents in educational, healthcare, legal, research, enterprise, and digital information management applications.

No. of Pages : 26 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127670 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Smart IoT Waste Bin with Embedded Waste Recognition, Automated Segregation, and Cloud-Based Analytics

(51) International classification	:B65F 1/14, B65F 1/00, G06Q 50/26, B65F 1/16, B65F 1/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mohan Babu University
Address of Applicant :IPR Cell, Mohan Babu University (Erstwhile Sree Vidyanikethan Engineering College), Sree Sainath Nagar, A. Rangampet, Tirupati, Andhra Pradesh, India - 517102 Tirupati Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. B. Rama Subba Reddy
2)Ms. Chilla Vanitha
3)Mr. D Sai Akshith Reddy
4)Mr. Doppala Vijay Lakshman
5)Mr. Chemallamudi Abhinav Karthik

(57) Abstract :
The present invention relates to an Intelligent IoT Smart Waste Management System configured to provide automated waste classification, multi-bin segregation, real-time monitoring, and intelligent waste collection using Internet of Things (IoT) technology. The system comprises an Arduino Uno controller, NodeMCU (ESP8266), infrared sensor, rain sensor, metal detection sensor, ultrasonic sensor, servo motor mechanism, wireless communication module, web-based monitoring dashboard, and automated notification module. The integrated sensors continuously detect the presence and type of waste, classify waste into appropriate categories, and automatically direct the waste into corresponding collection bins using servo-controlled mechanisms. The ultrasonic sensor continuously monitors the fill level of each bin, while the communication module transmits real-time alerts and operational data to a web-based dashboard for remote monitoring and efficient waste collection management. The proposed invention significantly improves waste segregation accuracy, reduces manual intervention, prevents waste overflow, enhances recycling efficiency, supports sustainable environmental management, and provides a scalable, cost-effective solution suitable for smart cities, educational institutions, commercial facilities, industries, and other public infrastructures.

No. of Pages : 26 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127671 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Blockchain-Based Provenance System for Allograft Transplant Management

(51) International classification	:G16H 10/60, G16H 40/20, H04L 9/06, H04L 9/32, G06F 16/23
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mohan Babu University
Address of Applicant :IPR Cell, Mohan Babu University (Erstwhile Sree Vidyanikethan Engineering College), Sree Sainath Nagar, A. Rangampet, Tirupati, Andhra Pradesh, India - 517102 Tirupati Andhra Pradesh India
(72)**Name of Inventor :**
1)Mrs. N Anusha Venkat
2)Ms. Mallisetti Hemasree
3)Ms. M. Harini
4)Mr. Kodali Mani Kumar
5)Mr. Kancharla Sumiyen

(57) Abstract :
The present invention relates to a blockchain-based provenance system for allograft transplant management that enables secure, transparent, and traceable monitoring of the complete transplantation lifecycle through a decentralized transaction management framework. The invention facilitates authenticated coordination among hospitals, tissue banks, organ procurement organizations, transplant surgeons, logistics providers, and regulatory authorities by recording every authorized operational activity within a blockchain network. The disclosed system incorporates role-based access control, cryptographic transaction verification, programmable workflow execution, and a hybrid data management architecture comprising blockchain-based transaction records and secure off-chain storage for confidential medical information. The invention supports patient registration, donor verification, donor-recipient matching, organ retrieval, transportation monitoring, transplantation confirmation, and post-transplant record management while maintaining an immutable provenance history for every transplantation event. The generated audit trail enhances transparency, prevents unauthorized modification of operational records, improves regulatory compliance, and facilitates secure information exchange among participating healthcare organizations. By reducing manual intervention, improving workflow efficiency, strengthening operational accountability, and enabling continuous lifecycle traceability, the present invention provides a practical, scalable, and technically reliable platform for modern allograft transplantation management. The invention is further adaptable for integration with existing healthcare information systems and may be extended to other regulated medical applications requiring secure provenance tracking, authenticated transaction management, and multi-organizational coordination.

No. of Pages : 25 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127678 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SECURE AND EXPLAINABLE LSTM-BASED HEALTHCARE INTRUSION DETECTION SYSTEM

(51) International classification	:G06N 3/04, G06N 3/08, G06F 21/55, H04L 29/06, G06N 3/0442
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR UNIVERSITY
Address of Applicant :ANANTHASAGAR, HASANPARTHY (M), WARANGAL URBAN, TELANGANA - 506371, INDIA Telangana India
(72)**Name of Inventor :**
1)KAMIGARI MOUNIKA
2)DR. SUDHA RANI V

(57) Abstract :
Disclosed herein is a a secure and explainable LSTM-based healthcare intrusion detection system (100) that comprises an input interface (102), a data acquisition unit (104), a processing unit (106), a memory unit (130), a communication interface (132), a breach prevention and response interface (134), and an alert output interface (136). The processing unit (106) includes a pre-processing module (108), a feature extraction module (110), an LSTM-based detection module (112), an anomaly detection module (114), an intrusion classification module (116), an explainability module (118), a privacy preservation module (120), a robustness assessment module (122), a threat assessment module (124), a breach prevention module (126), and an adaptive learning module (128). The system analyzes healthcare cybersecurity data, identifies anomalous activities, classifies potential intrusions, generates interpretable explanations, protects sensitive information, assesses threat severity, and facilitates preventive responses to reduce security breaches.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127679 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A MACHINE LEARNING-BASED DRIVER DROWSINESS DETECTION SYSTEM

(51) International classification	:G06K 9/00, G06V 40/16, B60W 40/08, A61B 5/18, G06V 10/764	(71)**Name of Applicant :** **1)SR UNIVERSITY** Address of Applicant :ANANTHASAGAR, HASANPARTHY (M), WARANGAL URBAN, TELANGANA - 506371, INDIA Telangana India (72)**Name of Inventor :** **1)MOHAMMAD SHAKEEL** **2)DR. SUDHA RANI V**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
Disclosed herein is a machine learning-based driver drowsiness detection system (100) that comprises an image acquisition unit (102), a data processing unit (104) having a pre-processing module (106), a facial landmark detection module (108), an eye closure analysis module (110), a blink dynamics analysis module (112), a head pose estimation module (114), a spatiotemporal feature fusion module (116), a machine learning classification module (118), an alertness assessment module (120), and a temporal window optimization module (122). The modules analyze facial image information, eye closure characteristics, blink frequency, blink duration, head pose angles, and temporal statistical characteristics to classify driver alertness and drowsiness. A memory unit (124) stores image information and processing results, an alert generation unit (126) generates warnings upon detecting driver drowsiness, a communication interface (128) facilitates information exchange with external devices, and a power supply unit (130) supplies electrical power to the system (100).

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127680 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : T-Shaped Nanomaterials-based Soil Sensor based on Fractional-Order Modelling for Agricultural Applications

(51) International classification	:G01N 27/22, G01N 33/24, G01R 27/26, G01N 27/00, G01V 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dayananda Sagar College of Engineering
Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr Shobha K R
2)Mr. Arya Suman Pattnaik
3)Dr. Sumit Swain

(57) Abstract :
The present invention describes a T-shaped epoxy-nanomaterial fractional capacitive sensor for electrical characterisation of soil at different moisture and ionic concentrations. The sensor consists of a T-shaped aluminium electrode mounted on a Teflon sheet, with an epoxy-based nanocomposite layer containing carbon nanotubes, graphene derivatives, metal-oxide nanoparticles, ferrite nanoparticles, or their combinations. The T-shaped electrode increases the electrode–soil contact area and permits measurement of the soil electrical response. The nanocomposite layer exhibits fractional-order capacitive behaviour represented by pseudo-capacitance Q and fractional-order parameter α. The sensor response is measured from impedance spectra over a selected frequency range, and the corresponding fractional-order parameters are obtained using an equivalent-circuit model. The measured Q and α values vary with soil moisture, ionic concentration, and dielectric properties. The sensor configuration is suitable for soil-moisture measurement, ionic-content assessment, and electrical characterisation of agricultural soil.

No. of Pages : 13 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127689 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ARTIFICIAL INTELLIGENCE-ENABLED MATERNAL PHYSIOTHERAPY WEARABLE SYSTEM FOR LABOUR PREDICTION AND REHABILITATION GUIDANCE

(51) International classification	:A61B 5/00, A61B 5/01, A61B 5/0205, A61B 5/11, A61B 5/1455
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY
(72)**Name of Inventor :**
1)KILANI KUSUMA
2)DR. P. V. HARI HARA SUBRAMANYAN
3)DR P. G. MAHESH KUMAR
4)DR PARTHASARATHY RANGANATHAN

(57) Abstract :
TITLE: ARTIFICIAL INTELLIGENCE-ENABLED MATERNAL PHYSIOTHERAPY WEARABLE SYSTEM FOR LABOUR PREDICTION AND REHABILITATION GUIDANCE APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention discloses an artificial intelligence-enabled biochemical maternal physiotherapy wearable system for labour prediction, maternal monitoring and personalized rehabilitation guidance. The system comprises a flexible wearable belt having integrated sensing modules for detecting uterine activity, maternal muscle activity, respiratory patterns, posture, movement, temperature and skin contact, together with a biochemical sensing module configured to analyse sweat biomarkers including lactate, cortisol, electrolytes and pH. An electronic processing unit performs signal conditioning, artefact reduction, normalization, synchronization and feature extraction, while a computational intelligence module fuses multimodal features to generate maternal-condition indices relating to fatigue, stress, exertion, movement efficiency and labour-related progression. A physiotherapy decision-support module compares the indices with individualized baseline values and generates guidance relating to breathing, relaxation, posture, positioning, movement, activity-rest modification and hydration. Feedback units provide guidance through vibration, visual indicators and wireless communication. The system further provides closed-loop reassessment, local memory and rechargeable power for operation with limited network connectivity.

No. of Pages : 34 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127690 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF PREBIOTIC HERBAL CANDY FOR DIGESTIVE AND ORAL WELLNESS

(51) International classification	:A23G 3/00, A23G 3/36, A23G 3/48, A23L 33/21, A61K 36/00, A61K 36/53, A61K 36/534, A61P 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)YATAKONA SURPRIYA
2)DR. SATHASIVAM SIVAMALAR

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF PREBIOTIC HERBAL CANDY FOR DIGESTIVE AND ORAL WELLNESS APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses a process of preparation of a prebiotic herbal health candy comprising the steps of: a. washing and drying at predetermined temperature, fresh mint (Mentha) leaves and fresh Karpuravalli (Plectranthus amboinicus) leaves, grinding the dried leaves, and extracting by aqueous extraction in distilled water, followed by filtration to obtain herbal extracts; b. heating jaggery with distilled water, followed by cooling and incorporating prebiotic fibers comprising inulin powder under continuous stirring to obtain a candy base; c. adding the herbal extracts into the candy base to form a mixture; d. homogenizing the mixture at predetermined RPM to obtain a homogenized mixture; e. casting the homogenized mixture into molds and allowing solidification to obtain solidified candies; and f. packaging the solidified candies under hygienic conditions to obtain prebiotic herbal health candies. The present invention also discloses a prebiotic herbal health candy, prepared by the process as described above.

No. of Pages : 21 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127691 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF HERBAL TOPICAL OINTMENT FOR BURN AND SCALD HEALING

(51) International classification	:A61K 36/886, A61P 17/02, A61K 9/06, A61K 36/896, A61K 36/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)A.YASMIN FATHIMAA
2)DR. PAVITHRA AMIRTKUMAR

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF HERBAL TOPICAL OINTMENT FOR BURN AND SCALD HEALING APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses a process for preparing an herbal topical ointment composition for burn and scald healing, comprising the steps of: a. washing fresh banana leaves, and subjecting to aqueous extraction with distilled water under continuous stirring, followed by filtration to obtain banana leaf extract; b. grinding fresh coconut kernel with distilled water and filtering to obtain coconut milk; c. melting beeswax at a predetermined temperature by double boiling to form a semi solid base; d. incorporating the banana leaf extract, aloe vera gel, the coconut milk, shea butter, and vitamin E into the molten beeswax under continuous stirring to form a mixture; e. subjecting the mixture to mechanical agitation, cooling, filling into sterile containers and storing under refrigeration to obtain topical herbal ointment composition. The present invention shall discloses an herbal topical ointment composition for burn and scald healing, prepared by the process as described above.

No. of Pages : 22 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127692 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EDGE-ENABLED IOT SYSTEM FOR REAL-TIME SOIL FERTILITY ANALYSIS AND NUTRIENT MANAGEMENT IN PRECISION AGRICULTURE

(51) International classification	:G01N 33/24, G06Q 50/02, A01B 79/00, A01B 79/02, G01N 21/359
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)G UMAMAHESWARI
2)V SIVASANKARI
3)M ANITHA

(57) Abstract :
TITLE: EDGE-ENABLED IOT SYSTEM FOR REAL-TIME SOIL FERTILITY ANALYSIS AND NUTRIENT MANAGEMENT IN PRECISION AGRICULTURE APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses to a soil fertility monitoring and nutrient management system. The system comprises a soil sensing subsystem having a multi-parameter soil sensor configured to measure volumetric water content, soil temperature and electrical conductivity, a nitrate ion-selective sensor for nitrate measurement, a pH sensor, and a multi-spectral sensor for obtaining spectral information from soil. The soil sensing subsystem is operatively connected to an STM32H743ZI microcontroller configured for signal acquisition, synchronization, signal conditioning, filtering, validation and threshold comparison. A Quectel EC200U-CN 4G LTE Cat 1 communication module transmits soil information to a cloud-based monitoring platform. The cloud-based monitoring platform stores historical soil records, provides graphical visualization through computing devices or mobile interfaces, and generates recommendations relating to fertilizer application, irrigation scheduling, soil amendments and corrective agricultural practices based on predefined agronomic thresholds. The system enables continuous soil monitoring, localized edge processing and nutrient management.

No. of Pages : 25 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127693 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-RETAINING DUAL BUCCOLINGUAL FLAP RETRACTOR WITH REVERSE-ACTION CONCAVE BLADES AND RATCHET LOCK

(51) International classification	:A61B 17/02, A61C 8/00, A61C 19/06, A61C 19/00, A61C 7/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL
(72)**Name of Inventor :**
1)DR. ADITIPPRIYA .B .S
2)DR. VIJAYALAKSHMI .R
3)DR. JENNIFER SUHASINI S
4)DR. JAIDEEP MAHENDRA
5)DR. BURNICE NALINAKUMARI .C

(57) Abstract :
TITLE: SELF-RETAINING DUAL BUCCOLINGUAL FLAP RETRACTOR WITH REVERSE-ACTION CONCAVE BLADES AND RATCHET LOCK
APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL
ABSTRACT The present invention discloses a dual buccolingual flap retractor configured for simultaneous retraction of reflected buccal and lingual mucoperiosteal flaps during periodontal, implant, and oral surgical procedures. The retractor comprises first and second concave atraumatic broad working ends configured to engage and support respective buccal and lingual mucoperiosteal flaps. The working ends are anatomically contoured and smooth for tissue support and are connected through shanks joined at a central pivot screw hinge. Ergonomic handles are operatively connected to the shanks, in which compression causes the working ends to open apart in a reverse-action manner for simultaneous bilateral flap retraction. A ratchet lock mechanism positioned between the handles selectively maintains the handles in a compressed position and retains the working ends at a selected opening. The retractor provides stable, self-retaining bilateral flap retraction, distributes retraction forces across tissue, preserves flap integrity, improves surgical visibility, and reduces manual support during surgical procedures.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127695 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A FLOORPLAN DESIGNER USING DEEP LEARNING TECHNIQUES

(51) International classification	:G06F 30/13, G06N 3/08, G06N 3/04, G06F 30/27, G06F 30/20	(71)**Name of Applicant :** **1)RNS Institute of Technology** Address of Applicant :RNS Institute of Technology, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098 Bangalore Karnataka India **2)Hemanth T** **3)Dhriti R** **4)Chinmayee S**
(31) Priority Document No	:NA	**5)Adith Prajwal k**
(32) Priority Date	:NA	**6)Nayana C P**
(33) Name of priority country	:NA	**7)Dr Suresh L**
(86) International Application No Filing Date	: :01/01/1900	(72)**Name of Inventor :** **1)Dr Suresh L**
(87) International Publication No	: NA	**2)Nayana C P**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**3)Adith Prajwal k** **4)Chinmayee S**
(62) Divisional to Application Number Filing Date	:NA :NA	**5)Dhriti R** **6)Hemanth T**

(57) Abstract :

TITLE OF INVENTION: A FLOORPLAN DESIGNER USING DEEP LEARNING TECHNIQUES FIELD OF INVENTION: COMPUTER SCIENCE ABSTRACT The architectural design industry remains a cornerstone of modern urban development, yet the traditional reliance on manual drafting and specialized Computer-Aided Design (CAD) software often results in high costs, significant delays, and limited design exploration for homeowners and developers. To address these inefficiencies, PlanCraft introduces an AI-powered solution that leverages Generative Adversarial Networks (GANs) — specifically a Conditional Wasserstein GAN with Gradient Penalty (cWGAN-GP) — to synthesize optimized, high-precision residential floorplans. By training deep learning models on comprehensive architectural datasets with advanced constraint-based learning, the system can rapidly generate diverse, structurally sound layouts that would otherwise require extensive human labor. The proposed system is implemented as a deep learning-based floor plan generation model that assists in the early stages of architectural design. The model accepts structured inputs such as plot dimensions, room requirements, adjacency preferences, and layout constraints, and generates optimized floor plan layouts based on learned architectural patterns. By leveraging deep learning techniques, the system aims to automate the traditionally time-consuming process of floor plan creation while maintaining spatial efficiency and functional room arrangements. The generated layouts are evaluated against user-defined constraints to ensure practicality and design consistency. Experimental results demonstrate the model's ability to produce diverse and feasible floor plan configurations, highlighting its potential as a decision-support tool for architects, designers, and homeowners. By integrating artificial intelligence into architectural planning, PlanCraft provides a foundation for future intelligent design systems capable of generating customized and efficient residential layouts.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127696 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TEXCRAFT AI: INTELLIGENT TEXTILE QUALITY ANALYSIS AND CREATIVE DESIGN GENERATION SYSTEM

(51) International classification	:G06N 3/04, G06N 3/08, G06V 10/764, G06T 7/00, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RNS Institute of Technology
Address of Applicant :RNS Institute of Technology, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098 Bengaluru Karnataka India
2)Vikas Patel K P
3)Sanjay J
4)Seeripi Ganesh Kumar
5)Suhas Varna
6)Vikas D H
7)Divyashree S
8)Dr.Sunitha K
(72)**Name of Inventor :**
1)Dr.Sunitha K
2)Divyashree S
3)Seeripi Ganesh Kumar
4)Suhas Varna
5)Vikas D H
6)Sanjay J
7)Vikas Patel K P

(57) Abstract :
TITLE OF INVENTION: TEXCRAFT AI: INTELLIGENT TEXTILE QUALITY ANALYSIS AND CREATIVE DESIGN GENERATION SYSTEM FIELD OF INVENTION: ARTIFICIAL INTELLIGENCE & MACHINE LEARNING Abstract TexCraft AI: Intelligent Textile Quality Analysis and Creative Design Generation System is an AI-powered platform that combines textile quality inspection and automated design generation within a single web-based system. It utilizes Convolutional Neural Networks (CNNs) for detecting and classifying fabric defects and Generative Adversarial Networks (GANs) for creating innovative textile patterns and designs. The system also incorporates pattern recognition techniques to identify fabric types and styles. Trained on a dataset of over 2,700 textile images, TexCraft AI enables users to upload fabric images, perform automated quality analysis, detect defects, classify patterns, and generate customized textile designs. By integrating quality assessment and creative design generation, the platform reduces manual inspection efforts, improves defect detection accuracy, minimizes production losses, and accelerates product development. TexCraft AI provides a comprehensive, intelligent, and scalable solution for modern textile manufacturing, quality assurance, and design innovation.

No. of Pages : 10 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127697 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DISHCOVERY – AI POWERED PERSONALIZED DIET PLANNER PLATFORM

(51) International classification	:G16H 20/60, G16H 10/60, G06Q 50/22, G16H 50/30, G16H 40/67	(71)**Name of Applicant :** **1)RNS Institute of Technology** Address of Applicant :RNS Institute of Technology, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098 Bangalore Karnataka India **2)Harshitha Padmanabha** **3)Chethan Mangle R** **4)Aakanksh S Kallihal** **5)Adithya Muralitharan** **6)Purnima S M** **7)Dr.Sathish Kumar S,**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr.Sathish Kumar S,**
(33) Name of priority country	:NA	**2)Purnima S M**
(86) International Application No	:	**3)Aakanksh S Kallihal**
Filing Date	:01/01/1900	**4)Adithya Muralitharan**
(87) International Publication No	: NA	**5)Chethan Mangle R**
(61) Patent of Addition to Application Number	:NA	**6)Harshitha Padmanabha**
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

TITLE OF INVENTION: DISHCOVERY – AI POWERED PERSONALIZED DIET PLANNER PLATFORM FIELD OF INVENTION: COMPUTER SCIENCE Abstract Dishcovery is a comprehensive AI-driven web application designed to deliver personalized nutritional guidance and meal planning. Leveraging user-provided health data including height, weight, age, and gender. The system calculates the Body Mass Index (BMI) to determine whether the user should gain, lose, or maintain their weight. The platform then tailors diet plans accordingly, enhancing accessibility to professional dietary advice without the need for direct consultation with a nutritionist. Dishcovery empowers users to personalize their dietary experience further by selecting from dietary categories such as vegan, vegetarian, eggetarian, or non-vegetarian, and by choosing preferred ingredients. The backend utilizes intelligent clustering algorithms and machine learning models trained on a calories and food composition dataset. These models cross-reference caloric needs based on BMI and lifestyle (e.g., sedentary, active) with food preferences, generating appropriate meal clusters. To foster ongoing user engagement and ensure flexibility, Dishcovery supports daily, weekly, and monthly meal planning. An embedded interactive nutrition chatbot enhances user interaction by providing real-time responses to queries related to nutritional content, ingredient benefits, and meal substitutions. Developed using the Flask framework and Python, Dishcovery features a user-friendly interface where users can create accounts, manage their health profile, and receive dynamic meal recommendations at the click of a button. The platform also accounts for allergies, work type, sleep patterns, hydration needs, and activity levels, creating a holistic and tailored approach to nutrition. Through smart automation and personalized care, Dishcovery not only promotes a balanced and health-conscious lifestyle but also democratizes nutritional expertise, making it affordable, accessible, and scalable across devices and demographics.

No. of Pages : 12 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202641127708 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A BIOCHAR-BASED AND MYCELIUM-BASED FILTRATE FOR WATER PURIFICATION AND METHODS OF MANUFACTURING THEREOF

(51) International classification	:C02F 1/28, B01J 20/20, B01J 20/28, C02F 3/10, B01J 20/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Karunya Institute of Technology and Sciences
Address of Applicant :Karunya Institute of Technology and Sciences, Karunya Nagar, Coimbatore Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)A K Priya
2)R Darshni

(57) Abstract :
The present invention relates to a biochar-based and mycelium-based filtrate for water purification, comprising an outer mesh bag [1] with a drawstring [1a] forming an outer containment structure; an inner fine-mesh sleeve [2] positioned within the outer mesh bag [1]; an outer mycelium-based filtration layer [3] disposed within the inner fine-mesh sleeve [2] and configured to provide depth filtration of suspended matter; a biochar-based adsorption layer [4] positioned downstream of the outer mycelium-based filtration layer [3] and configured to adsorb one or more dissolved contaminants; an inner mycelium-based filtration layer [5] positioned downstream of the biochar-based adsorption layer [4] and configured to provide fine filtration and bacteriostatic treatment; and a weight [6] coupled to the outer mesh bag [1] and configured to maintain the filtrate in a predetermined position in a water body. Further, the present invention relates to methods of manufacturing thereof. Figure 1

No. of Pages : 45 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127724 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF HERBAL CHEWABLE GUMMIES FOR RESPIRATORY AND DIGESTIVE DISORDERS

(51) International classification	:A61K 36/534, A61K 36/00, A61P 11/00, A61K 9/68, A61K 36/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI COLLEGE OF NURSING
(72)**Name of Inventor :**
1)DR. FABIOLA MERCY DHANARAJ

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF HERBAL CHEWABLE GUMMIES FOR RESPIRATORY AND DIGESTIVE DISORDERS APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF NURSING ABSTRACT The present invention discloses a process for preparing an herbal chewable gummy composition comprising the steps of: a. combining powdered betel leaf, powdered Mexican mint leaf, and pure honey and heating under continuous stirring to obtain an herbal mixture; b. dissolving pectin in purified water under stirring to form a gelling solution; c. blending the herbal mixture with the gelling solution at predetermined temperature to obtain a gummy matrix; d. pouring the gummy matrix into molds and allowing to set at predetermined to obtain herbal chewable gummies. The present invention also discloses a chewable herbal gummy composition, prepared by the process as described above.

No. of Pages : 19 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127725 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF FICIN-BASED CHEMOMECHANICAL CARIES REMOVAL GEL FOR SELECTIVE DENTIN EXCAVATION

(51) International classification	:A61Q 11/00, A61K 8/67, A61K 6/00, A61K 9/06, A61K 8/73
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL
(72)**Name of Inventor :**
1)DR. PREETHA ELIZABETH CHALY
2)DR. PADMA DEEPIKA K
3)DR. SHYAM S
4)DR. THARANI M

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF FICIN-BASED CHEMOMECHANICAL CARIES REMOVAL GEL FOR SELECTIVE DENTIN EXCAVATION APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL ABSTRACT The present invention discloses a process for preparing a ficin-based chemomechanical caries removal agent comprising dispersing ficin powder in distilled water under continuous stirring to obtain ficin solution. Carbopol is dissolved in distilled water under stirring to form a Carbopol gel, while amylopectin is dissolved in distilled water under heating to obtain an amylopectin solution. D-α-tocopherol acetate and propyl-p-hydroxybenzoate are added to the Carbopol gel under stirring to form a gel base. The ficin solution and amylopectin solution are then combined with the gel base under homogenization at a predetermined rotational speed. A green apple coloring agent is subsequently incorporated to obtain the ficin-based chemomechanical caries removal agent. The invention further discloses the ficin-based chemomechanical caries removal agent prepared by the aforementioned process. The resulting formulation provides a suitable gel-based composition containing ficin, a proteolytic enzyme, together with polymers and stabilizing components for application in chemomechanical caries removal procedures.

No. of Pages : 19 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127726 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF ANTIFUNGAL DENTURE ADHESIVE COMPOSITIONS

(51) International classification	:A61K 6/00, A61K 8/19, A61K 6/838, A61K 6/06, A61K 6/027	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India **2)MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)NADANA MERLIN DAYA .K**
(33) Name of priority country	:NA	**2)DEVI PARAMESWARI .B**
(86) International Application No Filing Date	: :01/01/1900	**3)ANNAPOORNI .H** **4)NARMADHA .D**
(87) International Publication No	: NA	**5)PINKY .A**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)SATHIYA SHRAVAN .R**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF ANTIFUNGAL DENTURE ADHESIVE COMPOSITIONS APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL ABSTRACT The present invention discloses a process for preparing a denture adhesive composition with antifungal activity, comprising: a. mixing Ganoderma powder with distilled water, incubating, and filtering to obtain a Ganoderma extract; b. dissolving cerium sulfate, in distilled water to obtain cerium precursor solution; c. adding the Ganoderma extract to the cerium precursor solution under continuous stirring and gradually increasing the pH of the reaction mixture using sodium hydroxide solution, and maintaining the reaction at predetermined temperature to obtain a white precipitate of cerium oxide nanoparticles; d. isolating the precipitate by settling, removing the supernatant, washing with distilled water, and drying to obtain cerium oxide nanoparticles; e. adding sodium carboxymethylcellulose, Polyvinylpyrrolidone, Gelatin, Glycerol, into the cerium oxide nanoparticles and adding distilled water under constant stirring to obtain a denture adhesive composition. The present invention also discloses a denture adhesive composition with antifungal activity, prepared by the process as described above.

No. of Pages : 20 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127727 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF ANTIDIABETIC HERBAL GUMMY

(51) International classification	:A61P 3/10, A61K 36/185, A61K 36/00, A61K 36/47, A61K 36/48
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)M.JAYASEELAN
2)DR. L. KARPAGAVALLI
3)DR.D.CHAMUNDEESWARI
4)B.GOWTHAM
5)R.KAMALI
6)A. EPCIBHA

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF ANTIDIABETIC HERBAL GUMMY APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses a process for preparing an antidiabetic herbal gummy composition comprising Gymnema sylvestre, Syzygium cumini, and Emblica officinalis extracts. The process comprises separately extracting Gymnema sylvestre leaves and Emblica officinalis fruits using ethanol by Soxhlet extraction and extracting Syzygium cumini seeds using distilled water by reflux extraction. The extracts are combined to form a herbal extract mixture. Methoxyl citrus pectin is dissolved in distilled water containing lemon juice concentrate under stirring to form a gelling solution, followed by addition of date syrup and inulin to obtain a sweetened gel base. The herbal extract mixture is incorporated into the sweetened gel base under stirring to form a herbal gel matrix. Vegetable glycerin and orange juice concentrate are subsequently added, followed by rosemary oil to obtain a blend. The blend is poured into moulds and allowed to set at a predetermined temperature to obtain antidiabetic herbal gummy composition.

No. of Pages : 27 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127728 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF NANOPARTICLE-INTEGRATED DENTURE ADHESIVE FOR DUAL RETENTION AND ANTIFUNGAL PROTECTION AGAINST CANDIDA BIOFILMS

(51) International classification	:A61K 6/00, A61K 6/838, A61K 6/884, A61K 6/06, A61K 6/02	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India **2)MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)NADANA MERLIN DAYA .K**
(33) Name of priority country	:NA	**2)DEVI PARAMESWARI .B**
(86) International Application No Filing Date	: :01/01/1900	**3)ANNAPOORNI .H** **4)NARMADHA .D**
(87) International Publication No	: NA	**5)PINKY .A**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)SATHIYA SHRAVAN .R**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

TITLE: A PROCESS OF PREPARATION OF NANOPARTICLE- INTEGRATED DENTURE ADHESIVE FOR DUAL RETENTION AND ANTIFUNGAL PROTECTION AGAINST CANDIDA BIOFILMS APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL ABSTRACT The present invention shall disclose a process of preparation of a bioactive denture adhesive composition configured to provide denture retention and inhibit Candida albicans colonization and biofilm formation, the process comprising: a. dispersing cerium oxide nanoparticles in distilled water by ultrasonication for 30 minutes to obtain nanoparticle dispersion; b. dissolving carboxymethyl cellulose, polyvinyl acetate, polyvinyl alcohol, hydroxyethyl cellulose, xanthan gum and glycerol in deionized water under continuous stirring to obtain polymeric adhesive matrix; c. introducing the nanoparticle dispersion into the polymeric matrix under continuous stirring to form a mixture; d. homogenizing the mixture at predetermined RPM to obtain homogenized formulation; e. adding glycerol as a moisture-retention agent and peppermint flavoring agent to the homogenized formulation under mechanical blending followed by adjusting the pH using sodium citrate to form an additive-formulation; and f. processing the additive-formulation into a paste by mechanical blending followed by vacuum drying to obtain bioactive denture adhesive.

No. of Pages : 25 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127736 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF HERBAL ANTIDIABETIC TEA BAG FORMULATION

(51) International classification	:A23F 3/00, A23F 3/34, A61P 3/10, A61K 36/00, A61K 36/24, A61K 36/48, A61K 36/899, A61K 36/54
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)GOWTHAM .B
2)DR. L. KARPAGAVALLI
3)DR. D. CHAMUNDESHWARI
4)R. KAMALI
5)M. JAYASEELAN
6)A. EPCIBHA

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF HERBAL ANTIDIABETIC TEA BAG FORMULATION APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention relates to an herbal antidiabetic tea bag formulation comprising Gymnema sylvestre leaf powder, Clitoria ternatea flower powder, Cymbopogon citratus leaf powder, Mentha piperita leaf powder, Cinnamomum verum bark powder, Glycyrrhiza glabra root powder, and Rosa damascena petal powder. The invention provides a preparation process involving shade drying the respective plant parts, pulverizing them individually, and sieving to obtain uniform powders. The obtained powders are blended by mechanical mixing to form a homogeneous herbal mixture. The mixture is subsequently packed into biodegradable filter-paper tea bags to facilitate convenient preparation and consumption as an herbal infusion. The formulation combines selected botanical ingredients traditionally associated with metabolic health, while offering ease of handling, portability, and consistent preparation. The disclosed process provides a practical approach for preparing a multi-herbal tea bag formulation intended to support dietary management of blood glucose levels.

No. of Pages : 28 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127737 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF PROTEIN- AND IRON-RICH NUTRITIONAL SNACK

(51) International classification	:A23L 33/00, A23L 33/185, A23L 33/105, A23L 7/10, A23L 25/00, A23P 30/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)KAMALI .R
2)DR. L. KARPAGAVALLI
3)DR. S. DHANALAKSHMI
4)EPCIBHA.A
5)PALLAVI .O
6)SOWMIYA.G

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF PROTEIN- AND IRON-RICH NUTRITIONAL SNACK APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses a process for preparing a functional nutritional bar composition comprising the following steps of: a. blending oat powder, whey protein isolate, pumpkin seed powder, halim (garden cress) seed powder, sesame seed powder, almond powder, moringa leaf powder, and amla fruit powder by mechanical blending to obtain a mixture; b. incorporating date paste and honey into the mixture followed by adding cocoa powder, cinnamon powder, and cardamom powder and mixing by planetary mixing to form a blend; c. moulding the blend into bar form under pressure and cold-setting the bars to obtain functional nutritional bar composition. The present invention also discloses a functional nutritional bar composition, prepared by the process as described above.

No. of Pages : 26 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127738 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF TARGETED PREOPERATIVE INTRAORAL ANTISEPTIC–ANAESTHETIC SPRAY FOR LOCALIZED MICROBIAL REDUCTION AND SITE PREPARATION

(51) International classification	:A61K 31/155, A61K 8/43, A61Q 11/00, A61K 9/12, A61P 31/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL
(72)**Name of Inventor :**
1)DR. ABIRAMI SAMPATH KUMAR
2)DR. GEETHA SRIDHARAN

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF TARGETED PREOPERATIVE INTRAORAL ANTISEPTIC–ANAESTHETIC SPRAY FOR LOCALIZED MICROBIAL REDUCTION AND SITE PREPARATION APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI AMMAL DENTAL COLLEGE AND HOSPITAL ABSTRACT The present invention relates to a targeted preoperative intraoral antiseptic–anaesthetic spray composition and a process for preparation thereof. The composition is formulated to provide localized antiseptic and anaesthetic action within the oral cavity prior to dental or intraoral procedures. The process comprises dissolving chlorhexidine gluconate in purified water under continuous stirring, followed by incorporation of cetylpyridinium chloride to obtain an antiseptic solution. Lidocaine hydrochloride is subsequently incorporated under controlled stirring to obtain a dispersion providing anaesthetic activity. The pH of the resulting dispersion is adjusted using sodium phosphate or citrate buffer, followed by addition of glycerol and hydroxypropyl methylcellulose to provide suitable formulation characteristics. The resulting formulation is filtered through a sterile membrane under aseptic conditions to obtain a sterile solution. The sterile solution is subsequently filled into a metered-dose spray device, thereby providing a targeted preoperative intraoral antiseptic–anaesthetic spray composition.

No. of Pages : 20 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127739 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS OF PREPARATION OF MEDICINAL FRUIT JELLY FORMULATION FOR NUTRACEUTICAL APPLICATIONS

(51) International classification	:A61K 36/815, A61K 36/81, A23L 21/10, A23L 21/15, A61K 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)M. KALAIMADHI
2)S. DHANALAKSHMI
3)D. CHAMUNDEESWARI
4)L. KARPAGAVALLI
5)K. SUDHAKAR
6)K. SHREE JAYARAM

(57) Abstract :
TITLE: A PROCESS OF PREPARATION OF MEDICINAL FRUIT JELLY FORMULATION FOR NUTRACEUTICAL APPLICATIONS APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses a process for preparing a medicinal fruit jelly formulation comprising Black Plum (Syzygium cumini) pulp and an aqueous extract of Kandankathiri (Solanum surattense). The process begins with extracting pulp from fresh, mature Black Plum fruits. Kandankathiri leaves are cleaned with potable water, dried, pulverized, and subjected to aqueous extraction using distilled water, followed by filtration. Ethanol is added to the filtrate for precipitation, and the precipitated alkaloid fraction is discarded to obtain an aqueous Kandankathiri extract. The Black Plum pulp and aqueous Kandankathiri extract are then mixed with sugar, potable water, and pectin to form a jelly mixture. The resulting mixture is subjected to control heating with continuous stirring to facilitate uniform incorporation and jelly formation. The heated mixture is subsequently transferred into sterilized containers, cooled, sealed, and stored under refrigeration to obtain the medicinal fruit jelly formulation.

No. of Pages : 21 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127744 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART ADAPTIVE MANUFACTURING FRAMEWORK USING REINFORCEMENT LEARNING AND MULTI-OBJECTIVE MATHEMATICAL OPTIMISATION FOR INDUSTRY 5.0 APPLICATIONS

(51) International classification	:G05B 13/04, G05B 13/02, G06N 20/00, G05B 19/418, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Jerry John
Address of Applicant :Director Academics, Yenepoya (Deemed to be University), Bengaluru Off Campus, Bengaluru, Karnataka Karnataka India
2)Dr. Nitin Rameshrao Chopde
3)Dr. Deepa V
4)Mr. Premanathan P
5)Dr. Chandrasekaran. G
6)J Sadhik Basha
7)Mr. Mahadeva Prasad
8)Dr. H.C. Nagaraj
9)Dr. T. T. Raman
10)Dr. Praveena B A
(72)**Name of Inventor :**
1)Dr. Jerry John
2)Dr. Nitin Rameshrao Chopde
3)Dr. Deepa V
4)Mr. Premanathan P
5)Dr. Chandrasekaran. G
6)J Sadhik Basha
7)Mr. Mahadeva Prasad
8)Dr. H.C. Nagaraj
9)Dr. T. T. Raman
10)Dr. Praveena B A

(57) Abstract :
A smart adaptive manufacturing framework, and method, using reinforcement learning and multi-objective optimization for Industry 5.0 applications is disclosed. A data-acquisition interface acquires operational data representing the state of a physical manufacturing system from a plurality of sensors. A reinforcement-learning agent determines, from the state, a control action for the controllable elements of the manufacturing system, such as machine speeds, feed rates, temperatures, and scheduling, so as to maximize a reward, and learns an effective control policy adaptively and without reliance upon a fixed model. A multi-objective optimization module balances competing production objectives, such as throughput, quality, energy consumption, material usage, and machine wear, characterizing their trade-offs as Pareto-optimal operating points and shaping the reward or constraining the actions according to a selected balance. A human-in-the-loop interface incorporates the input and oversight of a human operator in accordance with the human-centric principles of Industry 5.0, enabling the operator to adjust the balance of objectives and to approve or override control actions. A control interface applies the control action subject to safety constraints, and the agent is trained in a simulation before deployment. The framework adaptively controls the physical manufacturing system in real time so as to improve throughput, quality, energy and resource efficiency, and resilience, and reduce defects, scrap, and downtime.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127746 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD OF PREPARATION OF COLON-TARGETED DUAL POLYMERIC NANOPARTICLE FOR ENHANCED BIOAVAILABILITY AND CONTROLLED DRUG RELEASE

(51) International classification	:A61K 9/51, A61K 47/36, A61K 31/519, A61P 35/00, A61K 9/14	(71)**Name of Applicant :** **1)SRI RAMACHANDRA INSTITUTE OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :SRI RAMACHANDRA INSTITUTE OF HIGHER EDUCATION AND RESEARCH (DEEMED TO BE UNIVERSITY) NO.1, RAMACHANDRA NAGAR, PORUR CHENNAI CHENNAI TAMIL NADU INDIA 600116 CHENNAI Tamil Nadu India
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)DR DEEVAN PAUL A**
(32) Priority Date	:NA	**2)SAVEETHA ILANGOVAN**
(33) Name of priority country	:NA	**3)SWETHA ARUNACHALAM**
(86) International Application No Filing Date	: :01/01/1900	**4)YOGITHA MAHATMAJI SAWANT** **5)SANJAI BALAJI VILWANATHAN**
(87) International Publication No	: NA	**6)MEERAN MOHIDEEN NIZAM MOHIDEEN**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

TITLE: A METHOD OF PREPARATION OF COLON-TARGETED DUAL POLYMERIC NANOPARTICLE FOR ENHANCED BIOAVAILABILITY AND CONTROLLED DRUG RELEASE APPLICANT: SRI RAMACHANDRA INSTITUTE OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses a method of preparation of a colon targeted dual polymeric nanoparticle composition, comprising: a. dissolving Tofacitinib in dimethyl sulfoxide (DMSO) to obtain a drug solution; b. dissolving Eudragit S100 polymer in a mixture of dichloromethane and ethanol, followed by addition of Span 80, to obtain an organic phase; c. dissolving chitosan in acetic acid solution followed by adding citric acid and stirring followed by adjusting the pH obtain an aqueous phase; d. mixing the drug solution with the organic phase to form a mixture followed by adding the mixture dropwise into the aqueous phase under continuous stirring to obtain an oil-in-water emulsion; e. subjecting the oil in water emulsion to probe sonication and continuous stirring obtain nanoparticles suspension; f. centrifuging the nanoparticle suspension, washing the pellet with distilled water, and re suspending in aqueous medium to obtain purified nanoparticles; g. lyophilizing the purified nanoparticles to obtain colon targeted dual polymeric nanoparticle composition.

No. of Pages : 48 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127763 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVELOPMENT AND PERFORMANCE OF AIR PURIFYING CONCRETE USING TITANIUM DIOXIDE AND BIOCHAR

(51) International classification	:B01J 21/06, C04B 28/02, B01J 21/18, B01J 35/10, B01J 35/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)D. SAMUEL ABRAHAM
Address of Applicant :D. SAMUEL ABRAHAM ASSISTANT PROFESSOR, DEPARTMENT OF CIVIL ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND Technology , L&T BYPASS ROAD, COIMBATORE - 641062, TAMILNADU 9790687141 samuelabrahamcivil@siet.ac.in Tamil Nadu India
2)Agathiyan S
3)Dharun Sriram K
4)Misfar M
5)Priya Dharshini C
(72)**Name of Inventor :**
1)Samuel Abraham D
2)Agathiyan S
3)Dharun Sriram K
4)Misfar M
5)Priya Dharshini C

(57) Abstract :
Air pollution is a major environmental problem caused by vehicle emissions, industrial activities, and other human activities. This project focuses on the development of air-purifying green concrete using Titanium Dioxide (Ti02) and biochar as partial replacements for cement in M20 grade concrete. Titanium dioxide acts as a photocatalytic material that can help degrade certain pollutants under suitable light conditions, while biochar, produced from carbon-rich biomass, is incorporated to reduce cement consumption and investigate its influence on concrete properties. Three modified concrete mixes are considered: 2% TiO2 with 3% biochar, 2% TiO2 with 5% biochar, and 3% TiO2 with 5% biochar, based on the original cement mass. Conventional concrete is used as the control mix for comparison The experimental investigation is planned to evaluate the workability, compressive strength, water absorption, and durability of the concrete. Concrete cubes measuring 150 mm x -150 mm x 150 mm will be prepared and tested at 7 and 28 days of curing. A photocatalytic test may also be conducted under suitable light conditions to assess the degradation of a model pollutant such as methylene blue dye. The results will be compared with those of the control mix to identify the influence of TiO2 and biochar on concrete performance. Titanium dioxide is a photocatalytic material that can promote the degradation of certain pollutants when exposed to suitable light, while biochar is a carbon-rich material produced from biomass that may help reduce cement consumption and influence the properties of concrete. In this study, three modified concrete mixes are proposed: 2% TiO2 with 3% biochar, 2% TiO2 with 5% biochar, and 3% TiO2 with 5% biochar. These percentages are calculated based on the original cement mass. Conventional concrete without these materials is considered as the control mix for comparison. The main objective of this project is to identify a suitable combination of TiO2 and biochar that balances mechanical strength, durability, workability, and potential photocatalytic activity while reducing cement consumption. The findings may support the development of more sustainable construction materials. However, actual air-purification efficiency must be confirmed through suitable pollutant-gas testing, and the final concrete mix must be validated through an appropriate M20 mix design.

No. of Pages : 4 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127767 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CITIZENLEX: AN AI-BASED LEGAL LITERACY SYSTEM WITH LEGAL ACTION READINESS ASSESSMENT

(51) International classification	:G06Q 50/18, G06F 17/30, G06F 17/27, G06F 17/28, G06F 16/33	(71)**Name of Applicant :** **1)Dr.C.N.Vanitha** Address of Applicant :Professor, Department of Information Technology, Karpagam College of Engineering Tamil Nadu India **2)KPR Institute of Engineering and Technology** **3)Dr. S. Muthulakshmi** **4)Dr Malathy Sathyamoorthy** **5)Dr.C.Kumar** **6)Ms.Rithika K** **7)Ms. Vaishnavi B** **8)Mr.Vasikaran K**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	(72)**Name of Inventor :** **1)Dr. S. Muthulakshmi** **2)Dr Malathy Sathyamoorthy** **3)Dr.C.N.Vanitha** **4)Dr.C.Kumar** **5)Ms. Rithika K** **6)Ms.Vaishnavi B** **7)Ms.Vasikaran K**
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Legal information is widely available in India, but availability does not necessarily make that information useful to a person who has no legal training. Statutes, constitutional provisions, judgments, and procedural instructions often assume familiarity with legal terminology and formal processes. This paper presents CitizenLex, a citizen-focused digital platform designed to explain selected legal rights and possible remedies in accessible language while keeping the underlying legal sources visible. The proposed system combines natural language processing, retrieval augmented generation, and a structured legal knowledge base. Its main component, Legal Action Readiness Assessment (LARA), is designed to address a practical weakness of ordinary legal chatbots: an answer may sound complete even when the user's description leaves out facts that affect the next general legal step. LARA therefore identifies the information required by a supported legal pathway, compares it with the facts already provided, and asks a small set of targeted questions when important information is missing or inconsistent. The prototype is intended to run locally using a compact open-source language model together with retrieval, rather than depending on a paid cloud inference service. The paper presents the system architecture, knowledge representation, question-selection method, implementation plan, and evaluation methodology.

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127769 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FARMAI: INTELLIGENCE FARM MANAGEMENT SYSTEM

(51) International classification	:G06Q 50/02, G06Q 10/06, G06N 20/00, G06Q 10/10, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GUNASUNDHARI M
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BY-PASS, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)Guhan S
3)Jeeva Bharathi S
4)Manendhiran K
5)Mathan Prasath S
(72)**Name of Inventor :**
1)GUNASUNDHARI M
2)Guhan S
3)Jeeva Bharathi S
4)Manendhiran K
5)Mathan Prasath S

(57) Abstract :
ABSTRACT FarmAl is an intelligent Al-powered thrm management system designed to simplify and automate_agricultural_and_livestock_management activities through a unified web-based platform. The invention integrates artificial intelligence, machine learning, data analytics, and farm management functionalities to support efficient farm operations. It utilizes a Python-based backend, a responsive graphical user interface, and a local database for storing and managing farm-related information. The system incorporates a Linear Regression model to predict future milk production based on historical records and a knowledge retrieval mechanism to provide relevant farming information. An Al assistant enables users to access farm statistics, animal health information, vaccination details, feeding guidance, expenses, and scheduled tasks. The system also supports livestock and goat management, milk production tracking, expense categorization, task reminders, weather updates, and report generation in PDF, Excel, and CSV formats. Bilingual interaction in English and Tamil enhances accessibility for users. The proposed invention aims to improve farm productivity, simplify livestock management, support data-driven decision-making, and provide intelligent assistance through an integrated smart farm management platform.

No. of Pages : 7 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127773 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRICALLY CONDUCTIVE CONCRETE USING RECYCLED CARBON FIBER FROM END OF LIFE TIRES

(51) International classification	:C04B 16/06, C04B 14/38, C04B 18/16, C04B 20/10, C04B 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GOKUL A
Address of Applicant :DEPARTMENT OF CIVIL ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BYPASS ROAD, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)Haridharani S
3)Mohanaram ML
4)Siva S
5)Viswanatha Ragavedra P

(72)**Name of Inventor :**
1)GOKUL A
2)Haridharani S
3)Mohanaram ML
4)Siva S
5)Viswanatha Ragavedra P

(57) Abstract :

ABSTRACT The increasing demand for sustainable construction materials has created a need for innovative alternatives that improve the performance of conventional concrete while reducing environmental impact. This project focuses on the development of electrically conductive concrete using recycled carbon fiber recovered from end-of-life tires. The reuse of tire-derived materials provides an opportunity to reduce waste disposal and promote sustainable construction practices. Recycled carbon fiber is incorporated into concrete at different percentages to investigate its influence on the mechanical and electrical properties of concrete. The experimental investigation is planned to evaluate the workability, compressive strength, and electrical resistance of concrete containing different amounts of recycled carbon fiber. Conventional concrete without carbon fiber is used as the control mix for comparison. Concrete specimens will be prepared, cured, and tested at suitable curing ages, such as 7 and 28 days, to assess changes in strength and electrical performance. Electrical resistance measurements will be used to determine whether the addition of recycled carbon fiber improves the conductivity of concrete. The main objective of this project is to identify a suitable fiber content that balances mechanical strength, workability, and electrical conductivity. The conductive pathways formed by the carbon fibers may enable potential applications in smart infrastructure, self-heating pavements, de-icing systems, and structural health monitoring. The experimental results will help assess the feasibility of using recycled carbon fiber in conductive concrete. Overall, this study aims to combine waste material recycling with innovative concrete technology to support sustainable construction and future smart infrastructure development. Keywords: Electrically conductive concrete, recycled carbon fiber, end-of-life tires, compressive strength, electrical resistance, workability, sustainable construction, smart infrastructure

No. of Pages : 4 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127792 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LAWEASELM: AN EXPLAINABLE RETRIEVAL-AUGMENTED LEGAL INTELLIGENCE FRAMEWORK FOR AI-ASSISTED JUDICIAL DECISION SUPPORT

(51) International classification	:G06Q 50/18, G06F 16/35, G06F 16/33, G06F 16/338, G06F 16/332
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NARESH KUMAR YALLA
Address of Applicant :Flat no: 1403, Sai Raghava Towers, Jai Bharat Nagar, Hyder Nagar, Hyderabad, Telangana 500085, India Hyderabad Telangana India
(72)**Name of Inventor :**
1)Naresh Kumar Yalla
2)Dr.V. Kamakshi Prasad

(57) Abstract :
The present disclosure relates to a computer-implemented LawEaseLM: An Explainable Retrieval-Augmented Legal Intelligence Framework for AI-Assisted Judicial Decision Support for intelligent acquisition, retrieval, analysis, interpretation, verification, and presentation of legal information to support legal research and judicial decision-making workflows. The disclosed system comprises a Legal Data Acquisition Engine configured to collect legal information from authorized sources including court judgments, case records, statutory provisions, regulations, judicial precedents, legal repositories, citation databases, and legal information platforms. A Legal Document Ingestion and Preprocessing Engine validates, cleans, classifies, segments, and structures the acquired legal documents while preserving available source metadata, paragraph references, citations, and jurisdictional information. A Legal Knowledge Representation Engine organizes legal information into structured legal entities, hierarchical document representations, case relationships, statutory associations, and legal knowledge graphs. A Legal Knowledge Repository stores processed legal documents, metadata, semantic representations, and source-linked legal evidence. A Legal Indexing and Embedding Engine generates semantic embeddings and searchable indexes for legal documents, legal passages, statutory provisions, and precedent-related information. A Hybrid Legal Information Retrieval Engine combines semantic vector search, keyword-based retrieval, metadata filtering, citation-based retrieval, and legal knowledge graph traversal to identify legal information relevant to a submitted query. A Relevance Assessment and Evidence Selection Engine evaluates and ranks retrieved information according to semantic relevance, jurisdictional context, statutory applicability, citation relationships, document attributes, and other configured criteria. A Context Construction and Evidence Assembly Engine organizes selected legal passages, statutory provisions, precedent references, and associated source information into a structured context. A LawEaseLM Legal Intelligence Engine processes the legal query and retrieved context using retrieval-augmented generation and one or more legal-domain language models to generate contextual legal analysis and decision-support information. An Explainable Legal Output Generation Engine produces structured responses comprising relevant legal principles, supporting evidence, statutory references, precedent comparisons, and reasoning summaries. A Source Traceability and Legal Verification Engine associates generated legal statements with identifiable source documents, retrieved passages, and citation references to facilitate verification. A Judicial Review and Human-in-the-Loop Validation Engine enables authorized legal professionals and judicial decision-makers to review, evaluate, correct, and validate generated outputs before their use in legal workflows. A Governance, Security, and Model Lifecycle Management Engine manages access permissions, data protection, audit records, model versions, and retrieval configurations. The disclosed invention provides an explainable, evidence-linked, and adaptable legal intelligence framework intended to facilitate legal research, precedent discovery, statutory analysis, source verification, and human-supervised judicial decision support while preserving the authority of judicial decision-makers and the professional judgment of qualified legal practitioners

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127793 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : GOVTRACK360 - GOVERNMENT EMPLOYEE MANAGEMENT AND ADMINISTRATIVE WORKFLOW SYSTEM

(51) International classification	:G06Q 10/06, G06Q 10/10, G06Q 50/26, G06Q 10/00, G06Q 10/0631	(71)**Name of Applicant :** **1)KIRUBA S** Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BY-PASS, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India **2)Abisomam M** **3)Bhavadharani S** **4)Bhuvanashri SK**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)KIRUBA S** **2)Abisomam M** **3)Bhavadharani S** **4)Bhuvanashri SK**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT The present invention relates to a digital government workforce administration and workflow management system that integrates employee records, departmental operations, task allocation, and organizational monitoring within a unified platform. GovTrack 360 provides a structured solution for administrators, officers, and employees through secure authentication, controlled access, centralized data management, and functionality. The invention incorporates employee profile management, department administration, task assignment, workflow tracking, attendance and leave management, performance evaluation, rewards, notifications, reporting, and audit capabilities. By organizing these functions within a connected environment, the system improves visibility of administrative activities, supports coordination, and reduces dependence on fragmented records and manual procedures. Through its web-based interface and database architecture, GovTrack 360 enables authorized personnel to access workforce information, monitor responsibilities, and maintain consistent organizational records. The system addresses limitations in conventional administrative practices by connecting workforce management with operational workflows, thereby promoting transparency, accountability, efficiency, and scalable digital administration across government departments.

No. of Pages : 6 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127795 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : VEIL - A CRYPTOGRAPHIC SYSTEM FOR PRIVACY-PRESERVING SPATIAL QUERIES WITH VERIFIABLE ENCRYPTED INFERENCE FOR LOCATION PRIVACY

(51) International classification	:H04L 9/08, H04L 9/32, H04L 29/06, G06F 21/62, H04W 4/029	(71)**Name of Applicant :** **1)RNS Institute of Technology, Dr. Vishnuvardhan Road, R R Nagar Post, Channasandra, Bengaluru – 560098.** Address of Applicant :RNS Institute of Technology, Dr. Vishnuvardhan Road, R R Nagar Post, Channasandra, Bengaluru – 560098. Bengaluru Karnataka India **2)Mr. Manjunath G.S(Assistant Professor)** **3)Dr. Prakasha S(Associate Professor)** **4)Mr. T S Bhagavath Singh(Associate Professor)**
(31) Priority Document No	:NA	**5)Tripti Prasanna - 1RN23CS230**
(32) Priority Date	:NA	**6)Shanvi S Shetty - 1RN23CS190**
(33) Name of priority country	:NA	**7)Sinchana K - 1RN23CS205**
(86) International Application No	:	(72)**Name of Inventor :**
Filing Date	:01/01/1900	**1)Mr. Manjunath G.S(Assistant Professor)**
(87) International Publication No	: NA	**2)Dr. Prakasha S(Associate Professor)**
(61) Patent of Addition to Application Number	:NA	**3)Mr. T S Bhagavath Singh(Associate Professor)**
Filing Date	:NA	**4)Tripti Prasanna - 1RN23CS230**
(62) Divisional to Application Number	:NA	**5)Shanvi S Shetty - 1RN23CS190**
Filing Date	:NA	**6)Sinchana K - 1RN23CS205**

(57) Abstract :

Location-based services (LBS) such as navigation, food delivery, and ride-sharing require users to share their precise geographic coordinates to retrieve nearby points of interest. Sharing exact location data exposes sensitive personal information including home address, workplace, and daily routines, creating serious privacy risks. Existing solutions using encryption, anonymization, and differential privacy either introduce high computational overhead, reduce query accuracy, or fail to verify that the server returns correct results. The present invention, VEIL (Verifiable Encrypted Inference for Location Privacy), addresses these limitations through a unified cryptographic framework. The system employs Oblivious Transfer (OT) to encrypt user location inputs, Yao's Garbled Circuits to perform secure distance computation on encrypted coordinates without decryption, Voronoi diagrams constructed via Fortune's Algorithm for spatial indexing and result verification, and Merkle trees with SHA-256 hashing to provide cryptographic proof of result authenticity. The system enables users to submit private nearest-neighbour queries and receive verifiable, tamper-proof results without ever revealing their true location to the service provider. The invention has been validated on a real-world dataset of 4,565 geographic points-of-interest from Bengaluru, India.

No. of Pages : 9 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127809 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Behavior-Based Battery Health Prediction System

(51) International classification	:G01R 31/367, G01R 31/392, G01R 31/36, H01M 10/48, G01R 31/382	(71)**Name of Applicant :** **1)St. Joseph's College of Engineering** Address of Applicant :St. Joseph's College of Engineering, Chennai, India Chennai Tamil Nadu India **2)J.J.Mohamed Rafee** **3)S.Kowshick** **4)Mr. H.Prasad** (72)**Name of Inventor :** **1)St. Joseph's College of Engineering** **2)J.J.Mohamed Rafee** **3)S.Kowshick** **4)Mr. H.Prasad**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The CBMS have primarily report on voltage, current, temperature and SOC with little knowledge about degradation based on usage. This paper presents a Behavior Based Battery Health Prediction System (BBCHPS) which uses a Behavioral Temporal Degradation Transformer (BTDT) and an Adaptive Remaining-Life Bayesian Optimizer (ARLBO). BTDT is able to capture the temporal causality between discharge speed, loading behaviour, temperature response, charging frequency, charging depth and operation time while ARLBO adaptively update the battery health and remaining useful life prediction. The envisioned approach would map raw battery readings on a behaviour aware level to degradation representations to enable early warning signs of potential lifecycle end. On Battery level, evaluation is considered a lithium-ion battery splitting is done to reduce the information leakage as per the NASA lithium-ion battery aging data. The proposed methodology attains 98.25% accuracy which demonstrates the ability to provide reliable battery health prediction. It can be used for proactive maintenance and intelligent lifecycle management of electric vehicles, renewable-energy storage and portable battery applications.

No. of Pages : 10 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127811 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ADAPTIVE PORTABLE WATER PURIFICATION SYSTEM WITH SENSOR-CONTROLLED BIO-BASED FILTRATION AND CLOSED-LOOP ULTRAVIOLET-C DOSE VERIFICATION

(51) International classification	:C02F 1/00, C02F 1/32, C02F 1/28, C02F 9/00, C02F 1/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Karunya Institute of Technology and Sciences
Address of Applicant :Karunya Institute of Technology and Sciences, Karunya Nagar, Coimbatore Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)A K Priya
2)R Darshni

(57) Abstract :
The present invention provides an adaptive portable water purification system [100] with sensor-controlled bio-based filtration and closed-loop ultraviolet-c dose verification, comprising a container [101]; an opening [102]; a cap module [103], said cap module [103], comprises at least three water-quality sensors [105a] [105b] [105c] projected towards inside the container [101], so they stay in contact with water; a pod magazine [106] for holding plurality of individually moisture-sealed bio-based filter pods [107]; a trap door [108]; a dispensing actuator [109]; a piercing pin [111]; a seat [118] within the container [101] located at a water-intake region; a Ultra Violet-C disinfection source [112]; a UV photodiode [113]; a wireless module [117]; and a display device [114]. The adaptive portable water purification system [100] purifies water automatically in a closed loop based on the water quality in real time, then verifying disinfection without the user intervention. Referring to Figures 1 and 2

No. of Pages : 45 No. of Claims : 24

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127817 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE WEARABLE SOFT ROBOTIC MASSAGER WITH BIOSENSING FEEDBACK AND MACHINE LEARNING DRIVEN HYBRID ACTUATION SYSTEM

(51) International classification	:A61H 23/02, A61H 39/04, A61N 1/36, A61H 9/00, A61B 5/00	(71)**Name of Applicant :** **1)Dayananda Sagar Academy of Technology and Management** Address of Applicant :Dayananda Sagar Academy of Technology' and Management, Udayapura, Kanakapura road, opposite to Art of leaving, Bangalore, Karnataka - 560082 Karnataka India (72)**Name of Inventor :** **1)Dr. Srilatha Y** **2)Thaarmikaa Shree S** **3)Manasa J** **4)Z Farah Ibreez** **5)Deepthi N**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A smart wearable octopus massager system (100) comprises a flexible wearable body structure (001), microcontroller (002), tentacle members (003), servo actuators (004), tactile stimulation actuators (005), user input interface (006), and power supply assembly (007). Pressure sensing elements (008) detect pressure, and spatial contact mapping assembly (009), thermal an physiological sensing assembly (010), motion sensing assembly (011), and user-behaviour sensing assembly (012) detect body contact, body condition, movement, and behaviour. An artificial intelligence processing module (013) identifies an affected body region and corresponding tentacle member (003), determines a target pressure, and compares predicted and actual pressure. The microcontroller (002) adjusts the corresponding servo actuator (004) based on the pressure difference to increase, decrease, or maintain pressure, while other tentacle members (003) operate independently. The pressure difference is used to improve subsequent pressure control.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127819 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR EXTRACTING DATA FROM DOCUMENT IMAGES USING LAYOUT MATCHING AND SEMANTIC VERIFICATION

(51) International classification	:G06V 30/10, G06K 9/00, G06V 30/224, G06K 9/20, G06K 9/03
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Dinesh Singh Dhakar
Address of Applicant :Assistant Professor, School of Computers-CSE, Madanapalle Institute of Technology & Science (MITS) Deemed to be University Madanapalle, Annamaya District, Andhra Pradesh- 517325, India Andhra Pradesh India
2)Dr. Sachikanta Dash
3)Dr. Rabinarayan Panda
4)Smruti Ranjan Biswal
5)Bhaskar Mekala
6)Shazid Wahid Khandakhani
(72)**Name of Inventor :**
1)Dr. Dinesh Singh Dhakar
2)Dr. Sachikanta Dash
3)Dr. Rabinarayan Panda
4)Smruti Ranjan Biswal
5)Bhaskar Mekala
6)Shazid Wahid Khandakhani

(57) Abstract :
A system (100) comprises an image acquisition device (102), a memory (108) holding a template repository (110) and a processor (106), which binarises a document image, removes ruling lines and segments text blocks without character recognition, and compares an occupancy-grid layout signature with stored signatures. For the matched layout, each field region is located from the stored offset of its label block from an anchoring ruling line and from the adjacency of the value block to the label block, so that fields below tables of variable length are found correctly. Character recognition is performed only on the field regions, assembled into one composite image. Each value is verified against its semantic type and cross-field consistency rules; failing regions are re-recognised after enlargement and local thresholding, and values still failing are flagged. Badly matched documents are re-routed to full-page recognition, and new layouts are learned automatically from three similar documents.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127820 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-POWERED FOODGUARD FOR SMART REFRIGERATORS

(51) International classification	:F25D 29/00, F25D 11/00, G06Q 50/12, G05B 19/418, F25D 23/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M. PRIYANKA GANDHI
Address of Applicant :ASSISTANT PROFESSOR, ELECTRICAL AND ELECTRONICS ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L&T BY-PASS, CHINNIYAMPALAYAM POST, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)ABHIROOPA R R
3)ARVIND M
4)SREEMA M S
5)KAMAL G I
6)Dr. G. SUNDAR
(72)**Name of Inventor :**
1)M. PRIYANKA GANDHI
2)ABHIROOPA R R
3)ARVIND M
4)SREEMA M S
5)KAMAL G I
6)Dr. G. SUNDAR

(57) Abstract :
Food wastage is a major concern caused by improper food storage, forgotten expiry dates, and the lack of timely consumption decisions. Conventional smart refrigerators mainly provide expiry-date monitoring and basic alerts, but they do not intelligently determine which food should be consumed first. 1 he proposed AI-Based Smai 1- Food -Expiry-and-Waste Reduction System addresses this limitation by integrating IoT sensors, an ESP32 AI-based processing, and a user-friendly monitoring interface type. Sensors such as temperature, humidity, ammonia gas, door, and load cell sensors are used to monitor the refrigerator environment and spoilage-related indications in real time. Food information such as food, storage duration, added date, and expiry date is also recorded. The collected information is analyzed using an AI-based prioritization mechanism to determine the consumption priority of each food item. The system generates alerts such as "ConshMe: Today," "Use Within 2 Days," and "Safe for Later" through an LCD and mobile/web dashboard. Additionally, a recipe recommendation module suggests suitable recipes using food items approaching their expiry period. By combining real-time monitoring, intelligent consumption prioritization, an d recipe recommendation, the proposed system helps users utilize food efficiently, reduce avoidable food wastage, and achieve smarter refrigerator manfagernent."

No. of Pages : 9 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127826 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN AND DEVELOPMENT OF AN AUTOMATED MULTI-WHEEL VEHICLE FOR PRECISION IRRIGATION AND FERTILIZER APPLICATION

(51) International classification	:A01C 23/04, A01M 7/00, A01C 23/00, A01G 25/09, A01B 49/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Priyanka Rajan
Address of Applicant :8/29, 30, MARIYAN GARDENS, VELLALORE, COIMBATORE, TAMIL NADU, INDIA-641111 Tamil Nadu India
2)Mugesh M
3)Dharshini V
4)Gini P
5)Boomikartheeshwaran I
(72)**Name of Inventor :**
1)Priyanka Rajan
2)Mugesh M
3)Dharshini V
4)Gini P
5)Boomikartheeshwaran I

(57) Abstract :
ABSTRACT The present project relates to the design and development of an automated multi-wheel agricultural vehicle for precision irrigation and-fertilizer application in-crop-fields The system is developed to reduce manual labour, improve uniformity of liquid application, and provide safer operation during agricultural field activities. The proposed vehicle consists of a highclearance lightweight chassis, multi-wheel drive arrangement, high-torque DC gear motors, microcontrol ler-based control system, wireless remote control, overhead liquid reservoir, 12 V diaphragm pump, horizontal spray boom, and nine precision atomizing nozzles. The microcontroller controls the movement of the vehicle through H-bridge motor drivers based on wireless commands received from the operator. The diaphragm pump draws liquid from the reservoir and supplies it to the horizontal boom, where the nine evenly spaced nozzles distribute the liquid over the crop area. The main feature of the proposed system is the integration of the high-clearance multi-wheel vehicle and controlled liquid application mechanism into a single compact agricultural platform. The vehicle can be used for water, liquid fertilizer, and crop-protection liquid application. Its remote operation reduces the need for the operator to work directly within the application area, while the controlled boom and nozzle arrangement helps achieve more uniform coverage. The system is designed as a flexible platform that can be further developed with GPS/GNSS navigation, obstacle detection, IoT monitoring, computer vision, artificial intelligence, solar power, and variable-rate application technologies. The proposed vehicle therefore provides a practical foundation for improving the efficiency, safety, and precision of agricultural field operations.t and li

No. of Pages : 5 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127828 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Intelligent platform for resource discovery knowledge sharing and sustainable market linkage

(51) International classification	:G06Q 30/02, G06Q 10/06, G06N 5/02, G06Q 30/06, G06Q 50/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ms. Kavucika P.
Address of Applicant :Department of Artificial Intelligence and Data Science, KPR Institute of Engineering and Technology, Coimbatore Coimbatore Tamil Nadu India
2)Ms. Janavarshini PA
3)Ms. Prabhavathi D
4)Mrs. Kavya SP
(72)**Name of Inventor :**
1)Ms. Kavucika P.
2)Ms. Janavarshini PA
3)Ms. Prabhavathi D
4)Mrs. Kavya SP

(57) Abstract :
Intelligent platform for resource discovery knowledge sharing and sustainable market linkage Abstract: The present invention relates to an intelligent digital platform for resource discovery, knowledge sharing, and sustainable market linkage. An AI-based resource identification module, a knowledge-sharing interface, and a market-matching mechanism configured to connect resource providers, knowledge contributors, and potential users based on resource availability, domain-specific expertise, market demand, and sustainability parameters to optimize resource utilization, facilitate collaborative knowledge exchange, and promote sustainable commercial transactions.

No. of Pages : 7 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127829 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE WEARABLE SEIZURE MANAGEMENT SYSTEM WITH CONTINUAL LEARNING, PERSONALIZED THERAPY AND SECURE MEDICAL RECORDING

(51) International classification	:A61B 5/00, A61B 5/0205, A61B 5/11, G16H 50/20, A61B 5/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT
Address of Applicant :DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT, UDAYAPURA, KANAKAPURA ROAD, OPPOSITE TO ART OF LEAVING, BANGALORE, KARNATAKA-560082 Karnataka India
(72)**Name of Inventor :**
1)Priyanka V. Gudada
2)Dr. C. Nandini
3)Priyanka R
4)Chiranjeev Singh
5)Honmayee S P
6)Inarah Fathima Shariff

(57) Abstract :
A wearable intelligent epilepsy prediction, autonomous intervention, and adaptive learning system (100) is disclosed for continuous seizure monitoring and management. The system (100) comprises a wearable housing (001) accommodating a multimodal physiological sensing module (002) configured to collect physiological and contextual data of a user. An edge computing module (003) preprocesses the collected data, and an adaptive multimodal sensor fusion engine (004) combines 10 the data for analysis by a personalized continual-learning artificial intelligence engine (005) to predict seizure events. An explainable prediction engine (006) identifies factors associated with seizure prediction, while a confidence-adaptive emergency orchestration engine (007) determines an appropriate emergency response. A hybrid positioning module (008) identifies the user's location, an artificial intelligence therapeutic generation engine (009) provides personalized 15 therapeutic stimulation, and a post-seizure recovery assessment engine (010) evaluates user recovery. A federated learning synchronization module (011) updates the prediction model, a cryptographically secured medical event ledger (012) stores medical event information, an adaptive power optimization module (013) manages power consumption, a wireless communication module (014) transmits alerts, and a rechargeable power management module 20 (015) provides power for continuous system operation.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127834 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : “Laser Mic (Used for Spying)” - An Optoelectronic System for Remote Acoustic Sensing and Vibration Detection

(51) International classification	:G01H 9/00, G01N 29/24, G01H 11/06, G01H 11/08, H01S 3/13	(71)**Name of Applicant :** **1)CAMBRIDGE INSTITUTE OF TECHNOLOGY NORTH CAMPUS** Address of Applicant :KUNDANA, BENGALURU-562110 Kundana Karnataka India **2)Lokesh MV** **3)Pawan Kumar** **4)Prashanth T** **5)Kashinath** **6)Bhagya M** **7)Dr Prabha S Naik**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	(72)**Name of Inventor :** **1)CAMBRIDGE INSTITUTE OF TECHNOLOGY NORTH CAMPUS** **2)Lokesh MV** **3)Pawan Kumar** **4)Prashanth T** **5)Kashinath** **6)Bhagya M** **7)Dr Prabha S Naik**
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The present invention discloses a low-cost, non-contact optical acoustic sensing and remote vibration detection system. The system comprises a stabilized low-power continuous-wave laser module that projects a coherent beam of light onto a remote target reflective boundary, such as a single-pane glass window. Sound pressure fields cause the glass surface to act as a mechanical diaphragm, vibrating microscopically and introducing spatial trajectory changes into the reflected laser beam. A high-speed silicon photodiode is precisely positioned to capture the reflected beam variations, translating intensity modulations directly into an active analog signal stream. The photocurrent is conditioned via a high-gain transimpedance amplifier (TIA) combined with active bandpass noise-rejection filters to isolate the voice-frequency bandwidth. Finally, a digital acquisition interface linked to a processing device digitizes the voltage stream for real-time digital filtering, analysis display, and noise-free audio reproduction. The arrangement enables safe remote acoustic sensing in industrial or hazardous spaces without direct physical access or contact requirements.

No. of Pages : 8 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127842 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXPERIMENTAL INVESTIGATION OF SUSTAINABLE CONCRETE USING NANO SILICA AND NANO VANADIUM WITH WASTE TYRE RUBBER, PLASTIC WASTE AND E-WASTE

(51) International classification	:C04B 18/30, C04B 18/04, C04B 18/22, C04B 28/02, C04B 18/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Sathyakumar. N
Address of Applicant :DEPARTMENT OF CIVIL ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BYPASS ROAD, COIMBATORE -641062, TAMILNADU Tamil Nadu India
2)Jayaprakash. K
3)Kalai Selvam S
4)Nishandhini R
5)Veerakumar K

(72)**Name of Inventor :**
1)Dr. Sathyakumar. N
2)Jayaprakash. K
3)Kalai Selvam S
4)Nishandhini R
5)Veerakumar K

(57) Abstract :
The increasing demand for construction materials and the growing environmental concerns associated with cement production and waste disposal have encouraged the development of sustainable concrete. This project, entitled "Experimental Investigation of Sustainable Concrete Using Nano Silica and Nano Vanadium with Waste Tyre Rubber, Plastic Waste and E-Waste," investigates the potential of incorporating nanornaterials and waste materials into concrete to improve its performance while reducing environmental impact. In this study, nano-silica and nano-vanadium are incorporated into concrete to investigate their influence on strength, durability, and microstructural properties. Waste tyre rubber, plastic waste, and electronic waste (E-waste) are used as partial replacements for conventional aggregates. Nano-silica is expected to improve the microstructure of concrete through its pozzolanic activity and pore-filling effect, while nano-vanadium is investigated for its potential contribution to the mechanical and durability properties of the concrete. The experimental investigation includes the preparation of conventional concrete and modified concrete mixes with different replacement percentages. Tests such as workability, compressive strength, split tensile strength, flexural strength, water absorption, and density are conducted to evaluate the performance of the prepared specimens. The results are compared with those of conventional concrete to identify suitable combinations of nanomaterials and waste materials. The proposed approach aims to reduce the consumption of natural resources, promote waste utilization, and support environmentally responsible construction practices. The findings are expected to identify an appropriate mix proportion that balances mechanical performance, durability, and sustainability. The suitability of the developed concrete for structural applications will depend on the experimental results and applicable construction standards.

No. of Pages : 4 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127852 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Real-Time ESG Impact Assessment Dashboard

(51) International classification	:G06F 16/23, G06Q 50/00, G06F 16/21, G06Q 10/06, G06F 8/71
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VARANASI RAHUL
Address of Applicant :No 149, Sri Lakshmi venkateswara Nilaya, Opp JDS office Karnataka India
(72)**Name of Inventor :**
1)Rajesh Babu N
2)Pavithra V K
3)Shakeela A
4)Shruti B
5)Dr Usha Prabhu
6)Dr Shoba K B
7)Ashwini S
8)Dr C Srividhya

(57) Abstract :
A cloud-based system acquires environmental measurements and organisational records through network interfaces and converts them into version-identified canonical events. An event ledger prevents duplicate contributions and retains superseded records. A calculation service derives environmental impacts using applicable versioned factors and maintains dependencies linking source records and factors to indicators and aggregates. Upon receipt of a new or corrected event, affected dependent results are recalculated. A revision manifest references updated results and unchanged result versions and is atomically published after the updated results are stored. A dashboard client retrieves related values against the same manifest and presents environmental quantities, environmental, social and governance indicators, data freshness and coverage. Source records, factor versions and calculation history remain retrievable. The arrangement supports incremental processing of corrections and consistent presentation of derived results across heterogeneous data sources.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127857 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RISK-ADAPTIVE CYBER DEFENSE SYSTEM WITH DYNAMIC POST-QUANTUM CRYPTOGRAPHIC RESPONSE

(51) International classification	:H04L 9/08, H04L 9/40, G06N 10/00, H04L 29/06, G06F 21/57
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM University- AP
Address of Applicant :Neerukonda, Mangalagiri Mandal Guntur District-522240, Andhra Pradesh, India Guntur Andhra Pradesh India
(72)**Name of Inventor :**
1)DASARI, Karishma
2)Dr. Sarvani Anandarao

(57) Abstract :
A risk-adaptive cyber defense system (100) with dynamic post-quantum cryptographic response includes a traffic capture interface (102) and an artificial-intelligence-based intrusion detection engine (104) that outputs a classification and a confidence score for network traffic. A cryptographic threat indicator detection module (114) derives a cryptographic anomaly indicator from cryptographic protocol behavior. A risk assessment engine (106) fuses these outputs and the anomaly indicator into a continuously updated composite risk score that controls a cryptography controller (118) and a security policy engine (120). The cryptography controller selects a classical, hybrid, or post-quantum mode per session or asset by comparing the score with risk thresholds, and the security policy engine deploys policies to protected assets (122). A blockchain-based forensic evidence manager (124) records each security event, with the score and selected mode, on a distributed ledger, and a feedback module (126) updates the detection engine from the ledger. (FIG. 1)

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127884 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED MULTI-SOURCE DATASET AND AGENTIC ARTIFICIAL INTELLIGENCE-BASED SYSTEM FOR CROSS-CHANNEL FINANCIAL TRANSACTION MONITORING AND FRAUD DETECTION

(51) International classification	:G06Q 40/02, G06Q 20/40, G06N 20/00, G06Q 40/00, G07F 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TANUSHA B M(1RN23CS223)
Address of Applicant :RNS Institute of Technology, Dr. Vishnuvardhan Road, R R Nagar Post, Channasandra, Bengaluru – 560098. Bengaluru Karnataka India
2)SHREYA G BHAT(1RN23CS196)
3)Mr. Saad Qamar (Assistant Professor)
4)Dr. Sampada K.S (Professor)
5)SHREYA P S(1RN23CS198)
6)THANMAYEE A(1RN23CS227)
7)RNS Institute of Technology, Dr. Vishnuvardhan Road, R R Nagar Post, Channasandra, Bengaluru – 560098.

(72)**Name of Inventor :**
1)Dr. Sampada K.S (Professor)
2)Mr. Saad Qamar (Assistant Professor)
3)SHREYA G BHAT(1RN23CS196)
4)SHREYA P S(1RN23CS198)
5)TANUSHA B M(1RN23CS223)
6)THANMAYEE A(1RN23CS227)

(57) Abstract :

The present invention relates to the field of computer-implemented financial fraud detection systems and, more particularly, to an integrated multi-source transaction intelligence framework configured for cross-channel financial transaction monitoring, anomaly identification, fraud risk evaluation, and automated decision support using an Agentic Artificial Intelligence architecture. In one aspect, the invention provides a dataset integration framework configured to collect, standardize, normalize, and consolidate heterogeneous transactional data originating from a plurality of financial channels, including but not limited to Automated Teller Machine (ATM) systems, core banking systems, online and mobile banking platforms, and credit/debit card transaction systems, into a single unified analytical dataset. The framework preserves both common transactional attributes and channel-specific unique attributes within a coherent and extensible schema, thereby enabling interoperable, scalable, and cross-platform financial behavior analysis. In another aspect, the invention provides an Agentic AI-driven fraud analysis system comprising a plurality of autonomous and semi-autonomous intelligent software agents configured to collaboratively process the unified dataset. Said agents include, but are not limited to, a Transaction Processing Agent for ingestion and contextual parsing of transactional records, an Anomaly Detection Agent configured to detect statistically irregular transaction patterns using an Isolation Forest-based unsupervised machine learning model, a Risk Scoring Agent configured to compute fraud probability and severity scores based on behavioral, transactional, temporal, and contextual indicators, a Decision Agent configured to classify, prioritize, or escalate suspicious transactions based on predefined rules and learned intelligence, and a Communication and Coordination Agent configured to enable inter-agent interaction, alert generation, task orchestration, and response synchronization. The system is further configured to perform real-time or near-real-time fraud surveillance across multiple financial channels without dependency on fully labeled historical fraud datasets, thereby improving detection of novel, emerging, low-frequency, and previously unseen fraudulent patterns. The invention may further incorporate adaptive learning mechanisms, including periodic retraining, threshold refinement, agent-level performance feedback, and dynamic fraud rule evolution, to continuously enhance detection accuracy and resilience in changing transactional environments. Advantageously, the disclosed invention provides a modular, scalable, extensible, and intelligent fraud management framework capable of supporting cross-channel anomaly detection, fraud risk interpretation, autonomous decision support, and coordinated response generation, thereby improving operational efficiency, reducing false positives, and strengthening fraud prevention in modern digital financial ecosystems.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127886 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-INTEGRATED SMART LEARNING DESK SYSTEM FOR HOMEWORK RECOGNITION, PERSONALIZED TUTORING, ERGONOMIC MONITORING AND ENVIRONMENT OPTIMIZATION

(51) International classification	:G06Q 50/20, A47B 13/08, G09B 5/06, G06K 9/00, A47B 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)DR. INDU PURUSHOTHAMAN
2)DR. N. ARUNAGIRINATHAN
3)DR. C.VALLI NACHIYAR
4)DR. K. S SRIDEVI SANGEETHA
5)DR. T.THILAGAVATHI

(57) Abstract :
TITLE: AI-INTEGRATED SMART LEARNING DESK SYSTEM FOR HOMEWORK RECOGNITION, PERSONALIZED TUTORING, ERGONOMIC MONITORING AND ENVIRONMENT OPTIMIZATION APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT
The present invention relates to an integrated smart learning desk system for homework recognition, personalized tutoring, ergonomic assessment, and intelligent environmental optimization. The system comprises a desk body having a camera assembly, image-processing circuit, optical character recognition unit, homework assessment processor, tutoring processor, student learning-profile storage device, posture detection assembly, ergonomic alert device, environmental sensing assembly, environmental control device, central processing hardware unit, display screen, and communication hardware unit. The camera assembly captures handwritten or printed homework, which is processed and recognized for textual, mathematical, and diagrammatic content. The homework assessment processor evaluates responses, detects repeated errors, and identifies learning gaps, while the tutoring processor generates personalized explanations, hints, feedback, and practice questions. The posture detection assembly monitors posture, head position, viewing distance, and study duration. Environmental sensors monitor workspace conditions and enable corresponding adjustments. The central processing unit coordinates academic, ergonomic, and environmental data in real-time to provide personalized tutoring, ergonomic alerts, and adaptive environmental control, thereby creating an intelligent learning environment.

No. of Pages : 28 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127887 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTERACTIVE VOLUMETRIC DISPLAY SYSTEM FOR CONTACTLESS RESPIRATORY TRAINING, ADAPTIVE BREATHING GAMES AND HEALTH MONITORING

(51) International classification	:A61B 5/08, A63B 24/00, A61B 5/087, A63B 71/06, A61B 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)TREPHENA BLESSY R
2)DR. P. V. HARI HARA SUBRAMANYAN
3)DR P. G. MAHESH KUMAR

(57) Abstract :
TITLE: INTERACTIVE VOLUMETRIC DISPLAY SYSTEM FOR CONTACTLESS RESPIRATORY TRAINING, ADAPTIVE BREATHING GAMES AND HEALTH MONITORING APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses an interactive volumetric display system for adaptive breathing games and respiratory health monitoring with safety and infection analysis. The system comprises a volumetric projection subsystem having light-emitting modules, optical diffusion structures, projection guiding elements, and a volumetric projection window configured to generate floating visual objects. A respiratory sensing subsystem comprising a MEMS microphone, airflow sensor, and camera or optical module detects breathing-related parameters without physical contact. A processing and control subsystem comprising a microcontroller unit analyses respiratory parameters and controls visual responses according to breathing activity. Adaptive and guided breathing logic provides breathing interaction and pacing cues. A safety monitoring subsystem identifies unsafe respiratory patterns, while a respiratory pattern analysis subsystem identifies coughing, shallow breathing, irregular cycles, and infection-related abnormalities. Communication, power management, rechargeable battery module, protective housing and printed circuit board support integrated operation within an interaction region.

No. of Pages : 30 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127888 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MULTI-CHAMBER ADAPTIVE BALANCE DEVICE FOR LIFESPAN NEUROMOTOR TRAINING

(51) International classification	:A63B 26/00, A61H 1/02, A63B 23/035, A63B 22/16, A63B 21/008	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India (72)**Name of Inventor :** **1)POOJASREE .M** **2)DR. P. V. HARI HARA SUBRAMANYAN** **3)DR P. G. MAHESH KUMAR**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

TITLE: A MULTI-CHAMBER ADAPTIVE BALANCE DEVICE FOR LIFESPAN NEUROMOTOR TRAINING APPLICANT: MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH ABSTRACT The present invention discloses a multi-chamber adaptive balance device for directional instability rehabilitation and neuromotor training. The device comprises a non-spherical inflatable body having a rounded triangular, tri-lobed geometry with three outwardly convex lobes arranged about a central vertical axis. Three sealed inflatable chambers are disposed within the respective lobes and separated by flexible, air-impermeable partition walls, wherein each chamber is provided with an independently adjustable inflation/deflation valve. A superior surface comprises a central support region configured to support postural control and balance training, while an inferior stabilization region is configured to reduce uncontrolled rolling while permitting controlled deformation and instability. Differential pressurization of the chambers modifies regional compliance, stiffness and tilt tendency to generate controlled perturbations in anterior-posterior, mediolateral, diagonal and rotational directions. The device thereby provides controlled, graded and direction-specific instability for balance, postural control and neuromotor training across pediatric, neurological, geriatric and sports rehabilitation applications.

No. of Pages : 26 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127889 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PASSIVE VISUAL BIOFEEDBACK THERAPEUTIC ROLLER FOR CONTROLLED PRESSURE AND MOVEMENT-GUIDED PHYSIOTHERAPY

(51) International classification	:A61H 15/00, A61H 7/00, A61H 23/02, A61H 15/02, A63B 21/00	(71)**Name of Applicant :** **1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH** Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India **2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)PRIYANKA S**
(32) Priority Date	:NA	**2)DR. P.G.MAHESH KUMAR**
(33) Name of priority country	:NA	**3)DR. P. V. HARI HARA SUBRAMANYAN**
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

TITLE: PASSIVE VISUAL BIOFEEDBACK THERAPEUTIC ROLLER FOR CONTROLLED PRESSURE AND MOVEMENT-GUIDED PHYSIOTHERAPY APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention relates to rehabilitation devices, particularly a passive multi-parameter visual biofeedback therapeutic roller for controlled pressure and movement-guided physiotherapy. The therapeutic roller comprises a central cylindrical core, a cushioning and force-transmission layer, a mechanically responsive colour-change layer, a protective deformable surface layer, a textured contact surface, an externally visible indicator region, and detachable end caps. The cushioning and force-transmission layer transmits applied mechanical loading toward the mechanically responsive colour-change layer, which produces a reversible optical response corresponding to applied pressure and its distribution. The protective deformable surface layer protects the responsive layer while permitting visual observation of the optical response. The textured contact surface facilitates myofascial release, soft-tissue mobilisation, and muscle relaxation. The therapeutic roller provides passive, localised, and reversible visual feedback relating to pressure magnitude, distribution, duration, and direction during rolling and compression-based physiotherapy without electronic pressure sensing or external power.

No. of Pages : 23 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127890 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CLOSED-LOOP MULTI-SENSORY THUMB ILLUSION SYSTEM FOR CORTICAL MOTOR ACTIVATION WITHOUT PHYSICAL MOVEMENT

(51) International classification	:A61N 1/36, G06F 3/01, G09B 23/28, A61B 5/00, G09B 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH
Address of Applicant :NO.12, VEMBULIAMMAN KOIL STREET WEST K.K. NAGAR CHENNAI CHENNAI TAMIL NADU INDIA 600 078 Chennai Tamil Nadu India
2)MEENAKSHI COLLEGE OF PHYSIOTHERAPY
(72)**Name of Inventor :**
1)MAHMUDAA KN
2)DR. P. V. HARI HARA SUBRAMANYAN
3)DR. P.G.MAHESH KUMAR

(57) Abstract :
TITLE: CLOSED-LOOP MULTI-SENSORY THUMB ILLUSION SYSTEM FOR CORTICAL MOTOR ACTIVATION WITHOUT PHYSICAL MOVEMENT APPLICANT: 1. MEENAKSHI ACADEMY OF HIGHER EDUCATION AND RESEARCH 2. MEENAKSHI COLLEGE OF PHYSIOTHERAPY ABSTRACT The present invention discloses a closed-loop multi-sensory thumb illusion system for cortical motor activation without requiring equivalent physical thumb movement. The system comprises a wearable thumb interface, motor-intention sensors, physiological sensors, a directional micro-shear stimulation assembly, a visual feedback module, a force-feedback module, a control and processing unit, and a clinician and/or user interface. The motor-intention sensors detect voluntary motor attempts using surface electromyography electrodes and micro-movement sensors, while the physiological sensors monitor skin conductance and temperature. The control and processing unit coordinates direction-specific cutaneous shear stimulation with manipulated visual representations of thumb movement and optional force feedback to generate a perceived movement or effort that differs from actual thumb movement. The system adjusts stimulation parameters based on detected motor attempts and physiological responses while maintaining predefined safety limits. The invention provides coordinated multisensory feedback to support sensorimotor engagement and rehabilitation in individuals with absent, minimal, or impaired thumb movement.

No. of Pages : 33 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127898 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR TIME-LIMITED MAZE EXPLORATION BY AN AUTONOMOUS ROBOT USING A LEARNED ROUTE MAP

(51) International classification	:G06N 3/08, G06N 3/04, G01C 21/34, G01C 21/20, G01C 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. JESSY SUJANA G
Address of Applicant :Assistant Professor, Department of Computer Science(Emerging Technologies), SRM Institute of Science and Technology, Vadapalani, Chennai, Tamil Nadu, 600026 Tamil Nadu India
2)Akash S
3)Anou Zecharia W B
4) Gokul S
(72)**Name of Inventor :**
1)Dr. Jessy Sujana G
2)Akash S
3)Anou Zecharia W B
4)Gokul S

(57) Abstract :
.A micromouse must explore an initially unknown maze before perfbrming a timed fast run, while exploration is constrained by a limited time budget. This specification describes a learned route-weighted exploration system in. which a dilated convolutional neural network predicts the probability that each cell lies on the time-optimal route from a partially observed wall map. The predicted route probability adjusts the edge costs of a classical turn-aware planner, while turn-weighted flood-fill is used as a fallback whena complete exploration trip reveals no new wall information. In simulation, the method improved route quality over turn-weighted flood-fill on generated mazes under finite search budgets, but the advantage did not transfer to official competition mazes without remedies. The system is therefore intended as decision support for budget-constrained maze exploration; physical-robot performance requires further validation.

No. of Pages : 8 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127899 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPUTER-IMPLEMENTED SYSTEM AND METHOD FOR CAUSE DISCRIMINATION VALIDATION OF PERTURBATION-BASED MACHINE_LEARNING EXPLANATIONS

(51) International classification	:G06N 5/02, G06N 5/04, G06N 20/00, G06F 11/36, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. JESSY SUJANA G
Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING, SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, VADAPALANI, CHENNAI, TAMIL NADU, INDIA-600026 Tamil Nadu India
2)Aryan Murari
3)Abhinav Santhosh
4)Amitabh Thomas Saji
5)Ravi Vishnu D

(72)**Name of Inventor :**
1)DR. JESSY SUJANA G
2)Aryan Murari
3)Abhinav Santhosh
4)Amitabh Thomas Saji
5)Ravi Vishnu D

(57) Abstract :
ABSTRACT A computer-implemented method of evaluating a perturbation-driven feature-attribution explanation is used to determine causes of validation failure. The validation assessments are defined by intervention vectors, which represent an explainer seed, a target input, a neighborhood-sampling setup and a model-parameter condition that is either fixed or altered. A Diagnostic Discrimination Ledger keeps an outcome profile for each proposed cause, produced by using an associated fault operator and by performing the test multiple times. Candidates are not given tests which are considered invalid or whose profiles do not differentiate unresolved causes or whose interventions do not isolate a causal factor. Monitored results alter hypothesis validation by likelihoods adjusted for the intersection of stochastic sources with prior experiments; a result that falls outside all profiles suggests a residualcause hypothesis. A recommendation is made only if essential gates are cleared and both the criticalcause and residual support are below specified thresholds; else testing continues or the system does not act

No. of Pages : 26 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127900 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT BUS UTILIZATION PREDICTION FOR EFFICIENT GOVERNMENT PUBLIC TRANSPORTATION AND PASSENGER SERVICES

(51) International classification	:G06N 20/00, G01C 21/34, G06Q 50/30, G06Q 10/04, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Jessy Sujana G
Address of Applicant :Assistant Professor, Department of Computer Science(Emerging Technologies), SRM Institute of Science and Technology, Vadapalani, Chennai, Tamil Nadu, 600026 Tamil Nadu India
2)Sarangapani S
3)Gobikrishnaa S
4)Eamest Blessing B
(72)**Name of Inventor :**
1)Dr. Jessy Sujana G
2)Sarangapani S
3)Gobikrishnaa S
4)Eamest Blessing B

(57) Abstract :
A machine learning framework is described for predicting trip-level bus utilization in government public transportation. The framework uses route, schedule, calendar and weather attributes to estimate passenger use as a percentage of vehicle capacity. Linear regression, decision tree, random forest and gradient boosting models are compared, and random-forest tuning is explored with Taguchi and response-surface designs. In a prototype using 6,000 simulated trips, gradient boosting produced a mean absolute error of 5.04 percentage points, a root mean square error of 6.24, and an R2 of 0.81; crowding classes were predicted correctly for 88.1% of test trips. The outputs are intended to support fleet planning and advance passenger information. The reported results require validation with real transport-operator records.

No. of Pages : 9 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127901 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED AUTOMATED URBAN PARCEL MAPPING AND CADASTRAL FEATURE EXTRACTION SYSTEM USING DRONE IMAGERY

(51) International classification	:G06V 20/13, G06V 10/26, G06V 20/17, G06V 10/764, G06V 10/774
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Jessy Sujana G
Address of Applicant :Assistant Professor, Department of Computer Science(Emerging Technologies), SRM Institute of Science and Technology, Vadapalani, Chennai, Tamil Nadu, 600026. Tamil Nadu India
2)Putty Dhathri
3)Sivadharshini V
4)Mitushi Manoj
5)Sachitha G
6)Divya Prapanjani P A

(72)**Name of Inventor :**
1)Dr. Jessy Sujana G
2)Putty Dhathri
3)Sivadharshini V
4)Mitushi Manoj
5)Sachitha G
6)Divya Prapanjani P A

(57) Abstract :
Cadastral maps support property registration, land administration, and urban planning, but their preparation often depends on time-consuming manual interpretation and digitization. This paper presents Falcon-V;.-an AI-assisted workflow for extra-cting cadastral:relevant features from-highresolution - resolution drone or aerial imagery. The proposed pipeline combines semantic segmentation using U-Net or DeepLab, object detection using YOL0v8, and promptable mask refinement using Segment Anything ModerT (SAM 2). Image preprocessing and tiling prepare the imagery for inference; the resulting masks and detections are converted into spatial features and checked in a GIS environment. Topological checks, confidence information, and human review are intended to identify uncertain or inconsistent outputs. The workflow targets building footprints, visible boundary structures, and potential encroachment regions. The reported feature-level results indicate stronger detection performance for roof footprints and encroachment masks than for boundary walls. Since visible objects do not necessarily represent legally registered parcel boundaries, the output is treated as a preliminary mapping product requiring cadastral-reference comparison and surveyor verification.

No. of Pages : 10 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127902 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SKIN REGENERATION SPRAY COMPRISING COLLAGEN EXTRACTED FROM ELEUTHERONEMA TETRADACTYLUM FOR BURN INJURY MANAGEMENT

(51) International classification	:A61K 38/39, A61K 8/65, A61Q 19/00, A61Q 19/08, A61K 9/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mohan Kumar L
Address of Applicant :Assistant Professor, Department of Pharmacy Practice, Harsha College of Pharmacy, Nelamangala, Bangalore, India. Karnataka India
2)Pramod N
3)Shaik Azeem Ahmed
4)G.C Mamatha
5)Kempe Gowda N
6)Rajeeya Powar
7)Vidhyashree
8)Shiva Shankar
9)Kavya Sree G
10)Vinod Naik

(72)**Name of Inventor :**
1)Mohan Kumar L
2)Pramod N
3)Shaik Azeem Ahmed
4)G.C Mamatha
5)Kempe Gowda N
6)Rajeeya Powar
7)Vidhyashree
8)Shiva Shankar
9)Kavya Sree G
10)Vinod Naik

(57) Abstract :
The present invention relates to a topical spray composition comprising collagen extracted and purified from the skin and/or scales of the marine fish Eleutheronema tetradactylum. The composition comprises an aqueous vehicle containing approximately 0.5–1.0% w/v marine collagen, a humectant such as glycerin, an optional formulation aid such as polysorbate 20, a preservative system where required, and an acid or buffer for pH adjustment. The invention further relates to a process comprising extraction and purification of the collagen, controlled incorporation of the collagen into the aqueous vehicle, and filling of the resulting composition into a spray container. The formulation is intended for investigation as a topical biomaterial for supporting a hydrated wound environment and skin-regeneration processes. The formulation may be evaluated for collagen integrity, sprayability, delivered-dose consistency, stability, microbiological quality, skin compatibility, and in vitro wound-closure performance.

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127904 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART RESUME SCREENING PLATFORM

(51) International classification	:G06Q 10/10, G06Q 10/06, G06Q 10/00, G06F 17/30, G06F 16/951
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SARANYA M
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BY-PASS, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)Barath R
3)Deepak R G
4)Ajay P
5)Kavinraj S

(72)**Name of Inventor :**
1)SARANYA M
2)Barath R
3)Deepak R G
4)Ajay P
5)Kavinraj S

(57) Abstract :

ABSTRACT In the modern era of digital transformation and increasing recruitment demands, identifying suitable candidates from a large number of resumes has become both essential and challenging. SMART RESUME SCREENING PLATFORM aims to address this issue by providimg an intelligent, organized, and efficient approach to candidate screening and recruitment management. The system is designed to simplify the recruitment process by providing a centralized platform where recruiters can upload resumes, view candidate information, examine screening scores, and track application statuses. The platform utilizes a web-based architecture developed using React.js, Vite, and Tailwind CSS for the frontend, with Node.js and Express.js described as the backend technologies. The proposed screening workflow involves extracting relevant information from resumes, identifying technical skills, educational qualifications, professional experience, and projects, and comparing these details with job requirements. Based on predefined matching criteria, the system aims to calculate screening scores and rank candidates according to their suitability for specific job positions. Additionally, the platform provides an interactive recruiter dashboard that displays recruitment statistics, shortlisted candidates, applications under review, rejected candidates, and recent recruitment activities. Search and filtering functionalities enable recruiters to identify candidates efficiently and manage application statuses through a structured interface.

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127908 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT BASED AI IMPLEMENTED DRILLING MACHINE OPERATIONS

(51) International classification	:E21B 44/00, G06N 20/00, B23Q 17/09, B23Q 11/00, B23B 47/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SARANATHAN COLLEGE OF ENGINEERING
Address of Applicant :Venkateswara Nagar, Panjappur Tiruchirappalli Tamil Nadu India 620012 Tamil Nadu India
(72)**Name of Inventor :**
1) P. Dinesh Kumar
2) A. Sridevi
3)D. Mangalambigai
4)J. Ambika
5)T. Anita Dorothy
6)N. Muthu Aadhish
7)V.G. Pradeep Aanand
8)R. Bala Murugan
9)D. Abishiek
10)C. Praveen

(57) Abstract :
The present invention presents an AI-enabled drilling machine designed to improve productivity, accuracy, and operational safety. The system combines a drilling machine with 3ensors that continuously- monitor spindle speed, feed rate, vibration, temperature, motor current, torque, tool condition, and drilling depth. Sensor data is collected through a data acquisition unit and transmitted to an AT processing platform for analysis. Machine-learning algorithms identify patterns associated with tool wear, excessive vibration, overheating, and other abnormal operating conditions. By detecting these issues early, the system can alert operators, reduce the risk of equipment damage, and support timely maintenance. the AI platform can also recommend suitable drilling pararricters based on operating conditions, helping maintain consistent hole quality while minimizing unnecessary energy use and tool wear. A local display provides operators with real-time machine information, warnings, and recommendations. Remote monitoring and data storage allow supervisors to review performance, track maintenance history, and assess production trends. The system is intended to support predictive maintenance rather than relying solely on fixed schedules or unexpected breakdowns. Overall, integrating artificial intelligence and IoT monitoring into drilling operations can reduce downtime, improve process consistency, and enhance resource utilization. The proposed system offers a practical foundation for smarter, safer, and more efficient industrial drilling

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127923 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PROBIOTIC DELIVERY SYSTEM FOR FREEZE-DRIED MICROENCAPSULATED WEISSELLA CIBARIA MTCC 8292

(51) International classification	:C12N 1/20, C12R 1/01, A61K 35/745, A61K 35/74, A61K 9/50
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
Address of Applicant :Vidyagiri, Prasanthi Nilayam, Andhra Pradesh - 515134, India Prasanthi Nilayam Andhra Pradesh India
(72)**Name of Inventor :**
1)SUBBA, Sparsh Sumnirom
2)SUMANA, Araveti

(57) Abstract :
The present invention provides a strain-specific gastrointestinal survival of Weissella cibaria MTCC 8292, whereby the system comprises lyophilized freeze-dried Weissella cibaria MTCC 8292 microencapsulated in a matrix material comprising maltodextrin, skim milk powder, and gum Arabic. The probiotic delivery system of the present invention allows for enhanced gastrointestinal survival, cell viability, and stability of the encapsulated strain and its potential for application in probiotic delivery systems.

No. of Pages : 20 No. of Claims : 17

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127929 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EDUNEXAS-A CLOUD BASED STUDENT ECOSYSTEM PLATFORM

(51) International classification	:G06Q 50/20, G06Q 10/10, G06Q 10/06, H04L 29/06, G06F 21/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. R. P. S. Manikandan
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L&T BY-PASS, CHINNIYAMPALAYAM POST, COIMBATORE -641062 Tamil Nadu India
2)S. Vijayalakshmi
3)S. Magizhoviya
4)M. Poojitha
5)G. Preethi Roja
6)R. Roma

(72)**Name of Inventor :**
1)Dr. R. P. S. Manikandan
2)S. Vijayalakshmi
3)S. Magizhoviya
4)M. Poojitha
5)G. Preethi Roja
6)R. Roma

(57) Abstract :
EduNexas — Smart Campus Ecosystem Management System is a full-stack web-based platform designed to integrate academic, administrative, collaborative, and student-support activities into a centralized digital campus environment. The system provides authenticated student and administrator access and includes study-material sharing, a second-hand textbook marketplace, project team discovery, lost-and-found management, complaint tracking, events and clubs, notices, community discussions, notifications, and a campus helpdesk with ticket generation. The platform uses React with Vite for the frontend, Nodejs and Express.js for backend processing, and MongoDB Atlas with Mongoose for data storage. Security is implemented through college-domain email validation, JWT authentication, password hashing, role-based authorization, ownership checks, and controlled file uploads. The project team finder provides skill-based recommendations, while the lost-and-found module identifies potential matches using description similarity and generates notifications. The system reduces fragmented campus workflows, improves information accessibility, supports student collaboration, and ecosystem for cantos activities.

No. of Pages : 6 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127931 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HYALURONIC ACID COATED LIPOSOMES COENCAPSULATING PEMETREXED DISODIUM AND TETRAHYDROCURCUMIN

(51) International classification	:A61K 9/127, A61K 47/24, A61P 35/00, A61K 31/7048, A61K 31/122	(71)**Name of Applicant :** **1)JSS COLLEGE OF PHARMACY, OOTY – JSS Academy of Higher Education & Research** Address of Applicant :Post Box No. 20, Rocklands, Ootacamund, The Nilgiris - 643001, Tamil Nadu, India. OOTY Tamil Nadu India (72)**Name of Inventor :** **1)Archana Anebarassane** **2)Sambbhav G Shiyal** **3)D Maniraj** **4)C.K.Adhikesavan Balaji** **5)Bharati Pattanayak** **6)Rose Mary George K** **7)Roshni N** **8)Manimaran B** **9)Dr. Veera Venkata Satyanarayana Reddy Karri** **10)Dr. Gowthamarajan Kuppusamy**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a pharmaceutical liposomal composition coencapsulating pemetrexed disodium and tetrahydrocurcumin, and a process for its preparation. The liposomes comprise an aqueous core containing pemetrexed disodium and a lipid bilayer comprising soybean L-α-phosphatidylcholine, cholesterol and stearylamine, with tetrahydrocurcumin incorporated within the bilayer. Hyaluronic acid is non-covalently adsorbed onto the liposomal surface through electrostatic interaction between its carboxylate groups and positively charged amine groups of stearylamine. Preparation comprises lipid-film formation, hydration with an aqueous pemetrexed disodium solution, vesicle-size reduction, removal of unencapsulated drug and surface coating with hyaluronic acid. The composition enables incorporation of hydrophilic and lipophilic drugs within separate compartments of a single carrier and provides sustained release of both drugs. The composition is intended for CD44-directed delivery in non-small cell lung cancer and exhibits in vitro cytotoxic activity against A549 cells.

No. of Pages : 27 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127933 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI AFTERLIFE AND DIGITAL MEMORY: A HUMAN-CENTERED SYSTEM FOR ETHICAL DIGITAL LEGACY AND MEMORY RECONSTRUCTION

(51) International classification	:G06N 20/00, G06N 5/02, G06F 21/62, G06N 5/04, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.Shrinivas Sirdeshpande
Address of Applicant :Professor, Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning), AMC Engineering College, Bannerghatta Rd, Kalkare, Bengaluru, Karnataka 560083 Karnataka India
2)Dr Tamilarasan
3)Dr Ramani G
4)Bhuvan S
5)Thirthesh MJ
6)Niharikha N
7)Panchami

(72)**Name of Inventor :**
1)Dr.Shrinivas Sirdeshpande
2)Dr Tamilarasan
3)Dr Ramani G
4)Bhuvan S
5)Thirthesh MJ
6)Niharikha N
7)Panchami

(57) Abstract :
ABSTRACT OF THE INVENTION The present invention relates to a computer-implemented system and method for human-centered digital legacy preservation and memory reconstruction using artificial intelligence. The system receives authorized personal information including memories, stories, experiences, knowledge, profile attributes, communication characteristics, and other digital traces. A consent and identity module establishes the purpose of preservation, permitted data sources, authorized users, and lifecycle conditions. A memory and profile database stores the authorized information, while a backend retrieval module identifies relevant memories and context for a user request. An artificial intelligence response generation module uses the retrieved context to produce a personalized response while maintaining identity constraints and clear disclosure that the output is artificial. Conversation history provides continuity, and governance and security controls manage authentication, privacy, access, provenance, versioning, monitored evolution, and retirement. The system supports remembrance, family heritage, knowledge preservation, interactive memorialization, and controlled digital legacy management while maintaining human control over the representation.

No. of Pages : 13 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127937 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CODE COLLAB: A REAL-TIME COLLABORATIVE CODING PLATFORM

(51) International classification	:G06F 9/44, G09B 19/00, G09B 5/06, G09B 5/14, G06F 8/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SARANYA M
Address of Applicant :Sri Shakthi Institute of Engineering and Technology L&T Bypass Coimbatore Tamil Nadu India 641062 Tamil Nadu India
2)Anjali P
3)Deva Dharshini N
4)Diksha S
5)Harini B
(72)**Name of Inventor :**
1)SARANYA M
2)Anjali P
3)Deva Dharshini N
4)Diksha S
5)Harini B

(57) Abstract :
Collaborative software development plays a vital role in modern programming, enabling developers and students to work together on software projects from different locations. Traditional coding environments mainly support individual programming and often require separate tools for code sharing, communication, execufion, and debugging. The proposed system, "CODE COLLAB: A Real-Time Collaborative Coding Platform," introduces an interactive platform that allows multiple users to create or join a shared coding room and edit programs simultaneously. The platform supports five programming languages, including Java, JavaScript. Python, C, and C++, providing users with a common environment for multi-language programming and code execution. By integrating real-time code synchronization, room-based collaboration, and Al-assisted debugging, the system enables participants to instantly view and modify shared code while receiving suggestions for identifying and resolving programming errors. The invention addresses the limitations of conventional coding environments by providing a unified collaborative workspace for synchronous programming, lebugging, and learning. It Can be effectively utilized by students, developers, programming teams, educational institutions, and coding communities for efficient and interactive software development. interactive platform that would unite students, developers, and programming teams to collaborate on software development.. The application would allow its users to create or join a room with a specific room code where everyone can simultaneously write, edit, and modify the source code. The platform would -offer multi-language programming, including Java, iavaScript, Python, C, and C++. Additionally, there will be an option for collaborative programming and a debugging module, where the Al would be able to read. errors and suggest solutions or explain what error is. and how it could be resolved. The application will also allow the programmers to execute the code and set the output or error messages. With such a concept and an innovative idea, CODE -COLLA13 would revolutionize the programming world and allow people to collaborate in coding-, learn more about programming and bugs, and become better developers, all in a single online platform,

No. of Pages : 8 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127940 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI BASED INTERVIEW PREPARATION AND EVALUATION SYSTEM

(51) International classification	:G06Q 10/10, G06Q 10/06, G06N 20/00, G06N 5/02, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VIJAYALAKSHMI S
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BY-PASS, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)Adithya M
3)Asifa Thasnim M
4)Fran Delany A
5)Iniyaa M
(72)**Name of Inventor :**
1)VIJAYALAKSHMI S
2)Adithya M
3)Asifa Thasnim M
4)Fran Delany A
5)Iniyaa M

(57) Abstract :
ABSTRACT The AT Interview Preparation and Evaluation System is an intelligent platform designed to help students and job seekers improve their employability. It uses Artificial Intelligence and Generative Alto analyze resumes, extract candidate details, generate ATS-based scores, identify skill gaps, and recommend suitable jobs and learning paths. The system provides AI-powered mock interviews based on the candidate's resume and selected job role. It evaluates answers based on relevance, technical knowledge, communication, and grammar, and generates performance reports with scores, strengths, weaknesses, and feedback. The system also supports job application tracking, offering an integrated solution for personalized career guidance and interview preparation.

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641127941 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Method and System for Growth, Evolution, and Structural Remodelling in Tumour Spheroids

(51) International classification	:G06F 19/00, G01N 15/08, G06F 17/50, C12Q 1/68, G01N 33/50	(71)**Name of Applicant :** **1)Indian Institute of Technology Madras (IIT Madras)** Address of Applicant :The Dean, Industrial Consultancy & Sponsored Research [IC&SR], Indian Institute of Technology Madras, Sardar Patel Road, IIT P.O, Chennai- 600 036, Tamil Nadu, India Chennai Tamil Nadu India (72)**Name of Inventor :** **1)Kiran Raj M** **2)Tanishq**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A coupled computational framework, method, and system are disclosed for predicting spatiotemporal cellular and structural evolution in perfused and non-perfused three-dimensional tumour spheroids within microfluidic chip systems. The invention provides a computational formulation coupling species transport, concentration-dependent cellular kinetics, multi-state population transitions, and the evolution of construct geometry and microstructural transport properties. Spatially resolved viable, quiescent, and dead cellular populations are determined from governing transport and kinetic equations, wherein the evolving cellular-state distribution determines changes in geometry, porosity, permeability, tortuosity, and effective diffusivity through constitutive relationships. The resulting structural and transport properties are incorporated into the governing flow and species transport equations, establishing coupling between cellular evolution and the physical transport environment. The framework combines free-fluid and porous-medium flow formulations with transient species transport and cellular evolution in two-dimensional, axisymmetric, and three-dimensional configurations to predict nutrient availability, cellular viability, growth, structural remodelling, and therapeutic-species exposure under specified perfusion conditions.

No. of Pages : 30 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127952 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN LLM-POWERED LEGAL DOCUMENT ANALYSIS AND INTELLIGENT CASE RECOMMENDATION SYSTEM

(51) International classification	:G06Q 50/18, G06F 40/30, G06N 5/04, G06F 17/27, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Kiruba S
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L&T BY-PASS, CHINNIYAMPALAYAM POST, COIMBATORE -641062 Tamil Nadu India
2)Ajita K
3)Bangaiyarselvi B
4)Harshini K
5)Hemalatha S
(72)**Name of Inventor :**
1)Kiruba S
2)Ajita K
3)Bangaiyarselvi B
4)Harshini K
5)Hemalatha S

(57) Abstract :
The present invention relates to an LLM-Powered Legal Document Analysis and Intelligent Case Recommendation System, an AI-based platform designed to automate legal document understanding and assist legal professionals in case research and decision-making. The system utilizes Large Language Models (LLMs), Natural Language Processing (NLP), semantic search, and machine learning techniques to analyze legal documents such as judgments, petitions, contracts, statutes, and case files. It extracts key legal entities, identifies legal issues, summarizes lengthy documents, and recommends relevant precedent cases based on semantic similarity and contextual reasoning. The invention significantly reduces manual legal research time, improves document comprehension, enhances legal accuracy, and supports judges, lawyers, law firms, researchers, and law students through intelligent legal assistance.

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127958 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI BASED OS SECURITY THREAT DETECTION SYSTEM

(51) International classification	:G06N 20/00, H04L 29/06, G06F 21/55, G06F 21/56, G06F 21/57
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NISHMITHA K R
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BY-PASS, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)Asta Shirlyn P
3)Dhanya D
4)Haripriya M
5)Janani S

(72)**Name of Inventor :**
1)NISHMITHA K R
2)Asta Shirlyn P
3)Dhanya D
4)Haripriya M
5)Janani S

(57) Abstract :

ABSTRACT The Al Based OS Security Threat Detection System is an intelligent software platform designed to monitor operating system activities and detect potential security threats. It uses Artificial Intelligence and Machine Learning techniques to identify abnormal behavior such as malware activity, unauthorized access, suspicious processes, and abnormal network activities. The system analyzes system events, classifies detected threats, assigns severity levels, and generates real-time security alerts. A centralized dashboard allows users to monitor detected threats and security events. The proposed system helps in early threat detection, reduces manual monitoring, and improves overall operating system security.lized ca

No. of Pages : 8 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641127960 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR MULTI-PLANE RUNTIME CONTROL, GOVERNANCE, EVIDENCE LINEAGE AND VERIFICATION OF AI EXE-CUTION

(51) International classification	:G06F 21/57, H04L 12/24, G06F 9/445, G06F 8/71, G06F 21/56
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TekFrameworks Consulting Private Limited
Address of Applicant :House No. 2-91/23/C/402 Flat No.402, 4th Floor, Block C, Green Terraces, White fields, Kondapur, Madhapur, Shaikpet, Hyderabad, Telangana, India-500081 Hyderabad Telangana India
(72)**Name of Inventor :**
1)Arul Vignesh Arasan
2)Venkata Dakshinamurthy Kolluru
3)MaheshKumar Duvvarapu
4)Shilpa Kadam
5)Solomon J Ebenezer
6)Senthilkumar Soundrapandian
7)John Arul Prakash Sekar

(57) Abstract :
SYSTEM AND METHOD FOR MULTI-PLANE RUNTIME CONTROL, GOVERNANCE, EVIDENCE LINEAGE AND VERIFICATION OF AI EXE-CUTION ABSTRACT FOR THE INVENTION The present invention relates to a computer-implemented Enterprise Harness 100 comprising an Application Plane 101, a Management Plane 102, a Runtime Control Plane 103, and a Cognitive Engine Plane 104. The Application Plane 101 receives event requests from independently deployed enterprise applications 105, while the Management Plane 102 generates versioned runtime artifacts for application by corresponding runtime services. The Runtime Control Plane 103 provides enterprise-memory, policy, data-and-tool gateway, and telemetry runtimes for inline governance of live requests. A request envelope 113 associates a common correlation identifier 114 with enterprise identity, application identity, session context, policies and permitted resources. Cognitive engines 310 selectively process requests, while governance evidence and technical telemetry are correlated and stored fo

No. of Pages : 41 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128055 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT SMART CITY INFORMATION SYSTEM FOR DISASTER FORECASTING AND LOSS REDUCTION USING IoT SENSOR NETWORKS AND MACHINE LEARNING

(51) International classification	:G06N 20/00, H04W 4/38, G06Q 50/26, H04L 29/08, H04L 12/24	(71)**Name of Applicant :** **1)SHARMA, Monica** Address of Applicant :Associate Professor, School of Architecture, Dayananda Sagar Academy of Technology and Management, Kanakapura Main Road, Udayapura, Opp. Art of living, Bengaluru, Karnataka, India - 560082 Karnataka India **2)SAIDODDIN, Fathima Samana** **3)PANT, Piyush** **4)CHOUDHURY, Pradipta Roy** **5)DUBEY, Akriti** **6)Dayananda Sagar Academy of Technology and Management**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	(72)**Name of Inventor :**
(86) International Application No Filing Date	: :01/01/1900	**1)SHARMA, Monica** **2)SAIDODDIN, Fathima Samana**
(87) International Publication No	: NA	**3)PANT, Piyush**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**4)CHOUDHURY, Pradipta Roy** **5)DUBEY, Akriti**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention is related to an intelligent smart city information system for disaster forecasting and loss reduction using IoT sensor networks and machine learning. The proposed system comprises distributed multi-parameter IoT sensor nodes (101) connected to edge computing gateways (105) for local anomaly detection, a hybrid communication network (106) linking to a central cloud processing server (107) hosting multi-modal data fusion (108) and an ensemble machine learning prediction engine (109) that generates multi-hazard forecasts with risk scores and uncertainty estimates. A loss reduction optimization module (113) computes evacuation routes and resource plans, while an alert dissemination module (114) delivers targeted notifications. The system belongs to the technical field of smart city IoT and machine learning disaster management, providing technical advancement through hierarchical edgecloud processing, multi-hazard ensemble modeling, and integrated optimization over prior single-hazard or centralized systems, with principal use in urban multi-hazard forecasting and proactive loss mitigation. Figure 1.

No. of Pages : 12 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641128076 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ARTIFICIAL INTELLIGENCE-BASED INTELLIGENT TRAFFIC MANAGEMENT AND ACCIDENT PREVENTION SYSTEM

(51) International classification	:G08G 1/01, G08G 1/0967, G08G 1/16, G08G 1/04, G08G 1/005
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Arvind Mewada
Address of Applicant :Department of Computer Science and Engineering, National Institute of Technology Calicut, Kozhikode, Keralam-673601 Kerala India
2)Brajesh Kumar Umrao
3)Sushil Kumar Maurya
4)Anil Kumar
(72)**Name of Inventor :**
1)Arvind Mewada
2)Brajesh Kumar Umrao
3)Sushil Kumar Maurya
4)Anil Kumar

(57) Abstract :
Abstract ARTIFICIAL INTELLIGENCE-BASED INTELLIGENT TRAFFIC MANAGEMENT AND ACCIDENT PREVENTION SYSTEM The present invention relates to an artificial intelligence-based intelligent traffic management and accident prevention system (100) comprising a traffic data acquisition unit (101), an artificial intelligence-based processing unit (107), a traffic signal control unit (114), a communication network (115), a traffic control centre (116), and a user alert and information unit (117). The processing unit (107) analyses camera images and sensor measurements to detect vehicles and pedestrians, determine traffic density, vehicle speed, lane occupancy, and movement trajectories, and predict potential collision risks based on vehicle separation, relative speed, pedestrian proximity, and traffic signal status. A decision and control module (112) generates adaptive traffic signal instructions subject to predefined safety constraints. An alert and notification module (113) communicates hazardous conditions to authorised traffic operators and connected warning devices. The system supports adaptive traffic regulation, early hazard identification, coordinated incident notification, and improved road traffic monitoring. (FIG. 1)

No. of Pages : 19 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641128080 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Voice Based Email System for people with Visual Impairments.

(51) International classification	:H04L 12/58, G10L 15/26, G06Q 10/10, G06F 3/16, G10L 15/22	(71)**Name of Applicant :** **1)Dr.S.Tamilarasi** Address of Applicant :Professor and Head of the Department f Information Technology, St.Peters Institute of Higher Education and Research, Thiruvallur, Avadi, Chennai is 600054, Tamil Nadu Tamil Nadu India **2)Mrs.S.Priyadarshini** **3)Mr.M. Murugesan** **4)Mrs.A.Rebekah** **5)Ms. R.Kavitha** **6)Mrs.D.Priya**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr.S.Tamilarasi** **2)Mrs.S.Priyadarshini** **3)Mr.M. Murugesan** **4)Mrs.A.Rebekah** **5)Ms. R.Kavitha** **6)Mrs.D.Priya**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
Voice Based Email System for people with Visual Impairments. ABSTRACT Advent of computer technologies, many technological solutions have been implemented for the visually impaired so that they can utilize them and benefit from them. Considering it a key idea we have built a desktop application that will help blind people to send and read emails as ordinary people do. In this project, we describe the VMAIL system architecture for the Python platform that can be used by a Blind Person to access e-mails easily. The application uses a text-to speech, voice recognizer to facilitate sending and reading, and emails using a computer. The Internet has become one of the basic amenities for day-to-day living. Every human being is widely accessing knowledge and information through the internet. However, blind people face difficulties in accessing these text materials, also in using any service provided through the internet. The advancement in computer-based accessible systems has opened many avenues for the visually impaired across the globe in a wide way. Audio feedback-based virtual environments like screen readers have helped Blind people access internet applications immensely. We describe the Voicemail system architecture that can be used by a Blind person to access e-Mails easily and efficiently. In today's digital era, electronic mail (email) has become one of the most widely used forms of communication for both personal and professional purposes. It allows users to exchange information instantly and efficiently across the globe. However, for individuals with visual impairments, the process of reading, composing, and managing emails through conventional interfaces presents significant challenges. Most existing email systems are designed with visual interaction in mind, relying heavily on graphical user interfaces (GUIs), buttons, and text-based navigation. As a result, visually impaired users face considerable barriers when accessing these communication tools, thereby limiting their ability to fully participate in the digital world. According to the World Health Organization (WHO), more than 2.2 billion people globally suffer from some form of visual impairment or blindness. For these individuals, accessibility to technology is not just a convenience — it is a necessity for education, employment, and social inclusion. Although screen readers and accessibility features have improved over time, they still require a certain level of technical skill and are not always user-friendly. Many traditional email clients are not optimized to handle full voice-based control, leaving visually impaired users dependent on others for sending or reading emails.

No. of Pages : 15 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641128081 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Client-Side Web Application Filter for Preventing Web-Based Attacks

(51) International classification	:H04L 29/06, H04L 29/08, G06F 21/55, G06F 21/00, G06F 15/16	(71)**Name of Applicant :** **1)Dr.S.Tamilarasi** Address of Applicant :Professor and Head of the Department of Information Technology, St.Peters Institute of Higher Education and Research, Thiruvallur, Avadi, Chennai- 600054, Tamil Nadu Tamil Nadu India **2)Mr. M.Murugesan** **3)Mrs.A.Rebekah** **4)Mrs.S.Priyadarshini** **5)Mrs.V.Jeya Geetha**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr.S.Tamilarasi** **2)Mr. M.Murugesan** **3)Mrs.A.Rebekah** **4)Mrs.S.Priyadarshini** **5)Mrs.V.Jeya Geetha**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
Client-Side Web Application Filter for Preventing Web-Based Attacks ABSTRACT The Client-side Web Application Firewall (cWAF) is a security solution designed to protect client systems by monitoring and controlling web application traffic at the endpoint level. Unlike traditional server-side firewalls that focus on securing hosted applications, cWAF operates locally on the user's system, inspecting both incoming and outgoing HTTP and HTTPS traffic generated by web browsers and desktop applications. Its primary objective is to detect and prevent common web-based attacks such as Cross-Site Scripting (XSS) and SQL Injection (SQLi) before they impact the system. The system is built using a modular architecture that includes a traffic interception layer, a rule-based detection engine, and a monitoring dashboard. It is implemented using Python as the core programming language, with the mitmproxy framework enabling transparent interception and analysis of web traffic. A web-based interface developed using Flask and React allows users to monitor traffic activity, view logs, and manage configuration settings in real time. Functionally, cWAF operates as a local proxy that routes web traffic through an internal filtering engine. This engine analyzes requests using predefined security rules to identify suspicious patterns and block malicious content. All detected events are recorded in local storage using JSON or SQLite, enabling users to review logs and analyze system activity effectively. The proposed system provides an efficient and lightweight approach to client-side web security by combining real-time traffic inspection with user-friendly monitoring. By operating entirely within the client environment, cWAF enhances user privacy while providing an additional layer of protection against web-based threats. Overall, the system offers a practical and cost-effective solution for improving endpoint security in modern computing environments.

No. of Pages : 10 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128082 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED INTELLIGENT PREDICTION OF CHRONIC RENAL AND KIDNEY ABNORMALITIES DETECTION USING ADVANCED IMAGE PROCESSING AND MULTIMODAL PHYSIOLOGICAL DATA

(51) International classification	:G16H 50/20, A61B 5/00, G16H 30/40, A61B 5/055, G16H 30/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr M D Nandeesh

Address of Applicant :Associate Professor, Department of Electronics and Instrumentation Engineering, Ramaiah Institute of Technology , Bangalore , Karnataka. Karnataka India

2)Ms. Shashikala Rupesh Rathod

3)R Lalramdinmawia

4)Dr. Ranjana Santosh Kale

5)Dr. G. Appasami

6)Dr.T.M.Vijayalakshmi

7)Dr. Saravanan S

8)Dr. Rajinder Kumar

(72)**Name of Inventor :**

1)Dr M D Nandeesh

2)Ms. Shashikala Rupesh Rathod

3)R Lalramdinmawia

4)Dr. Ranjana Santosh Kale

5)Dr. G. Appasami

6)Dr.T.M.Vijayalakshmi

7)Dr. Saravanan S

8)Dr. Rajinder Kumar

(57) Abstract :

AI-BASED INTELLIGENT PREDICTION OF CHRONIC RENAL AND KIDNEY ABNORMALITIES DETECTION USING ADVANCED IMAGE PROCESSING AND MULTIMODAL PHYSIOLOGICAL DATA ABSTRACT An artificial intelligence (AI)-based system is disclosed for assisting in the prediction and detection of chronic renal and kidney abnormalities by integrating advanced medical image processing with multimodal physiological data analysis. The system receives renal imaging data, including ultrasound, computed tomography (CT), magnetic resonance imaging (MRI), or other clinically available images, together with physiological and laboratory parameters such as serum creatinine, estimated glomerular filtration rate (eGFR), blood urea, urine albumin-to-creatinine ratio, blood pressure, age, and relevant longitudinal measurements. An image-processing module performs image quality assessment, denoising, normalization, anatomical localization, kidney segmentation, and extraction of structural and textural features. A multimodal data-processing module validates, standardizes, and temporally aligns non-image data while identifying missing values and inconsistent measurements. Image-derived features and physiological features are combined by a multimodal learning architecture comprising one or more convolutional neural networks, feature encoders, temporal models, and a fusion layer. The resulting representation is supplied to one or more prediction models to estimate the likelihood of renal abnormalities, classify image patterns, and generate a risk score with associated confidence information. An explainability module identifies image regions and input variables contributing to the output, while a reporting module presents results for review by a qualified healthcare professional. The system may support longitudinal comparison, configurable thresholds, secure storage, and model-performance monitoring. The disclosed system is intended as a clinical decision-support aid and does not replace professional diagnosis or independently prescribe treatment.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128106 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DEPLOYABLE PERFORATED CANOPY FOR PROTECTING SOLAR PANEL ARRAYS AGAINST HAIL AND DOWNBURST LOADS

(51) International classification	:H02S 20/23, H02S 40/12, E04D 13/18, H02S 20/30, H02S 30/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)G. VINAYAGAMURTHY
2)UDIT SRIKANT RAO
3)CHINMAY SHASHIKANT PATIL

(57) Abstract :
The present disclosure relates to a protective system (100) for safeguarding ground-mounted and roof-mounted solar panel arrays (120) against hailstorms, downbursts, cyclonic winds, and snow. The protective system (100) includes a weather data unit (102), a manual input unit (104), a control unit (106), a power supply unit (108), a deployment drive unit (110), a telescopic mounting frame (112), and a perforated canopy (114) having an impact-absorbing foam layer (114-A) and a heating element (114-B). Upon receiving an indication of a weather event, the control unit (106) actuates the deployment drive unit (110) to extend the telescopic mounting frame (112) and deploy the perforated canopy (114) over the solar panel array (120). The perforations regulate airflow, reduce wind pressure and uplift, and cooperate with the impact-absorbing foam layer (114-A) to attenuate hail impact. The heating element (114-B) melts retained hail or snow before retraction.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128107 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BISTATIC GROUND PENETRATING RADAR FOR HIGH-RESOLUTION SUBSURFACE OBJECT DETECTION

(51) International classification	:G01V 3/12, G01S 13/88, G01S 13/89, G01V 3/38, G
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)PRIYANKA DAS
2)MANI RATNAM T S
3)M K CHARAN ARAVIND
4)SWAPNA S

(57) Abstract :
The present disclosure provides a system (102) and method (300) for detecting one or more buried objects using a bistatic ground penetrating radar (GPR). The system (102) includes a transmitting antenna, a spatially separated receiving antenna, a mechanism configured to move the antennas while maintaining a predetermined offset, a processor (202), and a memory (204). The processor (202) acquires reflected electromagnetic measurements from a plurality of scan locations, reconstructs a dataset from the acquired measurements, generates a matrix based on the reconstructed dataset, and identifies one or more buried objects based on the generated matrix. The disclosed bistatic configuration provides spatial diversity for subsurface imaging, enabling improved localization and detection of buried objects. The present disclosure provides enhanced subsurface imaging performance and improved detection accuracy for ground penetrating radar applications.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128108 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RESOURCE CONFIGURATION MANAGEMENT SYSTEM AND METHOD FOR A MULTI-COMPONENT CLOUD SERVICE

(51) International classification	:G06N 3/08, G06N 3/04, G06F 9/50, G06N 3/0464, G06N 3/042
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology Palakkad
Address of Applicant :Kanjikode West, Palakkad, Kerala - 678623, India. Palakkad Kerala India
2)Indian Institute of Technology Gandhinagar
(72)**Name of Inventor :**
1)S P, Abhinandan
2)KULKARNI, Sameer G
3)HEGDE, Adithya
4)GUPTA, Aman Kumar

(57) Abstract :
The present disclosure relates to a resource configuration management system (100) and a method (800) for a multi-component cloud service (116). The system (100) includes one or more processors (102) and memory (104). The one or more processors (102) generate a service graph (200) representing heterogeneous interconnected service components (116A, 116B, …, 116N) and process the service graph (200) using a graph neural network (GNN) (302) to generate a service-level embedding (304). Telemetry data (112) comprising resource utilization, latency, and budget consumption is processed with the service-level embedding (304) using an actor-critic reinforcement learning model trained using a reward function that rewards improved resource utilization and penalizes service-level objective violations and budget overruns. The system (100) selects, without prior knowledge of configuration capacities, a configuration action (114) comprising adding, removing, or maintaining resource allocation and applies the configuration action (114) to at least one service component.

No. of Pages : 47 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641128109 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR IN-SILICO DRUG DISCOVERY AND SCREENING

(51) International classification	:G16B 40/00, G16C 20/70, G16B 40/20, G16B 20/20, G16B 15/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Medxai Innovations Private Limited
Address of Applicant :No. 11, Joseph 2nd Street, Pazhavanthangal, Chennai - 600061, Tamil Nadu, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)SHEENA CHRISTABEL PRAVIN
2)REENA MONICA P.

(57) Abstract :
A system (100) and a method (400) for in-silico drug discovery and screening is disclosed. The system includes processors and input units that receive molecular inputs. The system has plurality of predictive units to generate predictive outputs from molecular inputs, and workflow orchestration unit that manages execution of predictive units, enforces dependency-aware data flow, and conditionally activates plurality of predictive units based on property of molecular inputs. The system uses transformation unit that derives secondary parameters from outputs of predictive units. Modeling unit generates quantitative biological model from secondary parameters. Genomic profiling unit generates personalized dosage recommendation by applying genotype-derived correction to quantitative biological model. Constrained inference unit generates scientifically grounded interpretation from aggregated outputs of predictive units and modeling unit. The interpretation is bounded by aggregated outputs and workflow orchestration unit enforces execution of constrained inference unit after aggregation of all outputs.

No. of Pages : 34 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128116 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PROCESS FOR PREPARATION OF COPPER MANGANESE SULFIDE (CuMnS) THIN FILMS FOR OPTOELECTRONICS

(51) International classification	:H01L 31/18, H01L 31/032, H01L 31/0747, H01L 31/072, H01L 31/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Technology, Tiruchirappalli
Address of Applicant :National Institute of Technology, Tiruchirappalli – 620015, Tamil Nadu, India Tamil Nadu India
(72)**Name of Inventor :**
1)M.C. Santhosh Kumar
2)ROY, Ronaldo
3)JOSEPH, Meril Maria

(57) Abstract :
The present invention relates to semiconductor materials and photovoltaic devices. Particularly, the present invention relates to copper manganese sulfide (CuMnS) semiconductor thin films, and a process for preparation thereof. The present invention also relates to the heterojunction diode and photovoltaic devices incorporating the CuMnS thin film as a semiconductor absorber layer. The process of the present invention enables precise control over film thickness, stoichiometry, and defect states, leading to improved charge transport and performance. Also, the process is cost-effective, scalable, and uses non-toxic absorber materials, making it suitable for next generation optoelectronic devices and solar cells.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641128124 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Passive Mechanically Decoupled Tilting System for Narrow Track Vehicles

(51) International classification	:B62D 17/00, B62D 7/18, B62D 7/16, B60G 3/20, B62D 9/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY MADRAS (IIT MADRAS)
Address of Applicant :The Dean, Industrial Consultancy & Sponsored Research (IC&SR), Indian Institute of Technology Madras, Sardar Patel Road, IIT Post, Chennai, Tamil Nadu, India, 600 036 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Guruprasad K R
2)Lohith GN
3)Pebbeti Pranavanath Reddy
4)Pranav Raj
5)Sachin Ramesh
6)Adithya Lakshminarayanan

(57) Abstract :
ABSTRACT Passive Mechanically Decoupled Tilting System for Narrow Track Vehicles A passive mechanically decoupled tilting system for a narrow-track vehicle comprises a tilting chassis, a parallelogram-based tilting linkage, a non-tilting suspension subframe, a steering-knuckle mechanism and a tilt-locking mechanism. The tilting linkage comprises substantially equal-length, parallel upper and lower wishbones configured to permit controlled chassis roll while substantially maintaining wheel orientation. The suspension subframe carries suspension coils and remains substantially non-tilting while the chassis tilts relative thereto, thereby substantially isolating suspension displacement from chassis tilting. The steering-knuckle mechanism independently accommodates steering and tilting rotations. The tilt-locking mechanism comprises a brake disc, a gravity-seeking pendulum having a cam profile, and a rocker arm carrying a roller follower and brake pad, whereby relative angular displacement actuates the brake pad to selectively restrict chassis tilting during low-speed or standstill conditions. Figure 6

No. of Pages : 37 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202641128125 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Adaptive Multi-Stage Battery Charging System and Method Using Electro-Thermal-Aging-Aware Multi-Objective Optimization

(51) International classification	:H02J 7/00, H01M 10/44, H02J 7/04, G01R 31/367, G01R 31/389	(71)**Name of Applicant :** **1)PRESIDENCY UNIVERSITY** Address of Applicant :Itgalpur, Rajanakunte, Bengaluru, Karnataka – 560 064, India Bengaluru Karnataka India (72)**Name of Inventor :** **1)A VIJAY KUMAR** **2)Dr V Joshi Manohar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT Adaptive Multi-Stage Battery Charging System and Method Using Electro-Thermal-Aging-Aware Multi-Objective Optimization An adaptive multi-stage battery charging system and method using electro-thermal-aging-aware multi-objective optimization are disclosed. The system comprises the following: a battery pack (1), a sensor acquisition unit (2) to acquire battery operating measurements, a state estimation module (3) to estimate state of charge, state of health, internal resistance, and battery temperature. An electro-thermal-aging model (4) represents electrical, thermal, and aging characteristics, while a multi-objective optimization module (5) determines optimized charging current levels and stage-transition conditions considering charging time, energy efficiency, temperature rise, and battery degradation. The optimized profiles are stored in a database (6) for selection by a charging controller (7), which regulates charging, monitors safety limits, controls stage transitions, initiates constant-voltage charging, and terminates charging at a specified cut-off current. Figure 1

No. of Pages : 31 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128126 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A portable bioimpedance measurement system for localized characterization of tissue and method thereof

(51) International classification	:A61B 5/053, A61B 5/0537, A61B 5/00, A61B 5/05, G01N 27/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology Madras (IIT Madras)
Address of Applicant :The Dean, Industrial Consultancy & Sponsored Research [IC&SR], Indian Institute of Technology Madras, Sardar Patel Road, IIT P.O, Chennai- 600036, Tamil Nadu, India Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Boby George
2)Shreenandan Sahu

(57) Abstract :
The invention discloses a bioimpedance measurement system (100) and method (1100) for characterization of a biological tissue region. The system (100) comprises a sensing unit (202) having a plurality of electrodes (304/1-304/8) arranged at multiple measurement locations, an electrode selection unit (402) configured to sequentially select one or more electrodes, an impedance measurement unit (410) configured to obtain impedance measurements over multiple excitation frequencies, and a processing unit (204) configured to generate location-specific impedance information. The system (100) may comprise a signal conditioning unit (406), a calibration unit (404), and a gain selection unit (408) for improving measurement reliability. The impedance measurements may be processed, compared, stored, and represented spatially to characterize variations in electrical properties of the biological tissue region. The system (100) may acquire measurements from multiple measurement locations without repositioning the biological tissue region and communicate measurement data to the user device (106) for analysis and visualization. To be published with: Figure 5a

No. of Pages : 46 No. of Claims : 18

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128130 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI ML based gesture to speech language translator

(51) International classification	:G06N 3/04, G06N 3/08, G10L 15/22, G06F 17/28, G10L 15/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)S. Gopinath
Address of Applicant :Professor and Head- Research & Extensions, Department of ECE, Karpagam Institute of Technology, Coimbatore - 641 105. Tamil Nadu India
2)Sanjay Karthick K
3)Dr.K.SIVAKAMI
4)Dr.K.SHEELA SOBANA RANI
5)Dr.S.PRAGADESWARAN
6)Ellakkiya A G
7)Hema shree M
8)Karthick E
9)Kaviya M
10)Manivel T
(72)**Name of Inventor :**
1)Dr.K.SIVAKAMI
2)Dr.K.SHEELA SOBANA RANI
3)Dr.S.PRAGADESWARAN
4)Ellakkiya A G
5)Hema shree M
6)Karthick E
7)Kaviya M
8)Manivel T
9)Sanjay Karthick K

(57) Abstract :
The AI-ML-based gesture to speech language translator is a pioneering initiative focused on facilitating effective communication between people with speech or hearing disabilities and the rest of the community by converting gestures into speech in real time. The AI-ML-based gesture-to-speech translator utilizes computer vision technology along with sensor-based gesture recognition and sophisticated machine learning techniques to accurately identify gestures of sign language and convert them into natural speech .Moreover, the accuracy and flexibility of the translation are ensured through the application of deep learning methods such as convolutional neural networks for gesture classification and recurrent neural networks for contextual mapping. The inclusion of the speech synthesis component also improves the usability of the translation tool by ensuring clear and human like output, thus providing an intuitive means of communication. The project is designed to be compatible with embedded systems and is light to deploy; therefore, it is highly portable and real-time responsive and accessible for use in various applications such as education, health, and workplaces. The socio-economic impact of the project is also notable for empowering differently-abled people and encouraging inclusiveness and breaking down communication barriers in day to day interactions. This paper seeks to examine the use of Artificial Intelligence (AI) and Machine Learning (ML) in the creation of an efficient gesture to speech translation system. The system will recognize human gestures, especially sign language, and translate them into speech. The system, therefore, seeks to improve communication between the hearing impaired and the rest of the community. The importance of the use of intelligent systems in the elimination of communication barriers is well addressed in this work.

No. of Pages : 11 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128150 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ARTIFICIAL INTELLIGENCE-BASED SURROGATE LIFE CYCLE ASSESSMENT

(51) International classification	:G06Q 50/02, G06Q 10/04, G06Q 10/06, G06N 20/00, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD
Address of Applicant :IIT Hyderabad Road, Near NH-65, Sangareddy, Kandi, Telangana, 502284, India. Hyderabad Telangana India
(72)**Name of Inventor :**
1)AMBIKA S
2)RAHUL KUMAR YADAV
3)JAYESH SINGH

(57) Abstract :
The present invention discloses an artificial intelligence-based surrogate life cycle assessment system 100 and a method 200 thereof. The system 100 includes an agricultural data input module 102 configured to receive agricultural production data and a field-emission calculation module 104 configured to determine field-level emission flows using applicable emission factors. An environmental impact prediction module 106 employs a surrogate model trained using an LCA reference dataset generated from life cycle inventories incorporating calculated field-level emission flows to predict environmental impact indicators without performing complete conventional LCA calculations for new agricultural scenarios. A multi-objective optimization module 108 evaluates candidate agricultural input combinations based on predicted environmental impact indicators and at least one of agricultural input cost, agricultural productivity, and carbon-credit potential. A recommendation engine 110 generates agricultural input or management recommendations based on the evaluated combinations. TO BE PUBLISHED WITH FIGURE 1

No. of Pages : 36 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128176 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SECURE DELIVERY ROBOT

(51) International classification	:B25J 5/00, G05D 1/02, B25J 11/00, B25J 9/16, B25J 19/02	(71)**Name of Applicant :** **1)P Brindha** Address of Applicant :8/46, Kamaraj Nagar, IInd Street, P.N.Road, Tirupur Tamil Nadu India **2)K. MITHUNESH**
(31) Priority Document No	:NA	**3)R. SATHISH**
(32) Priority Date	:NA	**4)K. SHANMUGAPRIYA**
(33) Name of priority country	:NA	**5)K. VEDHAHARSHINI**
(86) International Application No	:	(72)**Name of Inventor :**
Filing Date	:01/01/1900	**1)P Brindha**
(87) International Publication No	: NA	**2)K. MITHUNESH**
(61) Patent of Addition to Application Number	:NA	**3)R. SATHISH**
Filing Date	:NA	**4)K. SHANMUGAPRIYA**
(62) Divisional to Application Number	:NA	**5)K. VEDHAHARSHINI**
Filing Date	:NA	

(57) Abstract :

The present invention discloses a Secure Delivery Robot designed for autonomous and secure transportation of documents, files, books, parcels, and other lightweight articles within indoor environments such as educational institutions, offices, laboratories, and hospitals. The proposed system comprises a mobile robotic platform integrated with a ESP32, SSGM 800, ultrasonic sensors, wheel encoders, motor driver, DC geared motors, and rechargeable power supply. The robot utilizes predefined indoor maps and sensor-based navigation to determine suitable delivery routes, detect obstacles, and perform autonomous movement with minimal human intervention. A secure storage compartment equipped with an electronic locking mechanism protects the transported articles against unauthorized access. An One-Time Password (OTP) authentication system generates a unique access code for each delivery, which is communicated to the authorized recipient through a mobile application or communication interface. Upon reaching the designated destination, the robot verifies the entered OTP and activates the electronic locking mechanism only after successful authentication, thereby enabling secure retrieval of the delivered item. The mobile application facilitates delivery initiation, robot monitoring, and delivery-status notifications. Following successful delivery, the robot automatically returns to a predefined docking station. The proposed invention integrates autonomous indoor navigation, obstacle avoidance, wireless communication, and authenticated access control to provide a secure, efficient, and reliable solution for institutional delivery applications while reducing manual effort and minimizing unauthorized access.

No. of Pages : 24 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128219 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Intelligent Wheat Leaf Disease Diagnostic Device Using Dual-Branch Texture-Semantic Feature Fusion for Fine-Grained Disease Recognition

(51) International classification	:G06N 3/08, G06N 3/04, G06K 9/62, G06T 7/00, G06V 10/80
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M V S S BABU
Address of Applicant :Designation: Research Scholar Department: Department of ECE, Koneru Lakshmaiah Education Foundation, Vaddeswaram City: Vaddeswaram ,Pincode: 522302 ,Country: India Mobile No:9959950990 & Email: mvssbabu9@gmail.com Vaddeswaram Andhra Pradesh India
2)Sarat K Kotamraju
3)K. Ch Sri Kavya
(72)**Name of Inventor :**
1)M V S S BABU
2)Sarat K Kotamraju
3)K. Ch Sri Kavya

(57) Abstract :
The present invention relates to an intelligent wheat leaf disease diagnostic device utilizing dual-branch texture-semantic feature fusion for fine-grained disease recognition. The invention aims to provide an efficient, accurate, and automated approach for identifying and classifying wheat leaf diseases from digital images. The proposed device employs a dual-branch deep learning architecture in which one branch extracts discriminative textural features, including lesion patterns, spots, discoloration, and surface irregularities, while the other branch captures high-level semantic features, including lesion shapes, spatial relationships, and disease-specific visual characteristics. The extracted features are integrated through a feature fusion mechanism to generate a comprehensive representation of wheat leaf conditions, enabling improved discrimination between diseases exhibiting similar visual symptoms. The system incorporates image acquisition, preprocessing, feature extraction, feature fusion, and disease classification modules to facilitate automated diagnosis with minimal human intervention. The device is designed for potential deployment in agricultural fields, crop-monitoring systems, and laboratory environments to support timely disease identification and informed crop management. By combining complementary texture and semantic information, the invention provides a systematic approach to fine-grained wheat leaf disease recognition, with the potential to improve diagnostic reliability and assist farmers and agricultural professionals in implementing appropriate disease management practices.

No. of Pages : 8 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641128227 A
(19) INDIA
(22) Date of filing of Application :28/10/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : Portable Soil Monitoring and Agricultural Decision-Support System

(51) International classification	:G01N 33/24, G08C 17/02, G01D 21/02, A01G 9/24, G01N 27/22	(71)**Name of Applicant :** **1)R.M.K. College of Engineering and Technology** Address of Applicant :R.M.K. College of Engineering and Technology, Puduvoyal, Gummidipoondi Taluk Tiruvallur District, Tamil Nadu, India. Pin code: 601 206. Tiruvallur Tamil Nadu India (72)**Name of Inventor :** **1)Dr.INBA MALAR T M** **2)SREENIDHI K** **3)SREE RANJANI P**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The current invention provides a portable soil monitoring and agricultural decision support system to measure soil and environmental conditions at a cultivation site. The system includes a Printed Circuit Board (126), an ESP32 microcontroller (124), a soil moisture sensor (104), an NPK sensor (106), and a temperature and humidity sensor (108). RS–485 transceiver (120). OLED display (128) and power-management circuit. The soil moisture sensor senses the moist condition of the soil. The NPK sensor measures nitrogen, phosphorus, and potassium levels. Ambient temperature and humidity sensor obtains the sensor data and processes it from a main module, which then wirelessly sends this information to a remote computer platform using the ESP32 microcontroller. The harvested data can then be fed into a remote computing platform, which analyzes the farm and makes agricultural recommendations by means of a decision-support interface. These recommendations can be planting and irrigation schedules, nutrient and fertilizer use recommendations, crop choices, and soil management. This allows soil monitoring with a portable PCB-based configuration and provides information that can be easily interpreted by users for assessed practices suitable for cultivation.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128228 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : REMOTELY OPERATED CATAMARAN-TYPE WATERCRAFT WITH CONVEYOR-BASED WASTE COLLECTION AND LIVE VIDEO MONITORING SYSTEM

(51) International classification	:B63B 35/32, E02B 15/10, B63B 35/00, E02B 15/04, B63B 27/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Harish Nagaraj
Address of Applicant :79, Second Street Extension, Gandhipuram, Coimbatore - 641012 Tamil Nadu, India. Tamil Nadu India
(72)**Name of Inventor :**
1)Harish Nagaraj

(57) Abstract :
The present invention relates to a remotely operated watercraft for collecting floating solid waste from water bodies. The watercraft comprises a floating platform having two laterally spaced buoyant bodies joined by a support frame. The buoyant bodies define an open channel through which floating waste approaches an inclined conveyor assembly carried by the platform. The conveyor assembly comprises an endless belt supported about an upper roller and a lower roller. During operation, a lower intake portion of the belt is positioned at least partly below the waterline to engage floating waste entering the channel. A conveyor motor drives the belt to lift the waste towards an upper discharge portion, from which the waste is deposited into a collection receptacle. First and second propulsion motors, associated with respective sides of the platform, are independently controlled to propel and steer the watercraft by differential thrust. A receiver receives wireless commands from an operator, and a controller generates corresponding propulsion commands. The controller may stop the propulsion motors when a required control signal is invalid or unavailable for a predetermined period. Further, An onboard camera captures images of a water region forward of the conveyor intake. A processor encodes and transmits the images as a video stream for remote monitoring. The stream may include a source-side time indication and may be accessed through an outbound network relay and an authenticated viewing application. Separate regulated power branches may supply the conveyor motor and video-processing electronics. The watercraft is applicable to collecting floating waste from ponds, lakes, channels, and other water bodies.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128244 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ZK-MEDANALYTICS: A PRIVACY-PRESERVING AND VERIFIABLE MULTIHOSPITAL HEALTHCARE ANALYTICS PLATFORM USING ZERO KNOWLEDGE PROOFS, SECURE COMPUTATION AND ADAPTIVE PRIVACY CONTROLS

(51) International classification	:G16H 10/60, G06F 21/62, G16H 40/20, G06F 21/60, H04L 9/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Niranjan M
Address of Applicant :PRATHYUSHA ENGINEERING COLLEGE, ARANVOYAL KUPPAM, TIRUVALLUR, TIRUVALLUR-POONAMALLEE HIGH ROAD, TAMIL NADU, INDIA-602025 Tamil Nadu India
(72)**Name of Inventor :**
1)Devi R
2)Niranjan M
3)Amuthini K
4)Ashish N

(57) Abstract :
ABSTRACT OF THE INNOVATION The present invention discloses ZK-MedAnalytics: A Privacy-Preserving and Verifiable Multi- Hospital Healthcare Analytics Platform Using Zero-Knowledge Proofs, Secure Computation and Adaptive Privacy Controls. The invention provides a privacy-preserving approach for performing collaborative healthcare analytics across distributed medical datasets while minimizing the exposure of sensitive patient-level information. Conventional healthcare datasharing systems may require researchers or third-party services to access individual medical records, increasing the risk of unauthorized disclosure, patient re-identification, and inference attacks. The proposed platform integrates encrypted medical-data storage, secure computation, Zero- Knowledge Proof (ZI(P) generation and verification, patient consent management, role-based access control, adaptive query-risk assessment, differential privacy, and tamper-evident auditing. Healthcare data may be represented using the Fast Healthcare Interoperability Resources (FHIR) standard, allowing participating healthcare institutions to maintain their respective datasets within their controlled environments while supporting authorized collaborative analysis. Researchers can submit analytical queries involving population counts, statistical measurements, and cohort analysis without directly accessing the underlying patient records. An adaptive privacy-policy engine evaluates researcher authorization, research purpose, applicable consent, cohort size, query sensitivity, and potential information-disclosure risks before executing an analytical query. High-risk requests may be rejected, restricted, or transformed into safer queries. The platform further incorporates inference-attack detection to identify risks associated with repeated queries, query-differencing attacks, and small-cohort analysis. Differential privacy and other suitable output controls may be applied to reduce information leakage from released statistical results. The approved computation produces an analytical result associated with a cryptographic proof that enables an independent verifier to check specified computational requirements without receiving the underlying sensitive medical data. The proposed implementation may utilize Paillier-based encryption and Circom/snarkjs with Groth16 for supported encryption and proof-generation operations. A patient consent management mechanism supports permissions for specified data categories and research purposes, including enforcement of consent revocation for subsequent analytics operations. A tamper-evident audit and data-provenance mechanism records query execution, authorization decisions, consent status, dataset versions, computation versions, and proof-verification outcomes. Accordingly, the invention aims to facilitate trustworthy multi-hospital medical research and population-level analytics by integrating cryptographic verification, secure computation, adaptive privacy enforcement, and accountable healthcare data processing while minimizing unnecessary disclosure of confidential patient information.

No. of Pages : 19 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128245 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Multivariate Statistical Decision Support System Using Probability Models and Optimization Techniques for Complex Data-Driven Decision Making

(51) International classification	:G06N 5/04, G06Q 10/06, G06N 7/00, G06N 5/02, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. V.B.Srinivas
Address of Applicant :Associate Professor, Department of Mathematics, School of Engineering Venkatpur Hyderabad, Anurag University, Telangana - 500088 Telangana India
2)Dr. Ganesh D Kedar
3)Dr. T. Yugandhar
4)Mr. Narige Kiran
5)Mr. Ashok R
6)Dr. Y Raghunatha Reddy
7)Dr. Someshwar Siddi
8)Mr. A Ganga Prasad
(72)**Name of Inventor :**
1)Dr. V.B.Srinivas
2)Dr. Ganesh D Kedar
3)Dr. T. Yugandhar
4)Mr. Narige Kiran
5)Mr. Ashok R
6)Dr. Y Raghunatha Reddy
7)Dr. Someshwar Siddi
8)Mr. A Ganga Prasad

(57) Abstract :
[048] A computer-implemented multivariate statistical decision support system (100) receives heterogeneous timestamped observations and associates them with data-quality and uncertainty descriptors. A probabilistic inference engine (108) estimates a calibrated joint predictive distribution preserving dependence among decision-relevant variables. A scenario and constraint builder (110) generates weighted joint outcomes and technical operating constraints. An optimization engine (112) selects a feasible action by minimizing expected operating cost, tail risk and action-switching penalties while enforcing physical constraints. A stability evaluator selectively triggers recalculation in response to material changes and reduces oscillation of recommendations. An output interface (114) supplies the selected action, risk indicators, binding constraints and a provenance-linked audit record. The system supports auditable, uncertainty-aware control decisions in complex technical operating environments. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128246 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING-BASED SYSTEM FOR PREDICTING VOTER CITIZENSHIP BEHAVIOR FROM POLITICAL ENGAGEMENT PATTERNS

(51) International classification	:G06N 20/00, G06Q 10/04, G06Q 10/06, G06N 5/04, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr. Deepthi S

Address of Applicant :Assistant Professor Grade III, Department of CSE, Nitte Meenakshi Institute of Technology, NITTE(Deemed to be University), Bangalore, Karnataka, 560064, India. Karnataka India

2)Sudhakar Jangili

3)Dr. P. Chitra

4)S. Vinodhini

5)Dr. Arthi A

6)Dr. C. Ravichandran

7)Aparna Mohan

8)Riya Jacob

9)Athira M A

10)Dr. Pravin Kulurkar

11)Dr P Raja

12)Anthony Savio Herminio Da Piedade Fernandes

(72)**Name of Inventor :**

1)Dr. Deepthi S

2)Sudhakar Jangili

3)Dr. P. Chitra

4)S. Vinodhini

5)Dr. Arthi A

6)Dr. C. Ravichandran

7)Aparna Mohan

8)Riya Jacob

9)Athira M A

10)Dr. Pravin Kulurkar

11)Dr P Raja

12)Anthony Savio Herminio Da Piedade Fernandes

(57) Abstract :

A system utilizing machine learning and implemented on a computer is revealed for forecasting specific trends in civic participation based on recognized patterns of political and civic engagement. The system includes a data collection module, data verification module, temporal feature extraction unit, predictive machine-learning engine, confidence assessment module, privacy and governance overseer, aggregate reporting unit, and outcome assessment module. The system obtains approved civic-engagement records, verifies source authenticity and data integrity, and extracts temporal and contextual characteristics that illustrate engagement patterns and variations over specified observation periods. A trained model produces a forecast for a designated civic-participation metric and prediction timeframe. A confidence assessment system certifies, retains, or directs predictions that fail to meet established trustworthiness standards. Privacy controls limit processing and disclosure based on allowed purposes and relevant governance standards. Cumulative results enhance non-partisan access to civic information and planning for participation. Later approved results are matched against previous predictions to assess model effectiveness and allow for managed model revisions. The system differentiates between actual engagement and predicted future involvement, without needing to identify a person's preferred candidate or voting selection. FIG.1

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128248 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AEGIS AI: AUTONOMOUS NETWORK THREAT RESPONSE SYSTEM WITH BOUNDED MULTI-AGENT VERIFICATION AND EXPECTED-HARM DECISION OPTIMIZATION

(51) International classification	:H04L 29/06, H04L 9/40, G06N 5/04, G06F 21/55, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lochini U
Address of Applicant :Prathyusha Engineering College, Aranvoyal kuppam, Tiruvallur-Poonamallee high road, Tiruvallur, Tamilnadu-602025. Tamil Nadu India
(72)**Name of Inventor :**
1)Devi R
2)Lochini U
3)Ramya V R
4)Sivaranjani N

(57) Abstract :
The present invention discloses "Aegis AI: Autonomous Network Threat Response System with Bounded Multi-Agent Verification and Expected-Harm Decision Optimization," an artificial intelligence-based cybersecurity system• designed to detect network threats, investigate suspicious activities, validate security evidence, and determine appropriate response actions while minimizing unnecessary disruption to legitimate network services. The proposed_system integrates hybrid threat detection, threat intelligence enrichment, multiagent investigation, bounded confidence adjustment, evidence-grounding validation, and expected-harm-based decision optimization. Network-flow information, including network addresses, ports, protocols, timestamps, packet counts, byte counts, and DNS activity, is analyzed to identify potential threats such as port scanning, credential brute-forcing, denial-ofservice attacks, command-and-control communications, and data exfiltration. The multi-agent investigation mechanism employs Prosecutor, Skeptic, and Judge agents to examine supporting evidence, contradictory observations, and alternative explanations. A bounded confidence adjustment mechanism limits changes to the original detection confidence, while an evidence-grounding validation module checks AI-generated investigative claims against available network telemetry and detection records. The system further incorporates threat intelligence information and relevant MITRE ATT&CK technique mappings to support contextual threat assessment. An expected-harm-based decision optimization mechanism evaluates threat probability, evidence quality, asset criticality, expected security benefits, and potential operational disruption before selecting an appropriate response. Depending on predefined security policies and authorization requirements, the system may perform monitoring, traffic restriction, suspicious address blocking, host isolation, incident creation, or analyst escalation. Autonomous response actions are permitted only when the applicable confidence thresholds and authorization conditions are satisfied. The invention additionally incorporates an adaptive feedback mechanism that may use a Bayesian Beta-Bernoulli model to update threat-indicator reliability estimates based on analyst decisions and investigation outcomes. A cryptographically hash-chained audit mechanism records investigation findings, confidence adjustments, analyst approvals, response actions, and significant system events. A real-time monitoring dashboard presents security alerts, investigation evidence, risk assessments, and response activities. Accordingly, the proposed invention provides an integrated autonomous cyber-defense framework combining intelligent threat detection, evidence-grounded multi-agent verification, risk-aware response optimization, adaptive feedback learning, and tamper-evident auditing to improve the reliability, accountability, and operational safety of cybersecurity threat investigation and response while balancing threat mitigation against potential disruption to legitimate network services.

No. of Pages : 17 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128250 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT CYBERSECURITY DATA ANALYTICS FRAMEWORK USING AI-DRIVEN BEHAVIORAL MODELING, DYNAMIC RISK PREDICTION AND AUTONOMOUS DEFENSE IN DISTRIBUTED SYSTEMS

(51) International classification	:H04L 29/06, H04L 9/40, G06F 21/57, G06F 21/55, H04L 12/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PRATHYUSHA ENGINEERING COLLEGE
Address of Applicant :Aranvoyalkuppam Tiruvallur-Poonamallee high road Tiruvallur Tami1nadu India 602025 Tamil Nadu India
(72)**Name of Inventor :**
1)Sripriya N
2)Sneha
3)Guruvarshini
4)Jeevan Isnovar
5)K Vishnu Priya
6)Rama Matharasi
7)Banu Priya D
8)Beulah Elizabeth D
9)Dr. Sadhana C
10)Dr. Sujitha N
11)Anithalakshmi V

(57) Abstract :
The present invention discloses an Intelligent Cybersecurity_Data Analytics Framework Using AI-Driven Behavioral Modeling, Dynamic Risk Prediction, and Autonomous Defense in Distributed Systems. The invention integrates artificial intelligence, behavioral analytics, anomaly detection, and predictive risk assessment to improve cybersecurity monitoring and threat response across distributed computing environments. The proposed framework—collects and_analyzes_ security information from network traffic, system logs, authentication records, endpoints, applications, and cloud platforms. AI-driven behavioral modeling identifies deviations from normal activity patterns, while the dynamic risk prediction module estimates the likelihood and potential impact of cybersecurity incidents. A distributed threat correlation engine analyzes security events from interconnected systems to identify suspicious activities and potential coordinated attacks. Based on the assessed risk and predefined security policies, the autonomous defense mechanism supports appropriate responses, including restricting suspicious sessions, isolating affected devices, and blocking unauthorized communication. The framework also maintains security records to support incident investigation and defense evaluation. The invention aims to improve early threat detection, reduce response delays, and strengthen the security and operational resilience of distributed computing infrastructures.

No. of Pages : 15 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128251 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR QUANTUM MACHINE LEARNING-BASED PREDICTIVE ANALYTICS AND DECISION INTELLIGENCE PLATFORM

(51) International classification	:G06N 10/00, G06N 10/20, G06N 10/40, G06N 10/60, G06N 99/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mailam Engineering College
Address of Applicant :Mailam P.O, Tindivanam T.k, Villupuram Dist, Tamil Nadu, India-604304 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. S. Kalaivany
2)Mr. Gajendiran R
3)Mr. Loganathan K
4)Mrs. Ramya C
5)Ms. P. Archana Devi
6)Mrs. S. Sathiya
7)Mr. A. Karthikeyan
8)Mr. S. Ramesh
9)Ms. R. Aswini
10)Mrs. M. Lavanya
11)Mrs. M. Vijayalakshmi
12)Mrs. P. Sharmila

(57) Abstract :
A system and method constituting a quantum machine learning-based predictive analytics and decision intelligence platform is disclosed. A hybrid quantum-classical arrangement comprises a classical computing subsystem and a quantum computing subsystem with a quantum processing unit, accessed locally or as a cloud service. A data interface receives data of a technical or operational system, and a data encoding module encodes it into quantum states by a feature map, such as amplitude or angle encoding. A quantum machine learning module realizes a model at least partly on the quantum processing unit, such as a variational quantum circuit with trainable parameters or a quantum kernel evaluated for a classical kernel method, and executes it to compute a prediction, classification, or optimization; a training module trains it by a hybrid loop in which the quantum processor evaluates the model and the classical subsystem optimizes its parameters. An orchestration module divides the task, executing on the quantum processor the computations for which it offers advantage and managing access to it as a scarce resource. A decision intelligence module applies the result to inform or effect a decision upon the system, producing a concrete operational result.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128252 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXPLAINABLE ARTIFICIAL INTELLIGENCE FRAMEWORK FOR INTELLIGENT DETECTION OF ZERO-DAY CYBERSECURITY THREATS

(51) International classification	:G06N 20/00, H04L 29/06, G06F 21/55, G06F 21/56, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Priyaradhikadevi. T
Address of Applicant :Professor & Head Department of Computer Science and Engineering, Mailam Engineering College, Mailam P.O, Tindivanam T.K, Villupuram Dist, Tamil Nadu– 604304, India Tamil Nadu India
2)Mr. Mathavan. V
3)Mr. Vanakovarayan. S
4)Dr. Mohan. R
5)Mr. Srinivasan. D
6)Mr. Shanmuganathan. S
7)Mr. Arunkumar. R
8)Mrs. Vijayakumari. V
9)Mrs. Nandhini. E
10)Mrs. Malliga. G
11)Mr. Karankumar. B
12)Mr. Sridhar. S
13)Mr. Senthamizh Selvan. P
(72)**Name of Inventor :**
1)Dr. Priyaradhikadevi. T
2)Mr. Mathavan. V
3)Mr. Vanakovarayan. S
4)Dr. Mohan. R
5)Mr. Srinivasan. D
6)Mr. Shanmuganathan. S
7)Mr. Arunkumar. R
8)Mrs. Vijayakumari. V
9)Mrs. Nandhini. E
10)Mrs. Malliga. G
11)Mr. Karankumar. B
12)Mr. Sridhar. S
13)Mr. Senthamizh Selvan. P

(57) Abstract :
An explainable artificial intelligence framework for intelligent detection of zero-day cybersecurity threats is disclosed. A detection module comprising a machine learning model detects a zero-day threat, previously unknown and having no known signature, from monitored network and system data by its anomalous or malicious characteristics and behavior rather than a known signature, comprising an anomaly-detection or behavior-based model that learns the normal behavior or malicious characteristics and detects the novel threat by its deviation from the normal or resemblance to the malicious, which signature-based detection cannot do. An explanation module generates, for a detected zero-day threat, a human-interpretable explanation of the determination, identifying the features, behaviors, and characteristics of the data that led the model to determine the threat and their relation to the determination, by an explainability method such as feature attribution, rule extraction, or example-based explanation. The explanation, faithful to the model's actual determination, enables the analyst to trust the determination, verify whether it is a true threat or false alarm, and act upon it by understanding the threat's nature, overcoming the opacity of machine learning detection that would otherwise impede trust, verification, and action on novel threats.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128255 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN AUTONOMOUS CYBER THREAT DETECTION AND ADAPTIVE RESPONSE SYSTEM USING MACHINE LEARNING

(51) International classification	:G06N 20/00, H04L 29/06, G06F 21/55, G06F 21/57, H04L 12/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mrs. Sassirekha. S. M
Address of Applicant :Assistant Professor Department of Computer Science and Engineering, Mailam Engineering College, Mailam P.O, Tindivanam T.K, Villupuram Dist, Tamil Nadu- 604304, India Tamil Nadu India
2)Dr. Priyaradhikadevi.T
3)Mrs. Jayanthi P. R
4)Mrs. Lavanya. E
5)Mrs. Indra. E
6)Mrs. Pradeepa. S
7)Mrs. Selvapriya. G
8)Mrs. Sakthi Priya. P
9)Mrs. Aswini. S
10)Mrs. Thilagavathi. A
11)Ms. Hemalatha. K. P
12)Mrs. Prabha. D
13)Ms. Hani Bedcilla. J
(72)**Name of Inventor :**
1)Mrs. Sassirekha. S. M
2)Dr. Priyaradhikadevi.T
3)Mrs. Jayanthi P. R
4)Mrs. Lavanya. E
5)Mrs. Indra. E
6)Mrs. Pradeepa. S
7)Mrs. Selvapriya. G
8)Mrs. Sakthi Priya. P
9)Mrs. Aswini. S
10)Mrs. Thilagavathi. A
11)Ms. Hemalatha. K. P
12)Mrs. Prabha. D
13)Ms. Hani Bedcilla. J

(57) Abstract :
An artificial-intelligence-driven autonomous cyber threat detection and adaptive response system using machine learning is disclosed. A detection module comprising a machine learning model detects a cyber threat from monitored network and system data and assesses its severity and confidence. A response decision module autonomously selects a response action from among a plurality, isolation, blocking, containment, remediation, and alerting, in dependence upon the detected threat, its assessed severity and confidence, and the state and constraints of the network, selecting the response appropriate to the particular situation rather than executing a fixed predefined playbook. A response execution module executes the selected action upon the network under a graduated autonomy in which a low-risk action is executed automatically at machine speed, before a fast automated attack completes its damage, and a high-impact action is referred for human authorization, balancing speed against the risk of an over-aggressive response. An adaptation module observes the outcome, whether the response contained the threat and whether it caused adverse effects, and uses it, with the detection, to adapt the detection model and response policy, in embodiments by reinforcement learning, so the self-defending system improves over time where human-driven and predefined-automation defenses are too slow or static.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128259 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT-ENABLED FACIAL RECOGNITION ATTENDANCE SYSTEM FOR EDUCATIONAL AND CORPORATE INSTITUTIONS

(51) International classification	:G06V 40/16, G06K 9/00, G06N 20/00, G06K 9/62, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr. Banoth Ravi Kumar

Address of Applicant :Associate Professor & HoD, Department of Computer Science and Engineering, Siddhartha Institute of Technology and Sciences, (UGC-Autonomous), Narapally, Korremula Road, Ghatkesar, Medchal-Malkajgiri, Hyderabad, Telangana- 500088, India Telangana India

2)Mr. M. Sutharsan

3)Dr. T.S. Savita

4)Dr. S. Sujatha

5)Mr. Bodicherla Siva Sankar

6)Dr. Nellore Manoj Kumar

7)Dr. K. Abhinaya

(72)**Name of Inventor :**

1)Dr. Banoth Ravi Kumar

2)Mr. M. Sutharsan

3)Dr. T.S. Savita

4)Dr. S. Sujatha

5)Mr. Bodicherla Siva Sankar

6)Dr. Nellore Manoj Kumar

7)Dr. K. Abhinaya

(57) Abstract :

An IoT-enabled facial recognition attendance system and method for educational and corporate institutions is disclosed. A plurality of IoT-enabled capture devices arranged at locations of an institution acquire images of persons. A facial-recognition module comprising a trained model detects a face and identifies the person by comparing a derived facial representation with enrolled representations, the detection and derivation being performed locally upon the capture device or an associated edge-computing device so that a representation or identification, rather than the raw image, is transmitted, reducing bandwidth and exposure of sensitive data and enabling low-latency, connectivity-robust operation. An anti-spoofing module determines the liveness of the person, by analysing micro-movements, three-dimensional structure, skin texture, or responses to a prompt, so as to reject the presentation of a photograph or video. An attendance-recording module generates an attendance record, and a consolidation module compiles the records from the distributed capture devices into a tamper-resistant consolidated record, rendered tamper-evident by cryptographic means, accessible to the institution. A data-protection arrangement protects the facial images and derived biometric data throughout. The system provides automated, contactless, accurate, spoofing-resistant, and secure attendance recording, improving the accuracy, efficiency, and security of attendance recording in educational and corporate institutions.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128260 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING-BASED ADAPTIVE PREDICTIVE MAINTENANCE SYSTEM FOR INDUSTRIAL EQUIPMENT

(51) International classification	:G06N 20/00, G05B 23/02, G06Q 10/00, G06N 3/08, G06N 20/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM Institute of Science and Technology
Address of Applicant :Ramapuram Campus, Chennai, Tamil Nadu -600089 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. S. Nancy Lima Christy
2)Ms. S. Aarthi
3)Mr. C. Santhosh Kumar

(57) Abstract :
A machine learning-based adaptive predictive maintenance system and method for industrial equipment is disclosed. A data-acquisition interface acquires, from a plurality of sensors associated with an item of industrial equipment, sensor data representing both a condition of the equipment, such as vibration, temperature, acoustic emission, and current, and an operating condition, such as load and speed. A pre-processing and feature-extraction module extracts statistical and frequency-content features. A condition-assessment and prediction module comprising a trained machine-learning model analyses the features together so as to assess the condition, predict an impending failure, and estimate a remaining useful life with an uncertainty, the model being trained upon historical data recorded as equipment approached failure. An adaptation module adapts the model and the decision threshold to the individual item of equipment, by establishing a model of its normal behaviour, and to the changing operating condition, by conditioning the assessment upon the prevailing load and speed, so that a developing fault is detected as a departure from the individual equipment's normal behaviour for the prevailing condition. A maintenance-recommendation module generates a prioritized maintenance recommendation. The system predicts failures early and reliably while avoiding the false alarms and missed detections of fixed-threshold monitoring, reducing unplanned downtime and unnecessary maintenance and prolonging equipment life.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128263 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BEAD RING CONFIGURATION FOR REDUCTION OF LOCK RING DAMAGE TO A PNEUMATIC TYRE

(51) International classification	:B60C 15/024, B60B 21/12, B60B 21/02, B60C 15/02, B60B 21/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JK TYRE & INDUSTRIES LIMITED
Address of Applicant :JK TYRE TECH CENTRE, RAGHUPATI SINGHANIA CENTRE OF EXCELLENCE, PLOT NO. 437, HEBBAL INDUSTRIAL AREA, METAGALLI POST, MYSURU, KARNATAKA, INDIA 570016 MYSURU Karnataka India
(72)**Name of Inventor :**
1)H R, PRAVEEN KUMAR
2)E P, ABDUL JAFAR SHERIEF
3)CHAUBEY, RISHABH KUMAR

(57) Abstract :
BEAD RING CONFIGURATION FOR REDUCTION OF LOCK RING DAMAGE TO A PNEUMATIC TYRE A tyre-rim assembly (200) is disclosed comprising a pneumatic tyre (210) comprising a bead region (220) having a toe (222), a heel (224), and a bead core (224); a multi-piece rim (230) for mounting the pneumatic tyre (210) comprising a rim base (232), a flange (234) fixed relative to the rim base (232), a ledge (236) formed adjacent the flange (234), and a removable lock ring (238) fitted to the rim base (232) and configured to retain a bead region (220) against the ledge (236) and the flange (234); wherein the bead region (220) has a curing ring having curing ring heel diameter greater than a reference curing ring heel diameter for a conventional bead ring configuration resulting in enlarged effective contact area over which the heel (222) engages the ledge (236) forming an enlarged ledge contact-surface and consequently reduced localized contact pressure at the ledge region. Figure 2b

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128264 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : KNOWLEDGEBRIDGE: AN ADAPTIVE HYBRID RETRIEVAL-AUGMENTED GENERATION SYSTEM FOR RELIABLE AND CALIBRATED KNOWLEDGE -INTENSIVE QUESTION ANSWERING

(51) International classification	:G06F 16/332, G06F 17/27, G06F 17/30, G06N 5/02, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Theresa Cenate C F
Address of Applicant :block no.1, 100 Feet Rd, Vadapalani, Chennai, Tamil Nadu-600026 Tamil Nadu India
2)A.Anto Nimal
3)R. Chandhan Sayee
4)K. Sreelakshmi
(72)**Name of Inventor :**
1)Dr. Theresa Cenate C F
2)A.Anto Nimal
3)R. Chandhan Sayee
4)K. Sreelakshmi

(57) Abstract :
Hallucinations and unreliable answers are major challenges for knowledge-intensive question answering. This project presents Knowledge Bridge, an adaptive Retrieval-Augmented Generation system designed to improve answers produced by large language models. KnowledgeBridge retrieves relevant information before generating responses. It combines FAISS semantic search with BM25 keyword search to find passages matching a question's meaning and wording. A Cross-Encoder reranks passages to prioritize useful evidence. The system adjusts context size according to query complexity, selecting three, five, or eight chunks. If local evidence is insufficient, conditional source fallback retrieves information from external sources. KnowledgeBridge checks generated answers sentence by sentence using Calibrated Factual Grounding, which considers retrieval quality, semantic similarity, lexical overlap, source diversity, evidence coverage, and response coherence. These factors produce a confidence score showing how well an answer is supported by evidence. The system aims to reduce unnecessary external searches, improve factual grounding, and calibrate confidence. In evaluation, KnowledgeBridge achieved a 71.7% factual grounding rate, compared with 68.3% for a conventional RAG baseline. Average response latency decreased from 870 milliseconds to 765 milliseconds. KnowledgeBridge offers a practical approach to reliable question answering across different knowledge domains.

No. of Pages : 9 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128274 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT PEST DETECTION AND TARGETED BIO-PESTICIDE SPRAYING DRONE

(51) International classification	:A01M 7/00, B64D 1/18, B64C 39/02, B64D 47/08, G05D 1/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAJALAKSHMI ENGINEERING COLLEGE
Address of Applicant :RAJALAKSHMI ENGINEERING COLLEGE, THANDALAM, CHENNAI, TAMIL NADU, INDIA-602105 Tamil Nadu India
(72)**Name of Inventor :**
1)DR. P. KUMAR
2)Vignesh

(57) Abstract :
ABSTRACT The present invention discloses an Intelligent Pest Detection and Targeted Bio-Pesticide Spraying Drone designed to provide autonomous, precise and environmentally sustainable pest management in agricultural fields. The invention integrates an unmanned aerial vehicle (UAV) platform with advanced imaging sensors, artificial intelligence-based pest detection, IoT communication, cloud analytics and precision bio-pesticide spraying mechanisms to enable real-time identification and treatment of pest infestations. The drone is equipped with RGB cameras, multispectral cameras, thermal imaging sensors, GPS modules, inertial measurement units and environmental sensors for continuous acquisition of crop health and field data. An onboard edge processing unit performs image preprocessing, feature extraction and data analysis, while an AI-powered pest detection engine employing machine learning and deep learning algorithms identifies pest species, evaluates infestation severity and generates geo-tagged pest distribution maps. Based on the detected infestation patterns, a decision support module determines optimal spraying locations, bio-pesticide dosage requirements and flight trajectories. The system activates a precision spraying mechanism comprising smart pumps, electronically controlled valves and variable-rate nozzles to selectively apply bio-pesticides only to affected crop regions, thereby reducing pesticide consumption and minimizing environmental impact. The invention further incorporates cloud connectivity for historical data storage, predictive pest outbreak forecasting, Al model updating, performance analytics and remote monitoring through mobile and web-based dashboards. A closed-loop feedback mechanism continuously evaluates post-treatment crop conditions and updates detection and spraying strategies to improve operational efficiency and treatment effectiveness. The disclosed invention provides a technical solution for automated pest detection and targeted intervention, resulting in early pest identification, optimized bio-pesticide utilization, enhanced crop protection, improved agricultural productivity, reduced operational costs and sustainable farming practices through the integration of artificial intelligence, computer vision, IoT infrastructure, autonomous drone navigation and precision agricultural actuation technologies. Keywords: Intelligent Drone, Pest Detection, Bio-Pesticide Spraying, Precision Agriculture, Artificial Intelligence, Computer Vision, Deep Learning, IoT, Crop Health Monitoring, Autonomous Agricultural Systems.

No. of Pages : 10 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128275 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CARBON FOOTPRINT MONITORING SYSTEM FOR FARMS

(51) International classification	:G06Q 50/02, G06Q 10/04, G06N 20/00, G06Q 10/06, H04L 29/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAJALAKSHMI ENGINEERING COLLEGE
Address of Applicant : RAJALAKSHMI ENGINEERING COLLEGE, THANDALAM, CHENNAI-602105 9840573702 kumar@rajalakshmi.edu.in Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. P. Kumar Santhosh

(57) Abstract :
The present invention discloses an Intelligent Carbon Footprint Monitoring System for Farms designed to continuously monitor, estimate, predict and optimize greenhouse gas emissions associated with agricultural activities. The invention integrates IoT-enabled environmental sensors, farm operational monitoring devices, edge computing infrastructure, cloud analytics, artificial intelligence-based carbon estimation models and sustainability optimization mechanisms into a unified carbon management platform. The system collects real-time information relating to soil conditions, weather parameters, fuel consumption, energy utilization, irrigation activities, fertilizer application, livestock operations and crop growth characteristics. An edge processing unit performs preliminary data validation and analytics before transmitting processed information to a cloud-based carbon intelligence platform. Machine learning models estimate carbon dioxide, methane and nitrous oxide emissions while simultaneously evaluating carbon sequestration potential through vegetation and soil carbon analysis. The system generates carbon footprint scores, identifies emission hotspots, forecasts future environmental impacts, recommends emission reduction strategies and supports carbon-credit readiness assessment. Through continuous monitoring, predictive analytics and intelligent decision support, the invention provides a technical solution for sustainable agricultural carbon management, enabling improved environmental performance, regulatory compliance, resource optimization, climate resilience and environmentally responsible farming practices.

No. of Pages : 8 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641128276 A

(19) INDIA

(22) Date of filing of Application :28/10/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-POWERED SMART WATER CONSERVATION AND IRRIGATION SYSTEM

(51) International classification	:A01G 25/16, G06Q 50/02, G05B 13/04, G06N 20/00, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAJALAKSHMI ENGINEERING COLLEGE
Address of Applicant :RAJALAKSHMI ENGINEERING COLLEGE, RAJALAKSHMI NAGAR, THANDALAM, CHENNAI-602105 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. P. Kumar
2)Sai Ruthwik Vattigunta

(57) Abstract :
The present invention discloses an AI-powered smart water conservation and irrigation system utilizing reinforcement learning-based adaptive irrigation optimization for precision agriculture. The system integrates soil moisture sensors, environmental monitoring devices, weather forecasting services, crop growth stage analysis modules, cloud computing infrastructure, artificial intelligence prediction engines, reinforcement learning optimization modules and automated irrigation control units. Real-time agricultural field data including soil moisture, temperature, humidity, rainfall conditions and crop developmental stages are continuously collected and processed to estimate irrigation requirements. A reinforcement learning engine interacts with the agricultural environment through state-action-reward mechanisms and continuously learns optimal irrigation strategies for different crop types and soil conditions. Based on the optimized irrigation schedule, automated controllers regulate pumps, valves and water delivery systems to provide precise irrigation. The system operates through a closed-loop feedback architecture that continuously adapts to changing environmental conditions and agricultural requirements. The invention significantly reduces freshwater wastage, improves irrigation efficiency, enhances crop productivity and provides intelligent real-time water management. The disclosed system provides a technical solution to the technical problem of inefficient water utilization in agriculture through the integration of IoT sensing infrastructure, artificial intelligence, reinforcement learning and automated irrigation control mechanisms.

No. of Pages : 9 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128284 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PARTIAL REPLACEMENT OF FINE AGGREGATE USING SEWAGE SLUDGE

(51) International classification	:C04B 18/08, C04B 18/10, C04B 18/04, C04B 18/06, B09B 3/00	(71)**Name of Applicant :** **1)R.PRAVEEN** Address of Applicant :DEPARTMENT OF CIVIL ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BYPASS ROAD, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India **2)ARON.G** **3)ASHOK.S** **4)GOWTHAM.V** **5)NAVEEN KUMAR.M**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)R.PRAVEEN**
(33) Name of priority country	:NA	**2)ARON.G**
(86) International Application No	:NA	**3)ASHOK.S**
Filing Date	:NA	**4)GOWTHAM.V**
(87) International Publication No	: NA	**5)NAVEEN KUMAR.M**
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

Wastewater sludge (WWS) and sewage sludge serve as viable partial replacements for fine aggregates in concrete, primarily optimizing mechanical properties for non-structural applications such as paving blocks, kerbs, and interlocking bricks. Research indicates that replacing fine aggregates with 5% to 10% sludge generally maintains or enhances compressive-strength, with --optimal performance often-o-bserve-d at the 5% replacement level.Agricultural wastes such as bagasse ash (from sugarcane) and rice husk ash (from rice mills) are produced in large quantities and often cause environmental problems when disposed of improperly. Using these waste ashes in brick manufacturing helps reduce pollution and promotes sustainable construction. The rapid urbanization and industrialization across the globe have led to a substantial increase in the generation of sewage sludge from municipal wastewater treatment plants (WWTPs). The disposal of this sludge poses a significant environmental challenge, with concerns related to land contamination, greenhouse gas emissions, and secondary pollution from leachable heavy metals. Simultaneously, the construction industry's relentless demand for natural fine aggregates has intensified the depletion of sand resources, leading to ecological degradation of riverbeds and coastal ecosystems. The partial replacement of natural fine aggregates with treated sewage sludge in concrete offers a dual-purpose solution: it diverts waste from landfills while reducing the extraction of virgin natural resources, thereby contributing to a circular economy in construction. This study investigates the feasibility, mechanical performance, durability characteristics, and environmental safety of concrete produced by partially replacing fine aggregates with sewage sludge at varying levels (0%, 5%, 10%, 15%, and 20% by mass). The sewage sludge used in this investigation was sourced from municipal wastewater treatment plants and subjected to controlled drying and sieving to achieve a particle size distribution comparable to natural fine sand. The physical and chemical characterization of the sludge revealed a high Si02 content, moderate organic matter, and a low pozzolanic activity, indicating that the sludge primarily functions as a mineral filler rather than a reactive supplementary cementitious material. Concrete mixtures were designed with a water-cement ratio of 0.57 and a cement-to-fine aggregate-to-coarse aggregate ratio of 1:2:2, targeting a compressive strength of 25-27 MPa. Fresh concrete properties including slump, workability, and consistency were evaluated as per IS 1199 and IS 10262. Hardened concrete properties were assessed through compressive strength tests (IS 516), splitting tensile strength (IS 1605), flexural strength (IS 516), water absorption, acid resistance, sulfate resistance, and abrasion tests at curing ages of 7, 14, 28, 90, and 140 days. Environmental safety was evaluated using the Toxicity Characteristic Leaching Procedure (TCLP) as per US EPA Method 1311, measuring the leachable concentrations of heavy metals including lead (Pb), cadmium (Cd), chromium (Cr), nickel (Ni), copper (Cu), and zinc (Zn). Microstructural characterization was performed using Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), and Energy-Dispersive X-ray Spectroscopy (EDX) to understand the interfacial transition zone (ITZ) and hydration products.

No. of Pages : 4 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128292 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Weldless clamp-lock connection system for roofing frames and lattice assemblies

(51) International classification	:E04B 1/58, E04B 1/24, E04B 7/02, E04B 1/26, E04D 3/36	(71)**Name of Applicant :** **1)Bojan Gokul** Address of Applicant :No. 17, Vasantham Nagar, Kamsalapuram, Mariputhur, Murugambakkam Post, Madurantakam, Tamil Nadu, India – 603306. Tamil Nadu India
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Bojan Gokul**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The invention relates to a clamp-lock connection system for joining posts (14), rafters (15), purlins (22) and lattice members (13) of roofing frames without on-site welding or drilling of the members. Clamps (1, 2, 19, 20, 23, 25) of bent metal strip, shaped to the external profile of each member, wrap around the member and are fastened to a cooperating element by bolts. In a post-to-rafter connector, a bent connecting plate (16) sets the roof slope, and separate clamps (19, 20) secure the post and the rafter to it. In a rafter-to-purlin connector, a purlin clamp (25) is bolted to an angle member (24) carried by a lower U-clamp (23). In a lattice assembly, identically shaped clamps (1, 2) are used at both ends of each lattice member, and only the lower clamp (1) carries a tightening bolt (7) and pressure washer (9), so that a single tightening action secures both ends. Ref. Fig. 1

No. of Pages : 36 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128295 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MODULAR DUAL GLOVE COMMUNICATION SYSTEM FOR SIGN LANGUAGE TRANSLATION

(51) International classification	:G06F 3/01, G06F 3/16, A41D 19/00, G09B 21/00, G06F 3/0346
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)UNIVERSITY OF MADRAS
Address of Applicant :UNIVERSITY OF MADRAS CENTENARY BUILDINGS, CHEPAUK, CHENNAI 9600926907 registrar.unom@gmail.com Tamil Nadu India
(72)**Name of Inventor :**
1)Karthick S
2)Dr B Muthuraarnan
3)Dr A Stephan

(57) Abstract :
The present invention relates to a modular dual glove communication system for translating sign language gestures into text and synthesized speech. The system comprises a pair of wearable gloves having capacitive bend sensors (101) positioned dorsal surface of corresponding fmgers. Each capacitive bend sensor (101) comprises a PVA—Ti02 nanocomposite dielectric film positioned between aluminum electrode layers (102). The sensors (101) detect finger bending through capacitance variations produced by deformation of the dielectric layer. Each glove further comprises a three-axis accelerometer (104) for detecting hand orientation and movement. A Mode button (106) is electrically connected to the microcontroller to select a corresponding predefined operating mode. An electronic enclosure (202) houses a microcontroller, bluetooth module, Li-ion battery, and LCR/capacitance sensing circuit. The microcontroller processes sensor signals and recognizes predefined gesture patterns using rule-based conditional logic. Each glove wirelessly transmits recognized gesture information to a paired device. The device combines outputs from both gloves and displays corresponding translated text and converts the translated text into synthesized speech for audible communication (Figure 1).

No. of Pages : 14 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128297 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTERNAL PRESSURIZED AIRFLOW ROUTING DEVICE WITH SURFACE PROTECTION FOR GAS TURBINE BLADES

(51) International classification	:F01D 5/18, F01D 5/14, F02C 7/18, F01D 9/02, F01D 9/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NATIONAL INSTITUTE OF TECHNOLOGY -WARANGAL
Address of Applicant :National Institute of Technology Warangal, Warangal – 506004, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Choppari Kiranmai
2)Gangadari Bhargava Teja
3)Dr. Satyanand Abraham
4)Dr. K Kiran Kumar
5)Dr. Prasanth Anand Kumar Lam

(57) Abstract :
Title: AN INTERNAL PRESSURIZED AIRFLOW ROUTING DEVICE WITH SURFACE PROTECTION FOR GAS TURBINE BLADES The present invention discloses a pressurized air circulation device (12) for thermal management of a gas turbine blade. The device comprises a blade body (8) having a leading edge (10), a pressure-side outer wall surface (6) and a suction-side outer wall surface (7). A pressurized air inlet (1) supplies pressurized air at a flow rate of 0.00271 kg/s through six nozzles (3) into double-swirl channels (11) positioned along the spanwise direction of the blade body (8). The double-swirl channels (11) comprise opposing curved wall surfaces and an impingement cavity wall (9), which facilitate counter-rotating pressurized air vortices for heat transfer. Effusion holes (4) discharge a portion of the pressurized air onto the leading edge (10) to provide external film cooling. A connector (5), comprising curved and square connectors, thermally connects successive double-swirl channels (11) to transfer pressurized air between the channels while reducing flow separation. The system combines internal convective cooling and external film cooling to improve thermal management of the gas turbine blade.

No. of Pages : 15 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128310 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AMINO ACID–1,2,4-TRIAZOLE HYBRID COMPOUNDS AND PROCESS FOR PREPARATION THEREOF

(51) International classification	:C07D 249/08, C07D 249/12, A61K 31/4196, C12Q 1/48, C07D 249/14	(71)**Name of Applicant :** **1)JSS COLLEGE OF PHARMACY, OOTY – JSS Academy of Higher Education & Research** Address of Applicant :Post Box No. 20, Rocklands, Ootacamund, The Nilgiris - 643001, Tamil Nadu, India. OOTY Tamil Nadu India (72)**Name of Inventor :** **1)Dr. Gowramma Byran** **2)B. Hardha** **3)Balaji.N** **4)Dr. R. Kalirajan**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to amino acid–1,2,4-triazole hybrid compounds and a process for preparation thereof. The compounds comprise 6-amino-2-(3-amino-5-(substituted phenyl)-1H-1,2,4-triazol-4(5H)-yl)hexanoic acid derivatives designed for interaction with poly(ADP-ribose) polymerase-1 (PARP-1). Selected compounds A1–A10 are prepared by a one-pot reaction involving a substituted benzaldehyde, lysine and a semicarbazide reagent and are characterized by spectroscopic and analytical techniques. The compounds are evaluated by molecular docking, MM-GBSA binding-energy analysis, ADMET prediction, molecular-dynamics simulation and in-vitro cytotoxicity studies. Among the evaluated compounds, A2 exhibits an IC_{50} of 5.91 ± 1.98 µg/mL in the disclosed cell-based assay and is identified as a preferred compound of the series. The invention provides a structurally modifiable amino acid–1,2,4-triazole hybrid platform for development of PARP-1-targeting anticancer compounds.

No. of Pages : 25 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128318 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIFUNCTIONAL GRAPHENE NANOPLATELET-REINFORCED CONCRETE COMPOSITES FOR STRUCTURAL APPLICATIONS

(51) International classification	:C04B 18/08, C04B 28/02, C04B 28/04, C04B 14/02, C04B 20/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Technology-Warangal
Address of Applicant :National Institute of Technology Warangal, Warangal, Telangana, India-506004 Warangal Telangana India
(72)**Name of Inventor :**
1)Dr D. Ravi Prasad
2)SK. Lal Mohiddin

(57) Abstract :
TITLE: MULTIFUNCTIONAL GRAPHENE NANOPLATELET-REINFORCED CONCRETE COMPOSITES FOR STRUCTURAL APPLICATIONS The present invention relates to a method for preparing a graphene nanoplatelet (GNP)-reinforced concrete composite. The method comprises dispersing GNPs in water in the presence of a polycarboxylate ether superplasticiser and subjecting the dispersion to probe ultrasonication at 20 kHz and 300 W for 60 minutes to obtain a GNP suspension having a concentration of 6 mg/mL. The GNP suspension is incorporated into a concrete composition comprising cement, fly ash, silica fume, fine aggregate, coarse aggregate, and the superplasticiser. The concrete composition is blended with 0.5 wt% of the superplasticiser, and GNPs are incorporated at dosages of 0.025%, 0.050%, 0.075%, and 0.10% by weight of cement through mixing for 3 to 5 minutes to obtain the concrete composite. The optimum GNP dosage is identified as 0.0685% by weight of cement.

No. of Pages : 15 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128322 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMMON-GROUNDED SWITCHED-CAPACITOR MULTILEVEL INVERTER AND METHOD FOR DETERMINING CAPACITANCE VALUES THEREOF

(51) International classification	:H02M7/483, H02M7/5387
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF SCIENCE
Address of Applicant :CV Raman Road, Bangalore: 560012, Karnataka, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)Tapas Roy
2)Madhu Sudhan Rao Ravada

(57) Abstract :
COMMON-GROUNDED SWITCHED-CAPACITOR MULTILEVEL INVERTER AND METHOD FOR DETERMINING CAPACITANCE VALUES THEREOF A common-grounded switched-capacitor multilevel inverter (CGSCMLI) (100) comprises a first switched capacitor (C1), a second switched capacitor (C2), a flying capacitor (Cf), and power switches (S1-S9) connected with an input voltage source (102) and an output node (a). The first switched capacitor (C1), the second switched capacitor (C2), and the flying capacitor (Cf) are selectively charged to a source voltage (Vin), twice the source voltage (2Vin), and half of the source voltage (0.5Vin), respectively. The power switches (S1-S9) selectively connect the capacitors in a common-ground configuration to generate discrete voltage levels having a maximum positive voltage level of 2.5Vin. A method (300) determines capacitance criteria based on operating conditions of the CGSCMLI (100) and selects capacitance values for the capacitors to maintain voltage regulation, voltage boosting, and voltage balancing. The CGSCMLI (100) may exchange power between a direct-current energy source (102) and an AC utility grid (106). Fig. 1.

No. of Pages : 60 No. of Claims : 31

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128323 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BATTERY STRESS-BASED REGENERATIVE BRAKING INTENSITY CONTROL SYSTEM AND METHOD FOR ELECTRIC VEHICLES

(51) International classification	:B60L 7/18, B60L 11/18, B60L 7/10, B60L 7/14, B60L 15/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSG COLLEGE OF TECHNOLOGY
Address of Applicant :THE PRINCIPAL, PSG COLLEGE OF TECHNOLOGY, AVINASHI ROAD, PEELAMEDU, COIMBATORE-641004, TAMILNADU, INDIA 9952756485 principal@psgtech.edu Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.J.Niresh
2)Dr.S.Neelakrishnan
3)Ohm Pranav DV

(57) Abstract :
The increasing adoption of electric vehicles has highlighted the need for 340 regenerative braking systems that adapt to the real-time condition of the traction battery rather than applying a static or user-selectable braking intensity. In this work, a battery stress-aware regenerative braking intensity auto-adjustment system is presented, which continuously receives vehicle operational data, including hard acceleration events, hard braking events, and regenerative braking current, over a 345 Controller Area Network (CAN), and computes therefrom a running, continuously updating battery stress score reflecting accumulated electrical and thermal loading over the course of a trip. The system further obtains, from a Battery Management System (BMS), real-time battery temperature and, optionally, instantaneous current-draw data, and evaluates, at the onset of each braking event, whether the running 350 stress score and the battery temperature together indicate elevated stress. When elevated stress is indicated, the system reduces regenerative braking intensity and blends in additional mechanical friction braking, thereby limiting the charging current spike returned to the battery, and progressively restores full regenerative braking intensity as battery conditions normalise. The invention operates as a continuous, 355 closed-loop control system throughout the drive, re-evaluating the braking-intensity decision at every braking event rather than applying a fixed, one-time, or post-trip setting. The proposed architecture highlights a practical and extensible framework for protecting traction battery health while maximising energy recovery in electric vehicles.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128328 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Enabled Device for Secure Healthcare Data Management

(51) International classification	:G16H 10/60, G06N 20/00, G06F 21/62, G16H 40/20, G16H 50/70
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CVR COLLEGE OF ENGINEERING
Address of Applicant :CVR COLLEGE OF ENGINEERING, Vastu Nagar, Mangal Palli (V) Ibrahimpatnam (M) Ranga Reddy (Dist) Telangana 501510, India. Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr.Hemant Kumar Shastry

(57) Abstract :
The present invention relates to an AI-enabled device for secure healthcare data management. The device comprises a central processing unit and memory, a data collection module, a data processing and normalization module, an artificial intelligence processing engine, a security and access-control module, a communication interface and a user interface. The device receives healthcare information from authorized sources, including patient records, physiological parameters, diagnostic information, medication information and laboratory results. The data processing module validates, structures, normalizes and integrates the received information. The AI processing engine analyzes the processed data to identify patterns, anomalies, trends and data-quality conditions. The security module provides authentication, authorization, encryption and controlled data transfer. The device provides integrated healthcare summaries, alerts, analytical outputs and decision-support information to authorized users through an online interface or connected healthcare system.

No. of Pages : 9 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128348 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An AI-Based Business Resource Optimization System for Workforce Allocation, Inventory Planning, Production Scheduling, Demand Forecasting, Cost Analysis, and Operational Decision-Making

(51) International classification	:G06Q 10/06, G06N 20/00, G06Q 10/04, G06Q 10/08, G06Q 10/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CVR COLLEGE OF ENGINEERING
Address of Applicant :CVR COLLEGE OF ENGINEERING, Vastu Nagar, Mangal Palli (V) Ibrahimpatnam (M) Ranga Reddy (Dist.) Telangana 501510, India. Hyderabad Telangana India
(72)**Name of Inventor :**
1)Lalitha Suddapally

(57) Abstract :
The current invention offers an AI-Based Business Resource Optimization System to improve the efficient use and management of organizational resources. By combining artificial intelligence, machine learning, predictive analytics, and optimization techniques, the system assesses worker skills and availability, inventory levels, production capacity, historical and current demand, operational expenses, and resource utilization data. The system performs workforce allocation, inventory planning, production scheduling, demand forecasting, cost analysis, and data-driven recommendations for operational decision-making. The invention's integration of several operations into a single framework allows organizations to recognize resource requirements, optimize scheduling and allocation, decrease operational expenses and resource waste, boost productivity, and make timely and informed business decisions.

No. of Pages : 14 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128349 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : “INTEGRATED TAB BASED G.M COUNTING SYSTEM”

(51) International classification	:G05B 19/042, G06M 1/272, G01R 19/25, G06F 13/38, G06F 3/0488
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M/S. NUCLEONIX SYSTEMS PVT LTD
Address of Applicant :PLOT NO. 162A & B, IDA, PHASE-2, CHERLAPALLY, HYDERABAD, TELANGANA, INDIA - 500051 Telangana India
(72)**Name of Inventor :**
1)Mr. J. NARENDER REDDY

(57) Abstract :
ABSTRACT “INTEGRATED TAB BASED G.M COUNTING SYSTEM” integrates a G.M. detector with an electronic controller-based circuit module and an Android OS-based TAB with built-in software in JavaScript. The TAB works as a master unit and the microcontroller in the electronics module works as a slave unit connected wirelessly via Bluetooth. The G.M. Tutor software on the TAB replaces conventional printed handbooks by providing step-by-step experimental procedures, digital data acquisition, mathematical calculations, and automated graphical plotting. The electronic module comprises a PIC18LF26K42 microcontroller (U1), an I2C DAC (U26, MCP4725) for high-voltage programming (100V–1250V), a buffered ADC feedback loop (LM158) for high-voltage monitoring, and an LM193AD pulse-processing circuit for converting G.M. tail pulses into digital TTL pulses. Automated high-voltage step incrementing (`STEP HV`) enables acquisition and display of G.M. plateau characteristics directly on TAB. Refer Figure 1

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128350 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : An Optimized Hybrid Triple-Attention-Based Convolutional RoBERTa Model for Multilingual Sentiment Classification

(51) International classification	:G06N 3/04, G06N 3/08, G06K 9/62, G06F 17/27, G06F 16/35
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Kondapalli Sunanda
Address of Applicant :School of CS & AI SR University, Warangal Anantasagar, Hasanparthy, Warangal, Telangana 506371. Telangana India
2)Rupesh Kumar Mishra
(72)**Name of Inventor :**
1)Kondapalli Sunanda
2)Rupesh Kumar Mishra

(57) Abstract :
ABSTRACT [1] The present invention, titled “AN OPTIMIZED HYBRID TRIPLE-ATTENTION-BASED CONVOLUTIONAL RoBERTa MODEL FOR MULTILINGUAL SENTIMENT CLASSIFICATION,” relates to an advanced deep learning architecture for accurately classifying sentiment in multilingual textual data. The invention discloses an optimized hybrid neural network that integrates a Robustly Optimized Bidirectional Encoder Representations from Transformers (RoBERTa) model, convolutional neural network (CNN) layers, and a triple-attention mechanism to improve sentiment feature extraction, contextual representation, and classification performance across multiple languages. The invented system incorporates three complementary attention mechanisms designed to capture contextual dependencies, emphasize sentiment-bearing features, and identify important token-level and feature-level representations. The convolutional layers extract local linguistic patterns, including sentiment expressions, phrases, and contextual n-gram features, while the RoBERTa encoder generates contextualized representations of input text. The hybrid architecture combines these representations to improve sentiment discrimination across languages with different grammatical structures, vocabulary distributions, and linguistic characteristics.The invention further incorporates optimized feature fusion, attention-weighted representation learning, and a classification module to generate sentiment predictions from multilingual text. The proposed architecture is suitable for multilingual sentiment analysis of social media content, customer reviews, product feedback, online discussions, and other text-based applications. The system is designed to improve classification accuracy, contextual understanding, and robustness against linguistic variation, code-mixing, and differences in text length. Its comparative performance is evaluated using appropriate multilingual sentiment datasets and established evaluation metrics, subject to experimental validation.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128359 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SENTINELFUSION: A HYBRID MULTI-MODAL FRAMEWORK FOR MALWARE CLASSIFICATION AND ZERODAY DETECTION USING CNN, TRANSFORMER, AND XGBOOST WITH THREAT INTELLIGENCE INTEGRATION, CONTINUAL LEARNING, AND EXPLAINABLE Al

(51) International classification	:G06F 21/56, G06N 3/04, G06N 3/08, H04L 29/06, G06K 9/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. THERESA CENATE C F
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, 1, JAWAHARLAL NEHRU SALAI, CHENNAI, TAMIL NADU, INDIA-600026 Tamil Nadu India
2)Rupa R
3)Mohul Raj T
4)Brian Hamish E
(72)**Name of Inventor :**
1)DR. THERESA CENATE C F
2)Rupa R
3)Mohul Raj T
4)Brian Hamish E

(57) Abstract :
ABSTRACT The invention relates to a hybrid multi-modal framework for malware classification and zero-day detection. A convolutional neural network analyses a byte image of a malware sample, a Transformer analyses a sandbox-generated API call sequence, and a gradient-boosted tree model analyses engineered static features of a Portable Executable file. A confidence-aware fusion gate, which receives a per-branch availability mask, combines the branch embeddings so that a reliable prediction is produced even when one modality is unavailable. An open-set decision layer compares a novelty score computed from the fused embedding with a calibrated threshold to separate known malware families from previously unseen threats. A threat-intelligence module correlates the 2 sample hash and network indicators against STIX/TAXII feeds and returns either an immediate verdict or a reputation feature for the fusion gate. A continual learning controller updates the models using Elastic Weight Consolidation with replay and incremental boosting, and an explainability layer provides SHAP, Grad-CAM and attention-based explanations for every prediction.

No. of Pages : 14 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128363 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HOSPITAL ROOM WIRELESS NURSE ALERT SYSTEM BASED ON PATIENT MOVEMENT.

(51) International classification	:A61B 5/00, G08B 21/04, A61B 5/0205, A61B 5/11, A61B 5/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)J.J. COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :Ammapettai, Poolangulathupatti (PO), Tiruchirappalli, Tamil Nadu, India Tiruchirappalli Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. S SAGAYA MARY
2)Dr. P MATHIYALAGAN
3)Dr. S SUMITHRA
4)D HARINI
5)S YAKSHITHA RANGA SRI
6)R DHANALAKSHMI
7)E NIRANJANA DEVI

(57) Abstract :
The proposed system introduces a hospital room wireless nurse alert system based on patient movement using wireless sensor network (WSN) technology for ongoing health monitoring and emergency help. It tracks vital signs like heart rate, body temperature, and patient movement in real time. Sensors collect information's from the patients, and it is sent to the Arduino UNO for processing. A sudden motions or falls are detected by an MPU6050 to respond quickly in emergencies. From ESP8266 to CloudDaddy the processed data were sent, which enables doctors, nurses, and caregivers to monitor the patient's condition from a distance. The system stimulates alerts with a buzzer and shows information locally on an LCD if there are abnormal readings or a fall is detected. Using Proteus Simulation Software the circuit is tested, and using the Embedded C in the Arduino IDE the system is programmed. By providing real-time monitoring, it improves patient safety, which cuts down on manual work and allows for a quick response during emergencies. It is low cost, reliable, and works well for both hospital and home healthcare settings.

No. of Pages : 14 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128365 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Signal-Quality-Gated Contactless Vital-Sign Screening Station for Emergency Triage

(51) International classification	:A61B 5/00, A61B 5/0205, A61B 5/01, A61B 5/024, A61B 5/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mrs. S. P. Kavya
Address of Applicant :Assistant Professor II, Department of Artificial Intelligence and Data Science, KPR Institute of Engineering and Technology, Coimbatore Coimbatore Tamil Nadu India
2)Ms. Kavya Thirunavukkarasu
(72)**Name of Inventor :**
1)Mrs. S. P. Kavya
2)Ms. Kavya Thirunavukkarasu

(57) Abstract :
Signal-Quality-Gated Contactless Vital-Sign Screening Station for Emergency Triage Abstract The present invention relates to a Signal-Quality-Gated Contactless Vital-Sign Screening Station for Emergency Triage. The system comprises radar and thermal sensing units for acquiring data from a subject without direct physical contact. A processing unit analyses the acquired data to estimate physiological parameters, including heart rate and breathing rate. A signal-quality assessment unit evaluates the reliability of the acquired signals against predefined criteria. Based on this assessment, a decision unit permits suitable estimates to be displayed through an operator interface or prompts repeat measurement when the signal quality is insufficient. A local data and communication unit supports the storage and transfer of screening information. The invention aims to support preliminary emergency triage by providing a structured measurement-quality assessment workflow and reducing the presentation of unreliable physiological estimates. The station is intended for screening support and does not replace clinical examination or medical diagnosis.

No. of Pages : 10 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128425 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TETRANUCLEAR NICKEL–THIOLATE NANOCLUSTER AS AN ELECTROCATALYST FOR REDUCTION OF NITROGEN TO AMMONIA

(51) International classification	:C25B15/08, C25B1/27, C25B11/075, C25B11/073, C25B11/051, C25B11/04, C25B15/02, C25B9/65
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Science Education and Research Thiruvananthapuram
Address of Applicant :Maruthamala PO, Vithura, Trivandrum 695551, Kerala. Vithura Kerala India
(72)**Name of Inventor :**
1)Sukhendu Mandal
2)Kothandaraman Ramanujam
3)Pinaki Khan

(57) Abstract :
TITLE: TETRANUCLEAR NICKEL–THIOLATE NANOCLUSTER AS AN ELECTROCATALYST FOR REDUCTION OF NITROGEN TO AMMONIA The invention provides an electrocatalyst for the reduction of Nitrogen to Ammonia. The invention provides a tetranuclear nickel–thiolate nanocluster -[Ni4(C6H11S)8] of formula A as a homogeneous electrocatalyst for the reduction of Nitrogen to Ammonia. The invention also relates to a method of preparation of the tetranuclear nickel–thiolate nanocluster of formula-A. FORMULA -A

No. of Pages : 34 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128474 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR CONCRETE CORROSION DETECTION IN REINFORCED STRUCTURES

(51) International classification	:G01N 17/04, G01N 17/00, G01N 17/02, G01N 27/04, G01N 33/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. M. Nanthini
Address of Applicant :Assistant professor, Department of Civil Engineering, Meenakshi Sundararajan Engineering College, Kodambakkam, Chennai, Tamil Nadu. Tamil Nadu India
2)Mrs. U. Nirmalambal
3)Dr. M A. Raja
4)Mrs. M. Malini Gayathri
5)Dr M Ponni
6)Mrs. P Saranya
7)Mr. S. Saravanan
8)Mr. S. Vishnu Vardhan
(72)**Name of Inventor :**
1)Dr. M. Nanthini
2)Mrs. U. Nirmalambal
3)Dr. M A. Raja
4)Mrs. M. Malini Gayathri
5)Dr M Ponni
6)Mrs. P Saranya
7)Mr. S. Saravanan
8)Mr. S. Vishnu Vardhan

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR CONCRETE CORROSION DETECTION IN REINFORCED STRUCTURES The present invention relates to a system and method for detecting corrosion in reinforced concrete structures. The system comprises a plurality of sensors configured to measure corrosion-related parameters including half-cell potential, concrete resistivity, chloride concentration, temperature, and humidity. A data acquisition unit collects sensor measurements and transmits the data to a processing unit for analysis. The processing unit determines a corrosion indication based on predefined criteria and generates alerts corresponding to detected corrosion conditions. The system enables non-destructive monitoring, facilitates early identification of corrosion-prone regions, and supports structural maintenance and condition assessment through a monitoring interface. [Figure 1 is reference figure]

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128480 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR CLOSED - LOOP COMMODITY RISK - AWARE MANUFACTURING PLANNING AND INVENTORY ORCHESTRATION

(51) International classification	:G06Q 10/06, G06Q 10/08, G06Q 50/04, G06Q 30/02, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Karthiksai Chenna
Address of Applicant :SAP Lead, Department of IT, 1333 Locomotive Dr, Westfield, IN, 46074, Indiana, USA. U.S.A.
(72)**Name of Inventor :**
1)Karthiksai Chenna

(57) Abstract :
SYSTEM AND METHOD FOR CLOSED - LOOP COMMODITY RISK - AWARE MANUFACTURING PLANNING AND INVENTORY ORCHESTRATION
ABSTRACT The present invention discloses a computer-implemented system and method for closed-loop commodity risk-aware manufacturing planning and inventory orchestration. The system integrates commodity market information, supplier conditions, demand forecasts, production requirements, inventory positions, procurement data, and manufacturing constraints. A risk analysis engine generates commodity exposure and risk indicators, while an optimization engine determines risk-adjusted procurement, inventory, supplier, and production decisions. A closed-loop monitoring mechanism compares actual operational and market conditions with planned conditions and detects deviations. Upon detecting a significant deviation, the system dynamically recalculates risk and generates an updated manufacturing and inventory plan. Predictive analytics, scenario simulation, optimization, and feedback-based learning may be employed to improve planning decisions. The invention thereby enables continuous coordination of commodity risk, procurement, manufacturing, and inventory operations while reducing cost exposure, inventory inefficiency, supply disruption risk, and production interruptions.

No. of Pages : 15 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128485 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR PHYSICS-GUIDED SYNTHETIC Z-SCAN BEAM PROPAGATION AND DATASET GENERATION

(51) International classification	:G02F 1/35, G02B 27/09, G02F 1/39, G02F 1/37, G02F 1/355
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM University- AP
Address of Applicant :Neerukonda, Mangalagiri Mandal Guntur District-522240, Andhra Pradesh, India Guntur Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. Sakshi Choudhary
2)Dr. Ravi Kumar
3)Dr. Salla Gangi Reddy

(57) Abstract :
The present invention relates a system (100) for physics-guided synthetic Z-scan beam propagation and automated dataset generation. A beam source module (102) defines an input laser beam. A propagation engine (104) propagates an optical field through a virtual focusing element (118), a virtual nonlinear sample (120) translatable along a z-axis, a virtual aperture (122), and a virtual detector (124) using Fresnel diffraction-based field propagation preserving amplitude and phase. A nonlinear response module (106) returns a nonlinear phase distortion and a nonlinear amplitude attenuation, so diffraction and nonlinear phase distortion are simulated within one propagation event. A Z-scan acquisition module (108) computes open and closed aperture transmittance curves. A multimodal dataset generation module (110) pairs beam evolution images with the curves and metadata. A dataset structuring module (112) forms an AI ready dataset (3) stored in a data storage (114). Fig. 1

No. of Pages : 26 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128505 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A COLLIMATOR FOR AN X-RAY DEVICE

(51) International classification	:G21K 1/02, G21K 1/04, A61B 6/06, G03B 42/02, G01N 23/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PROGNOSYS MEDICAL SYSTEMS PVT LTD
Address of Applicant :Survey no. 168/1, Machohalli, Dasanpura Hobli Next to Vani Vidyalaya, Off Magadi Road Bengaluru, Karnataka, India - 560091 Karnataka India
(72)**Name of Inventor :**
1)Adarsha Kumar I B
2)ABHISHEKA HONAKERI

(57) Abstract :
ABSTRACT A COLLIMATOR FOR AN X-RAY DEVICE. A collimator (100) for an X-ray device comprises a common base (102) supporting first and second shutter assemblies (210, 212) on opposite faces. Each shutter assembly includes leadscrew (206, 208) having coaxial, oppositely threaded sections configured to rotate together and symmetrically displace a pair of shutter plates to define an X-ray aperture (230). A mirror frame (202) supports a mirror (204) and an LED (602) maintained at a fixed angle and distance, with the LED housed in an LED guide (110). The mirror frame is pivotably coupled to mirror guide block (228) slidable along mirror guide rods of an extended frame (108). Sliding and pivoting adjustments enable independent alignment of the light field with the X-ray field in two directions without dismantling the collimator. An adjustable laser holder enables centring of a laser crosshair on the X-ray aperture. Reference figure: FIG. 2

No. of Pages : 37 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128538 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING-BASED ENERGY CONSUMPTION PREDICTION AND RANGE-AWARE ROUTE PLANNING SYSTEM FOR ELECTRIC VEHICLES

(51) International classification	:G06N 20/00, G01C 21/34, B60L 53/66, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Santhosh T
Address of Applicant :Assistant Professor, Department of Information Science and Engineering, Bapuji Institute of Engineering Technology, Davanagere, Karnataka, 577004, India. Karnataka India
2)Nikhil Chauhan
3)Roshan Kumar
4)Ajit Singh
5)Saya
6)Mrs. Anuradha Sagar Nigade
7)Dr. Sharda Prakash Tondare
8)Dr Shankar Rakhamaji Madkar
9)Abdul Khadar Asundi
(72)**Name of Inventor :**
1)Dr. Santhosh T
2)Nikhil Chauhan
3)Roshan Kumar
4)Ajit Singh
5)Saya
6)Mrs. Anuradha Sagar Nigade
7)Dr. Sharda Prakash Tondare
8)Dr Shankar Rakhamaji Madkar
9)Abdul Khadar Asundi

(57) Abstract :
The present invention relates to the development of a machine learning based system to predict energy consumption and generate range aware driving directions for electric vehicles involving a vehicle telemetry interface, a charging network interface, historical telemetry data repository, multimodal sensor data acquisition component, vehicle state compiler, constrained route synthesis module, machine learning models, validation engine, energy evaluation engine, adaptive routing engine, and a longitudinal report generator. Vehicle states are compiled using a multi parameter vehicle state vector along with its associated uncertainty measures and exclusion structures based on both historical and real-time inputs. Physics informed machine learning models generate the candidate routes and these are validated in terms of their range feasibility and charger availability fingerprints against the exclusion structure to eliminate range failure possibilities. Consumption during the navigation process is evaluated to adaptively re-route the trajectory and charging points based on power consumption. FIG.1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128540 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : GRAPH THEORY-BASED DYNAMIC EMERGENCY VEHICLE ROUTE OPTIMIZATION AND TRAFFIC CLEARANCE SYSTEM

(51) International classification	:G08G 1/0968, G08G 1/01, G08G 1/00, G01C 21/34, G08G 1/0967
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Praveen N
Address of Applicant :Assistant Professor, Department of Aeronautical Engineering, Nitte Meenakshi Institute of Technology, NITTE (Deemed to be University), Bengaluru, Karnataka, 560064, India. Karnataka India
2)Dr. D. Kalamani
3)Hariharan M
4)Vasuki S
5)Thanmaya Jyothi B
6)Dr. S. Sree Priya
7)Dr. R. Sivasangari
8)R. Sinduja
(72)**Name of Inventor :**
1)Praveen N
2)Dr. D. Kalamani
3)Hariharan M
4)Vasuki S
5)Thanmaya Jyothi B
6)Dr. S. Sree Priya
7)Dr. R. Sivasangari
8)R. Sinduja

(57) Abstract :
The present invention pertains to a graph theory-based dynamic emergency vehicle route optimization and traffic clearance system (100) designed to enhance emergency transportation efficiency. The system incorporates a road network data acquisition module (101), real-time traffic monitoring module (102), graph construction and updating module (103), dynamic edge-weight calculation module (104), route optimization module (105), emergency vehicle tracking module (106), traffic clearance coordination module (107), communication and notification module (108), and monitoring and decision support interface (109). The system models a road network as a weighted graph and continuously computes route costs by integrating traffic density, estimated travel time, intersection delays, road accessibility, and road restrictions. Graph-based algorithms identify optimal routes and recalculate them in response to congestion or alterations in road conditions. The traffic clearance coordination module enables authorized emergency priority requests and synchronized traffic signal management. The system delivers adaptive route guidance, traffic surveillance, and decision support capabilities for enhancing emergency response effectiveness and transportation management operations. FIG.1

No. of Pages : 14 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128545 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED ACADEMIC RESEARCH ASSISTANT FOR KNOWLEDGE-GAP DISCOVERY AND RESEARCH-DESIGN RECOMMENDATION

(51) International classification	:G06N 5/02, G06Q 50/20, G06F 17/30, G06F 40/30, G06F 16/332
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. R. Ramya
Address of Applicant :Assistant Professor, Department of B.Com Finance & Taxation (Integrated with ACCA UK), Dwaraka Doss Goverdhan Doss Vaishnav College (Autonomous), Chennai, Tamil Nadu, India. Tamil Nadu India
2)Mrs. Jyoti Morbale
3)Dr. Prajakta Naregalkar
4)S. Hariprasath
5)Shirsath Kavita Bhanudas
6)Mrs. Devashree Shah
7)Mrs. Sonali Kishore Pawar
8)Dr. Suresh Kumar
9)Dr Babita Bantwan

(72)**Name of Inventor :**
1)Dr. R. Ramya
2)Mrs. Jyoti Morbale
3)Dr. Prajakta Naregalkar
4)S. Hariprasath
5)Shirsath Kavita Bhanudas
6)Mrs. Devashree Shah
7)Mrs. Sonali Kishore Pawar
8)Dr. Suresh Kumar
9)Dr Babita Bantwan

(57) Abstract :
The present invention relates to the development of an intelligent solution to assist researchers in finding the gaps in knowledge and suggesting appropriate research designs. This system includes research data input module (101), preprocessing and normalization module (102), knowledge representation module (103), and AI-based knowledge-gap discovery module (104) for processing academic papers, theses, datasets and research outcomes. Semantic and research trend analysis module (105) detects research trends, contradictions and potentialities. Research design recommendation module (106) develops appropriate research objectives, research questions, hypotheses, variables, methods and analysis approach. Research design evaluation module (107) ranks recommendations on the basis of their relevancy, feasibility and innovativeness. Interaction interface (108) delivers outputs, whereas continuous learning module (109) optimizes the recommendations based on researcher's feedback and new academic information. FIG.1

No. of Pages : 11 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128557 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BRAILLE AI: AN AI-POWERED BRAILLE AND AUDIO CONVERSION SYSTEM

(51) International classification	:G09B 21/00, G09B 21/02, G09B 21/04, G06V 30/10, G06F 3/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M. Buvana
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L&T BY-PASS, CHINNIYAMPALAYAM POST, COIMBATORE -641062 Tamil Nadu India
2)P. Brindhaa
3)S. Deepika
4)K.Deepika Rani
5)M. Harini
(72)**Name of Inventor :**
1)M. Buvana
2)P. Brindhaa
3)S. Deepika
4)K.Deepika Rani
5)M. Harini

(57) Abstract :
Braille ai: an ai-powered braille and audio conversion system is an assistive application that converts text from images and documents into Braille and speech. It uses OCR to extract content and provides both accessible outputs through a single platform. The system also includes secure login, administrative controls, and conversion history. By automating the conversion process, BRAILLE Al supports easier and more independent access to information for visually impaired users. It reduces the need for manual text conversion and external assistance. The system provides a simple way to process and access written content. Overall, BRAILLE Al combines modern technologies to improve accessibility and support independent information use. Braille ai: an ai-powered braille and audio conversion system is an assistive web-based platform developed to make written information easier to access for visually impaired users. It allows users to provide images or digital documents, identifies the text within them using OCR, and transforms the extracted content into Braille and spoken output. The system is developed using modern web technologies and integrates OCR, Braille conversion, and Text-to-Speech capabilities within a single platform. It also provides secure authentication, user management, administrative controls, and storage of previous conversion records for convenient access and organized operation. By combining multiple accessibility functions in one environment, BRAILLE AT reduces the need for manual conversion and external assistance. The system provides a simple and practical way for users to receive written information through tactile and audio formats, supporting more independent access to digital content.

No. of Pages : 6 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128564 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CLOUD DEPLOY: DEVOPS DEPLOYMENT PLATFORM

(51) International classification	:G06F 8/60, G06F 8/61, G06F 9/445, G06F 9/455, G06F 9/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. T. ESTHER
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BY-PASS, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)Partha sarathi K
3)Naveen K
4)Nesly Jeniston M
5)Poobesh M
(72)**Name of Inventor :**
1)DR. T. ESTHER
2)Partha sarathi K
3)Naveen K
4)Nesly Jeniston M
5)Poobesh M

(57) Abstract :
ABSTRACT A computer-implemented platform automates deployment of a software project. A user submits the project through a web interface by upload or as a Git repository address. A backend locates the project root from build-marker files, detects the technology from marker files in a fixed order, and writes a matching Dockerfile. It selects a container port, allocates a free host port, generates Kubernetes manifests from templates, and applies them using kubectl. It then triggers a parameterised Jenkins pipeline with the project path, name, TtecliNtogyToij portOplit401,01i41.aige isTbftiltarid con-Millert run A result 'page shows the container status and application address.

No. of Pages : 5 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128572 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR TEXT-INDEPENDENT SPATIAL-TOPOLOGICAL REASONING OVER DOCUMENTS

(51) International classification	:G06N 5/04, G06F 17/30, G06N 5/02, G06F 17/00, G06F 40/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SchemaMind LLP
Address of Applicant :303, RJ Homes, 5th A Cross, 3rd Main, LB Shasry Nagar, Bangalore, Karnataka, India 560017 Karnataka India
(72)**Name of Inventor :**
1)Shuaib Ahmad

(57) Abstract :
A computer-implemented system and method for text-independent spatial-topological reasoning over digital documents are disclosed. A source document is processed to identify document elements associated with identifiers and spatial, positional, visual, or structural attributes. Original content corresponding to the document elements is retained separately from a model-facing structural representation. The model-facing structural representation comprises the identifiers and selected attributes while withholding corresponding source textual content from a computational reasoning component. The computational reasoning component determines structural relationships and document topology from the representation and returns an identifier-based structural result. The identifiers are subsequently resolved against the retained source representation to retrieve corresponding original document content and generate a structured representation. The separation of structural inference from source-content retrieval permits structural reasoning to be performed without providing corresponding source textual content to the reasoning component, while deterministic identifier resolution permits the structured representation to be generated using the original source elements.

No. of Pages : 42 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128576 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ADAPTIVE TWO STAGE RETRIEVAL AUGMENTED GENERATION IN AUTOMATED INCIDENT RESPONSE

(51) International classification	:G06Q 50/20, G06Q 10/06, G09B 7/00, G06F 17/30, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)The Principal, SNS College of Technology

Address of Applicant :ASSOCIATE PROFESSOR, DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING, SNS COLLEGE OF TECHNOLOGY, SNS KALVI NAGAR, SATHY MAIN ROAD, NH209, VAZHIYAMPALAYAM, SARAVANAMPATTI, COIMBATORE, TAMILNADU-641035 9710928460 sumithragvskr@gmail.com Tamil Nadu India

(72)**Name of Inventor :**

1)S.Priyadarsini

2)Dr.M.Shobana

3)Dr.A.Sumithra

4)Dr.B.Vinodhini

5)K.K.Bhuvanesh

6)K.R.Deepisha

7)B.Dharshana

8)S.Dinesh

(57) Abstract :

This invention discribes an Al-powered student event management system for organising and managing academic and extracurricular activities in educational institutions. The system provides a single platform for event creation, event publishing, student registration, attendance monitoring, participation tracking, notifications, feedback management, and digital certificate generation. It allows students, organisers, and administrators to access and manage event-related information through respective interfaces. The system includes an Al analysis module that processes student profile information, registration history, attendance records, and participation patterns. Clustering techniques such as K-Means are used to find similarities in student participation behaviour, while an Al recommendation module uses the analysed information to provide personalised event suggestions. This enables students to discover events based on their interests and previous participation. The invention further includes an Al event assistant that uses Large Language Model (LLM) processing and Retrieval-Augmented Generation (RAG) to respond to natural-language queries using relevant event information. The integration of intelligent recommendations, participation analysis, conversational assistance, event administration, notifications, and certificate management provides an integrated platform for improving student engagement and simplifying institutional event coordination.

No. of Pages : 9 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128577 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SPORTIQ - AI-POWERED PLATFORM FOR SPORTS TALENT IDENTIFICATION AND ATHLETE PERFORMANCE ANALYSIS

(51) International classification	:G06Q 10/06, G06F 17/30, G06Q 10/10, G06Q 50/20, G06F 16/9535
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)The Principal, SNS College of Technology
Address of Applicant :SNS Kalvi Nagar, Sathy Main Road, NH-209, Vazhiyampalayam, Saravanampatti Post, Coimbatore, Tamil Nadu 641035 Tamil Nadu India

(72)**Name of Inventor :**
1)Dr.G.Vanitha
2)A.S.Arul Sowmiya
3)Dr.M.Shobana
4)Dr.A.Sumithra
5)K.Sanjeev
6)S.Santhosh
7)K.Shalini
8)S.Sivabharani

(57) Abstract :

SportIQ is an AI-powered platform that supports sports talent identification and analysis of athlete performance. Traditional talent evaluation relies heavily on manual observation, and athlete records-are-often-scattered; making-consistent comparison-arid-ongoing monitoring difficult. SportIQ provides a single web-based environment where athletes, coaches, scouts, and administrators work through role-based access. Athletes create profiles, select their sport, and submit performance information. The system analyses this information to produce a performance score, a potential level, and a summary of strengths and weaknesses, and it prepares personalised improvement recommendations. Coaches and scouts can compare athletes, review rankings, and search for players using structured performance indicators, while athletes can discover relevant events and opportunities. The platform stores assessment results so progress can be tracked over time, and training videos can serve as an additional source of performance information. The platform uses Next.js, React, TypeScript, and Tailwind CSS for the interface, Python and FastAPI for backend services, and PostgreSQL with SQLAIchemy for data storage. Together, these components connect user access, data collection, recommendations, and dashboards into a single workflow. SportIQ aims to reduce manual evaluation effort and help coaches and scouts make informed decisions, while keeping the final judgment with human evaluators.

No. of Pages : 9 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128620 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A HANDHELD FATIGUE SCREENING SYSTEM AND METHOD THEREOF

(51) International classification	:A61B 3/113, A61B 3/11, G06K 9/00, A61B 3/00, A61B 3/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD
Address of Applicant :IIT Hyderabad Road, Near NH-65, Sangareddy, Kandi, Telangana, 502284, India Hyderabad Telangana India
(72)**Name of Inventor :**
1)Vandana Sharma
2)Parishkrith Aravind
3)Kalapureddy Srinivasa Varun Reddy

(57) Abstract :
A method (600) for screening fatigue in a subject is disclosed. The method (600) includes providing illumination continuously to an eye. The method (600) includes continuously capturing a plurality of frames of the eye. The method (600) includes selectively illuminating the eye during a stimulus interval at a predetermined time for a predetermined interval of time followed by preprocessing the plurality of frames to generate an accepted pupil geometry. The method (600) includes calculating a pupil-area value associated with a timestamp corresponding to a frame from which the accepted pupil-boundary candidate is identified, identifying missing and implausible pupil-area samples to generate a conditioned pupil-response sequence. The method (600) includes determining that a measurement-quality condition is satisfied for the conditioned pupil-response sequence, calculating a plurality of PLR descriptors from the conditioned pupil-response sequence when the measurement-quality condition is satisfied, and generating a fatigue-screening indication based on the plurality of PLR descriptors.

No. of Pages : 32 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128649 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING-BASED ADAPTIVE FRAUD DETECTION SYSTEM

(51) International classification	:G06Q 20/40, G06N 20/00, G06Q 20/38, G06Q 40/04, G06Q 40/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Kumaraswamy Nekkalapudi
Address of Applicant :S/O Vishnudhararao Nekkalapudi B6 F1 HB quarters, HNo-6-3-841/58 Ameerpet colony, Hyderabad-500016 Telangana India
(72)**Name of Inventor :**
1)Kumaraswamy Nekkalapudi

(57) Abstract :
The present invention relates to a machine learning-based adaptive fraud detection system for identifying potentially fraudulent transactions across digital transaction environments. The system comprises a transaction data acquisition module, a preprocessing and feature extraction module, a machine learning-based fraud risk assessment module, an adaptive threshold management module, a transaction decision module, and a feedback and monitoring module. Transaction information is processed to generate contextual features, which are evaluated by a trained machine learning model to determine a fraud risk score. The score is compared with an applicable decision threshold to determine transaction approval, rejection, temporary holding, or additional authentication. Verified transaction outcomes and monitored performance indicators are utilized to support controlled model updating, threshold adjustment, and detection of changing transaction patterns. The system further supports model validation, version management, audit logging, and rollback procedures to facilitate adaptive fraud detection while maintaining traceability and operational control.

No. of Pages : 22 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641128652 A
(19) INDIA
(22) Date of filing of Application :28/10/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : CATIONIC CU(I)4-TRIRADICAL COMPLEX AND METHOD OF PREPARATION THEREOF

(51) International classification	:C07F1/08, C01G3/00, C07D327/00, C07D331/00, C07C323/09
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology Madras (IIT Madras)
Address of Applicant :The Dean, Industrial Consultancy & Sponsored Research [IC&SR], Indian Institute of Technology Madras, Sardar Patel Road, IIT P.O, Chennai- 600 036, Tamil Nadu, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Kartik Chandra Mondal
2)Christel Livia Mascarenhas
3)Sujit Das
4)Sonam Suthar

(57) Abstract :
ABSTRACT OF INVENTION The present disclosure relates to a copper(I)4–dithiolene complex represented by Formula (I) Formula (I) wherein Dip is . The present disclosure also relates to the method of preparation thereof.

No. of Pages : 30 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128688 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD OF CLOSED-LOOP AGRIVOLTAIC MICROCLIMATE REGULATION USING A TENDON-TILTED PHOTOVOLTAIC CANOPY

(51) International classification	:A01G 9/24, A01G 25/02, A01G 25/16, H02S 40/42, H02S 40/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)THIAGARAJAR COLLEGE OF ENGINEERING
Address of Applicant :Tirupparankundram, Madurai, Tamilnadu – 625015, India. Madurai Tamil Nadu India
(72)**Name of Inventor :**
1)S S. THANWIN
2)S. KUBERAESHWARAN
3)P M. CHANDRU
4)M. MOHAIDEEN AASIF AHMED
5)A. JAI SAKTHI SELVAM
6)M. MEENAKSHI DEVI
7)P.S. MANOHARAN
8)L. ASHOK KUMAR
9)P. DEEPAMANGAI

(57) Abstract :
Agrivoltaic System with Tendon-Tilted Canopy, Water-Based Panel Cleaning and Irrigation-Assisted Passive Cooling relates to a solar canopy combining tendon-driven tilt actuation with closed-loop irrigation for crop and photovoltaic yield. Fixed-tilt agrivoltaic canopies compromise crop yield through non-adaptive shading and leave panel-surface dust and elevated panel temperature unaddressed, reducing photovoltaic output. The system employs panel modules interconnected through a tendon-and-pulley mechanism for coordinated tilt adjustment using minimal drive motors, governed by a control unit based on solar incidence and crop-shading needs. A sensor-based irrigation subsystem applies drip irrigation beneath the panels, passively cooling them to raise conversion efficiency. Harvested rainwater, stored and filtered on-site, is reused for irrigation and for spraying the panel surfaces to remove dust, returning to storage for reuse in a closed hydrological loop. The system forms a dual-purpose installation for power generation, crop microclimate management and water conservation, suited to agrivoltaic installations.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128694 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Enabled Enterprise People Analytics Framework Using Leading and Lagging Indicators for Business Transformation and Sustainable Organizational Growth

(51) International classification	:G06N 20/00, G06Q 10/06, G06N 5/04, G06Q 10/10, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)Kiran Devarapally
2)Dr. Sunitha Purushottam Ashtikar

(57) Abstract :
The present invention relates to artificial intelligence-enabled enterprise people analytics using temporally related leading and lagging indicators. Source events are stored with event times, availability times, source versions, and subject identifiers. A task definition specifies observation and outcome windows and a prediction cutoff. A feature processor constructs snapshots using event versions available by the cutoff. A label-maturity processor withholds incomplete outcomes until window and source-completeness conditions are satisfied. Mature labels are associated with the corresponding snapshots for model training and evaluation. A lineage index identifies derived records affected by corrections while preserving historical inputs used for issued predictions. A model service generates versioned predictions and reconciles them with subsequently matured outcomes. The system supports reproducible analysis of workforce capabilities and enterprise transformation outcomes across asynchronously updated data sources.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128700 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LEADERSHIP-DRIVEN STRATEGIC TECHNOLOGY ADOPTION FRAMEWORK FOR EMERGING TECHNOLOGIES IN INDIAN AND GLOBAL ENTERPRISES

(51) International classification	:G06F 9/445, G06F 8/61, G06Q 10/06, G06F 8/71, G06F 8/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)Kamini Shiva Prasad
2)Dr. Tunguturi Srinivas

(57) Abstract :
The present invention relates to enterprise computing for controlled adoption of emerging technologies. A versioned dependency repository links technology components with target configurations and capability-evidence prerequisites. An assessment engine generates a staged deployment plan and prerequisite snapshot. An authorization service binds leadership approval to a digest of the plan and snapshot. Before dispatch, an orchestrator verifies authorization and compares referenced prerequisite versions with current versions. A material change causes dispatch to be withheld for reassessment. A validated deployment command carries an execution identifier. A telemetry evaluator correlates target observations with the command and deployment stage, enabling further rollout only when stage conditions are satisfied. A configured failure condition causes a hold or supported rollback operation. The system coordinates deployment state, authorization, and operational feedback across connected enterprise applications.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128715 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : NANOCOMPOSITE ELECTROCATALYST AND A PROCESS FOR ITS PREPARATION

(51) International classification	:C25B 11/04, C25B 1/04, C25B 11/091, C25B 11/054, C25B 11/061
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM UNIVERSITY
Address of Applicant :Amaravati, Mangalagiri, Andhra Pradesh-522502, India Guntur Andhra Pradesh India
(72)**Name of Inventor :**
1)CHAKRABORTTY, SABYASACHI
2)MANGALASSERI, ASWATHI
3)SIVAGURU, GOKUL
4)GHORUI, UDAY KUMAR

(57) Abstract :
ABSTRACT NANOCOMPOSITE ELECTROCATALYST AND A PROCESS FOR ITS PREPARATION The present disclosure relates to the field of electrolysis. Particularly, the present disclosure relates to a nanocomposite electrocatalyst and a process for the preparation its preparation, and an electrode obtained therefrom. The nanocomposite electrocatalyst of the present disclosure has lower over-potential in alkaline seawater, and has out-standing stability in alkaline seawater. The nanocomposite electrocatalyst of the present disclosure is cost-effective, and efficient in hydrogen generation. The electrode prepared by using the nanocomposite electrocatalyst is easy to fabricate. The process for the preparation of nanocomposite electrocatalyst is simple, and utilizes low-cost material and thereby reduces the overall expense and improves the economic feasibility of hydrogen production.

No. of Pages : 34 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128751 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-FUNCTIONAL ELECTRIC CHARGING SYSTEM AND METHOD FOR UNIFIED ENERGY MANAGEMENT THEREOF

(51) International classification	:H02J 7/35, H02J 3/32, H02J 3/38, B60L 53/51, H02J 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)AMRITA VISHWA VIDYAPEETHAM
Address of Applicant :Coimbatore Campus, Coimbatore- 641 112, Tamil Nadu, India. Tamil Nadu India
(72)**Name of Inventor :**
1)KUMAR, Anu Gopalakrishna
2)PISHARODY, Sindhu Madassery Pisharam Raghava
3)RAJENDRAN, Mohanrajan Sengodagounder
4)PARAMESWARAN, Sivraj

(57) Abstract :
The present invention discloses a multi-functional electric charging system 100 and a method 300 for unified energy management. The system 100 comprises a plurality of solar photovoltaic generation units 110 configured to generate electrical energy from renewable energy sources. A battery energy storage system 120 is coupled to the solar PV generation units 110 and a utility grid interface 130, the utility grid interface 130 being configured for bidirectional energy import and export. A plurality of service units 150 are each selectively operable in at least one of an EV charging mode or a battery swapping mode. A multi-functional convergence layer 140 logically integrates, within a shared physical infrastructure, an EV service module 141, an EV charging station service module 142, a battery swapping service module 143, a renewable energy service module 144, and a battery energy storage service module 145. A central control unit (CCU) 160 is communicatively coupled with the solar PV generation units 110, the BESS 120, the plurality of service units 150, and the utility grid interface 130, and is configured to execute a unified energy management control process that dynamically ranks and dispatches energy sources and energy sinks across a plurality of operational services in real time. Figure 1 is the representative figure.

No. of Pages : 40 No. of Claims : 17

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202641128759 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : VIBRASENSE: AI-Enabled Adaptive Vibration Isolation and Predictive Structural Protection System

(51) International classification	:F16F 15/02, F16F 15/00, F16F 15/03, G01P 15/00, F16F 7/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Sathishkumar Natesan

Address of Applicant :3-91/3-89a, Pallakadu, Akkaraipatti (Post), Mallasamudram (Via), Rasipuram (Taluk), Namakkal (District) Tamil Nadu India

2)RUDRESH RAJ

3)FRANCIS G

4)UJESH MANI R

5)SANTHOSH R

6)SABARISHWAR.A

7)SANTHOSH SL

8)NITHEN ESWAR.S

9)DARIO MASCARENHAS

10)VEDHISH B

(72)**Name of Inventor :**

1)RUDRESH RAJ

2)FRANCIS G

3)UJESH MANI R

4)SANTHOSH R

5)SABARISHWAR.A

6)SANTHOSH SL

7)NITHEN ESWAR.S

8)DARIO MASCARENHAS

9)VEDHISH B

(57) Abstract :

VibraSense is a proposed modular vibration-control assembly that combines tri-axial accelerometers, displacement sensing, operating-speed input, adjustable-stiffness isolation elements, a bounded electromechanical adjustment unit, and a supervisory controller. It seeks to detect harmful vibration patterns, distinguish transient excitation from persistent resonance, and adjust an isolation setting within verified mechanical limits. The system records before-and-after vibration response, prevents adjustment when measurement confidence is inadequate, and requests maintenance when a safe corrective response cannot be demonstrated. Performance and safety benefits remain subject to testing.

No. of Pages : 7 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128762 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYNCHRONIZED MULTI-STREAM 32-BIT HARDWARE-ALIGNED LINGUISTIC TENSOR ENCODING AND FACTORIZED ADDITIVE MEMORY ARCHITECTURE

(51) International classification	:G06F 9/30, G06F 9/38, G06F 9/45, G06F 8/41, G06F 9/308
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Tirja Consulting LLP
Address of Applicant :HD 109, WeWork Prestige Cube, Koramangala, Site No. 26, Laskar Hosur Road, Adugodi, Bangalore, Bengaluru Urban, Bengaluru Urban, Karnataka Bangalore Karnataka India

(72)**Name of Inventor :**
1)Shreos RoyChowdhury

(57) Abstract :
A hardware-software co-designed computing architecture for high-throughput processing of inflected natural languages on microprocessors and GPU accelerators is disclosed. The architecture packs seven grammatical and semantic dimensions into a single atomic 32-bit hardware register (Token32) or an aligned 6-stream 192-bit tensor (ATR1 v2). A factorized additive embedding memory system decomposes the 232 vocabulary space into factor tables of dimension dmodel summed elementwise at input, reducing physical accelerator parameter-storage memory consumption from over 4 Terabytes to approximately 218.43 Megabytes for input factor tables and under 275 Megabytes of static parameter storage for the complete disclosed model architecture while executing parallel linear output projections from the hidden representation to factor sub-vocabularies. Nominal compound constituent positions are actively projected as input embeddings into causal self-attention layers, and multi-task loss gradients are dynamically masked for unannotated task heads based on dataset provenance tags.

No. of Pages : 38 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128763 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A HIGH-THROUGHPUT GPU WARP-SHUFFLE COMPACTION KERNEL AND DYNAMIC IN-REGISTER ATTENTION MASKING SYSTEM

(51) International classification	:G06F 9/38, G06F 9/30, G06T 1/20, G06F 9/52, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Tirja Consulting LLP
Address of Applicant :HD 109, WeWork Prestige Cube, Koramangala, Site No. 26, Laskar Hosur Road, Adugodi, Bangalore, Bengaluru Urban, Bengaluru Urban, Karnataka Bangalore Karnataka India

(72)**Name of Inventor :**
1)Shreos RoyChowdhury

(57) Abstract :
A high-throughput GPU architecture for massive parallel signature search and constrained neural generation on streaming multiprocessors is disclosed. The kernel performs 128-bit vectorized reads, hardware Bloom filter intersection, and intra-warp ballot synchronization electing, conditioned on a non-empty qualification mask, a single thread per warp to execute a single atomic write, distributing write offsets via register shuffle instructions and bypassing atomic transactions for warps with zero qualifying threads, substantially reducing global memory atomic contention and serialization across GPU architectures. In-register dynamic sparse attention masking and in-loop memory logit clamping suppress syntactically invalid operations directly in GPU memory prior to softmax sampling.

No. of Pages : 39 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128764 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR PERMUTATION-INVARIANT GRAPH RETRIEVAL OF SANSKRIT USING PA?INIAN KARAKA LABELLED INJECTIVE-ARC MATCHING

(51) International classification	:G06F 17/30, G06F 17/27, G06F 16/901, G06F 16/2455, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Tirja Consulting LLP
Address of Applicant :HD 109, WeWork Prestige Cube, Koramangala, Site No. 26, Laskar Hosur Road, Adugodi, Bangalore, Bengaluru Urban, Bengaluru Urban, Karnataka Bangalore Karnataka India
(72)**Name of Inventor :**
1)Shreos RoyChowdhury

(57) Abstract :
A deterministic information retrieval system for Sanskrit is disclosed. The system includes a Pāṇinian Kāraka labelled injective-arc matching engine (GraphMatcher) that constructs directed dependency graphs from queries via an order-independent morphological parser and executes injective arc matching against precompiled verse graph signatures (TopicalPack), achieving mathematical permutation invariance (Score(π(Q),S)=Score(Q,S)) (108) across all surface word order permutations yielding unambiguous morphological and contextual resolution, and executing two-stage candidate-bounded retrieval across canonical classical corpora via structured signature repositories (TopicalPack). A multi-byte UTF-8 Indic fuzzy collision defense mechanism (OntologyManager) restricts approximate Levenshtein edit distance strictly to normalized pure-ASCII Romanized transliterations while enforcing exact byte equality on native Devanāgarī, preventing shared prefix bytes (\xE0\xA4) from producing false-positive matches in byte-level buffers.

No. of Pages : 38 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128776 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-BASED AUTOMATED KNOWLEDGE REPRESENTATION AND INSIGHT GENERATION SYSTEM

(51) International classification	:G06N 5/02, G06N 20/00, G06N 5/04, G06F 16/36, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)LINGALA THIRUPATHI
2)KATTA KARTHIK REDDY
3)JAGIRI SAI KIRAN GOUD
4)SANDALA DATTA SAI
5)SANGU MANIKESH

(57) Abstract :
Disclosed herein is an AI-based automated knowledge representation and insight generation system (100) that comprises a Dataset Input Interface (102), a Data Storage Unit (104), a Processing Unit (106), and a User Interface (126). The Processing Unit (106) comprises a Data Pre-processing Module (108), a Feature Analysis Module (110), a Statistical Analysis Module (112), an Outlier Detection Module (114), a Knowledge Representation Module (116), an Insight Generation Module (118), an Insight Prioritization Module (120), a Visualization Module (122), and a Report Generation Module (124). The system (100) receives and pre-processes structured datasets, analyzes attribute characteristics, performs statistical and machine-learning operations, and identifies anomalous observations. The Knowledge Representation Module (116) associates analytical findings with corresponding dataset attributes, pre-processing operations, and analytical methods. The Insight Generation Module (118) generates interpretable insights, while the Insight Prioritization Module (120) ranks the insights.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128777 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTOMATED INGREDIENT SAFETY ASSESSMENT SYSTEM

(51) International classification	:G16C 20/20, G16C 20/70, G06F 17/30, G06Q 50/04, G16C 20/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)CHENCHU SWETHA
2)GOLI VAISHNAVI
3)GEALLAPALLY RITHWIK KUMAR
4)YERUKALI ADITHYA

(57) Abstract :
Disclosed herein is an automated ingredient safety assessment system (100) that comprises an input interface (102), a chemical database interface (104), a processing unit (106), an output interface (122), a memory unit (124). The processing unit (106) comprises a pre-processing module (108), a text extraction module (110), an ingredient identification module (112), a chemical information retrieval module (114), a toxicity analysis module (116), a safety assessment module (118), a recommendation generation module (120). The system (100) is configured to receive product label images or manually entered ingredient information, extract ingredient text, identify and normalize ingredient names, retrieve corresponding chemical information, analyze available toxicity-related data. The system (100) further generates a numerical safety score based on identified risk indicators and produces associated safety warnings and precautionary recommendations. The output interface (122) presents the assessment results and facilitates generation of a savable report, thereby improving accessibility and interpretation of ingredient safety information.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128778 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A THREAD AI-BASED CUSTOM TAILORING SYSTEM

(51) International classification	:G06Q 30/06, G06F 17/50, G06Q 10/08, A41H 3/00, G06Q 50/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)KALAIARASI MANI
2)P. SINDHURA
3)RAJANALA PURUSHOTHAM

(57) Abstract :
Disclosed herein is a Thread AI-based custom tailoring system (100) that comprises a User Interface (102), an Input Acquisition Unit (104), a Communication Interface (106), a Processing Unit (108), a Memory Unit (130), a Database Unit (132), and an Authentication Unit (134). The Processing Unit (108) comprises a Pre-processing Module (110), a Design Generation Module (112), a Design Processing Module (114), a Visualization Module (116), a Measurement Processing Module (118), a Matching Module (120), a Recommendation Module (122), an Order Management Module (124), a Validation Module (126), and a Notification Module (128). The system (100) generates customized garment designs using generative artificial intelligence, provides interactive three-dimensional garment previews on virtual mannequins, maintains reusable digital measurement profiles, and ranks tailoring service providers based on pricing, trust ratings, geographical distance, and delivery speed using multi-criteria evaluation.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128779 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-POWERED MULTI-CLASS SKIN DISEASE CLASSIFICATION SYSTEM

(51) International classification	:G06V 10/774, A61B 5/00, G06V 10/764, G16H 50/20, G06N 20/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA 501301, INDIA Telangana India

(72)**Name of Inventor :**
1)MRS.CHENCHU SWETHA
2)NANDYALA JASHWANTH REDDY
3)DOMA SAMANVITHA
4)BANOTH HEMANTH NAYAK

(57) Abstract :
Disclosed herein is an AI-powered multi-class skin disease classification system (100) that comprises an image acquisition unit (102), an input interface (104), a memory unit (106), a processing unit (108), a communication interface (130), and a display unit (132). The processing unit (108) comprises an image pre-processing module (110), a data augmentation and class balancing module (112), a feature extraction module (114), a transfer learning module (116), a classification module (118), an ensemble prediction module (120), an explainability module (122), a training and optimization module (124), a performance evaluation module (126), and a result generation module (128). The system (100) utilizes MobileNetV2 and ResNet50 deep learning architectures to classify dermoscopic skin lesion images into predefined categories through ensemble prediction. A Grad-CAM-based heatmap identifies image regions contributing to the predicted classification, facilitating interpretable preliminary skin lesion screening and supporting teledermatology applications.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128780 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A REAL-TIME EDGE-BASED DUAL-MODEL VISION SYSTEM

(51) International classification	:G08G 1/01, G06K 9/00, G08G 1/017, G06V 20/58, G06V 20/56
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)K. SIVA KUMAR GOWDA
2)KOPPULA MAITHRI
3)PURGULA PAVAN KUMAR
4)V PRANEETH

(57) Abstract :
Disclosed herein is a real-time edge-based dual-model vision system (100) that comprises a visual data acquisition unit (102), a video frame acquisition unit (104), a processing unit (106), a memory unit (124), and a communication interface (126). The processing unit (106) comprises a pre-processing module (108), a vehicle detection module (110), a road-hazard detection module (112), a detection filtering module (114), a hazard validation module (116), a visualization module (118), an alert generation module (120), and a performance monitoring module (122). The system (100) processes live road video using separate detection models to identify vehicles and road-surface hazards, including potholes. The detection outputs are filtered and validated using predefined confidence, spatial, and size-based criteria. The system (100) provides real-time visual representations and conditional audio alerts while performing local inference to reduce processing latency and dependence on continuous cloud connectivity.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128781 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-POWERED INTELLIGENT TASK ALLOCATION AND WORKLOAD BALANCING SYSTEM

(51) International classification	:G06Q 10/06, G06Q 10/10, G06Q 10/00, G06F 9/50, G06F 9/48
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)K. SIVA KUMAR GOWDA
2)MOHAMMED ISMAIL AHAMED
3)KOTLA SAI PRAKASH RAO
4)G. SAI AKSHITH
5)C SWETHA

(57) Abstract :
Disclosed herein is an AI-powered intelligent task allocation and workload balancing system (100) that comprises a User Interface (102), an Input Interface (104), an Employee Profile Database (106), a Task Management Database (108), a Processing Unit (110), a Communication Interface (130), and a Memory Unit (132). The Processing Unit (110) comprises a Pre-processing Module (112), a Skill Extraction Module (114), an Effort Prediction Module (116), an Employee Matching Module (118), a Workload Balancing Module (120), a Fairness-Aware Recommendation Module (122), a Task Decomposition Module (124), an Allocation and Scheduling Module (126), and a Monitoring and Feedback Module (128). The system processes task descriptions to extract skill requirements, predict task effort, evaluate employee suitability, and assess workload capacity. The system generates fairness-aware allocation recommendations, facilitates task decomposition and managerial approval, and updates assignment and workload records based on task progress, thereby supporting capacity-aware scheduling and balanced task distribution.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128790 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADJUSTABLE PEDIATRIC TOOTH EXTRACTION FORCEPS APPARATUS

(51) International classification	:A61C 3/14, A61B 17/28, A61C 3/00, A61C 3/10, A61B 1/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)COORG INSTITUTE OF DENTAL SCIENCES
Address of Applicant :K. K. CAMPUS, MAGGULA VILLAGE, VIRAJPET, KODAGU (COORG) DISTRICT, KARNATAKA - 571218, INDIA Karnataka India
(72)**Name of Inventor :**
1)DR. PRIYA NAGAR
2)DR. BASAVARAJ ARAHUNASI

(57) Abstract :
Disclosed herein is an adjustable pediatric tooth extraction forceps apparatus (100), comprising, a handle assembly (102) provides a grip for holding and operating, a head portion (104) positioned at a distal end of the handle assembly (102) and is selectively oriented at a plurality of angular positions, an adjustable hinge (106) coupling the handle assembly (102) with the head portion (104) and permits selective angular adjustment of the head portion (104) relative to the handle assembly (102), a locking mechanism (108) associated with the adjustable hinge (106) and selectively secure the hear portion (104) at a selected angular position, a release button (110) operatively associated with the locking mechanism (108) and selectively engage and disengage the locking mechanism (108) to permit angular adjustment of the hear portion (104), a beak interface (112) at the head portion (104) and removably receive a plurality of interchangeable beak members having different tooth-engaging configurations.

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128791 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A DENTAL VISUALISATION DEVICE FOR OPERATIONAL ASSISTANCE AND METHOD THEREOF

(51) International classification	:A61C 19/04, A61B 1/06, A61B 1/247, A61C 1/08, A61B 1/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)COORG INSTITUTE OF DENTAL SCIENCES
Address of Applicant :K. K. CAMPUS, MAGGULA VILLAGE, VIRAJPET, KODAGU (COORG) DISTRICT, KARNATAKA - 571218, INDIA Karnataka India
(72)**Name of Inventor :**
1)DR. RADHIKA B
2)DR. ABHINIT SRIVATSAV
3)DR. AMIT WALVEKAR
4)DR. RASHMI S PATTANSHETTY
5)DR. SHAIK AMRULLA

(57) Abstract :
Disclosed herein is a dental visualization device (100) to provide guidance and operational assistance during a dental procedure, that comprises a handle assembly (102) to support components of the device (100) within an oral cavity, a mirror head (104) comprising a reflective surface (106), to provide an indirect field of view of a dental treatment region, an illumination assembly (108) to illuminate the dental treatment region, a camera assembly (110) to capture visual data, a laser reference assembly (112) to project a reference line onto the dental treatment region, a temperature sensing assembly (114) to sense a temperature of the reflective surface (106) and generate corresponding temperature data, a suction assembly (118) to generate an adjustable suction flow, a microcontroller (122) further comprising a data input module (124), a visual data processing module (128), an angular analysis module (130), a suction control module (134) and an output module (136).

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128793 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A METEOROLOGICAL DROUGHT-RISK ESTIMATION SYSTEM AND METHOD THEREOF

(51) International classification	:G01W 1/10, G06Q 50/26, G01W 1/02, G01W 1/00, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TIRUPATHI RAO PADI
Address of Applicant :DEPARTMENT OF STATISTICS, PONDICHERRY UNIVERSITY (A CENTRAL UNIVERSITY), PUDUCHERRY – 605014, INDIA Pondicherry India
2)MUHAMMED RAIS P K
(72)**Name of Inventor :**
1)TIRUPATHI RAO PADI
2)MUHAMMED RAIS P K

(57) Abstract :
A meteorological drought-risk estimation system (100) and method (200) are disclosed. An input device (102) acquires rainfall and temperature data, fills missing observations by imputation procedures selected by gap length and checks the data against physical limits. A processing assembly (110) calculates potential evapotranspiration, a moisture balance and a standardized moisture-balance index, classifies successive periods into Wet, Dry and absorbing Drought states, and forms a discrete-time transition probability matrix. A generator matrix estimation module (120) converts it into a continuous-time generator matrix, detects non-embeddability from a negative off-diagonal rate and corrects it. A three-state stochastic compartmental model with Wet and Dry compartments holding moisture-balance units takes transition and absorption rates from the valid generator matrix and immigration rates from observed moisture-balance growth. Exact mean, variance, covariance, skewness and kurtosis are calculated without moment-closure approximation and presented through an output interface (128).

No. of Pages : 66 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128795 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Herbal Medicated Mold for Steam-Based Decongestant Delivery

(51) International classification	:A61K 36/534, A61P 11/02, A61K 9/00, A61K 36/00, A61K 36/484	(71)**Name of Applicant :** **1)Dr. KVN Chenchu Lakshmi** Address of Applicant :Associate Professor, K L College Of Pharmacy, Koneru Lakshmaiah Education Foundation, Green Fields, Vaddeswaram, Guntur, Andhra Pradesh, India-522302 Guntur Andhra Pradesh India **2)KONERU LAKSHMAIAH EDUCATION FOUNDATION** (72)**Name of Inventor :** **1)KNV CHENCHU LAKSHMI** **2)JESTINA HEBRON** **3)BALUSUPATI SOWMYA** **4)SAJIDA FARHATH** **5)GUDURU BINDHU SRI** **6)M.V.N.A.SAI NEHA** **7)SOMANALLI ASHOK SOWMYA** **8)SAMUEL JYOTHILAL TANGUDU** **9)J. UDAY KIRAN REDDY**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT [0016] The present invention provides a herbal vaporising decongestant mold designed for temporary relief of nasal congestion and cold-related stuffiness. The mold is formed as a stable, layered solid unit that combines organic soy wax as the structural base, a eutectic mixture of camphor crystals and peppermint crystals as the active medicinal layer, and a dry top layer of citric acid mixed with sodium bicarbonate as the effervescent trigger. When the finished mold is placed in hot water, the outermost layer reacts to release carbon dioxide gas, producing controlled bubbling that opens the structure and allows the heat to convert the eutectic mixture into soothing aromatic vapours. These vapours disperse into the surrounding air and create a cooling sensation that helps clear the sensation of a blocked nose. The soy-wax foundation keeps the active ingredients protected and stable during storage while ensuring the product remains easy to handle and fully portable. Because the formulation relies on natural materials rather than synthetic carriers or petroleum bases, it is suitable for use across a wide age range, from infants to elderly persons, and presents a lower risk of irritation to sensitive skin or nasal passages. The layered construction and simple activation method distinguish the product from conventional vapour rubs, inhaler capsules, or gel formulations, offering a convenient, single-use alternative that requires only ordinary hot water and delivers consistent vapour release without specialised equipment. The invention thereby supplies an economical, environmentally considerate, and user-friendly means of obtaining short-term symptomatic comfort from nasal blockage.

No. of Pages : 7 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128809 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ADAPTIVE FUEL CONTROL OF A MOTORCYCLE

(51) International classification	:F02D 41/14, F02D 41/24, F02D 41/02, F02D 45/00, F02D 41/04	(71)**Name of Applicant :** **1)THIAGARAJAR COLLEGE OF ENGINEERING** Address of Applicant :THIRUPARANKUNDRAM, MADURAI, TAMIL NADU-625015 Madurai Tamil Nadu India (72)**Name of Inventor :** **1)P.S. MANOHARAN** **2)L. ASHOK KUMAR** **3)GOKUL SRINATH M** **4)SUNDHAR KARTHICK S**
(31) Priority Document No	:NA	**5)PRITHVI RAJ V**
(32) Priority Date	:NA	**6)MOHAMED FAROOK S**
(33) Name of priority country	:NA	**7)YOGESH T R**
(86) International Application No Filing Date	: :01/01/1900	**8)VASANTHAKUMAR S** **9)GURUPRAKASH R**
(87) International Publication No	: NA	**10)DHARANI N**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A fuel trim adaptation system (100) for a motorcycle engine (101) includes an electronic control unit (110) having a write-protected base calibration memory (112), an active trim memory (116), and a candidate trim memory (118) storing a candidate trim value received from a remote server (160). A zone classifier (120) assigns a present operating zone. In a closed-loop operating zone, a channel mapper (122) applies a trim value as a lambda bias scaling rich-to-lean and lean-to-rich switching delays of a lambda controller (108). An interleaved trial scheduler (124) alternates an active trim value and the candidate trim value over episodes in a balanced pseudo-random sequence. A zone ledger (126) accumulates fuel mass, distance, and residence time per operating bin. A trial evaluator (128) commits the candidate trim value upon a stratified fuel improvement ratio exceeding a commit threshold. Outcome records train a random forest outcome model (164) at the remote server (160).

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128814 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Enabled IoT Smart Pet Feeder with Behaviour-Adaptive Sound Communication and Automated Positive-Reinforcement Treat Dispensing

(51) International classification	:A01K 5/02, A01K 29/00, A01K 5/01, A01K 5/00, A01K 11/00	(71)**Name of Applicant :** **1)Akula Manoj Kumar** Address of Applicant :12-13-168, stno-17, tarnaka, secunderabad, telangana. pin : 500017 Telangana India **2)Kamigari Mounika** **3)Kosaraju Suneetha Krishna** **4)Dr.Ch.Naga Priyanka** **5)Gudimella Sai Ram** **6)B Naresh** **7)Malyala Prabhakar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	(72)**Name of Inventor :** **1)Kamigari Mounika** **2)Kosaraju Suneetha Krishna** **3)Dr.Ch.Naga Priyanka** **4)Gudimella Sai Ram** **5)B Naresh** **6)Malyala Prabhakar**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to an AI-Enabled IoT Smart Pet Feeder with Behaviour-Adaptive Sound Communication and Automated Positive-Reinforcement Treat Dispensing. The system integrates automated pet feeding, sensor-assisted behavioural observation, interactive audio communication, and controlled reward delivery within a unified electronic platform. The system comprises a food storage unit, a food-dispensing mechanism, a separate treat storage and dispensing assembly, sensors, an embedded processing controller, an audio output module, and an Internet of Things (IoT) communication interface. The controller processes sensor inputs and predefined operating parameters to coordinate feeding operations, generate appropriate auditory signals, and initiate treat dispensing according to configured reinforcement conditions. Artificial intelligence algorithms may be employed to analyse relevant behavioural indicators and select suitable system responses. The audio module provides feeding announcements, prerecorded voice prompts, or other configured sounds associated with detected conditions and training activities. The treat-dispensing mechanism delivers controlled quantities of treats subject to configurable limits and dispensing intervals. The IoT interface enables remote configuration of feeding schedules, food portions, sound preferences, and training parameters through a compatible connected device. The system may further incorporate food-level monitoring, dispensing verification, fault detection, and operational notifications. By integrating scheduled feeding, responsive sound communication, and controlled positive reinforcement, the invention provides a configurable technological platform for improving feeding consistency, supporting behavioural training, and facilitating intelligent pet-care management in domestic environments.

No. of Pages : 12 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128815 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN IN-SITU PROCESS OF RECOVERING LITHIUM CARBONATE FROM SPENT LITHIUM-ION BATTERIES AND THE PRODUCT THEREOF

(51) International classification	:C01D 15/08, C22B 26/12, H01M 10/54, H01M 10/0525, C01D 15/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Technology, Tiruchirappalli
Address of Applicant :National Institute of Technology, Tiruchirappalli – 620015, Tamil Nadu, India Tamil Nadu India
(72)**Name of Inventor :**
1)VADDEPALLI, Akhil
2)PUPPALA, Laxman Mani Kanta
3)KATAKAM, Siva Prasad
4)SHANMUGAM, Vasu

(57) Abstract :
The present disclosure provides a process for recovering lithium carbonate from a spent lithium-ion battery. A cathode composite material comprising a lithium transition metal oxide, a binder and embedded carbon black is separated from the cathode current collector. The composite material is heated under an inert atmosphere to effect in-situ carbothermal reduction, wherein the embedded carbon black is the major reducing agent with minor support from binder. The resulting reduced product comprises lithium carbonate and one or more water-insoluble transition metal species. Contacting the reduced product with water dissolves the lithium carbonate. An aqueous solution is separated from a solid residue comprising the water-insoluble transition metal species and evaporated to collect lithium carbonate. The process permits aqueous separation of lithium carbonate following reduction using carbon black present in the cathode composite material.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128818 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FLOWGUARD NEXUS: ADAPTIVE FOULING INTELLIGENCE AND CONSTRAINT-BASED CLEANING CONTROL FOR INDUSTRIAL HEAT EXCHANGERS

(51) International classification	:F28G 9/00, F28G 13/00, F28G 15/00, F28F 27/00, G05B 23/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sathishkumar Natesan
Address of Applicant :3-91/3-89a, Pallakadu, Akkaraipatti (Post), Mallasamudram (Via), Rasipuram (Taluk), Namakkal (District) Tamil Nadu India
2)SANYU XAVIER SATHISH XAVIER
3)SHANMUGAPRIYAN P
4)STAPHANSH SURESH S
(72)**Name of Inventor :**
1)SANYU XAVIER SATHISH XAVIER
2)SHANMUGAPRIYAN P
3)STAPHANSH SURESH S

(57) Abstract :
FlowGuard Nexus is a proposed supervisory system for industrial heat exchangers that combines temperature, pressure, flow and energy-use measurements to estimate performance drift consistent with fouling or flow restriction. Rather than treating any single sensor reading as proof of fouling, the system compares operating data with a context-adjusted baseline and reports an estimated condition with a confidence level. Where a site has an approved cleaning or operating-adjustment method, the controller may recommend or initiate a bounded action only after checking process permissives, equipment limits and operator authorization. It then evaluates the response against comparable operating conditions, records the outcome and escalates cases where the result is inconclusive or unsafe. The intended contribution is the integration of context-aware fouling estimation, confidence-gated action, independent interlocks and post-action verification. Performance, savings and reliability benefits are proposed hypotheses pending testing.

No. of Pages : 9 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128825 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Evolution of Digital Dashboard Indicators for Precise Fuel Volume Measurement in Passenger Vehicles

(51) International classification	:G01F 23/22, G01F 23/00, G01F 23/296, G01F 23/24, G01F 23/284	(71)**Name of Applicant :** **1)Sunil Manjeri** Address of Applicant :Kino B #1115 Brigade Orchards Near Shell Petrol Bank Devanahalli Bangalore 562110 Karnataka India (72)**Name of Inventor :** **1)Sunil Manjeri**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT [505] The present invention introduces an advanced digital dashboard fuel indicator system that integrates tank-geometry calibration, multi-sensor compensation, and adaptive signal filtering for precise determination and display of the volume of fuel remaining in a passenger vehicle, facilitating real-time volume computation, stable numerical presentation, and reliable low-fuel notification while maintaining seamless integration with existing vehicle electronic architectures and instrument clusters. [510] The comprehensive measurement framework employs a level sensing element, a tank-attitude sensing element, a temperature sensing element, and an embedded processing unit that converts raw sensor output into fuel volume using a stored tank-characterisation table, thereby overcoming the non-linear relationship between fuel height and fuel volume that is inherent in saddle-shaped, baffled, and irregular fuel tanks. [515] The integrated methodology combines slosh-rejection filtering with inclination correction and thermal-expansion compensation, converting noisy and motion-affected level readings into a stable volume estimate, and cross-validating the estimate against cumulative injector-based consumption so that the displayed value remains consistent during acceleration, braking, cornering, and travel on slopes. [520] The novel responsive architecture features an adaptive learning mechanism that refines the tank-characterisation table and sensor offset at each detected refuelling event, enabling the system to compensate for sensor ageing, tolerance variation, and tank deformation, while a display-conditioning module applies direction-aware damping and hysteresis so that the indicated quantity decreases smoothly and does not fluctuate. [525] The innovative design incorporates a diagnostic mechanism for detection of open-circuit, short-circuit, stuck-sensor, and implausible-signal conditions, and a derived-information module that calculates remaining range from the measured volume and a learned consumption rate, thereby providing the driver with accurate, trustworthy information in litres, in percentage, and in distance. [530] Implementation methodology emphasises low-cost realisation, in which the existing resistive or capacitive level sender is retained, in which computation is performed by the instrument cluster or an engine control unit, and in which calibration data is stored in non-volatile memory, so that the evolution from an analogue needle indicator to a precise digital volume indicator requires minimal additional hardware. [535] The system demonstrates adaptability across passenger vehicles of different tank shapes and capacities, including petrol, diesel, and hybrid vehicles, and maintains consistent measurement performance under diverse conditions such as hot and cold climates, hilly terrain, rough roads, and partial refuelling. [540] The developed framework enables dependable fuel-volume indication through a streamlined digital system, providing significant advantages over conventional analogue gauges and simple segmented bar displays through improved resolution, reduced fluctuation, accurate reserve warning, and an auditable record of sensor condition and calibration status.

No. of Pages : 15 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128828 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LEADERSHIP-DRIVEN CORPORATE SOCIAL RESPONSIBILITY GOVERNANCE FRAMEWORK FOR ENVIRONMENTAL SUSTAINABILITY IN IT AND ITES ORGANIZATIONS

(51) International classification	:G06Q 10/06, G06Q 10/10, G06Q 50/26, G06Q 10/00, G06F 16/248
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)L N Sivaprasadu Majji
2)Dr. Gurunadhan Goli

(57) Abstract :
The present invention relates to environmental evidence processing for corporate social responsibility governance in IT and ITES organizations. Acquisition interfaces receive environmental observations with source identifiers, measurement intervals, quantities, units, and versions. A boundary registry associates observations with defined projects. A reconciliation engine detects duplicate or overlapping evidence before approved aggregation. A calculation engine computes environmental results using versioned observations, factors, and calculation rules. An evidence manifest references the results and supporting versions, and leadership approval is bound to a digest of the manifest. A dependency index identifies results affected by corrected observations or changed calculation inputs. A report-export service withholds export when the approved digest differs from the current manifest or mandatory evidence conditions remain unresolved. The system supports traceable environmental reporting and preserves earlier approved states alongside subsequent corrections.

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128829 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEMS AND METHODS FOR DETERMINING CLEARANCE VALUE BETWEEN OVERHEAD CONDUCTOR AND VEGETATION IN TERRESTRIAL ENVIRONMENTS

(51) International classification	:G06T 17/05, G06N 20/00, G06V 20/13, G06T 7/00, G06V 10/764	(71)**Name of Applicant :** **1)Indian Space Research Organisation** Address of Applicant :Department of Space, Antariksh Bhavan, New BEL Road, Bangalore 560094, Karnataka, India Bangalore Karnataka India (72)**Name of Inventor :** **1)SINGHAL, Jayant** **2)SARAN, Sameer** **3)SRIVASTAV, Sushil Kumar** **4)CHAUHAN, Prakash** **5)KALORIA, Mahendra Kumar** **6)GARG, Uday Kumar** **7)CHAUHAN, Pradeep Singh** **8)SEERVI, Pradeep** **9)POPLY, Hitesh** **10)MISHRA, Anurag**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed herein is a method (1000) for determining a clearance value between an overhead conductor (108) and vegetation (114). The method (1000) includes receiving (1002) a monocular remote sensing image (502) of a terrestrial environment (116) from a remote sensing platform (104). A machine learning model (212) generates (1004) vegetation canopy height data (232) from the image (502). Ground elevation data is extracted (1006) from a terrain elevation model. Vegetation surface elevation data (602) is determined (1008) from the vegetation canopy height data (232) and the ground elevation data. A conductor elevation profile (402) is determined (1010) for each span (112) between consecutive towers (110). Local clearance values between the overhead conductor (108) and the vegetation (114) are determined (1012). The clearance value of each span (112) is determined (1014) as a minimum of the local clearance values.

No. of Pages : 49 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128833 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED INTRUSION PREVENTION FOR IoT DEVICES

(51) International classification	:H04L 29/06, G06N 20/00, G06F 21/55, H04L 12/24, H04L 12/26	(71)**Name of Applicant :** **1)S Narayanan** Address of Applicant :Assistant Professor Artificial Intelligence and Data Science Srm Trp Engineering College Srm Nagar, Mannachanallur (Tk), Tiruchirappalli, Tamil Nadu -621105 Tamil Nadu India **2)S P Kirthivasan** **3)P. Anitha** **4)R K Ananthi** **5)Dr.S.Sureshkumar** **6)Arunraj R** **7)S.Ramarajan**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)S Narayanan** **2)S P Kirthivasan** **3)P. Anitha** **4)R K Ananthi** **5)Dr.S.Sureshkumar** **6)Arunraj R** **7)S.Ramarajan**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

This invention describes an intrusion prevention system for Internet of Things (IoT) devices that uses artificial intelligence. The system watches device behaviour and network traffic, gathers the relevant data, cleans it, and pulls out useful features. An AI/ML detection engine then examines those features, spots unusual patterns, sorts suspected intrusions into categories, and judges how risky each one is. Depending on that risk level, the system acts on its own: it can block suspicious traffic, end unauthorised sessions, isolate a compromised device, limit which destinations a device may reach, or throttle traffic. An administrator is alerted to each event and to the action taken. A security event database keeps the network events, device behaviour, classifications, risk levels and mitigation records. A model update unit uses validated past data to refine the detection model as network conditions change. Together these parts give IoT networks continuous monitoring, AI-assisted detection, risk-based prevention and security that adapts over time. The system can run on an IoT gateway, an edge device, a server, or a mix of these, as the deployment requires.

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128841 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Machine Learning Based Intelligent Home Automation System for Predictive Appliance Control and Energy Optimization

(51) International classification	:G05B 15/02, G05B 19/418, H04L 12/28, G06N 20/00, G06Q 50/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. K. Manoharan
Address of Applicant :Megarali Street, Vellalore, Coimbatore -641 111. Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.Mohandass M P
2)Dr. Madhavi Jadhav
3)Mr. Ashok Kumar Shah
4)Dr Manoharan K
5)Dr. Shubhalaxmi Mohapatra
6)Mrs. N. Jayapratha
7)Mrs. S. Radhika
8)Dr. Shyam Viththalani
9)Dr. S Vijayakumar
10)Dr. S. Karthik
11)Dr. Prasoon Choudhary

(57) Abstract :
The present invention relates to a machine learning-based intelligent home automation system for predictive appliance control and energy optimization. The system comprises a sensor arrangement, an energy monitoring arrangement, a processing unit, a machine learning-based prediction engine, an appliance control unit, a communication interface, and a user interface. The system collects household operating parameters, appliance status information, and energy consumption data to identify usage patterns and predict future appliance requirements. Based on the predicted requirements, user preferences, occupancy conditions, and energy optimization objectives, the system determines suitable appliance operating schedules and control actions. The appliance control unit executes the selected actions through compatible switching devices or appliance interfaces. The system further provides energy monitoring, consumption reports, predictive insights, and notifications. Historical operating data may be used to update the predictive model and improve adaptation to changing household routines. The invention provides an integrated approach to intelligent appliance management, improved energy efficiency, and reduced unnecessary electricity consumption.

No. of Pages : 14 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128842 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HYBRID ENSEMBLE SYSTEM AND METHOD FOR STUDENT ACADEMIC PERFORMANCE PREDICTION WITH FACTOR-WISE SHAP ATTRIBUTION

(51) International classification	:G06Q 50/20, G06N 20/00, G09B 7/00, G09B 7/02, G06F 16/2457
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHRIST UNIVERSITY
Address of Applicant :Christ University, Dharmaram College Post, Hosur Road, Bengaluru - 560029, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Sumin Samuel Sybol
2)Dr. Shilpa Srivastava

(57) Abstract :
The invention relates to educational predictive computing. A processor executes instructions to validate original-format student records and invoke a single callable prediction object comprising an encoding pipeline and a hybrid model module that averages performance-level probabilities from different trained model types. Ordinary prediction requests and explanation queries use the same object. An explanation engine applies the SHAP permutation method by substituting original student parameter values with reference student record values across parameter orderings. Before evaluation, a record repairer reverses categorical numerical codes into valid categories and rounds rated parameters to whole numbers, enabling whole-pipeline attribution using valid records. Marginal probability contributions are averaged and aggregated into demographic, academic, socioeconomic and behavioural factor contributions separately for each performance level. Reports provide predicted performance levels and parameter-wise and factor-wise explanations for students and institutions.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128864 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Explainable Machine Learning-Based System for Aggressive Social-Media Content Prediction, Adaptive Exposure-Risk Thresholding, and Automated Regulation for Adolescent Psychological Wellness

(51) International classification	:G06N 20/00, G06Q 50/00, G06N 5/04, G06N 5/02, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. V. Sivaraj
Address of Applicant :156, 3rd Floor, Alagu Sanitary Building, Kalidass Road, Ramnagar Tamil Nadu India
2)SASIKALA DHAMODARAN
3)Dr. Jyoti Sharma
4)Dr Gaurav Agarwal
(72)**Name of Inventor :**
1)SASIKALA DHAMODARAN
2)Dr. Jyoti Sharma
3)Dr Gaurav Agarwal

(57) Abstract :
This predicted innovative system review about an artificial-intelligence-based system for discovering, predicting, classifying, and regulating exposure to aggressive or mentally harmful social-media content come across by adolescent users. This process makes use of machine learning based prediction algorithms combined with natural-language/text-derived aspects, social-media exposure features, and adolescent user-precise risk indicators to regulate an aggressive-content prospect and a related exposure-risk score. This research additionally gives an AI-driven adaptive exposure-risk threshold method that is used for classifying content or exposure as low, moderate, or high risk and is dynamically formed corresponding to predicted content harshness, cumulative exposure, exposure frequency, user-risk indicators, and predefined safety constraints, instead of depending exclusively on a static universal threshold. This innovation is envisioned to afford a technically executable method for intelligent organization of aggressive social-media exposure related to adolescent psychological-wellness risks.

No. of Pages : 16 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128867 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SHAPE: Student Harassment Analysis and Psychological Empowerment through Privacy-Aware Web-based Framework

(51) International classification	:G09B 19/00, G09B 7/00, G06Q 50/20, G09B 5/06, G09B 7/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mrs.A.M.SERMAKANI
Address of Applicant :ASSOCIATE PROFESSOR, Department of Information Technology, S.A. Engineering College, Veeraragavapuram, Chennai-77. Chennai Tamil Nadu India
2)Mr.P.SENTHIL
3)Ms.C.N.SABARISRI
4)S.A. ENGINEERING COLLEGE
(72)**Name of Inventor :**
1)Mrs.A.M.SERMAKANI
2)Mr.P.SENTHIL
3)Ms.C.N.SABARISRI

(57) Abstract :
ABSTRACT Student harassment is a serious concern in educational institutions that can affect students' emotional well-being, academic performance, confidence, and sense of safety. However, many incidents remain unreported because students may fear retaliation, social stigma, embarrassment, or negative consequences from revealing their identity. Traditional reporting mechanisms generally require students to disclose their personal details to faculty members, complaint committees, or institutional authorities. This can create hesitation and reduce the willingness of students to seek help. Manual handling of complaints may also result in delays and limited transparency in the response process. The motivation behind Breaking the Silence is to provide a safer and more accessible approach for students to report harassment and obtain appropriate support. The model proposes a web-based support system that prioritizes confidentiality and protects the identity of the student during the reporting process. The system connects students with trained psychologists who can review reported cases and provide professional guidance through confidential communication. Simulated within an NS-3 network environment across 50 nodes over a 1000-time-unit lifespan, the proposed framework achieved an accuracy of 87.5% and a precision of 88.1%, while maintaining a competitive communication round latency of approximately 340 ms. Crucially, SHAPE bridged the utility gap inherent to standard Differentially Private Federated Learning (DP-FL), recovering 8.9% in detection accuracy and 11.9% in precision. The proposed model also incorporates research-based safety concepts, recognizing that conventional reporting and reactive safety mechanisms may not always provide sufficient protection or timely support.

No. of Pages : 7 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128868 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CIRCULARFLOW AI : An Ai-Driven Multi-Agent System For Sustainable Production Orchestration And Zero-Waste Manufacturing

(51) International classification	:G06Q 10/06, G06Q 10/00, G06Q 50/04, G05B 19/418, G06Q 10/08	(71)**Name of Applicant :** **1)Ms.S.JULIA FAITH** Address of Applicant :ASSISTANT PROFESSOR Department of Information Technology, S.A.Engineering College (Autonomous), Poonamallee Avadi road, Veeraragavapuram, Chennai-60077 Chennai Tamil Nadu India **2)Mrs.T.VEENA** **3)Mrs.D.CHITHRA** **4)S.A. ENGINEERING COLLEGE** (72)**Name of Inventor :** **1)Ms.S.JULIA FAITH** **2)Mrs.T.VEENA** **3)Mrs.D.CHITHRA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT CircularFlow AI is an advanced Industry 5.0 Manufacturing Intelligence Platform designed to address production efficiency, machine reliability, operational costs, and environmental sustainability. Traditional manufacturing systems often rely on static scheduling and isolated decision-making, leading to production delays, machine downtime, and material wastage. This project combines Artificial Intelligence, Digital Twin Simulation, and Multi Objective Optimization to support intelligent manufacturing operations. The system utilizes a SimPy-based Discrete-Event Digital Twin to model machine operations, production queues, energy consumption, waste generation, and machine breakdowns. Eight specialized AI agents provide recommendations for production, maintenance, quality, sustainability, inventory, waste, recycling, and cost management. Retrieval-Augmented Generation (RAG) using ChromaDB ensures that recommendations are grounded in relevant manufacturing knowledge and standard operating procedures. The platform integrates NSGA-II and Google OR-Tools CP-SAT to optimize production throughput, energy consumption, waste, and delivery performance. A closed-loop validation engine evaluates proposed plans through simulation, while a Human-in-the-Loop (HITL) interface enables managers to review and approve decisions. By combining multi-agent intelligence, simulation, and optimization, CircularFlow AI aims to improve productivity, reduce environmental impact, and support sustainable circular manufacturing aligned with Industry 5.0 principles.

No. of Pages : 9 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128869 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NEUROSCAN: A DEEP LEARNING BASED SYSTEM AND METHOD FOR EARLY DETECTION AND STAGE CLASSIFICATION OF ALZHEIMER'S DISEASE FROM BRAIN MRI IMAGES

(51) International classification	:G06T 7/00, G06N 3/04, G06N 3/08, G06T 5/00, G01R 33/56
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mrs.K.VALARMATHI
Address of Applicant :ASSISTANT PROFESSOR, Department of Information Technology, S.A. Engineering College, Veeraragavapuram, Chennai-77. Chennai Tamil Nadu India
2)Mrs.A.SEETHA
3)Mrs.J.SUTHA MERLIN
4)S.A. ENGINEERING COLLEGE
(72)**Name of Inventor :**
1)Mrs.K.VALARMATHI
2)Mrs.A.SEETHA
3)Mrs.J.SUTHA MERLIN

(57) Abstract :
ABSTRACT The present invention discloses NeuroScan, an automated system and method for the early detection and stage classification of Alzheimer's Disease from brain MRI images. An uploaded MRI image is validated, resized to 224 × 224 pixels, de-noised, segmented and normalised, and is then classified by a compact custom Convolutional Neural Network having three convolution blocks of 16, 32 and 64 filters, dropout-regularised fully connected layers and a SoftMax output. The system reports the dementia stage (Non-Demented, Very Mildly, Mildly or Moderately Demented), a confidence score and the probability of each stage. Trained from scratch on about 59,000 MRI images, the network achieved 97.60% accuracy, 99.26% AUC, 97% recall and a loss of 0.091, clearly outperforming ResNet101, DenseNet121 and VGG16. The classifier is delivered as a secure, RESTful Flask application in a Docker container with input validation and safe file handling, enabling fast and consistent preliminary screening in clinical and research settings.

No. of Pages : 6 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641128870 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Enhancing Communication Skills in Children with Autism using AI- Powered Assistive Web Application

(51) International classification	:G09B 5/06, G09B 19/04, G09B 5/04, G09B 19/06, G09B 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.HIMA VIJAYAN
Address of Applicant :ASSISTANT PROFESSOR, Department of Information Technology, S.A. Engineering College, Veeraragavapuram, Chennai-77. Chennai Tamil Nadu India
2)Mrs.S.KRISHNAVENI
3)Mrs.S.ASHA
4)S.A. ENGINEERING COLLEGE
(72)**Name of Inventor :**
1)Dr.HIMA VIJAYAN
2)Mrs.S.KRISHNAVENI
3)Mrs.S.ASHA

(57) Abstract :
Autism Spectrum Disorder (ASD) can influence a child's communication skills, object recognition, and social interaction. While conventional speech therapy and Augmentative and Alternative Communication (AAC) tools offer valuable assistance, they may not always provide individualized learning experiences or maintain a child's interest over time. To overcome these limitations, this paper proposes an AI-powered web application that is designed to support both verbal and non-verbal activities to children through intelligent and interactive learning methods. The system incorporates real-time object recognition, text-to-speech functionality, and adaptive educational activities tailored to each child's individual needs. It also uses interactive games and AI-generated learning tasks to encourage participation and improve engagement. In addition, the application provides progress-monitoring features that enable parents and therapists to understand a child's development and identify areas requiring further support. Overall, the proposed solution is intended to complement existing therapeutic practices by offering an accessible, personalized, and engaging learning platform for children with ASD.

No. of Pages : 6 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128873 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED CRAB SHELL WASTE UTILIZATION AND THERMAL DISTILLATION SYSTEM FOR FRESHWATER RECOVERY

(51) International classification	:C02F 1/04, C02F 1/16, B01D 5/00, B01D 1/00, C02F 1/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :Coimbatore - 641407, Tamil Nadu, India Coimbatore Tamil Nadu India
2)Dr. S. JOE PATRICK GNANARAJ
3)Dr. BALASUBRAMANIAN SIVASAMY
(72)**Name of Inventor :**
1)Mr. Vanjith Vikram. A., (25ME060)
2)Mr. Aravindhan. K., (25ME008)
3)Mr. Dharrun. B., (26MEL04)
4)Mr. Praneeth Soundararajan Srinivasan, (26MEL07)

(57) Abstract :
The present invention relates to an integrated crab shell waste utilization and thermal distillation system for freshwater recovery. The system comprises a feed-water chamber, thermal evaporation unit, processed crab shell waste utilization unit, vapour passage, condensation unit and freshwater collection unit. Crab shell waste generated from seafood-processing activities is processed and incorporated into a designated functional section of the system for value-added utilization. Saline or contaminated water is thermally evaporated to generate water vapour, which is subsequently directed to the condensation unit for freshwater recovery. The proposed system integrates crab shell waste management with thermal water purification in a single configuration, thereby providing an approach for converting seafood-processing waste into a useful engineering resource while recovering freshwater from saline or contaminated water.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128877 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Adaptive personalized hemodynamic intelligence system using multi-modal physiological signal fusion and XAI

(51) International classification	:A61B 5/00, A61B 5/0205, A61B 5/01, A61B 5/02, A61B 5/021
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mr. Srivijaykrishna D.
Address of Applicant :M Tech Data Sciences, Department of Artificial Intelligence and Data Science, KPR Institute of Engineering and Technology, Coimbatore Coimbatore Tamil Nadu India
2)Mr. Aravinth S.
3)Dr. Saranya N.
4)Mrs. Sasikala C.
(72)**Name of Inventor :**
1)Mr. Srivijaykrishna D.
2)Mr. Aravinth S.
3)Dr. Saranya N.
4)Mrs. Sasikala C.

(57) Abstract :
The present invention relates to an Adaptive Personalized Hemodynamic Intelligence System (APHIS) for continuous cardiovascular monitoring using artificial intelligence and multimodal wearable sensors. The system integrates photoplethysmography, electrocardiography, skin temperature, and motion sensors to acquire synchronized physiological signals. A processing unit performs signal quality assessment, motion artifact reduction, feature extraction, and multimodal sensor fusion to generate a personalized physiological profile. An adaptive artificial intelligence model learns individual physiological baselines, dynamically updates deviation thresholds, and identifies deviations associated with cardiovascular stress and hemodynamic changes. Explainable artificial intelligence identifies contributing physiological features to improve interpretability. The system generates personalized hemodynamic scores, cardiovascular stress indices, and physiological trend indicators, transmitting the results to a connected device for monitoring and health recommendations, thereby supporting preventive healthcare and remote patient monitoring.

No. of Pages : 7 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128879 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ON- BOARD DIAGNOSTIC SYSTEM FOR MULTI STAGE BATTERY ASSET MANAGEMENT

(51) International classification	:G01R 31/36, B60L 11/18, H01M 10/42, H02J 7/00, H01M 10/48	(71)**Name of Applicant :** **1)Dr.C.N.Vanitha** Address of Applicant :Professor, Department of Information Technology, Karpagam College of Engineering Tamil Nadu India **2)Karpagam College of Engineering** **3)Ms. L.Jenifer Amla** **4)Mr.Shriniraj E K R** **5)Mr.Balasundar P** **6)Ms.Rithika M T** **7)Mr.Kathiravan J G**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Ms. L.Jenifer Amla**
(33) Name of priority country	:NA	**2)Dr.C.N.Vanitha**
(86) International Application No Filing Date	: :01/01/1900	**3)Mr.Shriniraj E K R**
(87) International Publication No	: NA	**4)Mr.Balasundar P**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**5)Ms.Rithika M T** **6)Mr.Kathiravan J G**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The rapid adoption of electric vehicles (EVs) is leading to an increasing number of used and aging batteries, creating a need for effective battery lifecycle management. A major challenge is the lack of easily accessible and organized information about a battery's identification, usage history, operating conditions, and present condition. This makes it difficult to evaluate used batteries for reuse, second-life applications, and recycling. The proposed Battery Passport System for Used EVs provides a digital platform for maintaining and managing important battery information throughout its lifecycle. The system collects parameters such as voltage, current, and temperature using a sensor module and processes the collected data through a controller. The processed information is stored and managed through a cloud and data platform, where battery-related information can be monitored through a user dashboard. The system also supports the presentation of battery health information such as State of Health (SoH) and Remaining Useful Life (RUL), along with charging and battery lifecycle information. The proposed invention improves battery traceability, transparency, and information management and helps users make informed decisions regarding the reuse, second-life utilization, and recycling of used EV batteries. Thus, the Battery Passport System contributes towards safer battery management and a more sustainable circular economy for electric vehicle batteries.

No. of Pages : 13 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128883 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : REMOTE LONGITUDINAL CHEMOTHERAPY TOXICITY STATE MANAGEMENT AND ADAPTIVE ALERTING

(51) International classification	:G16H 20/10, G16H 50/20, G16H 10/60, G16H 40/67, A61B 5/00	(71)**Name of Applicant :** **1)Amrita Vishwa Vidyapeetham** Address of Applicant :Amrita Vishwa Vidyapeetham, Health Sciences Campus, Amrita Institute of Medical Sciences, Elamakkara P.O., Kochi - 682041, Kerala, India. Kochi Kerala India (72)**Name of Inventor :** **1)TA, Dinesh**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The present disclosure provides a remote chemotherapy longitudinal toxicity state management system (100) for monitoring patient toxicity during chemotherapy treatment. The system (100) includes an input interface (102) for receiving patient observations comprising symptom severity ratings (102-2) and medication adherence events (102-4), and a memory (106) storing a patient baseline profile (106-2), a chemotherapy treatment schedule (106-4), and an intervention log (106-6). A state synthesizer (104-2) generates a feature vector (104-2A) and projects the feature vector (104-2A) onto a multidimensional contextual matrix (104-2B) to derive a toxicity state transition (104-2C) based on a trajectory deviation. An alert unit (110) generates a provider alert signal (110-2) based on adaptive alert thresholds, while a feedback controller (112) records clinician intervention actions (112-2) for subsequent toxicity evaluation. Unlike conventional threshold-based monitoring approaches, the system (100) supports longitudinal, context-based toxicity assessment and adaptive alert generation responsive to patient and intervention history.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202641128884 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PARALLEL PLATE CAPACITIVE SENSOR HAVING A GEOMETRICALLY PERTURBED ELECTRODE EDGE

(51) International classification	:G01N 27/22, G01R 27/26, G01N 33/00, G01N 27/00, G06F 3/044	(71)**Name of Applicant :** **1)Amrita Vishwa Vidyapeetham** Address of Applicant :Amrita Vishwa Vidyapeetham, Amritapuri Campus, Amritapuri, Clappana PO, Kollam - 690525, Kerala, India. Kollam Kerala India (72)**Name of Inventor :** **1)MENON, Sreedevi K.** **2)PRADEEP, Diya U.** **3)PILLAI, Bhadra S.** **4)R. Parvathy**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A capacitive moisture sensing device (100) includes a dielectric layer (102), a first conductive electrode (104) positioned on a first side of the dielectric layer (102), and a second conductive electrode (106) positioned on a second side of the dielectric layer (102). At least one of the first conductive electrode (104) and the second conductive electrode (106) includes a conductive region having a geometrically patterned perturbation formed by elongated recessed portions extending in different directions and defining opposing electrode-edge segments separated by gaps. The electrode-edge segments provide distributed fringing-field regions extending into material positioned adjacent to the electrode, thereby increasing interaction between an electric field and the material. The geometrically patterned perturbation can be provided on the first conductive electrode (104) opposite a substantially continuous second conductive electrode (106), or on both electrodes. Variations in dielectric properties associated with moisture content alter the capacitance response, enabling moisture sensing.

No. of Pages : 36 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128886 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PORTABLE NON-DESTRUCTIVE INSPECTION DEVICE FOR DETECTING INTERNAL SPOILAGE IN COCONUTS USING NEAR-INFRARED SENSING

(51) International classification	:G01N21/359, G01N33/02, G01J3/42, G01N21/17
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)ANAGH VASISHTHA
2)R KANNIGA DEVI
3)LENIN N C

(57) Abstract :
A portable non-destructive inspection device (100) detects internal spoilage in a coconut (110) using near-infrared sensing. A near-infrared illumination LED array (102) directs fixed-wavelength near-infrared light towards an intact coconut shell, and a photodiode sensor array (104) receives reflected near-infrared light and generates corresponding electrical signals. Signal conditioning circuitry (116), comprising an amplifier (116-A) and a filter (116-B), amplifies the electrical signals and reduces noise. A microcontroller (118) processes the conditioned reflected near-infrared signals and compares processed signal values with predetermined classification thresholds to determine a fresh, borderline, or spoiled condition. The microcontroller (118) generates an output control signal to actuate a corresponding visual indication of a status indicator LED arrangement (120). The device enables rapid inspection of internal coconut condition without cutting or damaging the coconut.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128913 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN AND IMPLEMENTATION OF AUTONOMOUS ROVER FOR TOMATO HARVESTING

(51) International classification	:B25J 9/16, A01D 46/30, B25J 13/08, G05D 1/02, G06T 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KCG COLLEGE OF TECHNOLOGY
Address of Applicant :DEPARTMENT OF MECHATRONICS, KCG COLLEGE OF TECHNOLOGY, KCG NAGAR, KARAPAKKAM, CHENNAI, TAMIL NADU, INDIA-600097 Tamil Nadu India
(72)**Name of Inventor :**
1)M.D. Guru Vikraman
2)M. Nishanth
3)Dr. K. Gobivel

(57) Abstract :
[008] ABSTRACT This project presents the design and development of an autonomous robot capable of navigating tomato fields and selectively harvesting ripe tomatoes. The system utilizes the advanced YOL0v8 deep learning model to analyze images captured by an onboard camera and identify the ripeness of tomatoes in real time. Based on maturity classification, the robot targets only those fruits that are ready for harvest. Once a ripe tomato is detected, a compact gripper is equipped with a cutting mechanism grasp and detaches the fruit, ensuring minimal harm to both the crop and the plant. By combining YOL0v8-based vision with robotic actuation, this work demonstrates a practical prototype for automated harvesting. The system is designed to reduce dependence on manual labor, improve picking accuracy, and enhance overall agricultural productivity. With the growing demand for precision farming and labor-efficient solutions, autonomous harvesting technologies are gaining importance. This study focuses on the development of an intelligent tomato-harvesting rover that integrates computer vision and robotic manipulation.F SRM INST

No. of Pages : 9 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128914 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI BASED PCB COMPONENT INSPECTION SYSTEM USING RASPBERRY PI AND YOLO

(51) International classification	:B25J 9/16, B25J 11/00, A61B 5/01, B25J 13/08, B25J 13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KCG COLLEGE OF TECHNOLOGY
Address of Applicant :KCG COLLEGE OF TECHNOLOGY, KARAPAKKAM, CHENNAI-600097 9600075082 iprcell@kcgcollege.com Tamil Nadu India
(72)**Name of Inventor :**
1)G. Grahesh
2)Dr. S. Gayathri Monicka
3)K. Rajesh

(57) Abstract :
KIRA is a fully automated intelligent robotic system designed for health monitoring, guidance, and interactive assistance in industrial and educational environments. The robot integrates multi-motor actuation, temperature sensing, embedded control units, and wireless communication. A dual mobile application architecture enables real-time robot control and interactive information display through an onboard tablet. Expressive visual features enhance user engagement. The invention reduces human effort, improves safety, and provides an efficient smart assistance solution.

No. of Pages : 9 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128936 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR REAL-TIME PLANT LEAF DISEASE PREDICTION USING CONVOLUTIONAL NEURAL NETWORK

(51) International classification	:G06V 10/764, G06N 3/04, G06N 3/08, G06V 10/774, G06V 10/82
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mantada Meghana
Address of Applicant :Siddhartha Academy of Higher Education, Deemed to be University, Vijayawada, Andhra Pradesh-520007. Andhra Pradesh India
(72)**Name of Inventor :**
1)Mantada Meghana
2)Gadi Ushanjali
3)Muthakana Kesavardan
4)Mohammad Imran
5)Gummadi Srinivasarao
6)J. Sivavara Prasad
7)K. Lakshmi Devi

(57) Abstract :
A system and method for real-time prediction of plant leaf disease are disclosed. A digital image of a leaf is received through a web application interface and preprocessed by resizing and normalising pixel values. A lightweight convolutional neural network, comprising two convolutional layers with ReLU activation each followed by max-pooling, a flatten layer, a dense hidden layer and a softmax output layer, extracts colour, texture and lesion features and classifies the leaf into a disease category such as corn common rust, potato early blight or tomato bacterial spot. The network is trained on augmented, labelled leaf images using an Adam optimiser and categorical cross-entropy loss, and achieves a test accuracy of about 99.44%. The predicted class and its probability are displayed instantly to the user in the web application, enabling early, w-cost, expert-independent diagnosis suitable for precision agriculture.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128943 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Intelligent Multimodal Medical Image Analysis System Using Transformer-Based Feature Fusion, BiFusion GAN, and Explainable AI

(51) International classification	:G06N 3/04, G16H 30/20, G16H 30/40, G06N 3/08, G06T 3/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Dakannagari Harith Reddy
2)Anurodh Kumar

(57) Abstract :
This invention describes a system for studying medical images and, where available, related clinical information. The system includes a data preparation unit, a transformer-based feature encoder, a BiFusion generative adversarial network (GAN) unit, a feature-combination unit, a prediction unit, an explanation unit, and an output interface. The input data are checked and prepared using suitable steps such as resizing, normalisation, and other required processing. The transformer-based encoder studies relationships between different parts of the input and creates feature representations for further use. The BiFusion GAN unit carries out the particular operation defined in the system, such as generating, improving, translating, or aligning images or features. The feature-combination unit joins the selected representations for the chosen task. The prediction unit then produces a classification or another task-related result. The explanation unit provides information linked to the result, which may include a saliency map, attention visualisation, or feature-attribution output. The output interface shows the result and its explanation so that a qualified healthcare professional can review them. The system is designed as a set of connected parts, with clear input types, processing steps, training settings, and testing procedures. It is intended to assist medical review and research, not to replace professional judgement. Its reliability and suitability for clinical use must be established through proper testing and validation.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128944 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Multimodal Federated Deep Learning for Early Detection of Diabetic Complications

(51) International classification	:G16H 10/60, G09B 19/00, G16H 40/67, G16H 50/20, G16H 30/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Annamalai University
Address of Applicant :Annamalainagar- 608 002 Tamilnadu Annamalainagar Tamil Nadu India
(72)**Name of Inventor :**
1)Mr. Edla Laxman
2)Dr. Nagarajan S
3)Dr. Kashi Sai Prasad

(57) Abstract :
This invention describes a system that may help identify possible diabetes-related complications early. It uses health information supported by the system, which may include medical images, laboratory results, patient records, or vital signs. Different types of information are checked and processed before being brought together for the selected task. The system works across participating medical sites. Each site trains a local model using records it is allowed to use and sends permitted model updates to a coordinating server. The server combines the updates using the method included in the system and may send the shared model back for further training. The trained model processes supported patient information and produces a result, such as a risk score or risk group for a specified diabetes-related complication. In this arrangement, the original patient records do not need to be sent to the coordinating server. Any extra privacy and safety measures depend on what has actually been included in the system. The system is intended to support research and review by qualified healthcare professionals, not to replace medical judgement. Its performance, safety measures, and suitability for healthcare use must be checked through proper testing.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202641128961 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT-BASED INTELLIGENT ENERGY MONITORING AND PREDICTIVE FAULT DETECTION SYSTEM FOR SMART ELECTRICAL NETWORKS

(51) International classification	:G06N 20/00, H02J 13/00, G01R 19/25, G01R 21/133, G01R 31/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM Institute of Science and Technology
Address of Applicant :SRM Institute of Science and Technology (Ramapuram Campus), Chennai- 600089, Tamil Nadu Tamil Nadu India
(72)**Name of Inventor :**
1)Mr. C. Santhosh Kumar
2)Dr. S. Nancy Lima Christy
3)Ms. S. Aarthi

(57) Abstract :
An IoT-based intelligent energy monitoring and predictive fault detection system and method for a smart electrical network is disclosed. A plurality of IoT sensor devices distributed throughout the network acquire electrical measurements, comprising voltage, current, power, frequency, and waveform or harmonic content, at a plurality of points. A data-acquisition and pre-processing arrangement, optionally comprising edge-computing devices, receives and pre-processes the measurements and extracts features. An energy-monitoring module determines the consumption, flow, and losses of energy within the network. A predictive fault-detection module comprising a machine-learning model, trained upon historical measurements including measurements preceding known faults, analyses the measurements so as to recognize the subtle patterns that precede an incipient fault, and predicts and localizes the fault to a region or component of the network, classifying its type and estimating its severity and expected time to failure, before the fault develops into a failure. An output module generates an energy-monitoring output and a graded fault-related alert enabling preventive action. By detailed distributed monitoring and machine-learning prediction, the system detects and localizes faults earlier than threshold-based protection and improves the reliability, safety, and efficiency of the physical electrical network.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128965 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MULTI-MODAL WIRELESS CONTROL SYSTEM FOR A FOUR-DEGREE-OF-FREEDOM ROBOTIC MANIPULATOR WITH PRIORITY BASED COMMAND ARBITRATION AND GESTURE-RESPONSIVE MOTION CONTROL

(51) International classification	:B25J 9/16, B25J 3/00, G06F 3/01, B25J 13/00, B25J 13/08	(71)**Name of Applicant :** **1)S. AYYAPPAN** Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF ECE, NPR COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS), NATHAM, DINDIGUL, TAMIL NADU, INDIA-624401 Tamil Nadu India (72)**Name of Inventor :** **1)S. AYYAPPAN** **2)A.Murali prasanth** **3)T.Sarvesh** **4)L.Raja kumar** **5)B.T.Balurathinam** **6)S.Arun Kumar** **7)S.Azhagu Selvam** **8)M.Bazir Ahamed** **9)R Gokul Krishnan**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A MULTI-MODAL WIRELESS CONTROL SYSTEM FOR A FOUR-DEGREE-OFFREEDOM ROBOTIC MANIPULATOR WITH PRIORITY-BASED COMMAND ARBITRATION AND GESTURE-RESPONSIVE MOTION CONTROL A wireless robotic manipulation apparatus includes a four-degree-of-freedom servoactuated arm, an embedded controller and a multi-source command management system. A browser interface generates manually selected joint targets; a vision-based gesture interface derives joint targets from detected hand landmarks; and a playback engine generates joint targets from recorded motion frames. A command arbiter selects actuator commands according to a predefined priority scheme including emergency-stop precedence. The controller constrains joint movement, converts accepted targets into pulse-width-modulated drive signals and supervises wireless communication to invoke a safe-hold response when required. The gesture interface can apply temporal filtering to suppress landmark-induced fluctuations, while compact wireless frames reduce control-message overhead. The apparatus provides a configurable platform for remote manipulation, repeated motion demonstrations and lightweight automation.CE NPR COLLEGE

No. of Pages : 10 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128966 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED UAV-BASED REAL-TIME TREE INFECTION DETECTION SYSTEM USING DUAL-PATH LIGHTWEIGHT VISION INFERENCE AND GEOSPATIAL MONITORING

(51) International classification	:G06V 20/17, G05D 1/10, G06V 20/13, H04N 7/18, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)S. AYYAPPAN
Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF ECE, NPR COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS), NATHAM, DINDIGUL, TAMIL NADU, INDIA-624401 Tamil Nadu India
(72)**Name of Inventor :**
1)S. AYYAPPAN
2)Dr.M.Ameena Banu
3)P.Jeyalakshmi
4)R.Arunachalam
5)A KEVIN AGATH
6)M LALITHA KISHORE
7)M GANESAN

(57) Abstract :
AN INTEGRATED UAV-BASED REAL-TIME TREE INFECTION DETECTION SYSTEM USING DUAL-PATH LIGHTWEIGHT VISION INFERENCE AND GEOSPATIAL MONITORING An integrated unmanned aerial vehicle (UAV) based real-time plant or tree infection detection system comprising a quadcopter carrying an imaging module and a wireless communication module, a processing unit executing a lightweight primary visual detector and a conditionally activated secondary visual detector, and a monitoring interface. Aerial image frames are transmitted to the processing unit, decoded, resized and normalized. A primary centroid-based heatmap detector generates localized infection candidates and a confidence value. When the confidence value satisfies a predetermined verification condition, a secondary object-detection model is invoked to verify or supplement the detection. The resulting infection class, image location, confidence, timestamp and geographic position are supplied to a monitoring dashboard. The dashboard provides annotated live imagery, classwise detection information, a geographic infection-density heatmap and a detection log. In an exemplary implementation, the models are INT8-quantised for reduced computational and memory requirements. The system provides an integrated technical arrangement for aerial precision-agriculture monitoring with conditional visual verification and geospatial reporting.

No. of Pages : 16 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128967 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A CLOUD-INTEGRATED RFID-BASED SMART ATTENDANCE MONITORING SYSTEM WITH MULTI-LAYER SECURITY, ANTI-PROXY VERIFICATION AND OFFLINE DATA SYNCHRONIZATION

(51) International classification	:H04L 29/06, H04L 9/32, G06K 7/10, H04L 29/08, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR. C. KANNIKA PARAMESHWARI
Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF ECE, NPR COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS), NATHAM, DINDIGUL, TAMIL NADU, INDIA-624401 Tamil Nadu India
(72)**Name of Inventor :**
1)DR. C. KANNIKA PARAMESHWARI
2)Dr.M.Ameena Banu
3)S.Ayyappan
4)A.Jeyan James Raj
5)K.Rishikesh
6)P.Thirumalainathan
7)O.G.Srinithiswari

(57) Abstract :
A CLOUD-INTEGRATED RFID-BASED SMART ATTENDANCE MONITORING SYSTEM WITH MULTI-LAYER SECURITY, ANTI-PROXY VERIFICATION AND OFFLINE DATA SYNCHRONIZATION The present invention relates to a cloud-integrated RFID-based smart attendance monitoring system comprising one or more RFID attendance nodes, an edge microcontroller, a cloud database and an application interface. Each attendance node obtains an identification value from an RFID card, performs local authentication and generates an attendance event associated with a timestamp, node identifier and attendance session. The attendance event is cryptographically protected before transmission to the cloud database. The system employs a multi-layer security arrangement comprising card-level authentication, encrypted payload protection, secure communication, database access control and application-level authorization. A three-layer anti-proxy mechanism performs spatial verification, temporal duplicate prevention and cryptographic integrity verification. During network unavailability, attendance events are retained in a local buffer and subsequently synchronized with the cloud database after communication restoration while preserving event timestamps. The node may further include an OLED display, programmable visual indicator, audible indicator and motion sensor for real-time feedback and power management. The system provides secure, scalable and resilient attendance monitoring for educational and institutional environments

No. of Pages : 25 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128977 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TB-SCOUT: AN AI-DRIVEN TUBERCULOSIS SCREENING SYSTEM COMBINING DEEP LEARNING WITH A SMART IOT IMAGING PROTOTYPE

(51) International classification	:G06T 7/00, A61B 6/00, G06N 3/08, G06N 3/04, G16H 50/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)P. JEYALAKSHMI
Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF ECE, NPR COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS), NATHAM, DINDIGUL DISTRICT, TAMIL NADU, INDIA-624401 Tamil Nadu India
(72)**Name of Inventor :**
1)P. JEYALAKSHMI
2)B.Dhivya Preethi
3)A.Janagan
4)N.Mani
5)K.Venkatesh
6)K.Virnal Kumar
7)K.Hemalatha
8)G.Thrishma Bharathi
9)R.Varsha

(57) Abstract :
TB-SCOUT: AN Al-DRIVEN TUBERCULOSIS SCREENING SYSTEM COMBINING DEEP LEARNING WITH A SMART IoT IMAGING PROTOTYPE A tuberculosis screening system combines a portable physical-film imaging assembly with machine-learning-based radiograph analysis. The assembly includes a camera module, a diffused backlight, a film-positioning enclosure, a capture control and a power supply. An image of a chest X-ray film is captured under controlled illumination and transmitted to a computing unit. The computing unit preprocesses the image and applies a trained convolutional neural network to generate a tuberculosis-suspicion classification or score. An explanation module generates a class-associated Grad-CAM saliency overlay, and a user interface presents the classification, confidence information and overlay for operator review. In an example implementation, an ESP32-CAM/0V2640 module communicates with a host application that performs preprocessing, inference and display. The system is intended as a screening or triage aid to support clinical review and confirmatory testing, particularly in settings with limited access to radiology infrastructure.

No. of Pages : 12 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128980 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RoomVoice: Voice-Driven In-Room Hospitality Service System with Code-Enforced Consent and Evidence Verified Post-Call Recovery

(51) International classification	:G06Q 50/12, H04M 3/42, H04M 15/00, H04M 3/493, G06Q 10/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Calvin Arulnathan Andrews
Address of Applicant :No. 36, 1st, main Kanakadasa Layout, Lingarajapuram, Bangolare North, Bengaluru, Karnataka - 560084 Karnataka India
(72)**Name of Inventor :**
1)Calvin Arulnathan Andrews

(57) Abstract :
A voice-driven service system for a hotel guest room is disclosed. A guest taps an in-room tablet and speaks, in any of a plurality of languages, to an artificial intelligence assistant that conducts the whole conversation without any member of the hotel staff taking part in the call. Each request becomes a ticket with a department, a timer and an owner on a staff console. A chargeable order is committed only when a server holds a consent token bound to that exact order, created from the guest's own confirmation. When the call ends, a second model compares the conversation with the tickets and names, with a pointer to the exact moment in the conversation, any request left without a ticket; program code validates each such item before a ticket is made. A single property file feeds the assistant, the validators and the on-screen menu. Guest data is erased at check out.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128983 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR STATISTICAL AND CLINICAL DATA-ENHANCED MORPHOLOGICAL IMAGE PROCESSING AND COLORECTAL ABNORMALITY RISK ASSESSMENT

(51) International classification	:G06T 7/00, G16H 30/40, G16H 50/20, G16H 30/20, G06T 7/11
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)N.A.Pappathi
Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF ECE, NPR COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS) NATHAM, DINDIGUL DISTRICT-624401, TAMILNADU, INDIA Tamil Nadu India

(72)**Name of Inventor :**
1)N.A.Pappathi
2)S.Sowmiyadevi
3)R.Arun
4)R.Jenifer
5)B.Maragathalakshmi
6)V.Santhiya
7)A.Sivaranjini

(57) Abstract :

The present invention relates to a computer-implemented system for processing colorectal medical images through integration of morphological imaging characteristics, statistical feature processing, and structured clinical parameters. The system comprises an image acquisition interface, preprocessing module, region-of-interest segmentation module, morphological feature extraction module, statistical feature processing module, clinical data processing module, multimodal feature-fusion module, classification module, and output interface. Medical images are processed to identify regions of interest and extract quantitative shape, texture, and intensity descriptors. Statistical techniques are applied to generate a reduced morphological feature representation, which is combined with encoded clinical parameters including demographic information, family history, and available biomarker measurements. The combined representation is processed using a trained computational classifier to generate an abnormality-risk assessment output. The invention provides a unified processing architecture for multimodal medical image analysis and structured computational assessment.

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128987 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT RIDER-SAFETY HELMET WITH ON-DEVICE ACCIDENT CLASSIFICATION, DROWSINESS ALERTING AND LONG RANGE EMERGENCY COMMUNICATION

(51) International classification	:G08B 21/06, A42B 3/04, A42B 3/30, G08B 21/02, G08B 21/04	(71)**Name of Applicant :** **1)S. SOWMIYADEVI** Address of Applicant :ASSISTANT PROFESSOR, DEPARTMENT OF ECE, NPR COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS), NATHAM, DINDIGUL, TAMIL NADU, INDIA-624401 Tamil Nadu India (72)**Name of Inventor :** **1)S. SOWMIYADEVI** **2)N.A.Pappathi** **3)V.Meenakshi Sundaram** **4)T.Geerthana** **5)R.Giruthika** **6)S.Preetha** **7)M.Sri Mithra**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

AN INTELLIGENT RIDER-SAFETY HELMET WITH ON-DEVICE ACCIDENT CLASSIFICATION, DROWSINESS ALERTING AND LONG-RANGE EMERGENCY COMMUNICATION An intelligent rider-safety helmet integrates inertial motion sensing, infrared eye-state monitoring, embedded machine-learning accident classification and long-range emergency communication. An embedded controller receives acceleration and angular-motion measurements and executes a compact locally stored inference model to distinguish probable accident events from ordinary riding disturbances. An infrared sensor detects rider eye-state behavior, and a vibration actuator provides a tactile warning when a drowsiness condition is identified. Upon accident detection, a LoRa transmitter communicates an event notification with available rider-location information to a remote receiving station. The receiving station may forward the notification to a designated caretaker through a cellular messaging module. A rechargeable battery with optional solar-assisted charging supports portable operation. The arrangement combines preventive rider feedback with locally determined accident detection and remote emergency notification. COLLE

No. of Pages : 13 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128994 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FRICTION STIR WELDING TOOL HAVING CYLINDRICAL, CONICAL OR THREADED PIN FABRICATED THROUGH WIRE ARC ADDITIVE MANUFACTURING

(51) International classification	:B23K 20/12, B23K 20/26, B23K 20/24, B29C 65/06, B23K 37/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Chennai Institute of Technology
Address of Applicant :CHENNAI INSTITUTE OF TECHNOLOGY Dr. A. Ramesh, Principal, Sarathy Nagar, Kundrathur, Chennai 600069, Tamil Nadu, India. CHENNAI Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. S. Mohan Kumar
2)Dr. V. Dhinakaran
3)Ms. M. Ashmita
4)Mr. R. Suba Velan
5)Mr. M. Mohith
6)Ms. J. Kanishkaa
7)Mr. A. Pechimuthu

(57) Abstract :
The invention is a friction-stir-welding tool comprising a substantially cylindrical shank, an enlarged shoulder and a centrally projecting cylindrical, conical or externally threaded pin arranged along a common longitudinal axis. The shank is configured for connection to a spindle or tool holder and transmits rotational torque and axial force. The shoulder extends radially beyond the shank and provides a workpiece-facing contact region for load distribution, frictional heat generation and containment of softened material. The pin penetrates the workpiece and mechanically stirs material along a joint line. The shank, shoulder and pin are integrally formed as a single metallic tool body by layer-by-layer wire-arc additive manufacturing using H13 tool-steel filler wire and a gas-metal-arc-welding power source. The deposited body may undergo controlled cooling, post-deposition heat treatment, machining and grinding to reduce residual stress and achieve desired dimensional accuracy, hardness, surface finish, concentricity and rotational runout. In one embodiment, the shank has an approximate 20-mm diameter and 40-mm height, the shoulder has an approximate 22-mm diameter and 25-mm height and the pin has an approximate 6-mm diameter and 5.7-mm length. In another embodiment, the pin is conical, tapering from an approximate 6-mm base diameter to an approximate 4-mm tip diameter at an approximate 10° taper angle. In a further embodiment, the pin is externally threaded, with an approximate 6-mm thread outer diameter, 1.5-mm thread pitch and 5.7-mm height. The tool provides a continuous torque-transfer path and is suitable for friction-stir welding of aluminium-alloy and workpieces while enabling penetration and balanced rotation.

No. of Pages : 40 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128995 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIMODAL SENSOR FUSION SYSTEM AND METHOD FOR NON-CONTACT DETECTION AND LOCALISATION OF UNDERGROUND WATER PIPE LEAKS

(51) International classification	:G01M 3/24, F17D 5/06, G01V 1/00, F17D 5/02, G01N 29/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Chennai Institute of Technology
Address of Applicant :Sarathy Nagar, Kundrathur, Chennai-600069, Tamil Nadu, India. CHENNAI Tamil Nadu India
(72)**Name of Inventor :**
1)IMMANLAZAR E
2)NANTHINI R
3)TAMILMANI.S
4)HARISH SANTHOSH.K
5)Mr. M. EZHILMARAN
6)Dr. JOLY. M

(57) Abstract :
A multimodal sensor fusion system for non-contact detection and localisation of leaks in buried water pipes is disclosed. The system comprises a ground-positioned sensor array having acoustic emission, vibration, pressure, soil-moisture and electromagnetic sensing; a resonant acoustic coupling unit having a piezoelectric resonator mounted on a ground-penetrating spike and tuned to a leak-related frequency band for passive amplification of ground-transmitted leak-induced stress waves; and a directional subsurface acoustic channel having a sleeve with acoustic metamaterial baffles and a geophone array for directional reception and ambient-noise attenuation. An edge processing unit extracts and fuses multimodal features using a cross-modal attention model and a physics-informed leak localisation network to generate a leak probability map and associated leak information. An adaptive sensor placement engine uses a signal-to-noise ratio map and leak probability map to reposition sensors when confidence is insufficient. The resulting leak coordinates, severity, flow estimate and confidence may be transmitted by a low-bandwidth wireless link to a remote monitoring dashboard. The system provides non-invasive, adaptive and continuous monitoring of buried water pipelines.

No. of Pages : 36 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128996 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADDITIVELY MANUFACTURED CARBON-FIBER-REINFORCED PLA COMPRESSION FIXTURE WITH SELF-ALIGNING CHAMFERED BASE AND STEPPED BORE

(51) International classification	:G01N 3/04, G01N 3/00, G01N 3/02, G01N 3/08, G01M 13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHENNAI INSTITUTE OF TECHNOLOGY
Address of Applicant :CHENNAI INSTITUTE OF TECHNOLOGY, Dr. A. Ramesh, Principal, Sarathy Nagar, Kundrathur, Chennai 600069, Tamil Nadu, India. CHENNAI Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. S. Mohan Kumar
2)Dr. V. Dhinakaran
3)Ms. M. Ashmita
4)Mr. R. Subavelan
5)Mr. P. Gunal

(57) Abstract :
A compression testing fixture (10) for a universal testing machine is formed as a single unitary monolithic body comprising a base disc (12) having a chamfered peripheral edge (14) for self-aligning seating within the testing machine; a central cylinder (16) integrally formed with and coaxial to the base disc (12), for receiving and transmitting compressive load; a coaxial stepped axial bore (18) comprising a counter-bore section (20) providing a positive seating feature for a fixture head and a through-hole section (22) for receiving and aligning a pin of a test arrangement; and a transverse hole (24) formed through a side wall of the central cylinder (16) for receiving a transverse pin. In a preferred embodiment, the fixture is fabricated from carbon fibre-reinforced polylactic acid by additive manufacturing and is validated as a functional, low-cost replacement for conventional metallic compression fixtures for testing polylactic acid, wood-polylactic acid and lower-strength composite specimens. A corresponding method of manufacture and method of use are also disclosed.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641128999 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GREEN SYNTHESIS AND STABILIZATION OF SULPHUR NANOPARTICLES USING EUCALYPTUS AND PONGAMIA LEAF EXTRACTS

(51) International classification	:A61K 36/61, C01B 17/02, B82B 3/00, C01B 17/027, A61K 9/14	(71)**Name of Applicant :** **1)Dr. Pallavi K.N.** Address of Applicant :Department of Agronomy, College of Agriculture, Rajendranagar, PJTAU, Hyderabad, Telangana - 500030 Hyderabad Telangana India **2)Dr. K.B. Suneetha Devi** **3)Dr. Y.S. Parameswari** **4)Dr. K. Pavan Chandra Reddy** **5)Dr. CH. Shilpa Chakra**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Dr. Pallavi K.N.**
(86) International Application No	:	**2)Dr. K.B. Suneetha Devi**
Filing Date	:01/01/1900	**3)Dr. Y.S. Parameswari**
(87) International Publication No	: NA	**4)Dr. K. Pavan Chandra Reddy**
(61) Patent of Addition to Application Number	:NA	**5)Dr. CH. Shilpa Chakra**
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The present invention relates to an eco-friendly process for synthesizing sulphur nanoparticles using leaf extracts of Eucalyptus spp. and Pongamia pinnata L. as biological reducing, stabilizing, and capping agents. The process comprises preparation of an aqueous leaf extract, formation of a sodium polysulphide precursor solution, mixing of the precursor with the leaf extract, followed by acid-induced precipitation of sulphur nanoparticles and subsequent purification. The synthesized nanoparticles exhibit particle sizes ranging from 30 to 40 nm, orthorhombic crystalline structure, irregular to granular morphology, and zeta potential values indicative of colloidal stability. Phytochemicals present in the plant extracts facilitate nanoparticle formation and prevent aggregation without the use of synthetic surfactants or hazardous stabilizers. The invention provides a simple, cost-effective, and sustainable method for producing stable sulphur nanoparticles suitable for agricultural, environmental, antimicrobial, and controlled-release applications.

No. of Pages : 29 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129008 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TASTEGRAPH AI BASED FLAVOR INTELLIGENCE AND RECIPE RECOMMENDATION PLATFORM

(51) International classification	:G06N 5/02, G06K 9/62, G06K 9/00, G06F 16/36, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India
(72)**Name of Inventor :**
1)DR.S.SHAHUL HAMMED
2)GAYATHRI G
3)SOUNDHARYA LAXHMI.V
4)HAFSAH JUMAANAH.A

(57) Abstract :

TITLE : TASTEGRAPH AI BASED FLAVOR INTELLIGENCE AND RECIPE RECOMMENDATION PLATFORM TasteGraph Al is an Artificial Intelligence platform for food innovation to support Food Micro, Small and Medium Enterprises (MSMEs) to craft innovative food products & recipe in an efficient manner. It saves time, cost and waste of materials on traditional trial and error methods. An electronic nose, camera, weighing module, and environmental sensors are used to collect ingredient data in the system. Collected data is analysed with Al methods such as aroma analysis, computer vision and compatibility prediction and then processed by ESP32 or Raspberry Pi. A TasteGraph knowledge graph relates ingredients, flavour, recipes and food products to create appropriate recipes, ingredients substitution and new product idea. The platform also assesses cost, nutritional value, quality, and profitability, as well as waste reduction. Advice and analytical knowledge are provided through a web-based MSME dashboard, facilitating speedy, savvy, and data-informed food product improvement.

No. of Pages : 18 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641129009 A
(19) INDIA
(22) Date of filing of Application :29/10/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART TRANSFORMER BASED ELECTRIC POLE SAFETY MONITORING SYSTEM

(51) International classification	:H02J13/00, G01R31/62, G16Y10/35, G16Y40/10, H02J13/00, G01R31/62, G16Y10/35, G16Y40/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL 624622, TAMILNADU Tamil Nadu India
(72)**Name of Inventor :**
1)C. PREETHI
2)H. SOFIA FARHANA
3)L.P DHANISKA SRI
4)A. BUSHRA

(57) Abstract :
The invention provides an intelligent, transformer-based electrical distribution monitoring system comprising an loT, LoRa communication and machine learning platform. The voltage, current, leakage current, temperature, tilt and vibration are monitored continuously by the pole nodes based on ESP32. The sensor data is sent by LoRa to a central control station on the transformer level where it is intelligently analyzed. Machine learning algorithms detect electrical and structural faults such as leakage or overload, overheating, and pole instability. The system estimates the severity of the fault, determines which poles are affected and estimates the energy loss. A dashboard with IoT capabilities visualizes the data in real time, sends alerts based on location and offers maintenance suggestions. It provides an integrated, scalable solution that allows coordinated monitoring, early fault detection, predictive maintenance, and efficient resource allocation, ultimately to improve electrical safety, operational reliability, energy efficiency, and overall distribution network performance, through proactive fault management.

No. of Pages : 20 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129010 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DIGITAL TWIN LITE RETROFIT PREDICTIVE MAINTENANCE OEE TRACKER FOR LEGACY MACHINES

(51) International classification	:G05B 19/418, G05B 23/02, G05B 19/042, H04L 29/08, G05B 19/406
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India
(72)**Name of Inventor :**
1)AADHITYA S
2)SHANMATHI G
3)SHREENITHI S
4)SOWMIYA N
5)VEERAMADHUBHALA VV

(57) Abstract :
A reirofit-based smart industrial machine monitoring system is provided for converting existing legacy industrial machines into smart and rnonitorable machines without requiring replacement of the existing equipment. The system comprises a plurality of sensors including vibration, temperature, current or power, sound, and rotational speed sensors for continuously collecting operational data from the machine. An IoT gateway acquires the sensor data and transmits the data to an edge processing unit and/or cloud-based monitoring platform. The processed data is analysed to identify abnormal operating conditions and changing machine behaviour. A Digital Twin Lite representation provides a simplified digital representation of the physical machine using selected operational and health parameters. The system further incorporates predictive maintenance functionality for identifying abnormal trends and generating early maintenance alerts before potential machine failure. An OEE monitoring unit evaluates machine production performance based on availability, performance, and quality parameters. A unified dashboard provides real-time visualization of machine health, sensor values, Digital Twin Lite status, maintenance alerts, and OEE information, thereby enabling continuous monitoring and datadriven maintenance of legacy industrial machines through a retrofit approach.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129014 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ELARA-INDIVIDUALIZED MULTI-MODAL ASSISTIVE FRAMEWORK FOR AUTISM SPECTRUM SUPPORT

(51) International classification	:A61B 5/16, A61B 5/00, A61M 21/00, G06K 9/00, A61B 5/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.V. MAGUDEESWARAN
2)Dr.S. BHARATH
3)N. THILSHANTH
4)D.SIVA SANKESH
5)S. VAISHNAVI
6)K.P. SRINITHI

(57) Abstract :

ELARA–Individualized Multi-Modal Assistive Framework for Autism Spectrum Support An individualized multi-modal assistive system for autism spectrum support is disclosed. The system comprises an ID card-type wearable unit worn by the user, which houses a camera module directed toward the person interacting with the user. The camera module sends captured images to a Raspberry Pi, which detects the face and classifies the facial expression using a locally executed model. The Raspberry Pi then delivers a simplified social cue to the user through a display or earphones, with no image data stored or transmitted. The system further comprises a wrist-worn wearable unit having a pulse sensor and a galvanic skin response (GSR) sensor connected to an ESP32 microcontroller, which transmits the signals to the Raspberry Pi through low-energy Bluetooth. The Raspberry Pi filters these signals and compares them with a user-specific baseline calculated during a calibration phase. When elevated stress is detected, the Raspberry Pi plays a calming sound through the earphones and starts a waiting period. A notification is sent to the caregiver application only if the stress persists, and is cancelled if the signals return to baseline. The system uses low-cost, easily available components, making it suitable for special schools, therapy centers, NGOs and home use, while reducing false alerts.-

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129015 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AEROBEE GUARD: DRONE-MOUNTED REPLACEABLE PRECISION SMOKE CARTRIDGE FOR SAFE HONEYBEE HIVE MANAGEMENT

(51) International classification	:A01K 47/06, A01K 51/00, A01K 47/00, A01K 47/02, G06Q 50/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL 624622, TAMILNADU Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. B.PUSHPAVANAM
2)ISHAANTH VEL S
3)SOWPARNIKA P S
4)YUVADHARSHINI K

(57) Abstract :
Beekeeping and honeybee colony management are important for honey production, agricultural pollination, and maintaining ecological balance. However, honeybee colonies often establish hives in difficult-to-reach or potentially hazardous locations, such as tall trees, electric poles, rooftops, bridges, building facades, and industrial structures. Inspecting and managing these colonies can expose beekeepers, rescue teams, and maintenance workers to bee stings, fall hazards, and other occupational risks.Conventional hive management generally involves direct human intervention using protective clothing, ladders, climbing equipment, and manual smokers. Traditional smokers are commonly used to produce smoke that can temporarily influence honeybee behaviour during hive inspection and management. However, these devices typically require manual operation and may produce inconsistent smoke output, excessive heat, and unnecessary fuel consumption.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129016 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SEA SHIELD AI: CONNECTIVITY-RESILIENT MARITIME SAFETY AND EMERGENCY RESPONSE SYSTEM FOR SMALL FISHING VESSELS

(51) International classification	:G08G 3/02, G08G 3/00, H04W 4/90, B63B 49/00, H04W 4/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. D. ARUN PRASAD
2)JAI SHALINI E
3)LAVANYA S
4)BAVYASRI R J
5)RATHI M

(57) Abstract :
ABSTRACT OF THE INVENTION SEA SHIELD AI: CONNECTIVITY-RESILIENT MARITIME SAFETY AND EMERGENCY RESPONSE SYSTEM FOR SMALL FISHING VESSELS The present invention relates to a connectivity-resilient maritime safety and emergency response system for small fishing vessels. The system comprises an onboard positioning module, motion sensing module, embedded processing unit, memory unit, communication module, manual emergency input, and local alert unit, operatively connected to a remote monitoring application. The positioning module determines the vessel's geographical position, while the motion sensing module monitors movement-related information. The memory unit stores predefined maritime boundary data and event records. The embedded processing unit locally evaluates position and movement information to identify boundary violations and vessel-related risks. The manual emergency input enables the user to generate an emergency alert. During communication failure, the system continues local safety monitoring, activates audible or visual warnings, and stores relevant event information. When connectivity is restored, stored records are synchronized with an authorized remote device, ensuring continuous safety monitoring and emergency response despite intermittent or unavailable communication connectivity.

No. of Pages : 27 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129017 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART MSME COPILOT AI-POWERED FACTORY ASSISTANT

(51) International classification	:G05B 23/02, G06N 20/00, G06N 5/04, G01H 1/00, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL, TAMIL NADU, INDIA-624622 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.B.KARTHIKA
2)AKSHAYA T
3)ANUDEEP T
4)JAYASRI S

(57) Abstract :
A low-cost AI-based predictive digital twin platform for MSME machinery predictive maintenance is revealed. The platform consists of an ESP32-class microcontroller edge gateway that streams sensor data to a cloud server via MQTT, low-cost IoT sensors (MEMS accelerometer vibration sensor, temperature sensor, and MEMS microphone sound sensor) mounted on the machine, a camera for periodic visual wear inspection, and an artificial intelligence pipeline that preprocesses the signals, extracts FFT/RMS spectral, temperature, acoustic, and visual features, detects anomalies against a learned per-machine baseline, classifies the likely fault type, and assigns severity as Low, Medium, High, or Critical.While a digital twin profile store maintains a live permachine record of faults, repairs, replacement parts, and operating hours that refines predictions over time, a maintenance decision module estimates maintenance urgency, remaining useful life, repair time, repair cost, and production-downtime loss and generates a repair-versus-replacement recommendation. T E

No. of Pages : 29 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641129022 A

(19) INDIA

(22) Date of filing of Application :29/10/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : CAMPUSGENT AI-AN AUTONOMOUS STUDENT SUCCESS AND PLACEMENT INTELLIGENCE PLATFORM

(51) International classification	:G06Q 50/20, G06Q 10/06, G06N 20/00, G06Q 10/10, G06N 5/04	(71)**Name of Applicant :** **1)DR. ESTHER T** Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L&T BY-PASS, CHINNIYAMPALAYAM POST, COIMBATORE -641062 6379082669 estherit@siet.ac.in Tamil Nadu India **2)Dharshan R** **3)Kamesh J** **4)Boopalan M** **5)Gopalakrishnan M**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)DR. ESTHER T**
(33) Name of priority country	:NA	**2)Dharshan R**
(86) International Application No	:NA	**3)Kamesh J**
Filing Date	:NA	**4)Boopalan M**
(87) International Publication No	: NA	**5)Gopalakrishnan M**
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The present invention relates to the design and development of CampusGent AT, an intelligent university operating and placement enablement Software-as-a-Service (SaaS) platform built using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The system orchestrates an ecosystem of 21 specialized cognitive artificial intelligence agents (covering student success, career recommendation, ATS resume intelligence, mock interview simulation, placement readiness scoring, and academic risk detection) integrated with real-time academic progression tracking and campus placement workflows. Users—comprising Students, Faculty, Training and Placement Officers (TP0s), and Institutional Administrators—interact through role-tailored reactive web interfaces. The Node.js and Express.js backend coordinates unified student intelligence snapshots, cognitive multi-agent orchestration, and persistent MongoDB document storage, secured by JSON Web Token (JWT) authentication with refresh token rotation and recursive NoSQL query sanitization. This platform demonstrates how full-stack web technologies and autonomous multi-agent AT merge to automate institutional workflows, streamline corporate CampusGent AT is an AI-driven university operating and placement intelligence platform designed to integrate academic monitoring, student career development, and recruitment management within a unified web application. The system uses a MERN-based architecture comprising MongoDB, Express.js, React.js, and Node.js to manage user information, academic records, and placement activities. Its role-based interfaces provide dedicated access for students, faculty members, placement officers, and administrators. The platform supports AI-assisted career recommendations, resume analysis, mock interview preparation, academic performance monitoring, and placement eligibility assessment. A backend orchestration module processes user requests and coordinates the relevant application services, while database integration enables the storage and retrieval of academic and recruitment information. Authentication, authorization, and input validation mechanisms help protect system resources. The platform aims to improve coordination among university stakeholders, support personalized student development, and streamline campus placement workflows.

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129026 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXPERIMENTAL INVESTIGATION OF SUSTAINABLE CONCRETE USING NANO SILICA AND NANO VANADIUM WITH WASTE TYRE RUBBER, PLASTIC WASTE AND E-WASTE

(51) International classification	:C04B 28/02, C04B 14/02, C04B 28/00, C04B 28/04, C04B 14/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Madheswaran.K
Address of Applicant :DEPARTMENT OF CIVIL ENGINEERING, SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BYPASS ROAD, COIMBATORE -641062, TAMILNADU Tamil Nadu India
2)Anand M
3)Kodeswaran R
4)Vasanth G
5)Dan Moses M
(72)**Name of Inventor :**
1)Dr. Madheswaran.K
2)Anand M
3)Kodeswaran R
4)Vasanth G

(57) Abstract :
Concrete is one of the most widely used construction materials in the civil engineering industry. However, the extensive consumption of natural river sand as fine aggregate has increased the demand for natural resources and created environmental concerns. Biochar, a carbon-rich material produced through the thermal decomposition of biomass under limited oxygen conditions, has potential for use as an alternative material in concrete. This project aims to experimentally investigate the strength and durability properties of concrete by using biochar as a partial replacement of fine aggregate. Biochar will be incorporated into the concrete mix at different replacement percentages, such as 0%, 5%, 10%, 15%, and 20% by mass of fine aggregate. A control mix without biochar will be prepared for comparison. The specimens will be cast, cured, and tested to evaluate their performance. The experimental investigation will include compressive strength testing and selected durability-related tests, such as water absorption and resistance to environmental exposure, depending on the available laboratory facilities. The results obtained from the biochar concrete specimens will be compared with those of conventional concrete to determine the influence of biochar on strength, water absorption, and durability-related performance. The study aims to identify a suitable biochar replacement level that provides acceptable concrete performance while potentially reducing the consumption of natural sand and promoting the beneficial utilization of biomass-derived materials. The findings may contribute to the development of sustainable construction materials and the effective utilization of biomass waste in concrete applications.

No. of Pages : 4 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129031 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT DIGITAL ASSISTANT USING AI AND NLP

(51) International classification	:G10L 15/22, G10L 15/26, G06F 3/16, G10L 15/18, G06F 16/332	(71)**Name of Applicant :** **1)DR. KAMALAVENI V** Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BY-PASS, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India **2)Hamdha S** **3)Hare Krishna Kumar S** **4)Lalish S K** **5)Mahaboob Mariam S M B**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)DR. KAMALAVENI V** **2)Hamdha S** **3)Hare Krishna Kumar S** **4)Lalish S K** **5)Mahaboob Mariam S M B**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	:NA :NA	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT The Intelligent Digital Assistant is an intelligent AI-powered system designed to automate computer operations=through- voice�and�text-based�interactions. - The -invention�integrates artificial intelligence, speech processing, computer automation, and natural language understanding within a unified platform. It utilizes Python-based backend processing, a graphical user interface, and local database storage for efficient operation. The system incorporates local Al models to generate intelligent responses and execute user commands. Face recognition provides an additional authentication mechanism, while optical character recognition enables text extraction from images. The assistant supports application launching, web browsing, communication tasks, and productivity automation. Its modular architecture enables the integration of multiple intelligent functionalities into a single system. The proposed invention aims to enhance human-computer interaction, operational efficiency, accessibility, and personalized digital assistance.

No. of Pages : 7 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641129038 A

(19) INDIA

(22) Date of filing of Application :29/10/2026

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI ASSISTED DIGITAL WELLNESS SYSTEM FOR REGULATING EMOTIONS USING REALTIME FACIAL EXPRESSION ANALYSIS

(51) International classification	:G06N 3/08, A61B 5/16, G06N 3/04, A61B 5/00, A61M 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)The Principal, Mepco Schlenk Engineering College
Address of Applicant :The Principal, Mepco Schlenk Engineering College, Mepco Schlenk Engineering College (P0), Sivakasi, Tamil Nadu-626005. Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. C Anitha
2)R.Kamaladevi

(57) Abstract :
The present invention discloses an Al assisted digital wellness system for regulating emotions using real-time facial expression analysis, intended to console the user by providing appropriate motivational videos, audio messages and relaxation guidance based on the detected facial expression. The proposed system comprises a Raspberry Pi interfaced with a camera module, display, speaker and Micro SD card memory. The camera captures the facial image of the user and transfers the image to the Raspberry Pi, where preprocessing, noise removal and face detection are performed. Facial features are extracted from the detected face, and a feature map is generated using a trained deep learning-based Convolutional Neural Network (CNN) algorithm. The extracted features are compared with trained patterns to predict the corresponding emotion of the user. Based on the predicted emotion, the system selects and provides suitable digital wellness content. When an anger emotion is identified, the system provides a voice message instructing the user to drink one cup of water, plays the user's two favourite melody songs and presents step-by-step guided meditation audio for ten minutes. After the specified interval, the user's facial expression is analysed again. If the anger emotion continues, the system plays the user's favourite comedy audio clips followed by a counsellor message explaining the negative effects of anger and providing guidance to reduce it. When a depression-associated facial expression is identified, the system plays three motivational videos delivered by experts to encourage selfconfidence, and subsequently provides breathing-exercise audio for three minutes and guided meditation audio for ten minutes. If the caregiver alert facility has been registered and enabled, an SOS message is transmitted to the caregiver regarding the detected emotional state. When a happy emotion is identified, the system plays an appreciation voice message fold enables the user, according to their preference, to write and share a message with friends. Unlike conventional systems that require users to manually select relaxation content, the proposed invention integrates facial-expression-based emotion prediction with. corresponding personalised audio-visual responses and repeated facial-expression assessment. The system provides an interactive digital wellness arrangement for encouraging relaxation, reducing negative thinking and supporting the management of emotional states through motivational content, music, breathing exercises, meditation and counselling guidance. The invention thereby offers an embedded, emotion-responsive solution for promoting a calm and positive user experience.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129050 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MULTIMODAL ARTIFICIAL INTELLIGENCE SYSTEM FOR CONTINUOUS SIGN LANGUAGE INTERPRETATION AND SPEECH G

(51) International classification	:G06N 20/00, G06K 9/00, G06K 9/62, G06N 3/08, G10L 15/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)P.N. CHENDILNADAN
Address of Applicant : STUDENT, DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING, MEENAKSHI COLLEGE OF ENGINEERING 12, VEMBULIAMMAN KOIL ST, ANNAJI NAGAR, KK NAGAR WEST, K. K. NAGAR, CHENNAI-600078, TAMILNADU, INDIA 7305601588 chendilnadan040@gmail.com Tamil Nadu India
2)KEVIN BENNETT A
3)HARISH Y
4)PAVITHRA D
5)NITHISH KUMAR P
6)MOHAN HARISH K
7)SRINATH S
8)SELVA KUMAR S
9)ROSHAN B R
10)LOKESHWARAN R
11)JIVITESH M
12)RISHIVANDAN S
13)SAI VISWA TEJA B
14)GARY MARCUS MANOGARAJ S
15)EASWAR RAM SRIVATS S
16)AHMAD RAJA M
17)ARAVIND V
18)KRISHITAA GOPINATH
19)RITHISH K
20)ANUSHKUMAR K

(72)**Name of Inventor :**
1)P.N. CHENDILNADAN
2)KEVIN BENNETT A
3)HARISH Y
4)PAVITHRA D
5)NITHISH KUMAR P
6)MOHAN HARISH K
7)SRINATH S
8)SELVA KUMAR S
9)ROSHAN B R
10)LOKESHWARAN R
11)JIVITESH M
12)RISHIVANDAN S
13)SAI VISWA TEJA B
14)GARY MARCUS MANOGARAJ S
15)EASWAR RAM SRIVATS S
16)AHMAD RAJA M
17)ARAVIND V
18)KRISHITAA GOPINATH
19)RITHISH K
20)ANUSHKUMAR K

(57) Abstract :
The invention facilitates natural and accessible communication between sign-language users and individuals who do not understand sign language by capturing and processing visual data of sign-language communication using one or more imaging devices. The system processes hand gestures, finger movements, facial expressions, head movements, body movements, posture, and other relevant non-manual cues. The extracted multimodal features are analyzed over continuous temporal sequences using artificial intelligence and machine learning techniques to recognize a sequence of signs. A contextual language processing module determines the intended linguistic meaning of the recognized sequence based on temporal, grammatical, semantic, and contextual relationships, and generates corresponding textual and/or synthesized speech output. A confidence assessment mechanism identifies uncertain interpretations and enables user confirmation or correction. The system may further provide adaptive and personalized recognition, multilingual processing including Indian Sign Language, privacy-conscious or offline processing, and bidirectional communication through visual or three-dimensional sign-language representations. The invention thereby provides an intelligent assistive communication bridge for natural and accessible interaction between sign-language and non-sign-language users.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129052 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ORAL IMAGE ACQUISITION SYSTEM

(51) International classification	:G06T 7/00, A61B 1/24, A61B 6/14, G16H 30/40, A61B 1/04	(71)**Name of Applicant :** **1)Amrita Vishwa Vidyapeetham** Address of Applicant :Amrita Vishwa Vidyapeetham, Coimbatore Campus, Coimbatore - 641112, Tamil Nadu, India. Coimbatore Tamil Nadu India (72)**Name of Inventor :** **1)MATHAVAN, Jayakumar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Embodiments of the present disclosure relate to a system (100) and a method (900) of oral image acquisition. The system (100) receives an intraoral image from an intraoral image capture device (500). A processor (102-2) of the system (100) computes an image quality score from one or more image-quality parameters and derives an uncertainty value from image features. Based on the image quality score and uncertainty value, the processor (102-2) generates an image acquisition instruction for an additional intraoral image. The intraoral image capture device (500) acquires the additional image according to the instruction. The processor (102-2) fuses the additional image with an accepted intraoral image to generate a fused image representation. The system (100) selects a transmission mode for the fused image representation based on image-analysis metadata and detected network-bandwidth conditions.

No. of Pages : 31 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129053 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE DEVICE FOR EMOTION STATE MONITORING AND ADAPTIVE RESPONSE

(51) International classification	:A61B 5/16, A61B 5/00, A61B 5/0205, A61B 5/11, A61B 5/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)HARISH R
2)KISHORE BHARATHI B
3)GOKUL R
4)TARUN VINOD PAI
5)SINDHU S
6)KABILAN G
7)V. PREMANAND

(57) Abstract :
A wearable device (102) is disclosed, including a control unit (106) having a processor (202) and a memory (204). The control unit (106) determines an emotional state of a wearer based on physiological signal data received from a physiological sensing unit (104), generates a feedback signal through a feedback unit (112), and monitors the physiological signal data over a time interval. The control unit (106) escalates the feedback signal when the physiological signal data continues to indicate the determined emotional state beyond the time interval, and de-escalates the feedback signal when the physiological signal data indicates a normalization thereof. The control unit (106) transmits, through a wireless communication unit (108), an alert and information corresponding to the determined emotional state to an external device (114), which provides an intervention recommendation.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129054 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIMODAL ASSISTANCE SYSTEM FOR VISUALLY IMPAIRED INDIVIDUALS AND METHOD THEREOF

(51) International classification	:G06F 3/01, G06T 19/00, G01C 21/36, G06V 40/20, G06T 7/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)JOEL J CHALAPPURAM
2)B. SALEENA
3)B.PRAKASH
4)S. HARINI

(57) Abstract :
A multimodal assistance system (100) may include a wearable assistive device (102) including an imaging unit (104), a processing unit (106), and a memory (108), secured to a body region such that the imaging unit (104) acquires image data corresponding to a substantially forward-facing field of view during movement; a visual perception engine (110) configured to process the image data to determine environmental information associated with surrounding environment; a context-aware assistance engine (112) configured to determine an assistance context based on environmental information, temporal information, movement, user input, and navigation information generated by a navigation engine (116); an adaptive assistance engine (114) configured to selectively activate, adapt, and/or prioritize one or more assistance functions, dynamically coordinate the assistance functions, and prioritize an assistance function associated with a higher-priority environmental or navigation information; and a feedback engine (118) configured to generate real-time assistance information through one or more output modalities.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129058 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FABRICATION AND EVALUATION OF DISSOLVING MICRONEEDLES PATCHES FOR TRANSDERMAL DELIVERY OF BIOMACROMOLECULES

(51) International classification	:A61M 37/00, A61K 9/70, A61K 38/00, A61K 9/00, A61K 47/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RAMAKRISHNA GARLAPATI
Address of Applicant :HINDU COLLEGE OF PHARMACY AMARAVATHI ROAD, MAIN ROAD, OPP A CONVENTION Andhra Pradesh India
2)Dr. Anusha Kota
3)Dr. Y. Ratna Sindhu
4)Dr. Indira Priyadarshini G
5)Dr. Mounika V
6)Adilakshmi CH
7)Shaik Sanjuda
8)Jayasri Sanam
9)Ponduri Gowtham
10)Yerragopu Nagasurekha
(72)**Name of Inventor :**
1)RAMAKRISHNA GARLAPATI
2)Dr. Anusha Kota
3)Dr. Y. Ratna Sindhu
4)Dr. Indira Priyadarshini G
5)Dr. Mounika V
6)Adilakshmi CH
7)Shaik Sanjuda
8)Jayasri Sanam
9)Ponduri Gowtham
10)Yerragopu Nagasurekha

(57) Abstract :
ABSTRACT The present invention relates to fabrication and evaluation of dissolving microneedles patches for transdermal delivery of biomacromolecules. The dissolving microneedle patch was fabricated and a sequential aqueous casting method for transdermal delivery of compatible proteins or peptides across the stratum corneum. The patch comprises a backing (20) carrying a plurality of microneedles, each having a distal protein-containing portion (40) and a compositionally distinct proximal support portion (30) positioned between the distal portion and the backing. The distal portion comprises polyvinylpyrrolidone, sodium hyaluronate, trehalose, and a protein or peptide incorporated within a dissolvable polymer matrix. The proximal support portion comprises polyvinylpyrrolidone and polyvinyl alcohol and provides structural support for the distal portion during handling and application. Fabrication comprises filling microneedle mould cavities with an aqueous active-containing formulation, partially drying the formulation, casting an aqueous support formulation, completing drying, and demoulding the resulting patch. The temperature of the active-containing product is controlled during processing according to the stability requirements of the selected protein or peptide. The differentiated architecture is designed to concentrate recoverable active substance within distal needle portions intended to penetrate the stratum corneum and enter the skin. Evaluation comprises regional content analysis to characterize active-substance distribution between the distal portions, proximal portions, and backing; assessment of retained biological activity following fabrication; mechanical integrity testing; determination of skin insertion performance; characterization of dissolution; and measurement of active-substance deposition in skin. Dose accounting relates initial patch content to skin deposition, residual patch content, and other recovered fractions to assess delivery performance and identify unrecovered active substance. Following application, dissolution of the inserted distal portions is intended to release the incorporated protein or peptide within the skin.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129061 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPLETE ENERGY CONSUMPTION FORECASTING

(51) International classification	:G06Q 50/06, G06N 20/00, G01R 22/06, G06F 17/18, G01R 22/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mayank Chauhan
Address of Applicant : SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L&T BY-PASS, CHINNIYAMPALAYAM POST, COIMBATORE -641062 8754971971 praveennm24ad@srishakthi.ac.in Tamil Nadu India
2)Praveen N M
3)Naveena R
4)Muthu Ragavan S
5)Monishwaran M
6)Rexy Chirstopher S

(72)**Name of Inventor :**
1)Mayank Chauhan
2)Praveen N M
3)Naveena R
4)Muthu Ragavan S
5)Monishwaran M
6)Rexy Christopher S

(57) Abstract :

This draft provides an overview of the proposed system for analysing electricity consumption to provide tailored suggestions. The system \yin take data regarding previous electricity usage and possibly information from bills to identify trends. Comparisons will be made based on the data. Peaks and troughs can be identified. This will allow the system to provide an explanation into the data and provide suggestions to increase energy awareness. The system could be implemented using a software pipeline. Data can be validated and pre-processed. Time based features can be created and statistical/machine learning techniques can be applied to identify trends. The comparisons and suggestions can then be provided in a user interface. The system could also include verification of data quality, model assessment and providing evidence for each suggestion. This document is an initial technical draft based on the concept of the project. The specifics of the proposed system will need to be confirmed including how it is implemented, the technical contribution, results from experimentation, application details and inventor details. There is no claim that any accuracy has been measured or that it is patentable.

No. of Pages : 8 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129065 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ENVIRONMENTAL MOUNTAIN-HAZARD MONITORING AND EARLYWARNING APPARATUS AND METHOD THEREOF

(51) International classification	:G08B 21/10, G08B 31/00, H04N 7/18, G06Q 50/26, G06T 7/00	(71)**Name of Applicant :** **1)T Pradeep** Address of Applicant :Associate Professor, Department of Civil Engineering, Kongu Engineering College, Perundurai-638060, Erode District, Tamil Nadu Erode Tamil Nadu India **2)Vivek S** **3)Dr S Mohan** **4)Lavanya Ganesan** **5)Satheeshkumar Seerangagounder** **6)Guruswamy Shyamala**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	(72)**Name of Inventor :** **1)T Pradeep** **2)Vivek S** **3)Dr S Mohan** **4)Lavanya Ganesan** **5)Satheeshkumar Seerangagounder** **6)Guruswamy Shyamala**
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT “ENVIRONMENTAL MOUNTAIN-HAZARD MONITORING AND EARLYWARNING APPARATUS AND METHOD THEREOF” An environmental mountain-hazard monitoring and early-warning apparatus (100) comprises an upstream observation node (110), a river-obstruction monitoring node (120) and a downstream warning node (130) coupled by a communication subsystem (140). The upstream observation node (110) includes an image-acquisition assembly (113), a physical observation-state sensor (115) and an image-observation eligibility module (118) that admits an image-derived physical-change record to a designated hazard-evidence path only upon camera observation-state correspondence, image-registration quality and image validity. The river-obstruction monitoring node (120) evaluates time-aligned measurements from first and second river-level sensors (121, 122) against a site-specific ordinary-flow baseline and generates an obstruction-stage event record. At the downstream warning node (130), an evidence qualification and transition gate (173) authorises hydraulic-event sampling upon validation of the obstruction-stage event record, while corroborated-event confirmation requires eligible image evidence. A local safety controller (171) activates a local warning output (133) independently of remote-event confirmation. Reference Fig. 1.

No. of Pages : 34 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129067 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A System and Method for Quantum-Enhanced Deep Learning-Based Brain Tumor Detection and Classification Using MRI Images

(51) International classification	:G06T 7/00, A61B 5/055, G06N 10/00, G06V 10/764, G06V 10/82
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Raguru Jaya Krishna, BMS Institute of Technology and Management
Address of Applicant :Associate Professor, Department of CSE, BMS Institute of Technology and Management, Doddaballapur Main Road, Avalahalli, Yelahanka, Bengaluru, Karnataka-560064, India Bangalore Karnataka India
2)Dr. D. Chitra, Panimalar Engineering College
3)P. Swapna Shankar, Geethanjali College of Engineering and Technology
4)Dr. S. Selvakanmani, St.Joseph's College of Engineering
5)Dr. Surabhi Shanker, K. R. Mangalam University Gurugram
(72)**Name of Inventor :**
1)Dr. Raguru Jaya Krishna, BMS Institute of Technology and Management
2)Dr. D. Chitra, Panimalar Engineering College
3)P. Swapna Shankar, Geethanjali College of Engineering and Technology
4)Dr. S. Selvakanmani, St.Joseph's College of Engineering
5)Dr. Surabhi Shanker, K. R. Mangalam University Gurugram

(57) Abstract :
The present invention relates to a system and method for quantum-enhanced deep learning-based brain tumor detection and classification using magnetic resonance imaging (MRI) images. The system comprises an MRI image input module, an image preprocessing module, a deep learning feature extraction module, a feature transformation module, a quantum processing module, a classification module, and an output generation module. The preprocessing module prepares MRI images for analysis, and the deep learning module extracts representative image features. The extracted features are transformed into a compact representation and encoded into quantum states for processing by a parameterized quantum circuit. Measurement outputs obtained from the quantum circuit are supplied to a classical classification component to generate a predicted brain tumor category and associated confidence information. A training and optimization module facilitates parameter optimization using labeled MRI datasets and a selected loss function. The system provides a hybrid computational framework for automated brain tumor analysis, with performance evaluated through appropriate classification metrics and comparison with conventional deep learning approaches.

No. of Pages : 11 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129072 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Ultrasound-Assisted Process for Preparing a Ternary Biopolymer Complex and Structured Plant-Based Protein Chunks

(51) International classification	:A23J 3/14, A23J 3/22, A23L 29/238, A23J 3/16, A23L 29/269
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T)
Address of Applicant :National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T), Pudukkottai Road, Thanjavur-613005, Tamil Nadu, India. Thanjavur Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. V. Hema
2)Er. Manojkumar B
3)Dr. Ashish Rawson
4)Dr. Sunil C. K
5)Er. Harish S
6)Dr. S. Shanmugasundaram

(57) Abstract :
The disclosure relates to a process for preparing a ternary biopolymer matrix from fenugreek malt protein, zein, and guar gum and using the matrix to prepare structured ready-to-cook protein chunks. Fenugreek malt protein is dissolved in distilled water at 1:10 w/v and adjusted to pH 11; zein is separately dissolved in 70% (v/v) ethanol at 1:10 w/v; and guar gum is dispersed in warm water. The preparations are mixed at respective weight percentages of 45%, 45%, and 10%, acidified to pH 4.5 using 1N hydrochloric acid, subjected to ultrasound processing at 45 kHz, heated in a water bath at 80°C for 3 hr, cooled to room temperature, neutralized to pH 7.0 using 1N sodium hydroxide, frozen at −40°C, and freeze-dried. The resulting matrix is combined with sodium alginate, salt, glucono-δ-lactone, methylcellulose, potato starch, sunflower oil, and water, molded, blast-frozen, immersed in 2% (w/v) calcium chloride at 4°C, blast-frozen again, and freeze-dried to obtain ready-to-cook protein chunks.

No. of Pages : 65 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129073 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EEG Signal Processing System for Coffee-Associated Hunger and Satiety Classification and Method of Operation

(51) International classification	:A61B 5/374, A61B 5/372, A61B 5/0476, A61B 5/00, A61B 5/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T)
Address of Applicant :National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T), Pudukkottai Road, Thanjavur-613005, Tamil Nadu, India. Thanjavur Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. V. Chandrasekar
2)Telanakula Hyndavi

(57) Abstract :
The present disclosure proposes an EEG signal processing system (100) and a method for processing resting-state electroencephalography recordings corresponding to pre-coffee-consumption and post-coffee-consumption conditions for subject-independent classification of hunger and satiety states. The EEG signal processing system (100) comprises a computing device (102) having a processor (104), a memory (106), and plurality of modules (108) comprising an EEG preprocessing module (110), a feature extraction module (112), a statistical analysis module (114), and a machine-learning classification module (116). The system (100) generates EEG feature vectors comprising relative spectral band power, spectral entropy, and frontal alpha asymmetry features, evaluates the EEG features with visual analogue scale measurements, and generates statistical feature-ranking information. The system (100) performs Leave-One-Subject-Out cross-validation and machine-learning classification to classify EEG feature data as representing a hunger state or a satiety state.

No. of Pages : 37 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129074 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Process for Preparing a Stable Hibiscus-Based Natural Food Colourant by Vacuum Oven Drying

(51) International classification	:A23L 5/00, A23L 5/40, A23L 5/43, C09B 61/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T)
Address of Applicant :National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T), Pudukkottai Road, Thanjavur-613005, Tamil Nadu, India. Thanjavur Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. M. Loganathan
2)Rajshri Naveen

(57) Abstract :
The present disclosure proposes a process for preparing a stable hibiscus-based natural food colourant by vacuum oven drying, utilizing Hibiscus rosa-sinensis flower petals as a natural source of colourant and subjecting the flower petals to vacuum oven drying under specified drying conditions to obtain a powder-based natural food colourant containing naturally occurring anthocyanin pigments with improved colour retention, reduced pigment degradation, and suitability for food colouring applications. The process comprises selecting red Hibiscus rosa-sinensis flowers, removing floral portions to obtain separated petals, washing the separated petals, removing surface water, vacuum oven drying, grinding, and sieving to obtain the hibiscus-based natural food colourant powder. The process utilizes red Hibiscus rosa-sinensis flower petals as a sustainable natural source for producing a value-added food colourant powder suitable for food colouring applications. The process reduces thermal degradation, oxidation, and browning of hibiscus-derived colourant material during conversion of flower petals into a stable powder form.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129077 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : THE MECHANICAL SOLAR TRACKING SYSTEM

(51) International classification	:H02S 20/32, F24S 50/20, F24J 2/54, G05D 3/12, F24S 30/425
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SARANATHAN COLLEGE OF ENGINEERING
Address of Applicant :VENKATESWARA NAGAR, PANJAPPUR, THIRUCHIRAPPALLI, TAMIL NADU, INDIA-620012 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. C. Pearline Kamalini
2)Dr. M. V. Suganyadevi
3)Dr. C. Krishnakumar
4)R. Sridhar
5)M. Lekkan
6)R. M. Krishnappan
7)R. Kesavarthini
8)S. Shivani
9)K. Vignesh
10)M. Najima Banu

(57) Abstract :
The present invention relates to,a mechanical solar tracking system for adjusting the orientation of a photovoltaic panel in accordance with the apparent movement of the sun. The system comprises a supporting base, a vertical support column, a panel mounting frame, a rotational shaft, a bearing assembly, a mechanical transmission unit and a drive mechanism. The drive mechanism transfers motion through the transmission unit to rotate the mounting frame about a predetermined axis. An angular adjustment linkage and travel-limit stops control the permitted movement, while a locking mechanism helps maintain the selected orientation. The drive may be implemented using a timed mechanical arrangement or another suitable drive configuration according to the intended application. The system can be configured for single-axis tracking and may be extended to dual-axis tracking by incorporating an additional rotational mechanism. The proposed arrangement aims to provide a serviceable mechanical alternative to fixed solar-panel mounting and may improve solar energy reception under appropriate operating conditions

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129087 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A GOVERNANCE-CONSTRAINED MULTI-AGENT APPARATUS

(51) International classification	:G06F 21/57, G06Q 10/06, G06N 5/04, G06N 5/02, G06F 9/45
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ALBATROZ (INDIA) PRIVATE LIMITED
Address of Applicant :DC-43, DEVELOPMENT CENTER, FOURTH FLOOR, TIDEL PARK COIMBATORE, ELCOSEZ, CIVIL AERODROME POST, COIMBATORE – 641014 Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mathan Bagavathiappan
2)Balaji Ragothaman
3)Gayathri Gupta Shatavelli

(57) Abstract :
A governance-constrained multi-agent apparatus (100) is disclosed. Said apparatus (100) broadly comprises: an ingestion and normalisation engine (102); a governance constraint compiler (104); a governance constraint vault (106); a orchestrator (108); a plurality of specialised agents (110); an action proposal generator (112); a pre-execution governance validation layer (114); an approval routing engine (116); a provenance tracker (118); a record connecting member (120); a recording member (122); a source artefact repository (124); and an audit log store (126). The disclosed apparatus (100) offers at least the following advantages: is simple; prevents unauthorised or non-compliant changes; offers controlled decision making; reduces computational and human rework; improved data integrity and governance compliance; end-to-end provenance and auditability; and/or the like. Figure to be Included is Figure 1.

No. of Pages : 42 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129089 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A COMPUTER-IMPLEMENTED SYSTEM AND METHOD FOR MACROECONOMIC REGIME RETRIEVAL AND REGIME-CONDITIONED MULTI-ASSET REAL-ASSET RISK ASSESSMENT

(51) International classification	:G06Q 50/16, G06Q 40/06, G06Q 10/06, G06Q 40/00, G06F 17/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Parimel Azhagan Kusala Kumaran
Address of Applicant :Research Scholar, Department of Computer Science & Engineering, Kalasalingam Academy of Research and Education, Krishnankoil, Virudhunagar (Dist), Tamil Nadu- 626126, India Tamil Nadu India
(72)**Name of Inventor :**
1)Parimel Azhagan Kusala Kumaran

(57) Abstract :
A computer-implemented system and method for assessing macroeconomic scenarios across listed and private real estate and infrastructure investments are disclosed. Versioned economic and asset records are filtered using point-in-time publication boundaries and transformed into aligned temporal features. A regime encoder generates a scenario representation, and a constrained retrieval engine ranks historically analogous windows using trajectory similarity, contextual compatibility and data-quality criteria. An asset-response engine estimates horizon-consistent response distributions with appraisal-freshness adjustments and uncertainty measures. A portfolio engine resolves direct and look-through holdings to avoid duplicate exposure and calculates aggregated sensitivity and downside measures. A provenance-linked output interface provides historical analogues, model assumptions and evidence-restricted explanations. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129091 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-ENABLED GEOSPATIAL SITE FEASIBILITY ANALYSIS SYSTEM AND METHOD FOR EVIDENCE-TRACEABLE SUITABILITY SCORING

(51) International classification	:G06Q 10/06, G06F 17/30, G06F 16/29, G06Q 30/02, G06F 16/2457
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Parimel Azhagan Kusala Kumaran
Address of Applicant :Research Scholar, Department of Computer Science & Engineering, Kalasalingam Academy of Research and Education, Krishnankoil, Virudhunagar (Dist), Tamil Nadu- 626126, India Tamil Nadu India
(72)**Name of Inventor :**
1)Parimel Azhagan Kusala Kumaran

(57) Abstract :
An AI-enabled geospatial site feasibility analysis system (100) and method (200) are disclosed. A natural-language property-selection request is converted into a validated typed spatial query plan and executed against harmonized, versioned parcel, infrastructure, market and regulatory datasets. Mandatory eligibility conditions are applied before normalized, weighted suitability scores are determined for eligible parcels. Regulatory provisions extracted from documents are linked to applicable geographic areas with source citations and effective-date information. A provenance ledger retains source identifiers, spatial operations, criterion values, weights and uncertainty indicators, enabling reproducible ranked maps and feasibility reports. The system distinguishes infrastructure proximity from confirmed service availability and flags unresolved geographic or documentary ambiguity. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129096 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A TRICHODERMA-BASED BIOFERTILIZER COMPOSITION AND METHOD FOR MASS PRODUCTION, FORMULATION, QUALITY CONTROL, AND ENHANCING CROP GROWTH

(51) International classification	:C05F 11/08, C12N 1/14, G01N 33/24, C12Q 1/04, C12Q 1/68
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. R. Nithyatharani
Address of Applicant :Assistant Professor, PG and Research, Department of Microbiology, Shrimati Indira Gandhi College, No:369, College Road, Near Chathram Bus Stand, Tiruchirappalli, Tamil Nadu- 620002, India Tamil Nadu India
2)Ms. S. Subashini
(72)**Name of Inventor :**
1)Dr. R. Nithyatharani
2)Ms. S. Subashini

(57) Abstract :
Soil samples were collected from the Thanjavur district and analyzed for various physicochemical properties including texture, bulk density, pH, electrical conductivity, organic carbon, macronutrients and micronutrients. The fungal isolate Trichoderma sp. was identified and characterized through phylogenetic analysis such as fatty acid profiling, amino acid analysis and PCR-RAPD fingerprinting. The biofertilizer potential of the isolate was evaluated by screening for phosphate solubilization, IAA production and cyanide production. Mass production of Trichoderma sp. was carried out using low-cost agrowaste substrates through submerged fermentation, and the biofertilizer components were analyzed using GC-MS. The efficiency of the biofertilizer was further evaluated using pot culture experiments with Vigna mungo. Growth parameters such as total biomass, root length, shoot length and number of nodules were recorded. Yield and biochemical parameters including number of seeds per pot, leghemoglobin content and seed protein were also analyzed. The results indicated that Trichoderma sp. significantly improved plant growth, nodulation and yield, demonstrating its potential as an effective biofertilizer for su promotion stainable agriculture. Keywords: Trichoderma sp., Biofertilizer, Vigna mungo, Soil physicochemical properties, Pot culture method, Plant growth. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129103 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SENSOR-FUSION REHABILITATION WALKER FOR VALIDATED GAIT, PHASE-LOCKED POSTURE AND SUPPORT-DEPENDENCE MONITORING

(51) International classification	:A61B 5/11, A61H 3/00, A61H 1/02, A61H 3/04, A61B 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. K. N. Apinaya Prethi
Address of Applicant :Civil Aerodrome Post, Coimbatore – 641014, Tamil Nadu, India Coimbatore Tamil Nadu India
2)Kalaiarasan T R
3)G. Sakthi
4)Sudharson D
5)Dr. Praveenkumar K S
(72)**Name of Inventor :**
1)Sudharson D
2)G. Sakthi
3)Kalaiarasan T R
4)Dr. K. N. Apinaya Prethi
5)Dr. Praveenkumar K S

(57) Abstract :
A rehabilitation monitoring system (100) for a four-pod walker (102) comprises pod contact/load sensors (112A-112D), bilateral handle-load sensors (110L, 110R), an inertial measurement unit (114), vertically separated ranging sensors (116U, 116L), and a processor (118). The processor synchronizes the sensor signals to identify an ordered walker-lift, advance, stable-plant, support-transfer and patient-progression sequence and registers a rehabilitation gait cycle only when predetermined inter-sensor concordance is satisfied. Posture is estimated during selected validated gait phases from patient-to-walker ranges with compensation for walker orientation, while support dependence and asymmetry are determined from bilateral loads. A confidence gate rejects low-confidence events. Validated metrics are compared with clinician-defined targets and a patient-specific baseline to generate constrained feedback and longitudinal records for a patient/caregiver device (130) and clinician device (140). FIG. 3.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129115 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN REAL-TIME CYBER THREAT DETECTION AND AUTOMATED INTRUSION PREVENTION SYSTEM FOR SECURE NETWORKS

(51) International classification	:G06N 20/00, H04L 29/06, G06F 21/55, H04L 9/40, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM Institute of Science and Technology
Address of Applicant :Ramapuram Campus, Chennai, Tamil Nadu -600089 Tamil Nadu India
(72)**Name of Inventor :**
1)Ms. S. Aarthi
2)Mr. C. Santhosh Kumar
3)Dr. S. Nancy Lima Christy

(57) Abstract :
An artificial-intelligence-driven real-time cyber threat detection and automated intrusion prevention system and method for a secure network is disclosed. A data-acquisition module acquires, in real time, network traffic and related data, and a feature-extraction module extracts features therefrom. A threat-detection module comprising at least one trained machine-learning model analyses the features so as to detect a cyber threat, including a previously unseen threat, an anomaly-detection model trained upon normal network behaviour detecting novel and zero-day attacks as departures therefrom, and produces a detection output comprising an indication, a classification, and a confidence measure. An automated-prevention module, upon detection, selects and applies an inline preventive action at an enforcement point, such as blocking, terminating, rate-limiting, isolating, or increased monitoring, so as to prevent or mitigate the intrusion in real time. A governance arrangement governs the action in dependence upon the confidence and the severity, applying a decisive action to a high-confidence or severe threat and a measured action to a lower-confidence threat, rendering actions reversible and subject to human oversight, so as to prevent threats while limiting disruption to legitimate traffic. The system detects novel threats and prevents intrusions automatically and in real time, improving the security and availability of the network.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129116 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DOCSENSE-AI POWERED DOCUMENT INTELLIGENCE USING RAG

(51) International classification	:G06F 17/30, G06F 17/27, G06F 16/338, G06F 16/33, G06F 16/332
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR.S.PRAKASH
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, L&T BY-PASS, COIMBATORE, TAMIL NADU, INDIA-641062 Tamil Nadu India
2)Nithesh B S
3)Naveen V
4)Krithick S
5)Pragadeeshwaran P
(72)**Name of Inventor :**
1)DR.S.PRAKASH
2)Nithesh B S
3)Naveen V
4)Krithick S
5)Pragadeeshwaran P

(57) Abstract :
ABSTRACT DocSense AT is an intelligent document assistant designed to help users efficiently access and understand digital documents. The system allows users to upload documents and ask questions using natural language. It extracts text from documents and divides the content into smaller meaningful sections. Relevant document information is retrieved based on the user's _query. The system uses Retrieval-Augmented Generation (RAG) to provide context-aware responses. Artificial Intelligence and Large Language Models are used to generate relevant answers. ◆ Source references are provided to improve transparency and help users verify information. The system reduces the time and effort required to manually search lengthy documents. DocSense Al provides a simple and interactive conversational approach to document understanding. It can be useful ◆for academic, business, research, technical, and personal documents. Overall, the system provides an efficient solution for intelligent document information retrievaliddle; p

No. of Pages : 6 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129124 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DOUBLE PRONGED NEURO ELECTRO SALIVARY STIMULATOR

(51) International classification	:A61N 1/40, A61N 2/00, A61N 1/36, A61N 2/04, A61N 2/02	(71)**Name of Applicant :** **1)Dr. Ramya Ramadoss., MDS PhD.,** Address of Applicant :1170, 14th Street, Ashok nagar, Chennai -600083, India Chennai Tamil Nadu India (72)**Name of Inventor :** **1)Dr. Ramya Ramadoss., MDS PhD.,** **2)Dr. Rajkumar Krishnan, BSc, MDS.,PhD.,**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a double pronged neuro electro salivary stimulator (100) comprising a first main housing (101) and a second main housing (102) interconnected by an ear-guided positioning assembly (103). A left flexible gooseneck arm (104) extends from the first main housing (101) and supports a left magnetic treatment head (106), while a right flexible gooseneck arm (105) extends from the second main housing (102) and supports a right magnetic treatment head (107). The flexible gooseneck arms (104, 105) permit independent adjustment of the positions and orientations of the magnetic treatment heads (106, 107). In an electromagnetic embodiment, each magnetic treatment head comprises an electromagnetic coil (118) associated with a magnetic core (119) and accommodated within a treatment head casing (124). An electronic control arrangement comprising a microcontroller (117) and magnetic field driver circuit (120) controls electrical excitation of the electromagnetic coils (118). The device provides a neck-supported arrangement having independently positionable magnetic treatment heads.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129126 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART HOUSEHOLD WASTE SEGREGATION AND PICKUP SYSTEM

(51) International classification	:G06Q 50/26, G06F 17/30, G06Q 10/00, G06Q 10/06, G06Q 50/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)S. KIRUBA
Address of Applicant :SriShakthi Institute Of Engineering and Technology L&T Bypass Coimbatore Tamil Nadu India 641062 Tamil Nadu India
2)M.Subalakshmi
3)S.Subika
4)DS.Theyjakshaya
5)N.Vishrudha
(72)**Name of Inventor :**
1)S. KIRUBA
2)M.Subalakshmi
3)S.Subika
4)DS.Theyjakshaya
5)N.Vishrudha

(57) Abstract :
This invention presents a role-based web application for coordinating household waste pickup, collector operations, and administrative monitoring through a unified digital platform. The system allows household users to identify a waste category, select a preferred collection date, and submit a pickup request without relying on phone calls or informal messaging, Submitted requests are retained as scheduled records and are made available to an authorized eolleetorthrough A dedicated dashboard. The collector cart view the requesting user, waste category, and scheduled date, complete the collection, and record the quantity collected in kilograms. An administrative dashboard consolidates registered users and pickup records, separates completed and pending requests, and presents category-wise collection statistics through a reporting, interface. The application is implemented as a web-based client-server system using geaet.js, _ Node.js, _Express.js, and _MongODB,_ with role-based access .controlling household, collector, and 1 L-adininistiafdr:dperatiiiiis. The described system introduces a role-based web environment for managing household waste pickup requests and collection 'actiVities.. A household user selects a waste category and preferred collection date Airripigh a dedicated request interface, after which the request is retained as a 'pending pickup record. A collector Views available collection tasks, completes the physical pickup, and records: the collected quantity in kilograms An administrator supervises the complete set of records :through -dashboards showing' users, total pickups; completed and pending requests, and category-wise waste statistics. The system is implemented using'Reactjs, Node js, fixpressjs, and MongoDB, and provides a unified workflow Connecting household request creation, collector execution, -status management, and administrative reporting.

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129143 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : The Impact of Green Recruitment Practices on Sustainability Outcomes in Indian IT Firms

(51) International classification	:G06Q 10/10, G06Q 10/06, G06Q 10/00, G06Q 50/26, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Arunkumar A
Address of Applicant :Associate Professor, Department of Management Studies, International Institute of Business Studies, Karnataka-562157, India. Bangalore Karnataka India
2)Santosh Ramdas Shivane
3)Kalaivani.S
4)Rajeshwari Devulapalli
5)Dr. Niharika Singh
6)Prof. Sardar Ramesh Jagdeo
7)Mrs.N.Manju
8)Dr.K.A.Arokiaraj
(72)**Name of Inventor :**
1)Dr. Arunkumar A
2)Santosh Ramdas Shivane
3)Kalaivani.S
4)Rajeshwari Devulapalli
5)Dr. Niharika Singh
6)Prof. Sardar Ramesh Jagdeo
7)Mrs.N.Manju
8)Dr.K.A.Arokiaraj

(57) Abstract :
The Impact of Green Recruitment Practices on Sustainability Outcomes in Indian IT Firms ABSTRACT The present invention relates to an integrated framework for evaluating and enhancing the impact of green recruitment practices on sustainability outcomes in Indian Information Technology (IT) firms. The proposed invention focuses on incorporating environmentally responsible recruitment strategies, including digital hiring platforms, paperless documentation, virtual interviews, and green competency-based candidate selection. The framework aims to reduce recruitment-related environmental impacts, optimize resource utilization, and promote sustainable organizational practices. It introduces a Green Recruitment Sustainability Index (GRSI) to assess environmental performance through measurable indicators such as paper consumption, carbon emissions, energy efficiency, and employee environmental awareness. Furthermore, the invention integrates sustainability monitoring mechanisms with existing Human Resource Management Systems (HRMS) to facilitate data-driven decision-making and continuous improvement. The proposed framework supports the identification and selection of environmentally conscious employees who contribute to organizational sustainability initiatives. By aligning recruitment processes with environmental, social, and economic sustainability objectives, the invention seeks to enhance operational efficiency, strengthen corporate environmental responsibility, and encourage sustainable workforce development. The framework is adaptable to various organizational structures and recruitment requirements within the Indian IT sector, providing a systematic approach to achieving measurable sustainability improvements and supporting long-term environmental performance and responsible human resource management practices across organizations.

No. of Pages : 16 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129150 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GITHUB REPOSITORY ANALYZER

(51) International classification	:G06F 9/44, G06F 17/30, G06F 11/36, G06F 8/20, G06F 9/445
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)T. SATHYA
Address of Applicant :SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, SRI SHAKTHI NAGAR, L&T BY-PASS, CHINNIYAMPALAYAM POST, COIMBATORE -641062 9698146506 tsathyait@siet.ac.in Tamil Nadu India
2)Sakthivel S
3)Sivamanikandan K
4)Sushilram R
5)Tharunkumar R

(72)**Name of Inventor :**
1)T. SATHYA
2)Sakthivel S
3)Sivamanikandan K
4)Sushilram R
5)Tharunkumar R

(57) Abstract :

The rapid growth of software development and open-source platforms has resulted in a large number of GitHub repositories containing valuable information about projects, development activity, collaboration, and software maintenance. However, manually examining multiple repository metrics such as stars, forks, contributors, open issues, programming languages, and repository activity can be time-consuming. Existing GitHub pages provide individual repository information, but users may need to examine multiple metrics separately to understand the overall condition of a repository. The "RepoLens — GitHub Repository Analyzer" system addresses these challenges by providing a centralized platform for analyzing GitHub repositories. Users can register and log in to the system, enter a GitHub repository URL, and obtain important repository information through the GitHub REST API. The system retrieves details such as repository name, owner, description, primary language, stars, forks, contributors, open issues, watchers, license, and programming languages. It also calculates a rule-based repository health score out of 100 based on selected repository metrics. The analyzed results are stored in a MongoDB database, allowing users to access their previous analysis through the history module. Overall, the system provides a simple and organized platform for evaluating GitHub repositories efficiently. . A—GitHub Repository Anily—zer is a webbed platfo-FniirEgiEn-ed to 51Ie—e-f, ofgariize, and evaluate GitHub repository information efficiently. The system allows users to enter a GitHub repository URL and retrieves repository details using the GitHub REST API. It provides information such as repository name, owner, primary language, stars, forks, contributors, open issues, watchers, license, and programming languages. The system also implements a rule-based repository health score to provide a quick evaluation of the repository based on selected metrics. Users can register and log in securely, analyze repositories, and access their previous analysis records through the history module. MongoDB is used to store user information and repository analysis history. By combining web technologies, API integration, database management, and rule-based evaluation, RepoLens provides a simple and organized solution for GitHub repository analysis.

No. of Pages : 6 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026

(21) Application No.202641129151 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED SYSTEM FOR INTEGRATING SPEECH RECOGNITION AND RESPONSE GENERATION

(51) International classification	:G10L 15/22, G10L 15/26, G10L 15/18, G10L 15/06, G10L 15/183
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)LINGALA THIRUPATHI
2)DONTHA ARPANA
3)ENNAMALLA SRILEKHA
4)ROYYALA CHANDANA
5)SHAIK MANEESHA

(57) Abstract :
Disclosed herein is an AI-based system (100) for integrating speech recognition and response generation. The system (100) comprises a user interface (102), an audio acquisition unit (104), a speech recognition unit (106), a processing unit (108), and an output unit (118). The processing unit (108) comprises a data pre-processing module (110), a request coordination module (112), an artificial intelligence interpretation module (114), and a response handling module (116). The system captures spoken input, converts the captured audio into text using a speech-to-text service, and transfers the text to a generative artificial intelligence model for response generation. The generated natural-language response is processed and presented to the user through the output unit (118) and user interface (102). The system facilitates conversational interaction by coordinating speech recognition, text processing, artificial intelligence-based response generation, and response presentation.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129152 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-BASED SYSTEM AND METHOD FOR STOCK MARKET ANALYSIS

(51) International classification	:G06Q 40/04, G06N 20/00, G06Q 40/00, G06Q 40/06, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)NYATHA VINAYKUMAR
2)GADDAM ANJUSHYA
3)ROHIT CHAND R N
4)GOLLAPALLI SANJANA

(57) Abstract :
Disclosed herein is an AI-based system (100) for stock-market analysis. The system (100) comprises a user interface (102), a data acquisition unit (104), a market-data interface (106), a data storage unit (108), a processing unit (110), and an output unit (126). The processing unit (110) comprises a data pre-processing module (112), a technical-analysis module (114), an artificial intelligence interpretation module (116), a recommendation-generation module (118), a validation module (120), a visualization module (122), and a presentation module (124). The system (100) retrieves stock-market data corresponding to a stock identifier, validates and organizes the retrieved data into a structured time series, and calculates a moving-average indicator and a price difference between a latest closing price and a corresponding moving-average value. Further, the system (100) generates a natural-language explanation using a pretrained large language model and presents the processed data, technical indicators, analysis results, and graphical representations through the output unit (126).

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129153 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A COMPOSITION AND METHOD FOR PREPARING COMPRESSED STABILIZED BRICKS

(51) International classification	:C04B 18/14, C04B 28/04, C04B 18/04, C04B 18/08, C04B 18/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)MD ARSHAD
2)D MANI
3)D HARSHITH RAO
4)SHRAVAN KUMAR

(57) Abstract :
ABSTRACT Disclosed herein is a composition (100) and a method (200) for preparing compressed stabilized bricks using ordinary Portland cement (OPC) (102), ground granulated blast-furnace slag (GGBS) (104), manhole sludge (106), and quarry dust (108). The composition (100) comprises 6% by weight of OPC, 9% by weight of GGBS, manhole sludge in an amount selected from 15-30% by weight, and quarry dust in a corresponding amount selected from 55%-70% by weight of a dry mixture, respectively. The method (200) comprises measuring and mixing the constituent materials in predetermined proportions, adding water to obtain a moldable mixture, introducing the mixture into a mould, compacting the mixture, and curing the compacted mixture to obtain a stabilized brick. The prepared bricks are evaluated for compressive strength after curing. The composition facilitates the utilization of manhole sludge and quarry dust in brick production while reducing reliance on conventional construction materials.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129160 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Development of a Paper-Based Colourimetric Sensor for Point-of-Care Detection of SGPT through Metal–Ligand Complex Formation

(51) International classification	:G01N 31/22, G01N 33/52, G01N 21/78, G01N 33/84, C12Q 1/52
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Amrita Vishwa Vidyapeetham
Address of Applicant :Coimbatore Campus, Coimbatore- 641 112, Tamil Nadu, India Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Arasalakshmi Kaleeswaran
2)Parakkal, Edachana Resmi
3)Thekkedath, Gopalakrishnan Satheesh Babu
4)Punathil, Vasu Suneesh

(57) Abstract :
The present invention relates to a paper-based colourimetric sensor for point-of-care detection of serum glutamic pyruvic transaminase (SGPT). The sensor comprises vertically stacked paper layers including a substrate layer containing alanine and α-ketoglutarate, a thiosemicarbazide-containing layer, and a detection layer containing a cobalt-citrate reagent, optionally together with a plasma separation membrane for whole-blood analysis. SGPT catalyses transamination of alanine and α-ketoglutarate to generate pyruvate. The pyruvate reacts with thiosemicarbazide to form pyruvate thiosemicarbazone, which interacts with the cobalt-citrate reagent to generate a cobalt–pyruvate thiosemicarbazone complex accompanied by a colour change from pink to reddish-brown. The colour response increases with SGPT concentration and can be visually interpreted or quantitatively analysed using optical imaging and image-processing software. In demonstrated embodiments, the sensor provided a response over approximately 10–210 U/L SGPT and was evaluated using serum and whole-blood samples. The device is fabricated using wax screen-printing and provides a compact, low-volume and potentially scalable platform for point-of-care SGPT testing.

No. of Pages : 25 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129163 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A BIOMETRIC-AUTHENTICATED AUTOMATIC RATION DISPENSING SYSTEM

(51) International classification	:G07C 9/00, G06K 9/00, G06F 21/32, G06Q 20/40, G07F 7/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)MILAN KUMAR MOTTA
2)DR. RAHUL PACHORI

(57) Abstract :
Disclosed herein is a biometric-authenticated automatic ration dispensing system (100) that comprises an Input Interface (102), a Processing Unit (104), a Fingerprint Sensor (124), an RFID Reader (126), a Load Cell and Amplifier (128), a Servo Motor and Dispensing Gate (130), a Hopper and Collection Container (132), a Keypad (134), a Liquid Crystal Display (136), a Wi-Fi Communication Module (138), a Power Supply Unit (140), a Cloud Platform (142), a Smartphone Monitoring Terminal (144), a Beneficiary Database (146), an Output Interface (148), and a Communication Interface (150). The Processing Unit (104) comprises a Pre-processing Module (106), an Authentication Module (108), a Quantity Selection Module (110), an Entitlement Verification Module (112), a Closed-Loop Weight Control Module (114), a Dispensing Control Module (116), a Transaction Recording Module (118), a Real-Time Monitoring Module (120), and a Consumption Tracking Module (122).

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129164 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ARTISANA INSIGHTS E-COMMERCE AND ANALYTICS SYSTEM FOR ARTISAN EMPOWERMENT

(51) International classification	:G06Q 30/06, G06Q 30/02, G06Q 10/08, G06Q 30/00, G06K 9/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)G. ANUSHA
2)S. RUPA
3)N. VARUDHINI
4)A. ABHINAV
5)DR.T. VENKAT NARAYANA RAO

(57) Abstract :
Disclosed herein is an Artisana Insights e-commerce and analytics system (100) for artisan empowerment that comprises a user interface (102), an authentication and access interface (104), a local database (106), a product image storage unit (108), and a processing unit (110). The processing unit (110) comprises a user management module (112) configured to manage user accounts; a product and inventory management module (114) configured to maintain product records; an order processing module (116) configured to manage transactions; a business analytics module (118) configured to analyze sales performance; a customer segmentation module (120) configured to group customers using K-Means clustering; an offline FAQ chatbot module (122) configured to provide predefined responses; a data synchronization module (124) configured to synchronize locally stored information; and a dashboard generation module (126) configured to present business insights. The system (100) facilitates offline-first e-commerce operations, improves data accessibility, and supports informed business decisions.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129165 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED ARTIFICIAL INTELLIGENCE-BASED HEALTHCARE MANAGEMENT SYSTEM

(51) International classification	:G06Q 50/22, G16H 40/20, G16H 10/60, G06Q 10/06, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)LINGALA THIRUPATHI
2)MANVITHA POLA
3)TEJASWI ARELLA
4)RUTHIWIK TADAKA
5)RASHMIKA REDDY KOTHA

(57) Abstract :
Disclosed herein is an integrated artificial intelligence-based healthcare management system (100) for providing healthcare-related information, predictive analysis, and service coordination. The system (100) comprises a user interaction unit (102), a user access management unit (104), an information processing unit (106), an information access interface (120), a communication and integration module (122), and an information management unit (124). The information processing unit (106) comprises an input analysis module (108), a predictive analysis module (110), an interactive assistance module (112), a service information and selection module (114), a scheduling and coordination module (116), and an output generation and presentation module (118). The system (100) processes user-provided input data using an artificial intelligence model to generate predictive information relating to possible health conditions, generates informational responses to user queries, facilitates healthcare service-provider selection, and coordinates scheduling requests. The system (100) further facilitates information exchange with external services and maintains relevant service-request information.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129166 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A RUNTIME ARTIFICIAL INTELLIGENCE AGENT GOVERNANCE AND SELF-CORRECTION SYSTEM

(51) International classification	:G06N 20/00, G06F 8/41, G06N 5/04, G06N 3/04, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)J V P UDAYA DEEPIKA
2)CHITHALURI. HEMANTH
3)BANDARI.VARUN KUMAR
4)T. VENKAT NARAYANA RAO
5)P. SRAVANTHI

(57) Abstract :
Disclosed herein is a runtime artificial intelligence agent governance and self-correction system (100) that comprises a host processor (102), an accelerator processor (104), a processing unit (106), and a dedicated context memory buffer (122), a system memory (124), and a network interface controller (126). The processing unit (106) comprises an agent profile composition module (108), a pre-processing module (110), an intent declaration module (112), a constraint evaluation module (114), a correction vector computation module (116), a context recomposition module (118), and a runtime identity management module (120). The system (100) further comprises a fragment repository (128), a blueprint registry (130), a constraint manifest object (132), a state ledger (134), an execution model interface (136), a memory management unit (138), and a communication interface (140).

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129167 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A WHATSAPP-BASED INTELLIGENT RESUME PARSING SYSTEM

(51) International classification	:G06Q 10/10, G06Q 10/06, G06V 30/10, G06F 17/27, G06K 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)NEELA SANTHOSHI
2)S. SWETHA

(57) Abstract :
Disclosed herein is a WhatsApp-based intelligent resume parsing system (100) that comprises a WhatsApp Communication Interface (102), a Resume Acquisition Unit (104), and a Processing Unit (106) comprising a Document Pre-processing Module (108), a Document Processing and OCR Module (110), a Multilingual NLP Module (112), a Resume Information Extraction Module (114), a Field Validation and Confidence Assessment Module (116), a Conversational Follow-up Module (118), a Candidate Profile Generation Module (120), a Job Matching and Ranking Module (122), and a Recruitment Workflow Automation Module (124). The system (100) further comprises a Memory Unit (126), a Candidate Profile Database (128), a Recruiter Interface (130), and a Notification Unit (132). The system (100) extracts and validates structured candidate information from multilingual resumes received through WhatsApp, requests missing or uncertain information through conversational interactions, generates standardized candidate profiles, and facilitates automated job matching, candidate ranking, and recruitment workflow management.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129180 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MODULAR EVAPORATIVE COOLING CARTRIDGE ASSEMBLY WITH DIFFERENTIAL MULTI-ZONE WATER DISTRIBUTION FOR AGRICULTURAL PRODUCE PRESERVATION

(51) International classification	:A23B 7/04, A01G 9/24, A01F 25/16, F24F 5/00, A23B 7/152
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GANDHAM YEDUKONDALU
Address of Applicant :Associate Professor, Department of Mechanical Engineering, Koneru Lakshmaiah Education Foundation, Greenfields, Vaddeswaram Andhra Pradesh India
2)Koneru Lakshmaiah Education Foundation
(72)**Name of Inventor :**
1)G. Yedukondalu
2)JASHWANDIKKA A.L
3)PUTHIREDDY MANICA
4)Pavani Chitrapu
5)Bondili Harshith Singh

(57) Abstract :
The present invention relates to a modular evaporative cooling system for agricultural produce preservation. The system comprises an insulated storage chamber, removable multilayer cooling cartridges containing coconut coir, sand and rice-husk ash, and a water distribution mesh. A three-zone hydration manifold with adjustable flow restrictors supplies different quantities of water to the upper, middle and lower cooling zones. Water is supplied from a reservoir by gravity or an optional DC pump. Air enters through a lower inlet, passes through the storage chamber containing ventilated produce crates, and exits through an upper plenum and passive solar chimney, with an optional DC fan to assist airflow. Excess water is collected through a drainage channel, filtered and stored in a recovery tank for reuse. Optional sensors and a microcontroller monitor temperature, relative humidity, cartridge moisture and water level. The modular assembly integrates differential water distribution, evaporative cooling, airflow management and water recovery for short-term agricultural produce preservation.

No. of Pages : 23 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129199 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A HERBAL VAGINAL WASH FORMULATION BASED ON PHYTOCHEMICAL EXTRACTS OF EMILIA SONCHIFOLIA AND AERVA LANATA, AND A PROCESS FOR THE PREPARATION THEREOF

(51) International classification	:A61K 36/185, A61K 36/00, A61Q 19/10, A61P 15/02, A61K 8/97
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.Revathy S
Address of Applicant :Dr.Revathy S, Assistant Professor, Department of Chemistry, St. Xavier's College for Women (Autonomous), Aluva, Kerala, India – Indian Kerala India
2)Dr. Stella KA
3)Dr. Newly Joseph
4)Dr. Linda Louis
(72)**Name of Inventor :**
1)Dr.Revathy S
2)Dr. Stella KA
3)Dr. Newly Joseph
4)Dr. Linda Louis
5)Theres Benny
6)Atlin Maria K.S

(57) Abstract :

ABSTRACT A HERBAL VAGINAL WASH FORMULATION BASED ON PHYTOCHEMICAL EXTRACTS OF EMILIA SONCHIFOLIA AND AERVA LANATA, AND A PROCESS FOR THE PREPARATION THEREOF The present invention relates to a herbal vaginal wash formulation and a process for its preparation. The formulation comprises a combined, water-free glycerin extract of Emilia sonchifolia and Aerva lanata, incorporated into an aqueous base containing sodium chloride and buffered with sodium bicarbonate, with the pH adjusted to about 4.0 to 4.5 using a suitable acidulant, so as to correspond to the natural physiological pH of the vagina. The plant sources and their principal flavonoid constituents (luteolin, quercetin, kaempferol, and rutin) were selected on the basis of molecular docking studies demonstrating favourable binding affinity towards protein targets (PDB ID: 6BIQ and 1EAG) associated with common vaginal pathogens, and the formulation and its constituent extracts were shown to exhibit significant, dose-dependent anti-inflammatory activity by the Human Red Blood Cell (HRBC) membrane stabilization assay, comparable to a standard anti-inflammatory drug. ADME evaluation further confirmed favourable drug-likeness of the principal constituents. The invention provides a safe, effective, and scientifically substantiated natural alternative to synthetic vaginal wash products, prepared by a simple and reproducible process suitable for scaled manufacture.

No. of Pages : 15 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129239 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PROFILE-ADAPTIVE MULTIMODAL DOCUMENT RENDERING WITH SYNCHRONIZED SENSORY OUTPUTS

(51) International classification	:G06V 30/10, G06F 17/24, G06V 30/413, G06F 17/21, G06V 30/414
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Amrita Vishwa Vidyapeetham
Address of Applicant :Amrita Vishwa Vidyapeetham, Mysuru Campus, #114, 7th Cross, Bogadi II Stage, Mysuru, Karnataka - 570026, India. Mysuru Karnataka India
(72)**Name of Inventor :**
1)JAYAKUMARI BALAKRISHNAN, Bipin Nair
2)KONGASSERY, Sreesankar
3)RATHEESH, Niranjan
4)PUTHIYOTILL, Anuranch

(57) Abstract :
The present disclosure provides a system (100) for profile-adaptive multimodal document rendering. Conventional accessibility arrangements impose a uniform presentation, whereby a form that can serve one category of user can hinder another category of user, and can propagate an extraction error into an output. The system (100) can receive a document image at an image capture device (102), can accept a profile signal at a profile selector (106), can derive a character sequence (302) at a text derivation unit (108), and can shape the character sequence at a parameter-setting backend (110) through a token-count limit (112) upon a first-state signal and a grapheme-spacing value (114) upon a second-state signal. A timing coupler (134) can drive an audio stream, a six-dot Unicode Braille matrix, and a word-level highlight (132). Unlike conventional arrangements, the system can resolve a disability conflict and can sustain a synchronized delivery for an accessible reading.

No. of Pages : 32 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129240 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM FOR MONITORING DEGRADATION IN HARDWARE SYSTEM AND METHOD THEREOF

(51) International classification	:G06N 3/08, G06N 3/04, G05B 23/02, G06N 20/00, G01R 31/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)RAVI PRAKASH DWIVEDI
2)RICHA DUBEY
3)RITAM PANDEY
4)HEMANT
5)SHIVAM YADAV
6)ANIKET RAJ

(57) Abstract :
The present disclosure discloses a system (100) for monitoring degradation in a hardware system. A sensing unit (102) having a plurality of sensors (102-1) acquires real-time signals. A response monitoring unit (104) applies controlled charge-discharge cycles and measures electrical response to determine a time constant (τ), wherein temporal variation indicates degradation parameters. A feature extraction unit (106) generates a feature vector and a time constant drift parameter. A processor (108) executes instructions stored in a memory (110) to detect anomalies, filter noise, estimate classification and severity, and generate degradation patterns using an anomaly detection model (108-1), an autoencoder model (108-2), an XGBoost model (108-3), and a Liquid Neural Network (108-4). A confidence-based fusion unit (108-5) combines outputs to generate an assessment report.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129269 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR INTELLIGENT ROUTE-AWARE TRAVEL ASSISTANT

(51) International classification	:G01C 21/34, G01C 21/36, H04W 4/029, H04W 4/02, H04W 4/021
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM UNIVERSITY
Address of Applicant :Amaravati, Mangalagiri, Andhra Pradesh-522502, India Guntur Andhra Pradesh India
(72)**Name of Inventor :**
1)ANANDARAO, SARVANI

(57) Abstract :
ABSTRACT The present disclosure envisages a system (100) for intelligent route-aware travel assistant. A reminder management module (110) enables a user to add reminders through a user interface (104), and a mapping engine (112) maps each reminder to a location category. A user positioning module (106) determines real time user location, and a route-determination module (114) determines a travel route to a destination location. A location identification module (116) determines a service location of the location category along or within a predetermined proximity of the travel route. A context aware decision engine (118) determines, based on the reminders and the real time user location relative to the service location, whether an alert is to be generated, and contextually validates the alert. An alert generation module (120) renders the validated alert. A habit learning module (122) learns user behaviour patterns, and a repository (124) stores user data and history.

No. of Pages : 30 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129270 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MICROWAVE-ASSISTED CO-PYROLYSIS SYSTEM AND PROCESS THEREOF

(51) International classification	:C10B 53/02, C10B 53/00, C10G 1/00, H05B 6/64, H05B 6/80
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM UNIVERSITY
Address of Applicant :Amaravati, Mangalagiri, Andhra Pradesh-522502, India Guntur Andhra Pradesh India
(72)**Name of Inventor :**
1)SELVASEMBIAN, RANGABHASHIYAM
2)POTNURI, RAMESH

(57) Abstract :
ABSTRACT The present disclosure relates to a microwave-assisted co-pyrolysis system (1000) and a microwave-assisted co-pyrolysis process for converting biomass feedstock and carbonaceous waste material into bio-oil, syngas and biochar. The microwave-assisted co-pyrolysis system (1000) comprises a feed preparation sub-system (100), a microwave-assisted co-pyrolysis reactor (200), a microwave irradiation sub-system (300), an inert atmosphere generation sub-system (400), and a product recovery sub-system (500). The microwave irradiation sub-system (300) supplies microwave energy to the microwave-assisted co-pyrolysis reactor (200), wherein recycled spent lithium-ion battery anode material functions as a microwave susceptor to facilitate volumetric heating of a feed mixture under an oxygen-deficient atmosphere. The process is simple, energy-efficient and sustainable, and provides improved microwave energy utilisation, enhanced thermal conversion efficiency, and increased recovery of bio-oil, syngas and biochar.

No. of Pages : 63 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129287 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DRY BIOCIDAL GEL-FORMING COMPOSITION AND METHOD FOR DECONTAMINATION, SOLIDIFICATION OF HUMAN AND ANIMAL BIOFLUIDS

(51) International classification	:C09K 3/32, C02F 1/50, A01N 25/04, A01N 25/10, A01N 59/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHERUBAL RAVEENDRAN MENON
Address of Applicant :TF804, PURVA MIDTOWN, 3RD A MAIN ROAD, KASTURINAGAR, BANGALORE-560016, KARNATAKA, INDIA BANGALORE Karnataka India
2)GEETHA RAVEENDRAN
(72)**Name of Inventor :**
1)CHERUBAL RAVEENDRAN MENON

(57) Abstract :
A dry, free-flowing biocidal and gel-forming composition for treating biological fluid spills on hard surfaces comprises, in a presently preferred finished formulation, 20 wt.% sodium dichloroisocyanurate (NaDCC), 70 wt.% superabsorbent polymer (SAP), 5 wt.% carboxymethyl cellulose (CMC), 3 wt.% hydrophobic fumed silica and 2 wt.% calcium sulfate. The SAP and CMC form a 75 wt.% hydrogel-forming component, while the hydrophobic fumed silica and calcium sulfate form a 5 wt.% rheology-modifying component. The composition is applied directly to a spill without addition of external liquid. Water present in the spill dissolves the NaDCC and hydrates the SAP/CMC component to form a cohesive gel, while the rheology-modifying component facilitates mechanical removal with reduced adhesion and smearing. In embodiments employing coated particles, the NaDCC-containing and SAP/CMC-containing fractions are separately coated to reduce premature interaction during storage and permit activation upon contact with the spill. FIG. 2

No. of Pages : 18 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129293 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI BASED HOSPITAL APPOINTMENT AND TRAFFIC OPTIMISED SCHEDULING SYSTEM

(51) International classification	:G16H 40/20, G06Q 50/22, G06Q 10/06, G06Q 10/02, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ms. Pamela Bera
Address of Applicant :Assistant Professor, Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning), AMC Engineering College, Bannerghatta Rd, Kalkare, Bengaluru, Karnataka 560083 Karnataka India
2)Mrs. Vinaya S Kavalgi
3)Mr. Sonam Kumar
4)Mrs. Ambili K
5)Mrs. Nallamilli Kavita Reddy
6)Dr. Swasti Patel
7)Mrs. Bhavya K Bharathan
8)Ms. Soumya K
9)Mr. Prashanth Pudoor
10)Ms. Sarika V
(72)**Name of Inventor :**
1)Ms. Pamela Bera
2)Mrs. Vinaya S Kavalgi
3)Mr. Sonam Kumar
4)Mrs. Ambili K
5)Mrs. Nallamilli Kavita Reddy
6)Dr. Swasti Patel
7)Mrs. Bhavya K Bharathan
8)Ms. Soumya K
9)Mr. Prashanth Pudoor
10)Ms. Sarika V

(57) Abstract :
ABSTRACT OF THE INVENTION The present invention relates to an AI-based hospital appointment and traffic optimised scheduling system for generating and dynamically managing a patient's healthcare journey. The system includes patient registration, symptom-based specialty recommendation, hospital and doctor selection, traffic acquisition, travel-time estimation, appointment optimization, departure-time calculation, dynamic route recommendation, ambulance services, dashboards, notifications, historical data, and continuous optimization. The invention particularly comprises a Time-Budget Engine configured to determine time requirements associated with travel, traffic delay, appointment, arrival and other healthcare activities; a Healthcare Plan Compiler configured to convert healthcare requirements and scheduling information into a structured healthcare plan; and a Cross-Hospital Plan Migration mechanism configured to migrate the healthcare plan to an alternative hospital when the original plan becomes unsuitable or infeasible. The system recalculates the affected time, appointment, travel and route parameters, thereby providing an integrated and adaptive healthcare journey planning system.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129311 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTOMATED TONGUE COATING ANALYSIS SYSTEM FOR HEALTH STATUS ASSESSMENT

(51) International classification	:A61B 5/00, G06T 7/00, G06T 7/11, G16H 30/40, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India
2)DR AKSHATHA SHETTY
3)DR. CHINMAI SHETTY
4)DR. URVASHI SHETTY
(72)**Name of Inventor :**
1)DR AKSHATHA SHETTY
2)DR. CHINMAI SHETTY
3)DR. URVASHI SHETTY

(57) Abstract :
Disclosed herein is an automated tongue coating analysis system (100) for health status assessment, comprising an image acquisition unit (102) and an adjustable holder (104). The system (100) comprising a processing unit (106) further comprising an input module (108) and an image assessment module (110) to reject or enhance the captured images. The processing unit (106) comprising a tongue segmentation module (112) to identify and isolate a plurality of tongue region from surrounding facial structures, and a feature extraction engine (114) to extract a set of hierarchical image features from each anatomical sub-region. The processing unit (106) comprising a fusion module (116) to assign dynamic confidence score to assign dynamic confidence score to each of the trained models and a classification module (118) to simultaneously determine two or more tongue coating characteristics. The processing unit (106) further comprising an interpretation engine (120) to correlate the classified tongue coating characteristics.

No. of Pages : 39 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129315 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BANANA STEM-FIBRE BASED BIODEGRADABLE THREAD COMPOSITION AND METHOD THEREOF

(51) International classification	:D02G 3/04, D02G 3/44, D02G 3/02, D01F 2/00, C08L 97/02	(71)**Name of Applicant :** **1)NITTE (DEEMED TO BE UNIVERSITY)** Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India **2)DR. RAKSHA JAIN**
(31) Priority Document No	:NA	**3)ISHWARYA**
(32) Priority Date	:NA	**4)DR. ABHILASH**
(33) Name of priority country	:NA	(72)**Name of Inventor :**
(86) International Application No Filing Date	: :01/01/1900	**1)DR. RAKSHA JAIN** **2)ISHWARYA**
(87) International Publication No	: NA	**3)DR. ABHILASH**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed herein is a banana stem fibre-based biodegradable thread composition (100), the composition (100) comprising a plant-derived cellulose fibre (102) provide fibre strength and flexibility to the biodegradable thread, an moisture-retaining agent (104) reduce rapid moisture loss and prevent brittleness within the biodegradable thread, a moisture stabilizing agent (106) regulate moisture content of biodegradable thread during storage and handling. The composition may include the plant-derived cellulose fibre (102) is a banana stem fibre form continuous strands suitable for ultra-thin biodegradable thread formation. The composition may include the moisture-retaining agent (104) is an aloe vera-derived gel provide a continuous treatment layer over the plant-derived cellulose fibre (102) and enhance adhesion of the aloe vera-derived gel onto the fibre surface. The composition may include the moisture stabilizing agent (106) is selected from glycerine, natural gums, cellulose-based stabilizers, citric acid and combinations thereof, configured to inhibit microbial deterioration of the treated biodegradable thread.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129316 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PHISHING UNIFORM RESOURCE LOCATOR (URL) DETECTION SYSTEM

(51) International classification	:H04L 29/06, G06N 20/00, H04L 9/40, G06F 16/955, G06F 21/55	(71)**Name of Applicant :** **1)NITTE (DEEMED TO BE UNIVERSITY)** Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India **2)S SHYAM KUMAR** (72)**Name of Inventor :** **1)S SHYAM KUMAR**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed herein is a phishing uniform resource locator (URL) detection system (100) that comprises a User Device (102), a Communication Interface (104), a Data Acquisition Unit (106), a Memory Unit (108), and a Processing Unit (110). The Processing Unit (110) comprises a Pre-processing Module (112), a Feature Extraction Module (114), a Temporal Segmentation Module (116), a Machine Learning Module (118), and an Evaluation Module (120). The system (100) further comprises a Display Unit (122), a Data Storage Unit (124), and an Output Interface (126). The Pre-processing Module (112) prepares acquired URL information for analysis, the Feature Extraction Module (114) extracts heuristic features, and the Temporal Segmentation Module (116) performs temporal analysis of the extracted heuristic features. The Machine Learning Module (118) classifies phishing URLs based on the extracted heuristic features and the temporal analysis, while the Evaluation Module (120) evaluates classification performance.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129320 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-ENABLED WEARABLE AUTOMATIC CPR AND EMERGENCY RESPONSE SYSTEM

(51) International classification	:A61H 31/00, A61B 5/00, A61N 1/39, A61B 5/0205, A61B 5/0245
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)S. Gopinath
Address of Applicant :Professor and Head- Research & Extensions, Department of ECE, Karpagam Institute of Technology, Coimbatore - 641 105. Tamil Nadu India
2)Mrs. R. Subbulakshmi
3)Twinaagash C
4)Varshini B
5)Shreyas Abiraman S
6)Nithin S
7)Udhayaragavan C
8)Ram S
9)Vinotha G
10)Rishwana Begam R
11)Dr. V. Rasikha
12)Dr. M. Sathiya
(72)**Name of Inventor :**
1)S. Gopinath
2)S. Gopinath
3)Twinaagash C
4)Varshini B
5)Shreyas Abiraman S
6)Nithin S
7)Udhayaragavan C
8)Ram S
9)Vinotha G
10)Rishwana Begam R
11)Dr. V. Rasikha
12)Dr. M. Sathiya
13)Mrs. R. Subbulakshmi

(57) Abstract :
CJack is an AI-enabled wearable emergency response system designed to provide rapid assistance during critical cardiac emergencies by combining physiological monitoring, automated chest-compression assistance, emergency communication, and real-time status feedback within a portable wearable platform. Sudden cardiac emergencies require immediate and consistent intervention, while professional medical assistance may be delayed during the critical initial period. Conventional automated CPR systems are generally standalone equipment intended for controlled clinical or emergency-service environments. This invention addresses these limitations through an integrated wearable system capable of monitoring physiological parameters, assessing emergency conditions, and providing controlled CPR assistance. The system incorporates an AD8232 ECG sensor for acquiring cardiac electrical signals, a MAX30102 pulse oximeter for measuring heart rate and blood oxygen saturation (SpO_2), and an MPU6050 accelerometer-gyroscope for detecting body movement, orientation, and motion conditions. A force and depth sensing arrangement monitors the mechanical chest-compression parameters. The collected sensor information is processed by an ESP32-based controller, which performs signal processing, multi-sensor assessment, emergency-state evaluation, and control of the compression mechanism. Upon confirmation of a critical emergency condition, the controller activates a motorized compression mechanism comprising a high-torque DC geared motor, eccentric cam/rotor arrangement, compression pad, and mechanical guide structure to provide controlled chest compressions. The compression force and depth are continuously monitored through feedback sensors, enabling adjustment of the actuation mechanism. The system further incorporates GPS and LoRa-based communication for transmitting emergency location and device status information to designated responders. A display and voice/status indication system provides real-time information regarding vital parameters, compression status, battery condition, and emergency state. An optional oxygen-assist arrangement may provide controlled airflow through a connected mask. The device is configured as a rapid-wrap wearable chest-support structure having adjustable straps, alignment markers, protective housing, integrated electronics, mechanical components, and a rechargeable power system. The system is intended to provide immediate CPR assistance and connected emergency support until trained medical personnel or emergency responders take over. The invention provides a compact and integrated approach to emergency preparedness by combining health monitoring, emergency assessment, CPR assistance, feedback control, location communication, and paramedic handover in a single wearable platform.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129326 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR ENABLING A CROSS-LAYER IDENTITY VIRTUALIZATION FOR A DEVICE

(51) International classification	:H04L 29/06, H04L 29/08, H04L 12/24, G06F 21/62, G06F 16/957	(71)**Name of Applicant :** **1)HASSAN S** Address of Applicant :C 94, KANDASAMY SALAI, JOTHINAGAR D COLONY, POLLACHI - 642001, TAMIL NADU, INDIA POLLACHI Tamil Nadu India (72)**Name of Inventor :** **1)HASSAN S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A system and method for enabling a cross-layer identity virtualization for a device is disclosed. The system virtualizes, one or more hypothetical devices with a common identity profile. System executes, browsing session configured to operate within an isolated deterministic container and derives, within the browsing session, fingerprint attributes of devices present in common identity profile for identity virtualization of each of devices. System maintains, consistency of fingerprint attributes among browser fingerprint data, WebRTC outputs, IP characteristics, time-zone, locale, and hardware-level attributes based on characteristics of devices stored in a master database. System dynamically switches, between browser engines present within a unified browser framework upon detection of inconsistencies in fingerprint attributes and generate, fingerprint profile comprising fingerprint attributes and updates, fingerprint profile based on released hardware, operating systems, or browser environment. System simulates, user interaction behaviors to maintain consistency between generated fingerprint profile and behavior patterns. FIG. 3

No. of Pages : 42 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129401 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Examining Objective Test Patterns and Approaches to Enhance Competence

(51) International classification	:G09B 19/00, G09B 7/00, G09B 7/02, A61B 5/00, G09B 7/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JOHN SUNIL MANOAH
Address of Applicant :S.A Engineering college (Autonomous),Dept of English, Veeraghavapuram ,Chennai-600077. Taminadu CHENNAI Tamil Nadu India
2)T SENTHIL KUMAR
3)S.A Engineering college
(72)**Name of Inventor :**
1)JOHN SUNIL MANOAH
2)T SENTHIL KUMAR

(57) Abstract :
The most popular method of assessment today, according to modernized assessors, is the objective type of exams. Since the sources of malpractice can be controlled, it gives the assessors 100% of satisfaction. A person's intended efforts and actions to attain or accomplish the purpose are referred to as their objective. The same meaning was dragged to name a specific type of testing that allows for easy measurement of the quality and quantity of an individual. The objective type of exam has got various advantages from both ends, with the objective type aiming to keep the substantial type in shorter format. In order to provide hypothetical strategies to get through the said format, we will present the system of objective type of questions and its exam pattern in this article.

No. of Pages : 6 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129416 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED EMPLOYABILITY READINESS SCORING AND PERSONALIZED SKILL-GAP RECOMMENDATION PLATFORM FOR UNDERGRADUATE STUDENTS

(51) International classification	:G06Q 50/18, G06N 20/00, G06Q 50/20, G06Q 10/06, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)N Sumedha Arya
2)Dr Salla Nityanth Kumar

(57) Abstract :
The present invention relates to an artificial intelligence-based undergraduate employability-readiness platform. A mapping engine associates evidence descriptions with canonical skills in a versioned ontology. An evidence processor checks source verification, evidence type, recency, and applicability before estimating proficiency. A readiness processor compares eligible evidence with a target-role profile and produces attainment together with evidence coverage and unresolved requirements. A recommendation planner distinguishes supported skill gaps from missing evidence, selecting learning resources or assessment actions accordingly. Resources are ordered using prerequisite relationships. Each plan references the role, ontology, and evidence versions used, and current versions are checked before assignment. Reassessment events update proficiency separately from learning completion. The platform provides traceable role-specific preparation and learning recommendations.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129418 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Self-Aware Recalibration System and Method for Calibrated EEG Seizure Detection

(51) International classification	:A61B 5/0476, A61B 5/374, A61B 5/00, A61B 5/372, G06N 3/08	(71)**Name of Applicant :** **1)Koneru Lakshmaiah Education Foundation** Address of Applicant :Green Fields, Vaddeswaram, Guntur, Andhra Pradesh, India- 522302 Vaddeswaram Andhra Pradesh India (72)**Name of Inventor :** **1)Mr. Kondanna Kanamaneni** **2)Dr. Venkata Raju K**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A computer-implemented system (100) for EEG seizure detection pre-processes EEG into windows, encodes each window by an attention-based encoder (108) and classifies it (112). A confidence-gated trigger computes a temperature-scaled confidence and accepts predictions at or above a threshold. For low-confidence windows, a self-aware recalibration module (114) retrieves the top-k nearest embeddings from an indexed memory bank (116), weights them by a normalised radial-basis-function kernel, performs a few weighted gradient updates of the classifier parameters on the support set, reclassifies the window with the adapted parameters and logs the result to the memory bank. An optional Reptile outer update is applied in offline training. The output (118) achieves high accuracy with substantially reduced calibration error.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129423 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A DIGITAL SYSTEM FOR TEACHING -LEARNING PERFORMANCE ASSESSMENT NEP / OUT COME BASED EDUCATION

(51) International classification	:G09B 7/00, G09B 7/02, G06Q 50/20, G09B 7/04, G06F 17/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr A. Mary Noya Leena
Address of Applicant :Principal, Dr.SNS College of Education, Saravanampatti, Coimbatore, Tamil Nadu, 641049, India. Tamil Nadu India
2)Mrs R. Amutha
3)Dr. G. Nirmaladevi
4)Dr Anil Trimbakrao Gaikwad
5)Dr. N. Prathiba Lakshmi
6)P. Tamilarasan
7)Prof. Vandana Vinayak Badage
8)R. Nirmala
9)Indulekha K V
10)Priyanka Giri
11)Dr. Dhanusha C
12)Vikas Pathak
(72)**Name of Inventor :**
1)Dr A. Mary Noya Leena
2)Mrs R. Amutha
3)Dr. G. Nirmaladevi
4)Dr Anil Trimbakrao Gaikwad
5)Dr. N. Prathiba Lakshmi
6)P. Tamilarasan
7)Prof. Vandana Vinayak Badage
8)R. Nirmala
9)Indulekha K V
10)Priyanka Giri
11)Dr. Dhanusha C
12)Vikas Pathak

(57) Abstract :
Academic institutions are increasingly utilising digital platforms, academic databases, spreadsheets, and online assessment tools to record students' performances and teaching-learning activities. Although these tools have enhanced the storage and retrieval of academic records, assessment records are usually not linked to the course, programme, and pre-set attainment targets. Outcome mapping and calculations are done manually and may be time-consuming, and overall marks may not easily indicate the learning outcomes requiring improvement. The present invention seeks to overcome the above shortcomings by providing Digital System for Teaching Learning Performance assessment (NEP-OBE). The system collects, compiles and processes the academic data related to attendance, examination result, assignment, practical assessment, project evaluation and course information. It maps assessment activities to predefined course outcomes and programme outcomes and generates performance indicators comprising a Course Outcome Attainment Score (COAS), Programme Outcome Attainment Index (POAI), Teaching Effectiveness Index (TEI), Student Learning Progress Score (SLPS), Assessment Coverage Index (ACI), and Continuous Improvement Priority Score (CIPS). A performance analysis module compares calculated attainment values with predefined targets and identifies outcomes requiring further academic attention. It uses computer science and machine learning to detect and identify patterns of outcomes across courses, assessments, and students. Reports of findings are communicated to academicians and instructional leaders via analytical reports and online dashboards for further investigation into student achievement and to take appropriate action. The above is a feedback loop representing the performance indicators and performance criterion values annotated in student records. The proposed invention is intended to improve consistency in academic assessment, reduce repetitive manual work, support clearer outcome-based reporting, and assist educational institutions in making informed decisions about teaching-learning improvement under NEP-aligned and OBE-based practices. FIG.1

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129427 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Hybrid Deep Learning Framework for Real-Time Anomaly Detection in High-Velocity Data Streams

(51) International classification	:G06N 3/04, G06N 3/08, H04L 29/06, H04L 12/24, H04L 12/26	(71)**Name of Applicant :** **1)Dr. V. Sivaraj** Address of Applicant :156, 3rd Floor, Alagu Sanitary Building, Kalidass Road, Ramnagar Tamil Nadu India **2)Dr. Nethravathi P.S** **3)Mrs. Vidyarani UA** **4)Mrs. Lavita Wilma Lobo** **5)Mrs. Latha KV** **6)Ms. Spoorthi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Dr. Nethravathi P.S**
(86) International Application No	:	**2)Mrs. Vidyarani UA**
Filing Date	:01/01/1900	**3)Mrs. Lavita Wilma Lobo**
(87) International Publication No	: NA	**4)Mrs. Latha KV**
(61) Patent of Addition to Application Number	:NA	**5)Ms. Spoorthi**
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The present invention relates to a hybrid deep learning framework for real-time anomaly detection in high-velocity data streams. The framework integrates complementary neural network architectures to learn temporal dependencies, nonlinear relationships, and evolving patterns in continuously generated data. Incoming data is processed through a stream preprocessing module, followed by a hybrid feature extraction and anomaly scoring mechanism. An adaptive thresholding unit identifies deviations from learned normal behavior and generates alerts when anomalous conditions are detected. The framework supports incremental model adaptation, reduced detection latency, and scalable deployment across distributed computing environments for industrial monitoring, cybersecurity, financial transactions, and Internet of Things applications.

No. of Pages : 10 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202641129428 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-ENABLED SYSTEM FOR PREDICTING TEACHER BURNOUT AND OPTIMIZING WORKLOAD IN HIGHER EDUCATION

(51) International classification	:G06Q 10/06, G06N 20/00, G09B 7/00, G09B 7/02, G
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. S. Ramasamy
Address of Applicant :Associate Professor, Department of Artificial Intelligence and Data Science, Hindusthan Institute of Technology, Malumichampatti, Coimbatore, Tamil Nadu, 641032, India. Tamil Nadu India
2)Pemma Radhika
3)Dr Nadeem Pasha
4)Dr Lalit Mohan Trivedi
5)D. Praveena
6)Dr. G. Prabhakaran
7)Dr. Ranjeet Singh
8)Dr. Irissappane Dhanusu Soubache
9)Gudise Anusha
10)Anvesh Perada
11)Dr. Prashant Mundeja
12)Dr. Muthukumar Subramanian
(72)**Name of Inventor :**
1)Dr. S. Ramasamy
2)Pemma Radhika
3)Dr Nadeem Pasha
4)Dr Lalit Mohan Trivedi
5)D. Praveena
6)Dr. G. Prabhakaran
7)Dr. Ranjeet Singh
8)Dr. Irissappane Dhanusu Soubache
9)Gudise Anusha
10)Anvesh Perada
11)Dr. Prashant Mundeja
12)Dr. Muthukumar Subramanian

(57) Abstract :
The innovation pertains to a system powered by AI that forecasts the risk of teacher burnout and optimizes their workload in higher education settings. The system collects approved documentation concerning instruction, evaluation, research, oversight, administration, and academic due dates. A module for normalization and feature extraction creates a workload representation that varies over time. A workload prediction engine forecasts upcoming demand, whereas a machine learning system evaluates the possible burnout risk linked to workload with relevant uncertainty data. A workload optimization engine that considers constraints assesses different task assignments based on staff availability, qualifications, teaching requirements, workload restrictions, and institutional policies. Candidate proposals are assessed and submitted to a designated reviewer prior to execution. A workload outcome tracking module assesses approved tasks against later observations and provides discrepancies for additional planning. Model governance, access management, and auditing functions facilitate regulated functioning. The system offers a unified framework for predicting workloads, assessing risks, allocating tasks under constraints, and managing academic workloads based on feedback, without considering model predictions as medical evaluations or automatic hiring choices. FIG.1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129433 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXPLAINABLE AI-BASED CYBERSECURITY FRAMEWORK FOR REAL-TIME ANOMALY DETECTION, RISK PREDICTION, AND THREAT RESPONSE

(51) International classification	:H04L 29/06, G06N 20/00, G06N 5/04, H04L 9/40, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. S. Vagdevi
Address of Applicant :Professor, Department of Computer Science and Information Systems, Work Integrated Learning Programmer (WILP), Birla Institute of Technology and Science, Pilani, Bengaluru – 560098, Karnataka, India Karnataka India
2)Mrs. Shruti B. Hiregoudar
3)Mr. Girish B. Shettar
(72)**Name of Inventor :**
1)Dr. S. Vagdevi
2)Mrs. Shruti B. Hiregoudar
3)Mr. Girish B. Shettar

(57) Abstract :
[052] An explainable artificial intelligence-based cybersecurity framework is disclosed for real-time anomaly detection, contextual risk prediction, and controlled threat response. The framework receives telemetry from network, endpoint, identity, application, and threat intelligence sources, normalizes the telemetry, and generates features for an anomaly detection engine. A risk prediction engine evaluates detected deviations with contextual factors including asset criticality, access privilege, potential impact, and correlated events. An explainability engine generates an event-specific explanation identifying contributing evidence and associates the explanation with the event and model version. A threat response controller selects, recommends, or executes a response according to risk thresholds, authorization requirements, and configurable security policies. An audit and feedback store records detection results, explanations, policy decisions, actions, and outcomes. Verified feedback may be used to refine baselines, thresholds, or models subject to validation and controlled deployment. The framework improves traceability and supports scalable cybersecurity management across on-premises, cloud, edge, and hybrid computing environments. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129444 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CIRCUIT BOARD HEAT PIPE SEALED AND CHARGED DURING REFLOW BY SOLDER-LOCKED BREAKABLE FLUID RESERVOIR

(51) International classification	:H05K 3/34, H05K 7/20, F28D 15/02, H01L 23/427, B23K 1/008
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BAPATLA SURENDRA BABU
Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India
2)IRUKUMALLI PRIYANKA
(72)**Name of Inventor :**
1)BAPATLA SURENDRA BABU
2)IRUKUMALLI PRIYANKA

(57) Abstract :
CIRCUIT BOARD HEAT PIPE SEALED AND CHARGED DURING REFLOW BY SOLDER-LOCKED BREAKABLE FLUID RESERVOIR A heat pipe inside a circuit board for cooling components is sealed and filled during reflow soldering, without a separate filling station. A laminated board (102) has a cavity (114) with an aluminium trihydroxide wick bed (202) and a port (212). A charge component (110), soldered over the port by a seal ring (302), holds a sealed breakable glass reservoir (308) of working fluid (316), fixed by solder locks (312) to a frame (304) of lower thermal expansion. During reflow the ring and locks melt, and water vapour from the wick bed pushes unwanted gas out through the melted ring. On cooling the ring and locks harden, closing the cavity; the glass then shrinks against the frame and breaks, releasing the fluid. A release index of 1.20 to 2.10 makes the reservoir break before room temperature but late enough to prevent blistering. Principal use: power-electronics and automotive circuit boards. FIG. 1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129452 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Grid-Aware EV Charging and Energy Optimization

(51) International classification	:G06Q 50/06, H02J 3/32, H02J 7/00, B60L 53/62, H02J 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)B Jyothi
Address of Applicant :Department of EEE, Koneru Lakshmaiah Education Foundation, Vaddeswaram, Guntur, AP, India Andhra Pradesh India
2)Koneru Lakshmaiah Education Foundation
(72)**Name of Inventor :**
1)B Jyothi
2)P. Sowmya Sri
3)P.Bhavana
4)B. Indrani

(57) Abstract :
The existing EV home energy management systems use smart charging, electricity tariffs, and V2G/V2H to coordinate EV charging with household energy demand. However, many existing systems involve complex optimization methods and require bidirectional charging hardware and effective battery management. The proposed system integrates household load forecasting, electricity tariff, EV State of Charge (SOC), and user requirements for intelligent energy management. An optimization-based method determines whether the EV should charge, remain idle, or discharge energy to the home/grid. The system maintains the required SOC while reducing electricity cost and peak household demand. The proposed system will be implemented using Python, Machine Learning, Pandas, NumPy, and optimization algorithms. A Raspberry Pi and smart-meter interface can be used for prototype-level implementation. The system aims to provide a simple and practical solution for integrating EVs with residential energy management.

No. of Pages : 15 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129474 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Deep Learning–Based System for Student Engagement Detection

(51) International classification	:G06N 3/08, G06N 3/04, G06K 9/00, G06K 9/62, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VEMU INSTITUTE OF TECHNOLOGY
Address of Applicant :P.Kothakota, Puthalapattu Mandal, Tirupati - Chittoor Highway, Chittoor (Dt), 517112, Andhra Pradesh, India Chittoor Andhra Pradesh India
(72)**Name of Inventor :**
1)Ms. J. Meena
2)Ms. P. Kavya
3)Ms. Y.Chandhana Devi

(57) Abstract :
The present invention discloses the various ways in which image data is used in the deep learning system to detect student engagement. The system classifies a student either as engaged or not based on their facial expressions and postures, amongst other visual features obtained from camera inputs. Therefore, a system that would use convolutional neural networks and MobileNet architecture would save manual extraction by eliminating this need for feature extraction. It supports both the physical classroom and online learning environment, enabling real-time monitoring of engagements. Preprocessing and transfer learning techniques may be followed so that even a small dataset can work effectively. The method herein described ensures efficient, yet scalable and automated mechanisms for enhancing educational monitoring and improving learning outcomes.

No. of Pages : 17 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129479 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Machine Learning–Based System for Detecting Fraudulent Social Media Accounts

(51) International classification	:G06Q 50/00, G06N 20/00, G06K 9/62, G06Q 30/02, G06F 16/9536
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VEMU INSTITUTE OF TECHNOLOGY
Address of Applicant :P.Kothakota, Puthalapattu Mandal, Tirupati - Chittoor Highway, Chittoor (Dt), 517112, Andhra Pradesh, India Chittoor Andhra Pradesh India
(72)**Name of Inventor :**
1)Ms. P. D. Santhi
2)Ms. V Nithya
3)Ms. V. Madhuri Reddy

(57) Abstract :
The invention introduces an intelligent system that implements methods concerned with detecting fraudulent social media accounts based on artificial intelligence and learning models. These models are used on the data analyzed from the social media users' profiles and behavior. The data is processed based on patterns that are associated with either fake social media profiles or behavior that may be considered fraudulent. The invention uses Random Forest and XGBoost models that are involved in the classification of the data and are capable of adapting to different patterns related to fraudulent activities. The invention is significantly useful when it comes to providing an efficient and safe solution in detecting fraudulent social media accounts.

No. of Pages : 17 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129492 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Intelligent Deep Learning–Based System and Method for Early Prediction of Brain Stroke Using Medical Imaging Data

(51) International classification	:G06N 3/04, G06T 7/00, G06N 3/08, A61B 6/00, A61B 6/03
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VEMU INSTITUTE OF TECHNOLOGY
Address of Applicant :P.Kothakota, Puthalapattu Mandal, Tirupati - Chittoor Highway, Chittoor (Dt), 517112, Andhra Pradesh, India Chittoor Andhra Pradesh India
(72)**Name of Inventor :**
1)Ms. S.Swarnalatha
2)Mr. Karanam Eswar
3)Ms. Koniki Jahnavi

(57) Abstract :
This invention reveals an intelligent deep learning-based system and method for early prediction of brain strokes using medical imaging data. The system uses a DenseNet-based convolutional neural network along with a Layer-wise Adaptive Moments (LAMB) optimizer to analyze brain CT scan images and accurately classify stroke conditions. The invention overcomes the limitations of traditional diagnostic methods by reducing reliance on manual interpretation and improving the speed and accuracy of diagnoses. With better feature extraction, adaptive optimization, and thorough preprocessing, the system achieves high predictive performance while reducing false classifications. The proposed solution aids real-time clinical decision-making and can fit into existing healthcare systems. This invention significantly improves early stroke detection, leading to better patient outcomes and less strain on healthcare resources.

No. of Pages : 22 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129493 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Real-Time Multimodal Emotion Detection System Using Machine Learning for Facial, Audio, Text, and Live Video Input

(51) International classification	:G10L 25/63, G06N 3/04, G06N 3/08, G06K 9/00, G06K 9/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VEMU INSTITUTE OF TECHNOLOGY
Address of Applicant :P.Kothakota, Puthalapattu Mandal, Tirupati - Chittoor Highway, Chittoor (Dt), 517112, Andhra Pradesh, India Chittoor Andhra Pradesh India
(72)**Name of Inventor :**
1)Ms. B.Gnaneswari
2)Ms. G.Vyshnavi
3)Ms. C Sowjanya

(57) Abstract :
The current invention introduces a real-time multimodal emotion recognition system based on machine learning algorithms to analyze and detect human emotions based on facial expressions, audio signals, text data, images, and live video feeds. It uses Deep Learning models like Convolutional Neural Networks and the Transformer Architecture to identify and classify emotional patterns in each mode. With the amalgamation of different modes of emotional data, there would be improved accuracy, efficiency, and practicality of the system compared to other singular mode analysis. It can conduct real-time processing, provide secure user interface functionality, and be deployable over different applications in the domain of mental health analysis, customer service assistance, and human-computer interaction.

No. of Pages : 20 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129494 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD, A PHOTOCATALYTIC NANOCOMPOSITE AND A SYSTEM FOR TREATMENT OF DYE-CONTAINING WATER

(51) International classification	:C02F 1/72, B01J 35/10, C02F 1/30, C02F 1/32, B01J 23/75
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Shilpa Susan Scaria
Address of Applicant :Department of Life Sciences, CHRIST(Deemed to be University), Hosur Road, Bengaluru, Karnataka, India - 560029 Karnataka India
(72)**Name of Inventor :**
1)Shilpa Susan Scaria
2)Kadanthottu Sebastian Joseph

(57) Abstract :
ABSTRACT A METHOD, A PHOTOCATALYTIC NANOCOMPOSITE AND A SYSTEM FOR TREATMENT OF DYE-CONTAINING WATER The present invention relates to a method, a photocatalytic nanocomposite and a system for treatment of dye-containing water. The nanocomposite comprises a cobalt oxide–zinc oxide heterojunction interior surrounded by a mesoporous silica shell having interconnected pores and an exterior metal-ion-binding region comprising covalently tethered iminodiacetate groups. Internal pore surfaces are substantially free of the tethered groups, spatially separating photocatalytic and metal-ion-binding regions. The pores permit transport of dye molecules, water, and dissolved oxygen toward the irradiated interior for photocatalytic dye degradation, while exterior groups capture released cobalt and zinc ions. The system comprises a treatment chamber, an irradiation source, a fluid-handling assembly, and a catalyst-retention assembly. A monitoring arrangement and controller may regulate irradiation, circulation, and water withdrawal according to measured treatment parameters. The method comprises contacting, irradiating, capturing released metal ions, and separating treated water while retaining the nanocomposite. Fig 3.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129495 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Data-Driven Machine Learning System for Personalized Heart Disease Prediction

(51) International classification	:G16H 50/20, G06N 20/00, G16H 50/70, G16H 50/30, G16H 10/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VEMU INSTITUTE OF TECHNOLOGY
Address of Applicant :IPR Cell, P.Kothakota, Puthalapattu Mandal, Tirupati - Chittoor Highway, Chittoor (Dt), 517112, Andhra Pradesh, India P. Kothakota Andhra Pradesh India
(72)**Name of Inventor :**
1)Ms. Chinthamakula Pavani
2)Ms. K. Lakshmi Priya
3)Mr. G.Tejesh Reddy

(57) Abstract :
According to the invention, there exists a data-driven machine learning system for personalized heart disease prediction. By using supervised machine learning algorithms, the system predicts the likelihood of heart disease in individuals based on clinical attributes. By using the Random Forest algorithm and the K-Nearest Neighbor algorithm, the system is capable of understanding complex associations between various clinical attributes. For this purpose, data preprocessing, attribute selection, and validation of the system enhance the accuracy of heart disease prediction. By using the system, personalized outcomes for heart disease prediction can be obtained with the help of a web interface.

No. of Pages : 20 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129505 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIDIMENSIONAL EMOTIONAL COMPETENCE ANALYTICS PLATFORM FOR PERSONALIZED LEARNING AND CULTURE DEVELOPMENT IN HEALTHCARE ORGANIZATIONS

(51) International classification	:G06Q 10/06, G06Q 50/22, G06N 20/00, G06Q 10/10, G16H 50/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)Narendra Tappa
2)Dr Salla Nityanth Kumar

(57) Abstract :
The present invention relates to a computing platform for emotional competence learning in healthcare organizations. Assessment records retain competence dimension, context, source type, and rubric version. A profile processor computes source-specific estimates and evidence sufficiency. A disagreement evaluator compares compatible sources and selects a clarification assessment when their difference exceeds a configured tolerance. Supported learning gaps are mapped to context-tagged resources and prerequisite sequences. Each proposed pathway references a profile snapshot, and supporting versions are checked before assignment through a learning application. Corrected evidence triggers reassessment of affected profiles and pathways. An aggregation service restricts organizational outputs using distinct-participant counts and overlapping-query controls. The platform preserves evidence provenance and separates individual learning records from permitted organizational summaries.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129507 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WOMAN SAFTEY SMART BUTTON

(51) International classification	:H04W 4/90, G08B 21/02, G08B 25/10, H04M 1/72457, G08B 25/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)FRANCIS XAVIER ENGINEERING COLLEGE
Address of Applicant :Tirunelveli - 627003, Tamil Nadu, India Tirunelveli Tamil Nadu India
(72)**Name of Inventor :**
1)AMALA MANUELA
2)Mr. C. MOHAN LAZARAS
3)Mr. M. SAKTHIVEL
4)Mr. G.A. GIRIDHARAN
5)Mr. S. GOKUL GANESH
6)Mr. D. MUKESH
7)Mr. M. YUVAN SANTHOSH
8)Mr. S. R. PREM

(57) Abstract :
This invention relates to a small, portable safety device that helps a person in danger call for help quickly, without needing to unlock or use a phone. The device includes a push button, a microcontroller, a GPS module, a GSM module, and a rechargeable battery. To request help, the user simply holds the button for eight seconds or presses it two to three times in quick succession. On detecting this action, the microcontroller reads the user's exact latitude and longitude from the GPS module. The GSM module then automatically sends an emergency SMS containing a map link of the location to a pre-saved guardian and the police authorities. Because the alert is sent as a normal SMS, no internet connection is required, and help can be requested even in areas without data coverage. The device is low cost, lightweight, easy to carry, and simple to operate in an emergency situation.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129508 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT WORKFORCE TRANSFORMATION ENGINE FOR ALIGNING LEADERSHIP DECISIONS, DIGITAL CAPABILITIES AND BUSINESS INNOVATION

(51) International classification	:G06Q 10/06, G06Q 10/10, G06F 8/41, G06F 9/44, G06F 9/45	(71)**Name of Applicant :** **1)SR University** Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India (72)**Name of Inventor :** **1)Somisetti Ganesh** **2)Dr Salla Nityanth Kumar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to an enterprise workforce transformation system, method, and computer-readable medium are disclosed for coordinating workforce capability development and digital workflow execution across heterogeneous application environments lacking a unified transaction boundary. A compiler parses a structured leadership intent into an executable directed graph of typed actions linked to targeted capability evidence—distinguishing assignment, completion, and demonstrated proficiency—and digital application requirements. Each action node specifies versioned preconditions, required completion evidence, and compensation or incident-hold procedures. Execution authorization across the graph is bound to a validated plan manifest. A durable step state machine continuously evaluates live dependency states, local transaction commits, and outgoing command persistence to prevent state drift. The system correlates target feedback and verified evidence back to individual action steps, keeping dependent actions blocked until all prerequisites are programmatically satisfied. In response to partial failures, the engine invokes reverse-dependency compensation logic or transitions affected branches into a guarded incident hold. Contextual changes in cohort membership, capability evidence, or application requirements trigger targeted dynamic reassessments of affected graph branches via reverse-dependency tracking. Successor plan versions retain execution history while reusing completed step evidence following compatibility validation.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202641129511 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING-BASED SYSTEM FOR EARLY IDENTIFICATION OF LEARNING DIFFICULTIES AND PERSONALIZED REMEDIAL SUPPORT IN HIGHER EDUCATION

(51) International classification	:G06N 20/00, G06Q 50/20, G09B 7/02, G09B 7/04, G09B 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr. S. Ramasamy

Address of Applicant :Associate Professor, Department of Artificial Intelligence and Data Science, Hindusthan Institute of Technology, Malumichampatti, Coimbatore, Tamil Nadu, 641032, India. Tamil Nadu India

2)Shilpa D V

3)Dr Nadeem Pasha

4)Dr Lalit Mohan Trivedi

5)D. Praveena

6)Dr. G. Prabhakaran

7)Dr. Ranjeet Singh

8)Dr. Irissappane Dhanusu Soubache

9)Gudise Anusha

10)Anvesh Perada

11)Dr. Prashant Mundeja

12)Prof. Gayathri Devi.S

(72)**Name of Inventor :**

1)Dr. S. Ramasamy

2)Shilpa D V

3)Dr Nadeem Pasha

4)Dr Lalit Mohan Trivedi

5)D. Praveena

6)Dr. G. Prabhakaran

7)Dr. Ranjeet Singh

8)Dr. Irissappane Dhanusu Soubache

9)Gudise Anusha

10)Anvesh Perada

11)Dr. Prashant Mundeja

12)Prof. Gayathri Devi.S

(57) Abstract :

The invention pertains to a machine learning-driven system for the early detection of learning challenges and tailored corrective assistance in higher education. The system obtains official academic records from educational platforms, verifies the information received, and produces attributes that reflect concept-level performance, persistent errors, and long-term learning progress. A machine learning system evaluates possible learning deficiencies and offers confidence data. A module for concept mapping links the identified patterns to course goals and prerequisite connections. A confidence-aware intervention decision system assesses if remedial assistance, extra assessment, or educator evaluation is suitable. A customized support system identifies and arranges suitable teaching materials, directed activities, and evaluations. A monitoring module for intervention-response assesses later performance and refreshes the representation of the learner's state. Updates to models and parameters are governed by established validation protocols. An educator portal, access restrictions, and monitoring features enable responsible management. The system offers a feedback-driven structure for aligning early identification of learning gaps with specific corrective actions and continuous educational assessment. FIG.1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129514 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-CRITERIA EMERGING TECHNOLOGY ADOPTION ENGINE USING ORGANIZATIONAL READINESS, RISK AND LEADERSHIP INDICATORS

(51) International classification	:G06Q 10/06, G06N 5/02, G06Q 30/02, G06N 5/04, G06Q 10/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)Kamini Shiva Prasad
2)Dr. Tunguturi Srinivas

(57) Abstract :
The present invention relates to multi-criteria evaluation and controlled pilot initiation for emerging technologies. Candidate configurations are assessed against a target environment using versioned readiness, risk, and leadership indicators. Mandatory constraints are evaluated before soft preference scoring, with failed and unknown conditions retained separately. Eligible indicators are normalized under stored bounds and combined using versioned weights. A sensitivity processor identifies rankings that change under permitted uncertainty or weight conditions and can request additional evidence. A decision manifest retains candidate configuration, input versions, constraints, and evaluation settings. Leadership authorization is bound to the manifest. Before pilot dispatch, a release service checks current dependencies and withholds execution when the approved state has materially changed. Pilot feedback supplies evidence for subsequent evaluations.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129519 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A STABILITY-INDICATING REVERSE-PHASE HPLC METHOD FOR THE DETERMINATION OF NISOLDIPINE

(51) International classification	:G01N 30/02, G01N 30/06, G01N 30/88, G01N 30/34, G01N 30/74
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)B V RAJU INSTITUTE OF TECHNOLOGY
Address of Applicant :Basic Science and Humanities, Vishnupur, Narsapur, Medak, Telangana 502313 Narsapur Telangana India
(72)**Name of Inventor :**
1)Dr.Patlola Madhusudhan
2)Dr.Sirisha Bandi
3)Dr.T.Mohan Kumar

(57) Abstract :
A reverse-phase high-performance liquid chromatography method is described for determination of nisoldipine in bulk drug and tablet dosage forms. Separation is performed on an ODS/C18 column (250 mm × 4.6 mm, 5 µm) using methanol and water (80:20, v/v) as an isocratic mobile phase at 1.0 mL/min. Detection is carried out at 240 nm. The reported retention time of nisoldipine is approximately 4.4 minutes and the run time is 8 minutes. Tablet samples are powdered, extracted with diluent, diluted as required, filtered through a 0.45 µm membrane and sonicated for 5 minutes. The source study reports linearity over 0.3–8 µg/mL and evaluates accuracy, precision, robustness and forced degradation under acid, alkaline, oxidative, thermal and photolytic conditions.

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129522 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN WORKFORCE EARLY-WARNING FRAMEWORK FOR PREDICTING ORGANIZATIONAL PERFORMANCE AND HUMAN CAPITAL RISKS

(51) International classification	:G06N 20/00, G06Q 10/06, G06N 5/04, G06N 5/02, G06F 11/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)Kiran Devarapally
2)Dr. Sunitha Purushottam Ashtikar

(57) Abstract :
The present invention is related to An artificial intelligence-driven workforce early-warning system and computer-implemented method are disclosed for monitoring cohort risk across multiple future horizons while handling missingness and support conditions. The system constructs a time-consistent feature snapshot from source event versions available prior to a prediction cutoff. Before generating risk estimates, the system evaluates source coverage, freshness, and model applicability, withholding unqualified predictions and generating a data-quality state when support conditions fail. When support passes, a model service estimates per-interval conditional hazards and cumulative horizon risk for defined outcome events, distinguishing incomplete follow-up from mature negative outcomes. A durable alert store manages lifecycle alert episodes using hysteresis, persistence, entry conditions, and lower exit thresholds, bypassing delivery suppression during escalated severity. Repeated supported risk observations correlate with active episodes, while review tasks track acknowledgement, action completion, and outcome verification. A lineage processor reconciles matured outcomes with earlier prediction records—retaining model versions, calibration versions, and feature snapshots—to track input drift, calibration deterioration, and increased abstention. Successor models are validated for schema compatibility before contributing to active episodes, ensuring auditable lineage across continuous operational deployment.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129529 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE ESG AND CSR DECISION-SUPPORT ARCHITECTURE FOR ENTERPRISE ENVIRONMENTAL SUSTAINABILITY MANAGEMENT

(51) International classification	:G06Q 10/06, G06Q 40/06, G06Q 40/00, G06Q 50/26, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)L N Sivaprasadu Majji
2)Dr. Gurunadhan Goli

(57) Abstract :
The present invention is related to a predictive environmental sustainability computing system and computer-implemented decision-support method are disclosed for evaluating portfolio scenarios and enforcing execution boundaries. The system acquires environmental observations, normalizing source intervals, quantities, units, and emission factor versions while reconciling duplicate or overlapping physical contributions—such as operational facility and CSR project boundaries—via a dedicated boundary registry to prevent double-counting. A model service generates baseline resource forecasts with explicit uncertainty representations using chronologically validated input snapshots restricted to observations available at the forecast origin. Candidate project scenarios are compiled using declared resource-change functions, temporal/cross-variable correlation models, and interaction rules that govern sequential application, mutual exclusivity, shared-cap constraints, or joint-response functions. Further, The system evaluates candidate portfolios against stored hard constraints across a reproducible, common-indexed scenario ensemble. Evaluated states are bound to an immutable authorization manifest encompassing source, boundary, factor, model, scenario, and constraint versions. The engine automatically withholds release of execution requests if material manifest dependencies change, marking dependent recommendations as stale while preserving historical approved states. Following deployment, post-implementation measurements are acquired under compatible boundaries to evaluate performance against retained forecasts using recorded adjustment rules, ensuring verifiable, auditable sustainability governance across enterprise operations.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129530 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A LINKPULSE ENGINE AUTONOMOUS WEB-TO-KNOWLEDGE AI ASSISTANT SYSTEM

(51) International classification	:G06N 5/02, G06N 5/04, G10L 15/26, G10L 15/22, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)DIVYASRI.K
2)BEENA.K
3)TAPASVI PANCHAGNULA
4)SHIVANI REDDY ALUMALLA
5)INDIRALA SURYA

(57) Abstract :
Disclosed herein is a LinkPulse Engine autonomous web-to-knowledge AI assistant system (100) that comprises an Input Interface (102), a Web Data Acquisition Unit (104), a Multimedia Acquisition Unit (106), a Communication Interface (108), a Memory Unit (110), and a Processing Unit (112). The Processing Unit (112) comprises a Data Extraction Module (114), a Speech-to-Text Conversion Module (116), a Data Pre-processing Module (118), a Semantic Chunking Module (120), an Embedding Generation Module (122), a Vector Indexing and Storage Module (124), a Semantic Retrieval Module (126), a Context Integration Module (128), a Response Generation Module (130), a Citation and Source Verification Module (132), a Knowledge Update and Synchronization Module (134), and a Conversational Interaction Module (136). The system (100) acquires and processes web and multimedia content, generates vector embeddings, retrieves semantically relevant information, and generates context-aware responses with source references, facilitating autonomous knowledge extraction and reliable conversational question answering.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129531 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A COMPOSITION AND METHOD FOR PREPARING LC³ BRICKS INCORPORATING WASTEWATER TREATMENT SLUDGE

(51) International classification	:C04B 28/04, C04B 18/04, C04B 14/10, C04B 18/30, C04B 18/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)G. ABHIJITH AMBEDKAR
2)D.RAMYA
3)S. PRUDHVI NARAYANA
4)S PADMAJA

(57) Abstract :
Disclosed herein is a composition (100) for preparing limestone calcined clay cement (LC³) bricks (114) incorporating wastewater treatment sludge. The composition (100) comprises ordinary Portland cement (102), limestone (104), calcined clay (106), wastewater treatment sludge (108), fine aggregate (110), and a polycarboxylate ether-based superplasticizer (112). The wastewater treatment sludge (108) is incorporated as a partial replacement of a reference constituent at a predetermined replacement level. The sludge comprises aerobic wastewater treatment sludge (116), anaerobic wastewater treatment sludge (118). The constituent materials are proportioned and mixed to obtain a cementitious mixture, which is shaped and cured to form the LC³ brick (114). The resulting bricks are evaluated for mechanical and durability properties and characterized using X-ray fluorescence, X-ray diffraction, and scanning electron microscopy. The evaluation facilitates comparison of sludge-incorporated formulations with a reference formulation to examine the influence of sludge type and replacement level and identify a suitable formulation for construction.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129532 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A MACHINE LEARNING-BASED CROP YIELD PREDICTION AND AGRICULTURAL ADVISORY SYSTEM

(51) International classification	:G06Q 50/02, G06N 20/00, A01C 21/00, G06Q 10/04, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)BEENA K
2)J.V. P UDAYA DEEPIKA
3)LATCHUBHUKTA BHARGAV
4)RACHAMALLA CHANDU
5)VIRAJITHA BEJJANKI

(57) Abstract :
Disclosed herein is a machine learning-based crop yield prediction and agricultural advisory system (100) that comprises an Input Interface (102), an Agricultural Data Acquisition Unit (104), a Processing Unit (106), a Memory Unit (126), a Communication Interface (128), and an Output Interface (130). The Processing Unit (106) comprises a Data Pre-processing Module (108), a Crop Yield Prediction Module (110), a Disease Risk Assessment Module (112), a Fertilizer Recommendation Module (114), a Weather Prediction Module (116), a Crop Recommendation Module (118), a Rule-Based Fallback Module (120), a Result Integration Module (122), and an Agricultural Advisory Generation Module (124). The system (100) utilizes machine learning algorithms and predefined agricultural rules to predict crop yield, assess disease risk, recommend fertilizers, forecast rainfall trends, and rank suitable crops. The Rule-Based Fallback Module (120) generates alternative recommendations when input information is incomplete or trained models are unavailable.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129533 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SECURE WEB-BASED MESSAGING SYSTEM

(51) International classification	:G06F 21/55, G06F 21/56, H04L 29/06, G06F 21/57, H04L 12/58
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)MRS. CHENCHU SWETHA
2)ACHANTA NAGENDRA KRISHNA SAI
3)ITHAGONI SOWMYA SRI
4)BATHALA MURALI
5)MRS. S. BHAVANA

(57) Abstract :
Disclosed herein is a secure web-based messaging system (100) comprising a User Interface (102), an Authentication and Identity Management Unit (104), a Secure Messaging Unit (106), a Real-Time Communication Unit (108), a File Preprocessing Unit (110), a Processing Unit (112), a Data Storage Unit (120), and a Security and Integration Unit (122). The Processing Unit (112) comprises a Ransomware Detection Module (114), a Risk Assessment and Decision Module (116), and a File Handling and Notification Module (118). The Ransomware Detection Module (114) analyzes suspicious file extensions, known ransomware signatures, entropy characteristics, and behavioral indicators. The Risk Assessment and Decision Module (116) calculates a risk score and classifies shared files according to configurable thresholds. The File Handling and Notification Module (118) controls file access and transmission based on the classification and generates security alerts.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129534 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEBCAM-BASED DRIVER DROWSINESS DETECTION SYSTEM

(51) International classification	:G06K 9/00, G08B 21/06, G06V 40/16, A61B 5/18, A61B 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)K BEENA
2)T VENKATNARAYAN RAO
3)T. SUMANTH
4)B. SANTHOSH KUMAR
5)K. SAI PRANAVI

(57) Abstract :
Disclosed herein is a webcam-based driver drowsiness detection system (100) for monitoring driver alertness and detecting drowsiness based on facial behaviour. The system (100) comprises an image acquisition unit (102), a processing unit (104), an alert generation module (120), a display unit (122), and a data storage unit (124). The processing unit (104) comprises a pre-processing module (106), a face detection module (108), a facial landmark localisation module (110), a feature extraction module (112), a calibration module (114), a temporal analysis module (116), and a drowsiness decision module (118). The system (100) processes facial images captured through a webcam to identify facial landmarks and determine an eye aspect ratio (EAR), a mouth opening ratio (MOR), and nose displacement. The system (100) establishes individual reference parameters and evaluates variations and persistence of facial features across consecutive images to determine a drowsiness condition.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129550 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXPLAINABLE AI FRAMEWORK FOR DETECTING WORKFORCE BIAS AND GENERATING INCLUSIVE ORGANIZATIONAL INTERVENTIONS

(51) International classification	:G06Q 10/06, G06N 20/00, G06N 5/04, G06N 5/02, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :H-No 3-140, Ananthasagar, Hasanparthy, Warangal - 506371, Telangana, India Telangana India
(72)**Name of Inventor :**
1)S Jyothsna
2)Dr. Tunguturi Srinivas

(57) Abstract :
The present invention relates to explainable artificial intelligence auditing of workforce decision systems. A snapshot builder reconstructs input, eligibility, score, review, and outcome records with their versions. A disparity processor qualifies comparable populations and sample sufficiency before calculating group metrics. An explanation engine uses the model version associated with a decision and records explanation fidelity. An intervention generator selects permitted workflow changes, and a replay service evaluates them against the fixed snapshot under configured constraints. Approval is bound to a manifest containing the candidate, input versions, and evaluation results. A release service withholds deployment when material dependencies have changed. Subsequent decision events are associated with the released intervention version for monitoring. The framework provides reproducible audit and intervention records while distinguishing statistical signals from causal or legal conclusions.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129585 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A FEDERATED LEARNING-BASED SYSTEM FOR IDENTIFYING SKIN DISORDERS

(51) International classification	:G06T 7/00, G06V 10/764, G06V 10/774, G06N 20/00, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)KOVVURI DIVYASRI
2)T VENKAT NARAYANA RAO
3)ARELLA TEJASWI
4)V.VIGNESH RAO
5)T. SHIVA KSHETRA

(57) Abstract :
Disclosed herein is a federated learning-based system (100) for identifying skin disorders, comprising an image input unit (102) configured to receive dermoscopic images depicting skin lesions, and a processing unit (104) comprising a preprocessing module (106), an object detection module (108), a classification module (110), and a local training module (112). The preprocessing module (106), the object detection module (108) and the classification module (110). A plurality of local client nodes (114) locally train model parameters using dermoscopic images and corresponding labels and transmit local model parameter updates to a federated learning server (116) without transmitting raw images. The server aggregates and redistributes global model parameters. An output interface (118) presents classification results, confidence information, and lesion localization, while a data storage unit (120) stores model parameters, configuration information, classification labels, and operational data.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129586 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ARTIFICIAL INTELLIGENCE-BASED SYSTEM FOR AUTOMATED EMERGENCY AND VIOLENCE DETECTION

(51) International classification	:G06K 9/00, G06V 10/764, G06N 3/08, G06V 20/40, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)G SESHU KUMAR
2)T VENKAT NARAYANA RAO
3)A. BHARGAVI HARSHINI
4)P. KHASIM
5)I. NISHANTH

(57) Abstract :
Disclosed herein is an artificial intelligence-based system (100) for emergency and violence detection, comprising a video acquisition unit (102), a frame preprocessing unit (104), a pose-keypoint extraction unit (106), a temporal feature analysis unit (108), a processing unit (110), an event management unit (124), an evidence storage unit (126), and an alert integration unit (128). The processing unit (110) comprises a temporal sequence analysis module (112), a violence classification module (114), an emergency classification module (116), a hybrid verification module (118), an adaptive video processing module (120), and a startup suppression module (122). The system extracts skeletal keypoints from video frames, analyses sequential movement patterns, and generates classification outputs for potential violent activities. The classification outputs are verified using deterministic rules, confidence thresholds. Upon event verification, the system generates event metadata, stores associated video clips or image snapshots, and transmits alerts to designated monitoring interfaces or notification endpoints.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129591 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NODE-LEVEL ADAPTIVE HYBRID CLASS BALANCING FOR SWARM LEARNING-BASED HISTOPATHOLOGICAL IMAGE CLASSIFICATION

(51) International classification	:G06N 20/00, G06V 10/764, G16H 30/40, G06K 9/62, G06V 10/774
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHRIST UNIVERSITY
Address of Applicant :Christ University, Dharmaram College Post, Hosur Road, Bengaluru-560029, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Ms. Varshini S
2)Dr. Fr. Jossy George
3)Dr. Kamal Upreti

(57) Abstract :
A system and method in distributed machine learning classify histopathological images as benign or malignant using peer-connected computing nodes. Each node determines local class frequencies and uses the same local class distribution for inverse-frequency weighted random sampling and class-weighted cross-entropy loss, jointly balancing training-batch formation and loss calculation. Processed images train compatible local image classifiers through backpropagation and parameter optimization. Corresponding model-weight tensors are exchanged peer-to-peer and combined by unweighted element-wise averaging before successive training rounds. Raw images, patient information, class counts, loss weights and sampling probabilities remain within their originating nodes, without a central coordinating server or pooled class distribution. Node-level hybrid balancing addresses differing local class imbalances while supporting collaborative histopathological image classification.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129593 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR URINE-BASED ESOPHAGEAL CANCER SCREENING USING AMPEROMETRIC BIOSENSORS AND FEDERATED LEARNING

(51) International classification	:C12Q 1/68, G01N 27/327, A61B 5/00, A61B 5/20, G01N 33/487
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHRIST UNIVERSITY
Address of Applicant :Christ University, Dharmaram College Post, Hosur Road, Bengaluru-560029, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Jewell Debnath
2)Dr. Fr. Jossy George

(57) Abstract :
A urine collection and sample-handling interface presents urine to an amperometric electrochemical biosensor comprising working, reference and counter electrodes, complementary nucleic-acid probes for miR-21 and miR-375, and an affinity-recognition element for 8-hydroxy-2-deoxyguanosine. Target binding alters an electrochemical reporting reaction under an applied working-electrode potential. A potentiostat, current-measurement circuit and analogue-to-digital converter acquire current-versus-time waveforms. Waveform-processing software applies baseline correction and smoothing and extracts peak current, time-to-peak, integrated waveform area and signal-to-noise ratio into sample-level feature records. Client computing nodes train a shared model while retaining patient-level records locally. A coordinating server updates the model through secure aggregation without recovering individual client contributions. The shared model produces a cancer/non-cancer classification presented through a screening interface.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129611 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A portable electrochemical water testing device with calibrated printed sensor interface

(51) International classification	:G01N 33/18, G01N 27/416, G01N 27/30, G01N 27/49, G01N 27/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
Address of Applicant :No.42 Vel tech Road Avadi, Chennai Tamil Nadu chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.V.Dhilipkumar
2)M. Menaka
3)Dr. G.R. Kangachidambaresan
4)Dr.M.Pandiyaraj
5)Dr.Muralidhar
6)Dr.P.Kumararaja

(57) Abstract :

Abstract A portable electrochemical water testing device with calibrated printed sensor interface The present invention relates to a portable electrochemical water testing device incorporating a printed electrochemical sensing strip, an electronic sensor-interface circuit, and an embedded calibration arrangement for determining pH of water samples. The sensing strip comprises a working electrode having an electrochemically responsive layer and a reference electrode configured to generate an electrical potential difference upon contact with a water sample. The sensor-interface circuit acquires the generated potential and communicates the corresponding electrical measurement to a microcontroller-based processing unit. The processing unit processes the acquired measurement and determines a calibrated pH value using stored calibration information derived from experimental characterization of the sensing strip. The calibration arrangement enables local interpretation of electrochemical measurements, while a communication interface facilitates display of the resulting values. The device provides a compact measurement arrangement suitable for aquaculture, agricultural water testing, and environmental monitoring.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129612 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A marine water quality monitoring buoy with multivariable sensor calibration

(51) International classification	:G01N 33/18, B63B 22/00, G01D 21/02, B63B 22/16, B63B 22/18	(71)**Name of Applicant :** **1)Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology** Address of Applicant :No.42 Vel tech Road Avadi, Chennai chennai Tamil Nadu India (72)**Name of Inventor :** **1)Dr.V.Dhilipkumar** **2)M. Menaka** **3)Dr. G.R. Kangachidambaresan**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Abstract A marine water quality monitoring buoy with multivariable sensor calibration The present invention discloses a marine water quality monitoring buoy incorporating multivariable sensor calibration for continuous monitoring of aquatic environmental conditions. The buoy comprises a floating structure, water quality sensors, an embedded processing unit, an inertial sensing unit, a solar-powered electrical arrangement, and wireless communication interfaces. The sensors measure pH, dissolved oxygen, oxidation-reduction potential, salinity, turbidity, and temperature. A multivariable calibration arrangement applies experimentally determined calibration relationships to compensate for temperature-dependent pH variations and temperature- and salinity-dependent dissolved oxygen variations. The processing unit acquires and processes sensor measurements and transmits the resulting data to a remote monitoring platform. The platform provides real-time visualization, historical analysis, and machine learning-assisted prediction of selected water quality parameters. The invention facilitates continuous monitoring and calibrated measurement of water quality parameters in marine aquaculture and related aquatic environments.

No. of Pages : 24 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129616 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NANO-ENGINEERED PHOTOCATALYTIC SELF-CLEANING COATING COMPOSITION AND METHOD FOR ENERGY-EFFICIENT BUILDING SURFACES

(51) International classification	:C09D 7/62, C09D 5/00, C09D 1/00, C09D 5/16, C09D 7/61	(71)**Name of Applicant :** **1)Dr. R.Priscilla** Address of Applicant :Associate Professor of Physics, Department of Physics, ADM College for Women, Nagapattinam, Tamil Nadu -611001 Tamil Nadu India **2)Mr. Goda Sreenivasulu Reddy** **3)Dr. SK. Mahammad Ali** **4)Miss Bhosale Shilpa Sanjay** **5)Dr.S.Surya** **6)Dr.T.Srinivasan** **7)Dr. M.C. Rao** **8)Mr. Swaroop Mallick** **9)Dr. Buddhaghosh Arjun Shingade**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	(72)**Name of Inventor :** **1)Dr. R.Priscilla** **2)Mr. Goda Sreenivasulu Reddy** **3)Dr. SK. Mahammad Ali** **4)Miss Bhosale Shilpa Sanjay** **5)Dr.S.Surya** **6)Dr.T.Srinivasan** **7)Dr. M.C. Rao** **8)Mr. Swaroop Mallick** **9)Dr. Buddhaghosh Arjun Shingade**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

[042] A nano-engineered photocatalytic self-cleaning coating composition and method for energy-efficient building surfaces are disclosed. The composition comprises visible-light-responsive modified anatase titanium dioxide nanoparticles, a silica-forming binder, a hydrophilic surface-modifying silane and a water- and/or alcohol-based carrier. An optional near-infrared-attenuating nanoparticle phase reduces solar near-infrared transmission. The coating is deposited on architectural glazing or facade substrates, optionally over an adhesion interlayer, and cured at a substrate-compatible temperature to form a porous silica-rich layer with photocatalytically accessible particles. Incident daylight assists degradation of susceptible organic surface contaminants, while a hydrophilic outer region promotes water-sheet removal of loosened deposits. A preparation and deposition method and a coated building article are also disclosed. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129626 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for Android Malware Detection Using Permission-Based and Behavioral Features

(51) International classification	:G06F 21/56, G06F 21/55, G06F 21/57, H04L 29/06, G06F 21/53
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr. Manoj T Joy

Address of Applicant :Professor & HOD, Department of Artificial Intelligence, Amal Jyothi College of Engineering, Kanjirappally, Koovappally(Po), Kottayam, Pin: 686518, Kerala, India Kerala India

2)Mathew George

3)Palak Pravinkumar Dave

4)Mr. Machindra Sanjay Pawar

5)Jimesh Kumar Mahesh Kumar Rana

6)Dr. Durgesh H Tupe

7)Dr. Barkha Joshi

8)Dr. Divvela Srinivasa Rao

9)Mr. Tejaskumar Patel

10)Harsh Yogendrakumar Kantawala

11)Dr. V S Bhagavan

(72)**Name of Inventor :**

1)Dr. Manoj T Joy

2)Mathew George

3)Palak Pravinkumar Dave

4)Mr. Machindra Sanjay Pawar

5)Jimesh Kumar Mahesh Kumar Rana

6)Dr. Durgesh H Tupe

7)Dr. Barkha Joshi

8)Dr. Divvela Srinivasa Rao

9)Mr. Tejaskumar Patel

10)Harsh Yogendrakumar Kantawala

11)Dr. V S Bhagavan

(57) Abstract :

The present invention relates to a system and method for detecting malicious Android applications through combined permission-based and behavioral analysis. The system comprises an application information acquisition module, a permission analysis module, a behavioral monitoring module, a feature extraction module, a permission-behavior correlation engine, a risk assessment module, and a security response module. Declared application permissions and available execution observations are processed to generate structured feature representations. The correlation engine evaluates relationships between requested permissions and observed activities to identify suspicious combinations and contextual deviations. A risk assessment mechanism generates a risk score and classifies the application according to predefined security criteria. The security response module generates detection reports and facilitates appropriate responses, including notifications, additional inspection, and supported remediation actions. The invention provides a contextual approach to mobile malware detection by combining permission-related evidence with observed application behavior.

No. of Pages : 23 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129630 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NON-CONTACT AC VOLTAGE SENSING AND MARKING DEVICE

(51) International classification	:G01R 19/155, G01R 19/00, G01R 29/08, G01R 15/14, G01R 29/12	(71)**Name of Applicant :** **1)NITTE (DEEMED TO BE UNIVERSITY)** Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India **2)PAWAN S POOJARY** **3)RAJATH N RAO** (72)**Name of Inventor :** **1)PAWAN S POOJARY** **2)RAJATH N RAO**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed herein is a non-contact AC voltage sensing and marking device (100), comprising a housing (102) configured to accommodate and provide structural support to the components of the device (100), a sensing antenna (104) detects an oscillating electric field without physical electrical contact and generate an induced electrical signal corresponding to the detected electric field, a transistor amplification arrangement (106) comprising a plurality of BC547 NPN transistors (108) amplifies the induced electrical signal, an indication unit (110) provides indication in response to the amplified signal, a power supply unit (112) associated with the housing (102) and configured to provide electric power to the components of the device (100), a writing assembly (114) marks the electrical conductor. The sensing antenna (104) comprises a copper coil and a rigid wire probe. The plurality of BC547 NPN transistors (108) are arranged in a Darlington configuration to provide high-gain amplification.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129631 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR AUTOMATED INTERVIEW SIMULATION AND EVALUATION

(51) International classification	:G06Q 10/10, G10L 15/22, G10L 15/26, G06F 3/16, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India
2)SUJAN S
3)K CHINMAYA SHARMA
4)DHANYA T KUNDER
5)MANJUNATHA A S

(72)**Name of Inventor :**
1)SUJAN S
2)K CHINMAYA SHARMA
3)DHANYA T KUNDER
4)MANJUNATHA A S

(57) Abstract :
Disclosed herein is a system (100) for conducting adaptive interview simulation, the system (100) comprising a user interface (102) integrated into a user device (104), the user interface (102) collect information of candidate, a capturing unit (106) capture visual data of a candidate, a voice input unit (108) receive spoken responses of a candidate, a communication network (110) facilitate data exchange, the processing unit (112) extract textual content from received documents, wherein the processing unit (112) comprises, a data input module (114), a data authentication module (116), a resume upload module (118), a data preprocessing module (120), a domain selection module (122), a question preparation module (124), an anti-cheat module (126), a real-time adaptive module (128), an evaluation module (130) and an output module (132).

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129632 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : HAND PHYSIOTHERAPY SYSTEM WITH LONGITUDINAL REHABILITATION MANAGEMENT

(51) International classification	:G16H 20/30, A63B 24/00, A61H 1/02, G16H 40/67, G16H 40/63
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India
2)VARADRAJ
3)SUMANTH
4)PRAVEENA KUMARI M K
(72)**Name of Inventor :**
1)VARADRAJ
2)SUMANTH
3)PRAVEENA KUMARI M K

(57) Abstract :
Disclosed herein is a hand physiotherapy system (100) with longitudinal rehabilitation management system, that comprises a user interface (102) integrated into a user device (104) the user interface (102) configured to present interactive therapeutic exercises and rehabilitation information, a capturing unit (104) configured to acquire image data representative of a patient's hand during performance of the therapeutic exercises, a communication network (106), a processing unit (108) coupled to the user interface (102) and the capturing unit (104) via the communication network (106), the processing unit (108) configured to process data, wherein the processing unit (108) further comprises a data input module (110), a patient authentication module (112), a data preprocessing module (114), a hand tracking module (116), a patient calibration module (118), a rehabilitation exercise module (120), an adaptive rehabilitation module (122), a movement stabilization module (124), a clinical assessment module (126), a recovery management module (128) and an output module (130).

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129633 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR EMERGENCY VEHICLE DISPATCH AND TRAFFIC-AWARE ROUTING BASED ON DYNAMIC HOSPITAL CAPACITY SCORING

(51) International classification	:G06Q 10/02, G16H 40/20, G08G 1/00, G08G 1/123, H04W 4/029	(71)**Name of Applicant :** **1)NITTE (DEEMED TO BE UNIVERSITY)** Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India **2)PRITHVIRAJ** **3)RAKSHA V SHETTY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)PRITHVIRAJ** **2)RAKSHA V SHETTY**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed herein is a system (100) for emergency vehicle dispatch and traffic-aware routing based on hospital capacity scoring, comprising a user interface (102) on a user device (104), a communication network (106), and a processing unit (108) comprising an input module (110) receiving incident and telemetry data, a preprocessing module (112) normalizing the location data, a real-time tracking module (114) generating a state representation, a destination scoring module (116) computing a destination score, a hospital identification module (118) identifying candidate hospitals, a route determination module (120) selecting a hospital and route, a capacity reservation module (122) reserving declared bed capacity, an in-transit re-scoring module (124) triggering route diversion, a logging module (126), an output module (128), and a database (130).

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129634 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR EARLY PREDICTION OF NEUROLOGICAL DISORDERS WITH INTEGRATED MENTAL HEALTH RISK ASSESSMENT

(51) International classification	:G16H 50/30, G16H 10/60, G16H 50/20, A61B 5/00, A61B 5/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India
2)DR. MANGALA SHETTY
3)CYOLLA MONTHI MENEZES
4)SPOORTHI B SHETTY
(72)**Name of Inventor :**
1)DR. MANGALA SHETTY
2)CYOLLA MONTHI MENEZES
3)SPOORTHI B SHETTY

(57) Abstract :
Disclosed herein is a system (100) for predicting neurological disorders and assessing mental health risk based on patient-related data, the system (100) comprising a user interface (102) integrated into a user device (104), the user interface (102) collect the patient related data, communication network (106) to facilitate data exchange, a processing unit (108) connected to the user interface (102) via the communication network (106), wherein the processing unit (108) comprises a data input module (110) to receive data, a data preprocessing module (112) to validate, normalize, and transforms the patient-related data to a processed dataset, a feature extraction module (114) extract the disease-relevant features, a prediction module (120) analyse the extracted feature data, a confidence score module (122) generate an automated risk evaluation for risk scores and corresponding confidence values, an interpretation module (124) provide the transparent and interpretated results, output module (126) present results through a user interface (102).

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129636 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI based Real Time Object Detection System

(51) International classification	:G06N 3/08, G06N 3/04, G06V 10/764, G06K 9/00, G06K 9/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr.V.Jayaraj

Address of Applicant :Dean – Electrical Sciences, Professor & HoD, Department of Electronics and Communication Engineering, Nehru Institute of Engineering and Technology, Coimbatore–641105. Tamil Nadu India

2)Dr. C J Raman

3)Balamurugan A

4)Manish Kumar E

5)Mohamed Kamaldeen Z

6)Dr. S Nagakishore Bhavanam

7)Aditya J

8)Dharunraj M K

9)Arunraj S

10)Balaguru S

(72)**Name of Inventor :**

1)Dr.V.Jayaraj

2)Dr. C J Raman

3)Balamurugan A

4)Manish Kumar E

5)Mohamed Kamaldeen Z

6)Dr. S Nagakishore Bhavanam

7)Aditya J

8)Dharunraj M K

9)Arunraj S

10)Balaguru S

(57) Abstract :

The present invention relates to an Artificial Intelligence (AI)-based real-time object detection system for identifying, classifying, locating, and tracking multiple objects from live video streams or captured images. The proposed system utilizes a camera module, image acquisition unit, preprocessing module, deep learning-based object detection model, object classification module, and real-time output visualization unit to detect objects with improved accuracy and reduced processing latency. The acquired video frames are processed using image enhancement techniques, resizing, normalization, and noise reduction to improve the quality of input data. A trained convolutional neural network (CNN) and object detection architecture are employed to extract visual features and identify multiple objects within a single frame. The system determines object categories, confidence scores, and bounding box coordinates for each detected object. A real-time tracking mechanism maintains object identities across consecutive frames and minimizes duplicate detections. An adaptive processing module dynamically adjusts image resolution, frame processing rate, and computational workload based on available hardware resources and environmental conditions. The detected information is displayed through a graphical interface with object labels, confidence values, and tracking information. The proposed invention can be implemented in surveillance systems, autonomous vehicles, industrial automation, traffic monitoring, healthcare assistance, smart retail, and intelligent robotic applications. The system provides an integrated, scalable, and computationally efficient solution for real-time visual object detection and monitoring.

No. of Pages : 18 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129641 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Design and Implementation of an IoT-Based Smart Classroom for Digital Education and Automated Learning Support

(51) International classification	:G06Q 50/20, G09B 5/08, H04L 29/08, G09B 5/14, G06K 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Narayana Engineering College(F1)
Address of Applicant :Narayana Engineering College, Gudur State: Andhra Pradesh-524101 Gudur Andhra Pradesh India
2)Dr. G.Srinivasulu Reddy
3)Dr. K.Vishwak sena Reddy
4)Dr. Rajesh Reddy
5)Kilari Kumar
6)Pathapati Omkar Reddy
7)Mr. M.Guru Prasad
(72)**Name of Inventor :**
1)Dr. G.Srinivasulu Reddy
2)Dr. K.Vishwak sena Reddy
3)Dr. Rajesh Reddy
4)Kilari Kumar
5)Pathapati Omkar Reddy
6)Mr. M.Guru Prasad

(57) Abstract :
The rapid advancement of the Internet of Things (IoT) has enabled the transformation of traditional classrooms into smart and interactive learning environments. This project proposes an IoT-based smart classroom device designed to enhance digital teaching and provide automated learning assistance. The system integrates various sensors, smart devices, and communication technologies to create a connected classroom ecosystem. The proposed system enables automated attendance using RFID or face recognition, real-time monitoring of classroom conditions such as temperature, lighting, and occupancy, and efficient control of electrical appliances. It also incorporates digital teaching tools such as smart boards and multimedia systems to improve student engagement. Furthermore, the system utilizes cloud-based platforms and data analytics to track student performance and provide personalized learning support. By automating routine tasks and enabling intelligent decision-making, the smart classroom system reduces manual effort, improves teaching efficiency, and enhances the overall learning experience. The solution also supports remote access and hybrid learning, making education more flexible and accessible. Despite challenges such as implementation cost and data privacy concerns, the proposed system demonstrates significant potential in modernizing educational infrastructure.

No. of Pages : 10 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129643 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Cross-Domain Federated Airspace Coordination for Resilient ATM, AAM and UTM/U-Space Interoperability

(51) International classification	:G08G 5/00, G08G 5/04, H04W 84/06, H04L 12/801, H04W 4/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mrs. Sri Sakthi. A
Address of Applicant :Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), KPR Institute of Engineering and Technology, Avinashi Road, Arasur, Coimbatore Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. Sri Sakthi. A
2)Mr. Dhamodharan M
3)Mr. Ahilesh kumar D
4)Mr. Ajay A T
5)Mr. Dhivakar N K

(57) Abstract :
Cross-domain federated airspace coordination for resilient ATM, AAM and UTM/U - space interoperability Abstract: The present invention discloses a Cross-Domain Federated Airspace Coordination (CDFAC) system and method for enabling resilient and interoperable coordination of heterogeneous aerial traffic across Advanced Air Mobility (AAM), Unmanned Traffic Management (UTM), U-Space, and Air Traffic Management (ATM) domains. The system comprises a federated interoperability layer, data standardization module, resilience assessment module, trajectory prediction and coordination module, cross-domain conflict detection and resolution module, and contingency management module. The system standardizes heterogeneous flight information and evaluates information freshness, communication delays, data completeness, message loss, service availability, and uncertainty to classify operational conditions as stable, degraded, or critical. Based on the assessed conditions, the system adaptively manages trajectories, resolves conflicts, coordinates vertiport sequencing, and initiates contingency responses, supporting fault-tolerant airspace coordination under normal and degraded operating conditions.

No. of Pages : 7 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129646 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NON-CONTACT THERMAL IMAGING SYSTEM AND METHOD FOR MONITORING AN INTRAVENOUS CANNULA SITE

(51) International classification	:A61B 5/01, A61B 5/00, G06T 7/00, A61M 5/44, G01J 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM University- AP
Address of Applicant :Neerukonda, Mangalagiri Mandal Guntur District-522240, Andhra Pradesh, India Guntur Andhra Pradesh India
(72)**Name of Inventor :**
1)KOPPARAPU, Jahanavi
2)DONEPUDI, Siva Padmavathi
3)PALLIPAMULA, Ratna Bhaskar
4)GADHE, Chandra Reddy
5)Dr. Jyothi Naik
6)Dr. Arun Babu Thirunavukkarasu
7)Dr. Nataraj Agrahara Rangashamiah
8)Dr. Karnam Anantha Sunitha
9)Dr. Anirban Ghosh

(57) Abstract :
A non-contact thermal imaging system (100) and method (200) for monitoring an intravenous cannula site are provided. An infrared thermal camera (102), spaced from a patient's limb, captures a thermal image covering a cannula insertion site and a reference region. A processing unit (104) includes sequentially connected modules: a preprocessing module (106); a region of interest selection module (108) that selects the insertion site and the reference region; a temperature extraction module (110) that extracts a cannula-site temperature and a reference temperature; a temperature difference calculation module (112) that subtracts the reference temperature from the cannula-site temperature; a threshold analysis module (114) that classifies the site as normal below a threshold or as possible early inflammation at or above it; and a decision support module (116) that outputs an indication based on the classification. A display unit (118) is coupled to the processing unit (104). (Figure. 1)

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129654 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MIND ASHER- AI POWERED MENTAL HEALTH CHAT AND THERAPY PLATFORM

(51) International classification	:G09B 19/00, A61M 21/00, G16H 20/70, G06Q 50/20, A61B 5/16	(71)**Name of Applicant :** **1)Srinivas University Institute of Engineering and Technology** Address of Applicant :Srinivas University Institute of Engineering and Technology, Mukka, Mangaluru – 574146, State of Karnataka, an Indian Institution Mangaluru Karnataka India (72)**Name of Inventor :** **1)Ms. Preethika Sequeira** **2)Mohammed Shareeq H** **3)Manisha**
(31) Priority Document No	:NA	**4)Meghana Acharya**
(32) Priority Date	:NA	**5)Megha**
(33) Name of priority country	:NA	**6)Abhinaya Ramesh PK**
(86) International Application No Filing Date	: :01/01/1900	**7)Fathimathul Shada PV** **8)Harshith TS**
(87) International Publication No	: NA	**9)Thaniya M**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT Mind-Asher: AI Powered Mental Health Chat and Therapy Platform Mental health is becoming a serious concern in today's fast-paced world, especially among students and working professionals who constantly deal with stress, pressure, and emotional challenges. Many people still hesitate to seek help due to stigma, lack of awareness, or limited access to professionals, which often leaves them struggling alone. MindAsher is designed to address this gap by offering a simple and accessible AI-powered mental health support platform. It provides features like guided meditation, breathing exercises, calming music, motivational content, and mood journaling to help users understand and manage their emotions. The platform also includes information about doctors and hospitals for those who need professional care. Built by using React.js, it ensures a smooth and organized user experience. An integrated AI chatbot interacts with users, offering supportive conversations and personalized suggestions. Overall, MindAsher promotes healthy coping habits and makes mental wellness support more approachable and convenient.

No. of Pages : 5 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129658 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Driven Multilingual Knowledge Retrieval Using MiniLM Semantic Embeddings

(51) International classification	:G06N 5/02, G06F 17/30, G06F 16/338, G06F 16/33, G06F 16/332	(71)**Name of Applicant :** **1)S.SANGEETHAPRIYA** Address of Applicant :S.A Engineering College (Autonomous),Dept of English, Chennai Tamil Nadu India **2)S.A. ENGINEERING COLLEGE** (72)**Name of Inventor :** **1)S.SANGEETHAPRIYA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The invention relates to an artificial-intelligence-enabled digital knowledge management system and method for preserving, organising, discovering and disseminating Indian Knowledge Systems (IKS) through multilingual and context-aware digital access. The system receives textual, oral, visual and practice-based knowledge records and associates them with structured metadata including language, location, knowledge domain, source, contextual description, permission status and community validation status. An AI processing module supports classification, semantic indexing and retrieval of records based on user queries expressed in supported languages. A multilingual access module supports language-based discovery and presentation, while a context and permission control module applies access conditions and displays source and contextual information to reduce decontextualised or unauthorised representation. A user interface enables learners, researchers and community members to discover knowledge through topic-based and natural-language queries. The system is intended to support responsible digital preservation and access to knowledge concerning ecology, sustainability, local innovation, culture and community practices. The proposed system combines AI-assisted discovery with contextual metadata, language accessibility and human or community validation in a single workflow.

No. of Pages : 6 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129659 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Voice Controlled Data Analytics & Visualization Tool

(51) International classification	:G10L 15/22, G10L 15/18, G10L 15/26, G06F 16/248, G06F 16/332
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Srinivas University , Institute of Engineering and Technology
Address of Applicant :Srinivas University Institute of Engineering and Technology, Mukka, Mangaluru – 574146, State of Karnataka, an Indian Institution Mangaluru Karnataka India
(72)**Name of Inventor :**
1)Swathi R
2)Shivani D Shetty
3)Vanshitha A V Puthran
4)Varsha M Baitmangalkar
5)Dhanya G
6)Vijesh
7)Dhanush
8)Nandan H

(57) Abstract :
ABSTRACT VOICE CONTROLLED DATA ANALYTICS & VISUALIZATION TOOL For individuals without technical knowledge of data analytics tools or query languages, performing data analysis can be challenging. This paper presents a voice-controlled data analytics and Visualization Tool that simplifies interaction with datasets using voice commands. The system integrates speech recognition, natural language processing (NLP), and visualization techniques to convert spoken input into executable queries. User voice commands are processed and translated into structured instructions for a data analytics engine, which retrieves relevant information from datasets. The results are then displayed as interactive visualizations such as bar charts, pie charts, or line graphs based on the user's request. The main objective of this system is to make data analysis more accessible to users without programming or analytical expertise. Experimental results show that the system can accurately interpret voice commands and generate meaningful visual outputs. The proposed approach improves usability, enhances user experience, and demonstrates the potential of voice-enabled interfaces in making data-driven decision-making more intuitive and accessible.

No. of Pages : 6 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129673 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXTERNALLY MOUNTABLE PASSIVE DYNAMIC VIBRATION ISOLATION

(51) International classification	:B23B 29/034, B23Q 11/00, B23B 29/02, F16F 15/02, F16F 7/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Amrita Vishwa Vidyapeetham
Address of Applicant :Amrita Vishwa Vidyapeetham, Coimbatore Campus, Coimbatore - 641112, Tamil Nadu, India. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)SUNDARAM, Saravanamurugan
2)KRISHNASWAMY, Rameshkumar
3)CHIDAMBARESH, Shravan
4)JAYAPRAKASH, Krishna Prakash

(57) Abstract :
The present disclosure provides a passive dynamic vibration control device (100) for mitigating machining chatter. The device (100) includes an absorber frame (102) defining a tool receiving recess (102-2A) that slidably receives a boring bar (110). A floating piece (108) and the absorber frame (102) constitute a damping medium (112) that dissipates vibrational excitation (110-4) via material damping. A spring-mass assembly (104) tuned to a fundamental frequency (110-2) comprises a mass unit (104-2) mounted between a tool-side spring (104-4) and a support-side spring (104-6). The assembly (104) generates a phase-opposed oscillatory response (104-8), which transmits an anti-resonant counter-force back through the damping medium (112) to suppress tool vibrations. Unlike invasive absorbers, this externally mountable 3D-printed design facilitates customizable damping and optimized placement along a tool length to enhance machining stability.

No. of Pages : 34 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129676 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PHYSIOTHERAPY MONITORING SYSTEM AND METHOD THEREOF

(51) International classification	:G16H 20/30, A61B 5/11, G16H 50/20, A61B 5/00, A63B 24/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)NATARAJAN B
2)TRILOK NATH PANDEY
3)PODALA KEERTHANA
4)ROHIT E
5)PRANESH UMASANKAR

(57) Abstract :
The present disclosure relates to a physiotherapy monitoring system (100) and method (300). The system (100) includes a sensor unit (102), a processor (104), and a memory (106). Sensor unit (102) captures motion data of patient performing physiotherapy exercise over physiotherapy exercise session. Segmentation unit (108) segments the motion data into a plurality of repetitions. Feature unit (110) determines a feature value for each repetition. A baseline unit (112) fits a time-series model to the feature values of preceding repetitions during the physiotherapy exercise session and in an absence of a previously trained model, to obtain a prediction interval for a subsequent repetition. An anomaly unit (114) identifies the subsequent repetition as an anomalous repetition when the feature value lies outside the prediction interval, and excludes the anomalous repetition to obtain a set of validated repetitions. An evaluation unit (116) determines a motion-quality parameter from the set of validated repetitions.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129677 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ASSESSING REAL-TIME FRESHNESS OF VEGETABLES USING WEARABLE DEVICE

(51) International classification	:G01N 33/02, A41D 19/00, A41D 19/015, A61B 5/00, A01F 25/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India.
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)HARSHA VARDHAN G
2)ANNEM SRAVAN KUMAR
3)OMANA J
4)INDIRA B

(57) Abstract :
A system 100 to assess freshness of vegetables includes a hand-worn glove 200 including a plurality of sensors 102, and each sensor 102 configure to assess a plurality of freshness parameters of target vegetable; one or more output modules 108 to alert user about decayed status of checked vegetables; and a microcontroller 104 in communication with the sensors 102, including a communication module 106, a database, one or more processors to perform operations to acquire freshness parameters of the vegetable from each sensor 102; compare each received freshness parameter with respective predefined freshness thresholds; classify freshness of the vegetable based on obtained freshness parameter based on comparison; transmit results over a Wi-Fi network 112 to display on a freshness dashboard 108-C of mobile device 114 held by an assessor; and generate alerts through audio and visual means about decayed status of the vegetable.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129678 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SPIRULINA-LOADED MICROSPHERE MUCOADHESIVE HERBAL GEL INCORPORATING NATURAL POLYMERS AND A PROCESS FOR PREPARATION THEREOF

(51) International classification	:A61K 36/886, A61K 47/36, A61K 35/748, A61K 9/16, A61K 9/00	(71)**Name of Applicant :** **1)Dr.M. Arunima** Address of Applicant :Bojjam Narasimhulu Pharmacy College For Women, Saidabad, Hyderabad, 500059,Telangana Telangana India (72)**Name of Inventor :** **1)Dr.M. Arunima** **2)Dr.P. Manichandrika** **3)Dr. Konda.Sri Chaya Reddy**
(31) Priority Document No	:NA	**4)Dr.R.Kusuma**
(32) Priority Date	:NA	**5)Edukulla Sushma**
(33) Name of priority country	:NA	**6)Morupoju Sushma**
(86) International Application No Filing Date	: :01/01/1900	**7)Gaini Madhri** **8)Maram Laxmi Prasanna**
(87) International Publication No	: NA	**9)Kowkuntla Dhanusha**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**10)Gaddey Ashwini**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

. ABSTRACT The present invention relates to a Spirulina-loaded microsphere mucoadhesive herbal gel incorporating natural polymers and a process for its preparation. Spirulina-loaded microspheres are prepared by ionotropic gelation using sodium alginate and calcium chloride and are incorporated into a gel matrix containing fenugreek mucilage, okra mucilage, Aloe vera gel, xanthan gum, chamomile oil, glycerin and pharmaceutically acceptable auxiliary ingredients. The natural polymer matrix provides viscosity and mucoadhesive characteristics, while the microsphere system provides sustained release of Spirulina. In an optimized embodiment, the gel contains 5% Spirulina-loaded microspheres, 1.5% fenugreek mucilage, 1.5% okra mucilage, 10% Aloe vera gel, 0.25% xanthan gum, 0.50% chamomile oil, 5% glycerin and 0.20% methyl paraben. The optimized formulation showed a pH of 6.12 ± 0.08, viscosity of 25,000 ± 350 cP, mucoadhesive strength of 34.5 ± 1.4 g, an Escherichia coli inhibition zone of 18.2 ± 0.8 mm and approximately 94.6% cumulative in-vitro release at 12 hours under the reported experimental conditions. The invention provides a plant-based localized controlled-release formulation platform.

No. of Pages : 16 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129679 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT HUMAN–AI COLLABORATIVE FRAMEWORK FOR LITERARY CRITICISM, CONTEXTUAL REASONING, AND PERSONALIZED PEDAGOGICAL DECISION-MAKING

(51) International classification	:G06N 5/04, G06N 5/02, G09B 19/06, G09B 19/00, G06F 16/332
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K. Nuthana Sai
Address of Applicant :Research Scholar, Dr. RVR NRIIT Deemed to be University, Agiripalli, Eluru, Pin: 521 212, Andhra Pradesh, India. Andhra Pradesh India
2)Dr. A. Neeraja Padma
3)Dr. V. Harileela
4)Dr. Sujata Sonthalia
5)Dr. G. Balreddy
6)Dr. V. Aparna Reddy
7)Dr. K. Santha Kumari
(72)**Name of Inventor :**
1)K. Nuthana Sai
2)Dr. A. Neeraja Padma
3)Dr. V. Harileela
4)Dr. Sujata Sonthalia
5)Dr. G. Balreddy
6)Dr. V. Aparna Reddy
7)Dr. K. Santha Kumari

(57) Abstract :
AN INTELLIGENT HUMAN–AI COLLABORATIVE FRAMEWORK FOR LITERARY CRITICISM, CONTEXTUAL REASONING, AND PERSONALIZED PEDAGOGICAL DECISION-MAKING Abstract: The present invention discloses an Intelligent Human–AI Collaborative Framework for Literary Criticism, Contextual Reasoning, and Personalized Pedagogical Decision-Making. The framework integrates Large Language Models with retrieval-augmented generation, a multi-layer Literary Knowledge Graph, stylometric Natural Language Processing, multi-lens critical interpretation, Explainable Artificial Intelligence, learner Digital Twins, and a fuzzy multi-criteria pedagogical decision engine within a unified architecture. Literary texts, socio-historical context archives, critical theory repositories, learner interpretations, and instructor objectives are processed to generate evidence-grounded, context-aware readings across feminist, postcolonial, psychoanalytic, Marxist, formalist, and reader-response lenses. A human-in-the-loop collaborative critique cycle allows learners and instructors to question, validate, or override machine interpretations, producing transparent consensus scores with explanations. Continuously updated learner profiles drive personalized learning pathways, adaptive tasks, and targeted feedback. The invention improves interpretive depth, critical thinking, contextual understanding, assessment transparency, and instructional efficiency in literature education at school and university levels while preserving human interpretive authority and academic integrity.

No. of Pages : 28 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129680 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NANOCARRIER-BASED THERAPEUTIC SYSTEM FOR TARGETED CARDIAC REGENERATION AND MYOCARDIAL REPAIR

(51) International classification	:A61P 9/10, A61K 38/18, A61K 31/573, A61K 47/34, A61K 9/51
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Mr. Akash Jakkannavar

Address of Applicant :Assistant Professor, Department of Pharmacognosy, JSS College of Pharmacy, Mysuru-570015, India. Karnataka India

2)Mr. Adel Ahmed

3)Ms. Subuhana Buhari

4)Ms. Singamsetty Sahasra Annapurna

5)Ms. Gaurika Chaudhari

6)Ms. Meera Nair

7)Dr. Mruthunjaya Kenganora

8)Mr. Shivadarshan J.

(72)**Name of Inventor :**

1)Mr. Akash Jakkannavar

2)Mr. Adel Ahmed

3)Ms. Subuhana Buhari

4)Ms. Singamsetty Sahasra Annapurna

5)Ms. Gaurika Chaudhari

6)Ms. Meera Nair

7)Dr. Mruthunjaya Kenganora

8)Mr. Shivadarshan J.

(57) Abstract :

The present invention discloses a nanocarrier-based therapeutic system engineered for targeted cardiac regeneration and myocardial repair following ischemic injury. The system comprises biodegradable polymeric nanocarriers formulated from poly(lactic-co-glycolic acid) and polyethylene glycol copolymers with a core-shell architecture, functionalized with cardiac-homing peptides for site-specific targeting of infarcted myocardial tissue. The nanocarrier encapsulates a synergistic combination of vascular endothelial growth factor, insulin-like growth factor 1, and dexamethasone, providing simultaneous pro-angiogenic, pro-survival, and anti-inflammatory therapeutic activities. A dual stimuli-responsive release mechanism incorporating pH-sensitive and reactive oxygen species-cleavable components enables spatiotemporally controlled drug liberation within the pathological microenvironment of the infarcted heart. Preclinical evaluation demonstrates significant improvements in left ventricular ejection fraction, reduc

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129682 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HYBRID DUAL-POWER INTELLIGENT AMPHIBIOUS RESCUE VEHICLE WITH AUTOMATIC MODE SWITCHING, SENSOR FUSION, SELF-BALANCING SYSTEM, AND EMERGENCY SOS COMMUNICATION ARCHITECTURE

(51) International classification	:B60F 3/00, B63H 21/17, B63H 21/21, G05D 1/00, B63G 8/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.Swetha

Address of Applicant :Student Department Of Electronics and communication, LOYOLA INSTITUTE OF TECHNOLOGY PALANCHUR CHENNAI 123, PALANCHUR,NAZRETHPET POST, CHENNAI-600123 Chennai Tamil Nadu India

2)A.Pragadheeshwari
3)Allen Jeffrey J
4)Anbazhagan A
5)Dr.K.R.Shanthy
6)Dr.V.Pushpa
7)Mrs.R.Umamaheswari
8)Mrs.B.Priyadharsini
9)Stephena E
10)Mr.M.Varadharajan
11)Mr.A.Manigandan
12)Dr.G.Sooryapriya
13)Mr.R.Sivakumar

(72)**Name of Inventor :**

1)K.Swetha
2)A.Pragadheeshwari
3)Allen Jeffrey J
4)Anbazhagan A
5)Dr.K.R.Shanthy
6)Dr.V.Pushpa
7)Mrs.R.Umamaheswari
8)Mrs.B.Priyadharsini
9)Stephena E
10)Mr.M.Varadharajan
11)Mr.A.Manigandan
12)Dr.G.Sooryapriya
13)Mr.R.Sivakumar

(57) Abstract :

ABSTRACT HYBRID DUAL-POWER INTELLIGENT AMPHIBIOUS RESCUE VEHICLE WITH AUTOMATIC MODE SWITCHING, SENSOR FUSION, SELF-BALANCING SYSTEM, AND EMERGENCY SOS COMMUNICATION ARCHITECTURE An intelligent amphibious rescue vehicle for operation on land and water is disclosed. The vehicle includes a sensor-fusion module with water-detection, ultrasonic depth, pressure, temperature, electrical-monitoring, and inertial sensors connected to a microcontroller-based control unit. The control unit processes sensor data to control transition between land and water modes and to support fault detection and stability control. A dual-power propulsion arrangement includes a primary diesel engine and a battery-powered electric backup motor. An inertial measurement unit monitors tilt and enables corrective action through propulsion adjustment or stabilizing structures. A buoyant hull incorporates multiple sealed compartments and foam-based flotation material. GPS and GSM modules support emergency SOS transmission using current or last-known location data. The vehicle is intended for flood rescue, disaster response, and emergency evacuation.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129689 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Secure and Energy-Efficient Cloud Data Protection System Using Elliptic Curve Cryptography

(51) International classification	:H04L 9/30, H04L 9/32, H04L 9/08, H04L 9/00, G06F 21/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VEMU INSTITUTE OF TECHNOLOGY
Address of Applicant :P.Kothakota, Puthalapattu Mandal, Tirupati - Chittoor Highway, Chittoor (Dt), 517112, Andhra Pradesh, India Chittoor Andhra Pradesh India
(72)**Name of Inventor :**
1)Ms. Thunga Haritha
2)Ms. Kuppan Venmathi
3)Ms. M. Pallavi

(57) Abstract :
The aim of the present invention is to provide an efficient and secure cloud data protection system that uses Elliptic Curve Cryptography to provide confidentiality, integrity, and controlled access to the data stored and transmitted within cloud systems. Due to the massive adoption of cloud technology, the existing data protection mechanisms like RSA and AES encounter major issues such as the size of the keys, computation cost, scalability, and power consumption due to the execution of complex calculations. These issues predictably face major concerns within the cloud system particularly while dealing with large-scale cloud systems and constrained scenarios. The proposed invention overcomes the above-mentioned drawbacks by incorporating the use of Elliptic Curve Cryptography, offering the same or even better security levels in smaller keys. It also promotes the encryption of data before storage, ensuring data security in case the cloud storage solutions are breached. It makes use of a role-based access control system whereby the request to gain access to the data can only be achieved by authorized persons. It uses elliptic curve mathematics in secure key exchange methods to establish the symmetric encryption keys required in the efficient encryption and decryption of data. The innovation also enables secure data sharing among data owners and authorized users. The innovation can be applicable within cloud computing environments and Internet of Things (IoT) due to its real-time secure data processing capabilities. On the whole, the invention offers a scalable, energy-conscious, and highly secured solution for protecting data in the cloud. It boosts the trustworthiness of cloud computing because of its strong encryption methods, efficiency, and controlled access to data, and hence contributes to the evolution of secure and trusted cloud infrastructure.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129712 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : QUICKBITE: A DEVOPS-DRIVEN FOOD DELIVERY PLATFORM USING DOCKER AND KUBERNETES

(51) International classification	:G06F 9/445, G06F 8/61, G06F 9/44, G06F 8/71, G06F 8/65
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)sanjay nivas g
Address of Applicant :1-26-42 Indhura nagar 1st block office madurai road aruppukottai,virudhunagar. Tamil Nadu India
2)Mrs.N.G.Dharaniya
(72)**Name of Inventor :**
1)Sanjay Nivas G
2)Yogeshkar P
3)Sunil Kumar R
4)Yuva Mugesh M
5)Mrs.N.G.Dharaniya

(57) Abstract :
Reliable and rapid software delivery plays a vital role in ensuring that an online service stays available and up to date. The QuickBite platform is a comprehensive DevOps-driven food delivery system developed to automate the path from source code to a running application and make releases dependable for every stakeholder. The platform combines a containerized React, Spring Boot and MySQL application, a Jenkins continuous integration pipeline, a Docker Hub image registry, Kubernetes orchestration and Argo CD GitOps synchronization to support the entire delivery journey. With automated builds, Git-based manifest management and continuous cluster synchronization, the QuickBite platform empowers development and operations teams to release changes consistently, trace every deployment and keep the running service aligned with its declared configuration—anytime, anywhere.

No. of Pages : 8 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129716 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Radioisotope production system for producing scandium radioisotopes and method therefor

(51) International classification	:G21G 1/10, H05H 6/00, A61N 5/10, G21G 4/08, G21G 1/00	(71)**Name of Applicant :** **1)INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD** Address of Applicant :IIT Hyderabad Road, Near NH-65, Sangareddy, Kandi, Telangana – 502284, India Sangareddy Telangana India (72)**Name of Inventor :** **1)Arun Kumar R M** **2)Bhuvanesh Ramakrishna**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The present disclosure relates to a laser-driven method and system for producing scandium radioisotopes for medical imaging applications. A high-intensity laser pulse irradiates a first target to generate an energetic proton beam. At least a portion of the proton beam is directed toward a boron-11 target, where proton-boron fusion generates alpha particles. The generated protons and alpha particles irradiate a calcium-containing target to produce scandium-44 and scandium-43 through the respective nuclear reactions ($^{44}Ca(p,n)^{44}Sc$) and ($^{40}Ca(\alpha,p)^{43}Sc$). The method enables the use of different incident-particle populations within a common laser-driven production framework. Illustrative estimates indicate end-of-bombardment activities of approximately 0.3 MBq for scandium-44 and 7.83 kBq for scandium-43. The disclosed approach provides a potential alternative or complementary pathway to conventional accelerator-based radioisotope production and may support positron emission tomography, radiopharmaceutical development, and future theranostic imaging applications through further experimental validation and optimization. Fig. 2

No. of Pages : 37 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129722 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Multimodal Artificial Intelligence System and Method for Interactive Processing, Translation, Visualization, and Educational Rendering of Classical Telugu Narrative Poems (Prabandhas)

(51) International classification	:G06N 20/00, G09B 5/06, G06N 3/08, G06N 3/04, G09B 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)D. Yuvasri
Address of Applicant :Department of Telugu Studies Sri Padmavati Mahila Visvavidyalayam Tirupati - 517502 Tirupati Andhra Pradesh India
2)Y. Bhavani Kumar
(72)**Name of Inventor :**
1)D. Yuvasri
2)Y. Bhavani Kumar

(57) Abstract :
A multimodal artificial intelligence system and method for interactive processing of classical Telugu narrative poems (prabandhas) is disclosed. The system receives a classical Telugu verse, performs automated metrical (chandassu) analysis, generates modern Telugu and English prose translations with word-level glosses, produces illustrative images and audio recitations, and supports text-grounded interactive querying through retrieval-augmented generation. The invention enables educational institutions and cultural platforms to make works such as Allasani Peddana's Manucharitra accessible to contemporary users while preserving scholarly accuracy. The system is deployable on cloud or edge devices.

No. of Pages : 11 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129723 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : STOCK PRICE PREDICTION USING TENSORFLOW

(51) International classification	:G06Q 40/04, G06N 3/08, G06N 3/04, G06Q 40/06, G06Q 40/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr Kavitha M
Address of Applicant :S.A. Engineering College, Veeraragavapuram, Chennai-77. Tamil Nadu India
2)Dr Muthukumarasamy S
3)S.A. Engineering college
(72)**Name of Inventor :**
1)Dr Kavitha M
2)Dr Muthukumarasamy S

(57) Abstract :
Stock trading remains a central activity in modern financial markets, and anticipating future price movements is a challenging yet important task. In this work, we explore a machine-learning framework designed to model and forecast stock price behavior using publicly available market data. Rather than relying solely on traditional approaches such as fundamental indicators, technical metrics, or classical time-series models, our method integrates a deep learning architecture capable of capturing nonlinear temporal patterns. The proposed system employs a Recurrent Neural Network built with Long Short Term Memory (LSTM) units, implemented in TensorFlow and Keras, to learn sequential dependencies from historical price data gathered through a Python-based data extraction tool using the Yahoo Finance API. The goal is to evaluate whether this neural architecture can produce more reliable short-term stock predictions compared to conventional strategies. The study outlines the model design, data preparation steps, and training methodology, ultimately demonstrating how machine-learning techniques can enhance the analysis of financial time-series data.

No. of Pages : 7 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129724 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Machine Learning Based Early Detection of Ongoing Cyber-Attack

(51) International classification	:G06N 20/00, H04L 29/06, G06F 21/56, G06F 21/55, H04L 9/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ms.RUBA SUDHA P
Address of Applicant :S.A. Engineering College, Veeraragavapuram, Chennai-77. Tamil Nadu India
2)Ms.G SARANYA
3)Ms.YOGA PREETHIKAA A M
4)S.A. Engineering College
(72)**Name of Inventor :**
1)Ms.RUBA SUDHA P
2)Ms.G SARANYA
3)Ms.YOGA PREETHIKAA A M

(57) Abstract :

Abstract— In the EXISTING SYSTEM, conventional cybersecurity and malware detection techniques are used to identify cyber threats, but they may not effectively detect evolving and ongoing cyber-attacks at an early stage. The existing approaches may require manual analysis and face challenges in handling large volumes of security data, changing attack patterns, false positives, and false negatives. In the PROPOSED SYSTEM, a Machine Learning Based Early Detection System is developed to identify and classify cyber threats using machine learning techniques. The system collects malicious and benign files, network information, system logs, and security events, followed by data preprocessing and feature extraction. A Random Forest algorithm is used to classify the input data as harmful or benign and support early detection of suspicious activities. In the MODIFICATION part of the project, the system is enhanced with continuous monitoring, threat intelligence, and automated alert and response mechanisms for improving cyber-attack detection and handling. The system can operate in real-time or batch-processing environments and can be updated to adapt to evolving cyber threats. The proposed Random Forest model achieved a reported accuracy of 99.64% during project evaluation, demonstrating its effectiveness in detecting and classifying cyber threats.

No. of Pages : 9 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129735 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A GENERALIZED ALMOST DISTRIBUTIVE FUZZY LATTICES APPROACH TO FACE RECOGNITION USING FUZZY MORPHOLOGICAL EROSION

(51) International classification	:G06K 9/00, G06T 7/00, G06K 9/68, G06K 9/46, G06K 9/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. T. Sangeetha
Address of Applicant :S.A. Engineering College, Veeraragavapuram, Chennai-77. Tamil Nadu India
2)Dr. S. Karpagam
3)Dr. S. Geetha
4)S.A. Engineering college
(72)**Name of Inventor :**
1)Dr. T. Sangeetha
2)Dr. S. Karpagam
3)Dr. S. Geetha

(57) Abstract :
With the assist of Generalized Almost Distributive Fuzzy Lattices (GADFL), this research provides a fuzzy morphological erosion for face recognition. Face recognition is a technique for identifying a person's face from a binary image. Because of the underlying nature of distributive fuzzy lattices, fuzzy morphological erosion is one of the most extensively utilised fuzzy techniques in image processing. As a result, the procedure is mostly used in the processing of various reactions of faces (binary picture) that have a partial ordering, resulting in a fuzzy lattice structure. Finally, based on the intensity value measurement of input photos, face detection has been performed. The training images were gathered from a variety of databases, including the Peak Signal - to - Noise Ratio, Signal - to - Noise Ratio and different Structural Similarity Index. The system worked effectively, detecting iris with an accuracy of 80%. This paper's software requirements are MATLAB software.

No. of Pages : 12 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129736 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Efficient Adsorptive Removal of Organic Pollutants from Reverse Osmosis Reject Water Using Cassia fistula Fruit Pod-Derived Activated Nanocomposite Carbon: A Comprehensive Kinetic Investigation

(51) International classification	:B01J 20/20, C02F 1/28, B01J 23/755, C02F 101/30, C02F 1/72	(71)**Name of Applicant :** **1)Dr.V. RAJESHBABU** Address of Applicant :S.A.ENGINEERING COLLEGE(AUTONOMOUS) POONAMALLEE-AVADI MAIN ROAD, CHENNAI-77 Tamil Nadu India **2)Ms.S.SUJA** **3)Ms.R.SHOBA** **4)S.A. ENGINEERING COLLEGE**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr.V. RAJESHBABU** **2)Ms.S.SUJA** **3)Ms.R.SHOBA**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT The present invention relates to a sustainable transition metal oxide–activated charcoal nanocomposite derived from Cassia fistula fruit pods (CFFP) for the treatment of reverse osmosis (RO) reject water containing organic pollutants. The invention provides a multifunctional nanocomposite comprising activated charcoal integrated with Nickel oxide (NiO), Copper oxide (CuO), Titanium dioxide (TiO_2) and Cobalt oxide (Co_3O_4), synthesised through a controlled chemical precipitation process to achieve enhanced adsorption and photocatalytic activity. The synthesized activated carbon and nanocomposite materials were characterized using X-ray diffraction (XRD), scanning electron microscopy (SEM), and UV visible spectroscopy to determine their crystallographic structure, surface morphology, and optical properties. The nanocomposite was evaluated for the removal of organic pollutants from RO reject water under batch operating conditions by investigating the effects of contact time, adsorbent dosage, initial pollutant concentration, and solution pH. Adsorption kinetics were analyzed using pseudo-first-order, pseudo-second-order, intraparticle diffusion, and Elovich kinetic models, with the pseudo-second-order model providing the best agreement with the experimental results. The synergistic integration of transition metal oxides with activated charcoal significantly enhanced adsorption efficiency, photocatalytic degradation, structural stability, and material reusability. The disclosed invention provides a cost-effective, scalable, and environmentally sustainable nanocomposite for advanced wastewater treatment while promoting the valorization of Cassia fistula fruit pod biomass into a high-value functional material suitable for industrial water purification applications.

No. of Pages : 19 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129744 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DIGITAL MEMORY DECAY: A TIME-DEPENDENT ENCRYPTION MODEL (TDEM) FOR SECURE CLOUD STORAGE

(51) International classification	:G06F 21/62, H04L 9/08, H04L 9/32, G06F 21/60, H04L 9/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RNS Institute of Technology
Address of Applicant :RNS Institute of Technology, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098 Bengaluru Karnataka India
2)Rithika Shetty
3)Sanjay N
4)Chethana R
5)Dr Manohar P
6)Dr Abhijith H V
(72)**Name of Inventor :**
1)Dr Abhijith H V
2)Dr Manohar P
3)Chethana R
4)Sanjay N
5)Rithika Shetty

(57) Abstract :
Abstract A computer-implemented system and method for enforcing temporal ephemerality on cloud-resident data using a Time-Dependent Encryption Model (TDEM) are disclosed. The system prevents indefinite ciphertext accessibility by linking data decryptability to a functional time-decay metric. A master cryptographic key is generated and immediately purged from system memory after being dynamically partitioned into a database fragment, a user identity fragment, and a temporal fragment derived from a time-windowed Key-Derivative Function (KDF). A temporal engine continuously updates a time-series HMAC seed to monitor fragment validity. If an authorized entropy refresh command is not executed within a predefined epoch, the temporal fragment undergoes mathematical decay.Consequently, reconstruction of the master cryptographic key becomes computationally infeasible within the intended operational framework, thereby rendering the encrypted data cryptographically inaccessible without requiring physical deletion of the stored ciphertext.

No. of Pages : 11 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129745 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A CYBER-PHYSICAL SMART CAMPUS SECURITY TESTBED FOR REAL-TIME DETECTION, SIMULATION AND MITIGATION OF IoT-BASED CYBER ATTACKS

(51) International classification	:H04L 29/06, H04L 9/40, H04W 12/122, G06F 21/57, H04W 12/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RNS Institute of Technology
Address of Applicant :RNS Institute of Technology, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098 Bengaluru Karnataka India
2)Shashwathi P
3)Mukund V
4)Nithya Shree R
5)Dev Kukreja
6)Latha P
7)Dr Swathi Darla
(72)**Name of Inventor :**
1)Dr Swathi Darla
2)Latha P
3)Nithya Shree R
4)Dev Kukreja
5)Mukund V
6)Shashwathi P

(57) Abstract :
ABSTRACT Smart campuses increasingly deploy Internet of Things (IoT) devices for automation, environmental monitoring, occupancy tracking and infrastructure management. However, such deployments introduce significant cybersecurity risks due to constrained devices, insecure communication channels and weak authentication mechanisms. The present invention discloses a cyber-physical smart campus security testbed capable of simulating, detecting and mitigating cyber-attacks in IoT environments. The system comprises ESP32-based sensing nodes integrated with multiple sensors, a Raspberry Pi based centralized controller, segmented wireless communication infrastructure, attack simulation modules and automated defense mechanisms. The platform enables controlled execution of Denial-of-Service attacks, Address Resolution Protocol spoofing, brute-force authentication attacks and Structured Query Language injection attacks while simultaneously evaluating layered defense strategies. The invention provides real-time monitoring dashboards, anomaly detection, traffic analysis and automated mitigation actions. The proposed system serves as a practical environment for cybersecurity research, education, attack-defense experimentation and evaluation of cyber resilience in smart campus deployments.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129746 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NEXORA ONE - A CYBER-SECURITY ENABLED SMART CONTROLLER for grid-connected Microgrid

(51) International classification	:H02J 3/38, G06N 20/00, H02J 13/00, H02J 7/35, H04L 29/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RNS Institute of Technology
Address of Applicant :RNS Institute of Technology, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098 Bangalore Karnataka India
2)Nandini S R
3)Madhiraju Manaswini
4)Pranav Ravi
5)Sidhanth Arya PB
6)Dhanraj
7)Yashaswini Nag M N
(72)**Name of Inventor :**
1)Yashaswini Nag M N
2)Dhanraj
3)Pranav Ravi
4)Sidhanth Arya PB
5)Madhiraju Manaswini
6)Nandini S R

(57) Abstract :
Abstract Energy infrastructure is critical to modern civilization, and the accelerating adoption of internet-connected microgrids has substantially expanded the attack surface of distributed power networks. Microgrids—compact self-sufficient systems combining solar generation, battery storage, and grid connectivity—depend on cloud-linked sensors and smart meters, exposing them to threats like sensor spoofing, load injection, meter disconnection, and switching attacks. Traditionally, operators monitored energy performance and cybersecurity through separate, manually inspected platforms, a paradigm unable to match the velocity of modern attacks. This project, NEXORA ONE, presents an automated unified platform integrating MATLAB-based physical simulation, the Isolation Forest machine learning algorithm, and a role-protected web dashboard deployed on Cloudflare Pages. The system fetches live Bengaluru weather data from the Open-Meteo API, simulates dispatch operations, flags seven rule-based physical anomaly categories, and statistically isolates cyberattack signatures through unsupervised learning. Anomalies and telemetry are persisted to InfluxDB Cloud and streamed live to operators via HiveMQ MQTT over WSS/TLS. Compared to conventional rule-based monitoring, the hybrid simulation–AI approach reduces false positives through dual-layer validation while delivering real-time situational awareness. Therefore, the platform functions as a deployable tool for utilities, campus grids, and rural microgrid operators to detect threats and respond promptly.

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129747 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : DISHARAKSHAK - A GPS-LESS NAVIGATION DEVICE FOR SOLDIER

(51) International classification	:G01C 21/16, G01C 22/00, G01S 19/47, G01S 5/14, G01C 21/20	(71)**Name of Applicant :** **1)RNS Institute of Technology** Address of Applicant :RNS Institute of Technology, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098 Bengaluru Karnataka India **2)Nidhi Vijay Shinde** **3)Harshini K** **4)Inchara R** **5)Mukund V** **6)Yashaswini Nag**
(31) Priority Document No	:NA	**7)Dhanraj**
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Dhanraj**
(86) International Application No	:	**2)Yashaswini Nag**
Filing Date	:01/01/1900	**3)Mukund V**
(87) International Publication No	: NA	**4)Inchara R**
(61) Patent of Addition to Application Number	:NA	**5)Harshini K**
Filing Date	:NA	**6)Nidhi Vijay Shinde**
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :
Abstract This invention relates to a wearable navigation system intended for soldiers operating in environments where Global Positioning System (GPS) signals are unavailable, unreliable, or intentionally disrupted. The proposed system, named Disha Rakshak, utilizes two foot-mounted inertial sensing modules and a belt-mounted processing hub to estimate user movement and position without relying on external infrastructure. Each foot module contains a BMI270 inertial measurement unit and an ESP32-C3 microcontroller for motion sensing and local processing. Sensor data is transmitted to the belt hub through ESP-NOW communication, where synchronization, step detection, heading estimation, and Zero Velocity Update (ZUPT) based drift correction are performed. The system processes movement data to reconstruct a soldier's path using dead reckoning, providing real-time navigation support even when GPS is unavailable. Designed to be lightweight, energy-efficient, and free from external infrastructure, this invention is tailored for military personnel operating in challenging environments such as tunnels, indoor facilities, dense urban areas, forests, or electronically contested zones.

No. of Pages : 7 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129748 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI AND IoT-BASED ASSISTIVE SYSTEM FOR ENHANCING AIR TRAFFIC CONTROL AND SAFETY

(51) International classification	:G08G 5/06, G08G 5/02, G08G 5/00, G08G 5/04, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RNS Institute of Technology
Address of Applicant :RNS Institute of Technology, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098 Bengaluru Karnataka India
2)Sukhmpreet Kaur
3)Sai Sankalp S
4)Rutu U D
5)Nandhitha V
6)Sanjith R S
7)Dr.Swathi Darla
8)Latha P
(72)**Name of Inventor :**
1)Latha P
2)Dr.Swathi Darla
3)Sanjith R S
4)Nandhitha V
5)Rutu U D
6)Sai Sankalp S
7)Sukhmpreet Kaur

(57) Abstract :
ABSTRACT The present invention relates to an intelligent Air Traffic Control (ATC) assistance system that integrates Internet of Things (IoT) sensing, cloud computing, and Artificial Intelligence (AI) to improve aircraft landing management, runway utilization, and operational safety. The system is designed to continuously acquire aircraft telemetry data through onboard sensing devices and wireless communication modules, enabling real-time monitoring of critical flight parameters such as aircraft speed, distance from runway threshold, altitude, fuel status, emergency conditions, and operational priorities. The collected data is transmitted to a cloud-based infrastructure where it is stored, synchronized, and processed through an AI-driven decision engine. The invention employs machine learning algorithms and intelligent rule-based analysis to evaluate runway occupancy, weather conditions, aircraft approach parameters, traffic density, and operational constraints. Based on this analysis, the system automatically generates decision-support recommendations including runway assignment, landing clearance advisories, aircraft prioritization, spacing alerts, descent guidance, and go-around recommendations. Unlike conventional ATC systems that rely primarily on continuous voice communication between pilots and controllers, the proposed invention introduces a digital advisory framework capable of delivering concise operational guidance directly to pilots and air traffic controllers through connected interfaces. This significantly reduces communication delays, minimizes controller workload, and enhances situational awareness during high-density traffic operations. The invention further incorporates a dynamic landing-priority mechanism that evaluates multiple aircraft simultaneously and determines optimal landing sequences based on parameters including fuel availability, emergency conditions, runway occupancy status, aircraft speed, and proximity to landing thresholds. The system continuously updates recommendations as new telemetry is received, thereby creating a real-time feedback loop for intelligent air traffic management. The proposed architecture provides a scalable and cost-effective framework for next-generation air traffic operations by combining IoT-based aircraft monitoring, cloud-integrated data exchange, and AI-assisted decision support. The invention improves runway safety, reduces operational risks, enhances efficiency, and supports the transition toward intelligent and semi-autonomous air traffic management systems.

No. of Pages : 8 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129763 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : "NEUROPROTECTIVE EVALUATION OF THE COMBINED EXTRACTS OF PHYLLANTHUS EMBLICA AND PSIDIUM GUAJAVA IN AN ALUMINUM CHLORIDE-INDUCED RAT MODEL OF ALZHEIMER'S DISEASE"

(51) International classification	:G01N 30/72, G01N 30/02, G01N 30/88, G01N 30/06, A61K 36/185
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Markapudi Santhi
Address of Applicant :Patent, Publications Andhra Pradesh India
2)Dr. Kasetti Ashok Babu
3)Dr.Samuel George Institute of Pharmaceutical Sciences
(72)**Name of Inventor :**
1)Markapudi Santhi
2)Dr. Kasetti Ashok Babu
3)Dr.Samuel George Institute of Pharmaceutical Sciences

(57) Abstract :
ABSTRACT: Alzheimer's disease (AD) is a progressive neurodegenerative disorder associated with memory impairment, cognitive decline, oxidative stress, and neuronal damage. The present study aims to evaluate the potential neuroprotective activity of combined fruit extracts of Phyllanthus emblica (Amla) and Psidium guajava (Guava) against aluminum chloride ($AlCl_3$)-induced experimental Alzheimer's disease in rats. The fruits were processed separately, and hydroalcoholic extracts were prepared using the Soxhlet extraction method. The extracts were characterized by liquid chromatography–mass spectrometry (LC–MS) to identify their principal phytochemical constituents, including relevant phenolic compounds where supported by analytical findings. The individual and combined extracts are evaluated for their effects on learning and memory, oxidative stress markers, antioxidant defenses, acetylcholinesterase activity, and brain histopathology. Donepezil is used as a reference drug for comparative evaluation. The study is intended to investigate whether the combined extracts demonstrate neuroprotective effects against $AlCl_3$-associated behavioral, biochemical, and histopathological alterations. The findings are expected to provide experimental evidence regarding the potential of amla and guava fruit extracts for further investigation in neuroprotective research related to Alzheimer's disease.

No. of Pages : 9 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129775 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MONOCLONAL ANTIBODY-BASED NANOPARTICLES COMPRISING COENZYME Q10 AS A MATRIX MATERIAL FOR TARGETED DRUG DELIVERY AND REDUCED ADVERSE EFFC IN BREAST CANCER TREATMENT

(51) International classification	:A61K 39/395, A61P 35/00, A61K 9/51, A61K 39/00, A61K 47/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)St Pauls College of Pharmacy, (Autonomous)
Address of Applicant :St Pauls College of Pharmacy, (Autonomous), Turkayamjal, Rangareddy district, Telangana 501510 Turkayamjal Telangana India
(72)**Name of Inventor :**
1)SNEHA THAKUR
2)KIRANMAI MANDAVA
3)VENKAT RATNAM DEVADASU

(57) Abstract :
ABSTRACT The present invention relates to monoclonal antibody-based nanoparticles (100) comprising Coenzyme Q10 (104) as a matrix material for targeted drug delivery and reduction of adverse effects in breast cancer treatment. The formulation comprises at least one monoclonal antibody (102) selected from Trastuzumab, Pertuzumab, and Bevacizumab, together with Coenzyme Q10 (104), Poloxamer (106), polyvinyl alcohol (PVA) (108), and trehalose (110). The nanoparticles are prepared using a double-emulsion (114) water-in-oil-in-water (W/O/W) (112) method to facilitate antibody encapsulation and sustained release. The formulation is characterized (126) by particle size, zeta potential, antibody content, pharmacokinetics, enhanced bioavialbility and physicochemical stability (130). The developed nanoparticles are intended to improve targeted delivery, provide sustained antibody release (128), and potentially reduce treatment-associated toxicity through the incorporation of Coenzyme Q10. a breast cancer patient-derived xenograft (PDX) model established from human clinical tumour samples exhibiting HER2 expression provides a platform for evaluating the formulation and investigating its therapeutic potential. Figure associated with Abstract is Fig. 1.

No. of Pages : 13 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129776 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NOVEL SCHIFF BASE COMPOUND TARGETING TOLL-LIKE RECEPTOR 4 FOR INFLAMMATION-ASSOCIATED DEPRESSION

(51) International classification	:C07D 471/04, A61K 31/4709, A61K 31/47, C07D 401/14, C07D 413/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)St Pauls College of Pharmacy, (Autonomous)
Address of Applicant :St Pauls College of Pharmacy, (Autonomous), Turkayamjal, Rangareddy district, Telangana 501510 Turkayamjal Telangana India
(72)**Name of Inventor :**
1)Dr.SOMNATH DE
2)Mr.NADEEMULLAH KHAN
3)Dr. KIRANMAI MANDAVA
4)Dr.E.NAVYA PRAVALA
5)Mrs.B.SUHASINI
6)Dr. ANUSHA KOMATI
7)Dr. VENU MADHAV KATLA
8)Mrs.Y. UMA RANI

(57) Abstract :
ABSTRACT The present invention relates to a novel Schiff base intermediate, N-[(1H-indol-3-yl)methylene]-quinolin-4-amine, comprising indole and quinoline moieties linked through an azomethine (C=N) bond for investigation as a TLR4-targeted candidate in inflammation-associated depressive disorders. The compound was synthesized with a yield of 80.44% and characterized using TLC, UV-Visible, and FTIR spectroscopy. Molecular docking against TLR4 (PDB ID: 3FXI) showed a Glide XP score of −5.146 kcal/mol, while molecular dynamics simulation supported protein–ligand complex stability. In-silico analysis indicated high gastrointestinal absorption, predicted blood–brain barrier permeability, and zero Lipinski violations. The compound exhibited concentration-dependent cytotoxicity in SH-SY5Y cells, with a CTC_{50} of approximately 248 µg/mL, and no major acute oral toxicity up to 2000 mg/kg. In an LPS+CUMS model, treatment improved behavioural and biochemical parameters and preserved neuronal morphology, supporting further investigation of its therapeutic potential. Figure associated with Abstract is Fig. 1.

No. of Pages : 30 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129777 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTEGRATED BIODIGESTER SYSTEM FOR IRON NANOPARTICLE-ASSISTED BIOGAS GENERATION AND MUSHROOM CULTIVATION

(51) International classification	:C02F 11/04, C02F 3/28, A01K 1/015, C12M 1/107, B01D 53/84
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)St Pauls College of Pharmacy, (Autonomous)
Address of Applicant :St Pauls College of Pharmacy, (Autonomous), Turkayamjal, Rangareddy district, Telangana 501510 Turkayamjal Telangana India
(72)**Name of Inventor :**
1)DR. SNEHA THAKUR

(57) Abstract :
ABSTRACT An integrated biodigester system (100) is disclosed for converting cattle waste into biogas and recovering bioresources through mushroom cultivation. An organic feed preparation unit (101) prepares cow dung and sheep litter, while a feed-ratio adjustment unit (102) adjusts their proportions. An iron nanoparticle dosing unit (103) introduces iron nanoparticles into the prepared feed before anaerobic digestion in a biodigester (104). An agitation arrangement (105) and a temperature-control arrangement (106) support digestion, while a biogas collection unit (107) collects the generated biogas. A digestate recovery unit (108) recovers the residual digestate, and an odor-control scrubber (109), comprising poultry litter and iron wool, reduces foul smell. A bioresource recovery arrangement (110) uses the recovered organic resources in a downstream mushroom cultivation process. The integrated system combines waste treatment, biogas generation, digestate recovery, and mushroom production to support agricultural waste utilization and resource recovery. The figure associated with abstract is Fig. 1.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129778 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NETWORK PHARMACOLOGY-GUIDED SELECTION OF TRADITIONAL ANTI-CANDIDA PLANTS AND DEVELOPMENT OF A CANDICUR-Q GEL FOR ORAL CANDIDIASIS

(51) International classification	:A61K 36/9066, A61P 31/10, A61K 9/06, A61K 36/00, A61K 31/12	(71)**Name of Applicant :** **1)St Pauls College of Pharmacy, (Autonomous)** Address of Applicant :St Pauls College of Pharmacy, (Autonomous), Turkayamjal, Rangareddy district, Telangana 501510 Turkayamjal Telangana India (72)**Name of Inventor :** **1)Dr. ANUSHA KOMATI** **2)MOMIN SHABHANA** **3)Dr. Kiranmai Mandava** **4)Dr.Somnath De** **5)Dr. Navya Pravala Ellandula** **6)Suhasini Boddu** **7)Sanjana Chirumamilla** **8)Deepika Devarapally** **9)Ramya Sri Gangapuram** **10)Harika Bathini**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT The present invention relates to a network pharmacology-guided herbal mucoadhesive gel (CandiCur-Q) for the management of oral candidiasis predominantly caused by Candida albicans. Anti-candidal medicinal plants, namely Psidium guajava, Mentha piperita, Ocimum sanctum, Azadirachta indica, Curcuma longa, Syzygium aromaticum, and Foeniculum vulgare, were selected from Charaka Samhita and analyzed using phytochemical databases and network pharmacology. Curcumin and Quercetin were identified as lead biomarkers based on drug-likeness and incorporated into the gel formulation. The optimized CandiCur-Q gel exhibited suitable physicochemical properties, including pH of 6.7 ± 0.5, viscosity of 1879 ± 18.5 cP, spreadability of 0.49 ± 0.02 gcm/sec, and swelling index of 0.8 mL. The formulation demonstrated inhibition of Candida albicans with MIC of 0.56 ± 0.12 mg/mL, zone of inhibition of 27 ± 0.85 mm, and biofilm inhibition of 72.80 ± 2.23%. Ex vivo studies demonstrated sustained release over 24 hours and satisfactory mucoadhesion, indicating potential as a topical herbal therapeutic for oral candidiasis. Figure associated with Abstract is Fig. 1.

No. of Pages : 22 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129779 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ANTIHYPERGLYCEMIC AND ANTIOXIDATIVE COOKIES COMPRISING ENRICHED FENUGREEK MICROGREENS

(51) International classification	:A21D 13/00, A21D 13/80, A23L 33/10, A23L 33/105, A23L 7/10, A23P 10/00, A21D 2/36
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)St Pauls College of Pharmacy, (Autonomous)
Address of Applicant :St Pauls College of Pharmacy, (Autonomous), Turkayamjal, Rangareddy district, Telangana 501510 Turkayamjal Telangana India
(72)**Name of Inventor :**
1)Dr. Anusha Komati
2)Dr. Kiranmai Mandava
3)Dr. Somnath De
4)Dr. Navya Pravala Ellandula
5)Divya Kaki
6)Madhuri Dadiga
7)Anusha Talari
8)Yashaswi Biruduraju

(57) Abstract :
ABSTRACT The present invention relates to a functional food composition (100) comprising fenugreek (Trigonella foenum-graecum) microgreen powder (104) and a cookie formulation thereof for supporting metabolic health. The fenugreek microgreens (102) comprise 41.25 ± 0.59 g/100 g protein, 37.67 ± 0.27 g/100 g dietary fibre, 4.81 ± 0.18 g/100 g fat, 13.06 ± 0.04 g/100 g ash and total phenolic content of 19.7 ± 0.1 mg GAE/g. The formulation preferably contains 5% fenugreek microgreen powder (104). At 5 mg/mL, the fenugreek microgreen cookie (112) exhibited 49.33 ± 1.4% hydroxyl radical, 29.28 ± 1.40% superoxide radical and 68.62 ± 2.5% nitric oxide radical scavenging activities, together with 57.33 ± 2.3% α-glucosidase inhibition and 66.66 ± 2.22% inhibition of advanced glycation end products. The composition further demonstrated modulation of glucose-stimulated insulin secretion in MIN6 pancreatic β-cells and improved postprandial glucose clearance at 500 mg/kg in an oral glucose tolerance model. The invention provides a nutritionally enriched functional food composition with demonstrated antioxidant, antidiabetic and antiglycation potential. Figure associated with Abstract is Fig. 1.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129809 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Load Based Smart Switching Hybrid Battery Management System and Method for Dynamic Source Selection Between Lithium-ion and Lead-acid Batteries with IoT-Enabled Monitoring

(51) International classification	:H02J 7/00, H01M 10/48, H01M 10/44, H01M 10/42, G01R 31/382
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Pokanati Veera Venkata Rama Rao
Address of Applicant :H. No.: 11-15-DC, Flat No.: 403 Shyam Heights I, Besides Water Tank, Road No.1 Doctors Colony, Saroornagar Telangana India
(72)**Name of Inventor :**
1)Pokanati Veera Venkata Rama Rao
2)Mr. Indrakanty Narasimha Swamy
3)Dr. Kandregula Rajesh Kumar

(57) Abstract :
A load based smart switching hybrid battery management system comprising a lithium-ion battery pack, a lead-acid battery, voltage sensing units, a temperature sensor, a load-demand sensing unit, a single change-over relay switching unit and a Wi-Fi enabled ESP32 microcontroller. The microcontroller estimates the state of charge of each battery from chemistry-specific voltage–SOC tables and connects the load to the lithium-ion pack during high load demand and to the lead-acid battery during low load demand. Selection uses a hysteresis band with adaptive thresholds, a minimum dwell time against relay chatter, and eligibility rules for temperature and state of charge. The relay defaults to the lead-acid battery through its normally-closed contact for fail-safe operation. Battery voltages, temperature, load, active source and alerts are shown on a local LCD and sent to the Blynk IoT cloud for remote monitoring, notification and mode control on a smartphone.

No. of Pages : 14 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129845 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INSIGHT AI

(51) International classification	:G06N 20/00, G06F 16/248, G06N 5/04, G06N 5/02, G06F 16/9038
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. P. Thilagarani
2)Samuktha S
3)Sulthana Fathima A
4)Sherlin Poustina R
5)Sharveshprannav S
6)Prathish K K
7)Mohamed Sahith K
8)Poovitha V
9)Mithul Sankar R K

(57) Abstract :
The present invention relates to an automated tabular data analytics system designed to perform data profiling, preprocessing, statistical analysis, machine-learning analysis, and artificial-intelligence-assisted interpretation. The system receives a tabular dataset and creates an identifiable processed dataset version. An evidence-generation engine computes analytical evidence for the selected dataset version. An evidence-context builder associates the evidence with the dataset-version identity and constructs a structured analytical context. A controlled artificial-intelligence gateway provides the structured context and a user query to a generative artificial-intelligence model, which generates an analytical interpretation. The system can present analytical evidence and interpretation through dashboards, visualizations, conversational interfaces, and reports. The association of dataset state, computed evidence, and artificial-intelligence interpretation provides a structured and traceable workflow for automated data analytics.

No. of Pages : 6 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129862 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPOSITION AND METHOD FOR AN ANNEALED NICKEL-DOPED TIN OXIDE THIN FILM FOR OPTICAL LIMITING APPLICATIONS

(51) International classification	:C01G 19/02, C23C 16/40, C01G 53/04, C03C 17/25, C03C 17/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NITTE (DEEMED TO BE UNIVERSITY)
Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India
2)DR RAGHAVENDRA BAIRY
3)ANUSHA DINAKAR RAO
(72)**Name of Inventor :**
1)DR RAGHAVENDRA BAIRY
2)ANUSHA DINAKAR RAO

(57) Abstract :
Disclosed herein is an annealed nickel-doped tin oxide thin film composition (100) for use in optical limiting applications, comprising tin chloride dihydrate (204) as a host metal-oxide precursor (104) and nickel chloride hexahydrate (206) as a dopant metal precursor (106), each dissolved in methanol (208) and deposited onto a glass substrate (202) by spray pyrolysis at a substrate temperature of 350°C to form an Sn1-xNixO2 thin film of thickness 500-600 nm. The as-deposited thin film is subjected to post-deposition annealing within a furnace atmosphere (214) maintained at 400°C or 500°C for 2 hours, producing a non-monotonic enhancement in crystallinity, defect density, and third-order nonlinear optical susceptibility, and yielding a composition (100) exhibiting reverse saturable absorption suitable for realization of an efficient optical limiter.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129863 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A WEARABLE SMART REHABILITATION BELT

(51) International classification	:A61B 5/11, A61H 1/02, A63B 24/00, A61B 5/00, A61H 3/00	(71)**Name of Applicant :** **1)NITTE (DEEMED TO BE UNIVERSITY)** Address of Applicant :6TH FLOOR, UNIVERSITY ENCLAVE, MEDICAL SCIENCES COMPLEX, DERALAKATTE, MANGALURU, KARNATAKA – 575018, INDIA Karnataka India **2)SWATHI.K** **3)NUTANA SHETTY**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)SWATHI.K**
(86) International Application No Filing Date	: :01/01/1900	**2)NUTANA SHETTY**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed herein is a smart rehabilitation system (100), comprising a wearable rehabilitation belt unit (102) further comprising an ergonomic belt (104). The system (100) comprising a plurality of sensors (106) configured to monitor load distribution on a user's body and detect motion of the user in real-time. The system (100) comprising a controller (108) configured to combine data from the sensors (106), compare the movement profile with predetermined rehabilitation parameters corresponding to a stage of fracture recovery, identify abnormal posture, excessive joint movement or imbalance, predict a potential fall or unsafe movement before impact occurs, and automatically increase or decrease mechanical support provided by the ergonomic belt (104) according to the identified movement. The system (100) comprising at least one remote monitoring unit (110) configured to periodically update rehabilitation thresholds according to recovery progress of the user.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129868 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Integrated AI-Based Personalized Interview Preparation, Simulation, Evaluation and Career Guidance System

(51) International classification	:G06Q 10/10, G06Q 10/06, G06N 20/00, G06Q 50/20, G06F 16/332
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. Mahalakshmi
2)Tharani Dharan S
3)Santhosh Kumar N
4)Saran Rajan K
5)Santhosh M
6)Sibi Rajan S
7)Shree Venkatesh Prasad G
8)Sakthi Kumar V
9)Sri Yogeshwaran R

(57) Abstract :
The present invention relates to an integrated Artificial Intelligence (AI)-based system for personalized interview preparation, simulation, evaluation, readiness assessment, answer coaching and career guidance. The system uses a Streamlit user interface and a FastAPI backend to coordinate resume processing, personalized question generation, mock interviews, answer evaluation, readiness scoring, preparation planning, AI mentoring and company-focused interview research. The system extracts text from an uploaded resume using a PDF processing engine and analyzes the extracted content to identify technical and professional skills. The identified profile is used as context for generating role-specific interview questions according to target role, difficulty and requested question count. A mock interview simulator presents categorized questions and collects candidate responses for structured evaluation. The evaluation engine produces scores and feedback across clarity, technical depth, relevance, STAR structure and communication. A readiness assessment module provides dimensions including Technical Skills, Behavioral Readiness, Communication, Domain Knowledge, Confidence Level and Problem Solving. An answer-coaching module generates improved responses and actionable preparation guidance. The system further includes a personalized preparation roadmap and seven-day plan, an AI mentor chatbot using resume context and conversation history, and a company research module generating interview-focused employer information. User authentication and local SQLite storage are provided, and a common OpenAIcompatible AI service layer supports configurable model access with module-specific fallback processing when an external AI service is unavailable.

No. of Pages : 8 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129898 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BILOSOME-ENCAPSULATED MAGNETIC CARBON DOT NANOFORMULATION FOR MULTIMODAL BIOMEDICAL IMAGING

(51) International classification	:A61K 49/18, A61K 49/06, A61K 49/12, A61K 49/00, A61K 49/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Chettinad Academy of Research and Education
Address of Applicant :Chettinad Hospital and Research Institute Chettinad Academy of Research and Education, Kelambakkam, Chengalpattu, Tamil Nadu-603103, India Tamil Nadu India
(72)**Name of Inventor :**
1)Agnishwar Girigoswami
2)Venkatakrishnan Kiran
3)Barath Kumar Balaji

(57) Abstract :
Disclosed is a multimodal imaging nanoformulation comprising bilosome-encapsulated magnetic carbon dots (BCDs) and a method for preparation thereof. The nanoformulation comprises magnetic carbon dots having a manganese ferrite-containing magnetic core and a carbonaceous shell, encapsulated within bilosomal vesicles comprising Span 20, cholesterol, and sodium cholate. The preparation method comprises hydrothermal synthesis of magnetic carbon dots using manganese chloride, iron (III) chloride hexahydrate, citric acid, and urea, followed by preparation of bilosomal vesicles through thin-film hydration and subsequent encapsulation through sonication. The resulting nanoformulation exhibits fluorescence emission, T2-weighted magnetic resonance imaging (MRI) contrast characteristics, improved colloidal stability, and cellular compatibility. The BCDs retain fluorescence characteristics and demonstrate concentration-dependent fluorescence intensity and T2-weighted magnetic resonance signal attenuation. The nanoformulation is suitable for dual-modal biomedical imaging using fluorescence imaging and T2-weighted MRI.

No. of Pages : 28 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129910 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Neuroimaging-Based System and Method for Alzheimer's Disease Stage Classification Using Structural and Deep Feature Representation

(51) International classification	:G16H 50/20, G06N 3/08, G06V 10/764, A61B 5/055, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Koneru Lakshmaiah Education Foundation
Address of Applicant :Koneru Lakshmaiah Education Foundation, Hyderabad-500075, Telangana, India Hyderabad Telangana India
(72)**Name of Inventor :**
1)Sridevi Balne
2)Dr. Mahesh Babu Arrama
3)Dr Hari Kunadharaju

(57) Abstract :
I / We Claim: 1. A computer-assisted neuroimaging system for classification of Alzheimer's disease and associated cognitive stages, comprising a neuroimaging input module configured to receive brain magnetic resonance imaging (MRI) data, a preprocessing module configured to prepare the MRI data, a hippocampal structural-processing module configured to generate a surface-mesh representation, a latent-feature representation module incorporating a variational autoencoder (VAE), a deep-learning module incorporating a pre-trained Inception Residual Network (IRN), and a classification module configured to generate a predicted category corresponding to a predefined cognitive stage. 2. The system as claimed in claim 1, wherein the preprocessing module is configured to perform one or more image-preparation operations selected from noise reduction, intensity normalization, quality control, and imaging-artifact mitigation to generate preprocessed MRI data for subsequent feature extraction. 3. The system as claimed in claim 1, wherein the hippocampal structural-processing module is configured to identify and segment a hippocampal region from the preprocessed MRI data and generate a surface mesh comprising a plurality of vertices and associated spatial relationships representing anatomical structural information. 4. The system as claimed in claim 3, wherein a spectral-matching module is configured to establish correspondence between vertices of structural representations of the hippocampal region to facilitate alignment and comparative analysis of anatomical shape characteristics. 5. The system as claimed in claim 1, wherein the latent-feature representation module comprises an enhanced variational autoencoder configured to encode structural information derived from the surface-mesh representation into a lower-dimensional latent representation for subsequent classification. 6. The system as claimed in claim 5, wherein a generative adversarial network (GAN)-based learning component is associated with the latent-feature representation module to support learning of structural representations, wherein the interaction between the variational autoencoder and the GAN-based learning component is determined by a configured training methodology. 7. The system as claimed in claim 1, wherein the deep-learning module comprises a pre-trained Inception Residual Network configured to learn neuroimaging features through transfer learning and fine-tuning using labelled MRI training data corresponding to a plurality of predefined cognitive-stage categories. 8. The system as claimed in claim 7, wherein the deep-learning module is configured to apply training-data augmentation comprising one or more of rotation, flipping, and intensity variation, together with layer-specific learning rates and adaptive learning-rate scheduling, for adjusting network parameters during training. 9. The system as claimed in claim 1, wherein the classification module is configured to classify the MRI data into one or more categories selected from Normal Control (NC), Early Mild Cognitive Impairment (EMCI), Late Mild Cognitive Impairment (LMCI), and Alzheimer's Disease (AD), using at least one of the latent representations generated by the variational autoencoder and deep features generated by the Inception Residual Network. 10. The system as claimed in claim 1, further comprising an evaluation module configured to evaluate classification performance using one or more measures selected from accuracy, precision, recall, sensitivity, specificity, F1-score, receiver operating characteristic area under the curve (ROC-AUC), confusion-matrix analysis, and computational time, wherein the evaluation module is configured to record performance results obtained according to a predefined validation or testing protocol.

No. of Pages : 30 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129911 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Smart Flood-Resilient Urban Drainage System with Predictive Water-Level Monitoring and Adaptive Flow Control

(51) International classification	:G06Q 50/26, G01W 1/14, E03F 1/00, G08B 21/10, G06Q 50/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Chaitanya (Deemed to be University)
Address of Applicant :Himayath nagar village, Moinabad mandal, Ranga Reddy District, Hyderabad, Telangana-500075, India. Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr. Venkateswarlu Gogana
2)Dr. Avuluri Vijaya Bhaskar
3)Dr. Polipati Ananyachandran
4)Mr. Modugu Naveen Kumar

(57) Abstract :
Title: Smart Flood-Resilient Urban Drainage System with Predictive Water-Level Monitoring and Adaptive Flow Control. The present invention relates to a smart flood-resilient urban drainage system (100) for predictive water-level monitoring and adaptive flow control. The system comprises a sensor acquisition module (101), a rainfall monitoring module (102), a data preprocessing module (103), a water-level and flow assessment module (104), a predictive flood analysis module (105), a drainage-capacity assessment module (106), an adaptive flow-control module (107), a drainage actuation assembly (108), a supervisory monitoring interface (109), a feedback and fault-detection module (110), and a data storage and communication module (111). The system receives water-level, flow-rate, and rainfall measurements to estimate future water levels and assess the available capacity of interconnected drainage sections. Based on predicted flood conditions and downstream capacity constraints, the adaptive flow-control module generates instructions for regulating gates, valves, pumps, and diversion structures. Feedback from sensors and actuators is used to evaluate control effectiveness and identify abnormal operating conditions, thereby supporting coordinated drainage management and flood-risk mitigation.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129932 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RETAILMIND AI

(51) International classification	:G06Q 10/08, G06Q 40/00, G06N 20/00, G06Q 30/04, G06Q 30/06	(71)**Name of Applicant :** **1)SNS College of Technology** Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India (72)**Name of Inventor :** **1)Mrs Mahalakshmi K** **2)Mohan Prasanth K** **3)Santhosh M** **4)Sharvesh R** **5)Vasim T** **6)Segan Paul J** **7)Srimathi K** **8)Vishnupriya B M** **9)Vrithisha C**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

RetailMind AI is an intelligent AI-based company-brain platform designed to receive, interpret, validate, and manage retail orders efficiently. It analyzes both structured cart orders and unstructured free-form customer messages, such as WhatsApp or Instagram style text. Using OpenAI's GPT models and Natural Language Processing (NLP), the system extracts product names and quantities from customer text. AI-based matching techniques compare the extracted items with the live product catalog and current stock levels. It automates order parsing, stock validation, Goods and Services Tax (GST) calculation, restock alerting, and invoice generation. A centralized React web dashboard provides store administrators with live orders, inventory status, sales metrics, and a conversational business-intelligence (BI) chat assistant. The system also suggests alternative products when a requested item is unavailable. By reducing manual work, it improves order-processing speed, accuracy, and consistency. Overall, RetailMind AI helps retail businesses convert customer requests into actionable orders and insights faster and more effectively.

No. of Pages : 4 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641129941 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SECURE FACE AUTH

(51) International classification	:G06F 21/32, H04L 9/32, G06K 9/00, G06F 21/31, H04L 29/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. P. Thilagarani
2)Pooja Sri S
3)Rinu Fathima M F
4)Tharun Ganesh P
5)Nikitha S
6)Ronald J
7)Sri Hari S
8)Santhosh A
9)Sabarish K

(57) Abstract :
The present invention relates to a Secure Face Auth system designed to provide secure access to protected applications and dashboards through multi-factor authentication. The system combines multiple verification methods including password authentication, Email OTP verification, eye-blink based liveness detection, and face recognition. Initially, the user provides a valid password. After successful password verification, an OTP is sent to the registered email address. The user must enter the correct OTP to proceed further. The system then activates the camera and performs eye-blink based liveness detection to verify that the user is a real person. After successful liveness verification, the system compares the live camera face with the registered face representation. Only when all authentication stages are successfully completed is the user allowed to access the protected dashboard. The system also supports JWT-based authenticated sessions and protected dashboard access. The invention provides a simple and layered approach to authentication by combining knowledge-based, email-based, liveness-based, and biometric verification methods.

No. of Pages : 5 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641129986 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PLASMA-CATALYTIC HYDROCARBON CRACKING SYSTEM WITH FEED-RESPONSIVE ADAPTIVE PLASMA CONTROL AND METHOD THEREOF

(51) International classification	:B01J 19/08, C10G 11/18, H05H 1/24, C10G 11/00, B01J 8/02	(71)**Name of Applicant :** **1)PRAYAGA VENKATA RAMAKRISHNA** Address of Applicant :BALAJI ENCLAVE, 4TH FLOOR, DOOR NO: 86-3-13/8. SHIRDISAIMARG, JN. ROAD, RAJAHMUNDRY, ANDHRA PRADESH, 533103, INDIA RAJAHMUNDRY Andhra Pradesh India (72)**Name of Inventor :** **1)PRAYAGA VAIBHAV NISCHAL**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention discloses a plasma-catalytic hydrocarbon cracking system (100) and a method (300) thereof. A feed property sensor (102e) measures a specific gravity of a hydrocarbon feedstock delivered to a non-thermal plasma reactor (104), wherein a pulsed electrical power subsystem (106) applies pulsed high-voltage discharges across a first electrode (104a) and a second electrode (104b) to generate a non-thermal plasma within a plasma discharge zone (104d). A catalyst bed (108a) comprising a supported bimetallic catalyst (108b) of nickel and a second metal is disposed within or downstream of the plasma discharge zone (104d). An adaptive process control subsystem (112) adjusts, during operation and in response to the measured specific gravity, a plasma electrical parameter and a flow rate through the plasma discharge zone (104d), whereby energy input and residence time are matched to the hydrocarbon feedstock. A product recovery subsystem (110) separates the products. FIG. 1

No. of Pages : 47 No. of Claims : 30

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641130002 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND A METHOD OF INTEGRATED ENZYMATIC AND THERMAL DOMESTIC WASTE TREATMENT

(51) International classification	:B09B 3/00, C05F 9/02, C02F 3/34, C12M 1/00, C12M 1/107
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED STUDIES (VISTAS)
Address of Applicant :VELAN NAGAR, PV VAITHIYALINGAM ROAD, PALLAVARAM-600117, CHENNAI, TAMIL NADU, INDIA Tamil Nadu India
(72)**Name of Inventor :**
1)Meenambiga Setti Sudharsan
2)Aarthi V
3)Vinod Kumar T
4)Chandrasekaran M
5)Deepak R

(57) Abstract :
Disclosed is a system (100) for integrated treatment of domestic waste through sequential enzymatic degradation and thermal neutralization. The system (100) comprises a Biological Decomposition and Conversion Chamber (102) having a degradation compartment (104), a door-type waste inlet (108), and a decomposition booster port (110) for introducing an enzyme cocktail (138) derived from Bacillus subtilis. The enzyme cocktail (138) comprises amylase, protease, cellulase, lipase, pectinase, and urease for degrading biodegradable constituents of domestic waste. A removable filter cartridge (112), an effluent outlet (114), and an effluent collection chamber (116) facilitate separation and collection of liquid or slurry generated during degradation. Residual waste is transferred to a High-Temperature Waste Neutralization Chamber (128) and thermally neutralized using a high-temperature neutralization torch (130). A slag collector tray (132) collects solid residue, while a wet scrubber (134) treats gaseous products before discharge through a syngas outlet (136), thereby enabling treatment of heterogeneous household waste. Fig. 1 is the reference figure.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641130003 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Naphthimidazole-Substituted Anthracene Fluorophore with Hydration-Dependent Emission Shift and A Process For Preparation Thereof

(51) International classification	:C09K 11/06, C07D 233/60, C07D 233/58, C07D 233/90, C07D 233/54
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology Hyderabad
Address of Applicant :IIT Hyderabad Road, Near NH-65, Sangareddy, Kandi, Telangana-502284, India. Hyderabad Telangana India
(72)**Name of Inventor :**
1)GANESAN PRABU SANKAR
2)ARAVIND KUMAR RENGAN
3)RAKESH KUMAR RAI
4)BIKASH LAHKAR
5)USHA GAVEL
6)AMITKUMAR RAMAKANT GIRBIDE
7)CHINMAYA SAHOO
8)GHANSHYAM TILAKWAR

(57) Abstract :
The present invention provides a compound of formula $C_{25}H_{16}N_2$, having the IUPAC name 1-(anthracen-9-yl)-1H-naphtho[2,3-d]imidazole (NAN) and its hydrated crystals NAN-H2O and the method of preparation thereof. The emission study showed that crystallized NAN emits green light upon excitation at 290 nm, whereas crystallized NAN-H_2O emits blue light upon excitation at 260 nm. An emission shift was also observed in the solution state, demonstrating the change in emission characteristics of NAN under different solution conditions.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641130049 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Nested Resonance Manifold system for Cross-Modal Agent Collaboration via Dynamic Spectral Binding

(51) International classification	:G06N 5/04, G06N 3/08, G06N 5/02, G06N 3/04, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sayyed Nagulmeera
Address of Applicant :DVR & Dr. HS MIC COLLEGE OF TECHNOLOGY, KANCHIKACHERLA, VIJAYAWADA, AP-521180 VIJAYAWADA Andhra Pradesh India
2)Dr Akbar Khan
3)Dr. Guntapalli Minni
4)Shareef Shaik
5)Nagul Shareef Shaik
6)Md.Shamshad Begum
7)Chaitanya Mokkapati
(72)**Name of Inventor :**
1)Sayyed Nagulmeera
2)Dr Akbar Khan
3)Dr. Guntapalli Minni
4)Shareef Shaik
5)Nagul Shareef Shaik
6)Md.Shamshad Begum
7)Chaitanya Mokkapati

(57) Abstract :
A Nested Resonance Manifold system enables cross-modal collaboration among heterogeneous artificial intelligence agents through dynamic spectral binding. Specialized agents, each processing a distinct data modality, project their internal states onto a hierarchy of nested geometric manifolds that separate coarse semantic structure from fine-grained residual details. A dynamic spectral binder continuously analyzes the frequency content and relative phase of these projected states and forms temporary resonance bonds only within spectral bands whose mutual coherence exceeds an adaptive threshold. Binding coefficients and manifold nesting weights are adjusted in real time by a binding controller that monitors energy balance and collaboration quality across layers. The resulting selective coupling allows agents with differing sampling rates, dimensionalities, and temporal horizons to coordinate without collapsing their representations into a shared latent space or relying on static message protocols. Private high-frequency components remain isolated, preserving specialization while productive interactions are reinforced. The architecture achieves faster convergence to coherent joint decisions, reduced cross-modal interference, and improved robustness under distribution shift compared with conventional attention-based or graph-message systems. The system is applicable to multi-sensor robotics, distributed diagnostic networks, collaborative content generation, and large-scale edge-agent coordination.

No. of Pages : 11 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641130050 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Valorization of Red Dragon Fruit Waste into Eco- friendly, Sustainable Granular Facewash Formulation as Dermocosmetic

(51) International classification	:A61Q 19/10, A61K 8/97, A61Q 19/00, A61K 8/02, A61K 8/73
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR.B.RAMYA KUBER
Address of Applicant :Professor, Institute of Pharmaceutical technology, Sri Padmavati Mahila Visvavidyalayam, Tirupati- 517502 Tirupati Andhra Pradesh India
(72)**Name of Inventor :**
1)DR.B.RAMYA KUBER
2)GANDAVARAPU. NANCY
3)JALLA. JYOTHIRMAYI
4)B. CHARISHMA NAGA SAI
5)K.N. MANEESHA
6)B. DHEERAJ
7)B. SAANVI SRI

(57) Abstract :
TITLE: Valorization of Red Dragon Fruit Waste into Eco- friendly, Sustainable Granular Facewash Formulation as Dermocosmetic The present invention relates to a process for preparing a granular herbal face-wash composition. The process comprises macerating dried dragon fruit peel powder with ethanol at a ratio of 1:3 (w/v) for 48 hours to obtain a first composition, followed by filtration through muslin cloth and Whatman filter paper. The resulting filtrate is treated at 40–45°C to obtain a semisolid composition, which is dried and stored in an amber-colored airtight container at 4°C. The dried composition is blended with soapnut powder, rice starch and talc to obtain a second composition. Gum tragacanth is dispersed in purified water to prepare a binder solution, which is gradually added to the second composition to obtain a wet mass. The wet mass is passed through a 60-mesh sieve to form wet granules, followed by drying at 40–45°C and sieving through a 60-mesh sieve. Lemongrass oil is subsequently added and gently stirred to obtain the granular herbal face-wash composition.

No. of Pages : 30 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641130072 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CUSTOMIZABLE BRAILLE DOT EMBOSSING DEVICE

(51) International classification	:G09B 21/00, B41J 3/32, G09B 21/02, G06V 30/10, B41J 2/525
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)S Akhil
Address of Applicant :205A, Railway Colony, Hemambika Nagar, Palakkad-678009, Kerala, India. Palakkad Kerala India
2)Jestin Jayas
3)Shinu Suresh K
4)Sreehari E
5)Mohammed Hadi Salim
6)Thanseeha Nasrin P M
7)S Ajal
8)Telson Lancy Crasta
(72)**Name of Inventor :**
1)S Akhil
2)Jestin Jayas
3)Shinu Suresh K
4)Sreehari E
5)Mohammed Hadi Salim
6)Thanseeha Nasrin P M
7)S Ajal
8)Telson Lancy Crasta

(57) Abstract :
A customizable braille dot embossing device, comprising a body 101 including at least an embossing head 102, a drive unit 103 for operating embossing head 102, and a control unit for controlling embossing, an integrated flatbed scanner 104 electrically connected to the control unit for scanning a printed document, an OCR (Optical Character Recognition) module for converting the scanned document into plain text, a translation engine provided with the control unit for converting the plain text into Braille data, a touchscreen 105 disposed on the body 101 for enabling a user to select a Braille dot diameter before embossing, and a plurality of replaceable embossing pins 201, each embossing pin 201 with a different diameter, the embossing head 102 comprises a pin holder and the embossing pins 201 are removably fitted in the pin holder so that a selected embossing pin 201 is installed in the embossing head 102 manually.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641130085 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR ASSISTING SAFE NAVIGATION OF A VEHICLE THROUGH MONITORING SURROUNDING REGIONS

(51) International classification	:B60W 30/095, G08G 1/16, B60W 50/14, B60W 30/09, G01S 13/931	(71)**Name of Applicant :** **1)PES University** Address of Applicant :Outer Ring Road, Banashankari 3rd Stage, Banashankari, Bengaluru-560085, Karnataka, India. Bengaluru Karnataka India (72)**Name of Inventor :** **1)LAKSHMI SATYA VANI NARAYANAM** **2)NANDITA R NADIG** **3)NANDYALA SREENIVASA LIKHITH CHANDRA** **4)SUKRUTHA D** **5)SUMAN KUMAR K R**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present disclosure relates to a system (102) and a method (1100) for assisting safe navigation of a vehicle through monitoring surrounding regions of the vehicle. A controller (104) receives object detection information and proximity information from a plurality of sensing devices to identify one or more objects within a predefined monitored region and determine their position and movement. The controller (104) processes proximity information to determine object distances and an effective monitoring boundary based on a selected vehicle configuration. Further, the controller (104) identifies object intrusion into an extended load boundary and predicts object movement into unmonitored regions. A risk level is determined based on object distance, position, movement, boundary intrusion, and predicted movement. The system (102) generates visual warnings based on the determined risk level, thereby improving driver awareness and assisting in collision-risk assessment.

No. of Pages : 71 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641130096 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR MANAGING APPLICATION USAGE OF A USER

(51) International classification	:H04W 4/24, H04L 29/08, G06F 11/34, G06F 11/30, H04W 4/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)AMRITA VISHWA VIDYAPEETHAM
Address of Applicant :Mysuru Campus, #114, 7th Cross, Bogadi II Stage Mysuru, Karnataka 570026, India. Karnataka India
(72)**Name of Inventor :**
1)THONIPARAMBIL, Sandeep Kumaran
2)KANNOLI, Sreesanth

(57) Abstract :
The present invention relates to a system and method for managing application usage of a user based on user-specific information and usage patterns. The system comprises a user device having a monitoring application and a target application, and a backend server communicatively coupled to the user device. The monitoring application receives user information and usage information associated with the target application and obtains one or more parameters indicative of application interaction and usage behavior. The backend server evaluates the parameters against usage guidelines to determine a trigger condition and, upon satisfaction thereof, provides a real-time intervention comprising one or more alerts, notifications, solutions, or recommendations. The system further analyzes historical usage information to identify usage patterns and generate periodic usage reports, thereby enabling personalized and adaptive management of application usage and facilitating a transition from passive or aimless usage toward mindful and productive usage. Figure 1 is the representative figure.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641130107 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR INTELLIGENT WEARABLE MARINE SAFETY BAND WITH HYBRID COMMUNICATION

(51) International classification	:H04B 11/00, G01S 5/18, B63C 9/20, G01S 19/47, G01S 5/22	(71)**Name of Applicant :** **1)Sir Issac Newton College of Engineering and Technology** Address of Applicant :Pappakoil, Nagapattinam - 611102, Tamil Nadu, India. Pappakoil Tamil Nadu India **2)Monisha. K** **3)Keerthika. S** **4)Jeyeswari. V**
(31) Priority Document No	:NA	**5)Mahalakshmi. A**
(32) Priority Date	:NA	**6)Vembu. S**
(33) Name of priority country	:NA	**7)Mahalakshmi. S**
(86) International Application No Filing Date	: :01/01/1900	(72)**Name of Inventor :** **1)Monisha. K**
(87) International Publication No	: NA	**2)Keerthika. S**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**3)Jeyeswari. V** **4)Mahalakshmi. A**
(62) Divisional to Application Number Filing Date	:NA :NA	**5)Vembu. S** **6)Mahalakshmi. S**

(57) Abstract :

The present invention discloses an intelligent wearable marine safety band system 5 providing hybrid above-water and underwater emergency localization. The SeaBand AI wearable device comprises an ESP32 microcontroller, GPS module, IMU, pressure sensor, LoRa wireless communication module, and underwater acoustic transmitter. During normal operation, GPS location and sensor data are transmitted to the Boat Hub through LoRa communication. Upon detection of a 10 submerged emergency condition where GPS and LoRa are unavailable, the underwater acoustic transmitter generates an identifiable emergency signal propagating through water to a hydrophone positioned below the boat waterline. A Raspberry Pi-based Boat Hub receives and processes the acoustic signal through noise reduction and direction estimation modules, providing rescue direction 15 guidance to operators. A non-paired architecture enables detection by any nearby Boat Hub without permanent pairing, supporting multi-boat coordination. The system provides improved emergency detection, reliable underwater localization, reduced rescue response time, and enhanced marine safety for fishermen and marine workers.

No. of Pages : 17 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641130135 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Multimodal Generative Al for Real-Time Synthetic Data Generation in Privacy Preserving Surveillance Systems

(51) International classification	:G06F 21/62, G06N 20/00, G06N 3/08, G06N 3/04, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.Balamuralikrishna Thati
Address of Applicant :DVR & Dr. HS MIC College of Technology, Kanchikacherla, NTR District, A.P.-521180 Vijayawada Andhra Pradesh India
2)Dr.D Varun Prasad
3)Nasaramma Kadiyam
4)V.Padmaja
5)Pasupuleti Nagendra
6)Vijayakumar nalluri
7)Dr. Chandini Mutta
(72)**Name of Inventor :**
1)Dr.Balamuralikrishna Thati
2)Dr.D Varun Prasad
3)Nasaramma Kadiyam
4)V.Padmaja
5)Pasupuleti Nagendra
6)Vijayakumar nalluri
7)Dr. Chandini Mutta

(57) Abstract :
The present invention discloses a multimodal generative artificial intelligence system engineered for the continuous production of synthetic data streams that enable effective real-time surveillance while strictly safeguarding individual privacy. Conventional surveillance infrastructures rely on the direct capture, storage, and transmission of identifiable visual, thermal, and acoustic recordings, thereby creating persistent risks of unauthorized access, regulatory non-compliance, and erosion of personal privacy. Existing anonymization techniques either operate offline on archived footage, process only a single sensory modality, or generate synthetic replacements that lack temporal and cross-modal coherence, rendering them unsuitable for live monitoring environments. To overcome these limitations, the invention introduces a fully integrated pipeline that ingests synchronized multimodal inputs comprising visible-light video, thermal imagery, audio signals, and optional environmental sensor readings. These inputs are time-aligned and projected by a joint encoder into a unified latent representation that captures structural layout, motion dynamics, and activity patterns while deliberately attenuating identity-bearing features. A privacy-conditioned generative core, guided by adaptive control tokens, synthesizes corresponding latent codes that are subsequently decoded into synthetic multimodal outputs. A consistency refinement stage continuously verifies and corrects spatial, temporal, and cross-modal alignment, ensuring that the generated streams faithfully reproduce scene evolution and event timing without reconstituting personal identifiers. The resulting synthetic data can be substituted in real time for original sensor feeds within live monitoring dashboards, archived for compliance purposes, or employed as high-fidelity training material for downstream analytic models. Because generation occurs continuously on edge-compatible hardware and is conditioned jointly across modalities, the system maintains low latency and high structural fidelity even under varying illumination, crowd density, and acoustic conditions. Quantitative evaluation confirms substantial reduction in measurable identity leakage relative to prior single-modality or batch-oriented approaches, together with retention of activity-detection accuracy sufficient for operational surveillance. The invention therefore furnishes a practical, privacy-preserving alternative to conventional recording systems and simultaneously supplies consistent multimodal synthetic datasets that enhance the robustness of secondary computer-vision algorithms. Its primary applications encompass public-space monitoring under stringent data-protection regimes, industrial activity logging without compromising worker identity, healthcare movement tracking that shields patient appearance, and large-scale generation of training corpora for privacy-aware analytic models.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641130149 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BUCCAL PATCHES OF ENALAPRIL AND HYDROCHLOROTHIAZIDE

(51) International classification	:A61K 9/70, A61K 31/549, A61P 9/12, A61K 47/38, A61K 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Nirmala College of Health Science
Address of Applicant :Dr Preetha S Panicker, Nirmala College of Health Science, Poolani-Puzhpagiri Rd, Meloor, Keralam, India, 680311. Meloor Kerala India
(72)**Name of Inventor :**
1)Sanjay B
2)Preetha S Panicker
3)Rosemol K John
4)Aadi Sathyan
5)Saranya Sasi
6)Remya Binu
7)Femi Joseph
8)Anju S Nair
9)Arun Francis
10)Sreelekha PP
11)Suni M
12)Manjusha M P

(57) Abstract :
The present invention relates to pharmaceutical formulations and, more particularly, to buccal patches comprising a combination of enalapril maleate and hydrochlorothiazide. The patches are prepared by a solvent-casting technique using hydroxypropyl methylcellulose (HPMC K15M) in combination with hydroxypropyl cellulose (HPC), polyvinyl pyrrolidone (PVP), or polyvinyl alcohol (PVA), with propylene glycol as a plasticizer. The formulation process involves dissolving the active pharmaceutical ingredients in dimethyl sulfoxide, incorporating the selected polymeric materials and other excipients, casting the resulting solution, drying the cast film, and cutting the dried film into patches. The prepared patches are evaluated for weight variation, thickness, surface pH, folding endurance, mucoadhesive strength, swelling index, drug content, tensile strength, percentage elongation at break, and in-vitro drug-release characteristics. Stability evaluation over six months showed no significant differences in the monitored appearance, drug content, swelling index, and drug-release characteristics under the investigated storage conditions.

No. of Pages : 22 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202641130167 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : APPARATUS FOR GENERATING THRUST USING ELECTROHYDRODYNAMIC AND ACOUSTIC PROPULSION UNDER ATMOSPHERIC CONDITIONS AND METHOD THEREOF

(51) International classification	:F03H 99/00, B64C 29/00, B64D 27/24, B64C 39/00, B64D 27/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)PRAYAGA VENKATA RAMAKRISHNA
Address of Applicant :BALAJI ENCLAVE, 4TH FLOOR, DOOR NO: 86-3-13/8. SHIRDISAIMARG, JN. ROAD, RAJAHMUNDRY, ANDHRA PRADESH, 533103, INDIA RAJAHMUNDRY Andhra Pradesh India
(72)**Name of Inventor :**
1)PRAYAGA VAIBHAV NISCHAL

(57) Abstract :
An apparatus (102) for generating thrust using electrohydrodynamic and acoustic propulsion under atmospheric conditions and method thereof is provided. Aerodynamic shroud hull (100) comprises intake region, intake throat, exhaust region, and internal airflow passage; propulsion assembly disposed at intake throat comprises structural frame (110), micro-needle emitters (120), an accelerating electrode (130), and high-voltage power supply to generate electrohydrodynamic thrust; a secondary piezoelectric acoustic array (140) is positioned above ionization region and to generate upward-moving toroidal acoustic pressure structures (150) for deflecting falling water droplets; sensing assembly configured to acquire rainfall intensity, wind velocity, and wind direction; control unit configured to control phase and amplitude of acoustic pressure waves for generating anti-phase acoustic pressure fields; and acoustic propulsion assembly (160) disposed at exhaust region for generating acoustic thrust, electrohydrodynamic thrust and acoustic thrust are coordinated based on environmental information to support stable lift and flight control during adverse atmospheric conditions. FIG. 1

No. of Pages : 50 No. of Claims : 11

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130168 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SELF-POWERED MOBILE ATMOSPHERIC PURIFICATION VEHICLE WITH INTEGRATED AIR COLLECTION, FILTRATION REGENERATION, POWER MANAGEMENT AND PURIFIED-AIR DISCHARGE ARCHITECTURE

(51) International classification	:B60H 3/06, B01D 50/00, B01D 46/00, B60P 3/00, B01D 47/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M/s PPS INNOVATIONS PRIVATE LIMITED
Address of Applicant :H.NO:1-6-110/1, BEEDI COLONY, METPALLY, JAGITIAL-505325, TELANGANA, INDIA METPALLY Telangana India
(72)**Name of Inventor :**
1)ALLADI PRABHAKAR

(57) Abstract :
The present invention relates to an automatic mobile air purification vehicle for outdoor atmospheric pollution control. The vehicle comprises a purification chamber mounted on a vehicle chassis and includes industrial air suction fans, air intake louvers, a multi-stage filtration assembly comprising a dust mesh filter, cyclonic separator, wet filtration pads, HEPA filter, and activated carbon filter, and rear-mounted purified-air discharge blowers. Polluted atmospheric air is collected, filtered, and discharged as purified air into the atmosphere. The invention further comprises a water-based self-cleaning filtration system, an onboard generator connected to the propulsion and purification systems, a battery charging arrangement, and optional solar panels for auxiliary electrical loads. The vehicle is capable of travelling up to approximately 55" km/h" and achieves optimum purification performance at approximately 28" km/h" . The invention provides economical and continuous outdoor air purification with reduced operational downtime and improved pollutant removal efficiency. (Fig No.1)

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130174 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SUBSTITUTED 3-ARYL-5,6,7,8-TETRAHYDROIMIDAZO[1,2-a]PYRAZIN-2-YL TRIFLUOROACETAMIDE COMPOUNDS AND THEIR SYNTHESIS

(51) International classification	:C07D 487/04, A61P 35/00, A61K 31/4985, C07D 471/04, A61K 31/4965
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ABDUL MAJEED KURUMBALA
Address of Applicant :199, 9th main, 4th cross, Duo city layout, Basapura main road, Bangalore, Karnataka, India Bangalore Karnataka India
2)KOODLUR SANNEGOWDA LOKESH
(72)**Name of Inventor :**
1)ABDUL MAJEED KURUMBALA
2)KOODLUR SANNEGOWDA LOKESH
3)SARIGAMA RAJESH
4)CHIDAMBARAM RISHINARADAMANGALAM RAMASWAMI

(57) Abstract :
Abstract: The present invention relates to development of new heterocyclic compounds for therapeutic applications. It particularly relates to the development of novel heterocyclic compounds with 5,6,7,8-tetrahydroimidazo[1,2-a]pyrazine core in which a trifluoroacetamido group at position 2 is combined with an aryl or heteroaryl group at position 3 of an N7-unsubstituted tetrahydroimidazo[1,2-a]pyrazine. It specifically relates to the development of novel compounds of substituted 3-aryl-5,6,7,8-tetrahydroimidazo[1,2-a]pyrazin-2-yl trifluoroacetamide or pharmaceutically acceptable salt thereof as potential anticancer agents. It also relates synthetic process for preparation of novel compounds of substituted 3-aryl-5,6,7,8-tetrahydroimidazo[1,2-a]pyrazin-2-yl trifluoroacetamide. It further relates to the development of pharmaceutical composition or formulation with synthesized novel compounds of substituted 3-aryl-5,6,7,8-tetrahydroimidazo[1,2-a]pyrazin-2-yl trifluoroacetamide or pharmaceutically acceptable salt thereof, and their use in the manufacture of medicaments for the treatment of cancer.

No. of Pages : 30 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641130175 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-CALIBRATED DYNAMIC FTP MODEL FOR REAL-TIME NET INTEREST MARGIN (NIM) PROTECTION IN RETAIL BANKS

(51) International classification	:G06Q 40/04, G06Q 40/00, G06N 3/04, G06Q 30/02, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GAYATRI PRAVEEN CHAVALI
Address of Applicant :RESEARCH SCHOLAR, DEPARTMENT OF COMMERCE AND MANAGEMENT STUDIES, COLLEGE OF ARTS AND COMMERCE, ANDHRA UNIVERSITY, VISAKHAPATNAM VISAKHAPATNAM Andhra Pradesh India
2)Dr. A. NARASIMHA RAO
(72)**Name of Inventor :**
1)GAYATRI PRAVEEN CHAVALI
2)Dr. A. NARASIMHA RAO

(57) Abstract :
AI-calibrated dynamic FTP model for real-time net interest margin (NIM) protection in retail banks. The present invention discloses an AI-calibrated dynamic Funds Transfer Pricing (FTP) system and method for real-time Net Interest Margin (NIM) protection in retail banking. The system includes a data ingestion engine for capturing live market feeds and transactional data streams, and an AI behavioral processor running neural networks to calculate real-time deposit decay vectors and asset prepayment probabilities. A dynamic yield curve calibration matrix integrates these behavioral vectors with market benchmarks to compute precise, risk-adjusted internal transfer rates at an account level. An automated NIM protection circuit monitors systemic interest margins via forward-looking simulations, automatically adjusting internal transfer credits and charges to mitigate margin compression. An API orchestration layer propagates these calibrated rates instantly to frontline loan origination and deposit platforms. A cryptographic ledger module records all operational metrics to ensure data lineage and regulatory compliance.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202641130178 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Wide band ClAlPc Photosensitizers for Photodynamic therapy in cancer therapeutics

(51) International classification	:A61K 41/00, A61P 35/00, A61K 31/409, A61K 47/48, C23C 14/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dayananda Sagar College of Engineering
Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. Chennemkeril Mathew Joseph

(57) Abstract :
In this invention, the use of Chloro Aluminium Phthalocyanine (ClAlPc) organic materials for photodynamic therapy in cancer therapeutics is focused. ClAlPc thin film layered structures were deposited using vacuum evaporation at room temperature and vacuum annealed post deposition at 50 oC and vacuum cooled to room temperature in a vacuum of 10-6 Torr. ClAlPc thin films (thickness-3000 Å) were deposited onto cleaned glass substrates for the optical absorption studies. All the ClAlPc films were deposited from a molybdenum boat filled with 99.99% pure ClAlPc source powder. Thickness of the films were measured using a digital quartz crystal thickness monitor. Rate of deposition for the ClAlPc films deposition was around 50 Å/s. All the ClAlPc films were studied using a uv-vis spectrophotometer to know the feasibility of ClAlPc as a photosensitizer in the visible region. Optical absorption spectra of room temperature deposited ClAlPc active layers were studied in the range 300-900 nm. ClAlPc thin films have a sharp peak at 760 nm for a room temperature deposited ClAlPc film with a band width of 375 nm (BW = 900-525 nm = 375 nm). It was found that ClAlPc films annealed post deposition at 50 oC had a widening of the spectra from 480 nm to 900 nm with a band width of 420 nm (BW = 900-480 nm = 420 nm) with more absorption which makes them useful for a wider range of wavelengths as a photosensitizer.

No. of Pages : 8 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130196 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Design and Development of an Intelligent Animatronic Student Companion Robot with Interactive Human-Robot Communication

(51) International classification	:B25J 11/00, G09B 5/06, B25J 9/16, G09B 5/04, B25J 13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Noorul Islam Centre for Higher Education
Address of Applicant :Noorul Islam Centre for Higher Education Kumaracoil-629180 Kanyakumari Dist, Tamil Nadu. Tamil Nadu India
(72)**Name of Inventor :**
1)Mr. Krishnan Unni J.S
2)Dr. A. Shajin Nargunam
3)Mr. APJMJ Sheik Saleem
4)Mr. U RAkhil Raj
5)Mr. Mohammed Habeeb Hassan
6)Mr. Abhinand Anilkumar

(57) Abstract :
Abstract The present invention relates to an intelligent animatronic student companion robot, designated as KALAM ROBO / KALAM SPARK, developed to facilitate interactive communication and educational engagement between students and a robotic companion. The invention integrates an animatronic mechanical structure, electronic control system, audio input and output components, and software-based intelligence to provide an expressive human–robot interaction experience. The system is configured to receive user inputs, process spoken queries, generate or select appropriate responses, and deliver audible communication coordinated with corresponding robotic movements. The robot may incorporate speech recognition, speech synthesis, voice modulation, conversational artificial intelligence, and configurable educational content delivery functionalities. It is further designed to support educational explanations, interactive storytelling, motivational communication, and student-oriented learning activities. The coordination of audio responses, character-based speech, and animatronic movements provides an integrated interface for engaging students through physical and conversational interaction. The invention offers a configurable robotic platform for educational institutions, learning environments, and student engagement applications.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/10/2026
(21) Application No.202641130197 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A system and Method for Personalized Knowledge Structure Reconfiguration Using Cognitive and Conceptual Inference Signals

(51) International classification	:G06Q 50/20, G06N 5/04, G09B 7/02, G09B 7/00, G06N 5/02	(71)**Name of Applicant :** **1)Dayananda Sagar College of Engineering** Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India (72)**Name of Inventor :** **1)Sai S Joshi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The invention relates to an artificial intelligence-based educational analytics system configured to dynamically model and reconfigure knowledge structures of individual learners. The system combines cognitive-state indicators, conceptual understanding metrics, and learner interaction data to construct a personalized knowledge graph that represents concept dependencies for a specific user. Unlike conventional educational systems that rely on static prerequisite relationships, the proposed invention continuously updates concept connections based on observed learning behavior, conceptual mastery patterns, and cognitive inference signals. The system receives learner interaction inputs including assessment performance, explanation quality, response latency, concept mastery indicators, and cognitive-state measures. These inputs are processed through a graph adaptation engine that identifies variations in individual learning pathways and modifies concept dependency relationships accordingly. The modified graph is utilized to generate personalized learning recommendations, alternative prerequisite sequences, and adaptive instructional interventions. The invention enables the discovery of individualized concept progression pathways and provides a computational framework for adaptive learning environments capable of evolving according to learner-specific knowledge acquisition patterns.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/10/2026
(21) Application No.202641130198 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FPGA Based Low Area Polynomial Multiplier for NTRU Cryptography

(51) International classification	:G06F 7/72, H04L 9/30, H04L 9/06, G09C 1/00, G06F 7/523
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K Gavaskar
Address of Applicant :Assistant Professor III, ECE Department, Kumaraguru College of Technology, Coimbatore, TamilNadu, India – 641049. Coimbatore Tamil Nadu India
2)S. N. Shivappriya
3)N. Prakash
4)S. Prasanna Kumaar
5)K. P. Thilagavathy
6)S. Sneka
(72)**Name of Inventor :**
1)K Gavaskar
2)S. N. Shivappriya
3)N. Prakash
4)S. Prasanna Kumaar
5)K. P. Thilagavathy
6)S. Sneka

(57) Abstract :
The present invention relates to a low-area sequential and pipelined polynomial multiplier architecture for NTRU cryptographic applications. The architecture is intended for FPGA-based implementation and is designed to perform cyclic polynomial multiplication with reduced hardware complexity. The proposed architecture comprises input polynomial registers, an FSM-based control unit, an address generator, a pipelined processing element, a modulo-q reduction unit and an output register. The input coefficients are processed sequentially, while the arithmetic hardware is reused for successive coefficient operations. Cyclic coefficient addressing is performed using modulo-N indexing. The processing element includes multiplication, partial-product registration, accumulation, modulo-q reduction and result storage stages. A MUX-based multiplication structure is used for suitable coefficient values such as −1, 0 and +1, reducing the requirement for conventional multiplication hardware. Pipeline registers are used to separate arithmetic operations and reduce the combinational critical path. In the described implementation, the architecture supports a polynomial size of N = 401 with signed 32-bit coefficient representation and uses q = 12289 for modulo reduction. The implementation using Verilog HDL and Xilinx Vivado on a Zybo Z7-10 FPGA reported 2356 LUTs, a critical path delay of 3.836 ns and an operating frequency of approximately 260.7 MHz The architecture provides a practical balance between FPGA resource utilization and computational performance. It is suitable for lightweight and resource-constrained cryptographic hardware in which reduced area and hardware complexity are important requirements for NTRU polynomial processing.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130202 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HAVEN PET-AI PET CARE ASSISSTANT

(51) International classification	:G06N 20/00, G06Q 50/22, A01K 29/00, G06N 5/04, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mr. MANIKANDAN M
2)Poovarasu S
3)Prajin B S
4)Ragul M
5)Naren kumar M
6)Ragul R
7)Mugesh B
8)Roshan M
9)Priyadarshan P

(57) Abstract :
The present invention relates to an AI-Powered Intelligent Pet Care Assistant system designed to enhance pet care activities through intelligent automation and data-driven analysis. The system facilitates efficient monitoring, organization, and management of pet routines and behavioral information. It incorporates artificial intelligence techniques to analyze pet activities and identify relevant behavioral patterns. Additionally, the system provides personalized recommendations and intelligent assistance based on pet profiles and historical information. Furthermore, the invention utilizes an AI chat assistant to provide natural-language interaction between pet owners and the system. The system also integrates agentic AI components to coordinate routine monitoring, behavior analysis, recommendation generation, and health reminders. The system integrates visualization tools to represent pet routines, reminders, and behavioral information in an intuitive and user-friendly manner. By combining monitoring, analysis, recommendation, and intelligent assistance capabilities within a unified platform, the invention reduces manual effort and complexity involved in daily pet care activities. Overall, it improves pet care management, supports informed decision-making, and provides meaningful assistance to pet owners.

No. of Pages : 5 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130203 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Launtree – Smart Laundry Management System

(51) International classification	:G06Q 10/08, G06Q 10/06, G06Q 30/06, G06Q 50/28, G06Q 10/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs Mahalakshmi
2)Sanjay P K
3)Prasanna M
4)Naveen V
5)Niranjan K
6)Rubakumar P
7)Sanjay Kumar S
8)Thamarai Selvan P

(57) Abstract :
The present invention relates to a Smart Laundry Management System designed to simplify and automate laundry and dry-cleaning operations through an integrated digital platform. The system connects customers, service providers, delivery agents, and administrators within a unified system for efficient management of laundry orders, pickups, processing, delivery, and tracking. The system enables customers to place laundry orders, select required services, schedule pickups, provide delivery details, and monitor the progress of their orders. A centralized order management mechanism maintains order information and synchronizes status updates across the different modules. The order progresses through defined stages including Order Confirmed, Pickup Pending, Picked Up, Washing, Ready, Out for Delivery, and Delivered. Furthermore, the invention integrates delivery and laundry-processing operations, wherein delivery agents manage pickup and delivery activities while service providers receive, process, and prepare laundry orders for final delivery. An administrative module provides centralized monitoring of customers, orders, laundry operations, and delivery activities. By combining order management, laundry processing, delivery coordination, customer tracking, and administrative monitoring within a unified platform, the invention reduces manual coordination and improves the efficiency, visibility, and organization of laundry service operations.

No. of Pages : 5 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130204 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Quickhire: Instant job marketplace

(51) International classification	:G06Q 10/10, G06Q 10/06, G06Q 10/00, G06Q 30/06, G06Q 30/00	(71)**Name of Applicant :** **1)SNS College of Technology** Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India (72)**Name of Inventor :** **1)Mr. MANIKANDAN M** **2)Murugesh R** **3)Naveen P** **4)Neha Priyadharshini M** **5)Nikashini S** **6)Nithish Kanna M** **7)Phrisha G** **8)Preshika S** **9)Remettha B**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to QuickHire, an AI-powered hyperlocal employment marketplace connecting job seekers with employers offering part-time, temporary, and task-based work. The system addresses challenges in job discovery, recruitment, and employment accessibility through user registration, profile management, job posting, location-based searching, application tracking, and skill-based matching. It enables workers to showcase their skills, experience, availability, and preferences, helping employers identify suitable candidates efficiently. QuickHire incorporates payment management with demo wallet and escrow functionality, real-time job tracking, and automated notifications to ensure reliable interactions between workers and employers. The platform simplifies recruitment, promotes transparent transactions, and improves access to local employment opportunities through an efficient digital marketplace.

No. of Pages : 5 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130205 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An AI-Powered Local Language Interpreter and Personalized Career Guidance Platform

(51) International classification	:G06Q 50/20, G09B 19/06, G09B 5/06, G06F 17/28, G09B 19/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Priyanka K
2)Sriram R
3)Rithik K K
4)Abdul Salam J
5)Aashik R
6)Ramesh Babu R
7)Dharanidharan S

(57) Abstract :
The proposed invention introduces an AI-powered digital platform that combines a local language interpreter with a personalized career guidance system, designed to reduce communication barriers caused by different local languages and regional dialects while helping students and fresh graduates choose the right career path. Conventional approaches often leave users unable to communicate clearly across languages and dialects, and leave learners dependent on generic advice, social media, or random online resources that lead to poor career decisions, wasted time, and skills that are not aligned with industry demand. The proposed system addresses these challenges by providing accurate, real-time speech and text translation across multiple languages and regional dialects, together with personalized career recommendations, skill gap identification, learning resource recommendation, and a complete roadmap to improve employability. AI-based analysis supports speech recognition, automatic language detection, real-time translation, dialect interpretation, text-to-speech output, skill and interest analysis, career path recommendation, personalized roadmap generation, and job preparation support. Through this integrated approach, the proposed system promotes seamless communication, accessible and affordable career guidance, and improved employability through a single intelligent platform.

No. of Pages : 4 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130207 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE SECONDARY CONTROL SYSTEM AND METHOD FOR COORDINATED REGULATION OF ISLANDED AC MICROGRIDS

(51) International classification	:H02J3/388, H02J3/38, H02J3/0014, G05B13/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM UNIVERSITY-AP
Address of Applicant :Neerukonda, Mangalagiri Mandal, Amaravati Guntur District, Andhra Pradesh-522240 Amaravati Andhra Pradesh India
(72)**Name of Inventor :**
1)Devarapalli Vimala
2)Naresh Kumar Vemula
3)Somesh Vinayak Tewari

(57) Abstract :
The present invention relates to a predictive secondary control system and method for coordinated regulation of islanded alternating current microgrids. The system comprises parallel inverter-based distributed generation units connected through respective line impedances, primary control arrangements, and a secondary control arrangement employing finite control set model predictive control. The secondary control arrangement generates voltage and frequency correction signals based on measured electrical quantities, predicted electrical responses, and cost function evaluation. The correction signals are applied to respective primary droop-control arrangements to coordinate active-power, reactive-power, voltage, and frequency regulation. The predictive controller determines inverter control actions by evaluating prospective switching states while considering computational delays and increased inductive characteristics of interconnecting line impedances. The system operates with a sampling interval of 10 microseconds and maintains regulated electrical responses under computational delays up to 100 microseconds. The invention facilitates improved dynamic regulation, reduced power oscillations, and enhanced operational robustness during load variations. Figure of Abstract – Fig. 1

No. of Pages : 58 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130208 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SEMICONDUCTOR-BASED GAS SENSOR ON NEGATIVE AROUND VERTICAL TUNNEL FIELD-EFFECT TRANSISTOR AND FABRICATION METHOD THEREOF

(51) International classification	:G01N 27/414, H01L 29/78, G01N 27/12, G01N 27/22, G01N 27/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM UNIVERSITY-AP
Address of Applicant :Neerukonda, Mangalagiri Mandal, Amaravati Guntur District, Andhra Pradesh-522240 Amaravati Andhra Pradesh India
(72)**Name of Inventor :**
1)Annavarapu Ratnavali
2)Piduguralla Kranthi
3)Dr. Patta Supraja

(57) Abstract :
The present invention discloses a semiconductor-based gas sensor (100) and a fabrication method thereof. The sensor integrates Negative Capacitance, a Gate-All-Around (GAA) architecture, a Vertical Tunnel Field-Effect Transistor (VTFET), and a T-shaped silicon-germanium (SiGe) pocket region (106) positioned at the source-channel interface. The architecture comprises a base substrate (102), a source region (104), an intrinsic channel region (108), a drain region (110), a ferroelectric hafnium zirconium oxide ($HfZrO_2$) dielectric layer (112), an interfacial silicon dioxide (SiO_2) dielectric layer (114), a control gate (116), a catalytic polar gate (118) and contact metal layer (120). The T-shaped SiGe pocket region (106) enhances band-to-band tunneling, while the ferroelectric layer (112) provides internal voltage amplification. This configuration achieves an ultra-steep subthreshold swing of 8.73 mV/dec and high ION/IOFF current ratios in the range of 1012 to 1013, making it highly suitable for ultra-low-power portable gas sensing applications. Figure of Abstract – Figure 1

No. of Pages : 29 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130228 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR THE PREPARATION OF AN ELECTRODE FILM

(51) International classification	:H01M 4/04, H01M 4/62, H01M 4/1391, H01M 10/0525, H01M 4/131
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)OLA ELECTRIC MOBILITY LIMITED
Address of Applicant :Regent Insignia, #414, 3rd Floor, 4th Block, 17th Main, 100 Feet Road, Koramangala, Bengaluru-560034, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)KUMAR, SHIVAM
2)MEKKAT, RAJESH
3)CHOUBEY, SHIKHAR

(57) Abstract :
ABSTRACT A PROCESS FOR THE PREPARATION OF AN ELECTRODE FILM The present disclosure relates to a process for the preparation of an electrode film. The process comprises processing a dry mixture powder comprising an active material, a conductive additive, and a binder through a plurality of calendering operations under controlled processing conditions to obtain a free-standing film, followed by laminating the free-standing film onto a metal current collector foil to obtain the electrode film. The process of the present disclosure facilitates the preparation of an electrode film having controlled structural and electrochemical characteristics suitable for use in energy storage applications, particularly lithium-ion batteries.

No. of Pages : 44 No. of Claims : 22

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/10/2026
(21) Application No.202641130229 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ELECTRODE COMPOSITION FOR DRY ELECTRODES

(51) International classification	:H01M 4/62, H01M 4/04, H01M 4/02, H01M 4/13, H01M 4/1391
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)OLA ELECTRIC MOBILITY LIMITED
Address of Applicant :Regent Insignia, #414, 3rd Floor, 4th Block, 17th Main, 100 Feet Road, Koramangala, Bengaluru-560034, Karnataka, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)KUMAR, SHIVAM
2)MEKKAT, RAJESH
3)CHOUBEY, SHIKHAR

(57) Abstract :
ABSTRACT AN ELECTRODE COMPOSITION FOR DRY ELECTRODES The present disclosure relates to an electrode composition for a dry. The electrode composition of the present disclosure comprises multi component conductive additive system and a dual-binder strategy. The electrode composition of the present disclosure exhibits a reduced percolation threshold, thereby enabling efficient electronic conduction at low conductive additive loading. The electrode composition of the present disclosure enables high active material loading while maintaining electrode processability.

No. of Pages : 38 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/10/2026
(21) Application No.202641130230 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR THE PREPARATION OF ELECTRODE COMPOSITION

(51) International classification	:H01M 4/04, H01M 4/62, H01M 10/0525, H01M 4/36, H01M 4/1391
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)OLA ELECTRIC MOBILITY LIMITED
Address of Applicant :Regent Insignia, #414, 3rd Floor, 4th Block, 17th Main, 100 Feet Road, Koramangala, Bengaluru-560034, Karnataka, India Bangalore Karnataka India
(72)**Name of Inventor :**
1)KUMAR, SHIVAM
2)MEKKAT, RAJESH

(57) Abstract :
ABSTRACT A PROCESS FOR THE PREPARATION OF ELECTRODE COMPOSITION The present disclosure relates to a process for the preparation of electrode composition and electrode composition obtained therefrom. The process comprises sequentially mixing an active material, multiple conductive materials, and binders in a plurality of stages under predetermined mixing speeds, temperatures, and time periods to progressively form a homogeneous mixture. The mixture is subsequently subjected to high-shear mixing until a target temperature is reached and is then cooled under controlled stirring conditions to obtain the electrode composition. The process enables solvent-free, energy-efficient, and industrially scalable production of a homogeneous, high-active-material-loading electrode composition.

No. of Pages : 54 No. of Claims : 19

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/10/2026
(21) Application No.202641130231 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR THE PREPARATION OF AN ELECTRODE FILM

(51) International classification	:H01M 4/04, H01M 4/62, H01M 4/1391, H01M 10/0525, H01M 4/139
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)OLA ELECTRIC MOBILITY LIMITED
Address of Applicant :Regent Insignia, #414, 3rd Floor, 4th Block, 17th Main, 100 Feet Road, Koramangala, Bengaluru-560034, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)KUMAR, SHIVAM
2)MEKKAT, RAJESH
3)CHOUBEY, SHIKHAR

(57) Abstract :
ABSTRACT A PROCESS FOR THE PREPARATION OF AN ELECTRODE FILM The present disclosure relates to a process for the preparation of an electrode film. The process eliminates the use of solvent, that reduces the energy required for condensation of solvent and the cost required for the condensation and recovery infrastructure. The process of the present disclosure offers the usage of electrochemically and thermally stable binder. The process of the present disclosure improves electrode density, active material mass loading, through-plane electronic conductivity, suppresses the rebound thickness and provides post-lamination dimensional stability. Further, the electrode film so obtained has high electrode density, high active material loading, low porosity and high dimensional stability. Further, the electrode film has improved peel strength and enhances electrochemical performance.

No. of Pages : 47 No. of Claims : 23

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130233 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRODE COMPOSITION AND A PROCESS FOR ITS PREPARATION

(51) International classification	:H01M 4/58, H01M 4/62, H01M 10/0525, H01M 4/36, H01M 4/1397
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)OLA ELECTRIC MOBILITY LIMITED
Address of Applicant :Regent Insignia, #414, 3rd Floor, 4th Block, 17th Main, 100 Feet Road, Koramangala, Bengaluru-560034, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)KUMAR, SHIVAM
2)MEKKAT, RAJESH
3)CHOUBEY, SHIKHAR

(57) Abstract :
The present disclosure relates to an electrode composition, and a process for its preparation. The composition comprises 90 wt.% to 97 wt.% of lithium iron phosphate active material, a conductive carbon comprising first conductive carbon, second conductive carbon and third conductive carbon, and binders comprising non-fibrillating binder and a fibrillating binder. The composition is suitable for solvent-free, energy-efficient, and industrially scalable lithium-ion battery manufacturing. The process of the present disclosure is energy-efficient, and commercially scalable g. The electrode of the present disclosure exhibits ultra-low through-plane electronic resistance and free-standing film integrity. The electrochemical cell of the present disclosure provides improved electrochemical utilization and enhanced cycling stability.

No. of Pages : 51 No. of Claims : 31

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :31/10/2026
(21) Application No.202641130234 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR THE PREPARATION OF A DRY ELECTRODE MIXTURE

(51) International classification	:H01M 4/62, H01M 4/04, H01M 4/36, H01M 4/1391, H01M 4/583
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)OLA ELECTRIC MOBILITY LIMITED
Address of Applicant :Regent Insignia, #414, 3rd Floor, 4th Block, 17th Main, 100 Feet Road, Koramangala, Bengaluru-560034, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)KUMAR, SHIVAM
2)MEKKAT, RAJESH

(57) Abstract :
ABSTRACT A PROCESS FOR THE PREPARATION OF A DRY ELECTRODE MIXTURE The present disclosure relates to a process for the preparation of a dry electrode mixture. The process of the present disclosure is a multi-stage sequential mixing process, that provides a solvent free dry mixture suitable for use as a cathode active material in dry electrode film fabrication. The process of the present disclosure provides a dry electrode mixture that achieves uniform dispersion of one or more conductive carbon additives having different morphologies, surface areas, and bulk densities within an LFP matrix. Further, the process of the present disclosure minimises carbon agglomeration and prevents encapsulation of carbon agglomerates by binder material.

No. of Pages : 48 No. of Claims : 29

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130235 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A DRY ELECTRODE MIXTURE

(51) International classification	:H01M 4/62, H01M 4/04, H01M 4/02, H01M 4/1391, H01M 4/1393
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)OLA ELECTRIC MOBILITY LIMITED
Address of Applicant :Regent Insignia, #414, 3rd Floor, 4th Block, 17th Main, 100 Feet Road, Koramangala, Bengaluru-560034, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)KUMAR, SHIVAM
2)MEKKAT, RAJESH
3)CHOUBEY, SHIKHAR

(57) Abstract :
ABSTRACT A DRY ELECTRODE MIXTURE The present disclosure relates to a dry electrode mixture for the fabrication of a dry electrode film. The present disclosure provides a dry electrode mixture that achieves uniform dispersion of one or more conductive carbons having different morphologies, surface areas, and bulk densities within an LFP matrix. Further, the dry electrode mixture of the present disclosure minimizes carbon agglomeration thereby preventing encapsulation of carbon agglomerates by binder material.

No. of Pages : 48 No. of Claims : 18

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130236 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A PROCESS FOR PREPARATION OF ELECTRODE COMPOSITION AND ELECTRODE COMPOSITION OBTAINED THEREFROM

(51) International classification	:H01M 4/04, H01M 4/62, H01M 4/02, H01M 4/1391, H01M 4/139
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)OLA ELECTRIC MOBILITY LIMITED
Address of Applicant :Regent Insignia, #414, 3rd Floor, 4th Block, 17th Main, 100 Feet Road, Koramangala, Bengaluru-560034, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)KUMAR, SHIVAM
2)MEKKAT, RAJESH

(57) Abstract :
ABSTRACT A PROCESS FOR PREPARATION OF ELECTRODE COMPOSITION AND ELECTRODE COMPOSITION OBTAINED THEREFROM The present disclosure relates to a process for the preparation of an electrode composition and an electrode composition obtained therefrom. The process enables in-situ mechanical de-bundling and substantially uniform distribution of SWCNTs without requiring solvent-assisted dispersion or jet milling. The resulting electrode composition exhibits a homogeneous conductive network, reduced electrical resistance, improved conductivity uniformity, enhanced structural integrity, and improved electrochemical performance, while providing an energy-efficient, cost-effective, and industrially scalable dry electrode manufacturing process.

No. of Pages : 56 No. of Claims : 27

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130247 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Context-Aware Device Usage Identification Using Sensor Transformation and Lightweight Knowledge Distillation

(51) International classification	:G06N 5/02, G06N 5/04, G06N 3/04, G06N 3/08, G06F 3/0484	(71)**Name of Applicant :** **1)Dr. Sreedhar Kumar S** Address of Applicant :Professor, Department of Information Science and Engineering, Sir M. Visvesvaraya Institute of Technology, Yelhanka, Bengaluru Bengaluru Karnataka India **2)Mr. Manvender Singh**
(31) Priority Document No	:NA	**3)Mr. Harsh Raj**
(32) Priority Date	:NA	**4)Mr. Manas Malkani**
(33) Name of priority country	:NA	**5)Dr. M. N. Thippeswamy**
(86) International Application No Filing Date	: :01/01/1900	**6)Sir. M. Visvesvaraya Institute of Technology** (72)**Name of Inventor :**
(87) International Publication No	: NA	**1)Dr. Sreedhar Kumar S**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**2)Mr. Manvender Singh** **3)Mr. Harsh Raj**
(62) Divisional to Application Number Filing Date	:NA :NA	**4)Mr. Manas Malkani** **5)Dr. M. N. Thippeswamy**

(57) Abstract :

Context-aware device usage identification using sensor transformation and lightweight knowledge distillation ABSTRACT The present invention relates to a system and method for context-aware device usage identification using hardware sensor feature transformation and lightweight knowledge distillation. The system comprises a Dynamic Hardware Normalization Layer configured to compensate for sensor variations across heterogeneous mobile devices, a feature transformation module, a knowledge distillation framework, and a lightweight student model for identifying micro-contextual device usage states and associated graphical user interface (GUI) contexts. The system performs contextual inference using passive sensor data without relying on privileged operating system accessibility or screen-reading APIs during production. An uncertainty-driven verification mechanism selectively requests user confirmation for ambiguous predictions, enabling improved reliability, cross-device adaptability, and reduced computational complexity, memory utilization, and energy consumption.

No. of Pages : 7 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130267 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT SYSTEM AND METHOD FOR MULTIMODAL ENGLISH WRITING ASSESSMENT AND ADAPTIVE FEEDBACK GENERATION USING NATURAL LANGUAGE PROCESSING

(51) International classification	:G09B 19/06, G06V 30/10, G06K 9/00, G06K 9/62, G10L 15/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. M. Sudha Menon
Address of Applicant :Assistant Professor, Department of English, TKR College of Engineering and Technology, Telangana -500097 Telangana India
2)Dr. Shirisha Deshpande
3)Gollapalli Sreedevi
4)Dr. Divvela Surendra
5)Dr. Sachin Arun Bhoyar
6)Dr. Sunita. M
7)Mrs. B.Swaroopa
8)Dr. Suman Devi
9)Koppuravuri Gurnadha Gupta
(72)**Name of Inventor :**
1)Dr. M. Sudha Menon
2)Dr. Shirisha Deshpande
3)Gollapalli Sreedevi
4)Dr. Divvela Surendra
5)Dr. Sachin Arun Bhoyar
6)Dr. Sunita. M
7)Mrs. B.Swaroopa
8)Dr. Suman Devi
9)Koppuravuri Gurnadha Gupta

(57) Abstract :
[055] An intelligent computer-implemented system and method assess English writing from typed text, handwriting images, and optionally associated speech evidence. Modality processors generate recognition hypotheses with source coordinates and confidence values. A cross-modal alignment engine constructs a provenance-preserving evidence graph, while linguistic analysis and rubric assessment modules derive criterion-specific judgments linked to evidence spans. A confidence calibration module distinguishes recognition uncertainty from learner language errors and selectively withholds, clarifies, or permits judgments. A feedback planner prioritizes personalized interventions using learner history and assessment confidence, and a semantic-preservation verifier screens suggested revisions for unintended changes in meaning. The system maintains auditable assessment records and updates longitudinal learner proficiency information. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130293 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GIS-BASED URBAN SPRAWL SYSTEM AND METHOD USING RELATIVE SHANNON ENTROPY AND CONCENTRIC BUFFERS

(51) International classification	:G06T 11/60, G06K 9/00, G06K 9/62, G06K 9/46, G06F 17/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Chitimala Venkata Koteswara Rao
Address of Applicant :Scholar, Department of Civil engineering, Vignan's Foundation for Science Technology and Research, Vadlamudi, Guntur-522213, Andhra Pradesh, India Guntur Andhra Pradesh India
2)Prof. S.S. Asadi
(72)**Name of Inventor :**
1)Chitimala Venkata Koteswara Rao
2)Prof. S.S. Asadi

(57) Abstract :
A geographic information system processes multi-temporal satellite imagery to quantify urban development concentration and dispersion. Composite images undergo projection, clipping, supervised classification and mixed-pixel correction through polygon-based raster editing to produce corrected land-use maps and built-up and non-built-up maps. Concentric analysis zones intersect a study boundary, with the study boundary, city centre and buffer distances maintained across observation dates. Built-up densities use clipped zone areas, while relative Shannon entropy is calculated from proportional shares of total built-up area and normalised by the natural logarithm of the number of zones. Successive outputs identify land conversion and changing growth patterns, enabling temporally consistent urban sprawl assessment. A common spatial data store retains geographically and temporally associated outputs.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130296 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Enabled Intelligent Water Quality Monitoring and Automated Contamination Detection System

(51) International classification	:G01N 33/18, G06N 20/00, G01N 27/06, G06Q 50/06, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)J.J College of Engineering and Technology
Address of Applicant :Ammapettai, Tiruchirappalli District, Tamil Nadu, Pin code: 620009 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr S Murugesan
2)Mrs T Keerthana Jerine
3)Mrs. V. Karpagam
4)Mrs. J. Lourdu Joanna
5)Mrs. K. Kaaviya

(57) Abstract :
The present invention relates to an artificial intelligence-enabled intelligent water quality monitoring and automated contamination detection system for continuously monitoring, analysing, and identifying variations in water quality across domestic, industrial, agricultural, and public water distribution environments. The proposed system comprises a plurality of water quality sensors configured to measure physical, chemical, and biological indicators, including pH, turbidity, temperature, electrical conductivity, total dissolved solids (TDS), dissolved oxygen, and oxidation-reduction potential. The collected sensor measurements are transmitted to a processing unit through an Internet of Things (IoT) communication module. An artificial intelligence-based analytical engine processes the acquired data using machine learning algorithms to identify abnormal water quality patterns, detect potential contamination events, and classify contamination levels. The system further incorporates a predictive analysis module for estimating possible contamination risks based on historical measurements and real-time sensor variations. When abnormal conditions are identified, an automated alert mechanism generates notifications through a mobile application, web dashboard, or centralized monitoring platform. An optional automated control unit activates water isolation valves, filtration mechanisms, or purification devices to prevent the distribution of contaminated water. The invention provides continuous monitoring, early contamination detection, predictive analysis, and automated response capabilities, thereby improving water safety, reducing manual inspection requirements, and supporting efficient water resource management.

No. of Pages : 14 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130297 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI BASED FAULT DETECTION IN AUTOMOTIVE ELECTRIC MOTORS

(51) International classification	:G06N 20/00, G01R 31/34, G06N 3/08, G01M 17/007, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)FRANCIS XAVIER ENGINEERING COLLEGE
Address of Applicant :Tirunelveli - 627003, Tamil Nadu, India Tirunelveli Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. M. J. AMALA
2)ATHITHIYA.M
3)BALA SUBRAMANIAN.M
4)BHUVANESH.S

(57) Abstract :
The present invention relates to an Artificial Intelligence (AI)-based fault detection system for automotive electric motors, particularly those used in electric vehicles. The system comprises a sensor unit (110), a data acquisition unit (120), a data processing unit (130), an AI-based fault detection module (140), a fault classification unit (150), an alert generation unit (160), a vehicle monitoring interface (170), and a data storage unit (180). The sensor unit monitors motor temperature, vibration, current, voltage, and rotational speed. The acquired measurements are processed through noise filtering, data normalization, missing-data handling, and feature extraction. A trained machine-learning model analyzes the processed data to identify abnormal motor operating conditions. The fault classification unit identifies potential faults, including bearing defects, winding insulation faults, overheating, electrical abnormalities, and excessive vibration. Upon detecting an abnormal condition, the system generates an alert through the vehicle monitoring interface. Historical sensor measurements and fault detection results are stored to support motor condition assessment, maintenance planning, and predictive maintenance, thereby supporting improved motor reliability and maintenance efficiency.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130304 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FREQUENCY-ADDRESSED FERROELECTRIC ACOUSTIC RESONATOR MEMORY WITH PILOT-TRACKED COMB READOUT FOR MULTIPLY-ACCUMULATE

(51) International classification	:G11C 11/22, G06N 3/063, G06N 3/04, G06G 7/60, G06N 3/08	(71)**Name of Applicant :** **1)BAPATLA SURENDRA BABU** Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India **2)IRUKUMALLI PRIYANKA**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)BAPATLA SURENDRA BABU** **2)IRUKUMALLI PRIYANKA**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The invention relates to analog in-memory computing for neural-network inference. Conventional crossbar memories need a word line, a row address and settling time for every stored weight. A memory apparatus (100) stores signed weights in N ferroelectric acoustic resonators (110) of different resonant frequencies on a common drive line (102) and sense line (104). Each sense transducer (114) has electrode segments (302), each fully poled in one of two opposite directions, the signed poled-area fraction forming the weight. A comb generator (120) applies one tone per resonator with amplitudes encoding an input vector, and a homodyne detector (130) multiplies the sense signal by a reference comb (132) and integrates it into one multiply-accumulate value. Adjacent resonant frequencies are spaced between $4f_0/Q$ and $2f_0/(3(N+1))$. Pilot resonators (116) correct frequency, phase and gain drift, and the drive and sense lines are fed and read at opposite ends to equalise path phase. FIG. 1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130309 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for Quality-Guided Evidence Acquisition Orchestration across Quantum Computing and Post-Quantum Cryptographic Environments

(51) International classification	:G06N 10/00, H04L 9/08, G06N 10/20, H04L 9/40, G06F 16/901
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Aditya University
Address of Applicant :Aditya University, ADB Road, Aditya Nagar, Surampalem, East Godavari-533437, Andhra Pradesh, India. Kakinada Andhra Pradesh India
(72)**Name of Inventor :**
1)Mr. Vilas Anil Chavan
2)Ms. Devaki Vilas Chavan
3)Dr. Devidas S. Bhagat
4)Dr. Mahipal Singh Sankhla

(57) Abstract :
The present disclosure proposes a system (100) for quality-guided orchestration of digital evidence acquisition across quantum computing and post-quantum cryptographic environments. A computing device (102) implements an environment profiling module (108), an evidence ecosystem graph modeling module (110), an acquisition method catalog and a quality profile store module (112), a quantum evidence acquisition quality index calculator module (114), an acquisition orchestration module (116), an acquisition execution manager module (118), and a provenance and explainability module (120). The system (100) determines environment attributes, constructs an evidence ecosystem graph, retrieves acquisition-method quality profiles, computes predicted quantum evidence acquisition quality index (QEAQI) values, and selects and sequences evidence-acquisition workflows subject to acquisition constraints. The system (100) validates integrity, maps acquired evidence, quantifies acquisition-induced changes, computes realized QEAQI values, and generates an evidence-acquisition provenance and soundness report.

No. of Pages : 50 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130310 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System for AI Evidence Acquisition Using Predicted and Realized Quality Index Values and Method Thereof

(51) International classification	:G06N 20/00, G06F 16/901, G06N 5/04, G06N 5/02, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Aditya University
Address of Applicant :Aditya University, ADB Road, Aditya Nagar, Surampalem, East Godavari-533437, Andhra Pradesh, India. Kakinada Andhra Pradesh India
(72)**Name of Inventor :**
1)Mr. Vilas Anil Chavan
2)Ms. Devaki Vilas Chavan
3)Dr. Devidas S. Bhagat
4)Dr. Mahipal Singh Sankhla

(57) Abstract :
The present disclosure proposes a system (100) and method for graph-based identification and multi-layer acquisition of digital evidence across artificial intelligence and autonomous intelligent ecosystems. An environment profiler (101) identifies evidence sources, and an AAEEG Modeler (102) constructs evidence-source relationships. An acquisition method catalog (103) maintains acquisition profiles. An AIEAQI Calculator (104) determines predicted acquisition quality scores for candidate plans generated by an acquisition orchestrator (105). The orchestrator (105) selects and sequences acquisition workflows based on predicted AIEAQI values and acquisition constraints. An execution manager (106) acquires digital evidence and records execution metadata. A provenance and explainability module (107) validates available integrity information, determines achieved evidence coverage, and identifies excluded sources and acquisition-induced changes. The calculator (104) determines realized AIEAQI values, and the module (107) generates a provenance and soundness report.

No. of Pages : 119 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130311 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Intelligent Autonomous Software Development System Using Collaborative Artificial Intelligence Agents for Automated Code Generation and Optimization

(51) International classification	:G06F 11/36, G06F 8/41, G06F 21/53, G06N 20/00, G06F 8/20	(71)**Name of Applicant :** **1)Dr. Vijayalakshmi Kakulapati** Address of Applicant :Professor, Department of CSE-Allied, Maturi Venkata Subba Rao (MVSR) Engineering College, Nadergul Village, Balapur Mandal (New), Hyderabad-501510, Telangana, India. Hyderabad Telangana India **2)Sree Keerthi Meghana Bhrugubanda** **3)Dr. Rajesh H. Kulkarni** **4)K.Padmini**
(31) Priority Document No	:NA	**5)M. Nagaraju**
(32) Priority Date	:NA	**6)Dr.A.Paramasivam**
(33) Name of priority country	:NA	(72)**Name of Inventor :**
(86) International Application No	:	**1)Dr. Vijayalakshmi Kakulapati**
Filing Date	:01/01/1900	**2)Sree Keerthi Meghana Bhrugubanda**
(87) International Publication No	: NA	**3)Dr. Rajesh H. Kulkarni**
(61) Patent of Addition to Application Number	:NA	**4)K.Padmini**
Filing Date	:NA	**5)M. Nagaraju**
(62) Divisional to Application Number	:NA	**6)Dr.A.Paramasivam**
Filing Date	:NA	

(57) Abstract :

An intelligent autonomous software development system (100) uses collaborative artificial intelligence agents to convert a natural-language requirement into tested, secure, optimized and documented software. A requirement analysis agent (102) structures the requirement, a planning agent (104) builds a task graph, and an orchestrator (120) assigns tasks to a code generation agent (106), testing agent (108), optimization agent (110), security agent (112) and documentation agent (114). Agents share context through a shared memory (130) and execute code in a sandboxed environment (140). A feedback loop (170) converts test failures, vulnerabilities and bottlenecks into corrective tasks until a quality score meets a threshold. A knowledge base (132) stores outcomes for continuous learning, reducing development time, effort and errors

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130313 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RARE-EARTH MODIFIED SILVER DIAMINE FLUORIDE COMPOSITION FOR REDUCING TOOTH DISCOLORATION

(51) International classification	:A61Q 11/00, A61K 8/21, A61K 8/19, A61K 6/00, A61K 8/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RNS Institute of Technology
Address of Applicant :RNS Institute of Technology, Dr. Vishnuvardhan Road Post, Channasandra, Rajarajeshwari Nagar, Bengaluru, Karnataka 560098, INDIA Bengaluru Karnataka India
2)Dr. Lalitha S Jairam
3)Dr. Akshay Arjun
4)Dr. Aishwarya Arun Koparde
5)Ms. Shruti Verma
6)Mr. Likhit Balse
7)Dr. Premakumar H. B.
(72)**Name of Inventor :**
1)Dr. Lalitha S Jairam
2)Dr. Akshay Arjun
3)Dr. Aishwarya Arun Koparde
4)Ms. Shruti Verma
5)Mr. Likhit Balse
6)Dr. Premakumar H. B.

(57) Abstract :
The present invention relates to a dental composition comprising Silver Diamine Fluoride (SDF) associated with rare-earth elements for reducing tooth discoloration following dental treatment. The composition involves selected rare-earth elements, namely lanthanum (La), neodymium (Nd), and praseodymium (Pr), evaluated at concentrations of 1%, 2%, and 4% in association with SDF. The formulations are investigated using extracted human primary teeth under in-vitro conditions, with plain SDF and artificial saliva serving as controls. Colour changes are quantitatively assessed using the CIELAB colour measurement system at 24 hours, 7 days, and 21 days. Among the investigated formulations, the 4% praseodymium-associated SDF group exhibits the lowest mean colour difference (ΔE) of 4.242 ± 0.296, compared with 35.461 ± 0.627 for plain SDF. The composition provides a potential approach for mitigating SDF-associated dental discoloration and improving the aesthetic acceptability of dental caries management.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :01/11/2026
(21) Application No.202641130322 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : QUANTUM COMPUTING BASED SECURE DATA ENCRYPTION AND COMMUNICATION SYSTEM

(51) International classification	:H04L 9/08, H04L 29/06, H04L 9/32, G06N 10/00, H04L 9/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KINGS ENGINEERING COLLEGE
Address of Applicant :Kings Engineering College, Chennai - Bangalore Highway, Opposite to HYUNDAI MOTOR, Sriperumbudur, Irungattukottai, Chennai - 602117, Tamil Nadu, India Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Mr. B. VIGNESHWARAN
2)Dr. M. VASUMATHI
3)Mrs. P. KOWSALYA DEVI
4)Dr. N. GOMATHI
5)Mrs. R. VIJAYALAKSHMI
6)Mrs. D. REJEES JENIFA
7)Mrs. KAWSHIKA M
8)Ms. RAKSHINI G
9)Mr. BERNAD SHAW D

(57) Abstract :
The present invention relates to a quantum computing based secure data encryption and communication system (100) for protecting digital information exchanged between authorized computing devices. The system comprises a quantum security gateway (110), a quantum key generation module (130), a quantum-assisted processing unit (140), a cryptographic encryption and decryption module (150), a secure key management module (160), an authentication module (170), a secure communication interface (180), and a data integrity verification module (190). The quantum key generation module establishes cryptographic key material through a quantum key distribution arrangement or an approved alternative mechanism. The encryption module protects digital information using authorized cryptographic keys, while the authentication module verifies communicating entities. The encrypted information is transmitted through a secure communication network and subjected to integrity verification before decryption and delivery to an authorized recipient. A communication monitoring module (200) identifies security events and supports key replacement, revocation, or session restriction according to predefined security policies. The system integrates quantum-assisted processing, quantum key management, and classical cryptographic operations to provide a coordinated secure communication architecture.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :01/11/2026
(21) Application No.202641130325 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SHUNT-WRITTEN OSMOTIC SELECTOR FOR LOAD-DIRECTION-DEPENDENT FAIL-SAFE ACTUATION ON SUPPLY LOSS

(51) International classification	:G01P 13/00, F03G 7/06, B01D 61/44, F16K 31/02, H01H 35/40	(71)**Name of Applicant :** **1)BAPATLA SURENDRA BABU** Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India **2)IRUKUMALLI PRIYANKA** (72)**Name of Inventor :** **1)BAPATLA SURENDRA BABU** **2)IRUKUMALLI PRIYANKA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
SHUNT-WRITTEN OSMOTIC SELECTOR FOR LOAD-DIRECTION-DEPENDENT FAIL-SAFE ACTUATION ON SUPPLY LOSS A fail-safe selector apparatus (100) chooses, on loss of the supply (142), the fail-safe action matching the direction of the preceding load, without stored electrical energy or a controller. Silver/silver-chloride electrodes (116, 118) of a two-chamber electrochemical cell (104) are connected across the current-sensing resistor (120) of the load line (122). The load current moves salt through a cation-exchange membrane (112) until the cell voltage equals the resistor voltage, so the stored state follows the sustained load. Water crosses a water-permeable membrane (114) toward the saltier chamber and displaces a spring-centred piston (126) and a slider (168). When the supply is lost, a released plunger (136) drives the slider through the nearest of three channels (304, 306, 308) of an entry block (134) into the first (144), default (148) or second (146) output member. The ridges (310) lie at 0.28 to 0.36 times the rated slider displacement. FIG. 1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130326 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Adaptive Retrieval-Augmented large language model framework for intelligent website security analysis and context-aware vulnerability assessment & Real Time Deep Vision Intelligence Architecture for Precision Human Facial Emotion Interpretation Through Adaptive Visual Representation Learning.

(51) International classification	:G09B 7/00, G09B 7/02, H04L 29/06, G09B 7/04, G06F 21/57	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(71)**Name of Applicant :**
1)Narsimha Reddy Engineering College
Address of Applicant :Survey No, 518, Hanuman Temple Rd, Maisammaguda, Kompally, Secunderabad, Hyderabad, Telangana 500100 Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr. Dabbu Murali
2)Dr Dileep Kumar Reddy .P
3)GUDURI RAJU KUMAR
4)EVURI KAVYA
5)DONIKELA RITHWIK RAM
6)GOVULA ANAND
7)BURRA RAHUL
8)GURRAM MANASA
9)JILLELLA ABHILASH
10)GOVINDA RISHIKESH

(57) Abstract :
The digital transformation of educational systems has significantly optimized administrative workflows and enhanced the user experience for educators and learners. However, the accumulation of sensitive personal data on educational websites has made them prime targets for cyber threats. Despite growing awareness of these security challenges, the technical roots of vulnerabilities within such platforms remain insuciently explored. To address this gap, we introduce LLM-Edu Attack Graph, a specialized tool designed to assist in vulnerability detection by leveraging large language models (LLMs). Rather than serving as a fully automated monitoring system, LLM-EduAttackGraph operates as a human-in-the loop assistant, combining expert knowledge with the analytical capabilities of LLMs to help identify potential penetration paths based on network fingerprint information. Using LLM Edu Attack Graph, we have so far identified 961 penetration vulnerabilities across educational websites in mainland China—a number that continues to grow as analysis progresses. These findings demonstrate the tool's practical value in augmenting cybersecurity research and efforts. An important aspect of human beings is their ability to display emotions. Emotions form the basis of intimate means of communications between one another. Recognition of emotions displayed via facial expression with the help of a computer could prove to be a very powerful too. In recent years, an increased number of intelligent systems are using facial emotion recognition to improve human interaction. These systems cause constant changes in their operation based on the emotion of humans. In this paper, we propose an architecture based on the convolutional neural network (CNN) for the facial recognition of emotions. In the implementation, we use the Facial Expression Recognition 2013 dataset (FER-2013). Through behavioral analysis, we show how different emotions seem to be sensitive to different parts of the face.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130327 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for AI-Based Multimodal Familiar Person Recognition and Memory Assistance

(51) International classification	:G06V 40/16, G06K 9/00, G06N 3/08, G06N 3/04, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)B.M.S College of Engineering
Address of Applicant :Bull Temple Road Basavanagudi, Bengaluru Karnataka – 560019 Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. K.R. Sudhindra
2)Dr. Surendra H H
3)Suprith Kumar K S
4)K M Skanda
5)K Sai Sri Harsha
6)Deeya Kaveramma MA

(57) Abstract :
ABSTRACT System and Method for AI-Based Multimodal Familiar Person Recognition and Memory Assistance A wearable, privacy-preserving multimodal cognitive assistance system and method are disclosed for recognising familiar individuals and assisting persons experiencing dementia or related cognitive impairment. A hardware acquisition subsystem (A) comprising a discreet wearable camera module (102) and embedded microphone (104) captures audio-visual data of an approaching individual and transmits it to an edge processing unit (108). A multimodal recognition software subsystem (C) performs face detection and alignment, facial and speaker embedding extraction, few-shot enrolment from minimal samples without model retraining, and cosine-similarity-based score fusion. An open-set recognition module (122) classifies an individual as recognised only where a fused confidence score meets a predetermined acceptance criterion, else as unknown, reducing false identification of strangers. A private feedback and safety subsystem (D) delivers relationship-based audio prompts through a Bluetooth earpiece (124) and generates caregiver alerts with optional human-in-the-loop verification (126), preserving patient dignity while enhancing recognition accuracy and safety. FIG. 1

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130328 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PROCESS FOR PREPARING ACID-FUNCTIONALIZED POROUS CARBON NANOSPHERES FROM SUGARCANE MOLASSES FOR SUPERCAPACITOR ELECTRODES

(51) International classification	:H01G 11/24, H01G 11/44, H01G 11/32, C01B 31/02, C01B 32/15
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)B.M.S. College of Engineering
Address of Applicant :Bull Temple Road, Basavanagudi, Bengaluru, Karnataka – 560019, Bangalore Karnataka India
(72)**Name of Inventor :**
1)Vandana Molahalli

(57) Abstract :
ABSTRACT PROCESS FOR PREPARING ACID-FUNCTIONALIZED POROUS CARBON NANOSPHERES FROM SUGARCANE MOLASSES FOR SUPERCAPACITOR ELECTRODES A process for preparing an acid-functionalized porous carbon nanosphere electrode material from sugarcane molasses is disclosed. Sugarcane molasses is dried, ground and sieved, and pyrolysed at about 800°C under an inert atmosphere to yield porous carbon nanospheres (SM8), which are treated with a mixed sulphuric-nitric acid solution under ultrasonication to yield acid-functionalized porous carbon nanospheres (SMF8) bearing oxygen-containing surface functional groups and a BET surface area increased from about 251.73 to about 422.43 $m^2 g^{-1}$. A working electrode fabricated from SMF8 on nickel foam exhibits a specific capacitance of 392.13 F g^{-1} at 0.1 A g^{-1} in a three-electrode configuration. A symmetric CR2032 coin-cell device assembled from the said electrodes exhibits a specific capacitance of 103.58 F g^{-1}, an energy density of 21.69 Wh kg^{-1}, a power density of 468.54 W kg^{-1}, and a capacitance retention of 95.12% with 83.27% coulombic efficiency after 10,000 charge-discharge cycles. Fig. 1

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641130329 A
(19) INDIA
(22) Date of filing of Application :01/11/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : PIPERAZINE PYRIDYL ETHER DERIVATIVES AND THEIR SYNTHESIS

(51) International classification	:A61P 31/10, A61K 31/495, A61K 31/496, C07D 213/75, A61P 31/04	(71)**Name of Applicant :** **1)ABDUL MAJEED KURUMBALA** Address of Applicant :199, 9th main, 4th cross, Duo city layout, Basapura main road, Bangalore-560100, Karnataka, India. Bangalore Karnataka India **2)KOODLUR SANNEGOWDA LOKESH** (72)**Name of Inventor :** **1)ABDUL MAJEED KURUMBALA** **2)KOODLUR SANNEGOWDA LOKESH** **3)SARIGAMA RAJESH** **4)CHIDAMBARAM RISHINARADAMANGALAM RAMASWAMI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
Abstract: The present invention relates to development of novel hybrid molecules of heterocyclic compounds. It particularly relates to the development of novel compounds of new ether derivatives incorporating two pharmacophore moieties, piperazine and pyridine, as a hybrid molecule. It specifically relates to the development of novel compounds of piperazine pyridyl ether derivatives or pharmaceutically acceptable salt thereof as potential anti-microbial, antibacterial, antifungal etc. It also relates synthetic process for preparation of novel compounds of piperazine pyridyl ether derivatives. It further relates to the development of pharmaceutical composition or formulation with synthesized novel compounds of piperazine pyridyl ether derivatives or pharmaceutically acceptable salt thereof. Formula I

No. of Pages : 30 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130330 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Interpretable Adaptive Intelligence Framework for Precision Anaemia Risk Profiling Through Transparent Hybrid Machine Learning Architecture & Transaction Graph Intelligence Framework for Adaptive Financial Laundering dicovery Using Machine Learning Driven Relationship Analytics

(51) International classification	:G06N 20/00, G16H 20/60, A61B 5/00, G16H 50/30, A61B 5/145
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Narsimha Reddy Engineering College
Address of Applicant :Survey No, 518, Hanuman Temple Rd, Maisammaguda, Kompally, Secunderabad, Hyderabad, Telangana 500100 Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr. Mujeeb Shaik Mohammed
2)Dr. P MOHAN
3)GUDURI RAMYA
4)CIRIPANGIRITVI K SWAMY
5)AAVULA KARTEEK
6)GAND LA AKSHARA
7)CHILUKA HARSHITH REDDY
8)GADDAM AJAY KUMAR
9)CHALLAGULLA HARSHITHA

(57) Abstract :
Low haemoglobin concentration or a decreased red blood cell count are hallmarks of anaemia, a common haematological condition that can have detrimental effects on health if it is not identified and treated right enough. Numerous factors, such as hereditary abnormalities, nutritional deficits, and chronic diseases, have an impact on its occurrence. The majority of current anaemia prediction systems depend on statistical techniques or traditional machine learning (ML) models combined with Explainable AI (XAI) approaches like SHAP and LIME. These methods offer interpretability, but their lack of algorithm-level optimization and very moderate prediction performance limit their usefulness in resource-constrained real-world healthcare settings.This study suggests a hybrid ensemblebased anaemia prediction system that uses a Voting Classifier mechanism to combine two potent base learners, Random Forest and XGBoost, in order to overcome these difficulties. The suggested approach ensures consistently excellent prediction performance by dynamically evaluating and choosing the best-performing estimator based on classification accuracy. According to experimental data, the hybrid model provides a dependable and scalable method for early anaemia identification, outperforming individual ML classifiers in terms of accuracy and resilience. In a variety of healthcare settings, this improved prediction framework may facilitate quick clinical decision-making and enhance patient outcomes. Money laundering poses a pervasive threat to the stability and integrity of global financial systems. Since traditional anti-money laundering (AML) methods predominantly rely on rule-based systems and statistical approaches, it has limitations to capture the intricate and interconnected relationships that is inherent in money laundering networks. In response to this challenge, this paper proposes an innovative approach to enhance money laundering detection through transactions. We begin by constructing network graphs from a large dataset of bank transactions. Drawing insights from language modeling and supervised learning, we transform these graphs into directed node representations that effectively encode these intricate relationships and community structures within the transaction network. Subsequently, we utilize Random Forest (RF) to predict suspicious behavior associated with money laundering. Additionally, we address the specific challenges posed by highly imbalanced classes in the context of money laundering detection through both oversampling and undersampling experiments to overcome these challenges. The predictive performance of the RF model with oversampling yielded an accuracy of 86%, whereas when undersampling was applied, the accuracy increased to 92%.

No. of Pages : 11 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130331 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Data Driven Predictive Analytics Architecture for Airline Passenger Satisfaction Assessment Using Supervised Classification Models with Balanced Evaluation of Accuracy Precision Recall and F1 Score & Autonomous Career Guidance Architecture Using Intelligent AI Agents for Personalized Student Development

(51) International classification	:G06N 20/00, G06Q 50/30, G06Q 10/06, G06K 9/62, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Narsimha Reddy Engineering College
Address of Applicant :Survey No, 518, Hanuman Temple Rd, Maisammaguda, Kompally, Secunderabad, Hyderabad, Telangana 500100 Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr KM Reena Bansal
2)Dr D. Bhadru
3)KASARLA RAKSHITHA
4)CHINTAKAYALA JOSHNA
5)BADDAM KOUSHIK REDDY
6)GANTAJI GANESH
7)GUMPULA ESHWAR
8)CHINTHALAPALLY SATHWIKA
9)B SIDDARDHA
10)GANTA SRIVARDHAN REDDY

(57) Abstract :
The Airline passenger satisfaction has become a major concern for airline operators and service management teams. In this paper, our focus is on developing a machine learning–based framework to predict passenger satisfaction in an accurate and reliable manner. The dataset used in this paper consists of airline passenger records sourced from Kaggle, collected from 129,880 passengers, containing passenger information, travel details, and multiple service quality ratings. The data provides several service-related characteristics such as in-flight service, seat comfort, cleanliness, online boarding, and baggage handling. Different supervised classification models are used in this study such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, KNN, XGBoost, and Gradient Boosting. Balanced data splitting is used to ensure the model is evaluated fairly. The results clearly show that Random Forest outperforms other models, achieving an accuracy of 95.83% and an F1-score of 95.15% in predicting airline passenger Career guidance is an important procedure that enables students to take up the right decisions concerning their academic and professional careers. Particularly, students after completion of their 10th and 12th grades face enormous difficulty in choosing a career due to the complexity and dynamic nature of the job market. To address challenges of this nature, this paper proposes CareerCompass, an AI-powered platform designed for personalised and data- driven support. Our system embeds the functionality of machine learning algorithms in analysing a student's distinctive profile comprising interests, academic performance, and aptitude assessments to offer customized career recommendations. The key feature of this platform is an integrated AI Agent, designed to work as a virtual counsellor, giving real-time guidance and responding to varied student queries in an interactive manner.

No. of Pages : 11 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130333 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SKIN LESION DETECTION AND DISEASE CLASSIFICATION USING DEEP LEARNING

(51) International classification	:G16H 50/20, G16H 30/40, G06N 3/08, G06V 10/764, G06N 3/04	(71)**Name of Applicant :** **1)S.Hrushikesava raju** Address of Applicant :Jyothi nilayam, Near SB Capital, Ippatam service road, Athmakur, Mangalagiri - 522503 Andhra Pradesh India **2)Dr. Suneetha Bulla** **3)KONERU LAKSHMAIAH EDUCATION FOUNDATION**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr. Suneetha Bulla** **2)R Poorna Chandra** **3)T Kavya** **4)A Sahasra** **5)K Sai Harshini**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A deep learning framework and method for analysing dermoscopic skin lesion images is disclosed. Separate modules find the lesion, name its disease category and show the image evidence behind that category, all in one workflow. A dataset preparation module divides an image set into training, validation and testing subsets. An annotation generation module automatically creates bounding-box annotations from image-level labels, so no manual annotation is needed. A lesion localization module, a single-class detection network of the YOLOv8 type, outputs a bounding box with a confidence score. A disease classification module, a network of the EfficientNetV2 type chosen by comparison with a basic convolutional network and a residual network, outputs one of several disease categories with a confidence value. An explainability module produces a Grad-CAM saliency map shown as a heatmap and an overlay. An integration module combines the outputs into one result. Each module can be trained, evaluated and replaced independently.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130339 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN EDGE-ENABLED QUANTUM-RESILIENT BLOCKCHAIN SYSTEM FOR CONFIDENTIAL AND TRUSTWORTHY DATA MANAGEMENT IN DECENTRALIZED CLOUD ECOSYSTEMS

(51) International classification	:H04L 9/08, H04L 9/32, H04L 9/06, H04L 9/00, G06F 16/182
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M. Kamatchi Saranya
Address of Applicant :Assistant Professor, Department of Information Technology, Agni College of Technology, Thalambur, Chennai, Tamil Nadu - 600130 Tamil Nadu India
2)Mrs. Akkanagamma
3)Mrs. Priyanka Ashutosh Deshpande
4)Mrs. Sucheta Prashant Devkare
5)Mrs. P. Poornima
6)Mr. Bhushan Malhari Borhade
7)Rajeshwari P
8)Mr. Mohammad Ashique Azad
9)Mrs. T. Vanitha
(72)**Name of Inventor :**
1)M. Kamatchi Saranya
2)Mrs. Akkanagamma
3)Mrs. Priyanka Ashutosh Deshpande
4)Mrs. Sucheta Prashant Devkare
5)Mrs. P. Poornima
6)Mr. Bhushan Malhari Borhade
7)Rajeshwari P
8)Mr. Mohammad Ashique Azad
9)Mrs. T. Vanitha

(57) Abstract :
An edge-enabled quantum-resilient blockchain system manages confidential data across decentralized cloud infrastructure using an edge trust coordinator, a post-quantum cryptographic agility engine, an encrypted object-fragment transformer, a policy-bound capability controller, a permissioned blockchain ledger, decentralized storage nodes, a possession-verification service and a recovery orchestrator. An object is authenticated, encrypted, erasure-encoded and placed across distinct failure domains. The ledger anchors object, policy, fragment-set and cryptographic-epoch commitments without publishing plaintext or content keys. Unpredictable challenges tied to finalized ledger checkpoints verify off-chain fragment availability. Time-limited, epoch-bound capabilities control retrieval, while verified fragments permit authenticated reconstruction and ledger-linked repair following storage failure. Cryptographic epoch transitions support algorithm migration and historical auditability. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130340 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-DRIVEN SECURE GEOGRAPHIC ROUTING FRAMEWORK FOR VEHICULAR AD HOC NETWORKS IN URBAN ENVIRONMENTS

(51) International classification	:H04W 84/18, H04W 40/24, H04W 4/40, H04W 12/06, H04L 12/721
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Dharani Kumari N.V
Address of Applicant :Assistant Professor, Department of MCA, Dr. Ambedkar Institute of Technology, Outer Ring Road Mallathalli Bangalore – 560056, Karnataka, India. Bangalore Karnataka India
2)Dr. Chethan Venkatesh
3)Indumathi S K
4)Anitha J
(72)**Name of Inventor :**
1)Dr. Dharani Kumari N.V
2)Dr. Chethan Venkatesh
3)Indumathi S K
4)Anitha J

(57) Abstract :
AN AI-DRIVEN SECURE GEOGRAPHIC ROUTING FRAMEWORK FOR VEHICULAR AD HOC NETWORKS IN URBAN ENVIRONMENTS Abstract The invention discloses an AI-driven secure geographic routing framework for vehicular ad hoc networks in urban environments. The framework combines vehicle location information, road topology, mobility prediction, wireless link assessment, trust evaluation, and threat detection to select secure packet-forwarding vehicles. A route selection engine ranks eligible candidates according to geographic progress, predicted connectivity, communication quality, security risk, and congestion. A forwarding controller supports adaptive next-hop selection, alternative routing, and bounded recovery during communication failures. A monitoring module evaluates delivery performance, route stability, security events, and resource overhead. The framework aims to improve routing reliability, security, and adaptability under dynamic urban traffic conditions.

No. of Pages : 16 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130341 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Adaptive Regression Intelligence Framework for Financial Outcome Forecasting Decision Support Within Modern Accounting Analytics Systems & Intelligent maternal Health Monitoring System Integrating Demographic Data Laboratory Results and Medical Histories with Machine Learning Algorithms for Risk Prediction

(51) International classification	:G06Q 30/02, G06N 20/00, G06Q 10/04, G06Q 10/06, G06Q 30/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Narsimha Reddy Engineering College
Address of Applicant :Survey No, 518, Hanuman Temple Rd, Maisammaguda, Kompally, Secunderabad, Hyderabad, Telangana 500100 Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr. SRILAKSHMI CHERUKURI
2)Mr. D Sravan Kumar
3)AVIDI HARISH
4)BANDI SREENIDHI
5)BONGU SHEKHAR
6)BANOTH ANIL KUMAR
7)KALLA DEEPIKA SAI
8)JARUPULA VAMSHI
9)AAKULA JASHWANTH SAI MUDIRA
10)BADAVATH SAI HARSHITH

(57) Abstract :
This study evaluates and optimizes sales prediction forecasting models through traditional time-series algorithms and machine learning algorithms, with a specific focus on dynamic models instead of static. The research analyzes four different forecasting models including Autoregressive Integrated Moving Average, Autoregressive Integrated Moving Average with Exogenous Variables along with Autoregressive Distributed Lag and Multi-Layer Perceptron. The forecasting models performance were evaluated through the assessment of Root Mean Square Error and Mean Absolute Error metrics. The results showed Multi-Layer Perceptron produces superior accuracy compared to traditional time-series models due to achieving the lowest Root Mean Square Error and Mean Absolute Error values. Results from Autoregressive Integrated Moving Average were satisfactory while Autoregressive Integrated Moving Average with Exogenous Variables and Autoregressive Distributed Lag showed poor performance. The Diebold-Mariano test statistically validates that Multi-Layer Perceptron is the superior model for sales forecasting. The results showed that Multi-Layer Perceptron can substantially improve sales forecasting precision, offering initiatives a more dependable instrument for decision-making and resources distribution. Policymakers and companies ought to incorporate modern machine learning methodologies, such as Multi-Layer Perceptron, to enhance sales forecasting, hence facilitating improved strategic planning and operational efficacy. Machine learning plays a main role in monitoring maternal health by detecting risk factors, best maternal care and enhanced decision making in clinical status. Machine learning methods could focuses on maternal datasets such as demographic data, laboratory results, medical history to determine risk factors like hypertension and gestational diabetes. Machine learning algorithms reviewed in this research are neural network, boosting methods, random forest, K-nearest neighbor, decision trees and hybrid model. Articles were chosen from recent six years for extensive search. Total number of research studies searched in the repository were 278. After filtering the research papers were 86 duplicates. Screened papers for this research were 192. Some papers are not included with inappropriate content were 92. Full text papers included in this study were 100. The most used machine learning algorithms for monitoring maternal health are random forest, followed by boosting methods, decision tree, hybrid methods, neural network and k-nearest neighbor. It was evident from the deep search of literature that, no single method is effective in monitoring maternal health. Every algorithm has its own strengths and weaknesses. It was evident from the literature that hybrid models have the best balance between robustness and predictive accuracy and make them suitable option for monitoring in real time and evaluating risk factors during pregnancy and overall supports in health outcomes of the mother and fetus. In Future, we can extend the study by developing a hybrid model by combining machine learning algorithms to monitor maternal health.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130345 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IONTO-POWERED DELIVERY: GEMFIBROZIL TRANSDERMAL PATCHES FOR LIPID MANAGEMENT

(51) International classification	:A61K 9/70, A61K 47/38, A61K 31/192, A61K 47/32, A61P 9/10	(71)**Name of Applicant :** **1)Dr. A Geethalakshmi** Address of Applicant :Professor & HOD, Department of Pharmaceutics, Harsha College of Pharmacy, Nelamangala, Bangalore-562123, Karnataka, India Karnataka India **2)Swathi R** (72)**Name of Inventor :** **1)Dr. A Geethalakshmi** **2)Swathi R**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a gemfibrozil polymeric transdermal patch and its preparation by solvent casting. The patch comprises gemfibrozil, methyl cellulose, Eudragit S100, hydroxypropyl methylcellulose K100M and polyvinylpyrrolidone, with dibutyl phthalate as plasticizer. A selected casting feed contains the drug and polymers in a mass ratio of 20:30:30:25:6.25, respectively, together with glacial acetic acid and methanol. The casting mixture is deaerated, cast onto a butter-paper-lined Petri plate, dried at room temperature for 24 hours, and cut into patches measuring 2 × 2 cm. The formulation is evaluated under passive conditions and with iontophoretic current of 0.4–0.6 mA. Goat-skin experiments report passive permeation of 85.86% at 24 hours and iontophoretic permeation of 94.61% at six hours. The selected formulation is characterized for film properties, drug content, skin retention, diffraction behaviour and storage performance. The patch provides a defined polymeric matrix for investigation of assisted gemfibrozil delivery under the disclosed experimental conditions. (Accompanied Figure No. 1-2)

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130346 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LEVERAGING AI TO MEASURE THE IMPACT OF ORGANIZATION CULTURE ON MOTIVATION AND RETENTION IN SERVICE SECTOR

(51) International classification	:G06Q 10/06, G06Q 10/00, G06Q 10/10, G06N 20/00, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Ms. Nirmala Gandla
2)Dr. Rajyalaxmi M
3)Dr. Geetha Manoharan

(57) Abstract :
An AI-based system measures the impact of organizational culture on employee motivation and retention in service organizations. A data acquisition module receives survey responses, feedback text, human-resource records, and service performance records. A culture profiling engine computes culture profile vectors for employee segments, a motivation estimation engine computes motivation indices, and a retention risk prediction engine predicts time-to-exit risk. A pathway decomposition unit separates the contribution of each culture dimension into a direct part and a motivation-mediated part. The system releases a retention alert and pathway report only when a calibration check is passed and the segment meets a minimum group size. A decision-support interface presents group-level results and suggested interventions (Fig. 1).

No. of Pages : 13 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130347 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IMPACT OF AI - DRIVEN LEADERSHIP PRACTICES ON ORGANIZATIONAL CULTURE: A STUDY WITH SPECIFIC REFERENCE TO INDIAN MANUFACTURING SECTOR

(51) International classification	:G06Q 10/06, G06Q 10/00, G06Q 50/04, G06Q 50/02, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Mr. Tejavath Kamal Nayak
2)Dr. Rajyalaxmi M
3)Dr. Geetha Manoharan

(57) Abstract :
An intelligent system assesses the impact of AI-driven leadership practices on organizational culture in manufacturing plants. A data acquisition module receives usage records of AI-based leadership tools, employee survey responses, and plant operational records. A leadership practice profiling engine derives exposure profiles for each team, and a culture index engine computes culture indices. A gated impact attribution engine selects matched comparison teams, estimates the culture change attributable to a leadership practice using a lag-aware hierarchical model, and releases the estimate only when covariate-balance, confounder-overlap, and stability checks are passed. Otherwise it withholds the estimate and records the reason. An explainability module identifies influential practice features, and a decision-support interface presents results with confidence information to plant management (Fig. 1).

No. of Pages : 15 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130348 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ENGAGEMENT THROUGH EMOTION: AN AI-DRIVEN STUDY OF EMOTIONAL INTELLIGENCE'S IMPACT ON JOB SATISFACTION AT ICICI AND SBI BANKS

(51) International classification	:G06Q 10/06, G06N 20/00, G06N 5/04, G06Q 10/10, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Rajitha Challa
2)Dr. Ramesh R
3)Dr. Geetha Manoharan

(57) Abstract :
An AI-driven system assesses the impact of emotional intelligence on job satisfaction and employee engagement in banking organizations. A data acquisition module receives emotional-intelligence assessment responses, survey responses, and branch and role records, and a consent and anonymisation module verifies consent and pseudonymises records. A profiling engine computes emotional-intelligence component scores. A peer-normalisation engine assigns each employee to a reference group defined by role family, branch category, and customer-load band, broadens the group when it is too small, and normalises the scores against it. A relationship modelling engine relates the normalised scores to satisfaction and engagement indices and attributes the relationship to individual components. A decision-support interface presents group-level results (Fig. 1).

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130349 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SIP AWARENESS AND INVESTMENT PREFERENCE OF WORKING PROFESSIONALS WITH REFERENCE TO TEACHING INDUSTRY USING ARTIFICIAL INTELLIGENCE

(51) International classification	:G06Q 40/06, G06N 20/00, G06Q 10/06, G06Q 40/04, G06Q 40/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Ms. G. Lavanya
2)Dr. Kafila
3)Dr. Geetha Manoharan

(57) Abstract :
The invention relates to an artificial-intelligence-based system for assessing Systematic Investment Plan (SIP) awareness and investment preferences among working professionals in the teaching industry. The system collects demographic, professional, financial, behavioural, investment-awareness and preference information through a structured digital interface. A preprocessing module cleans and transforms the collected data into analytical features. An artificial-intelligence engine evaluates the features to classify SIP awareness and investment preference. A recommendation module generates personalized financial-literacy content based on identified awareness gaps and stated preferences. A monitoring and feedback module periodically updates user profiles. The system provides an integrated, explainable and adaptive framework for SIP awareness assessment and preference analysis.

No. of Pages : 16 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130351 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LEARNING AS A CATALYST: HARNESSING QUALITY EDUCATION TO ADVANCE SUSTAINABLE DEVELOPMENGT IN HIGHER EDUCATION INSTITUITIONS

(51) International classification	:G06Q 50/20, G06Q 10/06, G06N 20/00, G09B 7/02, G09B 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Ms. Vedula Santhi
2)Dr. Gurunadham Goli
3)Dr. Geetha Manoharan

(57) Abstract :
The invention provides an intelligent learning and sustainability management system for higher education institutions. The system integrates educational, institutional, learner, faculty, research, infrastructure, and sustainability data; evaluates learning quality; maps learning outcomes to sustainable-development objectives; identifies performance gaps; and generates adaptive institutional interventions. A monitoring module measures implemented interventions and compares predicted outcomes with observed outcomes. A feedback module updates analytical and recommendation parameters according to measured results. A dashboard presents learning indicators, sustainability indicators, institutional performance, resource utilization, alerts, intervention status, and explainable recommendations. The invention establishes a continuous computational framework connecting quality education with measurable sustainable-development performance and adaptive institutional improvement.

No. of Pages : 15 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641130352 A
(19) INDIA
(22) Date of filing of Application :01/11/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN FRAMEWORK FOR QUALITY EDUCATION AND SDG IMPACT MEASUREMENT IN HIGHER EDUCATION

(51) International classification	:G06Q 50/20, G06Q 10/06, G06N 20/00, G06Q 10/10, G09B 7/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Ms. Vedula Santhi
2)Dr. Gurunadham Goli
3)Dr. Geetha Manoharan

(57) Abstract :
The invention provides an AI-driven framework for measuring quality education and Sustainable Development Goal (SDG) impact in higher education institutions. The framework acquires academic, administrative, student, resource, and sustainability data and processes the data into standardized performance indicators. An artificial-intelligence evaluation engine analyzes the indicators to identify performance gaps and predict educational and SDG-related outcomes. An intervention module generates recommended institutional actions, while a monitoring module collects observed outcomes and compares them with predicted outcomes and target values. An adaptive learning module uses deviations and intervention results to improve subsequent evaluations. A management dashboard presents indicators, predictions, interventions, alerts, comparisons, and explainable recommendations.

No. of Pages : 15 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130353 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-ASSISTED SYSTEM AND METHOD FOR INTERFERENCE-AWARE SPECTRUM ALLOCATION IN WIRELESS NETWORKS

(51) International classification	:H04W 16/14, H04W 72/08, H04W 72/04, H04W 16/10, H04W 24/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mrs. Shalini D
Address of Applicant :Assistant Professor, Department of CSE, Vignan's Insititute of Information Technology (A) Duvada, Visakhapatnam, Andhra Pradesh, India Andhra Pradesh India
2)Dr. D Mohan Reddy
3)Dr. G Reddy Hemantha
4)Mrs. Subhashini Kodur
5)Mrs. Saujanya Gudese
6)Dr. Abhijeet Thakur
7)Dr. Jami Venkata Suman
8)Dr. Meenakshi
(72)**Name of Inventor :**
1)Mrs. Shalini D
2)Dr. D Mohan Reddy
3)Dr. G Reddy Hemantha
4)Mrs. Subhashini Kodur
5)Mrs. Saujanya Gudese
6)Dr. Abhijeet Thakur
7)Dr. Jami Venkata Suman
8)Dr. Meenakshi

(57) Abstract :
[050] An AI-assisted system and method for interference-aware spectrum allocation in wireless networks are disclosed. Time-stamped radio measurements are fused to construct a channel-specific weighted interference graph. A prediction engine estimates future channel occupancy and interference risk together with uncertainty values. Candidate spectrum assignments are generated by considering predicted throughput, cross-node interference and radio reconfiguration cost. A constraint validator checks the assignments against spectrum-usage permissions, radio capabilities and operational limits. An allocation controller deploys a validated assignment using acknowledgements and configuration versioning, or retains a prevalidated fallback when deployment is unsuccessful. Post-allocation measurements are compared with predictions to adapt subsequent allocation cycles. The arrangement supports coordinated physical radio-resource management while reducing invalid configuration changes. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :31/10/2026

(21) Application No.202641130357 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CLOSED-LOOP PRECISION TISSUE SAMPLING AND PRESERVATION SYSTEM FOR MOLECULAR ANALYSES

(51) International classification	:A01N 1/02, G01N 1/31, B01L 3/00, G01N 1/28, C12M 1/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Aditya Institute of Technology and Management
Address of Applicant :K.Kotturu (Vill), Tekkali- 532201, Srikakulam (Dist), Andhra Pradesh, India. Andhra Pradesh India
2)Ramajogi Naidu Doddi
3)Bondala Ramakrishna
4)Dharmana Lokanadham
5)Muddada Venkatesh
6)Pavan Kumar Rejeti
7)Roopsandeep Bammidi
(72)**Name of Inventor :**
1)Ramajogi Naidu Doddi
2)Bondala Ramakrishna
3)Dharmana Lokanadham
4)Muddada Venkatesh
5)Pavan Kumar Rejeti
6)Roopsandeep Bammidi

(57) Abstract :
The present invention relates to a closed-loop precision tissue sampling and preservation system for collecting and preserving a tissue sample for molecular analysis. The system comprises a precision tissue sampling element configured to collect the tissue sample from a biological source, a sample receiving chamber configured to receive the tissue sample, and a controlled sample transfer mechanism configured to transfer the tissue sample through a defined transfer path. The sample receiving chamber retains a stabilization medium configured to stabilize the tissue sample following transfer. A sealing mechanism substantially encloses the tissue sample and the stabilization medium within the sample receiving chamber, while a sample identification element facilitates traceability of the tissue sample. The integrated arrangement reduces manual handling, sample loss, contamination, and operator-dependent variation, thereby facilitating consistent tissue sampling, preservation, identification, and downstream molecular analysis.

No. of Pages : 29 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130361 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ROTATION-PRESSURIZED LIQUID-CELL SEAM LOCK FOR TORSIONAL STIFFNESS OF FOLDABLE OPEN-SECTION ROTOR BLADE

(51) International classification	:B64C 3/56, B64C 27/467, B64C 27/32, B64C 27/46, B64C 27/51	(71)**Name of Applicant :** **1)BAPATLA SURENDRA BABU** Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India **2)IRUKUMALLI PRIYANKA** (72)**Name of Inventor :** **1)BAPATLA SURENDRA BABU** **2)IRUKUMALLI PRIYANKA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ROTATION-PRESSURIZED LIQUID-CELL SEAM LOCK FOR TORSIONAL STIFFNESS OF FOLDABLE OPEN-SECTION ROTOR BLADE A rotor blade (100) for unmanned aerial vehicles and fans has a shell (106) with a longitudinal slit forming a seam (108), so that the blade is soft and foldable at rest. Tooth strips (206, 208) are bonded to the overlapping lap edges (202, 204). A sealed bladder (212) along the seam holds a sodium polytungstate liquid (214) in cells (802) whose length decreases towards the tip (114). Rotation pressurizes the liquid, the bladder presses the lap edges together, the teeth (306) interlock and the open section becomes a closed torsion box whose torsional stiffness is 136 times that of the open section. The tooth flank angle lies between $40^\\circ$ and $65^\\circ$, so that the seam carries the nose-down flight moments, opens under a strike and never self-locks. The blade needs no pump, valve or electronics and returns to the open state at rest. FIG. 1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130363 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR AUTHENTICATED PROXIMITY-TRIGGERED REINFORCEMENT LEARNING TRAFFIC CONTROL FOR EMERGENCY VEHICLES

(51) International classification	:G08G 1/0965, G08G 1/07, G08G 1/087, G08G 1/01, G08G 1/0967
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)B.M.S. College of Engineering
Address of Applicant :Bull Temple Road, Basavanagudi, Bengaluru, Karnataka - 560019 Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. Sowmya K. S
2)Shreyas D. K
3)Somanath Mikali
4)Vishesh P. Gowda

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR AUTHENTICATED PROXIMITY-TRIGGERED REINFORCEMENT LEARNING TRAFFIC CONTROL FOR EMERGENCY VEHICLES A system and method for authenticated, proximity-triggered traffic signal control for emergency vehicles over a vehicular ad-hoc network are disclosed. On-board units (102) transmit signed status messages to a road side unit (202). A security verification module (212) admits only messages whose certificate and RSA-PSS signature are verified. A controller selection unit (210) computes the remaining distance of an authenticated emergency vehicle (106) to a protected intersection; within an activation threshold of 250 metres, a density-based adaptive controller (206) is suspended and a Double-DQN controller (208), executed locally at a fog controller (204), establishes a green-wave corridor, control reverting automatically once the vehicle has passed. A tiered RSA-4096/RSA-2048, RSA-OAEP and AES-256-GCM framework preserves real-time latency. A cloud layer (300) performs only offline retraining. In simulation, emergency-vehicle waiting time fell by approximately 79% and 49% relative to fixed-time and density-based control, with 96.02% packet delivery and 36.20 ms latency. Fig. 1

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130364 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Self-Adaptive Multiscale Forecasting Framework for Electrical Load Dynamics Using Temporal Dependency Modeling, Contextual Feature Fusion and Intelligent Ensemble Optimization & Context aware paragraph reliability verification architecture for hallucination identification correction within Network integrated large language intelligence system

(51) International classification	:G06Q 50/06, G06Q 10/04, G01R 22/06, G01R 21/133, G06Q 50/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Narsimha Reddy Engineering College
Address of Applicant :Survey No, 518, Hanuman Temple Rd, Maisammaguda, Kompally, Secunderabad, Hyderabad, Telangana 500100 Hyderabad Telangana India

(72)**Name of Inventor :**
1)Mrs. Jeevitha Ravula
2)Mr. Paindla Thirupathi
3)BOMMENA SOWMYA
4)GUGULOTH ESWAR
5)CHINNAVAR PAVAN KUMAR
6)ANUGU ROHITH
7)ENDLURI NIKITHA
8)GOPISHETTI AKHIL
9)B CHANDRA SHEKAR
10)CHALAMALLA VAISHANV KUMAR REDDY

(57) Abstract :

Electricity consumption predictions for a long period are critical in the institutions that distribute the electricity and governmental or private entities that supply the electricity. It guarantees optimum energy utilization and aids in making strategic decisions for improving the energy production quality. This need is especially important in nations like Iraq, which has suffered from energy crises for many years. This study uses daily household electricity consumption data acquired from the Ministry of Electricity in Iraq, namely the Rusafa area of Baghdad, from 2022 to 2024. Weather data for the same years was also included, which contains external weather factors such as temperature, humidity, and solar radiation that directly influence consumption patterns. This paper proposes a hybrid forecasting model that utilizes advanced deep learning architectures LSTM and CNN-based deep learning architectures for forecasting along with an upgraded stacked hybrid model that employs CNN, GRU, Stacked Bi-LSTM, and machine learning regressors, such as XGBoost Regressor, and LightGBM Regressor. These models are being trained to improve accuracy in the forecast and to improve energy acoustic production strategies. The 30 epochs were trained and evaluated on the proposed model using the mean relative absolute error (MAPE) and mean root mean square error (RMSE) to examine the prediction quality. Among all models tested, the best performance was achieved using LightGBM regressor in our hybrid model with MAPE and RMSE of periodic forecasts for the next spilled of time being 0.185155 and 0.094603, respectively. The results show the potential of hybrid modeling techniques for energy forecasts and electricity distribution systems optimization. Large Language Models (LLMs) are increasingly integrated into communication-centric and networked AI systems, supporting applications such as edge/cloud services, mobile assistants, and distributed collaborative platforms. However, their tendency to generate hallucinations (factually incorrect yet linguistically plausible content) poses significant risks to trust, reliability, and security in these environments. We propose SRF, a Sequential Revision Framework for paragraph-level hallucination detection and correction in pervasive and networked AI workflows. SRF decomposes text into sentences, generates factchecking queries, retrieves supporting evidence via web-scale APIs, and applies a semantic agreement mechanism to ensure globally coherent revisions. To evaluate factual correctness, we compare SRF against two baselines—(i) direct LLM correction (without external evidence) and (ii) sentence-level RARR—on the Natural Questions and SQuAD datasets using GPT-3.5, Mistral, and Gemini models. SRF yields up to a 15% improvement in factual accuracy, with minimal disruption to writing style and paragraph coherence. By enforcing fact-consistent and trustworthy LLM outputs, SRF strengthens the integration of AI/ML into communication networks, mobile computing, and distributed cloud/edge services, reducing vulnerabilities to misinformation propagation and adversarial manipulation.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :01/11/2026
(21) Application No.202641130366 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR AI-BASED EDGE-COMPUTING TREATMENT AND REUSE OF RO REJECT WATER

(51) International classification	:B01D 61/12, C02F 1/44, A01G 31/02, G06N 20/00, C02F 1/00	(71)**Name of Applicant :** **1)BMS College of Engineering** Address of Applicant :Bull Temple Road, Basavanagudi, Bengaluru, Karnataka - 560019 Bangalore Karnataka India (72)**Name of Inventor :** **1)Dr. Nalina V** **2)Ragavan R** **3)Ullas Gowda S** **4)Veeksha M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR AI-BASED EDGE-COMPUTING TREATMENT AND REUSE OF RO REJECT WATER An artificial-intelligence-based, edge-computing system and method for treatment and reuse of reverse-osmosis (RO) reject water in hydroponic cultivation and gardening is disclosed. A sensor unit continuously measures TDS, pH, temperature and humidity of the RO reject water and transmits the readings, via a microcontroller and a wireless MQTT link, to an edge-computing controller. The controller retrieves a predetermined target TDS and pH range for a user-selected plant type from a plant-specific database, and a regression-based machine-learning model predicts the corresponding target values. A decision engine dynamically computes, from the deviation between measured and predicted values, the required quantity of dilution water and nutrient dosing, and actuates a filtration, mixing and peristaltic dosing unit accordingly. Conditioned water is supplied to a hydroponic or garden cultivation unit in a closed loop, achieving a TDS control accuracy within approximately ±10 ppm, with a fail-safe halting delivery upon detection of unsafe parameters. FIG. 1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130370 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN, SYNTHESIS AND MECHANISM-GUIDED EVALUATION OF HETEROCYCLIC SCHIFF-BASE METAL COMPLEXES FOR POTENTIAL ANTIMICROBIAL AND ANTICANCER APPLICATIONS

(51) International classification	:C07F 15/00, C07F 15/02, G01N 33/574, G01N 33/50, G16C 20/64
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.R.HARIHARASUTHAN
Address of Applicant :S.A.ENGINEERING COLLEGE(AUTONOMOUS) POONAMALLEE-AVADI MAIN ROAD, CHENNAI-77 chennai Tamil Nadu India
2)S.A.ENGINEERING COLLEGE(AUTONOMOUS)
(72)**Name of Inventor :**
1)Dr.R.HARIHARASUTHAN

(57) Abstract :
The present disclosure relates to a proposed workflow for preparing and evaluating heterocyclic Schiff-base ligands and selected metal coordination complexes for preliminary antimicrobial and anticancer lead discovery. The workflow combines controlled ligand synthesis and metal complexation with complementary characterization, solution-stability assessment, computational analysis, standardized antimicrobial testing and in vitro cancer-cell evaluation. Candidate complexes may contain Ag(I), Cu(II), Zn(II), Fe(III) and/or Au(III), subject to chemical feasibility, safety, cost and facility access. We assess chemical identity, purity, coordination environment, metal content, and stability before biological interpretation. Comparative testing includes the corresponding free ligand, relevant metal-salt controls, vehicle controls and reference agents. For reproducibly active candidates, concentration response, preliminary selectivity and selected mechanistic assays may be investigated. The source material describes a proposed research program, the supplied proposal has not yet established compound-specific experimental results, a defined chemical structure, or patentability.

No. of Pages : 9 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130372 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD FOR INTERFACE-ENGINEERED VITRIMER COMPOSITE FOR MULTI-CYCLE REPROCESSING WITHOUT INTERFACIAL DEGRADATION

(51) International classification	:C08J 5/24, C08L 63/00, C08J 5/06, C08J 5/04, B32B 27/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Christ the King Engineering College

Address of Applicant :58/3, Cecilia Gardens, Onnipalayam Road, Karamadai, Coimbatore - 641104, Tamil Nadu, India. Coimbatore Tamil Nadu India

(72)**Name of Inventor :**

1)S. Dinesh Kirupha

2)M. Muthukrishnan

3)R. Hari Prasath

4)N. Manikandan

5)C. Prabhu

6)S. Giridharan

7)D. Domnic

8)J. Edison Thangaraj

9)S. Yamunarani

10)R. Suganya

11)M. T. Senthil Kannan

(57) Abstract :

The present invention discloses an interface-engineered vitrimer composite system for multi-cycle reprocessing without interfacial degradation. Dynamic covalent exchange functional groups including beta-hydroxy ester, disulfide, or imine units are installed onto reinforcing fibre surfaces through electrochemical anodization and silane coupling agent grafting, confirmed by X-ray photoelectron spectroscopy. Exchange-compatible fibre sizings incorporating dynamic epoxy oligomers and Lewis acid catalyst components are applied at 0.5 to 2.0 percent weight fraction. Epoxy-anhydride transesterification vitrimer matrix resin with zinc acetylacetonate catalysis is characterized by stress relaxation topology freezing temperature and gel fraction. Composite laminates undergo up to ten standardized hot-press reprocessing cycles above the topology freezing temperature. Mechanical property retention including interlaminar shear strength, fracture toughness, and flexural properties is evaluated at defined cycle intervals. Degradation mechanism analysis through fractographic microscopy informs refinement, providing improved multi-cycle reprocessability, eliminated interfacial stress accumulation, and sustainable composite end-of-life management.

No. of Pages : 18 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :01/11/2026
(21) Application No.202641130388 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Hybrid Multi-Modal Surveillance Robot Integrating Camera Visual Perception, Touch Sensing, and GPS-Based Intelligence

(51) International classification	:H04N 7/18, B25J 9/16, G05D 1/02, B25J 19/02, B25J 19/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sathyabama Institute of Science and Technology
Address of Applicant :Sathyabama Institute of Science and Technology, Jeppiaar Nagar, SH 49A, Chennai, Tamil Nadu 600119 Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Sathyabama Institute of Science and Technology
2)Ms.P.Gopika
3)Mr. Y. Rakul
4)Mr. Sabaris R

(57) Abstract :
The entitled “A Hybrid Multi-Modal Surveillance Robot Integrating Camera Visual Perception, Touch Sensing, and GPS-Based Intelligence” aims to develop an intelligent robotic system for effective surveillance, environmental monitoring, and security applications. Conventional surveillance systems often depend on fixed cameras and manual monitoring, which may limit their ability to cover different locations and respond to unexpected situations. To overcome these limitations, the proposed system integrates camera-based visual perception, touch sensing, GPS-based location intelligence, and chain-wheel mobility into a single robotic platform. The camera module enables the robot to capture visual information from its surroundings, supporting remote observation and surveillance activities. The touch sensor provides an additional layer of physical interaction detection by identifying unwanted contact with the robot during operation. When such contact is detected, the system can activate an alarm or send a notification to the user's mobile phone, depending on the communication facilities implemented. This feature helps the user become aware of possible interference with the robot while it is performing surveillance tasks. Furthermore, the GPS module provides geographical location information, which can assist in tracking the robot's position and supporting location-based monitoring.

No. of Pages : 8 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130389 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Frequency Selective Surface Based Linear-to-Linear Reflective Polarization Converter for Ku-Band Radar Applications

(51) International classification	:H01Q 15/24, H01Q 15/14, G02B 5/30, H01P 1/17, G02B 6/126
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Koneru Lakshmaiah Education Foundation
Address of Applicant :Green Fields, Vaddeswaram, Guntur, Andhra Pradesh, India- 522302 Vaddeswaram Andhra Pradesh India
(72)**Name of Inventor :**
1)P. Aruna Kumari
2)Madhavareddy Venkatanarayana
3)Immadi Govardhani

(57) Abstract :
A reflective linear-to-linear polarization converter (100) for the Ku-band comprises an FR-4 substrate (102), a copper ground layer (116) and a front-face array (120) of unit cells. Each cell has a central square patch (104) with a crossed diagonal slot (106), a square split-ring resonator (108) with gaps (110) on two adjacent sides, and a segmented octagonal ring (112) of triangular patches (114). The gaps make the cell mirror-symmetric about one diagonal only, so the principal-axis components of an incident linearly polarized wave are reflected with equal amplitude and a 180-degree phase difference, rotating the polarization by 90 degrees. Over 12.1-14.2 GHz the co-polarized reflection is below -10 dB.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130390 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : H-Shaped Coupling Structure Integrated Wideband Microstrip Bandpass Filter for C-Band Applications

(51) International classification	:H01P 1/203, H01P 1/205, H01P 7/08, H01P 1/213, H01Q 13/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Koneru Lakshmaiah Education Foundation
Address of Applicant :Green Fields, Vaddeswaram, Guntur, Andhra Pradesh, India- 522302 Vaddeswaram Andhra Pradesh India
(72)**Name of Inventor :**
1)Dommaraju Srilatha
2)Muzammil Parvez M
3)Habibulla Khan

(57) Abstract :
A compact microstrip wideband bandpass filter (100) for the C-band comprises a dielectric substrate (102), a copper ground plane (104) on its lower face, and on its upper face two tapered 50-ohm feed lines (108, 110) and an H-shaped coupling structure (112). The structure has two arms (114, 116) adjacent to the inner ends of the feed lines, joined by a cross member (118), forming quarter-wavelength resonant sections. On RT Duroid 5880 substrate of 0.508 mm the filter occupies 13 x 18.4 mm and provides a passband of 5.3-7.2 GHz (31.67 per cent fractional bandwidth), with S11 of -20.1 dB, insertion loss of 0.2 dB and group delay of 0.2 ns at 6 GHz, for satellite links and radar.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :01/11/2026
(21) Application No.202641130391 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Temporal Workload-Driven Hybrid Edge Intelligence System and Method for Proactive Task Offloading and Fair Scheduling

(51) International classification	:G06N 3/04, G06N 3/08, G06F 9/50, H04W 28/08, H04W 24/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Koneru Lakshmaiah Education Foundation
Address of Applicant :Green Fields, Vaddeswaram, Guntur, Andhra Pradesh, India- 522302 Vaddeswaram Andhra Pradesh India
(72)**Name of Inventor :**
1)B Swapna
2)Dr K Ravindranath

(57) Abstract :
An edge intelligence system (100) for mobile edge computing monitors task arrival rate, processor and memory utilization, bandwidth, queue length and deadlines at each edge node (200), and a recurrent neural network (106) converts the time-sequence into a workload-aware latent state (108). From this common state, jointly trained modules determine a continuous offloading proportion and target (112), bandwidth and computation allocation for anticipated demand (114), and fairness-aware scheduling priority using historical waiting time (116). A Gaussian-Process-based predictive optimization layer (118) guides exploration of the action space. The action is executed at the edge node or cloud resource (126), and measured latency, energy, fairness and utilization update the modules in a closed loop.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :01/11/2026

(21) Application No.202641130398 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Machine Learning-Based Adaptive Cybersecurity System for Real-Time Network Intrusion and Anomaly Detection

(51) International classification	:H04L 29/06, G06N 20/00, G06F 21/55, H04L 12/24, H04L 12/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)J.J College of Engineering and Technology

Address of Applicant :Ammapettai, Tiruchirappalli District, Tamil Nadu, Pin code: 620009 Tamil Nadu India

(72)**Name of Inventor :**

1)Ms.Sneha

2)Ms.R.Neelaveni

3)Mr.V.Praveen

4)Mr. S.Santhosh Kumar

5)Dr.T.Gurumekala

6)Mrs.T.Josephine Arockia Mary

7)Mrs.P.Sumathi

8)Ms.Radhika

(57) Abstract :

The present invention relates to a machine learning-based adaptive cybersecurity system for real-time network intrusion and anomaly detection. The invention provides an intelligent network security framework configured to continuously monitor network traffic, identify malicious activities, detect abnormal communication patterns and automatically respond to potential cyber threats. The proposed system comprises a network traffic acquisition module, data preprocessing module, feature extraction module, machine learning-based intrusion detection engine, anomaly identification module, adaptive learning module, threat classification module and automated response module. Network packets and communication metadata are collected from network interfaces and analysed using machine learning algorithms to distinguish normal network behaviour from suspicious or malicious activities. The system employs supervised and unsupervised learning techniques to identify known and previously unseen cyber threats, including denial-of-service attacks, port scanning, brute-force attacks, malware communication and unauthorized access attempts. An adaptive learning mechanism continuously updates the detection model based on newly observed traffic patterns and verified security incidents. Upon detecting a potential intrusion, the system generates security alerts, assigns threat severity levels and initiates suitable response actions, including suspicious IP blocking, session termination and traffic restriction. The invention improves real-time threat identification, reduces false-positive detection and provides an adaptive cybersecurity mechanism for protecting enterprise networks, institutional networks, cloud environments and Internet of Things infrastructures.

No. of Pages : 17 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130410 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT AI-BASED BUSINESS PERFORMANCE OPTIMIZATION SYSTEM

(51) International classification	:G06Q 10/06, G06N 20/00, G06N 5/04, G06Q 10/00, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Mr. Jagadeesh Gorrepotu

Address of Applicant :Assistant Professor, Department of Electrical and Electronics Engineering, Vignan's Institute of Engineering for Women (Autonomous), Kapujaggaraju Peta, Vadlapudi Post, Behind Visakhapatnam Special Economic Zone (VSEZ), Visakhapatnam, Andhra Pradesh, India - 530 049 Andhra Pradesh India

2)Dr. G. Keerti Marita

3)Dr. P. Hemalatha

4)Mr. Immanuel Prabaharan S

5)Mrs. S Sofia

6)Mrs. J. Gold Beulah Patturose

7)Dr. C. Meera

8)Mrs. D. Kalaiselvi

(72)**Name of Inventor :**

1)Mr. Jagadeesh Gorrepotu

2)Dr. G. Keerti Marita

3)Dr. P. Hemalatha

4)Mr. Immanuel Prabaharan S

5)Mrs. S Sofia

6)Mrs. J. Gold Beulah Patturose

7)Dr. C. Meera

8)Mrs. D. Kalaiselvi

(57) Abstract :

The present invention concerns an intelligent artificial intelligence-based business performance optimization system for optimizing business performance by means of advanced data analysis and automated optimization processes, which helps to increase the efficiency of the organization, operational productivity and the quality of decision-making. Traditional business management systems tend to be based on past performance reports and predetermined levels of business operation, restricting their ability to respond quickly to evolving market conditions and new business challenges. The proposed invention combines the concepts of machine learning, predictive analytics, explainable artificial intelligence and multi-objective optimization in a single decision support framework. It gathers and analyses diverse business information from sales, financial transactions, customer interactions, workforce activity and supply chain management systems to detect performance trends, operational inefficiencies and bottlenecks, and resource usage. The intelligent prediction module predicts the key performance indicators (KPIs) and recognizes possible deviations from pre-defined business goals. In addition, an adaptive optimization engine assesses different operating approaches based on profitability, cost reduction, productivity, the availability of resources and risks. The explainable AI module offers interpretable insights into factors influencing optimization recommendations, and a feedback-driven learning mechanism continually improves predictive models based on the actual business outcomes. It features real-time performance monitoring, scenario simulation, automated alerts and intelligent recommendation generation, enabling proactive management decisions. The purpose of the proposed invention is to increase the efficiency of the work, optimize the use of resources, decrease the unnecessary spending, increase the response of businesses and facilitate sustainable growth of the organization in the field of manufacturing, retail, banking, logistics and other areas where business activity is characterized by the use of data.

No. of Pages : 6 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130411 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-ENABLED MULTISPECTRAL IMAGE PROCESSING FRAMEWORK FOR INTELLIGENT ENVIRONMENTAL MONITORING

(51) International classification	:G06V 20/13, G06N 3/08, G06N 3/04, G06T 5/00, G01J 3/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sathyabama Institute of Science and Technology
Address of Applicant :Sathyabama Institute of Science and Technology, Chennai, Tamil Nadu, India - 600 119 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. R Sakthi Prabha
2)Dr. A. Divya
3)Ms. Rajshree A
4)Dr. M.R. Ebenezar Jebarani
5)Dr. S. Lakshmi
6)Dr. G. Jegan

(57) Abstract :
Monitoring techniques that are intellectually driven and accurate and timely for addressing environmental issues such as land degradation, water pollution, vegetation loss and climate change. It is challenging to monitor conventional and environmental modifications with traditional methods to analyse the complex multispectral data. The results of this study indicate that AI based multispectral image processing system is capable of delivering intelligent environmental monitoring using the help of sophisticated deep learning methods. It uses the framework for satellite, aerial and UAV images to gain useful and meaningful spectral and spatial data. To decrease noise, normalize spectral bands and improve data quality, image preprocessing techniques are performed on the data. A hybrid deep learning system consisting of CNN and attention mechanism is designed to improve the feature extraction and the environmental classification of the CNN. The main novelty of the primary research is the combination of adaptive spectrum and spatial features, cross temporal change detection and prediction with uncertainty within a single framework for reliable environmental assessment. The model is able to detect the relevant spectral bands and regions of the images using explainable artificial intelligence. The proposed system enables vegetation health assessment, land use classification, water body monitoring and finding environmental anomalies. It is effective when compared with the traditional machine learning and deep learning techniques. The purpose of the framework is to improve the quality of the change detection in the environment, quality and reliability of the predictions and the interpretability of the predictions. The proposed approach can be used in a variety of applications such as remote sensing, monitoring ecosystems, sustainable resource management and environmental protection.

No. of Pages : 8 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130412 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MARINE INTELLIGENCE PLATFORM FOR SMALL FISHING VESSELS

(51) International classification	:G01S 15/96, G01C 21/20, G08G 3/02, H04W 4/90, G01S 15/931
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)R.M.D. Engineering College
Address of Applicant :RSM Nagar, Gummidipoondi Taluk, Tiruvallur, Kavarapettai, Tamil Nadu, India - 601 206. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. PADMAVATHI N
2)Dr. N. Vini Antony Grace
3)Mr.Abishek C
4)Mr.Brayan Maximus J

(57) Abstract :
VAAYU is an integrated marine intelligence platform for small fishing vessels that consolidates navigation, sonar sensing, vessel awareness, weather information, maritime1boundary warnings, emergency communication, fishing assistance and voyage recording. A marine computing unit receives data from GNSS/GPS, sonar, motion sensors, communication interfaces and user inputs. A core engine validates, normalises and fuses the inputs into a consolidated marine state. An intelligence engine processes the state through navigation, safety, geofencing, vessel-awareness, weather and fishing functions. A decision-support stage generates alerts, route recommendations and action recommendations for presentation on a fisherman dashboard. In an emergency, the platform captures vessel location and attempts to transmit or relay distress information through available communication paths. Fishing assistance combines sonar observations, geographic coordinates, historical catch records and available environmental data to identify candidate fishing zones. Local storage records voyage tracks, sensor readings and events. Optional cloud and shore services provide synchronisation and fleet monitoring, while essential onboard processing remains available without continuous internet connectivity

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130413 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SECURE CLOUD STORAGE SYSTEM USING DATA PARTITIONING, RESOURCE ALLOCATION, AND TRANSFORMER-ENHANCED GRAPH ATTENTION NETWORK

(51) International classification	:G06F 9/50, G06F 16/901, H04L 29/08, G06F 21/62, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mr. Cinthuja Kaliyamoorthy
Address of Applicant :Research Scholar, Department of Computer Science and Engineering, Panimalar Engineering College, Poonamallee, Chennai – 600123, Tamil Nadu, India Chennai Tamil Nadu India
2)Dr. Sathya Preiya Vadamalai Muthu
(72)**Name of Inventor :**
1)Mr. Cinthuja Kaliyamoorthy
2)Dr. Sathya Preiya Vadamalai Muthu

(57) Abstract :
The present invention discloses a secure cloud data storage and retrieval system integrating adaptive data partitioning, intelligent resource allocation, multi-head graph attention, and Transformer-based contextual processing. The system comprises a data input interface, preprocessing module, partitioning engine, resource monitoring module, allocation engine, security module, graph construction module, learning modules, distributed storage layer, and retrieval engine. Incoming data is processed and divided into partitions according to data characteristics, workload, and storage requirements. Cloud resource parameters are monitored to select suitable storage and processing nodes. Relationships among data partitions, users, and cloud resources are represented using a graph, and a multi-head graph attention network generates relationship-aware representations. A Transformer processing module evaluates contextual relevance for retrieval and ranking. Authentication, authorization, encryption, and integrity verification protect stored information and retrieval operations. The system supports distributed storage, selective data access, adaptive resource management, and scalable cloud deployment, with potential improvements in storage utilization, retrieval efficiency, security, and operational cost.

No. of Pages : 22 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130415 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART SURVEILLANCE SYSTEM IN CLASSROOM

(51) International classification	:H04N 7/18, G08B 13/196, G06V 20/52, G06K 9/00, G06V 10/764	(71)**Name of Applicant :** **1)R.M.D. Engineering College** Address of Applicant :RSM Nagar, Gummidipoondi Taluk, Tiruvallur, Kavarapettai, Tamil Nadu, India - 601 206. Chennai Tamil Nadu India (72)**Name of Inventor :** **1)Dr.G.AMUDHA** **2)Ms. OVIYA G E** **3)Ms.PAVITHRA V** **4)Ms. SANJITHA S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An intelligent classroom surveillance system is disclosed for monitoring classroom environments and notifying authorised personnel of configured irregular activities. At least one CCTV or IP camera captures a video stream that is received and preprocessed by a video acquisition module. An edge-processing device or local server executes computer-vision or machine-learning analysis to generate outputs indicative of candidate activities. An event evaluation module assesses the outputs against configured conditions, which may include possible fighting, unauthorised entry, unusual student movement or a defined duration of faculty absence. Qualifying events are recorded with timestamps and associated camera information, and may be linked to relevant video segments or metadata. A notification module sends alerts through available SMS, email or mobile-application channels. A dashboard presents camera status, alerts and event history for authorised review. Optional audio input may be supplied by a microphone array where supported. Local processing, structured event logging and configurable alert rules provide an adaptable monitoring framework for classrooms and other authorised institutional spaces.

No. of Pages : 18 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130416 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART SOLAR POWERED GRASS CUTTING ROBOT WITH BATTERY CHARGING AND BLUETOOTH BASED MOBILE CONTROL SYSTEM

(51) International classification	:A01D 34/00, H02J 7/35, A01D 34/82, A01D 34/78, A01D 101/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)R.M.D. Engineering College
Address of Applicant :RSM Nagar, Gummidipoondi Taluk, Tiruvallur, Kavarapettai, Tamil Nadu, India - 601 206. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. P. Usha Rani

(57) Abstract :
The present invention relates to a smart solar-powered grass cutting robot with Bluetooth-based mobile application control. The system comprises a photovoltaic solar panel, rechargeable battery, Arduino Uno microcontroller, Bluetooth communication module, LCD display, motor drivers, DC motors, wheel assemblies, and a rotating cutting blade. The solar panel converts incident sunlight into electrical energy, which is supplied through a suitable charging arrangement to the rechargeable battery. The battery supplies electrical power to the electronic circuits and motors. The Bluetooth module receives operating commands from a compatible mobile application and transfers them to the microcontroller. The microcontroller generates control signals for the motor drivers to operate the cutting motor and locomotion motors. The cutting motor rotates the blade to cut grass, while the wheel motors enable movement of the robot. The integrated system facilitates convenient grass cutting, solar-assisted battery charging, programmable motor control, and reduced dependence on direct external electrical power in suitable lawn and garden environments.

No. of Pages : 21 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130419 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ORGANIC SEED COATING FORMULATION COMPRISING VERMICOMPOST, MORINGA LEAF POWDER AND BENEFICIAL MICROORGANISMS

(51) International classification	:A01C 1/06, A01N 63/00, A01N 63/20, A01N 63/38, A01N 65/08, C05F 11/08, C05F 9/04, A01P 21/00	(71)**Name of Applicant :** **1)E. KAVITHA** Address of Applicant :3/188, North Street, Anupperpalayam, Sadhumugai Post, Sathyamangalam Taluk, Erode District – 638 503, Tamil Nadu, India Sathyamangalam Taluk Tamil Nadu India (72)**Name of Inventor :** **1)E. KAVITHA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ORGANIC SEED COATING FORMULATION COMPRISING VERMICOMPOST, MORINGA LEAF POWDER AND BENEFICIAL MICROORGANISMS The invention provides an organic seed coating formulation comprising vermicompost; moringa leaf powder; a microbial consortium of Azospirillum, Phosphobacteria, Trichoderma viride and waste decomposer; tender coconut water; Pink Pigmented Facultative Methylotrophs (PPFM); and activated charcoal powder, wherein, per 100 g of vermicompost, it comprises 8–12 g moringa leaf powder, 8–12 ml Azospirillum, 8–12 ml Phosphobacteria, 4–6 ml Trichoderma viride, 4–6 ml tender coconut water, 3–5 ml waste decomposer, 1–2 ml PPFM and 5–10 g activated charcoal powder, mixed into a homogeneous moist mass. The vermicompost base carries the microorganisms on the seed without a synthetic binder, and the activated charcoal adsorbs toxins and excess moisture. The formulation is also provided as a liquid broth for seed coating and foliar application. Seeds coated with the formulation show improved germination, vigour, root growth and yield over untreated seed. A process for its preparation is also disclosed.

No. of Pages : 23 No. of Claims : 13

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130421 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DIFFERENTIALLY REFERENCED CAPILLARY RADIUS-LADDER CONTACT SENSOR WITH DOWNWASH-VENTILATED REGENERATION FOR UNMANNED AERIAL VEHICLES

(51) International classification	:B64C 39/02, G01N 27/22, B64D 45/00, G01N 27/06, G01N 27/416
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BAPATLA SURENDRA BABU
Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India
2)IRUKUMALLI PRIYANKA
(72)**Name of Inventor :**
1)BAPATLA SURENDRA BABU
2)IRUKUMALLI PRIYANKA

(57) Abstract :
DIFFERENTIALLY REFERENCED CAPILLARY RADIUS-LADDER CONTACT SENSOR WITH DOWNWASH-VENTILATED REGENERATION FOR UNMANNED AERIAL VEHICLES A landing-surface sensing apparatus (100) for an unmanned aerial vehicle has a probe (102) that touches the surface before the foot (114). Its cassette (104) holds capillary membranes (302) whose pore radius decreases in steps from group to group, each between two electrodes (304, 306). A measurement unit (122) reads, against sealed reference cells (206), which groups the surface liquid enters, showing open water, saturated, moist or dry surfaces; gated cells (208) under a gate mesh (210) identify free water. After contact the membranes drain through a throat sheet (310) into a nanoporous reservoir (312) of 25 nm to 55 nm median pore radius, which stays free of condensed water at 95% relative humidity and still empties the finest membrane. An evaporation layer (314) dries the reservoir into a chamber (316) ventilated by the rotor downwash. Use: autonomous landing of agricultural and delivery vehicles. FIG. 1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130435 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIO-BASED NON-THERMOPLASTIC ADHESIVE STICK

(51) International classification	:C09J 103/02, C09J 11/04, C09J 129/04, C09J 11/06, C08L 3/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. MANIKANDAN MURUGESAN
2)Ms. S. KAVYA
3)Mr. P. RAJARAJAN
4)Mr. A. MOHAMED UMAR
5)Mr. MANJUNATHAN ALAGARSAMY
6)Mrs. B. Sribharathi
7)N PRAGATHI

(57) Abstract :
BIO-BASED NON-THERMOPLASTIC ADHESIVE STICK This invention relates to a Bio-Based Non-Thermoplastic Adhesive Stick made from renewable and environmentally friendly materials. The adhesive formulation contains modified starch as the primary binder, polyvinyl alcohol (PVA), glycerol, citric acid, calcium carbonate ($CaCO_3$), water, and an optional preservative. It is produced through a simple cold-setting process involving preparation of a starch slurry, preparation of a PVA solution, uniform mixing, moulding, and natural curing. Unlike conventional thermoplastic adhesive sticks, bonding develops through water evaporation and molecular cross-linking between modified starch and PVA without requiring melting during use. Glycerol improves flexibility, citric acid supports cross-linking and bonding durability, and calcium carbonate reinforces the adhesive structure and improves dimensional stability. The resulting adhesive can bond paper, cardboard, wood, fabric, leather, bookbinding materials, corrugated packaging, and other porous substrates. The invention offers a simple, economical, biodegradable, and environmentally friendly alternative to conventional petroleum-based adhesive sticks for domestic, educational, office, packaging, and light industrial applications. Fig 1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130436 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GESTURE-CONTROLLED SMART LAPTOP STAND WITH INTEGRATED COOLING, ADJUSTABLE SUPPORT, USB-C POWER DELIVERY AND BACKUP POWER SYSTEM

(51) International classification	:G06F 1/20, G06F 1/16, H05K 7/20, A47B 21/03, A47B 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India

2)M.KUMARASAMY COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)Mr. Mohamed Asil

2)Ms.Prescena Chrsty J

3)Mr. Mohamed Abdul Haseeb

4)Ms. Ameera

5)Mrs. Swetha Bharathi

6)Mr.K.Sivakumar

7)PRIYADHARSHINI J

(57) Abstract :

ABSTRACT GESTURE-CONTROLLED SMART LAPTOP STAND A smart laptop stand includes an inclinable platform (1) supported by a foldable rear mechanism (13), with two front C-shaped clips (12) engaging only the laptop's lower front edge to prevent sliding without obstructing the screen or hinge. An active cooling subsystem (8, 9), comprising two fans and ventilation openings beneath the laptop's heat zone, is automatically controlled based on temperature. Two articulated actuator arms (3) extend from the platform to an attachment bracket (2) on the laptop lid, enabling controlled, gesture-driven positioning of the display via a camera/microphone module (5) and control processor (7). A USB-C Power Delivery port (10) allows the platform's internal battery (11), protected by a Battery Management System, to be charged or to supply backup power to the laptop. The integration of adjustable support, active cooling, non-intrusive retention, gesture-controlled actuation, and power backup into a single portable device improves ergonomics, thermal management, and laptop usability. Fig 1

No. of Pages : 31 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130437 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : OLOID-GEOMETRY PASSIVE GRAVITY-SEATING ANTI-SPILL CLOSURE FOR DRINKING VESSELS WITH SADDLE-PROFILE RECEIVER SEAT AND INTEGRAL RULED-SURFACE FLOW GUIDE

(51) International classification	:A47G 19/22, B65D 41/04, B65D 41/34, B65D 47/06, B65D 39/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India

2)M.KUMARASAMY COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)Mr. Mohamed Imran

2)Mrs. Sumathi

3)Ms. Kathrine Reena J

4)Mr. Mohamed Ansari Raja

5)Mr. Nandha

6)Mrs. S Ramya

7)KAMALITTA R

(57) Abstract :

ABSTRACT OLOID-GEOMETRY PASSIVE GRAVITY-SEATING ANTI-SPILL CLOSURE FOR DRINKING VESSELS WITH SADDLE-PROFILE RECEIVER SEAT AND INTEGRAL RULED-SURFACE FLOW GUIDE An oloid-geometry passive gravity-seating anti-spill closure (10) for drinking vessels comprises an oloid-geometry sealing element (20) formed as the convex hull of two congruent circles (22, 23) of radius R in mutually perpendicular planes, manufactured from food-grade silicone elastomer of Shore A hardness 40 to 60, with integral ruled-surface flow guide channels (21) as the natural concave surface regions of the oloid geometry; a saddle-profile receiver seat (30) geometrically derived from the oloid ruled surface contact locus providing a distributed sealing contact line (31); a closure body (40) with threaded connection section (41) and closure mouth (42); a flexible silicone tether element (50) retaining the oloid-geometry sealing element (20) within the closure body (40); and a vented vent passage (60) with silicone flap check valve (61). When the vessel tilts, gravity rolls the oloid-geometry sealing element (20) along its geometrically constrained straight-line rolling path into the saddle-profile receiver seat (30), sealing the closure mouth (42) without spring, elastic material, buoyancy, gas pressure, or electronics. Suction applied during intentional drinking partially unseats the oloid-geometry sealing element (20), opening the flow guide channels (21) to liquid flow. The oloid eliminates the spill failure mode of prior spherical sealing elements. Fig. 1

No. of Pages : 25 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130438 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADJUSTABLE ORAL BITE BLOCK WITH UNOBSTRUCTED WORKING PASSAGE AND KINEMATIC LOCKING MECHANISM

(51) International classification	:A61M 16/04, A61B 90/16, A61B 1/24, A61C 5/14, A61C 7/36
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr. Jayakumar S
2)Mr. Mohamed Abdul Haseeb
3)Ms. Kaviya Shree M
4)Mrs. S. Geetha
5)Mr.M. Jaganath
6)KAMALITTA R

(57) Abstract :
ABSTRACT ADJUSTABLE ORAL BITE BLOCK WITH UNOBSTRUCTED WORKING PASSAGE AND KINEMATIC LOCKING MECHANISM The present invention relates to an adjustable oral bite block (10) having a laterally positioned adjustment mechanism and an unobstructed working passage (22). The bite block comprises a base member (20) defining a longitudinal working passage, a movable bite member (40), and a drive assembly positioned laterally relative to the working passage. The drive assembly comprises a rotatable axle (34) and a cam (36) configured to displace the movable bite member through a follower (44). In a selected adjustment position, the follower is positioned beyond a dead-center position (48) relative to the rotational axis of the cam and is engageable with a mechanical stop (52) associated with the lateral housing (30). Occlusal loading applied to the movable bite member is transmitted through the follower toward the mechanical stop and housing, thereby resisting reverse displacement of the cam. The lateral positioning of the drive assembly maintains the working passage substantially unobstructed during adjustment and use. Fig. 1

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130439 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED TUBULAR BODY FILLING, INDEXING AND PRODUCTION MONITORING ASSEMBLY DEVICE

(51) International classification	:B65B 39/00, B65B 3/30, B65B 57/20, B65B 1/30, B65B 1/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)YAZHINI S
2)HARIHARAN BABU. V
3)Dr.S.DINESH
4)Dr.S.INFANCY VIMAL PRIYA
5)Dr.P.SAKTHI
6)SATHYA PRIYA B

(57) Abstract :
ABSTRACT AUTOMATED TUBULAR BODY FILLING, INDEXING AND PRODUCTION MONITORING ASSEMBLY DEVICE The present invention discloses an automated manufacturing and assembly device (10) configured to perform sequential operations including tubular body feeding, positioning, controlled filling with non-reactive material, automatic indexing, inspection, production counting, and product ejection through an integrated mechanical mechanism. The device comprises a tubular body (1), a hopper (2), a filling nozzle assembly (3), a rotary indexing plate (4), a base housing (5), and a drive motor (6). The drive motor (6) rotates the rotary indexing plate (4) to sequentially position the tubular body (1) beneath the filling nozzle assembly (3), while the hopper (2) continuously supplies material for controlled dispensing. A controller synchronizes the indexing and dispensing operations, receives inspection signals, automatically records production count, operating time, downtime, and rejection count, and transmits the production data to a computer-based monitoring system for displaying real-time hourly, daily, weekly, monthly, and yearly production statistics through a graphical user interface. The invention improves production consistency, reduces manual intervention, enhances process monitoring, and provides a modular automated manufacturing system suitable for industrial applications. Fig 1

No. of Pages : 21 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130440 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-ADJUSTING CONSTANT-FORCE TENDON SUTURE ANCHOR

(51) International classification	:A61B 17/04, A61F 2/08, A61B 17/00, A61B 17/68, A61B 17/064
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)M.KUMARASAMY COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mahalakshmi
2)Sumathi
3)Dr. R. SUJATHA
4)B. JULIET CELINE MARY

(57) Abstract :
ABSTRACT SELF-ADJUSTING CONSTANT-FORCE TENDON SUTURE ANCHOR The present invention discloses a self-adjusting constant-force suture anchor assembly for tendon-to-bone repair. This assembly is designed to handle controlled -operative tendon movement while keeping the suture tension the same. The assembly includes a bone-anchor body, a suture eyelet, a variable-profile compliant flexure an internal guide mechanism and a mechanical travel stop. The outer threaded bone-anchor body gives mechanical anchorage inside the target bone. The movable suture eyelet holds a suture connected to the target tendon tissue. This suture provides tensile fixation across the repair area. The profile compliant flexure is, inside the bone-anchor body and connects to the movable suture eyelet. This flexure changes shape in a controlled way to adjust the force response based on the movement of the attached tendon. The internal guide mechanism moves the eyelet along a path. This prevents the eyelet from rotating, tilting or getting stuck. The mechanical travel stop stops the eyelet from moving far. This prevents the flexure from being stretched much. The suture anchor assembly works without any sensors. It keeps the amount of tension during healing. This helps the repair not become too loose or too tight.

No. of Pages : 23 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130441 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXTERNAL GESTURE-CONTROLLED HINGE ACTUATOR

(51) International classification	:G06F 1/16, E05D 11/10, E05F 1/12, E05F 15/611, E05D 11/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr. Aditya
2)Mr. Mohamed Rifath A
3)Mr. Mohamed Imran
4)Mr. Hariharenbabu
5)Mr. T Praveen Kumar
6)Dr.S.Bharani Nayagi
7)Jenitha Princy A

(57) Abstract :
ABSTRACT EXTERNAL GESTURE-CONTROLLED HINGE ACTUATOR The present invention discloses an external gesture-controlled hinge actuator (10) for automatically operating the lid of a laptop computer without requiring replacement of the existing hinge of the laptop computer, so as to reduce the manual effort of the user. The device (10) comprises a housing (11) removably attached to the base of the laptop computer by means of a mounting assembly (12), adjacent to the existing hinge of the laptop computer. The device (10) further comprises a drive motor (13) operatively coupled with a gear reduction assembly and a coupling arm (16), wherein the coupling arm (16) is configured to transmit rotational movement to the lid of the laptop computer. A gesture sensor (17) senses a hand gesture of the user and transmits a corresponding signal to a controller (18), which classifies the said signal as an open command or a close command and actuates the drive motor (13). The controller (18) is further configured to receive a command by means of a microphone, a physical button or a wireless communication link. A lid angle sensor provides a closed loop angular position control of the lid, and a current sensor detects an obstruction to the movement of the lid and stops the drive motor (13) accordingly. A declutch mechanism permits the lid to be operated manually upon a loss of power supply to the device (10). The present invention accordingly reduces the manual effort and provides a touch free, precise and safe operation of the lid of the laptop computer. Fig.1

No. of Pages : 22 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130442 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : CLAMP-ON SMART UNDERGROUND OIL AND GAS PIPELINE MONITORING BAND

(51) International classification	:F17D 5/06, F17D 5/02, G01N 27/82, G01N 27/83, F17D 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. MANIKANDAN MURUGESAN
2)Mr. R. SRI SANJIT
3)Mr. S. EZHILAN
4)Mr. MANJUNATHAN ALAGARSAMY
5)Mrs N Srija
6)SUGANYA P

(57) Abstract :
ABSTRACT CLAMP-ON SMART UNDERGROUND OIL AND GAS PIPELINE MONITORING BAND The present invention relates to a clamp-on Smart Underground Oil/Gas Pipeline Monitoring Band for continuous, non-intrusive monitoring of an existing underground metallic oil or gas pipeline. The system comprises a protective band housing (1), a non-welding clamp assembly (2), a Magnetic Flux Leakage sensor and magnet assembly (3), a main printed circuit board (4) incorporating a microcontroller and sensor-fusion circuitry, a rechargeable battery pack (5), a solar panel and charging circuit (6), an acoustic or vibration sensor (7), a hydrocarbon or soil-vapour sensing element connected to a sampling probe (8), a temperature sensor (9), and a LoRa or wireless communication module (10). The band is mounted around the pipeline without welding, drilling, cutting, or permanently modifying the pipe. During operation, the Magnetic Flux Leakage system detects abnormal magnetic flux changes caused by cracks, corrosion, pitting, and metal loss; the acoustic or vibration sensor monitors abnormal sound and vibration produced by leakage, pressure disturbance, or crack activity; the hydrocarbon or soil-vapour sensing element detects oil-related chemical changes in the surrounding soil; and the temperature sensor records local temperature for supporting interpretation. The microcontroller compares and fuses the sensor signals to identify possible oil leakage, crack, corrosion, metal loss, or abnormal pipeline behaviour while reducing false alarms through multi-sensor confirmation. The processed condition, sensor data, and monitoring-node identification are transmitted through the LoRa or wireless communication module to a remote gateway or control station. The rechargeable battery supplies operating power and the solar panel recharges the battery for long-term autonomous operation. Multiple bands may be installed along a pipeline to compare node responses and estimate the approximate location of a leak or defect. Fig.1

No. of Pages : 15 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130443 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART CANE FOR OBSTACLE DETECTION AND REROUTING

(51) International classification	:A61H 3/06, G01C 21/16, H04N 7/18, G01C 21/20, G08B 21/02	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mr. MANJUNATHAN ALAGARSAMY**
(32) Priority Date	:NA	**2)Ms. LEENA JAYAVARSHINI PRABAHARAN**
(33) Name of priority country	:NA	**3)Mr. THIRUPPATHISAMY SEENIVASAN**
(86) International Application No Filing Date	: :01/01/1900	**4)Mrs.KIRUTHIKA G** **5)JAYA SHREE V**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT SMART CANE FOR OBSTACLE DETECTION AND REROUTING The present invention relates to an assistive navigation system for visually impaired users, mounted on a telescopic cane shaft (9) through a shaft mounting clamp (8). The system comprises a rotatable camera module head (1), vertical tilt servo motor (2), horizontal pan motor (3), multi-directional thumb joystick (4), main edge AI compute PCB board (4), GPS receiver, inertial measurement unit, rechargeable battery and private audio output port (5). The camera captures surrounding images, while the processing unit detects objects and determines whether they obstruct the user's walking corridor. When an obstacle is identified, the system evaluates available clearance and determines a suitable bypass direction. If required, peripheral scanning is performed to identify an alternate path. Navigation instructions are provided through private audio output and haptic feedback. The system thereby assists visually impaired users in identifying obstacles and navigating around them safely. Fig1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130444 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS OCEAN PLASTIC COLLECTION AND MARINE -LIFE PROTECTION

(51) International classification	:G05D 1/02, B63B 35/32, G05D 1/00, B63G 8/00, E02B 15/10	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Dr.S.MALATHI**
(86) International Application No	:	**2)J. JAYAPRAKASH**
Filing Date	:01/01/1900	**3)K.KAMALESH**
(87) International Publication No	: NA	**4)Dr.K.V. NITHYA**
(61) Patent of Addition to Application Number	:NA	**5)SATHYA PRIYA S**
Filing Date	:NA	**6)P.SARAVANAVEL**
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

ABSTRACT AUTONOMOUS OCEAN PLASTIC COLLECTION AND MARINE-LIFE PROTECTION The present invention discloses an Autonomous Ocean Plastic Collection and Marine-Life Protection System configured to detect, collect, and process floating plastic waste from oceans, seas, rivers, and coastal water bodies while protecting marine organisms from harm. The system comprises a vision and sensor module, an artificial intelligence processing engine, a marine-life detection and deterrence module, an autonomous navigation and propulsion controller, a robotic collection and sorting mechanism, a power management unit, and a remote communication and monitoring module. The invention enables continuous, unmanned, energy-efficient collection of plastic debris while identifying and safely deterring marine organisms from the collection zone, thereby providing a scalable and ecologically responsible solution to ocean plastic pollution. Fig1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130447 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-SENSOR SEED QUALITY ANALYSE SYSTEM

(51) International classification	:A01C 1/00, G01N 21/3563, A01C 1/02, G01N 21/3581, G06N 20/00	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)SHYAM THETCHANA GOPAL**
(32) Priority Date	:NA	**2)RAMKUMAR RAJAGOPAL**
(33) Name of priority country	:NA	**3)HARIHARAN BABU VENKATESWARA BABU**
(86) International Application No Filing Date	: :01/01/1900	**4)MRS.R.KEERTHANA** **5)N. PRAGATHI**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT MULTI-SENSOR SEED QUALITY ANALYSE SYSTEM The present invention relates to an AI-Based Smart Seed Quality Analyzer, which is a device that analyses seed quality and germination capability without harming seeds. This device consists of a seed hopper, a conveyor, a laser line and photodiode array sensor, fluorescence photodiode sensor, Terahertz (THz) sensor, resonance/vibration sensor, sensor-fusion machine learning model, Seed Viability Index (SVI), 7-inch LED display, and seed collection chambers. Hopper sends out individual seeds to be analysed through a conveyor. Laser and photodiode array analyses shape, defects, cracks, and density. Fluorescence photodiode analyses fungus presence, biochemical alterations, and stress indicators. THz analyses moisture, structure, density, and hidden characteristics of seeds, and the resonance/vibration sensor analyses cracks and hollowness in seeds. Using machine learning algorithms, all sensors' output is analysed to identify seed quality and germination capacity. Seed viability index (SVI) determines seed germination probability, and the result is displayed on the 7-inch LED display. Lastly, seeds are sorted into categories as GOOD, MEDIUM, OR REJECTED. Fig.1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130448 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : VITONE FUSION

(51) International classification	:G10H 3/18, G10D 3/18, G10D 1/02, G10H 1/00, G10H 1/32	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)R.SRI SANJIT**
(33) Name of priority country	:NA	**2)HARIHARAN KONDIN BALAKRISHNAN**
(86) International Application No Filing Date	: :01/01/1900	**3)S. EZHILAN**
(87) International Publication No	: NA	**4)MS. C. BAVITHRA**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**5)R. ANUPRIYA**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT VITONE FUSION The present invention relates to a hybrid smart violin pickup system for accurately capturing and reproducing the natural acoustic characteristics of a violin. The system comprises a pickup housing (1), a vibration isolation pad (2), a piezoelectric sensor module (3), a main printed circuit board (4) incorporating a MEMS vibration sensor, embedded microcontroller and signal processing circuitry, a rechargeable battery pack (5), and a lower housing base (6). The pickup is removably attached to the violin using an adjustable mounting clamp (2), rubber contact pad (3), clamp tightening knob (4), and locking bracket (5). During operation, the piezoelectric sensor detects string vibrations while the MEMS sensor detects body resonance. The embedded microcontroller performs intelligent sensor fusion and digital signal processing to generate a balanced, high-fidelity audio output with reduced noise and acoustic feedback. The processed signal is delivered through the audio output connector (6) and audio output cable (7) to external audio equipment, thereby improving sound quality, tonal accuracy, and performance reliability for professional musicians, studio recording, and live performances. Fig.1

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130449 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : THERMOSHADE ROOFING TILE AND METHOD OF FABRICATION THEREOF INCORPORATING PHASE CHANGE MATERIAL AND RECYCLED ALUMINIUM FOR PASSIVE SOLAR HEAT REGULATION

(51) International classification	:E04D 1/28, E04D 1/30, E04D 1/00, C04B 14/10, C04B 28/00	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. V.C. BHARATH SABARISH
2)M. JANANI
3)Dr S Lavanya
4)SUGANYA .B

(57) Abstract :
ABSTRACT THERMOSHADE ROOFING TILE AND METHOD OF FABRICATION THEREOF INCORPORATING PHASE CHANGE MATERIAL AND RECYCLED ALUMINIUM FOR PASSIVE SOLAR HEAT REGULATION The present invention discloses a passive thermal-regulating roofing tile comprising clay, white cement, paraffin wax phase change material, recycled aluminium scrap and sand. In an exemplary embodiment, the constituents are present in approximately 30%, 30%, 20%, 10% and 10% proportions respectively. The tile combines solar reflectivity, latent heat storage and thermal moderation to reduce and delay solar heat transfer through the roofing element. The PCM absorbs thermal energy during its phase transition and releases the stored energy during subsequent cooling, while the clay–sand matrix moderates heat transmission and the recycled aluminium contributes to the thermal response of the PCM-containing region. The invention further provides a non-fired, moulded and sun-dried fabrication method intended to preserve PCM functionality. The PCM may be retained within honeycomb and/or rectangular-strip configurations. The specification reports approximately 8–12°F lower surface temperature under the stated solar-testing conditions, together with reported flexural strength of approximately 10–15 MPa and water absorption of approximately 8–10%. The invention provides a passive, sustainable and potentially scalable roofing solution for reducing solar heat gain in buildings.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130450 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR REAL-TIME PERSONAL VOICE PROFILE TRANSFORMATION USING DIGITAL SIGNAL PROCESSING AND AUDIO ENHANCEMENT SYSTEM

(51) International classification	:G10L 21/0208, G10L 21/003, G10L 21/0364, G10L 21/02, G10L 21/007
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)ATULKRISHNA VALLIYEDATHU
2)SARAVANAN SANKAR
3)MS. C. BAVITHRA
4)R. ANUPRIYA

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR REAL-TIME PERSONAL VOICE PROFILE TRANSFORMATION USING DIGITAL SIGNAL PROCESSING AND AUDIO ENHANCEMENT SYSTEM The present invention relates to a real-time personal voice profile transformation system and method for converting live speech or singing into a selected target voice profile with substantially low latency. The system comprises a microphone and pre-amplifier, analog-to-digital converter, digital signal processing unit, memory, voice feature extraction engine, personal voice profile matching engine, artificial-intelligence-based voice transformation engine, intelligent audio enhancement stage, audio codec, and audio output interface. A reference audio recording is processed to generate and store a personal voice profile representing desired vocal characteristics. During operation, live speech or singing is captured and converted into digital audio data, subjected to signal conditioning and feature extraction, and matched with the stored voice profile. The AI voice transformation engine modifies selected vocal characteristics while substantially preserving the speech or singing content and temporal characteristics. The transformed audio is subsequently enhanced and delivered through wired or wireless audio interfaces for substantially real-time monitoring, amplification, or recording. Fig.1

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130451 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TRANSFORMER HOTSPOT EARLY WARNING SYSTEM

(51) International classification	:G01J 5/00, G08B 21/18, H01F 27/40, G01J 5/02, G08C 17/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)NIRMAL SIVAKUMAR
2)YOKESWARAN RAMADOSS
3)MOHAMED ZUBARULHAQ ABDUL RAHMAN
4)MOHANAGANESH BALAKRISHNAN
5)PRIYAN ANBALAGAN
6)PRASANTH ANBALAGAN
7)GOWTHAM SELVAKUMAR
8)KISHOORE PRASAD MANIMARAN
9)MRS.R.KEERTHANA
10)N. PRAGATHI

(57) Abstract :
ABSTRACT TRANSFORMER HOTSPOT EARLY WARNING SYSTEM The present invention relates to a Transformer Hotspot Early Warning System for continuously monitoring the thermal condition of electrical distribution transformers through a non-invasive sensing arrangement. The system comprises a collar ring mounted externally around the transformer body, multiple infrared thermal sensors for measuring surface temperature at different locations, a main control unit for processing thermal data, a display and interface for presenting real-time operating information, a communication antenna for wireless data transmission, and a mounting clamp for secure installation. The control unit continuously analyzes the temperature readings to identify abnormal thermal variations and predict hotspot formation before the occurrence of transformer failure. When the measured temperature exceeds a predetermined safe limit, the system generates immediate warning notifications and transmits the monitoring data to remote maintenance personnel for timely corrective action. The invention enables continuous condition monitoring without modifying the transformer structure, thereby improving transformer reliability, reducing maintenance costs, minimizing unexpected power interruptions, and supporting predictive maintenance in electrical power distribution networks. Fig 1

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130452 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TOPO LEARN BOX

(51) International classification	:G09B 7/02, G09B 7/04, G09B 7/00, G06Q 50/20, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)M.KUMARASAMY COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. S. MALATHI
2)G. S. DEEPTI SRI
3)Dr.S.Vanithamani
4)Dr. G. GABRIEL SANTHOSH KUMAR

(57) Abstract :
ABSTRACT TOPO LEARN BOX The present invention provides a Topo-Learn Box(10) configured to identify underlying prerequisite knowledge gaps in a learner's educational knowledge structure. The system receives learner performance parameters including response correctness, response time, repeated errors and question-level performance, and maps the parameters onto a topological representation of interconnected educational concepts. A persistent homology analysis module (17) analyses the learner-specific knowledge representation to identify persistent structural gaps and determine prerequisite concepts associated with observed learning difficulties. An adaptive learning route generation module (19) generates a personalized sequence of remedial concepts, questions and learning activities based on the identified gaps. A topological accelerator (18) may accelerate selected computational operations, while a local learning content database (20) enables substantially offline operation. The system thereby identifies root prerequisite deficiencies rather than merely detecting incorrect answers and dynamically provides an individualized learning pathway for improving the learner's conceptual understanding.

No. of Pages : 17 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130454 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE BENCH VISE WITH RACK-LOCK WITH ASSISTED JAW SYSTEM FOR MULTI-SHAPE CLAMPING UNDER HIGH FORCE OPERATIONS

(51) International classification	:B25B 1/24, B23Q 3/06, B25B 1/18, B25B 1/10, B23Q 1/76
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India

2)M.KUMARASAMY COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)ATULKRISHNA VALLIYEDATHU

2)VENKATRAMAN VIJAYAN

3)MOHAMED ZUBARULHAQ ABDUL RAHMAN

4)NIRMAL SIVAKUMAR

5)KISHOORE PRASAD MANIMARAN

6)KRISH ARVIND THOLKAPPIAN

7)MOHANA GANESH BALAKRISHNAN

8)VINO BALAJI MUKUNDAN

9)MS.C.BAVITHRA

10)D. JOTHI MANOHARI

(57) Abstract :

ADAPTIVE BENCH VISE SYSTEM WITH SHOCK-ABSORBING PAD AND PUSH-BUTTON LOCKING MECHANISM The present invention relates to an adaptive bench Vise system designed to provide rapid adjustment, secure locking, and vibration resistance during machining, drilling, milling, and fabrication operations. The system comprises a fixed jaw surface (1) and a sliding jaw (2) equipped with a shock absorbing rubber pad (3) to damp vibrations and absorb shocks, thereby maintaining stability and protecting workpieces. A rack (6) integrated with the sliding jaw, supported by a fixed rack support (7), enables smooth linear travel, while the main base surface (8) provides structural stability. A connecting handle (9) allows ergonomic manual control, and a push button locking system (10) with spring and rod arrangement ensures instant locking and automatic re engagement for safety. The invention offers significant advantages over conventional vises by combining speed of adjustment, reliable locking, and enhanced vibration resistance, making it suitable for industrial, workshop, and educational applications. Fig.1

No. of Pages : 20 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130455 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SPRING-POWERED SELF-WAVING HAND FAN

(51) International classification	:F04D 25/10, F03G 5/02, F03G 7/08, F03G 5/06, F03D 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India

2)M.KUMARASAMY COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)NIRMAL SIVAKUMAR

2)PARKUNAM NAGARAJ

3)MOHAMED ZUBARULHAQ ABDUL RAHMAN

4)MOHANAGANESH BALAKRISHNAN

5)PRIYAN ANBALAGAN

6)PRASANTH ANBALAGAN

7)GOWTHAM SELVAKUMAR

8)SANTHOSH KUMAR ARUMUGAM

9)MRS.T.PRIYADHARSHINI

10)JULIET CELINE MARY. B

(57) Abstract :

ABSTRACT SPRING-POWERED SELF-WAVING HAND FAN A swing-operated self-winding bidirectional fan device is disclosed for generating airflow or waving motion through a combination of manual swinging motion and stored mechanical energy. The device comprises a wing panel (1), a handle assembly (2), a pivot joint (3), a spring mechanism (4), a clutch and gear arrangement (5), a winding knob (6), and a shaft coupling arrangement (7). The wing panel (1) is rotatably connected to the handle assembly (2) through the pivot joint (3), while the shaft coupling arrangement (7) transmits manually generated motion to the wing panel through the clutch and gear arrangement (5). During operation, the swinging motion imparted by a user causes rotation of the wing panel and simultaneously stores mechanical energy within the spring mechanism (4). Upon cessation of the manual swinging motion, the stored energy is automatically released through the clutch and gear arrangement (5) to continue rotational movement of the wing panel (1). The device operates without batteries, electric motors, electronic circuits, or external power sources. The invention provides prolonged fanning or waving action, reduced user effort, improved portability, and a compact mechanical construction suitable for personal cooling, recreational activities, promotional events, and outdoor applications. Fig.1

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130456 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BACKHOE HYDRAULIC CYLINDER MONITOR

(51) International classification	:F15B 19/00, E02F 9/26, E02F 9/20, E02F 9/22, F15B 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MOHANA GANESH BALAKRISHNAN
2)YOKESWARAN RAMADOSS
3)MURALI KRISHNA PANNEER SELVAM
4)INFANTO WESELY REX JOHN VISUVASAM
5)MAHENDHIRAN CHINNAMANI
6)VINO BALAJI MUKUNDAN
7)NIRMAL SIVAKUMAR
8)MURUGESH GURUNATHAN
9)MRS.P.PRATHIBA
10)KIRTHIGA R

(57) Abstract :
ABSTRACT BACKHOE HYDRAULIC CYLINDER MONITOR The invention presents a smart sensor-based monitoring system capable of continuously monitoring a JCB hydraulic cylinder working condition as well as detecting abnormal hydraulic conditions at an early stage of the problem. The system includes a pressure sensor, a temperature sensor, an oil-level sensor, an ESP32-based control unit, a display unit, a communication unit, and an alert unit. The sensors are placed in appropriate parts of the hydraulic system in order to continuously receive data about hydraulic oil pressure, hydraulic oil temperature, and hydraulic oil level during operation of the equipment. The control unit collects and analyzes data from the sensors, and compares the received values against the values of the safe operating conditions in order to detect abnormal conditions such as high hydraulic pressure, high oil temperature, and low oil level. In the case when the system detects an abnormal condition, it generates a warning signal and can show the hydraulic information detected to operators or technical staff through an appropriate communication system. Collected data can also be stored in the database for future comparisons.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130459 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN AND DEVELOPMENT OF A DYNAMIC MULTI-SEGMENT ADAPTIVE VISE JAW SYSTEM

(51) International classification	:B25B 1/24, B25B 1/10, B25B 1/18, B25B 1/02, B25B 1/20	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)ATULKRISHNA VALLIYEDATHU**
(33) Name of priority country	:NA	**2)VENKATRAMAN VIJAYAN**
(86) International Application No Filing Date	: :01/01/1900	**3)DARUN KUMAR B**
(87) International Publication No	: NA	**4)AGILESH**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**5)ASWIN GANESH** **6)DR JASPER D**
(62) Divisional to Application Number Filing Date	:NA :NA	**7)DHAARAANI R**

(57) Abstract :

ABSTRACT DESIGN AND DEVELOPMENT OF A DYNAMIC MULTI-SEGMENT ADAPTIVE VISE JAW SYSTEM A dual bench vise jaw system for holding workpieces with irregular, curved, asymmetrical, or non-parallel surface geometries is disclosed. The system comprises a main vise jaw body block (7) featuring a central mounting aperture (5) for mechanical attachment, and an opposing secondary vise jaw block assembly (4). Both jaw blocks incorporate a series of dynamic, semi-circular gripping segments (1, 2, 6) seated within modular curved channels and retained by internal pin interfaces (3). When an irregular object is clamped, the dynamic contact segments (1, 2, 6) automatically rotate along their curved seats to conform to the exact surface profile of the workpiece. This dynamic movement converts single-point or line contact into multi-point surface area contact, redistributing clamping pressure evenly across the object profile. The system prevents workpiece slippage and surface crushing while mitigating shear stress on the retention pins, delivering high grip stability and enhanced durability across diverse work-holding applications. Fig.1

No. of Pages : 16 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130460 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ALL IN ONE WELD FIXTURE

(51) International classification	:B23K 37/04, B23K 37/053, B23K 37/047, B23Q 3/18, B23Q 3/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India

2)M.KUMARASAMY COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)VENKATESWARA BABU HARIHARAN BABU

2)VENKATRAMAN VIJAYAN

3)KISHOORE PRASAD MANIMARAN

4)UNA MAHESWARAN JAI AKASH

5)DR.SENTHIL KUMAR THIYAGARAJAN

6)RANI C

(57) Abstract :

ABSTRACT ALL-IN-ONE WELD FIXTURE The All-in-One Weld Fixture is an innovative mechanical fixture that has been developed to increase the accuracy and efficiency of the welding process. With conventional welding, different components require different fixtures, clamps, and other means of securing them in their respective locations. This increases the time taken to prepare for the welding process, the cost of production, and also increases the chance of dimensional error. In the suggested All-in-One Weld Fixture, the locating and supporting means have been incorporated into one fixture system. The fixture is equipped with an adjustable arrangement of the locating means and clamping means, which allows the workpiece to be positioned in the required location. The system allows maintaining the geometry of the components and minimises their movement and deformation during the welding process. The adjustable nature of the fixture ensures that it can be used for different sizes and shapes of the components. In this system, there is no need to change the fixtures and measure again and again during the welding process. It also helps to maintain consistency between the welded parts. The fixture can be easily manufactured using materials available in the market and manufacturing methods. It is cost-efficient and suitable for use in small-scale and medium-scale industries and workshops, as well as in education purposes.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130461 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART DOOR-MOUNTED INTELLIGENT REMINDER AND ITEM VERIFICATION SYSTEM

(51) International classification	:G08B 21/24, G07C 9/00, G06K 7/10, G08B 13/14, G06Q 10/08	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mr.Balaji M**
(32) Priority Date	:NA	**2)Mr. Abdul Rahman S**
(33) Name of priority country	:NA	**3)Ms.Azizya M**
(86) International Application No Filing Date	: :01/01/1900	**4)Dr.Velmurugan M** **5)Mr.Harish Raghavendar T**
(87) International Publication No	: NA	**6)Ms.Nabithrha R**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**7)Ms.Sarnika J R** **8)Mr.B.SUNDARA PANDIYAN**
(62) Divisional to Application Number Filing Date	:NA :NA	**9)Jagadeswari**

(57) Abstract :

ABSTRACT SMART DOOR-MOUNTED INTELLIGENT REMINDER AND ITEM VERIFICATION SYSTEM The present invention discloses a smart door-mounted intelligent reminder and item verification system (10) for assisting users in identifying essential items and scheduled activities before leaving a room or building. The system (10) comprises a system enclosure (20), a motion/Time-of-Flight (ToF) sensor (30), an ESP32 controller (40), a TFT display (50), an RFID reader (60) and a speaker (70). The motion/ToF sensor (30) detects a user approaching the door and signals the ESP32 controller (40) to activate a reminder interface on the TFT display (50). The RFID reader (60) scans for RFID tags (90) associated with predefined required items such as an identification card, keys or a laptop. If an expected item is detected, the system (10) provides a confirmation on the TFT display (50); if an expected item is missing, the TFT display (50) shows a warning message and the speaker (70) generates an audible alert. The present invention (10) reduces forgotten-item incidents and provides a practical, customizable smart-door assistance solution suitable for homes, hostels, offices, laboratories, classrooms, hotels and hospitals. Fig 1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641130463 A
(19) INDIA
(22) Date of filing of Application :02/11/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATIC STAND ON FLOOR CLEANING MACHINE

(51) International classification	:A47L 11/40, A47L 11/30, A47L 11/282, A47L 11/292, A47L 11/20	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)M.KUMARASAMY COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)KRISHNA AAKASH MURALI DHARAN
2)VENKATRAMAN VIJAYAN
3)DR.SHANMUGAVADIVEL G
4)SENEGA R

(57) Abstract :
ABSTRACT AUTOMATIC STAND ON FLOOR CLEANING MACHINE The present invention concerns a stand-on floor cleaning machine made to clean and maintain large floor surfaces. This floor cleaning machine brings together cleaning tasks into one mobile unit so that people do less hand-work and spend less time cleaning. The floor cleaning machine has a stand-on platform, a drive system, a rotating scrubber, a water and cleaner spray, a wastewater catcher, a suction and drying part and extra brushes to reach edges and corners. When the floor cleaning machine works it sprays liquid on the floor while the rotating scrubber lifts dust, dirt, stains and other grime. The floor cleaning machine then pulls the water out with a suction system and pushes it into a recovery tank leaving the floor relatively dry. An optional disinfecting feature can be added for hygiene. The floor cleaning machine is built to clean steadily continuously and easily while also cutting down worker fatigue. This floor cleaning machine suits floors in factories, warehouses, hospitals, offices, schools and other large indoor spaces giving a practical and versatile answer, to today's floor-cleaning needs.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130464 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED SMART HAIRBAND FOR WOMEN'S SAFETY

(51) International classification	:A45D 8/36, G08B 21/02, G08B 21/04, G08B 6/00, G08B 25/01	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Ms. A. THASNEEM FATHIMA** **2)Dr. M. VELMURUGAN** **3)Dr.Jasper D** **4)Elamathi E**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT AI-BASED SMART HAIRBAND FOR WOMEN'S SAFETY The present innovation relates to an AI-based smart hairband for women's safety, configured to provide a discreet and hands-free personal safety mechanism. The smart hairband integrates a stretch sensor, IMU sensor, touch sensor, processing and control unit, BLE communication module, vibration actuator, rechargeable battery, user input/cancellation interface and power management circuit within a wearable hairband structure. The processing and control unit receives and analyses a combination of stretching, movement and touch inputs to recognize a personalized user-defined safety gesture or possible distress condition. Upon detection of a possible distress condition, the vibration actuator provides silent tactile feedback to the user. The user can cancel an unintended event through the user input/cancellation interface. When the event is not cancelled according to a predetermined condition, the BLE communication module transmits the emergency event information to a connected smartphone. The smartphone can initiate an emergency notification and share relevant location information with designated trusted contacts, thereby enabling discreet and effective emergency assistance. Fig 1

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130466 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATIC RADIATOR TEST RIG

(51) International classification	:G01M 99/00, F01P 11/16, G01M 17/007, G01M 15/02, F01P 11/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)VIJAY KUMAR SELVAMURUGAN
2)SATHISH KUMAR GURUSAMY
3)SANJAI NAGARAJAN
4)RAGUL VAIRAVAN
5)KISHORE KUMAR MARIMUTHU
6)MR.P.ARUNKUMAR
7)ASHA G

(57) Abstract :
ABSTRACT AUTOMATIC RADIATOR TEST RIG The present invention relates to an Automatic Radiator Test Rig for testing and evaluating the thermal performance of automotive radiators under controlled operating conditions. The system comprises a radiator (1), a cooling fan (2), a coolant circulation pump and drive assembly (3), a coolant reservoir/tank (4) with a tank cap (5), and an integrated control and monitoring unit (6) having a display interface (7). The coolant is circulated through the radiator (1) by the pump assembly (3), while the cooling fan (2) provides airflow across the radiator to simulate operating conditions. The control and monitoring unit (6) enables the operator to control the test operation and observe the measured parameters through the display (7). The coolant reservoir (4) provides storage and supply of coolant, with the cap (5) providing access for filling and maintenance. The proposed test rig provides an automated and systematic method for evaluating radiator performance, reducing manual testing effort and improving repeatability and monitoring of test conditions. The invention is suitable for automotive radiator testing, performance evaluation, quality inspection, research, and educational applications. Fig.1

No. of Pages : 21 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130467 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED DUAL-FILAMENT SWITCHING FOR 3D PRINTING

(51) International classification	:B29C 64/209, B29C 64/118, B29C 67/00, B29C 64/321, B29C 64/295	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)PRAKASH KANNAN MANI** **2)VISHVA MUTHU SAMY** **3)VIGNESHKUMAR VIJAYAKUMAR** **4)MRS.R.ELANKEERTHANA** **5)PONNIVALAVAN M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AUTOMATED DUAL-FILAMENT SWITCHING FOR 3D PRINTING The present invention discloses a dual-filament switching mechanism (10) for a single-nozzle 3D printer, configured to print products using two different coloured filaments. The mechanism comprises two filament rolls (11), supporters (12), rollers with stepper motors (13), filament feeds (21, 22), a printer nozzle setup (18), filament heater (23), nozzle inlet (25), and nozzle (26). The system analyses the product design or G-code to determine the required filament colour and corresponding printing sequence. The selected roller feeds the required filament through its respective feeding path towards the common nozzle inlet (25), where the filament is heated and extruded through the nozzle (26). When the printing portion of one colour is completed, the corresponding roller rotates in the reverse direction to retract the remaining filament and provide a free path for the other filament. The second roller is then activated to feed the next coloured filament, and the same switching process is repeated according to the G-code. The rotational movement of each roller is also monitored to estimate filament consumption. The present invention provides controlled, automatic dual-colour printing through a single nozzle with filament feeding, retraction, switching and consumption monitoring. Fig 1

No. of Pages : 24 No. of Claims : 19

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130468 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GPS-BASED INTELLIGENT COLLEGE BUS TRACKING AND PROXIMITY-BASED SAFE HOLD SYSTEM

(51) International classification	:G08G 1/123, G06Q 50/20, G08G 1/127, H04W 4/029, H04W 4/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. V. C. BHARATH SABARISH
2)DEEPHIKA K
3)DHANYASRI P
4)SENTAMILSELVI M
5)Gayathri S

(57) Abstract :
GPS-BASED INTELLIGENT COLLEGE BUS TRACKING AND PROXIMITY-BASED SAFE HOLD SYSTEM The GPS-Based Intelligent College Bus Tracking and Proximity-Based Safe Hold System addresses a common problem faced by college students who miss their bus by a short distance or a few minutes. Conventional bus tracking systems provide only vehicle location information and lack a controlled mechanism to assist students who are close to the bus. The proposed system introduces a smart proximity-based hold mechanism combined with live bus monitoring, enabling authorized users to track bus movement and helping students determine whether they can reach it in time. An eligible student can send a hold request through the application, which is evaluated by the system and authorized transport staff. Upon approval, a dedicated device on the driver's dashboard displays an alert and the permitted hold duration, allowing the bus to wait at a suitable boarding point without the driver operating the app while driving. The system also limits repeated hold usage, improving coordination between students, drivers, and transport staff while reducing missed buses and maintaining safe, organized transportation.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130476 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART STOVE 360 – AI POWERED UNIVERSAL LPG GAS STOVE WITH INTELLIGENT COOKING CONTROL AND MULTI-SENSOR SAFETY SYSTEM

(51) International classification	:F24C 3/12, F24C 3/10, F23N 5/24, F24C 3/00, F24C 3/08	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India
(31) Priority Document No	:NA	**2)M.KUMARASAMY COLLEGE OF ENGINEERING**
(32) Priority Date	:NA	**3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(33) Name of priority country	:NA	(72)**Name of Inventor :**
(86) International Application No Filing Date	: :01/01/1900	**1)Dr. DEEPAN KUMAR MADHU** **2)HEMA VARSHINI VINOBA**
(87) International Publication No	: NA	**3)MR EZHILAN SENTHILKUMAR**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**4)MR SRI SANJIT RAJA** **5)Mrs.SHOBANA M**
(62) Divisional to Application Number Filing Date	:NA :NA	**6)Elamathi E**

(57) Abstract :

ABSTRACT SMART STOVE 360 – AI POWERED UNIVERSAL LPG GAS STOVE The present invention discloses a Smart Stove 360, an intelligent LPG gas stove configured for cooking assistance, real-time monitoring and enhanced cooking safety. The stove comprises one or more LPG burners, temperature sensor, flame sensor, gas-leak sensor, smoke or air-quality sensor, gas valve, electronic control unit, display unit, manual controls, automatic heat-control mechanism, automatic gas shut-off mechanism and wireless communication module. The control unit receives information from the sensors and analyses cooking conditions including normal cooking, boil-over, dry heating, excessive heating, flame failure and gas leakage. Based on the detected condition, the system can regulate burner heat, interrupt LPG supply and generate audible, visual or mobile notifications. The system further provides predefined cooking modes including milk, rice, dal, curry, fry, roast, oil heating and custom modes. Manual controls and mobile application-based control may be provided together with child-lock and emergency shut-off functions. The integrated architecture is intended to improve cooking convenience, fuel efficiency and safety by continuously monitoring cooking conditions and providing automated protective responses.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130477 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WIND-RESPONSIVE AUTOMATIC GRIP SMART CLOTH CLIP

(51) International classification	:D06F 55/00, A41F 1/00, D06F 53/04, D06F 53/00, D06F 57/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Ms.Sarnika JR
2)Ms.Nabithrha JR
3)Mr.Balaji M
4)Dr.Velmurugan M
5)Mrs.Swetha Bharathi
6)Mrs. KALAIVANI K
7)Karthik M

(57) Abstract :
ABSTRACT WIND-RESPONSIVE AUTOMATIC GRIP SMART CLOTH CLIP The present invention discloses a wind-responsive automatic grip smart cloth clip (10) for holding clothes securely on a clothesline during changing wind conditions. The smart cloth clip (10) comprises a clip body (20), an upper jaw (21), a lower jaw (22), a wind sensor (30), a control circuit (40), a rechargeable battery (50), a micro servo motor (60), a grip adjustment mechanism (70) and a spring (80). The wind sensor (30) detects the wind speed and supplies the measured value to the control circuit (40), which compares it with a predefined threshold level and, based on the comparison, commands the micro servo motor (60) to actuate the grip adjustment mechanism (70) so as to increase the gripping force of the jaws (21, 22). The spring (80) provides the normal gripping force and supports the return movement of the jaws (21, 22). The rechargeable battery (50) is chargeable through a micro USB charging port (51) and the micro servo motor (60) is configured to actuate only when required, thereby reducing power consumption. The present invention (10) reduces the accidental displacement of clothes during windy conditions. Fig 1

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130478 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE CLOSED-LOOP WATER PURIFICATION AND QUALITY MONITORING SYSTEM

(51) International classification	:C02F 1/28, G01N 33/18, C02F 1/44, C02F 1/42, C02F 1/00	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr V BANUPRIYA**
(33) Name of priority country	:NA	**2)SHARANYA S R**
(86) International Application No Filing Date	: :01/01/1900	**3)VASUDEVAN M** **4)TAMILSELVAN N**
(87) International Publication No	: NA	**5)NITHYA MEENAKSHI S**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)Mr. SANTHOSH KUMAR PALANIAPPAPILLAI CHOCKALINGAM** **7)FAZILA M**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Abstract ADAPTIVE CLOSED-LOOP WATER PURIFICATION AND QUALITY MONITORING SYSTEM The present invention relates to an adaptive closed-loop water purification and quality monitoring system for treating and monitoring contaminated water in rural, remote, and mining-affected areas. The system comprises a raw-water sensing arrangement configured to monitor selected water-quality parameters including pH, turbidity, total dissolved solids/electrical conductivity, and temperature, and a main control unit configured to determine an appropriate treatment pathway based on the sensed water condition. An automatically controlled treatment-routing manifold selectively directs water through one or more purification modules comprising sediment filtration, activated-carbon treatment, contaminant-specific adsorption or ion-exchange treatment, membrane-based purification, and UV-C disinfection. Following treatment, a post-treatment verification arrangement monitors selected water-quality parameters and provides a verification condition to the control unit. An outlet safety valve permits dispensing of treated water when the configured verification condition is satisfied and prevents dispensing when the condition is not satisfied, while enabling re-treatment or rejection of the water. The system may further include flow and pressure monitoring, water-level and leakage detection, local status indication, IoT communication, rechargeable battery, and solar power-management arrangements. Fig 1

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130479 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WATERPROOF AND BREATHABLE PADDY STORAGE AND TRANSPORTATION BAG

(51) International classification	:B65D 33/01, B65D 81/26, B65D 30/08, A01F 25/16, A01F 25/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. V. C. BHARATH SABARISH
2)NITHESWARAN S
3)SHANMUGANATHAN K
4)Mrs.S.KAYATHRI
5)Bhavani P

(57) Abstract :
ABSTRACT WATERPROOF AND BREATHABLE PADDY STORAGE AND TRANSPORTATION BAG The current invention relates to a rain resistant and breathable paddy storage and transport bag (10) that consists of an outer rain resistant layer (11), an air channel spacer layer (12), a waterproof vapor barrier membrane (13), an inner paddy contacting liner (14), a moisture periphery channel (15), moisture outlets (16), reinforced seams (17), reinforced bottom portion (18), and a closure mechanism (19). The outer layer (11) serves to protect against rain penetration. The spacer layer (12), located between the liner (14) and the membrane (13), creates an air passage for moisture vapor towards the periphery channel (15) where it is further guided to moisture outlets (16), equipped with protective flaps (20) limiting their exposure to the rain. The membrane (13) creates a barrier against liquid water allowing for controlled passage of vapor. The reinforced seams (17) and bottom (18) give mechanical strength to the bag for multiple uses. Fig 1

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130480 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HygroVent – A Passive Hygromorphic Ventilation Louver System for Post-Harvest Grain and Food Storage

(51) International classification	:A01F 25/14, A01F 25/22, A01F 25/16, F24F 3/14, F24F 7/013	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mrs. SHOBANA M** **2)AHAMED KHAN A** **3)BARANIDHARAN S** **4)JANA M** **5)PRADEEP V** **6)Dr. M. CHRISPIN DAS** **7)ATHITHYA A**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT HygroVent – A Passive Hygromorphic Ventilation Louver System for Post-Harvest Grain and Food Storage A passive, humidity-actuated ventilation device for grain and food storage enclosures is disclosed. The device comprises a bank of bilayer composite louver blades, each formed from an active hygroscopic layer bonded to a passive dimensionally stable layer, mounted within a frame suited for retrofit installation into an existing warehouse wall vent opening. Differential expansion between the bonded layers causes each blade to curl open as ambient relative humidity rises into the range associated with stored-grain spoilage risk, typically 65 to 75 percent, and to relax closed as humidity falls below that range. The device requires no electrical power, sensor or control circuitry, and offers a low-cost, reliable alternative to electronic humidity-control systems, particularly in regions with unreliable grid power. Fig 1

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130481 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A FLAME OVERSPILL THERMAL INDICATOR SYSTEM FOR REAL-TIMEQUANTIFICATION OF WASTED COOKING ENERGY IN OPEN-FLAME GAS STOVES

(51) International classification	:F24C 3/12, F24C 3/10, F24C 3/08, F24C 3/00, G01J 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India

2)M.KUMARASAMY COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)NIRMAL SIVAKUMAR

2)SATHISH KUMAR GURUSAMY

3)MOHAMED ZUBARULHAQ ABDUL RAHMAN

4)POOVARASAN

5)RITHAN VENKAT

6)MURALI KRISHNAN

7)MOHANAGANESH BALAKRISHNAN

8)MURUGESH

9)DR JEYAKUMAR

10)DR.B.KIRAN BALA

(57) Abstract :

A FLAME OVERSPILL THERMAL INDICATOR SYSTEM FOR REAL-TIME QUANTIFICATION OF WASTED COOKING ENERGY IN OPEN-FLAME GAS STOVES The present invention relates to a Flame Overspill Thermal Indicator System for real-time quantification of wasted cooking energy in open-flame gas stoves. The system comprises a sensor ring assembly, a plurality of IR sensors, a control housing, a display module, a USB port, a central flame probe, radial support arms, and a Type-C port. The sensor ring assembly is configured to surround a burner region and continuously monitor thermal radiation generated by the flame. The IR sensors and central flame probe acquire thermal information and transmit the data to a processing unit housed within the control housing. The processing unit analyzes temperature distribution patterns to detect flame overspill conditions and estimate energy losses resulting from inefficient combustion. The calculated information is displayed through the display module for user awareness. The invention enables real-time monitoring of burner efficiency, reduction of fuel wastage, improved cooking performance, and enhanced energy conservation in domestic and commercial cooking applications. Fig.1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130482 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI DRIVE COLD PLASMA BASED E-WASTE RECYCLING SYSTEM

(51) International classification	:B09B 3/00, H05H 1/24, C22B 7/00, B09B 5/00, H05H 1/46	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** Address of Applicant :The Principal, K.Ramakrishnan College of Engineering, NH-45, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr SIVARANJANI SUBRAMANIAN**
(33) Name of priority country	:NA	**2)SUBASRI KRISHNASAMY**
(86) International Application No	:	**3)SUBIKSHA THIRUCHELVAN**
Filing Date	:01/01/1900	**4)VIBUL SUBRAMANI**
(87) International Publication No	: NA	**5)VISHAL SENTHIL**
(61) Patent of Addition to Application Number	:NA	**6)Mr. H. RAMAKRISHNAN**
Filing Date	:NA	**7)THANUJA T**
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :
ABSTRACT AI DRIVE COLD PLASMA BASED E-WASTE RECYCLING SYSTEM Electronic waste is one of the fastest-growing waste streams, containing valuable metals, plastics, and other complex materials that are difficult to recover efficiently using conventional recycling methods. This project introduces an ai drive cold plasma-based e-waste recycling system, an intelligent and integrated recycling platform capable of material-dependent processing and resource recovery using cold plasma technology. The proposed system incorporates a waste feeding unit, shredding mechanism, material separation unit, AI-based gas selection system, cold plasma reactor, monitoring system, and material collection units for controlled processing of electronic waste.The AI-based control layer analyzes the identified material category and assists in selecting a suitable processing gas and operating condition for plasma generation. Different gases such as nitrogen, oxygen, argon, and helium can be selected according to the material characteristics and required treatment conditions. The cold plasma reactor generates energetic electrons, ions, radicals, and excited species through electrical discharge, enabling controlled interaction with the processed e-waste. For plastic-containing materials, plasma-assisted depolymerization can break polymer chains and generate useful gaseous products including hydrogen-rich syngas. For metallic fractions, plasma processing supports surface treatment and facilitates recovery of valuable constituents such as copper, aluminum, silver, gold, and other recoverable materials. Fig 1

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130483 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART INTERLOCKING BRICK

(51) International classification	:B09B3/35, B09B3/30, C04B22/04, B07B1/28, B03C1/005, B03C1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)PRIYAN ANBALAGAN
2)DR RAJKUMAR TAMILARASAN
3)MOHAMED ZUBARULHAQ ABDUL RAHMAN
4)SARANYA SRINIVASAN
5)NIRMAL SIVAKUMAR
6)GOWTHAM SELVA KUMAR
7)MOHANAGANESH BALAKRISHNAN
8)LOGESHKUMARAN ASHOKKUMAR
9)SURESH T

(57) Abstract :
ABSTRACT SMART INTERLOCKING BRICK The present invention relates to a magnetically identifiable brick comprising a magnetically responsive material recovered from machining waste generated during a lathe machining operation. The machining waste is collected and subjected to a suitable separation or treatment process to obtain a selected magnetically responsive fraction. The recovered material is prepared and incorporated into a binder or brick-forming composition in a controlled quantity before the mixture is moulded and processed to form the brick. The resulting brick contains the magnetically responsive material within at least a portion of its body and thereby produces a detectable magnetic response. A wireless magnetic scanning system comprising a magnetic sensing unit and processing controller is configured to detect the magnetic response of the brick. The detected signal may be processed to determine the presence or location of the brick and may optionally be compared with stored magnetic response information for identification. The invention provides an approach for utilizing selected machining waste as a functional component of a construction brick while enabling wireless and non-destructive detection of the resulting brick after manufacture and installation.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130484 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTOMATED SELF-SERVICE WEB-BASED PRINTING SYSTEM

(51) International classification	:G06Q 50/20, G06F 3/12, G06Q 30/00, G06Q 10/00, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. V. C. BHARATH SABARISH
2)NITHESWARAN S
3)ROHIT KRISHNA J
4)Mrs. M SHOBANA
5)Bhavani P

(57) Abstract :
ABSTRACT PRINTEASY – AN AUTOMATED SELF-SERVICE WEB-BASED PRINTING SYSTEM This invention relates to a self-service, web-based document printing system (10) for colleges and other organizations. The system includes a user device (11), a wireless Internet or Wi-Fi network (12), a web application (13), a document upload module (14) and a document analysis module (15) that analyzes the number of pages in the document being uploaded. Printing settings module (16) takes in printing settings, while cost calculation module (17) calculates the cost of printing. Online payment module (18) makes online payments, after which the QR code creation module (19) creates a QR code for the job. Schedule module (20) schedules when the printing will take place, and printer queue management module (21) manages the queue among the printers (22). Processor (23) manages all these activities, including analyzing, calculating cost, payment processing, scheduling, and releasing upon verification of QR code, while cancellation module (24) cancels the print job. Fig 1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130485 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SMART ASSISTIVE WALKING STICK WITH AUTOMATIC HEIGHT ADJUSTMENT, MULTI-ZONE OBSTACLE DETECTION AND DIRECTIONAL HAPTIC GUIDANCE

(51) International classification	:A61H 3/06, A45B 3/04, A45B 3/00, A45B 9/02, A61H 3/02	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)HARIHARAN BABU VENKATESWARA BABU**
(86) International Application No Filing Date	: :01/01/1900	**2)AKSHYASHRI BALAKRISHNAN** **3)ASHWINI .M**
(87) International Publication No	: NA	**4)ANUSHKA .M**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**5)Dr. DEEPAN KUMAR MADHU** **6)MANOJ S**
(62) Divisional to Application Number Filing Date	:NA :NA	**7)Elamathi E**

(57) Abstract :

ABSTRACT A SMART ASSISTIVE WALKING STICK WITH AUTOMATIC HEIGHT ADJUSTMENT, MULTI-ZONE OBSTACLE DETECTION AND DIRECTIONAL HAPTIC GUIDANCE The present invention discloses a smart assistive walking stick (10) for visually impaired persons. The walking stick comprises a telescopic stick body (20), height sensing unit (21), motorized height adjustment mechanism (22), mechanical locking mechanism (23), multi-zone obstacle sensing unit (24), animal detection unit (25), processing and control unit (26), haptic/vibration units (27), rechargeable battery (28), USB Type-C charging interface (29) and removable non-burning mosquito-repellent module (31). The height sensing unit determines the user's required stick height during setup and the motorized mechanism adjusts the telescopic body accordingly, followed by mechanical locking. The multi-zone sensing unit monitors left, front-left, centre, front-right and right zones for detecting obstacles. The processing and control unit determines the spatial position of detected objects, evaluates available walking directions and selects a safer available direction. Directional information is communicated to the user through distinguishable haptic/vibration patterns. The animal detection unit provides a corresponding warning upon detection of a nearby animal. The rechargeable power system enables portable operation and charging through USB Type-C. The invention provides an integrated assistive system combining height adjustment, environmental sensing, safer-direction selection and haptic guidance.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130486 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART PLANT PROTECTOR BELT

(51) International classification	:A01M 29/16, A01M 29/18, A01M 29/12, A01M 29/10, A01M 29/30	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
Address of Applicant :The Principal, K.Ramakrishnan College of Engineering, NH-45, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)M.KUMARASAMY COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)DR.R.BALAMURUGAN
2)MADHUMITHA M
3)MANASA T P
4)MADHUMITHA E
5)KAVIYA T
6)VIJAYA ARASU INBASEKARAN
7)ATHITHYA A

(57) Abstract :
ABSTRACT SMART PLANT PROTECTOR BELT The present invention relates to a Smart Plant Protector Belt with an Animal Feeding Module for protecting cultivated plants from damage caused by animal intrusion. The system comprises an adjustable plant protector belt, motion sensors, an audible warning unit, an organic scent spray mechanism, an animal feeding unit, level monitoring sensors, a smart notification system, and a solar power unit. When an approaching animal is detected, the system initially activates an audible warning. If the animal continues to approach, an organic repellent is released through spray nozzles positioned around the protected plant. The feeding unit provides suitable fodder at a designated location away from the crop to encourage the animal to move away from the protected plant. The system monitors the levels of organic repellent and fodder and provides refill alerts to the farmer when required. The solar power unit supports autonomous operation in agricultural areas. The invention provides an integrated, sustainable, and animal-friendly approach to crop protection by combining automatic detection, natural deterrence, alternative feeding, smart monitoring, and renewable energy. Fig 1

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130487 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SoC-Based Multimodal Wildfire Intelligence System with Predictive Risk Assessment, Hybrid Energy Harvesting and LoRa Communication

(51) International classification	:G08B 17/10, G01N 33/00, H04W 4/38, G08B 17/117, G08B 17/107	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr.VIMALNATH S** **2)PRIYADHARSHINI K** **3)JANESH G** **4)RANJITHKUMAR S** **5)RITHIS S** **6)PRIYADHARSHINI B** **7)HARINY P** **8)Mrs.MAHIMA R** **9)GANESH S**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT SOC-BASED MULTIMODAL WILDFIRE INTELLIGENCE SYSTEM WITH PREDICTIVE RISK ASSESSMENT, HYBRID ENERGY HARVESTING AND LORA COMMUNICATION The present invention provides a System-on-Chip (SoC) based multimodal wildfire intelligence system for early detection, predictive risk assessment and real-time communication of wildfire related conditions. The system includes a plurality of sensing modules configured to collect multimodal data including concentrations of gasses indicative of fire, environmental parameters and other parameters related to potential wildfire conditions. In one embodiment, the sensing arrangement comprises an NDIR based multi-gas sensing module for monitoring gasses including carbon monoxide (CO), carbon dioxide (CO2) and methane (CH4) along with environmental sensing elements for monitoring temperature, humidity and smoke.The sensed multimodal data is acquired and processed by an integrated processing architecture based on a System on Chip (SoC) comprising an interface for signal conditioning and data acquisition, and an embedded predictive risk assessment engine. The processing architecture conducts local real-time analysis of the acquired sensor parameters to identify variations, patterns and combinations indicative of potential wildfire initiation. The system computes a predictive wildfire risk value based on the processed multimodal information and classifies the monitored environment into one or more pre-defined risk levels. The system further includes a hybrid energy harvesting arrangement that is arranged to harvest electrical energy from a plurality of available energy sources, including a tree-trunk thermoelectric energy harvesting source. FIG 1

No. of Pages : 13 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130488 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RESQTRAK – Gas Triggered Mechanical Auto Extraction System

(51) International classification	:E02D 29/12, E03F 5/02, E03F 7/00, E03F 1/00, E02D 29/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs.K KALAIVANI
2)VISHNU V
3)NIRANJANA SRII S
4)JEEVAMANI S
5)Mr. B. PITCHIA KRISHNAN
6)GOBINATH P

(57) Abstract :
ABSTRACT RESQTRAK – Gas Triggered Mechanical Auto Extraction System The present invention describes a RESQTRAK Automatic Emergency Retrieval System developed to assist workers in hazardous confined spaces such as manholes, sewer chambers, septic tanks, drainage systems, and underground utility chambers. The system uses an H_2S-responsive silver (Ag) layer that forms silver sulfide (Ag_2S), producing a mechanical response in a coupled bilayer and causing it to bend or deform. This movement is transferred through a push rod, lever, and Bowden cable to release a surface latch. Once released, a preloaded spring drives the retrieval drum and winds the cable connected to the worker's full-body safety harness. A guide pulley or roller helps direct the worker toward the opening, while a stabilizing line and speed-control mechanism help reduce excessive movement during retrieval. The portable retrieval structure allows the system to be used at different confined-space locations, while a manual release mechanism provides an additional way to initiate retrieval. The invention combines hazard-responsive sensing, passive mechanical actuation, stored energy, guided retrieval, stabilization, and portability to support emergency worker extraction. Fig 1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130489 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-ANCHORING FLEXIBLE ROBOTIC APPARATUS WITH SURFACE-COUPLED TOMOGRAPHY AND RETRACTABLE PROTECTIVE EFFECTOR

(51) International classification	:B25J 9/16, B25J 19/02, B25J 11/00, B25J 15/06, B25J 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr. V. BANUPRIYA
2)PAVITHRA S
3)KARUNYA S
4)SHRI NITHIN K S
5)SUBASH K
6)Dr. S. SIVANANTHAN
7)PIRIYANGA D

(57) Abstract :
ABSTRACT SELF-ANCHORING FLEXIBLE ROBOTIC APPARATUS WITH SURFACE-COUPLED TOMOGRAPHY AND RETRACTABLE PROTECTIVE EFFECTOR The present invention relates to a self-anchoring flexible robotic apparatus for confined-space intervention in disaster and unstructured environments. The apparatus comprises a variable-stiffness continuum body having granular-jamming chambers that selectively stiffen portions of the body and engage surrounding rubble to form a temporary environmental reaction stator. A surface-coupled tomographic sensing arrangement uses electromagnetic and acoustic sensors together with micro-suction nodes to establish physical coupling with irregular structural surfaces. An onboard cognitive topography engine processes the acquired sensing information and evaluates a proposed debris movement before physical intervention. A flexible invaginated effector is stored within an internal protective cavity and pneumatically deployed when intervention is required. The effector comprises a distributed micro-adhesion matrix using micro-pneumatic suction and optionally electro-adhesion for interaction with irregular objects. After intervention, the effector is retracted into the protective cavity and the robotic body can return toward a compliant configuration.

No. of Pages : 12 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130490 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : T-ATISS: SYSTEM AND METHOD FOR NON-CONTACT SUB-SURFACE RAIL DEFECT DETECTION USING THERMO-ACOUSTIC LASER EXCITATION AND ACTIVE VIBRATION STABILIZATION

(51) International classification	:G01N 29/04, B61K 9/08, G01N 29/24, G01N 29/06, G01N 29/265	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr. V. BANUPRIYA** **2)PRIYADHARSHINI S** **3)REVIN GOLD ABRAHAM R** **4)SAHANAA V** **5)SHIRAVAN R** **6)Dr. SRINIVASAN D** **7)GOBINATH P**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT T-ATISS: SYSTEM AND METHOD FOR NON-CONTACT SUB-SURFACE RAIL DEFECT DETECTION USING THERMO-ACOUSTIC LASER EXCITATION AND ACTIVE VIBRATION STABILIZATION The present invention provides a fully integrated Thermo-Acoustic Track Integrity and Sub-Surface Scanner (T-ATISS) system and method for real-time, non-contact detection of surface and internal sub-surface rail defects during active, high-speed railway transit. Unlike conventional ultrasonic inspection systems that require direct mechanical contact or liquid coupling agents and are therefore limited to low operating speeds, the proposed system enables continuous track monitoring at commercial transit speeds. The system comprises a multi-physics sensor assembly mounted beneath a locomotive bogie. A high-energy pulsed laser generates non-ablative thermo-acoustic shockwaves within the rail material. When these waves encounter internal defects such as micro-cracks, transverse fissures, or thermal fractures, localized heat-flow disturbances are produced on the rail surface. A high-speed differential thermal camera captures these transient thermal variations to generate a subsurface defect profile. To minimize optical distortion caused by bogie vibrations, the sensor assembly incorporates an active electromagnetic stabilization mount controlled using a 6-DOF IMU and Hall-effect position sensors. The system dynamically adjusts electromagnetic forces to maintain sensor alignment and reduce vibration. An aerodynamic pneumatic lens protection system generates a continuous air curtain across the optical window, preventing brake dust, ballast particles, water, and mud from obstructing the camera. Thermal camera and 2D LiDAR data are processed by an onboard edge-AI computing unit using neural-network models combined with transient heat-diffusion analysis. The system estimates the location, depth, width, and severity of detected rail defects. Critical defects are automatically geotagged using GPS and transmitted through a cellular telemetry module to a centralized cloud maintenance platform, enabling real-time alerts, condition monitoring, and predictive maintenance. Fig 1

No. of Pages : 13 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130491 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART RAILWAY EDGE SAFETY AND MONITORING SYSTEM

(51) International classification	:B61B 1/02, B61L 23/04, B61L 23/00, B61L 27/00, G01S 13/931	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)Dr.SHEIKDAVOOD K** **2)HARISH PRANAV S** **3)SUBASHCHANDAR M** **4)DHARANI T** **5)DHARSHINI B** **6)KAMESH KUMAR J** **7)SUJIN INFANT ROY J** **8)Mr. B. VELUCHAMY** **9)GOBINATH P**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT SMART RAILWAY EDGE SAFETY AND MONITORING SYSTEM Railway platform gap accidents are a safety worry. They cause injuries and deaths when passengers fall or get stuck between a train and the platform. Current monitoring systems mainly use CCTV cameras and people watching. They have problems. These problems include lighting, weather, blind spots and slow human reaction.To solve these issues we propose a Train-Mounted Intelligent Platform Gap Safety Monitoring System. This system uses millimeter-wave radar and Time-of-Flight sensing technologies to detect hazards in time.The system has sensor nodes installed on the side of railway coaches. These nodes face the platform edge. Continuously monitor the train-platform interface. They detect people falls, entrapment incidents and unauthorized intrusions into zones. The sensor data is processed by an embedded controller. This controller performs sensor fusion and intelligent decision-making to improve detection accuracy and minimize alarms. When the system detects an event it instantly alerts the locomotive pilot and railway authorities. This enables emergency response and accident prevention.The system can also monitor crowd density near platform edges, track intrusions, water accumulation during weather and fire-related hazards. It does this through the integration of sensors. Unlike camera-based systems our solution preserves passenger privacy. Works well, in different environmental and lighting conditions. Fig 1

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130492 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PASSIVE TIME-INTEGRATING HYDROGEN SULFIDE(H2S) DOSIMETER INCORPORATING CONTROLLED-DIFFUSION COLORIMETRIC SENSING, ONE-TIME ACTIVATION AND IRREVERSIBLE EXPIRY INDICATION

(51) International classification	:G01T 1/02, G01N 31/22, G01N 21/78, G01T 1/10, G01T 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr. SIVANANDAM KALIANNAN
2)ANUDHARSAN RAVIKUMAR
3)HARISH ESWARAN
4)PRIDEESWARI ANANDARAJ
5)SRINIKETHAN BOOPATHYKUMAR
6)DEEKSHITHA SADHANAND
7)PRADEEP MAHENDRAN
8)Mr. B. PRAKASH
9)NAGULAN M

(57) Abstract :
ABSTRACT PASSIVE TIME-INTEGRATING HYDROGEN SULFIDE(H2S) DOSIMETER INCORPORATING CONTROLLED-DIFFUSION COLORIMETRIC SENSING, ONE-TIME ACTIVATION AND IRREVERSIBLE EXPIRY INDICATION The invention describes a passive, time-integrating hydrogen sulfide (H_2S) exposure dosimeter in the form of a flexible, disposable, sticker-type wristband designed to monitor cumulative H_2S exposure during an intended work period. The wristband incorporates a one-time activation mechanism, a validity indicator, a colorimetric H_2S sensing region, a visual exposure reference scale, a reference/blank region, and flexible adhesive backing. Before use, the dosimeter remains in an inactive and protected condition. A removable activation tab is provided to initiate the operating state when the worker begins the intended monitoring period. Once activated, the validity mechanism begins its predetermined operating period and is designed to progress from a valid condition through a warning condition to an irreversible expired condition. Removal of the wristband during the operating period does not reset the activation state or provide a new operating period, thereby preventing a previously activated dosimeter from being treated as a fresh device. The H_2S sensing region uses a multilayer passive construction comprising a gas-permeable protective layer, a controlled-diffusion layer, an H_2S-reactive colorimetric layer, a reflective substrate and a protective backing layer. Hydrogen sulfide present in the surrounding environment passes through the protective and diffusion-control layers and reaches the reactive colorimetric layer, where a chemical reaction produces a progressive and substantially irreversible colour change. The response is intended to represent the exposure history of the sensing material based on the concentration and duration of H_2S exposure within the validated operating range. A visual reference scale provided adjacent to the sensing region allows the developed colour to be compared with predetermined exposure-response categories. A separate reference or blank region provides a visual baseline for interpreting the sensing response, while a separation region maintains distinction between the sensing and reference portions. The sensing chemistry and diffusion structure can be calibrated experimentally under controlled H_2S concentrations, exposure durations, temperature and humidity conditions to establish the corresponding exposure ranges. Fig 1

No. of Pages : 15 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130493 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED SMART GLOVE SYSTEM FOR PREDICTIVE HAZARD ALERTING AND AUTONOMOUS RESCUE SIGNALLING

(51) International classification	:G08B 21/02, A41D 19/015, A41D 19/00, G08B 25/10, G08B 25/01	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)Dr.D.MERCY** **2)MADHUMITHA E** **3)VIDHYASRI S** **4)SARANYA K** **5)Ms. S. ANUSUYA** **6)KOWSALYA K**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AI-BASED SMART GLOVE SYSTEM FOR PREDICTIVE HAZARD ALERTING AND AUTONOMOUS RESCUE SIGNALLING The present invention discloses an AI-based smart glove system for monitoring workers operating in hazardous environments and providing predictive hazard alerting and autonomous rescue signalling. The system comprises a wearable glove, an ESP32 embedded controller, a plurality of sensors, local alert devices, a GPS receiver, a GSM communication module and a rechargeable power supply. The sensors acquire information relating to environmental and physical conditions including gas concentration, temperature, motion, finger movement and electrical-field-related conditions. The embedded controller processes the acquired sensor information and performs an intelligent or AI-based hazard analysis to identify potentially hazardous conditions. Upon detection of a hazardous condition, the system activates an audible buzzer and a vibration motor to alert the worker. The GPS receiver determines the geographical location of the worker, and the GSM module automatically transmits an emergency message containing available hazard and location information to an authorized remote recipient. The system thereby provides an integrated wearable architecture combining continuous sensing, intelligent hazard detection, immediate worker alerting, location determination and autonomous emergency rescue signalling. Fig 1

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130494 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Micro-Avionics Powered Bio-Harness for Active Kinetic Parasite Dislodgement in Free-Ranging Cattle Security

(51) International classification	:A01K 29/00, A01M 31/00, A01M 29/16, A01M 7/00, A01M 29/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr. RAMAKRSHINAN P
2)RAJESWARI GUNASEKARAN
3)PREETHIKA P E
4)PRADEEBA B
5)Mr. S. RAGHUVARAN
6)INFANT AJIN JOSE L

(57) Abstract :
ABSTRACT Micro-Avionics Powered Bio-Harness for Active Kinetic Parasite Dislodgement in Free-Ranging Cattle Security The present invention discloses an autonomous, solar-powered wearable smart collar and harness system designed for non-chemical insect dislodgement, active pest defence, and real-time livestock telemetry in grazing environments. Traditional livestock pest management relies heavily on toxic chemical sprays and manual dips, which cause environmental runoff, parasite resistance, and high labour overhead. The proposed system addresses these challenges through a self-sustaining wearable framework featuring a solar clad and a targeted micro-mist atomizer. operation, a bio-compatible subdermal sensor patch continuously monitors localized spasms, skin twitches, and physiological stress responses caused by insect bites. Sensor signals are evaluated in real time by an onboard TinyML AI microcontroller to confirm pest attack patterns. Upon detection, the hub triggers directional kinetic air-vortex cannons to fire high-velocity toroidal air pulses, physically blasting flies, bees, and ectoparasites off the animal's coat without chemical reliance. For high-stress encounters, a secondary collar-mounted micro-atomizer dispenses a fine burst of an organic calming compound near the facial region to soothe the animal and protect sensitive areas. Powered by harness-mounted flexible photovoltaic cells and a solid-state thin-film battery, the device maintains uninterrupted 24/7 operation. Integrated LoRaWAN and GPS modules continuously transmit real-time location data, battery health metrics, and spatial pest attack hotspots to a cloud database, accessible via an administrative web dashboard and mobile application. The system delivers a chemical-free, fully autonomous, and ecofriendly livestock defence solution that enhances animal welfare while eliminating manual labour and environmental toxicity. Fig 1

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130495 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NON-LINEAR DISTURBANCE COMPENSATION IN ELECTRO-HYDRAULIC SYS-TEMS USING EDGE-COMPUTED EXPONENTIAL FILTERING AND LOCALIZED HARDWARE ACTUATION

(51) International classification	:G05B 11/42, F15B 21/08, B62D 5/06, F15B 13/044, B62D 5/065
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.SHEIKDAVOOD K
2)RAVINDRA A
3)KIRUBANANDAN P
4)HARINI S
5)SHABANA FAREEN N
6)HARISH PRAVAV S
7)Mr. B. PITCHIA KRISHNAN
8)YOGIRAM B

(57) Abstract :
ABSTRACT NON-LINEAR DISTURBANCE COMPENSATION IN ELECTRO-HYDRAULIC SYS-TEMS USING EDGE-COMPUTED EXPONENTIAL FILTERING AND LOCALIZED HARDWARE ACTUATION A non-linear disturbance compensation system is disclosed, designed to mitigate high-frequency terrain shocks and actuator oscillation in heavy-duty electro-hydraulic steering applications. The system utilizes a custom-engineered printed circuit board (PCB) featuring strict physical zone isolation and dynamic copper trace routing to safely separate high-current MOSFET solenoid drivers from a sensitive dual-core microcontroller. The microcontroller locally executes an Exponential Moving Average (EMA) digital filter combined with dynamic Proportional-Integral-Derivative (PID) tuning. By heavily weighting the derivative gain during detected shock events, the system trans-forms the hydraulic cylinder into a rigid damper, completely rejecting high-frequency noise. Experimental testing demonstrates a near-instantaneous transient recovery time, a near-zero tracking error margin, and a greater than 50% reduction in cumulative hydraulic energy effort compared to conventional linear controllers. The localized edge-computation architecture elimin 1ates the requirement for bulky, expensive external industrial Programmable Logic Controllers (PLCs). Fig 1

No. of Pages : 9 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130496 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE MECHANICAL HEART VALVE WITH REAL TIME HEMODYNAMIC SENSING AND PROCESSOR-CONTROLLED LEAFLET ACTUATION

(51) International classification	:A61F 2/24, A61B 17/00, A61B 17/12, A61B 5/02, A61B 5/0215
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)MRS.SHARMILA ESWARAN
2)HARIHARAN GUNASEKARAN
3)RAGUL MANI
4)RANJITHKUMAR SIVAJI
5)VISHAL SANKAR
6)SRINIKETHAN BOOPATHYKUMAR
7)VIGNESH SEKAR
8)RAHUL CHANDRAMOHAN
9)Dr. RADHA N
10)GOPIKA P G

(57) Abstract :
ABSTRACT ADAPTIVE MECHANICAL HEART VALVE WITH REAL TIME HEMODYNAMIC SENSING AND PROCESSOR-CONTROLLED LEAFLET ACTUATION The invention is directed to a processor-controlled mechanical heart valve that provides an improved tilting disc occluder with a built-in sensing, computing and actuating system that transforms a passive pressure responsive mechanical heart valve into an active, adaptive, closed-loop system that also retains the ability to be used as a standard passive mechanical heart valve if the electronics in the system are turned off. An upstream pressure sensor, a downstream pressure sensor and a flow rate sensor constantly monitor the operating conditions of the heart valve and send this information over a data bus to a processor that has been programmed to calculate a desired leaflet trajectory that will minimize turbulence and shear stress while still meeting the physiological demands of the patient. This desired leaflet trajectory is sent to a micro-actuator that applies a supplemental force to the leaflet through a linkage arm that does not override the native pressure responsive movement of the leaflet. However, if power or the processor should fail, the leaflet would simply become a passive element. Thus, the electronics are always supplemental and the safety and reliability of the passive valve mechanism are never compromised. A power management unit and a micro battery that can be recharged by means of a wireless induction coil along with a telemetry unit and an antenna allow for the information from the sensors, the calculated desired leaflet trajectory and the activity of the micro-actuator to be sent to an external receiver for monitoring purposes. This allows for early detection of leaflet malfunction to occur. Fig 1

No. of Pages : 17 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130497 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT DUST SUPPRESSION SYSTEM FOR SMART CONSTRUCTION

(51) International classification	:B24B 55/06, B01D 50/00, B01D 47/06, B28D 7/02, B01D 47/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.NALINI NAGULSAMY
2)HARIKRISHNAN A
3)RANJANI G
4)SUDHARSAN.R
5)NISHANTH M
6)Miss SHAHINA SHALU SEBASTIAN
7)Mrs.M.Ruba

(57) Abstract :
ABSTRACT SILICA SHIELD: INTELLIGENT DUST SUPPRESSION SYSTEM FOR SMART CONSTRUCTION The SilicaShield – Intelligent Dust Suppression System is a portable dust control solution that cuts dust from work such as concrete cutting, tile cutting, marble and granite cutting, drilling, grinding and polishing. It brings together dust collection at the source, suction, particle separation, water mist suppression and fine particle filtration in one mobile unit. A dust collection hood sits close to the tool catching dust at the source and moving the dusty air through a flexible suction hose. A suction motor or blower keeps air moving and sends the collected air into the internal treatment area, where big dust particles are separated and dropped into a special dust bin. The system also uses a suppression mechanism and a HEPA filter to clean up the remaining fine dust. The water mist unit makes a controlled mist that tames suspended dust and the HEPA filter gives cleanup before clean air comes out. A dust sensor keeps checking the dust level. Sends information back to the control system. It has a display, control panel, LED status lights and an emergency stop button letting the operator see what is happening and control dust suppression. The monitoring system can also give alerts when something is wrong or when maintenance is needed. The whole unit sits in a portable case on wheels with a handle making it easy to move around the job site. By putting dust control stages together in one mobile platform SilicaShield gives a cleaner and more efficient way to manage dust right where it starts. This system keeps construction dust from spreading around the job area makes the air cleaner and lessens workers' exposure, to dust. So the SilicaShield offers a portable smart way to capture, suppress, filter, monitor and release clean air in construction settings. FIG 1

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130498 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE SYSTEM FOR EARLY PREDICTION OF EXERCISE-ASSOCIATED MUSCLE CRAMPS

(51) International classification	:A61B 5/11, A61B 5/00, A63B 24/00, A61B 5/22, A63B 71/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.S.MEIVEL
2)Kalaipriyan K,
3)Karanan S,
4)Kavinvarsan M,
5)Kishor K
6)Mr. B. PITCHIA KRISHNAN
7)PRIYADHARSHINI D

(57) Abstract :
ABSTRACT WEARABLE SYSTEM FOR EARLY PREDICTION OF EXERCISE-ASSOCIATED MUSCLE CRAMPS The present invention relates to an AI-Based Wearable System for Early Prediction of Exercise-Associated Muscle Cramps Using EMG, IMU, and TinyML. The system comprises an EMG sensing unit, an IMU sensing unit, an IoT processing board, a TinyML processing unit, a wireless communication unit, and an alert unit integrated into a wearable device. The EMG sensor acquires electrical muscle activity, while the IMU captures motion-related information during exercise. The acquired signals are processed by the IoT processing board for filtering, conditioning, and feature extraction. The extracted physiological and motion features are analyzed using an embedded TinyML model to identify patterns associated with an increased risk of muscle cramps. Upon detecting a predefined risk condition, the system generates an early warning through the wearable device or a connected wireless device. The invention provides continuous, real-time, and localized monitoring and prediction of exercise-associated muscle cramps using a compact wearable platform. Fig 1

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130499 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-EVOLVING ELECTROMAGNETIC METAMATERIAL INTERFACE FOR NON-CONTACT MAPPING AND ADAPTIVE REMEDIATION OF HIDDEN WATER CONTAMINANTS

(51) International classification	:G01V 3/12, G01N 33/18, C02F 1/48, G01N 22/00, C02F 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs KIRUTHIGA THANGAVEL
2)SRINITHI SAKTHI SARAVANA
3)SUBASRI KRISHNASAMY
4)VISNUPRIYA VIJAYAKUMAR
5)Dr.MURALIKRISHNAN P
6)JAFRIN ANISHKA JOHN FREDY

(57) Abstract :
ABSTRACT Self-Evolving Electromagnetic Metamaterial Interface for Non-Contact Mapping and Adaptive Remediation of Hidden Water Contaminants The present invention proposes a Self-Evolving Electromagnetic Metamaterial Interface for Non-Contact Mapping and Adaptive Remediation of Hidden Water Contaminants, an intelligent system that combines RF interrogation, reconfigurable metamaterials, artificial intelligence, and adaptive water treatment. The electromagnetic response obtained from each cycle is analyzed by an adaptive computational unit, which automatically modifies the configuration of the metamaterial interface and performs subsequent interrogations. The resulting multiple RF responses are combined to generate an electromagnetic response signature and contamination map representing the detected condition of the water. Based on the generated electromagnetic map, the system determines a corresponding contamination state and selects or modifies an appropriate remediation process. Following treatment, the water is subjected to RF re-interrogation, and the post-treatment electromagnetic response is compared with the pre-treatment response. If the desired improvement is not achieved, the system automatically modifies the treatment condition and initiates another treatment cycle. The principal novelty of the invention lies in the closed-loop self-evolving interaction between RF interrogation, reconfigurable metamaterial response, computational learning, contamination mapping, and adaptive remediation. Unlike a conventional fixed RF sensing system, the proposed system continuously changes its electromagnetic sensing configuration according to the information obtained from the water and subsequently uses the resulting information to control the treatment process. Fig 1

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130500 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS URGENT ROPE GUIDED ASSIST FOR MINING AND NAVIGATION EDGES

(51) International classification	:G01C 21/34, E21F 17/18, G06K 7/10, G01C 21/36, G08G 1/0968
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr. V. MARISELVAM
2)PRAWIN S
3)SANJAI S
4)SARANRAJ S
5)Mr. BALAKUMAR A
6)SHAM P

(57) Abstract :
ABSTRACT AUTONOMOUS URGENT ROPE GUIDED ASSIST FOR MINING AND NAVIGATION EDGES The suggested invention titled “Autonomous Urgent Rope-guided Assist for Mining Infrastructure & Navigation Edges” is a cost-effective solution aimed at minimizing the accident rate of heavy vehicles operating in open pit mines. Mine roads may experience issues associated with fog, dust, smoke, bad weather visibility, unexpected road obstructions, and unstable GPS communication. Thus, traditional navigation and safety techniques may become ineffective. In the proposed invention, the use of the modular Smart Rope with RFID tags attached to it is intended to allow identification of roadside objects in the local area. Vehicle's RC522 RFID Reader module identifies the segments of the rope near the vehicle, while the HC-SR04 ultrasonic sensor identifies obstacles on the way of the vehicle. Information provided by the sensors is processed by the ESP32 controller that allows the system to provide real-time safety decisions via status LEDs and piezo buzzer. When a critical obstacle is detected, the ESP32 controller triggers the emergency-stop module that is mounted in the prototype vehicle via L298N motor driver. The modular Smart Rope is easy to install, remove, relocate and reuse depending on changing road conditions in the mine. Fig 1

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130501 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE SMART MILK PRESERVATION CAN WITH NON-CONTACT VOLUME SENSING

(51) International classification	:B65D 81/38, B65D 51/24, B65D 81/18, G01F 23/284, G01F 23/296
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mohanraj S
2)Vijay Karur Balasubbiramani
3)Visva Pravin Sairam Subramanian
4)S Yogeswaran
5)Dr. G. KALPANADEVI
6)ABIRAMI R

(57) Abstract :
ABSTRACT ADAPTIVE SMART MILK PRESERVATION CAN WITH NON-CONTACT VOLUME SENSING The present invention relates to an adaptive smart milk preservation can with non-contact volume sensing and real-time quality monitoring. The system comprises a food-grade milk storage chamber, an insulated double-wall structure with a PUF insulation layer, reusable PCM or ice thermal cartridges, a non-contact ultrasonic or radar level sensor, a temperature sensor, a removable pH sensor port, an Electronic Control Unit, a display and indicator unit, a leak-proof lid with seal, ergonomic handles, a drain outlet, and a supporting stand. The thermal cartridges provide passive cooling to maintain milk at approximately 4°C to 8°C without continuous electrical power. The non-contact level sensor determines the quantity of milk in litres without contacting the milk. The temperature and pH sensors provide information to the ECU, which processes temperature, pH, milk level, and storage-time data to generate a milk quality risk indication. The display unit provides the user with milk quantity, temperature, pH, and preservation status, and the ECU generates warnings when monitored parameters deviate from predetermined reference conditions. The invention provides a compact, portable, hygienic, and reusable milk preservation and monitoring system suitable for rural milk collection and transportation. Fig 1

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130502 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ADAPTIVE WEARABLE SYSTEM FOR INTELLIGENT ENVIRONMENTAL SOUND DETECTION AND HAPTIC ALERT GENERATION

(51) International classification	:G08B 6/00, H04R 25/00, A61M 21/00, A61M 21/02, G08B 21/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112
(72)**Name of Inventor :**
1)Dr.SENTHIL KUMAR THIYAGARAJAN
2)HARISH PRANAV SENTHIL KUMAR
3)HEMANATHAN ANNAMALAI
4)HARISHWAR SUNDHAR
5)HARIPRASANTH KARUPPUSAMY
6)Dr .J.ROSELIN SUGANTHI
7)BHAVANISHA S

(57) Abstract :

ABSTRACT AN ADAPTIVE WEARABLE SYSTEM FOR INTELLIGENT ENVIRONMENTAL SOUND DETECTION AND HAPTIC ALERT GENERATION The present invention discloses the LIFE BEEP – a wearable system that senses significant environmental sounds and relays the information to the consumer via tactile and visual means. The system is beneficial to the target users (hearing impaired and/or elderly) and during the target conditions (sleeping or using in noise)It consists of the microphone, analog signal conditioning stage, filter stage, ESP32 processor, adaptable sound-detection module, event-priority evaluator, vibrational motor, indication LED/OLED, and is formed in a form of a compact wearable device. The microphone is used to pick up environmental sounds, which are then amplified with an operational amplifier (realized as IC 741/LM741), filtered in the next stage and supplied to the ESP32 processor to be assessed against the signal characteristics of the known events (car horns, doorbell ringing, alarm/siren sounds, crying baby, etc.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130503 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN OPTOELECTRONIC BIO-CUSHIONED WEARABLE PATCH FOR CONTINUOUS HEEL ISCHEMIA

(51) International classification	:A61B 5/00, A61B 5/11, A61B 5/1455, A61B 5/0205, A61B 5/0245
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.REVATHI G
2)ATITHI A R
3)BHARATHIPRIYA S
4)GIRIJA K
5)Mr.N.R.NAGARAJAN
6)MOUNISH M

(57) Abstract :
ABSTRACT AN OPTOELECTRONIC BIO-CUSHIONED WEARABLE PATCH FOR CONTINUOUS HEEL ISCHEMIA An optoelectronic bio-cushioned wearable patch for monitoring the development of heel ischemia and preventing pressure ulcers in an immobilized elderly person is disclosed. In particular, the present invention relates to a miniature patch-type device, which non-invasively monitors the state of the tissues through the use of optoelectronic sensors. Such a device can be placed on the heel, where pressure ulcers are most likely to develop. The patch includes light-emitting diodes, photodetectors, a microcontroller unit, a wireless communication module, a power supply, and a bio-cushioned layer. The optoelectronic sensors allow the patch to monitor the level of tissue oxygenation, blood perfusion, and microcirculation. These parameters are then processed by the microcontroller unit, which continuously analyzes them to identify early ischemic changes in real-time before they become apparent on the surface. A bio-cushioned layer is provided to help reduce pressure on the heel, alleviating the patient's discomfort. Notably, when the patch detects early signs of impaired tissue perfusion, it is able to transmit a warning message through the wireless communication platform to alert healthcare professionals in real-time. Thus, the present invention provides an effective solution to a long-standing issue, which is critical in terms of prevention and timely treatment. The miniature size and comfortable form factor allow it to be easily worn on the patient, while alleviating pressure on the heel. At the same time, the device allows for continuous monitoring of the tissue state in real-time.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130504 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART IV INFUSION TIME MONITORING AND NOTIFICATION SYSTEM

(51) International classification	:A61M 5/168, A61M 5/172, A61M 5/14, A61M 5/36, G16H 20/17
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
Address of Applicant :The Principal, K.Ramakrishnan College of Engineering, NH-45, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs. KIRUTHIKA GOVINDRAJ
2)KANIMOZHI NAGARAJAN
3)DIVYA KUMARAVEL
4)DEEPIKA RAJU
5)Dr. RADHA N
6)DHANUSH K

(57) Abstract :
ABSTRACT SMART IV INFUSION TIME MONITORING AND NOTIFICATION SYSTEM The invention relates to a Smart IV Infusion Time Monitoring and Notification System configured for continuous monitoring of intravenous fluid administration. The system integrates an infrared (IR)-based drop sensing arrangement and a load-cell-based weight sensing arrangement to monitor infusion activity and determine the remaining fluid quantity. The IR sensing arrangement detects individual drops passing through an IV drip chamber for determining the infusion drop rate, while the load-cell arrangement measures changes in the weight of an IV fluid container. An HX711-based signal acquisition circuit may be used for amplifying and digitizing the load-cell signal. An ESP32 or other suitable microcontroller receives the sensor information and processes the data to determine the infusion drop rate, fluid consumption, approximate remaining fluid quantity, infusion progress and estimated remaining infusion time. The processed information is provided through a local display and visual progress indicators. A unique device identification number may be associated with corresponding room and bed information through software configuration, thereby enabling identification of individual IV monitoring units. The system further provides wireless communication for transmitting infusion parameters and status information to a centralized monitoring dashboard, web interface, mobile interface or other authorized monitoring platform. The system may generate status indications corresponding to normal operation, abnormal or check conditions, near-completion conditions and completion of infusion. The monitoring unit may be powered by a regulated DC supply, rechargeable battery or portable power source. Fig 1

No. of Pages : 24 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130505 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF LEARNING MICROGRID ENERGY OPTIMIZATION AND BATTERY HEALTH PREDICTION SYSTEM

(51) International classification	:H02J 3/32, H02J 3/38, G06N 20/00, G06Q 50/06, H02J 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr. R. Dhivya
2)Navdeep K
3)Pavith S
4)Premnath J K
5)Ranjith Kumar M
6)Dr.S. MUTHUPERUMAL
7)Mr. RAJAVARMAN RAMARAJ

(57) Abstract :
ABSTRACT SELF LEARNING MICROGRID ENERGY OPTIMIZATION AND BATTERY HEALTH PREDICTION SYSTEM The Self-Learning Microgrid Energy Optimization and Battery Health Prediction System is an intelligent energy-management system designed to improve the efficiency, reliability, and sustainability of microgrids by integrating renewable energy generation, battery storage, IoT monitoring, and artificial intelligence. The system collects real-time parameters such as voltage, current, temperature, power generation, energy consumption, battery State of Charge (SOC), and operating conditions through distributed sensors. An IoT-enabled controller transmits the collected data to an intelligent processing module for analysis and prediction. Machine-learning algorithms analyze historical and real-time data to forecast energy demand, renewable energy availability, and battery performance. The system predicts important battery parameters including State of Health (SOH), degradation level, and Remaining Useful Life (RUL), enabling preventive battery management. Based on the predicted conditions, the energy optimization controller intelligently manages renewable power utilization, battery charging and discharging, grid interaction, and connected electrical loads. Smart load scheduling is performed according to load priority and available energy, thereby reducing peak demand and unnecessary grid dependency. A continuous feedback mechanism allows the system to learn from actual operating conditions and improve future predictions and control decisions. The proposed system therefore provides an adaptive and predictive approach to microgrid energy management while improving renewable-energy utilization, extending battery life, reducing energy losses, and enhancing overall system performance and reliability. The proposed system provides a compact, lightweight, cost-effective, and user-friendly point-of-care diagnostic platform suitable for hospitals, diagnostic centres, rural healthcare clinics, home healthcare, and resource-limited environments. FIG 1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130510 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT BASED SMART WEARABLE SHOCK DETECTION AND POWER ISOLATION SYSTEM

(51) International classification	:G08B 21/02, H02J 7/00, G08B 21/04, H02J 13/00, G08B 25/10	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)Dr S SATHISH KUMAR** **2)BARATH T** **3)BHARATH G** **4)DIVYA L** **5)DEEPATH M** **6)Dr.KANNAN VADIVELU** **7)Mr. MATHESWARAN PARAMASIVAM**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT IOT BASED SMART WEARABLE SHOCK DETECTION AND POWER ISOLATION SYSTEM A Proposed Smart Wearable Electrical Safety and Automatic Power Isolation System for enhancing additional safety for the electricians during fieldwork in industrial applications was described. The system comprises small wearable electronic unit and receiver /controller mounted near the electrical distribution. Wearable electronic unit comprises; Sensing circuit, ESP32 micro controller, Wireless communication unit, rechargeable battery and alert units. If a specified dangerous condition of current is found the sensed unit will send signal to ESP32 controller and accordingly the controller will generate emergency signal and transmit wireless to the receiver. The receiver unit is equipped with a wireless receiver and ESP32 controller to check the Emergency Signal and control through a relay of adequate rating, contractor or Shunt Trip Protection interface for powering the appliance. The 162 LCD is responsible for the status of the system, and optional GSM feature allowing sending Emergency SMS to the responsible personnel. GPS / GNSS can be utilized to send approximate position of the worker on the field. This proposed system integrates wearable components, embedded control, wireless communication, and industrial electrical safety into an all-in-one, modular structure, which operates and communicates with MCB, MCCB, RCCB/RCD, RCBO, and earthing protection. The prototype can be explained with isolated, current-limited low-voltage source, and further improvements can be used for more than one person identification, zones isolation, monitoring via the internet (IOT), industrial safety purposes. FIG 1

No. of Pages : 18 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130511 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART POWER GRID GUARDIAN

(51) International classification	:H02J 13/00, G01R 31/08, H02H 7/26, G01R 19/165, H02H 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)KIRUTHIKA S
2)ARSHATH M
3)BARATH M
4)BALADHANDAYUTHAPANI S
5)Mrs.K.Revathi

(57) Abstract :
ABSTRACT SMART POWER GRID GUARDIAN The Smart Power Grid Guardian is an intelligent monitoring and protection system designed to enhance the reliability, safety, and efficiency of modern electrical power grids. The system continuously monitors key electrical parameters such as voltage, current, power, frequency, and temperature using suitable sensors. A microcontroller processes the collected data and detects abnormal conditions or faults by comparing the measured values with predefined safety limits. Upon fault detection, the system automatically activates the protection mechanism to isolate the affected section and generates an alert. IoT connectivity enables real-time remote monitoring, data transmission, fault notification, and event logging. The proposed system provides faster fault response, reduces equipment damage and power interruptions, and supports efficient and reliable smart-grid operation. FIG 1

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130513 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE HYBRID POWER SYSTEM FOR BATTLE FIELD OPERATIONS

(51) International classification	:H02S10/12, H02S10/40, H02J7/35, G05F1/67, F03D9/25
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr BANUMATHI SEMBANAN
2)POORVIKA SASIKUMAR
3)LOHITH RAMESH
4)VINOTHKUMAR SIRINIVASAN
5)Dr P PRAMILA
6)Mr. MATHESWARAN PARAMASIVAM

(57) Abstract :
ABSTRACT PORTABLE HYBRID POWER SYSTEM FOR BATTLE FIELD OPERATIONS The present invention discloses a portable hybrid power system (10) for generating, storing and supplying electrical energy in remote and challenging locations. The system comprises a vertical axis wind turbine (1), wind turbine mount (2), foldable solar panels (3), MPPT hybrid controller (4), Battery Management System (5), lithium iron battery (6), inverter (7) and AC/DC output panel (8). The vertical axis wind turbine (1) and foldable solar panels (3) are configured to generate electrical energy from available wind and solar energy respectively. The MPPT hybrid controller (4) regulates the generated electrical energy and supplies the regulated power to the lithium iron battery (6) for storage. The Battery Management System (5) monitors and protects the lithium iron battery (6) during charging and discharging. The stored DC electrical energy is supplied directly through the DC output or converted into AC electrical energy by the inverter (7) and supplied through the AC output panel. The present invention provides a portable and dependable renewable power source for battlefield operations, remote mountainous villages, emergency situations and other off-grid locations where conventional electricity supply is unavailable or unreliable.

No. of Pages : 28 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130514 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT-BASED AUTOMATIC ARECA LEAF PLATE COUNTING AND PACKAGING AND MONITORING SYSTEM

(51) International classification	:B65B 57/20, G06M 1/272, B65B 57/18, G06M 7/00, B65B 57/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr SATHISH KUMAR S
2)BARATH T
3)DHANIKESH S
4)DIVYA L
5)Dr.S.Selvakumar
6)Mrs. JASMINE RAJENDRAN

(57) Abstract :
ABSTRACT IOT-BASED AUTOMATIC ARECA LEAF PLATE COUNTING AND PACKAGING AND MONITORING SYSTEM The proposed Smart Areca Leaf Plate Counting and Packaging Monitoring System is an IoT based automation system that can be used in the areca leaf plate manufacturing industries to solve the problems of manual plate counting and packaging. The conventional approach includes the manual counting of each areca leaf plate and arranged inside the packaging boxes. This leads to problems such as error in plate counting, incorrect quantity of areca leaf plates in packages, increases the workforce and difficult to maintain production records. The proposed system solves all these problems by using a capacitive proximity sensor to count the areca leaf plates to high accuracy, which are transported on a conveyor belt. The microcontroller ESP32 is used to process the sensor signal and calculate the total number of areca leaf plates. The system counts the areca leaf plates and automatically detects complete areca leaf plate packs of desired quantity (e.g., 50). For every detected pack, it increments the pack counter and resets the plate counter and announces the pack completion by a voice alert through a DFPlayer Mini and a speaker. The system provides a TFT display to provide the information like plate count, pack count, selected plate size and system status. It provides options for changing the plate size and desired quantity of areca leaf plate packs. Further, the system uses the inbuilt Wi-Fi of the ESP32 to upload the production data to an IoT cloud and view the data on a mobile or web dashboard. With this, the production managers can monitor, the total number of plates, total packs produced, daily production statistics and past production data. The proposed system provides a low-cost and reliable IoT based automation system for smart areca leaf plate industry with enhanced accuracy in counting, consistent packing, improved production and better data management. FIG 1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130515 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IoT-BASED SMART FLOWER WASTE MANAGEMENT AND AI-ASSISTED RESOURCE RECOVERY SYSTEM

(51) International classification	:G06N 20/00, G06Q 50/02, G06Q 50/26, G06Q 10/04, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.R.DHIVYA
2)ANBUSHELVAN P
3)DEEPAK A A
4)Dr. S. Dineshkumar
5)Mrs. VALLI PRIYADHARSHINI KRISHNAMURTHY

(57) Abstract :
ABSTRACT IOT-BASED SMART FLOWER WASTE MANAGEMENT AND AI-ASSISTED RESOURCE RECOVERY SYSTEM The present invention pertains to a system for intelligent prediction and optimization of flower-waste recovery in the IoT world by efficiently utilizing the flower-waste generated by temples, flower-shops, markets, marriage-halls, festivals and the like. The system consists of a flower-waste collection and separation unit, a sensor unit, an ESP32 based microcontroller, a wireless communication system, a cloud or database system, a machine-learning prediction system, a resource-recovery unit and a validation unit. Flow quality is determined using sensors which measure characteristics of the flower-waste batches collected, including weight, moisture, temperature and optionally pH The ESP32 collects the sensor data and sends it to a remote processing server via an IoT communication network. The measured properties and actual recovery values are recorded for the purpose of creating a dataset that is specific to the flower-waste. The measured characteristics are used in a machine-learning model, typically a Decision Tree, to determine a recovery pathway and/or a recovery outcome. The flower waste is then treated along the chosen route and the actual recovery is compared to the predicted recovery. The validated information can be incorporated into the dataset to be used for model improvement. The invention combines IoT monitoring, machine learning prediction, resource recovery and experimental validation for sustainable and data-driven flower waste management. FIG 1

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130517 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT BASED SOLAR HYBRID ELECTRIC GRAIN AND ONION DRYER

(51) International classification	:A23B 9/08, F26B 25/22, F26B 3/28, F26B 25/00, F26B 21/10	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)KIRUTHIKA S**
(32) Priority Date	:NA	**2)ARAVINDU K**
(33) Name of priority country	:NA	**3)BALA AKAS S**
(86) International Application No Filing Date	: :01/01/1900	**4)ADHIKESAVAN A** **5)Mrs .N.Nithya**
(87) International Publication No	: NA	**6)Mrs. VALLI PRIYADHARSHINI KRISHNAMURTHY**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT IOT BASED SOLAR HYBRID ELECTRIC GRAIN AND ONION DRYER The proposed IoT-Based Solar Hybrid Electric Grain and Onion Dryer is an innovative solution that combines renewable solar energy with electrical backup and IoT monitoring for efficient drying of agricultural produce.The system utilizes an ESP32 microcontroller to control the heater and blower fan based on real-time temperature and humidity readings. The data is uploaded to an IoT dashboard for remote monitoring and control.By using solar energy as the primary source, the system reduces dependency on electricity and ensures sustainable, eco-friendly, and uniform drying of crops. This invention is ideal for small-scale farmers and food processing industries seeking low-cost automation and energy efficiency. FIG 1

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130518 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED AMMONIA DETECTION AND NEUTRALIZING SYSTEM IN POULTRY FARMS

(51) International classification	:A01K 45/00, A01K 31/00, G05D 11/13, G01N 33/00, B01D 53/78	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 TRICHY Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)KIRUTHIKA S
2)RAMESH M
3)YOGAPRIYA B
4)YAASHIKA K
5)SUMANRAJ D
6)Dr.KANNAN VADIVELU
7)Mrs. UMA MAGESHWARI SWAMINATHAN

(57) Abstract :
Abstract AUTOMATED AMMONIA DETECTION AND NEUTRALIZING SYSTEM IN POULTRY FARMS An automated ammonia monitoring and neutralization system for poultry farms that includes an ammonia gas sensor set up to always sense ammonia concentration in a poultry farm environment, a control unit electrically connected to the ammonia gas sensor for operation, a neutralizing solution tank, and a spray mechanism that is fluidly connected to the neutralizing solution tank, where the control unit is capable of automatically turning on and adjusting the spray mechanism if the detected ammonia concentration goes beyond a preset threshold level thereby spraying a neutralizing solution which is proportional to the detected ammonia concentration in order to bring down ammonia levels and preserve a safe poultry house environment without manual intervention. FIG 1

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130519 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IoT-Based Smart Mining Worker Safety Monitoring and Alert System

(51) International classification	:E21F 17/18, G08B 21/16, G08B 21/02, G08B 21/18, H04W 4/38	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)Dr. HARIPRABHU MANOHARAN** **2)Mr.SHARVESH SENTHILKUMAR** **3)Mr.NANDHISH PERIYASAMY** **4)Mr.KESAVAN ADAIKKAPITCHAI** **5)Mrs. MAHESWARI MOHAN** **6)Mr. I. Infant raj**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT IoT-Based Smart Mining Worker Safety Monitoring and Alert System Mining environments are among the most hazardous workplaces, where workers may be exposed to methane accumulation, oxygen deficiency, abnormal temperature conditions, and sudden falls. The proposed IoT-Based Smart Mining Worker Safety Monitoring and Alert System addresses these risks through continuous multi-parameter sensing and automated hazard indication. The system uses an Arduino Nano to acquire and process data from an MQ-4 methane gas sensor, ME2-O2 oxygen sensor, DHT22 temperature sensor, and MPU6050 motion sensor. The NodeMCU ESP8266 provides Wi-Fi-based connectivity for IoT/cloud monitoring. Sensor readings are evaluated against predefined safety thresholds and the overall condition is classified as Safe, Warning, or Critical/Danger. When hazardous conditions are detected, the system activates the buzzer and LED indicators and presents relevant information on the LCD display. By combining environmental monitoring, worker fall detection, embedded processing, local warning mechanisms, and remote IoT connectivity, the proposed system supports continuous safety monitoring and faster awareness of hazardous conditions. The system is intended as a cost-effective and portable solution for improving worker safety and supporting supervision in mining environments. FIG 1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130521 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIO-INSPIRED AUTONOMOUS RESCUE UAV WITH INTEGRATED MISSION DECISION, TERRAIN FUSION AND ADAPTIVE PAYLOAD DEPLOYMENT SYSTEM

(51) International classification	:B64C 39/02, G05D 1/10, G05D 1/00, B64D 1/02, B64D 47/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr DHIVYA R
2)SIVAMANICKAM M
3)SRIDHAR S
4)SUDHARSHAN K
5)THIRUKUMARAN P S
6)Srikamali N
7)M.Subashree

(57) Abstract :
Abstract BIO-INSPIRED AUTONOMOUS RESCUE UAV WITH INTEGRATED MISSION DECISION, TERRAIN FUSION AND ADAPTIVE PAYLOAD DEPLOYMENT SYSTEM The present invention relates to VYOMRAKSHA, an autonomous rescue UAV designed for disaster-response applications. The system integrates sensor fusion, mission decision making, terrain assessment and adaptive rescue-payload deployment. A plurality of sensors provides information regarding position, altitude, obstacles, target conditions and battery status. The system processes this information to determine suitable mission actions such as search, approach, hold, reposition, deployment or return. A dedicated payload mechanism enables a rescue capsule to be released from an airborne position without requiring the UAV to land at the affected location. Payload release is performed only after evaluating deployment conditions, while unsuitable conditions trigger safe fallback actions. The invention provides a modular and adaptable architecture for low-cost autonomous aerial assistance during emergency situations.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130522 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-ENABLED IOT-BASED SMART SOLAR HOME ENERGY MANAGEMENT AND PREDICTIVE LOAD OPTIMIZATION SYSTEM

(51) International classification	:H02J 3/14, G06N 20/00, G05B 15/02, H02J 3/00, G06Q 50/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. K. Poongothai
2)Pavith S
3)Ranjith Kumar M
4)Kavitha P
5)DR. K. S. Guruprakash

(57) Abstract :
ABSTRACT AI-ENABLED IOT-BASED SMART SOLAR HOME ENERGY MANAGEMENT AND PREDICTIVE LOAD OPTIMIZATION SYSTEM The system integrates solar energy generation, IoT sensors, battery storage, household loads, and Artificial Intelligence (AI) to monitor and manage energy consumption in real time. IoT sensors continuously collect parameters such as solar voltage, current, battery status, power generation, and household energy consumption, while the data is transmitted to a centralized monitoring platform for analysis. The proposed system uses AI and machine learning techniques to analyze historical and real-time energy consumption patterns and predict future household loads. Based on the predicted demand and available solar energy, the system intelligently prioritizes and schedules electrical loads to reduce unnecessary energy consumption and maximize the use of renewable energy. When solar generation is insufficient, the system can optimize battery usage and minimize dependency on the conventional power grid. A web or mobile dashboard provides real-time visualization of energy generation, consumption, battery status, and predicted load requirements. The proposed solution aims to achieve efficient solar energy utilization, reduced electricity costs, improved battery management, and intelligent load optimization. By combining AI-based prediction with IoT-based monitoring and automated control, the system provides a scalable and sustainable approach for developing smart and energy-efficient homes.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130523 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT BASED SMART DAM WATER LEVEL MONITORING & GATE ALTER SYSTEM

(51) International classification	:G06Q 50/06, G01D 21/02, G01W 1/14, G06Q 50/26, G08B 21/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Ms .K.ANITHA
2)AKASH MA
3)DHINA K
4)LOGRATHAN R
5)Ms. Oviya P
6)R KAMALITHA

(57) Abstract :
ABSTRACT IOT – BASED SMART DAM WATER LEVEL MONITORING & GATE ALTER SYSTEM The present invention relates to an IoT-Based Smart Dam Water Level Monitoring and Gate Alert System for continuous and real-time monitoring of dam water level and environmental conditions. The system comprises water-level sensors, environmental sensors, an IoT controller, a data processing module, cloud communication, local data storage, gate control, alert and notification, user monitoring, and power supply modules. The IoT controller continuously collects water-level, temperature, humidity, rainfall, and water-pressure data and processes the collected information against predefined safety thresholds. When a warning or critical condition is detected, the system generates timely alerts through alarms, mobile applications, SMS, email, or push notifications. The system further enables automatic or remote operation of the dam gate based on monitored conditions. Real-time data may be transmitted to a cloud platform and displayed through a web dashboard or mobile application for remote monitoring by authorized users. Local storage provides backup of sensor readings for further analysis. The proposed invention reduces dependence on manual monitoring, facilitates early detection of abnormal dam conditions, enables faster emergency response, and enhances dam safety and flood-risk management. FIG 1

No. of Pages : 13 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130524 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUALSHIELD: A FAIL-SAFE HELMET–BIKE INTEGRATED EMERGENCY RESPONSE SYSTEM

(51) International classification	:A42B 3/04, B62J 27/00, G08B 21/02, A42B 3/30, A42B 3/22	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mr.P.MANIRAJ**
(32) Priority Date	:NA	**2)SOUNDHARYA R**
(33) Name of priority country	:NA	**3)SIBIDHARAN R**
(86) International Application No Filing Date	: :01/01/1900	**4)MADHUMITHA R** **5)Ms. Ameera H**
(87) International Publication No	: NA	**6)Mrs.Saranya.S**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT DUALSHIELD: A FAIL-SAFE HELMET–BIKE INTEGRATED EMERGENCY RESPONSE SYSTEM The proposed system is an intelligent safety device that would help detect accidents and provide emergency services to two-wheeler riders. The system is made up of a smart helmet and a smart bike component, which would continually communicate between themselves during the journey. The helmet component detects head collisions, vibrations, tilting of the head and movements of the rider, while the bike component detects sudden deceleration, tilting, vibrations, and accidents. The sensor data would be analyzed using the ESP32 microcontrollers, thus improving accident detection and avoiding false alerts. In case there is a potential accident, the helmet will start buzzing for 30 seconds, and the rider will be able to cancel the alert using the push-button. In case of neglect of the rider, the bike unit will automatically initiate the emergency process. The GPS component will locate the location of the rider, while the GSM component will transmit the information of the accident and the location to the emergency contacts. The fool-proof mechanism will make the available component take charge of the emergency process in case one of the components fails.

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130525 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Integrated Solar-Powered Automated Incense Processing System with Intelligent Drying, Robotic Handling and Smart Packaging

(51) International classification	:F26B 9/06, F26B 25/00, F26B 3/28, F26B 23/00, F26B 3/347
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr. P.MANIRAJ
2)KOWSALYA S
3)Mrs. MATHUMATHI MOOKKANDI
4)Dr.T.John Peter

(57) Abstract :
ABSTRACT Integrated Solar-Powered Automated Incense Processing System with Intelligent Drying, Robotic Handling and Smart Packaging Increasing demand for effective and efficient processing of agarbatti requires intelligent drying, robotic picking, and compact packaging systems that can be used effectively by rural women artisans who produce products from home. The current systems for drying and packing are heavily reliant on manual procedures under uncontrolled environmental conditions and may lead to ineffective drying processes, longer processing time, and additional physical efforts. To solve these issues, the proposed project will focus on an Intelligent Solar-Powered Robotic Drying and Compact Packing System for Agarbatti Products. Solar-powered drying chambers will ensure the drying under controlled conditions and reduce reliance on conventional electricity. Sensors will measure the temperature and humidity within the drying chamber, while ESP32 microcontroller will act as a main control system. Based on data received from sensors, the controller will regulate the heater and air flow and also determine the stage of drying process. Once the drying procedure is finished, robotic arm will automatically remove the dried sticks from the drying racks and place them on conveyor belt. The robotic system further performs arrangement and transfer of the sticks to a counting unit, where the required quantity is selected for each pack. The counted agarbatti sticks are then transferred to a compact packaging and sealing unit for final packing. A battery-supported solar power arrangement enables continued operation when solar availability is reduced. The complete system can be operated through a user-friendly control interface and can be developed as a compact, low-cost prototype suitable for home-based production. The proposed system reduces repetitive manual handling, improves drying consistency, supports uniform quantity packaging, and promotes solar-powered and sustainable agarbatti production. The system can be further extended for use by Self-Help Groups, rural women artisans, and small-scale manufacturing units, making it suitable for future intelligent and sustainable rural production technologies.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130526 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IoT-BASED INTELLIGENT POWER EXTENSION SYSTEM FOR ADAPTIVE APPLIANCE CONTROL AND STANDBY ENERGY REDUCTION

(51) International classification	:H01R 13/66, G05B 15/02, H02J 13/00, G05F 1/66, G06F 1/32	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr.Bharathi R** **2)Ashwin Khannaa R L** **3)Balaji S** **4)Kishore Balaji V** **5)Mohammed Ansari Raja A** **6)P.VIGNESHWARAN**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT IoT-BASED INTELLIGENT POWER EXTENSION SYSTEM FOR ADAPTIVE APPLIANCE CONTROL AND STANDBY ENERGY REDUCTION Unnecessary electricity consumption from electrical appliances connected to power extension boards may occur due to standby operation, prolonged inactivity, or appliances being unintentionally left powered. The present invention proposes a Context-Aware Adaptive Fuzzy Energy Management System for Multi-Socket Power Extension using appliance-specific electrical behaviour analysis and verified automatic power control. The system utilizes an ESP32-based embedded controller, electrical load sensors, a PIR-based user-presence sensor, inactivity tracking, individually controllable sockets, and an IoT communication interface. The system continuously monitors appliance electrical behaviour and generates an appliance-specific Energy Fingerprint based on repeated observations of load characteristics, low-power behaviour, operating duration, and electrical state transitions. The Energy Fingerprint is combined with real-time appliance load, user presence, and inactivity duration to determine a contextual appliance state. An adaptive fuzzy logic engine evaluates the contextual information and generates a decision confidence for maintaining power, continuing monitoring, alerting the user, or automatically disconnecting an individual socket. A predictive energy-waste estimation mechanism further estimates potential unnecessary energy consumption if an appliance continues operating.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130527 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-IOT BASED PLANT-LEVEL ROOT-ZONE PRECISION FERTILIZER INJECTION ROVER WITH CLOSED-LOOP SOIL VERIFICATION

(51) International classification	:A01C 21/00, A01C 23/04, A01C 23/00, A01G 25/16, A01B 79/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr .MANIRAJ PERUMAL
2)SHIYAM PONNUSAMY
3)ROOBAN MOHAN
4)PARKAVI RAJA
5)Mr. ADITYA S
6)Mrs.C.Chitra

(57) Abstract :
ABSTRACT AI-IOT BASED PLANT-LEVEL ROOT-ZONE PRECISION FERTILIZER INJECTION ROVER WITH CLOSED-LOOP SOIL VERIFICATION The present invention concerns a rover for root zone fertilizer precision injection in agriculture at plant level by artificial intelligence (AI) and internet of things (IoT). The components of the rover are: mobile drive system, camera for plant detection, a soil moisture sensor, an electrical conductivity sensor, a reservoir for fertilizers, a pump, a controlled valve, a flow sensor, and a retractable root-zone injector/electronic controller. When in operation the rover moves down the rows of crops and selects a particular plant via the image recognition system equipped with the camera. The measured local soil parameters are close to the plant of interest and sent to the controller for deciding upon a suitable micro-dose fertilizer. The retractable injector is then placed close to the root-zone of the selected plant and the predetermined fertilizer dose is injected in a controlled manner. They have a flow sensor to check the actual fertilizer application, ensuring that doses are accurate. Post-Treatment Soil verification is carried out after application by checking soil properties again. An IoT communication arrangement can record and transmit plant-specific soil measurements, dosage of fertilizers and the (actual) delivery and treatment. The invention hence provides an integrated closed-loop plant level fertilizer management solution integrating individual plant detection, local soil sensing, adaptive micro-dosing, target root-zone injection, delivery monitoring and post-treatment soil monitoring. Fig 1

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130528 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-CALIBRATING SMART IRRIGATION SYSTEM FOR GRAPE CULTIVATION USING SOIL MOISTURE AND LEAF ELECTRICAL CONDUCTIVITY FEEDBACK

(51) International classification	:A01G 25/16, G01N 33/24, A01G 27/00, A01G 25/02, G01N 27/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. S. Lavanya
2)Aruna G
3)Dharzhitha Thiyagaraj
4)Dayanidhi
5)P. PARTHASARATHY

(57) Abstract :
Abstract SELF-CALIBRATING SMART IRRIGATION SYSTEM FOR GRAPE CULTIVATION USING SOIL MOISTURE AND LEAF ELECTRICAL CONDUCTIVITY FEEDBACK Grapes cultivation requires precise irrigation, as both overwatering and water stress can significantly affect crop quality and yield. Conventional irrigation systems rely on fixed soil moisture thresholds, which often become inaccurate due to soil variability and sensor drift. This project proposes a Self-Calibrating Smart Irrigation System for Grapes Using Soil Moisture and Leaf Conductivity Feedback. The proposed system aims to integrate a soil moisture sensor, a leaf conductivity sensor, and an ESP32 microcontroller to monitor both soil conditions and the plant's physiological response. A closed-loop feedback mechanism is used to automatically calibrate the soil moisture sensor based on leaf feedback, enabling more accurate and adaptive irrigation decisions. The proposed approach aims to reduce water consumption, minimize manual calibration, and improve irrigation efficiency for high-value grapes cultivation. The complete workflow of the system is given below

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130529 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART POULTRY MONITORING SYSTEM WITH INDIVIDUAL FEED DISPENSING AND EGG-LAYING TRACKING

(51) International classification	:A01K 39/012, A01K 45/00, A01K 39/014, A01K 31/16, G01G 17/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. K.Ravikumar
2)Balashiva S
3)Ms. Kiruba Sri A
4)Mr.P.Kasthuri rengan

(57) Abstract :
ABSTRACT SMART POULTRY MONITORING SYSTEM WITH INDIVIDUAL FEED DISPENSING AND EGG-LAYING TRACKING The present invention discloses a conveyor-based smart poultry feeding and monitoring system for tracking the individual feed consumption and egg-laying performance of each hen on a poultry farm. The system comprises a feed hopper, an automatic feed dispenser with a weighing mechanism that dispenses a fixed quantity of feed into each of 'n' compartments of a conveyor assigned to individual hens, a leftover-feed weighing station employing a load cell to measure residual feed after a fixed period, and an egg detection sensor that identifies whether each hen has laid an egg. The recorded feed-consumption and egg-laying data for each hen are transmitted to a controller or software database, which computes individual feed intake, correlates it with egg output, and automatically flags hens showing low feed intake and no egg production for review or removal. By providing individual-hen-level performance analysis rather than flock-level data, the invention reduces feed wastage, improves feed efficiency, enables early identification of unproductive or unhealthy hens, and reduces manual labour and record-keeping in commercial poultry, layer and breeder farms.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130530 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DYNAMIC 360° REFLECTOR TRACKING TO REDUCE REAR IRRADIANCE MISMATCH IN BIFACIAL PHOTOVOLTAIC SYSTEMS

(51) International classification	:H02S 40/22, H02S 20/32, H01L 31/052, H01L 31/054, H01L 31/042
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)DR. T. HARIHARASUDHAN
2)BRINDHA S
3)SASMITA R
4)LINGESHWARAN A
5)Mrs. DEEPA RAMALINGAM
6)Dr.R.Sasikumar

(57) Abstract :
ABSTRACT DYNAMIC 360° REFLECTOR TRACKING TO REDUCE REAR IRRADIANCE MISMATCH IN BIFACIAL PHOTOVOLTAIC SYSTEMS The present invention relates to a Dynamic 360° Reflector Tracking System for reducing rear irradiance mismatch in bifacial photovoltaic (PV) systems. Bifacial PV panels generate electricity from both the front and rear surfaces, but the rear side often receives non-uniform reflected irradiance due to variations in sunlight, ground reflectivity and environmental conditions, which reduces bifacial gain and overall system efficiency. Existing methods, based on fixed reflectors or servo motor tracking, are unable to continuously optimize rear-side irradiance. The proposed system uses a front irradiance sensor as a reference and multiple rear irradiance sensors to measure reflected irradiance at different points. An ESP32 microcontroller compares the front and rear readings, calculates the irradiance mismatch, and determines the optimum reflector tilt angle. A piezoelectric tilt actuator, offering microscopic tilt adjustment, high precision, fast response and ultra-low power consumption, adjusts the reflector accordingly. The reflector surface is coated with eggshell-derived calcium carbonate ($CaCO_3$), providing a sustainable, non-toxic and highly diffusive reflective coating. Through this closed-loop, real-time differential control mechanism, the invention achieves uniform rear irradiance, resulting in higher bifacial gain, increased power output and improved overall efficiency, while remaining a cost-effective and eco-friendly solution suitable for residential, commercial and utility-scale bifacial photovoltaic installations.

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130531 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NEURO-SENSE: ADAPTIVE PERTURBATION-RESPONSE PROFILING SYSTEM FOR EARLY BEHAVIORAL SCREENING IN CHILDREN

(51) International classification	:G09B 19/00, G09B 7/00, G09B 7/02, A61B 5/16, G09B 5/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. S. Geeitha
2)M. Bavitheswari
3)V. Aishwarya
4)A. N. Loshiya
5)Mohamed Rifath A
6)Dr. S. TITUS

(57) Abstract :

Abstract NEURO-SENSE: ADAPTIVE PERTURBATION-RESPONSE PROFILING SYSTEM FOR EARLY BEHAVIORAL SCREENING IN CHILDREN Neuro-Sense is an intelligent embedded behavioral assessment system designed to support the early identification of atypical behavioral patterns in children. The proposed system uses a tabletop device that children interact with while completing structured visual tasks. A healthcare trained examiner guides the child through these tasks. Of relying on the child's ability to read or speak the system collects non-verbal data such, as how quickly the child responds, how accurate they are how many mistakes they make, how many attempts they take how consistent their responses are, how they handle switching between tasks and how their performance changes under controlled conditions. At the start Neuro-Sense gathers a behavioural baseline by tracking how the child performs during standardised tasks. Selected task conditions are then systematically modified, such as introducing visual distractors, changing task rules, altering stimulus positions, or increasing task complexity. The resulting behavioural changes and subsequent return toward baseline are measured and combined with repeated-trial and longitudinal session data to construct an individualized behavioral response profile. The collected data are transmitted through IoT connectivity to a secure digital platform for storage, analysis, and historical comparison. An AI-based analysis module evaluates the multidimensional behavioural features and identifies patterns that may indicate a need for further professional evaluation. The system presents the findings through a dashboard containing behavioural indicators, trends, and an interpretable screening summary.

No. of Pages : 17 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130532 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART TEXTILE WASTEWATER TREATMENT WITH CHLORINE-SAFE ELECTROCHEMICAL CONTROL

(51) International classification	:C02F 1/461, C02F 1/467, C02F 103/30, C02F 1/72, C02F 1/46
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs. K. RANI
2)DHEEPTHI G B
3)INDHU N
4)KEERTHIKA K
5)LATHIKHA R
6)Jayakumar S
7)Guru prakash K S

(57) Abstract :
Abstract SMART TEXTILE WASTEWATER TREATMENT WITH CHLORINE-SAFE ELECTROCHEMICAL CONTROL The present invention relates to a chlorine-safe electrochemical wastewater treatment system for the treatment of high-salinity textile wastewaters contaminated with various dyes and pollutants. Textile wastewaters with high salt contents require special considerations for their treatment by means of electrochemical methods, since improper control of the current can lead to the generation of undesirable chlorine. The chlorine-safe electrochemical wastewater treatment system for the treatment of textile wastewaters includes a wastewater sensing unit, an ESP32-based edge controller connected to a Pulse Width Modulation (PWM) power supply and an electro-oxidation core with titanium electrodes. The sensor unit was used to monitor the pH level and total dissolved solids (TDS) of the wastewater. The readings were processed by an ESP32-based edge controller that adjusted the power supply to the electro-oxidation core in terms of PWM. The chlorine-safe electrochemical wastewater treatment system for the treatment of textile wastewaters ensures the adaptation of the treatment process to the current properties of the wastewater and minimizes the risk of improper chlorine generation. The electro-oxidation core allows for dismantling the dye pollution mechanisms while minimizing the risk of electrode scaling and passivation due to the optimized pulse operation. The treated wastewater can be discharged into the reclamation system. The system also includes secure wireless telemetry to report relevant information to a cloud-based dashboard. Thus, the proposed solution can ensure efficient and smart treatment of textile wastewater with minimum chlorine safety risks.

No. of Pages : 7 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130533 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EcoShield AI: Intelligent Environmental Safety Device using Embedded AI and IoT

(51) International classification	:G01D 21/02, G06Q 50/26, G05B 19/418, G08B 21/18, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MRS. S. SUBHA
2)NATHIYA M
3)SUBHA R
4)THANUSIYA K S
5)SAKTHIVEL J
6)KASTHURI S

(57) Abstract :

Abstract EcoShield AI: Intelligent Environmental Safety Device using Embedded AI and IoT The present invention relates to an AI-assisted intelligent industrial environmental safety system for continuous, proactive, and real-time monitoring of multiple environmental parameters and early detection of potential industrial hazards. The system comprises an environmental sensing unit including MQ135, MQ2, DHT22, and water-level sensors, an ESP32 embedded controller, an artificial intelligence-based processing unit, wireless communication module, local alerting components, automated preventive control mechanisms, and mobile or web-based dashboard. The environmental parameters are continuously sensed and the corresponding sensor outputs are acquired by the ESP32 controller and processed using an embedded AI-based analysis mechanism. The processed sensor information is analysed through a multi-sensor environmental risk assessment mechanism, wherein combined environmental parameters are evaluated to identify abnormal patterns and potential hazardous conditions. An AI-based anomaly detection and risk prediction mechanism further analyses variations and trends in the sensor data to determine the likelihood and severity of developing environmental hazards. The resulting information is used to generate a dynamic Environmental Risk Score and corresponding risk level for intelligent safety decision-making. Based on the determined or predicted risk level, the system may automatically initiate preventive actions such as activating ventilation fans, exhaust systems, cooling mechanisms, pumps, or other connected safety devices, while simultaneously generating local and remote alerts. The environmental data, risk score, predicted trend, alert information, and preventive action status may be transmitted through Wi-Fi or other wireless communication to a mobile or web-based dashboard for real-time monitoring, historical analysis, and digital reporting. The proposed system provides a compact, scalable, cost-effective, and user-friendly industrial environmental safety platform suitable for factories, manufacturing plants, chemical industries, warehouses, processing units, industrial laboratories, and other industrial environments. The integration of multi-sensor environmental monitoring, embedded artificial intelligence, anomaly detection, risk prediction, adaptive risk assessment, automated preventive control, and IoT-based digital monitoring enables a proactive approach to industrial environmental hazard detection and prevention.

No. of Pages : 9 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130534 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LORANET: AN INFRASTRUCTURE-INDEPENDENT LORA MESH NETWORK WITH PIEZOELECTRIC SOS SIGNALING AND HYBRID ULTRASONIC VICTIM LOCALIZATION FOR DISASTER RESPONSE

(51) International classification	:H04W 4/029, H04W 84/18, G08B 21/02, H04W 4/90, G08B 21/06	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr.S.Sathiyanathan
2)Kesevan M
3)Jai Nithil R
4)Lisanth P
5)Kamalesh E S
6)Sarnika JR
7)Dr.T.M.Nithya

(57) Abstract :
ABSTRACT LORANET: AN INFRASTRUCTURE-INDEPENDENT LORA MESH NETWORK WITH PIEZOELECTRIC SOS SIGNALING AND HYBRID ULTRASONIC VICTIM LOCALIZATION FOR DISASTER RESPONSE The current literature has been mainly concerned with the problems in vehicle safety in terms of the following aspects: driver fatigue, distractions, accident detection, emergencies, and occupant presence. Numerous driving monitoring strategies use different technologies such as CNN, LSTM, YOLO, Transformers, multimodal, and computer vision. Meanwhile, accident response systems employ such sensors as accelerometers, GPS, GSM, and heart rate sensors. Nonetheless, these approaches have a narrow focus and are unable to integrate road conditions, vehicle conditions, driver's condition, and occupant safety continuously. This is where the revolutionary RVDP FUSION emerges as an Edge-AI solution that integrates monitoring of Roads, Vehicles, Drivers, and Passengers. What distinguishes the solution is that it is capable of bringing together different data streams and evaluating an overall safety-risk situation dynamically rather than detecting individual instances. The system can identify sleepy drivers, distracted drivers, road conditions, vehicle health, passengers in distress or even a possible medical emergency, and accidents among other things. From the knowledge obtained through the risk assessment process, the system can respond appropriately by sending alerts, vibrations, emergency calls, sharing GPS locations, and issuing emergency notifications. Through such a process, RVDP-FUSION is able to create a continual Safety Loop including Road, Vehicle, Driver, and Passenger in order to enable not only proactive risk management but also emergency response when needed. The key benefit provided by RVDP-FUSION is that it provides cross-domain safety intelligence at the edge, thus creating a safety ecosystem instead of fragmented monitoring solutions. Fig 1

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130535 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE EDGE-DRIVEN INTELLIGENT AQUATIC REVITALIZATION SYSTEM

(51) International classification	:G01N 33/18, C02F 1/00, G06Q 50/26, C12M 1/34, G06Q 50/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr.K.Sivakumar
2)Arul Kumaran T
3)Dharshikhan V R
4)Midhun V
5)Mr. Mohamed Asil
6)PRIYADHARSHINI J

(57) Abstract :
ABSTRACT ADAPTIVE EDGE-DRIVEN INTELLIGENT AQUATIC REVITALIZATION SYSTEM The present invention relates to an intelligent autonomous water-quality monitoring and algae management system, referred to as Aqua Sentry, for continuous monitoring and management of aquatic environments. The system integrates an embedded controller, water-quality sensing modules, an imaging unit, an AI-based early algae detection module, a hypoxia prediction module, an adaptive intake assembly, a pump mechanism, an algae and waste collection unit, and a controlled algae management mechanism. The sensing subsystem continuously monitors relevant water conditions, while the imaging unit captures aquatic images for detecting excessive algae accumulation. The embedded controller processes the acquired information and identifies abnormal algae growth and potential oxygen-depletion conditions. When algae accumulation exceeds a predefined threshold, the controller automatically operates the adaptive intake and pump assembly to collect algae and floating waste and transfer the collected material into a collection chamber. The system may further activate a controlled algae-suppression mechanism according to detected conditions. The proposed invention provides an autonomous, low-maintenance approach for real-time water monitoring, early hypoxia-risk identification, algae collection and aquatic environmental management, thereby reducing dependence on continuous manual inspection and intervention.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130536 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RFID–FSR SENSOR FUSION FOR AUTHENTICATED SEAT-BELT COMPLIANCE AND ANTI-BYPASS OCCUPANT SAFETY IN INTELLIGENT VEHICLES

(51) International classification	:B60R 22/48, G06K 17/00, B60R 21/015, G06K 7/10, G06K 19/077	(71)**Name of Applicant :** **1)M.Kumarasamy College of Engineering** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)Mrs C THILAGAVATHI** **2)ADITHYA M** **3)DIVAKAR V** **4)KAVIN N** **5)MANIKANDAN PRABHU C** **6)Nithika S** **7)Dr.R.Sasikumar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Abstract RFID–FSR SENSOR FUSION FOR AUTHENTICATED SEAT-BELT COMPLIANCE AND ANTI-BYPASS OCCUPANT SAFETY IN INTELLIGENT VEHICLES The present invention discloses an intelligent RFID–FSR Sensor Fusion System for authenticated seat-belt compliance and anti-bypass occupant safety in vehicles. The system integrates RFID authentication, Force Sensitive Resistor (FSR) sensing, MPU6050 motion monitoring, GPS-based location tracking, and emergency communication technologies to provide comprehensive seat-belt verification and occupant protection .The RFID subsystem verifies the authenticity of seat-belt buckles, while the FSR sensor confirms actual seat-belt usage by measuring the pressure exerted on the occupant. Vehicle movement and driving conditions are continuously monitored using accelerometer and gyroscope sensors. Data collected from all sensing modules are processed through a sensor fusion algorithm that classifies seat-belt status as genuine usage, improper fastening, tampering attempt, or critical safety condition .Upon detection of prolonged unsafe conditions, the system activates warning alerts and automatically transmits location-based emergency notifications using GPS and wireless communication technologies. The proposed invention offers a practical, low-cost, and highly reliable solution for improving occupant safety, preventing seat-belt misuse, and supporting emergency response operations in modern intelligent vehicles.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130537 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Smart Helmet with Real-Time Accident Detection and Emergency Alert System Using Embedded IOT

(51) International classification	:A42B 3/30, A42B 3/04, G08B 21/02, G08B 25/10, B60R 21/0132	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr. A. Syed Musthafa** **2)Navdeep K** **3)Nishanth P R** **4)Premnath J K** **5)Mahalakshmi K** **6)R.Nandhini**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Abstract Smart Helmet with Real-Time Accident Detection and Emergency Alert System Using Embedded IOT The present invention relates to an intelligent embedded IoT-based smart helmet system for real-time accident detection, automated emergency distress dispatch, and preventative vehicle ignition control. The system comprises an embedded sensing unit, core processing microcontroller, local feedback interface, GPS localization module, cellular/IoT communication module, and vehicle ignition interlock module. During vehicle operation, an accelerometer, gyroscope (MPU6050), and piezoelectric vibration sensor monitor kinematic forces to detect severe impacts and rollovers while distinguishing them from minor drops. When an accident threshold is breached, the microcontroller initiates an audio-visual countdown window via a buzzer and display, allowing an uninjured rider to manually cancel false alarms. When uncancelled, the GPS module resolves live coordinates, and the communication interface automatically transmits location-tagged SOS alerts via cellular SMS and IoT cloud networks to pre-stored emergency contacts and rescue teams. Furthermore, an integrated breath alcohol sensor (MQ-3) and wear sensor verify rider sobriety and helmet usage before enabling vehicle ignition through a wireless relay. The proposed system provides an integrated approach to collision detection, false-alarm mitigation, autonomous SOS routing, and secure vehicle operation. It reduces preventable road fatalities and provides a practical approach for improving two-wheeler safety. The system is designed to provide reliable crash verification, rapid emergency response, and centralized monitoring of safety functions.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130538 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT IOT-ENABLED AGRICULTURAL WATER MONITORING AND MANAGEMENT SYSTEM

(51) International classification	:A01G 25/16, G06Q 50/02, G01M 3/28, G06Q 50/10, G06Q 50/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs S.Kanimozhi
2)Navetha Sri R M
3)Roshan Begam J
4)Varshini S
5)Janani
6)K.Revathi

(57) Abstract :
Abstract AN INTELLIGENT IOT-ENABLED AGRICULTURAL WATER MONITORING AND MANAGEMENT SYSTEM The present invention relates to an IoT-based smart irrigation leak detection system configured for continuous monitoring and detection of water leakage, blockage, or abnormal flow conditions in agricultural irrigation pipelines. The system comprises a first YF-S201 flow sensor positioned at an inlet portion of the pipeline, a second YF-S201 flow sensor positioned at an outlet portion, an Arduino UNO controller, an ESP8266 Wi-Fi communication module, a relay-controlled solenoid valve, a buzzer, a display unit, and a Blynk-based monitoring interface. The two flow sensors continuously measure water flow and provide corresponding signals to the Arduino controller. During normal operation, the inlet and outlet flow measurements remain within an acceptable range. When leakage, blockage, pipeline break, or cut-off occurs between the two sensors, the outlet flow decreases significantly compared with the inlet flow or becomes zero. The Arduino identifies the abnormal condition by comparing the two flow measurements against a predefined threshold. The detected condition is communicated through the ESP8266 Wi-Fi module to the Blynk monitoring platform for real-time notification. The system may additionally activate a buzzer and display the detected condition locally and may control a solenoid valve through a relay to interrupt or regulate water flow. The proposed system provides a low-cost, automated, and remotely monitored solution for reducing water wastage and improving reliability of agricultural irrigation systems.

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130539 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ADAPTIVE LSTM - BASED SENSOR FUSION SYSTEM FOR SCAFFOLD JOINT HEALTH ASSESSMENT AND SAFE LIFE PREDICTION

(51) International classification	:G06N 3/04, G06N 3/08, G01M 99/00, G06Q 50/08, G06N 3/0442
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs C Thilagavathi
2)N Navishna
3)D Monisha
4)R Rasmitha
5)Nandhaa K G
6)R.ABIRAMI

(57) Abstract :

Abstract AN ADAPTIVE LSTM - BASED SENSOR FUSION SYSTEM FOR SCAFFOLD JOINT HEALTH ASSESSMENT AND SAFE LIFE PREDICTION Scaffold failures are among the leading causes of accidents in the construction industry, often triggered by joint damage, excessive loads, and structural instability. To address this challenge, we propose an Adaptive LSTM-Based Sensor Fusion System for Scaffold Joint Health Assessment and Safe Life Prediction. The system employs real-time monitoring through multiple sensors—load, vibration, and tilt—strategically installed at scaffold joints. These sensors capture critical structural parameters, which are then combined using sensor fusion techniques to provide a holistic view of joint conditions. An Adaptive Joint Health Index is calculated to quantify the health status of each joint on a scale, enabling precise evaluation of structural integrity. Historical sensor data and health scores are processed using a Long Short-Term Memory (LSTM) model, which is particularly effective for time-series analysis and predictive modeling. The LSTM model forecasts the remaining safe life of scaffold joints, allowing proactive interventions before failures occur. Based on predictive outcomes, the system generates intelligent maintenance recommendations such as inspection, repair, or replacement. This predictive approach ensures early fault detection, reduces the likelihood of scaffold collapse, and minimizes maintenance costs. By integrating AI with sensor fusion, the system enhances worker safety and operational efficiency. Furthermore, the proposed solution is scalable, cost-effective, and adaptable to diverse construction environments. It represents a significant advancement in smart construction safety management. With its innovative methodology and practical applicability, the project holds strong potential for patent development and future industrial adoption.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130540 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : EDGE-OPTIMIZED CYBER-PHYSICAL SYSTEM FOR BIOMECHANICAL INJURY

(51) International classification	:A61B 5/11, A61B 5/00, G16H 50/30, G16H 40/63, G16H 40/67
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)K V PRIYADHARSHINI
2)NIKITHAA S
3)NIVETHA K
4)PRASIKA S
5)Balaji M
6)R.Indradevi

(57) Abstract :
Abstract EDGE-OPTIMIZED CYBER-PHYSICAL SYSTEM FOR BIOMECHANICAL INJURY PREVENTION The present invention relates to an edge-optimized cyber-physical system for biomechanical injury prevention. The system comprises a biomechanical sensing assembly, a signal acquisition interface, an edge-processing unit, a memory unit, a feature extraction module, a risk-assessment module, a contextual analysis module, a feedback generation unit, an output interface, and a communication interface. The sensing assembly acquires biomechanical data corresponding to movement orientation, acceleration, angular velocity, joint position, applied force, pressure distribution, or movement duration. The edge-processing unit locally preprocesses the acquired data and extracts biomechanical features for evaluation against stored reference profiles and risk criteria. The contextual analysis module considers individual calibration, activity classification, repetitive movements, and accumulated biomechanical exposure to determine a biomechanical risk indicator. When an alert criterion is satisfied, the feedback generation unit activates a visual, audible, or haptic output to communicate corrective feedback to the user. The communication interface selectively transmits risk indicators and event information to an external monitoring platform. The system supports local monitoring and feedback during network interruptions, subject to available device resources, thereby reducing dependence on continuous remote processing for biomechanical risk assessment. Fig 1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130542 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FLOW TRIGGERED GSM ALERT SYSTEM FOR MUNICIPAL WATER SUPPLY ARRIVAL AND GROUND TANK LEVEL MONITORING

(51) International classification	:G01F 23/00, G08B 21/18, G05D 7/06, G08B 21/00, G08B 25/10	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)Ms.N.Valarmathi** **2)Abishek M** **3)Haris prabu R** **4)KrishnaKumar P** **5)Kumaraguru S** **6)Mr. Rohit S K** **7)R ANUPRIYA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Abstract FLOW TRIGGERED GSM ALERT SYSTEM FOR MUNICIPAL WATER SUPPLY ARRIVAL AND GROUND TANK LEVEL MONITORING This invention describes a GSM-based system for monitoring drinking water arrival and ground tank levels. It works without Wi-Fi or internet. The system uses a flow sensor at the tank inlet to detect when water from the corporation or municipality starts and stops flowing. A level sensor continuously tracks the water level inside the tank. A controller checks the flow data. Ensures that the flow stays above a set threshold for a certain time period before confirming a supply event. This helps avoid alarms from short bursts or sensor noise. The system classifies the tank level into states: empty/very low low normal high near-overflow and overflow. It sends SMS alerts for each condition using a GSM module. These alerts go to numbers that are registered with the system. The controller records when water supply starts and stops. It calculates the duration of each supply event and sends a final SMS with that duration when the supply ends. It also sends alerts if the supply lasts longer or shorter than expected helping to catch situations. To reduce overflow alerts the system uses a dual-condition check. Overflow warnings are only sent when both the level sensor and the flow sensor show overflow. This makes the system more reliable. It also detects flow rates and logs all events. These logs help users analyze past water supply patterns. The system sends alerts when power fails and when it comes back. It supports sending alerts to users at once. This invention is designed for homes and buildings that get water intermittently. It works without internet making it dependable in areas, with connectivity. It helps users get alerts manage water better and prevent overflow and waste.

No. of Pages : 18 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130543 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SETMARS - SMART EMERGENCY TRAFFIC MANAGEMENT AND ROUTING SYSTEM

(51) International classification	:G08G 1/0967, G08G 1/09, G08G 1/01, G08G 1/07, G08G 1/0968
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.S.Bharani Nayagi
2)Kavin Prabhu M
3)Sabari T
4)Shusil Singh K
5)Mr. Aditya
6)Jenitha Princy A

(57) Abstract :
ABSTRACT TITLE: SETMARS - SMART EMERGENCY TRAFFIC MANAGEMENT AND ROUTING SYSTEM ABSTRACT The present invention relates to an intelligent Smart Junction Controller (SJC-01) designed to enhance the monitoring, communication, and control of existing traffic signal infrastructure. The system integrates an ESP32 main controller, GPS module, 4G/Wi-Fi communication module, traffic signal interface module, power management module, surge protection circuit, LCD public notification display, and audible alert system within a compact and protected enclosure. The ESP32 controller processes information received from the connected traffic signal and communication modules, while the GPS module provides junction positioning information and the 4G/Wi-Fi communication module enables wireless connectivity with external monitoring or management systems. The traffic signal interface module facilitates communication between the controller and the existing traffic signal system. The LCD display provides public notifications, while the audible alert system provides additional warnings when required. The controller further incorporates a DC-DC converter, terminal block, and surge protection circuit for regulated power distribution and protection of the internal electronic components. The SJC-01 enclosure is configured for installation on an existing traffic signal pole using mounting brackets and clamps, thereby supporting integration with existing infrastructure. The proposed invention provides a compact, modular, reliable, and scalable smart junction platform for improved traffic signal communication, junction monitoring, and public notification.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130544 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Edge-AI-Enabled Multi-Spectral and Electrochemical Sensor Fusion System for Non-Destructive Milk Adulteration Screening

(51) International classification	:G01N 33/04, G01N 21/31, G01N 21/25, G01N 21/27, G01N 21/35
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.K.V. NITHYA
2)PRAVEEN S
3)SREESHANTH B
4)VELAYUDHAN SHREE R
5)Dr.S.MALATHI
6)SATHYA PRIYA S

(57) Abstract :
ABSTRACT An Edge-AI-Enabled Multi-Spectral and Electrochemical Sensor Fusion System for Non-Destructive Milk Adulteration Screening The present invention relates to an intelligent, portable milk quality analyzer and adulteration detection apparatus designated as SpectraLact, configured for rapid, non-destructive, and reagent-free milk purity evaluation directly at decentralized collection hubs. The apparatus comprises a base chassis integrating an optical sample chamber, an upper probe retention collar, an immersion sensor array including a combination pH probe and a digital temperature probe, a base-mounted turbidity sensor, a multi-wavelength narrow-band LED emitter array (Red, Green, Blue, and Infrared), a diametrically opposed BPW34 silicon PIN photodiode receiver, an on-device alphanumeric display, manual control buttons, an autonomous battery power pack, and an embedded ESP32 microcontroller. During operation, the microcontroller coordinates non-simultaneous pulsed optical excitation across the sample, captures transmitted light attenuation via the photodetector, measures particulate turbidity, acquires electrochemical parameters, and applies real-time mathematical temperature compensation to eliminate thermal drift. The standardized multi-sensor parameters are synthesized into a fused multidimensional feature vector that is evaluated directly by an onboard, quantized edge machine learning model executing locally on the microcontroller, enabling instantaneous discrimination between pure milk and adulterated specimens containing extraneous water, neutralizers, detergents, ionic salts, or thickening agents without requiring external computer resources or chemical consumables. Purity status, classification confidence percentage, and sample temperature are presented directly on the visual display, providing a portable, low-cost, and scalable analytical platform suitable for dairy cooperatives, quality enforcement agencies, and smallholder farming operations.

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130545 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ADAPTIVE THERMAL-MATRIX COOLING APPARATUS FOR PORTABLE COMPUTERS UTILIZING DYNAMIC SPATIAL HEAT MAPPING AND LOCALIZED MICRO-ACTUATION

(51) International classification	:G06F 1/20, H05K 7/20, G06F 1/16, F25B 21/02, H01L 23/467
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr.M. Jaganath
2)Jainivash R
3)Anuraga T
4)Dharanya T
5)Mr. Jayakumar S
6)KAMALITTA R

(57) Abstract :
ABSTRACT AN ADAPTIVE THERMAL-MATRIX COOLING APPARATUS FOR PORTABLE COMPUTERS UTILIZING DYNAMIC SPATIAL HEAT MAPPING AND LOCALIZED MICRO-ACTUATION The present invention discloses an intelligent, closed-loop thermal-matrix cooling apparatus engineered to resolve the structural inefficiencies of standard external thermal management systems for portable computers. The apparatus abandons indiscriminate, static airflow models in favor of a dynamic, precision-targeted thermal response architecture. Integrated beneath a highly conductive top deck is a flexible, high-density thermal sensing matrix utilizing a grid of SMD thermistors. This matrix continuously polls the resting host device, passing high-frequency data to an embedded logic controller. The controller executes a proprietary Adaptive Mapping Engine that renders a real-time virtual spatial heat map of the laptop chassis. Positioned directly beneath the sensing layer is a granular array of independent, variable-speed micro-fans equipped with motorized, vectorable vent vanes. Utilizing a Proportional-Integral-Derivative (PID) control loop, the microprocessor selectively activates, aims, and scales the RPM of exclusively those micro-fans located directly beneath active thermal hotspots. Fans situated beneath cool quadrants remain completely dormant. By dynamically tracking shifting heat signatures and routing airflow solely to critical exhaustion zones, the invention drastically maximizes cooling efficiency while simultaneously minimizing acoustic output and systemic power consumption.

No. of Pages : 15 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130546 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CryoCell THERMO GUARD

(51) International classification	:H01M10/6572, H01M10/613, H01M10/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs.PAVITHRA
2)DHANUSIA NATRAYAN
3)AHKSHAYA LOGANATHAN
4)DHARSHINI SUBRAMANIAN
5)PRAGADHEESHWARAN KARIKALAN
6)Mr. Prithvi Raj S
7)SOUNDARYA C

(57) Abstract :
Abstract CryoCell THERMO GUARD A smart self-cooling battery protection system (CryoCell THERMO GUARD) combines multi-point sensor fusion, edge predictive machine learning, and adaptive thermoelectric active cooling to optimize cell health and mitigate thermal runaway risks. Managed by an ESP32 microcontroller, the system monitors surface temperatures, operating current, bus voltage, airflow, and humidity parameters. When thermal trends indicate elevated heat generation, an on-board algorithm triggers tiered mitigation actions: modulating high-speed cooling fans, energizing a TEC1-12706 Peltier thermoelectric module, throttling charging currents, and activating local RGB/buzzer alerts. If critical temperatures exceed 60∘C, an independent hardware safety interlock physically disconnects the battery pack. Diagnostic telemetry is displayed on a local OLED screen and transmitted remotely via Bluetooth LE.

No. of Pages : 15 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130547 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT ADAPTIVE FARMING ASSISTANT

(51) International classification	:A01G 25/16, G06Q 50/02, G06N 5/04, G06N 20/00, G06N 5/02	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr. R. SUJATHA** **2)SHRAEMITHA B** **3)SRINITHI S** **4)VISHALI P** **5)Ms. Mahalakshmi** **6)B. JULIET CELINE MARY**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
Abstract INTELLIGENT ADAPTIVE FARMING ASSISTANT Agriculture requires timely and reliable decisions related to irrigation, soil conditions, weather changes and crop management. Although IoT and machine-learning technologies have improved agricultural monitoring, many existing systems provide predictions or automated actions without clearly explaining the reasoning behind them. This can make the recommendations difficult for farmers to understand and may reduce trust in automated decision-making. This paper proposes an Intelligent Farming Assistant that provides transparent, confidence-aware and farm-specific decision support. The proposed system collects field data such as soil moisture, temperature, humidity and rainfall and analyses the data using Exploratory Data Analysis (EDA). Simple and interpretable decision rules are then used to generate farming recommendations. Before performing an automatic action such as irrigation, the system evaluates the confidence of the recommendation. When confidence is low, the system asks the farmer for confirmation before taking any action. It does not act automatically in these cases. The system keeps track of changes happening in the field. It watches over things like weather, soil conditions and crop growth. When it detects a change, it updates the way it makes decisions. It also records what changed and the reason behind the change. This helps the system learn over time. The proposed approach aims to improve transparency, safety, personalisation, and farmer trust in intelligent farming systems. Index Terms— Intelligent Farming, Smart Agriculture, Precision Agriculture, Explainable Decision Support, Confidence-Aware Decision Making, Exploratory Data Analysis, Smart Irrigation Fig 1

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130548 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONTACTLESS MOUTH BLOW-ACTIVATED ASSISTIVE CONTROL SYSTEM

(51) International classification	:G10L 15/22, A61M 16/00, G10L 21/0208, A61B 5/08, A61B 5/087
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.P.SAKTHI
2)TRISHNA P S
3)VENNILA K
4)VINOTHA S
5)VISHMIDHA DHARANI S
6)YAZHINI S
7)SATHYA PRIYA B

(57) Abstract :
ABSTRACT CONTACTLESS MOUTH BLOW-ACTIVATED ASSISTIVE CONTROL SYSTEM The invention discloses a Contactless Mouth-Blow-Activated Assistive Control System designed for people with motor difficulties. This Contactless Mouth-Blow-Activated Assistive Control System uses a microphone that sits close to the user's mouth and is attached by a clip. The microphone listens for a mouth-blow. When the system hears a mouth-blow it boosts the signal cleans up noise and then checks it against a threshold that the user can adjust. If the cleaned signal passes the threshold the system declares a mouth-blow event. The Contactless Mouth-Blow-Activated Assistive Control System can count how many valid mouth-blow events happen and look at the timing, between them. By doing it can decide which assistive command to run. A single mouth-blow that is isolated is ignored, so that accidental activation does not happen. Once a command is chosen the Contactless Mouth-Blow-Activated Assistive Control System can turn on a relay that powers a fan, light, buzzer, indicator or nurse-call system. Because the threshold can be tuned the Contactless Mouth-Blow-Activated Assistive Control System can be set to the user's mouth-blow strength and the conditions in which it will be used. The Contactless Mouth-Blow-Activated Assistive Control System gives users a hands-free way to control devices without needing to press a button or use a mouth tube in the basic setup. Fig 1

No. of Pages : 23 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130549 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED HYBRID CHARGING OPTIMIZATION

(51) International classification	:B60L 11/18, B60L 8/00, H02J 7/35, B60L 53/51, H02J 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs N Srija
2)Shalini C
3)Sowmiya M
4)Sujithra G
5)Dr. MANIKANDAN MURUGESAN
6)SUGANYA P

(57) Abstract :
Abstract AI-BASED HYBRID CHARGING OPTIMIZATION The current invention is concerned with an AI-Based Hybrid Charging Optimisation system in relation to an electric vehicle, which includes the incorporation of solar energy production, dual battery storage systems on the vehicle, external grid electricity charging and energy management system. A solar panel fitted to the electric vehicle produces electrical energy that is used mainly for the operation of the vehicle whereas excess solar energy is stored in a first onboard battery. Electrical energy produced by an external electric vehicle charging station is stored in a second onboard battery. An AI-based energy management controller senses the presence of solar energy, electrical energy requirements and the status of charge of the two batteries in order to regulate energy transfer and battery charging. Where there is no sufficient solar energy, the stored solar energy in the first battery is utilized whereas the stored grid energy in the second battery comes into use when necessary. FIG 1

No. of Pages : 11 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130550 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RETROFITTABLE MULTI-PARAMETER EMERGENCY LOAD-RELIEF APPARATUS WITH SELF-LOCKING RETENTION AND EDDY-CURRENT-MEDIATED VERTICAL LOAD DECELERATION FOR SUSPENDED CEILING FANS

(51) International classification	:F04D 27/00, F04D 25/08, F04D 29/00, A62B 1/10, F04D 29/64	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mrs. B. Sribharathi** **2)Mr. M. Jaganath** **3)Thapaswini S** **4)Swathi S** **5)Dr. MANIKANDAN MURUGESAN** **6)N PRAGATHI**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT RETROFITTABLE MULTI-PARAMETER EMERGENCY LOAD-RELIEF APPARATUS WITH SELF-LOCKING RETENTION AND EDDY-CURRENT-MEDIATED VERTICAL LOAD DECELERATION FOR SUSPENDED CEILING FANS The present invention relates to a retrofittable emergency load-relief and controlled-descent system for an existing suspended ceiling fan. The system comprises a structural retrofit housing, multi-parameter sensing subsystem, embedded control unit, mechanical retention assembly, electrically actuated release assembly, cable spool assembly, load-bearing flexible cable, rotational braking assembly, landing-condition sensing subsystem, communication subsystem, local indication subsystem, and independent rechargeable energy-storage subsystem. The sensing subsystem acquires mechanical load, inertial, displacement, vibration, and cable-condition parameters associated with the suspended assembly. The embedded control unit processes and correlates the acquired parameters to determine a predefined emergency condition. During normal operation, the cable spool is mechanically restrained through a self-locking transmission arrangement. Upon determination of the emergency condition, the electrically actuated release assembly disengages the retention path, permitting controlled deployment of the load-bearing cable. The cable spool is connected to a braking system that uses electromagnetic force to slow down the spool by creating resistance between a spinning metal part and a magnetic field. This controls how fast the cable unwinds. A sensor also detects when the descent has finished. The system includes a separate rechargeable power source, local emergency signals, and wireless transmission of emergency status. This setup offers a combined solution that can be added to existing ceiling fan installations. It monitors conditions, holds the fan in place mechanically, controls its release, manages the slow descent, detects when it lands, and sends emergency alerts, all to improve safety.

No. of Pages : 14 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130551 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DROP2LIFE – SMART WATER MANAGEMENT AND COMPLAINT SYSTEM

(51) International classification	:G06Q 50/06, G06Q 10/06, G06Q 50/26, G06Q 10/10, G06Q 10/00	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)Mrs. S Ramya** **2)Praveenaa K** **3)Preethi V** **4)Shrinithi V** **5)Mr. Mohamed Imran** **6)KAMALITTA R**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT DROP2LIFE – SMART WATER MANAGEMENT AND COMPLAINT SYSTEM The present invention relates to Drop2Life – Smart Water Management and Complaint System, an integrated digital platform for water-service complaint handling, operational monitoring, and public communication. The system connects citizens, pump operators, technicians, panchayat officers, block officers, district authorities, and supervisory officials through role-based interfaces. Citizens can submit complaints concerning water leakage, damaged infrastructure, irregular water supply, or poor water quality through registered access or a guest option. Each complaint is assigned a unique ticket ID, associated with a district-to-street location hierarchy, routed to the responsible authority, assigned to a technician, and tracked through predefined status stages. Unresolved complaints can be escalated from panchayat to block and district levels, while notifications communicate important events. A pump-operator checklist records pump running hours, tank level, valve condition, and water-quality parameters including TDS, turbidity, and pH. In the present implementation, water-quality values are generated through software simulation and stored in a Firebase cloud database. The system also provides locality-specific water supply schedules and administrative monitoring. By integrating complaint management, location-based routing, technician assignment, escalation, scheduling, and simulation-based water-quality monitoring in one platform, the invention reduces manual record keeping, improves transparency and accountability, and supports more efficient water-service management in rural and semi-urban areas.

No. of Pages : 9 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130555 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FPGA SELF-HEALTH MONITORING OF BIODEGRADABLE SENSORS

(51) International classification	:G01D 21/02, A61B 5/00, G01N 33/00, G01N 27/416, A61B 5/145
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)DR.S.JEGADEESAN
2)R .SASMITHA
3)K . RUDRAN
4)T . SIVA SELVA DINESH
5)PARTHIBAN NATESAN
6)B. SENTHIL RAJA MANOKAR

(57) Abstract :
ABSTRACT FPGA SELF-HEALTH MONITORING OF BIODEGRADABLE SENSORS The present invention, titled "Self-Health Monitoring of Biodegradable Sensors," relates to the field of biodegradable electronics, smart sensor technology, and sustainable monitoring systems. The invention proposes an environmentally friendly and intelligent sensor monitoring system designed to evaluate the operational health, performance, and degradation of biodegradable sensors during their intended usage period. Conventional electronic sensors often contain non-biodegradable materials that may contribute to electronic waste and environmental pollution after disposal. Biodegradable sensors provide a potential alternative by using materials such as paper, cellulose, natural polymers, and other environmentally friendly substances. However, their electrical and physical characteristics may change due to moisture, temperature, mechanical stress, chemical exposure, and natural material degradation, which can affect their sensitivity, conductivity, resistance, and measurement accuracy. The proposed invention combines a biodegradable sensing element with an electronic monitoring circuit, microcontroller, power supply, and display or indicator unit to monitor the sensor's operational condition. The system is designed to observe relevant electrical characteristics of the biodegradable sensor and collect sensor data during operation. The microcontroller processes the collected information and evaluates changes in selected sensor parameters, such as resistance, conductivity, sensitivity, and signal stability. FIG 1

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130556 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT IOT-BASED SMART HOUSEHOLD SUSTAINABILITY MONITORING AND RECOMMENDATION SYSTEM

(51) International classification	:G06Q 50/06, G06Q 10/06, G05B 15/02, G01N 33/18, G06Q 50/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)DR S LAVANYA
2)SACHIN A
3)SRIRAMKRISHNA V
4)SUJAN P
5)SENTHI BALAJI S
6)Dr. V.C. BHARATH SABARISH
7)SUGANYA .B

(57) Abstract :
Abstract AN INTELLIGENT IOT-BASED SMART HOUSEHOLD SUSTAINABILITY MONITORING AND RECOMMENDATION SYSTEM The present invention relates to an IoT-based smart household sustainability monitoring system configured to monitor water quality, water consumption, electricity consumption, and indoor air quality through a single integrated platform. The system comprises a TDS sensor, water-flow sensor, PZEM-004T energy-monitoring module, MQ135 sensor, ESP32 microcontroller, wireless communication interface, database, web-based dashboard, EcoScore evaluation module, alert module, and artificial intelligence-based recommendation module. Sensor data acquired by the ESP32 is processed and transmitted through Wi-Fi to a centralized monitoring platform for real-time visualization and historical analysis. The EcoScore module combines normalized measurements associated with water, energy, and indoor air-quality conditions to provide an overall indication of household sustainability performance. The AI-based recommendation module analyzes current and historical sensor data to identify undesirable conditions, excessive resource consumption, and potential improvement areas and provides actionable recommendations to the user. The system further generates alerts when monitored parameters exceed predefined or configured conditions. Fig 1

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130557 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART POWER LINE FAULT DETECTION AND ISOLATION SYSTEM

(51) International classification	:G01R 31/08, G01R 31/02, H02H 7/26, G01R 31/58, H02J 13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)DR.S.JEGADEESAN
2)NARENDRAN.M
3)NITHILA.R
4)PRASANNA D.S
5)VIRUTHIKKESH.T
6)PITCHAIMANI T
7)VINCY LEENA F

(57) Abstract :
ABSTRACT SMART POWER LINE FAULT DETECTION AND ISOLATION SYSTEM The present invention relates to a Smart Power Line Fault Detection and Isolation System for detecting and locating faults in underground power cables. The system divides the cable into multiple sections using interval junction nodes and continuously monitors each section using sensors, logic circuits, and an ESP32-based communication module. When a fault occurs, the system identifies the affected cable section and sends the fault information to a monitoring unit. An isolation mechanism disconnects the faulty section from the healthy portions of the network. The system helps reduce fault detection time, unnecessary excavation, and maintenance effort while improving the safety and reliability of underground power distribution. FIG 1

No. of Pages : 14 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130558 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INVISIBLE DOOR BELL ALARM SYSTEM FOR TOUCH-FREE VISITOR DETECTION AND ALERT

(51) International classification	:G08B 3/10, G08B 13/19, H04M 11/04, G08B 13/196, G08B 13/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs.KIRUTHIKA GOVINJ
2)Mrs.SATHIYA AYYADURAI
3)DHARINEESHANBALAGAN
4)Mr. MANJUNATHAN ALAGARSAMY
5)JAYA SHREE V

(57) Abstract :
Abstract AN INVISIBLE DOOR BELL ALARM SYSTEM FOR TOUCH-FREE VISITOR DETECTION AND ALERT The present invention discloses an invisible door bell alarm system comprising a door-side unit (100) and a mobile phone unit (200). The door-side unit (100) comprises an infrared sensor (101) and an ultrasonic sensor (102), each concealed within a door frame, a controller (103) configured to apply a sensor-fusion and debounce logic to detection signals received from the infrared sensor (101) and the ultrasonic sensor (102), a buzzer (104) and a light-emitting diode (105) configured to provide a local audible and visual indication, and a wireless communication module (106) configured to transmit a status packet, over a wireless link (107), to the mobile phone unit (200). The mobile phone unit (200) comprises a wireless receiver (201), an alert application (202), and a display configured to present a notification (203). No button or actuator operable by a visitor is provided at any point of the door-side unit (100); the visitor's presence within a sensing zone is, of itself, sufficient to trigger the local and remote notification. The present invention thereby provides a touch-free, accessible and concealed alternative to a conventional push-button door bell. Fig 1.

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130559 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART IOT-BASED SYSTEM AND METHOD FOR SECURE REAL-TIME MONITORING, SPOILAGE RISK DETECTION, AND TRACEABILITY OF PERISHABLE AGRICULTURAL PRODUCT

(51) International classification	:G06Q 50/02, G06N 20/00, G06Q 10/08, G06Q 10/06, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.K.Sheikdavood
2)Gowtham N K
3)Vasudev sarvajith B S
4)Divani Shree S
5)Sivasankari S K
6)YAZHINI S
7)JOTHIMANOHARI D

(57) Abstract :
ABSTRACT SMART IOT-BASED SYSTEM AND METHOD FOR SECURE REAL-TIME MONITORING, SPOILAGE RISK DETECTION, AND TRACEABILITY OF PERISHABLE AGRICULTURAL PRODUCT AgroTrace Node is a smart tracking system designed to monitor agricultural and perishable products during transportation and storage. The system uses a small IoT device attached to the product package to continuously monitor temperature, humidity, ethylene gas, and package tampering. The collected data is processed by a microcontroller, which can identify abnormal conditions and estimate the risk of product spoilage using a small AI model. Based on the risk level, the device can increase or decrease the frequency of data collection to save battery power while maintaining proper monitoring .The system also provides secure data protection using a unique hardware identity, encryption, timestamps, and chained hashing. If the network is unavailable, the collected information is stored locally and transmitted automatically when the connection becomes available .The verified information can be recorded in a secure digital ledger, creating a traceable history of product conditions, tampering events, and custody changes. A dashboard provides users with real-time information and alerts when storage conditions exceed predefined limits .The system can also use federated learning to improve its spoilage prediction model without directly sharing raw shipment data. Overall, AgroTrace Node provides a simple and secure way to monitor product quality, detect tampering, reduce spoilage, and improve transparency across the agricultural supply chain. FIG 1

No. of Pages : 22 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130560 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A DEEP LEARNING-BASED AISLEVISION SYSTEM FOR AUTONOMOUS PRODUCT DETECTION AND RETRIEVAL

(51) International classification	:G05D 1/02, G06N 3/08, B25J 9/16, G06N 3/04, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**
1)Dr.S.VIMALNATH
2)MANSSHA YONAH A
3)NITHYA SRI R
4)PRIYANKA S
5)SANDHIYA R
6)Dr. V.C. BHARATH SABARISH
7)Ms. B. RAMA

(57) Abstract :
ABSTRACT A DEEP LEARNING-BASED AISLEVISION SYSTEM FOR AUTONOMOUS PRODUCT DETECTION AND RETRIEVAL The present invention describes a deep learning-based autonomous shopping trolley system, referred to as AisleVision, for intelligent product detection, navigation and robotic retrieval in supermarket environments. The system integrates an ESP32 controller, camera-based YOLOv8 product detection, AprilTag-based shelf localization, A* path planning, VL53L0X obstacle detection, RFID product verification and a 4-DOF robotic arm. Customer shopping requirements can be received through voice input or an OCR-based shopping list, while a product database provides product identity, price and shelf information. The trolley determines target shelf locations, plans an efficient route, navigates through supermarket aisles and detects obstacles during movement. At the target shelf, YOLOv8 identifies the requested product and provides its image position, which is used to determine the robotic-arm target through inverse kinematics. The robotic arm retrieves the product using an adaptive gripping operation, after which RFID identification provides an additional product verification layer. Retrieval feedback enables the system to accept a successful retrieval or initiate another attempt when required. For multiple products, the system can optimize the shelf-visiting sequence to reduce unnecessary movement. The system further supports automated billing and payment-controlled retrieval, wherein a digital payment QR can be transmitted to the customer's mobile communication channel and retrieval remains disabled until valid payment confirmation is received. The proposed architecture combines artificial intelligence, autonomous navigation, robotic manipulation, product verification and payment control to provide an integrated autonomous shopping solution. Fig 1

No. of Pages : 22 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130561 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Intelligent Smart Farming and Agricultural Management Platform

(51) International classification	:G06Q 50/02, G06N 20/00, G06Q 30/02, G06Q 10/06, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)DR.S.VANITHAMANI
2)SESHAAN B P
3)MOUNIKA R
4)RITHIKA P
5)Dr. S. MALATHI
6)Dr. G. GABRIEL SANTHOSH KUMAR

(57) Abstract :
Abstract Agriculture plays a vital role in the economy and food production, but farmers often face challenges such as crop diseases, inefficient farm management, lack of timely agricultural information, and limited awareness of government support schemes. These challenges can lead to reduced crop productivity and financial losses. Agri Nexus is an intelligent smart farming and agricultural management platform developed to address these issues using Artificial Intelligence and modern web technologies. The platform provides an integrated solution for farm management, crop management, AI-based crop disease detection, weather information, market price updates, government scheme recommendations, and smart farming guidance. The AI disease detection module enables farmers to upload crop leaf images for identifying diseases and provides information such as disease name, possible causes, symptoms, recommended treatment, and preventive measures. The platform also maintains digital records of farms, crops, and disease history, allowing farmers to monitor agricultural activities efficiently. Additionally, Agri Nexus offers access to weather forecasts, market prices, and government welfare schemes, enabling farmers to make informed decisions and improve productivity. The system is developed using React.js, Node.js, Express.js, MongoDB, and JWT Authentication, providing a secure, scalable, and user-friendly solution. Fig 1.

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130562 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE CHIP FOR A SENSOR TRUST EVALUATION

(51) International classification	:G01D 18/00, G06F 21/57, G06F 21/55, G06F 17/18, G06F 21/56	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr.P.SAKTHI** **2)GOKUL S** **3)KARTHIK G** **4)KAVIPRASATH G** **5)Dr.S.MALATHI** **6)Mr. A. ALBERT ALEXE**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT ADAPTIVE CHIP FOR A SENSOR TRUST EVALUATION The present invention discloses an adaptive chip (10) for evaluating the reliability of sensor data in real time. The chip receives sensor samples through an input interface (11), performs sampling and validity checking, conditions the samples using a preprocessing unit (14), and derives temporal quality features using a statistical feature unit (15). A drift-detection engine (16) compares present sensor behaviour with reference information stored in a baseline memory (17) to identify persistent change distinct from short-duration noise. A trust-evaluation engine (18) combines drift, noise, validity and optional cross-validation information to generate a bounded trust score and classify the sensor as trusted, degraded or untrusted. An adaptive logic core (19) adjusts selected thresholds, filtering parameters or baseline values during reliable operation and restricts or freezes adaptation when the sensor becomes unreliable, thereby preventing fault behaviour from being learned as normal. The decision and output-gating unit (24) supplies the conditioned sensor value together with trust or fault information to a downstream controller. The architecture is suitable for fixed-point FPGA or ASIC implementation and enables early drift detection with deterministic low-latency sensor-health awareness. FIG 1

No. of Pages : 24 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130563 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN RF-BASED HELMET AUTHENTICATION AND MOTORCYCLE STARTER INTERLOCK SYSTEM

(51) International classification	:B62H 5/20, A42B 3/04, B62J 99/00, A42B 3/30, B62J 6/00	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Mrs . KIRUTHIKA GOVINDRAJ**
(33) Name of priority country	:NA	**2)KAMALI PRAKASH**
(86) International Application No Filing Date	: :01/01/1900	**3)DHARSHIKA SUNDARA RAJAN** **4)JAYASRI RAMESH**
(87) International Publication No	: NA	**5)HARSHINI BASKARAN**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)Dr. MANIKANDAN MURUGESAN** **7)R.KAMALITTA**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT AN RF-BASED HELMET AUTHENTICATION AND MOTORCYCLE STARTER INTERLOCK SYSTEM The present invention discloses HelmLock, an RF-based helmet authentication and motorcycle starter interlock system comprising a helmet unit and a bike-side receiver unit. The helmet unit comprises a force-sensing resistor (FSR) pressure sensor (1) positioned within the top padding (8) of the helmet, for detecting the pressure applied by the rider's head when the helmet is worn, an LM358 comparator IC (2) and 10 kΩ potentiometer (3) for evaluating the sensed pressure against an adjustable reference level, and a 433 MHz RF transmitter module (4) for transmitting a coded authentication signal upon detection of the required pressure condition. The bike-side receiver unit comprises a 433 MHz RF receiver module (10) and decoding/signal processing section (11) for receiving and validating the said coded signal, and a transistor-driven 5V relay module (14) for selectively permitting a starter-enable path of the motorcycle through an output terminal (15) and output wires (17). Starter operation of the motorcycle is prevented unless valid helmet authentication is obtained, while the running condition of an already-started engine remains independent of subsequent loss of RF authorization, the relay (14) being associated with the starter-enable path rather than with continuous engine-running control. FIG 1

No. of Pages : 21 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130572 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FOOD CONDITION ASSESSMENT AND MONITORING SYSTEM

(51) International classification	:G01N 33/02, G06N 20/00, G16H 20/60, G06N 20/20, G06N 3/08	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)MRS.P.PRATHIBA** **2)DARSHINI M** **3)MIRTHIKA M** **4)SHYAM THETCHANA GOPAL** **5)Mohamed Faizal M K**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT FOOD CONDITION ASSESSMENT AND MONITORING SYSTEM The present invention relates to Food Trust AI, an integrated portable system for food condition assessment and time-sensitive redistribution. The system comprises an ESP32-based portable sensing device with multiple sensing modules for measuring food-related temperature, humidity, gas-related and non-contact temperature parameters, along with a GPS module for obtaining the assessment location. The acquired measurements are timestamped and transmitted to a cloud-based data storage and synchronization system. A trained XGBoost machine learning model processes the sensor-derived features to determine the food condition and estimate the remaining safe-consumption time. Based on these results, a redistribution decision module determines the urgency of redistribution. The system also uses geographical information of donors and available volunteers to identify or rank a suitable nearby volunteer based on location and operational eligibility. The device provides immediate local information through an OLED display and status indicators, while a role-based mobile application provides separate interfaces for donors, NGOs, volunteers and administrators. Thus, the invention connects on-site food sensing, timestamped data collection, machine learning prediction, safe-time estimation, redistribution prioritization and location-based volunteer coordination to support faster and more informed redistribution of surplus food. FIG 1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130573 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT IoT-ENABLED GPS TRACKING AND MULTI-SENSOR SECURITY SYSTEM FOR REAL-TIME MONITORING OF ELECTRONIC PARCELS

(51) International classification	:G06Q 50/28, G08B 13/196, G06Q 10/08, G08B 25/10, G08B 13/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MRS.R.ELANKEERTHANA
2)KAVI SHREE P
3)SHRI HARINI A
4)YUVAHARSHINI E
5)PRAKASH KANNAN MANI
6)Ponnivalavan M

(57) Abstract :
ABSTRACT SMART PARCEL GUARD: AN INTELLIGENT IoT-ENABLED GPS TRACKING AND MULTISENSOR SECURITY SYSTEM FOR REAL-TIME MONITORING OF ELECTRONIC PARCELS The present invention relates to an intelligent IoT-enabled Smart Parcel Guard system for real-time monitoring of electronic parcels, such as laptops, mobile devices, and computer components, during long-distance and international transportation, achieved through the integration of continuous GPS-based location tracking, multisensor physical and environmental monitoring, tamper/theft detection, camera-based visual verification, intelligent multi-sensor event correlation, and cloud-based remote monitoring within a single protective parcel enclosure. The system comprises an embedded controller, a GPS module, a camera module, an accelerometer, a vibration sensor, a tamper/theft sensor, a temperature sensor, a humidity sensor, a communication module, a rechargeable battery, a battery management system, and an optional vibration energy harvesting unit. The embedded controller continuously acquires and time/location-tags data from the GPS module and sensor suite, and performs event detection by correlating acceleration and vibration data to identify possible impact events, tamper sensor activation with camera capture to identify possible theft events, temperature and humidity readings to identify environmental risk events, and GPS movement patterns to identify possible route deviations. Collected data and detected event information are transmitted to a cloud platform through satellite communication as a primary mode for long-distance and ship-based transportation, with ship Wi-Fi or cellular communication available as an alternative mode, and with local data buffering during communication interruptions. The cloud platform provides user authentication, parcel registration, a unique tracking identifier, route history, sensor readings, security status, camera and event information, battery status, alert history, risk status, and a real-time monitoring dashboard with alert notifications. Taken together, these capabilities allow the invention to improve the traceability, condition monitoring, and security of valuable electronic parcels during transportation across road, port, and international maritime shipment environments. Fig 1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130574 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART WATER MANAGEMENT AND LEAK DETECTION SYSTEM FOR LARGE BUILDINGS

(51) International classification	:G06Q 50/06, G06Q 50/10, G06Q 50/26, G01M 3/28, G01M 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MRS.R.ELANKEERTHANA
2)VIKRAM S
3)LOHITH M
4)SAILESH K
5)VINO BALAJI MUKUNDAN
6)Gayathri S

(57) Abstract :
ABSTRACT Smart Water Management and Leak Detection System for Large Buildings The present invention relates to an IoT-based Smart Water Management and Leak Detection System designed for large buildings such as apartments, colleges, hospitals, and commercial complexes. The system uses water-level sensors and flow meters to continuously monitor tank levels and water consumption. An ESP32 microcontroller collects sensor data and communicates the information through Wi-Fi or IoT connectivity to a Java-based backend for processing and storage. The system automatically controls water pumps based on tank water levels and detects possible leakage by analysing abnormal water-flow and consumption patterns. House-wise or zone-wise water consumption can be monitored using individual flow meters. A web-based dashboard provides real-time information regarding tank status, water usage, leakage alerts, and billing information. The proposed system reduces water wastage, helps prevent tank overflow and pump damage, and supports fair and transparent water billing. FIG 1

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130575 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART MONITORING, REPORTING, AND VERIFICATION (MRV) BLUE CARBON ANCHORING SYSTEM

(51) International classification	:B63B 35/44, H02J 7/35, B63B 35/00, B63B 21/50, B63B 22/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MRS. P. PRATHIBA
2)KOWSIKA T
3)LEKA SRI K
4)POOVIZHI S
5)GOWTHAM SELVAKUMAR
6)Kirthiga R

(57) Abstract :
ABSTRACT SMART MONITORING, REPORTING, AND VERIFICATION (MRV) BLUE CARBON ANCHORING SYSTEM A smart monitoring, reporting, and verification (MRV) blue carbon anchoring system is disclosed for continuous, autonomous telemetry collection in coastal and shallow marine carbon sink environments. The system comprises a floating platform supported by buoyant corner pontoon members, an upper framework holding an inclined photovoltaic solar panel, an elevated GPS module, and an omnidirectional LoRa antenna. Centered on the frame is a sealed, waterproof electronics housing that encloses an ESP32 microcontroller, sensor interface circuitry, a solar charge management unit, and a rechargeable lithium-ion battery. Suspended beneath the floating structure is an automated sub-surface anchoring mechanism driven by a waterproof servo motor and articulated linkage. The mechanism actively drives an anchor rod with a fluked seabed tip into the marine sediment layer, preventing drift caused by tidal currents and wave motion. A modular environmental sensor array suspended alongside the anchor continuously samples pH, salinity, benthic sediment moisture, ambient temperature, and relative humidity. The ESP32 processor collects and validates these readings, attaches precise GPS coordinates and timestamps, and transmits the structured data frames via long-range LoRa telemetry to coastal gateways and cloud dashboards. This provides an autonomous, low-maintenance field station for certifying coastal carbon storage. FIG 1

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130576 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART SCREEN DISTANCE MONITORING AND ALERT SYSTEM FOR CHILDREN

(51) International classification	:H04N 7/18, G08B 21/02, H04N 5/765, H04N 5/445, H04N 21/4363
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr G SUBRAMANIAM
2)BAVANTHIGA SHRI U
3)INDHARAJITH M
4)JEEVA R
5)SUDHARSHAN TAMILARASAN
6)Kavitha M

(57) Abstract :
ABSTRACT SMART SCREEN DISTANCE MONITORING AND ALERT SYSTEM FOR CHILDREN The present invention is a Smart Screen Distance Monitoring and Automatic Video Blanking System that helps children watch screens safely. SafeView contains a HLK LD2450 24 GHz Radar Sensor (1) a 1080p USB Camera (2) an OLED Display (3) RGB LED Indicators (4) a Raspberry Pi 4 (4 GB) (5) a Speaker / Audio Alert (6) a Power Management / Interface Board (7) a Cooling Fan (8) HDMI IN (9) and HDMI OUT (10). The HLK LD2450 24 GHz Radar Sensor (1) watches the child's position all the time. The Raspberry Pi 4 (4 GB) (5) reads the radar data. Tells SafeView what to do. When the child is at a distance SafeView shows normal video. As the child moves closer SafeView shows a caution with the RGB LED Indicators (4) and the OLED Display (3). When the child is too close SafeView makes a sound with the Speaker / Audio Alert (6). If the child ignores the sound and stays close for a set time SafeView will replace the video with a black screen on HDMI OUT (10) while the TV stays on. The Raspberry Pi 4 (4 GB) (5) can send a message to a parent or guardian over the network. SafeView keeps checking the child's distance while the screen's black and will bring back normal video only after the child moves to the safe distance again. By combining distance checks step by step warnings, video blanking, guardian alerts and automatic screen return SafeView gives a full way to keep children safe, from screens. FIG 1

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130577 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Based Pre-Fire Prediction and Human-Aware Adaptive Acoustic Flame Suppression Using Multi-Modal Sensor Fusion

(51) International classification	:G08B 13/12, G08B 13/14, G08B 13/24, G08B 13/22, G08B 13/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.Senthil Kumar Thiyagarajan
2)Saravanan Vadivel
3)Sibikthan Nallipalayam Murugesan
4)Visva Pravin Sairam Subramanian
5)Sankara Narayanan Chinna Annamalai
6)VENKATESWARA BABU HARIHARAN BABU
7)Rani C

(57) Abstract :
ABSTRACT SMART ELECTRIC FENCE SYSTEM The present invention relates to a smart electric fence system for monitoring a protected perimeter and detecting unauthorized intrusion, fence tampering, electrical abnormalities, and power interruptions. The system comprises an electrically conductive fence assembly, an energizer unit, an electrical monitoring module, an intrusion detection module, a controller, a communication module, an alarm unit, and a power supply unit. The electrical monitoring module obtains electrical measurements associated with the fence, while the intrusion detection module detects physical disturbances or unauthorized access events. The controller processes the measurements and sensor signals, compares the processed information with predefined operating conditions, and classifies detected abnormal events. Upon identifying an abnormal condition, the controller activates a local alarm and transmits event information to an authorized remote monitoring device through the communication module. A backup power arrangement may maintain designated monitoring functions during a primary power interruption. The system further supports event recording and controlled energizer isolation under specified unsafe conditions, thereby facilitating real-time perimeter surveillance, fault identification, and improved security management. Fig 1

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130578 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI SMART POTHOLE DETECTION AND DRIVER ALERT SYSTEM

(51) International classification	:G08G 1/01, G06K 9/00, G06N 3/08, G06N 3/04, G06V 20/58
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**
1)MRS.R.KEERTHANA
2)ANGALAESWARI V
3)DHIVYADHARSHINI M
4)HARSHINI S
5)NIRMAL SIVAKUMAR
6)N. PRAGATHI

(57) Abstract :
ABSTRACT AI SMART POTHOLE DETECTION AND DRIVER ALERT SYSTEM The present invention relates to an AI Smart Pothole Detection and Driver Alert System designed for intelligent road monitoring and improved transportation safety through the integration of artificial intelligence-based pothole detection, computer vision, severity analysis, location tracking, real-time driver alerts, and data analytics within a single platform. The system is built around a camera-based road monitoring unit that includes an AI processing module, pothole severity analysis mechanism, location tracking system, driver alert mechanism, data storage system, and intelligent monitoring dashboard. The AI processing module continuously analyzes captured road images or video frames to detect potholes and determine parameters including pothole location, detection confidence, severity level, and distance from the vehicle. Based on the detected severity and distance, the system automatically generates audio or visual alerts to warn drivers about potential road hazards ahead. The location tracking system records the geographical position of detected potholes along with relevant detection information, enabling effective road condition monitoring and maintenance planning. Fig 1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130579 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BASED SMART ENERGY MONITORING AND CONSUMPTION PREDICTIONSYSTEM

(51) International classification	:G06Q 50/06, G01R 22/06, H02J 13/00, G01R 21/00, G01R 21/133	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)MRS.R.ELANKEERTHANA** **2)SAKTHI SABAREESWARAN P** **3)SANJU S** **4)SHAHUL HAMEED M J** **5)PRASANTH ANBALAGAN** **6)Sankari V**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT SMART ENERGY MONITORING AND CONSUMPTION PREDICTION SYSTEM The present invention relates to an intelligent, IoT-enabled smart energy monitoring and consumption prediction system designed for real-time measurement of electrical parameters and predictive forecasting of future energy consumption. The system comprises a DIN-rail mountable outer housing carrying a Wi-Fi / IoT antenna, an LCD display, sensor terminals, and power and load connectors, and internally housing an embedded ESP32 microcontroller, current and voltage sensing circuits, temperature and humidity sensors, a signal-conditioning module, a power supply module and a terminal block. The sensor module continuously measures voltage, current, power, energy, temperature and humidity parameters, which are processed by the embedded microcontroller, displayed locally on the LCD display, and transmitted through the Wi-Fi / IoT communication module to a cloud server. The cloud server hosts an artificial intelligence / machine learning based prediction engine that analyses the historical and real-time consumption data to forecast future energy consumption, which, together with real-time readings, is presented to the user through a web / mobile dashboard application capable of displaying consumption trends, predicted usage charts, and automated alerts upon detection of abnormal consumption or predicted threshold breach. The invention thereby provides a reliable, compact, scalable and intelligent energy monitoring platform suited to residential, commercial and industrial energy management and energy-saving applications. FIG 1

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130580 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI POWERED SMART STORAGE SYSTEM FOR FRESH VEGETABLES

(51) International classification	:F25D 29/00, H02J 7/35, F25D 11/00, A23B 7/04, A01F 25/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs.N.Sarmila Devi
2)Jude Mathew D
3)Mithun S N
4)Nagulan M
5)MADESWARAN KRISHNAKUMAR
6)Nagaraj P

(57) Abstract :
ABSTRACT AI POWERED SMART STORAGE SYSTEM FOR FRESH VEGETABLES The present invention relates to an AI-based solar-powered smart cold storage system designed for the efficient preservation of fresh vegetables and reduction of post-harvest losses. The system integrates an insulated storage chamber, solar power generation, rechargeable battery, environmental sensors, camera monitoring, AI-based processing, cooling system, ventilation system, and humidity-control system within a single platform. The temperature and humidity sensors continuously monitor the internal storage conditions, while the camera captures the condition of the stored vegetables for AI-based analysis. The AI processing module analyses the collected sensor and image data to identify vegetable health conditions, predict spoilage risk, and estimate the remaining shelf life of the stored vegetables. Based on the monitored conditions and predicted spoilage risk, the embedded IoT controller automatically regulates the cooling, ventilation, air circulation, and humidification units to maintain suitable storage conditions. The solar power subsystem supplies renewable electrical energy to the system and stores excess energy in a rechargeable battery for continuous operation. The system further provides real-time monitoring of temperature, humidity, vegetable condition, spoilage risk, shelf-life estimation, battery status, and system operation through an IoT-based monitoring dashboard. Alerts are generated when abnormal environmental conditions or increased spoilage risk are detected. Accordingly, the present invention provides an intelligent, adaptive, energy-efficient, and low-maintenance cold storage solution capable of extending vegetable shelf life, reducing spoilage and food wastage, and minimizing human intervention. FIG 1

No. of Pages : 11 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130581 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RESPONSE-DRIVEN CLOSED-LOOP ELECTROCHEMICAL FINGERPRINTING SYSTEM FOR ADAPTIVE AUTHENTICATION AND ADULTERATION SCREENING OF HERBAL EXTRACTS

(51) International classification	:G01N 27/26, G01N 27/416, G01N 27/49, G01N 27/327, G01N 33/487
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.SHANMUGAVADIVEL G
2)HARIHARAN S
3)DHARMESH V
4)BALA MURUGAN M
5)DHARANEESHWARAN A
6)KRISHNA AAKASH MURALI DHARAN
7)Senega R

(57) Abstract :
ABSTRACT RESPONSE-DRIVEN CLOSED-LOOP ELECTROCHEMICAL FINGERPRINTING SYSTEM FOR ADAPTIVE AUTHENTICATION AND ADULTERATION SCREENING OF HERBAL EXTRACTS The present invention generally relates to response driven closed-loop electrochemical fingerprinting system and method that can be used for authentication and adulteration screening of herbal extracts. The system is provided in the form of a compact and integrated device that can perform the chemical characterization of herbal extracts through electrochemical and gas sensing approaches. The system is configured to produce a characteristic response or electrochemical fingerprint from the sample under test that can be used for subsequent comparison and screening purposes. The proposed system comprises a sample handling and electrochemical sensing unit, an environmental and gas monitoring unit, a control and processing unit, and a power, display, and actuation unit. The sample is introduced into the system through a sample inlet and is guided into a chamber, wherein an electrochemical cell is provided to perform the sensing operation. A pump or a flow control mechanism is used to facilitate the flow of the sample or sample vapors through the system. The sensing unit is equipped with an NH3 sensor, whereas additional sensors for VOCs and TMA can be provided to record additional gas related responses. Additionally, temperature and humidity sensors can be provided to record the prevailing environmental conditions during the measurements. The responses of the electrochemical, gas, temperature, and humidity sensors can be combined to produce a multi-parameter response or electrochemical fingerprint of the herbal extract sample.

No. of Pages : 14 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130586 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART ELECTRIC FENCE SYSTEM

(51) International classification	:G10L 21/0208, H03G 3/34, G10L 15/20, G10L 21/02, H03G 3/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr S SIVARANJANI
2)THARUNIKHA P A
3)THAYA SRI V
4)TRISA P
5)VIJAYA SHREE R
6)RAJA MURUGAN
7)Mohamed Faizal M K

(57) Abstract :
ABSTRACT Adaptive Analog Noise-Level Responsive Suppression Control Circuit for Speech Enhancement The present invention discloses an Analog noise-level responsive suppression circuit for reducing unwanted background noise present in speech and audio signals. The circuit comprises an audio input unit, pre-amplifier, active filter, envelope detector, comparator, noise-level control circuit, and variable suppression stage. The audio input unit receives a speech signal containing both desired speech and unwanted background noise. The pre-amplifier increases the signal level, while the active filter processes the input signal to reduce unwanted frequency components. The envelope detector determines the signal-level information, and the comparator compares the detected level with a predetermined reference level. Based on the detected noise condition, the noise-level control circuit generates a control signal for controlling the variable suppression stage. The suppression level is varied according to the detected noise level, thereby reducing unwanted background noise while maintaining the desired characteristics of the speech signal. The proposed circuit provides a simple, compact, and low-complexity Analog approach for noise-level detection and suppression and may be integrated with speech communication and audio-processing systems.

No. of Pages : 12 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130587 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT MULTI-SENSOR MOBILE PLATFORM FOR RAILWAY SECURITY SCREENING

(51) International classification	:B61L 23/04, B61L 23/00, G05D 1/02, B61L 25/02, B61L 27/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr. M. RAMESH
2)GOWSHICK P T
3)HRISHINY S
4)KAVIYA K S
5)MOHAMED ZUBARULHAQ ABDUL RAHMAN
6)Mohamed Faizal M K

(57) Abstract :
ABSTRACT INTELLIGENT MULTI-SENSOR MOBILE PLATFORM FOR RAILWAY SECURITY SCREENING Current invention involves a portable mobile real time preliminary inspection system for railway security applications. This system essentially makes up a mobile robotic or handheld platform and it consists of a controller, chemical sensory module, environmental sensors, a camera, obstacle- detectors, wireless communicators, alert units, and a power supply. While visually inspecting the environment, the system gathers environmental and chemical data as well. The controller receives sensor input and decides if there is any sign of anomaly against the normal pattern using set threshold or pre-trained reference pattern. If an atypical pattern is noticed, the system gives a local and/ or remote alert and at the same time stores the record with the data of that particular sensory equipment and the time stamp. This record of data can be sent to the security staff monitoring screen. The new system is a portable, fully integrated and data recording unit which has made it much easier for the officials in charge of railway security to do preliminary screening and to get a headstart on warning so that they can intervene timely. FIG 1

No. of Pages : 13 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130588 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Smart Ambulance-Activated Rotating Speed Breaker System

(51) International classification	:B60Q 9/00, B60Q 5/00, B60Q 1/52, A61G 3/00, G08G 1/0967	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)CHOCKALINGAM ALAGAPPAN**
(33) Name of priority country	:NA	**2)KISHOR M**
(86) International Application No Filing Date	: :01/01/1900	**3)MOHAMED ZUBARULHAQ ABDUL RAHMAN** **4)Kirthiga R**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT Smart Ambulance-Activated Rotating Speed Breaker System The present invention relates to an intelligent ambulance-responsive rotating speed-breaker system configured to provide a substantially flat road surface for an approaching ambulance, thereby reducing sudden vertical impact, vibration, and discomfort experienced by patients and medical equipment during emergency transportation. The system comprises a sound detection unit, a microcontroller-based control unit, an electromechanical actuation unit, a rotatable speed-breaker assembly, vehicle detection sensors, warning indicators, and position detection means. The sound detection unit is configured to detect an ambulance siren from a predetermined distance and transmit a corresponding signal to the control unit. Upon confirmation of an approaching ambulance, the control unit activates the actuation unit to rotate the speed-breaker assembly from a normal raised position to a substantially flat position. Visual and audible indicators are simultaneously activated to alert nearby road users regarding the operation of the speed-breaker assembly. After the ambulance crosses the designated road section, vehicle detection sensors determine whether the passage area is clear. Upon confirmation that the area is clear, the control unit commands the actuation unit to return the speed-breaker assembly to its normal raised position. Position detection means may be provided to confirm the raised and flat configurations and to improve operational safety. The proposed invention provides an automated and adaptive road-safety solution that facilitates smoother ambulance passage while retaining the conventional speed-control function for ordinary vehicles. FIG 1

No. of Pages : 21 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130589 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART WATER QUALITY SURVEILLANCE FOR AGRICULTURE USING IoT IN INDUSTRIAL ZONES

(51) International classification	:G01N 33/18, A01G 25/16, G06Q 50/02, C02F 1/00, A01G 25/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. N. MAHENDRAN
2)S. DHARUN KUMAR
3)N. DHARUN
4)A. DHARUNKUMAR
5)S. DHAYANITHI
6)MR.BALAJI M
7)Geetha P

(57) Abstract :
ABSTRACT Smart Water Quality Surveillance For Agriculture Using IoT in Industrial Zones The increasing water pollution in agricultural zones near industrial areas poses a serious threat to crop productivity and soil health. The Noyal River, which passes through textile-industrial zones, carries harmful contaminants and dissolved salts that make the water unsuitable for direct irrigation. This project proposes a low-cost, IoT-based smart water quality surveillance system for farmers using a TDS sensor connected to an ESP8266 microcontroller and OLED display to monitor water quality in real time. When the system detects poor water quality (high TDS), the water is passed through a nano filtration unit to partially reduce TDS while retaining essential minerals required for plant growth. Nano filters are less expensive than RO systems and consume less energy, making them suitable for agricultural applications. The purified water can either be used directly or blended with stored water depending on the final TDS value displayed. This integrated approach of real-time monitoring and partial purification provides a practical, affordable, and sustainable solution for farmers to reuse polluted water safely, thus protecting soil fertility and improving agricultural productivity in industrially affected areas. Fig 1

No. of Pages : 16 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130590 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTONOMOUS ORBITAL DEBRIS DETECTION, TRACKING AND REMEDIATION SYSTEM

(51) International classification	:B64G 1/10, B64G 1/24, B64G 1/64, B64G 1/66, G05D 1/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MRS.R.ELANKEERTHANA
2)SURYA V M
3)ASWANTH M S
4)GURUNANDHAN G
5)ATULKRISHNA VALLIYEDATHU
6)Dhaaraani R

(57) Abstract :
ABSTRACT AN AUTONOMOUS ORBITAL DEBRIS DETECTION, TRACKING AND REMEDIATION SYSTEM The present invention relates to an autonomous orbital debris protection system designed for long-term detection, tracking, and mitigation of orbital debris risk, achieved through the integration of a coordinating Mothership spacecraft, a distributed pico-satellite detection and tracking swarm, autonomous capture spacecraft, and a set of interchangeable servicing payload modules within a single coordinated system. The Mothership comprises an onboard artificial-intelligence computing and flight-control unit, a power management and battery subsystem, communication and navigation/tracking antennas, a docking interface, a propulsion system, a modular payload bay, and a robotic arm for close-proximity capture and payload handling. The onboard computing unit continuously evaluates tracking data contributed by the pico-satellite swarm to identify objects on a predicted close-approach trajectory with a protected satellite, and autonomously selects an appropriate servicing payload, such as a magnetic capture module, a net capture module, an inspection module, an attitude and maneuvering module, or a repair and refuelling module, for the required intervention. Following execution of an intervention by the Mothership or an associated Space Remora spacecraft, the system verifies the resulting orbit and recalculates residual risk, so that the overall detection, assessment, execution, and verification workflow is recursive and closed-loop rather than a single isolated action. Taken together, these capabilities allow the invention to improve debris tracking coverage, reduce the cost and turnaround time of active debris remediation, and support the long-term sustainability of satellite operations, space agencies, and commercial orbital infrastructure. FIG 1

No. of Pages : 11 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130591 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE ANALOG NOISE-LEVEL RESPONSIVE SUPPRESSION CONTROL CIRCUIT FOR SPEECH ENHANCEMENT

(51) International classification	:A61B 5/00, A61B 5/024, A61B 5/344, A61B 8/00, A61B 8/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MR.P.ARUNKUMAR
2)HEMALATHA S
3)HEMAPRABHA A
4)INDHU SHREE E
5)JANANI C
6)VIJAY KUMAR SELVAMURUGAN
7)Asha G

(57) Abstract :
ABSTRACT MULTI-PARAMETER MATERNAL AND FETAL MONITORING SYSTEM FOR WATER BIRTH The present invention relates to a Multi-Parameter Maternal and Fetal Monitoring System for Water Birth, referred to as Aqua-Birth. The system is designed to provide continuous monitoring of both the mother and fetus during water birth, where conventional monitoring methods can be difficult to use because the mother is inside a water-birth tub. The proposed system provides a comfortable and less restrictive way of monitoring important maternal and fetal parameters while allowing the mother to move more freely during the procedure. The proposed system includes a maternal smart ring, maternal smart band, and multiple tub-integrated fetal monitoring units. The smart ring is used to monitor maternal ECG, heart rate, blood oxygen saturation, and movement. The smart band monitors maternal heart rate, blood oxygen saturation, temperature, movement, and wearing condition. It can also provide blood-pressure-related estimation or trend information using synchronized physiological signals and a calibrated estimation model. The collected information is processed locally by the respective microcontrollers. For fetal monitoring, multiple sensing zones are provided within the water-birth tub. Each fetal monitoring unit uses a 9-crystal Doppler ultrasound transducer to acquire fetal Doppler signals and determine fetal heart rate and movement-related information. The fetal monitoring units also include signal-quality processing to assess the received ultrasound signal and identify a suitable sensing zone for reliable monitoring. This arrangement allows fetal monitoring without requiring an electronic device to be attached directly to the fetus. FIG 1

No. of Pages : 13 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130592 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Fake Voter Detection Using Multiple Authentication And Data Storage Without Internet Connection

(51) International classification	:G07C 13/00, H04L 9/32, G06Q 50/26, G06K 7/14, G06Q 50/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.Jasper D
2)Prabhavathi B
3)Ram Kumar A
4)Danaseelan P
5)Ms. A. THASNEEM FATHIMA
6)Elamathi E

(57) Abstract :
ABSTRACT Fake Voter Detection Using Multiple Authentication And Data Storage Without Internet Connection This invention is a Fake Voter Detection and Offline Voter Verification System developed to provide a simple and secure method for verifying voters before allowing them to cast a vote. The system addresses voter impersonation and duplicate voting by combining multiple authentication methods into a single arrangement that can perform its core operations without requiring continuous Internet connectivity. At the identification level, the system links Aadhaar-based identity information with the Voter ID and generates a project-specific QR identifier for the registered voter. The QR code is scanned at the polling booth and checked against the locally stored voter database. The system also checks the voter's Voted/Not-Voted status before allowing the authentication process to continue. If the voter is unregistered, the QR code is invalid, or the voter has already voted, the system rejects the voting attempt. Fig 1

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130593 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A KITCHEN WASTE HEAT RECOVERY SYSTEM FOR WATER HEATING

(51) International classification	:F24C 15/20, F24C 13/00, F28D 21/00, F24C 15/32, F24H 1/00	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)Dr.G.R.Gopinath** **2)Mr.S.ADHITHYA** **3)Ms.M..KANISHKA** **4)Mr.VP.ABHIRAJ** **5)Mr.V.HARIKRISHNAN** **6)Dr. DEEPAN KUMAR MADHU** **7)Elamathi E**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT A KITCHEN WASTE HEAT RECOVERY SYSTEM FOR WATER HEATING The present invention relates to a kitchen waste heat recovery system for water heating. The system recovers a portion of the thermal energy normally rejected from a kitchen heat source during cooking operation. A heat collection hood and hot exhaust passage direct the recovered thermal energy toward a heat exchanger. Water is circulated through a separate passage of the heat exchanger, where thermal energy is transferred from the hot exhaust gases to the water without direct mixing. A circulation pump may be provided to control water flow through the heat exchanger. Temperature sensors monitor the inlet and outlet water temperatures and the available thermal condition. A controller regulates water circulation according to the measured temperature and heat availability. Thermal insulation is provided around selected components to reduce unwanted heat losses. The heated water is supplied to a storage tank or suitable point of use. The system thereby provides an integrated approach for recovering otherwise wasted kitchen thermal energy and utilizing it for water heating. Fig 1

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130594 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Tri-Modal Edge-AI Silent Speech Synthesizer Collar

(51) International classification	:A61B 5/00, A61B 5/11, G10L 15/22, G10L 21/0208, A61B 5/0205	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mrs.N.Sarmila Devi** **2)Mathivadhani D** **3)Monika S** **4)Pradeepa S** **5)Rithika S** **6)GEETHAN KUMAR SUBASH** **7)Senega R**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT Tri-Modal Edge-AI Silent Speech Synhesizer Collar The present invention relates to a wearable silent speech communication system configured to generate artificial speech from speech-related physiological activity without requiring conventional audible vocalization. The system is implemented as a neck-worn collar incorporating surface electromyography sensors, micro-ultrasound sensors and near-infrared sensors for acquiring complementary physiological information associated with speech-related activity. The system further includes a dual inertial measurement unit arrangement in which one unit detects movement near the throat and another provides a reference motion signal. The motion signals are differentially processed to reduce interference caused by head movement, swallowing and other non-speech-related movements. The acquired physiological signals are filtered, synchronized and combined using a multimodal artificial intelligence-based processing architecture. The processed signals are decoded into phonemes, words or predefined speech units and supplied to a speech synthesis module for generating corresponding synthetic speech. The speech output may be adapted using a personalized voice profile. The system can employ edge-based processing to support local speech decoding and reduce dependence on continuous cloud connectivity. The invention provides an integrated wearable architecture combining multi-modal physiological sensing, motion-artifact cancellation, AI-based silent speech decoding and synthetic speech generation for assistive and silent communication applications. FIG 1

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130595 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A DUAL-POINT CURRENT MISMATCH BASED ELECTRICAL MONITORING, TAMPER-DETECTION AND GSM ALERTING SYSTEM FOR DETECTING SUSPECTED POWER DIVERSION AND SUPPLY INTERRUPTION

(51) International classification	:G01R 19/165, G01R 19/25, G01R 31/40, G01R 19/00, H04W 12/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)SENTAMILSELVI M
2)MAHIL RAM E K
3)ABIJITH P
4)ARUN R
5)THARUN J
6)Dr. V. C. BHARATH SABARISH
7)Gayathri S

(57) Abstract :
ABSTRACT A DUAL-POINT CURRENT MISMATCH BASED ELECTRICAL MONITORING, TAMPER-DETECTION AND GSM ALERTING SYSTEM FOR DETECTING SUSPECTED POWER DIVERSION AND SUPPLY INTERRUPTION The invention relates to an electrical monitoring system comprising first and second current sensing units positioned at respective monitoring points of an electrical supply path, a processing unit, a supply-voltage monitoring unit, a tamper-detection unit, a memory unit and a GSM communication module. The processing unit determines a current mismatch between corresponding current measurements, evaluates the mismatch against a configurable threshold and persistence period, and processes supply-status and tamper information to classify abnormal conditions. The system identifies persistent current mismatch, suspected power diversion, supply interruption and monitoring-tamper events according to predetermined criteria. An event record containing available measurement values and event information is stored, and an event-specific alert is transmitted to a remote receiving device through the GSM communication module. The system facilitates remote monitoring of electrical supply conditions and reporting of suspected abnormal current paths and supply interruptions. Fig 1

No. of Pages : 26 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130596 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT SMART CITY INFORMATION SYSTEM FOR DISASTER FORECASTING AND LOSS REDUCTION USING IOT SENSOR NETWORKS AND MACHINE LEARNING

(51) International classification	:G06N 20/00, G06Q 50/26, G06Q 10/04, H04W 4/38, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr. U Sudhakar

Address of Applicant :Assistant Professor - SG, Department of Mechanical Engineering, School of Applied Engineering, GMRIT - Deemed to be University, Rajam, Vizianagaram, Andhra Pradesh, 532127, India. Andhra Pradesh India

2)Nangunuri Anjaneyulu

3)Bhujanga Reddy Bhavanam

4)Dr. Nagarajan L

5)E. Loganathan

6)Dr. Kavita Singh

7)Dr. Buddhaghosh Arjun Shingade

8)Lalitha Bhavani Bikkina

9)Dr. S J R K Padminivalli V

10)Dr. Pulicherla Siva Prasad

11)Soorya S

12)Dr Venkateswara Reddy Vennapusa

(72)**Name of Inventor :**

1)Dr. U Sudhakar

2)Nangunuri Anjaneyulu

3)Bhujanga Reddy Bhavanam

4)Dr. Nagarajan L

5)E. Loganathan

6)Dr. Kavita Singh

7)Dr. Buddhaghosh Arjun Shingade

8)Lalitha Bhavani Bikkina

9)Dr. S J R K Padminivalli V

10)Dr. Pulicherla Siva Prasad

11)Soorya S

12)Dr Venkateswara Reddy Vennapusa

(57) Abstract :

The present invention relates to an intelligent smart-city information system for disaster forecasting and loss reduction using Internet of Things (IoT) sensor networks and machine learning. The system comprises distributed sensor nodes, communication gateways, processing units, a data repository, a data-quality assessment component, a contextual data-fusion component, a machine-learning forecasting engine, a spatial exposure and vulnerability assessment component, and a loss-reduction decision-support engine. Sensor data is verified and linked to time, place, and pertinent contextual details, such as weather data, historical occurrences, geographic details, and infrastructure documentation. The prediction engine evaluates features of possible disasters, such as likelihood, intensity, progression pattern, and areas that may be impacted. The spatial evaluation element links anticipated hazards with vulnerable assets and indicators to create risk estimates specific to each location. The decision-support system prioritizes viable preventive or response suggestions based on expected outcomes, urgency, resource availability, and operational limitations. A notifying interface delivers pertinent forecasts and suggestions to authorized recipients. A feedback element assesses predictions against confirmed results and facilitates regulated model assessment and modification. The system offers a cohesive technical workflow that links urban sensing, disaster prediction, impact evaluation, and prioritized loss-reduction strategies. FIG.1

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130601 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : StrokeSense – ML based Intelligent Emergency Detection System

(51) International classification	:A61B 5/00, A61B 5/0205, A61B 5/0245, A61B 5/024, A61B 5/08	(71)**Name of Applicant :** **1)NEHA GOPAL N** Address of Applicant :Sri Bhaktanjeneyaswamy Nilaya, next to Indane gas station, kolthuru village, Karnataka India **2)Sakeena S K** **3)Chandana M** (72)**Name of Inventor :** **1)NEHA GOPAL N** **2)Sakeena S K** **3)Chandana M**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Cardiovascular diseases (CVDs), especially those resulting in a heart attack or stroke, are among the world's leading causes of death and disability. They must be identified and treated early, or else it could result in death. With the emerging developments in wearable technology, biomedical sensors and mobile computing we can now continuously monitoring parameters like heart beats, ECG, oxygenation and activity. This paper proposes a smart wearable health monitoring concept by consolidating multi-data sensors and data processing systems into a real-time blockage monitoring to detect early cardiac arrest or stroke events. In order to support patient safety, the intended system, specifically identifies an anomaly, and sends a notification to care-givers or emergency responders in a timely manner, using very low adaptability communication tools that enable transmitting fast, reliable data, which also contains the location of the patient’s GPS coordinates in order to access appropriate medical care as quickly as possible. By utilizing IoT Artificial Intelligence, and wearable sensing, this open and freely afforded territory solution is a transition from reactive treatment to proactive and preventative measures that can be dynamic and scalable to support both urbanizing and rural populations. Cardiovascular diseases cause approximately 179 million deaths annually around the world, 80% of which are attributed to the cardiovascular event of a heart attack or stroke and the cases of ischemic heart disease rose from 112 million to 255 million globally, with no change in standardized rates despite the increase in cases. After the pandemic, excess deaths due to CVD in U.S. were reported at 250,000 people, preceded by spikes in stroke, hypertension, and unknown arrhythmias. National statistics from India revealed that deaths because of heart attack increased by 12.5% in 2022, and over the past 2 years, age-standardized cases of heart attack reported in those aged older than 35 years have increased by approximately 5% each year

No. of Pages : 4 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130605 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Multi-Sensor Assisted Robotic Rover for Real-Time Snake Detection and Non-Lethal Deterrence

(51) International classification	:G05D 1/02, G01S 15/931, B25J 9/16, B25J 19/02, A01M 29/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr B Nagaraj
Address of Applicant :Rathinam Techzone Campus, Rathinam Technical Campus, Eachanari, Coimbatore - 641 021 Tamil Nadu India
2)ARUL RAJ S
3)JEEVAN A S
4)KARTHICK L S
5)DHARUN SANKAR M
6)RAMPRABHU B
7)AKILESH C
8)ATHIPATHY M
9)SARAH RUBY CHRISTINA
(72)**Name of Inventor :**
1)ARUL RAJ S
2)JEEVAN A S
3)KARTHICK L S
4)DHARUN SANKAR M
5)RAMPRABHU B
6)AKILESH C
7)ATHIPATHY M
8)SARAH RUBY CHRISTINA

(57) Abstract :
The invention presents a multi-sensor assisted robotic rover for real-time snake detection and non-lethal deterrence. The rover uses thermal, vibration, ultrasonic, camera, and environmental sensors for accurate detection and verification. Sensor data is processed using sensor fusion to improve detection reliability and reduce false detections. The autonomous rover can navigate through enclosed, cluttered, and low-light environments using obstacle avoidance. It provides non-contact deterrence through controlled sound, vibration, and light stimuli while enabling remote monitoring. The proposed system reduces human exposure to potentially dangerous snakes while supporting safe and wildlife-conscious monitoring.

No. of Pages : 10 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130608 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR IN-SITU MOISTURE-SOURCE IDENTIFICATION AND ADAPTIVE AIRFLOW CONTROL WITH REMEDIATION VERIFICATION

(51) International classification	:F25D 17/04, F25D 17/06, F25D 29/00, G05D 22/02, F24F 11/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY
Address of Applicant :Tiruvalam Road, Katpadi, Vellore, Tamil Nadu - 632014, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)RAGHAVAN R
2)MOHAMMED KAIF K
3)PRAVIN G

(57) Abstract :
A system (100) and method are disclosed for in-situ identification and remediation of a moisture source within an enclosed storage compartment (102). The system (100) comprises environmental sensors (112A-112N), a controllable airflow path, an airflow measurement device (114A-114N), and a controller (106). The controller (106) derives absolute humidity from temperature and relative humidity measurements, establishes first and second diagnostic air-exchange states having different measured airflow rates, and records corresponding time-dependent absolute-humidity responses. The controller estimates an effective moisture-source contribution using a moisture-balance relationship based on the measured airflow rates, absolute-humidity responses, an absolute-humidity difference between a storage zone and an available air source or sink, and temporal changes in absolute humidity. Based on the estimated contribution, the controller identifies a moisture condition and configures a remediation airflow path accordingly. Following remediation, the controller repeats the diagnostic states and compares pre-remediation and post-remediation moisture-source contributions to determine remediation status. FIG. 1

No. of Pages : 46 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130609 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR TRANSACTION-CONDITIONED TEMPORAL LAND-STATE RECONSTRUCTION AND SAFE REGISTRATION COMMITMENT

(51) International classification	:G06Q 10/06, G06Q 50/16, G06Q 10/08, G06Q 40/04, G06Q 20/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY
Address of Applicant :Tiruvalam Road, Katpadi, Vellore, Tamil Nadu - 632014, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)DR.K.UMA
2)S HARISH KUMAR
3)M KAARTHIK
4)JEEVITHA V

(57) Abstract :
A system (100) and method for transaction-conditioned temporal land-state reconstruction and safe registration commitment are disclosed. The system (100) comprises a processor (104) and a memory (106) storing instructions executable by the processor (104). The processor (104) receives land-related records from a plurality of land record sources (102) and normalizes the records into land events associated with parcel identifiers, land-state attributes, temporal information, source information, and evidence references. The processor (104) reconstructs a transaction-time land state based on effective temporal relationships, determines a transaction dependency set, and propagates uncertainty associated with unresolved conflicts, missing evidence, delayed records, or temporal inconsistencies. The processor (104) generates plausible transaction-time land states and evaluates the states using common registration validation rules to determine decision invariance. When registration decisions are not invariant, the processor (104) identifies a critical evidence set and controls an authoritative registration commitment based on decision stability and pre-commitment revalidation. FIG. 1

No. of Pages : 47 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130610 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : LIGNIN-INTEGRATED BIMETALLIC ZIRCONIUM-MANGANESE NANOCOMPOSITE AND PROCESS FOR PREPARATION THEREOF

(51) International classification	:C08H 7/00, B01J 23/34, C01G 45/02, C01G 45/00, A61K 36/03
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY
Address of Applicant :Tiruvalam Road, Katpadi, Vellore, Tamil Nadu - 632014, India Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)DR DHANYA VISHNU
2)VIJUNE LAWRENCE L

(57) Abstract :
The present disclosure relates to a process for preparing a lignin-integrated bimetallic zirconium-manganese nanocomposite. The process comprises preparing a Halymenia venusta seaweed extract by shade-drying red seaweed Halymenia venusta, grinding the shade-dried seaweed into a powder, and extracting the powder; preparing a lignin extract using ammonium hydroxide (NH_4OH) and formaldehyde (HCHO); and preparing a precursor solution by dissolving a zirconium precursor and a manganese precursor in distilled water. The Halymenia venusta seaweed extract and the lignin extract are added to the precursor solution and stirred at a predetermined stirring speed and temperature for a predetermined time. A pH of the stirred precursor solution is adjusted using an NH_4OH solution to form a precipitate. The precipitate is centrifuged, washed, and dried in a hot-air oven to obtain the lignin-integrated bimetallic zirconium-manganese nanocomposite. FIG. 1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130611 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR SAFETY-CONSTRAINED CANINE THREAT INTERVENTION WITH UNCERTAINTY-GATED ACTUATION AND HUMAN-EXCLUSION CONTROL

(51) International classification	:G08B 13/196, F41H 13/00, G08B 21/02, G08B 25/00, G08B 25/01
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY
Address of Applicant :Tiruvalam Road, Katpadi, Vellore, Tamil Nadu - 632014, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)N.S.NITHYA
2)MOHAMED NASURUDEEN M
3)JAMBUKUMAR S

(57) Abstract :
A safety-constrained canine threat intervention system (100) for controlling an intervention device (116) in a monitored environment (118) is disclosed. The system (100) comprises one or more imaging sensors (102), a processor (110), a memory (112), and an independent safety controller (114). The processor (110) detects and tracks a canine (104) and a human (106) across successive frames, determines persistent potentially threatening behaviour of the canine (104), and predicts a relative trajectory to determine time-to-contact, predicted minimum separation distance, and relative closing velocity. The processor (110) further determines an uncertainty value, predicts a human-exclusion envelope, and determines an intervention envelope associated with operation of the intervention device (116). The independent safety controller (114) generates an actuator-enable signal (124) only when predetermined threat, uncertainty, spatial separation, operating-validity, and system-health criteria are satisfied. Following intervention, subsequent frames are processed to determine whether to terminate, inhibit, or re-verify the intervention. FIG. 1

No. of Pages : 49 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130612 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : FOUR-PORT MULTIPLE-INPUT MULTIPLE-OUTPUT ANTENNA WITH PASSIVE METASURFACE REFLECTOR

(51) International classification	:H01Q 15/14, H01Q 1/38, H01Q 21/06, H01Q 1/48, H01Q 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY
Address of Applicant :Tiruvalam Road, Katpadi, Vellore, Tamil Nadu - 632014, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)DR INDRASEN SINGH
2)DR. DILIP KUMAR CHOUDHARY
3)JAYARAMAN G

(57) Abstract :
Embodiments herein provide a four-port multiple-input multiple-output (MIMO) antenna (100) comprising a dielectric substrate (102), four printed radiating elements (104-1, 104-2, 104-3, 104-4) electrically coupled to respective radio-frequency ports (106-1, 106-2, 106-3, 106-4), and a partial ground plane (120). The four printed radiating elements (104-1, 104-2, 104-3, 104-4) are arranged in mutually orthogonal orientations. The four-port MIMO antenna (100) further comprises a passive metasurface reflector (108) positioned above the four printed radiating elements (104-1, 104-2, 104-3, 104-4) with an air gap (114) therebetween. The passive metasurface reflector (108) comprises a periodic 3×3 arrangement of concentric triangular-ring conductive unit cells (110) formed on a dielectric material (112). The air gap (114) provides a selected vertical spacing such that electromagnetic fields reflected by the passive metasurface reflector (108) constructively interfere with radiation from the four printed radiating elements (104-1, 104-2, 104-3, 104-4), thereby enhancing radiation performance and inter-port isolation. FIG. 1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130613 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ADAPTIVE ASSESSMENT AND MANAGEMENT OF BOREWELL RESCUE OPERATIONS

(51) International classification	:G06Q 10/06, G06Q 50/26, G06Q 10/00, G06N 5/02, G16H 40/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY
Address of Applicant :Tiruvalam Road, Katpadi, Vellore, Tamil Nadu - 632014, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)DR.K.UMA
2)S HARISH KUMAR
3)VIGNESWARAN M
4)YOGALAKSHMI.S

(57) Abstract :
A system (100) for adaptive assessment and management of a rescue operation involving a person trapped inside a borewell is provided. The system (100) comprises a processor (116) and a memory (118), the processor receives incident evidence from a plurality of evidence sources and constructs and updates an incident state comprising rescue-relevant parameters associated with evidence source, acquisition time, confidence, freshness, and uncertainty status and identifies decision-critical uncertainties, determines evidence-acquisition actions, and updates the incident state using additional evidence. The processor further generates and evaluates candidate rescue pathways based on pathway requirements, assumptions, constraints, resource availability, expected rescue progress, and time-critical rescue requirements. During rescue execution, operational observations and physical-response evidence are monitored to determine whether pathway requirements or assumptions have become unsatisfied. Based on updated pathway feasibility, the processor initiates pathway reassessment to continue, modify, pause, or switch to an alternative rescue pathway while maintaining an evidence-linked record of rescue decisions and recovery-verification status. FIG. 1

No. of Pages : 68 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130614 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT-BASED SMART CAR ACCIDENT DETECTION AND EMERGENCY ALERT SYSTEM

(51) International classification	:G08G 1/16, G08B 21/02, B60W 50/14, G08G 1/01, G07C 5/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs. M SHOBANA
2)RENGANATHAN V
3)BARANI S
4)SANJAYSARUNATH B
5)Dr. V. C. BHARATH SABARISH
6)Bhavani P

(57) Abstract :
ABSTRACT Iot-Based Smart Car Accident Detection And Emergency Alert System An IoT-based smart car accident detection and emergency alert system is disclosed. Vehicle motion is continuously monitored using an accelerometer, vibration sensor, and optional motion sensors. Sensor data is acquired and processed by an embedded processing unit, which performs noise filtering and a preliminary threshold-based abnormal-event determination. When an abnormal event is detected, a Machine Learning classifier, including Logistic Regression and/or Decision Tree, evaluates the processed sensor pattern to distinguish a normal or non-accident event from an actual accident and thereby reduce false alarms associated with road irregularities or sudden braking. Upon confirmation of an accident, a GPS module retrieves the vehicle location and the system prepares an alert containing location and available event information such as time and vehicle identification. The alert is transmitted through GSM and/or Wi-Fi to a hospital, ambulance service, and emergency contact. Optional cloud integration supports real-time monitoring, storage, and analysis. The system provides automated accident reporting and faster communication with emergency responders.

No. of Pages : 22 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130615 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : URBAN FLOOD NOWCASTING SYSTEM

(51) International classification	:G06N 20/00, G06Q 50/26, G01W 1/10, G06Q 10/04, G01W 1/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MANOJ S
2)MADHUMITHA B
3)MAHESWARI S
4)SACHINBALA V
5)VIGNESHWARAN C
6)Dr. DEEPAN KUMAR MADHU
7)Elamathi E

(57) Abstract :
ABSTRACT Urban Flood Nowcasting System The present disclosure relates to an Urban Flood Nowcasting System, an AI-powered real-time system designed for predicting, monitoring, and communicating urban flood conditions. The system integrates rainfall, weather, temperature, humidity, water-level, drainage, and historical flood data through IoT sensors and environmental data sources. The collected data is processed through data cleaning, missing value handling, duplicate removal, filtering, and normalization mechanisms. Relevant predictive features including rainfall trends, water-rise rate, antecedent precipitation, soil saturation, and terrain features are generated and supplied to a machine learning model trained using historical flood events. The trained model processes live sensor telemetry to generate real-time flood predictions and potential inundation advance times. The system dynamically classifies urban areas into Low Risk, Medium Risk, and High Risk categories and displays the predicted conditions through a GIS-based monitoring dashboard. The system further provides localized early warning alerts to citizens and emergency authorities to support timely response. The proposed system provides an integrated architecture combining IoT sensing, AI/ML prediction, GIS visualization, risk analysis, and emergency alert mechanisms for scalable urban flood monitoring and nowcasting. Fig 1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130616 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Pole Integrated Edge Intelligent Adaptive Fault Discrimination and Autonomous Dual Path Distribution Feeder Reconfiguration System for Underground Supply Continuity Restoration

(51) International classification	:H02J 13/00, H02J 9/06, G01R 31/08, H02J 3/00, H02J 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)ADITHIYAN M
2)ABINAYAKI M
3)T.KIRUTHIGA
4)Dr.J.Sasidevi
5)GIRIJA J

(57) Abstract :
ABSTRACT POLE INTEGRATED EDGE INTELLIGENT ADAPTIVE FAULT DISCRIMINATION AND AUTONOMOUS DUAL PATH DISTRIBUTION FEEDER RECONFIGURATION SYSTEM FOR UNDERGROUND SUPPLY CONTINUITY RESTORATION The present invention concerns a system for monitoring an electrical feeder installed at a distribution pole and its automatic transfer in order to enhance continuity of power supply in distribution networks. The invention consists of a primary overhead electrical feeder, an underground backup electrical feeder, an electrical parameter measuring unit, an embedded controller, a communication unit, and an automatic switching device installed in a housing attached to a distribution pole. The measuring unit continuously measures electrical parameters of the primary electrical feeder and transmits them to the embedded controller to be evaluated with respect to predefined operating conditions. In case of any abnormal condition of the electrical feeder, the controller starts a switching procedure in which the primary feeder is disconnected from the load and then the underground backup electrical feeder is connected. Interlocking of the switching devices prevents connecting the two feeders simultaneously. The operating condition of the electrical feeder after transfer is verified by the measuring unit and communicated to the monitoring unit remotely. Fig 1

No. of Pages : 13 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130617 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART OVER-HEIGHT VEHICLE & OVERHEAD OBSTACLE COLLISION WARNING AND GPS DETECTION SYSTEM

(51) International classification	:G01S 15/931, G08G 1/16, B60Q 9/00, B60Q 1/52, G01S 15/08	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Mr.B.SUNDARA PANDIYAN**
(33) Name of priority country	:NA	**2)MANISH S**
(86) International Application No Filing Date	: :01/01/1900	**3)MADHAN S** **4)MANOJ KUMAR S**
(87) International Publication No	: NA	**5)MR.BALAJI M**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)Jagadeswari**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT Smart Over-Height Vehicle & Overhead Obstacle Collision Warning And Gps Detection System The present invention relates to a Smart Over-Height Vehicle and Overhead Obstacle Collision Warning System, belonging to the field of vehicular safety, embedded systems, and Internet of Things (IoT) enabled monitoring. The invention is mounted on a high-load or over-height vehicle and comprises an infrared (IR) sensor for detecting an approaching obstacle zone, an ultrasonic/Time-of-Flight (ToF) sensor for measuring real-time vertical clearance to an overhead obstacle, a non-contact power-line sensor for detecting live high-voltage overhead lines, and an LDR-based night-vision arrangement for reliable operation under all lighting conditions. An ESP32 embedded controller fuses the sensor data, compares it against a stored voltage-based safe-clearance reference table, and actuates a graduated tri-colour LED, buzzer, and voice-module warning to the driver, supplemented by an OLED status display. Upon detecting a hazardous clearance condition, the controller transmits a GSM-based SMS alert, together with GPS location data, to the vehicle owner or a control room, and continuously logs sensor data, clearance values, and location to a cloud-based IoT dashboard for real-time monitoring, historical analysis, and route-advisory purposes. Fig 1

No. of Pages : 11 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130618 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE MASTERMIND: REAL-TIME OBJECT DETECTION FOR AUTONOMOUS VEHICLES

(51) International classification	:G06K 9/00, G06K 9/62, G06V 10/764, G06N 3/04, G08G 1/017
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs. KALAIVANI K
2)AKSHAYA M S S
3)Ms.Sarnika JR
4)Karthik M

(57) Abstract :
ABSTRACT Smart Over-Height Vehicle & Overhead Obstacle Collision Warning And Gps Detection System The 'Machine Mastermind' is an advanced real-time object detection and environment perception system engineered for autonomous vehicles. Leveraging a highly optimized YOLO v7 deep learning architecture combined with the OpenCV computer vision library, the invention provides a high-fidelity, low-latency solution for identifying and classifying a wide range of road entities including pedestrians, vehicles, and traffic infrastructure. The system is characterized by its ability to perform high-speed inference on embedded edge devices, ensuring real-time response times of under 50ms. Key features include a robust preprocessing pipeline for adverse weather conditions, a sensor-fusion engine for precise 3D spatial mapping, and a temporal tracking module that ensures stable object identification. By prioritizing the detection of vulnerable road users and integrating directly with vehicle actuation systems, Machine Mastermind offers a comprehensive safety layer essential for the realization of fully autonomous transportation systems. Fig 1

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130619 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COLORLOCK

(51) International classification	:C04B 28/00, C04B 18/04, C04B 40/00, C04B 14/30, C04B 28/02	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)R.VETTURAYASUDHARSANAN** **2)MANGALAMUTHUU R M** **3)Dr. RADHA N** **4)DHANUSH SHIVANIV**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT COLORCLOCK The present invention is related to the use of a product called ColorLock that comprises a colored geopolymer interlocking block, wherein the material is obtained from appropriate industrial waste, and the material has been subjected to activation through an alkaline solution. An inorganic mineral pigment is used which is chosen from the list of oxide-based colourants like iron oxide, chromium oxide, cobalt blue, titanium dioxide or manganese oxide. The said pigment is directly mixed with the geopolymer binder material in an approximate proportion of 2% to 5% by weight of the binder material prior to the moulding operation. The mixture is then subjected to dry mixing, blending with pigment, activation by an alkaline solution, moulding of the interlocking block shape, compacting to eliminate the voids and curing of the same under ambient conditions to achieve the desired mechanical strength of the interlocking construction block. This interlocking construction block further comprises a central support rod running through the interior of the block and making communication with the mounting holes of the side walls. FIG 1

No. of Pages : 12 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130620 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PCB-Based Predictive Thermal Stress Monitoring System for Solar Inverter Power Semiconductors

(51) International classification	:G01R 31/26, H02M 1/32, H02M 7/48, H02M 7/5387, G01K 7/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs.D.PUSHPALATHA
2)SANJAY R
3)AATHITHYAN J
4)DHYANESHWARAN K
5)Mrs.P.Uma Maheswari
6)ALBAR RAHMAN A

(57) Abstract :
ABSTRACT PCB-BASED PREDICTIVE THERMAL STRESS MONITORING SYSTEM FOR SOLAR INVERTER POWER SEMICONDUCTORS The present invention proposes a PCB-based predictive thermal stress monitoring system for solar inverter power semiconductors such as MOSFETs and IGBTs. During normal inverter operation, these semiconductor devices generate heat due to switching and conduction losses, and excessive or continuous heating can reduce their performance and lifetime. To address this issue, the proposed system uses a dedicated PCB with temperature, current, and voltage sensors to continuously monitor the operating condition of the power semiconductor. The collected sensor data is processed to understand the relationship between electrical loading and temperature rise, allowing the system to identify abnormal thermal conditions and predict possible overheating at an early stage. When the thermal stress reaches a warning level, the system provides an indication through an LED, display, buzzer, or suitable monitoring interface. Thus, the proposed invention provides a compact and practical method for early thermal stress detection, helping improve the reliability, safety, and service life of solar inverter power semiconductor devices. Fig 1

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130621 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DIGITAL TWIN-BASED REVERSE INFERENCE SYSTEM FOR UNLABELED AIR POLLUTION SOURCE MAPPING

(51) International classification	:G06N 5/04, G01N 33/00, G01N 15/06, G06N 5/02, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs.SHOBANA M
2)INDHUJA K
3)SHALINI P
4)LATHIKA K
5)NARMADHA B
6)Dr. DEEPAN KUMAR MADHU
7)Elamathi E

(57) Abstract :
Abstract Digital Twin-Based Reverse Inference System for Unlabeled Air Pollution Source Mapping Air pollution is influenced by multiple sources such as vehicle emissions, industrial activities, crop-residue burning, dust, and changing weather conditions, making it difficult to determine the actual factors responsible for a rise in air pollution. Most existing air-quality systems mainly focus on monitoring or predicting AQI, but they provide limited information about the probable sources behind the observed pollution. To address this limitation, this project proposes an Explainable AI-based inverse air-quality framework that predicts AQI and estimates the contribution of different pollution sources from observed environmental conditions. The system integrates AQI measurements, meteorological data, satellite-derived fire information, and other relevant environmental parameters. A Deep Autoencoder is used to extract meaningful hidden patterns, while LSTM/GRU networks capture the temporal behaviour of pollution. XGBoost is then used to refine the AQI prediction. The proposed Inverse Inference Engine works backwards from observed AQI by using an Atmospheric Digital Twin to generate predicted conditions and iteratively updating source contributions through self-supervised gradient optimization. Fig 1

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130622 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART MOLASSES TO METHANE CONVERSION SYSTEM

(51) International classification	:C12P 5/02, C02F 11/04, B09B 3/00, C12M 1/107, C02F 3/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)MUKESH PANNEERSELVAM
2)BALAMURUGAN PANNEERSELVAM
3)SRINIVASAN PALANISAMY
4)LINGESHWARAN LOGANATHAN
5)MADHAN SELVAKUMAR
6)MAHALAKSHMI PONNUSAMY
7)Dr.M.MAHESWARI
8)HARI PRASATH M

(57) Abstract :
ABSTRACT SMART MOLASSES TO METHANE CONVERSION SYSTEM Solid waste management has turned into a critical environmental issue because of rapid urbanization, growing population, and industrial activities. Inadequate management of organic waste has caused environmental pollution, health issues, and greenhouse emissions. This project will discuss a sustainable approach to waste management by utilizing molasses, a byproduct of the sugar industry, to produce biogas. This process not only helps to manage waste but also generates clean energy, which is consistent with the waste-to-energy concept. The molasses used to produce biogas has a high concentration of organic compounds and sugars, which are suitable for microbial activity. Anaerobic digestion is a process that uses microorganisms to break down organic compounds in the absence of oxygen. This process produces biogas that is comprised mainly of methane and carbon dioxide. Methane has a high energy potential and is a clean fuel. The production of methane is a measure of efficiency. To make this process efficient and reliable, this project will be equipped with a sensor-based system. The maintenance of optimal temperature conditions helps to promote the effectiveness of microbial activity, while pH monitoring helps to maintain a constant environment for digestion. Pressure sensors are used to maintain gas storage and safe use, while methane sensors are used to analyze gas quality and yield in real-time. The use of these sensors helps to maintain real-time analysis of the process, improving efficiency while minimizing human intervention. The data collected can be used to analyze the process, improving efficiency in biogas production. Furthermore, the byproduct of this process can be used as fertilizer, promoting sustainable practices. FIG 1

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130623 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RFID-BASED SMART LIBRARY SHELF MONITORING SYSTEM

(51) International classification	:G06K 7/10, G06Q 10/08, G06K 17/00, G06K 7/00, G08B 13/14	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr. BALAMURUGAN R**
(32) Priority Date	:NA	**2)KIRUTHIKA S**
(33) Name of priority country	:NA	**3)KEERTHANA S**
(86) International Application No Filing Date	: :01/01/1900	**4)LOGAVARSHINI P** **5)KARTHIKA S** **6)Mrs.P.Uma Maheswari**
(87) International Publication No	: NA	**7)NAVEEN A**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT RFID-BASED SMART LIBRARY SHELF MONITORING SYSTEM An RFID-based intelligent library shelf monitoring system is disclosed for real-time identification of correctly placed, misplaced, and potentially missing books. The system associates individual library books with unique RFID tags and predefined shelf locations comprising row and column information. RFID readers positioned at designated shelf sections detect the physical location of books and transmit the identification information to a processing unit. The processing unit compares the detected location with the registered location stored in a library database. A green visual indicator is activated when the book is correctly positioned, while a red visual indicator is activated when the book is detected at an incorrect location. For misplaced books, the system records the actual detected location and detection timestamp and makes the information available through a digital monitoring interface, enabling rapid retrieval without manual searching of multiple shelves. The system further integrates book borrowing and returning status with scheduled shelf verification to identify books that are neither borrowed nor detected on monitored shelves. Such books are identified as potentially missing, and their last known location and detection time are provided to the librarian for further investigation. The proposed system thereby improves book-location accuracy, misplaced-book detection, missing-book identification, shelf organization, and overall efficiency of library management. Fig 1

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130624 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IRIS RECOGNITION IN TWINS ON MOBILE DEVICES USING DEEP LEARNING

(51) International classification	:G06K 9/00, G06F 21/32, G06N 3/08, G06V 40/19, G06V 40/18	(71)**Name of Applicant :** **1)Karpagam Academy of Higher Education** Address of Applicant :Pollachi Main Road, Eachanari Post Coimbatore Tamil Nadu India Tamil Nadu India **2)Karpagam Institute of Technology** **3)Sathish D** **4)Sakthivel V.P.V**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Sathish D** **2)Sakthivel V.P.V**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Iris recognition is widely recognized as one of the most reliable biometric authentication techniques due to the uniqueness and long-term stability of iris texture patterns. With the rapid proliferation of mobile devices, there is an increasing demand for secure, accurate, and user-friendly biometric systems that can operate effectively in unconstrained environments. This challenge becomes significantly more complex in the case of identical twins, where genetic similarity can lead to subtle biometric overlaps that may reduce the effectiveness of conventional recognition approaches. Addressing these challenges, this work presents a deep learning–based iris recognition framework specifically designed for mobile devices, with a focus on high-precision twin differentiation. The proposed system adopts a two-phase architecture built on the You Only Look Once (YOLO) deep learning framework to ensure real-time performance and robustness. In the first phase, iris and pupil regions are automatically detected and extracted from eye images captured using mobile cameras. This phase integrates three essential processes: iris localization, iris segmentation, and feature enhancement. Multiple iris localization strategies with varying computational complexity are introduced, enabling the system to balance accuracy, safety, and cost-effectiveness depending on the target mobile hardware. This design ensures reliable iris extraction even under challenging conditions such as variable illumination, occlusion, and non-frontal gaze. Iris segmentation is subsequently performed using both normalized and non-normalized image representations to evaluate their impact on recognition performance. Unlike traditional iris recognition systems that heavily rely on strict normalization procedures, the proposed approach investigates the necessity of normalization in deep learning–based iris identification. Experimental results demonstrate that high recognition accuracy can be achieved even with reduced normalization constraints, thereby simplifying preprocessing and improving computational efficiency for mobile deployment. Feature enhancement techniques are further applied to amplify fine-grained iris texture details, which are particularly critical for distinguishing between twins. In the second phase, the extracted iris Region of Interest (ROI) is processed using a deep learning–based classifier to generate robust and discriminative iris representations. The YOLO-based framework enables end-to-end learning and real-time processing, making the system suitable for practical mobile applications. Extensive performance evaluation is conducted to assess recognition accuracy, safety, and affordability. The proposed system achieves an overall recognition accuracy of 98.6%, with a near-zero Equal Error Rate (EER), indicating a high level of security and reliability. The results confirm that the proposed deep learning–based iris recognition system effectively addresses the limitations of traditional approaches in mobile environments and offers superior performance in twin recognition scenarios. By combining adaptive preprocessing, YOLO-based detection, feature enhancement, and efficient classification, the system provides a secure, accurate, and cost-effective biometric authentication solution. This work contributes to advancing mobile biometric security and demonstrates the feasibility of deploying high-performance iris recognition systems on resource-constrained devices.

No. of Pages : 15 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130627 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SENTINELNODE AI

(51) International classification	:H04L 29/06, G06F 21/55, H04W 12/12, G06F 21/57, H04W 12/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.ARUL A
2)MITHUN SARAVANAN
3)HARINI M P
4)HARIHARAN S
5)GOBIKA P
6)Mrs.M.Kirithika Devi
7)JAWAHAR K

(57) Abstract :
ABSTRACT SENTINELNODE AI The present invention discloses a cyber-physical IoT security system (10) comprising a microcontroller unit (20), a motion-sensing unit (30), a red LED (40), a green LED (41), a buzzer (50), a web dashboard (60) and a threat-classification module (70). The microcontroller unit (20) creates a fake wireless access point (21) and hosts a fake login portal (24), which captures unauthorized login attempts, source IP address, date and time as cyber-attack events. The motion-sensing unit (30), comprising an accelerometer (31) and a gyroscope (32), detects physical tampering of the device. The threat-classification module (70) classifies the recorded events into normal, suspicious or brute-force categories with a corresponding severity level, using lightweight rule-based logic that does not require cloud infrastructure. Upon detection of a cyber-attack event or a physical-tampering event, the red LED (40) is activated, the green LED (41) is deactivated, the buzzer (50) is actuated, and the web dashboard (60), hosted directly on the microcontroller unit (20), is updated in real time. The present invention thereby provides a unified, low-cost, embedded solution for simultaneous cyber and physical threat detection in IoT environments. Fig 1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130628 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BALL-BEARING HINGE ASSEMBLY FOR SWINGING GATES AND HEAVY DOORS

(51) International classification	:E05D 7/04, E05D 5/02, E05D 3/02, E05D 7/00, E05D 7/081
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(72)**Name of Inventor :**
1)BALAJI G
2)NANDHA KUMAR P
3)VETREVEL G K
4)RANJITH KUMAR G
5)SRI PRAANAUV V
6)Mr. A. BALAKUMAR

(57) Abstract :
ABSTRACT BALL-BEARING HINGE ASSEMBLY FOR SWINGING GATES AND HEAVY DOORS A hinge assembly for mounting between a metal door or gate and its supporting frame or post is disclosed. The assembly comprises a top end cap and a bottom end cap, each carrying a leaf by which the assembly is fastened, respectively, to the door and to the frame or gate post. A steel shaft extends between the top end cap and the bottom end cap and defines the axis of rotation of the assembly. A steel ball bearing is positioned at each of the two opposite ends of the steel shaft, one adjacent the top end cap and one adjacent the bottom end cap, so that the rotating surface of the top end cap turns on the ball bearings rather than on a plain metal-to-metal bearing surface. A rubber piece is fitted between the mating faces of the top and bottom end caps, and a grease fitting and grease nozzle on the top end cap allow lubricant to be introduced into and retained within the internal cavity that houses the shaft and the ball bearings. By supporting the rotating parts of the hinge on ball bearings rather than a plain pivot, and by retaining lubricant within a sealed cavity, the assembly is intended to provide increased freedom of rotation, greater load-carrying capacity, and reduced sticking and wear in comparison with conventional plate-type butt hinges. FIG 1

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130629 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTONOMOUS SUBSURFACE-TO-CELESTIAL COMMUNICATION FRAME WORK

(51) International classification	:H04B 7/185, H04W 84/06, H04B 10/118, H04L 12/26, H04L 12/24	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mr.M. Jaganath** **2)Archana S** **3)Lavanya R** **4)Maithraye N S** **5)MUGESH S** **6)Dr.J.Sasidevi**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AN AUTONOMOUS SUBSURFACE-TO-CELESTIAL COMMUNICATION FRAME WORK The BHUMI-NET framework is an autonomous, cross-medium communication system engineered to maintain life-saving telemetry links between blocked underground domains and celestial satellite networks during catastrophic infrastructure failures. At the subsurface edge, an operator's module fuses multi-modal biometric and motion sensor streams over an embedded TinyML engine, automatically calculating a unified Hazard Vector index to compile and fire a compressed 8-byte emergency payload even if the operator is rendered unconscious under rubble. Upon detecting an intermediate link collapse within 2 milliseconds, the mesh executes an automated 90-Degree Dynamic Polarization Phase Tuning routine, rotating the loop current to force the magnetic flux lines to couple with and bounce directly through continuous solid rock walls as a natural waveguide, completely bypassing tunnel debris. The surface gateway uses a LILYGO T-SIM7080G-S3 core to track background cellular mast health, autonomously booting into an internal 3GPP Release 17/18 Satellite Non-Terrestrial Network (NTN) stack during blackouts. It compensates for orbital Doppler frequency shifts to launch a 1.6 GHz Microwave L-Band carrier wave straight to Low Earth Orbit (LEO) satellite constellations, routing the alert to a backend layer powered by type-safe Scala Case Classes and Apache Spark Structured Streaming DataFrames to map sector-wise regional crisis indices on a live dashboard GUI within 5 milliseconds of space data landing.

FIG 1

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130630 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SAFEHOME SERVICES: A SECURE MOBILE-BASED HOME SERVICE MANAGEMENT SYSTEM WITH REAL-TIME GPS TRACKING AND OTP VERIFICATION

(51) International classification	:H04W 4/021, H04W 4/80, H04L 9/32, G06Q 10/06, H04L 29/06	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mrs. A. SELVANAYAGI** **2)SRIVELAN M** **3)SUDEEP K** **4)SUDHARSHANAN K S** **5)Dr R.ARULRAJ** **6)Mr. L. PANCHACHARAM**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT SAFE HOME SERVICES: A SECURE MOBILE-BASED HOME SERVICE MANAGEMENT SYSTEM WITH REAL-TIME GPS TRACKING AND OTP VERIFICATION The present invention relates to SafeHome Services, a secure mobile-based home service management system that integrates customer, worker and administrator interfaces with cloud-based services to provide safety throughout a home-service transaction. A service request is assigned to a worker, whose location is temporarily tracked during transit to the customer's premises. A Haversine-based distance calculation establishes a destination geofence, and a customer-controlled four-digit time-sensitive OTP is required before the worker can activate the service. When GPS information is unreliable, localized Bluetooth proximity may be used as a fallback condition. An SOS panic mechanism provides emergency notification to an administrator during an active service. The system further maintains service-event information and calculates a dynamic worker trust score from completed jobs, assigned jobs, customer rating, verified recent complaints and cancellations. The combined workflow links physical arrival, identity verification, emergency response and worker accountability in a single mobile platform, thereby improving the safety and traceability of on-demand home services. Fig1

No. of Pages : 16 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130631 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Physics-Informed Self-Calibrating Digital Twin System for Real-Time Thermal and Electrical Failure Prediction of PCB Systems

(51) International classification	:G06F 17/50, G06F 119/08, G06F 30/392, G06F 30/398, G01R 31/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)ABINAYAKI M
2)BABY SHALINI S
3)KAVIPRIYA N
4)KAMALI G
5)KARTHIKEYAN K
6)SURYA R
7)Mrs.P.Uma Maheswari

(57) Abstract :
ABSTRACT Physics-Informed Self-Calibrating Digital Twin System for Real-Time Thermal and Electrical Failure Prediction of PCB Systems The present invention provides a physics-informed, self-calibrating Digital Twin system for real-time thermal and electrical reliability monitoring and failure prediction of a printed circuit board (PCB). The system generates a virtual representation of a physical PCB using design parameters obtained from an electronic design automation platform such as Altium Designer, including trace length, trace width, conductor thickness, material properties, component placement, and PCB configuration. Real-time voltage, current, and temperature parameters are acquired from sensors associated with the physical PCB and supplied to a physics-based modeling module. The modeling module calculates electrical and thermal characteristics using mathematical relationships for resistance, power dissipation, temperature rise, and related operating parameters. The predicted values generated by the Digital Twin are continuously compared with corresponding measured values to determine prediction errors and identify abnormal operating conditions. A self-calibration mechanism adaptively updates selected Digital Twin parameters based on the deviation between predicted and measured values, thereby improving model accuracy under changing operating conditions. The system further identifies potential conditions such as excessive current, increased resistance, inadequate heat dissipation, component degradation, and thermal hotspots, and provides an early warning of potential PCB failure. The invention thereby integrates Altium-based PCB design data, real-time sensing, physics-based modeling, Digital Twin technology, self-calibration, and predictive fault analysis to improve PCB reliability and enable predictive maintenance.

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130632 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED PRECISION AGRO SPRAY ROVER WITH MOBILE APPLICATION FOR CROP DISEASE DETECTION, SPRAY RECOMMENDATION, AND AUTONOMOUS FIELD NAVIGATION

(51) International classification	:A01M 7/00, G06N 3/08, G06V 10/764, G06N 3/04, G06V 20/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)MAKANYADEVI KANDASAMY
2)DEEPANKUMAR S
3)DEEPAK RAJ T
4)MOHAMED ASHWATH N
5)Dr.M.MAHESWARI
6)P.MELVIN

(57) Abstract :
ABSTRACT AI-Based Precision Agro Spray Rover with Mobile Application for Crop Disease Detection, Spray Recommendation, and Autonomous Field Navigation The current invention is about an Artificial Intelligence Precision Agro Spray Rover integrated with a mobile application for the intelligent detection of crop diseases and the precision spray of pesticides. Crop images are taken using either the mobile phone or the rover's camera, and analyzed with the help of AI technology to detect the diseases, and also determine the degree to which they are severe. Following the result of the analysis, pesticides that will best work are recommended, along with the quantity that should be used. Once approved by the farmer, the autonomous rover uses GPS and obstacle sensors to drive across the farmland and selectively apply pesticides in areas where the crops are infected. Fig 1

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130633 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Integration of Blockchain Technology and Artificial Intelligence (AI) In Precision Agriculture to Avoid Food Wastage

(51) International classification	:G06Q 50/02, G06Q 10/06, G06F 16/27, G06F 16/23, H04L 9/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)DR.D.PRADEEP
2)MRS.E.NANDHINI PRIYA
3)R. BALAMURUGAN
4)ABINAYASRI S

(57) Abstract :
Abstract Integration of Blockchain Technology and Artificial Intelligence (AI) In Precision Agriculture to Avoid Food Wastag Integration of Blockchain technology and artificial intelligence (AI) in precision agriculture to sort out the challenges of food wastage and inefficient farming practices. This paper examines the advantages and disadvantages of each technology and proposes a comprehensive system for implementation at the village level. Through a detailed analysis, this research aims to provide insights into the potential impact of such integration on agricultural sustainability and farmer livelihoods. Due to the growing global population and increasing demand for food worldwide, as well as changes in weather conditions and water availability, artificial intelligence (AI) such as expert systems, natural language processing, speech recognition, and machine vision have changed the quantity but also quality of work in agriculture sector. Researchers and scientists are now moving towards the use of new IoT technologies in smart farming to help farmers use AI technology in the development of improved seeds, crop protection and fertilizers. This will improve the profitability of farmers and the overall economy of the country. In this context, farmers are increasingly adopting the use of sensors and soil sampling to collect data used by farm management systems for further investigation and analysis. This article contributes to the field by surveying AI applications in the agricultural sector. FIG 1

No. of Pages : 10 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130634 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART MODULAR INSULATION PANELS FROM TEXTILE AND COIR INDUSTRY WASTE

(51) International classification	:E04B 1/80, C04B 28/14, E04B 1/94, E04C 2/26, E04C 2/284
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(72)**Name of Inventor :**
1)DINESHKUMAR R
2)RISHVANTH S
3)SANJAY S
4)SIVAKAVIN S
5)ARAVINTHAN .R
6)Mrs.M.Kirithika Devi

(57) Abstract :
ABSTRACT SMART MODULAR INSULATION PANELS FROM TEXTILE AND COIR INDUSTRY WASTE The current invention is directed toward a Smart Modular Insulation Panel manufactured from recycled cotton textile waste and coir pith for use in construction applications. The cotton textile fibers undergo cellulase enzyme treatment to increase fiber-binder bonding and are then mixed with coir pith using a starch-guar gum natural binder. Borax-boric acid treatment is done to increase fire resistant properties, and gypsum coating is done to both surfaces for increased protection and fire-resistance. This composite panel is prepared using moulding, compression, and curing processes. DHT22 temperature and humidity sensors, linked to the ESP32 microcontroller, can monitor environmental conditions and send the data wirelessly to a monitoring cloud platform. The resulting panel is a thermal and acoustics insulator which facilitates the efficient use of agricultural and textile waste materials. FIG 1

No. of Pages : 14 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130635 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUAL-SENSOR FUSION-BASED LIVE WIRE HAZARD DETECTION AND AUTOMATIC ISOLATION SYSTEM FOR LV OVERHEAD CONDUCTORS

(51) International classification	:G01R 31/08, G01R 31/02, G01V 3/08, H02H 1/00, G01R 31/58
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.P.T.SIVAGURUNATHAN
2)HARSITHAASREE P
3)JANANI N
4)KALESWARI M
5)KAVISHKA B
6)Dr.J.Sasidevi
7)VISHNU VARTHAN P

(57) Abstract :
ABSTRACT DUAL-SENSOR FUSION-BASED LIVE WIRE HAZARD DETECTION AND AUTOMATIC ISOLATION SYSTEM FOR LV OVERHEAD CONDUCTORS A low-cost, dual-sensor fusion-based system is disclosed for detecting breakage of Low Voltage (LV) overhead electrical conductors and confirming whether the broken conductor remains live and hazardous. A sensor node deployed at each pole comprises an electric-field (capacitive) sensing unit with a comparator circuit to detect a live conductor lying near or on the ground, and a MEMS accelerometer (MPU6050) to detect the mechanical snap event characteristic of conductor breakage, as distinguished from wind-induced sway. A microcontroller executes a fusion logic routine that confirms a breakage event only when both signals are detected within a predetermined time window, substantially reducing false-positive alerts. Upon confirmation, the microcontroller actuates a relay to isolate the affected span and transmits an alert, via a LoRa radio module, to a remotely located gateway node, which forwards the alert to the utility via GSM or internet, thereby eliminating dependence on cellular coverage at the point of sensing. The invention is realizable at low per-node cost, rendering it suitable for large-scale deployment across rural and semi-urban LV distribution networks. Unlike existing high-impedance fault detection methods, which rely solely on electrical signatures analysed at the substation or feeder level, the present invention directly verifies the physical hazard condition at the point of failure. This per-span sensing approach further enables precise fault localization, unlike conventional feeder-level monitoring, which identifies only the general fault vicinity. Fig 1

No. of Pages : 13 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130636 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BREATHEBAND – CONTEXT-AWARE WEARABLE RESPIRATORY AND INHALER SAFETY MONITORING SYSTEM

(51) International classification	:A61B 5/08, A61B 5/00, A61B 5/087, A61M 16/00, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs. A. SELVANAYAGI
2)NIRMALA R
3)NISHANTHI A K
4)SHIVANI RAJALAKSHMI K
5)Dr. SRINIVASAN D
6)S.ANNISH

(57) Abstract :
ABSTRACT BREATHEBAND – CONTEXT-AWARE WEARABLE RESPIRATORY AND INHALER SAFETY MONITORING SYSTEM The present invention discloses an intelligent wearable respiratory monitoring and inhaler safety alert system (100), referred to as BreatheBand. The system comprises a wearable wristband (123), respiratory sensing unit (112), microcontroller (110), wireless communication unit (117), associated inhaler (113), alert unit, mobile application (118), and cloud server (119). The respiratory sensing unit (112) continuously detects the breathing activity of the user and provides the detected respiratory signals to the microcontroller (110) for determining normal or abnormal breathing conditions. The system further monitors the proximity of the associated inhaler (113) and generates an alert when the inhaler moves beyond a predetermined range from the user. When abnormal breathing is detected, the system provides an immediate alert through the buzzer (116), vibration motor (108), and/or LED indicator (110). When abnormal breathing and inhaler separation are detected together, a higher-priority safety notification is generated. The detected respiratory condition and inhaler availability status are displayed through the mobile application (118), enabling timely awareness and response by the user or authorised caregiver. Thus, the present invention provides an integrated wearable solution for real-time respiratory monitoring, inhaler availability detection, and timely safety alerting. Fig 1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130639 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for Generating Realistic Human Face Images Using a Deep Convolutional Generative Adversarial Network (DCGAN) with Integrated Text-to-Image Synthesis and Real/Fake Classification

(51) International classification	:G06N 3/04, G06N 3/08, G06K 9/62, G06V 10/764, G06K 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Malla Reddy (MR) Deemed to be University,
Address of Applicant :Malla Reddy(MR) Deemed to be University, Maisammaguda(H), Medchal Mandal, Medchal-Malkajgiri District, Telangana State-500100. ipr@mrdu.edu.in Medchal-Malkajgiri Telangana India
2)Amar Shravani

(72)**Name of Inventor :**
1)Amar Shravani
2)Dudekula Rabiya Begum
3)Konduru Vijay Gandhi
4)Kolipaka Dileep Reddy
5)K.Saiveena
6)Kokkirigadda Supriya
7)S.Murali Krishnamachari

(57) Abstract :

The present invention relates to an artificial-intelligence-based system and method for generating realistic and diverse synthetic human face images using a Deep Convolutional Generative Adversarial Network (DCGAN). The proposed system comprises a Generator network and a Discriminator network that are trained simultaneously through an adversarial learning process on a large-scale facial image dataset. The Generator receives a random noise vector as input and progressively transforms it into a synthetic human face image through a series of transposed convolutional layers, while the Discriminator, formed of convolutional layers, evaluates and classifies an input image as either a genuine (real) image drawn from the training dataset or a synthetic (fake) image produced by the Generator. Prior to training, the facial dataset is processed through a preprocessing pipeline comprising resizing, pixel-value normalization, augmentation and batching, thereby improving training stability and convergence. The system further incorporates a text-to-image synthesis module in which a Long Short-Term Memory (LSTM) network encodes a natural-language textual description into a semantic feature representation, which is thereafter decoded by a Convolutional Neural Network (CNN) into a corresponding visual output. An interactive user interface enables a user to upload a reference image, invoke generation of synthetic face images, view the Discriminator's real/fake classification of the displayed images, and submit textual descriptions for text-to-image generation. The performance of the system is quantitatively and qualitatively evaluated through visual inspection, training accuracy and loss monitoring, and standard generative-model metrics such as the Inception Score (IS) and Fréchet Inception Distance (FID). The invention provides an autonomous, dataset-driven mechanism for producing high-quality, diverse and non-identity-replicating human face images without manual graphic design, and finds application in virtual reality, gaming, entertainment, data augmentation for machine learning, identity anonymisation and other AI-based image-synthesis tasks.

No. of Pages : 13 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130641 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Real-Time Vision-Based System and Method for Contactless Human–Computer Interaction Using Hand-Landmark Gesture Recognition (HandoraX)

(51) International classification	:G06K 9/00, G06F 3/01, G06F 3/033, G06F 3/038, G06F 3/0488
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY
Address of Applicant :DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India Vijayawada Andhra Pradesh India
2)A. Ravi Kiran
3)K. Sri Naga Lasya
4)K. Mounika
5)K. Nandini
6)M. Naga Bala Vardhan
7)A. Cyril Denis

(72)**Name of Inventor :**
1)A. Ravi Kiran
2)K. Sri Naga Lasya
3)K. Mounika
4)K. Nandini
5)M. Naga Bala Vardhan
6)A. Cyril Denis

(57) Abstract :
ABSTRACT [0020] A real-time contactless human–computer interaction system is disclosed that employs computer vision and a landmark-based hand-tracking model to interpret hand gestures and convert them into computer operations. Live video frames from a webcam are mirrored and processed by a hand-detection module that extracts twenty-one hand landmarks. A landmark-processing module determines finger states and calculates distances between selected landmarks, and a rule-based gesture-recognition module compares them with predefined gesture patterns. Recognized gestures are mapped to operations in which index-finger movement controls a cursor, a thumb–index pinch generates a mouse click, the thumb–index distance controls system volume, and the thumb–pinky distance controls screen brightness. Smoothing and cooldown mechanisms stabilize continuous and discrete actions, and dynamic on-screen indicators provide real-time feedback. The entire pipeline runs on a general-purpose computer with an ordinary webcam, eliminating the need for wearable sensors or physical input devices, and supports accessibility, hygienic touch-free operation, smart environments and AR/VR interaction.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130642 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN AND IMPLEMENTATION OF A SECURE AND SCALABLE CIVIC WELFARE PLATFORM FOR DIGITAL GOVERNANCE

(51) International classification	:G06Q 50/26, G06Q 10/06, G06Q 10/10, G06Q 10/00, G06F 16/248
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs. A. SELVANAYAGI
2)SRIVELAN M
3)SUDEEP K
4)SUDHARSHANAN K S
5)Dr.J.ROSELIN SUGANTHI
6)Ms. M. FATHIMA LUHA

(57) Abstract :
Abstract DESIGN AND IMPLEMENTATION OF A SECURE AND SCALABLE CIVIC WELFARE PLATFORM FOR DIGITAL GOVERNANCE Disclosed is a networked civic issue management platform (100) that converts citizen reports into secure, location-aware municipal work orders. A citizen interface (110) receives complaint text, media and coordinates. A gateway (120) authenticates requests and applies role and rate controls. A triage engine (130) produces a category and urgency value from normalized text. A geospatial engine (140) clusters proximate reports, merges duplicates into a master issue while retaining reporter links, and generates hotspot data. A workflow controller (150) assigns the issue to a department, records state transitions, delivers real-time notices, and requires timestamped completion evidence. A datastore and cache subsystem (160) provides geospatial indexing, audit records and cached map snapshots. Citizen, field-worker and administrative interfaces (170) expose only authorized views. The arrangement reduces redundant records and map recomputation while improving secure dispatch, live visibility and verifiable closure of municipal complaints. Fig 1

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130643 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Intelligent Pesticide Sprinkling System Based on Infection Level Detection in Areca Palm for Precision Agriculture

(51) International classification	:A01M 7/00, G06K 9/00, A01G 13/00, G06T 7/00, G06K 9/62	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr. A. Syed Musthafa**
(33) Name of priority country	:NA	**2)Devatharshini R**
(86) International Application No Filing Date	: :01/01/1900	**3)Dhaniska V** **4)Kowsika K**
(87) International Publication No	: NA	**5)Mohamed Imran**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)M.Subashree**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT Intelligent Pesticide Sprinkling System Based on Infection Level Detection in Areca Palm for Precision Agriculture Effective disease management is essential for maintaining healthy areca palm growth and achieving better crop productivity. Areca palms are affected by various diseases that can reduce plant health and yield. Conventional disease identification often depends on manual visual inspection, which can be time-consuming, labour-intensive, and may lead to inaccurate disease identification. In addition, applying pesticides uniformly without considering the actual infection level can result in unnecessary pesticide usage, chemical wastage, and increased cultivation costs. To address these issues, an Intelligent Image-Based Areca Palm Disease Detection, Severity Assessment and Targeted Pesticide Spraying System is proposed to provide accurate and plant-specific disease management. The system uses camera images and colour information of areca palm leaves or affected plant areas as input parameters. The captured images are pre-processed to improve their quality before analysis. Machine Learning techniques, particularly CNN-based image analysis, are used to identify whether the plant is healthy or infected, determine the type of disease, and assess the infection severity as low, medium, or high. Based on the detected disease and severity level, the system determines the required pesticide or insecticide and checks its availability. If the required chemical is unavailable or insufficient, an alert is provided to the farmer through a mobile application or display for refilling. When the chemical is available, the system automatically prepares the required mixture and transfers it to the spraying unit. The pesticide is then sprayed selectively on the infected areca palm using a rail-based movement system, height adjustment mechanism, pumps, valves, and spray nozzles. Healthy plants are avoided to reduce unnecessary pesticide application. After spraying, the system automatically cleans the tank and nozzle to remove chemical residue before moving to the next plant. The proposed system aims to reduce manual labour, minimize pesticide wastage, improve disease detection and treatment accuracy, and support efficient and sustainable precision agriculture.

No. of Pages : 13 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130644 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Intelligent Time Slot Optimized Delivery System

(51) International classification	:G06Q 10/08, G06Q 30/06, G06Q 30/00, G06Q 50/28, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)ANANTHI S
2)SANJUSRI B
3)UMASANKARI M
4)VINOTHA P
5)Dr. P. MURALIKRISHNAN
6)PUNITHA

(57) Abstract :
ABSTRACT Intelligent Time Slot Optimized Delivery System The Intelligent Time Slot Optimized Delivery System is a full-stack logistics platform designed to improve order processing, delivery scheduling, agent assignment, weather-based rescheduling secure payment handling and real-time delivery tracking for modern e-commerce operations. The system enables customers to browse a categorized product catalog add items to a shopping cart select preferred two-hour delivery time windows from dynamically generated and availability-checked slot listings, complete transactions through UPI, Google Pay, Credit or Debit Card with 3D Secure OTP verification or Cash on Delivery and monitor their delivery progress through a live status dashboard with automatic reschedule notifications. The proposed intelligent scheduling engine evaluates customer preferences, available delivery slots, existing bookings, delivery capacity historical demand patterns and agent workload to recommend delivery time slots and prevent scheduling conflicts and overbooking. A weather integration component fetches meteorological data and enables delivery agents to reschedule unsafe deliveries, with automatic customer notification ensuring agent safety and goods integrity under adverse weather conditions. FIG 1

No. of Pages : 12 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130645 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SOLAR SYSTEM BASED LEGAL DOCUMENT SIMPLIFIER

(51) International classification	:G06Q 50/18, G06F 16/35, G06F 16/33, G06N 5/02, G06F 16/338
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)NANDHAKUMAR C
2)GOKUL K S
3)HARIHARAN C
4)VISHWANDH V
5)Mr P.VIGNESHWARAN
6)Mr. M. ARSHATH

(57) Abstract :
ABSTRACT AI-DRIVEN LEGAL DOCUMENT SIMPLIFIER SECURITY RECOMMENDATION FRAMEWORK The present invention, Solar system based legal document simplifier, is an AI-based system for simplifying legal documents and estimating litigation outcome probability for members of the general public, presented through a solar-system-themed graphical interface. The system comprises a document-ingestion module, a clause-extraction and classification engine, a plain-language explanation-generation module, a risk-flagging module, a case-description input and factor-extraction module, an outcome-probability estimation engine, and a solar-system-themed graphical presentation layer. An uploaded legal document is processed to identify and classify individual clauses, generate corresponding plain-language explanations, and flag potentially risky provisions. A user-provided description of a legal dispute is processed to extract relevant factors and generate an estimated probability of a favorable outcome, together with an explanation of contributing factors. The resulting document overview, clauses, risk flags, and outcome factors are presented as a central node and surrounding orbiting nodes whose visual attributes convey category, risk level, and relative importance, allowing a user without legal training to intuitively explore and understand complex legal information. The invention reduces reliance on costly professional legal consultation for an initial, plain-language understanding of a legal document or the general prospects of a described legal scenario. FIG 1

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130646 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVELOPMENT OF ECO-FRIENDLY PAVER BLOCK USING BAKELITE AND SEASHELL

(51) International classification	:E01C 5/22, E01C 5/00, E01C 5/06, E01C 5/20, C04B 28/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)SRINIVASAN PALANISAMY
2)MUKESH PANNEERSELVAM
3)SUVETHA GANESAMOORTHI
4)HARISH RAGUPATHI
5)PRANAV MARAN THIRUMARAN
6)RAHULKANDHASAMY
7)BALAKUMAR A
8)OVIYA R

(57) Abstract :
ABSTRACT DEVELOPMENT OF ECO-FRIENDLY PAVER BLOCK USING BAKELITE AND SEASHELL In the present age, waste management has become a critical global challenge faced by various regions in the world. Here, Plastic is significantly growing segment in the municipal solid waste. So, Our Project aims to develop eco-friendly paver blocks using seashell, which was available abundantly and waste material Bakelite as a partial replacement for natural aggregates. The Ordinary Portland Cement (OPC 43) was used as the primary binding material. The Seashells are collected from the coastal regions, cleaned and then crushed into fine aggregate size. The seashell was mixed in the 1:1 ratio of mussel shell and snail shell. The seashell was added in the ratio of 70:30 and 75:25, here the 70% is seashell and 30% is seashell. Bakelite waste is used as the partial replacement for coarse aggregate. And the Bakelite are collected from the plastic recycling shop and then crushed into the particle size of coarse aggregate. The Bakelite was added to the coarse aggregate into 4% and 6%. And then the prepared mix was moulded into the standard paver block sizes and allowed for curing for 28 days. The test for the paver block was conducted as per IS 15658:2006. As the result, the ratio of 70:30 of M-sand and seashell with 4% of Bakelite achieved the compressive strength of 42 MPa, water absorption of paver block was 3% and flexural strength of 5 MPa. And the ratio of 75:25 of M-sand and seashell with 6% of Bakelite showed the compressive strength of 48MPa, water absorption of 2.6% and the flexural strength of 6.5MPa. In both cases, the performance value meets the standard value as per BIS requirements. FIG 1

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130647 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MOBILE EYE TEST FOR VISION SCREENING

(51) International classification	:A61B 3/032, A61B 3/028, A61B 3/02, A61B 3/00, A61B 3/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)NANDHAKUMAR C
2)SIBEYA S
3)SUREKA B
4)SWETHA R
5)Mrs.M.Kirithika Devi
6)Mr. S. MOHAMED FAISAL

(57) Abstract :
ABSTRACT MOBILE EYE TEST FOR VISION SCREENING The present invention relates to a mobile-based vision screening system for performing preliminary eye screening using a smartphone or similar mobile device. The system provides a digital platform for conducting visual acuity, colour blindness, and contrast sensitivity tests through interactive visual interfaces and guided instructions. User responses are collected and processed by the application to determine a preliminary screening result, which is classified as Normal or Abnormal according to predefined evaluation criteria. The system automatically generates and stores a digital vision screening report together with the user's test history for future reference. The application is capable of supporting offline screening by storing relevant information locally when continuous internet connectivity is unavailable and enabling synchronization or communication of stored information when connectivity becomes available. In the event of a potentially abnormal screening result, the system provides a recommendation for further examination by a qualified eye-care professional. The invention provides a portable, accessible, and cost-effective approach for preliminary vision screening, particularly in rural, community, educational, and resource-limited environments. Fig1

No. of Pages : 11 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130648 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AEROLITE: A MODEL COMPRESSION PIPELINE FOR REAL-TIME ONBOARD UAV HEALTH MONITORING

(51) International classification	:G06N 5/02, G06N 5/04, G05B 23/02, G06N 20/00, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr. N. MOHANAPRIYA
2)Nureshma S
3)Suvitha R
4)Swathi S
5)Dr.P.Shanmuga Priya
6)A.SIVASUBRAMANIAN

(57) Abstract :
ABSTRACT AEROLITE: A MODEL COMPRESSION PIPELINE FOR REAL-TIME ONBOARD UAV HEALTH MONITORING Disclosed is a method and system (100) for compressing a trained, fault-conditioned, two-stage Integrated Vehicle Health Management (IVHM) model (106), comprising a diagnostic stage (202) and a Remaining-Useful-Life prognostic stage (204) conditioned thereon, for real-time execution on resource-constrained onboard computing hardware (104) of an unmanned aerial vehicle. A compression pipeline module (214) applies coordinated pruning (208), quantization (210), and knowledge distillation (212) across both stages so as to preserve the conditioning relationship between them, evaluates alternative technique orderings, recalibrates the compressed model's confidence output (406D), and benchmarks the compressed model on representative embedded hardware (406) to obtain measured latency and power values. The compressed model (106) is deployed onto the UAV's onboard computer (104) only upon simultaneously satisfying a predetermined accuracy floor and real-time latency and power constraints, and otherwise the compression is retuned and re-evaluated. The disclosed approach renders an accurate but otherwise undeployable IVHM model operationally usable in real time on constrained UAV hardware, while preserving the diagnostic-to-prognostic conditioning signal that distinguishes the underlying two-stage architecture. Fig 1

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130649 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SECURE CLOSED GROUP COMMUNICATION PLATFORM FOR DEFENCE PERSONNEL AND FAMILIES

(51) International classification	:H04L 29/06, H04L 9/40, H04W 12/06, H04L 9/32, H04W 4/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)PRABAVATHI S
2)ASHWANTH S
3)DHARUN KUMAR M
4)KISHORE S
5)Mr. SANTHOSH KUMAR PALANIAPPAPILLAI CHOCKALINGAM
6)PUNITHA MAM

(57) Abstract :
ABSTRACT SECURE CLOSED GROUP COMMUNICATION PLATFORM FOR DEFENCE PERSONNEL AND FAMILIES A secure closed group communication platform for defence provides controlled communication over a public or untrusted communication network. The platform comprises authorized user devices, registered user and device identities, a mandatory secure network gateway, an authentication and session-control server, a closed-group management server, a secure communication server, and an administrative command server. The gateway establishes an authenticated secure tunnel before protected communication is permitted. The authentication and session-control server verifies the user and device and performs session-level authorization for a requested group. The closed-group management server enforces membership and role permissions, while the communication server encrypts protected data. The administrative command server permits authorized revocation or termination of users, devices, groups and active sessions. Data-containment controls and audit functions reduce unauthorized disclosure and support incident investigation. Fig 1

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130650 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART MULTI DISC ROTARY TABLET FEEDER WITH AUTOMATED DOSE SCHEDULING AND DISPENSING VERIFICATION

(51) International classification	:B65G 1/04, B65G 47/90, A61J 7/00, G07F 11/16, B65G 1/137
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs. I KARTHIKA
2)H V DHARSHAN
3)S MOUNISHVAR
4)K S SUDHARSHANAN
5)Dr. RADHA N
6)Mr. R. DARSHAN

(57) Abstract :
ABSTRACT SMART MULTI DISC ROTARY TABLET FEEDER WITH AUTOMATED DOSE SCHEDULING AND DISPENSING VERIFICATION The present invention discloses an automated device (10) for product handling and dispensing. The device comprises a circular rotating assembly, plurality of product receiving compartments, drive mechanism, actuator assembly, transfer mechanism, conveyor track, output tray and a control panel. The plurality of product receiving compartments (20) are configured to receive and hold the products, wherein the circular rotating assembly (16) is operated by the drive mechanism (12) and drive motor (22) for positioning the selected product receiving compartment. The actuator assembly (26) is configured to operate the transfer mechanism (28) for transferring the selected product from the product receiving compartment towards the conveyor track (32). The conveyor track is configured to transport the selected product towards the output tray (30) for dispensing. The control panel (36) is configured to control the circular rotating assembly, drive mechanism, actuator assembly, transfer mechanism and conveyor track for performing the automated product handling and dispensing operation. The present invention provides automated product positioning, transfer, conveying and dispensing in an effective and controlled manner. Fig 1

No. of Pages : 24 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130651 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : BIOSIGNAL DEEPFAKE DETECTION

(51) International classification	:G06K 9/00, A61B 5/024, A61B 5/00, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)MAKANYADEVI KANDASAMY
2)DEEPIGA P
3)MADHUMATHI K
4)Mrs.MAHIMA R
5)Mr. L. PANCHACHARAM

(57) Abstract :
ABSTRACT BIOSIGNAL DEEPFAKE DETECTION A rapid advancement of deepfake technology has enabled the creation of highly realistic synthetic faces capable of impersonating individuals during live video interactions, posing significant threats to identity verification and digital security. Existing deepfake detection approaches primarily rely on visual artifacts; however, these methods are becoming increasingly ineffective as generation techniques continue to evolve. This project presents a real-time deepfake detection system based on biosignal analysis using Remote Photoplethysmography (rPPG). The proposed system analyzes subtle variations in facial skin color caused by blood flow to extract pulse signals from video streams. Unlike genuine human videos, deepfake content fails to replicate consistent physiological patterns, making it possible to identify manipulated media through irregular or absent pulse signals. The system operates entirely in software without requiring additional hardware and is designed for seamless integration as a browser-based solution. By combining real-time processing with physiological signal analysis, the proposed approach provides an efficient, non-intrusive, and reliable method for enhancing security in online communication platforms. The system aims to improve detection accuracy while minimizing false positives, ensuring reliable and trustworthy user authentication in real-time applications. Fig. 1

No. of Pages : 19 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130654 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Vasathi: An Intelligent AI-Based System for Rental House Search, Location-Aware Property Recommendation and Affordability Analysis

(51) International classification	:G06Q 50/16, G06Q 30/06, G06F 17/30, G06Q 10/00, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr RVR NRI Institute of Technology Deemed to be University.
Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India Vijayawada Andhra Pradesh India
2)N. Narasimha Rao
3)R. Rajeswari
4)U. Greeshma
5)P. Bhavitha
6)V. Durga Bhavani
7)M. Jayanth
(72)**Name of Inventor :**
1)N. Narasimha Rao
2)R. Rajeswari
3)U. Greeshma
4)P. Bhavitha
5)V. Durga Bhavani
6)M. Jayanth

(57) Abstract :
ABSTRACT [0030] An intelligent rental house finding system is disclosed for personalized rental property search. The system receives rental requirements such as location, budget, property type, number of bedrooms, amenities, furnishing status and availability, entered through filters or by voice, validates them, and filters stored property records. Each property is given a matching score from budget, location, property type, bedroom, amenities and availability matches, and the properties are ranked by relevance. The system identifies nearby schools, hospitals, supermarkets, offices and public transportation from the coordinates of a property, estimates distance and travel time, shows them on an interactive map, and classifies each property as within budget, slightly above budget or above budget. Property owners manage listings and enquiries through an owner dashboard, and an administrator manages users and listings. This project presents Vasathi, an AI-powered web application built with React.js, Node.js, Express.js and MongoDB. Traditional rental house searching is time-consuming because property information is scattered, may be incomplete or outdated, and must be compared manually. Vasathi connects property search and filtering, AI-based recommendation, location-based facility information, affordability analysis, voice search and owner and administrator management in a single platform, so that tenants can find and compare suitable rental houses in a simple and organized way.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130657 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Based Pre-Fire Prediction and Human-Aware Adaptive Acoustic Flame Suppression Using Multi-Modal Sensor Fusion

(51) International classification	:G10L 21/0208, A62C 99/00, A62C 3/00, G08B 17/10, G08B 17/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.Senthil Kumar Thiyagarajan
2)Saravanan Vadivel
3)Sibikthan Nallipalayam Murugesan
4)Visva Pravin Sairam Subramanian
5)Sankara Narayanan Chinna Annamalai
6)VENKATESWARA BABU HARIHARAN BABU
7)Rani C

(57) Abstract :
ABSTRACT ADAPTIVE ANALOG NOISE-LEVEL RESPONSIVE SUPPRESSION CONTROL CIRCUIT FOR SPEECH ENHANCEMENT The present invention is about an AI-Based Pre-Fire Prediction and Human-Aware Adaptive Acoustic Flame Suppression System Using Multi Modal Sensor Fusion. This system is designed to identify fires and respond in a controlled way. The system uses a group of sensors that can detect gases and chemicals that are released when a fire starts. It also monitors temperature, humidity and heat. These sensors collect data all the time. Send it to a small computer. This computer prepares the data checks the time and makes sure the sensors are working well. The data from all these sensors is given to an AI system that looks at both the chemicals and the heat to decide how risky a fire might be. When the risk is high enough another part of the system checks to see if there is really a fire. Then it checks if there are people nearby. If there are people the system makes sure it doesn't harm them. Based on what it finds the system uses a tool that uses sound to stop the fire. It keeps checking the fire and changes how it works as needed. The system keeps adjusting how it uses sound to control the fire. The system brings together fire detection, smart sensor systems checking for people and using sound to put out fires. It is meant for testing fires, in controlled situations. It is not meant to take the place of fire detectors or fire extinguishers. It is meant to work with them. FIG 1

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130658 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CIPHER SWITCH SECURE CHAT - DYNAMIC MULTI-ALGORITHM ENCRYPTION AND DECRYPTION FRAMEWORK FOR SECURE CHAT COMMUNICATION

(51) International classification	:H04L 9/08, H04L 29/06, H04L 9/32, H04L 9/30, H04L 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mr. S. PRAKASH
2)ABHI YAZHINI P
3)ABIRAMI M
4)ARUNTHATHI K J
5)Dr.P.Shanmuga Priya
6)Dr. R. JAIGANESH

(57) Abstract :
ABSTRACT CIPHER SWITCH SECURE CHAT A method and system for cryptographically agile secure chat communication uses a mapping table that associates user-facing numeric identifiers with independent cryptographic configurations, each specifying a symmetric encryption algorithm, an asymmetric key-exchange protocol, an authentication mechanism, and a key-derivation function. A sending client resolves a selected numeric identifier to its corresponding configuration, establishes a session key using the specified key-exchange protocol and key-derivation function, and encrypts the message using the specified symmetric algorithm and authentication mechanism. A receiving client resolves the same identifier to the same configuration to decrypt and verify the message. Because the numeric identifier is decoupled from the underlying algorithms, the mapping table can be rotated, diversified across sessions and devices, or migrated toward post-quantum and lightweight primitives without altering the user interface, mitigating single-point cryptographic failure and enabling gradual, coordinated cryptographic-agility across classical, lightweight, and post-quantum algorithm pools. quantum algorithm pools.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130659 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AEROGUARD – A COMPANION APP FOR REAL-TIME TOXIC GAS DETECTION & ALERT MONITORING

(51) International classification	:B64C 39/02, G01N 33/00, G08B 21/14, G05D 1/10, G08B 21/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs.Priya P
2)Hariharan C
3)Gokul K S
4)Ashwanth S
5)Mrs.B.Sathiya
6)S. NAVEEN SUNDAR

(57) Abstract :
ABSTRACT AEROGUARD – A COMPANION APP FOR REAL-TIME TOXIC GAS DETECTION & ALERT MONITORING Aerially, hazardous gases and environmental conditions are monitored by providing an intelligent drone based multi-gas detection, monitoring and hazard response system (100) over a designated geographical area. The system includes an aerial drone platform (102), a multi-sensor unit (104), an ESP32 based processing controller (106), a GPS and telemetry subsystem (108), a communication interface (110), a backend processing system (112), an intelligent analysis subsystem (114), a database (116), an alert management subsystem (118), and an AeroGuard mobile application (120). The multi-sensor unit includes gas sensors that are able to detect methane (CH_4), carbon monoxide (CO), hydrogen sulfide (H_2S) and ammonia (NH_3) as well as environmental sensors that measure temperature, humidity and pressure. The processing controller based on the ESP32 collects, preprocesses and links the sensor data with geographical and temporal data before sending the data to the back-end processing system. Intelligent analysis subsystem analyses gas measurements, environmental data, trends and geographical data to detect abnormal conditions and possible gas hotspots. The system can get more measurements from a known potential hotspot for re-sampling and re-analysis, and safety alerts can be issued for the system. The AeroGuard mobile application offers a real-time visualization of sensor measurements, drone status, geographical information, potential hotspots, alerts, missions and historical monitoring data. FIG 1

No. of Pages : 12 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130660 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PERMISSION ABUSE DETECTOR USING BEHAVIOURAL ANOMALY DETECTION

(51) International classification	:G06F 21/57, G06N 20/00, G06F 21/55, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.D. PRADEEP
2)JASWINPRATHAP NALLUSAMY
3)ARUL SELVAN K M
4)Dr. G. GABRIEL SANTHOSH KUMAR
5)S. ANNISH

(57) Abstract :
ABSTRACT PERMISSION ABUSE DETECTOR USING BEHAVIOURAL ANOMALY DETECTION The present invention discloses a Permission Abuse Detector — an on-device behavioural anomaly detection system (100) for Android mobile platforms — that transitions mobile application security from static permission validation to continuous intelligent runtime monitoring. The system employs a Behaviour Collector (120) and Preprocessing Layer (121) to gather and structure application usage data, which is transformed into an eleven-dimensional behavioural feature vector by the Feature Extraction Engine (131). The feature vector is subjected to a multi-layer hybrid Detection Engine (130) comprising a Rule Engine (132), an Adaptive Baseline Module (133) with risk-coupled EWMA learning governed by the novel relation $\alpha_t = \alpha_{base} \times (1 - \tanh(\beta \times R_t))$, a Consistency Gate (133B) enforcing deviation-proportional stability requirements, a Dual-Anchor Trust Controller (133C) bounding maximum baseline drift, an Anomaly Memory Retention unit (133D) persisting risk history across application updates, an Isolation Forest Machine Learning model (134) with contamination-aware retraining, a Trend Detector (135), and a Cross-App Correlation Engine (137). A Confidence Calculator (138) assesses detection reliability and a Score Aggregator (136) consolidates outputs using a weighted formula. The Alert and Notification Module (140) delivers real-time context-rich alerts. Working evaluation data demonstrates successful detection of a 10-day gradual poisoning attack on Day 18, with no baseline corruption, confirming the manipulation-resistance of the novel mechanisms. All processing is performed exclusively on-device, preserving user privacy. FIG 1

No. of Pages : 12 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130661 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED SMART LOGISTICS AND ACCESSIBILITY INTELLIGENCE PLATFORM FOR NORTH EASTERN REGION

(51) International classification	:G06Q 10/08, G01C 21/34, G06Q 10/06, G06Q 50/26, G06Q 10/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs.S.KAYATHRI
2)DHANISHKA C
3)MAHESWARI A
4)NIVETHA G
5)SUBITHA R
6)Dr. V. C. BHARATH SABARISH
7)Bhavani P

(57) Abstract :
ABSTRACT Ai-Based Smart Logistics And Accessibility Intelligence Platform For North Eastern Region The present disclosure relates to an AI-Based Smart Logistics and Accessibility Intelligence Platform for the North Eastern Region. The platform integrates shipment information with weather, road, terrain, incident and GPS data to support intelligent logistics route planning and monitoring. A data processing module cleans, validates and integrates the collected information. An AI-based risk engine evaluates potential risks including flooding, landslides, rainfall and road disruptions. An accessibility scoring module generates a route Accessibility Score, while a vehicle suitability module evaluates the compatibility of the vehicle and cargo with route conditions. A smart route engine evaluates multiple candidate routes based on accessibility, risk, ETA and vehicle suitability and provides a suitable route recommendation. During transportation, GPS tracking and incident monitoring enable real-time alerts. When route conditions change, the system recalculates risk and provides an alternative route. The platform provides centralized logistics monitoring and is intended to support resilient transportation operations in geographically challenging and disaster-prone regions. Important patent-drafting point Your strongest technical concept to emphasize is not simply “using AI for logistics,” because that is a broad area with substantial prior art. The specification should clearly explain the specific combination and interaction of: Multi-source NER data → AI disruption risk → Accessibility Score → Vehicle/Cargo Suitability → Route evaluation → Dynamic rerouting → Continuous monitoring. Fig 1

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130662 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI- NATIVE RESOURCE ALLOCATION SYSTEM FOR MULTI-AGENCY EMERGENCY RESPONSE

(51) International classification	:G06Q 50/26, G08B 17/06, G08B 17/10, G08B 17/00, G08B 17/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MS.C.BAVITHRA
2)AJEETHA E
3)DYUKSHA MOORTHI
4)GOWSHYA P G
5)ATULKRISHNA VALLIYEDATHU
6)D. JOTHI MANOHARI

(57) Abstract :
ABSTRACT AI-NATIVE RESOURCE ALLOCATION SYSTEM FOR MULTI-AGENCY EMERGENCY RESPONSE The present invention relates to an IoT-based integrated flood and forest fire monitoring and early warning system designed for continuous environmental monitoring. The system comprises distributed sensor nodes connected to ESP32-based controllers. The flood monitoring node measures water level using an ultrasonic sensing mechanism, while the forest-fire monitoring node monitors environmental parameters including temperature, humidity, smoke, gas and/or flame-related conditions. The ESP32 controller collects and processes sensor measurements and transmits the information to a centralized monitoring platform. The system analyzes the received data to identify abnormal conditions and generate appropriate warning alerts. A remote dashboard provides real-time visualization of sensor measurements, disaster status, locations, historical trends and alerts. The proposed invention combines environmental sensing, embedded processing, wireless communication, data storage, analysis and real-time monitoring to support early detection and improved awareness of flood and forest-fire conditions. FIG 1

No. of Pages : 10 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130663 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART FOOD STORAGE CONTAINER

(51) International classification	:G01N 33/02, G01D 21/02, G06Q 10/08, G06Q 50/12, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)MRS M SATHYA
2)NISHANTHINI G
3)SANGAVI P
4)DHARSHINI S
5)VARSHINI M
6)Mrs.M.Karthika
7)ANNISH.S

(57) Abstract :
ABSTRACT SMART FOOD STORAGE CONTAINER The present invention relates to a Smart Food Storage Container (1) configured to monitor the condition of stored food and estimate its spoilage risk. The container comprises a temperature sensor (3), humidity sensor (4), gas/VOC sensor (5) and weight sensing unit (6) for continuously or periodically collecting information related to the food and its storage environment. The collected sensor data is processed by an ESP32 controller (7) and analysed using an ML/data processing unit (8) to determine the spoilage-risk condition.Based on the estimated risk level, the system classifies the condition into Low Risk, Medium Risk or High Risk and controls a cooling unit (9) and ventilation mechanism (10) accordingly. A display unit (11) provides the monitored parameters and risk condition to the user, while an alert unit (12) provides an indication when a high spoilage risk is detected. The proposed invention enables continuous food-condition monitoring with reduced manual inspection and assists in improving food storage management and reducing avoidable food wastage. FIG 1

No. of Pages : 18 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130664 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Cyber Physical Based Hybrid Digital Twin Optimized Transformerless Charging Dock for EV Using IET

(51) International classification	:B60L 11/18, H02J 7/00, H02J 7/02, B60L 53/66, B60L 53/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)DINESH E
2)ANNAMALAI K
3)AAKASH V
4)AMARJEET KUMAR V
5)BHARATH ATHITHIYA S
6)BINU SHERLIN S
7)Dr.J.Sasidevi

(57) Abstract :
ABSTRACT Cyber Physical Based Hybrid Digital Twin Optimized Transformerless Charging Dock for EV Using IET The present invention relates to an intelligent transformer-less electric vehicle (EV) charging system designed to provide safe, efficient, and adaptive charging of EV batteries. The system receives electrical power from a three-phase AC supply and converts it into a controlled DC output using a transformer-less power electronic conversion stage. During charging, electrical and thermal parameters such as voltage, current, temperature, and battery state of charge are continuously monitored using sensors. A Raspberry Pi-based control unit processes the collected information and uses an artificial intelligence-based decision mechanism to determine suitable charging conditions and dynamically regulate the charging process. The system further incorporates a Digital Twin that represents the operating condition of the physical charging station and is continuously updated using real-time sensor data. The Digital Twin and AI-based monitoring mechanism assist in identifying abnormal operating conditions and predicting potential faults before they develop into critical conditions. Based on the detected condition, the charging controller can reduce the charging power, temporarily interrupt charging, or disconnect the charging supply through a protection mechanism. Charging events, operating parameters, and detected faults may also be recorded for monitoring and future analysis. The proposed system combines transformer-less power conversion, real-time sensing, adaptive charging control, artificial intelligence, predictive protection, and Digital Twin technology into a unified EV charging architecture, thereby providing a scalable approach for safer, more efficient, and reliable EV charging infrastructure. FIG 1

No. of Pages : 20 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130665 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR OFFLINE-AWARE GEOSPATIAL INCIDENT LIFECYCLE MANAGEMENT AND DYNAMIC SAFETY ROUTING FOR HILL INFRASTRUCTURE

(51) International classification	:G06Q 50/26, H04W 4/029, G08G 1/01, G06F 16/29, H04W 4/021	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)MAKANYADEVI KANDASAMY** **2)NITISH KHANNA S** **3)RAHUL KUMAR** **4)RAKESH N** **5)Dr.P.Shanmuga Priya** **6)Mr. M. KATHIRVEL**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT A SYSTEM AND METHOD FOR OFFLINE-AWARE GEOSPATIAL INCIDENT LIFECYCLE MANAGEMENT AND DYNAMIC SAFETY ROUTING FOR HILL INFRASTRUCTURE The proposed system relates to a system and method for geospatial incident lifecycle management and dynamic safety routing of hill infrastructure without the need for an online connection. The system allows users to report any infrastructure incident, such as a road damage, landslide, fallen tree, bridge damage, drainage failure or a road blockage, via a mobile application. When there is no network connectivity, incident information including images, description, GPS information, infrastructure category and timestamps is recorded and stored locally. Once connectivity is restored, the information is automatically synced with a central server without losing its geographical and temporal context. The system also includes geospatial incident correlation to identify if a synchronized report is a new incident, duplicate report, update or supporting evidence for an existing incident. Incidents are clustered into related incidents and processed in a dynamic lifecycle of verification, assignment, maintenance, resolution, expiry and reopening. FIG 1

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130666 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTEGRATED PASSIVE HYDRAULIC INLET CONDITIONING SYSTEM FOR REDUCING CAVITATION INCEPTION

(51) International classification	:F04D 29/66, F04D 15/00, F04D 29/44, F15D 1/00, F15D 1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.G.R.Gopinath
2)Mr. N Sethumadhavan
3)Mr. S Sharath
4)Mr. A Tharunsarwin
5)Mr. G Santhosh
6)Mr.Santhosh Kumar
7)Dr. R. JAIGANESH

(57) Abstract :
ABSTRACT INTEGRATED PASSIVE HYDRAULIC INLET CONDITIONING SYSTEM FOR REDUCING CAVITATION INCEPTION The current invention relates to an integrated passive hydraulic inlet flow conditioning system that improves the flow and pressure condition of hydraulic fluid prior to entering the hydraulic pump. The invention includes a primary inlet flow conditioning member, a flow guidance member, a pressure recovery member, and a pressure stabilization member connected in series along a single flow path of the hydraulic fluid. According to an exemplary embodiment of the invention, the primary inlet flow conditioning member is a perforated cylindrical member, the flow guidance member is a guide vane assembly, the pressure recovery member is a diffuser member, and the pressure stabilization member is a honeycomb member. The hydraulic fluid travels in sequence through all four members in such a way that flow condition generated in one member serves as an input condition to the next member in the sequence. Fig 1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130667 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMMERCIAL VEHICLE OVERLOAD DETECTION AND OWNER ALERT SYSTEM USING LOAD CELL AND ESP32

(51) International classification	:G01G 19/08, G01G 19/12, G01G 19/03, G08G 1/01, G01G 19/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.K.RAJU
2)Mr.A.ADHISWARAN
3)Mr.R C DHANUS VARMA
4)Mr.K.ARUN
5)Mr.U.NEGORIAN
6)Mrs.M.Manogiri
7)Mrs .T.SOUNDARYA

(57) Abstract :
ABSTRACT COMMERCIAL VEHICLE OVERLOAD DETECTION AND OWNER ALERT SYSTEM USING LOAD CELL AND ESP32 The present invention relates to a Commercial Vehicle Overload Detection and Owner Alert System using Load Cell and ESP32 for continuous monitoring and detection of excessive loading in commercial vehicles. The system employs multiple load cells strategically positioned on the vehicle chassis to measure the applied load. The sensor outputs are amplified using HX711 amplifier modules and processed by an ESP32 microcontroller to determine the total vehicle load. The measured load is compared with a predefined permissible load limit corresponding to the rated capacity of the vehicle. The loading status can be displayed to the driver, while visual and audible indicators provide appropriate warnings when the load approaches or exceeds the permissible limit. In the event of overloading, the system automatically transmits an alert containing relevant information such as vehicle identification, load status, and time of occurrence to the vehicle owner or fleet manager through a wireless communication interface. The system can additionally record load measurements, overload events, date and time, and other associated information for future analysis and fleet management. Multiple sensing points enable monitoring of load distribution and identification of significant variations in loading across the vehicle platform. The system may also be integrated with GPS tracking and cloud-based data storage for remote monitoring and historical analysis. By providing real-time load monitoring and automatic notification, the proposed system helps prevent unsafe overloading, reduce stress on tyres, brakes, suspension and chassis components, improve fuel efficiency, and minimize maintenance costs.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130668 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE SENSOR-BASED CRICKET BOUNDARY CATCH DETECTION DEVICE

(51) International classification	:A63B 71/06, A63B 24/00, G08B 21/02, G01P 15/00, G08B 21/04	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)MRS.SANGEETHA** **2)SURYA S M** **3)GANESHAN D** **4)VASANTHARAJAN R** **5)RITHICK A** **6)Mr. S. P. RICHARD** **7)Ms. M. VALARMATHI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT WEARABLE SENSOR-BASED CRICKET BOUNDARY CATCH DETECTION DEVICE The present invention is about a sensor-based catch detection and verification system for cricket, which helps confirm close or disputed catches near a boundary. The sensor-based boundary catch detection and verification system contains a sensing assembly, a wearable fielder sensing unit, wireless communication units, a receiver and processing unit, a display interface and a power supply. The wearable sensing unit is set to detect motion, acceleration, movement or impact from a fielder's catching action and to create a catch-event timestamp. The boundary sensing assembly is set to detect contact or pressure by the fielder at a boundary region and to create a boundary-contact timestamp. The event information is sent wirelessly to the receiver and processing unit, where the timestamps are lined up and compared to find the order of the catch event and the boundary contact event. The sensor-based boundary catch detection and verification system produces a decision-support indication based on that order. Shows the timing details on the receiver display, for an official to review. The sensor-based boundary catch detection and verification system adds evidence to normal video-based decisions helping cut doubt and make close boundary-catch checks faster and more reliable all without changing the cricket ball. FIG 1

No. of Pages : 15 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130669 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN AND DEVELOPMENT OF A NON-ELECTRICAL SOLAR TRACKING SYSTEM

(51) International classification	:F24S 50/20, F24J 2/54, H02S 20/32, G05D 3/12, F24J 2/40	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr. G. ARULMURUGAN**
(33) Name of priority country	:NA	**2)Ms.G.Ishwarya**
(86) International Application No Filing Date	: :01/01/1900	**3)Ms.P.Caroline** **4)Ms.P.Catherine**
(87) International Publication No	: NA	**5)Ms.M.Harini**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)Mr.G.N.Hariharasudhan** **7)Ms.M.Kanishka**
(62) Divisional to Application Number Filing Date	:NA :NA	**8)Mr. SUBRAMANIYA SIVA A** **9)Ms. A. AIYESHA SHITHIKKA**

(57) Abstract :

ABSTRACT DESIGN AND DEVELOPMENT OF A NON-ELECTRICAL SOLAR TRACKING SYSTEM The present invention proposes a non-electrical solar tracking system for automatically adjusting the angular position of a solar panel according to the apparent movement of the sun. The system uses a water tank (1), small water tank for weight balancing (2), water tube (3), tension spring (4), solar panel (5), rotating shaft (6), and panel holder (7). Water transfer between the tanks produces a changing gravitational load, which creates a mechanical moment about the rotating shaft. The tension spring (4) provides the required elastic force to control and stabilize the movement of the panel. As the spring extends or relaxes, its force changes according to the applied displacement, allowing the panel to move gradually toward a new equilibrium position. The system operates without motors, sensors, batteries, or electronic control units, providing a simple mechanical approach to solar tracking. The proposed arrangement can be used for low-cost, rural, agricultural, remote, and small-scale solar applications where simple construction and reduced electrical dependency are required.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130670 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART BUS FOOTBOARD LiDAR SAFETY INTERLOCK SYSTEM

(51) International classification	:G01S 17/931, B60Q 9/00, B60R 21/0134, B60R 3/02, B60R 21/015
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mr. VISWANATHKANI T
2)MANIKANDAN M
3)SARAVANAN AV
4)YASVANTHPALANI S
5)THIRUSELVAM K
6)Mrs.N.G.Gayathri
7)Mr. V. SURESHKUMAR

(57) Abstract :
ABSTRACT SMART BUS FOOTBOARD LIDAR SAFETY INTERLOCK SYSTEM Conventional bus safety mechanisms primarily rely on driver observation, mirrors, door sensors, or basic passenger warning systems and may not reliably detect passengers standing on or extending beyond the bus footboard during vehicle movement. The present invention provides a Smart Bus Footboard LiDAR Safety Interlock System configured to continuously monitor the bus entrance and footboard region using LiDAR sensing technology. The system generates spatial information representing persons and objects within a predefined three-dimensional safety region and classifies detected conditions according to passenger position, movement, occupancy duration, and location relative to the footboard and bus entrance. The system may define safe, boarding, alighting, footboard, and hazardous external zones and compare real-time LiDAR measurements with these virtual zones. The detected passenger behaviour is further correlated with vehicle operating parameters including vehicle speed, door status, movement status, and vehicle start request. When a passenger is detected in a hazardous footboard region while the vehicle is preparing to move or operating, the system evaluates the persistence and confidence of the condition before generating a safety response. Based on the detected condition, the system may generate audible or visual warnings and activate a safety interlock configured to prevent or restrict unsafe vehicle movement until the hazardous passenger condition is cleared. The system further incorporates object classification, temporal filtering, sensor-health monitoring, event memory, fail-safe operation, and recovery verification to reduce false alarms and improve reliability. Accordingly, the invention transforms conventional bus footboard monitoring from a manually observed safety process into an intelligent spatial safety and interlocking system capable of detecting hazardous passenger occupancy, evaluating risk, generating warnings, preventing unsafe vehicle movement, recording safety events, and verifying safe recovery in real time. FIG 1

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130671 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED MULTI-SENSOR FOG DETECTION, VISIBILITY ENHANCEMENT AND DRIVER SAFETY SYSTEM

(51) International classification	:B60W 50/14, G06K 9/00, G06V 20/58, G06V 20/59, G06V 10/764
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr. G. ARULMURUGAN
2)Ms.G.Ishwarya
3)Ms.P.Caroline
4)Ms.P.Catherine
5)Ms.M.Harini
6)Mr.G.N.Hariharasudhan
7)Ms.M.Kanishka
8)Mr. A. Jainulafdeen
9)Mr. R. POOVINKUMAR

(57) Abstract :
ABSTRACT AI-BASED MULTI-SENSOR FOG DETECTION, VISIBILITY ENHANCEMENT AND DRIVER SAFETY SYSTEM The present invention relates to an AI-based -sensor fog detection, visibility enhancement and driver safety system. This system is mounted on a vehicle. The system includes a camera that is placed at the part of the vehicle. The camera captures the road scene. There is a fog or visibility detection module that identifies low-visibility conditions. There is also an image enhancement module that uses an AOD-Net based dehazing model to improve the visibility of the captured image. An object detection module uses an SSD MobileNet based object detection model to detect vehicles, pedestrians and other road objects from the image. A radar or distance sensor independently detects obstacles. Gives distance information regardless of camera visibility. The system also has a sensor fusion module that combines camera-based object detection information with radar-based obstacle information. A camera cleaning mechanism includes a cyclone separator, an air filter, an air blower, an air pipe or air duct and a jet nozzle assembly. This mechanism supplies filtered air to the surface of the camera to remove moisture, water droplets and deposited particles. A processor or control unit coordinates the system operation. A warning or alert module gives a warning to the driver based on the combined obstacle detection output. The system provides fog detection, image enhancement, object detection, radar sensing, camera cleaning and driver warning for better road-obstacle perception, under low-visibility conditions. Fig 1

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130672 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TRANSFORMER SECURITY AND WIREMAN SAFETY

(51) International classification	:H01F 27/40, H02J 13/00, H02H 7/04, G08B 21/02, G08B 21/18	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)Dr. G. ARULMURUGAN** **2)Mr.M.Abishek** **3)Mr.T.Abishek** **4)Mr.A.Amarjith** **5)Mr.ARAVINTHKUMAR K** **6)Mr.P.Ranjith** **7)Ms. A. THASNEEM FATHIMA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT TRANSFORMER SECURITY AND WIREMAN SAFETY Transformer-related accidents, especially during rainy conditions, cause serious safety risks and loss of life. This invention presents a smart transformer safety system using sensors, Arduino, fingerprint authentication, and wireless communication. Rain and electrical sensors detect unsafe conditions and automatically disconnect power through a relay module. Authorized ON/OFF control is provided through fingerprint verification, ensuring secure operation of the transformer. The Arduino acts as the central controller for processing sensor and authentication inputs, while the display indicates the operating and safety status of the system. The NodeMCU ESP8266 enables wireless monitoring and safety alerts for quick response. The system helps prevent electrocution, reduces human dependency, and enhances transformer security and wireman safety through automated control and monitoring. Fig 1

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130673 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : OFFSPOT-Location Based Offer System

(51) International classification	:G06Q 30/02, H04W 4/029, H04W 4/02, G06Q 30/00, H04W 4/021
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs. A. KANIMOZHI
2)RHITHICKSEELAN A
3)MUSTAFA ASHIK M
4)PRANITHI SRIRAM G
5)Mrs.M.Karthika
6)Mrs. P. LATHA

(57) Abstract :
ABSTRACT OFFSPOT: LOCATION BASED OFFER SYSTEM One reason consumers have difficulty looking for nearby store discounts is that these details are so scattered. We created OFFSPOT to enable people to easily discover discounts at stores close to their homes. OFFSPOT is a location-based service mobile application that shows the deals available nearby. The user gets maps of the shops which highlight where there are deals. The feature of the shop information can be viewed through the application. The shop owners get the option to handle their deals via the app. We created a Flutter-based Firebase application. The main idea of the app is to give the customers a chance to locate discount places they can benefit from in their area. The app also aims at establishing a connection platform between shops and the people nearby through which they reach potential customers. The app's functionality can further be examined to see what aspects need improvement. The core is to check if the app successfully shows the users the deals and whether it is easy to use. FIG 1

No. of Pages : 12 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130674 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IV BOTTLE STAND WITH AUTOMATIC BLOOD BACKFLOW SHUTOFF MECHANISM

(51) International classification	:A61M 5/168, A61M 5/14, A61M 5/175, A61M 5/36, A61M 39/28	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)Mr.R.Karthick** **2)Mr.S.GOKULA KUMARAN** **3)Mr.T.ABISHEK** **4)Mr.P.HARIHARAN** **5)Mr.M.SATHISH KUMAR** **6)Mrs.N.G.Gayathri** **7)Mr. H.B MOHAMMED BILAL**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT IV BOTTLE STAND WITH AUTOMATIC BACKFLOW SHUT OFF MECHANISM The present invention relates to a non-electrical automatic IV(Intravenous) flow shut-off mechanism for preventing blood backflow when an IV bottle reaches a predetermined minimum weight. The mechanism comprises a bottle-support arrangement, slider, compression spring, connecting rod, crank, eccentric cam, tube support block, and manual shut-off/reset arrangement. As the IV fluid level decreases, the bottle becomes lighter, allowing the spring to move the slider upward and rotate the eccentric cam. At the predetermined minimum weight, the cam presses the IV tube against the support block, thereby stopping the IV fluid flow and helping to prevent blood backflow. The mechanism operates without electricity or electronic components and provides a simple, economical, reusable, and mechanically operated solution for automatic IV flow interruption.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130675 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : UNIVERSAL INTELLIGENT AUTOMATIC CHARGING SOCKET WITH MECHANICAL AND ELECTRICAL DETECTION

(51) International classification	:H01R 13/70, H01R 13/703, H01R 13/66, H02J 7/00, H01R 13/713
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.M.LOGANATHAN
2)Ms.M.KANISHKA
3)Mr. PRABHU A
4)Ms. A. AIYESHA SHITHIKKA

(57) Abstract :
ABSTRACT UNIVERSAL INTELLIGENT AUTOMATIC CHARGING SOCKET WITH MECHANICAL AND ELECTRICAL DETECTION The present invention relates to a mechanical safety charging socket that allows electrical power only when the plug is correctly and completely inserted. The mechanism uses a taper, slider, springs, switch actuating rod, and micro switch to provide mechanical interlocking. During correct insertion, the plug moves the taper and slider, which operates the switch actuating rod. The micro switch then closes the electrical path and allows charging. During incomplete or incorrect insertion, the required movement does not occur, so the micro switch remains OFF and charging is prevented. The mechanism works without electronic sensors or software. The spring arrangement automatically returns the components to their original position when the plug is removed. Thus, the invention provides a simple, low-cost, and mechanically operated charging safety system. FIG 1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130676 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IoT-Enabled Smart Retail Checkout System with Automated Product Unlocking and Tamper Detection

(51) International classification	:G06Q 20/20, G06K 19/06, G06Q 20/40, G06K 7/14, G06Q 20/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs. I KARTHIKA
2)HARSHAN R
3)JEEVA S
4)KISHORE S
5)Dr.R. MANIVASAGAM
6)Mr. L. PANCHACHARAM

(57) Abstract :
ABSTRACT AI-DRIVEN OEM DEVICE VULNERABILITY INTELLIGENCE AND EXPLAINABLE SECURITY RECOMMENDATION FRAMEWORK The proposed invention presents a Secure QR-Based IoT-Enabled Smart Security Tag System for Retail Automation that provides a secure and convenient self-checkout experience. Customers can scan QR codes attached to products using a mobile application, view product details, add items to a digital cart and complete the checkout process. The system verifies the payment through a centralized backend before sending an unlock command to the corresponding ESP32-based smart security tag. A tamper detection mechanism identifies unauthorized removal or damage of the security tag and generates real-time alerts for the administrator. The system maintains information related to products, orders, payments and tag status through a centralized database. An administrator dashboard provides real-time monitoring of transactions, security tags and tamper events. By integrating mobile technology, backend services, database management and IoT devices, the proposed system reduces manual billing operations, improves product security and provides a fast, convenient and contactless shopping experience. FIG 1

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130678 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUAL NODE SYNCHRONIZED WIRELESS WEARABLE SOMATOSENSORY STIMULATION

(51) International classification	:A61B 5/00, A61B 5/024, A61B 5/11, A61B 5/08, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs.M.GEETHA
2)SIVAGURUNATHAN L
3)VINOTH L
4)VETRIVEL T
5)Mrs.N.G.Gayathri
6)Mr. R.NAVEEN KUMAR

(57) Abstract :
ABSTRACT AeroVigil: Edge-Native Physio-Acoustic Digital Twin for Chrono-Respiratory Prediction Earlier detection of stroke and other similar brain vascular diseases will help doctors timely prescribe the right treatment. At present, however, numerous primary health care centers in resource-limited areas cannot afford the use of advanced equipment for accurate diagnostics. Sometimes, wearable devices such as smartwatches give false-positive warnings because of some body changes due to movements, fatigue, or stress, that look like signs of a disease, even when there is none. To address the above-mentioned problem, this paper presents NeuroSentinel-X – a wearable medical device that analyzes HRV (heart rate variability), tiny muscular tremors, and sEMG (electrical muscle signal) together and, thus, provides an in-depth picture of the physiological state of the body.Intelligent Physiological Validation represents a key element of this system, which determines the presence or absence of a physiological change in the body and whether they are permanent or just temporary. This element assesses the duration of change, consistency, and discrepancy of all physiological signals received. To do this, the system utilizes the TCMPCA (temporal cross-modal physiological consistency algorithm). This algorithm makes a decision about the relationship between the unusual physiological signals received from various sources: whether they relate to a single physiological problem or not because of movement or fatigue, etc.After the above-described process of assessment, the validated data goes through the process of risk assessment. This stage utilizes the TinyML (Tiny Machine Learning) technology that is processed on a small microchip ESP32 without connection to the internet. This ensures the privacy and security of the information. In conclusion, the presented system offers an affordable, rapid, and secure method for detecting stroke risk.

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130679 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE PERSONALIZED DRUG ANALYTICS: A MACHINE LEARNING FRAMEWORK USING RANDOM FOREST AND XGBOOST FOR ADVERSE DRUG REACTION PREDICTION AND VISUALIZATION

(51) International classification	:G16H 70/40, G16H 20/10, G16H 50/70, G16H 50/20, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr. N. MOHANAPRIYA
2)KANISHKAR. K
3)KAVIARASAN. M
4)MOHAMMAD ASRAR UL HAQUE AHANGER
5)Mrs. Kokila K
6)Dr .A.OORKALAN

(57) Abstract :
ABSTRACT Predictive Personalized Drug Analytics is a machine learning-based framework designed to predict Adverse Drug Reactions This project proposes an intelligent predictive system using Random Forest and XGBoost machine learning algorithms to analyze drug-related and patient-related data and identify patterns associated with possible adverse drug reactions. The system involves data collection, data preprocessing, feature selection, model training, prediction, and performance evaluation to generate reliable predictions based on the available dataset. Random Forest and XGBoost models are trained and compared using evaluation metrics such as accuracy, precision, recall, and F1-score to determine their predictive performance. The trained models analyze relevant input features and estimate the likelihood of adverse drug reactions. These visualizations help users explore adverse drug reaction patterns, compare the performance of both machine learning algorithms, and identify important factors that may influence drug safety. The personalized analytics approach aims to support more informed healthcare decisions by considering relevant patient and medication information available in the dataset. The proposed system can be useful for healthcare professionals and researchers in studying drug safety patterns and identifying potential adverse reactions at an early stage. Overall, this project aims to develop an efficient, user-friendly, and data-driven framework for adverse drug reaction prediction and visualization using Random Forest and XGBoost algorithms, contributing to improved drug safety assessment and healthcare decision support. The system is intended to support clinical decision-making and does not replace professional medical advice or diagnosis.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130680 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SELF-POWERED SERVER ROOM ENVIRONMENTAL MONITORING SYSTEM USING CPU WASTE HEAT RECOVERY AND ADAPTIVE THERMOELECTRIC ENERGY MANAGEMENT

(51) International classification	:H05K 7/20, G06F 1/20, G05D 23/19, H02N 11/00, F24F 11/00	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr. P. Manikandan**
(32) Priority Date	:NA	**2)Mr.S.Gokula Kumaran**
(33) Name of priority country	:NA	**3)Mr.S.Adhithya**
(86) International Application No Filing Date	: :01/01/1900	**4)Mr.P.Hariharan** **5)Mr.M.Sathish Kumar**
(87) International Publication No	: NA	**6)Dr.S.Saravanakumar**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**7)Mr.S.RAMKUMAR**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT A SELF-POWERED SERVER ROOM ENVIRONMENTAL MONITORING SYSTEM USING CPU WASTE HEAT RECOVERY AND ADAPTIVE THERMOELECTRIC ENERGY MANAGEMENT The present invention relates to a self-powered server room environmental monitoring system using CPU waste heat recovery. The system recovers a portion of the thermal energy generated by a server CPU during normal operation. A thermal spreader and adjustable thermal contact mechanism transfer the recovered heat to a Thermoelectric Generator (TEG). The TEG converts the temperature difference between its hot and cold sides into electrical energy through the Seebeck effect. A cooling arrangement maintains the cold side of the TEG at a suitable temperature for energy generation. The generated electrical energy is conditioned and stored in a supercapacitor, rechargeable battery, or suitable energy storage unit. The stored energy is utilized to operate temperature and humidity sensors and associated low-power monitoring components. An adaptive energy-management system adjusts sensor operation and communication according to the available stored energy. The system supports intermittent environmental monitoring while monitoring thermal and electrical parameters for safe operation. The invention thereby provides an integrated approach for utilizing server CPU waste heat to support self-powered environmental monitoring.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130681 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : NEUROSENTINEL-X: PHYSIONEURO FUSION ANALYTICS FRAMEWORK FOR EARLY CEREBROVASCULAR RISK ASSESSMENT

(51) International classification	:A61B 5/024, A61B 5/00, A61B 5/11, A61B 5/0245, A61B 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)NANDHAKUMAR C
2)SABAREESWARAN B
3)SANTHOSH KUMAR K
4)SARAVANA KUMAR S
5)Dr.P.Shanmuga Priya
6)Mr. R.NAVEEN KUMAR

(57) Abstract :
ABSTRACT NeuroSentinel-X:PhysioNeuro Fusion Analytics Framework for Early Cerebrovascular Risk Assessment Earlier detection of stroke and other similar brain vascular diseases will help doctors timely prescribe the right treatment. At present, however, numerous primary health care centers in resource-limited areas cannot afford the use of advanced equipment for accurate diagnostics. Sometimes, wearable devices such as smartwatches give false-positive warnings because of some body changes due to movements, fatigue, or stress, that look like signs of a disease, even when there is none. To address the above-mentioned problem, this paper presents NeuroSentinel-X – a wearable medical device that analyzes HRV (heart rate variability), tiny muscular tremors, and sEMG (electrical muscle signal) together and, thus, provides an in-depth picture of the physiological state of the body.Intelligent Physiological Validation represents a key element of this system, which determines the presence or absence of a physiological change in the body and whether they are permanent or just temporary. This element assesses the duration of change, consistency, and discrepancy of all physiological signals received. To do this, the system utilizes the TCMPCA (temporal cross-modal physiological consistency algorithm). This algorithm makes a decision about the relationship between the unusual physiological signals received from various sources: whether they relate to a single physiological problem or not because of movement or fatigue, etc.After the above-described process of assessment, the validated data goes through the process of risk assessment. This stage utilizes the TinyML (Tiny Machine Learning) technology that is processed on a small microchip ESP32 without connection to the internet. This ensures the privacy and security of the information. In conclusion, the presented system offers an affordable, rapid, and secure method for detecting stroke risk. FIG 1

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130682 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Visionx : Ai-Powered Multi-View Infrared-To-Rgb Image Reconstruction And Analysis

(51) International classification	:G06T 5/00, G06T 5/50, G06N 3/08, G06V 10/764, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(72)**Name of Inventor :**
1)Dr.B.PADMINI DEVI
2)KARTHIKEYAN M
3)KEVIN SIV RAJ M S
4)MADHAVAN C T P
5)Dr. M. Edwin Antony Raj,
6)Mrs. R. ESWARI

(57) Abstract :
ABSTRACT VisionX: AI-Powered Multi-View Infrared-to-RGB Image Reconstruction and Analysis The system collects infrared images from multiple viewpoints and applies image preprocessing techniques such as resizing, normalization, noise reduction, and alignment to improve image quality. An AI-based image reconstruction model learns the relationship between infrared and RGB image features to generate reconstructed images with improved visual details, colors, and structural information. The multi-view approach combines information from different viewing angles to improve scene understanding and support more comprehensive image analysis. After reconstruction, the system performs image analysis tasks such as object detection, feature extraction, The performance of the proposed system can be evaluated using suitable image quality metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and other relevant evaluation techniques. The main objective of VisionX is to develop an intelligent image reconstruction and analysis framework that improves the interpretability of infrared imagery and supports applications such as surveillance, robotics, industrial inspection, and environmental monitoring. Overall, the proposed system combines artificial intelligence, deep learning, and computer vision to transform infrared image information into more visually informative representations and enable effective multi-view image analysis.

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130683 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUDIUMSPEC: TRANSITION-AWARE TELEPHONY SYNTHETIC VOICE AUTHENTICATION USING GLOTTAL MICRO-JITTER, SPECTRAL GRADIENTS, AND ENERGY-GATED DECISION ROUTING

(51) International classification	:G10L 17/00, G10L 17/04, G10L 17/06, G10L 17/02, G10L 21/0208
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs. D.SATHIYA
2)E.NEVESHINI
3)P.PAVITHRA
4)G.PRIYA DHARSHINI
5)Dr.K.Dinesh
6)Mrs .T.SOUNDARYA

(57) Abstract :
ABSTRACT AUDIUMSPEC: Transition-Aware Telephony Synthetic Voice Authentication Using Glottal Micro-Jitter, Spectral Gradients, and Energy-Gated Decision Routing The system addresses the increasing risk of voice spoofing, impersonation, and telephone-based fraud caused by advanced voice synthesis technologies. It analyzes speech signals using acoustic features such as glottal micro-jitter, spectral gradients, and short-time energy to identify differences between natural and synthetic speech. Glottal micro-jitter measures small variations in vocal frequency, while spectral gradients capture changes in the frequency distribution of speech signals. Energy-gated decision routing evaluates the signal energy to determine the appropriate authentication pathway. The system begins by collecting speech audio from a telephone call and preprocessing it to reduce background noise and normalize signal levels. The processed audio is divided into short frames for feature extraction and analysis. Transition-aware analysis examines changes in acoustic characteristics between consecutive speech segments to identify unnatural patterns that may indicate synthetic voice generation. Machine learning techniques can then classify the voice as genuine, suspicious, or potentially synthetic based on the extracted features. The system generates an authentication result with a confidence score and displays the outcome through a user-friendly monitoring interface.The proposed system aims to improve voice authentication reliability, reduce telephone fraud risks, and provide an additional security layer for banking, customer service, and other telephony applications. Overall, AUDIUMSPEC combines acoustic signal processing, machine learning, and transition-aware analysis to support efficient and reliable detection of synthetic voices in telephone communication.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130684 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXPLAINABLE MULTIMODAL CORROBORATIVE COGNISANCE FOR INDOOR NAVIGATION UNDER DUAL SENSORY IMPAIRMENT

(51) International classification	:G01C 21/16, G01C 21/20, G01S 17/86, G01S 5/16, G01S 17/89
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs VANATHI PALANISAMY
2)HARISH ARUMUGAM
3)HARISH VISHWARAJA
4)KABEESH KANNAN GUNASEKARAN
5)Mrs.M.Manogiri
6)Dr .A.OORKALAN

(57) Abstract :
ABSTRACT Explainable Multimodal Corroborative Cognisance For Indoor Navigation Under Dual Sensory Impairment The present invention discloses an explainable multimodal corroborative cognisance system for indoor navigation under sensory impairment. The system comprises a wireless transceiver (1), camera (2), IMU (3), depth sensor (4), LiDAR (5), wheel encoder (6), sensor input group (7), sensor preprocessing module (8), multimodal corroboration engine (9), localization estimation module (10), drift monitor and detector (11), REST/IRES recovery engine (12), map and knowledge database (13), memory (14), storage (15) and user interface and feedback module (16). The system preprocesses multimodal observations and corroborates visual evidence (23), inertial evidence (24), geometric evidence (25), semantic evidence (26) and motion evidence (27). Localization is estimated and localization drift is detected using the drift monitoring architecture (48). A semantic anchor is selected using the environment model (56), semantic detector output (57), drift level (58) and localization state (59). The REST/IRES recovery engine (69) utilizes a failure vector (66), selected semantic anchor (67) and historical recovery data (68) for executing recovery. The recovery result is validated to generate recovery status (76) and updated localization (77). A wearable embodiment comprises a top cover (78), PCB holder (79), main PCB (80), camera module (81), IMU module (82), microphone (83), speaker (84), battery (85) and bottom cover (86), thereby providing an integrated system for explainable and reliable indoor navigation under degraded sensory conditions.

No. of Pages : 11 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130685 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ORS-SENSE: AN EDGE AI-BASED MULTI-SENSOR SYSTEM FOR REAL-TIME VERIFICATION OF ORAL REHYDRATION SOLUTION COMPOSITION

(51) International classification	:G01N 27/06, A61B 10/00, A61B 5/00, G01N 33/487, A61B 5/145	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr. M. MURUGESAN** **2)A.RAGAVI** **3)N.SUDESHNA** **4)A.VARSHIGAA** **5)Mrs.P.Uma Maheswari** **6)Dr .A.OORKALAN**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT ORS-SENSE: AN EDGE AI-BASED MULTI-SENSOR SYSTEM FOR REAL-TIME VERIFICATION OF ORAL REHYDRATION SOLUTION COMPOSITION The present invention discloses a portable ORS-Sense device (100) for verifying whether an Oral Rehydration Solution has been prepared within an acceptable concentration range. The device (100) comprises a sample chamber (104) with an electrode pair (116) for conductivity sensing, an optical window (118) with a light-emitting diode (120) and photodiode (122), a pH sensor (124), and a temperature sensor (126). A microcontroller (128) measures a source water sample in a BASELINE mode to store a baseline profile, and subsequently measures a prepared ORS sample in a TEST mode, comparing it against the baseline and a reference range to generate a result, with reasoning and an action recommendation, selected from SAFE, DILUTE and CONCENTRATED. The result is displayed on an OLED display (106), announced via a buzzer (134), and stored with a timestamp. The device (100) is a compact, rechargeable, hand-held device suitable for home, rural and field use. FIG 1

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130686 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PATIENT-GATED OPAQUE QR POINTER MECHANISM FOR LONGITUDINAL THERAPEUTIC EVIDENCE RECONSTRUCTION

(51) International classification	:G16H 10/60, G06K 19/06, G16H 20/10, G16H 40/20, G16H 50/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.D. PRADEEP
2)AJAY K M
3)HARISH K
4)DEEPAK M
5)Dr. G. Sumathi
6)Dr .A.OORKALAN

(57) Abstract :

ABSTRACT Patient-Gated Opaque Qr Pointer Mechanish For Longitudinal Therapeutic Evidence Reconstruction QR-MedPath is a research prototype designed to help doctors make safer medication decisions by providing a complete view of a patient's medical history. It considers important factors such as allergies, adverse drug reactions, organ function, drug interactions, existing diseases, and previous treatment failures. The system uses role-based login, a patient QR code, and OTP verification to securely identify the patient. The QR code contains only a Patient ID and does not store medical history or treatment recommendations. After verification, the doctor can access the patient's record and review potential medication-related risks. If a medication appears safe, QR-MedPath presents it as an option for the doctor to consider, while the final clinical decision remains with the doctor. The prototype runs locally using Node.js, SQLite, QR code generation, OTP verification, and simulated patient data. It is intended only for research and decision support and does not perform diagnosis, prescribing, dosage calculation, or medication dispensing. The study also considers Indian digital-health standards, drug-safety resources, data governance, and a replicable approach for evaluating the system.

No. of Pages : 16 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130687 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INTEGRATING ENVIRONMENTAL EXPOSOME DATA INTO HEALTHCARE DATA ANALYTICS: AN AI-DRIVEN CLIMATE RISK DASHBOARD

(51) International classification	:G16H 40/20, G16H 50/30, G16H 40/67, G06Q 10/06, G06Q 50/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)V RAJESHRAM
2)S. MONICA
3)P. PRANITHA
4)PRATHIKA CHANDRASEKARAN
5)Mrs.P.Uma Maheswari
6)Dr .A.OORKALAN

(57) Abstract :
ABSTRACT INTEGRATING ENVIRONMENTAL EXPOSOME DATA INTO HEALTHCARE DATA ANALYTICS: AN AI-DRIVEN CLIMATE RISK DASHBOARD Climate change and environmental pollution have become major factors affecting human health. Exposure to air pollution, extreme temperatures, humidity, and other environmental conditions can increase the risk of respiratory problems, heart diseases, and other health issues. The proposed project, “Integrating Environmental Exposome Data into Healthcare Data Analytics: An AI-Driven Climate Risk Dashboard,” aims to combine environmental exposure data with healthcare data to identify potential climate-related health risks. The system collects environmental information such as air quality index, temperature, humidity, and pollution levels from available environmental datasets or monitoring sources. Healthcare-related data, including relevant health indicators and disease patterns, can be analyzed alongside environmental information using data analytics techniques. Artificial Intelligence (AI) and Machine Learning (ML) algorithms are used to identify patterns, predict possible health risks, and understand the relationship between environmental conditions and health outcomes. The system presents the analyzed information through an interactive dashboard containing graphs, charts, maps, and risk indicators. The dashboard helps users understand environmental changes, identify areas with higher potential health risks, and monitor climate-related health trends. It can also provide alerts when environmental conditions reach predefined risk levels. The proposed system supports healthcare professionals, researchers, and public health authorities in making informed decisions and planning preventive measures. By integrating environmental exposome data with healthcare analytics, this project aims to improve health risk assessment, increase awareness of climate-related health impacts, and support the development of a healthier and safer environment.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130688 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD AND APPARATUS FOR PRECISION SCRIBING OF POLYCRYSTALLINE PEROVSKITE FILMS USING A FEMTOSECOND BESSEL BEAM

(51) International classification	:B23K 26/06, H01L 51/42, B23K 26/00, B23K 26/0622, B23K 26/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology, Hyderabad
Address of Applicant :IIT Hyderabad Road, Near NH-65, Sangareddy, Kandi, Telangana – 502284, India Hyderabad Telangana India
(72)**Name of Inventor :**
1)PROF. SAI SANTOSH KUMAR RAAVI
2)MR. RAJENDRA KUMAR CHALLA
3)PROF. SOMA VENUGOPAL RAO

(57) Abstract :
ABSTRACT METHOD AND APPARATUS FOR PRECISION SCRIBING OF POLYCRYSTALLINE PEROVSKITE FILMS USING A FEMTOSECOND BESSEL BEAM The present subject matter relates to a method (1200) and an apparatus (100) for forming a scribe (102) on a polycrystalline perovskite film (110). The method comprises providing the polycrystalline perovskite film (110) on a substrate (112), generating, from a laser beam (104), a Bessel beam (108) comprising one or more femtosecond laser pulses and having a central intensity lobe surrounded by one or more concentric rings, and directing and focusing the Bessel beam (108) at a predetermined targeted location on the polycrystalline perovskite film (110) at a predetermined laser irradiation condition sufficient to remove at least a portion of the polycrystalline perovskite film (110) to form the scribe (102). The Bessel beam (108) is generated by passing the laser beam (104) from a femtosecond laser source (202) through an axicon (106), and the polycrystalline perovskite film (110) is translated relative thereto on a dual-axis motorized stage (204). TO BE PUBLISHED WITH FIG. 1A

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130690 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Adaptive Semantic Shield Watermarking AI Text with a Secret Key and Idea Lattice

(51) International classification	:G06T 1/00, G06F 17/27, G10L 19/018, G06F 21/16, G06F 40/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NANDA KRISHNA MADDINENI
Address of Applicant :H.No: A/9, D.No.1-25-13 4th Lane, Near Post Office, S.V.N Colony, Pattabhipuram, Guntur, Andhra Pradesh-522006 Guntur Andhra Pradesh India
2)TENDI INNOVATION NEXUS FOUNDATION
(72)**Name of Inventor :**
1)NANDA KRISHNA MADDINENI

(57) Abstract :
As large language models generate more of the world's text, telling human writing from machine writing has become harder. Watermarking tries to solve this by secretly embedding a detectable signal in the AI's output. Yet most existing watermarks break when the text is rewritten, translated, or lightly edited, and attackers can sometimes plant a fake watermark on text the model never produced. Strengthening the mark usually lowers writing quality or opens new ways to attack it. This work presents Adaptive Semantic Shield, a new approach that builds the watermark inside the meaning structure of the text rather than on its surface words. A secret key k a private random value known only to the model owner and a public parameter n a small integer that sets the number of conceptual anchors is fed into a lattice-based pseudorandom function. The function uses the secret key to produce exactly nabstract anchors and turns them into a weighted directed hypergraph L_k=(Aⓜ,Eⓜ,w). The nodes of the hypergraph are the anchors; the edges carry secret requirements for how strongly one idea must support another, the order in which idea clusters should appear, and the density of supporting details that link them. During generation the model is guided so that the finished text realizes this secret lattice. Whenever a high-density edge remains under-realized, classifier-free guidance gently steers the model to add the needed supporting details without forcing unnatural word choices. At the same time the method still applies a lighter statistical signal on less important sentences, so two independent marks one semantic and one statistical protect the text. A detector that knows the secret key recovers the hypergraph from the finished document, measures how well the observed idea relationships match the secret lattice, and combines that score with the statistical signal. Because the lattice lives in abstract relationships rather than exact wording, ordinary rewriting or translation that keeps the same meaning leaves the mark intact. Experiments show that Adaptive Semantic Shield stays detectable after heavy paraphrasing, translation, and mixing with human text, while the generated writing remains natural and fluent. Forging the mark requires recreating the exact secret hypergraph controlled by k, which cannot be done by simple rewriting or limited queries to the model. The strength of the watermark automatically rises on the most important ideas and falls on secondary ones, shrinking the usual trade-off between detection power, text quality, and resistance to attack. The technique can be added to existing language models with only modest changes to the generation process and needs no expensive retraining of the full model.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130693 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CareerQuest - An AI-Powered Embedded Educational toy for Career Exploration

(51) International classification	:G09B 19/00, G09B 5/06, G06Q 50/20, G09B 7/02, G09B 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vishnu Women's University
Address of Applicant :Vishnupur, Kovvada, Bhimavaram, West Godavari District, Andhra Pradesh, 534202. Bhimavaram Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. T. Sudheer Kumar
2)Mr. B. Satya Krishna
3)A. Hema Sri
4)K. Vidya
5)G. Harini

(57) Abstract :
The present invention relates to an AI-powered embedded educational toy for career exploration configured to provide interactive career-oriented learning activities for children. The system comprises an embedded computing unit, display interface, camera module, audio output unit, user interaction interface and memory. The system enables a child to register, select an age group and explore a plurality of career domains through interactive missions. An artificial intelligence module generates or assists in generating age-appropriate career missions based on a selected career and user age group. The missions may include virtual activities such as multiple-choice questions, fill-in-the-blank activities and word-based tasks, as well as practical physical activities. Virtual responses are evaluated against expected answers, while practical tasks may be verified using an image captured by the camera module. The system provides experience points, badges, streaks and progress information based on successful mission completion. User information and learning progress may be maintained using a local database. The invention integrates artificial intelligence, embedded computing, interactive display, camera-based task verification and gamification into a unified educational platform for experiential career exploration.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130697 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING-ENABLED SYSTEM FOR EARLY EBOLA VIRUS DETECTION, TRANSMISSION-RISK PREDICTION, AND PREVENTIVE OUTBREAK CONTROL

(51) International classification	:G06N 20/00, G16H 50/80, G16H 10/60, G16H 40/20, G16H 50/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Dr. B. Kesava Rao

Address of Applicant :Assistant Professor, Department of ECE, Mahatma Gandhi Institute of Technology, Hyderabad, Rangareddy, Telangana, 500075, India. Telangana India

2)Sujith Sri Surya Ravi

3)Leela Kagithakara Vajravelu

4)Mathan Kumar Aiyakkannu

5)Manikandan Selvamani

6)Sujith Meenakshisundaram

7)Dr. Balamurali Venkatesan

8)Kesavarajan Govindarajan Karthikeyan

9)Dr. Sumit Kushwaha

(72)**Name of Inventor :**

1)Dr. B. Kesava Rao

2)Sujith Sri Surya Ravi

3)Leela Kagithakara Vajravelu

4)Mathan Kumar Aiyakkannu

5)Manikandan Selvamani

6)Sujith Meenakshisundaram

7)Dr. Balamurali Venkatesan

8)Kesavarajan Govindarajan Karthikeyan

9)Dr. Sumit Kushwaha

(57) Abstract :

The present invention provides a system and method utilizing machine learning for the early detection of the Ebola virus, prediction of transmission risks, and management of outbreaks are presented. This system includes an input data pipeline designed to consolidate diverse data streams related to healthcare-associated transmission, population movement patterns, metrics from conflicts and humanitarian crises social and behavioral insights, indicators of health system vulnerability, and information from the digital ecosystem. A data aggregation and machine learning ingestion engine processes, standardizes, and formulates uniform feature vectors from these various streams to facilitate predictive machine learning applications. The system produces real-time assessments of transmission risk to automate intervention strategies, enhance resource distribution, and reinforce protocols for containing outbreaks. FIG.1

No. of Pages : 11 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130698 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Smart IoT-Based Boiler Monitoring System Using Raspberry Pi for Real-Time Smoke Detection and Efficiency Prediction

(51) International classification	:G06Q 40/00, G06N 20/00, G06Q 10/06, G06Q 40/02, G06Q 30/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY
Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India. Vijayawada Andhra Pradesh India
2)Dr.J. Rajendra Prasad
3)G.Honey
4)A.V.S.Harikesh
5)G.Trisha Lakshmi
6)A.Lakshmi Akshaya
7)B.Dharma Teja
(72)**Name of Inventor :**
1)Dr.J. Rajendra Prasad
2)G.Honey
3)A.V.S.Harikesh
4)G.Trisha Lakshmi
5)A.Lakshmi Akshaya
6)B.Dharma Teja

(57) Abstract :
ABSTRACT [0025] A smart expense monitoring system is disclosed that records income and expense transactions, automatically assigns each transaction to a category by analysing its description and amount, and compares the resulting expenses with user-defined budgets and savings goals in real time. The system calculates total income, total expenses, savings, budget utilization and category-wise expenditure, issues a notification when spending approaches or exceeds a budget, and analyses historical transactions to identify spending patterns. An artificial-intelligence module generates personalized insights, and a conversational accountant answers natural-language questions using the user's own stored data. Results are presented on a dashboard with charts, summary cards and tables. The platform replaces manual expense recording with continuous, organized and secure monitoring that supports disciplined spending and improved savings habits.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130699 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : NextGen Smart College Transportation Management System

(51) International classification	:G08G 1/123, G06V 30/224, G06Q 50/30, G06K 7/14, G06K 7/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY
Address of Applicant :DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India. Vijayawada Andhra Pradesh India
2)Dr.M.Chaitanya Kishore Reddy
3)P.Varshini
4)P.Sri Harshini
5)S.Sravani
6)V.Chethana
7)P.Prem Sagar
(72)**Name of Inventor :**
1)Dr.M.Chaitanya Kishore Reddy
2)P.Varshini
3)P.Sri Harshini
4)S.Sravani
5)V.Chethana
6)P.Prem Sagar

(57) Abstract :
ABSTRACT [0032] An integrated system and method for managing a college bus service is disclosed. A route and tracking module stores each route as an ordered sequence of stops, maintains a current stop index for a bus, determines the current stop and next stop, and computes an estimated arrival time from the distance to the next stop and the bus speed; the position may be advanced by simulated movement along the route, so that no positioning hardware is needed. A bus-pass verification module decodes a QR code and extracts printed fields by optical character recognition from the same uploaded image, consolidates the information, compares it with the registered record in a database, checks the validity date, and classifies the pass as valid, mismatched, expired or invalid by predefined rules. A notification module communicates transportation events and administrator messages, and an assistance module generates weather-aware travel reminders using a locally hosted language model. All records are held in one database and served through a student, faculty and administrator web interface.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130702 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Next-Gen Campus Lost and Found: AI Smart Matching, Location Intelligence & Tracking, and Reward-Based Student Incentives

(51) International classification	:G06F 17/30, G06Q 10/00, G06Q 10/08, G06K 9/62, H04W 4/029	(71)**Name of Applicant :** **1)DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY** Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India. Vijayawada Andhra Pradesh India **2)A. Ravi Kiran** **3)N. Dharani** **4)N. Bhanu Prakash**
(31) Priority Document No	:NA	**5)N. Durga**
(32) Priority Date	:NA	**6)N. Sreenath**
(33) Name of priority country	:NA	**7)M.Srishanth**
(86) International Application No Filing Date	: :01/01/1900	(72)**Name of Inventor :** **1)A. Ravi Kiran**
(87) International Publication No	: NA	**2)N. Dharani**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**3)N. Bhanu Prakash** **4)N. Durga**
(62) Divisional to Application Number Filing Date	:NA :NA	**5)N. Sreenath** **6)M.Srishanth**

(57) Abstract :

ABSTRACT [0030] A campus lost and found system is disclosed for recording, matching, tracking and returning misplaced belongings. A user reports a lost or found item with a title, description, category, location, reward and image, and the item is stored with a status. A cognitive matching mechanism compares the title, description, category and location of lost and found records using string similarity and predefined weights of 40%, 30%, 20% and 10% to produce a matching score, and the location of each item is stored as latitude and longitude. A claimant submits a claim for a found item, and the item owner approves or rejects it. When the item is marked RETURNED, the owner may pay a reward through a payment gateway, the approved claimant receives 10 points, and a certificate is recognized when the points reach 50. The web-based workflow gives a centralized and transparent recovery process for a campus. This project presents the Next-Gen Campus Lost and Found platform, a web application built with React, Express.js, Prisma and PostgreSQL, with JWT authentication, Cloudinary image storage and Razorpay reward payment. Traditional lost and found handling depends on manual records and informal communication. The platform connects item reporting, cognitive matching, location tracking, claim management, item return, reward payment, points and certificate recognition in a single system, so that students and staff can recover belongings in a systematic way.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130703 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVELOPMENT OF HERBAL ANTIDIABETIC MEDICATED LOLLIPOPS CONTAINING SENNA AURICULATA: A NOVEL PATIENT COMPLIANT ORAL DELIVERY SYSTEM

(51) International classification	:A23G 3/48, A61K 36/482, A61P 3/10, A23G 3/36, A23G 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)DR. AGILANDESWARI DEVARAJAN

Address of Applicant :Professor cum HOD, Department of Pharmaceutics, MVM College of Pharmacy, Yelahanka, Bangalore, Karnataka, 560064, India. Karnataka India

2)DR. S.A. VADIVEL

3)MR. VIJAYANANDHAN V

4)MRS. ASHLY MERIN GEORGE

5)MRS. HARSHITA NEHRU

6)MR. MOHAMMAD AMIN YATOO

7)MRS. VINUTHA. S

8)MISS. NALAGATLA SUNANDA

9)MR. BHARGAV GA

10)MR. ABDULLAH

(72)**Name of Inventor :**

1)DR. AGILANDESWARI DEVARAJAN

2)DR. S.A. VADIVEL

3)MR. VIJAYANANDHAN V

4)MRS. ASHLY MERIN GEORGE

5)MRS. HARSHITA NEHRU

6)MR. MOHAMMAD AMIN YATOO

7)MRS. VINUTHA. S

8)MISS. NALAGATLA SUNANDA

9)MR. BHARGAV GA

10)MR. ABDULLAH

(57) Abstract :

The present invention relates to develop, optimize, and evaluate a novel herbal antidiabetic medicated lollipop containing Senna auriculata flower powder as a patient-friendly oral delivery system. Authenticated Senna auriculata flowers were shade-dried, powdered, and subjected to preformulation studies including organoleptic evaluation, solubility, pH, loss on drying, flow properties, Fourier Transform Infrared (FTIR) spectroscopy, and UV–Visible spectroscopy. The herbal lollipop was developed using isomalt and stevia, following conventional heating and moulding techniques. A full factorial design assessed the impact of Senna auriculata, stevia, and isomalt on formulation responses, identifying F4 as the optimized batch with a hardness of 10.5 kg/cm² and a disintegration time of 4.3 minutes. The final product had a smooth, caramel-brown appearance with a pH of 7.32, hardness of 6 kg/cm², and a disintegration time of 9 minutes 45 seconds at 37.2°C. The dissolution studies indicated a gradual release over 40 minutes. FTIR and UV–Visible analyses confirmed the herbal drug characteristics. The study validated the potential for a sugar-free herbal lollipop as an effective oral dosage form. FIG.1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130705 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A CORDIC-Based Hardware-Efficient ECG Processor for R-Peak Detection and Heart Rate Variability Analysis

(51) International classification	:A61B 5/00, A61B 5/352, A61B 5/361, A61B 5/363, A61B 5/366	(71)**Name of Applicant :** **1)Dayananda Sagar College of Engineering** Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India (72)**Name of Inventor :** **1)Dr Vimala P** **2)Abhay Surya Shankar** **3)Akshaykumar Tallur** **4)Bhavya Y** **5)Deepthi S R** **6)Dr. S. Thenmozhi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The demand for ECG signal processing using VLSI systems is increasing and such systems need to meet the demands of real time healthcare systems. The cardiac signal analysis system implemented using hardware achieves three essential functions of current medical devices, namely fast processing, accurate results and low power usage. The increasing requirement for continuous heart monitoring and accurate ECG signal analysis will enable early detection of heart diseases and enable the treating physicians to provide the required treatment to the patients in time. With increasing requirement of portable, energy efficient medical devices there is a need for improved hardware solutions that maintain high system performance while being energy efficient. The aim of this work is to design a hardware efficient ECG signal processor to accurately extract the relevant features of ECG signal i.e. R peaks and heart rate. Building a CORDIC based processing unit helps engineers run math functions on hardware. The system uses the CORDIC (Coordinate Rotation Digital Computer) algorithm to handle functions and signal processing tasks very well. This CORDIC (Coordinate Rotation Digital Computer) algorithm makes things run faster because the system uses little power and very little space on the hardware. The R peak detection works with no delay and works much better than older ways of doing things. The design shows it can save a lot of power because the design does not use multipliers.

No. of Pages : 24 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130706 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Role-Based Community-Driven Platform for Emergency Homemade Food Assistance and Doorstep Delivery

(51) International classification	:G06Q 50/12, G06Q 10/08, G06Q 30/06, G06Q 50/26, G06Q 50/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr RVR NRI Institute of Technology Deemed to be University.
Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India Vijayawada Andhra Pradesh India
2)Mrs. Ch. N. Santha Kumari
3)V. Divya Sri
4)R. Yogeswari
5)V. Bhanu
6)P. Hari Charan
7)U. Gunadeeswar
(72)**Name of Inventor :**
1)Mrs. Ch. N. Santha Kumari
2)V. Divya Sri
3)R. Yogeswari
4)V. Bhanu
5)P. Hari Charan
6)U. Gunadeeswar

(57) Abstract :
ABSTRACT [0030] An emergency food support platform is disclosed for coordinating persons who need food with local homemade food providers and community volunteers. The platform receives an emergency food request specifying the number of people, the number of days, the meals per day, the food category, special requirements and location, calculates the food quantity and cost from a food catalogue, and verifies online payment through a backend server. The request is matched with food partners by a rulebased score, accepted and prepared by a food partner, and delivered by a community volunteer who records a pickup proof and a doorstep delivery proof including a photograph and location coordinates. The progress of the request is tracked through a defined status lifecycle with notifications, and an administrator verifies participants and supervises requests, deliveries, wallets and service zones. The rolebased web workflow connects the food seeker, food partner, volunteer and administrator in a single process from request submission to delivery completion. This project presents Ma Aharam, a community emergency food support web application built with React, TypeScript, Firebase/Firestore and an Express backend, serving Vijayawada. Unstructured and manual arrangements for emergency food are slow and unverified. Ma Aharam connects the emergency request, food selection and cost calculation, payment verification, provider matching, volunteer delivery with proof, request tracking, rewards and administrative supervision in a single platform, so that emergency food can be requested, prepared and delivered in a systematic and traceable way.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130707 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TRANSPARENT POROUS MEDIUM COMPRISING A JAMMED OIL-IN-WATER EMULSION, PROCESS FOR PREPARING THE SAME, AND USES THEREOF

(51) International classification	:C12N 5/00, C12N 1/00, C12N 1/20, C12M 1/00, C12N 5/071
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Centre for Biological Sciences
Address of Applicant :Bellary Rd, Rajiv Gandhi Nagar, Kodigehalli Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Tapomoy Bhattacharjee
2)Nandhu Krishna Babu
3)Sreepadmanabh M

(57) Abstract :
A transparent porous medium comprises a jammed oil-in-water emulsion in which droplets of a water-immiscible oil, preferably a fluorinated oil such as perfluoroperhydrophenanthrene, stabilised by an emulsifier, are packed to form a self-supporting, solid-like matrix whose interconnected interstitial pores contain an aqueous culture medium. The refractive index of the oil matches that of the continuous phase without any refractive-index-matching agent being added, so the medium is optically transparent while its pores contain an unmodified culture medium. The medium is transport-confined, self-healing and mechanically stable under flow, permitting single-cell-resolution optical microscopy of organisms growing, moving and depositing within the pores. A kit, a process for preparing the medium, a process for recovering the oil, a perfusion device, and methods of observing organisms and of studying the deposition of microorganisms and other objects under flow are also disclosed.

No. of Pages : 39 No. of Claims : 30

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130710 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : EDGE-EEG COGNITIVE FATIGUE AND DROWSINESS INTERCEPTOR

(51) International classification	:A61B 5/374, A61B 5/0476, A61B 5/00, A61B 5/16, A61M 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)G Sekar
Address of Applicant :Associate professor Department of Electronics and communication engineering VSB College of engineering technical campus Kinathukadavu Tamil Nadu India
2)Dr. D Nageswari
3)Mr.A.S.Harinish
4)Mr.R.Gabriel
5)Mr.J.Kavinilavan
6)Mr.J.Joel Ebenezer
7)Mr.N.Lokeshwaran
8)Mr.K.Sivamuthu
9)Mr.T.Sakthivel
10)Mr.E.Sadham Hussain
11)Mr.M.Hariharan
12)Mr.M.Mohammed Imran
(72)**Name of Inventor :**
1)G Sekar
2)Dr. D Nageswari
3)Mr.A.S.Harinish
4)Mr.R.Gabriel
5)Mr.J.Kavinilavan
6)Mr.J.Joel Ebenezer
7)Mr.N.Lokeshwaran
8)Mr.K.Sivamuthu
9)Mr.T.Sakthivel
10)Mr.E.Sadham Hussain
11)Mr.M.Hariharan
12)Mr.M.Mohammed Imran

(57) Abstract :
Edge-EEG Cognitive Fatigue and Drowsiness Interceptor ABSTRACT OF THE INVENTION The present invention relates to an edge-based electroencephalography (EEG) system designed to detect cognitive fatigue and drowsiness at an early stage and provide timely warnings to prevent fatigue-related errors during attention-critical activities. The proposed system comprises an EEG acquisition module, signal-conditioning and amplification circuit, edge processing unit, power supply module, alert and notification module, and driver and actuation module. EEG electrodes placed at selected scalp locations acquire brainwave signals, which are amplified, filtered, and converted into digital data using suitable signal-conditioning circuits and an analogue-to-digital converter (ADC). A microcontroller, such as an ESP32 or STM32, processes the acquired signals locally, reducing dependence on continuous internet connectivity. The system extracts relevant features, including theta, alpha, and beta frequency-band power, relative frequency ratios, signal energy, and variations in signal complexity. To improve individual adaptability, the proposed Adaptive Multi-Window EEG Fatigue Index (AMEFI) combines these features with a baseline recorded during an alert state and evaluates changes across multiple time windows. Signal-quality assessment and noise rejection help distinguish sustained fatigue-related patterns from temporary disturbances caused by eye movements or body movements. The system operates through a battery-powered supply incorporating voltage regulation and protection circuitry, enabling portable and continuous monitoring. The proposed Persistence-Weighted Fatigue Interception Algorithm (PWFIA) evaluates the severity, duration, and consistency of the extracted EEG features before classifying the user's state as alert, fatigued, or drowsy. Unlike systems that generate warnings from a single abnormal measurement, the algorithm applies temporal smoothing, adaptive thresholds, and persistence-based decision rules to reduce false alarms and unnecessary interruptions. When fatigue or drowsiness is detected, the system activates a buzzer and displays the corresponding warning on an LCD; if the condition continues, a microcontroller-controlled relay activates a predefined warning or intervention device. A wireless communication module enables the transmission of fatigue status and event information to a monitoring device whenever connectivity is available, while local detection and warning functions remain operational independently. The principal proposed innovation is the integration of personalised baseline calibration, multi-window EEG feature evaluation, persistence-based fatigue assessment, and staged warning and intervention within a single edge-processing architecture. The system is intended for applications such as driver monitoring, industrial workplaces, and prolonged attention-demanding tasks, with performance evaluated through detection accuracy, false-alarm rate, response time, and reliability under different operating conditions. Keywords: EEG Monitoring, Cognitive Fatigue, Drowsiness Detection, Brain Signal Processing, Microcontroller, Alert System, Wireless Communication, Real-Time Monitoring.

No. of Pages : 5 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130711 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIPLE DISEASE PREDICTION USING MACHINE LEARNING

(51) International classification	:G16H 50/20, G16H 10/60, G16H 50/70, A61B 5/00, G06N 20/00	(71)**Name of Applicant :** **1)Karpagam Academy of Higher Education** Address of Applicant :Pollachi Main Road, Eachanari Post Coimbatore Tamil Nadu India Tamil Nadu India **2)Karpagam Institute of Technology** **3)C. Sasthi Kumar** **4)Sowmya A** (72)**Name of Inventor :** **1)C. Sasthi Kumar** **2)Sowmya A**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The increasing prevalence of chronic, neurological, and life-threatening diseases has intensified the demand for intelligent and automated healthcare diagnostic systems capable of early and accurate disease detection. Conventional medical diagnosis methods rely heavily on manual clinical evaluation and are often supported by isolated computational tools designed to analyze a single disease at a time. Such approaches lack scalability, interoperability, and the ability to consider the complex interactions among multiple physiological parameters. To address these limitations, this paper presents a unified multiple disease prediction system based on machine learning and deep learning techniques, deployed through a web-based application programming interface. The proposed system integrates multiple disease-specific predictive models within a single computational framework to analyze conditions including diabetes, heart disease, brain tumors, and Alzheimer's disease. Each disease module is trained using supervised machine learning algorithms and TensorFlow-based deep learning architectures on comprehensive clinical datasets. Unlike existing systems that utilize a limited number of input features, the proposed approach incorporates an expanded set of physiological, demographic, and hereditary parameters, such as glucose concentration, number of pregnancies, skin thickness, heart rate, and diabetes pedigree function, thereby enhancing predictive accuracy and robustness. A Flask-based Application Programming Interface (API) is employed as a centralized communication layer between end users and the predictive models. The API accepts patient health parameters along with the specified disease type, dynamically selects the appropriate trained model, and returns a diagnostic outcome indicating the predicted disease state. Model persistence is achieved using Python-based serialization techniques, enabling efficient model deployment and real-time inference without repeated retraining. The system architecture is designed to be modular and extensible, allowing additional disease prediction modules to be incorporated with minimal modification to the existing framework. This flexibility supports future expansion to include infectious diseases, general symptom-based diagnosis, and emerging healthcare challenges. The web-based nature of the system enables remote accessibility, making it suitable for integration with telemedicine platforms, hospital information systems, and mobile healthcare applications. Experimental evaluation demonstrates that the inclusion of additional clinical parameters improves prediction performance compared to conventional single-disease models with limited feature sets. The proposed system facilitates early disease detection, continuous patient monitoring, and timely clinical intervention, thereby contributing to reduced mortality rates and improved healthcare outcomes. In summary, this work introduces a scalable, efficient, and intelligent multi-disease prediction platform that consolidates diverse diagnostic capabilities into a single unified system. By combining advanced machine learning techniques, deep learning models, and API-driven deployment, the proposed approach provides a practical solution for modern healthcare decision-support systems and represents a significant advancement in automated medical diagnosis.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130719 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HAIR DRYER WITH MULTI-CARTRIDGE FLUID DELIVERY AND INDEPENDENT FORMULATION-RESPONSIVE MIST CONTROL

(51) International classification	:A45D 20/12, A45D 20/10, A45D 20/50, A45D 20/08, A45D 20/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)M.KUMARASAMY COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Jayakumar S
2)Mohamed Abdul Haseeb
3)Mohamed Imran
4)Geetha S
5)K. RANI
6)Guru prakash K S

(57) Abstract :
ABSTRACT HAIR DRYER WITH MULTI-CARTRIDGE FLUID DELIVERY AND INDEPENDENT FORMULATION-RESPONSIVE MIST CONTROL The present invention relates to a hair dryer having a plurality of integrated fluid cartridges and an independently controllable fluid-delivery system. The hair dryer comprises a primary airflow generating assembly, at least three fluid cartridges positioned on an upper portion of a housing, and separate first, second and third fluid conduits respectively associated with the cartridges. Each fluid conduit is independently controlled by a corresponding valve and is configured to transport a selected formulation toward a treatment-fluid outlet comprising an atomization nozzle. A side-mounted control interface permits independent selection and dosage adjustment of each cartridge. A fluid-driving assembly provides pressure for transporting the selected formulation, while a controller may regulate pressure, flow rate, valve opening duration or pump operation according to a selected dosage or formulation-dependent delivery profile. The treatment-fluid outlet is separate from a primary heated-air outlet, thereby permitting hair drying and selective application of one or more treatment formulations through separate fluid paths.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130724 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVELOPMENT OF WATER RESISTANCE BIODEGARBLE GARBAGE BAG

(51) International classification	:B65F 1/00, C08L 91/00, C08L 3/02, C09D 191/06, C08K 5/09	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)M.KUMARASAMY COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Mr.G.RAMACHANDRAN**
(33) Name of priority country	:NA	**2)Dr.S.SUJATHA**
(86) International Application No Filing Date	: :01/01/1900	**3)Mr.R.MUHILAN** **4)Mr.S.KESHAV RATHAN**
(87) International Publication No	: NA	**5)Mr.V.BOOPATHI RAJ**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)B.SATHIYA** **7)Dr. KARTHE M**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT DEVELOPMENT OF WATER RESISTANT BIODEGRADABLE GARBAGE BAG The present invention discloses a water-resistant biodegradable garbage bag developed using natural and biodegradable materials. The garbage bag comprises a cotton cloth , cassava starch , neem powder , lemon peel powder , glycerol , citric acid , distilled water , and beeswax/carnauba wax. The cassava starch, glycerol, neem powder, lemon peel powder, citric acid, and distilled water are mixed to form a substantially uniform biodegradable coating paste. The prepared coating paste is applied onto the cotton cloth and dried to form a biodegradable coated material. A beeswax/carnauba wax layer is subsequently applied to improve the water resistance of the developed garbage bag. The coated material is then cut and sealed to obtain the required garbage bag shape. The developed garbage bag provides improved flexibility, water resistance, odour-control characteristics, and biodegradability. The use of natural materials helps reduce dependence on conventional non-biodegradable plastic garbage bags and supports sustainable waste-management practices. The garbage bag is suitable for household waste collection, food waste disposal, dustbin applications, sanitation applications, and other suitable waste-management purposes. The present invention provides an eco-friendly and practical alternative to conventional plastic garbage bags while utilizing biodegradable and naturally derived materials. FIG 1

No. of Pages : 19 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130725 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART SOLAR SWITCH BOX COMPACT SOLAR-POWERED EMERGENCY POWER SYSTEM

(51) International classification	:H02J 7/35, H02J 9/06, H02J 7/00, H02S 40/38, H02J 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs. KALPANA VAIRAPERUMAL
2)Mrs. KEERTHANA SELLADURAI
3)Mr. ROHITH S
4)Mr. SARAN R
5)Dr.NALINI NAGULSAMY
6)Dr. R. LATHA

(57) Abstract :
ABSTRACT SMART SOLAR SWITCH BOX COMPACT SOLAR-POWERED EMERGENCY POWER SYSTEM The present invention relates to a Smart Solar Switch Box, which is a compact, portable, and integrated solar-powered electrical unit developed to provide a convenient source of electrical power for agricultural activities, night-time work, and emergency situations. The invention is designed to collect solar energy during daylight hours and store the generated electrical energy in an internal rechargeable battery so that the stored power can be used whenever required. By incorporating the major power-related components within a single box structure, the unit provides a practical alternative to carrying multiple separate power devices.An integrated solar panel is provided on the outer surface of the box for collecting solar energy and supplying the generated electrical energy to the charging section. A rechargeable battery stores the collected energy for later use when sunlight is unavailable or when the regular electrical supply is interrupted. A battery-level indicator is provided to allow the user to easily monitor the available charge level of the battery and understand the remaining stored energy before or during operation.The Smart Solar Switch Box therefore combines solar energy collection, battery storage, electrical power output, emergency lighting, battery monitoring, switching, and protection functions within a single compact structure. Fig 1

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130726 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : 3D PRINTER BED CLEANING DEVICE

(51) International classification	:B29C 64/35, B08B 1/04, B29C 64/20, B08B 1/00, B29C 67/00	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)M.KUMARASAMY COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)MOHAMED ZUBARULHAQ ABDUL RAHMAN** **2)YOKESWARAN RAMADOSS** **3)KAMEESWARAN PANCHATCHARAM** **4)HARISH KATHIRESAN** **5)KIRTI BHUSHAN MURUGESAN** **6)NIRMAL SIVAKUMAR** **7)PRASANTH ANBALAGAN** **8)CHOCKALINGAM ALAGAPPAN** **9)KIRTHIGA R**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT 3D PRINTER BED CLEANING DEVICE The present invention is related to a portable 3D Printer Bed Cleaning Device which is designed to clean and ready a build plate of a 3D printer using an integrated cleaning system. The components of the invention include a main body, storage tank, tank connecting base, rotary brush base, motor shaft, Nylon 6/6 brush, push-back wiper, hand gripper, microfiber cloth, metal scrubber body, metal scrubber brush, and respective power buttons. The storage tank has the ability to store the cleaning liquid and to discharge it through a nozzle that is located close to the rotary brush. The motorized Nylon 6/6 brush operates on the surface of the printer bed to wipe off dust, adhesive deposits, printing residues, and other types of contaminants. Metal scrubber performs mechanical cleaning actions on the build plate surface to get rid of stubborn residues. The push-back wiper is incorporated into the main body and is capable of being activated for performing the final cleaning actions. Microfiber cloth is attached to the wiper for absorbing excess cleaning liquid and for removing fine particles from the surface of the build plate. The hand gripper is used to handle the device during its operation. The integration of all mentioned actions into one portable device enables performing of liquid application, rotary brushing, intense scrubbing, and wiping.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130727 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ECOSOLUTIONS – A BIODEGRADABLE PACKAGING MATERIAL MADE FROM AGRICULTURAL WASTE THAT CHANGES COLOUR WHEN IT BEGINS TO DEGRADE

(51) International classification	:B65D 65/46, C08L 3/02, C08J 5/18, C08L 97/02, C08L 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. DEEPAN KUMAR MADHU
2)DEEPTI SRI GOPALAN
3)KARTHIKA MURUGANANTHAM
4)VIVETHA MANI
5)SWATHI M MARIMUTHU
6)Dr.G.R.Gopinath
7)Elamathi E

(57) Abstract :
ABSTRACT ECOSOLUTIONS – A BIODEGRADABLE PACKAGING MATERIAL MADE FROM AGRICULTURAL WASTE THAT CHANGES COLOUR WHEN IT BEGINS TO DEGRADE The present invention introduces Ecosolutions, describes an innovative packaging material that is made of rice husk powder, corn starch, aloe vera gel, glycerine, lemon juice, red cabbage extract, and water. The packaging material serves as a substitute for plastic and has an ability to change color during biodegradation process due to its unique components composition. No electronic sensors or additional energy is needed to observe the process of degradation.

No. of Pages : 10 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130728 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Smart Multimodal AI System and Method for Multi-Domain Interview Assessment Using Facial, Speech and Language Analysis

(51) International classification	:G10L 15/26, G06K 9/00, G06N 20/00, G10L 15/22, G10L 15/18	(71)**Name of Applicant :** **1)DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY** Address of Applicant :DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India. Vijayawada Andhra Pradesh India **2)Ch. Kiran Babu** **3)B. Jyoshna** **4)Ch. Nikhila** **5)K. Blessy Jones** **6)B. Geeth** **7)M. Harshitha**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	(72)**Name of Inventor :** **1)Ch. Kiran Babu**
(87) International Publication No	: NA	**2)B. Jyoshna**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**3)Ch. Nikhila** **4)K. Blessy Jones**
(62) Divisional to Application Number Filing Date	:NA :NA	**5)B. Geeth** **6)M. Harshitha**

(57) Abstract :

ABSTRACT [0028] A multimodal artificial-intelligence system and method for assessing interview performance is disclosed. A candidate selects an interview category and is presented with category-specific questions while a webcam and microphone record synchronised video and audio of each answer. A computer-vision module derives eye-contact, expression and posture measures from facial landmarks; a speech module derives a transcript, speaking pace, pauses and pitch variation; and a language module derives filler-word, grammar, vocabulary and answer-relevance measures from the transcript. The measures are normalised to a common scale and combined by predefined weights into an overall score, while the lowest-scoring factors are mapped to pre-written improvement suggestions. Scores are stored per user and session so that progress can be tracked across attempts. The system gives objective, repeatable feedback without a human interviewer, using ordinary consumer hardware.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130729 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : HIGH-PERFORMANCE SECURE FILE SHARING WITH DUAL SERVER ACCESS AND TIME-SENSITIVE RETRIEVAL

(51) International classification	:H04L 29/06, G06F 21/62, G06F 21/60, G06F 17/30, G06F 21/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Karpagam Academy of Higher Education
Address of Applicant :Pollachi Main Road, Eachanari Post Coimbatore Tamil Nadu India Tamil Nadu India
2)Karpagam Institute of Technology
3)Dr. K Ranjith Singh
4)A.Sudharsan
(72)**Name of Inventor :**
1)Dr. K Ranjith Singh
2)A.Sudharsan

(57) Abstract :
Cloud computing has become a foundational platform for data storage and collaborative information sharing due to its scalability, accessibility, and cost efficiency. However, the growing reliance on cloud infrastructures has intensified concerns related to data privacy, unauthorized access, key management, and prolonged exposure of sensitive information. Conventional cloud file sharing systems typically rely on single-authority access control models and lack effective mechanisms for enforcing time-bound access, thereby increasing vulnerability to insider threats, session hijacking, and unauthorized data disclosure. To address these limitations, the present invention introduces a high-performance secure file sharing system with dual server access control and time-sensitive retrieval capabilities. The proposed system enables secure and privacy-preserving data sharing by integrating encrypted cloud storage, dual-layer authorization, and owner-centric key management. Prior to storage, files are encrypted using symmetric key cryptography, ensuring confidentiality of data at rest. The encrypted files are stored on a cloud storage server, while the corresponding secret keys remain under the exclusive control of the file owner, thereby separating data storage from key custody and minimizing the risk of key compromise. Access to stored files is governed through a primary authorization server that authenticates users and regulates their interaction with file directories. Upon successful authentication, the primary server grants the requesting user a limited access window during which the user may view available file metadata and submit a file access request. If the user fails to initiate a valid request within the predefined time interval, the session is automatically terminated and redirected to the main access interface, preventing unauthorized browsing and prolonged session misuse. Once a file access request is generated, it is forwarded to the file owner for explicit approval. Upon acceptance, the system securely releases the corresponding secret key to the requesting user, enabling authorized decryption and file retrieval. Data access is granted only after successful validation by both the primary authorization server and the file owner, establishing a dual-server access control mechanism that enhances security and accountability. Additionally, the system incorporates a shared access authority module that supports anonymous or semi-anonymous request matching, reducing unnecessary exposure of user identity information while maintaining access traceability. Time-sensitive enforcement combined with dual authorization effectively mitigates replay attacks, session hijacking, and unauthorized data exploration. Overall, the proposed invention delivers a secure, scalable, and efficient cloud file sharing framework that strengthens data confidentiality, preserves owner control, and ensures controlled collaboration. The system is well suited for enterprise, academic, and distributed cloud environments where high security, privacy preservation, and controlled access are critical requirements.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130730 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SAFE MEDIHYDRA FLASK FOR DIABETIC PATIENTS HEALTH MONITORING

(51) International classification	:A61B 5/00, A61B 5/145, H02J 7/35, A61B 5/1455, G01N 33/49	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr. MANIKANDAN MURUGESAN**
(32) Priority Date	:NA	**2)Mr. MANJUNATHAN ALAGARSAMY**
(33) Name of priority country	:NA	**3)Mr. R. SRI SANJIT**
(86) International Application No Filing Date	: :01/01/1900	**4)Mr. S. EZHILAN** **5)Ms.M.SARANYA**
(87) International Publication No	: NA	**6)DR.S.KAVITHA**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT Safe MediHydra Flask for Diabetic Patients Health Monitoring The present invention discloses a Safe MediHydra Flask (10), being a portable smart flask configured to combine hydration and personal health monitoring in a single product. The device comprises a flask body (11) having a drinking-water chamber (12), a solar panel (13), an energy-management unit (14), a rechargeable energy-storage unit (15), a water-heating arrangement (16), a health-monitoring module (17), a blood test strip interface (18), a test-strip storage compartment (19), a display unit (20) and a control processor (21). The solar panel generates electrical energy from sunlight and the energy-management unit distributes the available energy to the rechargeable energy-storage unit and supported electronic functions. The water-heating arrangement is configured to use available energy for heating water. The health-monitoring module is configured to receive information from a blood test strip and provide blood glucose information and, according to the implemented sensing arrangement, additional information including blood oxygen (SpO_2), heart rate and blood pressure. The test-strip storage compartment stores unused test strips and the integrated display presents the processed health information to the user. The invention combines hydration, health monitoring, renewable-energy support, test-strip storage and user feedback in a portable personal healthcare companion intended particularly for diabetic patients and also for elderly persons, athletes, gym users, travellers and health-conscious users. The invention provides a compact integrated architecture and is intended to support convenient routine monitoring, subject to appropriate calibration, validation and applicable regulatory requirements. Fig.1

No. of Pages : 22 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130731 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : THERMOMIX SMART – INTELLIGENT HOT–COLD WATER HEATER WITH AUTOMATIC TEMPERATURE CONTROL

(51) International classification	:F25B 21/02, F25B 21/04, H01L 35/30, F24H 9/20, F24H 1/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India

2)M.KUMARASAMY COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)YAZHINI S

2)HARIHARAN BABU. V

3)Dr.S.DINESH

4)Dr.S.INFANCY VIMAL PRIYA

5)Dr.K.Sheikdavood

6)JOTHIMANOHARI D

(57) Abstract :

ABSTRACT THERMOMIX SMART – INTELLIGENT HOT–COLD WATER HEATER WITH AUTOMATIC TEMPERATURE CONTROL The present invention discloses a thermoelectric water heating and cooling system (10) configured to selectively provide heated or cooled water through a compact Peltier-module-based heat transfer mechanism. The system comprises electrical lead wires (1) configured to supply direct current to a thermoelectric module, a water storage chamber (2) configured to store water, a thermoelectric heat-exchange coil (3) positioned within the water storage chamber and thermally coupled to the thermoelectric module, and an insulated outer tank wall (4) configured to reduce heat loss and improve thermal efficiency. During operation, the thermoelectric module transfers heat through the heat-exchange coil, enabling either heating or cooling of the stored water depending on the direction of current flow. The helical heat-exchange coil increases the effective heat-transfer area, promoting efficient and uniform temperature distribution within the water storage chamber. The invention provides a compact, environmentally friendly, low-noise, and energy-efficient water heating and cooling system without requiring a conventional compressor-based refrigeration mechanism, making it suitable for domestic, commercial, and portable water temperature management applications

No. of Pages : 13 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130732 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-ADJUSTING RATIO MECHANISM

(51) International classification	:F16H 37/12, F16H 25/20, F16H 35/00, F16H 37/08, F16H 21/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr. SARAVANAKUMAR KALIMUTHU
2)G. S. DEEPTI SRI
3)Mr. MANJUNATHAN ALAGARSAMY
4)Dr.S.JEGADEESAN
5)R.KAMALITTA

(57) Abstract :
ABSTRACT SELF-ADJUSTING RATIO MECHANISM The present invention relates to a self-adjusting ratio mechanism for mechanically converting an input displacement into a predetermined output displacement. The mechanism comprises an input slider, a ratio/function selection plate, a compensation carriage, a differential linkage assembly, a mathematically profiled constraint track, a follower element and an output slider. The mathematically profiled constraint track represents a selected proportional or nonlinear displacement relationship, while the follower element mechanically follows the profile in response to movement of the input slider. The follower movement automatically repositions the compensation carriage and modifies the geometry of the differential linkage assembly, thereby producing a corresponding output displacement according to the selected relationship. The mechanism may further comprise guide tracks, a mechanical locking arrangement and calibration scales for maintaining and indicating the selected displacement relationship. The invention enables mechanical implementation of predetermined displacement relationships without continuous electronic calculation, electronic feedback or software control and is applicable to mechanical positioning, calibration, measurement, displacement conversion and motion-control systems. Fig.1

No. of Pages : 16 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130733 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PORTABLE AI-POWERED CIVIL MATERIAL QUALITY ANALYZER

(51) International classification	:G01N 33/38, G01D 21/02, G01N 5/04, G01N 21/84, G01N 21/3563
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. MANIKANDAN MURUGESAN
2)Mr. S. EZHILAN
3)Mr. R. SRI SANJIT
4)Mr. MANJUNATHAN ALAGARSAMY
5)Mrs . KIRUTHIKA GOVINDRAJ
6)R.KAMALITTA

(57) Abstract :
PORTABLE AI-POWERED CIVIL MATERIAL QUALITY ANALYZER The present invention discloses a portable AI-powered civil material quality analyzer for rapid field screening of construction materials. The analyzer comprises a main device (1) with a touch display (2), module slot (3), locking mechanism (4), controller (12), rechargeable battery (13), USB Type-C port (14), wireless communication arrangement (15) and data storage (16). Six interchangeable modules are provided for brick (6), sand (7), aggregate (8), cement (9), water (10) and steel/rebar (11). Each module contains material-specific sensing and sample arrangements. Depending on the selected module, the system can acquire image, moisture, temperature, weight, dimension, optical, electrochemical or non-destructive sensing information. The controller processes the acquired information using AI-assisted analysis and generates a material-specific quality score, suitability indication and digital report. The report can be displayed, stored or transferred using wired or wireless connectivity. The modular arrangement enables multiple civil material screening functions to be performed using one portable platform, thereby reducing the need for separate field devices and simplifying on-site quality documentation. Fig 1

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130734 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE WORKLOAD-AWARE THERMAL AND COOLING POWER OPTIMIZATION SYSTEM FOR GPU/CPU PROCESSORS WITH DIRECTIONAL COOLANT ROUTING, FAN ZONE CONTROL AND CHILLER-ASSISTED ESCALATION

(51) International classification	:G06F 1/20, G06F 1/32, G06F 11/30, G06F 9/50, G06F 1/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India

2)M.KUMARASAMY COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)Dr. V.C. BHARATH SABARISH

2)Ms. MONIKA SASIKUMAR

3)Ms. S. MAHESHAVARSHINI

4)Dr.D.DEVAHI

5)Dr.J.BENY

6)Dr.S.VIMALNATH

7)Ms. B. RAMA

(57) Abstract :

ABSTRACT PREDICTIVE WORKLOAD-AWARE THERMAL MANAGEMENT SYSTEM FOR PROCESSOR COOLING A predictive workload-aware thermal and cooling power optimization system (10) for GPU/CPU systems comprises a telemetry acquisition unit (20) providing utilization, temperature and power data; a workload forecasting module (30) predicting near-term GPU and CPU utilization; and a thermal impact prediction module (40) estimating per-processor temperature rise. A coolant routing decision unit (50) actuates GPU and CPU coolant control valves (51, 52) to preferentially direct coolant toward the processor predicted to operate hotter. A fan zone control unit (60) independently controls fan zones (61) based on predicted component activity. A thermal recheck module (70) evaluates temperature changes after actuation and, if cooling is insufficient, activates a cooling escalation unit (80) comprising a chiller (81) and water spray unit (82). A central cooling actuation controller (90) coordinates the pump, valves, fans and chiller, while a sensor feedback network (95) recalibrates forecasting using post-actuation data. The system anticipates thermal excursions, resolves asymmetric loading, and provides verified, iterative cooling escalation. Fig. 1

No. of Pages : 27 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130735 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MASSAGER SHOE WITH DETACHABLE LEG BAND

(51) International classification	:A61H 23/02, A43B 7/14, A43B 7/00, A43B 3/00, A61H 39/04	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)PITCHAIMANI T**
(33) Name of priority country	:NA	**2)RITHIKA S**
(86) International Application No Filing Date	: :01/01/1900	**3)ROSIKA P** **4)DR.S.JEGADEESAN**
(87) International Publication No	: NA	**5)VINCY LEENA F**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT MASSAGER SHOE WITH DETACHABLE LEG BAND The present invention relates to a Massager Shoe with Detachable Leg Band, a portable wearable massage system designed to provide convenient massage therapy to the foot and lower leg. The shoe incorporates integrated massage elements positioned in different zones such as the toe, arch and heel, while the detachable leg band extends the massage function to the ankle and calf region. The system provides multiple massage modes, including vibration, kneading and rolling, along with adjustable intensity and timer control. The device further incorporates heat therapy, rechargeable battery power, Bluetooth connectivity and mobile application control for convenient operation. A modular detachable insole facilitates easy cleaning, maintenance and replacement of internal components. A master–slave control arrangement can synchronize the operation of both shoes and the detachable leg band, while optional solar charging improves portability and energy efficiency. The proposed system provides a compact, wearable and customizable alternative to conventional stationary massagers. It is particularly suitable for elderly persons, athletes, office workers, travellers and individuals who spend long periods standing or walking. The invention combines footwear and leg-massage functionality into a single modular system, improving convenience, portability and ease of use. Fig 1

No. of Pages : 20 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130736 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LIGHTWEIGHT CHAOS-BASED IoT ENCRYPTION

(51) International classification	:H04L 9/00, H04L 9/08, H04L 9/30, H04L 9/06, H04L 9/32	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	**3)M.KUMARASAMY COLLEGE OF ENGINEERING**
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Dr.S.MALATHI**
(86) International Application No Filing Date	: :01/01/1900	**2)G. S. DEEPTI SRI** **3)Dr.P.SAKTHI**
(87) International Publication No	: NA	**4)A. ALBERT ALEXE**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT LIGHTWEIGHT CHAOS-BASED IoT ENCRYPTION The present invention provides a Lightweight Chaos-Based IoT Encryption System (10) configured to secure data generated by resource-constrained Internet of Things (IoT) devices. The system receives IoT data and generates a discrete chaotic sequence using secret initialization parameters through a chaotic sequence generation module (14). The generated sequence is subjected to modular transformation, permutation and diffusion operations to produce a cryptographic keystream. An encryption module (18) combines the cryptographic keystream with the IoT data to generate encrypted data for secure transmission. At a receiving device, synchronized secret parameters are used to regenerate the cryptographic sequence and perform corresponding inverse operations to recover the original data. The system employs computationally efficient arithmetic operations to reduce processing, memory and energy requirements, thereby enabling secure communication in battery-powered and resource-constrained IoT devices. The invention is applicable to agricultural sensors, smart meters, environmental monitoring systems, industrial sensors, wearable devices and other IoT applications requiring lightweight and efficient data security. Fig 1

No. of Pages : 15 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130737 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE MECHANICAL WORKPIECE CLAMPING FIXTURE FOR MICRO-EDM APPLICATIONS

(51) International classification	:B23H 11/00, B23H 7/26, B23H 9/14, B23Q 3/06, B23H 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)SHYAM THETCHANA GOPAL
2)RAJAGURU KADASARI
3)HARIHARAN BABU VENKATESWARA BABU
4)MRS.P.PRATHIBA
5)MOHAMED FAIZAL M K

(57) Abstract :
ABSTRACT ADAPTIVE MECHANICAL WORKPIECE CLAMPING FIXTURE FOR MICRO-EDM APPLICATIONS This invention is about an ADAPTIVE MECHANICAL WORKPIECE CLAMPING FIXTURE FOR MICRO-EDM APPLICATIONS. I designed this ADAPTIVE MECHANICAL WORKPIECE CLAMPING FIXTURE FOR MICRO-EDM APPLICATIONS to hold different shapes and sizes of workpieces during Micro-Electrical Discharge Machining.The fixture has a light base plate and a fixture body. It also uses six clamping arms that can move in a circle and six contact tips made of steel. These contact tips sit around the area where the workpiece goes. I used an adjustment screw on the side that connects to a pinion and scroll mechanism. This setup lets you move all six clamping arms at the time. When you turn the adjustment screw one way the arms move toward the center to grab and center the workpiece. If you turn the adjustment screw the way the arms move out so you can let go of the workpiece.The contact tips can also. Bend to fit different surfaces. Because the adjustment parts are on the side the top stays clear so the Micro-EDM electrode can move easily. This ADAPTIVE MECHANICAL WORKPIECE CLAMPING FIXTURE, FOR MICRO-EDM APPLICATIONS is an small tool that is very easy to use. It makes holding workpiece geometries during Micro-EDM machining operations much faster and safer. Fig.1

No. of Pages : 19 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130738 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PRIME-GRAPH FRAUD DETECTION SYSTEM

(51) International classification	:G06Q 40/00, G06Q 20/40, G06Q 40/02, G06Q 40/04, G06F 16/901
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India
2)M.KUMARASAMY COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)PARTHIBAN NATESAN
2)G. S. DEEPTI SRI
3)DR.S.JEGADEESAN
4)B. SENTHIL RAJA MANOKAR

(57) Abstract :
ABSTRACT PRIME-GRAPH FRAUD DETECTION SYSTEM The present invention provides a Prime-Graph Fraud Detection System for identifying anomalous financial transaction patterns. The system represents financial accounts as graph nodes and transactions as directed edges. Selected transaction characteristics are assigned predetermined prime-number representations. Transaction paths and cycles are analysed and converted into prime-factor signatures using the corresponding prime representations. The generated signatures are compared with historical behavioural signatures and predefined anomaly conditions to identify unusual transaction structures, repeated transfers, circular fund movements and coordinated transaction patterns. A fraud-risk score is generated based on graph characteristics and prime-factor signature deviations. According to the risk score, the system may generate an alert, request additional authentication, hold a transaction or escalate the transaction for investigation. The system provides a mathematical approach for analysing relationships among multiple financial transactions rather than relying solely on individual transaction characteristics. Fig 1

No. of Pages : 15 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130739 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SINGLE LAYER AIR MOISTURE COLLECTOR WITH SOLAR POWER

(51) International classification	:E03B 3/28, B01D 53/26, B01D 5/00, F24F 3/14, F24J 2/04	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** Address of Applicant :The Principal, K.Ramakrishnan College of Technology, Kariyamanickam Road, Samayapuram, Trichy, Tamil Nadu, India- 621112 Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr. MANIKANDAN MURUGESAN**
(33) Name of priority country	:NA	**2)Mr. MANJUNATHAN ALAGARSAMY**
(86) International Application No Filing Date	: :01/01/1900	**3)Ms. S. KAVYA**
(87) International Publication No	: NA	**4)Mr. P. RAJARAJAN**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**5)Mr. A. MOHAMED UMAR** **6)Dr. N. MAHENDRAN**
(62) Divisional to Application Number Filing Date	:NA :NA	**7)DR.S.MANIKANDAN**

(57) Abstract :

ABSTRACT SINGLE LAYER AIR MOISTURE COLLECTOR WITH SOLAR POWER The present invention describes a compact atmospheric water harvesting device (10) comprising a frame, a single hygroscopic layer, a condensation surface, a water collection channel, a solar energy arrangement and a control unit. The hygroscopic layer absorbs moisture from ambient air and, using solar thermal energy, undergoes regeneration to release the absorbed moisture as water vapour. The released vapour is directed toward the condensation surface, where it condenses into liquid water droplets. The water collection channel receives the condensed droplets and guides the collected water toward a storage tank. A processor with temperature and humidity sensors may monitor operating conditions, while photovoltaic power may supply energy to a fan, controller, indicator or optional battery. The single-layer arrangement reduces mechanical complexity and supports repeated moisture absorption, solar regeneration, condensation and water collection. The device is suitable for agricultural areas, greenhouses, rooftops, remote locations and water-scarce regions, enabling repeated operation under changing environmental conditions. FIG 1

No. of Pages : 27 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130745 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED FARMING ROBOT FOR CASUARINATREE MANAGEMENT AND HARVESTING

(51) International classification	:B25J 11/00, B25J 5/00, A01D 34/00, A01D 46/30, B62D 55/065
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)JASPER D
2)CHARULATHA K
3)ELAKKIYA U S
4)INDHUJA M
5)MRS. RAJENDRA KANNAMMAL GANAPATHY
6)Mr. K. SWAMINATHAN

(57) Abstract :
ABSTRACT AN AUTOMATED CASUARINA TREE CULTIVATION, MAINTENANCE AND HARVESTING ROBOT The present invention discloses an automated robotic device (10) for cultivation, maintenance and harvesting of Casuarina trees. The device comprises a mobile frame, a pair of crawler tracks (11, 12), a solar power unit (1), a control unit, a robotic arm assembly, sensors, a water/pesticide tank (9), and a harvesting and trimming mechanism. The crawler tracks (11, 12) are configured to provide stable movement of the device over agricultural and uneven terrain. The solar power unit (1) is configured to supply electrical power to the control unit and electrically operated components of the device. The robotic arm assembly comprises multiple articulated links (3, 4), rotary joints (5, 6, 8), and an end-effector (7), wherein the robotic arm is configured to perform trimming and harvesting operations at different positions and heights of the Casuarina tree. The sensors and camera (21) are configured to detect the tree, branches and surrounding conditions and provide information to the control unit for automated operation. The water/pesticide tank (9) is configured to store and supply water or agricultural treatment liquid through the associated delivery mechanism for plant maintenance. The device further comprises a weeding/ground-working mechanism and a dust or debris collection arrangement (10) for assisting in field maintenance. The control unit processes the sensor information and controls movement of the crawler tracks, robotic arm, trimming mechanism and spraying system. Accordingly, the present invention enables automated Casuarina tree cultivation and maintenance while reducing manual labour, improving operational safety and enabling efficient trimming, spraying, weeding and harvesting operations.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130746 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IoT-Based Smart Solar Energy Management System with Weather-Aware Intelligent Load Prioritization

(51) International classification	:H02J 7/35, H02J 3/38, H02J 13/00, H02S 40/38, H02J 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr. HARIPRABHU MANOHARAN
2)Ms.KAVIYA GUNASEKARAN
3)Mr.KESAVAN ADAIKKAPITCHAI
4)Mr.MAHENDRAN SUNDARARAJAN
5)Mrs. DEEPA RAMALINGAM
6)Dr Y. AROCKIYA JESURA

(57) Abstract :
ABSTRACT IoT-Based Smart Solar Energy Management System with Weather-Aware Intelligent Load Prioritization Power generated by solar energy systems may sometimes be inadequate in particular when the Sun is covered by clouds or the weather conditions are unfavorable or rainy which could result in not only the disruption of functioning of the key electrical loads but also in an unintentional depletion of the battery. As it uses IoT, real-time electricity monitoring, and intelligent weather-based control, the proposed system makes it possible to distribute the available solar energy very effectively. The system works with sensors and an ESP32 microcontroller for continuous monitoring of solar voltage current battery status, and connected electrical loads. Electrical loads are categorized as high, medium, and low priority by the system. A weather API delivers current weather and forecasts weather info which is categorized into different conditions like sunny, partly cloudy cloudy rainy and severe weather. Weather classification is combined with real-time electrical measurements to predict solar energy availability changes. Based on these conditions, the system controls load connected to the relays automatically. When there is low power or unpleasant weather, less priority loads are disconnected and critical loads are kept supplied with power.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130747 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN AND DEVELOPMENT OF AN AUTOMATED ROBOTIC WALL PAINTING SYSTEM USING HOIST AND HYBRID ROLLER-SPRAY TECHNOLOGY

(51) International classification	:E04F 21/08, B05B 13/04, B05B 13/02, E04F 21/12, B05B 15/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs. P. SASIREKHA
2)R. RAGAVAN
3)S. SRIDHAR
4)V. NITHISH
5)Mrs. VALLI PRIYADHARSHINI KRISHNAMURTHY
6)Ms.M.SUBASHREE

(57) Abstract :
ABSTRACT DESIGN AND DEVELOPMENT OF AN AUTOMATED ROBOTIC WALL PAINTING SYSTEM USING HOIST AND HYBRID ROLLER-SPRAY TECHNOLOGY The present invention is an Automated Robotic Wall Painting System Using Hoist and Roller-Spray Technology developed to make wall painting safer and easier, especially for large walls and building façades. In normal painting work, workers often have to use ladders, scaffolding, or suspended platforms to reach higher areas, which can make the work difficult and unsafe. The proposed system reduces this dependence on manual work by using a motorized hoist for vertical movement and motor-driven wheels for horizontal movement. It also uses a hybrid painting unit that combines spraying and rolling, allowing the paint to be applied quickly and then spread more evenly over the wall surface. The system uses an ESP32 microcontroller, ultrasonic sensors, a paint pump, load-cell-based paint monitoring, boundary detection, and IoT communication to control and monitor the painting process. The ultrasonic sensors help maintain a suitable distance between the painting head and the wall, while the boundary sensing arrangement helps avoid areas such as windows, doors, edges, and corners. The paint level is also monitored so that when the reservoir becomes insufficient, the pump and roller can be stopped and an alert can be provided to the operator. By combining these features into one system, the invention aims to reduce human involvement at elevated heights, improve the consistency of the painted surface, reduce unnecessary paint wastage, and make the overall wall painting process more controlled and efficient.

No. of Pages : 20 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130749 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMARTDRAIN: A SOLAR-POWERED IOT-BASED SYSTEM FOR TOXIC GAS DETECTION AND NEUTRALIZATION IN DRAINAGE NETWORKS

(51) International classification	:C02F 1/00, H02J 7/35, B01D 35/143, B01D 35/02, E03F 5/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)INDHUMATHI R
2)RAKSHAVIKASHINI K
3)RAGAVI S
4)SHARUKESH S
5)THARANESH K
6)Miss SOWMIYA VALAYAPATHY
7)Dr.T.M.Nithya

(57) Abstract :
ABSTARCT SMARTDRAIN: A SOLAR-POWERED IOT-BASED SYSTEM FOR TOXIC GAS DETECTION AND NEUTRALIZATION IN DRAINAGE NETWORKS The present invention relates to SMATDRAIN, a smart automated drainage, water filtration and management system. The system comprises a drainage pipe assembly, water-level and/or flow sensing arrangement, Arduino UNO-based control unit, pump/motor assembly, filtration unit, water storage section, solar panel and rechargeable battery. The sensing arrangement detects the water condition and transmits corresponding signals to the Arduino UNO. Based on a predetermined threshold, the Arduino automatically controls the pump to transfer collected drainage water through the filtration unit. The filtered water is directed to a storage section for appropriate reuse or disposal. The solar panel supplies electrical energy and charges the battery for continued operation. The invention provides automated drainage monitoring, pumping, filtration and renewable-energy-assisted water management and safety. FIG 1

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130750 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART MULTI-SENSOR RAILWAY HAZARD DETECTION AND PRIORITY ALERT SYSTEM

(51) International classification	:B61L 23/04, B61L 1/16, B61L 23/00, B61L 1/18, B61L 29/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.HARIPRABHU MANOHARAN
2)Mr.TARUN RAVIKUMAR ANITHA
3)Ms. PRIYADHARSHINI PARAMASIVAM
4)Mrs. MATHUMATHI MOOKKANDI
5)Mrs. M.KAVITHA

(57) Abstract :
ABSTRACT SMART MULTI- SENSOR RAILWAY HAZARD DETECTION AND PRIORITY ALERT SYSTEM This invention refers to a multi-sensor smart railway hazard detection and priority alert system that aims to improve safety and give early warnings for any temporary track hazards. Traditional railway safety measures mainly rely on signals, manual checks, and continuous monitoring, but these may not be very effective in responding to unexpected hazards such as ground shifting, track breaking, obstructions, and other track anomalies. The system consists of vibration sensors, IMU/tilt sensors, IR/laser sensors, a microcontroller-based control system, GPS, and LoRa communication for the continuous tracking and verification of hazards. The data from the sensors is analysed and classified to identify the kind of hazard and the level of risk. The LoRa communication continuously sending the hazard information to the remote control room without any influence by the train presence. If a train comes from the LoRa-based train detection device, the system can transmit the same information to the train via that vehicle's equipment, and provide an audio visual warning. Acknowledgement and retransmission mechanism, battery and health monitor for sensors and operation by solar power are supported by this system. The system helps the railways become more dependable, faster and overall safer in their operations. FIG 1

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130751 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART CLAYTHERM THERMAL ENERGY EFFICIENT DUAL-CHAMBER SANITARY NAPKIN DISPOSAL SYSTEM WITH BIO-BASED SMOKE PURIFICATION AND INTELLIGENT PROCESS MONITORING

(51) International classification	:A61F 13/15, A61L 2/04, A61F 13/472, A61F 13/84, F23J 15/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)KIRUTHIKA S
2)RAMESH M
3)YOGAPRIYA B
4)YAASHIKA K
5)SUMANRAJ D
6)Dr.KANNAN VADIVELU
7)Mrs. UMA MAGESHWARI SWAMINATHAN

(57) Abstract :
ABSTRACT SMART CLAYTHERM THERMAL ENERGY EFFICIENT DUAL-CHAMBER SANITARY NAPKIN DISPOSAL SYSTEM WITH BIO-BASED SMOKE PURIFICATION AND INTELLIGENT PROCESS MONITORING If used sanitary napkins are not disposed of properly they cause health, hygiene and environmental issues. Traditional ways of disposing and electric incinerators often use a lot of energy lose heat and make smoke and smell. To solve these problems the Smart ClayTherm Dual-Chamber Sanitary Napkin Disposal System gives a clean and energy-smart heat disposal method. This system has a heat chamber that heats used sanitary napkins and turns them into a little ash. A ClayTherm heat storage layer around the chamber keeps heat. Slowly releases it keeping the temperature steady and cutting heat loss. The gases that come out of the process go to a chamber for more treatment and cooling. After cooling the gases move through a bio based smoke filter that has biochar, coconut shell activated carbon and natural zeolite to cut smell and trap some pollutants. The ash falls into a tray that can be taken out for disposal. A microcontroller, a temperature sensor, an LCD, LEDs and a buzzer give smart process monitoring. The LCD shows System Ready Heating, Processing, and Smoke Purification. The LEDs show which stage is running and the buzzer sounds when the process finishes. Smart ClayTherm is made to shrink waste and use heat better by keeping heat in. The dual chamber layout gives a path for heat treatment and exhaust cleaning making the whole system work better. The removable bio based filter can be taken out cleaned and swapped when needed. The automated monitoring system also makes it easier, for users lets them see what is happening and keeps them safe. Smart ClayTherm can be put in schools, colleges, hostels, hospitals, offices and public washrooms where sanitary waste is made often. In short Smart ClayTherm mixes heat decomposition, clay heat storage bio smoke cleaning, ash grab and smart monitoring into one small and green sanitary napkin disposal way.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130752 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SOLAR-BASED ADAPTIVE SMPS CHARGER WITH INTELLIGENT SELF-LEARNING POWER REGULATION

(51) International classification	:H02J 7/35, H02J 7/00, G05F 1/67, H02J 3/38, H01M 10/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mr.S.Dineshkumar
2)Dharsan S
3)Dhivya M
4)Nigitha C
5)Dr. DEEPAN KUMAR MADHU,
6)Mrs.R.Nalini

(57) Abstract :
ABSTRACT SOLAR-BASED ADAPTIVE SMPS CHARGER WITH INTELLIGENT SELF-LEARNING POWER REGULATION The present invention describes a solar-based adaptive SMPS charger with intelligent self-learning power regulation for efficient and stable battery charging under varying solar irradiance conditions. The system comprises a solar photovoltaic panel, adaptive SMPS converter, ESP32 controller, voltage sensor, current sensor, temperature monitoring unit, battery, and wireless monitoring interface. The ESP32 continuously receives solar and charging parameters and dynamically regulates the SMPS converter according to the available solar power. An LPPT-inspired intelligent self-learning control mechanism adjusts the converter duty cycle and switching parameters to improve power utilization and maintain stable charging. The system further provides real-time monitoring through a mobile or web-based interface. By adapting to continuously changing solar conditions, the proposed system reduces power loss, improves charging efficiency, enhances reliability, and provides a practical and cost-effective alternative for intelligent solar battery charging application. FIG 1

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130753 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A LIGHTWEIGHT SEMI AUTO CABLE TIE TIGHTENING AND TRIMMING TOOL

(51) International classification	:B65B 13/22, B65B 13/02, B65B 13/28, B65B 13/18, B25B 25/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)CHOCKALINGAM ALAGAPPAN
2)KISHOR M
3)HARI PRASATH C
4)DHIVYALAKSHMI B R
5)MRS SURIYA MYNARBALAMURUGAN
6)Dr. S.NEELAKANDAN

(57) Abstract :
ABSTRACT A LIGHTWEIGHT SEMI AUTO CABLE TIE TIGHTENING AND TRIMMING TOOL A lightweight semi-automatic cable tie tightening and trimming tool is disclosed for electrical cable-management applications. Electrical control panels contain numerous wires that require secure bundling using cable ties, while conventional practice requires technicians to tighten each cable tie manually and then trim the excess portion using a separate tool. The repetitive process is time-consuming, physically demanding, difficult in confined panel spaces and capable of producing inconsistent cable-tie tension. The proposed tool is a rechargeable handheld device operated through a trigger-operated mechanism. The free end of a cable tie is introduced into a cable-tie feed and gripping arrangement and is pulled by a motor-driven tightening mechanism. A drive gear transfers motor output to the tensioning arrangement to tighten the cable tie around the wire bundle. The device is intended to reduce installation time, manual effort and operator fatigue while improving the consistency and finish of cable management. The project further proposes an embedded machine-learning enhancement capable of adapting tightening force based on cable-bundle size and material for consistent and damage-free fastening.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130754 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : INFUSION DYNAMICS SURVEILLANCE WITH AUTOMATED EVENT LOGGING

(51) International classification	:A61M 5/168, A61M 5/172, G16H 40/67, A61M 5/14, G16H 40/63	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)Dr.NALINI NAGULSAMY** **2)ABARNA SAKTHIVEL** **3)HARIVASANTH ANBAZHAGAN** **4)MANOJKUMAR RAVIKUMAR** **5)Mrs. KALPANA VAIRAPERUMAL** **6)Dr. R. LATHA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT INFUSION DYNAMICS SURVEILLANCE WITH AUTOMATED EVENT LOGGING Infusion Dynamics Surveillance with Automated Event Logging is an Internet of Things (IOT) based solution aimed at improving the safety, efficiency and accuracy of intravenous (IV) therapy. Currently, inpatient IV monitoring is done by hand and becomes inefficient and inaccurate when used with the multitude of patients nurses are asked to care for. This can cause empty saline bottles, incorrect infusion rates, leakage, and pressurisation. The system proposed involves the integration of various sensors, such as a load cell, used to measure fluid level, an IR sensor used to detect drip rate, a leakage sensor and a pressure sensor, used to monitor the IV conditions in real time. These sensors are then coupled to a Microcontroller(ESP32 in our case), which constantly analyzes the sensor data and identifies anomalies in the IV based on certain preset threshold values. Whenever an anomaly occurs, they provide quick warning signals through LCD, LEDs as well as voice announcers. Along with the alert system, an automatic safety feature has been introduced employing a servo motor to clamp the IV tube. This prevents blood flow backwards and reduces the risk to patients. Moreover, the IV-injected data is sent to ThingSpeak, the cloud platform where healthcare professionals can see it remotely. FIG 1

No. of Pages : 22 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130755 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RIDER SAFETY INTELLIGENT HELMET DETECTION AND SPEED REGULATION SYSTEM

(51) International classification	:A42B 3/04, A42B 3/30, G08G 1/052, B62J 27/00, G06K 9/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs. INDHUMATHI R
2)BHARATHRAGAVAN J
3)BIRUNDHA V
4)LITHIKA P
5)Dr. MANIKANDAN M
6)Mr. U. RAMANI

(57) Abstract :
ABSTRACT RIDER SAFETY INTELLIGENT HELMET DETECTION AND SPEED REGULATION SYSTEM The study propose the Road accidents especially related to two-wheelers have grown in recent years and are often caused by negligence in wearing helmets while riding. These situations often lead to severe head injuries or death. Traditional law enforcement, based on much manual observation by traffic authorities, cannot be effective and maintained over extended areas. This project deals with the design and implementation of an intelligent, automated helmet detection and speed-control system for electric two wheelers. The system proposed here uses a front-mounted camera, on the vehicle, which continuously snaps images of the rider's head region. These are processed in real-time with the help of artificial intelligence-based image classification methodologies to classify the riders with or without a helmet. This process of detection is done on a connected laptop or PC, which serves as the processing unit of the entire system. After detecting whether the rider is helmeted or not, the result is sent to a PIC microcontroller that controls the operating behavior of the vehicle. The hardware configuration consists of a rechargeable battery, voltage regulator, DC-DC buck converter, switching module, speed-measuring sensor, digital display, and interface circuitry. The speed sensor keeps track of the movement of the vehicle continuously to maintain accurate and stable control, while the digital display shows the real-time speed and the state of the helmet to the rider. The PIC microcontroller controls the speed of the DC motor driving the electronic bike through an L298N motor driver. If the system detects that a rider wears a helmet, then the vehicle functions normally, and the motor runs at full or user-defined speed. Similarly, in the case of detection without a helmet on the rider, the system automatically triggers and limits the bike's speed to less than 25 km/h. This intelligent control guarantees that safety is maintained without any effort from authorities or the rider themselves. It’s quite an effective and efficient way of managing the safety of a rider. The present work shows the possibility of embedding intelligent safety features in modern electric vehicles. In this respect, the present solution is a milestone on the path of reducing the severity of accidents.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130760 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART WATER QUALITY FLOATING MARKER WITH AUTOMATIC SENSOR CLEANING AND REAL-TIME MONITORING

(51) International classification	:G01N 33/18, B63B 35/00, B63B 22/00, C02F 1/00, B63B 35/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MRS.R.KEERTHANA
2)ASIN V
3)HARSHEETHAA K R
4)GAYATHRI R
5)SHYAM THETCHANA GOPAL
6)N. PRAGATHI

(57) Abstract :
ABSTRACT SMART WATER QUALITY FLOATING MARKER WITH AUTOMATIC SENSOR CLEANING REAL-TIME MONITORING The present invention relates to a floating water-quality monitoring device configured for continuous or repeated monitoring of water bodies. The device comprises a buoyant floating platform (101), a waterproof electronics enclosure (102), an ESP32-based controller (103), a power supply unit (104), and a plurality of water-quality sensors including a pH sensor (105), a turbidity sensor (106), and a temperature sensor (107). The device further comprises an automatic sensor-cleaning assembly including a servo motor (108) and a soft wiper or brush (109) positioned to contact the surfaces of the sensor probes. The floating platform is configured to remain on the water surface while the sensor probes are immersed at a predetermined measurement depth. The ESP32 controller acquires measurement data from the sensors and controls the servo motor to operate the wiper before or at predetermined intervals during the measurement cycle, thereby removing dirt, algae, sediment, and other deposits accumulated on the sensor surfaces. Following the cleaning operation and a stabilization period, the sensors obtain water-quality measurements, which are validated and processed by the controller. The controller may further convert the individual sensor measurements into normalized sub-scores and combine the sub-scores using a configurable weighted arithmetic or other project-defined aggregation method to generate a Water Health Score representing the overall condition of the monitored water. Fig 1.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130761 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : RURALCONNECT AI – An AI-Powered Smart Complaint Management System

(51) International classification	:G06Q 50/26, G06N 20/00, G06Q 10/06, G06Q 30/02, H04W 4/029
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs.T.PRIYADHARSHINI
2)MUKILAN M
3)SUBISESH S N
4)HARISHRAYAR G
5)NIRMAL SIVAKUMAR
6)JULIET CELINE MARY. B

(57) Abstract :
Abstract RURALCONNECT AI – An AI-Powered Smart Complaint Management System The present disclosure relates to RuralConnect AI, an AI-powered smart complaint management system intended to improve the reporting, processing, tracking, and resolution of public complaints in rural communities. The system enables citizens to submit complaints through web and mobile applications along with complaint descriptions, photographic evidence, and geographical location information. The proposed system includes an artificial intelligence-based analysis mechanism configured to automatically categorize complaints, predict their priority, and recommend the appropriate government department. Upon analysis, the complaint is routed through the concerned village officer, department officer, and assigned field worker for verification and resolution. The system further provides real-time complaint status tracking, notifications, and citizen feedback functionality. The proposed system supports Tamil and English languages and provides an integrated digital platform for improving transparency, reducing manual processes, enabling accurate complaint routing, and facilitating timely public service delivery in rural communities. Fig 1.

No. of Pages : 12 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130762 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED MENTAL HEALTH TRACKING, SCREENING AND EARLY RISK DETECTION SYSTEM

(51) International classification	:G16H 50/30, A61B 5/16, G16H 10/20, G16H 50/20, G16H 20/70
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MS.C.BAVITHRA
2)DARSHAN MP
3)ASWIN B
4)BHARANIDHARAN J
5)ATULKRISHNA VALLIYEDATHU
6)R. ANUPRIYA

(57) Abstract :
ABSTRACT AI-BASED MENTAL HEALTH TRACKING, SCREENING AND EARLY RISK DETECTION SYSTEM The present invention relates to an AI-assisted mental health tracking and screening system designed for preliminary mental health awareness, screening and historical monitoring. The system integrates structured PHQ-9 and GAD-7 questionnaire-based assessment, automated score calculation, risk classification, database storage, historical tracking and data visualization within a unified platform. A processing module receives user responses and calculates corresponding screening scores, which are analyzed according to predefined criteria to determine screening risk categories such as low, moderate and high risk. Assessment information including scores, assessment dates and risk classifications is stored in a database for subsequent retrieval and comparison. An interactive visualization dashboard presents the collected information through line charts, bar charts, doughnut charts, progress indicators, statistical cards and assessment history tables, thereby enabling users to observe changes in screening results over time. The proposed system provides an integrated, user-friendly and scalable platform for mental health screening, tracking, early risk identification and data visualization, while emphasizing that the generated screening results are not a substitute for professional medical diagnosis. Fig 1.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130763 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DEVELOPMENT OF A SUSTAINABLE COMPOSITE MANHOLE COVER USING WASTE HDPE, COCONUT SHELL POWDER, BAMBOO FIBRE, FINE AGGREGATE AND COARSE AGGREGATE

(51) International classification	:E02D 29/14, C04B 18/24, C04B 18/26, E02D 29/12, C04B 28/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)LOGESHKUMARAN ASHOKKUMAR
2)DHARSHANASUDHAKAR
3)SANTHIYA SENTHILKUMAR
4)VINITH ARIYAN
5)SUDHARSAN VENKATESAN
6)PRIYAN ANBALAGAN
7)Suresh T

(57) Abstract :
ABSTRACT DEVELOPMENT OF A SUSTAINABLE COMPOSITE MANHOLE COVER USING WASTE HDPE, COCONUT SHELL POWDER, BAMBOO FIBRE, FINE AGGREGATE AND COARSE AGGREGATE The present invention relates to a method for developing a sustainable composite manhole cover using waste High-Density Polyethylene (HDPE), coconut shell powder, bamboo fibre, cement, fine aggregate and coarse aggregate as a partial alternative or substitute to conventional cast iron, ductile iron and cement-based manhole covers. The invention comprises collection of raw materials, cleaning and drying, size reduction and processing including shredding of waste HDPE, grinding of coconut shell into powder and cutting of bamboo fibre, and material characterisation including moisture content, particle size analysis, specific gravity, density, melt flow index, bulk density, water absorption and cement properties. A mix design is prepared maintaining a key composite ratio, by weight, of waste HDPE to coconut shell powder to bamboo fibre of 70:20:10, together with cement, fine aggregate, coarse aggregate and water. The materials are dry-mixed and wet-mixed, moulded into a manhole cover mould, compacted using a vibration table or manual tamping, and cured in water for 7, 14 and 28 days. The cured cover is finished by trimming, surface finishing, drilling of lifting holes and final inspection, and is thereafter subjected to performance testing comprising load-carrying, flexural strength, impact resistance, water absorption, durability, surface wear, density and chemical resistance tests. The test results of control and trial mixes are analysed and compared to identify an optimum mix proportion, which is finalised to obtain the final composite manhole cover. The resulting composite manhole cover is lightweight, corrosion-resistant, of high strength, cost-effective, durable and eco-friendly, utilises waste and natural materials, and provides a sustainable partial alternative to conventional cast iron, ductile iron and cement-based manhole covers, thereby reducing environmental impact and dependence on conventional construction materials. FIG 1

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130764 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Passive Load-Adaptive Variable-Damping Conveyor Roller Using a Wedge-Actuated Expandable Friction Mechanism

(51) International classification	:B65G 39/09, B65G 39/02, B65G 39/07, B65G 39/04, B65G 39/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Dr.T.NITHYANANDAM
2)Mr.R.NITHISHKUMAR
3)Mr.P.DHANASHEELAN
4)Mr.S.GIRIDHARAN
5)Mr.G.KESAVAN
6)Dr. ILANGO R
7)Dr. S.HEMAVATHI

(57) Abstract :
ABSTRACT Passive Load-Adaptive Variable-Damping Conveyor Roller Using a Wedge-Actuated Expandable Friction Mechanism This invention pertains to a passive, self-tuning, variable damping conveyor roller that can be used in mechanical material handling and conveying systems for conveying packages, cartons, containers, and heavy industrial loads of various masses. In general, standard conveyor rollers exhibit relatively constant stiffness and damping properties, which can cause bounce and unstable operation of light packages and, at the same time, cause overloading, vibration, noise, bearing stress, and mechanical wear when heavy packages are conveyed. The current invention is related to the mechanically adaptive roller which can change its damping performance depending on the load of the roller itself without using any sensors, electronics, actuators, and power supply sources. The main components of the conveyor roller include the roller shell, inner housing, progressive spring mechanism, spring holder, load carrying plate, guide sleeve, cone expanding wedge, segmented friction ring, returning spring, retaining system, shaft, and bearing system. In case a package is put on or moves over the roller, the loading is carried through the load transfer plate to the progressive spring mechanism. With a relatively light loading, there will be less compression of the progressive spring and, thus, the cone expanding wedge is relatively away from the segmented friction ring and the roller functions with reduced damping and rotation resistances. However, with a relatively heavier loading, the greater compression of the progressive spring results in the axial movement of the cone expanding wedge. FIG 1

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130765 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-CLOSING REUSABLE SCREW WITH AUTOMATIC WALL

(51) International classification	:B23P 19/06, F16J 15/06, F16J 15/02, E04B 1/68, E04B 1/684	(71)**Name of Applicant :** **1)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** Address of Applicant :The Principal, K.Ramakrishnan College of Engineering, NH-45, Samayapuram, Trichy, Tamil Nadu, India- 621112 Karur Tamil Nadu India **2)M.KUMARASAMY COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Kirthiga R**
(33) Name of priority country	:NA	**2)Chithra G**
(86) International Application No Filing Date	: :01/01/1900	**3)Lakshnaasree SV**
(87) International Publication No	: NA	**4)Malavika A**
(61) Patent of Addition to Application Number	:NA	**5)Vidhya SV**
Filing Date	:NA	**6)MRS.P.PRATHIBA**
(62) Divisional to Application Number	:NA	**7)MOHANA GANESH BALAKRISHNAN**
Filing Date	:NA	

(57) Abstract :

ABSTRACT AN AUTOMATED SELF-SEALING SCREW ASSEMBLY The present invention discloses an automated assembly (11) for sealing wall holes after screw removal. The assembly comprises a frame 5 housing, expandable sealing insert, flexible sealing skirt, spring expansion portion, and a control unit. The expandable sealing insert (12) and flexible sealing skirt (14) are configured to close the wall cavity edges by processor-controlled expansion. The spring expansion portion (13) provides automatic adjustment to the sealing skirt for sealing wall holes of various 10 dimensions, thereby preventing loosening or instability of the assembly. The control unit (25) comprises a processor (26), proximity sensor (22), and camera (24). The processor is configured to detect and capture wall cavity images using the sensor and camera. The image processing and Internet of Things (IoT) modules are configured to identify the sealing condition of the 15 wall hole under environmental variations. The processor (26) operates the expandable insert and sealing skirt to close the cavity securely, while the paintable outer surface (16) ensures aesthetic preservation by blending with the wall finish. The present invention will automatically seal wall holes in an effective manner without manual intervention.

No. of Pages : 24 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130766 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI AND IOT BASED ERGONOMIC SMART FUEL DISPENSING AND AUTOMATED FUEL TYPE DETECTING AND BILLING SYSTEM

(51) International classification	:B67D 7/08, G06Q 20/14, G06Q 30/04, B67D 7/04, B67D 7/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)M.KUMARASAMY COLLEGE OF ENGINEERING

Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India

2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

(72)**Name of Inventor :**

1)MRS. P.PRATHIBA

2)KIRUTHIGA N

3)MONISHA S

4)NIKITHA R

5)PRIYAN ANBALAGAN

6)Karthik M

(57) Abstract :

ABSTRACT AI AND IOT BASED ERGONOMIC SMART FUEL DISPENSING AND AUTOMATED FUEL TYPE DETECTING AND BILLING SYSTEM The present invention relates to an AI and IoT based ergonomic smart fuel dispensing system for automated fuel-type detection and billing, achieved through the integration of artificial-intelligence-based vehicle verification, IoT-enabled controlled dispensing, ergonomic operator interaction, and secure digital billing within a single dispenser unit. The system comprises an ergonomically designed dispenser housing accommodating a camera-based optical character recognition module, an embedded controller, a fuel-type database, a solenoid-operated IoT nozzle control mechanism, a push button, a buzzer, one or more flow sensors, and a digital payment and billing module. Upon a vehicle's arrival, the camera captures the vehicle's number plate, and the embedded controller uses an artificial-intelligence-based recognition algorithm to identify the vehicle and retrieve its permitted fuel type from the fuel-type database. The controller verifies compatibility between the selected fuel type and the registered fuel type before enabling the corresponding nozzle path, with the push button and buzzer providing a dual-confirmation safety mechanism, and locks the nozzle path with a warning in the event of a mismatch. During dispensing, a flow sensor measures the fuel quantity delivered, and upon completion, the digital payment module processes a secure cashless transaction and instantly generates a digital receipt, while transaction data is relayed to a local or cloud-based operator dashboard for real-time monitoring. Taken together, these capabilities allow the invention to reduce wrong-fuel dispensing risk, shorten transaction time, eliminate manual billing errors, and improve operational visibility, resulting in a safer, ergonomic, and digitally traceable fuel dispensing platform suited to retail fuel stations, fleet depots, and industrial fuelling points. FIG 1

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130767 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART AI REFRIGERATOR AND AUTOMATIC GROCERY DETECTION AND REORDERING

(51) International classification	:F25D 29/00, G06K 9/00, G06T 7/00, H04N 5/232, G06K 9/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MS.C.BAVITHRA
2)DAKSHA V
3)DHANUSRI M
4)DHARUNIKA V
5)ATULKRISHNA VALLIYEDATHU
6)D. JOTHI MANOHARI

(57) Abstract :
ABSTRACT SMART AI REFRIGERATOR AND AUTOMATIC GROCERY DETECTION AND REORDERING The present invention relates to a Smart AI Refrigerator and Automatic Grocery Detection and Reordering System configured to automatically monitor, identify, and manage grocery products stored inside a refrigerator. The system primarily uses a wide-range AI camera installed inside the refrigerator to capture a broad view of the storage area and grocery items. The captured images are processed using an AI-based image recognition module to identify different grocery products based on their visual characteristics. The detected products are recorded in a digital inventory and updated using current and previous camera observations. A key feature of the invention is its ability to improve product monitoring when one grocery item is partially or completely hidden behind another item. Accordingly, the invention combines AI-based product recognition, inventory monitoring, hidden-product tracking, stock-condition detection, and grocery reordering support into a single smart refrigerator system, thereby reducing manual refrigerator checking and providing a convenient and efficient approach to household grocery management. Fig 1.

No. of Pages : 10 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130768 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI AND IOT INTEGRATED DUAL-CHAMBER FOOD PRESERVATION , QUALITY MONITORING AND SMART REDISTRIBUTION SYSTEM

(51) International classification	:F25D 29/00, F25D 17/04, F25D 11/00, F25D 17/06, G06F 3/041	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)Dr.K.RAJU** **2)Mr.A.ADHISWARAN** **3)Mr.R C DHANUS VARMA** **4)Mr.K.ARUN** **5)Mr.U.NEGORIAN** **6)Mrs.N.G.Gayathri** **7)Mr. N. HAARUN MEERAN**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AI AND IOT INTEGRATED DUAL-CHAMBER FOOD PRESERVATION, QUALITY MONITORING AND SMART REDISTRIBUTION SYSTEM The Ai and IoT Integrated Dual-Chamber Food Preservation, Quality Monitoring and Smart Redistribution System is designed to provide efficient storage, preservation, and controlled redistribution of food using two independently controlled storage chambers. The system consists of separate hot and cold environments to accommodate different types of food while maintaining suitable storage conditions. Temperature sensors, air circulation fans, door sensors, and a touch-screen control unit are incorporated for monitoring and controlling the operating conditions. The system also includes food shelf trays and a controlled food dispensing outlet to facilitate organized storage and redistribution. The intelligent control mechanism continuously monitors the chamber conditions and enables appropriate adjustments to maintain the required temperature. Ventilation and exhaust arrangements improve air circulation and thermal management. The proposed system aims to reduce food wastage, improve food preservation, enhance storage efficiency, and support safe and convenient food redistribution. Overall, the system provides a practical and intelligent approach for managing perishable and prepared food under controlled environmental conditions. FIG 1

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130769 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI BASED DISASTER INTELLIGENCE SYSTEM

(51) International classification	:G06Q 50/26, G08B 21/10, G06N 20/00, G06Q 10/06, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MS. C. BAVITHRA
2)SHYAM P
3)GOWSIKAN K
4)ELANGO M
5)R.SRI SANJIT
6)R. ANUPRIYA

(57) Abstract :
ABSTRACT AI BASED DISASTER INTELLIGENCE SYSTEM The present invention relates to an AI-Based Disaster Intelligence and Early Response Management System designed to improve disaster preparedness, early warning, risk assessment, and emergency response through the integration of Artificial Intelligence, environmental monitoring, geographical intelligence, and cloud-based decision support. The system collects real-time disaster-related information from environmental sensors and other available data sources and combines such information with historical disaster datasets for intelligent analysis. An Artificial Intelligence processing engine analyses the collected information to identify abnormal environmental patterns, estimate the probability of potential disaster events, and predict possible disaster severity and geographical impact. The system further provides intelligent emergency response support by identifying priority regions and assisting in response planning. All operational information, including environmental data, disaster predictions, risk levels, geographical locations, and emergency alerts, is displayed through a centralized cloud-based monitoring dashboard. Accordingly, the proposed invention provides a scalable, intelligent, data-driven, and proactive disaster management platform capable of improving early warning capabilities, reducing emergency response delays, supporting informed decision-making, and enhancing public awareness. Fig 1

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130776 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTOMATED MAIZE SEED SOWING SYSTEM WITH REAL-TIME SOIL ANALYSIS AND PRECISION SEED SPACING

(51) International classification	:A01C 7/20, A01C 7/08, A01C 7/10, A01C 7/04, A01B 79/00	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr. Karthe M**
(33) Name of priority country	:NA	**2)GIRUJASHRI K**
(86) International Application No Filing Date	: :01/01/1900	**3)SASMITHA S M** **4)SUCHITA P N**
(87) International Publication No	: NA	**5)TRISHNA P S**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)B.Sathiya** **7)Mr.G.RAMACHANDRAN**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AN AUTOMATED MAIZE SEED SOWING SYSTEM WITH REAL-TIME SOIL ANALYSIS AND PRECISION SEED SPACING The present invention concerns an Automated Maize Seed Sowing and Soil Condition Monitoring System for automation of maize seed sowing and monitoring of soil conditions during agricultural practices. The system consists of a rechargeable lithium-ion battery pack, Arduino Nano, soil sensor, motor driver, seed sowing mechanism, irrigation pump, relay module, LCD 16×2 display, motor/mechanism fault sensor, and NodeMCU/ESP8266. The soil sensor senses information about the soil moisture level and soil temperature from the area where sowing occurs and feeds the sensed information to the Arduino Nano. Depending upon the detected moisture level of soil, the controller controls the irrigation pump using the relay module. The seed sowing mechanism is driven by a motor for sowing maize seeds at predetermined time intervals during the sowing process. The motor/mechanism fault sensor is incorporated to detect any malfunction such as obstruction/jamming. The LCD 16×2 display gives information locally about the soil parameters and status of the operation.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130777 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUAL NODE SYNCHRONIZED WIRELESS WEARABLE SOMATOSENSORY STIMULATION

(51) International classification	:A61N 1/36, A61B 5/11, A61H 23/02, A61B 5/00, A61N 1/04	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr.S.Sujanthi**
(33) Name of priority country	:NA	**2)JANANI N**
(86) International Application No Filing Date	: :01/01/1900	**3)JANANI V** **4)SARANYA S**
(87) International Publication No	: NA	**5)Mrs.B.Sathiya**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**6)Mr. S. VIGNESHKANNAN**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT DUAL NODE SYNCHRONIZED WIRELESS WEARABLE SOMATOSENSORY STIMULATION The present invention is a bi-node synchronized wireless wearable somatosensory stimulation device (1) comprising a left wearable node (8) and a right wearable node (8), where each of the two nodes has its own microcontroller (3) and a multi-sensor fusion system of IMU (4), FSR (29), and PPG as well as a wireless communication unit (3), a vibration actuator (5), a display (2), and a battery with a charging module (6). The microcontroller (3) processes IMU, FSR, and PPG readings to calculate the amplitude and frequency of a tremor. The wireless module (3) allows to establish the Peer-to-Peer Low-Latency Inter-Node Wireless Timing Synchronization. Synchronization of the nodes is necessary for time referencing. The microcontroller establishes a comparison between the data from the left and right gees. After detecting the occurrence of a required tremor, both actuators (5) begin operation simultaneously and synchronized thus applying the bilateral stimulation and recording the corresponding responses for future process analysis.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130778 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Headlight Lifespan Monitoring and Predictive Maintenance Using IoT & IEA

(51) International classification	:B60Q 11/00, G01R 31/44, G07C 5/08, B60Q 1/00, B60Q 1/04	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Mr.R.PREMKUMAR**
(32) Priority Date	:NA	**2)Mr.V.AAKASH**
(33) Name of priority country	:NA	**3)Mr.J.AATHITHYAN**
(86) International Application No Filing Date	: :01/01/1900	**4)Mr.K.DHYANESHWARAN** **5)Mr.M.SHANNMUGAVEL**
(87) International Publication No	: NA	**6)B.Sathiya**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**7)Dr.K.PERUMAL**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT Headlight Lifespan Monitoring and Predictive Maintenance Using Iot & IEA The invention relates to a Headlight Lifespan Monitoring and Predictive Maintenance that continuously observes the operating condition of a headlight using electrical, thermal and time-based parameters. The system comprises an ACS712 current sensor for measuring headlight current, a W1209 temperature monitoring module with an NTC probe for monitoring temperature, a DS3231 real-time clock for recording operating time, and an ESP32 microcontroller for processing the monitored information. An OLED display provides real-time information such as current, temperature, operating hours and health status. A status LED and buzzer provide local warnings when abnormal conditions are detected. A SIM800L GSM module enables remote SMS notification to a registered mobile phone when maintenance is required. The system provides an early indication of abnormal headlight operation and supports condition-based maintenance instead of relying only on complete headlight failure.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130779 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE BRAKE SAFETY SYSTEM

(51) International classification	:B60T 7/22, B60W 30/09, B60W 30/095, B60W 50/14, B60W 10/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Mrs.D.Umamaheswari
2)Mr.ABISHEK M
3)Mr.ARAVINTHKUMAR K
4)Mr.AMARJITH A
5)Ms.THARUNYA M
6)Dr.J.Sasidevi
7)Mr. H. B MOHAMMED BILAL

(57) Abstract :
ABSTRACT PREDICTIVE BRAKE SAFETY SYSTEM The Predictive Brake Safety System is an intelligent vehicle safety system designed to predict potential collision risks and improve road safety by providing early warnings and automatic braking assistance. Nowadays, road accidents frequently occur due to overspeeding, driver distraction, sudden obstacles, and delayed braking. To overcome these problems, the proposed system uses sensors and intelligent control techniques to continuously monitor the vehicle's speed, distance from nearby obstacles, and surrounding conditions. The collected data is processed by a controller to identify potential collision risks and determine whether the vehicle can stop safely. When an obstacle is detected, the system calculates the risk based on the distance and speed of the vehicle. If a possible danger is identified, an alert is provided to the driver through a buzzer, warning light, or display. When the collision risk becomes critical, the system can activate automatic braking assistance to reduce the vehicle's speed or bring it to a stop, depending on the system design. Unlike traditional braking systems that mainly depend on the driver's reaction, the Predictive Brake Safety System aims to identify dangers in advance and help the driver respond quickly. The system can be implemented in cars, buses, trucks, and other vehicles to improve passenger safety and reduce the possibility of road accidents. Furthermore, the system can be enhanced by integrating Artificial Intelligence (AI), the Internet of Things (IoT), cameras, and advanced sensors to improve obstacle detection and collision prediction. Overall, the Predictive Brake Safety System provides an effective approach to developing safer and smarter vehicles by combining real-time monitoring, risk prediction, warning mechanisms, and automatic braking assistance.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130780 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMMERCIAL VEHICLE OVERLOAD DETECTION AND OWNER ALERT SYSTEM USING LOAD CELL AND ESP32

(51) International classification	:G01G 19/08, G01G 19/12, G01G 19/03, G08G 1/01, G01G 19/02	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)Dr.K.RAJU** **2)Mr.A.ADHISWARAN** **3)Mr.R C DHANUS VARMA** **4)Mr.K.ARUN** **5)Mr.U.NEGORIAN** **6)Mr.V.Gokula Krishnan** **7)Dr. S.HEMAVATHI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT COMMERCIAL VEHICLE OVERLOAD DETECTION AND OWNER ALERT SYSTEM USING LOAD CELL AND ESP32 The present invention relates to a Commercial Vehicle Overload Detection and Owner Alert System using Load Cell and ESP32 for continuous monitoring and detection of excessive loading in commercial vehicles. The system employs multiple load cells strategically positioned on the vehicle chassis to measure the applied load. The sensor outputs are amplified using HX711 amplifier modules and processed by an ESP32 microcontroller to determine the total vehicle load. The measured load is compared with a predefined permissible load limit corresponding to the rated capacity of the vehicle. The loading status can be displayed to the driver, while visual and audible indicators provide appropriate warnings when the load approaches or exceeds the permissible limit. In the event of overloading, the system automatically transmits an alert containing relevant information such as vehicle identification, load status, and time of occurrence to the vehicle owner or fleet manager through a wireless communication interface. The system can additionally record load measurements, overload events, date and time, and other associated information for future analysis and fleet management. Multiple sensing points enable monitoring of load distribution and identification of significant variations in loading across the vehicle platform. The system may also be integrated with GPS tracking and cloud-based data storage for remote monitoring and historical analysis. By providing real-time load monitoring and automatic notification, the proposed system helps prevent unsafe overloading, reduce stress on tyres, brakes, suspension and chassis components, improve fuel efficiency, and minimize maintenance costs. FIG 1

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130781 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Based Adaptive Exhaust Waste Heat Recovery and Intelligent Battery Charging System for Two-Wheelers

(51) International classification	:F01N 5/02, H02J 7/00, F02G 5/02, H01M 10/625, H01M 10/613
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(72)**Name of Inventor :**
1)Nandhakumar.S
2)Harish.D
3)Arun.N
4)Deepan.M
5)Karthick.M
6)Mr. T. VADIVELAN
7)Ms. L. NITHIYA SHRI

(57) Abstract :
ABSTRACT AI-Based Adaptive Exhaust Waste Heat Recovery and Intelligent Battery Charging System for Two-Wheelers The present invention relates to an AI-Based Adaptive Exhaust Waste Heat Recovery and Intelligent Battery Charging System for Two-Wheelers, designed to recover useful thermal energy from hot exhaust gases and convert it into electrical energy. The system comprises an exhaust header pipe, thermoelectric generator (TEG) modules, heat-transfer elements, an aluminum heat sink, temperature sensors, voltage and current sensors, an intelligent control unit, a DC-DC converter, a battery-management system and a rechargeable battery. During engine operation, heat from the exhaust gases is transferred through the exhaust pipe to the hot side of the TEG modules, while the heat sink dissipates heat from the cold side to maintain a temperature difference. The TEG modules thereby generate electrical energy through the thermoelectric effect. The generated electrical output is monitored and regulated using the sensing and power-conditioning system before being supplied to the rechargeable battery. The intelligent control unit adapts the charging operation according to exhaust temperature, generated voltage, generated current and battery condition. The system provides an externally mounted, compact arrangement for utilizing otherwise wasted exhaust heat and converting it into useful electrical energy for battery charging and suitable auxiliary electrical loads of a two-wheeler. Fig 1

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130782 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DUAL NODE SYNCHRONIZED WIRELESS WEARABLE SOMATOSENSORY STIMULATION

(51) International classification	:A61N 1/36, A61B 5/11, A61H 23/02, A61B 5/00, A61N 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(72)**Name of Inventor :**
1)Dr.S.Sujanthi
2)JANANI N
3)JANANI V
4)SARANYA S
5)Mrs.B.Sathiya
6)Mr. S. VIGNESHKANNAN

(57) Abstract :
ABSTRACT DUAL NODE SYNCHRONIZED WIRELESS WEARABLE SOMATOSENSORY STIMULATION The present invention is a bi-node synchronized wireless wearable somatosensory stimulation device (1) comprising a left wearable node (8) and a right wearable node (8), where each of the two nodes has its own microcontroller (3) and a multi-sensor fusion system of IMU (4), FSR (29), and PPG as well as a wireless communication unit (3), a vibration actuator (5), a display (2), and a battery with a charging module (6). The microcontroller (3) processes IMU, FSR, and PPG readings to calculate the amplitude and frequency of a tremor. The wireless module (3) allows to establish the Peer-to-Peer Low-Latency Inter-Node Wireless Timing Synchronization. Synchronization of the nodes is necessary for time referencing. The microcontroller establishes a comparison between the data from the left and right gees. After detecting the occurrence of a required tremor, both actuators (5) begin operation simultaneously and synchronized thus applying the bilateral stimulation and recording the corresponding responses for future process analysis.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130783 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART PEDIATRIC WEIGHING AND DUAL-MODE NUTRITIONAL ASSESSMENT SYSTEM

(51) International classification	:G01G 19/414, G16H 20/60, G06F 19/00, G01G 19/44, A61B 5/00	(71)**Name of Applicant :** **1)Niranjani S** Address of Applicant :F3, Sri Sai Flats, Sri Sai Avenue, Nemilichery, Chrompet, Chennai Tamil Nadu India (72)**Name of Inventor :** **1)Niranjani S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A smart pediatric weighing and nutritional assessment system comprises a weighing platform, housing, weight sensor, signal-conditioning circuit, analogue-to-digital converter, processor, memory, user interface, display, colour-coded indicator, mode selector and communication module. The processor validates a weight measurement obtained by direct weighing or by subtracting a caregiver-only weight from a combined caregiver-and-infant weight. A selected Gomez classification module or Indian Academy of Pediatrics (IAP) classification module uses chronological age and stored reference data to calculate weight-for-age percentage and assign a configured nutritional grade. The result and colour-coded indication are displayed, and the child-linked assessment record is stored locally and may be transmitted to an authorised health-worker database, including after network connectivity is restored.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130785 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : REINFORCEMENT LEARNING BASED OBSTACLE AVOIDANCE ROBOTIC SYSTEM AND METHOD THEREOF

(51) International classification	:B25J 9/16, G05D 1/02, B25J 13/08, G05D 1/00, B25J 19/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Amrita Vishwa Vidyapeetham
Address of Applicant :Amrita Vishwa Vidyapeetham, Bengaluru Campus, Kasavanahalli, Carmelaram P.O., Bengaluru - 560035, Karnataka, India. Bengaluru Karnataka India

(72)**Name of Inventor :**
1)ADUSUMALLI ABHINAYA, Deepika
2)BANDI, Harsha Dhruthija
3)KANDALA, Prapthi
4)SUJITH, Lekshmi
5)MOGENAHALLI RANGANATH, Rashmi

(57) Abstract :
An obstacle-avoidance robotic system (100) and method thereof are disclosed. The system (100) includes a chassis (102), and a sensor unit (104) mounted to the chassis (102). A memory (112) stores instructions and an on-policy SARSA Reinforcement Learning lookup Q-table trained off-line under simulated domain randomization incorporating sensor noise, wheel drift, and motor mismatch. A control unit (110) receives distance parameters from the sensor unit (104), discretizes the distance parameters into a 125-state environmental representation, evaluates the representation using the lookup Q-table retrieved from memory (112) to output a selected movement action, and drives a driving unit (106) via an H-bridge motor driver module (114) using a Proportional-Derivative (PD) controller receiving movement feedback from a set of wheel encoders (108) to maintain heading stability.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130786 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR MONITORING AN OCCUPANT CONDITION IN A VEHICLE

(51) International classification	:B60R 21/015, G01D 21/02, B60W 40/08, B60N 2/90, B60R 21/0132
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)KEVIN GEORGE
2)ADITYA SWAROOP
3)HARSH SINGHAL
4)SUKRITI

(57) Abstract :
The present disclosure relates to a vehicle occupant condition monitoring system (100) and method (400). The system (100) comprises a climate sensor arrangement (102), an inertial sensor arrangement (104), an acoustic sensor (106) and an embedded processor (108). The climate sensor arrangement (102) outputs a temperature signal and a humidity signal, the inertial sensor arrangement (104) outputs an acceleration signal, and the acoustic sensor (106) outputs an acoustic signal. A processing unit (110) derives, over a sampling window, a kinematic feature set and a sound level value, and forms a feature vector therefrom together with a temperature value and a humidity value. A classification unit (112) applies the feature vector to an artificial neural network executed within the embedded processor (108), and obtains a joint classification including an occupant comfort state and a safety state, without transmission of any of the said signals to a device external to the vehicle.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130788 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CONTROL SYSTEM, VEHICLE AND METHOD FOR TRAFFIC-REGIME-BASED ENERGY MANAGEMENT USING MULTI-WINDOW DRIVER-OPERATION AND SPEED ANALYSIS

(51) International classification	:B60L 50/15, B60W 10/06, B60W 30/18, B60W 10/08, B60R 16/03
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ASHOK LEYLAND LIMITED
Address of Applicant :No. 1, Sardar Patel Road Guindy Chennai 600 032 India Tamil Nadu India
(72)**Name of Inventor :**
1)Sivashankar S
2)Sahaya Surendira Babu P
3)Shalin Xavier S
4)Elakkia E
5)Manikanda Raja
6)Kaviya G
7)Frederick Royston J
8)Gowtham S

(57) Abstract :
ABSTRACT CONTROL SYSTEM, VEHICLE AND METHOD FOR TRAFFIC-REGIME-BASED ENERGY MANAGEMENT USING MULTI-WINDOW DRIVER-OPERATION AND SPEED ANALYSIS A control system (124) for a vehicle (100) having an electric traction machine (104), an energy-storage device (108), an engine (112) and a generator (114) includes a traffic-regime estimator (126) that determines, for first and second moving observation windows (312, 314) of different durations, a feature set (316, 318) of driver-operation and speed-distribution features, and determines congestion and highway confidence values. A city or congested-traffic state (504, 508) or a highway state (506) is accepted only when its confidence value satisfies an entry threshold (1004) for a persistence duration (1008) and a cross-window confirmation criterion (324). An energy-management controller (128) prioritizes electric-only propulsion in the accepted city or congested-traffic state, subject to battery protection, and, in the accepted highway state, operates the generator (114) based on a measured state of charge and a higher state-of-charge target (1102). A vehicle and a method are also provided. (FIG. 2)

No. of Pages : 40 No. of Claims : 15

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130804 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : METHOD FOR COMPARATIVE DETERMINATION OF SCHOTTKY BARRIER CHARACTERISTICS IN Au Ta2O5/n-GaN MIS SCHOTTKY DIODES

(51) International classification	:H01L 21/66, G01R 31/26, H01L 29/872, H01L 29/06, G01R 27/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)BAIREESETTY PRASANNA LAKSHMI

Address of Applicant :Falt no 201, Sri Lakshmi Nilayam, Sapthagiri enclave Road no 1, Pet bashherbad Telangana India

(72)**Name of Inventor :**

1)Ms.Smitha Shibu,Lecturer in Chemistry,niversity of Technology and Applied Sciences- Ibra Sultanate of Oman Pin code:400 State: North AL sharqia

2)G.SANDHYA, Assistant Professor,Department of Physics, TKR college of Engineering and Technology, Hyderabad Telangana.

3)Abha gupta, Lecturer in Physics, University of Technology and Applied Sciences- ibra Sultanate of Oman PIN code -400 State - North AL Sharqiyah

4)K. Satish Kumar, Assistant Professor, Department of Physics,Vaagdevi College of Engineering, Warangal Telangana-506005

5)VALLALA KARTHIK, Assistant Professor, Department of Physics, St.Martin's Engineering College , Dhulapally, Hyderabad.

6)M. Madhu, Assistant Professor, Department of Physics, School of Engineering, Anurag University, Hyderabad, Telangana 500088, India.

7)Kusuma Venugopal,Assistant Professor, Department of Physics, Vignan's Institute of Management and Technology for Women,Dist, Ghatkesar, malkajgiri, Kondapur, Padamatisaiguda, Telangana 501301.

8)Kavitha Donti, Assistant Professor, Vignan's Institute of Management and Technology for Women,Dist, Ghatkesar, malkajgiri, Kondapur, Padamatisaiguda, Telangana 501301.

9)PADIGELA SRINIVAS REDDY, ASSOCIATE PROFESSOR, Department of Physics,Vignan's Institute of Management and Technology for Women. Dist, Ghatkesar, malkajgiri, Kondapur, Padamatisaiguda, Telangana 501301.

(57) Abstract :

The invention relates to a comparative electrical-analysis procedure for determining Schottky barrier characteristics of Au/Ta2O5/n-GaN metal-insulator-semiconductor (MIS) Schottky diodes. The procedure obtains current-voltage (I-V) and capacitance-voltage (C-V) characteristics from the diode and derives barrier-height information independently from the two electrical measurements. The resulting values are placed in a comparative framework to examine the correspondence between the barrier characteristics obtained from the two approaches. The procedure may be applied to an as-deposited device and to devices subjected to controlled thermal treatment, thereby allowing changes in the extracted barrier characteristics to be evaluated.

No. of Pages : 9 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130805 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN AND IMPLEMENTATION OF SOLAR-POWERED INTELLIGENT BIRD REPELLENT SYSTEM FOR AGRICULTURAL CROP PROTECTION

(51) International classification	:A01M 29/16, A01M 29/06, A01M 29/32, A01M 29/10, A01M 29/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Mrs. P. SASIREKHA
2)AKHILESH R
3)KARUNYA SHREE S
4)SOLLAMUTHAN K M
5)SURYA MYNARBALAMURUGAN
6)Mr. K. SWAMINATHAN

(57) Abstract :
ABSTRACT DESIGN AND IMPLEMENTATION OF SOLAR-POWERED INTELLIGENT BIRD REPELLENT SYSTEM FOR AGRICULTURAL CROP PROTECTION The Smart Solar Powered Bird Deterrent and Crop Protection System, its purpose is to protect crops from bird in the agricultural fields. It is a more sophisticated and updated form of bird protection than existing methods which involve human observation and controls and they are less effective after a period of time due to recollection. The system detects motion by PIR sensor and then uses a camera to check that a bird is present.On the presences of bird, the system determines roughly the direction of the bird and moves the directional system to the right side of the bird detected. The system emits the sound, water-spray and wing-flapping deterrents and choose an effective counter-measure based on bird position, current energy of the battery, the previous deterrent result and safety. Systemized response of the bird is monitored and the result is stored for future auxiliary deliberation. The renewable energy source, solar energy, can charge the rechargeable battery as needed, feeding energy into the controller, sensors camera and deterrents. Certain high-energy actions can be restricted when the energy level in the battery drops, and the system can go into a low-power simulator mode when no bird is present. Human monitoring can restrict the spraying of water, whereas an accelerometer can respond to abnormal movement and potential damage. Communication over an IoT structure means the system can be remotely observed for the presence of the birds, the operation of the deterrents, the battery situation, the status of the solar charging, and the overall condition of the system.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130806 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-POWERED PEST INTRUSION DETECTION SYSTEM FOR PARKED VEHICLE

(51) International classification	:G08B 25/10, A01M 29/16, G08B 13/00, G08B 13/14, G08B 13/196
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr B RAJESH KUMAR
2)NIVETHA M
3)SRINITHI V
4)Miss SHAHINA SHALU SEBASTIAN
5)Mrs. S.RAJESWARI

(57) Abstract :
ABSTRACT SELF-POWERED PEST INTRUSION DETECTION SYSTEM FOR PARKED VEHICLE The invention relates to the construction of a self-powered pest intrusion detection system for parked vehicles that provides efficient detection of pests' activity. Parked vehicles become vulnerable to pest intrusion, including rats, squirrels, snakes, etc., which may damage the vehicle wiring, insulation, and other components. The invention aims to address these issues through the use of triboelectric sensors, multiple sensor nodes, automatic pest intrusion detection, active deterrent measures, energy harvesting, and remote notification via GSM technology in one device. The apparatus comprises a plurality of triboelectric sensors, a signal conditioning circuit, an ESP32 controller, a deterrent system, GSM communication circuitry, energy harvesting and power management system, battery and/or supercapacitor, and protection electronics box. The triboelectric sensors are placed in selected locations, such as the engine bay, wheel-arch, underbody, and trunk/rear portions of the parked vehicle. Thus, the vibrations and movements made by the pests can be detected. Electrical signals from the sensors are processed and analyzed in the ESP32 controller in order to find out whether there is an intrusion. Using several sensors helps to monitor various parts of the vehicle and locate unusual mechanical activity. In case of intrusion, the system triggers an LED flash, vibration motor, and alarm buzz to discourage the pest, whereas the GSM module sends an SMS alert to the vehicle owner. Furthermore, the triboelectric sensor nodes produce electricity from mechanical motion, which is harvested and controlled in order to charge the battery and supercapacitor for the functioning of the system. The system also offers automatic operation with less requirement for continuous human monitoring and is also adaptable based on the structure of various parked vehicles. The current invention offers a compact and efficient system to detect pests, discourage them, harvest energy, and send alerts to the vehicle owner, which will help reduce the problem of damage caused by the pests as well as continuous manual checking.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130807 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AUTONOMOUS CEILING CLEANING ROBOT

(51) International classification	:A47L 11/40, G05D 1/02, A47L 11/24, A47L 11/38, A47L 9/28	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)Dr BANUMATHI SEMBANAN** **2)ISWARYA PALANISAMY** **3)ASHWIN RAVI** **4)KARNIKA DURAISAMY** **5)MR ABISHEK CHRISTOPHER RAJUPILLAI** **6)Ruba M,**
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AN AUTONOMOUS CEILING CLEANING ROBOT This invention provides a robotic device (20), known as an autonomous ceiling cleaning robot, which is designed for cleaning the ceiling surface, monitoring the cleaning area in real time, detecting obstacles on its way and safe movement over the ceiling surface. The autonomous ceiling cleaning robot includes robot base frame (11) and tires (8) used to move the robot over the ceiling surface. The ultrasonic sensor (1) is designed to detect obstacles in front of the robot, while the camera module (2) is intended for taking photos and videos of the ceiling surface for further monitoring of the cleaning area. The IR sensor (7) is used to detect the ceiling surface and help identify the boundaries of the cleaning area. The STM32 controller (5) is intended to analyze the information received by sensors and control the robot's work on the basis of predetermined instructions. The ESP8266 Wi-Fi module (6) provides wireless communication and allows the user to send movement commands to the robot. The motor driver (15) is used to control the DC motor (4).The telescopic arm assembly consists of telescopic arm base (10), telescopic arm shoulder (12), telescopic arm elbow (13), and telescopic arm head (14) for placing the cleaning system at various places and angles. A servo motor (16) has been provided in order to provide a control for the motion of the telescopic arm assembly. A cleaning brush (3) mounted on the telescopic arm head (14) is designed for removing dust, d

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130808 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED INTELLIGENT MANURE MANAGEMENT AND NUTRIENT RECOMMENDATION SYSTEM FOR OPTIMIZING CROP NUTRIENT REQUIREMENTS AND AGRICULTURAL PRODUCTIVITY

(51) International classification	:A01C 21/00, G06Q 50/02, A01B 79/02, A01B 79/00, G06Q 10/06	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)Dr.K.Poongothai** **2)Bharathi R** **3)Bhumikasri V** **4)Deepika K** **5)Sakthivel J** **6)Dr.T.M.Nithya**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AI-BASED INTELLIGENT MANURE MANAGEMENT AND NUTRIENT RECOMMENDATION SYSTEM FOR OPTIMIZING CROP NUTRIENT REQUIREMENTS AND AGRICULTURAL PRODUCTIVITY Effective nutrient management is essential for maintaining soil fertility and achieving healthy crop growth. Manure is an important organic source of essential nutrients such as Nitrogen (N), Phosphorus (P), and Potassium (K), but its nutrient composition and availability can vary depending on its source, moisture content, decomposition, and storage conditions. Conventional manure application practices often rely on fixed or general quantities without considering the actual nutrient requirements of the soil and the nutrient content available in the manure. This can lead to over-application, nutrient wastage, increased cost, or insufficient nutrient supply to crops. To address this issue, an AI-Based Adaptive Manure Nutrient Management and Recommendation System is proposed to provide a more accurate and site-specific manure recommendation. The system considers soil NPK values, crop requirements, manure nutrient composition, and relevant manure conditions as input parameters. Manure nutrient values can be obtained from laboratory analysis or, for commercially packaged manure, from the nutrient information provided with the product. For stored manure, factors such as storage duration and moisture conditions can also be considered to estimate changes in nutrient availability. Machine Learning techniques are used to analyze these parameters and identify the relationship between soil nutrient requirements and available manure nutrients. Based on the analysis, the system estimates the suitable quantity of manure required for a particular field or crop instead of applying a fixed amount. The proposed system aims to improve nutrient-use efficiency, reduce unnecessary manure application and nutrient wastage, lower input costs, and support better soil fertility management.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130809 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Embedded Multi-Zonal Spatiotemporal Shot Trajectory Acquisition and Predictive Performance Analytics Framework for Handball Training Systems

(51) International classification	:A63B 71/06, A63B 63/00, A63B 24/00, A63B 69/00, A63B 69/36
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)MR. P MANIRAJ
2)MIDHUNBALAJI D
3)SUDHARSSAN P A
4)SOUNDHARYA R
5)Ms. Kavitha P
6)Dr. A. DURGADEVI

(57) Abstract :
ABSTRACT AN EMBEDDED MULTI-ZONAL SPATIOTEMPORAL SHOT TRAJECTORY ACQUISITION AND PREDICTIVE PERFORMANCE ANALYTICS FRAMEWORK FOR HANDBALL TRAINING SYSTEMS The present invention relates to an embedded sensing and performance analytics system integrated with a handball goal-post. The system comprises R-type sensors for spatial zone detection, C-type sensors positioned internally along a top pipe for additional zone detection, and S-type sensors positioned internally along a fixed straight rear pipe for shot-speed measurement. An electronic controller is housed inside the pipe structure and receives signals from the sensor groups. The controller processes the sensor signals to determine spatial and temporal characteristics of a handball shot, including detected zone and speed-related information. The sensor groups and controller are internally mounted so that the electronic components are not externally exposed. The system can correlate detected zone information with shot timing and speed to generate shot-event records and training-performance parameters. The acquired information can be stored and analysed to identify shooting patterns and performance trends and may further support predictive performance analytics for handball training. FIG 1

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130810 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IV Drip Monitoring System

(51) International classification	:A61M 5/168, A61M 5/14, A61M 5/172, A61M 5/36, A61M 5/00	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 trichy Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** **3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** (72)**Name of Inventor :** **1)Dr.R Sujatha** **2)S Sharvesh** **3)SM Sakthivel** **4)S Sreejith** **5)Mohamed Imran** **6)R.Nalini**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Abstract IV Drip Monitoring System The present invention relates to an automated intravenous (IV) drip monitoring system configured to monitor the remaining quantity and flow condition of an IV fluid container. The system comprises an ESP32 microcontroller, a load-cell-based weight measurement unit, a precision load-cell amplifier and analog-to-digital conversion unit, a drop detection sensor, an OLED display, and a wireless communication arrangement. The load cell measures the weight associated with the IV fluid container to estimate the remaining fluid quantity, while the drop detection sensor detects individual drops passing through an IV drip chamber and enables determination of the drip rate. The ESP32 processes the sensor information and displays relevant parameters including remaining fluid quantity, drip rate, and system status on the OLED display. The wireless communication arrangement enables monitoring information and warning notifications to be transmitted to a mobile device when predetermined conditions, such as low remaining fluid, absence of detected drops, or abnormal drip rate, are detected. The system provides an integrated, compact, and automated approach for continuous IV fluid monitoring and alert generation.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130811 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Sensor-Confidence and Energy-Aware Adaptive Sampling System and Method for Agro-Environmental Monitoring

(51) International classification	:H02J 7/00, G01N 33/24, G01W 1/14, G01D 21/02, G01W 1/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.S.Geeitha
2)S.Theshiya
3)M.Shanmitha
4)P.VijayaSrinithi
5)S.Subhiksha
6)Mohamed Rifath A
7)K.Revathi

(57) Abstract :

Abstract Sensor-Confidence and Energy-Aware Adaptive Sampling System and Method for Agro-Environmental Monitoring The present invention relates to a Sensor-Confidence and Energy-Aware Adaptive Sampling System and Method for Agro-Environmental Monitoring, comprising a field-deployed sensor arrangement (soil moisture sensors, temperature/humidity sensor, light intensity sensor, rain sensor, and voltage/current sensor) connected to an ESP32-microcontroller-based edge node housed in a waterproof enclosure. The edge node performs data acquisition, sensor validation and cross-validation, environmental change analysis, and computation of a sensor confidence value, and further monitors the energy status of a battery-based power management unit comprising lithium-ion cells, a charging and protection module, and a voltage regulator. The core inventive feature resides in a closed-loop adaptive sampling mechanism whereby the environmental change, sensor confidence, and energy status are jointly processed to dynamically select the next sensing interval from among a normal sampling mode, a verification sampling mode, and a rapid sampling mode, the outcome of which is fed back to the sensing stage. The system further computes Sensor Stability Index (SSI), Measurement Stability Index (MSI) and Fault/Sensor-confidence Index (FSI) values for anomaly detection, transmits processed data over a wireless link to a cloud server, and presents live monitoring data, historical data, alerts, the computed indices, and recommendations through a decision-support dashboard accessible via computing and mobile devices.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130812 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LEAFSORT –X ARDUINO BASED INTELLIGENT LEAF SEPERATION AND CLEANING SYSTEM

(51) International classification	:B07C 5/342, B07C 5/36, B07C 5/02, G05B 19/418, B07C 5/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.S.JEGADEESAN
2)K. PRANEETHA
3)A.SUJI
4)K.VASANTH
5)Mr. SARAVANAKUMAR KALIMUTHU
6)R.KAMALITTA

(57) Abstract :
ABSTRACT LEAFSORT –X ARDUINO BASED INTELLIGENT LEAF SEPERATION AND CLEANING SYSTEM The present invention is an innovative Arduino-based intelligent leaf separation, classification, and automated collection system developed for agricultural and industrial leaf-processing applications. The system combines an automated feeding mechanism, conveyor transportation assembly, multi-parameter sensing chamber, Arduino processing controller, intelligent classification engine, servo-actuated diversion mechanism, and dedicated collection chambers into a unified smart sorting platform. Leaves are individually analyzed based on characteristics including color, size, shape, moisture content, texture, maturity level, freshness, and health condition. The acquired data is processed in real time by the Arduino controller to determine the appropriate classification category. Based on the generated decision, servo-operated diversion gates automatically direct each leaf into its designated collection chamber. The system may further incorporate image-processing modules for disease detection, environmental monitoring sensors for operational optimization, and wireless communication modules for remote monitoring and agricultural analytics. By integrating intelligent sensing, automated decision-making, and real-time sorting capabilities, LEAFSORT-X is intended to reduce manual labor, improve classification accuracy, increase processing efficiency, minimize sorting errors, and support modern smart farming practices. The invention provides a cost-effective, energy-efficient, scalable, and automated solution for high-speed leaf separation and quality assessment applications. Fig 1

No. of Pages : 12 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130813 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : IV CANNULA PROTECTION AND PREDICTIVE MONITORING SYSTEM

(51) International classification	:A61M 5/168, A61B 5/00, A61M 5/172, A61M 5/14, A61M 5/158
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Ms.M.SARANYA
2)SASMITHA S M
3)SUBHAHARINI S
4)SUBHIKSHA E
5)SUMITHA P
6)Dr. MANIKANDAN MURUGESAN
7)DR.S.KAVITHA

(57) Abstract :
ABSTRACT IV CANNULA PROTECTION AND PREDICTIVE MONITORING SYSTEM The present invention proposes an intelligent IV Cannula Protection and Predictive Monitoring System for continuous monitoring of tissue surrounding an inserted IV cannula. Conventional monitoring mainly relies on periodic manual observation, which may delay identification of complications such as infiltration, blockage, infection, leakage, and cannula displacement. To overcome this limitation, the proposed system incorporates a smart protective sleeve equipped with pressure, temperature, and stretch sensors positioned around the cannula site. These sensors continuously collect tissue-related parameters and transmit the data to an ESP32 processing module. The sensor data are processed using multi-sensor data fusion and analysed through an AI-based monitoring algorithm. Based on the analysed information, a Tissue Stability Index (TSI) is generated to represent the overall tissue condition and classify it as Safe, Warning, or Critical. The system further identifies and predicts potential complications. In abnormal conditions, real-time wireless alerts are generated for healthcare professionals, enabling timely intervention and improving patient safety, comfort, and reliability during intravenous therapy.

No. of Pages : 16 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130814 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IV AI-OPTIMIZED BIOMIMETIC BIO-SPIRAL RICOCHET VORTEX SEPARATOR FOR TEXTILE MICROPLASTIC SEPARATION

(51) International classification	:B01D 21/26, C02F 1/38, B04C 5/103, B01D 17/02, B01D 63/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Karur Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
3)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
(72)**Name of Inventor :**
1)Dr.N.MAHENDRAN
2)KAVIYA T
3)SRIANU A
4)TANSI T
5)VISHMIDHA DHARINI M S
6)Dr. MANIKANDAN MURUGESAN
7)DR.S.MANIKANDAN

(57) Abstract :
ABSTRACT The present invention proposes a Bio-Spiral Ricochet Vortex Separator for Microplastic Removal for separating and collecting microplastic particles from contaminated water. Conventional microplastic removal methods mainly rely on filtration, membrane systems, or other treatment processes, which may involve clogging, filter replacement, and maintenance requirements. To overcome these limitations, the proposed system incorporates a vortex-based separation chamber with a tangential inlet and manta-ray-inspired inner spiral ribs. Contaminated water containing microplastic particles enters tangentially, generating a swirling vortex that influences particle movement through centrifugal and inertial effects. The moving microplastic particles interact with the internal spiral ribs, which modify their trajectories through a ricochet-type particle guidance mechanism and direct them toward a dedicated collection chamber. Comparatively cleaner water exits through the water outlet. The proposed system can further be evaluated using COMSOL Multiphysics to analyze fluid flow, pressure distribution, vortex formation, particle trajectories, separation efficiency, and pressure drop. Thus, the invention provides a bio-inspired, filter-free, compact, and potentially low-maintenance approach for microplastic separation from contaminated water.

No. of Pages : 22 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130816 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN EXPLAINABLE AI-BASED REAL-TIME EMERGENCY DETECTION AND ALERT SYSTEM

(51) International classification	:H04N 7/18, G08B 13/196, G06K 9/00, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)M.KUMARASAMY COLLEGE OF ENGINEERING
Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India
2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING
3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(72)**Name of Inventor :**
1)Mr. S. PRAKASH
2)PUNIT BADYAL
3)SAHIL KUMAR
4)YASHWANTH R
5)SANTHOSH M
6)Mrs.A.Farizan Begam
7)Dr. S. SUJATHA

(57) Abstract :
ABSTRACT An Explainable AI-Based Real-Time Emergency Detection and Alert System Using CCTV Traditional CCTV surveillance systems mainly depend on human operators to continuously monitor video footage, which can lead to delayed emergency identification due to fatigue, limited attention, or multiple cameras requiring simultaneous monitoring. To overcome these limitations, the proposed system integrates Artificial Intelligence (AI), computer vision, and Explainable Artificial Intelligence (XAI) techniques to identify potential emergencies in real time and provide understandable explanations for the generated alerts. The system captures live video streams from CCTV cameras and processes the frames using image preprocessing and deep learning techniques. Depending on the selected model and training dataset, the system can identify emergency situations such as fire, smoke, physical violence, accidents, falls, or unusual human activities. When a potential emergency is detected, the system analyzes the relevant visual features and generates an alert containing the emergency category, detection confidence, timestamp, and corresponding video frame or snapshot. Explainable AI techniques are incorporated to help users understand why the system generated a particular alert by highlighting relevant image regions or presenting the visual evidence that influenced the prediction. The alert information is displayed on a monitoring dashboard and can be communicated to authorized personnel through configured notification mechanisms.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130817 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : XAI-DRAINPULSE: SELF-POWERED UNDERGROUND SEWER MONITORING AND ROAD SINKHOLE PREDICTION USING TRAFFIC KINETICS AND EXPLAINABLE PHYSICS-INFORMED AI

(51) International classification	:H02N 2/18, G08G 1/01, G01V 1/00, G01V 1/52, G08G 1/0967
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)M.KUMARASAMY COLLEGE OF ENGINEERING

Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Trichy Tamil Nadu India

2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING

3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(72)**Name of Inventor :**

1)Mrs.S VANISRI

2)SATHANA M

3)SHRI YAZHINI A

4)SRI VARSHINI G

5)Mr. R. Pandiarajan

6)Mr .G.RAMACHANDRAN

(57) Abstract :

ABSTRACT XAI-DrainPulse: Self-Powered Underground Sewer Monitoring and Road Sinkhole Prediction Using Traffic Kinetics and Explainable Physics-Informed AI Rapid urbanization and aging underground sewer infrastructure have increased the risk of pipeline fractures, toxic bio-corrosion, sewer blockages, and catastrophic road sinkholes. Existing monitoring systems are mostly reactive, depend on battery-powered sensor nodes, and often produce false alarms due to black-box AI models. This paper proposes X-DrainPulse, a self-powered, 360° underground drainage health and subsurface hazard monitoring system. The system uses a PZT-5H Piezoelectric Kinetic Energy Harvester installed on the sewer pipe crown to convert vehicular traffic vibrations (10–60 Hz) into electrical energy. It generates 1.5–3.5 mW of power to continuously operate an IP68-rated multi-sensor edge node consisting of a MEMS Accelerometer, Ultrasonic Depth Sensor, Pressure Transducer, and H_2S Gas Sensor, eliminating the need for battery maintenance. To reduce false alarms caused by changing wastewater flow conditions, the collected sensor data is processed using a Coupled Hydro-Geomechanical Physics-Informed Neural Network with Graph Neural Networks (PINN-GNN). The model combines Saint-Venant equations for hydraulic flow and Boussinesq stress distribution for structural vibration analysis to distinguish normal flow variations from actual structural failures. As a result, the system can predict Subsurface Soil Voids (Sinkholes), Concrete Bio-Corrosion (H_2S), Hydraulic Flow Obstructions, and Structural Pipe Cracks up to 72 hours in advance. In addition, Explainable Artificial Intelligence (XAI) is integrated using SHAP values and physics informed residual metrics. Instead of providing only warning alerts, the Municipal Dashboard displays feature importance, physics equation boundary states, real-time energy harvesting metrics (mW), and actionable maintenance recommendations, enabling civil engineers to understand and trust the AI predictions.

No. of Pages : 15 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130818 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MODULAR CABIN SYSTEM FOR TERRACE BASED CULTIVATION

(51) International classification	:A01K 63/00, A01K 63/04, A01K 61/00, A01K 63/06, A01K 61/60	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)DR. K. RAMPRATHAP** **2)MRS. A. SANTHIYA** **3)V. MOUNIKA** **4)S. SURYA** **5)Dr.K.Dinesh** **6)Miss SOWMIYA VALAYAPATHY**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT Modular Cabin System For Terrace Based Cultivation" The present invention relates to a modular cabin system for terrace-based integrated cultivation, particularly a compact and portable agricultural system configured for integrated fish culture, Azolla cultivation, poultry rearing, and plant cultivation within a protected terrace environment. The system comprises a substantially enclosed modular cabin having a structural frame, transparent panels, a roof assembly, lockable castor wheels, and adjustable support members for facilitating installation, movement, positioning, and leveling on a terrace or supporting surface. The system comprises a fish tank having an upper Azolla cultivation layer and a lower fish-culture region separated by a mesh partition, a poultry cage positioned above a chicken waste collection tray, a fertilizer/nutrient tank, and a plant tray. The fish tank is configured with a water outlet connected to the chicken waste collection tray and a water inlet connected to the fertilizer/nutrient tank for receiving treated nutrient-rich water. The chicken waste collection tray is connected to the fertilizer/nutrient tank through a designated waste outlet pipe, thereby enabling collection and transfer of poultry waste for nutrient recycling. The fertilizer/nutrient tank is further connected to the fish tank and the plant tray for distributing the treated nutrient-containing liquid within the integrated system. The plant tray receives the nutrient-rich liquid for supporting plant growth without requiring a return connection from the plant tray to the fertilizer/nutrient tank.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130819 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED ROOT CHAMBER MONITORING SYSTEM FOR VERTICAL FARMING (AEROPONICS)

(51) International classification	:A01G 31/02, A01G 31/06, A01G 31/00, A01G 33/00, G06T 7/00	(71)**Name of Applicant :** **1)M.KUMARASAMY COLLEGE OF ENGINEERING** Address of Applicant :The Principal, M.Kumarasamy College of Engineering, Thalavapalayam, Karur, Tamil Nadu, India – 639113 Tamil Nadu India **2)K.RAMAKRISHNAN COLLEGE OF ENGINEERING** **3)K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY** (72)**Name of Inventor :** **1)Dr. PADMAYATHY. S** **2)Mr. CHANDRU. B**
(31) Priority Document No	:NA	**3)Mr. ASWIN. M**
(32) Priority Date	:NA	**4)Mr. ARUN. N**
(33) Name of priority country	:NA	**5)Mr. BALAJI. K**
(86) International Application No Filing Date	: :01/01/1900	**6)Mr. DEEPAN. M** **7)Gokula Krishnan V**
(87) International Publication No	: NA	**8)Dr. MANIKANDAN MURUGESAN**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT AUTOMATED ROOT CHAMBER MONITORING SYSTEM FOR VERTICAL FARMING (AEROPONICS) An automated root chamber monitoring system for vertical farming based on aeroponic cultivation is disclosed. The system comprises an active root chamber incorporating a concentric dual-cylinder distribution arrangement in which an inner passage supplies nutrient-rich solution to misting outlets and an outer passage supplies fresh air to the root zone. Environmental sensors and an internal camera continuously acquire root-zone information without requiring frequent disturbance of the plants. A processing unit analyses sensor data and root images, optionally using machine-learning or deep-learning techniques, to assess root health and detect abnormal conditions. A microcontroller-based control unit receives the resulting decision information and regulates nutrient pumping, misting intervals and airflow devices. The invention establishes a closed feedback loop of sensing, imaging, analysis, decision and actuation. The proposed system improves the ability to manage nutrient distribution, aeration and root-zone monitoring in compact aeroponic vertical farming systems and supports preventive, data-driven root health management.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130823 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Comparative Analysis Of Video Encryption And Decryption Techniques Detailed Description

(51) International classification	:H04N 7/167, H04N 21/2347, H04L 9/06, H04L 29/06, H04N 21/4405	(71)**Name of Applicant :** **1)Vishnu Women's University** Address of Applicant :Vishnupur Bhimavaram Bhimavaram Andhra Pradesh India (72)**Name of Inventor :** **1)Mr Kommisetti Murthy Raju** **2)Mr.Vemu Srinivas Rao** **3)P.Priyambica** **4)V.Vijaya** **5)T.Tarunika Sai Phanisri**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present work generally relates to a secure video encryption framework useful for protecting digital video data transmitted over public communication networks. The proposed framework, named Video Encryption using RSA, AES, and ECC, comprises cryptographic techniques designed to ensure confidentiality and secure transmission of multimedia content. The framework includes video processing, frame extraction, encryption, decryption, and performance evaluation modules. The overall system utilizes video frames as input and applies encryption algorithms to generate secure encrypted outputs while maintaining video quality after decryption. The presented framework is useful for analyzing and comparing the performance of three widely used cryptographic algorithms, namely RSA (Rivest–Shamir–Adleman), AES (Advanced Encryption Standard), and ECC (Elliptic Curve Cryptography). The input video is divided into individual frames, and each frame undergoes encryption and decryption using the selected algorithm. The performance of the framework is evaluated using quality and efficiency metrics such as Peak Signal-to-Noise Ratio (PSNR), Mean Square Error (MSE), encryption time, and decryption time.

No. of Pages : 26 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130826 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EduMesh: A Collaborative Platform for Intelligent Learning and Knowledge Exchange Ecosystem

(51) International classification	:G06Q 50/20, G09B 19/00, G09B 7/00, G09B 7/02, G09B 7/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY
Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India. Vijayawada Andhra Pradesh India
2)Dr. M. Venkateswara Rao
3)T. Mouli
4)S. Harshini
5)M. Praneetha
6)SK. Nagoorbee
7)S. Santhosh
(72)**Name of Inventor :**
1)Dr. M. Venkateswara Rao
2)T. Mouli
3)S. Harshini
4)M. Praneetha
5)SK. Nagoorbee
6)S. Santhosh

(57) Abstract :
ABSTRACT [0030] A collaborative learning and knowledge exchange system is disclosed for the discovery, matching and tracking of skills among learners, mentors and contributors. The system collects a profile of offered skills, desired skills, proficiency levels, interests, goals and availability for each user. A matching process compares the requirements of a learner with the capabilities offered by other users, and a personalized recommendation ranks collaborators and shared resources using skill relevance, proficiency compatibility, availability and interaction history. Selected collaborators are connected through a request, a scheduled session and recorded feedback, and the completed activities are tracked as progress that refreshes later recommendations. The web-based workflow gives members a connected ecosystem in which every user can learn, teach and contribute. This project presents EduMesh, a web application organized into interface, application-service, personalization, security and data layers, with a structured database for users, skills, matches, sessions, feedback, resources and progress. Conventional online learning is largely one-way and does not help a learner find a person who can teach a specific practical skill. EduMesh connects profile creation, skill matching, recommendation, collaborative exchange, shared knowledge and progress tracking in a single platform, so that members can learn, teach and contribute in a systematic way.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130833 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Visual Intelligence for Secure and Transparent Elections: An Intelligent System for Election Monitoring, Anomaly Detection and Audit-Based Transparency

(51) International classification	:G06F 21/55, G06F 17/30, G06F 11/30, H04L 12/24, G06Q 50/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr RVR NRI Institute of Technology Deemed to be University
Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India Vijayawada Andhra Pradesh India
2)Dr. B. Renuka Chowdary
3)SK. Tanveer Anjum
4)P. Rohini
5)S. Keerthi
6)MD. Riyaz

(72)**Name of Inventor :**
1)Dr. B. Renuka Chowdary
2)SK. Tanveer Anjum
3)P. Rohini
4)S. Keerthi
5)MD. Riyaz

(57) Abstract :
ABSTRACT [0030] A secure election intelligence system is disclosed for the monitoring of elections. The system authenticates users and controls access by role, collects election-related data and validates it, and converts it into charts, graphs, tables and dashboards. Anomalies and suspicious patterns are detected by comparing the data with predefined rules, thresholds or expected patterns, a risk level is assigned to security events, and alerts are generated. Election indicators are displayed on interactive geographic maps, key performance indicators are shown on a real-time monitoring dashboard, and important operations are stored in an audit trail from which transparency reports are produced. The detected patterns are indicators for review by authorized officials and not determinations of misconduct, so that the system supports transparency and accountability in the electoral process. This project presents Visual Intelligence for Secure and Transparent Elections, a web-based application built with React and Vite, Express.js, Prisma and PostgreSQL and JWT authentication. Traditional election management depends on manual verification, isolated records and limited visualization. The proposed system connects secure access, data validation, visual analytics, anomaly detection, geographic visualization, security risk assessment, alerts, audit trails and performance indicators in a single platform, so that election information can be monitored and understood in a systematic and transparent way.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130849 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI and IoT-Based Smart Warehouse Automated Guided Vehicle

(51) International classification	:G05D 1/02, G06Q 10/08, B65G 1/04, B65G 1/137, G06Q 50/28	(71)**Name of Applicant :** **1)Karthi Nallasivam** Address of Applicant :61/10, MOKKAIYAMPALAYAM, VILLARASAMPATTI POST, ERODE Tamil Nadu India **2)Dr.C.Sowmya Dhanalakshmi** **3)Divine Meckerth S** **4)Ragul H** **5)Ricwin.S**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Karthi Nallasivam**
(33) Name of priority country	:NA	**2)Dr.C.Sowmya Dhanalakshmi**
(86) International Application No Filing Date	: :01/01/1900	**3)Divine Meckerth S** **4)Ragul H**
(87) International Publication No	: NA	**5)Ricwin.S**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The AI and IoT-Based Smart Warehouse Automated Guided Vehicle (AGV) System is designed to automate material transportation and improve warehouse management through the integration of Artificial Intelligence, Internet of Things, and sensor-based technologies. The system enables autonomous navigation, obstacle detection, intelligent path selection, and real-time vehicle monitoring. It helps transport materials between designated warehouse locations with minimal human intervention while supporting efficient inventory management. The proposed invention aims to reduce manual labour, minimize operational errors, improve workplace safety, optimize transportation time, and increase overall warehouse productivity. This system provides an intelligent, reliable, and cost-effective solution for modern warehouses, manufacturing industries, logistics centers, and distribution facilities.

No. of Pages : 4 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130851 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ADAPTIVE INVENTORY REPLENISHMENT AND RETAIL SUPPLY CHAIN OPTIMIZATION USING ARTIFICIAL INTELLIGENCE AND REAL-TIME OPERATIONAL DATA

(51) International classification	:G06Q 10/08, G06N 20/00, G06Q 10/06, G06Q 10/04, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Rajesh Reddy Ellenki
Address of Applicant :Sr. Staff Software Engineer, Villa 127, Saket Bhustva, Gundlapochampally, Medchal, Hyderabad, Telangana, PIN-500015 Hyderabad Telangana India
(72)**Name of Inventor :**
1)Rajesh Reddy Ellenki

(57) Abstract :
SYSTEM AND METHOD FOR ADAPTIVE INVENTORY REPLENISHMENT AND RETAIL SUPPLY CHAIN OPTIMIZATION USING ARTIFICIAL INTELLIGENCE AND REAL-TIME OPERATIONAL DATA The present invention relates to a system (100) and method for adaptive inventory replenishment and retail supply chain optimization. The system (100) comprises inventory data sources (102), a data acquisition and synchronization module (104), an inventory state processing module (106), an artificial intelligence-based demand prediction engine (108), a replenishment optimization engine (110), a supply chain constraint evaluation module (112), a replenishment execution controller (114), a feedback and adaptive learning module (116), and a communication interface (118). The system (100) synchronizes inventory transactions, determines product-location inventory states, predicts future demand, and dynamically calculates replenishment quantities under operational constraints. Electronic replenishment instructions are transmitted to connected supply chain systems. Operational feedback enables adaptive updating of prediction and replenishment parameters, thereby providing closed-loop inventory control across distributed retail facilities. Figure of abstract: FIG. 1.

No. of Pages : 29 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130854 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR OPTIMIZING SECOND-LIFE BATTERY PACK COMPOSITION

(51) International classification	:H01M 10/42, H02J 7/00, G06Q 10/08, H01M 10/48, G01R 31/36
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Blubble Private Limited
Address of Applicant :7-A1, 3rd Cross Rd, near VR mall, Singayyanapalya, Krishnarajapuram, Bengaluru-560048, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Vignesh S
2)PIDAPA, Sudhindra Reddy

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR OPTIMIZING SECOND-LIFE BATTERY PACK COMPOSITION A system (100) for assigning second-life cells (110) to battery packs (138) includes a processor (102) and a memory (104) holding an inventory database (106) of per-cell records (108), a pack specification interface (114), a constraint enforcement module (118), an optimization engine (124), and a work order generation module (132). The constraint enforcement module (118) applies a mechanism compatibility matrix (120) to each cell's degradation mechanism class, passing compatible cells (110) as a feasible cell pool (122) to a greedy assignment unit (126), which minimizes within-pack variance of Warburg coefficient and capacity. A service duration estimator (128) computes projected service duration and, below a quality threshold, invokes an integer programming refinement unit (130). The work order generation module (132) outputs a work order (134) to a pack assembly station (136). (FIG. 1)

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130860 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : An Intelligent System and Method for Automated Helmet Detection, Safety Access Control and Accident Emergency Notification

(51) International classification	:A42B 3/04, G07C 9/00, A42B 3/30, G08G 1/16, G08G 1/017	(71)**Name of Applicant :** **1)VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY** Address of Applicant :VNR Vignana Jyothi Institute of Engineering & Technology Vignana Jyothi Nagar, Bachupally, Hyderabad – 500 118, Telangana, India. Hyderabad Telangana India
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Singamaneni Krishnapriya**
(33) Name of priority country	:NA	**2)Dr.K.Vijay Chandra**
(86) International Application No	:	**3)B.V.Seshu Kumari**
Filing Date	:01/01/1900	**4)Giriprasad Ambati**
(87) International Publication No	: NA	**5)Dr.L.Dharma Teja**
(61) Patent of Addition to Application Number	:NA	**6)P.Veeranjaneyulu**
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :
Road accidents involving two-wheeler riders continue to claim lives that could have been saved with something as simple and accessible as a helmet, and yet the challenge of ensuring consistent helmet usage has never been solved in a way that is reliable, scalable, and free from human error. This invention addresses that challenge directly by introducing a smart helmet detection and safety access control system that uses a combination of computer vision, artificial intelligence, embedded systems, and sensor technology to create a complete safety solution that works both inside a campus environment and out on the open road. At the entry gate, an ESP32-CAM module captures a real-time image of every incoming rider and passes it to a trained machine learning model that determines within seconds whether a helmet is being worn. Based on that result, an automated gate control mechanism either opens to allow entry or stays closed and displays a warning until the rider complies. Inside the campus, continuous monitoring ensures that riders who remove their helmets after crossing the gate are immediately identified and alerted, closing a behavioral gap that every manual system has historically failed to address. Running alongside this, a parallel accident detection system uses an accelerometer sensor to monitor rider motion at all times, and the moment a sudden impact is detected, the ESP32 microcontroller automatically sends an emergency alert to pre-saved contacts, ensuring that help reaches the rider as quickly as possible even if they are unable to call for assistance themselves. The system operates entirely on its own, requires no manual supervision, and delivers a level of consistency and reliability that no human-dependent process can match. Designed to be affordable, compact, and deployable across a wide range of environmentsfrom educational institutions to corporate campuses and public road checkpoints, this invention represents a meaningful and practical step forward in the effort to make two-wheeler riding safer for everyone.

No. of Pages : 11 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130861 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXTRUSION NOZZLE FOR OPTIMISING INTERLAYER ADHESION IN ADDITIVE MANUFACTURING AND METHOD THEREOF

(51) International classification	:B29C 64/209, B29C 64/118, B29C 48/30, B29C 64/112, B29C 67/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TVASTA MANUFACTURING SOLUTIONS PRIVATE LIMITED
Address of Applicant :OLD NO.345, NEW 172, BALA COMPLEX, RAJIV GANDHI SALAI, OMR, SHOLINGANALLUR, CHENNAI, TAMIL NADU, INDIA 600119 SHOLINGANALLUR Tamil Nadu India
(72)**Name of Inventor :**
1)BASIREDDY PARIVARTHAN REDDY
2)PRAGADEESWAR AP
3)S MOHAMED YOUSUF MATHAR

(57) Abstract :
The present invention discloses an extrusion nozzle for optimising interlayer adhesion in additive manufacturing and a method thereof. The extrusion nozzle (300) includes a nozzle body (302) defining an internal passage from an inlet end (304) to an outlet end (306), and a nozzle outlet (308) defining an outlet plane and an inner bore wall (310) for discharging printable material as an extrusion bead. Serrated teeth (312) are circumferentially distributed about the outlet end (306), each tooth extending axially beyond the outlet plane and positioned radially outward from the inner bore wall (310). A tooth-to-outlet geometry engages a trailing region of an extrusion bulge while maintaining clearance from leading and side regions, forming surface features (314) in an upper surface of the extrusion bead. The surface features (314) on the upper surface of a previously deposited bead interlock with a bottom surface of the extrusion bead, forming a mechanical interlock. FIG. 3A

No. of Pages : 47 No. of Claims : 17

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130862 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : FASTGEO: EFFICIENT GEOMETRIC RANGE QUERIES ON ENCRYPTED SPATIAL DATA

(51) International classification	:G06F 21/62, G06F 16/29, H04L 29/06, G06F 17/30, H04L 29/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Karpagam Academy of Higher Education
Address of Applicant :Pollachi Main Road, Eachanari Post Coimbatore Tamil Nadu India Tamil Nadu India
2)Karpagam Institute of Technology
3)P.Tamilselvan
4)Suriyakumar V
(72)**Name of Inventor :**
1)P.Tamilselvan
2)Suriyakumar V

(57) Abstract :
Abstract The rapid growth of location-aware technologies and geographic information systems has led to the generation of massive volumes of spatial data that are increasingly outsourced to public cloud infrastructures for scalable storage and processing. While cloud platforms provide significant economic and operational advantages, outsourcing sensitive geographic data introduces serious privacy and security concerns. Spatial datasets often contain highly sensitive information related to individual mobility patterns, critical infrastructure locations, or medical and epidemiological records. Conventional encryption techniques protect data confidentiality but prevent efficient query processing, forcing data owners to trade off between privacy and usability. In particular, executing geometric range queries—one of the most fundamental operations in spatial databases—over encrypted data remains a major technical challenge. This work presents FASTGEO, an efficient and privacy-preserving framework for executing geometric range queries on encrypted spatial data stored on untrusted cloud servers. FASTGEO is built upon a statistically searchable encryption model that allows an honest-but-curious server to process spatial queries without learning the plaintext data points, query regions, or their underlying geometric relationships. The framework is designed to minimize information leakage while maintaining practical performance suitable for real-world deployment. FASTGEO introduces a novel two-level encrypted spatial indexing structure. The first level performs coarse-grained encrypted spatial partitioning, enabling the server to rapidly eliminate large regions of irrelevant data during query execution. The second level applies fine-grained encrypted filtering to accurately determine whether individual encrypted data points lie within an arbitrary geometric query region. This hierarchical design significantly reduces computation and communication overhead, allowing FASTGEO to achieve sublinear search complexity over large-scale encrypted spatial datasets. Unlike many existing secure spatial query techniques that are limited to simple rectangular queries, FASTGEO supports arbitrary geometric range queries, including circular and polygonal regions, without revealing the shape, size, or location of the query. Secure query tokens generated by authorized clients enable precise matching against the encrypted index while preventing the server from inferring spatial distributions or distances. In addition, the framework supports dynamic updates over encrypted spatial data, allowing data points to be inserted, deleted, or modified efficiently without requiring complete re-encryption or index reconstruction. FASTGEO provides provable security guarantees under a statistically searchable encryption model, ensuring that the information exposed to the server during query processing is strictly bounded. Extensive experimental evaluation using real-world geographic datasets on an internet-based cloud platform demonstrates that FASTGEO achieves more than a 100× improvement in query execution time compared to existing secure spatial query approaches. By combining strong privacy protection, expressive query support, and high efficiency, FASTGEO offers a practical and scalable solution for secure spatial data management in cloud environments, enabling privacy-sensitive applications such as location-based services, smart cities, and medical geospatial analytics.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130864 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE WEARABLE SYSTEM AND METHOD FOR PROVIDING WILDLIFE SAFETY AND EMERGENCY RESPONSE

(51) International classification	:G08B 21/02, G08B 25/01, H04W 4/90, G08B 25/00, G08B 21/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)TANMAYI. M. K.
Address of Applicant :NO. 477, 3RD CROSS ROYAL COUNTY, JP NAGAR 8TH PHASE, BENGALURU, KARNATAKA, INDIA 560083 BENGALURU Karnataka India
2)KUMARASWAMY. M. S.
(72)**Name of Inventor :**
1)TANMAYI. M. K.

(57) Abstract :
An adaptive wearable system (100) for providing wildlife safety and emergency response is disclosed. The system comprises wearable structures (102) having a data acquisition unit (104) configured to acquire image data and multimodal data. A wildlife detection subsystem (204) detects wildlife and identifies species, while a wildlife analysis subsystem (206) performs confidence-aware multimodal confirmation, determines wildlife characteristics, and assesses threat levels using a Collaborative Intelligent Threat Index (CITI). A user-condition subsystem (208) detects abnormal movement conditions, and an emergency response subsystem (210) communicates alerts to designated recipients (132) independently of deterrent activation. A species-specific deterrent profile subsystem (212) and a deterrent control subsystem (214) determine species-adaptive deterrent strategies subject to electrical safety controls. A collaborative intelligent threat index subsystem (216) calculates species-weighted CITI values, shares contextual information among wearable structures, and generates individualized evacuation directions, safe movement corridors, or regroup points using user positions, wildlife movement, and historical geo-tagged encounters. FIG. 2

No. of Pages : 63 No. of Claims : 23

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130881 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A HYBRID MACHINE LEARNING APPROACH FOR MALICIOUS URL DETECTION USING XGBOOST AND BPNN

(51) International classification	:H04L 29/06, H04L 9/40, G06F 21/55, G06F 21/56, G06F 16/955	(71)**Name of Applicant :** **1)Karpagam Academy of Higher Education** Address of Applicant :Pollachi Main Road, Eachanari Post Coimbatore Tamil Nadu India Tamil Nadu India **2)Karpagam Institute of Technology** **3)Ms.Archana** **4)Saranya E** (72)**Name of Inventor :** **1)Ms.Archana** **2)Saranya E**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The rapid expansion of internet-based services has led to a significant rise in cyber threats, particularly phishing attacks that exploit malicious URLs to impersonate legitimate websites. These attacks aim to deceive users into revealing sensitive information such as login credentials, financial details, and personal data, posing severe risks to individuals and organizations. Traditional phishing detection methods, including blacklist-based and rule-based approaches, are increasingly ineffective against dynamically generated and previously unseen malicious URLs. As a result, there is a critical need for intelligent and adaptive detection mechanisms capable of identifying evolving phishing techniques in real time. This work presents a hybrid machine learning approach for malicious URL detection that integrates Extreme Gradient Boosting (XGBoost) and Back Propagation Neural Networks (BPNN) to improve detection accuracy and robustness. The proposed system analyzes both handcrafted data mining features and character-level embeddings extracted from URL strings, enabling the model to capture fine-grained character relationships as well as broader structural patterns commonly associated with phishing URLs. The dataset used for training and evaluation is organized in a structured comma-separated values (CSV) format, allowing efficient preprocessing and scalable experimentation. In the proposed framework, URLs are first preprocessed to normalize textual patterns and remove irrelevant symbols. A comprehensive feature extraction process is then performed, generating lexical features such as URL length, special character frequency, and token distribution, along with statistical attributes and character embedding representations. XGBoost is employed as an ensemble learning model to perform feature selection and preliminary classification by identifying the most discriminative features and handling non-linear relationships within the dataset. The selected features are subsequently passed to a Back Propagation Neural Network, which further refines the classification process by learning complex interactions through iterative weight updates and error minimization. To enhance predictive performance, the outputs of the XGBoost and BPNN models are combined using a hybrid decision strategy, resulting in improved generalization and reduced false positive rates. This integrated approach leverages the interpretability and efficiency of ensemble learning while benefiting from the deep pattern recognition capabilities of neural networks. Experimental evaluation demonstrates that the proposed hybrid model achieves superior performance compared to standalone machine learning classifiers in terms of accuracy, precision, recall, and F1-score. The proposed malicious URL detection system is suitable for deployment in real-time cyber security environments, including web browsers, enterprise security gateways, financial transaction systems, and cloud-based threat monitoring platforms. By combining character-level analysis with advanced machine learning techniques, the presented approach provides an effective, scalable, and adaptive solution for combating phishing attacks and enhancing web security in modern digital ecosystems.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130919 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A STATE-OF-CHARGE-ADAPTIVE SLIP-CONTROLLED BRAKING SYSTEM FOR LOCOMOTORY DEVICES

(51) International classification	:B60L 15/20, B60L 11/18, B60L 7/14, B60L 50/16, B60W 10/08	(71)**Name of Applicant :** **1)Indian Institute of Technology Madras (IIT Madras)** Address of Applicant :The Dean, Industrial Consultancy & Sponsored Research [IC&SR], Indian Institute of Technology Madras, Sardar Patel Road, IIT P.O, Chennai – 600 036, Tamil Nadu, India. Chennai Tamil Nadu India (72)**Name of Inventor :** **1)Srikanthan Sridharan** **2)Nikhil Suresh Chavan**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT A STATE-OF-CHARGE-ADAPTIVE SLIP-CONTROLLED BRAKING SYSTEM FOR LOCOMOTORY DEVICES A state-of-charge (SoC)-adaptive slip-controlled braking system for locomotory devices is disclosed. The system comprises: a battery pack (1); a voltage source inverter (2); a three-phase induction motor (3); a transmitting member (4); a plurality of wheels (5), and a controller. The controller determines the battery SoC and selects a target negative operating slip based on the determined SoC. The voltage source inverter (2) determines an excitation frequency corresponding to the selected target negative operating slip and regulates an excitation voltage to establish a target air-gap flux, thereby controlling braking energy distribution between regenerative energy returned to the battery pack (1) and internal losses within the three-phase induction motor (3). The controller employs a pre-characterized lookup table or control map for slip selection and scalar voltage-to-frequency (V/f) control, optionally using a proportional-integral-derivative (PID) control loop to track a reference speed profile. The internal losses comprise copper losses and core losses, enabling adaptive braking energy management without requiring an external braking resistor bank. FIG. 1

No. of Pages : 30 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130920 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR RECOGNIZING AND EXPRESSIVELY RENDERING MRIDANGAM STROKES WITH SHRUTI ADAPTATION

(51) International classification	:G06K 9/00, G06K 9/62, G10L 15/22, G06V 10/764, G10L 25/51
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MUSICLOUD STUDIO & TECHNOLOGY
Address of Applicant :1484, PURANDARAVITTALA, 21ST CROSS, 11TH MAIN, B BLOCK, SAHAKARANAGAR, BENGALURU, BENGALURU (URBAN), KARNATAKA - 560092, INDIA BENGALURU (URBAN) Karnataka India
(72)**Name of Inventor :**
1)RAMANUJAN MK

(57) Abstract :
The present invention discloses a system (100) and method (800) for recognizing and expressively rendering mridangam strokes with shruti adaptation. A sensor acquisition subsystem (110) acquires and calibrates signals through a plurality of channels disposed beneath a playing surface, grouping timestamped samples into a synchronized performance-event record. A stroke recognition subsystem (112) extracts features from the record and classifies a Mridangam stroke class. A shruti adaptation subsystem (114) receives a target shruti, determined from a manual selection or a shruti detection operation performed on an audio input, and generates a shruti-adapted stroke. An expression modelling subsystem (116) generates and applies an expression parameter to produce an expression-modulated stroke. A sound generation subsystem (118) renders the expression-modulated stroke into a percussion sound output preserving a traditional playing technique of the Mridangam. FIG. 1

No. of Pages : 48 No. of Claims : 16

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130922 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Dynamic Latch Comparator with Dual-Tail and Adaptive Tail Current Control

(51) International classification	:H03M 1/12, H03M 1/38, H03M 1/46, H03M 1/36, H03K 5/24	(71)**Name of Applicant :** **1)K. Gavaskar** Address of Applicant :Assistant Professor III, ECE Department, Kumaraguru College of Technology, Coimbatore, TamilNadu, India – 641049. Coimbatore Tamil Nadu India **2)S. N. Shivappriya** **3)N. Prakash** **4)S. Prasanna Kumaar** **5)K. P. Thilagavathy** **6)V. Ishwerya**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :** **1)K. Gavaskar** **2)S. N. Shivappriya** **3)N. Prakash** **4)S. Prasanna Kumaar** **5)K. P. Thilagavathy** **6)V. Ishwerya**
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a low-power dynamic latch comparator with reduced kickback noise for high-speed and low-power VLSI applications. The comparator is intended for use in Analog-to-Digital Converters (ADCs), particularly SAR ADCs, where power consumption, high comparison speed and accurate operation low are important. The proposed architecture is based on a Strong-ARM dynamic latch comparator implemented using 90 nm CMOS technology. The proposed comparator incorporates a dual-tail architecture, a preamplifier stage, adaptive tail current control and sampling switches. The dual-tail structure provides separate current paths for the input and latch stages, while the preamplifier amplifies small differential input signals before the regeneration process. The sampling switches provide isolation between the input stage and the regenerative latch during switching operation. The adaptive tail current technique controls the current according to the operating condition. A lower current level can be used for low-power operation, while additional current can be provided during high-speed operation to improve latch regeneration. The combination of these techniques helps reduce the effect of latch switching on the input signal and improves the overall comparator response. The proposed comparator is designed and simulated using the Cadence Virtuoso Design Environment. The reported simulation results show that the kickback noise is reduced from 10 mV to 5 mV and the propagation delay is reduced from 127.22 ns to 90.45 ns compared with the base-paper comparator. The layout is implemented in 90 nm CMOS technology and verified using Design Rule Check. The proposed architecture is suitable for high-speed ADCs, biomedical systems, portable electronics and other low-power mixed-signal VLSI applications

No. of Pages : 8 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130925 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN ADAPTIVE NUTRIENT AND WATER MANAGEMENT SYSTEM FOR HYDROPONIC CULTIVATION USING PREDICTIVE SENSOR FUSION

(51) International classification	:A01G 31/02, G06N 20/00, G06Q 50/02, A01G 31/06, A01G 31/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Krishnakumar A
Address of Applicant :Assistant Professor, Department of Computer Science, Kamalam College of Arts and Science, Anthiyur-642122, Tamil Nadu, India Tamil Nadu India
2)Dr. Kavipriya S
3)Dr. D. Vimal Kumar
4)Dr. M. Savitha
5)Nandhini N
6)Dr. Puneeta Singh
(72)**Name of Inventor :**
1)Dr. Krishnakumar A
2)Dr. Kavipriya S
3)Dr. D. Vimal Kumar
4)Dr. M. Savitha
5)Nandhini N
6)Dr. Puneeta Singh

(57) Abstract :
The present invention discloses an AI-driven adaptive nutrient and water management system (100) for hydroponic cultivation using predictive sensor fusion. The system (100) comprises a sensor fusion module (110) integrating pH, electrical conductivity, temperature, dissolved oxygen, and water level sensors to acquire real-time nutrient solution parameters. A predictive analytics engine (120) employing machine learning models processes fused sensor data to forecast nutrient depletion trends and water consumption patterns. A nutrient dosing module (130) executes precision dispensing of nutrients based on predictive outputs, while a water management module (140) regulates water supply and recirculation. A central processing unit (150) orchestrates communication between all modules, and a user interface module (160) provides real-time monitoring and control. The system adaptively adjusts nutrient concentrations and water levels in a hydroponic growing unit (170) to optimize plant growth while minimizing resource wastage.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130930 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Intelligent Power Hub for EV Charging: An AI-Enabled Asymmetric Hybrid Modular Multilevel Converter

(51) International classification	:H02J1/102, B60L53/00, B60L53/60, H02J7/02, H02J105/37
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)A PANDIAN
Address of Applicant :Department of EEE, Koneru Lakshmaiah Education Foundation, Vaddeswaram, Guntur, Andhra Pradesh -INDIA Andhra Pradesh India
2)GUDISE HARITHA
3)Koneru Lakshmaiah Education Foundation
4)P. Srinivasa Varma
(72)**Name of Inventor :**
1)A PANDIAN
2)GUDISE HARITHA

(57) Abstract :
An AI-enabled four-port electric vehicle charging system based on an Asymmetric Hybrid Modular Multilevel Converter (AHMMC) for efficient and reliable grid-to-vehicle power conversion. The AHMMC operates as the front-end AC–DC converter, interfacing the three-phase utility supply with a regulated DC link, while independent DC–DC converter stages supply four EV charging ports simultaneously. A dynamic power allocation strategy is employed to distribute the available charging power among the connected EVs according to their individual power requirements and operating conditions. The system further incorporates an AI-assisted digital twin for real-time state estimation, capacitor health assessment, remaining useful life prediction, and semiconductor fault detection. The proposed architecture offers high conversion efficiency, low harmonic distortion, modular scalability, bidirectional power-flow capability, and improved fault tolerance. Hence, the developed system provides an intelligent, scalable, and condition-aware solution for future multi-port EV fast-charging infrastructure.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130932 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADOPTION OF ARTIFICIAL INTELLIGENCE AND ITS IMPACT ON JOB SATISFACTION, STRESS, AND WORK-LIFE BALANCE AMONG PRIVATE ENGINEERING FACULTY

(51) International classification	:G06Q 10/06, G06Q 50/20, G06N 20/00, G06Q 10/04, G06Q 50/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Ms. Ummehani
2)Dr. Geetha Manoharan

(57) Abstract :
The present invention relates to an artificial intelligence-based faculty wellbeing assessment and predictive management system for private engineering institutions. The system integrates AI adoption measurement, job satisfaction evaluation, occupational stress prediction, and work-life balance assessment through intelligent data processing and machine learning techniques. It collects faculty-related information concerning AI utilization, teaching workload, technological competency, institutional support, and professional responsibilities. An analytical engine generates multidimensional wellbeing indices and predicts potential workplace challenges. An adaptive recommendation module provides suggestions for workload optimization, AI training, institutional support, and digital working policies. The invention promotes responsible AI adoption and sustainable faculty wellbeing.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130933 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ARTIFICIAL INTELLIGENCE DRIVEN CHANGES IN ACADEMIA AMONG ENGINEERING EDUCATORS FOR BETTER GLOBAL EXPOSURE

(51) International classification	:G06Q 50/20, G06N 20/00, G06Q 10/06, G06Q 10/10, G06F 16/332
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Ms. Ummehani
2)Dr. Geetha Manoharan

(57) Abstract :
The present invention relates to an Artificial Intelligence-driven academic transformation system for enhancing global exposure among engineering educators. The system integrates intelligent academic profiling, machine learning-based competency assessment, international opportunity recommendation, research collaboration matching, and AI-assisted teaching innovation. Natural language processing identifies research interests and supports academic knowledge exchange, while predictive analytics generates personalized professional development pathways. The invention further incorporates continuous feedback, academic performance monitoring, and intelligent recommendation mechanisms to evaluate international participation, research engagement, and teaching transformation. The proposed system provides an integrated technological framework for strengthening engineering educators' global academic visibility, professional competencies, and international research collaboration.

No. of Pages : 20 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130934 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI INTEGRATED SMART ENERGY MANAGEMENT AND ADAPTIVE CONTROL SYSTEM FOR HIGH-EFFICIENCY ELECTRIC DRIVES IN INDUSTRIAL APPLICATIONS

(51) International classification	:G05B 13/04, G06N 20/00, G05B 23/02, G05B 13/02, H02P 29/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Dr. K Ayyar
2)Dr. Geetha Manoharan

(57) Abstract :
The present invention relates to an AI integrated smart energy management and adaptive control system for high-efficiency electric drives in industrial applications. The system comprises intelligent sensors, a data acquisition unit, an artificial intelligence processing engine, an energy optimization module, an adaptive motor controller, a predictive maintenance subsystem, and a centralized monitoring dashboard. The system collects electrical, mechanical, and environmental parameters and processes the information using machine learning algorithms to predict energy requirements, evaluate motor performance, and identify abnormal operating conditions. An adaptive control mechanism dynamically regulates motor speed, torque, voltage, current, and operating frequency according to industrial workload requirements. The energy optimization module coordinates electrical consumption while maintaining predefined operating constraints. The predictive maintenance subsystem identifies potential equipment deterioration and generates maintenance alerts. The invention provides integrated intelligent monitoring, adaptive control, energy optimization, and equipment reliability management for improving industrial electric drive efficiency and supporting sustainable manufacturing operations.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130935 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIONIC HAND SYSTEM WITH BIOMETRIC TENDON-LIGAMENT ACTUATION AND A METHOD THEREOF

(51) International classification	:B25J 15/00, A61F 2/58, A61F 2/68, B25J 15/10, A61F 2/54
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM University- AP
Address of Applicant :Neerukonda, Mangalagiri Mandal Guntur District-522240, Andhra Pradesh, India Guntur Andhra Pradesh India
(72)**Name of Inventor :**
1)GELEJA, Seth Job
2)Dr. Anuj Pradeep Deshpande

(57) Abstract :
A bionic hand system (100) features a plurality of finger assemblies (102) with phalanx segments (104) connected in series by pivot joints (106). A dual-action movement system utilizes inextensible tendon strings (110) routed along each finger to transmit pulling forces for active flexion, paired with elastic ligament strings (114) that connect adjacent segments to store potential energy during flexion and release the stored energy to bias the phalanx segments toward an extended position. Motion is driven by independently controllable stepper motors (116) housed within an arm enclosure (118), utilizing pulleys (120) mounted on motor shafts to convert rotational movement into linear tendon displacement. Additionally, a thumb assembly (122) integrates a servo motor (124) for carpometacarpal joint positioning and a stepper motor (116) for phalangeal flexion, enabling complex thumb opposition. A central microcontroller (126) coordinates all actuation via dedicated motor driver modules (128). (FIG. 6)

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130937 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI DRIVEN OPTIMIZATION SYSTEM ON CONTROL AND ENERGY FOR ELECTRIC DRIVE SYSTEMS IN SUSTAINABLE INDUSTRIAL APPLICATIONS

(51) International classification	:G05B 13/04, G06N 20/00, G05B 23/02, H02J 13/00, G05B 13/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :AnanthaSagar (V), Hasanparthy (M),Warangal, Telangana - 506371 Telangana India
(72)**Name of Inventor :**
1)Dr. K Ayyar
2)Dr. Geetha Manoharan

(57) Abstract :
The present invention relates to an AI-driven optimization system for control and energy management of electric drive systems in sustainable industrial applications. The system integrates electric motors, power electronic converters, sensor networks, data acquisition modules, machine learning algorithms, adaptive controllers, and intelligent energy management mechanisms. Real-time operational parameters are analyzed to predict motor operating conditions, optimize speed and torque regulation, minimize unnecessary energy consumption, and identify equipment abnormalities. The system further incorporates predictive maintenance, energy performance evaluation, and sustainability monitoring through an integrated supervisory platform. The invention improves electric drive efficiency, operational reliability, and energy utilization in modern industrial environments.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130939 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR ADAS FUNCTIONING BASED ON REAL-TIME FRONT SCENARIO

(51) International classification	:G06K 9/00, G06V 20/56, G06V 20/58, G06V 10/764, B60W 30/095
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VELLORE INSTITUTE OF TECHNOLOGY, CHENNAI
Address of Applicant :Vandalur - Kelambakkam Road, Chennai, Tamil Nadu - 600127, India. Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)G. HANNAH GRACE
2)T. SAM DAVIS
3)SYEDHU HASEEN

(57) Abstract :
An advanced driver assistance system 100 for a vehicle 102 includes a forward facing camera 104; a controller 108 including a database, one or more processors to perform operations to receive real-time video frames 106 of the road; pre-process video 106 and produce bounding boxes 406, class labels, and confidence scores using selected object detection model; computes a class-weighted count for road objects 404 from cumulative bounding-box 406 area relative to frame area 408; computes dense optical flow between consecutive frames 106 and extracts angular and magnitude variance; apply machine learning models to derive a lane-confidence score; computes a unified complexity score 208 to compare complexity score 208 for the current mode to lock the active model and unloads or pauses the prior model to activate target model; normalize detections from the currently active model to the standardized output to log the transition event and publish downstream ADAS modules.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130951 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Quizora –Intelligent Quiz Creation and Assessment System

(51) International classification	:G09B 7/02, G09B 7/04, G09B 7/00, G06F 17/27, G06F 17/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr RVR NRI Institute of Technology Deemed to be University.
Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India Vijayawada Andhra Pradesh India
2)Dr.Chaitanya Kishore Reddy Maddireddy
3)D.Charishma
4)K.Keerthana
5)K.Venkata Divya
6)K.Vamsi Swaroop
7)D.Lokesh
(72)**Name of Inventor :**
1)Dr.Chaitanya Kishore Reddy Maddireddy
2)D.Charishma
3)K.Keerthana
4)K.Venkata Divya
5)K.Vamsi Swaroop
6)D.Lokesh

(57) Abstract :
ABSTRACT [0028] An intelligent quiz generation system is disclosed for creating assessments automatically from study content. The system validates and preprocesses content supplied by an instructor, ranks its sentences, and extracts keywords and named entities as concepts. A fine-tuned transformer-based sequence-to-sequence model generates multiple-choice, true/false and fill-in-the-blank questions from each concept and its source sentence, and low-quality candidates are discarded. Distractors that are related to but not synonymous with the correct answer are generated for multiple-choice questions, and each question is classified as easy, medium or hard. A quiz satisfying the requested question count and difficulty distribution is assembled and presented to the instructor for review, editing and export, while generation data and edits are logged for refining the models. The format-agnostic, web-based workflow reduces authoring time and effort and provides consistent, difficulty-calibrated quizzes for classrooms, online courses and self-paced learning. This project presents Quizora, an AI-driven web application built with Python, spaCy/NLTK, the Hugging Face Transformers library and a Flask-based interface. Traditional quiz authoring is manual, subjective in its difficulty and does not scale with large classrooms and online learning. Quizora connects concept extraction, multi-format question generation, distractor generation, difficulty classification, constrained quiz assembly and instructor review in a single platform, so that high-quality quizzes can be produced from study material within minutes.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130955 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING-BASED FAKE NEWS DETECTION AND MISINFORMATION ANALYSIS USING NATURAL LANGUAGE PROCESSING

(51) International classification	:G06N 20/00, G06F 17/27, G06F 16/951, G06F 17/30, G06F 16/906
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY

Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India. Vijayawada Andhra Pradesh India

2)A. Ravi Kiran

3)Yeluri Sivakasi Reddy

4)Ravinuthala Rojesh

5)Tommandaru Sandeep

6)Valiveti Tejaswi

7)Mallarapu Dhanush

8)Pragada Sri Sai Varshitha

(72)**Name of Inventor :**

1)A. Ravi Kiran

2)Yeluri Sivakasi Reddy

3)Ravinuthala Rojesh

4)Tommandaru Sandeep

5)Valiveti Tejaswi

6)Mallarapu Dhanush

7)Pragada Sri Sai Varshitha

(57) Abstract :

ABSTRACT [0030] A machine learning and natural language processing based system is disclosed for detecting fake news and analyzing misinformation in online information. The system accepts news text, a claim or a news URL, validates and preprocesses the input, and identifies the main factual claim and its keywords. Evidence is retrieved from fact-checking databases and trusted sources, the reliability of the sources is evaluated, and the claim is compared with the evidence to determine supporting and contradicting evidence. A credibility score is calculated from the supporting evidence, contradicting evidence, source reliability and claim similarity, and the information is classified as True, False or Suspicious/Unverified. The verdict, credibility score, reliability level, reasons and evidence are presented through a web interface, giving users a real-time and evidence-based method of verifying online information. This project presents a machine learning-based fake news detection and misinformation analysis system built using natural language processing and a web-based architecture with HTML, CSS, JavaScript, Node.js and Express.js. Traditional fact-checking requires users to search several websites manually, and dataset-based fake news detection cannot always verify new claims. The proposed system connects claim identification, evidence retrieval, trusted source verification, credibility scoring and result explanation in a single platform, so that users can understand the reliability of online information and make informed decisions before sharing it.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130961 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR PREDICTIVE RESOURCE PROVISIONING FOR TRANSACTION-PROCESSING SERVICES

(51) International classification	:H04L 12/24, G06F 9/50, H04L 12/911, H04L 12/26, H04L 29/08	(71)**Name of Applicant :** **1)Dr. Andrews Thomas** Address of Applicant :Bishop Abraham Memorial College, Thuruthicad, Mallappally, Pathanamthitta District, Kerala – 689597 Mallappally Kerala India **2)Sreeresh D.** **3)Dr. Biju Thomas** **4)Divya Kochukoshy** **5)Sajan P. Sam**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Dr. Andrews Thomas**
(33) Name of priority country	:NA	**2)Sreeresh D.**
(86) International Application No Filing Date	: :01/01/1900	**3)Dr. Biju Thomas** **4)Divya Kochukoshy** **5)Sajan P. Sam**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A system (100) for predictive provisioning of computing resources includes a memory (102) and one or more processors (104) implementing a signal interface (106), an edge scoring node (112), a demand forecaster (118), a capacity controller (120), and a resource-orchestrator interface (126). The signal interface (106) receives live session interaction data (108), including server-observed session events (130), over an out-of-band signal path ahead of a transactional request path (110). The edge scoring node (112) computes transactional intent scores for active user sessions (114), and the demand forecaster (118) aggregates the scores into a predicted arrival rate. The capacity controller (120) determines target capacity of a backend service (124), and the resource-orchestrator interface (126) commands a resource orchestrator (128) to activate or allocate already-staged capacity before predicted requests arrive, reducing provisioning delay and improving networked computing infrastructure operation.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130962 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ACCIDENT PREDICTION AND PREVENTION SYSTEM WITH AI AND IoT FOR VEHICLES ON MOUNTAIN ROADS

(51) International classification	:B60W 30/095, B60W 50/14, B60W 50/00, G08G 1/16, B60W 30/09
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ASHOK RAJ
Address of Applicant :ASSOCIATE PROFESSOR, DEPARTMENT OF MECHANICAL ENGINEERING, J.J.COLLEGE OF ENGINEERING AND TECHNOLOGY, TIRUCHIRAPPALLI, TAMILNADU,INDIA Tamil Nadu India
2)Mr. S. Prathap
3)Mrs. P. Vinothini
4)Mr.R. Satheesh kumar
5)Dr. T. Jayakumar
6)Mrs.V. Vinothini
7)Mrs. V. Anusuya
8)Mr. V. Sivaganesan
(72)**Name of Inventor :**
1)ASHOK RAJ
2)Mr. S. Prathap
3)Mrs. P. Vinothini
4)Mr.R. Satheesh kumar
5)Dr. T. Jayakumar
6)Mrs.V. Vinothini
7)Mrs. V. Anusuya
8)Mr. V. Sivaganesan

(57) Abstract :
The present invention provides an Artificial Intelligence (AI)- and Internet of Things (IoT)-based accident prediction and prevention system for heavy vehicles driving on mountain roads. A set of vehicle-mounted and road-context sensors provide vehicle speed, inertial motion, load, braking condition, gradient, curvature and environmental information to an onboard edge processor. A sensor-fusion engine generates validated features for a trained machine-learning module which estimates skid risk, rollover risk, collision risk, and risk of degraded brake performance, and determines context dependent safe speed advice. A safety controller can provide bounded speed, torque or braking assistance requests (via compatible and authorized interfaces) as well as graded visual, audible or haptic driver alerts (with safety interlocks). In times of disconnection, the edge processor continues to provide important predictive and warning capabilities, and (optionally) the IoT module reports on an incident or diagnostic information to a remote platform. The system can support earlier recognition of hazards and context-aware preventative responses for heavy vehicles on segments of the winding and steep mountain roads.

No. of Pages : 15 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130966 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR AI-OPTIMIZED POST-QUANTUM SECURITY PROTOCOLS IN 6G COMMUNICATION NETWORKS

(51) International classification	:H04L 9/08, G06N 10/00, H04L 29/06, H04L 9/40, H04L 9/32	(71)**Name of Applicant :** **1)G.Pullaiah College of Engineering and Technology (Autonomous)** Address of Applicant :G. Pullaiah College of Engineering and Technology, Kurnool-518452. Andhra Pradesh Andhra Pradesh India **2)Dr.N.Mahendra** **3)N.Somanna** **4)Dr.P.Suman Prakash** **5)Dr M.Janardhan**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Dr.N.Mahendra**
(86) International Application No	:	**2)N.Somanna**
Filing Date	:01/01/1900	**3)Dr.P.Suman Prakash**
(87) International Publication No	: NA	**4)Dr M.Janardhan**
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

ABSTRACT SYSTEM AND METHOD FOR AI-OPTIMIZED POST-QUANTUM SECURITY PROTOCOLS IN 6G COMMUNICATION NETWORKS A system and method for AI-optimized post-quantum security protocols in 6G communication networks are disclosed. The invention integrates an artificial intelligence security orchestration engine with post-quantum cryptographic mechanisms to provide adaptive protection against quantum-enabled cyber threats. The system includes a network analytics module, threat intelligence module, post-quantum cryptographic repository, policy enforcement controller, and key lifecycle management system. Operational data including network conditions, device capabilities, traffic characteristics, latency requirements, and security indicators are continuously analyzed to determine suitable quantum-resistant cryptographic protocols for individual communication sessions. The system dynamically manages cryptographic selection, key generation, key rotation, and security policy enforcement while maintaining communication efficiency. Hybrid deployment capabilities support gradual migration from conventional cryptography to post-quantum cryptographic infrastructures. The invention is applicable to 6G telecommunications, edge computing, autonomous systems, Internet of Things ecosystems, cloud environments, and intelligent digital communication platforms.

No. of Pages : 19 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641130967 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN REAL-TIME ANOMALY DETECTION AND ALERT SYSTEM

(51) International classification	:G06N 20/00, H04L 12/24, H04L 12/26, G06F 11/30, H04L 29/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)G Usharani
Address of Applicant :Assistant Professor, Department of CSE- AI&ML, GMRIT Deemed to be University, GMR Nagar, Rajam, Srikakulam, Andhra Pradesh Andhra Pradesh India
2)Dr.P.Panimalar
3)Kushagra Shukla
4)Arivumani Samson S
5)Aarthi D
6)M.Pushpalatha
7)Venkatesan S
8)Dr.Abhijit Janardan Patankar
9)Abirami S
(72)**Name of Inventor :**
1)G Usharani
2)Dr.P.Panimalar
3)Kushagra Shukla
4)Arivumani Samson S
5)Aarthi D
6)M.Pushpalatha
7)Venkatesan S
8)Dr.Abhijit Janardan Patankar
9)Abirami S

(57) Abstract :
AI-DRIVEN REAL-TIME ANOMALY DETECTION AND ALERT SYSTEM ABSTRACT This invention relates to an artificial intelligence-based system for detecting abnormal events in real time within operational, industrial, information-technology, network, infrastructure, and other monitored environments. The system receives data streams from one or more heterogeneous sources, including sensors, machines, applications, servers, network devices, transaction platforms, and monitoring databases. A data acquisition module collects and timestamps incoming observations, while a preprocessing module validates, normalizes, filters, and organizes the observations into time-aligned feature records. An intelligent analysis engine evaluates the records using one or more machine-learning, statistical, rule-based, and time-series analysis techniques to identify deviations from learned or configured normal operating patterns. The system may establish baseline behaviour for individual devices, users, processes, locations, or groups and continuously update such baselines as operating conditions change. A context and correlation module combines detected deviations with operating state, historical events, asset information, and related signals to estimate an anomaly score, severity, confidence, and potential impact. When a configurable threshold or decision condition is satisfied, an alert generation module creates a prioritized notification containing the event time, affected source, anomaly category, measured deviation, confidence level, and recommended response. Notifications may be delivered through a dashboard, mobile device, email, messaging service, or an application programming interface. Feedback from authorized users and verified outcomes may be used to refine detection models, thresholds, and response rules. The system further supports event logging, visualization, audit trails, role-based access, and integration with external monitoring or incident-management platforms. By combining continuous data acquisition, adaptive anomaly analysis, contextual correlation, and automated alert delivery, the disclosed system enables earlier identification of abnormal conditions, reduces dependence on manual monitoring, and supports timely operational decisions. The system is configurable for deployment on edge devices, on-premises infrastructure, cloud platforms, or a hybrid environment.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641130996 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR CAPACITY-WEIGHTED PRE-ASSEMBLY STATE-OF-CHARGE BALANCING OF BATTERY CELLS

(51) International classification	:H01M10/42, H01M10/48, H02J7/82, H02J7/52
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Blubble Private Limited
Address of Applicant :7-A1, 3rd Cross Rd, near VR mall, Singayyanapalya, Krishnarajapuram, Bengaluru-560048, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Vignesh S
2)PIDAPA, Sudhindra Reddy

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR CAPACITY-WEIGHTED PRE-ASSEMBLY STATE-OF-CHARGE BALANCING OF BATTERY CELLS A pre-assembly cell balancing system (100) includes a balancing station (102) having a housing (104), charge/discharge ports (106) on the housing (104) each receiving a battery cell (108), an identifier reader (110) at each port (106), a voltage measurement unit (112), a charge controller (114), a data store (118) of per-cell records (120) holding screened capacity and AC internal resistance, and a processing unit (116) connected to these components. The processing unit (116) computes, for each cell (108), a target state-of-charge that is higher for below-average capacity cells and lower for above-average capacity cells, applies an internal-resistance correction, clamps the target within a safe window, and commands the controller (114) to charge or discharge the cell (108) before series assembly into a battery pack (130). A corresponding method (200) is also provided. (FIG. 1)

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131014 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ENVIRONMENT TRIGGERED GENE ACTIVATION SYSTEM FOR PRECISION PLANT CONTROL

(51) International classification	:G05B 13/02, G05B 13/04, A01G 7/00, A01H 5/00, A01G 7/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Supriya Lethakula
2)Dr. Banoth Madhu
3)Dr. Narkhede Gopal Wasudeo

(57) Abstract :
An environment triggered gene activation system for precision plant control, comprises of a one or more environmental sensors 101 configured to continuously acquire environmental data, a data processing unit 102 receives, filter, and pre-process the environmental data, an artificial intelligence (AI) control unit 103 analyse the pre-processed environmental data, determine, based on the analysis, , a stimulus delivery module 104 deliver the at least one stimulus to the plant in accordance with the activation strategy, wherein the at least one stimulus is selected from the group consisting of light, a chemical agent, heat, and combinations thereof, a feedback unit 105 configured to monitor a response from the plant indicative of expression of the at least one target gene, the system operates as a real-time, closed-loop adaptive system such that the AI control unit 103 iteratively optimizes the activation strategy based on the feedback data and subsequent environmental data.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131015 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : QUANTUM-OPTIMIZED HYBRID ENERGY DISPATCHER SYSTEM FOR GRID PLANNING AND REAL-TIME DISPATCH

(51) International classification	:G06Q 50/06, G06Q 10/04, G06Q 10/06, H02J 3/00, G06N 10/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Dr. Sathyavani Bandela

(57) Abstract :
A quantum-optimized hybrid energy dispatcher system for grid planning and real-time dispatch comprising a forecast integration module 101 to ingest real-time and predictive data about renewable generation, demand, and grid conditions, a QUBO encoding engine 102 to translate a grid planning problem into a quantum-friendly optimization formulation, the grid planning problem selected from generation scheduling, energy storage dispatch, transmission routing, and microgrid reconfiguration, a quantum-classical hybrid solver 103 to use quantum computational resources to explore candidate solutions of the formulated optimization problem while using classical computing resources to validate feasibility, resolve constraints, and refine solutions and a real-time dispatch controller 104 to output actionable reconfiguration commands to grid-connected assets to reconfigure the power grid in response to the optimized solution and the grid-connected assets includes at least one of generators, energy storage units, microgrid controllers, and inverters.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131016 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : NEURAL DIAGNOSTIC CAP FOR MONITORING AND DETECTION OF NEURODEGENERATION

(51) International classification	:A61B 5/00, A61B 5/0476, A61B 5/08, A61B 5/11, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Amrutham Nagajyothi
2)Dr. C Madan Kumar

(57) Abstract :
A neural diagnostic cap for monitoring and detection of neurodegeneration, comprising a hemispherical panel 101 to cover a calvaria of a user, a plurality of EEG electrodes 102 to detect a real-time brainwave signal of the user, an impedance monitoring sensor 103 to analyze a contact quality of the EEG electrodes 102, a neurological monitoring module 201 to track an electrical activity of a brain under an environment, a physiological vital parameter monitoring module 202 in an ear region of the hemispherical panel 101, a respiratory monitoring module 203 to track a breathing pattern of the user, an autonomic stress and motion monitoring module 204 to detect an internal behavioral change in the user, and an interactive cognitive assessment module comprising an artificial intelligence (AI) camera 205 to execute facial expression recognition.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131017 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : HYBRID COMPOSITE BATTERY ENCLOSING DEVICE

(51) International classification	:H01M 10/625, H01M 10/613, H01M 10/653, H01M 10/6555, H01M 10/6551	(71)**Name of Applicant :** **1)SR University** Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India (72)**Name of Inventor :** **1)Dr. Pulla Sammaiah**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A hybrid composite battery enclosing device for an electric vehicle, comprises a monolithic body defining a battery receiving cavity configured to house one or more battery modules, the composite material comprises 40–60 wt% PEEK matrix, 30–50 wt% carbon fiber reinforcement, and 5–15 wt% titanium diboride particles,an outer TiB_2-enriched layer configured for enhanced thermal conductivity and corrosion resistance, a carbon fiber-dominant structural core layer configured for high stiffness and impact resistance, an inner thermally resistant barrier layer configured to mitigate propagation of thermal runaway events, at least one thermally conductive pathways formed by dispersion of TiB_2 nanoparticles within the PEEK matrix to distribute heat away from battery modules, a plurality of integrated cooling channels embedded within the body and formed as part of the composite structure, embedded metal inserts for mounting the body to a vehicle frame while maintaining electrical isolation.

No. of Pages : 16 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131018 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SEMANTIC COMMUNICATION SYSTEM FOR TRANSMITTING INFORMATION IN MULTILINGUAL AND MULTICULTURAL DISTRIBUTED ENVIRONMENTS

(51) International classification	:G06F 17/27, G06N 5/02, G06F 17/28, G06F 17/30, G06N 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Bura Vijay Kumar
2)Dr. Bhavana Jamalpur

(57) Abstract :
A semantic communication system for transmitting information in multilingual and multicultural distributed environments, comprises a semantic encoder 101 to extract contextual meaning from input data instead of transmitting raw bits or symbols, a knowledge base 102 storing linguistic, cultural, and contextual information, the knowledge base 102 comprises multilingual language models, cultural interpretation mappings, and domain-specific ontologies, a semantic compressor 103 to remove redundant or non-essential data based on extracted contextual meaning, a communication interface 104 to transmit semantically encoded data over a communication channel, and a semantic decoder 106 to reconstruct intended meaning at a receiver end using a corresponding knowledge base 102, the system reduces bandwidth consumption, minimizes semantic ambiguity, and improves communication efficiency under noisy channel conditions, a user adaptation module to personalize reconstructed output based on user preferences, cultural background, or contextual parameters, a noise adaptation module 105 to compensate for channel distortions using semantic redundancy modeling.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131019 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE DEVICE FOR MONITORING GASTRIC MOTILITY AND PROVIDING GASTROINTESTINAL THERAPY

(51) International classification	:A61N 1/36, A61N 1/05, A61B 5/00, A61N 1/04, A61N 1/372
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Srinivas Rathula
2)Johnson Kolluri

(57) Abstract :
A wearable device for monitoring gastric motility and providing gastrointestinal therapy, comprising a flexible belt 101, a fit management arrangement including inflatable air bladders 102 and at least one miniature pump 103 for inflating/deflating the air bladders 102 to maintain contact pressure, a fit sensing arrangement including strain sensors 104 for detecting belt 101 deformation and contact pressure providing fit feedback, a gastric monitoring sensor suite 105 including an EGG sensor array 105a for measuring gastric electrical activity and an acoustic sensor array 105b for measuring digestion related abdominal sounds, a therapeutic stimulation set including a TES electrode matrix 106 for delivering electrical stimulation, a GSR sensor array 107 and at least one thermoelectric module 108 for applying heating and cooling to an abdominal region, an impedance spectroscopy set including electrodes 109 to measure abdominal impedance, a plurality of haptic actuators 110 to apply controlled pressure pulses.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131020 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE HEALTH RISK MONITORING DEVICE FOR FITNESS GUIDANCE

(51) International classification	:A61B 5/00, A61B 5/0205, A63B 24/00, G16H 40/67, A61B 5/11
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Miriappalli Santoshi
2)Syed Nageena Parveen
3)Shaik Thaherbasha

(57) Abstract :
A wearable health risk monitoring device for fitness guidance, comprising, a wearable body 101 to cover an upper body and legs of a user, a body covering the upper body and legs, the device combines user information and uploaded medical reports to generate a digital health profile and personalized thresholds, a display panel 103 to receive uploaded medical reports, machine learning extracts clinical indicators and supports personalized workout, activity and diet recommendations while restricting exercises associated with identified risks, an alert module provides graded haptic and interface warnings for threshold breaches or excessive activity, vital monitoring module 104 for monitoring physiological data, a therapeutic module delivers regulated heating, cooling , vibration or muscle stimulation in response to detected discomfort, and embedded piezoelectric transducers 106 harvest movement energy to help power the device through a rechargeable battery configured with the wearable body.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131021 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIFUNCTIONAL ANTENNA

(51) International classification	:H01Q 1/38, H01Q 9/04, H01Q 1/48, H01Q 1/24, H01Q 1/36
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Dr. Pulla Sammaiah

(57) Abstract :
A multifunctional antenna, comprises a composite substrate 101 comprising a polyether ether ketone (PEEK) matrix reinforced with carbon fibers and titanium diboride (TiB_2) particles, at least one radiating element 102 disposed on an upper surface of or embedded within the composite substrate 101, the radiating element 102 being electrically conductive to generate electromagnetic radiation in response to an applied radio frequency signal, a ground plane 103 disposed on an opposing surface of composite substrate 101 relative to the radiating element 102, the composite substrate 101 positioned between the radiating element 102 and the ground plane 103 to function as a dielectric medium for electromagnetic wave propagation, a feed line 104 electrically coupled to the radiating element 102 to transmit the RF signal thereto, the composite substrate 101 mechanically supports the radiating element 102 and ground plane 103 while functioning as a dielectric layer enabling controlled impedance matching and radiation performance.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131022 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CLIMATE ADAPTIVE SYSTEM FOR COASTAL AND RIVERINE URBAN SHORES

(51) International classification	:G01W 1/10, G01W 1/00, G01W 1/02, G01C 13/00, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Kande Vamsi Krishna
2)G Shyamala

(57) Abstract :
A climate adaptive system for coastal and riverine urban shores, comprising a structural mounting pole 101 via a plurality of vibration-dampening brackets 102, a weather monitoring module 105 to detect weather conditions of surroundings, a satellite data acquisition module to extract precipitable water vapour from tropospheric signal delays, a hydrological and aquatic behaviour monitoring unit 107 to capture real-time hydrological data comprising water elevation, surface current velocity, and water turbidity to evaluate hydrological stress and upstream water movement, a controller to establish a DEBP (Dynamic Environmental Baseline Profile) by learning location-specific seasonal baseline patterns over an initial 90-day adaptation window, and continuously compute a Composite Climate Risk Score based on weighted multi-modal deviations from the DEBP, a predictive simulation module to simulate infrastructure failure risks and generate a Vulnerability Priority Matrix, a multi-modal validation module executed by the controller.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131023 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : GASTROINTESTINAL ASSESSMENT SYSTEM WITH MULTI-MODAL NON-INVASIVE ABDOMINAL MAPPING

(51) International classification	:A61B 5/00, A61B 5/07, A61B 34/20, A61B 1/04, A61B 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Gadipe Sunitha
2)Dr. Bhavana Jamalpur

(57) Abstract :
A gastrointestinal assessment system with multi-modal non-invasive abdominal mapping, comprising a cuboidal housing 101, an imaging assembly comprising a camera 103 and a LiDAR module 104 to capture a visual information of the user and to obtain a spatial information of an abdominal surface of the user, an articulated robotic arm 105 to extend toward the user related spatial coordinates, a distal sensing assembly 106 to contact the abdominal surface, a body-surface gastric mapping (BSGM) electrode array 107 to acquire a plurality of gastric electrical signals, a plurality of acoustic transducer 108 to capture internal gastrointestinal acoustic signatures, a central processing unit to filter a noise and a motion-related disturbance from the acquired gastric electrical signals and the acoustic signatures, an external device 109 to gather a user symptom information and generate a risk score against a predefined disease pattern.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131024 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Vignantra Ai – An AI Academic Assistance Platform for Engineering Students

(51) International classification	:G06Q 50/20, G06Q 10/10, G09B 7/00, G06Q 10/06, G09B 5/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR RVR NRI INSTITUTE OF TECHNOLOGY (DEEMED TO BE UNIVERSITY)
Address of Applicant :DR RVR NRI INSTITUTE OF TECHNOLOGY (DEEMED TO BE UNIVERSITY), Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India Vijayawada Andhra Pradesh India
2)Ch. Naga Santha Kumari
3)Bh. Trisha Sri
4)G. Ramana
5)D. Keerthana
6)D. Badrinath
7)M. Lokesh
(72)**Name of Inventor :**
1)Ch. Naga Santha Kumari
2)Bh. Trisha Sri
3)G. Ramana
4)D. Keerthana
5)D. Badrinath
6)M. Lokesh

(57) Abstract :
ABSTRACT [0023] A context-aware academic assistance platform for engineering students is disclosed. A student is authenticated by a signed token, and a server validates each request and selects one of a plurality of academic modules, such as chat, assignment, examination planning, project guidance, code debugging, documentation, presentation, research, résumé and study planning. A prompt builder concatenates the module template, stored student context comprising branch, year, subject and goal, an output-format constraint and the student's task, and an AI service layer submits the prompt to a generative language model. The response is displayed in the module and the academic activity is recorded against the student. A real-time channel displays the number of connected students, and an error-handling layer and an in-memory database fall back handle unavailable services. The platform integrates academic workflows that are otherwise spread across separate tools under one authenticated account with shared context, supporting progress tracking and reuse of the same backend by further modules or clients.

No. of Pages : 11 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131031 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE HEALTH MONITORING AND DISEASE PREDICTING DEVICE

(51) International classification	:A61B 5/00, A61B 5/0205, A61B 5/1455, A61B 5/01, A61B 5/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Dr. Shanker Chandre

(57) Abstract :
A wearable health monitoring and disease predicting device, comprising a wearable member 101 consists of a neck support collar 102, a padded chest panel 103, a sensing module 104 to continuously measure multiple vital parameters and detect early physiological symptoms of metabolic disorders like diabetes and thyroid dysfunction, a health tracking sensor suite 105 to determine physiological parameters of user, a monitoring module integrated with the panel 103 comprises a camera 109, an electronic nose (E-Nose) volatile organic compound (VOC) sensor array 110, and a multispectral optical sensor 111, to continuously monitor the user's immediate environment and track intake parameters during meal times, a plurality of strain sensor patches 112 to detect changes in physical dimensions, a user interface inbuilt within a computing unit works in sync with the computing unit through a communication module, to virtually visualize historical trends, tracking whether the user's health is improving or deteriorating daily.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131032 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ENERGY-EFFICIENT ZONAL CLIMATE CONTROL SYSTEM

(51) International classification	:F24F 11/00, F24F 11/02, F24F 3/044, F24F 11/79, F24F 1/00	(71)**Name of Applicant :** **1)SR University** Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India (72)**Name of Inventor :** **1)Prasanna Mandala** **2)Dr. Shanker Chandre**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An energy-efficient zonal climate control system comprises a duct assembly 101 with a plurality of grill units 102 disposed throughout an enclosure 103 to deliver thermally regulated air to comfort zones, a servo damper 104 in each grill unit 102 to regulate volumetric flow rates, a central HVAC unit 105 connected to a branching duct network 106 with a centrifugal blower fan 107 to deliver primary airflow, a Peltier module array 108 in the central HVAC unit 105 to generate the thermally regulated air, an air filter 109 to remove dust and particles, a heat-dissipation heat sink 110 with an exhaust fan 111 to remove waste heat, and a control unit with a MCU configured to dynamically adjust servo dampers 104, blower speed, and Peltier cooling output based on combined thermal demand, zone-priority scores, and predefined Service-Level Agreement (SLA) limits.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131033 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM FOR MULTI-DISEASE RETINAL IMAGE CLASSIFICATION AND METHOD THEREOF

(51) International classification	:G06N 3/08, G06N 3/04, G06V 10/764, G06T 7/00, G06V 10/82
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)K.Saresh
2)Dr. S.Jamalaiah

(57) Abstract :
A system for multi-disease retinal image classification and method thereof comprises comprising an image preprocessing module 101 configured to receive and perform quality enhancement on an input fundus image, a convolutional neural network (CNN) feature extraction module 102 processes the preprocessed fundus image to extract high-order spatial and structural feature representations associated with retinal anatomical structures and pathological lesions, a liquid neural network (LNN) module 103 dynamically modifies the feature representations using continuous-time adaptive neuron dynamics to account for patient-specific and imaging condition variations, a hybrid feature fusion layer 104 fuses the CNN-derived feature representations and LNN-modified feature representations into a unified multidimensional feature representation, a multi-class classification engine 105 evaluates the unified multidimensional feature representation and outputs a diagnostic prediction comprising one or more retinal disease classifications and associated confidence scores.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131034 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED FIELD PREPARATION DEVICE FOR CROP MANAGEMENT

(51) International classification	:A01C 5/06, A01B 49/06, A01C 7/08, A01C 7/20, A01C 7/06	(71)**Name of Applicant :** **1)SR University** Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India (72)**Name of Inventor :** **1)Nallagondla Jyothi** **2)Dr. S.Jamalaiah**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
An automated field preparation device for crop management comprises a body 101 for supporting automated agricultural operations, a plurality of motorized reconfigurable wheels 102 for movement across varying field terrains, a user interaction module for receiving field boundaries and crop-specific requirements, a navigation and mapping module for mapping field boundaries and detecting obstacles, a pneumatic soil probe arrangement 104 for assessing zone-wise soil conditions, a soil treatment assembly for preparing and applying suitable treatment materials, a furrow forming assembly for forming crop-specific furrows or ridges, a seed sowing assembly for controlled seed release and coating and precise seed placement at selected depths and spacing, a control unit for analyzing soil condition data and controlling agricultural operations based on selected crop requirements, thereby enabling automated field preparation, efficient soil treatment, accurate seed sowing, and improved agricultural resource utilization across different field zones.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131035 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RESPIRATORY ASSESSMENT SYSTEM FOR MONITORING LUNG HEALTH AND WORKPLACE COMPLIANCE

(51) International classification	:A61B 5/08, A61B 5/087, A61B 5/00, A61B 5/091, G16H 40/63
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Kasana Gottu Srinivas
2)Dr. Srinivas Samala
3)Dr. Sudhansu Sekhar Singh

(57) Abstract :
A respiratory assessment system for monitoring Lung health and workplace compliance, comprising a control box 101 in an enclosure 102, having a display module 103 disposed thereon, a spirometry unit 104 to analyze a lung capacity of a user and measure an inhalation capacity upon a secured contact, an AI camera 105 to analyze a facial expression of the user, a microphone 106 to detect a voice command from the user and analyze a user audio wave, a microcontroller to analyze the user audio wave to detect an acoustic signature indicative of a lung condition, an IoT module to transmit a user data to a centralized cloud database in real-time, a base platform 107 comprising a vertical telescopic rod 108 attached to the base platform 107 comprising a horizontal plate 109 with a cushioned pad 110 to adjust in a configuration upon detecting the user.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131036 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS SPEARMINT CROP MONITORING AND PRECISION MANAGEMENT SYSTEM

(51) International classification	:B64C 39/02, G01N 33/24, G05D 1/10, G05D 1/00, G06V 20/13
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)O. Divya Jyothi
2)Johnson Kolluri
3)B Sankara Babu

(57) Abstract :
An autonomous spearmint crop monitoring and precision management system, comprises an UAV 101, a positioning unit associates observations with field coordinates, an imaging assembly 102 captures canopy images and spectral signatures used to detect plant condition and growth stage, an articulated probe reach a root zone soil region at a sampling point, a root zone sensing suite measure nitrogen, phosphorus and potassium availability in soil near a plant, a root zone moisture sensing unit measure soil moisture substantially, a penetration resistance sensing module measure force during probe insertion, an ambient sensing unit establish a temperature-exposure record, at least one leaf condition sensing unit measure leaf reflectance/chlorophyll characteristics, a nutrient delivery assembly 106 deliver a selected formulation for plant nutrition, an edge processing unit and a field digital twin store for storing a field map divided into sections and associating each section with at least plant health.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131037 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : GARDEN MAINTENANCE DEVICE FOR WEED REMOVAL

(51) International classification	:A01M 21/02, A01B 39/18, A01M 21/04, A01D 34/00, A01M 21/00	(71)**Name of Applicant :** **1)SR University** Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India (72)**Name of Inventor :** **1)Pilligundla Niharika** **2)Dr. Shanker Chandre**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A garden maintenance device for weed removal, comprising a mobile platform 101 for traversing over a garden surface, a navigation module 102 comprising a GPS (Global Positioning System), a depth camera and a plurality of IR (Infrared) sensors for movement along a pre-defined inspection route, a visual inspection module 103 mounted on an inverted U-shaped frame 104 comprising a hyperspectral camera, an RGB camera, a depth sensor, a thermal imaging camera and an illumination unit for identifying plant species and weeds, a pulling assembly mounted through a twelve-bar linkage arrangement 116 comprising a pair of C-shaped adjustable jaws 109 with serrated inner gripping surfaces for extracting detected weeds and a spraying assembly comprising a pressurized fluid reservoir 113 connected to an electronic nozzle 114 for targeted treatment.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131038 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MEDICAL PROFESSIONAL WELLNESS MONITORING AND WORKLOAD SUPPORT SYSTEM

(51) International classification	:A61B 5/00, A61B 5/024, A61B 5/0205, A61B 5/0245, A61B 5/11
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India
(72)**Name of Inventor :**
1)Dr. Vandana Kandukuri
2)Prof. Damarla Ramesh Babu
3)Dr. D Rajashekar

(57) Abstract :
A medical professional wellness monitoring and workload support system, comprising a wearable band 101, a physiological sensing suite 102 and at least a PPG sensor for heart rate and heart rate variability related measurements and a GSR/EDA sensor for skin conductance related measurements, a movement and environment sensing unit 103 and including at least an IMU and at least one ambient sensor selected from noise sensor and light sensor for monitoring movement, ambient noise and light, an edge artificial intelligent (AI) controller and a memory storing a user baseline profile, a local intervention module and at least a silent vibration actuator 108 and a visual indicator 109 for notifying user, a user interface and a computing unit for user status display and for delivering intervention guidance, and a hospital management server and a plurality of the bands to receive band summaries and generate staffing or workload recommendations.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131039 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : POLLUTION PROTECTION AND BREATHING ASSISTANCE MASK

(51) International classification	:A62B 18/02, A62B 18/08, A62B 23/02, A62B 7/10, A41D 13/11	(71)**Name of Applicant :** **1)SR University** Address of Applicant :Ananthasagar, Hasanparthy (PO), Warangal-506371, Telangana, India. Warangal Telangana India (72)**Name of Inventor :** **1)Dr. V. Gokula Krishnan** **2)Dr. N. Venkatesh**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A pollution protection and breathing assistance mask, comprising, a wearable body 101 having facial padding, a pair of adjustable straps 104 for securing the body 101 on a user, a verification sensor 111 for determining whether the body 101 is positioned properly on user's face, an airflow control assembly 105 for regulating intake airflow, a replaceable multi stage filtration cartridge for gaseous contaminants, a breathing response sensor suite for monitoring at least one of airflow rate, respiration intensity, tidal volume, coughing event, or breathing pattern, a fragrance dispensing assembly 201 for introducing fragrance into an airflow region, a plurality of LEDs (Light Emitting Diode) 108 on a periphery of the body 101 and an ambient light sensor for enabling the LED light set under low light conditions, a location module for associating detected exposure with a geographic location and for transmitting location during an emergency to an authorized individual.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131042 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Intelligent Wearable-Based Attendance Authentication System

(51) International classification	:G06K 9/00, G06Q 50/20, G06V 40/16, G06Q 10/10, G06Q 10/06	(71)**Name of Applicant :** **1)Dayananda Sagar College of Engineering** Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India (72)**Name of Inventor :** **1)Bhagyashree Kulkarni** **2)Aarathi S**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to an artificial intelligence (AI)-powered wearable attendance management system, herein referred to as 4Sight, designed to automate classroom attendance through real-time facial recognition while minimizing manual intervention. The proposed system integrates a wearable device equipped with an ESP32-CAM module that continuously captures live video streams of classroom participants. The captured visual data are processed using a computer vision engine developed with Python and OpenCV, enabling accurate real-time face detection, recognition, and automated attendance recording. The generated attendance records are securely transmitted to a centralized processing backend, which synchronizes the data with a React-based web dashboard. The dashboard provides an intuitive interface for viewing attendance records, monitoring attendance trends through graphical visualizations, generating analytical reports, and exporting data for administrative purposes. Unlike conventional attendance systems that rely on RFID cards, fingerprint scanners, or manual entry, the proposed wearable solution offers a contactless, hands-free, and minimally disruptive approach suitable for educational environments. The system is designed to function efficiently in low-connectivity conditions, ensuring uninterrupted attendance management even in resource-constrained settings. Its lightweight architecture, portability, and affordability make it suitable for deployment across classrooms of varying sizes.

No. of Pages : 12 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131044 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Digital Twin Based Intelligent Micro-Climate Monitoring and Irrigation Management System for Small Farms

(51) International classification	:G06Q 50/02, A01G 9/24, G06F 30/20, G06F 119/08, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :VNR Vignana Jyothi Institute of Engineering & Technology Vignana Jyothi Nagar, Bachupally, Hyderabad – 500 118, Telangana, India. Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr.K.Vijay Chandra
2)Dr.B.V. Seshu Kumari
3)Dr.Ravikanth Motupalli
4)Dr.V. Priyanka Brahmaiah
5)Dr.E.Shiva Prasad
6)Anjaneyulu Nelluru

(57) Abstract :
The present invention relates to a micro-climate digital twin system developed for small farms using distributed sensor clusters. The system is designed to monitor environmental conditions such as soil moisture, temperature, humidity, and light intensity in real time. Multiple sensor nodes are placed across different areas of the farm to capture localized micro-climate variations. The collected sensor data is transmitted to a central server through a wireless communication network. The server processes the incoming data and creates a digital twin model of the farm. This digital twin acts as a virtual representation of the actual farm environment. The model continuously updates based on live sensor readings. The system also includes an AI-based prediction module that analyzes both real-time and historical environmental data. Based on this analysis, the system generates recommendations for irrigation and crop management. Alerts are automatically sent when abnormal environmental conditions are detected. Farmers can monitor farm data using mobile and web dashboards. The interface displays sensor values, alerts, and graphical visualization of farm conditions. The digital twin also supports simulation of environmental scenarios. Farmers can understand the effect of irrigation decisions before applying them in the field. The invention helps in improving crop productivity by maintaining optimal environmental conditions. Water usage is optimized by providing accurate irrigation suggestions. The system reduces manual monitoring efforts and saves time. The invention is cost-effective and suitable for small-scale farmers. It provides zone-wise monitoring and remote accessibility. The system promotes smart and sustainable farming practices. Thus, the invention provides an intelligent micro-climate monitoring and digital twin-based decision support system for small farms.

No. of Pages : 10 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131045 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTONOMOUS WORKER SAFETY SYSTEM FOR MINING ENVIRONMENT

(51) International classification	:E21F 17/18, A61B 5/00, A61B 5/01, A61B 5/0205, A61B 5/024	(71)**Name of Applicant :** **1)JAIN (Deemed-to-be University)** Address of Applicant :Jain Global Campus, Jakkasandra Post, Kanakapura Road, Kanakapura Taluk, Ramnagar District, Bangalore –562112, Karnataka, India. Bangalore Karnataka India (72)**Name of Inventor :** **1)Dr. Muralidhar L B**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An autonomous worker safety system for mining environment, comprising a mobile assembly 109 provided with a collapsible pole 101 for maneuvering thereof, a fingerprint scanner 102 positioned on the pole 101 for authenticating an assigned worker, an artificial intelligence-based imaging camera 103 for capturing visual health indicators, and a contactless physiological monitoring unit comprising a remote photo-plethysmography sensor, infrared temperature sensor, and thermal imaging sensor for monitoring physiological parameters without physical contact, a storage chamber 104 having a motorized door 105 stores a quadcopter unit comprising an imaging unit 106, multiple motorized grippers 107, and electromagnetic coupling for forming a load-bearing aerial platform and securing worker, a control unit analyzes visual and physiological data, generates an emergency risk score, transmits worker location and health information, deploys the quadcopter unit for aerial evacuation upon exceeding a predefined threshold, and transmits reports to a user interface on a wirelessly linked computing unit.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131046 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WALL PANEL SYSTEM FOR PROVIDING THERMAL INSULATION AND SOUND DAMPING

(51) International classification	:E04B 1/82, E04B 1/84, E04B 2/74, E04F 13/08, E04B 2/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JAIN (Deemed-to-be University)
Address of Applicant :Jain Global Campus, Jakkasandra Post, Kanakapura Road, Kanakapura Taluk, Ramnagar District, Bangalore – 562112, Karnataka, India. Bangalore Karnataka India
(72)**Name of Inventor :**
1)Ramachandra N.U
2)Dr. Shivakumara D.C
3)Anagha. S
4)Divyansh Gupta
5)Nagamani. R
6)L.A.Ponmukilan
7)Tannistha Banerjee

(57) Abstract :
A wall panel system for providing thermal insulation and sound damping, comprises a track 101 mounted over a wall substrate, a panel module 102 removably fitted on the track 101, the panel module 102 includes a breathable front layer 102a for allowing sound waves to pass into the panel, a thermal acoustic core 102b for providing thermal resistance and sound absorption, a moisture indicating matrix layer 102c to show a visible change upon humidity or moisture rise, a structural backing plate 102d carrying an interlock portion for locking the panel module 102 to the track 101, and the interlock portion allowing removal and replacement of a selected panel module 102 without removing adjacent panel module 102 and without breaking a seal between neighboring panel module 102.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131047 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ASSISTANCE DEVICE FOR INSURANCE ASSERTION

(51) International classification	:G06Q 40/08, G16H 10/60, G16H 40/67, G16H 40/20, G06Q 40/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JAIN (Deemed-to-be University)
Address of Applicant :Jain Global Campus, Jakkasandra Post, Kanakapura Road, Kanakapura Taluk, Ramnagar District, Bangalore – 562112, Karnataka, India. Bangalore Karnataka India
(72)**Name of Inventor :**
1)Jyothi Shanbhag
2)Awab Asim Elhaj Abdelgader
3)Dhanusri Balamurali Swarnalakshmi
4)Awab Ahmed Osman
5)Dikshant Mehta
6)Harsh Vardhan Agarwal

(57) Abstract :
An assistance device for insurance assertion, comprising, a platform 101 having a housing 102, an imaging unit 103 to monitor patient facial expressions for patient identity verification, an input means to initiate an insurance application, a speaker 105 to prompt the patient and directing the patient to upload requisite documentation, through the input means, wherein upon successful document ingestion and processing, the control unit synthesizes these data streams to construct a complete electronic profile, subsequently transmitting the structured application file directly to designated life insurance enterprise to facilitate streamlined underwriting and policy issuance, an RGB light 108 integrated with the housing 102 glow a red LED, on detection of patient has specific errors or major superior deficiencies which cause the claim to be rejected and glow green light on detection of documents are in order and the claim has a sufficient instance of approval.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131048 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SITE-DIRECTED ELECTROCHEMICAL SENSING COMPOSITE FOR SIMULTANEOUS DETECTION OF MERCURY AND CADMIUM IONS AND DEVICE THEREOF

(51) International classification	:G01N 27/30, G01N 27/327, G01N 27/416, B01L 3/00, G01N 27/49
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JAIN (Deemed-to-be University)
Address of Applicant :Jain Global Campus, Jakkasandra Post, Kanakapura Road, Kanakapura Taluk, Ramnagar District, Bangalore – 562112, Karnataka, India. Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. Chandan Hunsur Ravikumar
2)Prof. R. Geetha Balakrishna
3)Dani George

(57) Abstract :
A site-directed electrochemical sensing composite for simultaneous detection of mercury and cadmium ions, comprises: a conductive heterostructure comprising MXene and graphene oxide, and cysteine immobilized on the conductive heterostructure. The present invention further relates to a paper-based electrochemical sensing device, comprising: a microfluidic paper substrate defining at least one sample receiving zone and at least one detection zone interconnected by one or more microfluidic channels, a working electrode, a counter electrode and a reference electrode disposed on the paper substrate, and a site-directed electrochemical sensing composite deposited on the working electrode, wherein the sensing device is configured to simultaneously discriminate and electrochemically detect mercury ions and cadmium ions in an aqueous sample.

No. of Pages : 32 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131049 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AIR PURIFYING SYSTEM

(51) International classification	:B01D 53/14, B01D 53/62, F25B 21/02, B01D 53/78, B01D 53/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JAIN (Deemed-to-be University)
Address of Applicant :Jain Global Campus, Jakkasandra Post, Kanakapura Road, Kanakapura Taluk, Ramnagar District, Bangalore – 562112, Karnataka, India. Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. Shrinivas R Patil
2)Shriraksha S P

(57) Abstract :
An air purifying system comprises a chamber 102 having a vacuum unit 101 for drawing ambient air, a Peltier unit 103 for cooling the resulting gas mixture, and a gas sensor for detecting gases therein, an iris hole 104 directs the mixture through a conduit 106 into a receptacle 105 containing an amine scrubber unit 107 having first sprayers 108 connected to multi-sectioned storage tank storing amine and alkaline liquids and activated carbon for carbon-dioxide removal, iris valves direct processed gas to storage unit 112 and separated CO2 to first storage bag 113 having conversion unit for producing and releasing O2, wet scrubber having second sprayers 115 and carbon filters removes sulphur, which is directed to second storage bag 119. Processed gas enters vessel 118 containing zeolite-based pressure swing adsorption unit 120, iris apertures release processed gas into the environment and direct separated nitrogen into a third storage bag 123.

No. of Pages : 30 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131050 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM FOR MANAGING EMERGENCY VEHICLE PASSAGE THROUGH A NETWORK OF SIGNALIZED INTERSECTIONS

(51) International classification	:G08G 1/01, G08G 1/0968, G08G 1/00, G08G 1/0965, G08G 1/07
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JAIN (Deemed-to-be University)
Address of Applicant :Jain Global Campus, Jakkasandra Post, Kanakapura Road, Kanakapura Taluk, Ramnagar District, Bangalore – 562112, Karnataka, India. Bangalore Karnataka India
(72)**Name of Inventor :**
1)Swapna H R
2)Jayadeep S G
3)Ankita Baid
4)Saachi Shetty
5)Muthuvikraam M
6)Gargi Bhardwaj

(57) Abstract :
A system for managing emergency vehicle passage through a network of signalized intersections, comprises a GPS (Global Positioning System) tracking unit 101 to receive position data of an emergency vehicle 102, a traffic management computing module 201 to determine a route of the emergency vehicle 102 to a destination and to identify a plurality of intersections along the route, a plurality of imaging units 103 to obtain image data representing traffic conditions associated with the respective intersections, a computer vision processing module 202 to process the image data and determine a physical clearance state of a designated approach lane to be traversed by the emergency vehicle 102, a traffic signal controller interface 206 to communicate with a traffic signal controller 207 associated with the upcoming intersection, and a signal authorization module 205 withholds green authorization while the approach lane is blocked and authorizes green upon computer-vision-verified physical clearance.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131051 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : POROUS ADSORBENT MATERIAL FOR DYE ABSORPTION AND A METHOD FOR PREPARATION THEREOF

(51) International classification	:B01J 20/20, B01J 20/30, B01J 20/28, C02F 1/28, B01J 20/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)JAIN (Deemed-to-be University)
Address of Applicant :Jain Global Campus, Jakkasandra Post, Kanakapura Road, Kanakapura Taluk, Ramnagar District, Bangalore –562112, Karnataka, India. Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. Usha Nellur
2)Dr. Akshaya K Samal
3)Dr. Manav Saxena

(57) Abstract :
A porous adsorbent material for dye absorption and a method for preparation thereof,the material comprises a carbonaceous, potassium activated carbon material having: a specific surface area greater than 1000 m^2/g, a hierarchical pore structure comprising both micropores and mesopores, and surface functional groups capable of π-π stacking and hydrogen bonding with aromatic dye molecules and the preparation method comprises of: washing onion peel with deionised water, drying washed onion peel, crushing and sieving dried onion peel to obtain particles, carbonizing sieved onion peel under a nitrogen atmosphere to obtain an onion peel biochar (OPBC), mixing OPBC with potassium hydroxide, activating mixture under a nitrogen atmosphere to obtain a chemically activated carbon, washing activated carbon with 2 M hydrochloric acid for 20-28 hours and then washing with deionized water until a filtrate pH of about 7 is reached, and drying washed activated carbon to obtain onion peel-derived activated carbon (OHPBC).

No. of Pages : 27 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131061 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Driver Drowsiness Detection System Using CNN

(51) International classification	:G08B 21/06, A61B 5/18, G06K 9/00, G08G 1/16, B60W 50/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)A PANDIAN
Address of Applicant :Department of EEE, Koneru Lakshmaiah Education Foundation, Vaddeswaram, Guntur, Andhra Pradesh -INDIA Andhra Pradesh India
2)Bandaru Alekhya
3)Koneru Lakshmaiah Education Foundation
(72)**Name of Inventor :**
1)A PANDIAN
2)Bandaru Alekhya
3)Saladi Sri Anjana Jyothika
4)S Ravi Teja

(57) Abstract :
Driver drowsiness poses a major threat to road safety and is one of the most important reasons for fatal road traffic accidents in the world – making continuous driver monitoring a key requirement for enhancing road safety. The problem with the present invention is the Driver Drowsiness Detection System developed by Raspberry Pi 5, USB webcam and an active buzzer using open source Python libraries. The system continuously gathers, in real time, images of the driver's face and makes analysis to recognize the signs of driver fatigue before they can turn into hazardous driving situations. The system in the proposed will detect the drowsiness of the eyes by extracting six coordinates of the eye landmarks using a pre-trained dlib facial landmark detector and calculating Eye Aspect Ratio (EAR). The driver is continually monitored across subsequent video frames to determine whether the driver is becoming drowsy or not. The Mouth Aspect Ratio (MAR) is calculated from 12 landmark points in the mouth area and determined when the person is yawning. It takes eight frames to detect the movement in the mouth in order to reduce the false detection rate caused by normal speech, and the system uses a speaking filter to detect a yawning event. The proposed system is implemented using a lightweight Convolutional Neural Network (CNN) and exclusively with the NumPy library, without any other high-level deep learning frameworks such as TensorFlow or PyTorch.The proposed system uses a lightweight Convolutional Neural Network (CNN) using the NumPy library and not any other high-level deep learning frameworks such as Tensorflow or PyTorch. The image of the eye is captured in a 16×16 grayscale image and is processed by the CNN to check whether the driver's eye is open or closed. By using the EAR values calculated in the network, the network continually refines and improves its prediction, so there is no need to have a labelled training set, and the computational complexity is low, suitable for embedded hardware. The system automatically detects the light level of each frame captured to provide reliable working in varying lighting conditions. If the brightness falls below the set value, the image is brightened by the use of Contrast Limited Adaptive Histogram Equalization (CLAHE) algorithm, which enhances the brightness of the image and makes better the detection of facial landmarks and eye-state analysis in low light intensity. When the driver is drowsy, the system triggers progressive warning sounds after 1.5 seconds, 3 seconds and 5 seconds of eye closure using an active buzzer connected to GPIO Pin 18 and after 5 seconds of eye closure a continuous buzzer alarm is played. Similarly, a confirmed yawning occurrence will also result in a specific warning alert. When running on the Raspbian operating system, the entire system runs at about 7–10 frames per second on the Raspberry Pi platform with good performance both in normal light and low-light conditions. The proposed invention combines the use of facial landmark detection, EAR, MAR, lightweight CNN based on NumPy, image enhancement with CLAHE and a progressive warning system with GPIO in a compact embedded platform. The system provides a low cost, mobile, non-invasive and online solution to accurately identify driver drowsiness and driver yawning, thereby boosting driver awareness and enhancing road safety.

No. of Pages : 15 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131066 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Smart Composite Machining Optimization using Generative AI

(51) International classification	:G06N 3/08, G06N 3/04, G06N 20/00, G05B 19/4097, G05B 19/4099
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)G ASHWIN PRABHU
Address of Applicant :No. 11, Thirumagal Nagar, II Street, Karthick Avenue, Flat No. F1, First Floor, "Sai Guru Appartments", Chitlapakkam Tamil Nadu India
2)Dr. K. P. VASANTHAKUMAR
3)Dr. ALUGONDA RAJ KUMAR
4)Dr. K. VELAVAN
5)Dr. J. M. PRABHUDASS
6)Dr. V. VENKATA RAMANA
7)Dr. M. SAKTHIVEL
8)Mr. K. BALAMANIKANDASUTHAN
9)Dr. RITTIN ABRAHAM KURIEN
(72)**Name of Inventor :**
1)Dr. K. P. VASANTHAKUMAR
2)Dr. ALUGONDA RAJ KUMAR
3)Dr. K. VELAVAN
4)Dr. J. M. PRABHUDASS
5)Dr. V. VENKATA RAMANA
6)Dr. M. SAKTHIVEL
7)Mr. K. BALAMANIKANDASUTHAN
8)Dr. RITTIN ABRAHAM KURIEN

(57) Abstract :
The machining of advanced composite materials, particularly Carbon Fiber Reinforced Polymer (CFRP) composites, presents significant challenges due to their anisotropic properties, abrasive reinforcement, rapid tool wear, delamination, and poor surface integrity. This study proposes a Generative Artificial Intelligence (GenAI)-driven smart machining optimization framework to enhance machining efficiency, dimensional accuracy, and sustainability. The proposed methodology integrates Generative Adversarial Networks (GANs), machine learning-based predictive models, and multi-objective optimization algorithms to identify optimal machining parameters for CFRP composites. A representative experimental framework considers cutting speeds of 50–200 m/min, feed rates of 0.05–0.25 mm/rev, and depths of cut of 0.5–2.0 mm. These ranges are selected to investigate the trade-offs between material removal rate, cutting forces, thermal effects, and surface quality. The GenAI model generates synthetic machining datasets to supplement limited experimental observations, while predictive algorithms estimate surface roughness, tool wear, and delamination characteristics. A multi-objective optimization strategy aims to minimize surface roughness below 1.5 µm, reduce the delamination factor below 1.10, and improve material removal rate by 20–30% compared with conventional parameter selection. Additionally, the proposed framework targets a 15–25% reduction in tool wear, a 10–20% decrease in specific energy consumption, and predictive accuracy exceeding 95%, subject to experimental validation. These targets are motivated by the need to balance machining productivity with composite structural integrity and manufacturing cost. Real-time sensor data, including cutting force, vibration, and temperature, are incorporated to support adaptive parameter adjustment and intelligent process monitoring. The proposed approach provides a foundation for autonomous machining decision-making, reduced experimental trial requirements, and improved process reliability. Overall, the integration of Generative AI with composite machining offers a promising pathway toward data-driven, energy-efficient, and sustainable smart manufacturing systems for aerospace, automotive, and high-performance engineering applications.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131068 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System for Linkage-Conditioned Multimodal Fusion and Capability-Gated Coronary Anatomical Assertion Control

(51) International classification	:G16H 50/20, G16H 50/70, G06N 5/04, A61B 5/361, A61B 5/363
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vellore Institute of Technology
Address of Applicant :Vellore Institute of Technology, Tiruvalam Road, Katpadi, Vellore-632014, Tamil Nadu, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)Dr M Rajasekhara Babu
2)Dhanush Konduru

(57) Abstract :
The present disclosure proposes a system (100) for linkage-conditioned multimodal cardiac evidence processing and capability-gated coronary anatomical assertion control. A computing device (102) processes provenance-verified cardiac evidence to generate modality-specific prediction outputs. A linkage-conditioned multimodal fusion module (124) executes trained fusion for established same-subject co-measurement linkage or reliability-weighted aggregation for unlinked observations, identifying unvalidated aggregation. A progressive evidence inference module (126) updates predictions upon receipt of observations. A coronary anatomical localization module (128) generates candidate subject-, vessel-, and segment-level assertions. A capability-gated anatomical assertion module (130) authorizes assertions only upon satisfying predictive-capability, probability-threshold, parent-vessel, anatomical-consistency, and reference-label-granularity conditions, and refuses assertions when requirements or capability information are unavailable. A shared coronary anatomy registry (132) and three-dimensional coronary digital-twin visualization module (134) display authorized categorical regions and image-derived positions on population-schematic anatomy without unsupported positional precision.

No. of Pages : 126 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131069 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A System and Method for Batch-Optimized GPU Acceleration of Post-Quantum V2X Key Exchange

(51) International classification	:H04L 9/08, H04W 12/041, H04W 4/40, H04L 9/32, H04W 12/0431
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vellore Institute of Technology
Address of Applicant :Vellore Institute of Technology, Tiruvalam Road, Katpadi, Vellore-632014, Tamil Nadu, India. Vellore Tamil Nadu India
(72)**Name of Inventor :**
1)Arun Manoharan
2)Ujwal Kumar N L

(57) Abstract :
The present disclosure proposes a batch-optimized GPU acceleration system (100) for post-quantum vehicle-to-everything (V2X) key exchange in over-the-air (OTA) update infrastructures. The system (100) comprises a computing device (102) having a host processor (104) and a host memory (106), communicatively coupled to a plurality of vehicles (130) through a network (126). A plurality of modules (108) coordinates sequential host-processor cryptographic operations and parallel GPU-based polynomial arithmetic associated with a module-lattice-based key encapsulation mechanism (ML-KEM). The plurality of modules (108) comprises a batch preparation module (110), a data-transfer module (112), a GPU execution-grid configuration module (114), a GPU memory management module (116), a parallel polynomial computation module (118), a host-side cryptographic processing module (120), a V2X payload framing module (122), and a key-exchange validation module (124). The system (100) facilitates accelerated cryptographic processing, computational bottleneck identification, and consistent key-exchange results across multiple vehicle sessions.

No. of Pages : 81 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131070 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Method for Determining Colon-Reaching Prebiotic Fractions of Coffee By-Products by Gastrointestinal Digestion and Fermentation

(51) International classification	:A23F 5/24, C12Q 1/34, G01N 33/02, A23F 5/02, A61K 35/745
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T)
Address of Applicant :National Institute of Food Technology, Entrepreneurship and Management – Thanjavur (NIFTEM-T), Pudukkottai Road, Thanjavur-613005, Tamil Nadu, India. Thanjavur Tamil Nadu India
(72)**Name of Inventor :**
1)R. Vidyalakshmi
2)Atava Lokesh

(57) Abstract :
The present disclosure proposes a method for determining colon-reaching prebiotic fractions of coffee by-products using simulated in vitro gastrointestinal digestion and probiotic fermentation for evaluating gastrointestinal resistance, colon-targeted delivery, selective microbial utilisation, and short-chain fatty acid production of coffee by-product-derived prebiotic fractions. A coffee by-product-derived test material is subjected to simulated oral, gastric, and intestinal digestion using α-amylase, gastric lipase and protease, and amyloglucosidase and bile salts, respectively. A gastrointestinal digestion-resistant fraction is isolated through ethyl alcohol precipitation and filtration and corrected for protein content and ash content. The resulting colon-reaching prebiotic fraction is subjected to in vitro fermentation using at least one selected probiotic microorganism. A resulting fermentation sample is analysed by gas chromatography-mass spectrometry to determine short-chain fatty acids associated with microbial utilisation of the colon-reaching prebiotic fraction.

No. of Pages : 38 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131074 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : INDUSTRIAL POLLUTION MONITORING SYSTEM

(51) International classification	:G08B 21/20, G06Q 50/26, G08B 21/18, G06Q 10/06, G08B 25/10	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(71)**Name of Applicant :**
1)Cambridge Institute of Technology North Campus
Address of Applicant :Cambridge Institute of Technology North Campus, Bangalore-562110, Karnataka, India Bangalore Karnataka India
2)Dr. Tejaswini C
3)Dr. Sendamarai P
4)Ravi N
5)Keshava N
6)Dr. Sugumar S J
7)Dr. Yoganandini A P
8)Tanuja G
9)Ashwini N
(72)**Name of Inventor :**
1)Cambridge Institute of Technology North Campus
2)Dr. Tejaswini C
3)Dr. Sendamarai P
4)Ravi N
5)Keshava N
6)Dr. Sugumar S J
7)Dr. Yoganandini A P
8)Tanuja G
9)Ashwini N

(57) Abstract :
To Design and develop a low-cost real-time monitoring, alert to workers, notified data of polluted area has been checked for the immediate surroundings. A various parameter such as Air Quality, Temperature & Humidity, and Sound intensity sensors with PIC controller which collects and upload data into the cloud using ESP8266 Wi-Fi module. Then transmit to the cloud platform using MQTT protocol to perform Digital Dashboard on Smartphone checks anomaly notification and alert to the user through a app. If the industry doesn't take any step to reduce the pollution within a certain period, the pollution control board will shut down the industrial electricity power until they pay a penalty amount of polluting the city according to the government rules and regulation. Industrial pollution poses a serious threat to environmental sustainability and human health due to the continuous emission of harmful gases from industrial processes. This project presents the design and implementation of an Industrial Pollution Monitoring System that continuously monitors air quality and automatically takes corrective action when pollution levels exceed permissible limits. The system employs gas sensors to detect various hazardous and combustible gases present in industrial environments. A microcontroller serves as the core processing unit, collecting sensor data and analyzing pollution levels in real time. An interface is used to reduce wiring complexity and improve system efficiency. When gas concentrations rise beyond predefined thresholds, a relay is activated to control an exhaust fan, ensuring effective ventilation and immediate reduction of pollutant concentration. To enhance safety and monitoring capabilities, a GSM module is integrated to send instant alert messages to authorized personnel, while a GPS module provides accurate location information of the affected industrial area. This system offers a reliable, cost-effective, and automated solution for industrial air quality monitoring. The proposed model helps industries ensure worker safety, comply with environmental regulations, and minimize the impact of air pollution, with scope for future expansion through cloud-based monitoring and data analytics.

No. of Pages : 7 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131082 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : System and Method for Sustainable Cold Pothole Patching Mixture Comprising Reclaimed Asphalt Pavement Aggregates and Tyre Pyrolysis Oil-Based Cutback Bitumen

(51) International classification	:C04B 26/26, C08L 95/00, E01C 7/35, E01C 7/18, E01C 23/09
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dayananda Sagar College of Engineering
Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India
(72)**Name of Inventor :**
1)Raghavendra Mahesh
2)Dr. Doma Hemanth Kumar
3)Dr. Shiva Kumar G
4)Sridhar. H. N

(57) Abstract :
The present invention relates to a sustainable cold pothole patching mixture and a method of preparing the same, for use in routine and emergency repair of bituminous pavements. The mixture comprises processed Reclaimed Asphalt Pavement (RAP) aggregates coated with a cutback binder prepared by blending a viscosity-graded paving-grade bitumen (VG-30) with tyre pyrolysis oil (TPO), derived from the pyrolysis of end-of-life tyres, as a viscosity-reducing diluent, together with a minor proportion of an organosilane-based anti-stripping additive. In one embodiment, the cutback binder comprises about 70% paving-grade bitumen and about 30% tyre pyrolysis oil by mass, together with about 0.1% anti-stripping additive by mass of the binder. The RAP aggregates, being of rough, porous and angular surface morphology, mechanically interlock with one another and permit penetration of the cutback binder into surface pores, while the residual aged binder present on the RAP aggregate surface is chemically compatible with the TPO-modified fresh binder. The resulting cold patching mixture exhibits enhanced Marshall stability, cohesion, adhesion, abrasion resistance and resistance to moisture-induced damage as compared to a corresponding mixture prepared using conventional virgin aggregates, while simultaneously reducing material cost by valorising two categories of waste material, namely reclaimed pavement material and end-of-life tyres, thereby providing a technically effective, economically viable and environmentally sustainable solution for pothole maintenance.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131112 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A System and Method for Multi-Sensor Satellite Data Fusion and Explainable Machine Learning-Based High-Resolution Coastal Eutrophication Risk Mapping and Prediction

(51) International classification	:G01S 13/90, G06N 20/00, G06N 3/08, G06N 3/04, G01W 1/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (M), Warangal Urban, Telangana - 506371, India. Warangal Telangana India
(72)**Name of Inventor :**
1)G. Shyamala
2)S. Vivek
3)Siddharth M
4)Naveen Arasu Anbarasu
5)Amitha P Sunil
6)Vadivel M

(57) Abstract :
A System and Method for Multi-Sensor Satellite Data Fusion and Explainable Machine Learning-Based High-Resolution Coastal Eutrophication Risk Mapping and Prediction ABSTRACT 1. The present invention discloses an integrated system and method for high-resolution coastal eutrophication risk mapping and prediction using multi-sensor satellite data fusion and explainable machine learning techniques. The system comprises satellite data acquisition, preprocessing, multi-sensor data fusion, environmental feature extraction, machine learning-based risk assessment, explainable artificial intelligence, spatiotemporal prediction, and geospatial visualization modules. Optical, thermal, and Synthetic Aperture Radar (SAR) satellite observations are integrated with environmental and oceanographic parameters to derive eutrophication-related indicators, including chlorophyll-a concentration, sea surface temperature, turbidity, and nutrient-related variables. Advanced machine learning algorithms analyze complex spatial and temporal relationships among these indicators to generate high-resolution eutrophication risk maps and forecast potential eutrophication events. Explainable artificial intelligence techniques, including SHAP and LIME, identify influential environmental factors contributing to predicted risks, thereby enhancing model transparency and interpretability. The invention further incorporates automated hotspot identification, risk classification, uncertainty assessment, and early warning mechanisms to facilitate timely environmental interventions. The proposed system addresses limitations of conventional monitoring approaches by improving data integration, spatial coverage, predictive capabilities, and decision-support functionality. The invention provides a scalable technological framework for coastal water quality monitoring, marine ecosystem protection, and sustainable coastal environmental management.

No. of Pages : 15 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131113 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ANTICOUNTERFEIT TAG AND A PROCESS OF MANUFACTURING THEREOF

(51) International classification	:G06K 19/077, G08B 13/24, G09F 3/02, G06K 19/06, G06K 19/07	(71)**Name of Applicant :** **1)INDIAN INSTITUTE OF SCIENCE** Address of Applicant :Indian Institute of Science, Bangalore, Karnataka 560012, India Karnataka India (72)**Name of Inventor :** **1)PRIYADARSINI, Sushree Sangita** **2)DASGUPTA, Subho**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
ABSTRACT AN ANTICOUNTERFEIT TAG AND A PROCESS OF MANUFACTURING THEREOF An anticounterfeit tag comprising an energy storage unit, a visual indicator unit, a stimulus-responsive element, a resistance element, a node element, and a 5 switching element. The stimulus-responsive element to produce an electrical response upon exposure of the stimulus-responsive element to one or more stimuli. The switching element comprising a control input and a current path. The resistance element connected in series with the stimulus-responsive element. The node element located between the stimulus-responsive element and the 10 resistance element. The switching element is actuated by the electrical response received at the control input such that electrical energy is supplied from the energy storage unit through the current path to the visual indicator unit. In response to the electrical energy supplied to the visual indicator unit, a visually perceptible change is produced in the visual indicator unit, thereby authenticating the anticounterfeit 15 tag itself and the product having the anti-counterfeit tag on it. Fig. 3a 20 43

No. of Pages : 64 No. of Claims : 27

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131114 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Based Goal Achievement Using LLM-Based Personalized Planning, Multimodal Evidence Verification, and Adaptive Replanning

(51) International classification	:G06Q 10/06, G09B 19/00, G06Q 10/00, G06F 17/30, G
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Nallamilli Kavita Reddy
Address of Applicant :Assistant Professor, Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning), AMC Engineering College, Bannerghatta Rd, Kalkare, Bengaluru, Karnataka 560083 Karnataka India
2)Ambili K
3)MS. Pamela Bera
4)Nagavarshini B R
5)Likish R
6)Manvith H Shetty
7)Nisarga P
8)Nisarga S
(72)**Name of Inventor :**
1)Nallamilli Kavita Reddy
2)Ambili K
3)MS. Pamela Bera
4)Nagavarshini B R
5)Likish R
6)Manvith H Shetty
7)Nisarga P
8)Nisarga S

(57) Abstract :
Abstract: Achieving long-term personal and professional goals, such as developing new skills, improving productivity, or preparing for a career, requires more than receiving recommendations; it requires structured planning, consistent execution, reliable progress tracking, and adaptation to changing circumstances.

No. of Pages : 9 No. of Claims : 1

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131116 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COLORIMETRY-BASED OPTICAL SCANNING SYSTEM FOR NON-INVASIVE SCREENING OF URINARY TRACT INFECTIONS

(51) International classification	:A61B 5/00, G01J 3/46, G01J 3/50, G01N 21/78, A61B 5/1455	(71)**Name of Applicant :** **1)B.M.S College of Engineering** Address of Applicant :Bull Temple Road Basavanagudi, Bengaluru Karnataka – 560019 Bangalore Karnataka India (72)**Name of Inventor :** **1)Monisha H M** **2)Nethra Sureshkumar** **3)Rida Rabeea Fatima** **4)Shama**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT COLORIMETRY-BASED OPTICAL SCANNING SYSTEM FOR NON-INVASIVE SCREENING OF URINARY TRACT INFECTIONS A colorimetry-based reusable handheld optical scanning system for non-invasive screening of urinary tract infections (UTIs) in wearables is disclosed. The system comprises a barcode-scanner-form-factor device incorporating three TCS34725 RGB colour sensors individually aligned in fixed spatial correspondence with leukocyte, nitrite, and pH reagent pads of a urinalysis strip embedded in a diaper. A PCA9548A I²C multiplexer enables multi-sensor integration with an ESP32 microcontroller, which executes Euclidean distance-based colour classification by direct comparison of the measured RGB values against a calibrated reference RGB dataset, without intermediate normalisation. Classified results are transmitted via Wi-Fi HTTP POST to a Flask backend executing UTI risk determination, storing data in MongoDB, and updating a web-based caregiver dashboard with automated alerts. Validated at 93.33% accuracy across fifteen test cases, the system enables continuous, non-invasive, biochemical UTI monitoring without embedding electronics in disposable wearables. Fig. 2

No. of Pages : 50 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131119 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR IOT-ML BASED PREDICTIVE MAINTENANCE WITH SELF-DIAGNOSING MONITORING INFRASTRUCTURE

(51) International classification	:G06N 20/00, G05B 23/02, G06F 11/30, G06N 5/04, H04L 12/24	(71)**Name of Applicant :** **1)B.M.S College of Engineering** Address of Applicant :Bull Temple Road, Basavanagudi, Bengaluru, Karnataka - 560019 Bangalore Karnataka India (72)**Name of Inventor :** **1)Tejashwini A Gondhale** **2)Pavanaj S Achar** **3)Manu K N** **4)Sanjay C** **5)Vinay S P**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT SYSTEM AND METHOD FOR IOT-ML BASED PREDICTIVE MAINTENANCE WITH SELF-DIAGNOSING MONITORING INFRASTRUCTURE A low-cost IoT-ML system monitors an external device via temperature, humidity, voltage, current, vibration and timestamp sensors interfaced to a monitoring controller. A datasheet-extraction module parses uploaded manufacturer datasheets into a validated specification record that is applied in common by a health-score engine, a machine-learning module and a control module. The health-score engine computes a bounded, penalty-weighted health score both for the monitored device and, independently, for the monitoring controller itself. A Random Forest Regressor predicts device health from a feature vector normalised using the said record, and outputs a prediction-confidence score derived from ensemble-tree variance. The control module classifies operating condition against user-configurable thresholds drawn from the same record, driving a visual indicator and a protective relay without firmware modification. The system achieved a health-prediction R^2 of 0.9996, a 0% false-alarm rate, and datasheet-extraction accuracy of approximately 90.9%, at a hardware cost of approximately ₹2,161 per unit. Fig. 1

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131120 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT SYSTEM FOR SATELLITE-DERIVED COASTAL NUTRIENT DYNAMICS ASSESSMENT AND SPATIOTEMPORAL EUTROPHICATION PREDICTION USING EXPLAINABLE DEEP LEARNING

(51) International classification	:G06Q 10/04, G06N 3/08, G06N 3/04, G06Q 10/06, G06Q 50/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR University
Address of Applicant :Ananthasagar, Hasanparthy (M), Warangal Urban, Telangana - 506371, India. Warangal Telangana India
(72)**Name of Inventor :**
1)G. Shyamala
2)S. Vivek
3)G. Vimala
4)Shanmuharajan M. B
5)S. P. M. Kannan
6)Vadivel M

(57) Abstract :
AI DRIVEN MULTI-AGENT SYSTEM FOR AUTONOMOUS PROJECT MANAGEMENT AND RESOURCE ALLOCATION ABSTRACT The invention presents an intelligent system for assessing coastal nutrient dynamics and predicting eutrophication using satellite-derived observations and explainable deep learning. The system integrates multispectral imagery, field measurements, hydrological records, and meteorological data through quality-controlled preprocessing and spatial-temporal harmonisation. A calibrated nutrient estimation engine derives georeferenced indicators for nitrogen, phosphorus, and related water-quality conditions. A spatiotemporal prediction model combines spatial features with historical sequences to forecast eutrophication probability, severity, and geographical progression. An explainability module identifies influential variables and generates interpretable attribution maps, while uncertainty indicators communicate prediction reliability. A decision-support interface displays nutrient distributions, risk hotspots, forecast trends, and monitoring priorities, supporting evidence-based coastal assessment, targeted sampling, and proactive ecosystem management using independent field observations and documented model validation.

No. of Pages : 15 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131121 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Intelligent Hybrid Quantum-Classical Machine Learning System and Method for Early Prediction of Cardiovascular Diseases

(51) International classification	:G16H 50/30, G16H 50/20, A61B 5/00, G16H 10/60, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Anurag University
Address of Applicant :Venkatapur, Ghatkesar, Medchal–Malkajgiri district, Hyderabad - 500 088, Telangana, India. Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr. N Suresh Rao
2)Mr. Dingari Acharya Kiran kumar
3)Mr. Komitla srihari
4)Mrs. T Veda Reddy
5)Mr. Dammu Venkata Ravi Kumar
6)Dr. A. Radha Krishna
7)Mr. P. Shivashankar
8)Dr. B.Srinivasulu

(57) Abstract :
The present invention focuses on the development of a hybrid quantum-classical machine learning framework for the early identification of cardiovascular disease risk from patient clinical records and health-related information. The framework employs a Hybrid Quantum Support Vector Machine (HQSVM), combining quantum-based feature representation with classical machine learning techniques to support cardiovascular risk prediction. Its architecture incorporates dedicated modules for patient data collection, data preprocessing, relevant clinical feature selection, quantum encoding, disease prediction, model optimization, result interpretation, and risk evaluation. Patient attributes, including age, blood pressure, cholesterol level, blood glucose, maximum heart rate, and electrocardiographic measurements, are analyzed to determine clinically relevant patterns associated with cardiovascular risk. The selected attributes are transformed into quantum-state representations through parameterized quantum circuits, and the corresponding measurement results are supplied to the classical SVM classifier for prediction. The performance of the proposed HQSVM approach is assessed against conventional SVM, Random Forest, and XGBoost models using standard evaluation measures, including accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC). Experimental evaluation on the selected test dataset records a prediction accuracy of 92.65% for the proposed HQSVM model.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131122 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Cardiovascular Disease Prediction System Using Ensemble Learning

(51) International classification	:G16H 50/20, G16H 10/60, G06N 20/00, G16H 50/30, G16H 50/70	(71)**Name of Applicant :** **1)Chaitanya Bharathi Institute of Technology** Address of Applicant :Vidya Nagar, Proddatur, Y.S.R. Kadapa (Dist.), Andhra Pradesh-516360 Proddatur Andhra Pradesh India (72)**Name of Inventor :** **1)KHAJAMAINUDDIN S M** **2)BANDELA RAJESH** **3)T.ARSHAD** **4)MADAM OBULESU** **5)ELLURU UMADEVI** **6)PILLACHENUGALLA SRUTHI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
Cardiovascular Diseases require accurate early detection to mitigate global mortality rates. Addressing the limitations of manual diagnosis, we developed a cloud-deployed Clinical Decision Support System using Machine Learning. A Random Forest Classifier trained on the UCI Heart Disease dataset predicts risk with approximately 88 percent accuracy. Uniquely, the system integrates secure SMTP email authentication and live cloud hosting, providing a scalable, accessible solution for preliminary health screening.

No. of Pages : 9 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(21) Application No.202641131123 A
(19) INDIA
(22) Date of filing of Application :02/11/2026
(43) Publication Date : 06/11/2026

(54) Title of the invention : Adaptive Spatial-light Reconstruction System for Generating Digital Imagery on Passive Surfaces

(51) International classification	:G06T 5/50, G06T 15/50, G06T 5/00, G06T 7/80, G06T 7/00	(71)**Name of Applicant :** **1)Suresh Kumar Chittanoori** Address of Applicant :U11 21-23 Lane St Wentworthville NSW 2145 Australia Australia **2)Dharani Juluri** **3)Srivas Chittanoori** **4)Srihan Chittanoori**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	(72)**Name of Inventor :** **1)Suresh Kumar Chittanoori** **2)Dharani Juluri** **3)Srivas Chittanoori** **4)Srihan Chittanoori**
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The present disclosure proposes an adaptive spatial-light reconstruction system (100) comprising a plurality of spatially distributed, independently controllable light sources (102), a computing device (104), a processor (106), a memory (108), and functional modules (110) including a calibration module (112), an optical response module (114), an image-processing module (116), a spatial-light optimisation module (118), and light-source control module (120). The system (100) determines optical interaction characteristics between the spatially distributed, independently controllable light sources (102) and regions of a passive surface, generates an optical response matrix, converts a digital image into target illumination characteristics, and determines collective operating parameters for reconstructing the image through reflected or diffused illumination. A feedback and error-correction module (122) and imaging sensor (124) enable reconstruction-error correction. The adaptive spatial-light reconstruction system (100) provides many-to-many light-source contribution modelling, adaptive recalibration following physical or optical changes, and regeneration of operating parameters using updated optical-response information.

No. of Pages : 112 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131126 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Bedaquiline-Loaded Oral Proliposomal Composition for Pulmonary Macrophage-Targeted Delivery in the Management of Tuberculosis

(51) International classification	:A61K 9/127, A61P 31/06, A61K 47/26, A61K 31/4704, A61K 47/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dayananda Sagar University
Address of Applicant :College of Pharmaceutical Sciences, Dayananda Sagar University, Devarakaggalahalli, Harohalli Kanakapura Road, Ramanagara Dt., Karnataka, India, 562112. Harohalli Karnataka India
(72)**Name of Inventor :**
1)Vanaja Kenchappa
2)Rayadurgam Naveen

(57) Abstract :
The present invention relates to a Bedaquiline -loaded proliposomal pharmaceutical composition for oral administration in the management of tuberculosis and a process for preparing the same. The composition comprises Bedaquiline, a phospholipid, cholesterol, and a pharmaceutically acceptable solid carrier selected from mannitol, trehalose, or combinations thereof. The proliposomal composition is prepared by a solvent evaporation technique to obtain a free-flowing powder capable of forming liposomal vesicles upon hydration. The optimized composition exhibits an average vesicle size of approximately 554.8 nm, a zeta potential of −17.45 mV, an entrapment efficiency of 76.3%, and a reconstitution time of approximately 52 seconds in phosphate-buffered saline at pH 7.4. The composition demonstrates cumulative in vitro drug release of approximately 90.02% over 24 hours in phosphate buffer at pH 7.4. The invention provides a lipid-based proliposomal drug delivery composition suitable for further development into solid oral dosage forms for tuberculosis management.

No. of Pages : 28 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131127 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : 5G NETWORK SECURITY MONITORING DEVICE WITH COMPACT GEOMETRIC CONFIGURATION

(51) International classification	:H04L 29/06, H04L 12/26, H04W 12/12, H04W 12/08, H04L 9/40
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahesh. D. S
Address of Applicant :Assistant Professor, Department of Electronics and Communication Engineering, SDM Institute of Technology, Ujire, Dakshina Kannada, Karnataka-574240, India. Dakshina Kannada Karnataka India
2)Dr. Mohan Naik R
3)Dr. Amith K Jain
4)Dr. Antheesh. R
5)Raghuveera Pandith. T. S
6)Niranjan Mamadapur
7)Radhakrishna
8)Amaresha Hebbar. A
(72)**Name of Inventor :**
1)Mahesh. D. S
2)Dr. Mohan Naik R
3)Dr. Amith K Jain
4)Dr. Antheesh. R
5)Raghuveera Pandith. T. S
6)Niranjan Mamadapur
7)Radhakrishna
8)Amaresha Hebbar. A

(57) Abstract :
5G NETWORK SECURITY MONITORING DEVICE WITH COMPACT GEOMETRIC CONFIGURATION ABSTRACT The present invention relates to a 5G network security monitoring device designed with a compact geometric configuration for efficient deployment in space-constrained communication environments. The device integrates multiple sensing, processing, and communication modules within a structurally optimized housing to continuously observe 5G network traffic, radio-frequency activity, access behavior, and security events. A built-in processing unit analyzes collected data to identify abnormal traffic patterns, unauthorized access attempts, signal interference, protocol anomalies, and potential cyber threats in real time. The geometric arrangement of internal components improves space utilization, thermal management, signal reception, and operational stability while reducing overall device dimensions. The device may support edge-based analytics, encrypted data transmission, configurable alert generation, and secure communication with centralized network management platforms. It can be deployed at base stations, edge nodes, enterprise networks, private 5G installations, and critical infrastructure locations. By combining compact construction with continuous security surveillance, the invention provides a practical solution for improving network visibility, accelerating threat detection, supporting rapid incident response, and strengthening the reliability and resilience of modern 5G communication systems, without requiring extensive installation space or complex infrastructure.

No. of Pages : 12 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131128 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RESETTABLE OPTICAL OVERPOWER LIMITER USING LIGHT-HEATED CAPILLARY THROATS AND LATCHING REFLECTIVE GAS CHAMBERS

(51) International classification	:G02B 6/42, G02F 1/313, G02B 6/26, G02F 1/015, G02B 6/122	(71)**Name of Applicant :** **1)BAPATLA SURENDRA BABU** Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India **2)IRUKUMALLI PRIYANKA** (72)**Name of Inventor :** **1)BAPATLA SURENDRA BABU** **2)IRUKUMALLI PRIYANKA**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

RESETTABLE OPTICAL OVERPOWER LIMITER USING LIGHT-HEATED CAPILLARY THROATS AND LATCHING REFLECTIVE GAS CHAMBERS A passive optical overpower limiter (100) blocks and records sustained overpower without electronics. Liquid-filled chambers (110, 112, 114, 116) cross an input waveguide (104) beyond the critical angle of a core-to-gas interface. Throats (120, 122, 124, 126), each thinner than its chamber, join a sealed gas space (118) to the chambers. A tap waveguide (108) heats an absorber pad (210) at the first throat (120); above the threshold the surface tension falls, gas fills the first chamber, the chamber latches and totally reflects the light into a dump waveguide (128). A gate coupler (130) on that dump waveguide heats the next throat, so the number of latched chambers records the peak power band. A reference vent (134), whose thickness is 1.026 to 1.047 times that of the first throat, sets the pressure difference and holds the threshold within 0.5 dB from -40 to $+85\text{ }^\circ\text{C}$. Light resets the limiter. FIG. 1

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131133 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI based Patient Health Monitoring System

(51) International classification	:A61B 5/00, A61B 5/0205, G16H 50/30, A61B 5/01, A61B 5/1455
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.Ancy S
Address of Applicant :Associate Professor, Department of Artificial Intelligence and machine learning, St.Joseph's College of Engineering, Chennai-119, Tamilnadu, India. Tamil Nadu India
2)Dr. S. Mohan
3)Dr. Vasujadevi Midasala
4)Durga R
5)Mathaneeswaran S
6)Gautham K
7)Balaji S
8)Prakash Raj M
9)Anushanth R S
10)Karthikeyan V
(72)**Name of Inventor :**
1)Dr.Ancy S
2)Dr. S. Mohan
3)Dr. Vasujadevi Midasala
4)Durga R
5)Mathaneeswaran S
6)Gautham K
7)Balaji S
8)Prakash Raj M
9)Anushanth R S
10)Karthikeyan V

(57) Abstract :
The present invention relates to an Artificial Intelligence-based patient health monitoring system for continuous monitoring and intelligent assessment of patient physiological conditions. The system comprises multiple biomedical sensors configured to acquire heart rate, body temperature, blood oxygen saturation, blood pressure, and respiratory rate. A data acquisition unit processes the acquired measurements and transmits the physiological data through a wireless communication module to an artificial intelligence processing unit. The AI processing unit employs machine learning algorithms to analyse real-time and historical health measurements, identify abnormal physiological variations, and generate patient-specific health risk assessments. A cloud-based storage module maintains patient health records, while a mobile application and healthcare dashboard provide remote access to monitoring information. An automated notification module generates alerts and communicates abnormal health conditions to authorized healthcare professionals and caregivers. The system facilitates continuous patient monitoring, early identification of potential health abnormalities, predictive health assessment, and remote healthcare supervision. The proposed invention provides an integrated and intelligent healthcare monitoring solution suitable for hospitals, intensive care units, elderly care facilities, and home-based patient monitoring environments.

No. of Pages : 13 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131134 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR AUTOMATED DATABASE QUERY OPTIMIZATION USING RUNTIME FEEDBACK

(51) International classification	:G06F 17/30, G06F 16/2455, G06F 16/2453, G06F 11/36, G06F 16/25
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. Meerasharif Sheik
2)Mantripragada Tilak
3)AVSM Ganesh
4)SNVSSST Murthy
5)Kadali S V Swamy

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR AUTOMATED DATABASE QUERY OPTIMIZATION USING RUNTIME FEEDBACK A system and method for database execution engines and feedback-driven query planning are disclosed. A processor receives normalized query fingerprints, parameter selectivity classes, operator row counts, elapsed times, memory use, spill volume and database snapshot identifiers. The processor is configured to attribute runtime feedback to a query and execution context, identify an operator-level estimation error, evaluate a semantics-preserving candidate under a resource budget, and promote the candidate only when comparable executions satisfy a guarded improvement test. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131135 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR CONTEXT-AWARE INFORMATION RETRIEVAL AND DYNAMIC RESULT RANKING

(51) International classification	:G06F 17/30, G06F 16/2457, G06F 16/33, G06F 16/332, G06F 16/9535
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Andhra Pradesh India
(72)**Name of Inventor :**
1)KOYYALA SRAVANI
2)Boggavarapu naga Lakshmi Prasanna
3)KONE SAI SRI MANASA
4)Mrs. BONAM SRIRAMA DURGA LAKSHMI

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR CONTEXT-AWARE INFORMATION RETRIEVAL AND DYNAMIC RESULT RANKING A system and method for computerized search and context-sensitive retrieval are disclosed. A processor receives query text, explicit task context, session identifiers, document versions, access-control labels, retrieval scores and interaction events. The processor is configured to generate a permission-filtered candidate set, derive a bounded context representation, rerank candidates using context relevance and freshness, and selectively recompute affected scores when an authorized context or document-version event occurs. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131136 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR PREDICTIVE FAILURE DETECTION IN COMPUTER SERVER INFRASTRUCTURE

(51) International classification	:G06F 11/07, G06F 11/30, G06F 11/34, G06F 11/00, H04L 12/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Andhra Pradesh India
(72)**Name of Inventor :**
1)Vinjamuri S N C H Dattu
2)Kamani Venkata Chandran
3)Mattaparthi Lakshmi Priyanka
4)Valavala Nalini Devi Prasanna

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR PREDICTIVE FAILURE DETECTION IN COMPUTER SERVER INFRASTRUCTURE A system and method for server telemetry processing and predictive infrastructure maintenance are disclosed. A processor receives processor temperatures, fan speeds, corrected memory errors, storage health counters, power readings, workload indicators and maintenance-event records. The processor is configured to align multi-source telemetry, condition expected behaviour on workload and hardware configuration, aggregate persistent residuals into a failure-risk estimate with an uncertainty state, and authorize a graded maintenance action only when evidence and service constraints permit. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131137 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR DETECTING MALICIOUS IOT DEVICES USING COMMUNICATION SIGNATURES

(51) International classification	:H04L 29/06, H04L 9/40, H04L 12/26, H04L 12/24, H04W 12/122
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Andhra Pradesh India
(72)**Name of Inventor :**
1)T S S PHANI
2)S S V S S R S SARMA ADITHE
3)P JAYASRI LAKSHMI
4)RYALI SATISH KUMAR

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR DETECTING MALICIOUS IOT DEVICES USING COMMUNICATION SIGNATURES A system and method for network security for Internet of Things installations are disclosed. A processor receives packet timestamps, packet lengths, communication direction, protocol identifiers, destination categories, connection outcomes and gateway port associations. The processor is configured to construct a device communication signature over overlapping observation windows, compare the signature with a versioned baseline for the device role and operating context, and require persistence or independent signature evidence before assigning a malicious-device state. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131138 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR SECURE IMAGE ENCRYPTION USING ADAPTIVE KEY GENERATION

(51) International classification	:H04L 9/08, H04L 9/32, G06F 21/60, H04L 9/30, H04L 9/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Andhra Pradesh India
(72)**Name of Inventor :**
1)Thota naga Lakshmi bhavani
2)Y S N CHANDRESWARI
3)BONTHU SUBRAHMANYAM
4)P.B.L.APARNA Kolla PrasannaLakshmi Venkata Bala

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR SECURE IMAGE ENCRYPTION USING ADAPTIVE KEY GENERATION A system and method for authenticated encryption and cryptographic image protection are disclosed. A processor receives image bytes, image identifiers, authenticated tenant identifiers, key-version references, cryptographically random salt, unique encryption nonces and security policy parameters. The processor is configured to select a key-generation policy for an image transaction, derive a per-image key from a protected master key and fresh context using a cryptographic key derivation function, encrypt the image with authenticated encryption, and authenticate the context before releasing plaintext. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131139 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR INTELLIGENT LOAD BALANCING ACROSS HETEROGENEOUS COMPUTING NODES

(51) International classification	:G06F 9/50, G06F 9/48, G06F 9/46, H04L 29/08, G06F 15/173
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Andhra Pradesh India
(72)**Name of Inventor :**
1)Adabala P V D Kumar
2)Suresh Sappa
3)Dr K VIjay Kumar
4)SNVSSS Trinadha Murthy
5)AVSM Ganseh

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR INTELLIGENT LOAD BALANCING ACROSS HETEROGENEOUS COMPUTING NODES A system and method for distributed task scheduling across heterogeneous computing resources are disclosed. A processor receives task resource requirements, processor and accelerator capabilities, available memory, queue estimates, locality information, reservation tokens and task execution feedback. The processor is configured to filter nodes by hard task constraints, estimate completion cost with node-specific runtime feedback and telemetry uncertainty, reserve capacity atomically on the selected node, and reconsider placement only when a hysteresis and migration-cost test is satisfied. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131140 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR ADAPTIVE CYBERATTACK DETECTION AND NETWORK-ISOLATION CONTROL

(51) International classification	:H04L 29/06, H04L 9/40, G06F 21/55, H04L 12/26, H04L 12/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Amalapuram Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr. Meerasharif Sheik
2)D P Sesha Ratnam
3)G.Tatayya Naidu
4)Ch. Suma
5)Ch shanthi
6)M.Surya kiran

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR ADAPTIVE CYBERATTACK DETECTION AND NETWORK-ISOLATION CONTROL A system and method for network intrusion detection and verified isolation control are disclosed. A processor receives network-flow records, authentication events, endpoint identity mappings, topology dependencies, baseline versions and enforcement acknowledgments. The processor is configured to correlate attack evidence across network and identity events, adapt baselines only with accepted benign observations, select the narrowest policy-compliant isolation scope, and verify both enforcement state and observed traffic before declaring containment. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131141 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR DETECTING DEEPFAKE MEDIA USING MULTIMODAL CONSISTENCY ANALYSIS

(51) International classification	:H04N 21/442, H04N 21/845, H04N 21/81, H04N 21/2343, H04N 21/44
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Andhra Pradesh India
(72)**Name of Inventor :**
1)G.Geetha Rani
2)M.Yashvanth
3)K.santhoshi Lakshmi
4)K Sravani
5)KVD Padmavathi
6)K.Venkatesh

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR DETECTING DEEPFAKE MEDIA USING MULTIMODAL CONSISTENCY ANALYSIS A system and method for audio-visual media integrity assessment are disclosed. A processor receives video frames, audio samples, presentation timestamps, media-container metadata, modality-quality estimates and available provenance records. The processor is configured to align available media modalities, measure quality-qualified cross-modal and temporal consistency, fuse only usable evidence with explicit uncertainty, and produce a localized suspected-manipulation or indeterminate result rather than treating absent evidence as authenticity. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202641131142 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR AUTOMATED DATA QUALITY ASSESSMENT AND ERROR CORRECTION

(51) International classification	:G06F 17/30, G06F 16/11, G06F 16/25, G06F 16/21, G06F 16/23
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bonam Venkata Chalamayya Institute of Technology and Science
Address of Applicant :Amalapuram, Andhra Pradesh. Andhra Pradesh India
(72)**Name of Inventor :**
1)Dr V.Ravi Kishore
2)Dr K Srinivas
3)Dr K Vijay Kumar
4)Dr. V Anurag
5)M.Venkata Rao

(57) Abstract :
ABSTRACT SYSTEM AND METHOD FOR AUTOMATED DATA QUALITY ASSESSMENT AND ERROR CORRECTION A system and method for data pipeline integrity and controlled record correction are disclosed. A processor receives input records, schema versions, validation constraints, reference-data snapshots, source provenance, correction candidates and downstream dependency metadata. The processor is configured to identify quality violations with provenance, generate evidence-supported correction candidates, apply a candidate only when confidence and cross-record constraints permit, and commit a versioned correction atomically while quarantining unresolved records. Intermediate results identify the observation and reference configuration. A controlled output interface provides the assessment or command to a corresponding consumer. An event record store retains supporting evidence, configuration versions and available execution feedback. Incomplete evidence is distinguished from an accepted assessment.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131146 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : USB CHARGING SECURITY HUB FOR CHARGING WITH SELECTIVE DATA-LINE ISOLATION

(51) International classification :H02J 7/00, H02J 7/02, G06F 1/26, H01R 13/66, G06F 13/38

(31) Priority Document No :NA

(32) Priority Date :NA

(33) Name of priority country :NA

(86) International Application No :

Filing Date :01/01/1900

(87) International Publication No : NA

(61) Patent of Addition to Application Number :NA

Filing Date :NA

(62) Divisional to Application Number :NA

Filing Date :NA

(71)**Name of Applicant :**

1)K.S.R. College of Engineering

Address of Applicant :KSR Kalvi Nagar, Tiruchengode - 637 215, Namakkal (Dt.), Tamil Nadu, India Tiruchengode Tamil Nadu India

(72)**Name of Inventor :**

1)Mr.K.Manikandan

2)Ms.K.Sudha

3)Mrs.G.Vinoly

4)Ms.T.Divya

5)Ms.R.Manoja

6)Mr.M.Hariomprakaas

7)Mr.M.Azhagesan

8)Mrs.N.Nandhini

9)Mrs.G.Angelpriya

10)Mr.P.Thirumurugan

11)Ms.D.Navasindhu

12)Mr.K.Anandarajan

13)Ms.M.Irfanshirin

14)Mr.K.Prakash

(57) Abstract :

A USB charging security hub (100) may be provided for charging an electronic device while restricting USB data communication. The hub (100) may have a USB input interface (102) for receiving power, a USB output interface (104) for connecting the electronic device, and a charging power path (106) that may connect the VBUS and GND conductors of the interfaces to supply charging power. A data-line control section (108) may be configured to selectively isolate the D+ and D- data lines, so that charging can continue without normal USB data communication. A sensing circuit (110) may monitor electrical activity on a data line, and an activity indication section (112) may be configured to provide a visual indication by a light emitting diode (114) and an audible indication by a buzzer (116) when a predefined electrical condition is detected. The hub (100) may be compact and low-cost.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131148 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : PIEZOELECTRIC CONTACT SPRING APPARATUS ISOLATING AND ENERGIZING A BATTERY AUTHENTICATION CIRCUIT AGAINST POWER ANALYSIS

(51) International classification	:G06F 1/26, H01M 2/10, H02J 7/00, H01M 10/44, H01M 10/42
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BAPATLA SURENDRA BABU
Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India
2)IRUKUMALLI PRIYANKA
(72)**Name of Inventor :**
1)BAPATLA SURENDRA BABU
2)IRUKUMALLI PRIYANKA

(57) Abstract :
PIEZOELECTRIC CONTACT SPRING APPARATUS ISOLATING AND ENERGIZING A BATTERY AUTHENTICATION CIRCUIT AGAINST POWER ANALYSIS An authentication apparatus (100) for a battery pack (150) protects the key of a cryptographic circuit (136) against power analysis and glitch injection through the pack terminals. Piezoelectric bimorph contact springs (106) join the circuit to fixed terminals (104) while a hold-down bar (118) presses them closed. A mechanical stroke of an actuator (114) releases the bar; the springs rise to a free gap (302), leaving the circuit floating, and their ringing charges a reservoir capacitor (128) through a rectifier (126). The key memory (138) is supplied only from the reservoir through a key gate (140), and the circuit through a blocking diode (134). A purge resistor (132) with a time constant of 0.30 to 0.70 times the open dwell time set by a timing escapement (122) empties the reservoir before the contacts close again. The apparatus is used in battery packs of unmanned aircraft and portable equipment. FIG. 1

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131150 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : LOW-POWER DUAL-LOOP FAST-TRANSIENT LOW-DROPOUT VOLTAGE REGULATOR WITH LOAD-TRACKING ZERO COMPENSATION AND DYNAMIC CASCODE BIASING

(51) International classification	:G05F 1/575, G05F 1/56, G05F 3/26, G05F 3/24, G05F 1/59	(71)**Name of Applicant :** **1)SOPHROSYNE TECHNOLOGIES PRIVATE LIMITED** Address of Applicant :First Floor, 1585/1, 20th Main Rd, Agara Village, 1st Sector, HSR Layout, Bengaluru - 560102, Karnataka, India Karnataka India (72)**Name of Inventor :** **1)Soman Purushothaman** **2)Manish Srivastava**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT LOW-POWER DUAL-LOOP FAST-TRANSIENT LOW-DROPOUT VOLTAGE REGULATOR WITH LOAD-TRACKING ZERO COMPENSATION AND DYNAMIC CASCODE BIASING Embodiments relate to a low-power dual-loop fast-transient low-dropout voltage regulator (100) comprising a slow/DC feedback loop (110) and a fast feedback loop (120). The fast feedback loop (120) includes a source-coupled MOS transistor arrangement sensing VOUT, a common-gate amplification arrangement, and a PMOS pass transistor (MPP) for regulation of output-voltage variations. A load-current-tracking zero compensation circuit (130) coupled to a fast-loop control node (VFL) includes variable resistance and capacitance for varying a compensation zero with load condition. A load-dependent dynamic cascode bias circuit (140) is coupled to VFL and includes a compensation network for stabilizing a bias-control path and controlling a cascode transistor (MPPCAS) coupled with the PMOS pass transistor (MPP) to maintain saturation voltage margin and improve power-supply rejection at VOUT. The common-gate amplification arrangement may include a PMOS current-mirror load having a magnitude varying inversely with a change in VOUT. The regulator (100) may further include a class-AB push-pull NMOS arrangement. FIG. 1

No. of Pages : 30 No. of Claims : 16

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131152 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE TRUST-WEIGHTED RFID IDENTITY AND PHYSICAL ENTRY CORRELATION SYSTEM FOR SECURITY GATES

(51) International classification	:G07C 9/00, G06K 7/10, G08B 13/14, G08B 13/24, G06K 19/07
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.S.R. College of Engineering
Address of Applicant :KSR Kalvi Nagar, Tiruchengode - 637 215, Namakkal (Dt.), Tamil Nadu, India Tiruchengode Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs.N.Nandhini
2)Mrs.G.Angelpriya
3)Ms.R.Manoja
4)Mr.K.Manikandan
5)Mrs.K.Sudha
6)Mrs.G.Vinoly
7)Mrs.T.Divya
8)Mr.M.Azhagesan
9)Mrs.P.Deepalakshmi
10)Mr.P.Gowtham
11)Mr.S.Dinesh
12)Mr.I.Navin Kumar
13)Mr.G.Dhanush
14)Mr.P.Deepak

(57) Abstract :
A system (100) for a security gate (190) may be provided for correlating an authenticated RFID identity with physical entry. The system (100) may have an RFID reader (102) that reads a credential (104) and generates an identity event, and physical sensors (106a, 106b, 108, 110, 112) that may provide person count, movement direction, spatial RFID observations, gate state and entry timing. An edge controller (120) may comprise an adaptive sensor trust engine (130) that computes event-specific trust values and adaptive weights for each sensing source, an evidence conflict resolver (150) that limits unreliable evidence, an identity-physical correlation engine (140) that determines whether the physical entry matches the identity event, and a decision engine (170) that may select an access response using risk and confidence scores. The system (100) may be configured to detect tailgating and credential misuse and to operate locally at the edge.

No. of Pages : 21 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131153 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR DETECTING COVERT COMMUNICATION IN AIR-GAPPED ENVIRONMENTS

(51) International classification	:G06N 20/00, H04W 4/38, H04B 17/318, H04B 17/00, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.S.R. College of Engineering
Address of Applicant :KSR Kalvi Nagar, Tiruchengode - 637 215, Namakkal (Dt.), Tamil Nadu, India Tiruchengode Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs.P.Deepalakshmi
2)Mr.M.Azhagesan
3)Mr.K.Manikandan
4)Mrs.G.Vinoly
5)Ms.T.Divya
6)Dr.K.Sudha
7)Mrs.G.Angelpriya
8)Ms.R.Manoja
9)Mr.M.Hariomprakaas
10)Mr.V.S.Hariragavan
11)Mr.G.Dharsin
12)Mr.D.Ragul
13)Ms.K.Mogitha
14)Mr.T.Chandru

(57) Abstract :
A system (100) for detecting covert communication from an air-gapped environment (10) may be provided. The system (100) may include a sensing unit (102) having sensors (104, 106, 108, 110) that may capture electromagnetic emissions, optical signals, acoustic signals or power consumption variations emanating from the environment (10). A signal conditioning unit (112) may filter and amplify the captured signals, and a feature extraction module (120) may extract features from the conditioned signals. A classification module (122) having a machine learning classifier (124) may be configured to classify the signal behaviour as normal or as indicative of covert communication. An alert unit (126) may be configured to generate an alert for review when covert communication is suspected. The system (100) may thereby function as a continuous monitor that identifies hidden communication attempts in isolated environments where network-based security tools are ineffective.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131154 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR DETECTING DEVICE TAMPERING AND INTRUSION WITH CYBER-RISK ASSESSMENT

(51) International classification	:G08B 13/16, H04L 29/06, G06F 21/55, G08B 29/18, G08B 25/00	
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(71)**Name of Applicant :**
1)K.S.R. College of Engineering
Address of Applicant :KSR Kalvi Nagar, Tiruchengode - 637 215, Namakkal (Dt.), Tamil Nadu, India Tiruchengode Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs.T.Divya
2)Mrs.G.Vinoly
3)Dr.K.Sudha
4)Ms.R.Manoja
5)Mr.K.Manikandan
6)Mrs.P.Deepalakshmi
7)Mr.M.Hariomprakaas
8)Mrs.G.Angelpriya
9)Mrs.N.Nandhini
10)Mr.N.Kishoor Kumar
11)Mr.V.Akshay
12)Mr.R.Thiyagesh
13)Ms.B.Gopika
14)Mr.T.Dharun

(57) Abstract :
A device tampering and intrusion detection system (100) may be provided. The system (100) may have a sensing unit (104) comprising a vibration sensor (106) that detects physical disturbance of a monitored asset (102) and a motion sensor (108) that detects human presence around the asset. A controller (110) may be configured to detect tampering and intrusion events, to monitor its communication with a remote server (124) so as to detect cyber-risk events such as communication failure, abnormal sensor behaviour and unauthorised control attempts, and to assign a risk level from a plurality of predefined risk levels. A communication module (116) may transmit an alert indicative of the risk level to a remote user device (126), which may also arm or disarm the system (100) and review an event log. The system (100) may function as a cost-effective security solution for ATMs, banks, offices and industrial facilities.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131155 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A WIRELESS SECURITY MONITORING APPLIANCE FOR REAL-TIME DETECTION OF UNKNOWN DEVICES AND METHOD THEREOF

(51) International classification	:G08B 13/196, H04N 7/18, G08B 25/10, G08B 25/01, G08B 25/00	(71)**Name of Applicant :** **1)K.S.R. College of Engineering** Address of Applicant :KSR Kalvi Nagar, Tiruchengode - 637 215, Namakkal (Dt.), Tamil Nadu, India Tiruchengode Tamil Nadu India (72)**Name of Inventor :** **1)Mr.M.Hariomprakaas** **2)Ms.R.Manoja** **3)Mrs.T.Divya** **4)Dr.K.Sudha** **5)Mrs.G.Vinoly** **6)Mr.M.Azhagsan** **7)Mrs.N.Nandhini** **8)Mrs.P.Deepalakshmi** **9)Mr.K.Manikandan** **10)Ms.V.Sashmitha** **11)Mr.S.Naveen** **12)Mr.C.S.Muhammed Nazar** **13)Ms.R.Sanjudharani** **14)Ms.S.Gowthami**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No	:	
Filing Date	:01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

A wireless security monitoring appliance (100) may be provided as a portable, low-cost miniature Security Operations Center. The appliance (100) may have a controller (102) with a wireless transceiver (104), a memory (118) storing known device identifiers (120), a display (106), a visual indicator (108), an audible indicator (110) and a user input element (112). The controller (102) may be configured to repeatedly scan the surrounding wireless environment, compare identifiers of detected devices with the known device identifiers (120) and classify unmatched devices as unknown. On such classification, a warning may be presented on the display (106) while the visual indicator (108) and audible indicator (110) are concurrently activated, and the alert may be reset through the user input element (112). The appliance (100) may function as a real-time, low-power tool for detecting unauthorised devices and rogue networks. A corresponding method (200) is also disclosed.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131160 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR AUTONOMOUS PREDICTIVE ORCHESTRATION OF CLOUD COMPUTING RESOURCES USING ARTIFICIAL INTELLIGENCE IN LARGE-SCALE RETAIL PLATFORMS

(51) International classification	:G06F 9/50, G06N 20/00, H04L 12/24, G06N 5/04, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Venkateswara Rao Movva
Address of Applicant :Software Engineer III, 2-85 Ravipadu, Pedanandipadu Mandal, Guntur District, Andhra Pradesh, PIN-522015 Guntur Andhra Pradesh India
(72)**Name of Inventor :**
1)Venkateswara Rao Movva

(57) Abstract :
SYSTEM AND METHOD FOR AUTONOMOUS PREDICTIVE ORCHESTRATION OF CLOUD COMPUTING RESOURCES USING ARTIFICIAL INTELLIGENCE IN LARGE-SCALE RETAIL PLATFORMS The present invention relates to an autonomous predictive cloud resource orchestration system (100) and method for managing distributed computing resources. The system (100) comprises a telemetry acquisition module (104), workload prediction engine (106), resource optimization engine (108), policy and constraint validation module (110), orchestration controller (112), infrastructure execution interface (114), performance feedback module (116), and anomaly detection and recovery module (118). The workload prediction engine (106) predicts future computational demand using artificial intelligence. The resource optimization engine (108) determines resource allocation configurations by evaluating predicted workloads, resource provisioning delays, and application service dependencies. The orchestration controller (112) coordinates infrastructure reconfiguration operations through the infrastructure execution interface (114). The performance feedback module (116) enables closed-loop adjustment of resource allocation decisions. Figure of abstract: FIG. 1.

No. of Pages : 27 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131161 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : EXPLAINABLE AI-ENABLED ELECTRIC VEHICLE CHARGING STATION USING COMPUTER VISION

(51) International classification	:B60L 53/66, B60L 11/18, B60L 53/65, B60L 53/14, B60L 53/30	(71)**Name of Applicant :** **1)JAYARAJU GADDALA** Address of Applicant :Rgukt ongole campus rk valley idupalapaya vempalli kapda Andhra Pradesh India **2)Dr. A. Sudhakar** (72)**Name of Inventor :** **1)Dr. A. Sudhakar** **2)M. Anitha** **3)Ch. Deepika**
(31) Priority Document No	:NA	**4)Mareswaramma Pilli**
(32) Priority Date	:NA	**5). D. Ritwik Prakash**
(33) Name of priority country	:NA	**6)A. Aruna Anil Kumar**
(86) International Application No Filing Date	: :01/01/1900	**7)T. Swati** **8)Md. Eliaz**
(87) International Publication No	: NA	**9). O. Teja Sri**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**10)Gadamsetti Sudhir**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Abstract of the Invention An explainable artificial intelligence (XAI)-enabled electric vehicle (EV) charging station system (100) uses computer vision to monitor a charging area. A camera (102) captures frames of the charging bay. An edge processing unit (104) preprocesses the frames and executes a trained computer-vision detection module (106) that detects the vehicle and charging-related visual conditions, such as vehicle position, bay occupancy, connector or cable position, and a person or obstacle in a monitored region. An explainability module (108) associates each detection with a confidence value and an explanation, such as a bounding box, highlighted region, heat map or reason label. A charging control interface (110) evaluates the detection against configured safety rules and sends an enable, inhibit, pause, warning or operator-attention indication to the EVSE controller (114). The frame, detection and explanation are shown on a display (112), and selected events are stored in local storage (116), reducing dependence on cloud connectivity

No. of Pages : 16 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131165 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT-DRIVEN SMART ENERGY GRID WITH REAL-TIME DEMAND AND SUPPLY OPTIMIZATION FOR URBAN SUSTAINABILITY

(51) International classification	:G06Q 50/06, H02J 13/00, H02J 3/32, H02J 3/38, H02J 3/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)FRANCIS XAVIER ENGINEERING COLLEGE
Address of Applicant :Tirunelveli - 627003, Tamilnadu, India Tirunelveli Tamil Nadu India
(72)**Name of Inventor :**
1)M. Robinson
2)Abishek U
3)Anton Abinesh B
4)Balakrishna M
5)Balavenkatesh S
6)David Samson A
7)Harish G K

(57) Abstract :
The present invention relates to an IoT-driven smart energy grid for real-time demand and supply optimization in urban electrical distribution networks. The system comprises an IoT sensor unit (100), a data acquisition unit (110), a communication unit (120), a data processing unit (130), a demand forecasting module (140), a renewable energy monitoring module (150), an energy storage management module (160), a demand-supply optimization module (170), a load control unit (180), and a monitoring and reporting unit (190). The IoT sensor unit acquires electrical parameters relating to electricity generation, consumption, distribution, and storage. The acquired measurements are processed to estimate electricity demand, determine renewable energy availability, and assess energy storage conditions. The demand-supply optimization module evaluates demand, available supply, and electrical operating constraints to determine energy allocation and load management instructions. The load control unit executes authorized instructions, while the monitoring and reporting unit records grid operating conditions and energy utilization information.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131166 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MACHINE LEARNING-ENABLED INTELLIGENT DIGITAL TWIN SYSTEM FOR SUSTAINABLE SMART CITIES

(51) International classification	:G06N 20/00, G06Q 10/06, G06Q 50/26, G06Q 10/00, G06Q 10/04	(71)**Name of Applicant :** **1)KINGS ENGINEERING COLLEGE** Address of Applicant :Kings Engineering College, Chennai - Bangalore Highway, Opposite to HYUNDAI MOTOR, Sriperumbudur, Irungattukottai, Chennai - 602117, Tamil Nadu, India Chennai Tamil Nadu India (72)**Name of Inventor :** **1)Dr. D C Jullie Josephine** **2)Dr. P. Udayasankaran** **3)Mr. I John R Dinakaran** **4)Mrs. Jennifa J** **5)Mrs. G Christina Mercy** **6)Mrs. M Divya** **7)Mrs. D Bharathi** **8)Ms. P. Divya Bharathi** **9)Ms. Chitra S** **10)Mrs. J Juanola Pearl**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a machine learning-enabled intelligent digital twin system (100) for sustainable smart cities. The system comprises a distributed urban data acquisition module (110), a data validation and normalization module (130), a geographical infrastructure mapping module (140), a digital twin engine (150), a machine learning analytics module (160), a predictive maintenance and demand forecasting module (170), a scenario simulation module (180), a sustainability assessment module (190), and an intelligent decision-support interface (200). Information received from urban sensors, connected devices, infrastructure systems, and municipal databases is validated and associated with corresponding physical urban assets to update their digital representations. Machine learning models analyse current and historical information to generate predictions relating to infrastructure demand, equipment conditions, resource utilization, and environmental parameters. The scenario simulation module evaluates alternative urban operating arrangements, while the sustainability assessment module determines selected environmental and resource-efficiency indicators. The system presents the results to authorized municipal personnel and may communicate approved operational instructions through a municipal infrastructure control interface (210). The invention provides an integrated digital environment for monitoring, predicting, simulating, and supporting sustainable management of interconnected urban infrastructure.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131176 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AMMONIUM CARBAMATE GAS ACTUATOR WITH SELECTIVE AMMONIA SORBENT FOR LOW-TEMPERATURE BATTERY CURRENT INTERRUPTION

(51) International classification	:H01H 37/76, H01H 37/32, H01H 37/54, H01M 10/0525, H01M 10/0567
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BAPATLA SURENDRA BABU
Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India
2)IRUKUMALLI PRIYANKA
(72)**Name of Inventor :**
1)BAPATLA SURENDRA BABU
2)IRUKUMALLI PRIYANKA

(57) Abstract :
AMMONIUM CARBAMATE GAS ACTUATOR WITH SELECTIVE AMMONIA SORBENT FOR LOW-TEMPERATURE BATTERY CURRENT INTERRUPTION A current interruption apparatus (120) for a sealed lithium-ion cell (100) actuates its pressure-actuated current interrupt device (108) at $78.7\,^\circ\text{C}$ without overcharge. A hermetic reservoir (204), thermally coupled to the electrode stack (104), holds a charge (206) of ammonium carbamate ($\text{NH}_2\text{COONH}_4$) behind a burst element (208) whose burst pressure is 1.8 to 4.4 times the carbamate dissociation pressure at $60\,^\circ\text{C}$, so the dissociation pressure fixes the release temperature. The charge decomposes into a gas-gland chamber (210), sealed from the electrolyte (114), where a solid acid sorbent bed (212) chemisorbs the ammonia. The remaining carbon dioxide exceeds the dissociation-limited pressure and drives a pressure-responsive member (216) against the current interrupt device (108). Without the sorbent, actuation occurs only at $93.2\,^\circ\text{C}$. The apparatus needs no power or maintenance, keeps gas out of the electrolyte, and actuates the device if the reservoir leaks. FIG. 1

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131177 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Solar-Powered IoT-Enabled Cutting Machine with Ultrasonic Obstacle Avoidance and Remote Monitoring

(51) International classification	:H02J 7/35, A01D 34/00, A01D 34/68, A01D 34/82, A01D 34/63
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)DR RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY

Address of Applicant :Dr RVR NRI Institute of Technology Deemed to be University, Agiripalli, Pothavarappadu, Vijayawada Rural, Andhra Pradesh, India. Vijayawada Andhra Pradesh India

2)Ogeti Naga Venkata Pavan Bhagavan Kumar

3)Gorantla Durga Prasad

4)Latike Lokesh

5)Atla Devi Sri Charan

6)Neelapala Hari Govardhan

7)Lukka Geetha Krishna

8)Koduru Charan Teja

9)Arepalli Rakesh

(72)**Name of Inventor :**

1)Ogeti Naga Venkata Pavan Bhagavan Kumar

2)Gorantla Durga Prasad

3)Latike Lokesh

4)Atla Devi Sri Charan

5)Neelapala Hari Govardhan

6)Lukka Geetha Krishna

7)Koduru Charan Teja

8)Arepalli Rakesh

(57) Abstract :

ABSTRACT [0033] An IoT based solar cutter (10) is disclosed for fuel-free cutting of grass and vegetation. The cutter comprises a wheeled chassis (12) carrying a photovoltaic panel (20) on an elevated support frame (22), a rechargeable battery pack (24) charged through a charging and power regulation circuit (46), a cutting motor (36) with a cutter blade (38) at the front end, traction motors (34), first and second motor drivers (30, 32), an ultrasonic sensor (40) and a Wi-Fi enabled microcontroller unit (28). The microcontroller unit (28) receives commands from a user device (44) through a cloud server (42), controls the traction and cutting motors, stops and turns the cutter when the ultrasonic sensor (40) detects an obstacle nearer than a threshold distance, and sends status information such as battery condition and obstacle distance to the user device (44). The cutter runs on solar energy and is supervised remotely.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131179 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AUTOMATED ROBOTIC SYSTEM FOR HIGH RISE GLASS FAÇADE CLEANING

(51) International classification	:A47L 1/02, A47L 11/38, E04G 23/00, B25J 11/00, A47L 11/40	(71)**Name of Applicant :** **1)Venkatanarayanan P S** Address of Applicant :891 / S.L.C-13, NORTH HOUSING UNIT, SELVAPURAM, Tamil Nadu India **2)Mr. V. VARATHARASAN** **3)Mr. U. LIBAN** **4)Mr. K. NITHISH** **5)Mr. K. NITHISHWARAN**
(31) Priority Document No	:NA	**6)Mr. L.P. PRAJWEL**
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Venkatanarayanan P S**
(86) International Application No	:	**2)Mr. V. VARATHARASAN**
Filing Date	:01/01/1900	**3)Mr. U. LIBAN**
(87) International Publication No	: NA	**4)Mr. K. NITHISH**
(61) Patent of Addition to Application Number	:NA	**5)Mr. K. NITHISHWARAN**
Filing Date	:NA	**6)Mr. L.P. PRAJWEL**
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :
The present invention relates to an Automated Robotic System for High-Rise Glass Facade Cleaning, designed to automate the cleaning of vertical glass surfaces in high-rise buildings while reducing human involvement and occupational hazards. The proposed system comprises a structural frame, an adhesion mechanism, a motorized movement arrangement, and an integrated cleaning mechanism incorporating brushes or wipers and a water-spraying unit. The adhesion mechanism enables the robot to maintain contact with vertical glass surfaces, while the motorized arrangement facilitates controlled movement across the facade. The cleaning unit removes accumulated dust, dirt, and other surface contaminants through coordinated cleaning and water-spraying operations. The system is remotely operated, allowing users to control its movement and cleaning functions from a safe location. Suitable safety and monitoring arrangements may also be incorporated to improve operational stability and reliability. The invention aims to provide a practical, compact, and cost-effective solution for high-rise glass facade maintenance by improving cleaning efficiency, reducing manual labour, and minimizing the risks associated with conventional cleaning methods.

No. of Pages : 10 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131180 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Hydro Whale Edge-AI Biomimetic Flapping Hydrokinetic Energy Converter

(51) International classification	:F03B 17/06, F03B 13/26, F03B 13/00, F03B 13/10, F03B 3/12
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Venkatanarayanan P S
Address of Applicant :891 / S.L.C-13, NORTH HOUSING UNIT, SELVAPURAM, Tamil Nadu India
2)Dr.G.Yuvaraj
3)Mr. R. MAHESH
4)Mr. P.S. EZHIL
5)Mr. K. MANIKANDAN
6)Mr. M. NISHANTH

(72)**Name of Inventor :**
1)Venkatanarayanan P S
2)Dr.G.Yuvaraj
3)Mr. R. MAHESH
4)Mr. P.S. EZHIL
5)Mr. K. MANIKANDAN
6)Mr. M. NISHANTH

(57) Abstract :
The present invention relates to a Biomimetic Flapping-Wing Hydrokinetic Energy Conversion System named HydroWhale, designed to harness the kinetic energy of flowing water and convert it into useful mechanical and electrical energy through a whale-flipper-inspired wing mechanism. The proposed system comprises a biomimetic flapping wing, supporting frame, pivot arrangement, mechanical transmission mechanism, and a power take-off unit connected to an electrical generator. The flapping-wing mechanism utilizes hydrodynamic forces generated by flowing water to produce oscillating movement, while the mechanical transmission arrangement transfers this movement to the power generation unit. The system may incorporate sensors and a microcontroller-based monitoring arrangement to measure operating parameters such as wing displacement, oscillation frequency, water-flow rate, voltage, and current. An IoT-based monitoring system may also be integrated to support data acquisition, transmission, and remote observation of operating conditions. The invention aims to provide an alternative and practical approach to hydrokinetic energy conversion through biomimetic design, prototype fabrication, and experimental evaluation, contributing to the development of small-scale renewable energy applications.

No. of Pages : 12 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131181 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIO-KEY WHICH HAS A BIOMETRIC AUTHORIZATION AND LOCK ACCESS SYSTEM WITH PORTABLE CREDENTIAL VAULT AND CONTROLLED POWER TRANSFER

(51) International classification	:G07C 9/00, G06F 21/32, E05B 49/00, G06K 9/00, H04L 9/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Third Eye Locks Private Limited
Address of Applicant :18/21, Tiruneermalai Main Road, Nagalkeni, Chromepet Chennai Tamil Nadu India
(72)**Name of Inventor :**
1)J Jayavasanth
2)Pernetti Chandra Sekhar
3)Sekar H

(57) Abstract :
The present invention relates to a biometric authorization and lock access system comprising a portable authorization device referred to as a Bio-Key and one or more compatible electronic locks. The Bio-Key includes a fingerprint sensor, processor, secure memory, rechargeable battery, wireless communication interfaces and a USB Type-C connector. During enrollment, a user's fingerprint is captured and a biometric profile is generated and stored exclusively within the Bio-Key. Authorization credentials, access rights, lock records, security certificates and related access information are also stored within the Bio-Key, thereby forming a portable credential vault. When access to a lock is required, the Bio-Key performs local fingerprint authentication and, upon successful verification, releases authorization information for communication to the lock. The lock validates the received authorization information and executes a lock or unlock operation. The user's biometric profile is neither transmitted to nor stored within the lock. In certain embodiments, the Bio-Key communicates with a mobile application and a remote server through Bluetooth and Wi-Fi for registration, credential updates and access management. Operational power may additionally be supplied from the Bio-Key to the lock, wherein power transfer is enabled only after successful biometric authentication within the Bio-Key and authorization validation within the lock. The invention enables a single Bio-Key to securely operate multiple locks while maintaining user-controlled biometric ownership and eliminating biometric data storage within the lock.

No. of Pages : 18 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131191 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Method for Fabricating Drug-Loaded Nanohydroxyapatite and Polymer Nanofiber Coatings on 316L Stainless Steel Implants

(51) International classification	:A61L 27/34, A61L 27/54, A61L 31/10, D04H 1/728, D01D 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)MS.S.JAYACHITRA
Address of Applicant :Associate Professor Dept of Chemistry, S.A. Engineering college,.Chennai Chennai Tamil Nadu India
2)Ms. V. O. SANGEETHA
3)S.A.Engineering College (An Autonomous Institution)
(72)**Name of Inventor :**
1)MS.S.JAYACHITRA
2)Ms. V. O. SANGEETHA

(57) Abstract :
The present invention relates to a VAN-loaded nHAP and collagen–PVA bilayer coating developed on 316L SS for potential biomedical implant applications. The coating comprises an nHAP base layer deposited by EPD, followed by a VAN-containing collagen–PVA nanofiber layer fabricated using electrospinning. The combination of ceramic and polymeric materials is intended to provide a bioactive surface with antibacterial functionality. The developed coating is characterized using suitable analytical techniques to evaluate its structural, morphological, chemical, and surface properties. Its biological performance is investigated through antibacterial assays against selected bacterial strains and in-vitro cell viability and cell-migration studies. The proposed coating system aims to combine the functional properties of nHAP, collagen, and PVA with the antibacterial action of VAN, offering a potential surface-modification approach for metallic implants intended for bone-contacting applications.

No. of Pages : 5 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131195 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Graph-Aware Cyber Range Compiler with Failure-Localized Targeted Regeneration

(51) International classification	:G06T 19/00, G09B 5/06, G06F 3/01, G06K 9/00, G06F 3/0481	(71)**Name of Applicant :** **1)Mahendra Engineering College** Address of Applicant :Mahendra Engineering College, Mahendirapuri, Mallasamudram, Namakkal, Tamilnadu, India. Pin: 637503. Tamil Nadu India (72)**Name of Inventor :** **1)Dr.A.RAJAMURUGAN** **2)Dr.A.NEELA MADHESWARI**
(31) Priority Document No	:NA	**3)Dr.B.KALAISELVI**
(32) Priority Date	:NA	**4)Ms.R.KALAIVANI**
(33) Name of priority country	:NA	**5)Mr.R.BALAMURUGAN**
(86) International Application No	:	**6)Mr.M.SAMPATH**
Filing Date	:01/01/1900	**7)Ms.S.SHURITHI**
(87) International Publication No	: NA	**8)Mr.R.DINESH**
(61) Patent of Addition to Application Number	:NA	**9)Mr.S.GOKULNATH**
Filing Date	:NA	**10)Mr.S.CHANDRU**
(62) Divisional to Application Number	:NA	**11)Mr.S.AJAY**
Filing Date	:NA	

(57) Abstract :

An augmented reality system and method for enriching printed or digital textbooks is disclosed. A portable device stores a content pack comprising a page recognition index, page layout map, overlay assets and variants, and assessment items. A recognition module identifies textbook pages without printed markers and computes page pose. An adaptive content selector selects overlay variants according to a learner profile, device capability profile and connectivity state. A rendering engine places the selected variants and adjusts rendering level of detail according to runtime parameters. Assessment, interaction and logging operate on the device and remain available offline. A synchronisation manager transmits learning events and receives integrity-checked content updates when connectivity is restored. Teacher authoring, accessibility validation and concept-level analytics are supported, while camera images remain on the device.

No. of Pages : 21 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131205 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUGMENTED REALITY TEXTBOOK OVERLAY SYSTEM WITH ADAPTIVE MULTIMEDIA CONTENT ENRICHMENT AND INTERACTIVE LEARNING INTEGRATION

(51) International classification	:G06T 19/00, G09B 5/06, G06F 3/01, G06K 9/00, G06F 3/0481	(71)**Name of Applicant :** **1)Mahendra Engineering College** Address of Applicant :Mahendra Engineering College, Mahendirapuri, Mallasamudram, Namakkal, Tamil Nadu, India. PIN: 637503. Namakkal Tamil Nadu India (72)**Name of Inventor :** **1)Dr.A.RAJAMURUGAN** **2)Dr.A.NEELA MADHESWARI** **3)Dr.B.KALAISELVI** **4)Ms.R.KALAIVANI** **5)Mr.R.BALAMURUGAN** **6)Mr.M.SAMPATH** **7)Ms.S.SHURITHI** **8)Mr.T.DINESH** **9)Mr.M.MANOJ** **10)Mr.K.DHIVAGAR** **11)Mr.A.MONISH**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An augmented reality system and method for enriching printed or digital textbooks is disclosed. A portable device stores a content pack comprising a page recognition index, page layout map, overlay assets and variants, and assessment items. A recognition module identifies textbook pages without printed markers and computes page pose. An adaptive content selector selects overlay variants according to a learner profile, device capability profile and connectivity state. A rendering engine places the selected variants and adjusts rendering level of detail according to runtime parameters. Assessment, interaction and logging operate on the device and remain available offline. A synchronisation manager transmits learning events and receives integrity-checked content updates when connectivity is restored. Teacher authoring, accessibility validation and concept-level analytics are supported, while camera images remain on the device.

No. of Pages : 19 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131209 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT BASED IV BAG MONITORING AND SWITCHING SYSTEM

(51) International classification	:A61M 5/168, A61M 5/14, A61M 5/172, A61M 5/00, H04W 4/14	(71)**Name of Applicant :** **1)SARAVANA KUMAR M N** Address of Applicant :308/253, PLOT 35, SECOND STREET, AGAN NAGAR Tamil Nadu India **2)Dr. SARAVANA KUMAR M N** **3)Dr. PRABHADEVI NARAYANAN** **4)Dr. SARATH BABU VENKATESAN** (72)**Name of Inventor :** **1)Dr. SARAVANA KUMAR M N** **2)Dr. PRABHADEVI NARAYANAN** **3)Dr. SARATH BABU VENKATESAN**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An IoT-based IV bag monitoring and switching system is described for real-time tracking of fluid administration and automated flow control. The system uses a load cell attached to an IV bag stand and an HX711 amplifier to measure the weight of the fluid. An ATmega328P microcontroller processes the weight signals. Converts them into percentage levels. These levels are then sent over Wi-Fi using an ESP8266 module to a remote IoT server. The system includes a multi-tier alert mechanism. At 40% level a visual LED indicator lights up. When fluid drops to 20% the system sends an SMS via GSM. At 10% remaining it makes a phone call. Activates a buzzer alarm. An optical sensor also monitors the IV line for air bubbles. If the system detects a bag or air bubbles the microcontroller triggers a solenoid valve, through a relay module. This action immediately seals the IV tubing. This prevents air embolism. Stops blood backflow. The system ensures safety by responding quickly to critical conditions.

No. of Pages : 10 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131210 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WEARABLE ASSISTIVE DEVICE AND METHOD FOR DISPLAYING CAPTION TEXT SPATIALLY ASSOCIATED WITH SPEECH SOURCES

(51) International classification	:G10L 15/22, G06F 3/01, G10L 15/26, H04R 3/00, G06F 3/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)VANTESH AND GINNI TECHNOLOGIES PRIVATE LIMITED
Address of Applicant :7-1-25/A, B-505, IMPERIAL PRESTIGE, LEELANAGAR, AMEERPET, HYDERABAD – 500016, TELANGANA, INDIA HYDERABAD Telangana India
(72)**Name of Inventor :**
1)VANTESH BAKSHI

(57) Abstract :
The present invention discloses a wearable assistive device and a method for displaying caption text spatially associated with one or more speech sources. A direction estimation subsystem (128) estimates an acoustic direction of arrival from inter-channel time differences. A gaze estimation subsystem (130) determines a gaze direction and a gaze confidence value. A steering control subsystem (132) determines a control direction combining gaze and acoustic directions, weighted by the gaze confidence value and shifting toward the acoustic direction of arrival as confidence decreases. A beamforming subsystem (134) generates a spatially filtered speech signal. A speaker diarization subsystem (136) associates speech segments with anonymous speaker tracks. A speech recognition subsystem (138) generates caption text. A caption rendering subsystem (140) renders the caption text at discrete display zones in each of a pair of optical display lenses (108), selected by the control direction or the acoustic direction of arrival. FIG. 1

No. of Pages : 107 No. of Claims : 26

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202641131215 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ANTIBACTERIAL GAUZE INFUSED WITH AEGLE MARMELOS (VILVAM) LEAF AND ANANAS COMOSUS (PINEAPPLE) CROWN EXTRACTS MADE FROM BANANA PSEUDOSTEM FIBER

(51) International classification	:A61L 15/28, A61L 15/44, A61F 13/00, A61L 15/46, A61L 15/42	(71)**Name of Applicant :** **1)Madhankumar C** Address of Applicant :1/x kg thottam, sellappampalayam,kabulipalayam,kovil palayam post, pollachi taluk,642110 Tamil Nadu India **2)Mr. Avinash K - Rathinam Technical Campus** **3)Ms. Jaivarshini E - Rathinam Technical Campus** **4)Ms. Hinduja G - Rathinam Technical Campus** **5)Ms. Keerthana L - Rathinam Technical Campus** **6)Ms. Nivedha D - Rathinam Technical Campus** **7)Mr. Raguraj S - Rathinam Technical Campus**
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Mr. Avinash K - Rathinam Technical Campus**
(33) Name of priority country	:NA	**2)Ms. Jaivarshini E - Rathinam Technical Campus**
(86) International Application No	:	**3)Ms. Hinduja G - Rathinam Technical Campus**
Filing Date	:01/01/1900	**4)Ms. Keerthana L - Rathinam Technical Campus**
(87) International Publication No	: NA	**5)Ms. Nivedha D - Rathinam Technical Campus**
(61) Patent of Addition to Application Number	:NA	**6)Mr. Raguraj S - Rathinam Technical Campus**
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

ABSTRACT The present invention describes a biodegradable antibacterial wound dressing prepared using banana pseudo stem fiber and plant extracts. The dressing consists of natural banana pseudo stem fibers infused with extracts of Aegle marmelos (Vilvam) leaves and Ananas comosus (pineapple) crown. Banana pseudo stem, an agricultural by-product, is used as the main fiber material for preparing the gauze. The pseudo stem is processed to separate the fibers, which are cleaned, treated to remove unwanted non-cellulosic materials, washed, dried, and formed into a gauze or braided fibrous structure. Separately, Aegle marmelos leaves and Ananas comosus crowns are cleaned and processed to prepare aqueous extracts containing naturally occurring bioactive compounds. The prepared plant extracts are infused into the banana pseudo stem fiber gauze so that the bioactive compounds are retained within the fiber matrix. Phytochemical screening of the Aegle marmelos leaf and Ananas comosus crown extract shows the presence of alkaloids, flavonoids, tannins, saponins, terpenoids, steroids and phenolic compounds. These phytochemicals are associated with the antibacterial activity of the extracts. The developed herbal-infused gauze shows antibacterial activity against selected microorganisms associated with wound contamination. The combined use of the two plant extracts provides antibacterial activity, while the banana pseudo stem fiber provides a biodegradable and plant-based material for carrying the extracts. The invention provides a wound dressing made from renewable plant materials and agricultural biomass. It can be used as an alternative plant-based material for wound dressing applications where antibacterial activity and biodegradability are desired.

No. of Pages : 15 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131221 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SECURE EMBEDDED COMMUNICATION SYSTEM WITH FAULT DETECTION AND SELF-RECOVERY FOR REAL-TIME INDUSTRIAL AUTOMATION

(51) International classification	:H04L 29/06, H04L 9/32, G05B 19/418, H04L 12/24, G05B 19/05
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.S.Bharathiraja
Address of Applicant :Assistant Professor, Department of Electronics and Communication Engineering, A.V.C College of Engineering, Mayiladuthurai-609305. Email: bharathiavcraja@gmail.com Tamil Nadu India
2)Nivedita V S
3)Dr.Anlin Sahaya Infant Tinu M
4)Sendil Vadivu D
5)Dr.M.Saravanan
6)Rajan S

(72)**Name of Inventor :**
1)Dr.S.Bharathiraja
2)Nivedita V S
3)Dr.Anlin Sahaya Infant Tinu M
4)Sendil Vadivu D
5)Dr.M.Saravanan
6)Rajan S

(57) Abstract :

A secure embedded communication system (100) comprising a data acquisition unit (102), an embedded communication interface (104), a security authentication unit (106), an encryption and decryption unit (108), a communication monitoring unit (110), a fault detection unit (112), a fault diagnosis unit (114). The data acquisition unit (102) receives industrial operating data, sensor measurements, machine parameters, equipment status information, and communication signals from interconnected industrial devices and controllers. The embedded communication interface (104) establishes communication between industrial devices, programmable logic controllers, sensors, actuators, and industrial computing units. The security authentication unit (106) verifies device identities and access authorization, and the encryption and decryption unit (108) encrypts outgoing data and decrypts incoming data to protect communication confidentiality and integrity. The communication monitoring unit (110) monitors network connectivity, transmission latency, packet loss, communication response time, and data transfer status. The fault detection unit (112) identifies communication abnormalities and transmission failures, and the fault diagnosis unit (114) analyses detected abnormalities and determines possible causes of communication failures. The invention enhances communication reliability, data security, and fault detection in real-time industrial automation. The invention facilitates automated fault diagnosis and self-recovery, thereby reducing communication downtime and improving operational continuity.

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131245 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Artificial Intelligence-Based for Ambulance Patient Detection and Siren Access Verification Using YOLOv8-Based Computer Vision with Patient Localization, Confidence-Based Decision Logic and GPS-Enabled Pickup Verification

(51) International classification	:G16H 40/20, G16H 40/67, G16H 10/60, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Prof. Ambili K
Address of Applicant :Assistant Professor, Department of Computer Science and Engineering(Artificial Intelligence and Machine Learning), AMC Engineering College, Bannerghatta Rd, Kalkare, Bengaluru, Karnataka 560083 Karnataka India
2)Dr. Swasti Patel
3)Prof. Bhavya Bharatan K
4)Prof. Vinaya S Kavalgi
5)Mohammed Touseef
6)R Karthik
7)Merin Shibu
8)Mahek J

(72)**Name of Inventor :**
1)Prof. Ambili K
2)Dr. Swasti Patel
3)Prof. Bhavya Bharatan K
4)Prof. Vinaya S Kavalgi
5)Mohammed Touseef
6)R Karthik
7)Merin Shibu
8)Mahek J

(57) Abstract :
ABSTRACT OF THE INVENTION The present invention relates to an artificial intelligence-based system and method for ambulance patient verification and siren access management. An ambulance unit captures an image of the stretcher area and provides GPS location information. A server-side gateway forwards the image to a patient detection engine comprising image pre-processing, YOLOv8-based person detection, patient-presence classification, patient localization, and verdict generation. A siren access decision module determines whether a patient is detected on the stretcher. If no patient is initially detected, the module verifies an active emergency case and patient location, enables GPS-based pickup tracking where applicable, and evaluates patient presence after arrival. The system generates an access-granted or access-denied status and provides relevant information through driver and police dashboards. An SQLite-based data storage module maintains emergency cases, captured images, AI verdicts, access decisions, GPS history, and fake-access reports. The system provides an integrated workflow for visual patient verification and location-assisted emergency-trip validation.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131246 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART EV FLEET INTELLIGENCE AND PREDICTIVE MAINTENANCE PLATFORM

(51) International classification	:G06N 20/00, G07C 5/08, G07C 5/00, G06Q 10/00, G06Q 50/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. Poonkodi P
2)Ashwin S
3)Hiruthuvanthika B
4)Rahul K
5)Vishal J

(57) Abstract :
The present invention relates to a Smart EV Fleet Intelligence and Predictive Maintenance Platform designed to monitor electric vehicle performance, analyze vehicle telemetry data, and identify potential maintenance requirements using Artificial Intelligence (AI) and Machine Learning (ML). The proposed system integrates vehicle telemetry collection, data validation, database management, predictive analysis, fleet monitoring, and automated alert generation to support efficient electric vehicle fleet management. The system collects vehicle parameters, including battery percentage, battery temperature, motor temperature, vehicle speed, and charging status, through a Python-based telemetry simulator. The collected data is transmitted to a FastAPI-based backend, validated, and stored in a MySQL database using SQLAlchemy. The stored telemetry data is processed through data preprocessing and feature extraction techniques to support machine learning-based predictive maintenance using algorithms such as Random Forest and Logistic Regression. The platform provides a centralized fleet dashboard to monitor vehicle status, analyze performance information, and identify abnormal operating conditions. The system is also designed to support automated maintenance alerts and fleet reports through workflow automation. By integrating telemetry monitoring, predictive analysis, and centralized fleet management, the proposed platform aims to identify potential vehicle issues earlier, reduce unexpected breakdowns, improve maintenance planning, and enhance the operational efficiency and reliability of electric vehicle fleets.

No. of Pages : 7 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131247 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-POWERED HYBRID E-COMMERCE MARKETPLACE

(51) International classification	:G06Q 30/08, G06Q 30/00, G06Q 30/06, G06Q 30/02, G06Q 40/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. R. Saranya
2)Surjith. K
3)Surya Prakash K
4)Navin Prabhu G

(57) Abstract :

This paper presents an AI-powered hybrid e-commerce marketplace designed to bring Fresh/New, Used, and Refurbished products into one platform while keeping each category clearly separated for browsing and comparison. The marketplace supports three transaction modes across all categories: fixed price, negotiable price, and auction. Its proposed AI assistance focuses on four functions: seller-side product description generation, buyer-side product questions and summaries, a trust score based on verified-purchase reviews, and guided auction and negotiation support. The product description generator uses facts entered by the seller to prepare a clear listing draft that the seller can review and edit. On the buyer side, a product Q&A assistant answers questions using the available listing information and should indicate when a detail is missing rather than inventing facts. The proposed trust score uses verified-purchase reviews only; a trusted badge may be considered when the average rating reaches a proposed threshold of 4.2 out of 5, subject to review and validation. For auctions, sellers set a starting price and a maximum duration of 24 hours, while buyers enter their maximum bid. Proxy bidding raises a buyer's bid only by the minimum increment needed to remain competitive and never exceeds the buyer's stated maximum; buyers are notified when they are outbid. The negotiation assistant compares available market-value signals and uses product condition, photos, and description details to suggest a reasonable offer range with a short rationale. These functions aim to make product information clearer, transactions more flexible, and pricing decisions easier to understand. The marketplace is proposed as a unified shopping and selling experience that addresses common challenges in new and second-hand commerce: incomplete listings, uncertainty about condition, difficulty judging fair value, and the need for flexible payment or bidding choices. The system is a project proposal; performance, usability, and trust outcomes must be validated through implementation and user testing.

No. of Pages : 9 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131248 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ZERO-TRUST API KEY EXPOSURE MITIGATOR FOR CI/CD PIPELINES

(51) International classification	:H04L 9/32, H04L 29/06, G06F 21/62, G06F 21/31, G06F 21/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Eazhil. K. M
2)Sagadevan K
3)Vashitha M
4)Vinisha S
5)Balaji N

(57) Abstract :
The present invention relates to a secure code scanning system configured to automatically identify and prevent the exposure of sensitive credentials in source code. The proposed system provides an integrated security platform combining source code scanning, regex-based secret detection, automatic redaction, Git pre-push validation, and secure credential management. The system scans source code for predefined sensitive information, including API keys, passwords, access tokens, and other confidential credentials. Regex-based patterns are used to identify potential secrets within source files. When a sensitive credential is detected, the system flags the security issue and replaces the exposed value with a safe placeholder such as REDACTED , according to the configured scanning workflow. The system integrates a Git pre-push hook that automatically executes the security scanner before code is uploaded to GitHub. This helps developers identify and address exposed credentials before pushing their code to a remote repository. The scanning mechanism is designed to reduce accidental disclosure of sensitive information during software development and deployment. The invention provides an integrated workflow in which source code is scanned, potential secrets are identified using predefined patterns, detected credentials are flagged or redacted, and code is checked before being pushed to a remote repository. The system aims to reduce credential leakage, improve software development security, and provide a structured mechanism for protecting sensitive information throughout the development workflow.

No. of Pages : 6 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131249 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MOTOLINK: AI-ENABLED SMART TWO-WHEELER SERVICE AND SPARE PARTS MARKET PLACE

(51) International classification	:G06Q 10/02, G06Q 30/06, G06Q 50/30, G06Q 30/02, H04W 4/021
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. K. Kalaivani
2)MIRTHUL SHANJAY T
3)MOHAN RAJ R
4)AJAY P

(57) Abstract :
MotoLink is an AI-enabled web platform that connects two-wheeler owners with verified mechanics, service centers, spare-parts dealers, and roadside assistance providers. It brings essential vehicle services together in one convenient place. Using location-based service discovery and AI-supported recommendations, MotoLink helps users find suitable services, compare providers, book maintenance appointments, purchase spare parts, and request emergency roadside assistance. Users can manage bike details, review provider profiles and ratings, track bookings, and view service history through a browser-based dashboard. Providers can manage business profiles, services, prices, availability, and customer requests. MotoLink integrates user management, service discovery, booking, spare-parts shopping, provider management, and roadside assistance into one workflow. It aims to reduce the time and effort needed to maintain a two-wheeler and improve access to reliable local services.

No. of Pages : 5 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131250 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-DRIVEN ENTERPRISE DECISION INTELLIGENCE PLATFORM USING PREDICTIVE ANALYTICS AND EXPLAINABLE AI

(51) International classification	:G06Q 30/02, G06N 20/00, G06Q 10/06, G06N 5/04, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048 Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. K. Kalaivani
2)Akash S
3)Mouleeswaran P S
4)Syed Wajid A
5)Balamanikandan C

(57) Abstract :
The present invention relates to an Artificial Intelligence (AI)-based Enterprise Decision Intelligence Platform (EDIP) configured to automatically analyze business data and generate predictive insights, explanations, and actionable recommendations. The proposed system provides a unified decision intelligence platform combining data integration, predictive analytics, customer analytics, anomaly detection, Explainable AI (XAI), recommendation generation, application programming interfaces (APIs), a Retrieval-Augmented Generation (RAG) chatbot, and an interactive business intelligence dashboard. The system receives raw business data from configured data sources and processes the information for analytical purposes. The established analytical functions include forecasting future revenue and product demand, segmenting customers and predicting churn risk, detecting unusual transaction patterns, and identifying inventory-related risks. The system evaluates the processed data using analytical and AI-based techniques to generate predictions and insights. Explainable AI provides information regarding the factors influencing the generated predictions, while the recommendation engine suggests relevant actions based on the analyzed business information. The generated insights and recommendations are presented through a unified dashboard accessible to authorized users through a web-based interface. The dashboard provides business performance indicators, revenue projections, demand forecasts, inventory monitoring, stockout risk identification, and prioritized restocking recommendations. A RAG-based decision assistant enables users to submit business-related questions in natural language and retrieve relevant information from the platform's available data. The API layer facilitates communication between the backend analytical modules and the dashboard. The invention provides an integrated workflow in which raw business data is transformed into structured information, analyzed using predictive and AI-based techniques, evaluated to generate explainable insights, and converted into actionable business recommendations. The platform supports business performance monitoring, demand analysis, revenue forecasting, inventory management, and operational decision-making across different business domains. The system is intended to reduce dependence on repeated manual analysis by specialized data teams and provide a structured mechanism for enabling faster, more accessible, and data-driven business decision-making.

No. of Pages : 8 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131251 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-ASSISTED GOVERNANCE, RISK AND COMPLIANCE (GRC) SYSTEM FOR AUTOMATED CONTROL-EVIDENCE MAPPING, POLICY DOCUMENT RETRIEVAL AND CITATION-GROUNDED DOCUMENTATION GAP ANALYSIS

(51) International classification	:G06N 20/00, G06N 5/02, G06N 5/04, G06F 16/951, G06F 16/248
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. K. Kalaivani
2)DHARANI B
3)ABARNA S
4)SUMAIYA FATHIMA M
5)SOWNDARYA V
6)DIVYA SRI S

(57) Abstract :
The present invention relates to an Artificial Intelligence (AI)-assisted Governance, Risk and Compliance (GRC) system, referred to herein as Aegis GRC, configured to maintain a library of security and compliance controls, link supporting evidence to each control, ingest policy documents, retrieve relevant policy passages, and explain potential documentation gaps with traceable source citations. The system provides a unified compliance platform combining user authentication with role-based permissions, a control library, an evidence register, policy document ingestion, passage segmentation, relevance-based retrieval, rule-based gap evaluation, citation generation and validation, persistent relational storage, and a browser-based compliance dashboard. An authorized user creates a control, for example a privileged access review control, and attaches evidence records with a truthful status such as Missing, Pending or Available. Policy documents are added either as text or as uploaded files and are segmented into passages that retain their source title and paragraph or page locator. When gap analysis is requested for a control, the system retrieves passages relevant to the control, applies a relevance threshold so that unrelated text is not treated as evidence, and compares what the policy requires against the evidence actually attached to the control. The system distinguishes between policy design evidence, which shows that a requirement is documented, and operating evidence, which shows that the requirement was actually carried out. A control is not reported as satisfied merely because a policy exists. Each statement in the gap result is tied to a validated citation comprising the source title, locator and excerpt. Where no adequate source passage is found, the system expressly reports insufficient evidence rather than producing an unsupported conclusion. The invention provides an integrated workflow in which controls, evidence and policy documents are converted into a structured, persistent compliance record, and in which documentation gaps are identified and explained with verifiable references. The system is intended to reduce scattered, spreadsheet-based compliance tracking, shorten audit preparation, and reduce the risk of unsupported or fabricated compliance statements.

No. of Pages : 8 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131252 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MODULAR ADAPTIVE TRANSMISSION CONTROL ASSEMBLY FOR MULTI-SPEED INDUSTRIAL DRIVE

(51) International classification	:F16H 61/02, F16H 61/04, F16H 59/72, F16H 61/00, F16H 61/686
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.S.LOGESWARI
Address of Applicant :Assistant professor, Department of Computer Application, Dr. M.G.R Educational Research Institute, Chennai - 600095, Tamil Nadu, India Tamil Nadu India
2)Dr. H. SELVI
3)ASHWANTH R
4)R.SHAMILE
5)B.UMAMAHESWARI
6)SOWMIYA BHARANI R
7)Dr.R.S.VETRIVEL
8)Dr.M.GOKILADEVI
9)S.SOWDEESWARI
(72)**Name of Inventor :**
1)Dr.S.LOGESWARI
2)Dr. H. SELVI
3)ASHWANTH R
4)R.SHAMILE
5)B.UMAMAHESWARI
6)SOWMIYA BHARANI R
7)Dr.R.S.VETRIVEL
8)Dr.M.GOKILADEVI
9)S.SOWDEESWARI

(57) Abstract :
The present invention relates to a Modular Adaptive Transmission Control Assembly for Multi-Speed Industrial Drive, designed to regulate the speed and torque of industrial drive systems under varying operating conditions. The assembly comprises a motor drive unit, a plurality of transmission modules having different transmission ratios, a sensing unit, an adaptive controller, and a transmission engagement mechanism. The sensing unit acquires operating parameters including rotational speed, load, torque, temperature, and vibration. The adaptive controller processes the acquired parameters and determines a suitable transmission mode according to the required operating conditions. Control signals are generated to selectively engage the appropriate transmission module and coordinate motor operation to achieve the desired output speed and torque. A feedback mechanism enables continuous monitoring and adjustment of operating performance. The modular configuration facilitates maintenance, replacement, and reconfiguration of transmission components according to specific industrial requirements. The assembly may further incorporate fault detection and protective control functions to improve operational reliability. The invention provides a flexible approach to multi-speed industrial drive control, with potential benefits in transmission adaptability, energy management, maintenance, and operational efficiency across various industrial applications.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131254 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR KNEE OSTEORTHRITIS SEVERITY PREDICTION

(51) International classification	:G06T 7/00, A61B 6/00, G16H 30/20, A61B 6/03, G06T 7/11
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)G. VENKATA KISHORE
2)B. VIDHYA SREE
3)A. MALLIKA REDDY
4)MD. ABDUL AZEEZ
5)MR. K. SIVA KUMAR GOWDA

(57) Abstract :
Disclosed herein is a system (100) for predicting knee osteoarthritis severity and estimating joint space width from knee X-ray images (130). The system (100) comprises a user interaction unit (102), an image acquisition and storage unit (104), a radiologist review unit (106), an image preprocessing unit (108), an ensemble classification unit (110), a grade-based routing unit (112), a segmentation unit (114), a joint space width estimation unit (116), a clinician review unit (118), a report generation unit (120), and a data management unit (122). The image preprocessing unit (108) generates a preprocessed image (132), which is processed by the ensemble classification unit (110) using a plurality of trained convolutional neural network models to determine an aggregated severity result (134). The grade-based routing unit (112) determines eligibility for joint space width estimation. The clinician review unit (118) facilitates review the results, the report generation unit (120) generates a clinician-approved report (140).

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131255 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE FUZZY LOGIC SYSTEM FOR PHOTOVOLTAIC POWER TRACKING

(51) International classification	:G05F1/67, G05B13/02, H02J3/38, H02M3/158, H02J101/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)DR. C. BHARGAVA
2)M. MANJARI

(57) Abstract :
Disclosed herein is an adaptive fuzzy logic system (100) for photovoltaic power, comprising a Photovoltaic Unit (102), a Measurement Unit (104), a Processing Unit (106), a DC–DC Boost Converter (114), and a Battery Storage Unit (116). The Processing Unit (106) comprises a First Sugeno Fuzzy Inference Module (108), a Second Sugeno Fuzzy Inference Module (110), and a Control Module (112). The First Sugeno Fuzzy Inference Module (108) estimates a reference maximum power point voltage based on solar irradiance and temperature parameters. The Second Sugeno Fuzzy Inference Module (110) determines a duty-cycle correction based on voltage error and change in voltage error derived from measured photovoltaic voltage. The Control Module (112) generates a control signal to regulate the DC–DC Boost Converter (114), thereby adjusting the photovoltaic operating point toward the maximum power point. Electrical energy delivered through the converter is supplied to the Battery Storage Unit (116) for storage.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131256 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ADAPTIVE NEURAL NETWORK-BASED PHOTOVOLTAIC POWER CONTROL SYSTEM

(51) International classification	:G05F 1/67, H02J 3/38, G06N 3/08, G06N 3/04, H02J 7/35
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)DR. C. BHARGAVA
2)P THARUN

(57) Abstract :
Disclosed herein is an adaptive neural network-based photovoltaic power control system (100) for maximum power point tracking under varying electrical and environmental conditions. The system (100) comprises a Parameter Acquisition Unit (102), an Input Data Processing Unit (104), a Processing Unit (106), a DC–DC Boost Converter (118), and a Data Storage Unit (120). The Processing Unit (106) comprises a Parameter Analysis Module (108), a Maximum Power Point Prediction Module (110), a Reference Voltage Generation Module (112), a Converter Control Module (114), and a Tracking Performance Evaluation Module (116). A trained feed-forward Artificial Neural Network having a 9–20–10–1 architecture predicts a reference maximum power point voltage. The Converter Control Module (114) generates a duty-cycle control signal to regulate the DC–DC Boost Converter (118). The Tracking Performance Evaluation Module (116) evaluates tracking performance using MSE, RMSE, MAE, and MAPE

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131257 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A CONTROL SYSTEM FOR SECURE CLOUD DATA STORAGE

(51) International classification	:H04L 9/32, H04L 9/08, G06F 21/62, H04L 29/06, G06F 21/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)K. MOUNIKA
2)DR. T. VENKAT NARAYANA RAO
3)MRS. CHENCHU SWETHA
4)MR. K. SIVA KUMAR GOWDA
5)MRS. S. BHAVANA

(57) Abstract :
Disclosed herein is a control system (100) for secure cloud data storage, comprising a user device (102), a data owner interface (104), a decentralized verification unit (106), a processing unit (108), a cloud storage unit (126), and an access policy and permission data store (128). The processing unit (108) comprises an authentication verification module (110), an anonymous authentication module (112), an authorization policy evaluation module (114), a replay-protection module (116), a revocation management module (118), a cryptographic key management module (120), an encryption and decryption module (122), and a data operation management module (124). The system verifies authentication evidence through decentralized verification, validates request freshness, evaluates operation-specific access permissions, and checks credential validity. The system further encrypts data for storage and regulates authorized data operations based on authentication, authorization, and revocation conditions, thereby facilitating privacy-preserving authentication and controlled access to encrypted data in a cloud storage environment.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131258 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A HYBRID LEXICON AND MACHINE LEARNING SYSTEM

(51) International classification	:G06N 20/00, G06F 17/27, G06N 3/08, G06F 40/216, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)ADUVALA SRINIVAS
2)BHAGAVATHULA VENKATA MANEESH
3)GANNOJU SANTHOSH
4)KOTHAMASU MANE
5)DR T VENKAT NARAYANA RAO

(57) Abstract :
Disclosed herein is a hybrid lexicon and machine learning system 100 for political security threat prediction, comprising a data acquisition unit 102, a data validation unit 104, a processing unit 106, and a data storage unit 126. The processing unit 106 comprises a text pre-processing module 108, a lexicon analysis module 110, a preliminary label generation module 112, a feature generation module 114, a model training module 116, a model evaluation and selection module 118, a threat prediction module 120, an output and review module 122, and a vocabulary and model update module 124. The system 100 receives and validates political textual content, identifies emotion-related indicators using an emotion lexicon, generates preliminary classification labels, and creates numerical feature representations using term frequency-inverse document frequency and optionally lexicon-derived emotion features. The predicted results are presented for human review, and validated feedback supports controlled updates to the lexicon, classification policy, and classifier.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131260 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-Enabled Smart Campus Security and Energy Management System

(51) International classification	:G06Q 50/20, G06K 19/07, G06K 7/10, G07C 9/00, G08B 13/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Easwari Engineering College
Address of Applicant :Bharathi Salai, Ramapuram, Chennai – 600 089. Tamil Nadu India Chennai Tamil Nadu India
2)SRM Institute of Science and Technology, Ramapuram Campus
(72)**Name of Inventor :**
1)Abisha A
2)Sankrisha S
3)Dr.K.Hema Priya

(57) Abstract :
The present invention relates to an AI-Enabled Smart Campus Security and Energy Management System designed to improve campus safety and reduce energy consumption. The system integrates Artificial Intelligence and IoT technologies with RFID authentication, PIR-based occupancy detection, automated lighting and fan control, and AI-enabled CCTV monitoring. An Arduino Nano verifies RFID credentials and automatically unlocks the door for authorized users. PIR sensors detect classroom occupancy and control lights and fans automatically. When no motion is detected, the lights and fans are turned off and the door is locked after a configured interval. Unauthorized RFID access activates a buzzer and sends alerts to authorities. AI-enabled CCTV cameras monitor campus activities and detect suspicious activities. The system reduces electricity wastage, minimizes manual intervention, improves security, enables real-time monitoring, and increases operational efficiency. It provides a scalable and intelligent solution for modern educational institutions. Number of words: 141

No. of Pages : 17 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131262 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DOMINO CLOCKING FOR MULTISTAGE CHARGE PUMP

(51) International classification	:H02M 3/07, G11C 16/30, H03L 7/089, G11C 16/06, G11C 11/4074	(71)**Name of Applicant :** **1)Biprangshu Saha** Address of Applicant :A904, Glen Gate , House of Hiranandani, Devinagar Karnataka India (72)**Name of Inventor :** **1)Biprangshu Saha**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

DOMINO CLOCKING FOR MULTISTAGE CHARGE PUMP A multi-stage charge pump system and operating method are disclosed for generating positive or negative output voltages in integrated circuits without requiring auxiliary clock level-shifters or high-voltage capacitors. The system includes a first charge pump stage driven by primary complementary clock signals to produce an initial boosted voltage and self-generated, level-shifted complementary clock waveforms on internal capacitive nodes. Each subsequent stage in an N stage cascade has clock input terminals directly connected to the internal capacitive nodes of the immediately preceding stage. By propagating clock signals directly through internal nodes in a domino arrangement, clock amplitude swings are maintained at Vdd across all stages while DC potential offsets increment sequentially. Consequently, maximum terminal-to-terminal voltage across any stage capacitor remains bounded by Vdd, eliminating dedicated level-shifting circuitry, reducing dynamic power consumption, and saving silicon area.

No. of Pages : 13 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131267 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DESIGN, SYNTHESIS AND INVITRO EVALUATION OF BISCHALCONE DERIVATIVES AS AN ANTITUBERCULAR AGENTS

(51) International classification	:C07C 49/84, C07C 45/00, C07C 45/67, C07D 333/24, C07D 333/38
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)G BUELA PRIYANKA
Address of Applicant :Guru Nanak Institutions Technical Campus, School of Pharmacy, Ibrahimpatnam, Hyderabad, Telengana, 501506. Telangana India
2)Rayini Venkata Sai Mounica
3)Dr.Yadala Prapurna Chandra
4)Dr Venugopalaiah Penebaka
5)Nimma Vijaya Rekha
6)Dr M.Neelima
7)Guduru Rajeswari
8)Mrs Puchalapalli Sailaja
9)Dr B Karuna Devi
10)Mrs K.Kavya Sudha
11)Ramalingeswara Rao Kasula
(72)**Name of Inventor :**
1)G BUELA PRIYANKA
2)Rayini Venkata Sai Mounica
3)Dr.Yadala Prapurna Chandra
4)Dr Venugopalaiah Penebaka
5)Nimma Vijaya Rekha
6)Dr M.Neelima
7)Guduru Rajeswari
8)Mrs Puchalapalli Sailaja
9)Dr B Karuna Devi
10)Mrs K.Kavya Sudha
11)Ramalingeswara Rao Kasula

(57) Abstract :
Using aromatic and heterocyclic aldehydes with alkyl substituted cyclic ketone, a series of new 2,6-bis ((E) benzylidene)-4-methylcyclohexan-1-one derivatives were created at room temperature over the course of 18 to 48 hours with alcohol acting as a solvent and a base acting as a catalyst. After the reaction is finished, final compounds are recrystallized with alcohol to generate a series of final chemicals. Additionally, the creation of a number of new Bischalcone derivatives with heterocyclic rings is completed by the spectral characterization. The Alamar Blue Assay (MABA) method was used to perform the biological evaluation for antitubercular activity and the compounds showed promise as powerful and innovative inhibitors for the development of new antitubercular therapies. Additionally, compounds 3a, 3b, 3c and 3g exhibit high binding energies with the target protein Mur E enzyme of the Mur pathway, Molecular Docking verified the uncompetitive inhibition of M.tb Mur E enzyme.

No. of Pages : 16 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131268 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A REAL-TIME CROWD MONITORING SYSTEM

(51) International classification	:G06V 20/52, G06K 9/00, G06V 10/764, G06N 3/08, G06V 40/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)DR KUSUMALATHA KARRE
2)SRI HARSHA SUNKARA
3)K. SHALINI
4)S. PRAVALIKA
5)K. DAYAKAR

(57) Abstract :
Disclosed herein is a real-time crowd monitoring system (100) that comprises a Video Acquisition Unit (102), a Processing Unit (104), a Memory Unit (126), a Communication Interface (128), and a Display Unit (130). The Processing Unit (104) comprises a Pre-processing Module (106), a Person Detection Module (108), a Crowd Counting Module (110), a Distance Calculation Module (112), a Density Heatmap Generation Module (114), a Threshold Evaluation Module (116), a Frame Annotation Module (118), an Alert Management Module (120), an Evidence Capture Module (122), and a Trend Analysis Module (124). The system (100) utilizes a YOLOv8 nano model to detect persons, determine crowd size, identify proximity violations, and generate a crowd density heatmap. The system (100) evaluates configurable thresholds to detect overcrowding, generates notifications while preventing repeated alerts during the same episode, and stores timestamped annotated frames as evidence.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131269 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-POWERED PRIVACY-PRESERVING INTELLIGENT ASSESSMENT AND LEARNING SYSTEM

(51) International classification	:G09B 7/02, G09B 7/00, G09B 7/04, G06Q 50/20, G06F 21/62
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)G. VENKATA KISHORE
2)KASOJU SPOORTHY
3)NALLU MANWAL REDDY
4)BOBIDAY KOUSHAL
5)MR. K. SIVA KUMAR GOWDA

(57) Abstract :
Disclosed herein is an AI-powered, privacy-preserving intelligent assessment and learning system (100) that comprises an Input Interface (102), a Document Acquisition Unit (104), a User Authentication Unit (106), a Processing Unit (108), a Memory Unit (136), a Communication Interface (138), a Display Unit (140), an Output Unit (142), and a Database Unit (144). The Processing Unit (108) comprises a Pre-processing Module (110), a Text Segmentation Module (112), a Question Generation Module (114), a Validation Module (116), a Difficulty Classification Module (118), an Approval Module (120), an Assessment Management Module (122), an Evaluation Module (124), an Anomaly Detection Module (126), an Analytics Module (128), a Feedback Module (130), an Engagement Management Module (132), and a Notification Module (134). The system generates and validates questions from academic documents using a locally deployed language model, facilitates administrator-approved assessments, detects anomalous assessment behaviour, provides personalized learning analytics while maintaining assessment-related data on local infrastructure.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131270 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED ACADEMIC EVALUATION AND SEMANTIC PLAGIARISM DETECTION SYSTEM

(51) International classification	:G06F 17/30, G06Q 50/20, G06F 17/27, G06F 40/30, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Address of Applicant :YAMNAMPET, GHATKESAR, HYDERABAD, TELANGANA - 501301, INDIA Telangana India
(72)**Name of Inventor :**
1)MRS. SANGERS BHAVANA
2)MR. SOMA VARSHITH KUMAR
3)MR. RAMUNIGARI SAKETH GOUD
4)MR. UPPULA SANNITH KUMAR
5)DR.T. VENKAT NARAYANA RAO

(57) Abstract :
Disclosed herein is an AI-based academic evaluation and semantic plagiarism detection system (100) comprising a User Interface Unit (102), an Application Server Unit (104), a Document Preprocessing Unit (106), a Processing Unit (108), an Academic Knowledge Repository (120), a Vector Database Unit (122), a Structured Data Management Unit (124), a Document Storage Unit (126), a Security and Access Control Unit (128), and a Communication and Notification Unit (130). The Processing Unit (108) comprises an Embedding Generation Module (110), a Contextual Retrieval Module (112), a Retrieval-Augmented Evaluation Module (114), a Semantic Plagiarism Detection Module (116), and a Results Management Module (118). The system processes academic submissions, retrieves relevant assessment criteria and reference materials, generates evaluation scores and feedback using a large language model, and identifies potential semantic overlap through vector similarity analysis. The system generates structured assessment reports to support academic evaluation, feedback generation, and academic integrity review.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131272 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM FOR AI-ENABLED ALTERNATIVE-DATA-BASED FINANCIAL ASSESSMENT AND FINANCIAL INCLUSION

(51) International classification	:G06Q 40/00, G06Q 40/02, G06Q 40/06, G06Q 20/40, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NAIDU SUNETHRI
Address of Applicant :SR UNIVERSITY, ANANTHASAGAR, HASANPARTHY, WARANGAL URBAN, TELANGANA - 506371, INDIA Telangana India
2)SR UNIVERSITY
(72)**Name of Inventor :**
1)NAIDU SUNETHRI

(57) Abstract :
Disclosed herein is a system (100) for ai-enabled alternative-data-based financial assessment and financial inclusion that comprises a user computing device (102), a communication interface (104), a data acquisition interface (106), and a processing unit (108). The processing unit (108) comprises a registration and consent management module (110), a data collection module (112), a data processing module (114), a feature engineering module (116), an AI credit scoring module (118), a risk prediction module (120), a fraud detection module (122), a financial recommendation module (124), an API integration module (126), and a government scheme eligibility module (128). The system (100) further comprises a memory unit (130), a secure data storage unit (132), a network gateway device (134), a lender computing device (136), a display unit (138), and an input unit (140). The system (100) processes consent-based alternative data to generate a dynamic credit score, determine a risk profile, detect anomalous patterns, generate financial recommendations, facilitate institutional access, and determine financial inclusion scheme eligibility.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131273 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DIGI PREP DENTAL SYSTEM – Precision Perfected: Embracing Digital Dentistry for Tooth Preparation

(51) International classification	:A61C 1/08, A61C 13/00, A61C 9/00, A61C 19/04, A61C 13/08	(71)**Name of Applicant :** **1)SRM Dental College (Ramapuram)** Address of Applicant :Bharathi Salai, Ramapuram, Chennai, Tamil Nadu 600089 Chennai Tamil Nadu India **2)Easwari Engineering College** (72)**Name of Inventor :** **1)Dr Ramya.S** **2)Mr. S.Kiran Kumar** **3)Ms. S.Stefina Rachel** **4)Mr. R.Rajalokesh** **5)Ms. S.R.Priyadharshini**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a DIGI PREP DENTAL SYSTEM for digitally guided tooth preparation. The system comprises an airotor handpiece equipped with an angulation sensor, an electronic processing circuit and a digital display. The sensor detects changes in handpiece orientation during tooth preparation and the processed information is displayed in real time to guide the operator. The system may additionally provide feedback relating to preparation depth, detect deviations from predetermined preparation parameters and record procedural data. The invention is intended to improve the precision, accuracy and consistency of tooth preparation, reduce the risk of iatrogenic damage and facilitate efficient clinical and educational workflows. The system may be used for crown and bridge preparation, inlays, onlays, veneers, implant site preparation, endodontic access preparation, orthodontic preparations and custom abutment preparation. Total Number of Words in Abstract: 129

No. of Pages : 11 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131274 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Compact Portable Stereolithography 3D Printing Apparatus with Arduino-Controlled Z-Axis Movement and UV Photopolymerization for Dental and Maxillofacial Applications

(51) International classification	:A61C 13/15, A61C 9/00, A61C 13/00, B29C 67/00, B29C 64/129
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM Dental College (Ramapuram)
Address of Applicant :Bharathi Salai, Ramapuram, Chennai, Tamil Nadu 600089 Chennai Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Dr Ramya C
2)Dr Praveena A S
3)Dr J Brintha Jei
4)Dr Muthukumar B

(57) Abstract :
The present disclosure relates to a compact, portable, and economical stereolithography (SLA) 3D printing apparatus for customized dental and maxillofacial prosthetic fabrication. The invention integrates Arduino-controlled Z-axis movement, UV photopolymerization, and digital 3D printing technology into a compact system. The apparatus incorporates an Arduino Pro Mini, NEMA 8 stepper motor, lead screw, linear guide, resin vat, monochrome LCD masking system, and UV LED light source for controlled layer-by-layer fabrication. The system processes digital 3D models into successive layers, precisely positions the build platform, and selectively cures photopolymer resin to produce customized components. The invention provides a portable, efficient, and economical alternative to conventional 3D printing systems, with applications in dental prostheses, surgical guides, and maxillofacial prosthetic fabrication Total Number of Words: 140

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131275 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ENGINEERED BIMODAL SENSING ANTIBODY FRAGMENT (CETUXIMAB-DFD) AND LOGIC GATED SYNTHETIC NOTCH CAR T-CELL ARCHITECTURE FOR OVERCOMING EGFR DRUG RESISTANCE IN ONCOLOGY

(51) International classification	:C07K 16/28, A61K 39/00, A61P 35/00, A61K 39/395, C07K 16/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM DENTAL COLLEGE (RAMAPURAM)
Address of Applicant :Bharathi Salai, Ramapuram, Chennai, Tamil Nadu 600089 Chennai Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)S.N.SRI VARSHAA
2)T. Dinesh Kumar
3)Nagarathinam AE
4)K. Rajkumar

(57) Abstract :
Head and neck squamous cell carcinoma causes nearly a million annual cases, with epidermal growth factor receptor (EGFR)-targeted therapies like cetuximab often failing due to variants like R521K, S492R, G465R, and G33S/N56K. To bypass this resistance with minimal off-tumour toxicity, an engineered anti-EGFR single-chain variable fragment (scFv), designated CTX DFD, and a logic-gated synthetic notch (SynNotch) CAR-T cell therapeutic modality is disclosed. The CTX DFD scFv remodels somatic EGFR mutations into precision anchoring triggers via a bimodal sensing architecture, utilizing localized anchorage at arginine 468 (Mechanism A), and coulombic pull (Mechanism B). In silico modelling yielded 99.2% Ramachandran favourability, and TAP profiling confirmed the sequence stability. The thermodynamic analysis revealed CTX DFD scFv superseded native cetuximab in binding against EGFR variants, while COPASI simulation provided the functional proof linking binding affinity to signalling velocity, validating a safe and variant-specific therapeutic for HNSCC that overcomes clinical resistance with minimal off-tumour toxicity. Total Number of Words in Abstract:150

No. of Pages : 9 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131289 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Real-Time AI-Based Kidney Function Monitoring and Predictive Intervention System for Dialysis Reduction

(51) International classification	:A61B 5/00, A61M 1/16, G16H 40/67, A61B 5/145, A61B 5/021
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Dr. S. Suresh
2)Dr. V. Gokula Krishnan

(57) Abstract :
This invention is about a smart healthcare monitoring system that can continuously monitor kidney function and, through early intervention, can potentially decrease the reliance on dialysis. The components of the whole system are a multi-parameter sensor module, a data acquisition interface, an embedded processing unit, and an artificial intelligence-based predictive analytics engine. Initially, the system monitors and collects various physiological and biochemical parameters like hydration levels, blood pressure, heart rate, and approximate kidney function indicators. The gathered data is being processed and analyzed in real-time through machine learning techniques in order to detect any decline in kidney function before it occurs. According to the forecast results, the system sends out warnings and suggests necessary actions such as adjusting hydration levels, offering nutritional advice, and possibly scheduling a medical examination. It offers a preventive and real-time approach towards managing chronic kidney diseases through the continuous monitoring functionality combined with predictive intelligence, which ultimately leads to an improvement in patient health and reduction in the frequency of dialysis.

No. of Pages : 8 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131290 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Human-AI Collaborative Financial Decision Support System with Adaptive Rationale, Selective Override and Cognitive Offloading Control

(51) International classification	:G06Q 40/00, G06N 20/00, G06Q 10/06, G06Q 40/06, G06Q 40/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Ruthvika Reddy A
2)M. Meera

(57) Abstract :
ABSTRACT The present invention discloses a Human-AI Collaborative Financial Decision Support System for financial forecasting, risk assessment, advisory, valuation, and related analytical tasks. The 10 system integrates financial data acquisition, bias-controlled preprocessing, machine learning analytics, transparent evidence generation, algorithmic self-contradiction detection, human contextual input, adaptive collaboration interfaces, selective human override, decision synthesis, and audit feedback. The system presents interpretable model evidence including feature attribution, sub-scores, diagnostic information, and uncertainty indicators, while enabling an 15 authorized human evaluator to add qualitative soft information and selectively intervene when evidence supports an override. The adaptive interface can preserve selected causal or mechanism gaps to reduce excessive cognitive offloading and maintain active human evaluation. The system thereby provides a traceable human-AI decision workflow in which computational scale and pattern recognition are combined with contextual judgment, evidence-based oversight, and 20 auditable final decision synthesis.

No. of Pages : 10 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131291 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Experimental Investigation on Behaviour of Cold-Formed Steel Built-Up Columns Composed of Lipped Channels with Spacers

(51) International classification	:E04B 1/24, E04C 3/32, E04B 1/58, E04B 1/30, E04C 3/04	(71)**Name of Applicant :** **1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS** Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
(31) Priority Document No	:NA	**2)Easwari Engineering College**
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Dr.S.D.Kumar**
(86) International Application No	:	**2)Mr.Mohammed Shuaib Z**
Filing Date	:01/01/1900	**3).Dr.A.Mathivanan**
(87) International Publication No	: NA	**4)Dr.R.Suthan**
(61) Patent of Addition to Application Number	:NA	**5)Dr.M.Vikneshwaran**
Filing Date	:NA	**6)Dr.A.Venkatakrishna**
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

Abstract Cold-formed steel members are widely adopted in modern construction owing to their high strength-to-weight ratio and fabrication efficiency. The present investigation examines the structural behaviour of cold-formed steel built-up columns composed of lipped channel sections connected with spacers under axial compression. Experimental testing and finite element analysis were conducted to evaluate load-carrying capacity, deformation characteristics, and failure modes.

No. of Pages : 7 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131292 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Arduino-Driven Intelligent Bus Stop Monitoring system

(51) International classification	:G08G 1/123, G08G 1/127, G06Q 50/30, G08G 1/133, G08G 1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Akash D
2)Dr. Nagarajan S
3)Prathin Kanth K
4)Swagath R
5)Raghavan S P

(57) Abstract :
ABSTRACT : The transportation network is an important component of the urban infrastructure and adherence to the rules on the bus stops is required for safe functioning. The buses which do not stop at bus stops are a nuisance to the commuters, among other things. It is suggested here an Automated Bus Stop Monitoring System to monitor whether a bus has reached the bus stop in question, if it stops there for the specified amount of time and record all the violations. In this system, the key controller used was the Arduino UNO board. Ultrasonic sensor will determine whether the bus is present at the bus stop and for how long it will stay there. RFID technology will help in identifying each individual bus. Real Time Clock module will record the dateand time of each event. When a bus is found to repeat the same violation more than three times in one day, it is marked as being absent on that particular day. A GSM module is also used to send SMS alert whenever the violations repeat themselves more than three times in one day. The proposed system is economical and practical.

No. of Pages : 6 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131293 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : WAYPOINT: A SMART CAMPUS NAVIGATION KIOSK AND PERMISSION-CONTROLLED INDOOR STAFF LOCATOR

(51) International classification	:G01C 21/20, G06Q 50/20, H04W 4/33, G06Q 10/02, H04W 4/024
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)P. Monish Kumar
2)Rehan Ansari
3)J.Gowthamy
4)Dr. Mageshkumar.N

(57) Abstract :
Abstract A campus navigation system is proposed comprising public touchscreen kiosks, a mobile application, a campus map database, an indoor route engine, an identity service and a permission-controlled staff-location service. The map represents campus buildings, floors, rooms, corridors, stairs, lifts, accessible paths, departments and event venues as navigable locations and connections. A user may search for a destination at a kiosk or through a mobile client. The route engine provides a highlighted map and ordered indoor directions. Staff location may be estimated from signals emitted by a tag associated with a staff identity and detected by fixed readers assigned to known campus zones. The system records the time and source of each observation and distinguishes a current estimate from an uncertain or older observation according to a validation method that must be defined and tested. Before staff-location information is disclosed, the system checks the requester's permissions and the staff member's sharing settings. If location evidence is insufficient, the system may show the assigned office or a clearly labelled last-verified location rather than claim that the person is currently there. A temporary route reference may allow a user to continue a kiosk-selected route on a mobile device. Kiosk session data is cleared after use. The proposed platform is intended to support navigation for students, staff and visitors. The novelty and inventive step of the overall combination and each proposed claim remain to be assessed against prior art.

No. of Pages : 12 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131294 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Adaptive Energy Monitoring and Optimization System for Solar Applications

(51) International classification	:H02S50/10, H02S20/32, H02J7/35
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS

Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India

2)Easwari Engineering College

(72)**Name of Inventor :**

1)DESHNA R

2)Dr.Jacintha.V

3)LOGANATHAN.M

4)SABAREESH S

5)SIVAAKILAN M

(57) Abstract :

As far as the approach currently used for design is concerned, in this situation, the emphasis will be made on the creation of energy monitoring and optimization module for utilization in solar applications. Therefore, in the given situation, it will be crucial not only to be able to control the process of energy production but also regulate the process of energy consumption. In other words, from the point of view of functional capabilities, the current design will include such elements as photovoltaic cell, solar tracker, and energy monitoring and optimization module incorporated in the design. With the help of the given module, the process of monitoring important factors in relation to energy production and energy consumption will become possible. This will enable making the necessary conclusions concerning solar radiation in the given place and regulating the work of solar tracker.

No. of Pages : 11 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131295 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HUMAN IDENTIFICATION, FACE RECOGNITION & TAG IDENTIFICATION USING SMART SYSTEMS

(51) International classification	:G07C 9/00, G06K 9/00, H04L 29/08, G07C 9/37, G06K 19/077
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Dr. R. Suthan
2)Dr. S. D. KUMAR
3)Dr. N. PARTHIPAN
4)Dr. S. SATHEESH KUMAR
5)Dr. D. SARUKASAN
6)Mr. G. GUGAN
7)Dr. D. PREMKUMAR
8)Dr. S. RAJKAMAL
9)Mr. G. GUGAN
10)Mr. A. KIRAN KUMAR

(57) Abstract :
Abstract The present invention relates to a smart human identification system that integrates human presence detection, face recognition, tag-based identification, impact sensing, cloud-based data storage, and alert communication within a single portable unit. The system utilizes a microcontroller-based processing unit with wireless connectivity to acquire sensor data, authenticate individuals, and transmit information to a remote cloud server in real time. The invention provides a reliable, contactless, and automated identification solution suitable for security, attendance monitoring, access control, and safety applications.

No. of Pages : 8 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131296 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Neutrosophic Over Soft Complex Locally Closed Set-Based Intelligent Accident Prevention and Road Safety Monitoring System

(51) International classification	:G06N 3/08, G06N 3/04, G08G 1/01, G08G 1/04, G08G 1/017
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS
Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India
2)Easwari Engineering College
(72)**Name of Inventor :**
1)Dr. Yamini Parthiban
2)Dr. Raghavendran Prabakaran

(57) Abstract :
ABSTRACT OF THE INVENTION: The present invention concerns an intelligent accident prevention and road safety monitoring system based on a Neutrosophic Over Soft Complex Locally Closed Set framework which integrates environmental sensors and Artificial Neural Networks (ANN). The system is intended for use at low visibility sites such as blind curves, hilly roads, tunnels etc. The collected environmental information from the visibility sensor, moisture sensor, fog sensor, temperature sensor and the vehicle detection sensor is converted to neutrosophic over soft complex representations to deal with uncertainty and indeterminate condition. The proposed framework assesses the environmental risk and provides processed information to an ANN prediction module for real time accident risk assessment. If these dangerous conditions occur, the system will trigger warning lights and audible alarms and also trigger wireless systems to warn approaching vehicles. The invention increases the accuracy of prediction, increases safety of the roads, decreases the likelihood of accidents, and offers an effective method of solving the problems of intelligent transportation systems, smart infrastructure and connected vehicle applications in uncertain environmental conditions.

No. of Pages : 13 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131297 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DIVA: A DECISION-IMPACT VALUE ACQUISITION SYSTEM FOR AUTONOMOUS SPACECRAFT OBSERVATION SELECTION

(51) International classification	:B64G 1/10, B64G 1/24, G06N 7/00, G06N 5/02, G05D 1/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS

Address of Applicant :SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, RAMAPURAM CAMPUS Tamil Nadu India

2)Easwari Engineering College

(72)**Name of Inventor :**

1)Farhan Khan S

2)Shahil Raj

3)Ritisha Garnaik

4)R Shanmuga Priya

(57) Abstract :

Abstract A computer-implemented autonomous spacecraft observation-selection system and method is disclosed, comprising a mission-state representation, a candidate observation generation engine, a counterfactual outcome simulation engine, a future-policy generation engine, a policy-divergence evaluation engine, and a resource-constrained observation selector. The system evaluates candidate observations not solely according to information gain or target priority, but according to the expected extent to which possible observation outcomes alter subsequent spacecraft observation policies and mission decisions. For each candidate observation, the system generates a plurality of hypothetical observation outcomes, propagates each outcome into a corresponding future mission state, generates a future observation policy for each state, and computes a quantitative divergence between the resulting policies and a current or reference policy. The policy divergence is weighted by predicted outcome probability and mission consequence to obtain a decision-impact value. The system may further incorporate decision-relevant uncertainty, counterfactual future utility, expected regret reduction, opportunity cost, and spacecraft resource constraints. Following execution of a selected observation, the actual observation result is incorporated into the mission state and the observation-selection process is repeated, thereby forming a closed-loop autonomous planning architecture capable of adapting observation priorities as mission conditions evolve.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131299 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ADAPTIVE RESOURCE ALLOCATION SYSTEM FOR CLOUD-EDGE COMPUTING

(51) International classification	:G06F 9/50, H04L 12/24, H04L 12/911, H04L 12/26, H04L 67/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR UNIVERSITY
Address of Applicant :ANANTHASAGAR, HASANPARTHY (M), WARANGAL URBAN, TELANGANA - 506371, INDIA Telangana India
(72)**Name of Inventor :**
1)KURRA NALINI
2)DR. ADHYASHA SAHU

(57) Abstract :
Disclosed herein is an adaptive resource allocation system (100) for cloud–edge computing that comprises a data acquisition unit (102), a communication interface (104), a resource monitoring unit (106), a resource repository (108), and a processing unit (110). The processing unit (110) comprises a pre-processing module (112), a workload prediction module (114), a resource assessment module (116), a resource allocation module (118), an energy optimization module (120), an SLA management module (122), a performance evaluation module (124), and a feedback and adaptation module (126). The pre-processing module (112) standardizes acquired workload information, and the workload prediction module (114) predicts future resource requirements using machine learning. The resource allocation module (118) allocates computing resources across cloud and edge environments based on predicted workloads and resource availability. The energy optimization module (120) evaluates energy consumption, while the SLA management module (122) assesses service compliance.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131300 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN ADAPTIVE MULTI-RESOURCE CLOUD WORKLOAD PREDICTION AND ALLOCATION SYSTEM

(51) International classification	:G06F 9/50, H04L 12/24, G06N 20/00, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR UNIVERSITY
Address of Applicant :ANANTHASAGAR, HASANPARTHY (M), WARANGAL URBAN, TELANGANA - 506371, INDIA Telangana India
(72)**Name of Inventor :**
1)KURRA NALINI
2)DR. ADHYASHA SAHU

(57) Abstract :
Disclosed herein is an adaptive multi-resource cloud workload prediction and allocation system (100) that comprises a data acquisition unit (102), a resource monitoring unit (104), a workload data repository (106), a communication interface (108), a memory unit (110), a processing unit (112), and a cloud infrastructure (132). The processing unit (112) comprises a pre-processing module (114), a residual anchoring module (116), a feature extraction module (118), a temporal learning module (120), an attention refinement module (122), a workload prediction module (124), a provisioning decision module (126), a resource allocation module (128), and a performance evaluation module (130). The system (100) employs a residual-anchored deep learning framework to predict multi-resource workloads across future time intervals and proactively allocate cloud resources based on predicted requirements, thereby improving resource utilization and reducing under-provisioning in dynamic cloud computing environments.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131304 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Structural, Optical and AFM Studies of Organometallic Guanidinium Chlorochromate (GCC) Single Crystal

(51) International classification	:G02F 1/355, C30B 7/00, G02F 1/361, G02F 1/37, C30B 29/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. SUVITHA A
Address of Applicant :Associate Professor, Dept of Physics, S.A Engineering college (Autonomous),AVADI MAIN ROAD, VEERAGHAVAPURAM, CHENNAI Chennai Tamil Nadu India
2)Dr. R.GANESAN
3)S.A Engineering college (Autonomous),
(72)**Name of Inventor :**
1)Dr. SUVITHA A
2)Dr. R.GANESAN

(57) Abstract :
Nonlinear optical single crystals of Guanidinium Chlorochromate [GCC] are grown by slow evaporation solution growth technique using water as the solvent. Purity of crystals is increased by the method of recrystallization. The solubility of the material is measured at various temperatures in de-ionized water. The grown crystals are characterized by single crystal XRD for obtaining unit cell parameters. The UV-vis-NIR spectrum of GCC shows less absorption and good transmittance in the entire visible region enabling its use in optical application. Also, the Kurtz powder second harmonic generation test shows that the compound is a potential material for optical second harmonic generation. The growth mechanism and surface features of as grown crystal are analyzed through chemical etching studies. AFM analysis was used to investigate the surface topography, roughness, and nanoscale morphological features of the sample.

No. of Pages : 12 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131306 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Effect of Magnetic Field and Heat Source on Natural Convection of Silver Nanofluid in a Trapezoidal Enclosure: A Numerical Approach

(51) International classification	:G06F 17/50, G01N 25/20, G06F 19/00, G06F 119/08, G01N 25/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr KUMARESAN E
Address of Applicant :Associate Professor Department of Mathematics(H&S) S.A. Engineering College, Veeraragavapuram, Chennai Chennai Tamil Nadu India
2)Dr DIVYA P
3)S.A Engineering College (Autonomous)
(72)**Name of Inventor :**
1)Dr KUMARESAN E
2)Dr DIVYA P

(57) Abstract :

The present study investigates natural convection heat transfer of silver-water nanofluid inside a trapezoidal enclosure with a diamond-shaped obstacle at its centre. The study examines the effects of magnetic field and internal heat generation on fluid flow and temperature distribution. Silver nanoparticles are mixed with water to improve heat transfer. Numerical simulations are carried out using COMSOL Multiphysics 6.1. The Galerkin weighted residual method is used for numerical analysis. Important parameters such as Rayleigh number, Hartmann number, nanoparticle concentration and fluid viscosity are studied. The results show that increasing the Rayleigh number improves heat transfer. Increasing the magnetic field reduces fluid motion. Streamlines and isotherms are used to analyse fluid flow and temperature changes. This study helps in developing efficient thermal management and heat transfer systems.

No. of Pages : 8 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131308 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A NOVEL COMPUTATIONAL METHOD FOR ARC DOMINATION OPTIMIZATION IN DIRECTED GRAPHS AND ITS APPLICATIONS IN NETWORK DESIGN

(51) International classification	:H04L 12/24, G06F 17/50, H04W 16/18, H04L 12/28, H04L 12/26	(71)**Name of Applicant :** **1)R. ANBUNATHAN** Address of Applicant :S.A. Engineering College (Autonomous), Poonamallee Avadi Main Road, Veeraragavapuram, Chennai-600077. Tamil Nadu India **2)M. LAKSHMI** **3)V. SEETHALAKSHMI** **4)S.A. ENGINEERING COLLEGE** (72)**Name of Inventor :** **1)R. ANBUNATHAN** **2)M. LAKSHMI** **3)V. SEETHALAKSHMI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Directed graphs are widely used to represent communication networks, transportation systems, electronic circuits, and other interconnected systems in which relationships between components have a specific direction. Efficiently selecting a minimum set of arcs to achieve the required domination or coverage is an important optimization problem in directed graph theory. This invention proposes a novel computational method for arc domination optimization in directed graphs, with the objective of identifying an effective arc-dominating set while reducing computational effort and improving network coverage efficiency. The proposed method represents a directed network as a digraph (D = (V, A)), where (V) is the set of vertices and (A) is the set of directed arcs. It incorporates graph-structural analysis, candidate arc selection, domination verification, and iterative optimization to identify a feasible arc-dominating set according to the specified domination criteria. The method can be designed to accommodate different classes of directed graphs and network constraints, including connectivity requirements, coverage conditions, and restrictions on the selection of arcs. Its computational framework can also be extended using heuristic or metaheuristic optimization techniques to explore alternative solutions for larger network instances. Potential applications include communication network monitoring, network resource allocation, circuit testing, and the design of efficient directed network structures. The proposed approach provides a systematic computational framework for studying and solving arc domination problems and offers a foundation for evaluating solution quality, computational complexity, and practical applicability across different network configurations. The novelty and performance of the method are to be established through formal mathematical characterization, algorithmic analysis, comparative experiments, and validation on representative directed network models.

No. of Pages : 8 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131316 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A HERBAL HAIR AND SCALP SERUM COMPOSITION ENRICHED WITH OYSTER MUSHROOM EXTRACT AND A METHOD FOR PREPARATION THEREOF

(51) International classification	:A61K 8/97, A61Q 5/00, A61Q 5/12, A61Q 7/00, A61K 8/9728
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mrs. Poornam S
Address of Applicant :Chief Executive Officer, ORISH TECH, Chatrapatti, Madurai Tamil Nadu India
2)Mr. Munishwaran S
3)Dr. Rajesh Nallaiah
4)Dr. P. Prabhavathi
5)Dr. A. Saranya
6)Dr. G. Krishnaveni
7)Dr. A. Thavaselvi
(72)**Name of Inventor :**
1)Mrs. Poornam S
2)Mr. Munishwaran S
3)Dr. Rajesh Nallaiah
4)Dr. P. Prabhavathi
5)Dr. A. Saranya
6)Dr. G. Krishnaveni
7)Dr. A. Thavaselvi

(57) Abstract :
A herbal hair and scalp serum composition and a preparation method are disclosed. The composition includes 0.2–5.0 wt% dried-solids equivalent of a standardized Pleurotus ostreatus extract, a humectant system, a rheology modifier or film former, a preservative system and an aqueous carrier adjusted to pH 4.8–6.2. Mushroom biomass is authenticated, extracted, clarified and characterized for extract solids and an analytical marker or fingerprint before incorporation into a low-residue topical carrier. The formulation is intended for scalp conditioning and for further evaluation of possible DHT-related biological activity; such activity and hair-regeneration efficacy are not established by the disclosure alone. Accompanied Drawing [FIGS. 1-2].

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131322 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-POWERED VEHICLE MAINTENANCE AND ASSISTANCE SYSTEM

(51) International classification	:G06N 20/00, G06Q 10/00, G07C 5/08, G07C 5/00, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)P. Revathi
2)Dr. P. Sumathi
3)Divagar J M
4)Jevesh M M
5)Johan Ricky Franklin
6)Kabilan M
7)Kavinila S
8)Kaviya V
9)Kumari Masha Ivanovna V
10)Lochana R
11)Madhushri S

(57) Abstract :
This project presents SmartVahaan, an AI-Powered Vehicle Maintenance and Assistance System designed to help vehicle owners understand their vehicle condition and manage maintenance requirements more effectively. The system integrates Artificial Intelligence (AI), AI-assisted vehicle analysis, predictive maintenance, image-based issue analysis, and location-based garage discovery to analyze vehicle information, identify potential maintenance requirements, provide maintenance, modification, and replacement recommendations, and assist users with vehicle-related queries. SmartVahaan allows users to enter vehicle details such as model, fuel type, city, distance traveled, driving patterns, and usage conditions, and also supports vehicle image analysis for identifying visible issues. By combining intelligent vehicle analysis, personalized maintenance recommendations, vehicle history, AI assistance, and nearby garage discovery with navigation support, the proposed system provides a centralized and convenient platform for improving vehicle maintenance awareness and assisting vehicle owners in taking appropriate maintenance actions.

No. of Pages : 6 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131323 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AUGMENTED REALITY TEXTBOOK OVERLAY SYSTEM WITH ADAPTIVE MULTIMEDIA CONTENT ENRICHMENT AND INTERACTIVE LEARNING INTEGRATION

(51) International classification	:G06F 17/30, H04L 9/32, G06F 21/62, G06K 9/00, G06F 16/583
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mahendra Engineering College
Address of Applicant :Mahendra Engineering College, Mahendirapuri, Mallasamudram, Namakkal, Tamil Nadu, India. PIN: 637503. Tamil Nadu India
(72)**Name of Inventor :**
1)Mr.R.BALAMURUGAN
2)Dr.A.NEELA MADHESWARI
3)Dr.B.KALAISELVI
4)Dr.A.RAJAMURUGAN
5)Ms.R.KALAIVANI
6)Mrs.D.MOHANAPRIYA
7)Mrs.S.VIJAYA
8)Mr.T.Vishal
9)Mr.V.PRAVEEN
10)Mr.V.VISHAL
11)Mr.C.ABISHEK

(57) Abstract :
The sharing of images across social media, websites, and anonymous forums has increased image misuse, impersonation, and identity theft. Existing protection methods are reactive, fragmented, and dependent on manual reporting. Dragon Wipe addresses this with an automated framework that unifies image fingerprinting, perceptual hashing, and internet monitoring in a single platform. Users upload images to establish ownership, and the system generates visual fingerprints that stay resilient to resizing, cropping, compression, and minor edits. These fingerprints are continuously compared against publicly accessible online images to detect identical or visually similar copies. When misuse is detected, the system generates verifiable evidence, including similarity scores, cryptographic hashes, timestamps, and source identifiers. Users can then initiate reporting, takedown requests, or deletion without technical or legal expertise, and every incident is securely logged for integrity and traceability widespread. A key innovation is the Cyber Alert Protocol. It aggregates misuse logs into periodic alert bundles, classifies them by severity, and securely transmits them to designated cybercrime authorities, turning isolated incidents into actionable intelligence. Built on privacy-by-design principles, the system monitors only public content and user-consented images. Its modular architecture supports personal, organizational, and governmental deployment, with scope for future deepfake detection and AI-driven threat analysis.

No. of Pages : 29 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131325 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A CLEANING COMPOSITION COMPRISING MACROCARPAL-A

(51) International classification	:C11D 3/48, C11D 3/20, C11D 3/382, C11D 1/94, C11D 1/83	(71)**Name of Applicant :** **1)Badrunnisa S** Address of Applicant :Naaz Biolabs, Research Incubation Centre, Krupanidhi Degree College, Bengaluru- 560035, Karnataka, India Bengaluru Karnataka India **2)Aarav Marwaha** **3)Tessy Anu Thomas** **4)Vinitha Ramanath Pai**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	(72)**Name of Inventor :**
(33) Name of priority country	:NA	**1)Badrunnisa S**
(86) International Application No Filing Date	: :01/01/1900	**2)Aarav Marwaha** **3)Tessy Anu Thomas**
(87) International Publication No	: NA	**4)Vinitha Ramanath Pai**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Disclosed herein is a cleaning composition comprising Macrocarpal A derived from Eucalyptus tereticornis (0.1-1.0% w/w), at least one functional agent (0.5-5.0% w/w), a plant-derived non-ionic surfactant (0.1-2.0% w/w), a solubilizer (0.5-5.0% w/w) and an aqueous carrier, and ethanol (60–70% w/w). The composition optionally further comprises a plant-derived non-ionic surfactant (0.1–2.0% w/w). The composition exhibits antibacterial activity against Staphylococcus aureus, Pseudomonas aeruginosa and Enterococcus faecium, as determined by well diffusion assay, and inhibits biofilm formation by these organisms. The composition further inhibits the virulence factor LasB elastase and reduces pyocyanin production by Pseudomonas aeruginosa. The composition is suitable for use as a floor cleaner, glass cleaner, toilet-seat disinfectant spray or a mosquito repellent spray.

No. of Pages : 24 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131331 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPUTATIONAL EVALUATION OF BIOACTIVE PHYTOCONSTITUENTS FROM WITHANIA SOMNIFERA (ASHWAGANDHA) AGAINST INFLAMMATORY TARGETS: INSIGHTS FROM DENSITY FUNCTIONAL THEORY (DFT) AND MOLECULAR DOCKING STUDIES

(51) International classification	:A61K 36/81, G01N 33/50, A61K 36/185, G01N 33/68, A61K 36/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.R KUMUTHA
Address of Applicant :S.A Engineering college (Autonomous),Dept of Physics, Veeraghavapuram ,Chennai-600077. Taminadu Chennai Tamil Nadu India
2)S.A Engineering College
(72)**Name of Inventor :**
1)Dr.R KUMUTHA

(57) Abstract :
ABSTRACT [0001] Ayurvedic formulations heavily rely on multi-component, multi-target botanical extracts, where active phytoconstituents drive therapeutic efficacy. This project investigates the structural reactivity, electronic properties, and binding interactions of primary bioactive markers from Withania somnifera (Ashwagandha)—specifically Withaferin A, Withanolide D, and Withanone—against key inflammatory protein targets (such as Cyclooxygenase-2 [COX-2] and Tumor Necrosis Factor-alpha [TNF-α]).To evaluate molecular stability and reactive sites, [0002] Density Functional Theory (DFT) calculations are performed at the B3LYP/6-31++G(d,p) level of theory to map Frontier Molecular Orbitals (HOMO-LUMO) and global reactivity descriptors. Subsequently, molecular docking simulations are executed to estimate binding affinities, hydrogen-bonding networks, and hydrophobic interactions within the active pockets of the target proteins. The integration of DFT and docking provides atomic-level validation of the anti-inflammatory mechanisms claimed in traditional Ayurvedic texts, establishing a robust computational framework for rationalizing herbal drug design. KEY WORDS Density Functional Theory (DFT), B3LYP/6-31++G(d,p), HOMO-LUMO.

No. of Pages : 4 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131344 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Explainable AI for Intrusion Detection Systems: LIME and SHAP Applicability on Multi- Layer Perceptron

(51) International classification	:G06N 3/04, G06N 3/08, H04L 29/06, H04L 9/40, G06N 20/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Malla Reddy(MR) Deemed to be University
Address of Applicant :Malla Reddy(MR) Deemed to be University, Maisammaguda(H), Medchal Mandal, Medchal-Malkajgiri District, Telangana State-500100. ipr@mrdu.edu.in Medchal-Malkajgiri Telangana India
2)Geddam Varahgiri
(72)**Name of Inventor :**
1)Geddam Varahgiri
2)Janapareddi Sanyasamma
3)Dr.Satharla Mahipal
4)Garrepally Pranavi
5)Uppitarla Shireesha
6)Vikram Veeranjaneyulu
7)Varikuntla Balakrishna

(57) Abstract :
The present invention discloses an explainable artificial intelligence (XAI)-based intrusion detection system and method for enhancing cybersecurity through accurate and interpretable network threat detection. The proposed system integrates a Multi-Layer Perceptron (MLP) classification model with Explainable Artificial Intelligence techniques, including Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP), to provide transparent and reliable classification of network traffic. The system acquires network traffic data, performs preprocessing operations comprising data cleaning, feature encoding, normalization, and feature selection, and subsequently classifies the processed data into normal or malicious traffic using the trained MLP model. To overcome the black-box limitations of conventional deep learning-based intrusion detection systems, the invention generates both local and global explanations by identifying the contribution and significance of individual network features influencing each prediction. The framework further incorporates visualization and reporting modules that present feature importance, prediction rationale, and intrusion alerts in an interpretable manner, thereby assisting cybersecurity professionals in validating model decisions and improving incident response. The proposed architecture is extensible and supports integration with additional machine learning models, ensemble classifiers, and real-time monitoring environments. By combining high-performance intrusion detection with explainable decision-making, the invention enhances transparency, trustworthiness, operational reliability, and practical deployment of AI-driven cybersecurity systems across enterprise networks, cloud infrastructures, Internet of Things (IoT) ecosystems, and other critical digital environments.

No. of Pages : 10 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131355 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Dynamic Predictive Harvest Pooling with Uncertainty-Bounded Commitment and Automatic Shortfall Re-allocation

(51) International classification	:G06Q 50/02, G06Q 10/08, G06Q 30/06, G06Q 30/00, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Vidyavardhaka College of Engineering
Address of Applicant :Vidyavardhaka College of Engineering P.B. No.206, Kannada Sahithya Parishath Rd, III Stage, Gokulam, Mysuru, Karnataka 570002 Mysuru Karnataka India
(72)**Name of Inventor :**
1)Srinivasa B L
2)Shobhitha Poojar N
3)Keerthana K
4)Prof. Noor Fathima F

(57) Abstract :
The invention is a smart supply system that lets buyers of farm produce book crops from many small farms before the crops are harvested, without the risk of being promised more than the farms can deliver. Today a hotel, retailer or food processor that needs, say, 2,000 kg of tomatoes on a fixed date usually buys only after harvest, or signs a contract with one farm. A single small farm is too small and too uncertain: rain, heat or pests can cut its harvest at any time, and the buyer finds out only on delivery day. The invention solves this problem. For every crop it forecasts the harvest as a range, a cautious amount, a most-likely amount and a best-case amount, and allows a farmer to promise only the cautious amount. It combines the cautious amounts of many farms into one “harvest pool” that fills the buyer’s order. Whenever a farm’s forecast drops, a farmer withdraws, or new supply appears, the system calculates the exact gap, takes it only from the most recent bookings and refills it from other farms automatically, recording every change. Produce counts as delivered only after it is weighed and graded at a collection hub, and every weighed harvest is used to make future forecasts more accurate. Key benefits: buyers can book supply early and reliably; small farmers find a market before harvest without over-promising; shortfalls are repaired automatically instead of being discovered on delivery day; and every quantity is traceable from farm to buyer. In simulated tests, when half of the farms lost 30–70% of their crop, the invention still delivered 99.0% of demand, against 83.5% for a fixed allocation.

No. of Pages : 15 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131387 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : HEALTH INSURANCE PRICE PREDICTION USING MACHINE LEARNING

(51) International classification	:G06Q 40/08, G06Q 40/00, G06N 20/00, G06N 5/04, G06Q 30/02	(71)**Name of Applicant :** **1)Lords Institute of Engineering and Technology (Autonomous)** Address of Applicant :Lords Institute of Engineering and Technology, Survey No. 32, Near Police Academy, Appa Junction, Himayath Sagar, Hyderabad, Telangana - 500091, India Hyderabad Telangana India (72)**Name of Inventor :** **1)Dr. K. Nagi Reddy** **2)Dr. Karunakar Reddy** **3)Dr. Srikanth** **4)Ms. M. Neelima** **5)Mr. Yellaiah Ponnam** **6)Mr. Syed Juber** **7)Mrs. Devarakonda Latha** **8)Ms. Swaroopa**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

Health insurance is a critical component of financial planning, providing individuals and families with a safety net against unexpected medical expenses. The rising cost of healthcare has made it essential for individuals to predict potential insurance charges accurately, enabling them to choose suitable insurance plans. Given the complexity of calculating health insurance premiums, which depend on multiple factors such as age, lifestyle, and health status, machine learning offers a promising solution for automating and optimizing this process. This project, titled “Health Insurance Price Prediction using Machine Learning,” aims to develop a system that can accurately predict health insurance premiums based on personal and lifestyle attributes. The system is built using Python as the coding language, with a user-friendly web interface developed using HTML, CSS, and JavaScript. The Flask web framework integrates the front end and backend, allowing seamless interactions for users. We aim to build the predictio

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131396 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : DATA-DRIVEN METHOD FOR INDIVIDUAL CELL LIFE CYCLE PREDICTION OF ELECTRIC VEHICLE BATTERY CELLS

(51) International classification	:H01M 10/48, G01R 31/36, H01M 10/42, H02J 7/00, G01R 31/382
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.C.N.Vanitha
Address of Applicant :Professor, Department of Information Technology, Karpagam College of Engineering Tamil Nadu India
2)Karpagam College of Engineering
(72)**Name of Inventor :**
1)Ms.L.Jenifer Amla
2)Dr.C.N.Vanitha
3)Mr.Guruprasath Haridoss
4)Mr.M.Hemachandran
5)Mr.R.Naveen kumar

(57) Abstract :
The present invention generally relates to the fields of early prediction of individual cell degradation using voltage, current, temperature, and cycle data of Electric vehicle battery packs. Battery pack consist of multiple lithium-ion cells whose performance gradually decreases due to repeated charging and discharging, temperature, current, and operating conditions. Since individual cells can degrade at different rates, early identification of abnormal degradation is important for maintaining battery pack reliability and safety. Thus, a lithium-ion battery pack is monitored using a battery cell monitoring IC/AFE, NTC temperature sensors, and a current sensor connected to an STM32 Nucleo development board. The collected data is processed using Python, Pandas, and NumPy, and Random Forest Regression is implemented using Scikit-learn to predict the degradation trend of individual cells. The extracted battery features are used to identify degradation patterns and estimate the future health of each cell. Matplotlib is used to visualize battery parameters and degradation trends, while Stream lit is used to develop a monitoring dashboard. The proposed invention combines real-time battery monitoring with machine-learning-based prediction to support early detection of cell degradation and improved battery pack management.

No. of Pages : 12 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131403 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SUSTAINABLE CEMENTITIOUS ROOFING SYSTEM WITH INTEGRATED THERMAL AND STRUCTURAL MONITORING

(51) International classification	:E04D 11/02, E04D 11/00, E04D 12/00, G01N 25/18, G01N 25/20	(71)**Name of Applicant :** **1)IFET COLLEGE OF ENGINEERING** Address of Applicant :IFET Nagar, Gangarampalayam, Villupuram-605108 Viluppuram Tamil Nadu India (72)**Name of Inventor :** **1)Mrs. P. Priyanka** **2)Mr. N. Vaithianathan** **3)Ms. R. Ramya** **4)Ms. S. Thamizhini**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a sustainable cementitious roofing system incorporating electronic monitoring of thermal and structural conditions. The system comprises a cementitious roofing sheet (102) formed from a cement–sand matrix (104) incorporating marble powder (106), silica fume (108), and Aloe vera gel (110). A temperature sensor (112) arranged in thermal association with the roofing sheet (102) measures temperature variations and communicates the measurements to a microcontroller-based monitoring unit (120). A wireless communication module (122) transmits the processed measurements to a remote monitoring platform for continuous observation of thermal behaviour. The system may further incorporate a moisture sensor (114), strain sensor (116), and heat-flux sensor (118) for monitoring additional physical conditions. The arrangement combines a modified cementitious roofing composition with remote thermal and structural-condition monitoring for building roofing applications.

No. of Pages : 29 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131404 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTIFUNCTIONAL INTERLOCKING PAVER BLOCK WITH RAINWATER RETENTION AND PHOTOCATALYTIC SURFACE

(51) International classification	:E01C 11/22, E01C 5/04, E01C 5/06, E01C 5/00, E03F 5/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)IFET COLLEGE OF ENGINEERING
Address of Applicant :IFET Nagar, Gangarampalayam, Villupuram-605108 Viluppuram Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. P. Priyanka
2)Mr. N. Vaithianathan
3)K.Apinaya
4)S.Divyadharshan

(57) Abstract :
The present invention relates to a multifunctional interlocking concrete paver block for stormwater management, passive cooling and environmental pollutant mitigation. The paver comprises an interlocking concrete paver body (102), a photocatalytic surface layer (104), water-entry openings (106), internal water-flow channels (108), rainwater retention cavities (110), a bio-retention zone (112) containing filtration and moisture-retaining media (114), drainage and infiltration outlets (116), and interlocking engagement profiles (118). Rainwater entering through the water-entry openings (106) is directed through the internal water-flow channels (108) into the rainwater retention cavities (110), temporarily stored, filtered through the bio-retention zone (112) and gradually discharged through the drainage and infiltration outlets (116). Retained moisture supports passive evaporative cooling, while the photocatalytic surface layer (104) facilitates light-assisted pollutant degradation and rainfall-assisted self-cleaning. The paver is applicable to pedestrian pathways, institutional campuses, landscaped areas and low-traffic pavements.

No. of Pages : 30 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131405 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : VEHICLE-MOUNTED SYSTEM FOR EXCAVATED MATERIAL TRANSPORTATION MONITORING AND COMPLIANCE DETECTION

(51) International classification	:G01G 19/08, G01S 19/42, G01C 21/34, G08G 1/123, G08G 1/01	(71)**Name of Applicant :** **1)IFET COLLEGE OF ENGINEERING** Address of Applicant :IFET Nagar, Gangarampalayam, Villupuram-605108 Viluppuram Tamil Nadu India (72)**Name of Inventor :** **1)Mr. N.Sivanantham** **2)Mrs. R. Revathy** **3)Mr. M. Ravichandran** **4)Ms. V. Abinaya** **5)Ms. K. Dhiviyadarshiny**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to a vehicle-mounted monitoring system (102) for monitoring transportation of excavated materials. The system (102) comprises a microcontroller (104), a satellite-based positioning module (106), a load-cell sensor (108) connected through a load-cell amplifier module (110), a wireless communication module (112), a trip-detection input switch (120), and an alerting arrangement comprising a liquid crystal display unit (114), an audible buzzer (116), and a vibration motor (118). The microcontroller (104) acquires geographical coordinates, determines transported-load measurements, and increments a transportation trip count in response to trip-detection input transitions. The acquired information is compared with predefined geographical, load, and trip-count limits to identify transportation-related violations. Upon detecting a violation, the system generates corresponding local warnings and transmits violation information together with geographical coordinates to a remote recipient. The system facilitates monitoring of excavated-material transportation activities in mining, quarrying, and construction operations.

No. of Pages : 25 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131406 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ELECTRONIC IDENTIFICATION AND LIFE-CYCLE TRACKING SYSTEM FOR MODIFIED AUTOCLAVED AERATED CONCRETE BLOCKS

(51) International classification	:G06K 7/10, G06K 19/077, G06K 17/00, G06K 19/07, G06Q 10/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)IFET COLLEGE OF ENGINEERING
Address of Applicant :IFET Nagar, Gangarampalayam, Villupuram-605108 Viluppuram Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. P. Priyanka
2)Mr. N. Vaithianathan
3)S. Sivapriya
4)S. Gowsalya

(57) Abstract :
The present invention relates to an electronic identification and life-cycle tracking system for ceramic waste–ultra-high-molecular-weight polyethylene modified autoclaved aerated concrete blocks (102). The system comprises an embedded radio-frequency identification tag (104) enclosed within a protective tag encapsulation (106), a radio-frequency identification reader (110), and a cloud server (112) comprising a digital material passport database (114) and a life-cycle tracking platform (116). The identification tag (104) is embedded before curing and associated with a unique identification code corresponding to an individual block (102). The protective encapsulation (106) protects the tag during curing while permitting subsequent electronic identification. A machine-readable quick-response code (108) provides an alternative identification mechanism. The digital material passport stores material composition, manufacturing parameters, curing information, and quality-test results, while the life-cycle tracking platform maintains transportation, installation, in-service, and end-of-life records, enabling block-specific traceability and construction-material management.

No. of Pages : 24 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131407 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ADAPTIVE ENERGY MANAGEMENT SYSTEM FOR HYBRID SOLAR AND WIND POWER GENERATION

(51) International classification	:H02J 3/38, H02J 3/32, H02J 7/35, H02J 1/10, H02J 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)IFET COLLEGE OF ENGINEERING
Address of Applicant :IFET Nagar, Gangarampalayam, Villupuram-605108 Viluppuram Tamil Nadu India
(72)**Name of Inventor :**
1)Dr. A. Devi
2)A. Abinaya

(57) Abstract :
The present invention relates to an adaptive energy management system for hybrid renewable energy installations. The system comprises a solar photovoltaic array (102), a wind turbine generator (106), respective power converters (104, 108), a common DC bus (110), a lithium-ion battery pack (114) coupled through a bidirectional DC–DC converter (112), a controlled switching unit (116), and a DC–AC inverter (118) supplying an electrical load (120). Sensing units (122, 124, 126, 128) monitor renewable power generation, battery state of charge, and load demand. A microcontroller (130) executes a multi-threshold finite-state control mechanism to supply renewable power directly to the load, direct surplus power towards battery charging, or enable battery-assisted supply when renewable generation is insufficient and battery charge exceeds a predetermined threshold. The arrangement coordinates power distribution while restricting unnecessary battery cycling and excessive discharge. A wireless communication interface (132) enables remote monitoring and load control.

No. of Pages : 26 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131408 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ENHANCING LIFETIME PREDICTION IN SEQUENTIAL RELIABILITYSTRUCTURES VIA NONLINEAR INFERENCE

(51) International classification	:G06F 17/50, G06F 19/00, G06F 17/18, G06Q 10/06, G05B 13/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)K.GLORY PRASANTH
Address of Applicant :S.A. Engineering College, Veeraragavapuram, Chennai-77. Tamil Nadu India
2)S.SELVI
3)M.RAMALATHA
4)S.A. Engineering college
(72)**Name of Inventor :**
1)K.GLORY PRASANTH
2)S.SELVI
3)M.RAMALATHA

(57) Abstract :
Evaluating k-out-of-n systems is critical in reliability theory to optimize mission planningandminimize downtime. This paper advances sequential order statistics by transitioningfromthree-component configurations to more robust four-component (1,4) and (2,4) systems underlocation-scale families. Unlike traditional independent models, our sequential approachcaptures the cascading stress transferred to surviving components immediately followinganindividual unit failure. By modeling these structures under gamma-exponential mixturedistributions, we derive explicit closed-form expressions for the reliability function andMeanTime Between Failure (MTBF), proving that a four-component architecture extends systemlongevity by up to 12%. Furthermore, we develop the Minimum risk equivariant estimator(MREE) under squared error loss. Rigorous Monte Carlo simulation protocols demonstratethat the proposed MREE framework consistently outperforms naive moment-basedestimation, achieving a 15% to 45% reduction in Mean Squared Error (MSE) across varioussample sizes and providing a precise toolkit for industrial maintenance scheduling

No. of Pages : 7 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131431 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Novel intermediate process and its use in the preparation of PI3K/mTOR dual inhibitor

(51) International classification	:A61K 31/445, C07D 211/58, C07D 401/12, C07D 401/06, A61K 31/4545	(71)**Name of Applicant :** **1)SSV Pharma Private Limited** Address of Applicant :SSV Sai Emerald, H.No.2-91/b/12 and H.no.2-91/b/13, Sy.no.21/P, Madhapur, Khanamet(V), Ranga Reddy Dist, Telangana, 500081 Hyderabad Telangana India (72)**Name of Inventor :** **1)VENKATESWARA RAO GORANTLA** **2)SAI PHANENDRA GORANTLA** **3)SHYAMSUNDER CHENUMALLA** **4)SUDHARSHAN REDDY KARNATI** **5)MOHAN KRISHNA KUMAR VEERAMSETTI**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The present invention related to an improved process for the preparation of (4-Aminophenyl) (4-(dimethylamine) piperidin-1-yl) methanone compound of formula-I.
Formula-I

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131440 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RESISTIVE RANDOM-ACCESS MEMORY DEVICE WITH TELEMETRY WEIGHTING FOR PULSE CONTROL

(51) International classification	:G11C 13/00, G06F 11/10, G11C 11/00, G11C 11/56, G11C 11/16
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Amrita Vishwa Vidyapeetham
Address of Applicant :Amrita Vishwa Vidyapeetham, Bengaluru Campus, Kasavanahalli, Carmelaram P.O., Bengaluru - 560035, Karnataka, India. Bengaluru Karnataka India

(72)**Name of Inventor :**
1)SIDDIREDDY, Kapileswarreddy
2)HEGDE, Ganapathi

(57) Abstract :

The present disclosure provides a resistive random-access memory device (100). A telemetry unit (106) collects a switching parameter (202) of a resistive cell (102-2), a cycling parameter (204) of a memory array (102) and a residual error (206) remaining after error correction (104). An aggregation unit (108) windows the cycling parameter (204) and a projection unit (110) forms a token sequence (304). A weighting unit (112) derives a weight of a token from a relation to a remaining token, whereupon a control unit (114) selects a mitigation action (316) and a pulse generation unit (116) sets a width of a programming pulse (318). Unlike conventional controllers that track one indicator against a fixed threshold, the device (100) ranks indicators by prevailing conditions, achieving a failure prediction accuracy improvement of 35.9 percent ± 1.6 percent over static fault-injection baselines and providing BIST-based early warning 320 cycles to 490 cycles before ECC saturation onset, with controller area and power overheads of 4 percent and 5 percent respectively, suiting in-memory computation, storage-class memory, automotive and medical equipment.

No. of Pages : 38 No. of Claims : 12

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131465 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : REAL-TIME AGRICULTURAL MONITORING SYSTEM

(51) International classification	:G06Q 50/02, A01B 79/00, H04N 7/18, G06T 7/00, G06Q 10/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BELLARE GIS CONSULTANCY PRIVATE LIMITED
Address of Applicant :CHANDRA COMPLEX, P.O & VILLAGE BELLARE, SULLIA TALUKU, DAKSHINA KANNADA - 574212, KARNATAKA, INDIA Karnataka India
(72)**Name of Inventor :**
1)DR NAVEENCHANDRA B
2)ADYA HOSAAVALIKE

(57) Abstract :
Disclosed herein is a system (100) for real-time monitoring of agricultural field to visualize and analyse the data that comprises a user device (102) configured to receive and display the data, a user interface (104) connected to the user device (102) to receive and interpret real-time information, a camera unit (106) to capture the surrounding environment, a sensor unit (108) to acquire physical, physiological and wireless environmental data, a GPS unit (110) to locate the farm monitoring points, a communication network (112) to facilitate transmission of acquired data, a processing unit (114) connected to the user device (102) via the communication network (112) to process data, wherein the processing unit (114) further comprises a data input module (116), a data processing module (118), an image processing module (120), an agriculture assessment module (122), a real-time alert module (124), a recommendation module (126), and an output module (134).

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131486 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : RescueConnect: Eliminating Critical Latency in Emergency Dispatch

(51) International classification	:H04W 4/90, H04M 11/04, G08B 25/01, H04M 3/51, G08B 25/10	(71)**Name of Applicant :** **1)Mahendra Engineering College** Address of Applicant :Mahendra Engineering College, Mahendirapuri, Mallasamudram, Namakkal, Tamilnadu, India. Pin: 637503. Namakkal Tamil Nadu India (72)**Name of Inventor :** **1)Ms.R.KALAIVANI** **2)Dr.A.NEELA MADHESWARI** **3)Dr.B.KALAISELVI**
(31) Priority Document No	:NA	**4)Dr.A.RAJAMURUGAN**
(32) Priority Date	:NA	**5)Mr.R.BALAMURUGAN**
(33) Name of priority country	:NA	**6)Ms.R.JANANIPRIYA**
(86) International Application No Filing Date	: :01/01/1900	**7)Mr.K.S.ARUN** **8)Mr.Y.POOVENDHAN**
(87) International Publication No	: NA	**9)Mr.V.NISHANTH**
(61) Patent of Addition to Application Number Filing Date	:NA :NA	**10)Mr.M.NITHISH** **11)Ms.B.RITHIGA**
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

RescueConnect is an emergency-dispatch system developed to reduce the delay in providing medical help during accidents and other emergencies. In many emergency situations, people have to spend valuable time explaining the incident over a phone call, which can slow down the response. RescueConnect solves this problem by allowing bystanders to report an emergency instantly through a WebApp or WhatsApp without downloading any application. The system captures real-time images, the exact GPS location, and other important information and sends them directly to the emergency call center. This helps the dispatcher understand the situation quickly and arrange the appropriate ambulance. Even when there is no internet connection, the system can use SMS or USSD to send the essential location details. The captured images are stored only for a limited time and are automatically deleted after 60 minutes to protect privacy. Overall, RescueConnect aims to make emergency reporting faster, simpler, and more reliable while helping emergency teams respond efficiently.

No. of Pages : 21 No. of Claims : 2

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131493 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT ADAPTIVE LEARNING SYSTEM FOR DYNAMIC PERSONALIZED EDUCATION

(51) International classification	:G06Q 50/20, G09B 5/14, G09B 7/04, G09B 7/02, G09B 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)KRISHNENDU PRASANTH
Address of Applicant :NITTE (DEEMED TO BE) UNIVERSITY, NITTE INSTITUTE OF PROFESSIONAL EDUCATION, DEPARTMENT OF COMPUTER APPLICATION, MANGALURU, KARNATAKA, INDIA Karnataka India
2)RUDRANARAYAN
3)KISHORE S
4)DR AMRITHKSLA M SHETTY
(72)**Name of Inventor :**
1)KRISHNENDU PRASANTH
2)RUDRANARAYAN
3)KISHORE S
4)DR AMRITHKSLA M SHETTY

(57) Abstract :
Disclosed herein is an intelligent adaptive learning system (100) for dynamic personalized education that comprises a learner interface (102), a learner-state management unit (104), a curriculum preparation system (106), a knowledge retrieval system (108), a data storage unit (110), a communication interface (112), and a processing unit (114). The processing unit (114) comprises an instructional decision module (116), a context construction module (118), a prompt generation module (120), an AI teacher communication module (122), an assessment and evaluation module (124), a response validation module (126), an adaptive learning control module (128). The curriculum preparation system (106) generates structured concept packages from approved educational resources, and the knowledge retrieval system (108) retrieves curriculum-relevant information. The processing unit (114) determines instructional actions, constructs learner-specific instructional contexts, generates personalized educational responses. Learner responses are evaluated, learner-state information is updated, generated responses are validated before delivery, enabling controlled lesson progression and closed-loop adaptive learning.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131500 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPOSITE MATERIAL SYSTEM WITH ENHANCED MECHANICAL AND THERMAL PROPERTIES

(51) International classification	:C08J 5/04, C08K 7/02, C08J 5/24, C08K 7/14, C08K 9/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Meenakshi Sundararajan Engineering College (Autonomous)
Address of Applicant :Meenakshi Sundararajan Engineering College (Autonomous) , Kodambakkam, Chennai, Tamil Nadu 600024 Tamil Nadu India
(72)**Name of Inventor :**
1)Dr.S.V.Saravanan
2)Dr. Raja M A
3)Mr. Brithivi raj .G
4)Mr. K. Rajkumar
5)Dr. S. Kamatchisankaran
6)Mr. Iyyanar K
7)Dr. T. Prabakaran
8)Mr. Daniel Asirvatham A

(57) Abstract :
ABSTRACT COMPOSITE MATERIAL SYSTEM WITH ENHANCED MECHANICAL AND THERMAL PROPERTIES 5 10 The present invention relates to a composite material system comprising a polymer matrix, reinforcing fibres, thermally conductive fillers, an interfacial coupling agent, and optionally functional additives. The reinforcing fibres improve load-bearing performance, while the thermally conductive fillers facilitate heat transfer through the matrix. The coupling agent promotes interfacial compatibility between selected constituents. Manufacture includes matrix preparation, filler dispersion, fibre incorporation, and consolidation by moulding or curing. The composite may be formed into structural panels, housings, automotive components, and thermal management components. [Figure 1 is reference figure]

No. of Pages : 18 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131515 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A RESOURCE-EFFICIENT MULTI-DOMAIN EEG SIGNAL PROCESSING AND CLASSIFICATION SYSTEM

(51) International classification	:G01K 1/14, G01K 7/02, G01K 13/02, G01K 13/00, G01N 25/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR UNIVERSITY
Address of Applicant :ANANTHASAGAR, HASANPARTHY (M), WARANGAL URBAN, TELANGANA, 506371, INDIA Telangana India
(72)**Name of Inventor :**
1)PADAKANTI SWAPNA
2)RAVICHANDER JANAPATI

(57) Abstract :
Disclosed herein is performance measuring system (100) that comprises a device (102) and an enclosure panel (104) attached to the device (102) via an electrical wire duct (106). The device (102) includes a mounting rim (108) having an outer side flange (110) and a spindle. The device (102) also includes a heating assembly (114) fixed at the outer side flange (110) of the mounting rim (102). The heating assembly (114) having a set of mounting brackets (116), tubular heating elements (118), thermocouples (120), a slip ring (122), and an adapter (124). The enclosure panel (104) includes a data logger (128) and a temperature controller (126) configured to control the heating of the mounting rim (108) and a tyre bead area to a pre-set temperature for specific heating degree and time leading to the deformation in bead material bonding and bead pull-out.

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131516 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : AN AI-BASED MULTILINGUAL VOICE ADVISORY AND RENEWAL-RETENTION PREDICTION SYSTEM

(51) International classification	:G10L 15/22, G06N 20/00, G10L 15/26, G06Q 10/06, G10L 15/18
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR UNIVERSITY
Address of Applicant :ANANTHASAGAR, HASANPARTHY (M), WARANGAL URBAN, TELANGANA - 506371, INDIA Telangana India
(72)**Name of Inventor :**
1)DR. JYOTI PRAKASH RATH
2)PROF. (DR.) DAMARLA RAMESH BABU
3)PROF. (DR.) SOUMENDRA DAS

(57) Abstract :
Disclosed herein is an AI-based multilingual voice advisory and renewal-retention prediction system (100) that comprises a Data Acquisition Unit (102), a Communication Unit (104), a Voice Interaction Unit (106), a Display Unit (108), a Processing Unit (110), and a Memory Unit (128). The Processing Unit (110) comprises a Pre-processing Module (112), a Prediction Module (114), a Reason-Code Generation Module (116), a Voice Advisory Module (118), a Guidance Module (120), a Response Capture Module (122), an Analytics Module (124), and a Model Updating Module (126). The system (100) processes policyholder information to predict policy lapse risk before the renewal due date, identifies contributing factors, and generates personalized multilingual voice advisory in regional languages. The system (100) further provides claim-utilisation guidance, records policyholder responses and renewal outcomes, evaluates retention metrics, and updates prediction models using recorded data, thereby supporting improved policy renewal, benefit utilisation, and retention among rural health insurance policyholders.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131522 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PREDICTIVE MECHANICAL FAILURE PREVENTION SYSTEM USING INTEGRATED SENSOR NETWORKS

(51) International classification	:G05B 23/02, G06Q 10/00, G06Q 10/06, G06F 11/07, G06Q 50/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. L. MYTHILI
Address of Applicant :Head & Assistant Professor, Department of Commerce and Professional Accounting ,Theni Kammavar Sangam College of Arts and Science, Koduvilarpatti, Theni Dt -625534, Tamil Nadu, India. Tamil Nadu India
2)Dr. S. JEYALAKSHMI
3)Dr. R. VAIRAM
4)Dr. P. RAMYA
5)S. KANNAN
6)A. SUGANYADEVI
7)R. NANTHINI
8)R. NAGANANDHINI
9)M. SARAN
10)S. SURUTHI
(72)**Name of Inventor :**
1)Dr. L. MYTHILI
2)Dr. S. JEYALAKSHMI
3)Dr. R. VAIRAM
4)Dr. P. RAMYA
5)S. KANNAN
6)A. SUGANYADEVI
7)R. NANTHINI
8)R. NAGANANDHINI
9)M. SARAN
10)S. SURUTHI

(57) Abstract :
Unexpected mechanical failures in rotating and reciprocating machinery cause costly unplanned downtime, safety hazards, and shortened equipment life. Conventional strategies, whether reactive (run-to-failure) or time-based preventive maintenance, either allow breakdowns to occur or replace healthy components too early. This project presents a Predictive Mechanical Failure Prevention System that uses an integrated network of sensors to monitor machine health continuously and forecast faults before they develop into failures. A dashboard visualizes live readings, health indicators, and trend analysis, and the system issues automated alerts via SMS, email, or an on-site alarm when predefined thresholds or predicted failure patterns are detected, enabling maintenance to be scheduled before breakdown occurs. Experimental validation on a test rig showed that multi-sensor data fusion improves fault detection accuracy and reduces false alarms compared with single-sensor monitoring. The proposed system offers a low-cost, scalable, and reliable solution for condition-based maintenance. It can reduce downtime and maintenance costs, improve operational safety, and extend equipment life, making it suitable for manufacturing plants, power generation, automotive, and other Industry 4.0 applications.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131525 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SELF-ADAPTIVE AI POWER MANAGEMENT SYSTEM FOR IMPLANTABLE NEURAL INTERFACE SYSTEM

(51) International classification	:A61N 1/05, A61N 1/36, A61N 1/372, A61N 1/378, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SR UNIVERSITY
Address of Applicant :ANANTHASAGAR, HASANPARTHY (M), WARANGAL URBAN, TELANGANA, 506371, INDIA Telangana India
(72)**Name of Inventor :**
1)DR.TATIRAJU.V.RAJANI KANTH
2)DR. SANDIP BHATTACHARYA

(57) Abstract :
Disclosed herein is a self-adaptive AI power management system for implantable neural interfaces (100) comprising a neural electrode array (102), a neural signal acquisition unit (104), a signal processing unit (106), and a processing unit (108) having a pre-processing module (110), a feature extraction module (112), a workload prediction module (114), an energy demand prediction module (116), an energy availability estimation module (118), a confidence estimation module (120), an optimization module (122), a performance monitoring module (124), an adaptive learning module (126). The system further comprises a programmable neural stimulation unit (128), a stimulation electrode array (130), a power allocation, control unit (132), an energy supply and storage unit (134), a safety and fallback control unit (136), a sensing unit (138), a communication unit (140), an external programmer unit (142). The system predicts energy demand, optimizes power allocation, enforces safety constraints, continuously adapts power management based on monitored energy consumption.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131549 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : CYBER THREAT DETECTION USING ONE-CLASS MACHINE LEARNING

(51) International classification	:G06N 20/00, H04L 29/06, H04L 9/40, G06Q 10/06, G06F 21/55
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. K. Kalaivani
2)ABINITHI S
3)GIRITHARAN A
4)SANGAMESHWAR A
5)AJAY A

(57) Abstract :
The present project relates to a cybersecurity system designed to identify unusual and potentially suspicious employee activities using context-aware user behavior analytics and One-Class Machine Learning. The proposed system provides an integrated security monitoring platform combining role-based authentication, employee activity monitoring, activity-log storage, contextual data collection, machine-learning-based anomaly detection, risk classification, alert generation, and browser-based dashboards. The system records employee activities such as login, logout, profile update, password change, file upload, and file download. The collected activity records may include employee role, department, work shift, approved permissions, previous user behavior, device information, IP address, session duration, login/logout time, failed login attempts, and file upload/download counts. The recorded data is stored in MongoDB and prepared for analysis using a machine-learning pipeline. The Isolation Forest algorithm is used to identify unusual patterns in employee activity. Based on the configured detection logic, the system categorizes activity into Low, Medium, and High risk levels. Relevant alerts and activity information are presented through the HR and CEO/Admin dashboards to support investigation and security monitoring. The proposed system aims to reduce dependence on manual inspection of activity logs and provide a structured mechanism for identifying, recording, and reviewing potentially suspicious employee behavior. An anomaly indicates a need for investigation and does not, by itself, establish malicious intent.

No. of Pages : 6 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131550 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ScopeSync: AI-Based Client Requirement Analyzer

(51) International classification	:G06N 20/00, G06F 16/35, G06N 5/02, G06N 5/04, G06F 16/33
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs.K.Kalaivani
2)Cheithanya S
3)Monisha M
4)Prahadeesh K
5)Shobini M

(57) Abstract :
The present invention relates to an Artificial Intelligence (AI)-based intelligent client requirement analysis system configured to automatically analyze, interpret, and organize client requirements provided in textual format. The proposed system provides a unified platform combining client input processing, Natural Language Processing (NLP), AI-based requirement analysis, requirement classification, ambiguity identification, requirement prioritization, and automated report generation. The system receives client requirements through a user interface in the form of text documents or manually entered descriptions. It analyzes the textual content to identify key requirements, understand the client's needs, and extract important information related to the proposed project. The system uses AI and NLP techniques to categorize requirements into predefined categories, such as functional requirements, non-functional requirements, technical requirements, and business requirements. It can also identify incomplete, unclear, or ambiguous statements that may require further clarification from the client. When the requirements are analyzed, the system generates a structured summary of the client's expectations and organizes the extracted information for easier understanding. The system can prioritize requirements based on their importance and relevance to the project, helping development teams identify essential features and project objectives. Based on the analyzed information, the system generates a requirement analysis report that can be reviewed and used during the software development process. An authorized user can access the ScopeSync application through a web browser using a computer or compatible mobile device. The dashboard provides user authentication, client requirement submission, document processing, AI-based analysis, categorized requirement summaries, ambiguity identification, and access to generated reports. The analyzed results can be stored in a configured database or exported in a suitable document format for future reference and project planning. The invention provides an integrated workflow in which client-provided information is converted into structured requirements, the extracted information is analyzed using AI and NLP techniques, and the analyzed results are organized into meaningful categories and reports. The system is intended to reduce manual effort, minimize misunderstandings between clients and development teams, improve requirement clarity, and support more efficient and accurate software project planning.

No. of Pages : 8 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131551 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Intelligent Endpoint Security Against Keylogger Attacks

(51) International classification	:G06N 20/00, G06F 21/56, H04L 9/40, H04L 29/06, G06F 21/55
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. R. Saranya
2)AKASH RAMKUMAR S
3)SARAN KUMAR R
4)THARUN KUMAR A
5)THIRUPPATHI K M

(57) Abstract :
The present invention relates generally to an Artificial Intelligence (AI)-enabled and Machine Learning (ML)- powered endpoint cybersecurity platform configured to bridge the critical security gap between static signature-based antivirus tools and modern, zero-day keystroke logging malware. The proposed system provides a unified, local-first, real-time endpoint protection architecture combining continuous process telemetry monitoring, low-level operating system keyboard hook detection, behavioral context analysis, deterministic rule-based pre-filtering, behavioral feature extraction, supervised and anomaly-based machine learning threat detection, instant user alerting, risk-tier threat classification, automated malicious application blocking, and tamper-evident forensic security logging. The system deterministically monitors running processes and operating system input application programming interfaces (APIs), including global message hook registrations and asynchronous key-state polling loops, prior to algorithmic classification. For active processes exhibiting keyboard monitoring indicators, an explainable behavioral analysis engine constructs high-dimensional feature representations across process execution states, hidden window parameters, background file write frequencies, registry persistence modifications, and outbound network socket connections, executing machine learning inference to produce normalized threat probability scores alongside explicit diagnostic behavioral flags. The platform additionally integrates a deterministic rule-based engine enforcing cryptographic whitelist verification and known indicator-of-compromise matching, an instant user notification subsystem delivering actionable desktop alerts, and an automated containment controller capable of unhooking compromised input chains and terminating malicious processes in real time. Operational reliability and minimal false positive rates are guaranteed via a hybrid detection workflow that correlates low-level keyboard interception with secondary post-capture exfiltration behaviors before sensitive credentials can be compromised. The invention establishes an integrated, deterministic, and intelligent endpoint defense workflow transforming raw process and API telemetry into verified threat signals, eliminating signature dependency, preventing silent credential theft, neutralizing unknown keyloggers prior to data leakage, and providing continuous forensic security visibility for individual users, developers, and enterprises.

No. of Pages : 8 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131552 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Al Hyperlocal Maintenance & Emergency Platform

(51) International classification	:G06Q 50/26, G06N 20/00, G06Q 10/06, G06Q 10/00, G06N 3/08
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SNS College of Technology
Address of Applicant :SNS College of Technology, SNS Kalvi Nagar, Sathy Main Road, NH209, Vazhiyampalayam, Saravanampatti, Coimbatore, Tamil Nadu 641048. Coimbatore Tamil Nadu India
(72)**Name of Inventor :**
1)Mrs. K. Kalaivani
2)DAVID NICODAMAS A
3)JAIAKASH KUMAR S
4)KALIDHASS S
5)MADHAN KUMAR M

(57) Abstract :
Timely maintenance and rapid emergency response are critical pillars of modern urban living and community well-being, directly impacting public safety, infrastructure longevity, and daily convenience. However, conventional maintenance and emergency service systems often suffer from operational fragmentation, delayed response times, and inefficient dispatch mechanisms that rely heavily on manual coordination and lack localized situational awareness. To overcome these limitations, the proposed project introduces an AI-driven Hyperlocal Maintenance & Emergency Platform that integrates artificial intelligence with geospatial intelligence to provide rapid, localized incident triage, resource allocation, and service dispatch. The system employs advanced machine learning models capable of analyzing real-time multimodal inputs such as emergency distress signals, user incident descriptions, and proximity metrics to accurately prioritize requests and assess urgency levels. Using intelligent matching and dynamic routing algorithms, the platform autonomously connects users to the nearest certified technicians, civic service personnel, or emergency responders, significantly minimizing response latency. Over time, the platform leverages reinforcement learning and user feedback loops to refine service quality, predict recurring localized infrastructure failures, and optimize resource deployment. Unlike traditional fragmented dispatch methods, this AI-powered hyperlocal solution offers an accessible, transparent, and scalable service network suitable for a wide range of applications, including critical residential repairs, utility breakdowns, and urgent civic emergencies. By bridging the gap between localized community needs and immediate professional interventions, this project highlights the transformative potential of artificial intelligence in modernizing urban maintenance into an agile, highly reliable, and life-enhancing ecosystem that strengthens community resilience and infrastructure reliability.

No. of Pages : 9 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131565 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : IOT-BASED MULTI-PARAMETER SAFETY HELMET WITH PER-SENSOR FALSE ALARM VALIDATION

(51) International classification	:A42B 3/04, A42B 3/30, G08B 21/04, A42B 1/24, A42B 3/28
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)B.M.S. College of Engineering
Address of Applicant :Bull Temple Road, Basavanagudi, Bengaluru, Karnataka - 560019 Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. Sowmya K S
2)Dr. Nalina V
3)Dr. Anitha H M
4)B. Darshan
5)Bhoomika A P
6)Ankur Sai M
7)Neichelle Alvina Mendonca

(57) Abstract :
ABSTRACT IOT-BASED MULTI-PARAMETER SAFETY HELMET WITH PER-SENSOR FALSE ALARM VALIDATION A wearable IoT safety helmet is disclosed for hazardous industrial environments. A sensing layer monitors gas concentration, temperature, humidity, heart rate, blood-oxygen saturation, fall or impact motion and ambient light. A processing layer independently evaluates the gas concentration, temperature, heart rate, blood-oxygen saturation and fall or impact motion parameters against a respective threshold or confirmation condition. Gas concentration, temperature, heart rate and blood-oxygen saturation, together with fall or impact motion, are each evaluated using a predefined threshold or confirmation condition and a five-second confirmation period to distinguish a sustained hazardous condition from a temporary or transient reading. Confirmed hazardous conditions trigger the corresponding alert and communication functions. The system further incorporates an on-site display, cloud-based IoT monitoring, electronic-mail alerts, an SOS switch, ambient-light-based operation and a photovoltaic charging subsystem with a rechargeable lithium-ion cell. The independent parameter-specific operation enables a hazard in one parameter to be detected without requiring corroboration from another sensor. Fig. 1

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131568 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : KISANSAHAY-MITHRA: A FARMER-FRIENDLY SMART SYSTEM

(51) International classification	:A01G 25/16, G06Q 50/02, G05B 15/02, G05B 19/042, H04W 4/80	(71)**Name of Applicant :** **1)BTP Madhav** Address of Applicant :Department of ECE, Koneru Lakshmaiah Education Foundation, Vaddeswaram, AP, India Andhra Pradesh India **2)Koneru Lakshmaiah Education Foundation** (72)**Name of Inventor :** **1)Thotakura Yasaswini Aparna** **2)Akkaladevi Meenakshi** **3)Raya Vikhyath** **4)Kolla Bhargavi** **5)V Viswanath Shenoi** **6)B T P Madhav**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The invention delivers automated irrigation, real-time monitoring, and farmer support through a mobile application with voice command capabilities. The system uses an ESP32 microcontroller, soil and weather sensors, automated irrigation valves, and a mobile application. Components connect via Wi-Fi and Bluetooth. Field trials demonstrated a 20% reduction in water use and a 15% increase in crop yields compared to traditional farming methods. The system uses a modular IoT architecture that combines hardware sensors and wireless communication to automate irrigation. Farmers use a mobile application that supports voice commands and works offline.

No. of Pages : 17 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131572 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MICROBIAL COMMUNICATION-RESPONSIVE THERAPEUTIC INTERFACE FOR WOUND MANAGEMENT

(51) International classification	:A61L 15/44, A61F 13/00, A61L 15/28, A61L 15/46, A61L 2/18	(71)**Name of Applicant :** **1)IFET COLLEGE OF ENGINEERING** Address of Applicant :IFET Nagar, Gangarampalayam, Villupuram-605108 Viluppuram Tamil Nadu India (72)**Name of Inventor :** **1)Dr. M. Jebastin Sonia Jas** **2)Dr. P. Jacquline Rosy** **3)Mrs. P. Pravina** **4)Mrs. R. Ezhilarasi**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The present invention relates to a microbial communication-responsive therapeutic interface (120) for wound infection management and tissue regeneration. The interface comprises a protective outer layer (102), a microbial communication sensing layer (104), a quorum-sensing recognition layer (106) incorporating recognition elements (116), a predictive decision layer (108), a therapeutic nano-reservoir layer (110), a regenerative biomaterial layer (112), and a wound-contact interface (114). The recognition elements (116) detect bacterial quorum-sensing molecules and generate a corresponding detection signal. The predictive decision layer (108) assesses infection-risk conditions based on the detection signal and generates a therapeutic activation signal upon satisfaction of a predetermined threshold. A nano-reservoir gate structure (118) regulates localized release of antimicrobial and regenerative agents in response to the activation signal. The integrated arrangement enables microbial activity-dependent therapeutic delivery while supporting tissue regeneration in wound-care applications.

No. of Pages : 25 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131581 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : LINEZOLID-LOADED POLYMERIC BEAD AND PROCESS OF PREPARATION THEREOF

(51) International classification	:A61K 47/36, C08L 5/04, A61K 9/50, A61K 9/16, B01J 13/02	(71)**Name of Applicant :** **1)Amrita Vishwa Vidyapeetham** Address of Applicant :Amrita Vishwa Vidyapeetham, Bengaluru Campus, Kasavanahalli, Carmelaram P.O., Bengaluru - 560035, Karnataka, India. Bengaluru Karnataka India (72)**Name of Inventor :**
(31) Priority Document No	:NA	**1)GADAGI, Aishwarya G**
(32) Priority Date	:NA	**2)Athira K**
(33) Name of priority country	:NA	**3)BELLIRAJ, Siva Kumar**
(86) International Application No Filing Date	: :01/01/1900	**4)REDDY, Giridhar S**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
The present disclosure relates to a linezolid-loaded polymeric bead comprising sodium alginate, lignosulfonic acid and linezolid, wherein the sodium alginate and lignosulfonic acid form a polymeric matrix that is ionically crosslinked using a multivalent cation to form beads, and wherein linezolid is encapsulated within the ionically crosslinked polymeric matrix. The polymeric bead may optionally comprise one or more inorganic fillers selected from hydroxyapatite, silver nanoparticles and combinations thereof. The disclosure further provides a process for preparing the linezolid-loaded polymeric beads comprising preparing a polymer suspension comprising sodium alginate and lignosulfonic acid, incorporating linezolid and optionally one or more inorganic filler suspensions therein, and introducing the resulting mixture into a crosslinking medium comprising a multivalent cation to form crosslinked beads, followed by separation, washing and drying. The disclosed polymeric beads provide controlled release of linezolid and are useful for pharmaceutical and antimicrobial applications.

No. of Pages : 43 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131582 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : NANOFILLER-REINFORCED POLYMERIC COMPOSITE FILM AND PROCESS FOR PREPARATION THEREOF

(51) International classification	:C08J 5/18, C08K 3/34, C08K 3/36, C08K 3/04, C08K 3/22
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Amrita Vishwa Vidyapeetham
Address of Applicant :Amrita Vishwa Vidyapeetham, Bengaluru Campus, Kasavanahalli, Carmelaram P.O., Bengaluru - 560035, Karnataka, India. Bengaluru Karnataka India
(72)**Name of Inventor :**
1)VENUGOPAL, Veena
2)KANNAN CHERUVAKKODAN HOUSE, Hanima
3)BELLIRAJ, Siva Kumar
4)MATHUR, Parul

(57) Abstract :
The present disclosure relates to a nanofiller-reinforced polymeric composite film having one or more polymers; and one or more nanofillers, wherein a total amount of the one or more nanofillers is a range of 2 to 15 wt% based on the total weight of the polymeric matrix. The present disclosure also relates to a process for preparing a nanofiller-reinforced polymeric composite film comprising dissolving one or more polymers in water to obtain a polymer mixture; stirring the polymer mixture using magnetic stirrer to obtain a homogeneous polymer solution; adding one or more nanofillers thereto in an amount ranging from 2 to 15 wt% based on the weight of the polymers to obtain a nanofiller-containing suspension; subjecting the suspension to sonication to obtain a uniformly dispersed composition; and casting the homogeneous composition onto a substrate followed by drying to obtain the nanofiller-reinforced polymeric composite film. The nanofiller-reinforced polymeric composite film is suitable for microwave absorption and electromagnetic interference shielding applications.

No. of Pages : 40 No. of Claims : 17

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131588 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An Artificial Intelligence Based System and Method for Real-Time Financial Fraud and Phishing Threat Detection Using a Hybrid Multi-Model Ensemble with Contextual Verification, Explainable Risk Scoring and Feedback-Driven Retraining

(51) International classification	:G06Q 20/40, G06N 3/08, G06N 20/00, G06Q 40/00, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**

1)Prof. Ramakrishna Reddy Badveli

Address of Applicant :Assistant Professor, Department of Computer Science and Engineering (Artificial Intelligence and Machines Learning), AMC Engineering College, Bannerghatta Rd, Kalkare, Karnataka 560083 Karnataka India

2)Prof. Pamela Bera

3)Prof. Chadalawada Anul Kumar

4)Rajesh R

5)Shashank S

6)Mehral Maryam Shoaib

7)Madhumitha G

(72)**Name of Inventor :**

1)Prof. Ramakrishna Reddy Badveli

2)Prof. Pamela Bera

3)Prof. Chadalawada Anul Kumar

4)Rajesh R

5)Shashank S

6)Mehral Maryam Shoaib

7)Madhumitha G

(57) Abstract :

ABSTRACT OF THE INVENTION The present invention relates to a computer-implemented system and method for real-time detection of financial fraud and phishing threats. On receiving a transaction, the system first performs customer-specific contextual verification, namely a pre-authorisation match and a known-payee whitelist check, to approve legitimate payments at a low score, and determines device trust. Remaining transactions are converted into behavioural features and scored by a supervised classifier, an autoencoder trained only on legitimate data, and a Long Short-Term Memory network. A weighted ensemble score, adjusted by merchant-category risk and device trust, is mapped to a risk level and an action of approving, flagging, or blocking. Each decision is accompanied by ranked risk factors, Shapley-value attributions, and a plain-language reason. A phishing module analyses URLs and messages using lexical features, edit-distance brand-spoofing detection, and TF-IDF text features. Rate-limited alerts are issued, and analyst feedback with drift detection drives gated model retraining.

No. of Pages : 16 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131600 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : TRI-GENERATIVE SMELTER OFF-GAS VALORIZATION SYSTEM FOR GREEN HYDROGEN, OXYGEN AND SOLID SULFUR PRODUCTION

(51) International classification	:C01B 17/04, B01D 53/50, B01D 53/14, C01B 17/05, C25B 1/04	(71)**Name of Applicant :** **1)Dr T Sree Renga Raja** Address of Applicant :Professor, Dept. of Electrical and Electronics Engineering, Uninversity College of Engineering Nagercoil Tamil Nadu India **2)Ajithra J** **3)Thirukarthick T** (72)**Name of Inventor :** **1)Dr T Sree Renga Raja** **2)Ajithra J** **3)Thirukarthick T**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An integrated pyrometallurgical off-gas valorization system and process converts sulfur dioxide off-gas from flash smelting furnaces (100) into green hydrogen, industrial oxygen, and solid elemental sulfur cakes. The system comprises a decoupled slipstream gas conditioning train with a chemisorption guard bed (118), a vapor-fed SDE electrolyzer stack (200) incorporating non-leaching covalently cross-linked membranes and in-situ potential pulse stripping, a high-temperature bayonet cracking reactor (300) indirectly heated by clean gas from a dry centrifugal slag granulator (304A) over native copper-iron spinel catalyst pellets (306), an oxygen recycle train (600–612) operating at an over-stoichiometric ratio of 1.3:1 to 1.8:1 with a VPSA purge unit (612) recovering high-purity industrial argon (614) and recycling 98.5% O_2 to burner lances (102), a warm catalytic sulfur solidification circuit (400, 500) casting solid sulfur cakes (416), and an exergy cascade steam turbine power island (700–710) generating 12 MWe to 18 MWe autothermally.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131629 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Auto ML Pipelines for Scalable and Interpretable Big Data Analytics

(51) International classification	:G06N 20/00, G06F 9/50, G06N 5/04, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CH G V N PRASAD
Address of Applicant :Professor, Department of CSE, Sri Indu College of Engineering and technology, Hyderabad Hyderabad Telangana India
2)DR. K. BALA CHANDRA REDDY
3)CH RAJESH
4)MRS B HIMA BINDU
5)SMT. P. SHASIKALA
6)YARLA KAPIL DEV
7)SAI SANTHOSHI
8)A SANDEEP
(72)**Name of Inventor :**
1)CH G V N PRASAD
2)DR. K. BALA CHANDRA REDDY
3)CH RAJESH
4)MRS B HIMA BINDU
5)SMT. P. SHASIKALA
6)YARLA KAPIL DEV
7)SAI SANTHOSHI
8)A SANDEEP

(57) Abstract :
An automated machine learning pipeline system and method are disclosed for performing scalable and interpretable analysis on large-volume data sets. The system receives input data in batch or streaming form and automatically constructs optimized processing sequences that balance predictive performance with computational efficiency and explanatory transparency. Core components include a data ingestion and partitioning module that divides large data sets into manageable shards while preserving statistical properties, a pipeline synthesis engine that generates candidate sequences of preprocessing, feature construction, and modeling operations under resource constraints, a distributed execution coordinator that maps pipeline stages onto available computing nodes for parallel processing, an interpretability generator that produces global and local explanations of model behavior in both quantitative and natural-language forms, and a multi-objective evaluator that ranks candidates according to accuracy, resource consumption, and explanation quality. Intermediate results and configuration metadata are maintained in a shared repository to support reproducibility and incremental refinement. The selected pipeline is packaged for deployment and continues to generate predictions accompanied by updated explanations when new data arrive. The approach reduces end-to-end processing time for multi-terabyte data sets relative to sequential automation methods while producing explanations of higher fidelity than typical post-hoc techniques. The resulting pipelines are suitable for applications requiring both high-throughput analytics and accountable decision support.

No. of Pages : 11 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131678 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : The novel Block chain-Enabled Secure Data Sharing for Decentralized Cloud Infrastructures

(51) International classification :H04L 9/06, H04L 9/32, H04L 9/08, H04L 29/06, G06F 16/23

(31) Priority Document No :NA

(32) Priority Date :NA

(33) Name of priority country :NA

(86) International Application No :

Filing Date :01/01/1900

(87) International Publication No : NA

(61) Patent of Addition to Application Number :NA

Filing Date :NA

(62) Divisional to Application Number :NA

Filing Date :NA

(71)**Name of Applicant :**

1)Dr. D. Bhanu Mahesh

Address of Applicant :Associate Professor Department of Computer Science and Engineering - Data Science Geethanjali College of Engineering and Technology, Cheeryal (V),Keesara (M), Medchal (D), Hyderabad, Telangana - 501301 Hyderabad Telangana India

2)Dr. G Prabhakar Raju

3)Dr.Eslavath Ravi

4)Dr. M. VIJAYA KANTH

5)S. VINAYKUMAR

6)Godi Madhavi

7)V Kalyani

8)K. Radharani

(72)**Name of Inventor :**

1)Dr. D. Bhanu Mahesh

2)Dr. G Prabhakar Raju

3)Dr.Eslavath Ravi

4)Dr. M. VIJAYA KANTH

5)S. VINAYKUMAR

6)Godi Madhavi

7)V Kalyani

8)K. Radharani

(57) Abstract :

ABSTRACT [0013] A blockchain-enabled system for secure data sharing within decentralized cloud infrastructures is disclosed. The system addresses the vulnerabilities of centralized key management and opaque access histories by distributing both storage and authorization functions across independent nodes. Data owners encrypt and fragment content into threshold-protected shards that are dispersed among autonomous storage nodes, while programmable access policies and cryptographic commitments are recorded on a lightweight blockchain ledger maintained by rotating validators. Requesters submit credentialed requests that undergo distributed policy evaluation; upon consensus, short-lived capability tokens are issued, enabling selective retrieval of shards only from authorized storage nodes. Every significant event—policy publication, authorization decision, and data release—is immutably logged, producing a complete provenance trail. The architecture employs adaptive routing guided by on-chain reputation metrics and avoids single points of failure or trust. Experimental evaluation under multi-region workloads demonstrates reduced retrieval latency relative to fully on-chain storage approaches and enhanced resistance to collusion compared with conventional centralized schemes. The invention supports collaborative applications in scientific research, supply-chain coordination, and multi-institutional healthcare data exchange while preserving organizational autonomy and data confidentiality.

No. of Pages : 7 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131679 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Causal Inference Frameworks for Bias Mitigation in Observational Data Studies

(51) International classification	:G06N 5/04, G06N 5/02, G16H 50/20, G06F 16/901, G16H 50/70
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)J.Hari Krishna
Address of Applicant :Deparment of Computer Applications, School of Science, Siddhartha Academy of Higher Education, Deemed to be University Kanuru,vijayawada vijayawada Andhra Pradesh India
2)Dr.B.RATNAKANTH
3)Dr U Rakesh
4)P.G.VARNA KUMAR REDDY
5)M. Mani Kumar Reddy
6)Dr.C.Sivanandha reddy
7)Dr.P.Prathusha
8)S.Prudhvi Raj
(72)**Name of Inventor :**
1)J.Hari Krishna
2)Dr.B.RATNAKANTH
3)Dr U Rakesh
4)P.G.VARNA KUMAR REDDY
5)M. Mani Kumar Reddy
6)Dr.C.Sivanandha reddy
7)Dr.P.Prathusha
8)S.Prudhvi Raj

(57) Abstract :
ABSTRACT [0013] A causal inference framework and method are provided for mitigating bias in observational data studies. The system receives observational datasets containing treatment indicators, outcomes, and covariates, and constructs or refines a directed acyclic graph encoding causal relationships through domain knowledge and data-driven constraints. Valid adjustment sets are identified via d-separation criteria to block non-causal paths. Propensity scores or balancing weights are estimated using regularized models that incorporate interactions, after which outcome regressions are performed on the adjusted sample to obtain causal effect estimates. A sensitivity analysis module quantifies residual bias by varying the strength of hypothetical unmeasured confounders and selection mechanisms, generating bounds on the estimates. Bootstrap and simulation diagnostics assess stability and coverage. The modular architecture integrates graph construction, bias detection, multi-stage estimation, and sensitivity evaluation into a coordinated pipeline, producing adjusted estimates with substantially reduced bias and improved reliability compared with unadjusted or single-method approaches. The framework is applicable to healthcare outcome evaluation, educational policy analysis, labor-market studies, and environmental exposure research, enabling more trustworthy causal inferences from non-experimental data.

No. of Pages : 8 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131705 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : VICTIM-ROW FLOATING CHARGE COLLECTOR MEMORY ARRAY FOR READ-DISTURBANCE, FAULT-INJECTION AND AUTHENTICATION SENSING

(51) International classification	:G11C 11/4091, G11C 11/401, G11C 16/26, H01L 27/108, G11C 11/408
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BAPATLA SURENDRA BABU
Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India
2)IRUKUMALLI PRIYANKA
(72)**Name of Inventor :**
1)BAPATLA SURENDRA BABU
2)IRUKUMALLI PRIYANKA

(57) Abstract :
VICTIM-ROW FLOATING CHARGE COLLECTOR MEMORY ARRAY FOR READ-DISTURBANCE, FAULT-INJECTION AND AUTHENTICATION SENSING The invention relates to dynamic random access memory and to read disturbance (RowHammer, RowPress) that activation counters detect late. An n-type collector pocket (218) beneath the storage node junction (212) of each victim cell collects electrons released by the aggressor row; a collector line (108) in the isolation trench (216) integrates them as a dose. A reset transistor (302) gated by the victim sub-wordline (304) clears the dose whenever the victim row is restored, and a comparator (312) with an event latch (316) raises an alert naming the victim row. The collector pocket is spaced from the access-transistor sidewall (210) by 1.05 to 1.40 times its depletion width, which keeps the transistor body tied and the victim leakage intrinsic. The alert falls at 10.9 percent of the victim failure dose for every aggressor on-time. The collectors also detect particle strikes and laser attacks and provide self-test and authentication. FIG. 1

No. of Pages : 33 No. of Claims : 14

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :03/11/2026

(21) Application No.202641131713 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Secure Bluetooth RFID Scanner with Acknowledged Delivery and Power-Safe Flash Record Storage

(51) International classification	:G06K 7/00, G06K 7/10, G06K 17/00, G06K 19/07, H04L 9/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BTP Madhav
Address of Applicant :Department of ECE, Koneru Lakshmaiah Education Foundation, Vaddeswaram, AP, India Andhra Pradesh India
2)Challapally Kranthi Kumar
3)Koneru lakshmaiah Education Foundation
(72)**Name of Inventor :**
1)Challapally Kranthi Kumar
2)B T P Madhav
3)P Rakesh Kumar

(57) Abstract :
An RFID scanner system (100) comprises a microcontroller (102), an RFID reader (104), a flash memory (108) and a Bluetooth Low Energy radio (114). Each tag identifier is first stored in flash as a 32-byte record with a record number, scan time, CRC and status byte, and is then sent in encrypted form to a mobile application (118). The application returns an acknowledgement containing the record number, and the status becomes Sent, No ACK or Offline. Status values are chosen so that every update only clears bits, so no flash erase is needed; this gives power-safe storage and reduced flash wear. The link uses passkey pairing, AES-128-GCM encryption and a new session for each connection. Repeated reads are ignored and the reader recovers automatically. A password-protected HTTPS interface allows record access and Firmware Over-The-Air update with automatic rollback.

No. of Pages : 9 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131725 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AN INTELLIGENT CUSTOMER SENTIMENT ANALYSIS AND BUSINESS DECISION SYSTEM

(51) International classification	:G06N 20/00, G06Q 30/02, G06Q 30/00, G06N 3/08, G06N 3/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
Address of Applicant :Bachupally, Survey No. 288, Nizampet Road, Kukatpally, Hyderabad, Telangana Telangana India
2)Dr. Gotlur Karuna
3)Dr. Ediga Poornima
4)Dr. Sanjeeva Polepaka
5)Dr. M. Shamila
6)M.Santoshi Kumari
7)K.Saritha
8)P. Deepthi
9)P. Nethrasri
10)A Samatha
(72)**Name of Inventor :**
1)Dr. Gotlur Karuna
2)Dr. Ediga Poornima
3)Dr. Sanjeeva Polepaka
4)Dr. M. Shamila
5)M.Santoshi Kumari
6)K.Saritha
7)P. Deepthi
8)P. Nethrasri
9)A Samatha

(57) Abstract :
The invention relates to an intelligent customer sentiment analysis and business decision system comprising a data acquisition module, a data preprocessing module, a sentiment analysis module, a customer insight generation module, a business decision analysis module, and a decision output module. The data acquisition module receives customer interaction data from multiple sources, including customer reviews, textual feedback, social media content, survey responses, and customer service communications. The data preprocessing module removes noise, normalizes textual content, and extracts linguistic features. The sentiment analysis module employs a trained machine learning model to determine sentiment polarity and intensity. The customer insight generation module correlates the determined sentiment with customer attributes, product attributes, service attributes, and business performance indicators. The business decision analysis module evaluates and prioritizes recommended business actions against predefined business objectives and operational parameters. The decision output module presents the prioritized actions through a user interface. Figure 1

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131730 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SYSTEM AND METHOD FOR LEARNING-UNIT-CENTRIC EXECUTABLE DIGITAL TEXTBOOKS WITH CONTEXT-BOUNDED MACHINE ASSISTANCE

(51) International classification	:G06N 3/08, G09B 7/02, G09B 7/00, G06N 3/04, G09B 7/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHRIST UNIVERSITY
Address of Applicant :Christ University, Dharmaram College Post, Hosur Road, Bengaluru-560029, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Sunita Kumar
2)K Tarun Balaji
3)Dr. Haritha P
4)Dr. Roy Mathew
5)Dr. Praveen Babu
6)Dr. Kalyani V

(57) Abstract :
The invention relates to computer-implemented educational technology for executable digital textbooks organised around learning units and concept graphs. A content repository stores learning-unit descriptors with persistent concept identifiers, prerequisite links, modality-tagged content, executable activity specifications, and concept-scoped error taxonomies. Learner code is executed in an isolated constrained environment to generate structured execution telemetry, which is normalised into error signatures and used to assemble a bounded context payload from graph-bounded content and learner-specific evidence. An assistance governor computes a permitted disclosure level for a generative language model, thereby providing pedagogically constrained assistance while suppressing over-disclosure. Execution, assistance, assessment, and intervention evidence are persisted to update concept-indexed competence vectors, support cohort-level error aggregation, and trigger remedial interventions with post-intervention measurement. The system advances over existing digital textbooks by combining executable assessment, graph-bounded retrieval, disclosure-controlled machine assistance, and closed-loop remediation.

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131731 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR SKILL-BASED EMPLOYEE RE-ENGAGEMENT USING LINKED DEVELOPMENT PATHWAYS AND RETENTION GOVERNANCE

(51) International classification	:G06Q 10/06, G06Q 10/10, G06F 16/28, G06N 20/00, G06F 16/2457
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CHRIST UNIVERSITY
Address of Applicant :Christ University, Dharmaram College Post, Hosur Road, Bengaluru-560029, Karnataka, India Bengaluru Karnataka India
(72)**Name of Inventor :**
1)Dr. Rangapriya S
2)Dr. Priya P

(57) Abstract :
The invention concerns computer-implemented employee re-engagement using a shared backend and shared relational database linking shared candidate records through a common candidate identifier. An HRIS connector imports records for explainable candidate scoring, skill-gap identification and fairness auditing, excluding gender, age and marital status from scoring. Career-break duration informs skill-refresh needs rather than penalising discontinuous employment. A Dynamic Tripartite Engagement Matrix integrates concurrent adaptive microlearning, peer mentorship and paid micro-gigs through shared data pipelines, using activity from each pathway to update recommendations or progression in another. An Analytics & Reporting Service updates a Career Readiness Index from shared pathway records. A Bond Governance Service records development expenditure and a time-capped retention bond linked to development progress and measured readiness. A user interface layer supports returner progression and placement assessment.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131733 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Pharmaceutical Topical Composition of Ficus religiosa Extract for Promoting Angiogenesis and Wound Healing

(51) International classification	:A61P 17/02, A61K 36/605, A61K 9/107, A61K 36/185, A61K 36/60
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Sindhu A J
Address of Applicant :Department of Pharmacognosy, College of Pharmacy, Madras Medical College, Chennai-600003 Chennai Tamil Nadu India
2)B Kumudhaveni
(72)**Name of Inventor :**
1)Sindhu A J
2)B Kumudhaveni

(57) Abstract :

The present invention relates to an herbal nanoemulgel composition comprising Ficus religiosa Linn. leaf extract for promoting angiogenesis-mediated wound healing and managing diabetic foot ulcers. The composition comprises 2–10% w/w of Ficus religiosa leaf extract, black seed oil, Tween 80, Span 60, propylene glycol, Carbopol 940, triethanolamine, and distilled water. The leaf extract is incorporated into a nanoemulsion dispersed within a Carbopol 940 gel matrix to obtain a homogeneous topical formulation. The composition is formulated to promote wound contraction, re-epithelialization, granulation tissue formation, collagen deposition, and neovascularization. Experimental studies demonstrated antioxidant and angiogenic activities of the leaf extract. In vivo studies using diabetic rat wound models demonstrated improved wound contraction, increased hydroxyproline content, and enhanced tissue regeneration following topical application of the nanoemulgel. The invention provides a plant-derived topical composition for supporting wound repair and facilitating the management of diabetic foot ulcers.

No. of Pages : 29 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131734 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A SUSTAINABLE CLEANING BAR COMPOSITION DERIVED FROM LEAF ASH FOR DISHWASHING APPLICATIONS

(51) International classification	:C11D 9/38, C11D 9/50, C11D 9/60, C11D 10/04, C11D 9/26
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.G.S.GAYATHRI
Address of Applicant :Associate Professor Dept of Chemistry, S.A Engineering College, Chennai Tamil Nadu India
2)Mr.D.VEDAMANICKAM
3)S.A. ENGINEERING COLLEGE, CHENNAI (An Autonomous Institution)
(72)**Name of Inventor :**
1)Dr.G.S.GAYATHRI
2)Mr.D.VEDAMANICKAM

(57) Abstract :
The present invention discloses an eco-friendly and biodegradable dishwashing bar formulated using purified burnt leaf ash as a key cleaning component. The invention provides a sustainable solution for the effective utilization of dry leaf waste generated from residential, agricultural, and public areas, which is otherwise discarded or openly burned, causing environmental pollution. By converting leaf waste into a value-added cleaning product, the invention promotes waste management, resource recovery, and environmental sustainability. The dishwashing bar comprises purified burnt leaf ash, a natural soap base, sodium bicarbonate, citrus peel extract, aloe vera gel, natural binding agents, and essential oils. The burnt leaf ash functions as a natural cleansing and mild abrasive material that assists in the removal of grease, stains, and food residues from kitchen utensils. The citrus peel extract enhances degreasing and deodorizing performance, while aloe vera gel improves skin compatibility and user comfort. Essential oils impart antimicrobial properties and a pleasant fragrance to the final product. The method of preparation involves collecting dry leaves, subjecting them to controlled combustion, collecting and purifying the resulting ash through sieving, and blending the purified ash with the remaining ingredients to form a homogeneous mixture. The mixture is subsequently molded into desired shapes and cured to obtain a solid dishwashing bar suitable for household and commercial cleaning applications. The invention offers numerous advantages including reduced dependence on synthetic detergents, biodegradability, low production cost, effective cleaning performance, and environmentally responsible utilization of biomass waste. The formulation minimizes chemical contamination of water bodies and supports the development of green consumer products. Furthermore, the invention contributes to the circular economy by transforming an abundant waste material into a useful and marketable cleaning product.

No. of Pages : 5 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131738 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYNTHESIS, GROWTH AND CHARACTERISATION OF NLO MATERIALS: PHENYLALANINE HYDROBROMIDE (PHB)

(51) International classification	:C07C 229/36, C07C 227/40, C07C 227/42, C07C 227/18, A61K 49/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr.R.RAJA
Address of Applicant :Associate Professor, Department of PHYSICS, S.A. Engineering College, Veeraragavapuram, Chennai Chennai Tamil Nadu India
2)Dr.C.SENTHAMIL SELVI
3)S.A Engineering College
(72)**Name of Inventor :**
1)Dr.R.RAJA
2)Dr.C.SENTHAMIL SELVI

(57) Abstract :
This work describes Phenylalanine Hydrobromide (PHB), a unique non-linear optical organic crystal produced from aqueous solution by slow evaporation, as well as its thermal, optical and dielectric properties. The structure and form of the generated crystals are determined using single-crystal XRD. The material forms Orthorhombic crystal system with the P212121 space group, and the cell parameters are a = 5.562 Å, b = 21.645 Å, and c = 25.970 Å. The volume is V =3126.51 (Å). The (h k l) values are assigned using data from powder X-ray diffraction. The ^{1}H and ^{13}C NMR spectra support the FT-IR analysis in identifying the functional groups present in the sample. The PHB's optical transmittance has been measured, and the crystal exhibits a wide transparency range in the visible spectrum. The band gap was discovered to be 4.2 eV. TGA tests revealed that the PHB crystal is thermally stable up to 182°C. Microhardness and dielectric tests of the compound were also performed. The NLO properties of the aforementioned material were investigated using SHG conversion efficiency. The second harmonic generation (SHG) of LCHC is 1.1 times that of potassium dihydrogen phosphate (KDP). Keywords: Semi-organic, single crystal, non-linear optical material, second harmonic generation, microhardness, dielectric and thermo gravimetric analysis.

No. of Pages : 8 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131739 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Sustainable Interlocking Blocks Using Biomedical Incineration Ash as Partial Replacement of Cement

(51) International classification	:C04B 18/08, C04B 28/02, C04B 18/06, C04B 18/10, E04B 2/02	(71)**Name of Applicant :** **1)Dayananda Sagar College of Engineering** Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India (72)**Name of Inventor :** **1)Dr. Neethu Urs** **2)Yasmeen Taj** **3)Dr. Meghashree M** **4)Dr. R Shanthi V** **5)Dr. Tahera**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention relates to sustainable interlocking construction blocks incorporating Biomedical Incineration Ash (BMIA) as a partial replacement for cement. Conventional construction materials are associated with significant carbon emissions and environmental impacts, while the increasing generation of biomedical waste and the disposal of ash generated from its incineration is adding to the environmental challenges. The invention provides a method for utilizing BMIA in the manufacture of interlocking blocks, thereby reducing cement consumption and use of waste material from going to the landfill. The invention includes collecting and processing BMIA, characterizing its physical and chemical properties, and incorporating the ash as a partial cement replacement in a controlled proportion during the production of interlocking blocks. Blocks incorporating different replacement levels are prepared and evaluated for water absorption, compressive strength, and durability characteristics.

No. of Pages : 10 No. of Claims : 3

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131744 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Individual Metal Oxide Materials for Electrical Resistance-Based Humidity Detection

(51) International classification	:G01N 27/12, G01N 27/04, H01C 7/00, G01N 27/22, C04B 35/453	(71)**Name of Applicant :** **1)Dr SHUNMUGAPRIYA B** Address of Applicant :Assistant Professor Department of Physics (H&S) S.A. Engineering College, Veeraragavapuram, Chennai Chennai Tamil Nadu India **2)Dr KAMESHWARAN R** **3)Dr SURESH A** **4)S.A Engineering College** (72)**Name of Inventor :** **1)Dr SHUNMUGAPRIYA B** **2)Dr KAMESHWARAN R** **3)Dr SURESH A**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A resistive humidity-sensing device includes a substrate, interdigitated tin electrodes and an individual metal-oxide sensing layer positioned between the electrodes. The sensing layer comprises one oxide selected from zinc oxide (ZnO), cerium oxide (CeO_2) and lanthanum oxide (La_2O_3), with each oxide prepared and evaluated separately. The oxide powder is prepared from a corresponding metal precursor using double-distilled water, methoxyethanol and monoethanolamine, followed by stirring at 70 °C, drying at 80 °C and annealing at 1100 °C. A paste is formed using the individual oxide powder, ethylene glycol and ethanol, deposited between the electrodes and dried at room temperature. Sensor resistance is measured under controlled relative-humidity conditions to evaluate the electrical response of each individual oxide sensing layer. Dated this 31st Day of October 2026 Signature of the Applicant(s) Dr. SHUNMUGAPRIYA B Dr. KAMESHWARAN R Dr. SURESH A S.A. ENGINEERING COLLEGE

No. of Pages : 9 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131757 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BREWSTER-ANGLE POLARIZATION-NULL GLINT PHOTOMETER MEASURING LEADING-EDGE EROSION OF ROTATING WIND TURBINE BLADES

(51) International classification	:G01N 21/21, G01J 4/04, G01J 4/00, G01N 21/47, G01N 21/55
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BAPATLA SURENDRA BABU
Address of Applicant :303, DELIGHT HOMES, JAYA PRAKASH NAGAR ROAD, ITHANAGAR, TENALI Andhra Pradesh India
2)IRUKUMALLI PRIYANKA
(72)**Name of Inventor :**
1)BAPATLA SURENDRA BABU
2)IRUKUMALLI PRIYANKA

(57) Abstract :
BREWSTER-ANGLE POLARIZATION-NULL GLINT PHOTOMETER MEASURING LEADING-EDGE EROSION OF ROTATING WIND TURBINE BLADES
A passive glint photometer (100) measures leading-edge erosion of wind turbine blades (138) while the rotor (136) turns, using sunlight (144) reflected from the leading edge (140). An objective lens (208) images the blade onto a slot-shaped field stop (210); a polarizing beam splitter (214) divides the light between s-channel (216) and p-channel (218) photodetectors. A sun-sighting tube (106) fixed at 58 to 76 degrees to the optical axis holds the reflection near the Brewster angle of the topcoat, where the p-channel receives no specular light and serves as a simultaneous reference for the diffuse light of the blade. The splitter and the slot are fixed to the plane of the tube and the optical axis. A difference stage (908) removes the diffuse light, a pulse-duration stage (912) measures the glint duration, which lengthens with surface roughness, and a ratio stage (914) measures gloss, free of scintillation and blade dirt. FIG. 1

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :03/11/2026
(21) Application No.202641131759 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PDF Malware Detection And Analysis Using Python

(51) International classification	:G06F 21/56, H04L 29/06, G06F 40/143, G06F 21/00, G06F 21/55
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lords Institute of Engineering and Technology (Autonomous)
Address of Applicant :Survey No. 32, Near Police Academy, Appa Junction, Himayath Sagar, Hyderabad, Telangana 500091, India Hyderabad Telangana India
(72)**Name of Inventor :**
1)Dr. Veerabhadrababu
2)Ms. B. Naga Lakshmi
3)Mrs. Khutaija Abid
4)Mr. Md Mateen Ahmed
5)Mr. Somanath Rao Kattekola
6)Ms. Neha Hasan
7)Ms. Molugu Shilpa

(57) Abstract :
Portable Document Format (PDF) files have become one of the most widely adopted formats for digital document exchange due to their portability, platform independence, and support for rich interactive content; however, they have also emerged as a significant vector for malware distribution in recent years. Modern attackers increasingly exploit advanced PDF capabilities such as embedded JavaScript, automatic execution triggers, and complex internal object structures to evade traditional security mechanisms. Extensive research across academia and industry has explored techniques for detecting malicious PDF files using static inspection and intelligent analysis methods, demonstrating promising effectiveness with limitations hovering around them. Consequently, a large and continuously expanding body of literature now addresses various strategies for identifying and mitigating PDF-based threats. Nevertheless, the diversity of detection approaches and feature engineering methods makes it difficult to determine which techniques are most suitable for real-time deployment. To address this challenge, the present work proposes a secure PDF malware detection framework implemented using Python and the Flask web framework, supported by PyPDF for structural parsing and ReportLab for automated report generation. The system performs safe static structural analysis combined with signature-based evaluation and dynamic risk scoring to classify documents based on threat severity, The system incorporates a blockchain-inspired integrity mechanism to protect generated scan reports from tampering. Each analysis result is hashed using the SHA-256 cryptographic algorithm and appended to a sequential hash chain. The hash of the previous record is included in the computation of the next record, creating a tamper-evident ledger of scan results. Any modification to stored reports results in a hash mismatch during verification, allowing the system to detect unauthorized alterations. This study examines the complete detection workflow, key structural threat indicators, and performance considerations necessary for dependable PDF malware identification. Overall, the proposed framework provides a practical, efficient, and scalable solution capable of supporting cybersecurity practitioners in strengthening proactive defenses against evolving PDF-based attacks. The proposed system enhances reliability, transparency, and security by ensuring that malware analysis results remain tamper-proof and trustworthy. Additionally, its scalable design supports efficient real-time detection.

No. of Pages : 17 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :04/11/2026

(21) Application No.202641131767 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Machine Learning based Approaches in the Education Sector to Impart the Quality among Students & Dynamic Content Recommendation Considering Cache State and Preference Using Reinforcement Learning

(51) International classification	:H04L 29/08, G06F 15/16, H04L 12/26, H04N 21/258, H04N 21/2343
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Narsimha Reddy Engineering College
Address of Applicant :Survey No, 518, Hanuman Temple Rd, Maisammaguda, Kompally, Secunderabad, Hyderabad, Telangana 500100 Hyderabad Telangana India
(72)**Name of Inventor :**
1)Mrs. LINGAMPALLY SUNANDA
2)Mrs. THATIPALLI RUPA RANI
3)MARISHETTY KRANTHI KUMAR
4)NIMMA KEERTHANA
5)KESANAKURTHI VENKATESWARLU
6)PALDE SUJITH
7)KYATHAMWAR SHIRISHA
8)NEELAKANTARAO SIVA
9)KILARU JATHIN REDDY
10)PUTTALA SWAPNA

(57) Abstract :
In recent years, with the spread of services that handle large amounts of content, the importance of content recommendation systems has increased. At the same time, many content providers utilize CDNs (Content Delivery Networks) because delays in content delivery and low throughput lead to an increase in service churn rates. However, when using CDNs in combination, issues arise because the content that should be recommended differs from the user's perspective and the network's perspective. Therefore, if content recommendation does not simultaneously consider both of these factors, there is a risk of degraded delivery quality. While previous studies have considered both factors, most of them assumed static environments. In this paper, we propose a recommendation method that aims to improve both user satisfaction and cache hit rate by utilizing reinforcement learning in an environment where user content preferences change dynamically. The concept of Machine learning (ML) algorithms can be significantly applied in the area of academic research, which has the capability to impact teaching and learning processes, and research by offering a platform for individualized learning, responsive evaluation, and also providing information on student achievement, advancement, and learning practices. ML is transforming the way in which educational institutions work, communicate knowledge, and educate students. In the present era, young minds are being exposed to a wide range of global concepts. This would help the students to tune their mindset to the technologically advanced world of tomorrow. Due to the breakthrough in technology, machine learning has influenced the education sector by enabling facilities such as transcription, text-to-speech, and personalized learning. By applying ML, it is possible to determine whether a student is at risk of failing out or not graduating on time. These kinds of issues can be framed as binary classification problems, where the solution has two possible outcomes: the student either graduated on time or not. The proposed ML algorithms aim to find students who are at risk of not graduating in time by applying the concept of prediction from a set of training data. In this paper, the performance of supervised machine learning algorithms such as SVM, RBF, and Naïve Bayes algorithms is analysed by applying adaptive learning patterns to predict the students who are at risk by dynamically monitoring the progress of students. The proposed machine algorithms aim at identifying the learning gaps and also to create a personalized learning path that every student receives instruction tailored to their individual needs. The experimental investigation revealed that the SVM algorithm in combination with the RBF kernel attained a higher accuracy rate of 96% compared to other ML models. It can also be analysed from the experiment that the proposed algorithm also improved the learnability of the students by 10%.

No. of Pages : 9 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :04/11/2026
(21) Application No.202641131770 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-BASED INTELLIGENT WORKFORCE MANAGEMENT SYSTEM FOR TALENT PREDICTION AND DYNAMIC ROLE ALLOCATION FOR PRODUCTIVITY OPTIMIZATION AND FINANCIAL GROWTH

(51) International classification	:G06Q 10/06, G06Q 10/10, G06Q 10/00, G06Q 10/04, G06Q 40/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DR NAGELLA VENKATA RAMANA
Address of Applicant :Department of Management Studies, Madanapalle Institute of Technology and Science Deemed to be University, Madanapalle, Annamayya, Andhra Pradesh, India. Andhra Pradesh India
2)GIRIJA VASUMATHI G G
3)DR. S. PRASATH
4)DR. SAGAR CHINTAMANRAO KHURSANGE
5)MURUGAN K
6)DR. ARUNA S
7)R. PAZHSNISAMY
8)DR. RAMESH PETTELA
9)N. ANANDHA PRIYA
10)K. PRIYADHARSHINI
11)DR. R. MARISAKTHI
12)DR. B. UJWALA
(72)**Name of Inventor :**
1)DR NAGELLA VENKATA RAMANA
2)GIRIJA VASUMATHI G G
3)DR. S. PRASATH
4)DR. SAGAR CHINTAMANRAO KHURSANGE
5)MURUGAN K
6)DR. ARUNA S
7)R. PAZHSNISAMY
8)DR. RAMESH PETTELA
9)N. ANANDHA PRIYA
10)K. PRIYADHARSHINI
11)DR. R. MARISAKTHI
12)DR. B. UJWALA

(57) Abstract :
The present invention relates to an AI-based intelligent workforce management system for talent prediction, dynamic role allocation, productivity optimization, and financial growth. The system includes a workforce-data gathering element, a talent-profile generator, a workload and skill-needs forecasting engine, a constraint-sensitive allocation optimizer, a productivity and financial assessment engine, and a feedback mechanism. Records of authorized employee capabilities, task specifications, project timelines, availability details, and cost metrics are verified and transformed into structured formats. The prediction engine evaluates future workload and skill needs, whereas the allocation optimizer produces viable role assignments based on capability, eligibility, availability, workload, and task-dependency limitations. Candidate configurations are assessed based on operational and financial metrics, encompassing anticipated completion, resource usage, staffing expenses, and possible costs associated with delays. Recommendations are given to authorized users along with supporting details and relevant constraint alerts. A flexible evaluation system reassesses impacted tasks when workload or availability shifts. Confirmed operational results are contrasted with projections to facilitate regulated model assessment and revision. The invention provides an integrated computational workflow for workforce planning and allocation decision support while maintaining traceability, access control, and appropriate human review. FIG.1

No. of Pages : 22 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :04/11/2026
(21) Application No.202641131772 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Flyback Converter System with Active Disturbance Rejection Control for Reduced Output Capacitance

(51) International classification :H02M3/335
(31) Priority Document No :NA
(32) Priority Date :NA
(33) Name of priority country :NA
(86) International Application No :
Filing Date :01/01/1900
(87) International Publication No : NA
(61) Patent of Addition to Application Number :NA
Filing Date :NA
(62) Divisional to Application Number :NA
Filing Date :NA

(71)**Name of Applicant :**
1)Dayananda Sagar College of Engineering
Address of Applicant :Shavige Malleswara Hills, Kumaraswamy Layout, Bangalore Bangalore Karnataka India
(72)**Name of Inventor :**
1)Dr. Priyanshu Kumar
2)Devaraj
3)Bharat Sai
4)Kashyap Hande P
5)Kevin David
6)Dr. Anubhav Kumar Pandey
7)Dr. G Manikanta

(57) Abstract :
The present invention discloses a flyback converter system employing an Active Disturbance Rejection Control (ADRC)-based control scheme for regulating output voltage under variable operating conditions. The system incorporates an Extended State Observer (ESO) configured to estimate and compensate lumped disturbances including load variation, input voltage fluctuations, and parasitic effects in real time. The controller dynamically adjusts the duty ratio based on disturbance estimation to maintain stable output voltage. Due to this active disturbance compensation, the proposed system enables operation with reduced output capacitance while maintaining acceptable voltage regulation and transient performance. Comparative evaluation demonstrates that the proposed ADRC-controlled system achieves improved disturbance rejection, faster settling, and stable operation under reduced capacitance conditions, where conventional PI or PID-controlled systems exhibit increased voltage deviation and degraded performance.

No. of Pages : 17 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202643128745 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SOLID FORMS OF FINERENONE AND PROCESS FOR THE PREPARATION THEREOF

(51) International classification	:C07D 471/04, A61K 31/4375, C07D 471/14, A61P 9/10, A61K 31/4745
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filed on	:202541085279 :01/01/1900
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)CSD PHARMA PRIVATE LIMITED
Address of Applicant :Plot 17, 1st floor, Mayuri Nagar, Opp: Bus Body Building Unit, Miyapur, Hyderabad - . Telangana India Hyderabad Telangana India
(72)**Name of Inventor :**
1)SATYANARAYANA REDDY .MULLANGI
2)SRINIVASA REDDY KONDAPALLY
3)SRIDHAR REDDY CHILUKURI
4)SURESH BABU NANDAM

(57) Abstract :
The diastereomeric salt of 4-(4-cyano-2-methoxyphenyl)-5-ethoxy-2, 8,-dimethyl-l, 4- dihydro-1, 6-naphthyridine-3 –carboxylic acid with (1R)-Camphor-10-sulfonic acid represented by the compound of formula-IX, is used for the preparation of Finerenone compound of formula-I (or) its acid addition salts thereof. Formula-IX

No. of Pages : 29 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :28/10/2026
(21) Application No.202644128817 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Porous Cathodes with Immobilized Bromine Complexing Agents for Zinc-Bromine Batteries

(51) International classification	:H01M 10/36, H01M 4/62, H01M 4/24, H01M 4/04, H01M 4/58	(71)**Name of Applicant :** **1)Golden Gate Battery, Inc.** Address of Applicant :ROSEMAR CT, San Jose, California, USA 95127 U.S.A. (72)**Name of Inventor :** **1)Saroj Kumar Sahu** **2)Dhanya Sreekumar** **3)Febeena Shameer** **4)Aina Babu** **5)Aswathy C Asokan** **6)Nila C M Mathai** **7)Prima Francis** **8)Muhammed Salim Edachannur**
(31) Priority Document No	:19/825,132	
(32) Priority Date	:30/09/2026	
(33) Name of priority country	:U.S.A.	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
A porous positive electrode structure of an electrochemical cell comprises a bromine complexing agent, an electronically conductive carbonaceous material, and a plurality of electrolyte-accessible pores. A method of manufacturing a positive electrode of a zinc-bromine electrochemical cell comprising the steps of preparing a slurry; applying the slurry to a porous carbonaceous fibrous substrate; at least partially drying the applied slurry; and removing at least a portion of a soluble pore-forming material. A method of operating an electrochemical cell comprising the steps of electrochemically generating bromine during charging; complexing at least a portion of the bromine; retaining at least a portion of a complexed bromine; and electrochemically reducing at least a portion of the retained bromine.

No. of Pages : 67 No. of Claims : 20

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202644128821 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : POROUS CONDUCTIVE COMPOSITE ELECTRODES AND ELECTRODE ASSEMBLIES FOR ELECTROCHEMICAL CELLS

(51) International classification	:H01M 4/66, H01M 4/04, H01M 4/62, H01M 4/13, H01M 4/1391	(71)**Name of Applicant :** **1)GOLDEN GATE BATTERY Inc.** Address of Applicant :ROSEMAR CT, San Jose, California, USA 95127 U.S.A. (72)**Name of Inventor :** **1)Saroj Kumar Sahu** **2)Dhanya Sreekumar** **3)Aswathy C Asokan** **4)Nila C Mathai** **5)Prima Francis** **6)Muhammed Salim Edachannur**
(31) Priority Document No	:19/824,456	
(32) Priority Date	:29/09/2026	
(33) Name of priority country	:U.S.A.	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

An electrode assembly comprises a metallic current collector, an electrically conductive polymer-based intermediate layer, and a porous conductive composite electrode. The porous conductive composite electrode comprises a polymeric binder, one or more electrically conductive fillers, and an interconnected pore network formed by removal of a sacrificial pore-forming material from a precursor conductive composite. A method of manufacturing a corrosion-isolated porous electrode assembly comprises providing a metallic current collector, applying an electrically conductive polymer-based intermediate layer, providing a precursor conductive composite comprising at least one sacrificial pore-forming material, bonding the precursor conductive composite to the electrically conductive polymer-based intermediate layer, and removing at least a portion of the at least one sacrificial pore-forming material.

No. of Pages : 62 No. of Claims : 28

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :15/10/2025
(21) Application No.202531036263 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-MODE TRAP FOR ADULT MOSQUITO MANAGEMENT

(51) International classification	:A01M0001020000, A01M0001040000, A61K0039120000, A61P0031140000, A01M0001100000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Zoological Survey of lndia
Address of Applicant :M-Block, New Alipore, Kolkata-700053, West Bengal, lndia West Bengal India
(72)**Name of Inventor :**
1)Devi Shankar Suman

(57) Abstract :
Mosquitoes are the vectors of various deadly diseases such as malaria, filaria, dengue, chikungunya and various arbovirus diseases to humans and other animals. Vector-borne disease control depends on the management of mosquito populations. The inventions provicje a technoiogy that cietects anci coiiects mosquitoes eificientiy using multiple modes, including designs - shapes, covers, fan positions, and cues - heat, chemical lure, light and colour alone and in combination. The device is controlled with a microprocessor and sensors to control the trap components. The use of multiple cues catches significantly more mosquitoes than the single cue. The device is eco-friendly and safe to other non-target organisms as it does not release any harmful substances in the environment.

No. of Pages : 15 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :09/10/2025
(21) Application No.202531097264 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Cationic Peptides

(51) International classification	:A61P0033020000, A61P0033000000, A61P0031040000, A23K0050750000, A61P0033140000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ELMENTOZ RESEARCH PRIVATE LIMITED
Address of Applicant :DCB-107, AREA OF 1326 SQ.FT., LOCATED AT CYBER CITY, CHANDRASEKHARPUR , BHUBANESWAR, Orissa, India - 751024. BHUBANESWAR Orissa India
(72)**Name of Inventor :**
1)JAYA SHANKAR DAS
2)PADMAJA MOHANTY

(57) Abstract :
The present invention relates to novel synthetic cationic peptides and optimized for application in animal health and nutrition. The peptides are particularly useful in livestock feed formulations for the prophylaxis and control of coccidiosis caused by Eimeria species, a major parasitic disease affecting poultry and other livestock. The invention encompasses peptide sequences exhibiting potent antimicrobial and antiparasitic properties, providing an effective and sustainable alternative to conventional chemotherapeutic agents. The invention lies within the domains of microbiology, parasitology, veterinary medicine, peptide biotechnology, and animal nutrition, and offers broad potential for enhancing animal health, improving feed efficiency, and managing protozoal infections in livestock.

No. of Pages : 25 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :13/10/2025

(21) Application No.202531098712 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A WIRE ARC ADDITIVE MANUFACTURING (WAAM) PROCESS FOR ULTRASONIC VIBRATION-ASSISTED LIQUID-BASED COOLING SYSTEM AND THE SYSTEM TO ENSURE EFFICIENT THERMAL MANAGEMENT

(51) International classification	:B23K0009040000, G06Q0010060000, A01N0001020000, G01N0001280000, A61D0019020000	(71)**Name of Applicant :** **1)INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD** Address of Applicant :Indian Institute of Technology (Indian School of Mines) Dhanbad, Dhanbad- 826004, Jharkhand, India Dhanbad Jharkhand India
(31) Priority Document No	:NA	(72)**Name of Inventor :**
(32) Priority Date	:NA	**1)Ankit Raj**
(33) Name of priority country	:NA	**2)Raushan Kumar**
(86) International Application No	:	**3)Peeraj Majumder**
Filing Date	:01/01/1900	**4)Alok Kumar Das**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The proposed Wire arc additive manufacturing (WAAM) process for Ultrasonic vibration-assisted liquid-based cooling system and the system for in situ cooling offers a promising hybrid approach for robotic WAAM process. The methodology begins with WAAM deposition in open air for the initial layer sequence, to establish a strong metallurgical bond with the substrate and ensure stable arc operation. Once this base is formed, water is gradually introduced into the fabrication chamber (108) until the deposited part is partially submerged. To maintain effective cooling, the water is circulated continuously: heated water exits through an outlet (102) while fresh cold liquid simultaneously enters through an inlet (109). The robotic WAAM torch (106) and shielding gas nozzle remain positioned above the waterline throughout the process to prevent water from contacting the arc, thereby preserving arc stability and shielding effectiveness. After the water level is stabilized at the required clearance, an immersed ultrasonic transducer (104), fixed to the chamber wall or base, is activated to introduce high-frequency ultrasonic Vibrations (103) into the liquid inside the chamber (108). The combined effect of continuously rising liquid levels, circulating flow, and ultrasonic Vibration (103) is expected to accelerate heat removal, break up dendritic structures during solidification, and promote heterogeneous nucleation, thereby refining the grain structure in the final part. [fig 1]

No. of Pages : 20 No. of Claims : 16

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :20/10/2025
(21) Application No.202531101208 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A MATRIX TYPE TRANSDERMAL PATCH

(51) International classification	:A61K 9/70, A61K 9/51, A61K 31/46, A61K 31/439, A61K 47/32, A61K 47/44, A61P 1/08, B82Y 5/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)SARKAR, Ashim Kumar
Address of Applicant :Babupara (Near town club), PO/Dist.-Alipurduar, West Bengal-736121 Alipurduar West Bengal India
2)GHOSH, Shyamoli
(72)**Name of Inventor :**
1)SARKAR, Ashim Kumar
2)SHAH, Pranav
3)DEWANGAN, Hitesh Kumar

(57) Abstract :
Abstract: The present invention relates to a matrix type transdermal patch, comprising: scopolamine acts as a pharmaceutically active agent encapsulated in solid lipid nanoparticles (SLNs); granisetron acts as a pharmaceutically active agent encapsulated in solid lipid nanoparticles (SLNs); a polymer matrix formed by acrylic pressure-sensitive adhesive or Polyisobutylene adhesive; a permeation enhancer; a plasticizer; backing made of a flexible and impermeable material; and release liner.

No. of Pages : 23 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :23/10/2025

(21) Application No.202531102077 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : ANTIPARASITIC SEQUENCE-AGNOSTIC CATIONIC PEPTIDE COMPOSITION FOR BROILER FEED, AND METHOD OF PREPARATION THEREOF

(51) International classification	:A61P0031040000, A61P0033020000, A23K0050300000, A23K0020189000, A23K0050750000
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ELMENTOZ RESEARCH PRIVATE LIMITED
Address of Applicant :DCB-107, AREA OF 1326 SQ.FT., LOCATED AT CYBER CITY, CHANDRASEKHARPUR , BHUBANESWAR, Orissa, India - 751024. BHUBANESWAR Orissa India

(72)**Name of Inventor :**
1)JAYA SHANKAR DAS.
2)PADMAJA MOHANTY

(57) Abstract :
Disclosed herein is an antiparasitic peptide composition for broiler feed and method of its preparation. The method comprises dissolving sequence-agnostic cationic (synthetic) peptide and sodium alginate in deionized water at a weight ratio of 1:2 to 1:4, followed by introducing the resulting precursor solution into a 0.1–0.2 M calcium chloride gelling bath at a flow rate of 0.5–1.0 mL/min to form calcium alginate-coated peptide particles. The particles are subsequently washed, coated with chitosan using an aqueous chitosan solution, collected by centrifugation, washed, and freeze-dried to obtain chitosan-coated calcium alginate peptide microspheres. 2% w/w microspheres are dry-blended with maltodextrin to obtain the peptide composition. For feed application, this composition is incorporated into the broiler feed at 0.25–5.0% by weight of the total feed.

No. of Pages : 30 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/12/2025

(21) Application No.202531133067 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A visible light-responsive Cu-mZrO2 and TiO2 heterojunction for enhanced photocatalytic activity, and a process for the preparation thereof

(51) International classification	:H10D 62/82, B01J 35/39, C25B 11/087, B01J 37/00, A01N 25/24
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD
Address of Applicant :Dhanbad - 826004, Jharkhand, India Dhanbad Jharkhand India
(72)**Name of Inventor :**
1)Ravi Kumar
2)Dr. Abdur Raheem
3)Prof. Suman Dutta

(57) Abstract :
The present invention relates to a visible light-responsive Cu-mZrO2 and TiO2 heterojunction for enhanced photocatalytic activity, and a process for the preparation thereof. The invention belongs to the field of photocatalysis and discloses a novel process for the preparation of a highly efficient photocatalyst composite comprising Cu-doped monoclinic ZrO2 (CuZ) and TiO2 through the hydrothermal method. A key innovation of this process lies in the hydrothermal synthesis of highly crystalline CuZ, integrated within a composite structure with pre-synthesized pristine TiO2 nanoparticles, achieved without requiring any post-synthesis high-temperature calcination. This advancement simplifies the overall synthesis process and significantly reduces energy consumption, making it a more efficient and scalable approach for photocatalyst fabrication. The resulting photocatalyst exhibits enhanced hydrogen production via seawater splitting under light irradiance. In addition to hydrogen generation, composite materials can be used as a potential photocatalyst in other photocatalysis processes like pollutant degradation and CO2 reduction. The improved photocatalytic performance of the synthesized photocatalyst is attributed to the formation of a heterojunction, which facilitates efficient separation of photogenerated charge carriers and reduces the band gap energy. This invention provides a cost-effective and scalable approach for developing multifunctional photocatalysts aimed at addressing global energy and environmental challenges.

No. of Pages : 30 No. of Claims : 5

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :05/08/2026

(21) Application No.202631094753 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : APPARATUS AND METHOD FOR ENERGY HARVESTING FROM SURFACE MOTION OF A BODY OF WATER

(51) International classification	:H02N 2/18, F03B 17/06, F03B 13/18, F03B 17/02, F03B 13/14
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)RATHOR, Ashwani Singh
Address of Applicant :124, Co Operative Colony, Bokaro Steel City, Jharkhand - 827001, India. Bokaro Jharkhand India
(72)**Name of Inventor :**
1)RATHOR, Ashwani Singh

(57) Abstract :
The present disclosure pertains to an apparatus (100) for energy harvesting from surface motion of a body of water. The apparatus (100) includes an impact housing (102) comprising an inverted cup-shaped body defining an internal impact surface (110), piezoelectric transducers (108) mechanically coupled to the internal impact surface (110), a buoy (104) configured to float on the body of water and flexible couplings (106) to maintain the buoy (104) and the impact housing (102) as a connected assembly while permitting relative movement therebetween. The apparatus (100) includes an impact member (112) coupled to the buoy (104) and an electrical circuit. The movement of the buoy (104) in response to surface motion of the body of water produces repeated relative movement between the buoy (104) and the impact housing (102), causing the impact member (112) to repeatedly engage the internal impact surface and apply compressive loading to the piezoelectric transducers (108) to generate electrical energy.

No. of Pages : 40 No. of Claims : 24

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :05/08/2026

(21) Application No.202631094924 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Centrifugal force based Potato grading Machine.

(51) International classification	:B07B7/08, B07B13/16, B07B13/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr Rajendra Prasad Central Agricultural University
Address of Applicant :Pusa, Samastipur, Bihar, India Samastipur Bihar India
(72)**Name of Inventor :**
1)Jyoti
2)Vishal Kumar
3)Anupam Amitabh
4)P.K. Pranav

(57) Abstract :
A centrifugal force based potato grading machine. The grading machine comprising of a rotating disc (2) which receives power through a vertical shaft (5) attached to an electric motor (4) through belt and pulley system (6). The disc (2) is supported on a frame (10). The shaft (5) is supported at both ends by square flanged bearings. The disc rotates at a controlled speeds between 100-140 rpm generating centrifugal force across disc regions to prevent potatoes from settling uniformly near the disc center. Potatoes are fed to the disc through feeding tray and hopper (1). The disc's rotation facilitates the outward movement of the potatoes towards a clearance ring (8) of adjustable height and concentrically positioned around the disc and having four openings corresponding to four size grades. The machine is capable of separation of potatoes into four specific grades such as grade 1: 3 to 4cm; grade 2:4-5.5cm; grade 3:5.5-7cm; and grade 4:7 – 9cm. There are four outlet trays (3) each corresponding to a specific grade. Fig. 2A

No. of Pages : 18 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :09/08/2026

(21) Application No.202631096199 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Compact Low Power CCII Based LIF Neuron with Memristive Synapse Exhibiting Synaptic Plasticity

(51) International classification	:G06N 3/063, G06N 3/04, G06N 3/08, G06F 15/18, G06N 3/049
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Technology, Jamshedpur ,
Address of Applicant :National Institute of Technology, Jamshedpur , NIT Campus, P.O. RIT, Adityapur, Jamshedpur, Jharkhand 831014, India National Institute of Technology, Jamshedpur, Jamshedpur Jharkhand India 831014 2021rsec007@nitjsr.ac.in Jharkhand India
(72)**Name of Inventor :**
1)Mr. Abhishek Kumar
2)Dr. Akhilesh Kumar
3)Dr. Ravi Pushkar
4)Dr. Raushan Kumar
5)Mr. Rahul Kumar Gupta
6)Dr. Amit Prakash

(57) Abstract :
ABSTRACT OF THE INVENTION Biological neural systems perform information processing through the collective operation of neurons and synapses, where learning is achieved by activity dependent modification of synaptic strength. Emulating these mechanisms in electronics circuit requires compact, low power and scalable implementations of spiking neurons and adaptive synapses. This paper presents a compact low power second generation current conveyor (CCII) based leaky integrate and fire (LIF) neuron with a memristive synapse exhibiting synaptic plasticity. The proposed neuron operates in the current domain and employs a CCII to realize weighted summation of presynaptic inputs, leaky membrane integration and threshold-based spike generation using a low power comparator and a PMOS based reset mechanism. Both resistor and memristor are used to represent synaptic weights at the neuron inputs, while a memristive device is employed as the synapse in the learning model to provide non-volatile representation of synaptic strength. In addition, In addition, spike timing dependent plasticity (STDP) is implemented, where the variation of synaptic strength is governed by the relative timing difference between presynaptic and postsynaptic spikes. A two-neuron system composed of a presynaptic and a postsynaptic neuron interconnected through a memristor as synapse is used to demonstrate long term potentiation (LTP) and long-term depression (LTD). During LTP, the memristor conductance increases, whereas during LTD the conductance decreases which demonstrates synaptic plasticity similar to biological systems. The proposed circuit is designed and simulated in Cadence Virtuoso using 0.18µm CMOS technology to verify its operational performance. Simulation results validate correct LIF operation, stable learning behaviour and low power consumption in 23 µW which confirming the suitability of the proposed neuron-synapse architecture for neuromorphic computing systems. Keywords; Neuron, Synapse, Memristor, LTP, LTD, CCII.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/08/2026
(21) Application No.202631102835 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A NON-SLIPPERY DUAL-TEXTURE ADAPTIVE YOGA MAT

(51) International classification	:B60N 3/04, B28B 21/66, G10L 15/065, H01C 17/22, H01C 17/30
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ADAMAS UNIVERSITY
Address of Applicant :BARASAT-BARRACKPORE ROAD, BARBARIA, P.O JAGANNATHPUR, 24 PARGANAS (NORTH), KOLKATA-700126, WEST BENGAL, INDIA West Bengal India
(72)**Name of Inventor :**
1)DR. DEBABRATA SARKAR
2)DR. ABHISEK PANDA

(57) Abstract :
ABSTRACT Disclosed herein is a non-slippery dual-texture adaptive yoga mat (100) configured to provide enhanced practitioner grip, moisture management, cushioning, floor adhesion, and compact portability. The yoga mat (100) comprises a top microfiber grip layer (102) having a microfiber ergonomic grip layer (114) with herringbone micro-ridges (116) and moisture-management microchannels (118), an intermediate cushioning layer (104) comprising a thermoplastic elastomer (TPE) cushioning material (120) having a density-gradient configuration and internal moisture-routing ducts (122), and a bottom anti-slip layer (106) comprising a plurality of micro-suction cups (124) configured to generate localized partial-vacuum adhesion with a supporting surface. The yoga mat (100) further comprises postural alignment guides (108), reinforced fold regions (110) incorporating thermoplastic polyurethane (TPU)) hinge strips for an accordion tri-fold configuration, and a magnetic closure assembly (112) comprising embedded magnetic snap pairs for retaining the folded configuration. The integrated construction provides multidirectional grip, perspiration management, adaptive cushioning, and convenient storage and transportation.

No. of Pages : 31 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :25/09/2026
(21) Application No.202631115150 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A METHOD FOR GENRE-CALIBRATED MULTI-PASS STRUCTURAL DIAGNOSIS OF A COMPLEX DOCUMENT

(51) International classification	:G06F 17/30, G06F 17/27, G06F 17/00, G06F 17/21, G16H 50/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Purusarth Labs Private Limited
Address of Applicant :Flat No. 201, 2nd Floor, N3/273, IRC Village, Nayapalli, Khorda, Bhubaneswar-751012,, Odisha, India Bhubaneswar Orissa India
(72)**Name of Inventor :**
1)Badal Patnaik
2)Sambit Rath

(57) Abstract :
A METHOD FOR GENRE-CALIBRATED MULTI-PASS STRUCTURAL DIAGNOSIS OF A COMPLEX DOCUMENT The present invention provides a method for genre-calibrated, multi-pass structural diagnosis of a complex document (102). The method comprises determining a genre category of the complex document and configuring a fixed ordered plurality of diagnostic passes having one or more genre-specific parameters associated with the determined genre category. The ordered plurality of diagnostic passes is executed on the complex document (102) to generate respective diagnostic findings. The diagnostic findings generated by different diagnostic passes are aggregated, and relationships among the diagnostic findings are identified, including causal relationships. A finding that does not correspond to any of the diagnostic passes is classified as a reserved finding. A consolidated diagnostic output (144) representing a structural diagnosis of the complex document (102) is generated based on the aggregated diagnostic findings and the identified relationships. FIG.1

No. of Pages : 30 No. of Claims : 17

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/10/2026
(21) Application No.202631118278 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : THERMO-RESILIENT BOAR SEMEN DILUENT FOR PRESERVTION AND TRANSPORTATION AND METHOD THEREOF

(51) International classification	:A01N 1/02, A61D 19/02, A61D 19/04, C12N 5/076, A01N 1/00	(71)**Name of Applicant :** **1)Dr. Biswajyoti Borah** Address of Applicant :Department of Animal Biotechnology, College of Veterinary Science, Khanapara, Assam Veterinary and Fishery University Guwahati-781022, Assam, India Guwahati Assam India **2)Dr. Mithun M D** **3)Dr. Chandan Kumar Gautam** **4)Dr. Sangeeta Baro**
(31) Priority Document No	:NA	**5)Dr. Dipak Deka**
(32) Priority Date	:NA	**6)Dr. Niranjan Kalita**
(33) Name of priority country	:NA	(72)**Name of Inventor :**
(86) International Application No	:	**1)Dr. Biswajyoti Borah**
Filing Date	:01/01/1900	**2)Dr. Mithun M D**
(87) International Publication No	: NA	**3)Dr. Chandan Kumar Gautam**
(61) Patent of Addition to Application Number	:NA	**4)Dr. Sangeeta Baro**
Filing Date	:NA	**5)Dr. Dipak Deka**
(62) Divisional to Application Number	:NA	**6)Dr. Niranjan Kalita**
Filing Date	:NA	

(57) Abstract :
The present invention provides for a boar semen diluent capable of preserving sperm viability during preservation, storage and transportation and reduces strict temperature requirements during storage and artificial insemination procedure. Further, the boar semen diluent comprises buffering agent, osmotic stabilizer, membrane-protective compound, antioxidant and antimicrobial agent. Furthermore, the present invention provides for a method for diluting boar semen using the thermo-resilient diluent facilitate preservation of semen quality and a method for transporting diluted boar semen without dependence on stringent temperature-controlled conditions. Figure 1

No. of Pages : 33 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :05/10/2026
(21) Application No.202631118825 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : PLANT-BASED IMPROVED ANTI-VENOM AGAINST CATFISH STING

(51) International classification	:A61K 36/00, A61K 36/8962, A61K 36/47, A61K 36/886, A61P 39/02, A61K 9/06, A61K 47/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)DULAL KUMAR DE
Address of Applicant :MIDNAPORE COLLEGE, DEPERTMENT OF BOTANY, RAJABAZAR , MIDNAPORE,WEST BENGAL,INDIA PIN-721101 West Bengal India
2)TARAKNATH JANA
3)SARAT JANA
(72)**Name of Inventor :**
1)DULAL KUMAR DE
2)TARAKNATH JANA
3)SARAT JANA

(57) Abstract :
The present invention of three ethno-medicinal plant species, Allium cepa L. and Allium sativum L. belonging to family Amaryllidaceae and Euphorbia neriifolia L. belonging to Euphorbiaceae used for the treatment of poison by cat fish sting.The composition of the anti-venom against cat fish sting consists of following ingredients combined in the stated proportions: Allium sativum L. (bulb) - 30- 80%, Allium cepa L. (bulb) - 40-70%, Euphorbia neriifolia L. (Latex)- 20-30 % , Aloe vera (L.) Burm. f. (leaves)- 15-60% and Polyethylene glycol (as general base media) - 40%. The herbal anti-venom composition is used for treatment of cat fish sting.

No. of Pages : 10 No. of Claims : 4

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :06/10/2026

(21) Application No.202631119260 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : An adaptive tapioca and wax treatment process to reduce cracking, improve polishing and adhesion of bamboo sticks used for incense/ dhoop products.

(51) International classification	:B27K 5/00, B24B 29/00, B27K 5/02, B27K 5/04, B24B 31/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Ardhendu Kumar Koley
Address of Applicant :Ramkrishnabati, kadamtala, Dankuni, Mrigala West Bengal India
(72)**Name of Inventor :**
1)Ardhendu Kumar Koley

(57) Abstract :
The invention relates to an adaptive treatment and polishing process for bamboo sticks intended for incense/ dhoop products. The process uses mechanical polishing, wax and boiled tapioca applied while the bamboo sticks continuously move in a polishing machine. In one embodiment, approximately 40 kg of bamboo sticks are processed in a two row arrangement. Wax is applied from the beginning and continued during treatment. After about 45 minutes, hot boiled tapioca mixture is progressively applied to the moving sticks together with wax. A representative mixture comprises about 400gm tapioca with about 4 litres water adjustable according to bamboo type and thickness. Treatment is continued for about three hours with duration adjustable according to seasonal weather and ambient moisture. The process is intended to improve surface smoothness, reduce cracking and improve adhesion of incense/ dhoop masala. In reported observations cracking reduced from about 10% to about 0.5% . Treated sticks are sorted and packed promptly to help preserve colour and quality.

No. of Pages : 3 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :06/10/2026
(21) Application No.202631119569 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : Electricity-Independent Microfluidic Cartridge for Magnetic Enrichment and Lysis of Mycobacterium Tuberculosis

(51) International classification	:B01L 3/00, C12Q 1/68, C12N 1/06, C12Q 1/6806, G01N 33/569
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Rajdeep Paul
Address of Applicant :Palpara, Sanbandha, Bankura, West Bengal, 722180 West Bengal India
2)Dr. Kuldeep Singh
3)Dr. Harsha Gupta
4)Dr. Akshaya Khattri
(72)**Name of Inventor :**
1)Dr. Rajdeep Paul
2)Dr. Kuldeep Singh
3)Dr. Harsha Gupta
4)Dr. Akshaya Khattri

(57) Abstract :
Electricity-Independent Microfluidic Cartridge for Magnetic Enrichment and Lysis of Mycobacterium Tuberculosis This invention describes a single-use microfluidic cartridge for the preparation of Mycobacterium tuberculosis from saliva The apparatus comprises a closed cartridge defining a specimen-receiving reservoir, a microfluidic transfer pathway, and a reduced-volume lysis chamber. A conditioning formulation reduces bacterial viability and specimen viscosity while preserving surface structures for affinity capture. Superparamagnetic microbeads bearing mechanistically distinct ligands selectively bind the bacilli. An externally operated magnetic slider maintains a predetermined standoff and bounded traverse velocity to transport bead–bacillus complexes along the pathway. A passive capillary barrier retains bulk saliva while permitting magnetically driven complex passage. The complexes are de-watered and concentrated from a millilitre-scale specimen into the lysis chamber, wherein complementary lytic constituents sequentially disrupt different mycobacterial envelope layers. The apparatus thereby releases amplifiable nucleic acid, increases target-genome equivalents per unit volume, reduces co-transferred amplification inhibitors, and operates without an electrically powered centrifuge, pump, magnetic separator or purification instrument.

No. of Pages : 35 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :08/10/2026

(21) Application No.202631120526 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A NANO-HERBAL WOUND-HEALING GEL COMPRISING HIGHLY SOLUBLE 98% NANO-CURCUMIN, ALOE VERA, TRIDAX PROCUMBENS L., AND EUCALYPTUS OIL FOR ENHANCED TISSUE REPAIR AND REGENERATION.

(51) International classification	:A61K 36/00, A61K 36/9066, A61K 36/886, A61K 36/28, A61K 36/61, A61K 9/06, A61K 9/51, A61P 17/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)AMIT VIKRAM
Address of Applicant :FLAT NO.H-108, JAGANNATH PALACE APARTMENT, ACHARYA VIHAR, NEAR WATER TANK, BHUBANESWAR. ODISHA-751013 Orissa India
2)MANI SHANKAR PANDA
3)B. B. KAR
(72)**Name of Inventor :**
1)AMIT VIKRAM
2)MANI SHANKAR PANDA
3)B. B. KAR

(57) Abstract :
The present invention relates to a nano-enabled polyherbal topical gel for wound-management applications. The composition comprises highly soluble nano-crrcumin standardiied to approximatety SSrl, cur-c-uminoids, Aloe vera' Tridax procumbens L. extract, eucalyptus oil, and one or more. phur-aceutically acceptable gelfonning and stabilizing excipients. The nano-curcumin is incorporated to enhance dispersion and topical delivery of curcuminoids, while the botanical components provide complementary phytochemical functionalities within a hydrophilic gel matrix. The formulation is designed to provide a stable, homogeneous and skin-compatible wound interface and to support biological proi".."s associated with wound repair, including modulation of the wound microenvironment, cellular proliferation, extracellular-matrix deposition, and re-epithelialization. The invention further "n"ompuss"i methods for preparation, nano-sizing, incorporation, homogenization, standardization, characterization and quality contri oithe composition. The resulting nano-herbal gel provides a scalable platform for further in-vitro, ex-vivo, preclinical and clinical evaluation for wound-care applications.

No. of Pages : 11 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :26/10/2026
(21) Application No.202631127575 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : NUTRACEUTICAL COMPOSITION AND A METHOD FOR PREPARATION THEREOF

(51) International classification	:A61K 36/9066, A23L 33/105, A61K 9/48, A23L 33/00, A23L 33/10
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Centurion University of Technology and Management, Odisha
Address of Applicant :Village Alluri Nagar, R. Sitapur, Uppalada, Paralakhemundi, Gajapati Dist, Odisha-761211, India. Gajapati Orissa India
2)Centurion University of Technology and Management, Andhra Pradesh
(72)**Name of Inventor :**
1)Dr. Yashaswi Nayak
2)Dr. Sanjib Mohanty
3)Dr. Bikash Ranjan Sahoo
4)Debashish Mishra

(57) Abstract :
The present invention relates to a nutraceutical composition comprising 35-55 wt.% of Momordica charantia powder, 5-15 wt.% of soya lecithin, 10-30 wt.% of rice bran powder, 10-30 wt.% of flaxseed powder, 5-15 wt.% of turmeric extract standardized to 90-95% curcuminoids and 0.5×10^9 to 5×10^9 CFU of Lactobacillus rhamnosus, wherein the composition is formulated as an orally administrable capsule configured to provide neuroprotective activity, gut microbiota modulation, antioxidant activity and metabolic health support. A method for preparation of nutraceutical composition comprises the steps: selecting Momordica charantia powder, rice bran powder, flaxseed powder, turmeric extract standardized to 95% curcumin and soya lecithin, blending the selected ingredients to obtain a homogeneous powder blend, incorporating Lactobacillus rhamnosus having a viable count of 2×10^9 CFU into the homogeneous powder blend under aseptic conditions, filling the resulting blend into capsules, and optionally applying an enteric protective coating to the capsules.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :26/10/2026

(21) Application No.202631127707 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SELF-LEARNING SOLAR-PHOTOVOLTAIC PANEL MAINTENANCE SYSTEM AND METHOD WITH ONLINE MICRO-TRAINING, DIAGNOSTIC MICRO-CLEANING AND INTELLIGENT-ADAPTIVE CLEANING

(51) International classification	:H02S 40/10, H02S 50/10, H02S 50/15, G06N 20/00, G06Q 50/06
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Mr. Sudeep Samanta
Address of Applicant :11B/11, Prabhuram Sarkar Lane, Tangra, Kolkata, West Bengal, India, Pin-700015 Kolkata West Bengal India
2)Mr. Samrat Hazra
3)Ms. Arpita Sarkar
4)Mr. Rajdeep Samanta
(72)**Name of Inventor :**
1)Mr. Sudeep Samanta
2)Mr. Samrat Hazra
3)Ms. Arpita Sarkar
4)Mr. Rajdeep Samanta

(57) Abstract :
The invention provides a self-learning solar photovoltaic panel (102) maintenance system and method featuring online micro-training, diagnostic micro-cleaning, and intelligent adaptive cleaning. A sensing node (104) samples panel electrical operating parameters, temperature, and solar irradiance, while an environmental transceiver retrieves ambient data. A central processing unit (110) recursively calibrates an unsoiled baseline power characteristic using bounded gradient updates to dynamically compute an instantaneous soiling index. Future soiling trends are projected via an adaptive weighted sum incorporating ambient parameters to trigger a diagnostic micro-cleaning traversal over a localized sub-region limited to 5% to 10% of the panel area. By evaluating immediate electrical power recovery from this localized operation, the system derives a surface dust adhesion coefficient to dynamically map optimal traverse velocity, liquid flow rate, and sweeping passes. A full-surface cleaning sequence is selectively executed only when projected photovoltaic electrical energy yield gains exceed calculated total cleaning energy consumption by a predetermined multiplicative energy threshold.

No. of Pages : 34 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :27/10/2026
(21) Application No.202631127883 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : System for child growth monitoring and personalized nutrition management

(51) International classification	:G16H 20/60, G16H 40/67, G16H 10/60, G06Q 50/22, G16H 40/20	(71)**Name of Applicant :** **1)Swami Vivekananda University** Address of Applicant :Swami Vivekananda University, Telinipara, Barasat - Barrackpore Rd, Bara Kanthalia, Kolkata, WestBengal – 700121, India. West Bengal India (72)**Name of Inventor :** **1)Dr. Jagannath Ghosh** **2)Ms. Shampa Dhara** **3)Mr. Sayan Patra** **4)Mr. Rohit Manna** **5)Ms. Sarmistha Bera** **6)Mr. Samrat Ganguly** **7)Ms. Sampi Ghara** **8)Dr. Souvik Tewari**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
System for child growth monitoring and personalized nutrition management A system for child growth monitoring and personalized nutrition management is described herein. The system consisting of mobile application interface, centralized computing backend and a database management system, multi-role user interface, a child profile management module, an anthropometric calculation engine, a QR code generation module, an AI-driven nutritional processor, an immunization management module, a location-based service module, an offline data management module. The system is designed to help Anganwadi workers, ASHA workers, healthcare professionals, nutritionists, parents, and carers with systemic child health, nutrition, growth, and immunization monitoring.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202631128032 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GRAVITY-DRIVEN SELF-REGULATING NANOFLUID BASED TEMPERATURE CONTROL OF SOLAR PHOTOVOLTAIC MODULE AND THERMAL ENERGY HARVESTING SYSTEM

(51) International classification	:H02S 40/44, F24S 50/20, F24J 2/40, H02S 40/42, F24J 2/46	(71)**Name of Applicant :** **1)Birla Institute of Technology, Mesra, Ranchi** Address of Applicant :Birla Institute of Technology, Mesra, Ranchi, Jharkhand, Pin Code: 835215 Jharkhand India (72)**Name of Inventor :** **1)Dr. Sushil Kumar Dhiman** **2)Izhar Sabri**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present disclosure relates to a system and method for controlling gravity-driven nanofluid circulation to regulate the temperature of a solar photovoltaic module and harvesting thermal energy. The system comprises a segmented heat-extraction plate (104) with gradient microchannels (106), an elevated reservoir (108) containing aluminium-oxide and deionised-water nanofluid, a hydrostatic-head regulator (110), zone-selective flow-control valves (116), temperature sensors (112), a processor (114), a flow-balancing manifold (118), an inclination-compensation assembly (120), a hydraulic-pulse chamber (122) and a stratified thermal-storage unit (124). The processor senses spatial temperature variations and cooperatively controls the hydrostatic head and zone-specific hydraulic resistance to direct increased nanofluid flow to hotter regions. The system reduces localised hot spots, prevents reverse circulation, decreases pumping energy, resuspends nanoparticles, enhances photovoltaic electrical efficiency and enables simultaneous module cooling and thermal-energy recovery.

No. of Pages : 25 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :27/10/2026

(21) Application No.202631128114 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MOISTURE-RETAINING AUTONOMOUS-CURING CONCRETE COMPOSITION AND METHOD OF PREPARATION

(51) International classification	:C04B 28/04, C04B 40/00, C04B 40/02, C04B 24/38, C04B 111/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)National Institute of Technology Durgapur
Address of Applicant :Mahatma Gandhi Avenue, Durgapur, West Bengal - 713209, India. Durgapur West Bengal India
(72)**Name of Inventor :**
1)SAHA, Suman
2)CHOWDHURY, Sharwar Ahmed
3)BISWAS, Sayan

(57) Abstract :
The present disclosure provides an internal autonomous-curing concrete composition and a process for preparing the composition. The composition includes 12.0 to 20.0 wt % of a binder including Ordinary Portland Cement, 0.06 to 0.40 wt % of a polyethylene glycol-based autonomous-curing agent, 68.0 to 82.0 wt % of fine and coarse aggregates, and 5.5 to 10.0 wt % of water. The curing agent can be dissolved in water to form a liquid curing solution, which can then be incorporated with the binder and aggregates. The curing agent can retain internal moisture within the concrete composition to support cement hydration in the absence of external water curing. An exemplary composition can exhibit a 28-day compressive strength of 41.25 MPa, a split tensile strength of 2.66 MPa, and a flexural strength of 6.37 MPa. Unlike conventional curing practices that rely on continuous external water application, the disclosed composition supports internal moisture retention during cement hydration.

No. of Pages : 24 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202631128226 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Autonomous Privacy-Preserving Personal Safety System and Method with Staged Sensor Fusion and Offline Alert Relay

(51) International classification	:G08B 21/02, G08B 21/04, G08B 25/01, A61B 5/00, A61B 5/0205
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Institute of Engineering and Management, School of University of Engineering and Management
Address of Applicant :Institute of Engineering and Management, Newtown, Kolkata, School of University of Engineering and Management Kolkata Newtown West Bengal India
(72)**Name of Inventor :**
1)Ahana Sen
2)Anirban Basak
3)Anwesha Banik
4)Mainak Ghosh

(57) Abstract :
A wearable safety system (100) for women and children detects threats without user action. Low-power motion and physiological sensors (120) run continuously and compute a distress score against a personal baseline; a microphone (131) is powered up only when this score crosses a wake threshold. Audio is converted on the device into non-reconstructible features and deleted. A hybrid stage fuses the first-tier score, a learned acoustic classifier and rule-based context into a threat score. Above an alert threshold, a silent haptic cue opens a cancel window; a duress gesture appears to cancel but transmits a flagged alert. The encrypted alert is sent by cellular link or, without connectivity, by BLE/LoRa broadcast relayed store-and-forward by peer devices that cannot decrypt it, with duplicate suppression and hop limiting.

No. of Pages : 18 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202631128305 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SYSTEM AND METHOD FOR REAL-TIME OIL SPILL MONITORING AND REMEDIATION USING MODIFIED WATER HYACINTH BIOSORBENT

(51) International classification	:G01N 33/18, C02F 1/40, E02B 15/10, E02B 15/04, C02F 3/32
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Borkha Mech
Address of Applicant :Department of Petroleum Technology, Dibrugarh University Dibrugarh Assam India
2)Dr. Deepjyoti Mech
3)Arnavraj Baruah
4)Ipsita Deka
5)Abhishek Dutta

(72)**Name of Inventor :**
1)Dr. Borkha Mech
2)Dr. Deepjyoti Mech
3)Arnavraj Baruah
4)Ipsita Deka
5)Abhishek Dutta

(57) Abstract :
The present invention relates to a system and method for real-time monitoring and remediation of oil spills on water surfaces using a chemically modified water-hyacinth biosorbent integrated with image processing, hydrodynamic sensing and feedback-controlled deployment. The biosorbent is prepared by subjecting water-hyacinth petioles to washing, drying, alkaline treatment with sodium hydroxide, and hydrophobic and oleophilic modification using stearic acid dissolved in ethanol, followed by drying, grinding and sieving. An imaging unit acquires successive images of an oil-contaminated water surface, which are processed to determine the oil-slick area and boundary characteristics. A hydrodynamic sensing unit measures parameters representative of water movement, including flow velocity and wave motion. A processor integrates image-derived and sensor-derived data to determine temporal changes in oil-slick geometry and predict its subsequent movement, spreading, fragmentation and spatial extent. An optimization module determines biosorbent deployment parameters, including quantity, location and timing, based on the predicted behaviour of the oil slick. Following deployment, continuous image acquisition and hydrodynamic monitoring enable dynamic adjustment and redeployment of the biosorbent until a predetermined remediation criterion is satisfied. The disclosed invention provides an integrated approach to adaptive oil-spill management using biodegradable biomass-derived material and real-time environmental monitoring.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202631128390 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A CLOSED-LOOP THERMOCHEMICAL ENERGY HARVESTING APPARATUS WITH DYNAMIC FLUX-MODULATED CONVERSION

(51) International classification	:F24V 30/00, F25B 21/02, H01L 35/30, B01J 19/24, B01J 8/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Brainware University
Address of Applicant :398, Ramkrishnapur Rd, near Jagadighata Market, Barasat, Kolkata, West Bengal 700125 West Bengal India
(72)**Name of Inventor :**
1)Mr. Krishnendu Ghosh
2)Mr. Susmit Chakraborty
3)Mr. Tanajit Manna
4)Dr. Partha Mitra
5)Mr. Soumyajit Pramanik
6)Mr. Paragkanti Chattopadhyay
7)Ms. Sohini Sengupta

(57) Abstract :
A closed-loop thermochemical energy harvesting apparatus (200) includes an exothermic reactor (202) having a catalytic bed (206) supplied by a controllable reactant feed mechanism (208). A thermoelectric module (210) transfers heat from the reactor to an active heat rejection sub-assembly (216) while providing direct-current output to a synchronous boost converter (226). A digital supervisory controller (242) evaluates a module thermal operating condition using terminal electrical measurements and dynamic internal resistance. The controller adjusts converter switching for power extraction, regulates reactant dosing relative to a programmed hot-face temperature ceiling, and varies coolant circulation in relation to module loading and heat transfer. An auxiliary start-up circuit (234) supports initiation from low module voltage. The apparatus coordinates reaction heat generation, electrical loading and cold-side heat rejection. FIG. 1.

No. of Pages : 21 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202631128723 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A THEORETICAL APPROACH TO MODELING LOW-DISPERSION FEW-MODE FIBER FOR NEXT-GENERATION OPTICAL NETWORKS

(51) International classification	:G02B 6/028, G02B 6/036, G02B 6/00, G02B 6/032, H04B 10/2525	(71)**Name of Applicant :** **1)C. V. Raman Global University** Address of Applicant :Bidyanagar, Mahura, Janla, Bhubaneswar, Odisha 752054 Bhubaneswar Orissa India (72)**Name of Inventor :** **1)Dr. Hiranmayee Mohapatra**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

A few-mode optical fiber with a simplified composite two-q graded refractive index profile is disclosed for eliminating intermodal dispersion among the first four to six guided modes. The core has a refractive index n1 on the axis and the cladding has a refractive index n2, the core profile being described by a composite two-exponent function with constant c1, exponents q1 and q2. In an embodiment, c1 is 0.7099, q1 is 0.51, q2 is 2.4099, n1 is 1.46 and n2 is 1.45. A numerical method solves the scalar wave equation for LP0m and LPl1 modes using a Riccati substitution and Runge-Kutta integration, with an inverse substitution at field zeros, to obtain the propagation constant and normalised group delay. An optimisation routine equalises the group delays of LP01, LP11, LP02 and LP21 modes at normalised frequency V equal to 6.99, giving zero intermodal dispersion for long-distance optical networks.

No. of Pages : 15 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :28/10/2026

(21) Application No.202631128891 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Self-Adjusting Welding Torch Positioning System

(51) International classification	:B23K 37/02, B23K 9/127, B23K 9/12, B23K 37/047, B23K 37/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Swami Vivekananda University
Address of Applicant :Swami Vivekananda University, Telinipara, Barasat - Barrackpore Rd, Bara Kanthalia, Kolkata, WestBengal – 700121, India. West Bengal India
(72)**Name of Inventor :**
1)Dr. Samrat Biswas
2)Mr. Soumka Bose
3)Dr. Ranjan Kumar
4)Mr. Suman Kumar Ghosh

(57) Abstract :
Self-Adjusting Welding Torch Positioning System A self-adjusting multi-axis welding torch positioning system described herein. The system consisting of a base bed (14), a workpiece (10), a joint line (12), surface irregularities (11), guide rails (15), a travelling carriage (20), rail blocks (21), uprights (22), a cross-beam (23), a carriage drive motor (24), an encoder (25), a lateral slide (30), a lateral lead screw (31), a lateral stepper motor (32), a vertical column (40), a vertical carriage (41), a vertical lead screw (42), a vertical stepper motor (43), a tilt stage (50), an arcuate guide plate (51), a tilt carriage (52), a tilt stepper motor (53), a centre of curvature (54), a welding torch (70), a torch body (71), a gas nozzle (72), a tungsten electrode (73), an electrically insulating feeler (74), an arc (80), a spring-biased floating head (60), lower and upper plates (61, 62), guide rods (63), a floating plate (64), compression springs (65), a displacement sensor (66), stop collars (67), a sensing unit (90), first and second laser displacement sensors (91, 92), differential inductive proximity probes (93a, 93b), a sensor arm and bracket (94), a heat shield (95), a sensor head plate (96), a controller unit (100), a microcontroller (102), a sensor interface (102a), a collision-protection and re-centring routine (102b), motor drivers (103), a human-machine interface (104), an emergency stop (105), limit switches (106), an arc voltage and current sensor (107), a power-source interface (108), a data logger (109), and a gas tungsten arc welding power source (110). A self-adjusting multi-axis welding torch positioning system keeps the electrode-to-workpiece gap constant while an arc weld or clad is deposited on a flat, curved or uneven metallic component.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202631128912 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A Hybrid Renewable Energy Generation System Using Piezoelectric and Thermoelectric Effect for Low-Power IoT Devices

(51) International classification	:H02N 2/18, H02J 3/38, H02J 7/34, H01L 41/113, H02J 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Pritam Kumar Singh
Address of Applicant :, Ph.D (IIT ISM, Dhanbad), Assistant Professor, Department of Physics, Degree College Tundi, Binod Bihari Mahto Koyalanchal University, Dhanbad, Jharkhand, India – 828109. Jharkhand India
2)Sujeet kumar
3)Dr. Indrajit Kumar
4)Sr. Prof. (Dr.) Ram Kumar Singh
(72)**Name of Inventor :**
1)Dr. Pritam Kumar Singh
2)Sujeet kumar
3)Dr. Indrajit Kumar
4)Sr. Prof. (Dr.) Ram Kumar Singh

(57) Abstract :
ABSTRACT [505] The present invention introduces an advanced hybrid renewable energy generation system that integrates piezoelectric mechanical-to-electrical conversion with thermoelectric heat-to-electrical conversion and adaptive power-management mechanisms for comprehensive self-powering of low-power IoT devices, facilitating simultaneous multi-source energy harvesting, dynamic power conditioning, and robust storage management while maintaining seamless device integration and operational reliability for consistent sensor-network applications. [510] The comprehensive harvesting framework employs complementary transduction modules and energy-efficient conditioning protocols, utilising embedded rectification arrays, maximum-power-point-tracking (MPPT) circuitry, and energy-storage elements to ensure stable voltage output, enhanced energy availability, and optimal operational autonomy while maintaining continuous power-monitoring capabilities across variable ambient conditions. [515] The integrated methodology combines vibration-based and thermal-gradient-based acquisition techniques with intelligent power-combining systems, leveraging variable-amplitude mechanical signals and variable thermal-gradient indicators to optimise harvesting procedures and charging workflows for maximum energy yield and minimal conversion loss during critical low-power operations. [520] The novel hybrid architecture features engineered high-efficiency piezoelectric and thermoelectric components with specialised impedance-matching protocols, enabling complex multi-source power combination while ensuring output consistency and performance optimisation across various deployment configurations without compromising system reliability. [525] The innovative design incorporates strategic protection mechanisms for enhanced circuit safety and energy security, utilising optimised rectification and regulation systems and adaptive storage technology to ensure stable power delivery while maintaining functionality across diverse environmental conditions and load scenarios. [530] Implementation methodology emphasises scalable module integration and efficient conversion sequences, implementing complementary harvesting measures and power-management algorithms to achieve superior energy determination, enhanced autonomy, and battery-dependency reduction while ensuring technological simplicity during device operation. [535] The system demonstrates exceptional adaptability through comprehensive integration of dual-transduction protocols and intelligent power technologies, validating its effectiveness across various IoT configurations and deployment scenarios while maintaining consistent harvesting performance and operational efficiency under diverse conditions. [540] The developed framework enables sustainable and reliable self-powering of IoT devices through a streamlined, dual-source energy-harvesting system, providing significant advantages over single-source and battery-only approaches through complementary harvesting mechanisms, adaptive power-management protocols, and improved energy autonomy while maintaining superior operational reliability during critical sensing applications.

No. of Pages : 15 No. of Claims : 8

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :29/10/2026
(21) Application No.202631129041 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : SYNCHRONIZED MULTIMODAL SLEEP-SIGNAL RECORDER WITH QUALITY-GATED EVENT ARCHIVING

(51) International classification	:A61B 5/0488, A61B 5/00, A61B 5/0476, A61B 5/296, A61B 5/0478
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Brainware University
Address of Applicant :398, Ramkrishnapur Rd, near Jagadighata Market, Barasat, Kolkata, West Bengal 700125 West Bengal India
(72)**Name of Inventor :**
1)Mr. Alankar Das Dalal
2)Dr. Ranadip Kundu

(57) Abstract :
The invention relates to a sleep-signal recording apparatus (100) having electroencephalographic, electrooculographic and electromyographic sensor interfaces (110, 112, 114) coupled to a common-clock acquisition unit (120). A controller (130) evaluates signal quality and sleep-event indicators, maintains a time-indexed rolling buffer (140), and causes an archive (150) to retain pre-event and post-event signal windows when preset event and quality conditions are satisfied. A report interface (160) accepts a voluntary post-awakening account and associates the account with the retained signals through a synchronized event identifier. An access-control unit (170) protects recorded data. The quality-gated archival arrangement selectively retains synchronized physiological signal windows and associated event metadata, thereby reducing unnecessary non-volatile storage operations while maintaining temporal correspondence between recorded sleep events and voluntary post-awakening reports. The apparatus is suitable for consent-based sleep research and documentation, without purporting to reproduce visual dream imagery or autonomously diagnose a disorder. FIG. 1 is the representative figure.

No. of Pages : 16 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202631129294 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : A Riboflavin-based fluorescent dye composition for DNA staining

(51) International classification	:G01N 27/447, G01N 21/64, G01N 33/58, G01N 33/533, G01N 33/48
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Indian Institute of Technology Guwahati
Address of Applicant :Indian Institute of Technology Guwahati, North Guwahati, Kamrup, Assam, 781039 Guwahati Assam India
(72)**Name of Inventor :**
1)Arpan Kotal
2)Ritwik Banerjee
3)Rajiv K. Kar

(57) Abstract :
A Riboflavin-based fluorescent dye composition for DNA staining The present invention relates to a riboflavin-based fluorescent composition comprising riboflavin and ethidium bromide (EtBr) for enhanced DNA staining and visualization during agarose gel electrophoresis. The composition reduces EtBr usage while maintaining DNA band visibility and improving cytocompatibility. Optimized dual-dye formulations demonstrate enhanced fluorescence intensity and DNA band visualization under ultraviolet illumination following electrophoresis at 80 V for 70 minutes. Spectroscopic analyses, including UV–visible, fluorescence, and FTIR spectroscopy, indicate non-covalent interactions between riboflavin and EtBr, supported by zeta potential measurements. Cell viability assays using HEK-293 cells demonstrate approximately 99% viability with riboflavin at 15 μM, while combined formulations exhibit up to 74.83% cell viability, indicating reduced cytotoxicity compared with EtBr alone. Live/dead staining further supports the improved cytocompatibility of the formulations. The invention provides a fluorescent DNA-staining composition and associated method for improving DNA visualization while reducing EtBr-associated laboratory exposure and environmental concerns.

No. of Pages : 30 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202631129336 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : GRID CONNECTED PHOTOVOLTAIC INVERTER SYSTEM WITH REDUCED SWITCHING FREQUENCY AND REDUCED SWITCHING LOSS

(51) International classification	:H02J3/38, H02M7/48, H02M7/5387, H02M1/08, H02M1/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Siksha 'O' Anusandhan (Deemed to be University)
Address of Applicant :J-15, Khandagiri Square, Bhubaneswar-751030, Odisha, India. Bhubaneswar Orissa India
(72)**Name of Inventor :**
1)Dr. Asim Kumar Dey
2)Dr. Swagat Pati
3)Dr. Tapas Kumar Mohapatra

(57) Abstract :
A grid connected photovoltaic inverter system with reduced switching frequency and reduced switching loss, comprises a photovoltaic DC (Direct Current) source 101, a DC-DC converter 102 coupled to the photovoltaic DC source 101 for providing a regulated DC link voltage on a DC link, a DC link capacitor coupled across the DC link, a single phase full bridge voltage source inverter 103 coupled to the DC link and comprising four power switches arranged as two inverter legs, an L-type filter inductor 104 coupled between an AC output of the inverter and a utility grid connection 105, at least one current sensor coupled to measure an inverter/grid AC current, at least one voltage sensor coupled to measure at least a grid voltage and/or DC link voltage, a digital controller 108 coupled to receive sensed current and voltage signals and to generate gate pulses for the four power switches.

No. of Pages : 30 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202631129338 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : COMPOSITE BOARD AND METHOD FOR PRODUCTION THEREOF

(51) International classification	:C08L 97/02, B27N 3/04, D21H 11/20, B27N 1/00, B27N 3/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Siksha 'O' Anusandhan (Deemed to be University)
Address of Applicant :J-15, Khandagiri Square, Bhubaneswar-751030, Odisha, India. Bhubaneswar Orissa India
(72)**Name of Inventor :**
1)Dr. Ranjan Kumar Mallick
2)Dr. Ashutosh Das

(57) Abstract :
A composite board comprising rice husk particles, wastewater-derived cellulosic fibres originating from pulp and paper mill wastewater containing suspended solids, the cellulosic fibres being mechanically interlocked and entrapped within capillaries and surface asperities of the rice husk, and a urea formaldehyde resin binder, and a method for producing the composite board, comprising the steps of pre-treating rice husk, packing the pre-treated rice husk into a fixed bed column, passing pulp and paper mill wastewater through the fixed bed, causing in situ mechanical deposition and fibre-fibre interlocking of wastewater-derived cellulosic fibres within the silica-rich rice husk, recovering the enriched rice husk from the fixed bed, blending the dried enriched rice husk with urea formaldehyde resin to obtain a homogeneous mixture, forming the mixture into mats, and hot pressing the pre-dried mats in a mould under compressive pressure to consolidate the composite and obtain the composite board.

No. of Pages : 19 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :29/10/2026

(21) Application No.202631129447 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : BIODEGRADABLE AEROGEL FOR WASTEWATER TREATMENT AND GREEN ENERGY PRODUCTION

(51) International classification	:B01J 35/10, C02F 1/72, C02F 1/30, B01J 35/00, B01J 27/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)NATIONAL INSTITUTE OF TECHNOLOGY, ROURKELA
Address of Applicant :National Institute of Technology, Rourkela, Sector-1, Odisha, India Rourkela Orissa India
(72)**Name of Inventor :**
1)HRITANKHI TRIPATHY
2)MAHENDRA CHINTHALA
3)ARVIND KUMAR

(57) Abstract :
ABSTRACT BIODEGRADABLE AEROGEL FOR WASTEWATER TREATMENT AND GREEN ENERGY PRODUCTION The present disclosure relates to a biodegradable photocatalytic aerogel for wastewater treatment and photocatalytic chemical generation. The aerogel comprises a cellulosic material, a polymeric material, a pore-forming and/or crosslinking material, and a photocatalytically active metal sulfide incorporated within a three-dimensional interconnected porous matrix. In one embodiment, the aerogel comprises carboxymethyl cellulose, polyvinyl alcohol, calcium carbonate and indium sulfide (In_2S_3), and is prepared by aqueous mixing, gel formation and freeze-drying. The porous aerogel facilitates contaminant adsorption and visible-light-driven photocatalytic degradation of wastewater contaminants. The aerogel further facilitates photocatalytic generation of ammonium/ammonia and hydrogen peroxide. The monolithic, lightweight and recoverable structure facilitates separation from treated water and repeated use. The disclosure further provides processes for preparing and methods for using the biodegradable photocatalytic aerogel for wastewater remediation and photocatalytic chemical generation.

No. of Pages : 47 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202631129617 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : "A Novel Process for Manufacturing of Bioactive Moringa Leaf Extract-Enriched Functional Paneer"

(51) International classification	:A23C 19/00, A23C 19/06, A23C 19/076, A23L 33/10, A23L 33/105, A61K 36/00, A61K 36/185
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Lavlesh Kumar
Address of Applicant :M.Tech Student, Department of Dairy Chemistrty, F/O Dairy Technology West Bengal University of Animal and Fishery Sciences, Mohanpur Campus, Nadia West Bengal India 741252 Mohanpur West Bengal India
2)Dr. Chaitali Chakraborty
3)Dr. Reshma Sinha Ray
4)Dr. Pavel Rout
5)Dr. Shreya Saha
6)Ms. Trisha Roy
(72)**Name of Inventor :**
1)Lavlesh Kumar
2)Dr. Chaitali Chakraborty
3)Dr. Reshma Sinha Ray
4)Dr. Pavel Rout
5)Dr. Shreya Saha
6)Ms. Trisha Roy

(57) Abstract :
The present invention relates to a process for manufacturing bioactive Moringa leaf extract-enriched functional paneer by incorporating Moringa oleifera leaf extract into milk prior to acid coagulation. The process comprises cleaning, washing, drying and powdering Moringa leaves, preparing a food-compatible leaf extract and filtering the extract, followed by filtration and heat treatment of milk. The Moringa leaf extract is incorporated into the treated milk at a concentration of 9–11% and uniformly mixed before coagulation. The milk is subsequently coagulated using aqueous citric acid at a controlled coagulation temperature of 80–85°C and a citric acid concentration of 1–2%. The resulting curd is separated from whey, drained, transferred into a mould or hoop, pressed, cooled, cut and packaged under refrigerated conditions. The controlled incorporation of Moringa leaf extract provides paneer enriched with bioactive constituents including phenolic compounds, flavonoids, saponins and tannins, together with an enhanced mineral profile and antioxidant characteristics. The process further provides functional paneer having acceptable physicochemical, textural and sensory characteristics, thereby providing a standardized method for producing a plant-bioactive-enriched dairy food product.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202631129846 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : SMART BRIDGE HEALTH MONITORING SYSTEM

(51) International classification	:G01M 5/00, G01B 11/16, G06F 19/00, G06F 17/14, G06N 3/08	(71)**Name of Applicant :** **1)HARINANDAN TUNGA** Address of Applicant :Department of CSE, RCC Institute of Information Technology, Kolkata 700015 West Bengal India **2)DIPANKAR MAJUMDAR** **3)SHRIZA BAIDYA** **4)SEKH MOTIER RAHAMAN**
(31) Priority Document No	:NA	(72)**Name of Inventor :** **1)DIPANKAR MAJUMDAR**
(32) Priority Date	:NA	**2)HARINANDAN TUNGA**
(33) Name of priority country	:NA	**3)SHRIZA BAIDYA**
(86) International Application No Filing Date	: :01/01/1900	**4)SEKH MOTIER RAHAMAN**
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

ABSTRACT The present invention provides an autonomous, IoT-enabled structural health monitoring system for continuous surveillance of bridge infrastructure. The system measures strain using strain gauges in a Wheatstone Bridge circuit with an HX711 24-bit ADC, three-axis vibration using an ADXL345/MPU6050 accelerometer, surface temperature anomalies using an MLX90640 infrared thermal camera, and ambient conditions using a BME280 sensor. An ESP32 dual-core microcontroller executes Fast Fourier Transform analysis, Hooke's Law stress derivation, thermal anomaly detection and corrosion risk modelling, and computes a composite Health Index HI = 0.4S + 0.3V + 0.2T + 0.1E. The Health Index is mapped to a four-tier alert classification — Normal, Advisory, Warning and Critical — and transmitted via MQTT over Wi-Fi or LoRa radio to a remote monitoring dashboard.

No. of Pages : 20 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202631129875 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : AI-CONTROLLED AUTONOMOUS SEWER CLEANING ROBOT

(51) International classification	:E03F 9/00, B08B 9/049, E03F 7/10, B08B 9/051, B08B 1/04
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)ANISH PANDEY
Address of Applicant :Kalinga Institute of Industrial Technology (KIIT) Deemed to be University, School of Mechanical Engineering, Campus-8, Bhubaneswar-751024, India Orissa India
2)Mubashshera Mirza
3)Mohammed Salman Baig
4)Adeitia Kalyann Boniface
5)Aman Kumar
6)Manhar Mukul
7)Santosh Ramakrishnan
8)Duraibabu Lakshmipathy
9)Aditi Agrawal
10)Rahul Ashok Jeyalakshmi
11)Partha Pratim Bhattacharjee
12)Jitam Saha
13)Sampurna Majumder
14)Sifat Bhatia
15)Tunan Shikder Any
(72)**Name of Inventor :**
1)ANISH PANDEY
2)Mubashshera Mirza
3)Mohammed Salman Baig
4)Adeitia Kalyann Boniface
5)Aman Kumar
6)Manhar Mukul
7)Santosh Ramakrishnan
8)Duraibabu Lakshmipathy
9)Aditi Agrawal
10)Rahul Ashok Jeyalakshmi
11)Partha Pratim Bhattacharjee
12)Jitam Saha
13)Sampurna Majumder
14)Sifat Bhatia
15)Tunan Shikder Any

(57) Abstract :
An AI-Controlled Autonomous Sewer Cleaning Robot is disclosed. The robot comprises a robot housing (104), a cleaning shaft (101), a cleaning brush (102), a camera (103), a traction arrangement comprising sprocket wheel (105) and grip pad (106), a helical cleaning element (107), and an electronic box (108). The cleaning shaft carries the cleaning brush for engaging and dislodging deposits within a sewer conduit. The traction arrangement facilitates movement and engagement of the robot with the internal surface of the conduit, while the camera acquires visual information about the operating environment. The electronic box houses control components configured to coordinate movement and cleaning operations. In an AI-enabled configuration, the control components process image data to generate control outputs associated with movement or cleaning of the robot. The robot integrates mechanical sewer cleaning with environmental monitoring and intelligent control.

No. of Pages : 45 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202631129933 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Electromechanical System for Synchronized Bandha (Ikat) Weft Yarn Preparation with Variable Drum and Indexing Control

(51) International classification	:D03D 47/34, D03D 47/36, D03D 51/00, D03D 47/38, D03D 47/27
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Bhuliaa Handloom
Address of Applicant :Plot No. 1311, Khata No. 786/518, Near Gayatri Temple, Sahala, Dunguripali, Sonepur, Odisha, 767023 Sonepur Orissa India
(72)**Name of Inventor :**
1)Gopabandhu Meher
2)Rupananda Meher

(57) Abstract :
An integrated electromechanical system (100) is disclosed for the synchronized preparation and marking of weft yarn Lees for Bandha (Ikat) handloom weaving. A cylindrical preparation drum (124), formed by a longitudinal shaft (108), wheel assemblies (110) and longitudinal members (114), has an adjustable working circumference through adjustable members (116) and is collapsible through pivoted members (120) for yarn loading and doffing. A drive (128, 130) rotates the drum, and a Lee-indexing transmission (132) converts predetermined drum rotation into a predetermined, repeatable Lee pitch, with selectable 1-Karia and 2-Karia modes through an intermittent motion converter (134). A PLC (140) counts validated Lee events via a feedback sensor (138) and automatically stops the drum at each marking position, and a mechanically assisted doffing mechanism (146) assists unloading.

No. of Pages : 28 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202631129934 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : Integrated Multilingual Voice-First Kiosk System for Non-Diagnostic Administrative Intake with Resilient Offline Synchronization

(51) International classification	:G06Q 20/18, H04L 29/06, H04L 9/32, G06F 3/16, G06Q 20/20
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Affinity Minds Private Limited
Address of Applicant :184/4047 Naranpur, Play Ground, Belgachhia, Barang, Cuttack 754005, Odisha, India. Cuttack Orissa India
(72)**Name of Inventor :**
1)Soren Singh Dary
2)Chirag Sharma
3)Priya Gupta

(57) Abstract :
An integrated kiosk system (100) is disclosed comprising a tamper-resistant kiosk terminal (102) with a touchscreen (104), far-field microphone array (106), speaker (108), document capture window (110, 112) and printer (114), and an edge compute module (116) executing a local workflow runtime (118), local speech pipelines (122) and a multilingual assistance layer (146), whereby users conduct voice-first, non-diagnostic administrative intake and service routing in local languages. During network outages, session events are stored in an encrypted local queue (120) with unique identifiers, and upon reconnection a synchronization agent (128) transmits signed payloads to backend processors (130) whose idempotent pipeline (132) deduplicates events, preventing duplicate records. Administrative image processing (124) decodes ABHA/QR cards and documents for workflow support only, and security logic (126, 140) binds consent, device and facility identity and audit metadata to every transaction, with routing (138) to registration, queue, appointment, referral or teleconsultation outcomes.

No. of Pages : 26 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/10/2026

(21) Application No.202631129935 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : VAJRA — Modular Electric Agricultural Weeder with Shaft-Gear Drive and Adaptive Control System

(51) International classification	:A01B 39/18, A01M 21/02, A01B 39/20, A01D 34/68, A01D 34/00	(71)**Name of Applicant :** **1)Evart Engineering and Automobiles LLP** Address of Applicant :Plot No. 465/1480, Lane No-1, Gopinath Nagar,Near Samal Sahi, Uttara, Bhubaneswar, Odisha – 751002, India. Bhubaneswar Orissa India (72)**Name of Inventor :** **1)Dr. Smitanjali Rout** **2)Chinmaya Ranjan Moharana** **3)Satish Ranjan Dhar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :

The present invention discloses a modular, battery-operated electric agricultural weeder (100) comprising a main frame (102), an adjustable handle assembly (104), a rechargeable battery pack (110), an electronic motor controller (112) and a BLDC electric motor (114) coupled to a rotary weeding wheel (120) carrying soil-engaging weeding blades (122) through an enclosed shaft-gear drive having a drive shaft (116) and a gear assembly (118), providing positive, slip-free torque transmission free of belts and chains. An adaptive control system (124), comprising a load sensor (126) and a speed sensor (128), automatically regulates motor torque and speed according to soil and weed-load conditions, preventing stalling, protecting against overload and conserving battery energy. The principal modules are detachably secured to the frame through modular mounting interfaces (136), permitting field servicing and reconfiguration. The weeder eliminates fuel consumption and exhaust emissions, reduces labor and drudgery, and is suited to small and marginal farmers.

No. of Pages : 23 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :30/10/2026
(21) Application No.202631129978 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : A phytobiotic fish feed supplement for enhancing reproductive performance in Anabas testudineus

(51) International classification	:A23K 50/80, A23K 10/30, A23K 1/16, A23K 1/18, A23K 20/174
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Dr. Rajendra Prasad Central Agricultural University
Address of Applicant :Pusa Samastipur Bihar India 848125 Samastipur Bihar India
(72)**Name of Inventor :**
1)Dr. Shivendra Kumar
2)Dr. Maneesh Kumar Dubey
3)Dr. P. P. Srivastava
4)Dr. R. K. Brahmchari
5)Dr. Tanushri Ghorai

(57) Abstract :
ABSTRACT A phytobiotic fish feed supplement for enhancing reproductive performance in Anabas testudineus This invention relates to the development of novel phyto-stimulants formulation as aquatic animal feed supplements comprising varying ratios of Ashwagandha root powder, Shatavari root powder, and Ashoka bark powder. The developed formulation described herein has application in aquaculture for improving the gonadal indices, antioxidant activity, reproductive related hormone (VTG, E2, and testosterone), and gene expression (Cyp19a and IL-6), gonadal histology biomarkers, and immunomodulatory effect under captivity condition.

No. of Pages : 25 No. of Claims : 6

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :01/11/2026
(21) Application No.202631130336 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : INTELLIGENT AI GOVERNANCE AND COMPLIANCE FRAMEWORK FOR ENTERPRISE HEALTHCARE CRM PLATFORMS

(51) International classification	:G06Q 10/06, G06F 21/62, G06N 20/00, G06N 5/04, G06N 5/02
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Susil Kumar Sahu
Address of Applicant :Solution Engineer Executive Advisor (Enterprise Salesforce Architect) 1204 Dutcess ct, chattanooga Tennessee 37421. U.S.A.
(72)**Name of Inventor :**
1)Susil Kumar Sahu

(57) Abstract :
INTELLIGENT AI GOVERNANCE AND COMPLIANCE FRAMEWORK FOR ENTERPRISE HEALTHCARE CRM PLATFORMS ABSTRACT The invention discloses an Intelligent AI Governance and Compliance Framework for Enterprise Healthcare CRM Platforms. The framework integrates healthcare data classification, contextual policy evaluation, authorization enforcement, artificial intelligence monitoring, compliance analytics, and tamper-evident audit storage. Incoming CRM events are assessed using user identity, information sensitivity, processing purpose, consent conditions, and configurable risk indicators. Based on the assessment, operations are permitted, restricted, denied, or escalated for human review. Privacy controls include masking, data minimization, controlled disclosure, and retention enforcement. The monitoring module examines AI interactions and generated outputs for policy violations and unauthorized information exposure. An analytics engine identifies recurring exceptions and recommends corrective measures. The framework supports interoperability with existing healthcare applications while improving governance traceability, accountability, privacy protection, and regulatory oversight.

No. of Pages : 15 No. of Claims : 7

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :02/11/2026

(21) Application No.202631130625 A

(43) Publication Date : 06/11/2026

(54) Title of the invention : MULTI-TIERED DISEASE RISK FUSION SYSTEM FOR FIELD-SPECIFIC CROP DISEASE RISK ESTIMATION

(51) International classification	:G06Q 50/02, G06Q 10/06, G01W 1/10, G01W 1/00, G06T 7/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)BRAINWARE UNIVERSITY
Address of Applicant :398, Ramkrishnapur Rd, near Jagadighata Market, Barasat, Kolkata, West Bengal 700125 KOLKATA West Bengal India
(72)**Name of Inventor :**
1)Dr. Suvojit Bose
2)Dr. Soham Hazra
3)Dr. Soham Bachaspati
4)Ms. Sanchari Ghosh
5)Mr. Akshaya Kumar Paine
6)Mr. Sujay Patra

(57) Abstract :
A multi-tiered computer-implemented system (100) estimates crop disease risk for a defined agricultural field. An environmental risk engine (130) derives an environmental risk estimate from temperature, relative humidity and crop-disease rules. A satellite-index input module (140) derives a spectral stress estimate from normalised Structure Insensitive Pigment Index (SIPI) and/or Plant Senescence Reflectance Index (PSRI) measurements over usable pixels within the field boundary. An input-alignment and quality-control module (150) associates the estimates with the field and analysis period, assigns source-quality values and selects full fusion, weather-only, satellite-only or insufficient-data operation. A risk fusion module (160) combines qualifying estimates by quality-dependent weighting and a concordance contribution to produce a classified risk score. An output module (170) retains the assessment components, confidence indicator and selected mode. FIG. 1 represents the system architecture.

No. of Pages : 27 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202631130756 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : 3D-printed CeO2-doped 1393 bioactive glass scaffold for bone regeneration

(51) International classification	:A61L 27/56, A61L 27/10, A61L 27/54, A61F 2/28, A61L 27/12	(71)**Name of Applicant :** **1)Swami Vivekananda University** Address of Applicant :Swami Vivekananda University, Telinipara, Barasat - Barrackpore Rd, Bara Kanthalia, Kolkata, West Bengal – 700121, India. West Bengal India (72)**Name of Inventor :** **1)Dr. MD Ershad** **2)Dr. Ranjan Kumar**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
(33) Name of priority country	:NA	
(86) International Application No Filing Date	: :01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number Filing Date	:NA :NA	
(62) Divisional to Application Number Filing Date	:NA :NA	

(57) Abstract :
3D-printed CeO_2-doped 1393 bioactive glass scaffold for bone regeneration A 3D-printed CeO2-doped 1393 bioactive glass scaffold for bone tissue regeneration is described herein. The scaffold consisting of a three-dimensional body fabricated from a bioactive glass composition consisting of silicon dioxide SiO2, calcium oxide CaO, sodium oxide Na2O, potassium oxide K2O, phosphorus pentoxide P2O5, magnesium oxide MgO, and cerium oxide CeO2 integrated within a 1393B3 matrix. The scaffold body consisting of a plurality of extruded, sintered struts arranged layer-by-layer defining an interconnected porous network having 50% to 70% porosity configured to facilitate vascularization and cellular ingrowth. The integrated cerium species exhibit a dual Ce3+}/Ce4+ redox-active state configured to scavenge reactive oxygen species (ROS), mitigate local oxidative stress, and provide antibacterial activity at a bone defect site. The scaffold exhibits enhanced compressive mechanical strength, sustained bioactivity, and accelerated osteogenesis suitable for orthopedic and load-bearing bone defect repair.

No. of Pages : 22 No. of Claims : 9

(12) PATENT APPLICATION PUBLICATION
(19) INDIA
(22) Date of filing of Application :02/11/2026
(21) Application No.202631130993 A
(43) Publication Date : 06/11/2026

(54) Title of the invention : System and method for resolving disagreement between human and machine instructions

(51) International classification	:G06F 21/57, G06F 11/36, G06F 9/44, G06F 11/07, G06F 9/455
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

(71)**Name of Applicant :**
1)Swami Vivekananda University
Address of Applicant :Swami Vivekananda University, Telinipara, Barasat - Barrackpore Rd, Bara Kanthalia, Kolkata, West Bengal – 700121, India. West Bengal India
(72)**Name of Inventor :**
1)Dr. Liton Mallick
2)Dr. Ranjan Kumar
3)Mr. Dipak Sarkar

(57) Abstract :
System and method for resolving disagreement between human and machine instructions A system for resolving disagreement between human and machine instructions is described herein. The system consisting of one or more processors and a memory, an instruction interface (10), a compatibility rule store (20), a versioned target-state store (30), an execution gateway (40), a disagreement detector (50), a transactional hold controller (60), an authenticated reviewer interface (70), a resolution controller (80), an authorisation store (90), a review notification component (100) and an event logger (110). The system is intended to prevent execution of unresolved or obsolete instructions.

No. of Pages : 27 No. of Claims : 10